Informal Notes on Mathematics

reorganized English version

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Read it, absorb it, and forget it.

— P. R. Halmos

Preface

This book is a reorganized edition of my informal mathematical notes, written across a long stretch of time from middle school to the present. In that sense, *INM* is not only a mathematical document, but also a personal record. It preserves the way a subject first appeared to me, the way a computation was once hard, the way a theorem gradually stopped being a name and became a tool, and the way one topic kept opening a door into another.

The original notes were not planned as a textbook. They were accumulated from exercises, self-study, lectures, fragments of reading, exam preparation, conversations, and attempts to explain ideas to myself. Some entries begin with very elementary problems: set identities, inequalities, Euclidean geometry, congruences, derivatives, determinants, or multivariable integrals. Other entries move toward more advanced objects: quotient spaces, exact sequences, cyclotomic fields, exterior algebra, Hilbert spaces, differential forms, de Rham cohomology, Fourier expansions, operator methods, topos-flavored foundations, mathematical physics, and the mathematics around machine learning. The purpose of this edition is to let these layers coexist rather than to hide one behind the other.

A recurring intention in the reorganization is to connect elementary techniques with the higher mathematics they quietly anticipate. A Venn diagram is also Boolean algebra and measurable-event logic. A gcd computation is also ideal arithmetic in a principal ideal domain. A determinant is also the action on the top exterior power. A line integral is also the pullback of a differential form. Fourier coefficients are also Hilbert-space coordinates. These connections are not meant to make the elementary material look more sophisticated than it is; they are meant to show why the elementary material is worth keeping. The first calculation is often the most honest place where a later abstraction can be seen.

The present volume arrangement follows the main themes of the original notes. The book is divided into Foundations, Arithmetic and Algebra, Combinatorics, Linear Algebra, Calculus and Inequalities, Multivariable Analysis, Fourier and ODEs, Geometry, Topology, and Research Vignettes. The last part is deliberately kept for material that is exploratory, cross-disciplinary, or difficult to classify cleanly: small mathematical-physics notes, AI-related mathematical explanations, and research-flavored digressions. I prefer to keep such pieces visible, because they record moments when mathematics was not a syllabus but a landscape.

There is also a memorial reason for making this collection. A notebook from one's earlier years is full of unevenness: some pages are naive, some are overambitious, some are clumsy, and some still contain the excitement of first contact. To reorganize these notes is not to erase that unevenness, but to give it form. The book marks a path from contest-style manipulation and classroom calculus toward algebra, analysis, topology, geometry, and the modern languages that now feel unavoidable. It is a record of mathematical growth, but also of the persistence of certain questions: what is the right structure, what is invariant, what is being measured, and why should one care?

This edition is therefore best read as a working companion rather than a polished authority. Definitions and theorems are included where they clarify the original notes, but the spirit remains informal. I have tried to preserve computations, examples, and small motivations, while adding enough surrounding structure to make the notes reusable. Some chapters are closer to textbook exposition; others are closer to annotated memory.

That mixture is intentional. It is the shape of the archive.

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I

Sets and Foundations

Part I: Contents

Sets, Classes, and the First Warnings About Size

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§1.1 Why classes enter before functions

The point of this note is to set up the basic language of sets, classes, indexed families, and functions. The original PDF begins very much in the style of a reading note: it is following the opening sections of a book, but it also pauses at the places where the distinction between a set and a proper class starts to feel less innocent.

Definition 1.1.1. A **set** is a class which can be an element of another class. A class which is not a set is called a **proper class**.

This is not meant to be the whole foundation of set theory, but it explains the working convention of the note: some collections are small enough to be members of other collections, while others are too large and must remain proper classes.

Caution

The original note says that the distinction between sets and proper classes is "not too clear". Informally this is understandable, but formally this distinction is exactly what prevents the usual paradoxes. In a class theory such as NBG, sets are precisely the classes that may occur as elements of classes; proper classes may not be elements.

§1.2 Russell's class as the first warning

The first serious example is Russell's construction. Consider

$$M = \{X \mid X \text{ is a set and } X \notin X\}.$$

At first sight this is a natural class: many sets are not members of themselves. For instance, the set of all books is not itself a book.

Proposition 1.2.1

The class M is not a set.

Proof. Assume, for contradiction, that M is a set. Then it makes sense to ask whether $M \in M$. By the defining condition of M,

$$M \in M \iff M \notin M,$$

which is impossible. Hence M must be a proper class. ∎

This is the first place where the note is not merely listing definitions. The example shows why unrestricted comprehension cannot simply produce a set from every condition.

§1.3 Power sets and indexed families

Definition 1.3.1. The **power set axiom** asserts that for every set A, the class

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

of all subsets of A is itself a set. It is also common to write 2^A for this power set.

The next construction is an indexed family. Given a family of classes or sets

$$\{A_i \mid i \in I\},$$

its union and intersection are written

$$\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\},$$

and

$$\bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}.$$

Remark 1.3.2 — The original note explicitly points out that the indexing collection I need not be finite or countable. This is a useful mental shift: the notation $\{A_i\}_{i \in I}$ is not just a long list. It may be a genuinely infinite family.

Caution

If I is allowed to be a proper class, then extra care is needed. In ordinary ZFC, one usually speaks of set-indexed families. In a class-theoretic setting, class-indexed families can be discussed, but one must check separately when their unions or intersections are sets.

If A and B are classes, the relative complement of A in B is

$$B \setminus A = \{x \mid x \in B \text{ and } x \notin A\}.$$

If all classes under discussion are subsets of a fixed universe U, then $U \setminus A$ is often denoted A^c.

§1.4 Set identities

The original note lists the standard identities and leaves the proof as easy. A cleaned-up version is:

Proposition 1.4.1

Let A, B, and B_i be subsets of a fixed universe U. Then

$$A \cap \left(\bigcup_{i \in I} B_i \right) = \bigcup_{i \in I} (A \cap B_i),$$

$$A \cup \left(\bigcap_{i \in I} B_i \right) = \bigcap_{i \in I} (A \cup B_i),$$

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \qquad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Moreover, $A \cup B = B \cup A$, $A \subseteq B$ means every element of A lies in B, and $A \cap B = A$ when $A \subseteq B$.

Proof. All of these follow by chasing membership. For instance, an element x lies in $A \cap (\bigcup_i B_i)$ exactly when $x \in A$ and $x \in B_i$ for at least one i, which is exactly the condition that $x \in \bigcup_i (A \cap B_i)$. The De Morgan laws are proved similarly after replacing "there exists" by "for all" under negation. □

§1.5 Images, inverse images, and functions

Let $f : A \to B$ be a function. If $S \subseteq A$, the image of S under f is

$$f(S) = \{ b \in B \mid b = f(a) \text{ for some } a \in S \}.$$

If $T \subseteq B$, the inverse image of T is

$$f^{-1}(T) = \{ a \in A \mid f(a) \in T \}.$$

Example 1.5.1

For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, we have

$$f(\mathbb{R}) = [0, \infty), \qquad f^{-1}(\{9\}) = \{-3, 3\}.$$

A map $f : A \to B$ is **injective** if $f(a) = f(a')$ implies $a = a'$. It is **surjective** if $f(A) = B$. It is **bijective** if it is both injective and surjective.

Proposition 1.5.2

For composable maps $A \xrightarrow{f} B \xrightarrow{g} C$:

(i) if f and g are injective, then $g \circ f$ is injective;

(ii) if f and g are surjective, then $g \circ f$ is surjective;

(iii) if $g \circ f$ is injective, then f is injective;

(iv) if $g \circ f$ is surjective, then g is surjective.

§1.6 One-sided inverses

The last theorem in the original note is about recognizing injective and surjective maps through one-sided inverses.

Theorem 1.6.1

Let $f : A \to B$ be a function and assume $A \neq \varnothing$.

(i) f is injective if and only if there is a map $g : B \to A$ such that $g \circ f = 1_A$.

(ii) f is surjective if and only if there is a map $h : B \to A$ such that $f \circ h = 1_B$, provided the required choices of elements from the fibres of f can be made.

Proof. For (i), suppose f is injective. For $b \in f(A)$, there is a unique $a \in A$ with $f(a) = b$. Choose once and for all an element $a_0 \in A$. Define

$$g(b) = \begin{cases} a, & b \in f(A) \text{ and } f(a) = b, \\ a_0, & b \notin f(A). \end{cases}$$

Then $g(f(a)) = a$ for all $a \in A$. Conversely, if $g \circ f = 1_A$, then $f(a) = f(a')$ implies

$$a = g(f(a)) = g(f(a')) = a',$$

so f is injective.

For (ii), if h exists and $f \circ h = 1_B$, then every $b \in B$ equals $f(h(b))$, so f is surjective. Conversely, if f is surjective, each fibre $f^{-1}(b)$ is nonempty. Choosing $h(b) \in f^{-1}(b)$ gives a right inverse. □

Caution

The last choice of one element from each fibre is exactly where the Axiom of Choice may enter. The original note itself flags this point in a footnote, and it will reappear in the later note on choice and cardinal numbers.

§1.7 Set-level bookkeeping behind functions

The calculations in this note are elementary only after the ambient set has been fixed. Russell's class shows why the notation

$$\{x \mid P(x)\}$$

cannot be used without asking whether the collection is actually a set. Once a set A is available, the usual constructions are safe: subsets live in $\mathcal{P}(A)$, indexed unions and intersections are formed from a family $(A_i)_{i \in I}$, and membership proofs reduce identities such as De Morgan's laws to quantifier manipulations.

For functions the same bookkeeping reappears in a more operational form. The inverse image $f^{-1}(T)$ is always defined for $T \subseteq B$, while the image $f(S)$ depends on the part of the domain actually hit. Injectivity is detected by a left inverse $g \circ f = 1_A$; surjectivity is detected by a right inverse $f \circ h = 1_B$. The construction of g for an injection uses only a single default element of A, whereas the construction of h for a surjection asks for one chosen element from each fibre $f^{-1}(b)$. That fibre-by-fibre choice is the exact point to remember when the later notes discuss the Axiom of Choice and cardinal comparison.

Relations, Products, and the First Appearance of Universal Properties

N-20210206

§2.1 Relations and equivalence classes

This note continues the foundational setup, moving from sets and functions to relations, products, and the integers. The original title says "Intergers"; I will silently correct this to *integers*.

Definition 2.1.1. For sets or classes A and B, their Cartesian product is

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

A **relation from A to B** is a subclass of $A \times B$. A **relation on A** is a subclass of $A \times A$.

Caution

The original note has the phrase "a subset of R is a relation on $A \times B$", which is clearly a slip. The intended statement is that a relation is a subset of $A \times B$.

Definition 2.1.2. A relation $R \subseteq A \times A$ is an **equivalence relation** if it is:

(i) reflexive: $(a, a) \in R$ for all $a \in A$;

(ii) symmetric: $(a, b) \in R$ implies $(b, a) \in R$;

(iii) transitive: $(a, b), (b, c) \in R$ implies $(a, c) \in R$.

When R is an equivalence relation, we write $a \sim b$ for $(a, b) \in R$, and define the equivalence class of a by

$$[a] = \{x \in A \mid x \sim a\}.$$

The quotient class is

$$A/R = \{[a] \mid a \in A\}.$$

Proposition 2.1.3

If R is an equivalence relation on A, then:

(i) $[a] \neq \varnothing$ for every $a \in A$;

(ii) $A = \bigcup_{a \in A} [a] = \bigcup_{C \in A/R} C$;

(iii) $[a] = [b]$ if and only if $a \sim b$.

Lemma 2.1.4

For any $a, b \in A$, either $[a] \cap [b] = \varnothing$, or $[a] = [b]$.

Proof. If $x \in [a] \cap [b]$, then $x \sim a$ and $x \sim b$. By symmetry and transitivity, $a \sim b$, and hence any element equivalent to a is equivalent to b, and conversely. Thus $[a] = [b]$. ☐

§2.2 Partitions

Definition 2.2.1. Let A be a nonempty class. A family $\{A_i \mid i \in I\}$ of subclasses of A is a **partition** of A if

(i) $A_i \neq \varnothing$ for every $i \in I$;

(ii) $\bigcup_{i \in I} A_i = A$;

(iii) $A_i \cap A_j = \varnothing$ whenever $i \neq j$.

Remark 2.2.2 — This is one of the most useful ways to remember equivalence relations: an equivalence relation cuts A into disjoint blocks, and those blocks form a partition. Conversely, a partition defines an equivalence relation by declaring two elements equivalent exactly when they lie in the same block.

§2.3 Indexed products and projections

The note then generalizes finite Cartesian products. For a set-indexed family $\{A_i \mid i \in I\}$, define

$$\prod_{i \in I} A_i$$

to be the set of functions

$$f : I \to \bigcup_{i \in I} A_i$$

such that $f(i) \in A_i$ for every $i \in I$.

When $I = \{1, 2, \ldots, n\}$, this agrees with the usual product

$$A_1 \times A_2 \times \cdots \times A_n,$$

because such a function is the same information as the ordered n-tuple

$$(f(1), f(2), \ldots, f(n)).$$

Remark 2.3.1 — The original note pauses over the case $I = \{1, 2\}$. A point of $A_1 \times A_2$ may be written as a pair (a_1, a_2), but it may also be regarded as a function f with $f(1) = a_1$ and $f(2) = a_2$. This is a small but important shift: products are not just lists; they are choice-like functions over an index set.

For each $k \in I$, the **canonical projection**

$$\pi_k : \prod_{i \in I} A_i \to A_k$$

is defined by

$$\pi_k(f) = f(k).$$

Example 2.3.2
If $I = \mathbb{R}$ and $A_i = \mathbb{C}$ for all i, then an element of $\prod_{i \in I} A_i$ is a function $f : \mathbb{R} \to \mathbb{C}$. For example, $f(x) = ix$. The projection at $k \in \mathbb{R}$ returns

$$\pi_k(f) = f(k) = ik.$$

§2.4 The first universal property

Here the original note becomes excited: "Appears! Universal property!" This is worth preserving. Products are not only sets of tuples; they are characterized by how maps into them behave.

Theorem 2.4.1 (Universal property of products)
Let $\{A_i \mid i \in I\}$ be a family of sets. There is a set D, equipped with maps

$$\pi_i : D \to A_i \qquad (i \in I),$$

such that for every set C and every family of maps $f_i : C \to A_i$, there is a unique map

$$f : C \to D$$

with

$$\pi_i \circ f = f_i$$

for every $i \in I$. Moreover, D is unique up to a unique bijection compatible with the projections.

Proof. Take $D = \prod_{i \in I} A_i$. Given maps $f_i : C \to A_i$, define

$$f(c)(i) = f_i(c).$$

Then $f(c) \in \prod_i A_i$, and $\pi_i(f(c)) = f_i(c)$. Uniqueness follows because a point of the product is determined by all of its coordinates. □

Remark 2.4.2 — The original note says that the proof is a little long and should maybe be postponed until systematic study of category theory. The cleaned-up proof above is short because we already know the concrete product. The deeper point is that the same pattern will later define products in arbitrary categories, where elements and coordinates may no longer be available.

§2.5 Integers and induction

The note then turns to the natural numbers and the integers through Peano's axioms. In modern notation the Peano axioms say, roughly:

(i) 0 is a natural number;

(ii) every natural number has a successor;

(iii) 0 is not the successor of any natural number;

(iv) distinct natural numbers have distinct successors;

(v) any property true of 0 and inherited by successors is true of all natural numbers.

The fifth axiom is the principle of mathematical induction.

Theorem 2.5.1 (Induction and strong induction)
Let $S \subseteq \mathbb{N}$ with $0 \in S$. If either

(i) $n \in S$ implies $n+1 \in S$ for all $n \in \mathbb{N}$, or

(ii) $\{0, 1, \ldots, n-1\} \subseteq S$ implies $n \in S$ for all $n \in \mathbb{N}$,

then $S = \mathbb{N}$.

Caution

The original note proves this by taking the least element of $\mathbb{N} \setminus S$. That proof is clean if the well-ordering principle for $\mathbb{N}$ is already available. If one is deriving the theory strictly from Peano's axioms, one should first justify the well-ordering principle or prove the statement directly by induction.

§2.6 Division, gcd, and congruence

Theorem 2.6.1 (Division algorithm)
If $a, b \in \mathbb{Z}$ and $a \neq 0$, then there exist unique integers q, r such that

$$b = aq + r, \qquad 0 \leq r < |a|.$$

Definition 2.6.2. For $a \neq 0$, we write $a \mid b$ if there exists $k \in \mathbb{Z}$ such that $ak = b$. Otherwise we write $a \nmid b$.

Definition 2.6.3. The positive integer c is the greatest common divisor of $a_1, a_2, \ldots, a_n$ if

(i) $c \mid a_i$ for every $1 \leq i \leq n$;

(ii) whenever $d \in \mathbb{Z}$ and $d \mid a_i$ for every i, then $d \mid c$.

It is denoted $(a_1, a_2, \ldots, a_n)$.

Definition 2.6.4. Let $m > 0$. If $a, b \in \mathbb{Z}$ and $m \mid (a - b)$, then a is congruent to b modulo m, written

$$a \equiv b \pmod{m}.$$

Theorem 2.6.5

Let $m > 0$ and $a, b, c, d \in \mathbb{Z}$.

(i) Congruence modulo m is an equivalence relation on $\mathbb{Z}$, with precisely m equivalence classes.

(ii) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \qquad ac \equiv bd \pmod{m}.$$

(iii) If $ab \equiv ac \pmod{m}$ and $(a, m) = 1$, then $b \equiv c \pmod{m}$.

Proof. The original note leaves this as an exercise. The key point is divisibility. For example,

$$m \mid (a - b), \qquad m \mid (c - d)$$

imply

$$m \mid (a + c) - (b + d).$$

For multiplication, write

$$ac - bd = c(a - b) + b(c - d).$$

For cancellation, $ab \equiv ac \pmod{m}$ means $m \mid a(b - c)$. If $(a, m) = 1$, Euclid's lemma gives $m \mid (b - c)$, hence $b \equiv c \pmod{m}$. □

Choice, Well-Ordering, and the Arithmetic of Cardinal Size

N-20210207

§3.1 Choice enters the story

This note begins exactly where the previous chapter warned us: sometimes a construction requires choosing one element from each set in a family. The original note states the Axiom of Choice in product form.

Axiom 3.1.1 (Axiom of Choice)
The product of a family of nonempty sets indexed by a nonempty set is nonempty.

Remark 3.1.2 — Equivalently, if S is a set, then there is a choice function defined on the collection of all nonempty subsets of S:

$$f(A) \in A$$

for every nonempty $A \subseteq S$. This is the alternate version recorded in the original note.

§3.2 Partially ordered sets

Definition 3.2.1. A **partially ordered set** is a nonempty set A together with a relation $\leq$ which is reflexive, transitive, and antisymmetric. Antisymmetry means that

$$a \leq b \text{ and } b \leq a \quad \Longrightarrow \quad a = b.$$

Two elements $a, b \in A$ are **comparable** if either $a \leq b$ or $b \leq a$. If every pair of elements is comparable, then the order is a **total order** or **linear order**.

Example 3.2.2
Let $A = \mathcal{P}(\{1,2,3,4,5\})$, ordered by inclusion. Then $C \leq D$ means $C \subseteq D$. This is a partial order, but not a total order: for example, $\{1,2\}$ and $\{2,3\}$ are not comparable.

Let $(A, \leq)$ be a partially ordered set. An element $a \in A$ is **maximal** if whenever $c \in A$ is comparable with a and $a \leq c$, then $c = a$. An **upper bound** of a nonempty subset $B \subseteq A$ is an element $d \in A$ such that $b \leq d$ for every $b \in B$. A nonempty subset $B \subseteq A$ is a **chain** if it is totally ordered by the inherited order.

Theorem 3.2.3 (Zorn's lemma)
Let A be a nonempty partially ordered set. If every chain in A has an upper bound in A, then A has a maximal element.

Caution

The original note says "local maximum". The standard term is *maximal element*. It need not be a greatest element: there may be no element above all elements of A.

§3.3 Well-ordering

Definition 3.3.1. A partially ordered set A is **well ordered** if every nonempty subset of A has a least element.

Proposition 3.3.2 (Well-ordering principle)

Every nonempty set A admits a total ordering under which it is well ordered.

Remark 3.3.3 — The original note notes that this extends mathematical induction from $\mathbb{N}$ to arbitrary well-ordered sets. Conceptually this is the beginning of transfinite induction, even if the note does not yet develop ordinals.

§3.4 Equipollence and cardinal numbers

Definition 3.4.1. Two sets A and B are **equipollent**, written $A \sim B$, if there exists a bijection $A \to B$.

Definition 3.4.2. If a set A is equipollent to

$$I_n = \{1, 2, \dots, n\}$$

for some $n \in \mathbb{N}$, or to $I_0 = \varnothing$, then A is finite. Otherwise A is infinite.

Definition 3.4.3. The **cardinal number** of a set A, denoted $|A|$, is its equivalence class under equipollence. It is a finite or infinite cardinal according as A is finite or infinite.

Caution

Strictly speaking, the "equivalence class of all sets bijective with A" is usually too large to be a set. A formal treatment either uses representatives such as initial ordinals, or works in a class theory where such collections are proper classes. The original note is using the usual informal convention.

If A and B are disjoint sets with $|A| = \alpha$ and $|B| = \beta$, define

$$\alpha + \beta = |A \cup B|.$$

Similarly, cardinal multiplication is defined by

$$\alpha\beta = |A \times B|.$$

§3.5 Cantor's theorem

Theorem 3.5.1 (Cantor)
For every set A,
$$|A| < |\mathcal{P}(A)|.$$

Proof. The map $a \mapsto \{a\}$ is injective from A into $\mathcal{P}(A)$, so $|A| \leq |\mathcal{P}(A)|$.

Suppose for contradiction that $f : A \to \mathcal{P}(A)$ is surjective. Define
$$B = \{a \in A \mid a \notin f(a)\}.$$
Since $B \subseteq A$, surjectivity gives $a_0 \in A$ with $f(a_0) = B$. Then
$$a_0 \in B \iff a_0 \notin f(a_0) = B,$$
which is impossible. Hence no surjection $A \to \mathcal{P}(A)$ exists, and so $|A| < |\mathcal{P}(A)|$. □

This proof deliberately echoes Russell's paradox from the first chapter. In one case the self-reference shows that a class is too large to be a set; in the other, diagonalization shows that the power set is strictly larger than the original set.

§3.6 Comparing cardinals

Theorem 3.6.1 (Cantor–Bernstein)
If A and B are sets such that $|A| \leq |B|$ and $|B| \leq |A|$, then
$$|A| = |B|.$$

Theorem 3.6.2 (Trichotomy for cardinals)
The class of cardinal numbers is linearly ordered by $\leq$. If α and β are cardinal numbers, then exactly one of the following holds:
$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha.$$

Remark 3.6.3 — The original note calls this the Law of Trichotomy and mentions that, in the proof, one orders a family of functions by extension. This is one of the standard places where Zorn's lemma enters the comparison of cardinalities.

§3.7 Infinite cardinals and finite disturbance

The final part of the original note deliberately becomes a theorem list after the writing process was interrupted. I keep the statements, but the better way to read them is through explicit encodings: pairing functions and binary strings let us put our hands on the size of these infinite sets.

Theorem 3.7.1

Every infinite set has a denumerable subset. In particular,

$$\aleph_0 \leq \alpha$$

for every infinite cardinal α, assuming the usual choice principles being used in this chapter.

Lemma 3.7.2

If A is an infinite set and F is a finite set, then

$$|A \cup F| = |A|.$$

In particular,

$$\alpha + n = \alpha$$

for every infinite cardinal α and finite cardinal n.

Theorem 3.7.3

If $\alpha \leq \beta$ and β is infinite, then

$$\alpha + \beta = \beta.$$

Theorem 3.7.4

If $0 < \alpha \leq \beta$ and β is infinite, then

$$\alpha\beta = \beta.$$

In particular,

$$\aleph_0 \cdot \aleph_0 = \aleph_0,$$

and if B is finite then

$$\aleph_0 \cdot |B| = \aleph_0$$

provided $B \neq \varnothing$.

Theorem 3.7.5

Let A be a set and, for each integer $n \geq 1$, let

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ factors}}.$$

If A is finite, then

$$|A^n| = |A|^n,$$

and if A is infinite, then

$$|A^n| = |A|.$$

Moreover,

$$\left| \bigcup_{n \in \mathbb{N}} A^n \right| = \aleph_0 |A|.$$

Corollary 3.7.6

If A is an infinite set and $\mathcal{F}(A)$ denotes the set of all finite subsets of A, then

$$|\mathcal{F}(A)| = |A|.$$

§3.8 Concrete encodings behind the arithmetic

The equality

$$\aleph_0 \cdot \aleph_0 = \aleph_0$$

is not just a theorem name. A direct bijection is the Cantor pairing function

$$\pi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}, \qquad \pi(m,n) = \frac{(m+n)(m+n+1)}{2} + m.$$

It enumerates lattice points by diagonals. For instance,

$$\pi(0,0) = 0, \qquad \pi(1,0) = 2, \qquad \pi(0,1) = 1.$$

To invert it, start with $k \in \mathbb{N}$. Find the unique integer $s \geq 0$ such that

$$\frac{s(s+1)}{2} \leq k < \frac{(s+1)(s+2)}{2}.$$

Then

$$m = k - \frac{s(s+1)}{2}, \qquad n = s - m.$$

This recovers a unique pair (m,n), so π is a bijection.

There is an equally concrete encoding of finite subsets of $\mathbb{N}$. If

$$S = \{s_1 < s_2 < \cdots < s_k\} \subseteq \mathbb{N}$$

is finite, define

$$f(S) = \sum_{i=1}^{k} 2^{s_i},$$

with $f(\varnothing) = 0$. Binary expansion is unique, so two different finite subsets have different codes. Hence

$$|\mathcal{F}(\mathbb{N})| \leq \aleph_0.$$

The reverse injection is the singleton map

$$\mathbb{N} \hookrightarrow \mathcal{F}(\mathbb{N}), \qquad n \mapsto \{n\}.$$

By Cantor–Bernstein,

$$|\mathcal{F}(\mathbb{N})| = \aleph_0.$$

This is the finite-subset theorem in a form one can actually compute: a finite subset is just the support of the 1's in a binary expansion.

Gap

The general statement $|\mathcal{F}(A)| = |A|$ for every infinite set A still uses the comparison machinery above, and with arbitrary A one must pay attention to the choice principles being used. But for the countable model case, the pairing function and the binary encoding already show the mechanism rather than merely quoting the cardinal-arithmetic slogan.

Rethinking Set Theory Through Functions Rather Than Membership

N-20220131

§4.1 Why this translation/note matters

This note began as a translation and reading note on Tom Leinster's article *Rethinking Set Theory*. The article won the Chauvenet Prize, and the reason is easy to feel while reading it: it is not trying to make set theory more technical, but to ask why the set theory used by working mathematicians feels so different from the official ZFC story.

The guiding question is simple:

Question

If most mathematicians manipulate sets safely without remembering the ZFC axioms, what principles are they actually using?

The answer explored here is Lawvere's Elementary Theory of the Category of Sets, usually abbreviated ETCS. The important point is that ETCS is not an attempt to replace set theory by category theory. It is another axiomatization of set theory, written in a language closer to everyday mathematical practice.

§4.2 The discomfort with ZFC

ZFC treats every element of every set as itself a set. Formally, this is convenient. But in ordinary mathematics it is strange. If I ask what the elements of π are, or what the elements of a real number are, most mathematicians would say that the question is meaningless.

In ZFC, however, every object has been coded as a set, so asking for the elements of a real number is formally allowed. This does not mean ZFC is wrong. It means that ZFC uses the word "set" differently from how mathematicians often use it in practice.

Remark 4.2.1 — The point is not merely that real numbers are encoded arbitrarily. The deeper point is linguistic: ZFC makes membership primitive, while ordinary mathematics often treats functions, products, subsets, and inverse images as the more natural operations.

A typical ZFC axiom such as foundation says that every nonempty set has an element disjoint from it. This makes formal sense only because all elements are also sets. But when the set under discussion is something like $\mathbb{R}$, the ordinary mathematical intuition does not ask whether $\pi \cap \mathbb{R} = \varnothing$.

§4.3 Three misunderstandings about ETCS

The original article emphasizes three misunderstandings.

(i) ETCS is not a competitor to set theory; it is a form of set theory.

(ii) ETCS does not require more sophisticated mathematics than ZFC. It can be stated elementarily, even though its origin is categorical.

(iii) ETCS is not circular merely because it is inspired by categories. ZFC says, roughly, "there are things called sets and a relation called membership." ETCS says, roughly, "there are things called sets and things called functions, and functions compose." Both are first-order theories.

Caution

It is easy to overstate the category-theoretic flavor. The point here is not to presuppose the whole theory of categories. The point is to take functions and composition as basic, because that is often closer to actual mathematical use.

§4.4 Elements as special functions

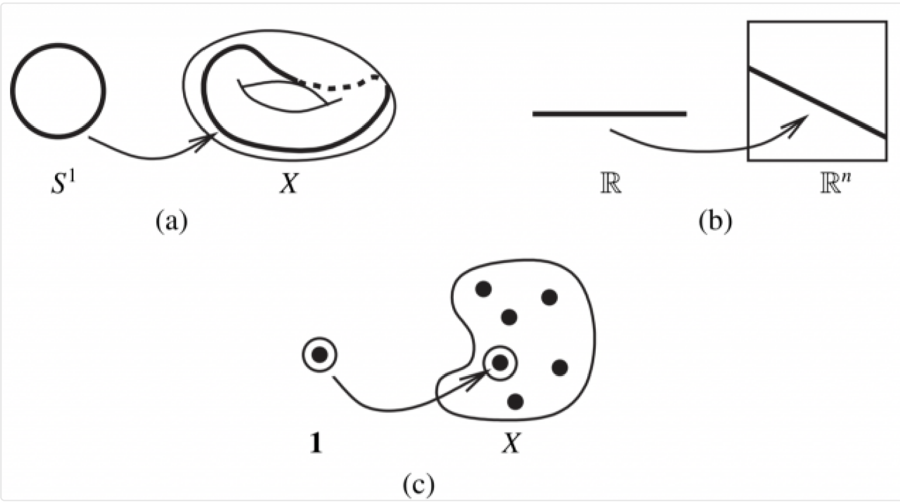


Figure 4.1: The motivating picture: elements can be represented as maps from a singleton.

The key idea is to define elements using functions. Suppose there is a singleton set

$$1 = \{\bullet\}.$$

Then an element of a set X can be identified with a function

$$1 \to X.$$

Indeed, such a function is determined by the image of $\bullet$, and that image is an element of X.

At first this looks like a formal trick, but it is a very general mathematical pattern:

- a loop in a topological space X is a continuous map $S^1 \to X$;
- a sequence in X is a function $\mathbb{N} \to X$;
- a solution of equations in a ring can be represented by a suitable homomorphism out of a quotient ring;
- a point of a space is often best treated as a map from a test object.

Remark 4.4.1 — This is the first philosophical shift: instead of saying that functions are built out of elements, we may say that elements are special cases of functions.

§4.5 The informal ETCS-style principles

The note lists the principles in an informal table. Rewritten cleanly, they say roughly:

(i) functions compose associatively and have identity functions;

(ii) there is a singleton set;

(iii) there is an empty set;

(iv) a function is determined by its effect on elements;

(v) products $X \times Y$ exist;

(vi) function sets Y^X exist;

(vii) inverse images of elements or subsets exist;

(viii) subsets of X correspond to functions $X \to \{0, 1\}$;

(ix) the natural numbers form a set;

(x) every surjection has a right inverse, i.e. a choice principle.

These principles are striking because they sound like the operations used constantly in ordinary mathematics. They do not begin by asking what the elements of elements are.

§4.6 Subsets via characteristic functions

A subset $A \subseteq X$ may be represented by its characteristic function

$$\chi_A : X \to \{0, 1\}, \qquad \chi_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Thus the power set of X is expressed as a function set

$$2^X.$$

This fits the ETCS point of view perfectly: subsets are not introduced by a membership relation alone, but through maps into a classifier of truth values.

§4.7 Products, exponentials, and inverse images

The ordinary constructions of mathematics appear directly:

$$X \times Y$$

represents pairs; the exponential

$$Y^X$$

represents functions from X to Y; and inverse images let us pull subsets back along functions. These are exactly the operations that mathematicians use constantly when building structures.

> **Remark 4.7.1** — In this sense, ETCS feels less like an encoding scheme and more like a formalization of actual mathematical behavior.

§4.8 Choice as right inverses

The choice principle in this language says that every surjection has a right inverse. If

$$f : X \twoheadrightarrow Y$$

is surjective, then there exists

$$g : Y \to X$$

with

$$f \circ g = \mathrm{id}_Y.$$

This is exactly the idea of choosing one element from each fibre of f.

> **Caution**
>
> This formulation is often more transparent than the usual family-of-nonempty-sets version, but it is the same choice principle in a categorical dress.

§4.9 What I take from the article

The main lesson is not that ZFC is useless or wrong. ZFC is powerful and foundationally successful. The lesson is that there is a mismatch between the official membership-first universe of ZFC and the function-centered practice of most mathematics.

ETCS suggests that one can axiomatize set theory using:

sets, functions, composition,
products, function spaces, subsets,
natural numbers, choice.

That language feels closer to how I actually use sets.

> **Question**
>
> Is ETCS only a more convenient presentation of ordinary mathematics, or does it also change how one should teach foundations? The article strongly suggests the latter, and this is the part I find most provocative.

§4.10 Translation note

This chapter is a reconstruction of a translation/reading note rather than a verbatim translation. I keep the mathematical thread: why ZFC feels alien in ordinary practice, how ETCS starts from functions, and why elements can be treated as maps from a singleton.

§4.11 ETCS intuition: membership is not primitive practice

One of the central insights of ETCS is that working mathematicians rarely use the membership relation as their main tool. They use functions. A set is understood by the maps into it, the maps out of it, its products, its subsets, and its quotients.

For example, instead of defining an ordered pair through nested braces, one can characterize the product $A \times B$ by its universal property. Instead of treating a subset as a special kind of element relation, one can treat it through its characteristic map to a subobject classifier. This does not deny membership; it changes which notions are primitive.

§4.12 Element reasoning inside categorical set theory

A common worry is that categorical set theory loses elements. It does not. In a category of sets, an element of A can be represented as a map

$$1 \to A$$

from a terminal object. Thus element reasoning is recovered as morphism reasoning from the one-point set.

This is a typical categorical move: replace internal membership statements by external mapping properties. The result often looks more abstract, but it is closer to how structures are transported by functions.

§4.13 Why this matters philosophically

ZFC gives a reductionist foundation: everything is encoded as a set. ETCS gives a structural foundation: sets are objects in a category satisfying axioms that support ordinary set operations. The philosophical difference is that ETCS does not ask what a set is made of at the lowest membership level; it asks what role sets play in mathematics.

This is why Leinster's article is compelling. It shows that the official foundation and the working foundation are not identical. ZFC is powerful as a background theory, but ETCS more closely describes the everyday language of functions, products, quotients, and subsets.

A First Comparison of ZFC, NBG, KM, and New Foundations

N-20220205

§5.1 Why there are several axiomatizations of set theory

This note compares several axiomatic systems for set theory: ZFC, NBG, Kelley–Morse set theory, and Quine's New Foundations. The underlying question is not merely historical. Different systems express different compromises between intuition, rigor, large collections, and self-reference.

§5.2 The main systems

The main systems in this note are:

ZFC. Zermelo–Fraenkel set theory with Choice, the standard foundation for most modern mathematics.

NBG. Von Neumann–Bernays–Goedel set theory, which distinguishes sets from classes.

KM. Kelley–Morse set theory, a stronger class theory than NBG.

NF. Quine's New Foundations, which uses stratification to control comprehension.

§5.3 ZFC

ZFC is built from several familiar axioms: extensionality, empty set, pairing, union, power set, foundation, replacement, infinity, and choice.

> **Remark 5.3.1 —** ZFC's strength is that it is compact, standard, and sufficient for almost all ordinary mathematics. Replacement and Infinity allow one to build complicated structures such as ordinals, cardinals, and the continuum.

> **Caution**
>
> ZFC does not directly treat proper classes as objects of the theory. This is often fine, but it becomes awkward when discussing objects such as the class of all sets, large categories, or global constructions in category theory.

§5.4 NBG

NBG introduces classes as first-class objects while still distinguishing sets from proper classes.

Definition 5.4.1. In NBG, every set is a class, but not every class is a set. A proper class is a class that is too large to be a set.

This makes it possible to speak about the universe of all sets as a proper class. It also makes certain category-theoretic discussions feel more natural, since large categories can be treated using classes.

> **Remark 5.4.2** — NBG is conservative over ZFC for statements purely about sets. In ordinary set-level mathematics, it usually proves the same theorems as ZFC. Its advantage is expressive convenience when classes are involved.

§5.5 Kelley–Morse set theory

KM extends the class-theoretic viewpoint further. It allows stronger comprehension principles for classes and therefore gives a richer theory of proper classes.

> **Remark 5.5.1** — KM can be thought of as a stronger language for discussing global class-sized structures. It is useful when NBG's class comprehension is too weak for the intended argument.

The tradeoff is that KM is logically stronger and less minimal. It may be more expressive, but it also introduces more foundational weight.

> **Question**
>
> Is there a clean presentation of KM that remains beginner-friendly while preserving its extra expressive power? This is still unclear in the note, and it is worth returning to after learning more model theory and class theory.

§5.6 New Foundations

Quine's New Foundations takes a more radical route. It uses a stratification condition on formulas to control paradoxes while allowing constructions that look very different from ZFC.

> **Caution**
>
> The original wording describes NF as allowing self-reference in a loose way. More precisely, NF restricts comprehension to stratified formulas. The point is not that arbitrary self-reference is allowed, but that the theory draws the safety boundary differently from ZFC.

NF differs from ZFC in several ways:

(i) ZFC uses Foundation to rule out membership cycles; NF uses stratification.

(ii) ZFC builds sets through iterative hierarchy intuition; NF modifies the logic of allowable comprehension.

(iii) NF has subtle consistency questions, and variants such as NFU are often easier to handle.

§5.7 How to compare the systems

The comparison can be organized along three axes.

§5.7.i Expressive power

ZFC is clean for ordinary set-level mathematics. NBG and KM are better suited for talking about proper classes and large structures. NF explores a different response to self-reference.

§5.7.ii Logical strength

NBG is conservative over ZFC for set statements. KM is stronger than NBG. NF and its variants sit in a different part of the foundational landscape and should not be casually compared by intuition alone.

§5.7.iii Use in practice

For most everyday mathematics, ZFC is enough. For category theory or discussions involving universes and proper classes, NBG-style language is often more comfortable. KM becomes relevant when one truly needs stronger class principles.

§5.8 What remains unclear

Gap

The original note contains several broad claims about conservativity and consistency that need more precise model-theoretic formulation. In particular, KM should not be casually described as conservative over ZFC in the same way NBG is. A later revision should add exact statements about relative consistency and conservativity.

Question

Which foundation is best for actual mathematical practice: a minimal set theory such as ZFC, a class theory such as NBG, a stronger class theory such as KM, or a category-inspired system such as ETCS? The answer probably depends on what one wants to do.

§5.9 Where the idea leads

ZFC is the mainstream baseline. NBG keeps the same set-level mathematics while adding proper classes. KM strengthens the class theory. NF changes the handling of comprehension and self-reference. The deeper lesson is that foundations are not only about avoiding paradoxes; they also shape what kinds of mathematical objects feel natural to discuss.

§5.10 Why classes are introduced

The main pressure on ZFC is that some collections are too large to be sets. The collection of all sets, the collection of all groups, or the collection of all ordinals cannot be sets without recreating paradoxes. Class theories introduce a formal distinction:

sets are classes that can be elements of other classes,
proper classes are too large to be elements.

In NBG, classes are first-class objects of the language, but quantification over classes is restricted in a way that keeps the theory conservative over ZFC for set-level statements. KM allows stronger class comprehension and is therefore stronger.

This explains the practical difference. ZFC can simulate classes by formulas, but NBG and KM allow one to talk about classes directly. For category theory this is convenient: the category of all sets is not small, but it can be treated as a class-sized category.

§5.11 New Foundations and stratification

Quine's NF tries a different repair of naive comprehension. Instead of restricting comprehension to subsets of already existing sets, it allows comprehension only for stratified formulas. A formula is stratified if one can assign type levels to variables so that membership $x \in y$ raises type by one, while equality $x = y$ keeps type the same.

For example, $x \notin x$ is not stratified, because it would require the type of x to be one higher than itself. Thus Russell's paradox is blocked. But a formula like $x = x$ is stratified, so NF has a universal set.

This is philosophically striking: ZFC avoids a universal set; NF permits one by controlling the syntax of comprehension.

§5.12 Comparing foundational choices

The systems differ not only in strength but in taste. ZFC is minimal enough to serve as a common background for ordinary mathematics. NBG is convenient when large classes are used explicitly. KM is stronger and supports more class-level reasoning. NF preserves a universal-set intuition but pays for it with syntactic restrictions.

The useful conclusion is not that one system is simply correct and the others wrong. A foundation is a formal environment. Different environments make different constructions easy or hard.

Vectors, Ordered Fields, and the First Axioms of Real Numbers

N-20220710

§6.1 Plane vectors

The first page contains exercises on plane vectors. A vector in the plane may be written

$$\vec{a} = (x, y) = x\mathbf{i} + y\mathbf{j}.$$

Addition and scalar multiplication are coordinatewise:

$$(x, y) + (z, w) = (x + z, y + w), \qquad \lambda(x, y) = (\lambda x, \lambda y).$$

The length and dot product are

$$|(x, y)| = \sqrt{x^2 + y^2}, \qquad (x, y) \cdot (z, w) = xz + yw.$$

Thus the angle between two nonzero vectors satisfies

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|\,|\vec{b}|} = \frac{xz + yw}{\sqrt{x^2 + y^2}\sqrt{z^2 + w^2}}.$$

If

$$\vec{x} = (\cos\alpha, \sin\alpha), \qquad \vec{y} = (\cos\beta, \sin\beta),$$

then

$$\cos\theta = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta).$$

This gives a vector proof of the cosine subtraction formula.

§6.2 Field axioms

Axiom 6.2.1 (Field axioms)
A field is a set F with operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group, $(F \setminus \{0\}, \cdot)$ is an abelian group, and multiplication distributes over addition.

The second page moves to the algebraic structure of the real numbers. The field axioms are:

(F1) $x + (y + z) = (x + y) + z$;

(F2) $x + y = y + x$;

(F3) there exists 0 such that $0 + x = x$;

(F4) for each x, there is $-x$ such that $x + (-x) = 0$;

(F5) $x(yz) = (xy)z$;

(F6) $xy = yx$;

(F7) there exists $1 \neq 0$ such that $1x = x$;

(F8) for $x \neq 0$, there is x^{-1} such that $xx^{-1} = 1$;

(F9) $x(y + z) = xy + xz$.

The note emphasizes that a "space" in mathematics is usually a set equipped with extra structure. A field is not merely a set; it is a set equipped with addition, multiplication, identities, inverses, and compatibility laws.

§6.3 Ordered fields

The order axioms include:

(O1) Transitivity: if $x \leq y$ and $y \leq z$, then $x \leq z$.

(O2) Antisymmetry: if $x \leq y$ and $y \leq x$, then $x = y$.

(O3) Totality: for all x, y, either $x \leq y$ or $y \leq x$.

(O4) Compatibility with addition: $x \leq y \Rightarrow x + z \leq y + z$.

(O5) Compatibility with multiplication: $0 \leq x$ and $0 \leq y \Rightarrow 0 \leq xy$.

The Archimedean axiom says that for every $x > 0$ and every y, there exists a positive integer n such that

$$nx \geq y.$$

A typical consequence is that every interval $(0, a)$ with $a > 0$ is nonempty: choose n so large that $n > a^{-1}$, then $1/n < a$.

§6.4 Elementary consequences of the order axioms

Proposition 6.4.1 (Consequences of an ordered field)
In an ordered field, addition preserves order and multiplication by a positive element preserves order. In particular, squares are nonnegative:

$$x^2 \geq 0.$$

The page records several exercises:

(i) $x \geq 0$ iff $-x \leq 0$, and $y > 1$ implies $0 < 1/y < 1$.

(ii) $1 > 0$ and $-1 \neq 1$.

(iii) If $x \leq y$ and $a \leq 0$, then $ax \geq ay$.

(iv) If $a \leq b$ and $x \leq y$, then $a + x \leq b + y$, with equality iff $a = b$ and $x = y$.

(v) If $a < x$ for every $a < x$, then $x \leq y$; this is a typical contradiction argument using an interval between two distinct real numbers.

(vi) $x^2 \geq 0$ for every real x.

(vii) If $a^2 < a$, then $0 < a < 1$.

(viii) If nonzero x, y have the same sign, then

$$(x+y)^2 > (x-y)^2.$$

§6.5 Embedding $\mathbb{Q}$ into $\mathbb{R}$

The note then proves that $\mathbb{R}$ contains a copy of $\mathbb{Q}$. Define for integers n

$$\iota(n) = \underbrace{1 + \cdots + 1}_{n \text{ times}}$$

for $n > 0$, $\iota(0) = 0$, and $\iota(n) = -\iota(-n)$ for $n < 0$. Then define

$$\iota\left(\frac{p}{q}\right) = \frac{\iota(p)}{\iota(q)}.$$

One must check that this is well-defined: if $p/q = s/t$, then $pt = qs$, so

$$\iota(p)\iota(t) = \iota(q)\iota(s),$$

and therefore

$$\frac{\iota(p)}{\iota(q)} = \frac{\iota(s)}{\iota(t)}.$$

This embedding preserves addition, multiplication, and order.

§6.6 Nested intervals

The final part introduces the nested interval axiom. If

$$I_n = [a_n, b_n]$$

is a descending sequence of closed intervals with real endpoints, then

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

implies

$$\bigcap_{n \geq 1} I_n \neq \varnothing.$$

Together with the field, order, and Archimedean axioms, this is one way to characterize the real numbers.

§6.7 Ordered fields and the real line

An ordered field is a field equipped with an order compatible with addition and multiplication. Compatibility means

$$a < b \Rightarrow a + c < b + c, \qquad 0 < a,\ 0 < b \Rightarrow 0 < ab.$$

The rational numbers form an ordered field, but they are not complete. The real numbers are obtained by adding the missing limits, for example through Dedekind cuts or Cauchy sequences.

This is why early vector and coordinate exercises naturally lead to axioms of the real numbers. Geometry needs coordinates, coordinates need arithmetic, and analysis needs completeness.

§6.8 Vectors as algebraicized geometry

A plane vector (x, y) packages a displacement into two real numbers. Vector addition corresponds to composing displacements; scalar multiplication corresponds to stretching. The algebraic laws of vectors are therefore not arbitrary axioms. They express basic geometric operations in coordinate form.

§6.9 The conceptual payoff

The note moves from computational vector exercises to the axiomatic construction of the real line. The important conceptual shift is that familiar formulas are only the visible part of an axiomatic structure. A rigorous real-number theory must explain why rational numbers embed, why intervals contain points, and why irrational numbers such as $\sqrt{2}$ can exist.

§6.10 Ordered fields and what they cannot prove

An ordered field supports algebra and order, but not necessarily completeness. The rationals satisfy the ordered-field axioms but have gaps. Thus many analytic theorems need more than order and algebra.

§6.11 Archimedean property and density

In the real numbers, the Archimedean property says no positive infinitesimal exists inside $\mathbb{R}$: for every x, some integer n satisfies $n > x$. This leads to density of rationals and helps connect arithmetic approximations with the geometry of the line.

§6.12 Completeness as the real-number axiom

The supremum property, nested intervals, Cauchy completeness, and Dedekind completeness are equivalent ways of saying that the real line has no gaps. This is the feature that turns ordered algebra into analysis.

§6.13 From ordered algebra to the real line

Vectors need linear structure; ordered fields need compatibility between addition, multiplication, and order; the real numbers need completeness. The early axioms are not formal decoration: each one supports a different layer of geometry and analysis.

Completeness, Supremum, Dedekind Cuts, and Metric Spaces

N-20220713

§7.1 Large and small positive real numbers

The page begins with a simple but important fact: for every positive real A, there is a positive real $M > A$; for every positive real a, there is a positive real $\varepsilon < a$. For instance,

$$M = A + 1, \qquad \varepsilon = \frac{a}{2}.$$

This is the first appearance of the ε-language of analysis.

§7.2 The geometric view of the real line

We think of $\mathbb{R}$ as a line, where order is represented by left and right. This picture is helpful, but the note emphasizes that all conclusions must ultimately come from axioms and rigorous reasoning, not from the drawing itself.

A set $X \subseteq \mathbb{R}$ is bounded above if there exists $A \in \mathbb{R}$ such that

$$x \le A \quad \text{for all } x \in X.$$

Such an A is an upper bound. Similarly, X is bounded below if there exists a such that

$$a \le x \quad \text{for all } x \in X.$$

§7.3 Supremum property

The least upper bound property says: if $X \subseteq \mathbb{R}$ is nonempty and bounded above, then the set of upper bounds has a least element. This least upper bound is denoted

$$\sup X.$$

Equivalently, $M = \sup X$ iff:

(i) $x \le M$ for all $x \in X$;

(ii) for every $\varepsilon > 0$, there exists $x \in X$ such that

$$x > M - \varepsilon.$$

The note sketches a proof of the supremum property from the nested interval principle by bisecting intervals and selecting the half that still contains upper bounds or points of the set.

The dual notion is the infimum:

$$\inf X.$$

One always has

$$\inf X \le \sup X$$

for a nonempty bounded set.

§7.4 Nested intervals imply completeness

Let $X \subseteq \mathbb{R}$ be nonempty and bounded above. One can construct nested intervals

$$I_n = [a_n, b_n]$$

so that a_n is not an upper bound of X, while b_n is an upper bound, and

$$b_n - a_n \le 2^{-n}.$$

The nested interval axiom gives a point

$$M \in \bigcap_{n=1}^{\infty} I_n.$$

The shrinking length forces M to be the least upper bound of X. If M were not an upper bound, some $x \in X$ would lie above it; for large n, the right endpoint b_n would be too close to M, contradiction. If M were not least, a smaller upper bound would contradict the construction of the a_n's.

§7.5 Dedekind cuts

A Dedekind cut is a subset $X \subset \mathbb{Q}$ such that:

(i) $X \neq \varnothing$ and $\mathbb{Q} \setminus X \neq \varnothing$;

(ii) if $x \in X$ and $y < x$, then $y \in X$;

(iii) X has no greatest element.

A rational number p/q defines the cut

$$X_{p/q} = \{x \in \mathbb{Q} : x < p/q\}.$$

An irrational such as $\sqrt{2}$ is represented by

$$X_{\sqrt{2}} = \{x \in \mathbb{Q}_{>0} : x^2 < 2\} \cup \mathbb{Q}_{\le 0}.$$

Thus real numbers can be constructed from rational cuts.

§7.6 Metric spaces

A metric space is a set X equipped with a function

$$d : X \times X \to \mathbb{R}_{\ge 0}$$

satisfying:

(i) $d(x, y) = 0$ iff $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, z) \le d(x, y) + d(y, z)$.

The usual metric on $\mathbb{R}$ is
$$d(x, y) = |x - y|.$$
On $\mathbb{R}^q$, common metrics are:
$$d_2(x, y) = \left(\sum_{n=1}^{q} |x_n - y_n|^2 \right)^{1/2},$$
$$d_1(x, y) = \sum_{n=1}^{q} |x_n - y_n|,$$
and
$$d_\infty(x, y) = \max_{1 \le n \le q} |x_n - y_n|.$$
The note remarks that d_2 requires Cauchy–Schwarz for the triangle inequality, d_1 is the Manhattan metric, and d_∞ is often called the Chebyshev metric.

A final example is the discrete metric:
$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$
In the discrete metric, open balls are especially simple:
$$B_r(x) = \begin{cases} \{x\}, & 0 < r \le 1, \\ X, & r > 1. \end{cases}$$
The note points out that boundary points are not included in an open ball, explaining why the case $r = 1$ gives $\{x\}$, not all of X.

§7.7 Subspaces and density

If $Y \subseteq X$ and X has metric d, then Y inherits the subspace metric
$$d_Y(y_1, y_2) = d(y_1, y_2).$$
A subset $Y \subseteq X$ is dense if for every $x \in X$ and every $\varepsilon > 0$, there exists $y \in Y$ such that
$$d(x, y) < \varepsilon.$$
This formalizes the idea that Y comes arbitrarily close to every point of X.

§7.8 Supremum as least upper bound

The supremum of a set $A \subseteq \mathbb{R}$ is not merely a maximum. It is the least number s such that every element of A is at most s. A maximum must belong to A; a supremum need not.

For example,
$$\sup(0, 1) = 1,$$
but $(0, 1)$ has no maximum. This distinction is exactly what makes completeness subtle.

§7.9 Dedekind cuts and completeness

A Dedekind cut separates rationals into a lower part and an upper part with no gap. Irrational numbers appear as cuts not produced by a rational boundary. Completeness says that every such cut corresponds to a real number.

This is the philosophical point of constructing $\mathbb{R}$: the real line is obtained by filling all rational gaps.

§7.10 Looking ahead

The note moves from the order-completeness of $\mathbb{R}$ to the metric-space language of distance and density. Supremum, Dedekind cuts, nested intervals, and metrics are different ways of making precise the intuition that the real line has no holes.

§7.11 Supremum as order-completeness

The supremum property says every nonempty bounded-above set has a least upper bound. This is the order-theoretic heart of the real line. It is what makes monotone convergence and intermediate-value reasoning possible.

§7.12 Cauchy completion and metric completion

A metric space can be completed by adding equivalence classes of Cauchy sequences. Two Cauchy sequences are equivalent if their distance tends to zero. The real numbers can be constructed as the completion of $\mathbb{Q}$ under the usual metric.

§7.13 Dedekind cuts versus Cauchy sequences

Dedekind cuts construct reals by filling order gaps; Cauchy completion constructs reals by adding missing limits of Cauchy sequences. The two constructions produce isomorphic complete ordered fields, but they emphasize different structures: order versus metric.

§7.14 Completeness as the no-gap principle

Completeness is the bridge from algebraic number systems to analysis. Supremum, nested intervals, Dedekind cuts, density, and metric completion are not separate facts; they are equivalent ways of ruling out gaps.

Completed Decimals, Hyperfinite Truncations, and Nonstandard Analysis

N-20230225

§8.1 One notation can hide three different objects

The expression

$$0.999\ldots$$

is often discussed as though there were only one possible mathematical object behind it. In fact, three nearby constructions must be separated.

For an ordinary natural number n, the finite decimal with n nines is

$$d_n = \sum_{k=1}^{n} 9 \cdot 10^{-k} = 1 - 10^{-n}.$$

It is strictly less than 1. The real infinite decimal is the limit

$$d = \lim_{n\to\infty} d_n,$$

so it equals 1. A third object becomes available in nonstandard analysis: if H is an infinite hyperinteger, one may form the hyperfinite sum

$$d_H = \sum_{k=1}^{H} 9 \cdot 10^{-k}.$$

It is still a finite sum inside the hyperreal universe, but it has more terms than every standard finite truncation.

The familiar disagreement is therefore usually not about algebra. It is about which of these three objects the notation is intended to denote.

§8.2 The real decimal is a completed limit

The real-number definition is direct. Since

$$d_n = 1 - 10^{-n},$$

we have

$$|1 - d_n| = 10^{-n}.$$

Given $\varepsilon > 0$, choose N so large that $10^{-N} < \varepsilon$. Then for $n \geq N$,

$$|1 - d_n| < \varepsilon.$$

Hence

$$\boxed{0.999\ldots = \lim_{n\to\infty} d_n = 1.}$$

The standard subtraction argument says the same thing more compactly. If $x = 0.999\ldots$, then

$$10x = 9.999\ldots,$$

and subtraction gives $9x = 9$. This computation is valid because both decimals denote completed real limits. It is not a calculation with an unfinished string.

The uniqueness of decimal expansion fails only at terminating endpoints. More generally,

$$0.a_1a_2\cdots a_{m-1}(a_m - 1)999\ldots = 0.a_1a_2\cdots a_m000\ldots$$

whenever $a_m > 0$. This is not a defect of the real line; it is the usual nonuniqueness of positional notation at boundary points of adjacent decimal intervals.

§8.3 An ultrapower turns sequences into numbers

To make "larger than every ordinary integer" precise, begin with all real sequences

$$(x_1, x_2, x_3, \ldots) \in \mathbb{R}^{\mathbb{N}}.$$

Fix a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. Two sequences are declared equivalent when they agree on a set belonging to $\mathcal{U}$:

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad \{n : x_n = y_n\} \in \mathcal{U}.$$

The hyperreal field is the quotient

$$^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathcal{U}.$$

Addition and multiplication are performed coordinatewise before passing to equivalence classes.

Ordinary reals embed as constant sequences:

$$r \longmapsto [r, r, r, \ldots].$$

The class

$$H = [1, 2, 3, \ldots]$$

is larger than every standard integer m, because

$$\{n : n > m\}$$

is cofinite and therefore belongs to every nonprincipal ultrafilter. Thus H is an infinite hyperinteger. Its reciprocal

$$\eta = \frac{1}{H} = [1, 1/2, 1/3, \ldots]$$

is positive but smaller than every positive standard real number.

This construction does not adjoin a mysterious symbol by decree. It packages asymptotic behavior of sequences into algebraic elements of a larger ordered field.

§8.4 Infinitesimal closeness and standard part

A hyperreal x is infinitesimal when

$$|x| < r$$

for every positive standard real r. We write

$$x \approx y$$

when $x - y$ is infinitesimal. A hyperreal is finite when its absolute value is bounded by some standard integer.

Every finite hyperreal is infinitely close to a unique real number. This real number is its standard part:

$$\mathrm{st}(x) \in \mathbb{R}, \qquad x \approx \mathrm{st}(x).$$

Uniqueness is easy. If $x \approx r$ and $x \approx s$ for real numbers r, s, then $r - s$ is a real infinitesimal. The only real infinitesimal is 0, so $r = s$.

Existence uses completeness of $\mathbb{R}$. For finite x, consider the standard set

$$A = \{r \in \mathbb{R} : r < x\}.$$

It is nonempty and bounded above. Let $a = \sup A$. If $x - a$ were appreciably positive or negative, the least-upper-bound property would be contradicted. Therefore $x \approx a$.

The standard-part map is not defined on infinite hyperreals, and it is not an order-field homomorphism on all of ${}^*\mathbb{R}$. On finite hyperreals, however, it respects addition and multiplication:

$$\mathrm{st}(x + y) = \mathrm{st}(x) + \mathrm{st}(y),$$

$$\mathrm{st}(xy) = \mathrm{st}(x)\,\mathrm{st}(y).$$

It is the operation that turns an infinitesimally accurate hyperreal computation back into an ordinary real answer.

§8.5 Transfer explains why finite algebra still works

The transfer principle says, roughly, that every first-order statement true in the ordered field $\mathbb{R}$ remains true in its hyperreal extension. In particular, polynomial identities, order laws, and statements quantified over finitely many elements transfer.

For every natural number n,

$$\sum_{k=1}^{n} 9 \cdot 10^{-k} = 1 - 10^{-n}.$$

This is a first-order statement about finite sums. Transfer therefore gives, for every hypernatural H,

$$\sum_{k=1}^{H} 9 \cdot 10^{-k} = 1 - 10^{-H}.$$

If H is infinite, then 10^{-H} is a positive infinitesimal. Consequently,

$$d_H < 1, \qquad d_H \approx 1, \qquad \mathrm{st}(d_H) = 1.$$

There is no contradiction with the real identity. The real decimal is the completed limit 1; the hyperfinite decimal is a different hyperreal number infinitesimally below 1. Standard part sends the latter to the former.

§8.6 Internal sets are the domain of transfer

Transfer cannot be applied indiscriminately to every subset of the hyperreals. The relevant distinction is between internal and external objects.

An internal set is represented by a sequence of standard sets. For example, if

$$A_n = \{1, 2, \dots, n\},$$

then the class $[A_n]$ is the hyperfinite interval

$$\{1, 2, \dots, H\}$$

for $H = [n]$. Although it contains more elements than any standard finite set, it behaves internally like a finite set: transferred finite-sum and finite-counting statements apply to it.

By contrast, the set of all infinitesimals

$$\mu(0) = \{x \in {}^*\mathbb{R} : x \approx 0\}$$

is external. So is the set of all finite hyperreals. These sets are mathematically legitimate subsets, but they are not themselves produced by the ultrapower in the internal way required by transfer.

A useful test comes from overspill. Suppose an internal set $A \subseteq {}^*\mathbb{N}$ contains every standard natural number. Then A must contain some infinite hypernatural. To see the idea, write

$$A = [A_n].$$

For each standard m, membership of m means that $m \in A_n$ for $\mathcal{U}$-many n. The internal statement "$1, \dots, m \in A$" holds for every standard m, and transfer forces it to hold up to some nonstandard bound.

Now consider the set of standard natural numbers inside ${}^*\mathbb{N}$. It contains every standard natural but no infinite hypernatural, so overspill shows that it cannot be internal. The same phenomenon explains why one cannot transfer a statement that quantifies over "all infinitesimals" as though that collection were an internal finite-scale object.

§8.7 Derivatives become standard parts of infinitesimal quotients

Let $f : \mathbb{R} \to \mathbb{R}$ and let ${}^*f$ denote its hyperreal extension. A standard function is differentiable at a precisely when there is a real number L such that for every nonzero infinitesimal η,

$$\frac{{}^*f(a+\eta) - {}^*f(a)}{\eta} \approx L.$$

Then

$$f'(a) = \operatorname{st}\left(\frac{{}^*f(a+\eta) - {}^*f(a)}{\eta}\right).$$

For $f(x) = x^2$,

$$\frac{(a+\eta)^2 - a^2}{\eta} = 2a + \eta,$$

so

$$f'(a) = \operatorname{st}(2a + \eta) = 2a.$$

The condition must hold for every nonzero infinitesimal, not merely one convenient choice. This is the nonstandard form of the ordinary limit requiring all sufficiently small increments to give the same first-order behavior.

For $f(x) = |x|$ at 0, a positive infinitesimal gives quotient 1, while a negative infinitesimal gives -1. No single real standard part works for all infinitesimal directions, so the function is not differentiable at 0. The usual left-right failure appears without introducing two separate limit symbols.

§8.8 Hyperfinite sums recover the Riemann integral

Let f be continuous on $[a, b]$, choose an infinite hyperinteger H, and set

$$\Delta x = \frac{b-a}{H}, \qquad x_i = a + i\Delta x.$$

The hyperfinite Riemann sum is

$$S_H = \sum_{i=0}^{H-1} {}^*f(x_i)\Delta x.$$

Continuity on a compact interval implies uniform continuity. Hence replacing each infinitesimal cell by any other point in the same cell changes the sum only infinitesimally. The standard part is therefore independent of the hyperfinite sampling choice, and

$$\boxed{\int_a^b f(x)\,dx = \operatorname{st}(S_H).}$$

For $f(x) = x$ on $[0, 1]$, take $x_i = i/H$. Then

$$S_H = \sum_{i=0}^{H-1} \frac{i}{H}\frac{1}{H} = \frac{H(H-1)}{2H^2} = \frac{1}{2} - \frac{1}{2H}.$$

Since $1/H$ is infinitesimal,

$$\operatorname{st}(S_H) = \frac{1}{2}.$$

The phrase "add infinitely many infinitesimal rectangles" has become an ordinary finite-sum identity transferred to a hyperfinite index set.

§8.9 Loeb measure turns counting into a standard measure

The hyperfinite grid also explains how a genuinely standard measure can emerge from internal counting. Let

$$\Omega_H = \left\{0, \frac{1}{H}, \dots, \frac{H-1}{H}\right\}.$$

For an internal subset $A \subseteq \Omega_H$, define the internal counting probability

$$\nu_0(A) = \frac{|A|}{H}.$$

This value is a finite hyperreal. Taking standard part gives

$$\nu(A) = \operatorname{st}\nu_0(A).$$

On the algebra of internal subsets, ν is finitely additive. The Loeb construction applies the usual measure-extension machinery to obtain a countably additive standard measure on a larger sigma-algebra.

Consider the interval $[0,t) \subseteq [0,1]$ with standard $0 \le t \le 1$. Its grid approximation contains

$$\lfloor tH \rfloor$$

points, so

$$\nu_0([0,t) \cap \Omega_H) = \frac{\lfloor tH \rfloor}{H} \approx t.$$

Therefore its Loeb measure is t, exactly the ordinary length of the interval.

Thus the same hyperfinite idea that distinguishes d_H from the completed decimal also builds a bridge from finite counting to continuum measure. Hyperfinite objects are not completed real limits, but after standard part and measure completion they can encode the standard analytic structures those limits describe.

Foundations of Mathematics: Set Theory, Category Theory, and Type Theory

N-20230710

§9.1 The question of foundations

This chapter is a reconstructed version of an expository article comparing three major foundational languages: set theory, category theory, and type theory. The point is not to declare a single winner. Rather, I want to understand how each framework answers the same basic questions:

(i) What are mathematical objects?

(ii) What counts as equality?

(iii) What kind of reasoning is allowed?

(iv) How does computation enter mathematics?

(v) How does one organize structures and transformations between them?

Set theory emphasizes membership; category theory emphasizes morphisms and universal properties; type theory emphasizes construction, judgment, and computation. The most interesting modern developments, especially univalent foundations and homotopy type theory, no longer keep these worlds separate.

§9.2 Set theory

§9.2.i From naive comprehension to axioms

Modern set theory begins with the intuition that mathematical objects can be collected into sets. In naive set theory, one might try to form

$$\{x \mid P(x)\}$$

for any property P. Russell's paradox shows that unrestricted comprehension is impossible: the class

$$R = \{x \mid x \notin x\}$$

leads to contradiction if it is treated as a set.

The response is axiomatization. Zermelo–Fraenkel set theory with Choice, ZFC, restricts set formation and supplies a controlled universe in which ordinary mathematics can be developed.

§9.2.ii ZFC as the standard background

The familiar axioms include extensionality, pairing, union, power set, infinity, replacement, foundation, separation, and choice. Their roles may be summarized as follows:

Extensionality. A set is determined by its elements.

Pairing, union, power set. These build new sets from old ones.

Infinity. This guarantees an infinite set, usually giving the natural numbers.

Replacement. Definable images of sets are sets.

Foundation. Membership is well-founded and circular membership is excluded.

Choice. Products of nonempty families are nonempty; equivalently, every surjection has a right inverse.

> **Remark 9.2.1 —** The strength of ZFC is not that it matches everyday mathematical language perfectly. Its strength is that it gives a stable formal universe large enough for most mathematics and disciplined enough to avoid naive paradoxes.

§9.2.iii Applications and limitations

Set theory supports analysis, algebra, topology, measure theory, model theory, and essentially all ordinary mathematical constructions. It also enables transfinite methods: ordinals, cardinals, large cardinals, forcing, independence, and descriptive set theory.

But ZFC has philosophical and practical limitations. It represents everything as sets, even when this representation feels artificial. A group is not experienced as a nested set; it is experienced as a structure with operations. A topological space is not merely a set of ordered pairs defining open sets; it is a space with continuous maps. This motivates category theory.

§9.3 Category theory

§9.3.i From objects to morphisms

Category theory changes the focus from internal membership to external relations. A category consists of objects and morphisms, with associative composition and identities. In this language, mathematical structures are understood by the maps they admit.

Definition 9.3.1. A category $\mathcal{C}$ consists of objects, morphism sets $\mathcal{C}(X,Y)$, identity morphisms id_X, and composition

$$\mathcal{C}(Y,Z)\times\mathcal{C}(X,Y)\to\mathcal{C}(X,Z),$$

satisfying associativity and identity laws.

> **Example 9.3.2**
> **Set**, **Grp**, **Top**, **Ring**, and $\mathbf{Vect}_k$ are standard categories. Each has a different notion of structure-preserving map.

§9.3.ii Universal properties

The central categorical idea is the universal property. Products, coproducts, pullbacks, pushouts, quotients, tensor products, free objects, and completions can all be characterized by mapping properties.

For example, a product $X \times Y$ is characterized by

$$\begin{tikzcd} & Z \arrow[d, dashed, "\langle f,g\rangle"] \arrow[ddl, bend right, "f"'] \arrow[ddr, bend left, "g"] & \\ & X \times Y \arrow[dl, "\pi_X"] \arrow[dr, "\pi_Y"'] & \\ X & & Y. \end{tikzcd}$$

This diagram says that giving a map into a product is the same as giving compatible maps into its two factors.

> **Remark 9.3.3** — Universal properties explain why the same construction appears in many different subjects. They reveal that the construction is not about the elements of a particular model, but about a structural role.

§9.3.iii Functors and natural transformations

Definition 9.3.4. A functor $F : \mathcal{C} \to \mathcal{D}$ sends objects to objects and morphisms to morphisms, preserving identities and composition.

Definition 9.3.5. A natural transformation $\eta : F \Rightarrow G$ between functors $F, G : \mathcal{C} \to \mathcal{D}$ assigns to each object X a morphism $\eta_X : F(X) \to G(X)$ such that for every $f : X \to Y$,

$$\begin{tikzcd} F(X) \arrow[r, "F(f)"] \arrow[d, "\eta_X"'] & F(Y) \arrow[d, "\eta_Y"] \\ G(X) \arrow[r, "G(f)"'] & G(Y) \end{tikzcd}$$

commutes.

This square is one of the simplest and most important categorical diagrams. It encodes the idea that a construction is canonical, not dependent on arbitrary choices.

§9.3.iv Categorical foundations

Category theory can function as a language for organizing mathematics, but it can also become foundational. ETCS axiomatizes sets categorically. Topos theory generalizes set-like universes. Higher category theory and infinity-categories extend the categorical language to homotopical and derived contexts.

> **Remark 9.3.6** — Category theory does not merely replace set theory. It clarifies invariance. It asks not only what an object is made of, but how it transforms and how it participates in universal constructions.

§9.4 Type theory

§9.4.i Judgments and types

Type theory begins from judgments such as

$$a : A,$$

meaning that a is a term of type A. Instead of putting every object into one universe of sets, type theory stratifies objects by the types they inhabit.

> **Remark 9.4.1** — The phrase "$a \in A$" in set theory and the judgment "$a : A$" in type theory look similar but behave differently. Membership is a proposition; typing is a judgment of well-formedness.

§9.4.ii Dependent types

Dependent type theory allows types to depend on terms. If A is a type and $B(x)$ is a type depending on $x : A$, then one can form:

$$\prod_{x:A} B(x), \qquad \sum_{x:A} B(x).$$

The product type corresponds to universal quantification, while the sum type corresponds to existential quantification.

§9.4.iii Curry–Howard

The Curry–Howard correspondence identifies propositions with types and proofs with terms:

$$\text{proposition} \leftrightarrow \text{type}, \qquad \text{proof} \leftrightarrow \text{term}.$$

For example,

$$A \times B$$

corresponds to conjunction, while

$$A \to B$$

corresponds to implication.

This makes type theory naturally computational. A proof is not merely a static certificate; it may carry an algorithm.

§9.4.iv Constructivity and computation

Classical set theory freely uses excluded middle and choice. Type theory is often constructive: to prove existence, one should construct a witness. This is not only a philosophical choice. It is what makes type theory suitable for proof assistants such as Coq, Lean, Agda, and Isabelle/HOL.

§9.5 Homotopy type theory and univalence

Homotopy type theory interprets types as spaces, terms as points, and identity proofs as paths. Higher identity proofs become higher homotopies. In this interpretation, equality is no longer merely a binary relation; it has structure.

The univalence axiom says, roughly, that equivalent structures may be identified:

$$(A = B) \simeq (A \simeq B).$$

This expresses a structuralist principle: isomorphic or equivalent objects should be interchangeable in mathematics.

Remark 9.5.1 — Univalence is one of the most striking attempts to synthesize set-theoretic rigor, categorical invariance, and type-theoretic computation. It turns the informal mathematical habit of identifying isomorphic objects into a formal principle.

§9.6 Comparative analysis

Framework	Primitive emphasis	Equality	Strength
Set theory	membership	extensional equality	universal coding power
Category theory	morphisms	isomorphism/ equivalence	structural invariance
Type theory	judgments/terms	identity types	computation and construction

Set theory is excellent as a common foundational background. Category theory is excellent for organizing structures and transformations. Type theory is excellent when proofs, programs, and constructions are meant to interact.

§9.7 Modern applications

§9.7.i Proof assistants

Proof assistants make foundations operational. Lean's mathlib, Coq's constructive type theory, Agda's dependent types, and Isabelle's higher-order logic show that foundational choices affect how mathematics is formalized.

§9.7.ii Computer science

Type theory underlies programming language theory, dependent types, formal verification, compiler correctness, and proof-carrying code. Category theory appears in semantics, monads, functional programming, and compositional systems. Set theory remains important in logic, model theory, semantics, and databases.

§9.7.iii Geometry and higher structures

Modern geometry often uses categorical and type-theoretic ideas: stacks, sheaves, derived categories, infinity-categories, and higher topos theory. Foundations are no longer merely a safety net beneath mathematics; they become active tools inside research.

§9.8 Toward a unified perspective

A mature view should not treat the three foundations as enemies. They answer different needs:

(i) ZFC gives a robust global universe;

(ii) category theory reveals structural patterns and invariance;

(iii) type theory connects proof, construction, and computation;

(iv) HoTT and univalent foundations attempt to synthesize equivalence-based mathematics with formal computation.

§9.9 Conclusion

The foundational landscape is plural. Set theory gives a powerful ontology of membership. Category theory gives a language of structure and transformation. Type theory gives a constructive and computational account of mathematical reasoning. The deepest modern lesson is not that one of them eliminates the others, but that mathematics becomes clearer when one understands which foundational lens is being used and why.

§9.10 Equality in the three languages

The three foundations treat equality differently. In set theory, equality is extensional:

$$A = B \iff \forall x\,(x \in A \leftrightarrow x \in B).$$

In category theory, sameness is usually replaced by isomorphism or equivalence. Two groups may not be literally equal as sets, but if they are isomorphic, group theory treats them as the same structure.

Type theory makes equality internal. A statement $a = b$ is itself a type, and a proof of equality is a term of that type. Homotopy type theory then interprets such proofs as paths. This explains why univalence is so conceptually strong: it aligns equality of types with equivalence of structures.

§9.11 Universal properties versus encodings

A set-theoretic construction often gives a concrete encoding. A categorical construction gives a role. For example, a product is not fundamentally a set of ordered pairs; it is an object receiving projections and satisfying a universal mapping property.

This matters foundationally because encodings are arbitrary but universal properties are invariant. Kuratowski's ordered pair and other encodings are useful inside ZFC, but no working mathematician cares which coding was used once the universal property is available.

§9.12 Computation and formal proof

Type theory is not only a philosophical alternative. It is the foundation behind many formal proof systems. A proof is a term, and checking a proof becomes type checking. Dependent types allow mathematical statements to be expressed with high precision: a proof of existence contains a witness, and a proof of a universal statement behaves like a function.

This computational character is one reason type theory feels different from classical set theory. It is not only about what objects exist, but also about how they are constructed and verified.

II

Categories and Logic

Part II: Contents

A Gateway to Dualities: Space, Algebra, and the Dualizing Object

N-20210721

§10.1 Starting point

In July 2021, a table titled “Some dualities” appeared in our Geek College chat. It was shared by Dr. Keyao Peng, and at first glance it looked like a compact list of examples. The table had three columns: a geometric object, an algebraic object, and a mysterious third entry called the dualizing object or gauge object.

The table became one of the first places where I felt that modern mathematics was not merely a collection of topics, but a repeated pattern. Geometry and algebra were not standing separately. They seemed to be two languages for the same structure. The first row, Stone spaces versus Boolean algebras, was the entry point that most immediately caught my attention. It led to my later attempt to write about Boolean algebra, but that earlier writing mostly explained the algebraic operations. It did not yet explain the missing spatial half: why logic should have a topology.

This note rewrites that summer question in a more systematic form. The aim is not merely to list several dualities. The aim is to understand the mechanism behind the table:

a space is studied through its observations,
an algebra is studied through its models or points.

The third column of the table is the key. The dualizing object is the interpreter through which the two sides talk to each other.

§10.2 The table is not a list

Here is the table in a cleaned-up form.

Geometry	Algebra	Dualizing object / gauge object
Stone spaces	Boolean algebras	the Boolean values $\mathbf{2} = \{0, 1\}$
compact Hausdorff spaces	commutative C^*-algebras	the complex numbers $\mathbb{C}$
affine schemes	commutative rings	the affine line $\mathbb{A}^1$
locales	frames	the Sierpiński space $\mathbb{S}$
topoi	logoi	the topos **Set** of sets
∞-topoi	∞-logoi	the ∞-topos of ∞-groupoids

The table should not be read as six unrelated examples. It is an architectural diagram. The left column contains objects that behave like spaces. The middle column contains algebraic or logical presentations of those spaces. The right column contains the object used to measure, test, or observe them.

The shortest slogan is:

geometry is recovered from algebra, and algebra is recovered from geometry.

But this slogan is still too vague. The real mechanism is contravariant homming into a dualizing object.

§10.3 The dualizing object

Let Ω be a dualizing object. A geometric object X gives an algebra of Ω-valued observations:

$$X \longmapsto \mathrm{Hom}(X, \Omega).$$

A point of $\mathrm{Hom}(X, \Omega)$ is a way of asking a question about X, with answer living in Ω.

Conversely, an algebraic object A gives a geometric space of Ω-valued models:

$$A \longmapsto \mathrm{Hom}(A, \Omega).$$

A point of this new space is a way to interpret the algebra A inside the measuring object Ω.

The reason a duality reverses arrows is already visible here. If

$$f : X \to Y$$

is a map of spaces, then precomposition gives

$$\mathrm{Hom}(Y, \Omega) \longrightarrow \mathrm{Hom}(X, \Omega), \qquad \varphi \longmapsto \varphi \circ f.$$

So a map $X \to Y$ produces a map in the opposite direction on the algebras of observations. This elementary reversal is the seed of the whole table.

> **Remark 10.3.1** — The third column is therefore not an accessory. It tells us what kind of observation is allowed. Changing the dualizing object changes the kind of geometry that can be seen.

§10.4 Stone duality as the first complete example

The first row is the best place to begin because it is the most logical and the most concrete:

$$\text{Stone spaces} \quad \longleftrightarrow \quad \text{Boolean algebras}, \qquad \Omega = \mathbf{2} = \{0, 1\}.$$

Here $\mathbf{2}$ is the space of truth values. A map into $\mathbf{2}$ is a yes-or-no observation.

§10.4.i Boolean algebras as propositional logic

A Boolean algebra is an algebraic model of classical propositional logic.

Definition 10.4.1 (Boolean algebra)**.** A Boolean algebra is a structure

$$B = (B, \vee, \wedge, ', 0, 1)$$

where $\vee$ and $\wedge$ are join and meet, 0 and 1 are bottom and top, and each element $a \in B$ has a complement a'. It satisfies the usual associativity, commutativity, absorption, distributivity, and complementation laws.

The example to keep in mind is the power set $\mathcal{P}(X)$. Then

$$A \vee B = A \cup B, \qquad A \wedge B = A \cap B, \qquad A' = X \setminus A.$$

So Boolean algebra is the algebra of regions that can be combined by union, intersection, and complement.

§10.4.ii Stone spaces as spaces with Boolean topology

A Stone space is a topological space whose topology is controlled by clopen sets.

Definition 10.4.2 (Stone space)**.** A Stone space is a compact Hausdorff totally disconnected space. Equivalently, it is a compact Hausdorff space with a basis of clopen subsets.

A subset is clopen if it is both closed and open. In ordinary connected spaces, clopen sets are rare. In a Stone space, clopen sets are abundant enough to form the logical basis of the space.

For any Stone space X, the clopen subsets form a Boolean algebra:

$$\mathrm{Clop}(X).$$

The Boolean operations are exactly the set operations:

$$U \vee V = U \cup V, \qquad U \wedge V = U \cap V, \qquad U' = X \setminus U.$$

This gives the direction

$$X \longmapsto \mathrm{Clop}(X).$$

§10.4.iii Why 2 is the dualizing object

The set $\mathbf{2} = \{0, 1\}$ is given the discrete topology. A continuous map

$$\chi : X \to \mathbf{2}$$

is the characteristic function of a clopen subset of X. Indeed,

$$\chi^{-1}(\{1\})$$

is clopen because $\{1\} \subset \mathbf{2}$ is clopen.

Thus

$$\mathrm{Hom}_{\mathbf{Top}}(X, \mathbf{2}) \cong \mathrm{Clop}(X).$$

This is the first half of the row: the algebra of logic on a Stone space is the algebra of truth-valued continuous observations.

§10.4.iv Ultrafilters as points

Now start from a Boolean algebra B. A point of the Stone space of B is not given in advance. It must be reconstructed from the algebra.

Definition 10.4.3 (Filter and ultrafilter)**.** A filter $F \subseteq B$ is a subset such that $1 \in F$, $0 \notin F$, it is closed under finite meets, and it is upward closed. An ultrafilter is a maximal proper filter.

An ultrafilter is a complete truth assignment. For every proposition $a \in B$, exactly one of a and a' belongs to the ultrafilter. So an ultrafilter decides every proposition in a way compatible with Boolean logic.

Let

$$S(B) = \{\mathfrak{u} : \mathfrak{u} \text{ is an ultrafilter on } B\}.$$

For each $a \in B$, define

$$U_a = \{\mathfrak{u} \in S(B) : a \in \mathfrak{u}\}.$$

The sets U_a form a basis of clopen subsets. This turns $S(B)$ into a Stone space.

The map

$$a \longmapsto U_a$$

preserves Boolean operations:

$$U_{a\wedge b} = U_a \cap U_b, \qquad U_{a\vee b} = U_a \cup U_b, \qquad U_{a'} = S(B) \setminus U_a.$$

Theorem 10.4.4 (Stone representation theorem)
For every Boolean algebra B, the map

$$B \longrightarrow \mathrm{Clop}(S(B)), \qquad a \longmapsto U_a$$

is an isomorphism of Boolean algebras.

This is the missing geometric half of Boolean algebra. A Boolean algebra is not only a symbolic calculus of propositions. It is the algebra of clopen regions of a canonically reconstructed space.

§10.5 What Stone duality says in one sentence

Stone duality says:

logic has a topology.

The propositions are clopen subsets. The truth assignments are points. The Boolean algebra is the algebra of observable yes-or-no regions of the Stone space.

The dualizing object **2** explains both directions:

$$X \longmapsto \mathrm{Hom}(X, \mathbf{2}) \cong \mathrm{Clop}(X),$$

$$B \longmapsto \mathrm{Hom}(B, \mathbf{2}) \cong S(B).$$

On the geometric side, maps to **2** are clopen predicates. On the algebraic side, homomorphisms to **2** are truth assignments, equivalently ultrafilters.

This is what I had not fully seen in 2021. I understood the algebraic rules of Boolean algebra, but I had not yet seen that the rules reconstruct a space.

§10.6 Gelfand duality as a continuous analogue

The second row replaces truth-valued observations by complex-valued observations:

$$\text{compact Hausdorff spaces} \quad \longleftrightarrow \quad \text{commutative } C^*\text{-algebras}, \qquad \Omega = \mathbb{C}.$$

If X is compact Hausdorff, then $C(X)$, the algebra of continuous complex-valued functions on X, is a commutative unital C^*-algebra. This is the direction

$$X \longmapsto C(X) = \mathrm{Hom}(X, \mathbb{C})$$

in a suitably interpreted sense.

Conversely, if A is a commutative unital C^*-algebra, its spectrum is the set of characters

$$\mathrm{Spec}(A) = \mathrm{Hom}_{C^*\text{-alg}}(A, \mathbb{C}).$$

These characters are the points of the reconstructed compact Hausdorff space.

Theorem 10.6.1 (Gelfand duality)

The category of compact Hausdorff spaces is contravariantly equivalent to the category of commutative unital C^*-algebras. In particular,

$$A \cong C(\mathrm{Spec}(A))$$

for commutative unital C^*-algebras A.

The analogy with Stone duality is direct:

Stone duality	Gelfand duality
2-valued observations	$\mathbb{C}$-valued observations
clopen predicates	continuous functions
ultrafilters	characters
Boolean logic	commutative functional analysis

§10.7 Affine schemes and the affine line

The third row is algebraic geometry:

$$\text{affine schemes} \quad \longleftrightarrow \quad \text{commutative rings}, \qquad \Omega = \mathbb{A}^1.$$

The affine line $\mathbb{A}^1$ is the object that represents regular functions. For an affine scheme X, maps

$$X \to \mathbb{A}^1$$

are regular functions on X. Thus the ring of functions on X is recovered as

$$\mathrm{Hom}(X, \mathbb{A}^1).$$

Conversely, a commutative ring R gives the affine scheme

$$\mathrm{Spec}\, R,$$

whose points are prime ideals of R. The slogan is the same: an algebraic space is reconstructed from its algebra of functions.

This row is close in spirit to Gelfand duality, but the geometry is algebraic rather than analytic. The measuring object is no longer $\mathbb{C}$ as a topological field of values, but $\mathbb{A}^1$ as the universal carrier of regular functions.

§10.8 Locales and the Sierpiński space

The fourth row is

$$\text{locales} \quad \longleftrightarrow \quad \text{frames}, \qquad \Omega = \mathbb{S}.$$

A locale is a point-free space. Instead of starting with points and then defining open sets, one starts with the lattice of opens.

Definition 10.8.1 (Frame)**.** A frame is a complete lattice L in which finite meets distribute over arbitrary joins:

$$a \wedge \bigvee_i b_i = \bigvee_i (a \wedge b_i).$$

For a topological space X, the lattice $\mathcal{O}(X)$ of open subsets is a frame. Locale theory studies this frame as the primary object.

The dualizing object here is the Sierpiński space $\mathbb{S}$, with two points $\{0,1\}$, where $\{1\}$ is open but $\{0\}$ is not. The key fact is:

$$\mathrm{Hom}(X, \mathbb{S}) \cong \mathcal{O}(X).$$

A continuous map $X \to \mathbb{S}$ is exactly the characteristic function of an open subset of X.

This makes the third-column idea completely visible. The Sierpiński space is not merely a two-point curiosity. It is the classifier of open sets.

§10.9 Topoi, logoi, and higher versions

The last rows are much larger in scope:

$$\text{topoi} \quad \longleftrightarrow \quad \text{logoi}, \qquad \Omega = \mathbf{Set},$$

and

$$\infty\text{-topoi} \quad \longleftrightarrow \quad \infty\text{-logoi}.$$

A topos can be read as a generalized space, often presented as a category of sheaves. It is a place where mathematics can be carried out internally. A logos is the corresponding logical or algebraic presentation.

The row is harder than Stone duality, but the analogy is still useful:

$$\text{space of models} \quad \longleftrightarrow \quad \text{logic/theory of that space}.$$

The topos of sets **Set** acts as a reference universe. For ∞-topoi, ordinary sets are replaced by spaces or ∞-groupoids, so that higher homotopical information is retained rather than collapsed.

For this note, the details of logoi are less important than the direction of generalization. The table begins with Boolean truth values and ends with entire universes of mathematics as measuring objects.

§10.10 The formal meaning of duality

Categorically, a duality between categories $\mathcal{C}$ and $\mathcal{D}$ is an equivalence

$$\mathcal{C} \simeq \mathcal{D}^{op}.$$

Equivalently, there are contravariant functors

$$F : \mathcal{C} \to \mathcal{D}, \qquad G : \mathcal{D} \to \mathcal{C}$$

which recover the original objects up to natural isomorphism.

In many rows of the table, the functors are built by homming into a dualizing object. This is why the third column matters. It is the object that converts a space into its algebra of observations and an algebra into its space of models.

So the statement is not simply:

algebra is the dual of geometry.

A more precise statement is:

for suitable categories, a chosen measuring object
produces a contravariant equivalence.

The geometry/algebra distinction is then a matter of viewpoint. A space can be treated through its functions. An algebra can be treated through its spectrum of points.

§10.11 What I should have understood in 2021

The table's first row was already enough to contain the lesson. A Boolean algebra is not only a formal calculus of $\vee$, $\wedge$, and complement. It has a Stone space of complete truth assignments. Conversely, a Stone space is not only a topological object. It has a Boolean algebra of clopen predicates.

The right summary is:

points are models or truth assignments,
propositions are regions,
the dualizing object tells us what kind of question is being asked.

In Stone duality, the question is yes-or-no, so the dualizing object is $\mathbf{2}$. In Gelfand duality, the question is complex-valued measurement, so the dualizing object is $\mathbb{C}$. In locale theory, the question is openness, so the dualizing object is the Sierpiński space. In topos theory, the question becomes internal mathematics itself.

This is the missing half of the 2021 Boolean algebra study. I had understood the syntax of Boolean operations, but not yet the topology of truth assignments. The real lesson of the table is:

space is recovered from its observations,
logic is recovered as the topology of its truth assignments.

Boolean Algebras, Boolean Rings, and the First Hint of Stone Duality

N-20210804

§11.1 Boolean algebra beyond computing

This note began from a small but common misunderstanding: Boolean algebra is often first seen through computer science, so it is easy to think that it is mainly a computational convention rather than a mathematical structure. I want to push against that impression.

Boolean algebra belongs simultaneously to logic, set theory, topology, measure theory, ring theory, and later to categorical duality. The original motivation was a comparison between Stone duality for Boolean algebras and broader dualities involving topological or logical structures. I also had Heyting algebras in mind, although at this stage it is better not to start there.

> **Remark 11.1.1** — The planned route is deliberately elementary: first collect the algebraic vocabulary, then introduce Boolean rings and Boolean algebras, and only later move toward Stone spaces, Stone duality, and perhaps Heyting algebras. The point is not to pretend that Boolean algebra is difficult at the entry level, but to show that it has more mathematical depth than its first appearance suggests.

§11.2 A first definition via Heyting algebras

One compact definition is the following.

Definition 11.2.1. A Boolean algebra is a Heyting algebra satisfying the law of excluded middle. Equivalently, for every element x,

$$\neg\neg x = x,$$

or equivalently

$$x \vee \neg x = \top.$$

> **Caution**
>
> This definition is elegant but not beginner-friendly. If one has not learned Heyting algebras, it hides the concrete content. So in the rest of this chapter I reconstruct Boolean algebras through groups, rings, lattices, subsets, and characteristic functions.

§11.3 A small amount of algebra

Let S be a nonempty set. A binary operation on S is a map

$$S \times S \to S.$$

Depending on context we may denote the value on (a, b) by ab, $a + b$, $a - b$, $a * b$, or $a \circ b$.

Definition 11.3.1. A semigroup is a nonempty set G with an associative binary operation:

$$a(bc) = (ab)c.$$

If there is an element $e \in G$ such that $ae = ea = a$, and every element $a \in G$ has an inverse a^{-1} with

$$aa^{-1} = a^{-1}a = e,$$

then G is a group. If $ab = ba$ for all $a, b \in G$, then G is an abelian group.

I will not need the full theory of group homomorphisms, kernels, images, cosets, normal subgroups, or quotient groups here, but the rough memory is useful: a quotient group G/N consists of cosets of a normal subgroup $N \lhd G$, with multiplication

$$(aN)(bN) = abN.$$

Definition 11.3.2. A ring R is a set with addition and multiplication such that $(R, +)$ is an abelian group, multiplication is associative, and multiplication distributes over addition:

$$a(b + c) = ab + ac, \qquad (b + c)a = ba + ca.$$

If multiplication is commutative, the ring is commutative. If there is an element $1 \in R$ with $1a = a1 = a$, the ring has a unit.

Definition 11.3.3. A lattice is a partially ordered set in which finite meets and finite joins exist. We write

$$p \wedge q, \qquad p \vee q$$

for meet and join.

> **Remark 11.3.4** — This algebra review is intentionally minimal. The important point is that Boolean algebra can be approached from two sides: as a special ring and as a special lattice.

§11.4 Boolean rings

The smallest nontrivial ring with unit is

$$\mathbb{F}_2 = \{0, 1\},$$

with addition and multiplication modulo 2:

$$\begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \qquad \begin{array}{c|cc} \cdot & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}$$

Every element satisfies $p^2 = p$. This condition is the key.

Definition 11.4.1. A Boolean ring is a ring R in which every element is idempotent:

$$x^2 = x \qquad (x \in R).$$

Some definitions also require a unit, but the idempotence condition is the essential feature.

Proposition 11.4.2

Every Boolean ring satisfies $x + x = 0$ for every $x \in R$, and every Boolean ring is commutative.

Proof. Since $x + x$ is idempotent,

$$x + x = (x + x)^2 = x^2 + xx + xx + x^2 = x + x + 2x^2.$$

Thus $2x^2 = 0$, and since $x^2 = x$, we get $2x = 0$.

Next,

$$x + y = (x + y)^2 = x^2 + xy + yx + y^2 = x + y + xy + yx.$$

So $xy + yx = 0$. But every element is its own additive inverse, hence $xy = yx$. □

§11.5 Examples of Boolean rings

The two-dimensional example is

$$\mathbb{F}_2^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

with coordinatewise operations:

$$(p_0, p_1) + (q_0, q_1) = (p_0 + q_0, p_1 + q_1),$$

$$(p_0, p_1)(q_0, q_1) = (p_0q_0, p_1q_1).$$

The zero and unit are

$$(0, 0), \qquad (1, 1).$$

The same construction gives $\mathbb{F}_2^n$.

Another standard example is a ring of sets.

Example 11.5.1

Let $\mathcal{R}$ be a class of sets such that if $E, F \in \mathcal{R}$, then

$$E \cup F \in \mathcal{R}, \qquad E - F \in \mathcal{R}.$$

Define

$$E + F = (E \setminus F) \cup (F \setminus E),$$

that is, symmetric difference, and define

$$EF = E \cap F.$$

Then the set-theoretic laws match the Boolean ring laws.

Remark 11.5.2 — This is also why Boolean algebra naturally appears in measure theory. Collections of measurable sets, especially modulo null sets, behave like Boolean algebras or Boolean rings.

§11.6 Boolean algebras directly

Let X be a set. The power set $\mathcal{P}(X)$ has union, intersection, and complement:

$$P \cup Q, \qquad P \cap Q, \qquad P^c = X \setminus P.$$

This is the concrete model for Boolean algebra.

Definition 11.6.1. A Boolean algebra is a nonempty set A equipped with two binary operations $\vee, \wedge$, a unary operation $(')$, and elements $0, 1$, satisfying the identities modeled on union, intersection, and complement. In particular,

$$p \wedge 0 = 0, \quad p \wedge 1 = p, \quad p \wedge p' = 0, \quad p \wedge p = p,$$

$$p \vee 1 = 1, \quad p \vee 0 = p, \quad p \vee p' = 1, \quad p \vee p = p,$$

$$(p')' = p, \qquad (p \wedge q)' = p' \vee q', \qquad (p \vee q)' = p' \wedge q',$$

together with commutativity, associativity, and distributivity.

Caution

One sentence in the original note has an obvious slip: it says, in effect, that every Boolean algebra can be turned into a Boolean ring and every Boolean algebra can be turned into a Boolean algebra. The intended statement is that Boolean algebras and Boolean rings with unit are two equivalent presentations of the same structure.

§11.7 Characteristic functions and the meaning of 2^X

Every subset $P \subseteq X$ corresponds to a characteristic function

$$\chi_P : X \to \mathbb{F}_2, \qquad \chi_P(x) = \begin{cases} 1, & x \in P, \\ 0, & x \notin P. \end{cases}$$

This gives a natural bijection

$$\mathcal{P}(X) \cong 2^X.$$

Thus writing the power set as 2^X is not merely a notational habit: a subset is the same thing as a $0/1$-valued function.

Under this identification, pointwise operations on functions become set operations:

$$P + Q = (P \cap Q^c) \cup (P^c \cap Q),$$

which is symmetric difference, and

$$PQ = P \cap Q.$$

The zero and unit are

$$0 = \varnothing, \qquad 1 = X.$$

§11.8 Moving between the two structures

From a Boolean ring R with unit, define

$$p \wedge q = pq, \qquad p \vee q = p + q + pq, \qquad p' = p + 1.$$

These operations make R a Boolean algebra.

Conversely, from a Boolean algebra A, define

$$p + q = (p \wedge q') \vee (p' \wedge q), \qquad pq = p \wedge q.$$

This makes A a Boolean ring.

> **Remark 11.8.1 —** The formulas are not magic. In the subset model, addition is exactly symmetric difference, multiplication is intersection, and complement is symmetric difference with the whole universe.

§11.9 The logical shadow

The operations $\wedge, \vee, '$ correspond to "and", "or", and "not". The elements 0 and 1 correspond to false and true. This is why Boolean algebra is the algebra of classical propositional logic.

> **Remark 11.9.1 —** This also explains why Heyting algebras are the natural next object: Boolean algebras model classical logic, while Heyting algebras model intuitionistic logic. Stone duality then brings topology into the same picture.

§11.10 Filters, ultrafilters, and Stone spaces

The next natural step is to introduce filters, ultrafilters, Stone spaces, and the Stone representation theorem. This does not replace the dictionary between Boolean rings, Boolean algebras, sets, and logic; it explains why that dictionary has a topological shadow.

Let B be a Boolean algebra. A filter $F \subseteq B$ is a collection of propositions regarded as "large" or "accepted". It satisfies three basic rules:

$$1 \in F, \qquad 0 \notin F,$$

if $a, b \in F$, then $a \wedge b \in F$, and if $a \in F$ and $a \leq b$, then $b \in F$. Thus a filter is closed under strengthening by consequence and under finite conjunction.

An ultrafilter is a maximal proper filter. Equivalently, for every $a \in B$, exactly one of a and a' belongs to the ultrafilter. This makes an ultrafilter look like a complete truth assignment: every Boolean proposition is judged either true or false, compatibly with the Boolean operations.

The Stone space of B, denoted $S(B)$, is the set of all ultrafilters on B. For each $a \in B$, define

$$U_a = \{\mathfrak{u} \in S(B) : a \in \mathfrak{u}\}.$$

These subsets form a basis of clopen sets. The Boolean operations are reflected topologically by

$$U_{a \wedge b} = U_a \cap U_b, \qquad U_{a \vee b} = U_a \cup U_b, \qquad U_{a'} = S(B) \setminus U_a.$$

So the algebra B is represented inside the topology of $S(B)$ as the Boolean algebra of clopen sets.

Stone's representation theorem says, in one common form, that every Boolean algebra is isomorphic to the Boolean algebra of clopen subsets of its Stone space. The finite example is easy to remember: if $B = \mathcal{P}(X)$, then ultrafilters are just choices of points of X, and the Stone space recovers X with the discrete topology. For infinite Boolean algebras, non-principal ultrafilters appear, and the resulting Stone space is compact, Hausdorff, and totally disconnected.

§11.11 Boolean rings from Boolean algebras

Given a Boolean algebra, define addition by symmetric difference

$$A + B = (A \setminus B) \cup (B \setminus A)$$

and multiplication by intersection

$$AB = A \cap B.$$

Then every element satisfies $A^2 = A$, so the resulting ring is Boolean. Conversely, a Boolean ring gives a Boolean algebra by defining meet as multiplication and join by

$$a \vee b = a + b + ab.$$

This equivalence explains why logic, sets, and algebra keep producing the same identities.

§11.12 Stone duality as a first hint

Stone duality says that Boolean algebras correspond to certain topological spaces, namely Stone spaces. The rough idea is that algebraic operations encode clopen subsets of a space. Thus Boolean logic can be represented geometrically. This is one of the earliest examples of algebra and topology reflecting each other.

Linear Logic: Resources, Dialogue, Narrative, and Geometry

N-20221005

§12.1 First orientation

This note is reconstructed from a ten-page PDF titled *The Geometry of Dialogue and the Logic of Narrative: A Taste of Linear Logic*, dated 2022-10-05. The original text was not a conventional mathematical note. It was closer to a reflective synthesis: a recent immersion into linear logic, several remarks attributed to Dr. Keyao Peng, an interest in game design and emergent narrative, and a set of visual fragments from Ludics, sequent calculus, and Geometry of Interaction.

The background matters. The original memo was written at the moment when logic stopped looking like a static taxonomy of truth values and began to look like a theory of interaction. The guiding words were:

interaction, resource, game, narrative.

The reconstruction below keeps this origin, but changes the form. Instead of an essay moving from metaphor to metaphor, this note is organized as a usable mathematical notebook: definitions first, then small calculi, then code-like models, and finally the more philosophical interpretation.

> **Remark 12.1.1 —** The original PDF repeatedly tried to connect three worlds: a child's fable, a philosophical debate, and high-level topology or geometry of interaction. The stable mathematical bridge between them is linear logic's treatment of resources and duality.

§12.2 From truth to resources

Classical logic treats propositions as reusable truths. If A is available, then one may use A as often as needed. This is encoded by the structural rules:

weakening: ignore a hypothesis, contraction: duplicate a hypothesis.

Linear logic removes these structural rules by default. A hypothesis is a resource. If it is consumed by an inference, it is no longer freely available.

Definition 12.2.1 (Linear implication)**.** The linear implication

$$A \multimap B$$

means: given one resource of type A, one may consume it to produce one resource of type B. It is not the same as the classical implication $A \to B$, because A is not automatically reusable.

The exponential modality ! marks reusable resources. A useful slogan is:

$$A \to B \quad \text{behaves like} \quad !A \multimap B.$$

The ordinary implication first makes A reusable, then spends it linearly.

classical reading	linear reading	informal meaning
$A \to B$	$!A \multimap B$	reusable assumption produces B
$A \otimes B$	tensor	both resources are present together
$A\&B$	additive conjunction	one must be ready for either choice
$A \oplus B$	additive disjunction	one side is chosen by the prover

The essential change is not symbolic. It is operational. Linear logic asks what a proof *does* to its premises.

§12.3 Narrative as a resource pipeline

The PDF's clearest concrete example is the story of the Three Little Pigs. The story is treated as a multiset-rewriting system. Characters and materials are resources. Events are linear implications.

Let the initial state be

$$\Delta_0 = \{\texttt{pig}, \texttt{pig}, \texttt{pig}, \texttt{straw}, \texttt{sticks}, \texttt{bricks}, \texttt{wolf}\}.$$

The story rules are:

$$\begin{aligned}
r_1 &: \texttt{pig} \otimes \texttt{straw} \multimap \texttt{straw_house},\\
r_2 &: \texttt{pig} \otimes \texttt{sticks} \multimap \texttt{stick_house},\\
r_3 &: \texttt{pig} \otimes \texttt{bricks} \multimap \texttt{brick_house},\\
r_4 &: \texttt{wolf} \otimes \texttt{straw_house} \multimap \texttt{wolf},\\
r_5 &: \texttt{wolf} \otimes \texttt{stick_house} \multimap \texttt{wolf},\\
r_6 &: \texttt{wolf} \otimes \texttt{brick_house} \multimap \texttt{brick_house}.
\end{aligned}$$

The canonical ending of the fable is the sequent

$$\Delta_0 \vdash \texttt{brick\_house}.$$

This says: starting from the initial resources, there is a linear proof whose final surviving resource is the brick house.

Example 12.3.1 (A code-level model of the fable)

The following pseudo-Python is often clearer than a large proof tree.

```
from collections import Counter

state = Counter({
    "pig": 3,
    "straw": 1,
    "sticks": 1,
    "bricks": 1,
    "wolf": 1,
})

rules = [
    (("pig", "straw"),        "straw_house"),
    (("pig", "sticks"),       "stick_house"),
    (("pig", "bricks"),       "brick_house"),
    (("wolf", "straw_house"), "wolf"),
    (("wolf", "stick_house"), "wolf"),
    (("wolf", "brick_house"), "brick_house"),
]

def apply(rule, state):
    inputs, output = rule
    new_state = state.copy()
    for x in inputs:
        if new_state[x] <= 0:
            return None
        new_state[x] -= 1
    new_state[output] += 1
    return +new_state

story = [rules[0], rules[1], rules[2], rules[3], rules[4], rules[5]]
for rule in story:
    state = apply(rule, state)

assert state == Counter({"brick_house": 1})
```

This is exactly the resource-sensitive reading: a pig used to build the straw house is consumed; it cannot simultaneously remain an unused pig. The wolf survives the destruction of the straw and stick houses, but not the encounter with the brick house.

The proof tree in the original PDF should therefore be read as a storyboard. Each inference step is not only a logical step but a plot event. The proof search is the narrative search.

§12.4 Tensor, additives, and branching stories

The tensor $A \otimes B$ means that both resources are available together. In the fable,

$$\texttt{pig} \otimes \texttt{straw}$$

means that a pig and straw must both be present for the event r_1 to fire.

Additives model choice. The connective $A \& B$ means that the environment may choose which side to test; the proof must be ready for both. This is useful for games and branching narratives.

A compact narrative grammar is:

linear-logic object	narrative reading
Δ	current multiset of actors and props
$A \otimes B$	co-presence of resources
$A \multimap B$	event consuming A and producing B
$A \& B$	branch where the environment chooses
$A \oplus B$	branch where the system chooses
$!A$	persistent rule or reusable condition

This explains why linear logic is attractive for game design. It tracks what is actually available after each action.

§12.5 Dialogue as a game

The second major theme of the PDF is Dr. Peng's comment that linear logic opened a "new world" because logic ascends to the level of interaction, or even to the level of a game. In game semantics:

$$\text{propositions are games,} \qquad \text{proofs are strategies.}$$

A proof of A is not merely a certificate that A is true. It is a strategy for the Proponent to win the game associated with A, no matter how the Opponent plays.

Definition 12.5.1 (Game-semantical slogan)**.** A formula specifies a protocol of interaction. A proof specifies a winning strategy in that protocol.

The cut rule has a dynamic meaning. If one proof produces A and another proof consumes A, cutting them together makes the two strategies interact. Cut elimination is the execution of that interaction until no internal communication remains.

$$\begin{gathered}\text{proof of } A \\ \Downarrow \\ \text{strategy producing moves of type } A \\ \Downarrow \\ \text{cut with a counter-strategy} \\ \Downarrow \\ \text{interaction / normalization / play}\end{gathered}$$

§12.6 Schopenhauer's stratagem as a ludic interaction

The PDF's second example uses a Schopenhauer-style debate. The Opponent says:

> "He is a child, so you must make allowance for him."

The Proponent answers:

> "Precisely because he is a child, I must correct him; otherwise he will persist in bad habits."

The content of the debate is less important than its structure. The Proponent accepts one premise and flips the other.

A code-like representation is:

```
O = {
    "P1": "he is a child",
    "P2": "one must make allowance for children",
    "claim": "do not correct him",
}

P = {
    "accept": "P1",
    "negate": "P2",
    "counterclaim": "because he is a child, correct him",
    "support": "otherwise he will persist in bad habits",
}

def retorsio_argumenti(opponent, proponent):
    assert proponent["accept"] == "P1"
    return {
        "kept": opponent["P1"],
        "reversed": opponent["P2"],
        "new_claim": proponent["counterclaim"],
    }
```

This is close to the Ludics interpretation. A debate is a tree of possible moves. A winning strategy is a path through that tree that answers every possible continuation by the opponent.

§12.7 Ludics: loci instead of signifieds

The PDF's phrase "the signifier chain without the signifier" should not be read merely as literary theory. In Girard's Ludics, formulas are replaced by loci: addresses at which interaction takes place.

Definition 12.7.1 (Locus, informal)**.** A locus is an address in a design. Instead of saying that a proof manipulates formulas with fixed semantic content, Ludics says that designs interact at addresses.

The meaning of a proposition is therefore not hidden behind the symbol as a static signified. Meaning is generated by interaction: by how a design responds to its dual.

This gives a cleaner version of the PDF's Lacanian metaphor:

meaning is not stored inside a sign;
meaning appears through interaction with its negation.

§12.8 Orthogonality and negation

A central operation in Ludics and linear logic is orthogonality. Very roughly, two designs are orthogonal if their interaction terminates properly.

Definition 12.8.1 (Orthogonality, notebook version)**.** Write $D \perp E$ when the interaction between designs D and E converges successfully. Then the orthogonal of a collection A is

$$A^{\perp} = \{E : \forall D \in A,\ D \perp E\}.$$

A behavior is often recovered by biorthogonal closure:

$$A = (A^{\perp})^{\perp}.$$

This is the mathematically controlled form of the PDF's claim that a proposition is defined by its negation. One does not understand A by staring at A alone. One understands A through the class of all counter-designs with which it can interact.

§12.9 Geometry of interaction

Geometry of Interaction is the diagrammatic and dynamical side of the same idea. A proof is not a static derivation tree but a flow of information. The PDF's single most useful visual artifact is a colored morphism diagram from Melliès-style topologies.

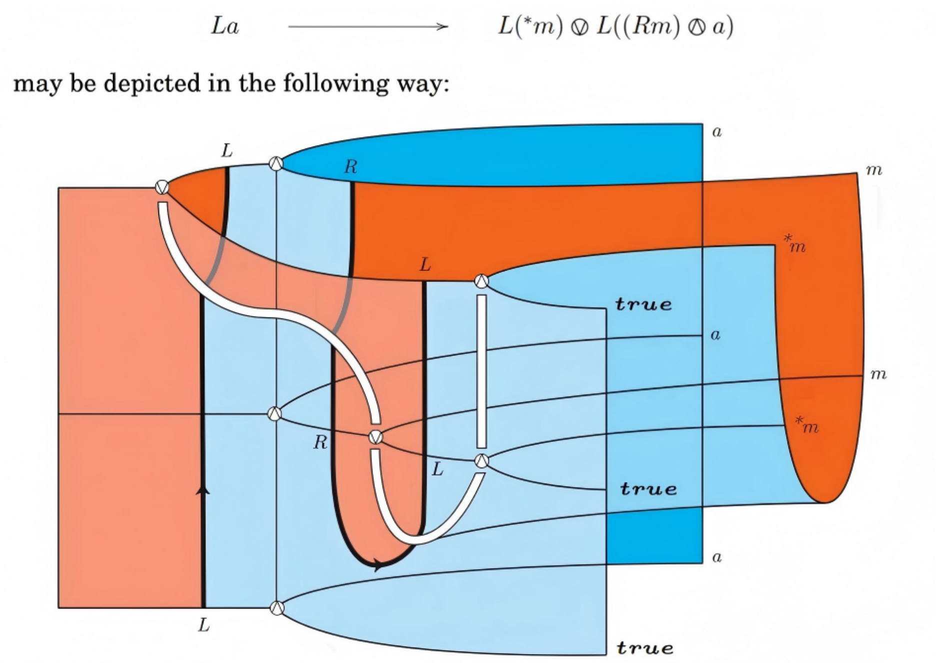


Figure 12.1: Geometry-of-Interaction morphism.

The visible morphism has the form

$$La \longrightarrow L(^{*}m) \otimes L((Rm) \otimes a).$$

For a first reading, the notation is less important than the picture. The colored ribbons behave like wires. They route input types into output types, bend around each other, and meet at nodes. This is why the original PDF compares human interaction to Feynman diagrams: logical equivalence becomes a kind of topological deformation of processes.

Remark 12.9.1 — The diagram should not be treated as a decorative metaphor. In Geometry of Interaction, cut elimination is modeled by the circulation of information through a network. The graphical form is an attempt to make proof normalization visible.

§12.10 From narrative to social relations

The original PDF ends by extending the same logic to narrative generation and social relations. This is speculative, but it has a precise core. A social or narrative process can be modeled as a resource transformation system:

$$\text{inputs} \otimes \text{labor/process} \multimap \text{outputs}.$$

The important constraint is that resources do not duplicate for free. If a theory of narrative or society assumes arbitrary reuse of agents, time, attention, or material goods, it is silently using weakening and contraction. Linear logic forces these assumptions into the open.

A minimal production rule looks like:

```
production_rule = {
    "inputs":  ["labor", "wood", "tools"],
    "output":  "table",
    "logic":   "labor tensor wood tensor tools -o table",
}
```

The same pattern covers the fable rule

$$\texttt{pig} \otimes \texttt{bricks} \multimap \texttt{brick\_house}.$$

This is why the PDF's jump from the Three Little Pigs to social production is not arbitrary. Both are resource-sensitive narratives.

§12.11 The general pattern

The reconstructed message is simple:

$$\begin{gathered}\text{logic is not only truth classification}\\ \Downarrow\\ \text{a proof is a resource-sensitive process}\\ \Downarrow\\ \text{a process can be read as a game}\\ \Downarrow\\ \text{a game can be read as dialogue}\\ \Downarrow\\ \text{dialogue can be visualized as Geometry of Interaction.}\end{gathered}$$

The first version of the PDF used large philosophical language: signifier chains, narrative, game, topology, dialectics, social relations. The cleaned-up mathematical version keeps the ambition but anchors it in four concrete devices:

$$A \multimap B, \qquad A \otimes B, \qquad A^{\perp}, \qquad A = (A^{\perp})^{\perp}.$$

Linear logic is the logic of finite resources; Ludics is the logic of interaction at addresses; game semantics is the logic of strategies; Geometry of Interaction is the logic of information flow. Together they explain why a proof can look like a story, a debate, or a diagram of motion.

Category Theory Continued and the Question Topology

N-20221227

§13.1 More examples of categories

This note continues the category-theory examples. Abelian groups with group homomorphisms form **Ab**. Modules over a ring A form $\mathbf{Mod}_A$. Rings with unital ring homomorphisms form **Rings**. Topological spaces with continuous maps form **Top**, whose isomorphisms are homeomorphisms.

A partially ordered set $(S, \geq)$ can be regarded as a category: the objects are elements of S, and there is exactly one morphism $x \to y$ iff $x \geq y$. Reflexivity gives identities, transitivity gives composition, and antisymmetry ensures isomorphisms are trivial.

§13.2 Subcategories and functors

A subcategory $\mathcal{A} \subseteq \mathcal{C}$ has some objects and some morphisms of $\mathcal{C}$, closed under identities and composition. A functor

$$F : \mathcal{C} \to \mathcal{D}$$

sends objects to objects and morphisms to morphisms, preserving identities and composition:

$$F(1_A) = 1_{F(A)}, \qquad F(g \circ f) = F(g) \circ F(f).$$

The diagram in the original page is a standard functoriality square: the image of a composable path under F is still composable, and the composite maps to the composite.

§13.3 Forgetful functors

A forgetful functor forgets structure. For example,

$$\mathbf{Grp} \to \mathbf{Set}$$

sends a group to its underlying set and a homomorphism to its underlying function. Similarly,

$$\mathbf{Top} \to \mathbf{Set}$$

forgets open sets, and $\mathbf{Vect}_k \to \mathbf{Set}$ forgets vector-space operations.

The note's warning is that a forgetful functor is not innocent: once structure is forgotten, many questions cannot be answered anymore. A bijection of underlying sets need not be an isomorphism of groups, vector spaces, or topological spaces.

§13.4 A topology generated by a distance-like question

The second half moves into topology. The page considers a collection of subsets and asks what topology it generates. Given a basis-like family $\mathcal{B}$, the topology generated by it consists of arbitrary unions of finite intersections of elements of $\mathcal{B}$.

The displayed rational-looking function and its graph on the page are used as a reminder: topology is not only about familiar Euclidean open intervals. One can impose a topology by declaring certain sets to be basic neighborhoods.

§13.5 Posets as categories, with an example

A partially ordered set becomes a category in a very economical way. There is a unique morphism $x \to y$ exactly when $x \leq y$. Identity morphisms come from reflexivity $x \leq x$, and composition comes from transitivity: if $x \leq y$ and $y \leq z$, then $x \leq z$.

This example is important because it shows that a category need not look like a collection of algebraic objects and homomorphisms. A category can encode logic or order. Meets and joins in a poset become limits and colimits in the corresponding category.

§13.6 Functorial thinking

A functor is not merely a map between categories. It is a structure-preserving translation. If $F : \mathcal{C} \to \mathcal{D}$, then for every composable pair

$$A \xrightarrow{f} B \xrightarrow{g} C$$

we require

$$F(g \circ f) = F(g) \circ F(f).$$

This condition says that doing a construction after composing maps gives the same result as first applying the construction to each map and then composing.

Examples clarify the point. The fundamental group construction is a functor from pointed topological spaces to groups. Homology is a functor from spaces to graded abelian groups. The forgetful functor from groups to sets forgets multiplication but still sends homomorphisms to functions.

§13.7 Generated topology

The topology part of the note concerns how open sets can be generated from a chosen collection. If $\mathcal{B}$ is intended as a basis, two conditions are expected: every point lies in some basis element, and intersections of basis elements are locally refined by basis elements. Then every open set is a union of basis elements.

If one starts with a subbasis $\mathcal{S}$, the generated topology consists of arbitrary unions of finite intersections of elements of $\mathcal{S}$. This explains the phrase "generated by": we first force the chosen sets to be open, then close under the axioms of topology.

§13.8 Why category theory appears beside topology

Topological constructions are full of universal properties. Product topology is the coarsest topology making the projection maps continuous. Quotient topology is the finest topology making the quotient map continuous. These statements are categorical: they characterize objects by how maps into or out of them behave.

Thus the note's shift from categories to topology is natural. Topology is not only a study of open sets; it is a study of spaces through continuous maps, and that is exactly categorical thinking.

§13.9 The larger pattern

Category theory and topology meet through structure-preserving maps. A functor preserves categorical structure; a continuous map preserves open-set structure. A forgetful functor deliberately discards structure. This is why the note naturally moves from categories to topology: both ask what data must be preserved for two objects to be considered the same.

§13.10 Posets as categories and logic

A poset viewed as a category has at most one morphism between any two objects. In this setting, categorical statements become order-theoretic statements. A product is a meet, a coproduct is a join, an initial object is a least element, and a terminal object is a greatest element.

For example, in the poset of open subsets of a space ordered by inclusion, intersections behave like meets and unions behave like joins. This is why topology and order theory are so close: open sets form a lattice, and categorical language packages its universal properties.

§13.11 Forgetful functors and free constructions

A forgetful functor loses structure, but it often has a left adjoint that freely adds structure. For instance,

$$U : \mathbf{Grp} \to \mathbf{Set}$$

forgets the group operation. Its left adjoint sends a set S to the free group $F(S)$. The adjunction says that a set map

$$S \to U(G)$$

corresponds uniquely to a group homomorphism

$$F(S) \to G.$$

This is the categorical form of "define a homomorphism out of a free object by choosing images of generators."

§13.12 Initial topology generated by questions

The topology generated by a family of maps can be understood as an initial topology. Suppose we have functions

$$f_i : X \to Y_i$$

where each Y_i is already topological. The initial topology on X is the coarsest topology making all f_i continuous. Its subbasis is

$$f_i^{-1}(U), \qquad U \subseteq Y_i \text{ open}.$$

This matches the note's "topology generated by a question" intuition: each map f_i is a way of observing X, and the generated topology is exactly the open-set structure forced by those observations.

§13.13 Product topology as an initial topology

The product topology is the initial topology for the projection maps

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y.$$

A map $h : Z \to X \times Y$ is continuous iff the coordinate maps $\pi_X \circ h$ and $\pi_Y \circ h$ are continuous. This is a categorical universal property, not a visual fact about rectangles.

The rectangle basis $U \times V$ appears because

$$U \times V = \pi_X^{-1}(U) \cap \pi_Y^{-1}(V).$$

Thus the usual construction of product open sets is forced by the initial-topology viewpoint.

§13.14 Quotient topology as a final topology

Dually, quotient topology is a final topology. For a quotient map

$$q : X \to Y,$$

the topology on Y is the finest topology making q continuous, and a map $f : Y \to Z$ is continuous iff $f \circ q : X \to Z$ is continuous.

Thus product and quotient are opposite-looking constructions:

initial topology	maps into known spaces	coarsest topology
final topology	maps from known spaces	finest topology

This is the precise categorical version of the topology examples in the note.

§13.15 Functors from topology to algebra

The note mentions that the fundamental group is functorial. More generally, algebraic topology studies functors such as

$$\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}, \qquad H_n : \mathbf{Top} \to \mathbf{Ab}.$$

A continuous map of spaces induces a homomorphism of groups or abelian groups. This turns geometric questions into algebraic ones.

The power of this move is that algebraic invariants can obstruct homeomorphisms. If two spaces have non-isomorphic fundamental groups or homology groups, they cannot be the same topological space.

§13.16 Initial and final structures in topology

The category-theory material is not separate from topology. Posets show that categories can encode order and logic. Functors express structure-preserving translation. Generated topologies, products, and quotients are governed by universal properties. Algebraic topology then uses functors to extract algebra from spaces. The common theme is: understand objects through the maps they admit.

Category Theory Continued: Posets, Groups as One-Object Categories, Products, and Monomorphisms

N-20230313

§14.1 What this note continues

This note continues the elementary category-theory reading. The source pages are not just a list of definitions. They contain several examples whose purpose is to change one's intuition about what an arrow may be. In ordinary algebra courses, a morphism is often a function preserving structure. Category theory is more flexible: arrows can be order relations, group elements, inclusions, or constructions whose nature depends on the category.

The main examples are:

$$\text{posets as categories}, \quad \text{groups as one-object categories},$$
$$\text{products of categories}, \quad \text{monomorphisms and epimorphisms}.$$

§14.2 Posets are categories

Let $(P, \leq)$ be a partially ordered set. We can build a category, still denoted P, as follows. The objects are the elements of P. For two objects $p, q \in P$, define

$$\mathrm{Hom}(p,q) = \begin{cases} \{*\}, & p \leq q, \\ \varnothing, & p \not\leq q. \end{cases}$$

Thus there is at most one arrow from p to q. The identity arrow exists because $p \leq p$. Composition exists because order is transitive: if $p \leq q$ and $q \leq r$, then $p \leq r$.

This example is surprisingly important. It shows that arrows in a category need not be functions. In the poset category, the arrow $p \to q$ is the fact that $p \leq q$.

§14.3 A small poset example

Suppose P has four objects a, b, c, d with relations

$$a \leq b, \qquad a \leq c \leq d, \qquad a \leq d,$$

and identities at every object. The corresponding category has arrows exactly for these order relations. In a Hasse diagram, upward paths encode arrows, but the category contains the composite arrow as well. For example, from $a \leq c$ and $c \leq d$, the composite gives the arrow $a \to d$.

The handwritten note emphasizes the point: the same symbol $\leq$ can be treated as an arrow. This is not a metaphor; it satisfies the category axioms.

§14.4 Isomorphisms in a poset category

An isomorphism in a category is an arrow

$$f : A \to B$$

for which there exists

$$g : B \to A$$

such that

$$g \circ f = \mathrm{id}_A, \qquad f \circ g = \mathrm{id}_B\,.$$

In a poset category, an arrow $p \to q$ means $p \le q$. An isomorphism between p and q would require arrows both ways:

$$p \le q, \qquad q \le p.$$

By antisymmetry of a partial order, this forces $p = q$. Therefore no two distinct objects of a poset category are isomorphic.

This explains the source's short handwritten answer, "Obviously." It is obvious only after translating categorical isomorphism back into the order axioms.

§14.5 Groups as one-object categories

A group G can be regarded as a category $\mathcal{G}$ with one object $*$. The arrows from $*$ to itself are the elements of G:

$$\mathrm{Hom}_{\mathcal{G}}(*, *) = G.$$

Composition of arrows is group multiplication. The identity arrow is the identity element $e \in G$. Inverses of arrows are group inverses.

Thus a group is exactly a category with one object in which every arrow is an isomorphism. This is not a trick. It says that a group can be understood as the collection of all symmetries of one object, with composition as the operation.

§14.6 Why this example matters

If one first learns category theory, a group may look like a special category: one object, only reversible arrows. This reverses the usual learning order. The note quotes the idea that a group is what one gets by collecting all symmetries of an object. From the categorical viewpoint, a group is the extreme case where there is only one object and every transformation is invertible.

The slogan is:

$$\text{monoid} = \text{one-object category}, \qquad \text{group} = \text{one-object groupoid}.$$

A groupoid is a category in which every arrow is invertible.

§14.7 Products of categories

Given categories $\mathcal{C}$ and $\mathcal{D}$, their product category $\mathcal{C} \times \mathcal{D}$ has objects

$$(C, D), \qquad C \in \mathcal{C},\ D \in \mathcal{D},$$

and morphisms
$$(f,g):(C,D)\to(C',D')$$
where $f:C\to C'$ is a morphism in $\mathcal{C}$ and $g:D\to D'$ is a morphism in $\mathcal{D}$. Composition is componentwise:
$$(f',g')\circ(f,g)=(f'\circ f,\ g'\circ g).$$
The identity on (C,D) is $(\mathrm{id}_C,\mathrm{id}_D)$.

This is the categorical version of taking a product structure coordinate by coordinate.

§14.8 The opposite category

For any category $\mathcal{C}$, the opposite category $\mathcal{C}^{op}$ has the same objects as $\mathcal{C}$, but every arrow is reversed:
$$f:A\to B \text{ in } \mathcal{C} \quad\longleftrightarrow\quad f^{op}:B\to A \text{ in } \mathcal{C}^{op}.$$
Composition is reversed accordingly. The opposite category is a formal way to turn statements about maps out of an object into statements about maps into an object.

For posets, passing to the opposite category corresponds to reversing the order. If P is viewed as a category, then P^{op} is the poset with
$$p\le_{op} q \quad\Longleftrightarrow\quad q\le p.$$

§14.9 Monomorphisms and epimorphisms

In ordinary set theory, an injective function is one that does not identify distinct elements. Category theory cannot always talk about elements, so it defines a left-cancellation property instead.

A morphism $f:X\to Y$ is a monomorphism if for every pair of arrows $g,h:Z\to X$,
$$f\circ g=f\circ h \quad\Longrightarrow\quad g=h.$$
It is an epimorphism if for every pair of arrows $g,h:Y\to Z$,
$$g\circ f=h\circ f \quad\Longrightarrow\quad g=h.$$
In **Set**, monomorphisms are exactly injective functions, and epimorphisms are exactly surjective functions. In other categories, this may fail, which is why the categorical definitions are more fundamental.

§14.10 The lesson of the note

The note's guiding idea is that category theory asks us to stop identifying arrows with functions. In a poset, arrows are inequalities. In a group-as-category, arrows are group elements. In a product category, arrows are pairs of arrows. In an opposite category, arrows are formally reversed. Monomorphism and epimorphism then express injective-like and surjective-like behavior without mentioning elements.

The transition is from element-based thinking to arrow-based thinking. This is the first real shift in category theory.

Free Constructions, Adjunctions, Concrete Categories, and Group Actions

N-20230314

§15.1 A free object solves an extension problem

Let S be a set. The free group $F(S)$ consists of reduced words

$$s_1^{\varepsilon_1} s_2^{\varepsilon_2} \cdots s_n^{\varepsilon_n}, \qquad s_i \in S, \quad \varepsilon_i \in \{1, -1\},$$

with multiplication given by concatenation followed by cancellation of adjacent pairs ss^{-1} and $s^{-1}s$. The empty word is the identity.

There is a canonical function

$$\eta_S : S \to U(F(S)), \qquad s \longmapsto [s],$$

where $U : \mathbf{Grp} \to \mathbf{Set}$ forgets the group structure.

The point of $F(S)$ is not merely its word construction. It satisfies a universal property.

Theorem 15.1.1 (Universal property of the free group)
For every group G and every function $f : S \to U(G)$, there is a unique group homomorphism

$$\widetilde{f} : F(S) \to G$$

such that

$$U(\widetilde{f}) \circ \eta_S = f.$$

Proof. Define

$$\widetilde{f}\,(s_1^{\varepsilon_1} \cdots s_n^{\varepsilon_n}) = f(s_1)^{\varepsilon_1} \cdots f(s_n)^{\varepsilon_n}.$$

Cancelling ss^{-1} in a word does not change this value, so the formula is well defined. Concatenation becomes multiplication, hence $\widetilde{f}$ is a homomorphism. Any homomorphism agreeing with f on the generators must take every word to the displayed product, proving uniqueness. □

The free vector space has the same pattern. For a set S, let

$$k^{(S)} = \{a : S \to k : \operatorname{supp}(a) \text{ finite}\}.$$

A function $S \to U(V)$ extends uniquely to a linear map $k^{(S)} \to V$. The construction changes, but the universal problem is identical.

§15.2 The universal property is a natural Hom-set bijection

The extension theorem gives a bijection

$$\Phi_{S,G} : \operatorname{Hom}_{\mathbf{Grp}}(F(S), G) \longrightarrow \operatorname{Hom}_{\mathbf{Set}}(S, U(G)),$$

with

$$\Phi_{S,G}(\varphi) = U(\varphi) \circ \eta_S.$$

Its inverse sends a function $f : S \to U(G)$ to its unique extension $\widetilde{f}$.

This bijection is natural in both variables. If $a : S' \to S$, then

$$\Phi_{S',G}(\varphi \circ F(a)) = \Phi_{S,G}(\varphi) \circ a.$$

Indeed, both functions send s' to $\varphi([a(s')])$.

If $b : G \to H$ is a group homomorphism, then

$$\Phi_{S,H}(b \circ \varphi) = U(b) \circ \Phi_{S,G}(\varphi).$$

Thus the family $\Phi_{S,G}$ is not a collection of accidental bijections. It is a natural isomorphism

$$\boxed{\mathrm{Hom}_{\mathbf{Grp}}(F(-), -) \cong \mathrm{Hom}_{\mathbf{Set}}(-, U(-)).}$$

This is the adjunction

$$F \dashv U.$$

§15.3 Unit and counit encode the same adjunction

The unit of the adjunction is the natural transformation

$$\eta : \mathrm{Id}_{\mathbf{Set}} \Rightarrow UF,$$

whose component η_S inserts each element as a one-letter word.

The counit is

$$\varepsilon : FU \Rightarrow \mathrm{Id}_{\mathbf{Grp}}.$$

For a group G,

$$\varepsilon_G : F(U(G)) \to G$$

evaluates a word in the underlying elements of G:

$$\varepsilon_G(g_1^{\varepsilon_1} \cdots g_n^{\varepsilon_n}) = g_1^{\varepsilon_1} \cdots g_n^{\varepsilon_n} \quad \text{in } G.$$

The triangle identities are

$$\varepsilon_{F(S)} \circ F(\eta_S) = \mathrm{id}_{F(S)},$$

and

$$U(\varepsilon_G) \circ \eta_{U(G)} = \mathrm{id}_{U(G)}.$$

The second identity sends $g \in G$ to the one-letter word $[g]$, then evaluates it back to g. For the first, a generator $[s] \in F(S)$ is sent by $F(\eta_S)$ to the one-letter word whose letter is itself the word $[s]$; evaluation removes the outer layer. Since both sides are homomorphisms and agree on generators, they agree everywhere.

The Hom-set bijection and the unit–counit description determine one another. For example, given $f : S \to U(G)$, its adjunct is

$$F(S) \xrightarrow{F(f)} F(U(G)) \xrightarrow{\varepsilon_G} G.$$

§15.4 Full and faithful describe maps on arrows, not objects

For a functor $T : \mathcal{C} \to \mathcal{D}$, each pair X, Y gives

$$T_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(T(X), T(Y)).$$

The functor is faithful if every $T_{X,Y}$ is injective, full if every $T_{X,Y}$ is surjective, and fully faithful if both hold.

The forgetful functor

$$U : \mathbf{Grp} \to \mathbf{Set}$$

is faithful: two group homomorphisms with the same underlying function are equal. It is not full: an arbitrary function between underlying sets need not preserve multiplication.

The inclusion

$$I : \mathbf{Ab} \hookrightarrow \mathbf{Grp}$$

is fully faithful. A homomorphism between abelian groups is already a morphism in **Ab**, so no arrows are lost or added.

Fullness and faithfulness do not assert that every object of $\mathcal{D}$ is in the image. That additional condition, up to isomorphism, is essential surjectivity. A fully faithful and essentially surjective functor is an equivalence of categories.

§15.5 A concrete category includes a chosen faithful underlying-set functor

A concrete category is a pair

$$(\mathcal{C}, U), \qquad U : \mathcal{C} \to \mathbf{Set} \text{ faithful.}$$

Groups, rings, vector spaces, and topological spaces carry familiar choices of U. Faithfulness ensures that morphisms can be distinguished by their underlying functions.

The functor U is part of the structure. Saying that an object "has underlying elements" is not enough to specify how arrows are represented as functions. A category may admit several inequivalent concrete realizations, and a categorical construction should not silently depend on one unless it is named.

For **Top**, the usual forgetful functor is faithful but not full: many set maps are not continuous. The same pattern appears in **Grp**. Concreteness records access to elements; it does not say that all set-theoretic functions are allowed morphisms.

§15.6 A group is a one-object category

Given a group G, form the category BG with one object $*$ and

$$\mathrm{Hom}_{BG}(*, *) = G.$$

Composition is group multiplication, and every arrow is invertible.

A functor

$$X : BG \to \mathbf{Set}$$

chooses a set $X(*)$, written simply X, and for each $g \in G$ a function

$$X(g) : X \to X.$$

Functoriality says

$$X(e) = \mathrm{id}_X, \qquad X(gh) = X(g) \circ X(h).$$

With the convention

$$g \cdot x = X(g)(x),$$

these are exactly the axioms of a left G-action:

$$e \cdot x = x, \qquad (gh) \cdot x = g \cdot (h \cdot x).$$

Therefore

$$\boxed{\mathrm{Fun}(BG, \mathbf{Set}) \cong G\text{-}\mathbf{Set}.}$$

A functor $BG \to BH$ is exactly a group homomorphism $G \to H$, because there is only one possible object map and functoriality is preservation of multiplication and identity.

§15.7 Natural transformations are equivariant maps

Let $X, Y : BG \to \mathbf{Set}$ be two actions. A natural transformation

$$\alpha : X \Rightarrow Y$$

has one component

$$\alpha_* : X \to Y.$$

For each $g \in G$, naturality is the commutative square

$$\begin{tikzcd}
X \arrow[r, "X(g)"] \arrow[d, "\alpha_*"'] & X \arrow[d, "\alpha_*"] \\
Y \arrow[r, "Y(g)"'] & Y.
\end{tikzcd}$$

Elementwise,

$$\alpha_*(g \cdot x) = g \cdot \alpha_*(x).$$

Thus natural transformations between action functors are exactly equivariant maps. The category G-**Set** is therefore not merely analogous to a functor category; it is one.

The same idea gives representations. Functors

$$BG \to \mathbf{Vect}_k$$

are linear representations of G, and natural transformations are intertwining linear maps.

§15.8 An adjunction generates a monad by adding and flattening structure

From $F \dashv U$ one obtains the endofunctor

$$T = UF : \mathbf{Set} \to \mathbf{Set}.$$

The unit is $\eta : \mathrm{Id} \Rightarrow T$. The multiplication

$$\mu = U\varepsilon_F : T^2 \Rightarrow T$$

flattens a word whose letters are themselves words into one word.

The free-monoid version is especially transparent. Let

$$T(S) = \coprod_{n \geq 0} S^n$$

be the set of finite lists in S. Then

$$\eta_S(s) = [s],$$

and

$$\mu_S([[s_{11}, \ldots, s_{1k_1}], \ldots, [s_{m1}, \ldots, s_{mk_m}]])$$

is the single concatenated list

$$[s_{11}, \ldots, s_{1k_1}, \ldots, s_{m1}, \ldots, s_{mk_m}].$$

The monad laws say that inserting singleton brackets and then flattening does nothing, and that flattening three nested levels is independent of the order of flattening. These are the unit and associativity laws of substitution.

Free constructions, group actions, and monads are therefore connected by one recurring mechanism:

$$\text{universal extension} \longleftrightarrow \text{natural Hom-bijection} \longleftrightarrow \text{unit/counit and substitution.}$$

The abstraction is useful because it records not only what objects are built, but why every map out of them is forced.

Topos Theory: Categorical Preliminaries, Internal Logic, and the Geometry of Sets

N-20240211

§16.1 Why topoi enter the story

A topos is one of the places where several lines of mathematics meet: category theory, logic, geometry, sheaf theory, and foundations. I want to regard it not merely as an abstract generalization of set theory, but as a way of making precise the idea that "sets may vary over a space" and that logic may depend on the ambient universe in which it is interpreted.

The route of this chapter is therefore:

$$\begin{gathered}\text{categories} \to \text{limits and universal properties}\\ \to \text{cartesian closure} \to \text{subobject classifiers}\\ \to \text{elementary topoi} \to \text{sheaf topoi and internal logic.}\end{gathered}$$

§16.2 Categories

Definition 16.2.1. A category $\mathcal{C}$ consists of:

(i) a class of objects $\mathrm{Ob}(\mathcal{C})$;

(ii) for each pair of objects X, Y, a collection $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ of morphisms from X to Y;

(iii) a composition operation

$$\mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X, Z),$$

written $(g, f) \mapsto g \circ f$;

(iv) for each object X, an identity morphism $\mathrm{id}_X : X \to X$.

These data satisfy associativity and identity laws:

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad f \circ \mathrm{id}_X = f, \quad \mathrm{id}_Y \circ f = f.$$

Example 16.2.2

The category **Set** has sets as objects and functions as morphisms. The category **Grp** has groups and group homomorphisms. The category **Top** has topological spaces and continuous maps.

Remark 16.2.3 — The categorical viewpoint is not that objects become unimportant, but that objects are understood through their maps. This is exactly the shift that later makes generalized set theories possible.

§16.3 Universal properties and limits

The most important categorical habit is to define objects not by their internal elements, but by how maps into or out of them behave.

Definition 16.3.1. A terminal object in $\mathcal{C}$ is an object 1 such that for every object X, there is a unique morphism

$$X \to 1.$$

Dually, an initial object is an object 0 such that for every X, there is a unique morphism

$$0 \to X.$$

In **Set**, a singleton is terminal and the empty set is initial.

Definition 16.3.2. A product of X and Y is an object $X \times Y$, equipped with projections

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y,$$

such that for every object Z and every pair of maps $f : Z \to X$, $g : Z \to Y$, there exists a unique map $\langle f, g\rangle : Z \to X \times Y$ making the diagram commute:

$$\begin{tikzcd}
 & Z \arrow[d, dashed, "\langle f,g\rangle"] \arrow[ddl, bend right, "f"'] \arrow[ddr, bend left, "g"] & \\
 & X \times Y \arrow[dl, "\pi_X"] \arrow[dr, "\pi_Y"'] & \\
X & & Y.
\end{tikzcd}$$

Definition 16.3.3. An equalizer of parallel morphisms $f, g : X \to Y$ is a morphism $e : E \to X$ such that

$$f \circ e = g \circ e,$$

and such that any other map $h : Z \to X$ with $f \circ h = g \circ h$ factors uniquely through e:

$$\begin{tikzcd}
Z \arrow[r, dashed, "!"] \arrow[rr, bend left, "h"] & E \arrow[r, "e"] & X \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Y
\end{tikzcd}$$

Definition 16.3.4. A pullback of $f : X \to Z$ and $g : Y \to Z$ is an object $X \times_Z Y$ fitting into a cartesian square

$$\begin{tikzcd}
X \times_Z Y \arrow[r] \arrow[d] & Y \arrow[d, "g"] \\
X \arrow[r, "f"'] & Z
\end{tikzcd}$$

with the universal property that any object mapping compatibly to X and Y factors uniquely through the pullback.

> **Remark 16.3.5** — Products, equalizers, and pullbacks are not isolated gadgets. They are instances of limits. A limit is a universal cone over a diagram; it is the categorical way to say that an object satisfies all the compatibility conditions encoded by the diagram.

§16.4 Exponentials and cartesian closure

In set theory, the set of functions from X to Y is denoted Y^X. Categorically, this is characterized by an adjunction.

Definition 16.4.1. A category with finite products is cartesian closed if for every pair of objects X, Y, there exists an exponential object Y^X and a natural bijection

$$\mathrm{Hom}_{\mathcal{C}}(Z, Y^X) \cong \mathrm{Hom}_{\mathcal{C}}(Z \times X, Y).$$

The corresponding evaluation map is

$$\mathrm{ev} : Y^X \times X \to Y.$$

A morphism $Z \to Y^X$ is the same thing as a morphism $Z \times X \to Y$. This is currying in categorical form:

$$\begin{tikzcd}
Z \times X \arrow[rr, "\tilde{f}"] \arrow[dr, "\hat{f} \times \mathrm{id}_X"'] & & Y \\
& Y^X \times X \arrow[ur, "\mathrm{ev}"'] &
\end{tikzcd}$$

Example 16.4.2

In **Set**, Y^X is the ordinary set of functions $X \to Y$. The bijection

$$\mathrm{Hom}_{\mathbf{Set}}(Z, Y^X) \cong \mathrm{Hom}_{\mathbf{Set}}(Z \times X, Y)$$

says that a Z-indexed family of functions $X \to Y$ is the same thing as a function of two variables $Z \times X \to Y$.

§16.5 Subobjects and the subobject classifier

A subset $A \subseteq X$ may be represented by its characteristic function

$$\chi_A : X \to \{0, 1\}.$$

The categorical version replaces $\{0, 1\}$ by an object Ω, called the subobject classifier.

Definition 16.5.1. A subobject classifier in a category with a terminal object is an object Ω, together with a map

$$\mathsf{true} : 1 \to \Omega,$$

such that for every monomorphism $m : A \hookrightarrow X$, there is a unique characteristic morphism $\chi_A : X \to \Omega$ making the square a pullback:

$$\begin{tikzcd}
A \arrow[r] \arrow[d, "m"'] & 1 \arrow[d, "\mathsf{true}"] \\
X \arrow[r, "\chi_A"'] & \Omega.
\end{tikzcd}$$

In **Set**, $\Omega = \{0, 1\}$, true picks 1, and χ_A is the ordinary characteristic function.

Remark 16.5.2 — This is one of the decisive steps toward topos theory. A topos has an internal object of truth values, so logic becomes an internal structure of the category rather than an external metalanguage imposed from outside.

§16.6 Elementary topoi

Definition 16.6.1. An elementary topos is a category $\mathcal{E}$ satisfying:

(i) it has finite limits;

(ii) it is cartesian closed;

(iii) it has a subobject classifier.

This definition is strikingly short. It says that a category behaves enough like **Set** to support finite constructions, function spaces, and a logic of predicates.

Example 16.6.2
The category **Set** is an elementary topos. More generally, for any small category $\mathcal{C}$, the presheaf category

$$\widehat{\mathcal{C}} = \mathbf{Set}^{\mathcal{C}^{op}}$$

is an elementary topos.

Example 16.6.3
If X is a topological space, the category $\mathrm{Sh}(X)$ of sheaves on X is a topos. It behaves like a category of sets varying continuously over X.

§16.7 A toy presheaf topos

The smallest example that is not simply **Set** is already useful. Let

$$\mathbf{2} = (0 \to 1)$$

be the category associated to the two-element poset. Consider the presheaf category

$$\widehat{\mathbf{2}} = \mathbf{Set}^{\mathbf{2}^{op}}.$$

Every presheaf category is a topos, so this is a genuine elementary topos.

An object $P \in \widehat{\mathbf{2}}$ is exactly the data

$$P(1) \xrightarrow{r} P(0),$$

because the arrow $0 \to 1$ in $\mathbf{2}$ becomes a restriction map $P(1) \to P(0)$ in $\mathbf{2}^{op}$. It is safest to think of this as a two-stage set: elements visible at stage 1 restrict to elements visible at stage 0. It is not automatically an equivalence relation or a quotient; it is simply a function carrying information from the higher stage to the lower stage.

The subobject classifier is made of sieves. With this convention, the sieves on 0 are

$$\Omega(0) = \{\varnothing, \{id_0\}\},$$

while the sieves on 1 are

$$\Omega(1) = \{\varnothing, \{u : 0 \to 1\}, \{id_1, u\}\}.$$

Thus the three-valued part occurs at stage 1, and the restriction map

$$\Omega(1) \longrightarrow \Omega(0)$$

is

$$\varnothing \mapsto \varnothing, \qquad \{u\} \mapsto \{id_0\}, \qquad \{id_1, u\} \mapsto \{id_0\}.$$

The element $\{u\}$ is the interesting third truth value: not true at stage 1 itself, but true after restricting along $0 \to 1$.

Here is a characteristic map in this toy universe. Let

$$P(1) = \{a, b\}, \qquad P(0) = \{*\},$$

with both a and b restricting to $*$. Define a subpresheaf $Q \hookrightarrow P$ by

$$Q(1) = \{a\}, \qquad Q(0) = \{*\}.$$

The characteristic morphism $\chi_Q : P \to \Omega$ sends

$$\chi_Q(1)(a) = \{id_1, u\}, \qquad \chi_Q(1)(b) = \{u\},$$

and

$$\chi_Q(0)(*) = \{id_0\}.$$

So a is fully true, while b is not true at stage 1 but becomes true after restriction to stage 0. This is the point at which internal logic stops being two-valued. In $\mathbf{Set}^{\mathbf{2}^{op}}$, a proposition may have a stage-dependent truth value; truth is not merely "yes" or "no", but is tracked through the restriction structure.

§16.8 Grothendieck topoi and sheaves

A Grothendieck topos is, roughly, a category equivalent to sheaves on a site. A site is a category equipped with a notion of covering. Thus Grothendieck topoi generalize sheaves on topological spaces.

> **Remark 16.8.1 —** This is the geometric origin of topos theory. Instead of studying a space only through its points, one studies its sheaves. In many situations, the sheaf category remembers more robust information than the point set itself.

In algebraic geometry, this becomes essential. The étale topos of a scheme, for example, packages arithmetic and geometric information that is invisible from the underlying topological space alone.

§16.9 Internal logic

Every elementary topos carries an internal higher-order intuitionistic logic. Subobjects of an object X behave like predicates on X, and the subobject classifier Ω behaves like the object of truth values.

> **Remark 16.9.1 —** The internal logic of a general topos is usually intuitionistic, not necessarily classical. The law of excluded middle
>
> $$P \vee \neg P$$
>
> need not hold internally. Thus moving from **Set** to a general topos also changes the logic that is naturally available.

For a sheaf topos, this has a geometric meaning: truth may be local. A statement may hold on an open cover without having a single global witness. This is exactly the kind of phenomenon that sheaf theory is designed to express.

§16.10 Power objects

In set theory, $\mathcal{P}(X)$ is the set of subsets of X. In a topos, the analogue is the power object

$$\Omega^X.$$

Indeed, by cartesian closure and the subobject classifier,

$$\mathrm{Hom}_{\mathcal{E}}(B, \Omega^X) \cong \mathrm{Hom}_{\mathcal{E}}(B \times X, \Omega),$$

which classifies subobjects of $B \times X$. Thus Ω^X parametrizes subobjects of X, just as 2^X parametrizes subsets of X in ordinary set theory.

§16.11 Topos as generalized universe of sets

A topos can be read in two complementary ways:

(i) as a generalized universe of sets, where one can do mathematics internally;

(ii) as a generalized space, where sheaves and local data become fundamental.

This double reading is the source of its power. In one direction, topos theory generalizes set-theoretic foundations. In the other direction, it turns geometry into logic.

§16.12 A high-level comparison

Setting	Truth values	Subsets/predicates
Set	$\{0, 1\}$	characteristic functions $X \to 2$
Boolean topos	Boolean algebra object	classical internal logic
General topos	Ω	subobjects classified by $X \to \Omega$
Sheaf topos	local truth values	predicates varying over a space

Remark 16.12.1 — From a higher viewpoint, a topos is not merely a category satisfying three axioms. It is a world in which mathematics can be interpreted. The shift from sets to topoi is analogous to the shift from Euclidean geometry to geometry relative to a transformation group: the ambient universe becomes variable.

§16.13 Further directions

Several directions naturally follow:

(i) Lawvere–Tierney topologies, which internalize sheafification;

(ii) geometric morphisms, the correct maps between topoi;

(iii) classifying topoi, which classify models of geometric theories;

(iv) Grothendieck topoi in algebraic geometry;

(v) elementary topoi as foundations for constructive mathematics.

Question

The most important question for me is not only what a topos is, but what it means to do mathematics inside a topos. Once truth, subsets, and function spaces become internal categorical objects, foundations stop being a fixed background and become part of the geometry.

§16.14 Subobjects and characteristic maps

In **Set**, a subset $A \subseteq X$ is classified by its characteristic function

$$\chi_A : X \to \{0, 1\}.$$

A subobject classifier generalizes this. In a topos, there is an object Ω and a map

$$\mathsf{true} : 1 \to \Omega$$

such that every subobject $A \hookrightarrow X$ is obtained as a pullback of true along a unique characteristic map $\chi_A : X \to \Omega$.

This is the categorical replacement for membership predicates. A subobject is not merely a subset; it is something classified by a truth-value object internal to the topos.

§16.15 Why logic becomes intuitionistic

In a general topos, the object Ω of truth values need not behave like the two-element Boolean algebra. As a result, the internal logic is generally intuitionistic. In particular, the law of excluded middle

$$P \vee \neg P$$

may fail internally.

This is not a defect. It means that truth can vary with geometric or contextual information. In a sheaf topos, a proposition may be true locally on some open sets without being globally decidable.

§16.16 Sheaves as variable sets

A sheaf on a topological space assigns data to each open set, with restriction maps and gluing conditions. One can think of a sheaf as a set varying over the space. The topos of sheaves on X therefore behaves like a universe of variable sets over X.

This is the geometric intuition behind topoi. They are not arbitrary abstract categories; they generalize the category of sets in a way that remembers variation, locality, and gluing.

Kernels, Cokernels, Images, and Coimages in Abelian Categories

N-20251109

§17.1 The setting: maps with zero maps

The uploaded note studies a morphism

$$f : A \to B$$

from the categorical viewpoint. The words domain and codomain mean exactly what they mean for an ordinary function:

$$A = \text{domain}, \qquad B = \text{codomain}.$$

For linear maps one may think of A as the input space and B as the declared output space. The actual values reached by f may form a smaller object inside B; that smaller object is the image.

The categorical definitions below need a zero object and zero morphisms. A zero object is simultaneously initial and terminal. Once a category has a zero object 0, every pair of objects X, Y has a distinguished zero morphism

$$0_{X,Y} : X \to 0 \to Y.$$

This lets equations such as $f \circ k = 0$ make sense without mentioning elements.

The most familiar examples are:

$$\mathbf{Vect}_k, \qquad \mathbf{Ab}, \qquad R\text{-}\mathbf{Mod}.$$

These are abelian categories. In them, kernels, cokernels, images, and coimages behave like the objects from linear algebra and module theory.

§17.2 Kernel: the universal object killed by a map

For a morphism $f : A \to B$, a kernel of f is a morphism

$$k : K \to A$$

such that

$$f \circ k = 0,$$

and universal with this property: whenever $k' : K' \to A$ also satisfies $f \circ k' = 0$, there is a unique morphism $u : K' \to K$ with

$$k' = k \circ u.$$

In diagram form:

$$\begin{tikzcd}
K' \arrow[d, dashed, "u"'] \arrow[dr, bend left, "k'"] & & \\
K \arrow[r, "k"'] & A \arrow[r, "f"'] & B.
\end{tikzcd}$$

The condition is that both routes from K' to B are zero after applying f.

In vector spaces this recovers

$$\operatorname{Ker}(f) = \{a \in A : f(a) = 0\}.$$

The categorical definition says something stronger than just listing elements: the kernel is the best, largest, and universal way to factor all inputs that become zero.

In an abelian category, the kernel morphism $k : K \to A$ is a monomorphism. It is therefore legitimate to think of K as a subobject of the domain A, though the invariant datum is the arrow k, not merely the object K.

§17.3 Cokernel: the universal quotient killing a map

The cokernel is the dual construction. A cokernel of $f : A \to B$ is a morphism

$$c : B \to C$$

such that

$$c \circ f = 0,$$

and universal with this property: whenever $c' : B \to C'$ satisfies $c' \circ f = 0$, there is a unique morphism $u : C \to C'$ with

$$c' = u \circ c.$$

The diagram is:

$$\begin{tikzcd}
A \arrow[r, "f"] & B \arrow[r, "c"] \arrow[dr, bend right, "c'"'] & C \arrow[d, dashed, "u"] \\
& & C'.
\end{tikzcd}$$

In vector spaces or modules,

$$\operatorname{Coker}(f) = B/\operatorname{Im}(f).$$

Thus the cokernel measures what remains in the codomain after the actual output of f has been collapsed to zero.

In an abelian category, the cokernel morphism $c : B \to C$ is an epimorphism. It is therefore the categorical version of a quotient object of the codomain.

§17.4 Image: the part of the codomain actually reached

In linear algebra the image is

$$\operatorname{Im}(f) = \{f(a) : a \in A\} \subseteq B.$$

In an abelian category, one defines the image without element language by first taking the cokernel

$$c : B \to \operatorname{Coker}(f)$$

and then taking the kernel of that cokernel:

$$\operatorname{Im}(f) = \operatorname{Ker}(B \xrightarrow{c} \operatorname{Coker}(f)).$$

So the image is represented by a monomorphism

$$i : \mathrm{Im}(f) \to B.$$

The diagram is:

$$\mathrm{Im}(f) \overset{i}{\hookrightarrow} B \xrightarrow{c} \mathrm{Coker}(f).$$

The equation $c \circ i = 0$ says that everything in the image is killed by the quotient map c. The universal property says that it is the largest subobject of B killed by c.

This gives a useful slogan:

$$\mathrm{Im}(f) = \text{the part of the codomain not visible to the cokernel.}$$

§17.5 Coimage: the effective part of the domain

The coimage is dual to the image. Begin with the kernel

$$k : \mathrm{Ker}(f) \to A.$$

Then define the coimage as the cokernel of the kernel:

$$\mathrm{Coim}(f) = \mathrm{Coker}\big(\mathrm{Ker}(f) \xrightarrow{k} A\big).$$

So it is represented by an epimorphism

$$q : A \to \mathrm{Coim}(f).$$

The diagram is:

$$\mathrm{Ker}(f) \overset{k}{\hookrightarrow} A \overset{q}{\twoheadrightarrow} \mathrm{Coim}(f).$$

In vector spaces this is the quotient

$$\mathrm{Coim}(f) = A / \mathrm{Ker}(f).$$

It is the domain after removing precisely the part that f sends to zero. In that sense, it is the effective input object of f.

§17.6 The standard factorization

In an abelian category, every morphism $f : A \to B$ factors through its coimage and image:

$$A \xrightarrow{q} \mathrm{Coim}(f) \xrightarrow{\overline{f}} \mathrm{Im}(f) \xrightarrow{i} B.$$

Here:

- $q : A \to \mathrm{Coim}(f)$ is the cokernel of $\mathrm{Ker}(f) \to A$, hence an epimorphism;
- $i : \mathrm{Im}(f) \to B$ is the kernel of $B \to \mathrm{Coker}(f)$, hence a monomorphism;
- the middle map $\overline{f} : \mathrm{Coim}(f) \to \mathrm{Im}(f)$ is an isomorphism in an abelian category;
- the original morphism is recovered by

$$f = i \circ \overline{f} \circ q.$$

A compact diagram is:

$$
\begin{tikzcd}
\operatorname{Ker}(f) \arrow[r, hook, "k"] & A \arrow[r, two heads, "q"] \arrow[rrr, bend left, "f"] & \operatorname{Coim}(f) \arrow[r, "\overline{f}", "\cong"'] & \operatorname{Im}(f) \arrow[r, hook, "i"] & B \arrow[r, two heads, "c"] & \operatorname{Coker}(f).
\end{tikzcd}
$$

This is the categorical form of the first isomorphism theorem:

$$A/\operatorname{Ker}(f) \cong \operatorname{Im}(f).$$

The uploaded note's phrase that coimage is effective input and image is actual output is exactly this factorization: q discards useless input, i embeds the actual output into the declared codomain, and $\overline{f}$ identifies these two descriptions.

§17.7 A vector-space sanity check

Let $T : V \to W$ be a linear map. Then

$$\operatorname{Ker}(T) = \{v \in V : T(v) = 0\}, \qquad \operatorname{Coker}(T) = W/\operatorname{Im}(T).$$

The coimage is

$$\operatorname{Coim}(T) = V/\operatorname{Ker}(T),$$

and the induced map

$$\overline{T} : V/\operatorname{Ker}(T) \to \operatorname{Im}(T), \qquad [v] \mapsto T(v),$$

is an isomorphism. It is well-defined because if $v - v' \in \operatorname{Ker}(T)$, then $T(v) = T(v')$. It is injective because $T(v) = 0$ means $[v] = 0$, and it is surjective by definition of image.

For a matrix $A : k^n \to k^m$, this says that the rank is the dimension of both sides of the middle isomorphism:

$$\dim(k^n/\operatorname{Ker} A) = \dim \operatorname{Im} A = \operatorname{rank} A.$$

This is rank-nullity in categorical clothing.

§17.8 A module example: multiplication by an integer

Consider the group homomorphism

$$f : \mathbb{Z} \to \mathbb{Z}, \qquad f(x) = nx.$$

If $n \neq 0$, then

$$\operatorname{Ker}(f) = 0, \qquad \operatorname{Im}(f) = n\mathbb{Z}, \qquad \operatorname{Coker}(f) = \mathbb{Z}/n\mathbb{Z}.$$

The coimage is

$$\operatorname{Coim}(f) = \mathbb{Z}/0 \cong \mathbb{Z}.$$

The middle isomorphism is

$$\mathbb{Z} \xrightarrow{\sim} n\mathbb{Z}, \qquad x \mapsto nx.$$

This example is useful because the image is literally a subgroup of the codomain, while the coimage is a quotient of the domain. They look different as subobject and quotient, but in an abelian category they are canonically isomorphic through f.

§17.9 Exactness language

Kernel and image are also the grammar of exact sequences. A sequence

$$X \xrightarrow{g} Y \xrightarrow{h} Z$$

is exact at Y when
$$\mathrm{Im}(g) = \mathrm{Ker}(h)$$
as subobjects of Y. The equality means that the elements, vectors, or generalized inputs that h kills are exactly those that came from g.

In categorical language, exactness replaces element chasing by universal properties. For example, saying that
$$\mathrm{Ker}(f) \to A \to B$$
is exact at A is precisely saying that the first arrow is the kernel of f. Saying that

$$A \to B \to \mathrm{Coker}(f)$$

is exact at B is precisely saying that the second arrow is the cokernel of f, after passing through the image.

§17.10 Warnings: what depends on being abelian

The statements above should not be read as valid in every category. Several layers are being used.

- A category must have a zero object and zero morphisms before kernels and cokernels can even be formulated in this form.
- A category may have kernels and cokernels but still fail to make every monomorphism a kernel or every epimorphism a cokernel.
- The canonical map $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ need not be an isomorphism outside abelian categories.

Thus the beautiful factorization

$$\mathrm{Coim}(f) \cong \mathrm{Im}(f)$$

is not a formal tautology. It is one of the defining strengths of abelian categories.

For intuition, **Set** is not the right environment for this theory: it has no zero object in the needed sense. Pointed sets have a zero object and kernels in a pointed sense, but they are not abelian. Vector spaces, abelian groups, and modules are the safe models.

§17.11 Synthesis: from element chasing to universal properties

The original note's definitions can be compressed into the following table:

object	categorical construction	linear algebra model	where it lives
$\mathrm{Ker}(f)$	universal map into A killed by f	$\{a : f(a) = 0\}$	subobject of A
$\mathrm{Coker}(f)$	universal quotient of B killing f	$B/\mathrm{Im}(f)$	quotient of B
$\mathrm{Im}(f)$	$\mathrm{Ker}(B \to \mathrm{Coker}(f))$	$\{f(a) : a \in A\}$	subobject of B
$\mathrm{Coim}(f)$	$\mathrm{Coker}(\mathrm{Ker}(f) \to A)$	$A/\mathrm{Ker}(f)$	quotient of A

The standard factorization

$$A \twoheadrightarrow \mathrm{Coim}(f) \xrightarrow{\sim} \mathrm{Im}(f) \hookrightarrow B$$

turns a morphism into three conceptual pieces: quotient out the part of the input killed by f, identify the remaining effective input with the subobject actually reached in B, and then include that subobject in the codomain.

For vector spaces this says exactly the familiar rank–nullity picture: $A/\ker f$ is isomorphic to $\operatorname{im} f$. For modules and abelian groups the same statement is the first isomorphism theorem. The categorical formulation is useful because it records which part of the argument is universal and which part depends on being in an abelian category: the comparison map $\mathrm{Coim}(f) \to \mathrm{Im}(f)$ is an isomorphism in the safe models listed above, but it should not be assumed in an arbitrary category with kernels and cokernels.

III

Number Theory Basics

Part III: Contents

Recurrences as Orbits: Linear Dynamics, Normalization, and Reversible Transformations

N-20220717

§18.1 A recurrence generates an orbit

A first-order recurrence

$$a_{n+1} = F(a_n)$$

is the iteration of a map $F : X \to X$. Once an initial value a_0 is chosen, the sequence is the forward orbit

$$a_0, \quad F(a_0), \quad F^2(a_0), \quad \ldots.$$

This point of view separates two questions. The recurrence rule determines the dynamics, while the initial value selects one orbit. A closed formula is an explicit description of the iterate F^n.

Not every recurrence is reversible. If F is not injective, different states can merge and no inverse iteration is possible. This distinction will later separate semigroup actions, which describe forward iteration, from group actions, which describe reversible transformations.

Higher-order recurrences can also be written as first-order systems by enlarging the state. For example,

$$a_{n+2} = 5a_{n+1} - 6a_n$$

becomes

$$\begin{pmatrix} a_{n+2} \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}.$$

Thus recurrence theory is already a form of discrete dynamical systems and linear algebra.

§18.2 Arithmetic progressions are translation orbits

An arithmetic progression satisfies

$$a_{n+1} = a_n + d.$$

The update map is the translation

$$T_d(x) = x + d.$$

Since

$$T_d^n(x) = x + nd,$$

we obtain

$$\boxed{a_n = a_0 + nd.}$$

With indexing beginning at 1, this becomes

$$a_n = a_1 + (n-1)d.$$

The sum of the first n terms follows by pairing terms from opposite ends:

$$\begin{aligned} S_n &= a_1 + a_2 + \cdots + a_n, \\ S_n &= a_n + a_{n-1} + \cdots + a_1. \end{aligned}$$

Every column sums to $a_1 + a_n$, so

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = na_1 + \frac{n(n-1)}{2}d.}$$

Arithmetic progressions convert additive relations among indices into additive relations among terms. If

$$i_1 + \cdots + i_k = j_1 + \cdots + j_k,$$

then

$$\begin{aligned} a_{i_1} + \cdots + a_{i_k} &= ka_1 + (i_1 + \cdots + i_k - k)d \\ &= ka_1 + (j_1 + \cdots + j_k - k)d \\ &= a_{j_1} + \cdots + a_{j_k}. \end{aligned}$$

The equal-index-sum rule is therefore a direct consequence of translation dynamics, not an isolated trick.

§18.3 Geometric progressions are scaling orbits

A geometric progression satisfies

$$a_{n+1} = qa_n.$$

The update map is the scaling

$$S_q(x) = qx,$$

whose iterates are

$$S_q^n(x) = q^n x.$$

Hence

$$\boxed{a_n = a_0 q^n}$$

or, with indexing beginning at 1,

$$a_n = a_1 q^{n-1}.$$

For $q \neq 1$, write

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}.$$

Then

$$qS_n = a_1 q + a_1 q^2 + \cdots + a_1 q^n.$$

Subtracting gives

$$(q-1)S_n = a_1(q^n - 1),$$

so

$$\boxed{S_n = a_1 \frac{q^n - 1}{q - 1}.}$$

The exceptional case $q = 1$ must be handled separately and gives $S_n = na_1$.

If $q \neq 0$, scaling is reversible, with inverse $S_{1/q}$. If $q = 0$, all states collapse to zero after one step. The same-looking recurrence can therefore represent either a group action or a genuinely irreversible map depending on the parameter.

§18.4 Affine recurrences become geometric after shifting a fixed point

Consider

$$a_{n+1} = ra_n + b.$$

The update map

$$T(x) = rx + b$$

is affine. If $r \neq 1$, it has a unique fixed point L satisfying

$$L = rL + b,$$

namely

$$L = \frac{b}{1-r}.$$

Subtracting the fixed point gives

$$a_{n+1} - L = r(a_n - L).$$

The shifted sequence is geometric, so

$$\boxed{a_n = L + r^n(a_0 - L).}$$

Equivalently,

$$a_n = r^n a_0 + b\frac{1-r^n}{1-r}.$$

If $r = 1$, the recurrence is arithmetic:

$$a_n = a_0 + nb.$$

For example,

$$a_{n+1} = 2a_n + 1, \qquad a_0 = 0.$$

The fixed point is $L = -1$, so

$$a_n + 1 = 2^n(a_0 + 1) = 2^n,$$

and therefore

$$\boxed{a_n = 2^n - 1.}$$

The first values

$$0, 1, 3, 7, 15, \ldots$$

confirm the formula.

The fixed-point form also gives the stability classification. When $|r| < 1$, every orbit converges to L. When $|r| > 1$, every orbit except the fixed point moves away exponentially. When $r = -1$, nonfixed orbits oscillate with period two. Thus a closed formula also reveals the qualitative dynamics.

§18.5 Variable forcing gives a discrete variation-of-constants formula

For the nonautonomous recurrence

$$a_{n+1} = ra_n + b_n,$$

repeated substitution gives

$$\begin{aligned} a_1 &= ra_0 + b_0, \\ a_2 &= r^2 a_0 + rb_0 + b_1, \\ a_3 &= r^3 a_0 + r^2 b_0 + rb_1 + b_2. \end{aligned}$$

The pattern is

$$\boxed{a_n = r^n a_0 + \sum_{k=0}^{n-1} r^{n-1-k} b_k.}$$

This is the discrete variation-of-constants formula. A forcing term introduced at time k is propagated through the remaining $n-1-k$ iterations by the factor r^{n-1-k}.

If $b_k = b$ is constant, the sum becomes geometric and recovers the affine fixed-point formula. If the forcing has a special shape, such as $b_k = c\lambda^k$, the sum reveals resonance when $\lambda = r$.

§18.6 Normalization should match the coefficient growth

The recurrence

$$a_n = na_{n-1} + 1, \qquad a_1 = 1,$$

does not become simpler after subtracting a constant. The coefficient n suggests dividing by $n!$. Define

$$b_n = \frac{a_n}{n!}.$$

Then

$$b_n = \frac{na_{n-1}+1}{n!} = \frac{a_{n-1}}{(n-1)!} + \frac{1}{n!} = b_{n-1} + \frac{1}{n!}.$$

The normalized recurrence telescopes:

$$b_n = 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

Hence

$$\boxed{a_n = n!\left(1 + \frac{1}{2!} + \cdots + \frac{1}{n!}\right).}$$

For example,

$$a_1 = 1, \quad a_2 = 3, \quad a_3 = 10, \quad a_4 = 41,$$

and the formula gives

$$4!\left(1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24}\right) = 41.$$

The method is structural: multiplication by n points toward factorial normalization, just as multiplication by a constant points toward geometric normalization.

§18.7 Second-order recurrences are matrix powers

Consider

$$a_{n+2} = 5a_{n+1} - 6a_n.$$

Set

$$v_n = \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}.$$

Then

$$v_{n+1} = Cv_n, \qquad C = \begin{pmatrix} 5 & -6 \\ 1 & 0 \end{pmatrix},$$

so

$$v_n = C^n v_0.$$

The characteristic polynomial is

$$\det(\lambda I - C) = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3).$$

The eigenvalues 2 and 3 explain the two exponential solutions. Thus

$$a_n = A2^n + B3^n.$$

Take

$$a_0 = 0, \qquad a_1 = 1.$$

The initial conditions give

$$A + B = 0, \qquad 2A + 3B = 1,$$

so $A = -1$ and $B = 1$. Therefore

$$\boxed{a_n = 3^n - 2^n.}$$

Indeed,

$$0, 1, 5, 19, 65, \ldots$$

satisfies the recurrence.

The trial solution $a_n = \lambda^n$ is therefore not a guess without justification. It searches for eigenvectors of the shift operator, or equivalently eigenvectors of the companion matrix.

If the characteristic polynomial has a repeated root λ, the matrix may have a Jordan block and the second independent solution becomes

$$n\lambda^n.$$

For example,

$$a_{n+2} = 2a_{n+1} - a_n$$

has characteristic polynomial $(t-1)^2$, so

$$a_n = A + Bn.$$

Arithmetic progressions are exactly this repeated-root case.

§18.8 Generating functions turn shifts into multiplication

The same recurrence can be solved by a generating function. Let

$$A(z) = \sum_{n\geq 0} a_n z^n$$

for the initial data $a_0 = 0$, $a_1 = 1$. Multiply

$$a_{n+2} - 5a_{n+1} + 6a_n = 0$$

by z^{n+2} and sum over $n \geq 0$. The shifted sums are

$$\sum_{n\geq 0} a_{n+2} z^{n+2} = A(z) - z,$$

$$\sum_{n\geq 0} a_{n+1} z^{n+2} = zA(z),$$

and

$$\sum_{n\geq 0} a_n z^{n+2} = z^2 A(z).$$

Hence

$$A(z) - z - 5zA(z) + 6z^2 A(z) = 0,$$

so

$$A(z) = \frac{z}{1 - 5z + 6z^2} = \frac{z}{(1-2z)(1-3z)}.$$

Partial fractions give

$$A(z) = \frac{1}{1-3z} - \frac{1}{1-2z}.$$

Since

$$\frac{1}{1-cz} = \sum_{n\geq 0} c^n z^n,$$

we recover

$$a_n = 3^n - 2^n.$$

The characteristic roots appear as poles of the generating function. Matrix powers, exponential trial solutions, and rational generating functions are three views of the same linear recurrence.

§18.9 Forward iteration is a semigroup action

The nonnegative integers form a monoid under addition: they have an associative operation and an identity element 0, but they do not contain additive inverses. The iterates of any map $F : X \to X$ satisfy

$$F^{m+n} = F^m \circ F^n, \qquad F^0 = \mathrm{id}_X\,.$$

Therefore every recurrence

$$a_{n+1} = F(a_n)$$

defines an action of $(\mathbb{N}, +)$ on X:

$$n \cdot x = F^n(x).$$

This language says exactly that advancing by n steps and then by m steps is the same as advancing by $m+n$ steps.

No inverse is required. For example,

$$F(x) = x^2$$

on $\mathbb{R}$ identifies x and $-x$ after one step. Once their orbits merge, the previous state cannot be recovered from the current one. The recurrence still has perfectly good forward iterates, but there is no globally defined F^{-1}.

If F is bijective, negative iterates are available and the action extends from $\mathbb{N}$ to $\mathbb{Z}$:

$$(-n) \cdot x = F^{-n}(x).$$

This is the precise boundary at which groups enter. A group action describes transformations that can be composed and undone; a monoid action describes forward evolution whether or not it is reversible.

Definition 18.9.1 (Group)**.** A group is a set G with an associative binary operation, an identity element e, and an inverse g^{-1} for every $g \in G$. If the operation is commutative, the group is abelian.

The phrase "binary operation on G" already includes closure:

$$* : G \times G \to G.$$

The axioms say that compositions are unambiguous, doing nothing is allowed, and every permitted transformation can be reversed.

§18.10 First-order linear recurrences live inside the affine group

For $r \neq 0$ and $b \in \mathbb{R}$, define the affine transformation

$$T_{r,b}(x) = rx + b.$$

Composing two such maps gives

$$\begin{aligned} T_{r,b}(T_{s,c}(x)) &= r(sx + c) + b \\ &= rsx + (rc + b), \end{aligned}$$

so

$$T_{r,b} \circ T_{s,c} = T_{rs,rc+b}.$$

The inverse is obtained by solving $y = rx + b$:

$$T_{r,b}^{-1}(y) = \frac{y-b}{r} = T_{1/r,-b/r}(y).$$

Thus all affine transformations with nonzero slope form a group.

Translations and scalings sit inside this group:

$$\tau_b = T_{1,b}, \qquad \sigma_r = T_{r,0}.$$

They satisfy

$$\tau_b \tau_c = \tau_{b+c}, \qquad \sigma_r \sigma_s = \sigma_{rs}.$$

However, they do not commute in general. Conjugating a translation by a scaling gives

$$\begin{aligned}\sigma_r\tau_c\sigma_r^{-1}(x) &= \sigma_r\tau_c(x/r)\\ &= \sigma_r(x/r+c)\\ &= x+rc\\ &= \tau_{rc}(x).\end{aligned}$$

Scaling changes the size of a translation. This interaction is the reason the affine group is a semidirect product rather than a direct product:

$$\operatorname{Aff}(1,\mathbb{R}) \cong \mathbb{R} \rtimes \mathbb{R}^\times.$$

The multiplication law

$$(b,r)(c,s) = (b+rc, rs)$$

records exactly how the scaling component r acts on the translation component c.

Now return to the recurrence

$$a_{n+1} = ra_n + b.$$

Its sequence is the orbit of a_0 under the cyclic subgroup generated by $T_{r,b}$. The fixed-point trick from the first half of the note has a group-theoretic meaning. If $r \neq 1$, let

$$L = \frac{b}{1-r},$$

so $T_{r,b}(L) = L$. Translate the fixed point to the origin using

$$\tau_{-L}(x) = x - L.$$

Then

$$\begin{aligned}\tau_{-L}\circ T_{r,b}\circ\tau_L(x) &= \tau_{-L}(r(x+L)+b)\\ &= rx + rL + b - L\\ &= rx,\end{aligned}$$

because $(1-r)L = b$. Hence

$$\tau_{-L}T_{r,b}\tau_L = \sigma_r.$$

The affine recurrence is not merely similar to a geometric progression after a clever substitution; its update map is conjugate to a pure scaling. Taking powers gives

$$T_{r,b}^n = \tau_L\sigma_r^n\tau_{-L},$$

and therefore

$$T_{r,b}^n(x) = L + r^n(x-L).$$

This is exactly the closed formula derived earlier.

§18.11 Homogeneous coordinates turn affine iteration into matrix algebra

Affine maps are not linear on the one-dimensional vector space because they do not necessarily preserve the origin. They become linear after adding one extra coordinate. Represent $x \in \mathbb{R}$ by the homogeneous vector

$$\begin{pmatrix} x \\ 1 \end{pmatrix}$$

and define

$$M_{r,b} = \begin{pmatrix} r & b \\ 0 & 1 \end{pmatrix}.$$

Then

$$M_{r,b} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} rx + b \\ 1 \end{pmatrix}.$$

Matrix multiplication reproduces affine composition:

$$M_{r,b} M_{s,c} = M_{rs,rc+b}.$$

Thus the affine group is realized as a subgroup of $GL_2(\mathbb{R})$.

The fixed-point conjugacy becomes an ordinary matrix factorization. Let

$$P_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}, \qquad D_r = \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}.$$

Since $(1 - r)L = b$,

$$M_{r,b} = P_L D_r P_L^{-1}.$$

Therefore

$$M_{r,b}^n = P_L D_r^n P_L^{-1} = \begin{pmatrix} r^n & (1 - r^n)L \\ 0 & 1 \end{pmatrix}.$$

Substituting $L = b/(1 - r)$ gives

$$\boxed{M_{r,b}^n = \begin{pmatrix} r^n & b\dfrac{1 - r^n}{1 - r} \\ 0 & 1 \end{pmatrix}} \qquad (r \neq 1).$$

Applying this matrix to $(a_0, 1)^T$ recovers the affine recurrence formula.

The case $r = 1$ is different because there is no finite fixed point to translate to the origin. Now

$$M_{1,b} = I + N, \qquad N = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

The binomial theorem truncates:

$$(I + N)^n = I + nN.$$

Hence

$$M_{1,b}^n = \begin{pmatrix} 1 & nb \\ 0 & 1 \end{pmatrix},$$

which gives

$$a_n = a_0 + nb.$$

Arithmetic progressions are therefore governed by a unipotent Jordan block, while affine recurrences with $r \neq 1$ are governed by the two eigenvalues r and 1. The matrix model explains why the formulas split exactly at $r = 1$.

For the concrete recurrence

$$a_{n+1} = 2a_n + 1, \qquad a_0 = 0,$$

we have $L = -1$ and

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus

$$M^n = \begin{pmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{pmatrix},$$

and the upper-right entry is precisely the orbit value $a_n = 2^n - 1$.

§18.12 Dihedral symmetry is affine geometry modulo n

Label the vertices of a regular n-gon by $\mathbb{Z}/n\mathbb{Z}$. A rotation by b vertices is

$$i \longmapsto i + b,$$

while a reflection followed by a rotation has the form

$$i \longmapsto -i + b.$$

Thus every symmetry can be written uniformly as

$$i \longmapsto \varepsilon i + b, \qquad \varepsilon \in \{1, -1\}, \quad b \in \mathbb{Z}/n\mathbb{Z}.$$

This is an affine transformation over the cyclic ring $\mathbb{Z}/n\mathbb{Z}$.

Compose two such maps:

$$(b, \varepsilon)(c, \delta)(i) = \varepsilon(\delta i + c) + b = (\varepsilon\delta)i + (b + \varepsilon c).$$

Hence

$$(b, \varepsilon)(c, \delta) = (b + \varepsilon c, \varepsilon\delta).$$

The sign acts on translations by either preserving or reversing them. Therefore

$$D_{2n} \cong C_n \rtimes C_2,$$

where the nontrivial element of C_2 acts on C_n by inversion.

Let

$$r(i) = i + 1, \qquad s(i) = -i.$$

Then

$$r^n = 1, \qquad s^2 = 1,$$

and

$$\begin{aligned} srs(i) &= sr(-i) \\ &= s(-i + 1) \\ &= i - 1 \\ &= r^{-1}(i). \end{aligned}$$

Thus

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ srs = r^{-1} \rangle.$$

Every element is uniquely either r^k or sr^k, so the group has $2n$ elements.

For a regular pentagon, the ten maps are

$$i \mapsto i + b \quad \text{and} \quad i \mapsto -i + b, \qquad b \in \mathbb{Z}/5\mathbb{Z}.$$

The formula makes the geometry easy to compute. For example,

$$(2,-1)(3,1) = (2-3,-1) = (4,-1) \pmod 5.$$

A reflection centered differently is obtained because the preceding rotation changes the translation parameter with a sign.

The dihedral group is therefore not an unrelated second example of a group. It is another affine semidirect product, now over a finite cyclic space rather than the real line.

§18.13 Direct products and semidirect products encode different kinds of coupling

If transformations on two coordinates are genuinely independent, the correct construction is a direct product. Suppose

$$x_{n+1} = F(x_n), \qquad y_{n+1} = G(y_n).$$

The combined update is

$$(x,y) \longmapsto (F(x), G(y)),$$

and the two components can be iterated separately. If F and G are invertible, their transformation groups act through

$$(g,h)\cdot(x,y) = (g\cdot x, h\cdot y).$$

The multiplication law

$$(g_1,h_1)(g_2,h_2) = (g_1g_2, h_1h_2)$$

contains no cross-term because neither component modifies the other.

The affine and dihedral groups are different. In

$$(b,r)(c,s) = (b+rc, rs),$$

the scaling r changes the later translation c. In

$$(b,\varepsilon)(c,\delta) = (b+\varepsilon c, \varepsilon\delta),$$

the reflection sign changes the direction of the later rotation. These cross-terms are the signature of a semidirect product: one subgroup acts on the other.

An isomorphism is a bijective homomorphism

$$\varphi : G \to H$$

that preserves multiplication. The maps

$$b \longmapsto \tau_b$$

and

$$(b,\varepsilon) \longmapsto (i \mapsto \varepsilon i + b)$$

are useful because they replace abstract group elements by transformations whose composition can be seen directly.

This completes the bridge from recurrences to groups without identifying the two subjects. Every recurrence gives a forward $\mathbb{N}$-action. Bijective update rules extend to $\mathbb{Z}$-actions. First-order linear recurrences sit inside the affine group, where shifting a fixed point is conjugacy and closed forms are power formulas. Polygon symmetries form a finite affine semidirect product. The common structure is not repetition in the vague sense, but the algebra of composing transformations and deciding whether their effects are independent, interacting, or reversible.

Polygon Motions and Roots of Unity: Modular Hops, Chords, and Vanishing Sums

N-20220723

§19.1 The ant on a decagon is a modular-arithmetic problem

Label the vertices of a regular decagon by

$$0, 1, \ldots, 9$$

in counterclockwise order. Moving forward by k vertices sends a position j to

$$j + k \pmod{10}.$$

Only the residue class of the total displacement matters.

Suppose an ant makes ten jumps of length 1, then ten jumps of length 2, and continues in this way through ten jumps of length 10. One block of ten jumps of length k changes the vertex label by

$$10k \equiv 0 \pmod{10}.$$

Therefore every block returns the ant to its starting vertex. If it begins at vertex 1, it finishes at vertex 1.

The point is not specific to ten sides.

Proposition 19.1.1 (A full block on a regular polygon)
On a regular N-gon, any block of N equal jumps returns to the starting vertex.

Proof. If each jump advances by k vertices, the total displacement is Nk, and

$$Nk \equiv 0 \pmod{N}.$$

□

For an arbitrary list of jump lengths $k_1, \ldots, k_m$, the final vertex is

$$j_0 + k_1 + \cdots + k_m \pmod{N}.$$

A geometric hopping problem has become addition in the cyclic group $\mathbb{Z}/N\mathbb{Z}$.

§19.2 The greatest common divisor determines the orbit

Repeated jumps of a fixed length k produce the orbit

$$j, \quad j + k, \quad j + 2k, \quad \ldots \pmod{N}.$$

The ant returns after the least positive integer t satisfying

$$tk \equiv 0 \pmod{N}.$$

Write

$$d = \gcd(N, k), \qquad N = dN_0, \qquad k = dk_0,$$

with $\gcd(N_0, k_0) = 1$. The condition $N \mid tk$ becomes

$$N_0 \mid tk_0.$$

Since k_0 is invertible modulo N_0, the smallest possible t is N_0. Hence the orbit length is

$$\boxed{\frac{N}{\gcd(N, k)}.}$$

On a decagon, jumps of length 4 visit

$$0, 4, 8, 2, 6, 0.$$

There are five distinct vertices because

$$\frac{10}{\gcd(10, 4)} = 5.$$

Jumps of length 3 visit all ten vertices because 3 is coprime to 10.

Thus a fixed jump length visits every vertex exactly when

$$\gcd(N, k) = 1.$$

This is the first appearance of a general principle: multiplication by k permutes residue classes modulo N exactly when k is invertible modulo N.

§19.3 Roots of unity place the polygon on the complex plane

Let

$$\zeta_N = e^{2\pi i/N}.$$

The N-th roots of unity are

$$\mu_N = \{1, \zeta_N, \zeta_N^2, \ldots, \zeta_N^{N-1}\}.$$

They satisfy

$$z^N = 1$$

and form the vertices of a regular N-gon on the unit circle.

Vertex j is represented by ζ_N^j. A jump of k vertices is multiplication by

$$\zeta_N^k : \qquad \zeta_N^j \longmapsto \zeta_N^{j+k}.$$

After m such jumps, the point is

$$\zeta_N^{j+mk}.$$

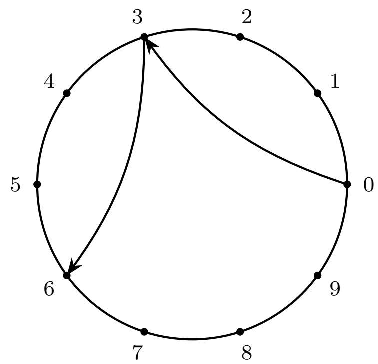
Returning to the starting point means

$$\zeta_N^{mk} = 1,$$

which is the same congruence

$$mk \equiv 0 \pmod{N}.$$

The complex and modular descriptions are two coordinate systems for the same cyclic motion.



Multiplication by ζ_{10}^3 advances three vertices.

Figure 19.1: A regular decagon encoded by the tenth roots of unity.

§19.4 Power maps have a visible kernel and image

For an integer m, consider the power map

$$p_m : \mu_N \to \mu_N, \qquad p_m(z) = z^m.$$

In exponent coordinates it is multiplication by m:

$$\zeta_N^j \longmapsto \zeta_N^{mj}.$$

The kernel consists of exponents satisfying

$$mj \equiv 0 \pmod{N}.$$

There are exactly $\gcd(m, N)$ such residue classes. Therefore

$$|\ker p_m| = \gcd(m, N).$$

Since $|\mu_N| = N$, the image has size

$$|\operatorname{im} p_m| = \frac{N}{\gcd(m, N)}.$$

Consequently, the following conditions are equivalent:

$$\begin{aligned} &\gcd(m, N) = 1, \\ &p_m \text{ is injective,} \\ &p_m \text{ is surjective,} \\ &p_m \text{ permutes the vertices of the regular } N\text{-gon.} \end{aligned}$$

For example, squaring permutes the vertices of a regular pentagon, but not those of a regular hexagon. On μ_6, both 1 and -1 square to 1, so the map has a nontrivial kernel.

This power-map viewpoint simultaneously explains orbit lengths, vertex permutations, and many elementary root-of-unity sums.

§19.5 The sum of the vertices is zero

Let $\zeta = \zeta_N$ with $N > 1$. The sum of all vertices is

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{N-1}.$$

Multiplying by $\zeta - 1$ gives

$$(\zeta - 1)(1 + \zeta + \cdots + \zeta^{N-1}) = \zeta^N - 1 = 0.$$

Since $\zeta \neq 1$,

$$\boxed{1 + \zeta + \cdots + \zeta^{N-1} = 0.}$$

Geometrically, the centroid of a regular polygon centered at the origin is the origin.

More generally, for any integer m,

$$\sum_{j=0}^{N-1} \zeta^{mj} = \begin{cases} N, & N \mid m, \\ 0, & N \nmid m. \end{cases}$$

If $N \nmid m$, the ratio ζ^m is not 1, so the sum is a finite geometric series:

$$\sum_{j=0}^{N-1} \zeta^{mj} = \frac{(\zeta^m)^N - 1}{\zeta^m - 1} = 0.$$

If $N \mid m$, every summand equals 1.

The case $d = \gcd(m, N) > 1$ has a useful geometric interpretation. The list

$$1, \zeta^m, \zeta^{2m}, \ldots$$

traces a smaller regular polygon with N/d vertices, each vertex repeated d times when j runs through a complete set of residues modulo N. Its vector sum is still zero unless the polygon has only one vertex.

§19.6 Orthogonality is a vanishing polygon sum

For integers r, s,

$$\begin{aligned} \sum_{j=0}^{N-1} \zeta_N^{rj} \overline{\zeta_N^{sj}} &= \sum_{j=0}^{N-1} \zeta_N^{(r-s)j} \\ &= \begin{cases} N, & r \equiv s \pmod{N}, \\ 0, & r \not\equiv s \pmod{N}. \end{cases} \end{aligned}$$

The vectors

$$(1, \zeta_N^r, \zeta_N^{2r}, \ldots, \zeta_N^{(N-1)r})$$

are therefore mutually orthogonal for distinct residues $r \pmod{N}$.

This is the elementary core of the discrete Fourier transform. Fourier orthogonality is not an unrelated analytic miracle: it is the statement that a nontrivial regular polygon has vector sum zero.

As a quick application, let $c_0, \ldots, c_{N-1}$ be complex numbers and define

$$\widehat{c}_r = \sum_{j=0}^{N-1} c_j \zeta_N^{-rj}.$$

Then orthogonality gives the inversion formula

$$c_j = \frac{1}{N} \sum_{r=0}^{N-1} \widehat{c}_r \zeta_N^{rj}.$$

The proof is only a double sum followed by the vanishing identity above.

§19.7 Chord lengths depend only on modular distance

The distance between vertices a and b on the unit circle is

$$|\zeta_N^a - \zeta_N^b|.$$

Factor out ζ_N^b, whose absolute value is one:

$$|\zeta_N^a - \zeta_N^b| = |\zeta_N^{a-b} - 1|.$$

Using

$$e^{i\theta} - 1 = 2ie^{i\theta/2} \sin \frac{\theta}{2},$$

we obtain

$$\boxed{|\zeta_N^a - \zeta_N^b| = 2 \left| \sin \frac{\pi(a-b)}{N} \right|.}$$

Thus chord length depends only on the difference $a - b \pmod N$, exactly as rotational symmetry predicts.

For a unit circumradius decagon, the side length is

$$2 \sin \frac{\pi}{10},$$

the chord spanning two edges has length

$$2 \sin \frac{2\pi}{10},$$

and opposite vertices are distance

$$2 \sin \frac{5\pi}{10} = 2.$$

The same modular difference that controls hopping also controls Euclidean distance on the polygon.

§19.8 The roots of $z^N - 1$ form the polygon

The polynomial

$$z^N - 1$$

has the N roots

$$1, \zeta_N, \ldots, \zeta_N^{N-1}.$$

They are distinct because the derivative

$$Nz^{N-1}$$

does not vanish at a root of unity. Hence

$$z^N - 1 = \prod_{j=0}^{N-1} (z - \zeta_N^j).$$

Complex conjugation reflects the polygon across the real axis:

$$\overline{\zeta_N^j} = \zeta_N^{-j} = \zeta_N^{N-j}.$$

Therefore nonreal roots occur in conjugate pairs. If N is odd, 1 is the only real root. If N is even, the real roots are 1 and -1.

For instance,

$$\omega = \frac{-1 + i\sqrt{3}}{2} = e^{2\pi i/3}$$

is a nontrivial cubic root of unity, so

$$\omega^3 = 1$$

and

$$1 + \omega + \omega^2 = 0.$$

This single equilateral-triangle identity is enough to solve a basic polynomial-divisibility problem.

§19.9 One elementary divisibility test

Let

$$P_n(x) = x^{2n} + x^n + 1$$

and

$$Q(x) = x^2 + x + 1.$$

The roots of Q are ω and ω^2, where $\omega^3 = 1$ and $\omega \neq 1$. Since the roots are distinct,

$$Q(x) \mid P_n(x)$$

exactly when both roots of Q are roots of P_n.

At $x = \omega$,

$$P_n(\omega) = \omega^{2n} + \omega^n + 1.$$

If $n \equiv 0 \pmod 3$, then $\omega^n = 1$ and

$$P_n(\omega) = 3 \neq 0.$$

If $n \equiv 1$ or $2 \pmod 3$, then ω^n is one of the two nontrivial cubic roots, and

$$1 + \omega^n + \omega^{2n} = 0.$$

The same argument applies to ω^2. Therefore

$$\boxed{x^2 + x + 1 \mid x^{2n} + x^n + 1 \iff 3 \nmid n.}$$

The proof is elementary: polynomial divisibility is checked on the two vertices of an equilateral triangle, and exponent reduction is arithmetic modulo three.

Arithmetic and Geometric Sequence Exercises

N-20220803

§20.1 A trigonometric opening example

The first exercise concerns a triangle ABC such that

$$\lg(\sin A), \quad \lg(\sin B), \quad \lg(\sin C)$$

form an arithmetic progression, and A, B, C themselves also form an arithmetic progression. Since

$$A + B + C = \pi,$$

if A, B, C are in arithmetic progression, then

$$B = \frac{\pi}{3}.$$

The logarithmic condition means

$$2\lg(\sin B) = \lg(\sin A) + \lg(\sin C),$$

so

$$\sin^2 B = \sin A \sin C.$$

Because $\sin B = \sqrt{3}/2$, we get

$$\sin A \sin C = \frac{3}{4}.$$

Using $A + C = 2\pi/3$, one can analyze

$$\cos(A - C) = \cos A \cos C + \sin A \sin C.$$

The computation in the page leads to the conclusion that the triangle is equilateral. The underlying idea is that two independent arithmetic-progressions conditions force symmetry.

§20.2 Logarithmic sequences

If

$$a_n = 10 + \lg 2^n,$$

then

$$a_n = 10 + n \lg 2.$$

Hence

$$a_{n+1} - a_n = \lg 2,$$

so $\{a_n\}$ is an arithmetic progression.

This is a typical use of logarithms: they turn powers into multiples. A geometric pattern in 2^n becomes an arithmetic pattern after applying lg.

§20.3 Recovering terms from partial sums

If S_n is the sum of the first n terms of a sequence, then

$$a_n = S_n - S_{n-1}.$$

The note uses this in problems involving two arithmetic progressions $\{a_n\}$, $\{b_n\}$ whose sums S_n, T_n satisfy

$$\frac{S_n}{T_n} = \frac{2n}{3n+1}.$$

For arithmetic progressions,

$$S_n = \frac{n}{2}(a_1 + a_n),$$

but the cleaner trick is to use

$$a_n = S_n - S_{n-1}, \qquad b_n = T_n - T_{n-1}.$$

The page obtains

$$\frac{a_n}{b_n} = \frac{4n-2}{6n-2} = \frac{2n-1}{3n-1}.$$

§20.4 Sums and recurrence relations

A common type of problem gives a relation between S_n and a_n. For example,

$$S_n = 2a_n - 2.$$

Then

$$a_{n+1} = S_{n+1} - S_n = 2a_{n+1} - 2 - (2a_n - 2),$$

so

$$a_{n+1} = 2a_n.$$

Thus the sequence is geometric. This is a useful principle: whenever a problem relates a term to a partial sum, subtract consecutive equations.

§20.5 Geometric progression products

If $\{a_n\}$ is a geometric progression, then

$$a_m^2 = a_{m-r}a_{m+r}.$$

This appears in several exercises. It follows from

$$a_n = a_1 q^{n-1}.$$

For example,

$$a_6^2 = a_5 a_7.$$

The note uses such identities to compute missing terms and ratios from partial information.

§20.6 A harmonic-looking sequence

One page studies

$$a_n = \frac{1}{1+2} + \frac{1}{2+3} + \cdots + \frac{1}{n+n+1}$$

or similar telescoping sums. The key transformation is

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Telescoping is the additive analogue of cancellation in products. When the denominator factors into consecutive terms, always look for a difference of reciprocals.

§20.7 A final synthesis

This note is mostly exercise-driven, but the underlying methods are stable: logarithms convert multiplication into addition, partial sums are inverted by differences, arithmetic progressions have linear terms, geometric progressions have exponential terms, and telescoping makes hidden cancellation explicit. These are exactly the same patterns that later become finite-difference calculus.

§20.8 Sequences as discrete dynamical systems

An arithmetic progression and a geometric progression are both generated by repeatedly applying a simple transformation. For an arithmetic progression,

$$a_{n+1} = a_n + d,$$

so the update is translation. For a geometric progression,

$$a_{n+1} = qa_n,$$

so the update is scaling. This explains why logarithms turn geometric progressions into arithmetic ones:

$$\log a_{n+1} = \log q + \log a_n.$$

Multiplicative iteration becomes additive iteration after applying log.

This is the same structural idea behind linear recurrences. A sequence is not only a list of numbers; it is often an orbit under an update rule.

§20.9 Partial sums and finite differences

If $S_n = a_1 + \cdots + a_n$, then

$$a_n = S_n - S_{n-1} = \Delta S_{n-1}.$$

So recovering terms from partial sums is finite differentiation. Conversely,

$$S_n = S_0 + \sum_{k=1}^{n} a_k$$

is finite integration.

For an arithmetic progression $a_n = a_1 + (n-1)d$, the first difference is constant:

$$\Delta a_n = d.$$

For a quadratic sequence, the second difference is constant. This is why finite-difference tables detect polynomial sequences, just as derivatives detect polynomial degree in calculus.

§20.10 Generating functions for sequence exercises

A sequence (a_n) can be packaged into a generating function

$$A(x) = \sum_{n\geq 0} a_n x^n.$$

Arithmetic and geometric progressions have especially simple forms. For example,

$$\sum_{n\geq 0} q^n x^n = \frac{1}{1-qx}.$$

A linear sequence produces a rational generating function with denominator $(1-x)^2$. Recurrences translate into algebraic equations for generating functions.

This turns many sequence problems into algebra: instead of manipulating terms one by one, encode the whole sequence and operate on the generating function.

§20.11 Telescoping as a coboundary

A telescoping sum has the form

$$a_k = F(k+1) - F(k).$$

Then

$$\sum_{k=m}^{n} a_k = F(n+1) - F(m).$$

Only the boundary terms survive. In this language, a telescoping term is a discrete derivative or coboundary.

The identity

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

is exactly this pattern with $F(k) = 1/k$. The cancellation is not accidental; it is the discrete fundamental theorem of calculus.

§20.12 Second-order recurrences and characteristic roots

Some sequence problems go beyond arithmetic and geometric progressions. A linear recurrence such as

$$a_{n+2} = pa_{n+1} + qa_n$$

is solved by trying $a_n = \lambda^n$. This gives the characteristic equation

$$\lambda^2 = p\lambda + q.$$

When the roots are distinct, the solution has the form

$$a_n = A\lambda_1^n + B\lambda_2^n.$$

Thus geometric progressions are the eigenvectors of the shift recurrence. This is the bridge from elementary sequences to linear algebra and dynamical systems.

§20.13 Discrete calculus of elementary sequences

The exercises are unified by discrete calculus. Logarithms linearize multiplication, finite differences invert partial sums, telescoping records boundary cancellation, generating functions encode whole sequences, and linear recurrences are controlled by characteristic roots. The elementary tricks are the visible form of a general principle: choose the representation in which the sequence's update rule becomes simple.

Divisibility, Congruences, and the Habit of Proving Necessity

N-20220913

§21.1 From guessing to proving

The note opens with a useful warning: in number theory, guessing examples is not enough. One must separate necessity and sufficiency. If a condition is necessary, it must follow from the assumption. If it is sufficient, it must imply the desired conclusion. Many false solutions prove only one direction.

The first example asks for a prime p of the form

$$p = x(x+1) - 4, \qquad x \in \mathbb{N}_+.$$

Since $x(x+1)$ is always even,

$$p = x(x+1) - 4$$

is even. If p is prime, then the only even prime is 2. Hence

$$p = 2, \qquad x(x+1) = 6,$$

so

$$x = 2.$$

This is a clean example of using parity as a necessary condition before solving.

§21.2 Divisibility

For integers a, b, with $a \neq 0$, we write

$$a \mid b$$

if there exists $q \in \mathbb{Z}$ such that

$$b = aq.$$

The page lists standard properties:

(i) $a \mid b \iff -a \mid b \iff a \mid -b \iff |a| \mid |b|$;

(ii) if $a \mid b$ and $b \mid c$, then $a \mid c$;

(iii) if $a \mid b$ and $a \mid c$, then $a \mid bx + cy$ for all integers x, y;

(iv) if $m \neq 0$, then $a \mid b \iff ma \mid mb$;

(v) if $a \mid b$ and $b \mid a$, then $b = \pm a$;

(vi) if $b \neq 0$ and $a \mid b$, then $|a| \leq |b|$.

The handwritten note points out that property (iii) is often the most useful. It lets us combine divisibility relations linearly.

§21.3 Example: reducing the dividend

Suppose

$$x^2 + 1 \mid x^3 - x, \qquad x \in \mathbb{Z}_+.$$

Modulo $x^2 + 1$, we have

$$x^2 \equiv -1,$$

so

$$x^3 - x = x(x^2) - x \equiv -x - x = -2x.$$

Thus

$$x^2 + 1 \mid 2x.$$

For $x > 1$,

$$x^2 + 1 > 2x,$$

so divisibility is impossible unless $2x = 0$, which cannot happen for positive x. Checking $x = 1$, we get

$$1^3 - 1 = 0,$$

so $x = 1$ works.

This example illustrates the handwritten strategy: reduce the large expression by using the divisor itself, then use size estimates to force a small list of cases.

§21.4 Example: polynomial divisibility in one variable

For divisibility such as

$$x^2 + x + 1 \mid x^4 - 2,$$

one uses

$$x^2 + x + 1 = 0 \implies x^3 - 1 = (x-1)(x^2 + x + 1) \equiv 0.$$

Thus $x^3 \equiv 1$, and

$$x^4 - 2 \equiv x - 2.$$

So the divisor must divide $x - 2$. This leaves only very small candidates, because for large $|x|$, the quadratic divisor is too large in absolute value to divide the linear remainder.

The general method is Euclidean division in disguise: replace the dividend by its remainder modulo the divisor.

§21.5 Congruences

Congruence notation compresses divisibility:

$$a \equiv b \pmod{m}$$

means

$$m \mid a - b.$$

This turns many divisibility problems into arithmetic with remainders. The note emphasizes that congruences can be added, subtracted, and multiplied:

$$a \equiv b \pmod{m}, \quad c \equiv d \pmod{m} \Longrightarrow a + c \equiv b + d, \quad ac \equiv bd \pmod{m}.$$

§21.6 A large power sum

The last problem asks to prove a divisibility statement involving

$$1^{1983} + 2^{1983} + \cdots + 1983^{1983}.$$

The intended thought is pairing and congruence. When the exponent is odd,

$$(m-k)^{1983} \equiv -k^{1983} \pmod{m}.$$

So terms can be paired to cancel modulo a suitable modulus. To prove divisibility by

$$1 + 2 + \cdots + 1983 = \frac{1983 \cdot 1984}{2},$$

one decomposes the modulus into factors and proves divisibility factor by factor, using pairing and residue classes.

The handwritten solution is valuable because it records a strategy rather than just a theorem: for large power sums, first look for symmetry $k \leftrightarrow m-k$, then reduce to congruences.

§21.7 A wider interpretation

Divisibility is the first place where algebraic thinking becomes economical. Instead of expanding huge expressions, we work modulo the divisor and keep only remainders. This is exactly the same principle behind quotient rings such as

$$\mathbb{Z}/m\mathbb{Z}$$

and polynomial quotients

$$F[x]/(f).$$

The note's small examples are therefore early versions of modular algebra.

§21.8 Divisibility as ideal containment

In $\mathbb{Z}$, the statement $a \mid b$ is equivalent to

$$b \in (a),$$

where $(a) = a\mathbb{Z}$ is the ideal generated by a. Divisibility reverses inclusion of principal ideals:

$$a \mid b \iff (b) \subseteq (a).$$

This explains why linear combinations are central. If $a \mid b$ and $a \mid c$, then every integer linear combination $bx + cy$ lies in (a).

§21.9 GCD and LCM as ideal operations

For integers a, b, Bezout's theorem says

$$(a) + (b) = (\gcd(a, b)).$$

The ideal generated by all linear combinations of a and b is generated by their greatest common divisor. Similarly,

$$(a) \cap (b) = (\operatorname{lcm}(a, b)).$$

Thus gcd and lcm are not just computational devices; they are sum and intersection of ideals in $\mathbb{Z}$.

§21.10 Congruences as quotient rings

The congruence
$$a \equiv b \pmod{n}$$
means that a and b define the same element of the quotient ring
$$\mathbb{Z}/n\mathbb{Z}.$$
The projection
$$\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$$
forgets the exact integer and remembers only its residue class. Its kernel is $n\mathbb{Z}$, so this is the first isomorphism theorem in its simplest form:
$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/\ker \pi.$$
Modular arithmetic is therefore quotient algebra, not merely a remainder shortcut.

§21.11 Units and Bezout inverses

An integer a has a multiplicative inverse modulo n exactly when
$$\gcd(a, n) = 1.$$
By Bezout, there exist integers x, y such that
$$ax + ny = 1.$$
Reducing modulo n gives
$$ax \equiv 1 \pmod{n},$$
so x is the inverse of a modulo n. This is the concrete connection between gcd computations and units in $\mathbb{Z}/n\mathbb{Z}$.

§21.12 Valuations as prime coordinates

For nonzero integers, $v_p(n)$ denotes the exponent of the prime p in n. Divisibility becomes coordinatewise comparison:
$$a \mid b \iff v_p(a) \le v_p(b) \text{ for all primes } p.$$
Gcd and lcm become coordinatewise minimum and maximum:
$$v_p(\gcd(a, b)) = \min(v_p(a), v_p(b)), \qquad v_p(\operatorname{lcm}(a, b)) = \max(v_p(a), v_p(b)).$$
This turns divisibility into geometry on the exponent lattice of prime factorizations.

§21.13 Chinese remainder theorem as local-to-global

Theorem 21.13.1 (Chinese remainder theorem)
If $m_1, \dots, m_r$ are pairwise coprime positive integers, then

$$\mathbb{Z}/(m_1 \cdots m_r)\mathbb{Z} \cong \prod_{i=1}^{r} \mathbb{Z}/m_i\mathbb{Z}.$$

If

$$n = \prod_i p_i^{e_i},$$

then the Chinese remainder theorem gives

$$\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{e_i}\mathbb{Z}.$$

So a congruence problem modulo n can often be solved prime-power by prime-power and then recombined.

This is a first local-to-global principle: understand arithmetic at each prime separately, then assemble the global answer.

§21.14 Polynomial divisibility as the same story

Over a field k, the ring $k[x]$ has Euclidean division:

$$f = qg + r, \qquad \deg r < \deg g.$$

Therefore the methods of integer divisibility have polynomial analogues. In particular,

$$x - a \mid f(x) \iff f(a) = 0.$$

Working modulo a polynomial $g(x)$ means working in the quotient ring $k[x]/(g)$. The earlier examples that reduce $x^4 - 2$ modulo $x^2 + x + 1$ are exactly quotient-ring computations in elementary disguise.

§21.15 Divisibility as quotient algebra

Divisibility problems become clearer when the divisor is treated as a structure rather than an obstacle. Ideals organize divisibility, quotient rings organize congruence, valuations localize divisibility at primes, and Euclidean division transfers the same ideas to polynomials. The elementary habit "reduce by the divisor" is the first step toward quotient algebra.

Divisibility by Finite Differences and Pairing

N-20220918

§22.1 A general odd-power divisibility principle

The page begins with a useful theorem: for odd r, the sum

$$1^r + 2^r + \cdots + k^r$$

has a strong divisibility property by the triangular number

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

The natural proof idea is pairing. Pair j with $k+1-j$. Since r is odd,

$$j^r + (k+1-j)^r$$

is divisible by $k+1$, because

$$k+1-j \equiv -j \pmod{k+1}.$$

A second argument handles the factor k or the remaining factor according to parity. The handwritten note records the right high-level method in blue: pairing, oddness, and treating the whole expression rather than each term separately.

§22.2 A divisibility problem with 512

The main worked example is:

$$\text{prove that } 512 \mid 3^{2n} - 32n^2 + 24n - 1, \qquad n \in \mathbb{N}_+.$$

Set

$$f(n) = 3^{2n} - 32n^2 + 24n - 1.$$

The note uses finite differences. Since $512 \mid f(1)$, it is enough to show

$$512 \mid f(n+1) - f(n)$$

for all n. Compute

$$\begin{aligned} f(n+1) - f(n) &= 3^{2n+2} - 32(n+1)^2 + 24(n+1) - 1 \\ &\quad - 3^{2n} + 32n^2 - 24n + 1 \\ &= 8 \cdot 3^{2n} - 64n - 8 \\ &= 8(3^{2n} - 8n - 1). \end{aligned}$$

Thus it remains to prove

$$64 \mid 3^{2n} - 8n - 1.$$

Define
$$h(n) = 3^{2n} - 8n - 1.$$
Then
$$\begin{aligned} h(n+1) - h(n) &= 3^{2n+2} - 8(n+1) - 1 - 3^{2n} + 8n + 1 \\ &= 8 \cdot 3^{2n} - 8 \\ &= 8(3^{2n} - 1). \end{aligned}$$
So it is enough to know
$$8 \mid 3^{2n} - 1.$$
But $3^2 = 9 \equiv 1 \pmod{8}$, so
$$3^{2n} \equiv 1 \pmod{8}.$$
Therefore $64 \mid h(n)$, and hence $512 \mid f(n)$.

§22.3 Why finite differences work

The solution is essentially induction, but written in a more structural way. If $m \mid f(1)$ and $m \mid f(n+1) - f(n)$ for every n, then
$$f(n) = f(1) + \sum_{j=1}^{n-1} (f(j+1) - f(j))$$
is divisible by m. This is the discrete analogue of recovering a function from its derivative.

The page is valuable because it shows a common trick for expressions involving n both in an exponent and a polynomial. Direct induction may be messy, but finite differences reduce the polynomial part and often reveal a smaller congruence.

§22.4 An alternating harmonic numerator

The second problem is:
$$\frac{p}{q} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1318} + \frac{1}{1319},$$
where p, q are coprime positive integers, and one must prove
$$1979 \mid p.$$
The note rewrites the alternating sum by separating odd and even denominators:
$$1 + \frac{1}{3} + \cdots + \frac{1}{1319} - 2\left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{1318}\right).$$
After simplification, this becomes a sum of paired fractions in which a factor 1979 appears in the numerator. The important observation is that 1979 is prime and is coprime to the denominators appearing after cancellation, so if the rational number is written in lowest terms, the factor 1979 cannot disappear into the denominator. Hence it must divide p.

§22.5 The next abstraction

Both problems use the same philosophy: do not expand everything. For power sums, pair symmetric terms. For congruence sequences, take finite differences. For rational sums, reorganize the denominator structure so that a prime factor becomes visible. These are all examples of choosing the right invariant before calculating.

§22.6 Finite differences as discrete derivatives

The finite difference operator

$$\Delta f(n) = f(n+1) - f(n)$$

is the discrete analogue of differentiation. If $m \mid f(1)$ and $m \mid \Delta f(n)$ for every n, then

$$f(n) = f(1) + \sum_{j=1}^{n-1} \Delta f(j)$$

is divisible by m. This is induction written as a discrete fundamental theorem of calculus.

For a polynomial of degree d,

$$\Delta^{d+1} f = 0.$$

So finite differences lower polynomial complexity in exactly the way derivatives do.

§22.7 The binomial basis and integer-valued polynomials

The binomial polynomials

$$\binom{x}{0}, \quad \binom{x}{1}, \quad \binom{x}{2}, \quad \dots$$

are adapted to finite differences:

$$\Delta \binom{x}{k} = \binom{x}{k-1}.$$

A classical theorem says that every integer-valued polynomial has a unique expansion

$$f(x) = \sum_k c_k \binom{x}{k}, \qquad c_k \in \mathbb{Z}.$$

This basis is often better than the monomial basis $1, x, x^2, \dots$ when proving divisibility for all integers.

The reason is simple: $\binom{n}{k}$ is automatically integral for integer n, and finite differences act on it cleanly.

§22.8 Pairing as group symmetry

In odd-power sums, the pairing

$$j \longleftrightarrow m - j$$

is an action of a two-element symmetry group on residue classes modulo m. If r is odd, then

$$(m-j)^r \equiv -j^r \pmod{m},$$

so each orbit contributes zero modulo m, except possibly fixed points.

Thus the cancellation is not a lucky trick. It is orbit cancellation under the involution $j \mapsto m - j$. Similar symmetry arguments appear in character sums and finite group actions.

§22.9 P-adic valuation and lifting intuition

For a prime p, $v_p(N)$ records the exponent of p dividing N. Divisibility by p^k becomes the inequality

$$v_p(N) \geq k.$$

In problems such as proving divisibility by $512 = 2^9$, the real task is to track how many factors of 2 are gained at each finite-difference step.

The deeper principle is lifting. A congruence modulo p may sometimes be lifted to a congruence modulo higher powers p^k. Hensel lifting is the systematic version of this idea. Even when the theorem is not invoked, the intuition is valuable: prime-power divisibility is controlled locally and incrementally.

§22.10 Alternating harmonic sums and denominator control

For rational sums, the numerator after reduction depends on which prime factors can survive cancellation with the denominator. If a prime p appears in the numerator of a common-denominator expression and is coprime to the denominator after simplification, then p must remain in the reduced numerator.

This is a valuation statement. Writing a rational number as A/B, one compares $v_p(A)$ and $v_p(B)$. Reduction cancels the smaller valuation from both sides. If $v_p(B) = 0$ and $v_p(A) > 0$, then the reduced numerator is still divisible by p.

§22.11 Discrete summation by parts

There is also a discrete analogue of integration by parts. For sequences a_n, b_n, one form is

$$\sum_{n=m}^{N-1} a_n \Delta b_n = a_N b_N - a_m b_m - \sum_{n=m}^{N-1} b_{n+1} \Delta a_n.$$

This technique turns products inside sums into boundary terms plus a simpler sum. It is the natural extension of telescoping and finite differences.

§22.12 Discrete calculus and local divisibility

The note's divisibility tricks are forms of discrete calculus and local arithmetic. Finite differences lower complexity, telescoping keeps boundary terms, pairing exploits symmetry, valuations measure prime-power divisibility, and rational-sum arguments track which primes can cancel. The correct invariant is usually not the whole expression, but its difference, orbit pairing, or valuation.

Oriented Boundary Sums, Polynomial Remainders, and Congruence Classes

N-20221013

§23.1 Three short computations hide three different kinds of structure

A coordinate-area formula, a polynomial remainder problem, and a modular-exponent problem can appear on the same exercise sheet without sharing one master theorem. What they do share is a style of simplification: each calculation throws away information that the question does not need.

For a polygon, the relevant data are the oriented contributions of its edges. For a polynomial modulo $x-a$, all multiples of $x-a$ become invisible and only the value at a remains. For an integer modulo m, numbers differing by a multiple of m are treated as the same residue class. The useful lesson is therefore not that every "remainder" is secretly identical, but that one should identify the map that preserves exactly the requested invariant.

§23.2 A determinant is the signed area of two position vectors

For vectors

$$p = (x_1, y_1), \qquad q = (x_2, y_2),$$

define

$$\det(p, q) = x_1 y_2 - x_2 y_1.$$

Its absolute value is the area of the parallelogram spanned by p, q, while its sign records orientation. Thus the oriented area of the triangle with vertices $0, p, q$ is

$$\frac{1}{2} \det(p, q).$$

Let a polygon have cyclically ordered vertices

$$p_i = (x_i, y_i), \qquad i = 1, \ldots, n, \qquad p_{n+1} = p_1.$$

Triangulating from the origin suggests the signed sum

$$\frac{1}{2} \sum_{i=1}^{n} \det(p_i, p_{i+1}).$$

If the origin lies outside the polygon, some triangles point with the opposite orientation. This is not a defect: the negative pieces cancel the extra positive pieces, leaving the polygon's oriented area.

§23.3 The shoelace formula is a boundary integral evaluated edge by edge

Green's theorem gives an origin-free expression for the oriented area of a positively oriented region D:

$$\mathrm{Area}(D) = \frac{1}{2}\int_{\partial D}(x\,dy - y\,dx).$$

For a polygon, parametrize the edge from p_i to p_{i+1} by

$$r_i(t) = p_i + t(p_{i+1} - p_i), \qquad 0 \le t \le 1.$$

Writing $r_i(t) = (x(t), y(t))$, a direct calculation gives

$$\begin{aligned}\frac{1}{2}\int_{p_i}^{p_{i+1}}(x\,dy - y\,dx) &= \frac{1}{2}\int_0^1 (x(t)y'(t) - y(t)x'(t))\,dt\\ &= \frac{1}{2}(x_i y_{i+1} - x_{i+1}y_i).\end{aligned}$$

Summing over the edges yields

$$\boxed{\mathrm{Area}(D) = \frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1}y_i)\right|.}$$

The familiar crossing columns are therefore a mnemonic for a discrete line integral, not the reason the formula is true.

§23.4 Translation and orientation provide immediate consistency checks

The shoelace sum should not depend on where the coordinate origin is placed. Translate every vertex by a fixed vector t. Then

$$\det(p_i + t, p_{i+1} + t) = \det(p_i, p_{i+1}) + \det(t, p_{i+1} - p_i).$$

After summing around the closed polygon, the extra term vanishes because

$$\sum_{i=1}^{n}(p_{i+1} - p_i) = 0.$$

Thus the signed area is translation invariant even though each individual determinant uses position vectors from the origin.

Reversing the cyclic order changes every edge orientation and therefore changes the sign of the sum. The absolute value in the ordinary shoelace formula forgets this orientation. For a self-intersecting closed polygonal path, the same sum computes signed area with winding multiplicity; overlapping regions may cancel rather than simply add.

§23.5 A complete polygon calculation

Take the vertices

$$(3,4),\ (5,11),\ (12,8),\ (9,5),\ (5,6)$$

in the displayed cyclic order. The signed double area is

$$\begin{aligned} 2A = {} & 3 \cdot 11 - 5 \cdot 4 + 5 \cdot 8 - 12 \cdot 11 \\ & + 12 \cdot 5 - 9 \cdot 8 + 9 \cdot 6 - 5 \cdot 5 \\ & + 5 \cdot 4 - 3 \cdot 6 \\ = {} & -60. \end{aligned}$$

Hence

$$A = 30.$$

The negative sign only says that the listed orientation is clockwise. Reversing the order would give +60 without changing the ordinary area.

§23.6 Evaluation removes every multiple of $x - a$

Let k be a field and $f \in k[x]$. Polynomial division gives unique $q \in k[x]$ and $r \in k$ such that

$$f(x) = (x - a)q(x) + r.$$

Substituting $x = a$ kills the divisible part:

$$f(a) = r.$$

Thus the remainder theorem is the statement

$$f(x) \equiv f(a) \pmod{x - a}.$$

This is naturally encoded by the evaluation homomorphism

$$\mathrm{ev}_a : k[x] \longrightarrow k, \qquad f \longmapsto f(a).$$

Its kernel is the ideal $(x - a)$, so the first isomorphism theorem gives

$$k[x]/(x - a) \cong k.$$

The quotient notation is useful because it says exactly what has been discarded: two polynomials become equal when their difference is divisible by $x - a$.

§23.7 Remainders modulo several linear factors are interpolation data

Suppose $f(x)$ leaves remainder 99 modulo $x - 19$ and remainder 19 modulo $x - 99$. The remainder modulo

$$(x - 19)(x - 99)$$

has degree at most one, so write

$$R(x) = ux + v.$$

It must satisfy

$$R(19) = 99, \qquad R(99) = 19.$$

Hence

$$19u + v = 99, \qquad 99u + v = 19,$$

and therefore

$$u = -1, \qquad v = 118.$$

Thus

$$\boxed{R(x) = 118 - x.}$$

This is not a separate trick from interpolation. A polynomial of degree at most one is uniquely determined by its values at two distinct points. In Lagrange form,

$$R(x) = 99\frac{x-99}{19-99} + 19\frac{x-19}{99-19},$$

which simplifies to the same answer.

§23.8 The polynomial Chinese remainder theorem explains uniqueness

Because $19 \neq 99$, the ideals $(x-19)$ and $(x-99)$ are comaximal. The Chinese remainder theorem gives

$$k[x]/((x-19)(x-99)) \cong k[x]/(x-19) \times k[x]/(x-99).$$

Using evaluation on each factor,

$$k[x]/((x-19)(x-99)) \cong k \times k.$$

A residue class modulo the quadratic is therefore exactly a pair of values

$$(f(19), f(99)).$$

The linear remainder found above is the unique degree-<2 representative of the pair $(99, 19)$.

More generally, for distinct $a_1, \ldots, a_n$,

$$k[x]/\prod_{i=1}^{n}(x-a_i) \cong k^n,$$

and the inverse map is Lagrange interpolation. The elementary remainder theorem has grown into a structural statement without changing the computation it was designed to perform.

§23.9 Integer congruence is another quotient arithmetic

For integers, write

$$a \equiv b \pmod{m}$$

when $m \mid (a-b)$. Arithmetic then takes place in the quotient ring

$$\mathbb{Z}/m\mathbb{Z}.$$

Addition and multiplication descend because replacing an integer by another representative of the same class does not change the resulting class.

Division is more delicate. From

$$ca \equiv cb \pmod{m}$$

one may cancel c only when its residue class is a unit, equivalently when

$$\gcd(c, m) = 1.$$

For example,

$$2x \equiv 2 \pmod 6$$

has the two solutions

$$x \equiv 1, 4 \pmod 6.$$

Cancelling 2 would incorrectly discard one of them. Quotient arithmetic preserves addition and multiplication automatically, but inverse operations require invertibility.

Small residue sets give quick impossibility tests. Squares satisfy

$$n^2 \equiv 0, 1 \pmod 3$$

and

$$n^2 \equiv 0, 1, 4 \pmod 8.$$

A proposed square lying outside these lists is impossible before any large-number calculation begins.

§23.10 Finite unit groups turn huge powers into short cycles

If p is prime and $p \nmid a$, Fermat's theorem says

$$a^{p-1} \equiv 1 \pmod p.$$

The reason is that the nonzero residue classes form the finite group

$$(\mathbb{Z}/p\mathbb{Z})^\times$$

of order $p-1$. Lagrange's theorem makes the exponent of every element divide the group order.

To compute

$$2^{1000} \pmod{13},$$

use

$$2^{12} \equiv 1 \pmod{13}$$

and reduce the exponent:

$$1000 = 83 \cdot 12 + 4.$$

Therefore

$$2^{1000} \equiv 2^4 = 16 \equiv 3 \pmod{13}.$$

The quotient ring has retained precisely the residue information while discarding the enormous integer itself.

The three parts of the chapter now end differently. The polygon formula converts a closed boundary into oriented area. Polynomial evaluation converts divisibility by linear factors into interpolation data. Modular arithmetic converts integers into residue classes and exposes finite cycles. Their common practical habit is to choose the correct invariant map before expanding the calculation.

Integer Arithmetic: Euclidean Descent, Bézout Identities, and the Euler Function

N-20221030

§24.1 Descent is the algorithmic form of well-ordering

Many arguments about integers begin with a simple impossibility: a strictly decreasing sequence of nonnegative integers cannot continue forever. This is the computational face of the well-ordering principle.

Theorem 24.1.1 (Well-ordering principle)
Every nonempty subset of $\mathbb{Z}_{>0}$ has a least element.

Induction, minimal-counterexample arguments, and the Euclidean algorithm all use the same structure. One assumes that some positive integer with a desired property exists, chooses the least such integer, and then produces a smaller one unless the process has already reached its terminal case.

The division algorithm is the first important example.

Theorem 24.1.2 (Division algorithm)
Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There exist unique integers q, r such that

$$a = qb + r, \qquad 0 \leq r < |b|.$$

To see existence, consider all nonnegative integers of the form

$$a - k|b|, \qquad k \in \mathbb{Z}.$$

This set is nonempty: for sufficiently negative k, the number is positive. Let r be its least element, so

$$r = a - q|b|$$

for some integer q. If $r \geq |b|$, then

$$r - |b| = a - (q+1)|b|$$

would be a smaller nonnegative element of the same set, a contradiction. Hence $0 \leq r < |b|$. A sign adjustment replaces $|b|$ by b.

For uniqueness, suppose

$$a = qb + r = q'b + r'$$

with both remainders in $[0, |b|)$. Then

$$(q - q')b = r' - r.$$

The right side has absolute value less than $|b|$, while a nonzero multiple of b has absolute value at least $|b|$. Therefore both sides vanish.

The theorem is more than a formal version of long division. It supplies a remainder whose size is strictly smaller than the divisor, and that strict decrease is what makes Euclidean descent possible.

§24.2 The gcd is preserved when a remainder replaces a number

For integers a, b, not both zero, the greatest common divisor $d = \gcd(a, b)$ is the unique positive integer satisfying

$$d \mid a, \qquad d \mid b,$$

and every common divisor of a, b divides d.

If

$$a = qb + r,$$

then

$$\gcd(a, b) = \gcd(b, r).$$

Indeed, a number divides both a and b exactly when it divides both b and

$$r = a - qb.$$

The set of common divisors has not changed.

This invariant turns division into an algorithm. Starting with a, b, repeatedly replace the larger pair by the divisor and remainder:

$$\begin{aligned} a &= q_0 b + r_0, \\ b &= q_1 r_0 + r_1, \\ r_0 &= q_2 r_1 + r_2, \\ &\vdots \end{aligned}$$

with

$$b > r_0 > r_1 > r_2 > \cdots \geq 0.$$

Well-ordering forces the remainders to reach zero. The last nonzero remainder is the gcd because the common-divisor invariant survives every step.

The important object is not the displayed chain of divisions but the transformation

$$(a, b) \longmapsto (b, a - qb),$$

which preserves the ideal of integer linear combinations as well as the gcd.

§24.3 A complete Euclidean calculation

Take

$$a = 57970, \qquad b = 10353.$$

Repeated division gives

$$\begin{aligned} 57970 &= 5 \cdot 10353 + 6205, \\ 10353 &= 1 \cdot 6205 + 4148, \\ 6205 &= 1 \cdot 4148 + 2057, \\ 4148 &= 2 \cdot 2057 + 34, \\ 2057 &= 60 \cdot 34 + 17, \\ 34 &= 2 \cdot 17. \end{aligned}$$

Hence

$$\gcd(57970, 10353) = 17.$$

The same computation contains more information than the gcd. Reading the equations backward expresses 17 as an integer linear combination of the original numbers:

$$17 = 2057 - 60 \cdot 34.$$

Since

$$34 = 4148 - 2 \cdot 2057,$$

we obtain

$$17 = 121 \cdot 2057 - 60 \cdot 4148.$$

Now use

$$2057 = 6205 - 4148$$

to get

$$17 = 121 \cdot 6205 - 181 \cdot 4148.$$

Then

$$4148 = 10353 - 6205$$

gives

$$17 = 302 \cdot 6205 - 181 \cdot 10353.$$

Finally,

$$6205 = 57970 - 5 \cdot 10353,$$

so

$$\boxed{17 = 302 \cdot 57970 - 1691 \cdot 10353.}$$

This backward pass is the extended Euclidean algorithm. It computes coefficients, not merely a divisor.

§24.4 Bézout's identity is an ideal statement

For integers a, b, consider the set

$$(a, b) = \{ax + by : x, y \in \mathbb{Z}\}.$$

This is an ideal of $\mathbb{Z}$: it is closed under addition and under multiplication by arbitrary integers.

Let d be the least positive element of this ideal. Dividing a by d, write

$$a = qd + r, \qquad 0 \leq r < d.$$

Because both a and d lie in the ideal, so does

$$r = a - qd.$$

Minimality of d forces $r = 0$, so $d \mid a$. Similarly, $d \mid b$. Conversely, every common divisor of a, b divides every linear combination $ax + by$, hence divides d. Therefore

$$d = \gcd(a, b).$$

We have proved Bézout's identity:

$$\boxed{\gcd(a, b) = ax + by}$$

for suitable integers x, y.

In ideal notation,

$$(a) + (b) = (\gcd(a, b)).$$

The gcd is the positive generator of the sum of the two principal ideals.

This viewpoint explains why the extended Euclidean algorithm is so useful: it is a constructive proof that a particular ideal is principal. More generally, a Euclidean domain is an integral domain equipped with a size function that permits division with smaller remainder. Euclidean descent then produces gcds and Bézout identities. The rings

$$\mathbb{Z} \qquad \text{and} \qquad k[x]$$

for a field k are the standard examples.

Every Euclidean domain is a principal ideal domain, and every principal ideal domain is a unique factorization domain. The converses do not all hold. The elementary integer algorithm is therefore the visible beginning of a hierarchy of algebraic properties.

§24.5 Coprimality is exactly invertibility modulo n

Suppose

$$\gcd(a, n) = 1.$$

Bézout gives integers x, y such that

$$ax + ny = 1.$$

Reducing modulo n,

$$ax \equiv 1 \pmod{n}.$$

Thus x is a multiplicative inverse of a modulo n.

Conversely, if a has an inverse modulo n, then

$$ax \equiv 1 \pmod{n}$$

for some x, so

$$ax + ny = 1$$

for an integer y. Any common divisor of a, n must divide 1, and therefore

$$\gcd(a, n) = 1.$$

Hence

$$[a] \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \gcd(a, n) = 1.$$

The extended Euclidean algorithm is the computational heart of inversion in the quotient ring.

For example, the identity

$$17 = 302 \cdot 57970 - 1691 \cdot 10353$$

can be divided by 17 after first reducing the numbers:

$$1 = 302 \cdot 3410 - 1691 \cdot 609.$$

Thus

$$302 \cdot 3410 \equiv 1 \pmod{609},$$

so 302 is the inverse of 3410 modulo 609, or equivalently the inverse of its residue class 365.

This note uses modular inverses to reveal the connection between Bézout and unit groups. The systematic solution of linear congruences and simultaneous congruence systems belongs to the following arithmetic note.

§24.6 Prime divisibility makes factorization rigid

A prime p is an integer greater than 1 whose only positive divisors are 1 and p. The key lemma is not merely the definition but Euclid's divisibility property.

Proposition 24.6.1 (Euclid's lemma)

If p is prime and

$$p \mid ab,$$

then

$$p \mid a \qquad \text{or} \qquad p \mid b.$$

If $p \nmid a$, then $\gcd(p, a) = 1$. Bézout gives

$$px + ay = 1.$$

Multiplying by b,

$$pbx + aby = b.$$

Both terms on the left are divisible by p, so $p \mid b$.

This short proof shows how uniqueness of prime factorization grows from Bézout's identity.

Theorem 24.6.2 (Fundamental theorem of arithmetic)

Every integer $n > 1$ can be written as a product of primes, and the product is unique up to reordering.

Existence follows by minimal counterexample. If the least integer without a prime factorization were composite, it would split into two smaller positive integers, both of which factor by minimality.

For uniqueness, suppose

$$p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

Then p_1 divides the product on the right. Euclid's lemma says it divides one q_j, and since both are prime they are equal. Cancel this factor and continue.

Writing

$$n = \prod_p p^{v_p(n)},$$

the exponent $v_p(n)$ is the p-adic valuation of n. Gcd and lcm become coordinatewise minimum and maximum:

$$v_p(\gcd(a, b)) = \min\{v_p(a), v_p(b)\},$$

$$v_p(\operatorname{lcm}(a,b)) = \max\{v_p(a), v_p(b)\}.$$

Therefore, for positive a, b,

$$\gcd(a,b)\operatorname{lcm}(a,b) = ab.$$

Prime factorization turns divisibility into arithmetic on exponent vectors.

§24.7 Euler's function counts the unit group

Euler's totient function is

$$\varphi(n) = \#\{1 \le a \le n : \gcd(a,n) = 1\}.$$

The preceding section gives its structural meaning:

$$\boxed{\varphi(n) = \left|(\mathbb{Z}/n\mathbb{Z})^\times\right|.}$$

For a prime power p^k, the nonunits are exactly the multiples of p. Among the p^k residue classes, there are p^{k-1} such multiples. Hence

$$\varphi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right).$$

If m, n are coprime, the Chinese remainder theorem gives a ring isomorphism

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

An element of a product ring is a unit exactly when each component is a unit. Therefore

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times,$$

and

$$\varphi(mn) = \varphi(m)\varphi(n) \qquad \text{when } \gcd(m,n) = 1.$$

Thus φ is a multiplicative arithmetic function. If

$$n = \prod_{p \mid n} p^{v_p(n)},$$

then

$$\boxed{\varphi(n) = n \prod_{p \mid n}\left(1 - \frac{1}{p}\right).}$$

For example,

$$360 = 2^3 \cdot 3^2 \cdot 5,$$

so

$$\varphi(360) = 360\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = 96.$$

This is simultaneously an inclusion–exclusion count and the order of a finite group.

§24.8 Euler's theorem is Lagrange's theorem in disguise

If $\gcd(a, n) = 1$, then $[a]$ belongs to the finite group

$$G = (\mathbb{Z}/n\mathbb{Z})^{\times},$$

whose order is $\varphi(n)$. Lagrange's theorem implies that the order of $[a]$ divides $|G|$, so

$$\boxed{a^{\varphi(n)} \equiv 1 \pmod{n}.}$$

There is also a proof that looks elementary but contains the same group action. Let

$$r_1, \ldots, r_{\varphi(n)}$$

be a reduced residue system modulo n. Multiplication by a permutes these residues because $[a]$ is a unit. Hence

$$ar_1, \ldots, ar_{\varphi(n)}$$

represent the same residue classes in a different order. Taking products,

$$a^{\varphi(n)} r_1 \cdots r_{\varphi(n)} \equiv r_1 \cdots r_{\varphi(n)} \pmod{n}.$$

The product of the r_i is a unit and can be cancelled, giving Euler's theorem.

The group-theoretic proof explains both the exponent and the permutation. The elementary proof displays the action concretely.

The exponent $\varphi(n)$ is always valid but is not usually minimal. The least positive exponent m satisfying

$$a^m \equiv 1 \pmod{n}$$

for every unit a is the exponent of the group, called the Carmichael function $\lambda(n)$. Group structure refines the coarse bound supplied by group order.

§24.9 Counting elements by their exact order gives a divisor identity

A useful identity is

$$\boxed{\sum_{d|n} \varphi(d) = n.}$$

One proof counts the integers

$$1, 2, \ldots, n$$

according to their gcd with n.

Fix a divisor $d \mid n$. The integers k satisfying

$$\gcd(k, n) = \frac{n}{d}$$

are exactly those of the form

$$k = \frac{n}{d} a$$

with

$$1 \le a \le d, \qquad \gcd(a, d) = 1.$$

There are $\varphi(d)$ of them. Summing over all divisors partitions the n integers and gives the identity.

There is also a group-theoretic interpretation. In a cyclic group of order n, the number of elements of exact order d is $\varphi(d)$ for every $d \mid n$. Counting all group elements by their order gives the same sum.

The identity is an early example of arithmetic convolution. For arithmetic functions f, g, define the Dirichlet convolution

$$(f * g)(n) = \sum_{d \mid n} f(d) g(n/d).$$

Let $\mathbf{1}(n) = 1$ and $\mathrm{id}(n) = n$. Then

$$\mathbf{1} * \varphi = \mathrm{id}\,.$$

The convolution identity can be inverted using the Möbius function μ, which satisfies

$$\mu * \mathbf{1} = \varepsilon,$$

where $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$. Therefore

$$\varphi = \mu * \mathrm{id},$$

or explicitly,

$$\varphi(n) = \sum_{d \mid n} \mu(d) \frac{n}{d}.$$

For squarefree d, $\mu(d) = (-1)^k$ when d has k prime factors, so this formula is inclusion–exclusion written as an algebraic inverse.

This is the natural point where elementary counting begins to resemble analytic number theory: multiplicative functions form an algebra under Dirichlet convolution, and Euler products arise because prime-power data determine multiplicative functions.

The route from Euclidean descent to arithmetic convolution is continuous rather than a stack of unrelated viewpoints. The step

$$(a, b) \longmapsto (b, a - qb)$$

preserves common divisors and the generated ideal; back-substitution turns it into a Bézout identity and computes modular inverses. Euclid's lemma then converts coprimality into uniqueness of factorization, prime powers determine the unit count $\varphi(n)$, the Chinese remainder theorem explains multiplicativity, and counting by exact divisors leads to Dirichlet convolution and Möbius inversion. The later language does not replace the algorithm: it records the structures the same descent has been preserving from the beginning.

Congruences and Integer Linear Algebra: From Residue Classes to Smith Normal Form

N-20221031

§25.1 A remainder is a coordinate on a quotient

The division algorithm assigns to every integer a a unique remainder

$$r \in \{0, 1, \ldots, n-1\}$$

when divided by a positive integer n. Two integers have the same remainder exactly when their difference is divisible by n:

$$a \equiv b \pmod{n} \quad \iff \quad n \mid (a-b).$$

Congruence is therefore not approximate equality. It is equality in the quotient ring

$$\mathbb{Z}/n\mathbb{Z}.$$

The remainder is a chosen representative, or normal form, for the residue class.

Addition and multiplication descend to the quotient because replacing an integer by a congruent one does not change the resulting class:

$$a \equiv a', \qquad b \equiv b' \pmod{n}$$

implies

$$a + b \equiv a' + b', \qquad ab \equiv a'b' \pmod{n}.$$

The quotient remembers arithmetic while forgetting multiples of n.

This chapter begins where the preceding Euclidean-arithmetic note ended. Bézout identities and modular inverses will be used as tools, but the main questions are now different: which residue classes can a linear map reach, how do several moduli fit together, and what replaces row reduction when equations must be solved over the integers rather than over a field?

§25.2 A linear congruence is an image problem

Consider

$$ax \equiv b \pmod{n}.$$

Multiplication by a defines a group homomorphism

$$m_a : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}, \qquad [x] \longmapsto [ax].$$

The congruence asks whether $[b]$ lies in the image of this map.

Let

$$d = \gcd(a, n).$$

Every value ax is divisible by d modulo n, so solvability requires

$$d \mid b.$$

Conversely, suppose $b = db'$, and write

$$a = da', \qquad n = dn'.$$

Then $\gcd(a', n') = 1$, so a' has an inverse modulo n'. The reduced congruence

$$a'x \equiv b' \pmod{n'}$$

has a solution, and hence so does the original one.

Thus

$$\boxed{ax \equiv b \pmod{n} \text{ is solvable } \iff \gcd(a,n) \mid b.}$$

The kernel explains the number of solutions. We have

$$ax \equiv 0 \pmod{n}$$

exactly when

$$n' \mid x.$$

Modulo $n = dn'$, this gives the d classes

$$0, n', 2n', \ldots, (d-1)n'.$$

Therefore

$$|\ker m_a| = d.$$

By the finite-group version of rank–nullity,

$$|\operatorname{im} m_a| = \frac{n}{d}.$$

Whenever $[b]$ lies in the image, its fibre is a coset of the kernel and contains exactly d solutions modulo n.

This is the structural form of the elementary theorem: divisibility decides existence, and the kernel decides multiplicity.

§25.3 A complete congruence calculation

Solve

$$14x \equiv 8 \pmod{30}.$$

Since

$$\gcd(14, 30) = 2$$

and $2 \mid 8$, solutions exist. Divide the entire congruence and modulus by 2:

$$7x \equiv 4 \pmod{15}.$$

The inverse of 7 modulo 15 is 13, because

$$7 \cdot 13 = 91 \equiv 1 \pmod{15}.$$

Therefore

$$x \equiv 13 \cdot 4 \equiv 52 \equiv 7 \pmod{15}.$$

This describes one class modulo 15, but the original modulus was 30. Lifting back gives

$$x \equiv 7 \qquad \text{or} \qquad x \equiv 22 \pmod{30}.$$

There are exactly two solutions, as predicted by

$$d = \gcd(14, 30) = 2.$$

A direct check gives

$$14 \cdot 7 = 98 \equiv 8 \pmod{30},$$

$$14 \cdot 22 = 308 \equiv 8 \pmod{30}.$$

The calculation illustrates the general procedure:

(i) compute $d = \gcd(a, n)$;

(ii) test whether $d \mid b$;

(iii) divide by d, producing a coefficient invertible modulo n/d;

(iv) solve the reduced congruence;

(v) lift its single class modulo n/d to d classes modulo n.

§25.4 The Chinese remainder theorem glues local coordinates

Suppose m, n are coprime. The map

$$\Phi : \mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z},$$

$$[x]_{mn} \longmapsto ([x]_m, [x]_n)$$

is a ring homomorphism.

Its kernel consists of classes divisible by both m and n. Coprimality implies that such an integer is divisible by mn, so the kernel is zero. Both rings contain mn elements, hence the injective map is also surjective.

Therefore

$$\boxed{\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.}$$

This is the Chinese remainder theorem. It does not merely promise a solution to simultaneous congruences; it says that arithmetic modulo a product of coprime moduli is equivalent to independent arithmetic in each factor.

The isomorphism can be made constructive. Choose Bézout coefficients

$$um + vn = 1.$$

Then

$$e_m = vn$$

satisfies

$$e_m \equiv 1 \pmod{m}, \qquad e_m \equiv 0 \pmod{n},$$

while

$$e_n = um$$

satisfies the opposite conditions. The solution to

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}$$

is

$$x \equiv ae_m + be_n \pmod{mn}.$$

The elements e_m, e_n are orthogonal idempotents:

$$e_m^2 = e_m, \qquad e_n^2 = e_n, \qquad e_m e_n = 0, \qquad e_m + e_n = 1$$

in $\mathbb{Z}/mn\mathbb{Z}$. They project onto the two ring factors.

§25.5 A three-modulus reconstruction

Solve

$$x \equiv 2 \pmod{3}, \qquad x \equiv 3 \pmod{5}, \qquad x \equiv 2 \pmod{7}.$$

The moduli are pairwise coprime, so there is a unique solution modulo

$$3 \cdot 5 \cdot 7 = 105.$$

Set

$$N_1 = 35, \qquad N_2 = 21, \qquad N_3 = 15.$$

We need inverses of these partial products in the corresponding moduli:

$$\begin{aligned} 35 &\equiv 2 \pmod{3}, & 2^{-1} &\equiv 2 \pmod{3}, \\ 21 &\equiv 1 \pmod{5}, & 1^{-1} &\equiv 1 \pmod{5}, \\ 15 &\equiv 1 \pmod{7}, & 1^{-1} &\equiv 1 \pmod{7}. \end{aligned}$$

Hence

$$x \equiv 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 \pmod{105}.$$

The right side is

$$140 + 63 + 30 = 233,$$

so

$$\boxed{x \equiv 23 \pmod{105}.}$$

Indeed,

$$23 \equiv 2 \pmod{3}, \qquad 23 \equiv 3 \pmod{5}, \qquad 23 \equiv 2 \pmod{7}.$$

The reconstruction is a weighted sum of local answers. Each weight equals 1 in one component and 0 in all the others.

§25.6 When moduli are not coprime

Consider

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}.$$

If a solution exists, then

$$a - b = (a - x) + (x - b)$$

is divisible by

$$d = \gcd(m, n).$$

Thus a necessary condition is

$$a \equiv b \pmod{d}.$$

It is also sufficient. Write

$$a - b = dc, \qquad m = dm', \qquad n = dn'$$

with $\gcd(m', n') = 1$. Searching for $x = a + mt$, the second congruence becomes

$$m't \equiv -c \pmod{n'},$$

which is solvable because m' is invertible modulo n'.

Therefore

$$\boxed{\begin{matrix} x \equiv a \pmod{m}, \\ x \equiv b \pmod{n} \end{matrix} \text{ is solvable } \iff a \equiv b \pmod{\gcd(m, n)}.}$$

When solvable, the solution is unique modulo

$$\operatorname{lcm}(m, n).$$

The coprime CRT is the clean product decomposition. The general version is a compatibility-and-gluing theorem: local data must agree on the overlap measured by the gcd.

§25.7 Integer row reduction needs a stricter notion of invertibility

Over $\mathbb{R}$, Gaussian elimination allows multiplication of a row by any nonzero real number. Over $\mathbb{Z}$, this can destroy the integer lattice. Dividing a row by 2, for example, is invertible over $\mathbb{Q}$ but not as a map

$$\mathbb{Z}^m \longrightarrow \mathbb{Z}^m.$$

The correct coordinate changes are unimodular matrices: integer matrices U with

$$\det U = \pm 1.$$

Their inverses also have integer entries. The elementary unimodular operations are:

(i) exchange two rows or two columns;

(ii) add an integer multiple of one row or column to another;

(iii) multiply a row or column by -1.

These operations change bases of the domain or codomain lattice without changing the underlying homomorphism up to integer coordinates.

This is the essential difference between linear algebra over a field and linear algebra over $\mathbb{Z}$. Rank records the dimension after rational or real scalars are allowed, but it forgets divisibility information inside the lattice.

For example, the maps

$$[2] : \mathbb{Z} \to \mathbb{Z} \qquad \text{and} \qquad [3] : \mathbb{Z} \to \mathbb{Z}$$

both have rank 1, yet their images have indices 2 and 3, and their cokernels are

$$\mathbb{Z}/2\mathbb{Z} \qquad \text{and} \qquad \mathbb{Z}/3\mathbb{Z}.$$

An integer normal form must remember these indices.

§25.8 Smith normal form diagonalizes divisibility

Theorem 25.8.1 (Smith normal form)
For every integer matrix $A \in \mathbb{Z}^{m\times n}$, there exist unimodular matrices

$$P \in GL_m(\mathbb{Z}), \qquad Q \in GL_n(\mathbb{Z})$$

such that

$$PAQ = \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0),$$

where

$$d_i > 0, \qquad d_1 \mid d_2 \mid \cdots \mid d_r.$$

The nonzero numbers d_i are uniquely determined by A.

The algorithm repeatedly uses gcd and Bézout operations on rows and columns. A nonzero entry is moved to the upper-left corner. Euclidean combinations reduce it until it divides every entry in its row and column. One then clears that row and column and continues on the remaining submatrix. If the divisibility chain fails, adjacent diagonal entries are adjusted by another Bézout step.

Thus Smith reduction is the matrix-sized descendant of the Euclidean algorithm. A single gcd becomes a sequence of invariant factors.

The determinantal divisors provide an intrinsic description. Let Δ_k be the positive gcd of all $k \times k$ minors, with $\Delta_0 = 1$. Then

$$d_k = \frac{\Delta_k}{\Delta_{k-1}}.$$

In particular,

$$d_1 = \gcd\{a_{ij}\},$$

and for a full-rank square matrix,

$$d_1 d_2 \cdots d_n = |\det A|.$$

The invariant factors are therefore encoded in the gcds of minors.

§25.9 A complete Smith reduction

Take

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}.$$

Apply the integer row operation

$$R_2 \leftarrow R_2 - 3R_1 :$$

$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 4 \\ 0 & -4 \end{bmatrix}.$$

Then

$$C_2 \leftarrow C_2 - 2C_1$$

gives

$$\begin{bmatrix} 2 & 0 \\ 0 & -4 \end{bmatrix}.$$

Finally change the sign of the second row:

$$\boxed{PAQ = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.}$$

One choice is

$$P = \begin{bmatrix} 1 & 0 \\ 3 & -1 \end{bmatrix}, \qquad Q = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Both determinants are -1 or 1, so both matrices are unimodular.

The invariant factors are $2, 4$. They also follow from minors:

$$\Delta_1 = \gcd(2, 4, 6, 8) = 2,$$

$$\Delta_2 = |\det A| = 8,$$

so

$$d_1 = 2, \qquad d_2 = \frac{8}{2} = 4.$$

Over $\mathbb{Q}$, the matrix is simply invertible. Over $\mathbb{Z}$, the diagonal entries reveal exactly how its image sits inside $\mathbb{Z}^2$.

§25.10 Integer systems reduce to divisibility tests

Consider

$$Ax = b, \qquad A \in \mathbb{Z}^{m\times n}, \quad x \in \mathbb{Z}^n.$$

If

$$PAQ = D$$

is Smith normal form, set

$$y = Q^{-1}x, \qquad c = Pb.$$

Since P, Q are unimodular, x is integral exactly when y is integral. The system becomes

$$Dy = c.$$

For

$$D = \operatorname{diag}(d_1, \ldots, d_r, 0, \ldots, 0),$$

integer solvability is equivalent to

$$d_i \mid c_i \qquad (1 \le i \le r)$$

and

$$c_i = 0 \qquad (i > r).$$

The normal form separates the problem into independent one-dimensional divisibility conditions. This is the matrix analogue of the criterion

$$ax \equiv b \pmod n \quad \Longleftrightarrow \quad \gcd(a, n) \mid b.$$

For the example matrix, solving

$$Ax = b$$

reduces after multiplication by P to

$$2y_1 = c_1, \qquad 4y_2 = c_2.$$

The right-hand side must therefore land in the sublattice

$$2\mathbb{Z} \oplus 4\mathbb{Z}.$$

Smith form does not merely decide solvability; it displays the obstruction.

§25.11 The cokernel records what the map cannot reach

For an integer matrix

$$A : \mathbb{Z}^n \longrightarrow \mathbb{Z}^m,$$

define the cokernel

$$\operatorname{coker} A = \mathbb{Z}^m / A\mathbb{Z}^n.$$

Two target vectors represent the same cokernel class when their difference lies in the image of A. The group measures the failure of surjectivity over the integer lattice.

Smith normal form gives

$$\operatorname{coker} A \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_r\mathbb{Z} \oplus \mathbb{Z}^{m-r}.$$

For

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix},$$

we obtain

$$\operatorname{coker} A \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}.$$

Its order is

$$2 \cdot 4 = 8 = |\det A|.$$

Geometrically, the image lattice has index 8 in $\mathbb{Z}^2$. A fundamental parallelogram for the image has area 8, matching the determinant.

This is the finite-index lattice meaning of the determinant. Over $\mathbb{R}$, $|\det A|$ is a volume-scaling factor. Over $\mathbb{Z}$, for a nonsingular integer matrix, it is also the number of residue classes left after quotienting by the image.

§25.12 Finite abelian groups inherit the same normal form

Every finitely generated abelian group has a presentation

$$G \cong \mathbb{Z}^m / A\mathbb{Z}^n$$

for some integer matrix A. Smith normal form yields

$$G \cong \mathbb{Z}^{m-r} \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_r\mathbb{Z},$$

with

$$d_1 \mid d_2 \mid \cdots \mid d_r.$$

This is the invariant-factor form of the classification theorem for finitely generated abelian groups.

The CRT reappears inside this classification. If a, b are coprime, then

$$\mathbb{Z}/ab\mathbb{Z} \cong \mathbb{Z}/a\mathbb{Z} \oplus \mathbb{Z}/b\mathbb{Z}.$$

Factoring each invariant factor into prime powers gives the elementary-divisor form of the same group.

Thus CRT and Smith normal form are not separate tricks. Both decompose arithmetic objects into independent pieces, one by splitting coprime moduli and the other by diagonalizing an integer homomorphism.

§25.13 Congruence systems are Smith problems in finite disguise

A system

$$Ax \equiv b \pmod{n}$$

can be rewritten as an integer equation

$$Ax - ny = b$$

with an auxiliary integer vector y. Equivalently,

$$\begin{bmatrix} A & -nI \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = b.$$

Smith normal form of this enlarged integer matrix gives complete solvability conditions and parametrizes all solutions.

Alternatively, reduce A by unimodular row and column operations and interpret those operations modulo n. The system becomes a collection of scalar congruences

$$d_i z_i \equiv c_i \pmod{n}.$$

Each one is decided by

$$\gcd(d_i, n) \mid c_i.$$

The one-variable congruence theorem is therefore not an isolated result. It is the first diagonal component of a general normal-form theory for homomorphisms between finitely generated modules.

Over a field, rank and reduced row echelon form answer most elementary questions about a linear map. Over $\mathbb{Z}$, division is restricted, so the lattice retains divisibility information that rank cannot see. The scalar congruence theorem, CRT, unimodular reduction, Smith normal form, cokernels, and the classification of finitely generated abelian groups are therefore successive stages of one problem: deciding what an integer homomorphism can reach and what discrete obstruction remains. The passage from

$$ax \equiv b \pmod{n}$$

to

$$d_i z_i \equiv c_i \pmod{n}$$

is the matrix form of the same image-and-kernel calculation, while the invariant factors record the obstruction permanently in the quotient group.

Prime Numbers, Composite Numbers, Divisor Functions, Congruences, and Number-Theoretic Exercises

N-20221119

§26.1 Prime and composite numbers

The first page defines prime and composite numbers. An integer $a > 1$ is prime if its only positive divisors are 1 and a. If $a > 1$ is not prime, it is composite.

A basic observation is that if a is composite, then it has a prime divisor not exceeding $\sqrt{a}$. Indeed, if

$$a = pq, \qquad 1 < p \leq q,$$

then

$$p^2 \leq pq = a,$$

so $p \leq \sqrt{a}$. This justifies trial division up to $\sqrt{a}$.

The note then mentions the sieve method. To find primes up to 50, begin with $2, 3, 5, 7$ as sieving primes because

$$\sqrt{50} < 8.$$

Cross out multiples of these primes. The remaining numbers are prime.

§26.2 Euclid's theorem

Theorem 26.2.1 (Euclid's theorem)

There are infinitely many prime numbers.

Euclid's theorem says that there are infinitely many primes. Suppose the primes were

$$p_1, p_2, \ldots, p_k.$$

Let

$$N = p_1 p_2 \cdots p_k + 1.$$

No p_i divides N, since division by p_i leaves remainder 1. But every integer $N > 1$ has a prime divisor. This prime divisor is not among the listed primes, contradiction.

The handwritten version uses exactly this remainder idea: the product of all known primes plus one creates a new prime factor.

§26.3 Prime divisibility

The page records several prime-divisibility facts. If p is prime and

$$p \mid ab,$$

then

$$p \mid a \quad \text{or} \quad p \mid b.$$

More generally, if

$$p \mid a_1 a_2 \cdots a_n,$$

then p divides at least one factor. This is proved by induction from the two-factor statement.

The note also records the contrapositive: if p divides none of the factors, then it does not divide their product.

§26.4 Fundamental theorem of arithmetic

Theorem 26.4.1 (Fundamental theorem of arithmetic)
Every integer greater than 1 has a unique prime factorization, up to the order of the prime factors.

Every integer $n > 1$ can be written uniquely as

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where the p_i are distinct primes and $\alpha_i > 0$. The order of factors is the only ambiguity.

The note uses this factorization to compute divisor functions. If

$$d \mid n,$$

then

$$d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}, \qquad 0 \le \beta_i \le \alpha_i.$$

Therefore the number of positive divisors is

$$\tau(n) = \prod_{i=1}^{k} (\alpha_i + 1).$$

The sum of positive divisors is

$$\sigma(n) = \prod_{i=1}^{k} (1 + p_i + p_i^2 + \cdots + p_i^{\alpha_i}) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

§26.5 Perfect numbers and square numbers

A number n is perfect if the sum of its positive divisors is $2n$:

$$\sigma(n) = 2n.$$

The note lists the first examples:

$$6, \quad 28, \quad 496, \quad 8128, \ldots$$

Another important fact is:

$$n \text{ is a square} \quad \Longleftrightarrow \quad \tau(n) \text{ is odd}.$$

If $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, then n is a square iff every α_i is even. This is equivalent to every $\alpha_i + 1$ being odd, so their product $\tau(n)$ is odd.

§26.6 Exponent of a prime in $n!$

The note records Legendre's formula. If p is prime, then the exponent of p in $n!$ is

$$\sum_{j\geq 1}\left\lfloor \frac{n}{p^j}\right\rfloor .$$

This counts multiples of p, then multiples of p^2, then multiples of p^3, and so on. For example, numbers divisible by p^2 contribute an extra factor of p beyond the first count.

§26.7 Congruence exercises

The printed pages contain exercises on congruences. The basic definition is

$$a \equiv b \pmod m \quad\Longleftrightarrow\quad m \mid (a-b).$$

The operations of addition and multiplication are compatible with congruence:

$$a\equiv b,\ c\equiv d \pmod m \quad\Longrightarrow\quad a+c\equiv b+d,\quad ac\equiv bd \pmod m.$$

A typical problem asks for a digit or a remainder of a large power. The method is to find the period of powers modulo m. For example, powers of 2 modulo 10 cycle through

$$2, 4, 8, 6.$$

Once the period is known, the exponent can be reduced modulo the period.

§26.8 Broader perspective

This note is a transition from divisibility to arithmetic structure. Prime factorization turns every positive integer into exponent data. Divisor counting and divisor sums become products over primes. Congruences turn equality into equality up to remainders. Together, these tools make large integer problems finite and computable.

§26.9 Prime exponents and valuations

For a prime p, the exponent $v_p(n)$ turns divisibility into arithmetic:

$$a \mid b \quad\Longleftrightarrow\quad v_p(a) \leq v_p(b) \text{ for all } p.$$

Legendre's formula gives

$$v_p(n!) = \sum_{k\geq 1}\left\lfloor \frac{n}{p^k}\right\rfloor .$$

It counts multiples of p, then multiples of p^2, and so on.

§26.10 Divisor functions as multiplicative functions

If $n = \prod p_i^{e_i}$, then

$$d(n) = \prod_i (e_i+1), \qquad \sigma(n) = \prod_i \frac{p_i^{e_i+1}-1}{p_i-1}.$$

These formulas hold because choosing a divisor is choosing each prime exponent independently. This is multiplicativity coming from unique factorization.

§26.11 Congruences and finite groups

Reduced residue classes modulo n form the unit group $(\mathbb{Z}/n\mathbb{Z})^\times$. Euler's theorem is the statement that every element of this finite group has order dividing the group size.

§26.12 Prime coordinates and finite unit groups

Prime factorization supplies coordinates, divisor functions multiply across primes, factorial valuations count prime powers, and congruence problems become group theory in finite unit groups.

§26.13 Arithmetic functions and convolution

The divisor-counting functions in the original note are examples of arithmetic functions, functions

$$f : \mathbb{N} \to \mathbb{C}.$$

They form an algebra under Dirichlet convolution:

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d).$$

The identity element is

$$\varepsilon(n) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For example, if $1(n) = 1$ for all n, then the divisor-counting function is

$$\tau = 1 * 1.$$

The sum-of-divisors function is

$$\sigma = \mathrm{id} * 1, \qquad \mathrm{id}(n) = n.$$

§26.14 Möbius inversion in number theory

The number-theoretic Möbius function is

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^k, & n \text{ is a product of } k \text{ distinct primes}, \\ 0, & p^2 \mid n \text{ for some } p. \end{cases}$$

It is the inverse of the constant function 1 under Dirichlet convolution:

$$\mu * 1 = \varepsilon.$$

Therefore if

$$F(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d)F(n/d).$$

This is Möbius inversion on the divisibility poset.

§26.15 Dirichlet series and Euler products

To an arithmetic function f one may attach a Dirichlet series

$$D_f(s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}.$$

Dirichlet convolution corresponds to multiplication of Dirichlet series when convergence is justified:

$$D_{f*g}(s) = D_f(s)D_g(s).$$

For the constant function 1,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}.$$

The Euler product is unique factorization written analytically.

The elementary prime and divisor computations in the note therefore open the door to analytic number theory: primes are encoded in the analytic behavior of $\zeta(s)$ and related series.

§26.16 P-adic valuations and multiplicativity

If

$$n = \prod_p p^{v_p(n)},$$

then multiplicative functions are determined by their values on prime powers. For example,

$$\tau(n) = \prod_p (v_p(n)+1), \qquad \sigma(n) = \prod_p \frac{p^{v_p(n)+1}-1}{p-1}.$$

This is why divisor-function formulas are products over primes. Elementary prime factorization is already a coordinate system for arithmetic functions.

§26.17 Euler products and prime independence

The Euler product

$$\zeta(s) = \prod_p (1-p^{-s})^{-1}$$

comes from multiplying the geometric series

$$1 + p^{-s} + p^{-2s} + \cdots$$

for every prime p. Expanding the product chooses an exponent for each prime, hence chooses a positive integer.

This is unique factorization written as analysis. Elementary divisor formulas are multiplicative because the exponent choices at different primes are independent.

§26.18 Möbius inversion over primes

The Möbius function $\mu(n)$ performs inclusion–exclusion over the set of primes dividing n. If n is squarefree with k prime factors, then $\mu(n) = (-1)^k$. If a square divides n, it contributes zero because repeated prime factors break the simple subset structure.

Thus number-theoretic Möbius inversion and set-theoretic inclusion–exclusion are the same operation on different posets: divisors ordered by divisibility versus subsets ordered by inclusion.

Congruence Classes, Complete Residue Systems, and Euler's Function

N-20221126

§27.1 Congruence

Let m be a positive integer and $a, b \in \mathbb{Z}$. We say that a and b are congruent modulo m, written

$$a \equiv b \pmod{m},$$

if

$$m \mid (a-b).$$

This definition is just divisibility language with a new notation. It allows us to work with remainders rather than whole integers.

§27.2 Basic properties

The note lists the standard properties:

(i) Reflexivity: $a \equiv a \pmod{m}$.

(ii) Symmetry: if $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.

(iii) Transitivity: if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

(iv) Compatibility with addition and multiplication:

$$a \equiv b,\ c \equiv d \pmod{m} \quad \Rightarrow \quad a+c \equiv b+d, \quad ac \equiv bd \pmod{m}.$$

(v) If $a \equiv b \pmod{m}$, then

$$a^n \equiv b^n \pmod{m}$$

for every positive integer n.

The last two rules are what make congruence useful: one can replace a large number by a small remainder and keep computing.

§27.3 Residue classes

A residue class modulo n is a set

$$M_i = \{x \in \mathbb{Z} : x \equiv i \pmod{n}\}.$$

The integers split into n disjoint classes:

$$M_0, M_1, \ldots, M_{n-1}.$$

This is a partition of $\mathbb{Z}$. The note records two important consequences:

(i) among any $n+1$ integers, two are congruent modulo n;

(ii) if $(i, n) = 1$, then every integer in M_i is coprime to n.

The first is the pigeonhole principle; the second follows because congruent numbers have the same gcd with n.

§27.4 Complete and reduced residue systems

A complete residue system modulo n is any set of n representatives, one from each residue class. The standard example is

$$0, 1, 2, \ldots, n-1.$$

A reduced residue system modulo n consists of one representative from each residue class coprime to n. The number of such classes is Euler's function

$$\varphi(n).$$

If

$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$

then

$$\varphi(n) = n\left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right).$$

In particular,

$$\varphi(p) = p - 1, \qquad \varphi(p^a) = p^a - p^{a-1}.$$

§27.5 Example: last digit

To compute the last digit of 15^{200}, work modulo 10. Since

$$15 \equiv 5 \pmod{10},$$

we get

$$15^{200} \equiv 5^{200} \equiv 5 \pmod{10}.$$

Thus the last digit is 5.

The note's method is to reduce early. Do not expand powers; replace the base by its residue.

§27.6 Example: modulo 17

To compute

$$15^{100} \pmod{17},$$

notice

$$15 \equiv -2 \pmod{17}.$$

Then

$$15^{100} \equiv (-2)^{100} = 2^{100} \pmod{17}.$$

Since

$$2^4 = 16 \equiv -1 \pmod{17},$$

we have

$$2^8 \equiv 1 \pmod{17}.$$

Therefore

$$2^{100} = 2^{96} 2^4 \equiv 1^{12}(-1) \equiv 16 \pmod{17}.$$

§27.7 Example: reducing a long expression

For an expression such as

$$2001 \cdot 2002 + 2003 \cdot 2004 + 2005 \cdot 2006$$

modulo 45, reduce every factor modulo 45 first. The arithmetic becomes small. The main habit is:

$$\text{large expression} \quad \longrightarrow \quad \text{small residues.}$$

This is the same principle behind all modular computation.

§27.8 A polynomial congruence by induction

The note also includes a divisibility proof for an expression depending on k, of the form

$$12^{2k+1} + 36^k + 5^k + 3.$$

The standard method is induction or modular periodicity. One chooses the modulus, checks the base case, and then shows the expression changes by a multiple of the modulus when k increases. Equivalently, reduce each base modulo the modulus and use periodicity of powers.

§27.9 Quadratic residues modulo 8

A consolidation exercise proves that a perfect square modulo 8 is congruent to

$$0, \ 1, \quad \text{or} \quad 4.$$

Indeed, every integer is congruent modulo 8 to one of

$$0, 1, 2, 3, 4, 5, 6, 7.$$

Squaring gives residues

$$0, 1, 4, 1, 0, 1, 4, 1.$$

Thus only $0, 1, 4$ occur. This is a powerful small filter in number theory problems.

§27.10 Where the idea leads

Congruence classes are quotient thinking. Instead of remembering the whole integer, we remember only its class in

$$\mathbb{Z}/n\mathbb{Z}.$$

Complete residue systems are representatives of all classes; reduced residue systems are the invertible classes. Euler's function counts the units. This note is therefore the elementary entrance to the ring $\mathbb{Z}/n\mathbb{Z}$ and its unit group.

§27.11 Residue systems as representatives

A complete residue system modulo n is a choice of one representative from each class in $\mathbb{Z}/n\mathbb{Z}$. A reduced residue system is a choice of representatives from the unit group

$$(\mathbb{Z}/n\mathbb{Z})^\times.$$

Thus the difference between complete and reduced systems is the difference between the whole quotient ring and its invertible elements.

§27.12 Euler function through prime powers

For a prime power,

$$\varphi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right).$$

Using the Chinese remainder theorem gives

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

§27.13 Orders and exponent reduction

If a is a unit modulo n, its order is the smallest d with $a^d \equiv 1 \pmod{n}$. Exponent reduction is really reduction modulo the order of a, not blindly modulo $\varphi(n)$.

§27.14 Residues, units, and exponent cycles

Residue classes form quotient rings, reduced residue systems form unit groups, Euler's function counts units, and exponent tricks are statements about orders in finite groups.

§27.15 Residue systems as groups of units

A reduced residue system modulo n is the unit group

$$(\mathbb{Z}/n\mathbb{Z})^\times.$$

Its order is Euler's function $\varphi(n)$. Euler's theorem says that if $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

This is simply Lagrange's theorem applied to the finite group of units.

Thus the elementary reduced-residue computations are group theory in disguise.

§27.16 CRT decomposition of unit groups

If m, n are coprime, then

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

Taking units gives

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

This explains the multiplicativity of φ.

For prime powers, the structure of the unit group is more subtle. For odd primes p, the group $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is cyclic of order $p^{k-1}(p-1)$. Powers of 2 require separate treatment.

§27.17 Primitive roots and characters

A primitive root modulo n is a generator of the unit group. When the unit group is cyclic, every unit is a power of one element. This converts multiplicative congruence problems into exponent arithmetic.

Dirichlet characters are homomorphisms

$$\chi : (\mathbb{Z}/n\mathbb{Z})^\times \to \mathbb{C}^\times$$

extended to all integers by setting $\chi(a) = 0$ when $\gcd(a, n) > 1$. They are Fourier characters on the finite abelian group of units and are central in Dirichlet's theorem on primes in arithmetic progressions.

§27.18 Residue systems are calculations inside finite groups

The note's residue tables, reduced systems, and Euler-function formulas are not just modular arithmetic. They are computations inside finite groups. The next levels are group decomposition, primitive roots, and characters.

§27.19 Primitive roots as logarithms in finite groups

When $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic with generator g, every unit has the form g^k. This turns multiplication into addition of exponents modulo the group order. In other words, a primitive root supplies a discrete logarithm coordinate.

This is powerful but also limited: not every modulus has primitive roots. The classification says primitive roots exist for

$$n = 1, 2, 4, p^k, 2p^k$$

where p is an odd prime. This explains why some modular arithmetic problems become simple after choosing a primitive root, while others require decomposing the unit group into several cyclic factors.

§27.20 Characters as Fourier analysis on unit groups

Dirichlet characters are not arbitrary functions; they are group homomorphisms from the unit group to complex roots of unity. Orthogonality of characters gives filters for congruence classes. This is the multiplicative analogue of using roots of unity to filter residues modulo n.

The elementary reduced-residue system is therefore the domain on which finite harmonic analysis is performed.

Cyclotomic Pullbacks: Exact Orders, Galois Conjugacy, and Polynomial Divisibility

N-20230127

§28.1 A plausible generalization fails for a precise reason

Write

$$H_m(x) = 1 + x + x^2 + \cdots + x^{m-1} = \frac{x^m - 1}{x - 1}.$$

The elementary problem

$$x^2 + x + 1 \mid x^{2n} + x^n + 1$$

is the case

$$H_3(x) \mid H_3(x^n).$$

For $m = 3$, the answer is simply $3 \nmid n$. It is tempting to guess that, for general m,

$$H_m(x) \mid H_m(x^n) \quad \text{whenever} \quad m \nmid n.$$

The guess is false.

Take $m = 6$ and $n = 2$. Although $6 \nmid 2$, the number -1 is a root of

$$H_6(x) = 1 + x + \cdots + x^5,$$

because

$$1 - 1 + 1 - 1 + 1 - 1 = 0.$$

But

$$H_6((-1)^2) = H_6(1) = 6,$$

so -1 is not a root of $H_6(x^2)$. Therefore

$$H_6(x) \nmid H_6(x^2).$$

The failed guess looked only at the order m itself. The polynomial H_m, however, contains every nontrivial m-th root of unity, including roots whose exact orders are proper divisors of m. In the counterexample, the troublesome root has order 2, and $2 \mid n$. The correct question is therefore not merely whether m divides n, but how the map

$$z \longmapsto z^n$$

changes the exact order of every root of unity involved.

§28.2 Taking powers changes order by a greatest common divisor

Let ζ be a root of unity of exact order d. Thus

$$\zeta^d = 1$$

and no smaller positive power of ζ equals 1.

Theorem 28.2.1 (Order after taking a power)
For every positive integer n,

$$\operatorname{ord}(\zeta^n) = \frac{d}{\gcd(d,n)}.$$

Proof. Set

$$g = \gcd(d,n), \qquad d = gd_0, \qquad n = gn_0,$$

where $\gcd(d_0, n_0) = 1$. Then

$$(\zeta^n)^{d_0} = \zeta^{gn_0d_0} = \zeta^{n_0d} = 1,$$

so the order of ζ^n divides d_0.

Conversely, if $(\zeta^n)^r = 1$, then $d \mid nr$. After dividing by g,

$$d_0 \mid n_0r.$$

Since d_0 and n_0 are coprime, Euclid's lemma gives $d_0 \mid r$. Hence no smaller positive exponent than d_0 can kill ζ^n. □

This formula explains several phenomena at once. If $\gcd(d,n) = 1$, taking the n-th power preserves exact order. If $d \mid n$, it collapses the root all the way to 1. Between these extremes, the order decreases by exactly the common factor.

For example, let ζ have order 12. Then

$$\operatorname{ord}(\zeta^8) = \frac{12}{\gcd(12,8)} = 3.$$

Thus the eighth-power map sends primitive twelfth roots to primitive cubic roots. It neither preserves all twelve vertices nor collapses them to one point; it folds the twelve-cycle onto a three-cycle.

§28.3 The correct divisibility criterion

The roots of $H_m(x)$ are precisely the nontrivial m-th roots of unity. Indeed,

$$(x-1)H_m(x) = x^m - 1,$$

and $x = 1$ is the only m-th root removed by the quotient.

Now let $\zeta \neq 1$ satisfy $\zeta^m = 1$. Then

$$H_m(\zeta^n) = 0$$

exactly when ζ^n is still a nontrivial m-th root of unity. It is automatically an m-th root because

$$(\zeta^n)^m = (\zeta^m)^n = 1.$$

The only possible failure is $\zeta^n = 1$.

Theorem 28.3.1 (General geometric-sum divisibility)
For positive integers m, n,

$$H_m(x) \mid H_m(x^n) \quad \iff \quad \gcd(m,n) = 1.$$

Proof. Assume first that $\gcd(m,n) = 1$. Every root ζ of H_m has exact order $d > 1$ dividing m. Therefore $\gcd(d,n) = 1$, and the order formula gives

$$\operatorname{ord}(\zeta^n) = d > 1.$$

Hence $H_m(\zeta^n) = 0$. Every root of H_m is a root of $H_m(x^n)$, and the roots of H_m are simple, so divisibility follows.

Conversely, suppose

$$g = \gcd(m,n) > 1.$$

Choose a primitive g-th root of unity η. Since $g \mid m$, the number $\eta \neq 1$ is a root of H_m. Since $g \mid n$,

$$\eta^n = 1,$$

and therefore

$$H_m(\eta^n) = H_m(1) = m \neq 0.$$

Thus divisibility fails. $\square$

The condition is stronger than $m \nmid n$ because every nontrivial divisor of m matters. Coprimality guarantees that the power map is an automorphism of the entire cyclic group of m-th roots, not merely that a primitive m-th root avoids collapsing to 1.

§28.4 The exact common factor is stronger information

When full divisibility fails, the two polynomials may still share a substantial factor. Exact orders identify it completely.

Let ζ have exact order d, where $d \mid m$ and $d > 1$. Then ζ is a root of $H_m(x^n)$ precisely when

$$\zeta^n \neq 1.$$

By the order formula, this is equivalent to

$$d \nmid n.$$

Thus the roots shared by $H_m(x)$ and $H_m(x^n)$ are exactly the roots whose exact orders divide m but do not divide n.

To turn this root description into a polynomial formula, group roots by exact order. Denote by $\Phi_d(x)$ the polynomial whose roots are the primitive d-th roots of unity. Then

$$H_m(x) = \prod_{\substack{d \mid m \\ d > 1}} \Phi_d(x).$$

Consequently,

$$\boxed{\gcd\big(H_m(x), H_m(x^n)\big) = \prod_{\substack{d \mid m,\ d>1 \\ d \nmid n}} \Phi_d(x).}$$

The gcd is taken monic in $\mathbb{Q}[x]$.

For $m = 6$ and $n = 4$,

$$H_6(x) = \Phi_2(x)\Phi_3(x)\Phi_6(x).$$

The relevant orders are $2, 3, 6$. Since $2 \mid 4$ but $3 \nmid 4$ and $6 \nmid 4$,

$$\gcd\big(H_6(x), H_6(x^4)\big) = \Phi_3(x)\Phi_6(x).$$

Using

$$\Phi_3(x) = x^2 + x + 1, \qquad \Phi_6(x) = x^2 - x + 1,$$

we obtain

$$\gcd\big(H_6(x), H_6(x^4)\big) = x^4 + x^2 + 1.$$

The factor is neither trivial nor all of H_6. It records exactly which order strata survive the fourth-power map.

§28.5 Why one primitive root forces an entire rational factor

A root test over $\mathbb{C}$ becomes a divisibility statement over $\mathbb{Q}$ only because primitive roots come in algebraically inseparable families.

Let

$$\zeta_d = e^{2\pi i/d}$$

be a primitive d-th root of unity. Its powers

$$\zeta_d^a, \qquad a \in (\mathbb{Z}/d\mathbb{Z})^\times,$$

are exactly the other primitive d-th roots. The polynomial

$$\Phi_d(x) = \prod_{a \in (\mathbb{Z}/d\mathbb{Z})^\times} (x - \zeta_d^a)$$

is the d-th cyclotomic polynomial.

A fundamental theorem says that $\Phi_d(x)$ lies in $\mathbb{Z}[x]$ and is irreducible over $\mathbb{Q}$. Therefore it is the minimal polynomial of ζ_d over $\mathbb{Q}$. If a rational polynomial $P(x)$ satisfies

$$P(\zeta_d) = 0,$$

then polynomial division by the minimal polynomial forces

$$\Phi_d(x) \mid P(x).$$

Indeed, write

$$P(x) = Q(x)\Phi_d(x) + R(x), \qquad \deg R < \deg \Phi_d.$$

Evaluating at ζ_d gives $R(\zeta_d) = 0$. Minimality implies $R = 0$.

The field

$$\mathbb{Q}(\zeta_d)$$

is the cyclotomic field generated by ζ_d. Every automorphism fixing $\mathbb{Q}$ is determined by

$$\zeta_d \longmapsto \zeta_d^a$$

for a unit a mod d, and every such choice occurs. Thus

$$\mathrm{Gal}(\mathbb{Q}(\zeta_d)/\mathbb{Q}) \cong (\mathbb{Z}/d\mathbb{Z})^\times.$$

If $P \in \mathbb{Q}[x]$ vanishes at one primitive root, applying these automorphisms shows that it vanishes at every Galois conjugate primitive root. The minimal-polynomial proof and the Galois proof are two descriptions of the same rigidity.

For $d = 5$, the primitive roots are

$$\zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4,$$

and

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1.$$

A rational polynomial cannot vanish at ζ_5 alone; it must contain the whole four-root orbit as a factor.

§28.6 Cyclotomic polynomials separate roots by exact order

Every m-th root of unity has a unique exact order d dividing m. Partitioning the roots of $x^m - 1$ by exact order gives

$$\boxed{x^m - 1 = \prod_{d|m} \Phi_d(x).}$$

The factors are pairwise coprime because their roots have different exact orders.

Taking degrees yields

$$m = \sum_{d|m} \deg \Phi_d.$$

Since the primitive d-th roots are indexed by the units modulo d,

$$\deg \Phi_d = \varphi(d),$$

so the degree identity becomes the familiar divisor sum

$$\sum_{d|m} \varphi(d) = m.$$

The arithmetic identity counts elements of a cyclic group by exact order, while the polynomial identity groups roots by the same invariant.

For $m = 12$, the divisors are

$$1, 2, 3, 4, 6, 12.$$

The factorization is

$$\begin{aligned} x^{12} - 1 &= \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x) \\ &= (x-1)(x+1)(x^2+x+1)(x^2+1) \\ &\quad \cdot (x^2 - x + 1)(x^4 - x^2 + 1). \end{aligned}$$

The degrees

$$1 + 1 + 2 + 2 + 2 + 4 = 12$$

match the twelve roots. The factor Φ_{12} contains exactly the roots of order twelve, not all roots whose orders merely divide twelve.

This exact-order separation is what the original geometric-sum problem needs. The polynomial H_m contains every Φ_d with $d \mid m$, $d > 1$, and the power map acts differently on each such stratum.

§28.7 Substitution pulls cyclotomic factors back along a power map

To understand $\Phi_r(x^n)$, ask which roots x have the property that x^n has exact order r. Suppose x has exact order s. The order formula says

$$\frac{s}{\gcd(s,n)} = r.$$

Write $s = rd$. The condition becomes

$$\gcd(rd, n) = d.$$

This forces $d \mid n$, and after writing $n = dq$, it becomes

$$\gcd(r, q) = 1.$$

Therefore the possible exact orders of roots of $\Phi_r(x^n)$ are

$$rd \quad \text{with} \quad d \mid n \quad \text{and} \quad \gcd\left(r, \frac{n}{d}\right) = 1.$$

Theorem 28.7.1 (Cyclotomic pullback formula)

For positive integers r, n,

$$\boxed{\Phi_r(x^n) = \prod_{\substack{d \mid n \\ \gcd(r, n/d)=1}} \Phi_{rd}(x).}$$

Proof. The root analysis above shows that both sides have exactly the same roots, grouped by the exact order rd. All these roots are simple because $x^{rn} - 1$ has no repeated roots in characteristic zero. Both sides are monic, so equality follows. □

Take $r = 3$ and $n = 4$. Every divisor d of 4 satisfies

$$\gcd\left(3, \frac{4}{d}\right) = 1,$$

so

$$\Phi_3(x^4) = \Phi_3(x)\Phi_6(x)\Phi_{12}(x).$$

Indeed,

$$x^8 + x^4 + 1 = (x^2 + x + 1)(x^2 - x + 1)(x^4 - x^2 + 1).$$

The three factors correspond to roots of exact orders $3, 6, 12$, all of which become primitive cubic roots after taking fourth powers.

By contrast,

$$\Phi_2(x^4) = x^4 + 1 = \Phi_8(x).$$

Only $d = 4$ contributes because

$$\gcd(2, 4/d) = 1$$

holds only when $d = 4$. An eighth root of exact order eight becomes -1 after the fourth power, while roots of orders two or four collapse to 1 instead.

§28.8 Möbius inversion reconstructs the exact-order factor

The identity

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

expresses an "order divides n" object as a product of "exact order d" objects. Möbius inversion reverses this divisor aggregation.

Recall the Möbius function:

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & n \text{ is divisible by a square}, \\ (-1)^k, & n \text{ is a product of } k \text{ distinct primes}. \end{cases}$$

Its key property is

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Multiplicative Möbius inversion gives

$$\boxed{\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}.}$$

Although negative exponents appear, the total expression is a polynomial because it is obtained by inverting the already established cyclotomic factorization.

The cancellation can be checked factor by factor. Substitute

$$x^d - 1 = \prod_{e|d} \Phi_e(x)$$

into the right-hand side. The exponent of a fixed $\Phi_e(x)$ becomes

$$\sum_{\substack{d|n \\ e|d}} \mu(n/d).$$

Writing $d = eq$, this is

$$\sum_{q|n/e} \mu\left(\frac{n/e}{q}\right),$$

which equals 1 when $e = n$ and 0 otherwise. Every unwanted exact-order factor cancels.

For $n = 12$, the nonzero values of $\mu(12/d)$ occur for

$$d = 2, 4, 6, 12.$$

Thus
$$\Phi_{12}(x) = \frac{(x^{12}-1)(x^2-1)}{(x^6-1)(x^4-1)}.$$
Using
$$x^{12}-1 = (x^6-1)(x^6+1)$$
and
$$x^4-1 = (x^2-1)(x^2+1),$$
we obtain
$$\Phi_{12}(x) = \frac{x^6+1}{x^2+1} = x^4-x^2+1.$$
The formula extracts the primitive twelfth roots from all lower-order roots by inclusion–exclusion on the divisor lattice.

§28.9 The root plots have an exact formula, not merely a pattern

Consider
$$H_m(x^n) = 1 + x^n + x^{2n} + \cdots + x^{(m-1)n}.$$
Using the geometric-series identity,
$$H_m(x^n) = \frac{x^{mn}-1}{x^n-1}.$$
Hence its roots are the mn-th roots of unity that are not n-th roots of unity.

Let
$$\xi = e^{2\pi i/(mn)}.$$
Every root has the form
$$\xi^q, \qquad 0 \le q < mn,$$
subject to
$$m \nmid q.$$
Indeed,
$$(\xi^q)^n = e^{2\pi i q/m} = 1$$
exactly when $m \mid q$. Therefore
$$\boxed{\operatorname{Roots}(H_m(x^n)) = \left\{e^{2\pi i q/(mn)} : 0 \le q < mn,\ m \nmid q\right\}.}$$
There are $mn - n = n(m-1)$ such roots, matching the degree.

For $m = 6, n = 4$, the roots are the twenty fourth roots of unity except
$$1, i, -1, -i,$$
which are precisely the fourth roots of unity. A plot therefore shows a regular twenty-four-gon with every sixth vertex removed. The missing pattern is not numerical noise; it is the denominator x^4-1 in
$$\frac{x^{24}-1}{x^4-1}.$$

The exact orders present in the plot can be read from the cyclotomic pullback formula. Since

$$H_m(x^n) = \prod_{\substack{r|m \\ r>1}} \Phi_r(x^n),$$

each factor $\Phi_r(x^n)$ splits into the order strata rd permitted by

$$d \mid n, \qquad \gcd(r, n/d) = 1.$$

Thus the visual rings of angles, the algebraic factorization, and the arithmetic of exact orders are the same data in three forms.

Over $\mathbb{R}$, conjugate roots pair into quadratic factors. If $\theta \notin \{0, \pi\}$, then

$$(x - e^{i\theta})(x - e^{-i\theta}) = x^2 - 2\cos\theta\, x + 1.$$

For example,

$$\Phi_5(x) = \big(x^2 - 2\cos(2\pi/5)x + 1\big)\big(x^2 - 2\cos(4\pi/5)x + 1\big).$$

The individual quadratic coefficients lie in $\mathbb{Q}(\sqrt{5})$, while their product lies in $\mathbb{Z}[x]$. This is the real shadow of the Galois orbit of primitive fifth roots.

§28.10 Resultants distinguish sharing a root from containing a factor

The divisibility question asks whether every root of H_m remains a root after the substitution $x \mapsto x^n$. A weaker question asks whether the two polynomials share at least one root. The resultant detects exactly this weaker condition.

If $P(x)$ is monic with roots $\alpha_1, \dots, \alpha_r$, define

$$\operatorname{Res}(P, Q) = \prod_{j=1}^{r} Q(\alpha_j).$$

Equivalently, the resultant is the determinant of the Sylvester matrix of P and Q. The product formula shows

$$\operatorname{Res}(P, Q) = 0 \quad \iff \quad P, Q \text{ have a common root over } \mathbb{C}.$$

Apply this to

$$P(x) = H_m(x), \qquad Q(x) = H_m(x^n).$$

If $m \nmid n$, choose a divisor-order stratum that survives. In fact, a primitive m-th root ζ satisfies

$$\zeta^n \neq 1$$

whenever $m \nmid n$, so

$$H_m(\zeta^n) = 0.$$

Thus the resultant is zero.

If $m \mid n$, then every nontrivial m-th root ζ satisfies

$$\zeta^n = 1,$$

so

$$H_m(\zeta^n) = H_m(1) = m.$$

Since H_m has $m-1$ roots,

$$\operatorname{Res}\bigl(H_m(x), H_m(x^n)\bigr) = m^{m-1}.$$

Therefore

$$\boxed{\operatorname{Res}\bigl(H_m(x), H_m(x^n)\bigr) = \begin{cases} 0, & m \nmid n, \\ m^{m-1}, & m \mid n. \end{cases}}$$

This result sharply separates three levels of interaction:

condition	polynomial consequence
$m \mid n$	$\gcd(H_m, H_m(x^n)) = 1,$
$1 < \gcd(m, n) < m$	a proper nontrivial common factor may remain,
$\gcd(m, n) = 1$	$H_m(x) \mid H_m(x^n).$

The resultant detects whether the gcd is nontrivial; the exact-order factorization determines the gcd itself; the power-map automorphism criterion determines when the entire divisor survives.

The original factorization problem has therefore grown without changing its central question. A false congruence guess forced us to track exact orders. Exact orders forced a decomposition into cyclotomic factors. Rational divisibility forced us to understand minimal polynomials and Galois conjugacy. Pulling those factors through $x \mapsto x^n$ led to a precise factorization theorem, while Möbius inversion reconstructed each exact-order piece. Finally, the resultant clarified the difference between one surviving orbit and all surviving orbits. Each new object entered because the previous language could no longer state the next question precisely.

IV

Arithmetic Geometry

Part IV: Contents

From Diophantine Inequalities to Arithmetic Deformation

N-20211007

§29.1 The question is an inequality, not a slogan

Fesenko's survey presents inter-universal Teichmueller theory under another name: arithmetic deformation theory. This name is useful because it puts the reader's attention on the mathematical target before the new language appears. The target is not first of all a mystical "inter-universal" philosophy. It is a family of Diophantine inequalities in which one wants to control the arithmetic size of a curve, an elliptic curve, or a rational point.

The basic shape is always the same. One side measures size: height, discriminant, or a quantity built out of bad reduction. The other side measures the places where the object is allowed to be complicated: conductor, ramification, different, or a divisor of excluded points. The problem is to prove that the first kind of size cannot grow too fast compared with the second kind of size.

Definition 29.1.1 (Number field and absolute Galois group)**.** A number field is a finite extension $K/\mathbb{Q}$. If K^{alg} is an algebraic closure of K, its absolute Galois group is

$$G_K = \mathrm{Gal}(K^{\mathrm{alg}}/K),$$

regarded as a profinite topological group.

The appearance of G_K is not decoration. A large part of the story leading to IUT asks how much arithmetic geometry can be recovered from groups of symmetries. In ordinary algebraic geometry one starts with equations and then studies automorphisms or fundamental groups. Anabelian geometry tries to reverse this direction: from a suitable fundamental group, recover the geometric or arithmetic object.

§29.2 A first chain of rigidity results

The first rigidity theorem in this note is about number fields alone.

Theorem 29.2.1 (Neukirch–Ikeda–Uchida, schematic form)

Let K_1 and K_2 be number fields. An isomorphism of topological groups

$$G_{K_1} \cong G_{K_2}$$

comes from a unique field-theoretic identification of the corresponding algebraic closures, in a way compatible with the two number fields. Thus, for number fields, the absolute Galois group remembers the field.

This theorem should be read as a warning against the naive view that a Galois group is only an auxiliary invariant. In the number-field case, the group is so rigid that the field can be reconstructed from it.

The next theorem connects curves with a surprisingly small amount of ramification.

Theorem 29.2.2 (Belyi)

Let C be an irreducible smooth projective algebraic curve over a field of characteristic zero. Then C can be defined over $\overline{\mathbb{Q}}$ if and only if there exists a finite morphism

$$C \longrightarrow \mathbb{P}^1$$

which is ramified over at most three points of $\mathbb{P}^1$.

Such maps are called Belyi maps. For this note, the important point is not the proof but the direction of thought. Curves over number fields, coverings of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, and Galois/fundamental-group data are not separate topics. They are different faces of the same arithmetic geometry.

Remark 29.2.3 (Why this belongs in the IUT entrance) — Fesenko emphasizes that versions of Belyi maps are used later in arithmetic deformation theory and its application to Diophantine geometry. This is why the story does not begin directly with theta-links. Before one reaches theta-links, one has already entered a world where curves, number fields, and fundamental groups are designed to remember one another.

§29.3 From Mordell finiteness to effective inequalities

Faltings' theorem gives a finiteness result.

Theorem 29.3.1 (Faltings–Mordell)

If C is a smooth projective curve of genus > 1 over a number field K, then $C(K)$ is finite.

Finiteness is a qualitative statement. The effective Mordell problem asks for more: not only that rational points are finite, but that their heights can be bounded in terms of computable arithmetic data. In the background of Fesenko's survey, this effective direction sits beside several closely related conjectures:

effective Mordell, Szpiro, *abc*, Frey, Vojta for hyperbolic curves.

These are not identical statements, but Fesenko presents them as a closely connected cluster. The important organizing idea is that they all seek an inequality strong enough to control Diophantine complexity.

§29.4 The Szpiro inequality as a concrete target

The Szpiro conjecture is a good first model because it is phrased directly in terms of elliptic curves and bad reduction. Let E_K be an elliptic curve over a number field K. In the split multiplicative situation, the bad fibers of the minimal regular proper model have type I_{n_v}. Here v runs over nonarchimedean valuations of K, $k(v)$ is the residue field, and n_v is the number of components in the bad fiber.

Definition 29.4.1 (Conductor-size and discriminant-size, split multiplicative model)**.** In the split multiplicative case, the logarithmic discriminant-size and conductor-size are, schematically,

$$\sum_{v \text{ bad}} n_v \log |k(v)|, \qquad \sum_{v \text{ bad}} \log |k(v)|.$$

The first expression counts bad fibers with multiplicity n_v. The second remembers only the bad places themselves.

The conjectural Szpiro inequality then has the form

$$\sum_{v \text{ bad}} n_v \log |k(v)| \le c + (6+\epsilon) \sum_{v \text{ bad}} \log |k(v)|,$$

for every $\epsilon > 0$, with a constant c depending on K and ϵ. Equivalently, using the absolute norms of the minimal discriminant and the conductor, one writes

$$N(\operatorname{Disc} E_K) \le c' N(\operatorname{Cond} E_K)^{6+\epsilon}.$$

> **Remark 29.4.2** (The meaning of the exponent) — The number 6 is not a random constant inserted into the conjecture. In Fesenko's discussion it is connected with the geometric Szpiro-type inequality for elliptic surfaces and with the comparison between the arithmetic and geometric sides of the problem. For the present note, the safe takeaway is that Szpiro is an inequality bounding discriminant-size by conductor-size.

A very simple reading is this: the conductor records where the elliptic curve behaves badly; the discriminant records how badly it behaves there. Szpiro predicts that "how badly" cannot be much larger than a fixed power of "where badly".

§29.5 Vojta's inequality and the hyperbolic curve viewpoint

The Vojta form is closer to the geometry of punctured curves.

Definition 29.5.1 (Hyperbolic curve)**.** Over a field of characteristic zero, a hyperbolic curve is a smooth projective geometrically connected curve of genus g with r points removed, such that

$$2 - 2g - r < 0.$$

Examples include $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and an elliptic curve with the origin removed.

The condition is not merely topological decoration. Hyperbolic curves have nonabelian algebraic fundamental groups, and these groups are the kind of objects anabelian geometry can hope to use for reconstruction.

Let C be a smooth proper geometrically connected curve over a number field K, and let D be a reduced effective divisor on C such that $C \setminus D$ is hyperbolic. Vojta-type inequalities compare a height attached to $\omega_C(D)$ with terms measuring ramification and contact with D:

$$h_{\omega_C(D)}(x) \le c + (1+\epsilon)\big(\log\text{-diff}_C(x) + \log\text{-cond}_D(x)\big).$$

Here log-diff measures the arithmetic complexity of the field of definition of x, while log-cond measures intersection with the divisor D.

Remark 29.5.2 (Why Belyi maps reappear) — Fesenko notes that Belyi maps reduce the Vojta conjecture for such (C, D) to the special case

$$C = \mathbb{P}^1, \qquad D = [0] + [1] + [\infty].$$

Thus the same three-punctured projective line that appears in Belyi's theorem also acts as a universal testing ground for the Diophantine inequality.

§29.6 Arithmetic fundamental groups

For a geometrically integral scheme X over a perfect field K, there is a fundamental exact sequence

$$1 \longrightarrow \pi_1^{\mathrm{geom}}(X) \longrightarrow \pi_1(X) \longrightarrow G_K \longrightarrow 1,$$

where

$$\pi_1^{\mathrm{geom}}(X) = \pi_1(X \times_K K^{\mathrm{alg}}).$$

For schemes over number fields or related fields, $\pi_1(X)$ is often called an arithmetic fundamental group.

Definition 29.6.1 (Anabelian reconstruction, guiding form)**.** Anabelian reconstruction is the idea that, for suitable schemes X, one can recover the arithmetic or geometric object from its algebraic fundamental group together with the projection

$$\pi_1(X) \longrightarrow G_K.$$

The slogan is not that every space is determined by its fundamental group, but that certain highly nonabelian arithmetic schemes are rigid enough for this to happen.

In the IUT entrance, this matters because the later theory needs structures that survive when ordinary ring-theoretic comparison is no longer available. Fesenko summarizes the transition as follows: arithmetic deformation theory is related to anabelian geometry, but uses a different package of tools, including mono-anabelian geometry, nonarchimedean theta-functions, monoid-theoretic categories, and deconstruction/reconstruction of ring structures.

§29.7 The local elliptic curve model: Tate curves and q-parameters

The first truly arithmetic-deformation object in the survey is not the *abc* equation itself. It is an elliptic curve with split multiplicative reduction and its local q-parameters.

Let E_F be an elliptic curve over a number field F. Suppose that, at a bad nonarchimedean valuation v, the curve has split multiplicative reduction. Let F_v be the completion of F at v. Then locally one has a Tate uniformization

$$E_F(F_v) \cong F_v^\times / \langle q_v \rangle,$$

where $q_v \in F_v^\times$ is the local q-parameter and $\langle q_v \rangle$ is the cyclic subgroup generated by q_v.

Definition 29.7.1 (The idele q_{E_F})**.** Define an idele $q_{E_F} \in \mathbb{A}_F^\times$ by setting its component to be

$$(q_{E_F})_v = \begin{cases} q_v, & v \text{ a bad split multiplicative valuation,} \\ 1, & v \text{ archimedean or good reduction.} \end{cases}$$

after choosing the local q_v at each bad valuation.

The valuation of q_v records the number n_v of components of the corresponding bad fiber. Therefore the degree of this idele is not an artificial invariant: it packages the left-hand side of the Szpiro inequality.

§29.8 Normalised degree as log-volume

Fesenko uses ideles to define a normalised degree. Let k be a number field and $\mathbb{A}_k^\times$ its idele group. For $\alpha \in \mathbb{A}_k^\times$, the non-normalised degree is

$$\deg_k(\alpha) = -\log|\alpha|,$$

where $|\alpha|$ is the module by which multiplication by α scales Haar measure on the adele ring. The normalised degree is

$$\deg(\alpha) = \frac{1}{[k:\mathbb{Q}]}\deg_k(\alpha).$$

This normalisation is stable under finite field extension, which is why the same symbol deg can be used after passing to larger number fields.

> **Remark 29.8.1** (Why log-volume enters) — Because of the minus sign, $-\deg(\alpha)$ is naturally interpreted as a log-volume. In the later IUT inequality, the left side is written in terms of $-\deg q_E$. This is why the problem of bounding Szpiro-type data becomes a problem about bounding log-volumes.

For the elliptic curve above, Fesenko identifies the goal as giving a suitable upper bound on

$$\deg q_{E_F}.$$

This is the cleanest first target of arithmetic deformation theory in the survey.

§29.9 Why ordinary scheme-theoretic geometry is not enough

The next move is the point where the theory stops looking like ordinary algebraic geometry. Fix a prime $l > 3$, relatively prime to the bad valuations and to the relevant local valuation numbers. For a bad valuation v, Fesenko records a monoid-theoretic transformation on the submonoid of $F_v^\times$ generated by units and q_v:

$$q_v \longmapsto q_v^{m^2}, \qquad u \longmapsto u \quad (u \in \mathcal{O}_{F_v}^\times),$$

where

$$1 \le m \le \frac{l-1}{2}.$$

The element $q_v^{m^2}$ is later viewed as a special value of a nonarchimedean theta-function.

Proposition 29.9.1 (The first deformation warning)

The transformation

$$q_v \mapsto q_v^{m^2}, \qquad u \mapsto u$$

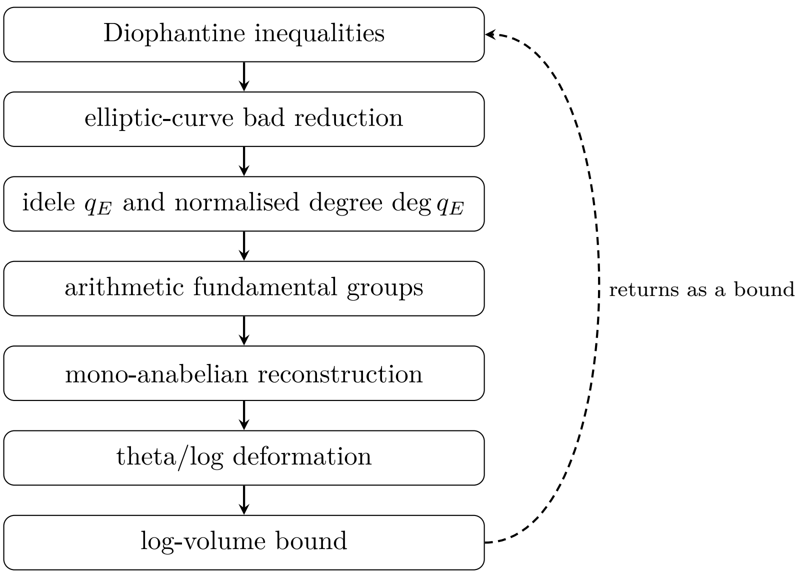
should be read as monoid-theoretic rather than ring-theoretic. It is not a morphism of schemes and does not preserve the full ring structure.

Proof. The point is conceptual rather than computational. A ring morphism must respect both multiplication and addition. The displayed transformation is defined on multiplicative data: the local units and the generator q_v of the Tate period subgroup. It is designed to manipulate multiplicative/theta data, not to preserve addition in F_v. Thus it belongs to the monoid-theoretic side of the theory. □

This is the first reason the word "deformation" is appropriate. The ring structure is not transported intact. It is partly dismantled so that certain multiplicative and theta-theoretic information can be moved, after which one needs reconstruction tools.

§29.10 The route from deformation to inequality

The full theta-link belongs to later notes. But the route can already be drawn.



The diagram is not a proof. It is a navigation map. Starting from Diophantine inequalities, one passes to elliptic curves and their bad reduction. The bad reduction is encoded by q_E. IUT then introduces a deformation machine involving arithmetic fundamental groups, mono-anabelian reconstruction, theta-links, log-links, and log-theta-lattices. The intended output is a log-volume inequality which returns to the original Diophantine problem.

Theorem 29.10.1 (Main IUT inequality, schematic form)

In Fesenko's account, the main theorem of IUT gives a bound on log-volumes of the form

$$-\deg q_E \leq -\deg \Theta_E,$$

where the right-hand side is the log-volume of a union of possible images of theta-data after applying the theta-link, subject to specified indeterminacies.

The statement above is deliberately schematic. It records the mathematical role of the theorem, not the construction of the objects. The important point for this first note is that the theorem does not directly say "abc is true". It gives a bound on a deformation volume. A further computation of the right-hand side is then used to obtain a bound of the desired Diophantine kind.

Remark 29.10.2 (The bridge to later notes) — The words theta-link, log-link, log-theta-lattice, indeterminacy, and multiradiality all belong to the mechanism behind Θ_E. They should not be treated as decorative terminology. They are the devices by which the theory tries to compare arithmetic data after the ordinary ring structure has been deconstructed.

§29.11 What this first note establishes

The first entrance to IUT should therefore be remembered through four questions.

1. What is being bounded? The initial concrete target is $\deg q_E$, which packages the bad-reduction side of a Szpiro-type inequality.
2. What is rigid? Absolute Galois groups and arithmetic fundamental groups can, in suitable settings, remember number fields or curves.
3. What is allowed to vary? The ordinary ring structure is no longer transported as a single indivisible object; multiplicative and additive aspects are separated.
4. Why is advanced machinery needed? Once ordinary scheme morphisms are not enough, one needs mono-anabelian reconstruction, Kummer-type methods, nonarchimedean theta-functions, and log-volume control.

This is already a mathematical picture, but not yet the full theory. The next note should begin where this one stops: with the moment when the map involving $q_v^{m^2}$ becomes part of the theta-link, and when the lost additive structure has to be handled through log-links and log-shells.

From Algebraic Number Theory to the First Shadow of Langlands

N-20220123

§30.1 Why this note starts with algebraic number theory

This note was written as preparation for a seminar on the Langlands program. Since the seminar was expected to involve algebraic number theory, I first collect the basic vocabulary: number fields, algebraic integers, ideals, local fields, modular forms, L-functions, and finally the first shadow of the Langlands correspondence.

> **Remark 30.1.1** — This is not a proof of the Langlands program. It is a personal attempt to see why many different objects—number fields, ideals, local completions, modular forms, Galois representations, and automorphic representations—might belong to one story.

§30.2 Number fields and algebraic integers

Definition 30.2.1. A number field is a finite extension $K/\mathbb{Q}$. Equivalently, K is a field that is a finite-dimensional vector space over $\mathbb{Q}$. A basic example is a quadratic field

$$K = \mathbb{Q}(\sqrt{d}),$$

where d is a nonsquare integer.

Definition 30.2.2. Let K be a number field. An element $\alpha \in K$ is an algebraic integer if it satisfies a monic polynomial with integer coefficients:

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0, \qquad a_i \in \mathbb{Z}.$$

The algebraic integers in K form a ring, denoted $\mathcal{O}_K$.

The reason for introducing algebraic integers is that they play in K the role played by ordinary integers in $\mathbb{Q}$. Unique factorization of elements may fail, but that failure leads to a deeper replacement: factorization of ideals.

§30.3 Ideals and factorization

Definition 30.3.1. An ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ is an additive subgroup such that for every $\alpha \in \mathfrak{a}$ and every $\gamma \in \mathcal{O}_K$,

$$\gamma\alpha \in \mathfrak{a}.$$

Every nonzero ideal of $\mathcal{O}_K$ factors uniquely into prime ideals. Thus even when element factorization fails, ideal factorization remains well behaved.

Remark 30.3.2 — This is the first conceptual jump: instead of forcing unique factorization of elements, one changes the objects being factored. Ideals see the arithmetic more cleanly.

§30.4 Quadratic fields and class number

For $K = \mathbb{Q}(\sqrt{d})$, the ring of integers is

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}[\sqrt{d}], & d \equiv 2, 3 \pmod 4, \\ \mathbb{Z}\left[\dfrac{1+\sqrt{d}}{2}\right], & d \equiv 1 \pmod 4. \end{cases}$$

Definition 30.4.1. The class group of K is the group of nonzero fractional ideals modulo principal fractional ideals. Its order is the class number h_K.

If $h_K = 1$, then every ideal class is principal. If $h_K > 1$, nonprincipal ideal classes exist, and element factorization can fail.

Example 30.4.2
For

$$K = \mathbb{Q}(\sqrt{-5}), \qquad \mathcal{O}_K = \mathbb{Z}[\sqrt{-5}],$$

one has $h_K = 2$. The familiar failure of unique factorization is

$$6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5}).$$

Caution

The original calculation of the ideal factorization of (6) is too compressed and drops one conjugate prime. A corrected form is

$$(2) = \mathfrak{p}_2^2, \qquad (3) = \mathfrak{p}_3\mathfrak{p}_3',$$

where

$$\mathfrak{p}_2 = (2, 1+\sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1+\sqrt{-5}), \quad \mathfrak{p}_3' = (3, 1-\sqrt{-5}).$$

Thus

$$(6) = \mathfrak{p}_2^2\mathfrak{p}_3\mathfrak{p}_3'.$$

§30.5 Localization, completion, and p-adic numbers

A recurring idea in number theory is the comparison between local and global information. Fix a prime p. Completing $\mathbb{Q}$ under the p-adic metric gives

$$\mathbb{Q}_p.$$

For a number field K, one can complete at a prime ideal $\mathfrak{p}$, producing a local field $K_{\mathfrak{p}}$.

Remark 30.5.1 — The local–global theme is one of the first hints of Langlands. Global arithmetic objects often decompose into local pieces, and the Langlands program repeatedly asks how local and global data determine each other.

Definition 30.5.2 (Hensel's lemma, informal version)**.** Hensel's lemma says that, under suitable nondegeneracy hypotheses, a root of a polynomial equation modulo p can be lifted to a root over the p-adic integers or p-adic field.

This resembles Newton iteration: an approximate solution is improved step by step in a complete space.

Example 30.5.3 (A small Hensel lifting computation)
Consider

$$f(x) = x^2 - 2$$

over the 7-adic integers. Modulo 7, one has

$$3^2 - 2 = 7 \equiv 0 \pmod 7,$$

so $x_0 = 3$ is a root mod 7. Also

$$f'(x) = 2x, \qquad f'(3) = 6 \not\equiv 0 \pmod 7.$$

Hensel's lemma therefore says that this root lifts uniquely to a 7-adic root.
To lift from mod 7 to mod 49, write

$$x = 3 + 7t.$$

Then

$$f(3 + 7t) = (3 + 7t)^2 - 2 = 7 + 42t + 49t^2 \equiv 7(1 + 6t) \pmod{49}.$$

We need $1 + 6t \equiv 0 \pmod 7$. Since $6^{-1} \equiv 6 \pmod 7$, this gives

$$t \equiv -6 \equiv 1 \pmod 7.$$

Thus the lifted root modulo 49 is

$$x_1 = 3 + 7 = 10, \qquad 10^2 - 2 = 98 \equiv 0 \pmod{49}.$$

The computation shows the Newton-like nature of Hensel lifting: a root modulo p is corrected by one p-adic digit at a time.
In Python, the first lifting step can be written as:

```
p = 7
x = 3
# lift x from mod p to mod p^2 for f(x)=x^2-2
t = (-(x*x - 2)//p * pow(2*x, -1, p)) % p
x = x + p*t
print(x, (x*x - 2) % (p*p))   # 10, 0
```

§30.6 Modular forms and L-functions

Modular forms are analytic functions with strong symmetry. A typical transformation law is

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z),$$

for matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

in a suitable discrete subgroup such as $\mathrm{SL}_2(\mathbb{Z})$. The integer k is the weight.

If a modular form has Fourier expansion

$$f(z) = \sum_{n\geq 0} a(n)e^{2\pi inz},$$

then one can attach an L-function

$$L(f,s) = \sum_{n\geq 1} \frac{a(n)}{n^s}.$$

Analytic continuation and functional equations of such functions encode arithmetic information.

Example 30.6.1 (Ramanujan's Delta function)

The modular discriminant is

$$\Delta(z) = q\prod_{n=1}^{\infty}(1-q^n)^{24} = \sum_{n\geq 1}\tau(n)q^n, \qquad q = e^{2\pi iz}.$$

It is a modular form of weight 12. The coefficients $\tau(n)$ form the Ramanujan tau function, whose growth properties were predicted by Ramanujan and later proved by Deligne.

§30.7 First understanding of Langlands

The Langlands program is often described as a unifying theory linking number theory, representation theory, geometry, and analysis. A safer first slogan is

$$\text{Galois-side symmetry} \quad \leftrightarrow \quad \text{automorphic-side symmetry},$$

with L-functions as a major bridge.

On one side are Galois representations, which encode actions of Galois groups on vector spaces. On the other side are automorphic representations, which organize modular forms and their higher-dimensional analogues.

Caution

It is too imprecise to say simply that an automorphic representation is a homomorphism from a group to a general linear group. That phrase is a useful beginner's memory of representation theory, but automorphic representations are representations of adelic groups on spaces of functions. For this note, the essential idea is that representation theory becomes a language for arithmetic.

§30.8 Local and global correspondences

Local Langlands studies local fields such as $\mathbb{Q}_p$ and finite extensions. Global Langlands studies number fields and their adele rings. The local theory is more complete in many settings, while the global theory contains many deep conjectures.

Caution

A naive statement such as

$$\{\rho : G_F \to \mathrm{GL}_n(\mathbb{C})\} \longleftrightarrow \{\text{automorphic representations}\}$$

is only a shadow of the real statement. For GL_n, local Langlands relates suitable Langlands parameters to irreducible admissible representations of $\mathrm{GL}_n(F)$.

§30.9 Cross-field intuitions

The original train of thought then branches into several speculative but useful directions.

§30.9.i Category theory

Category theory offers a language for comparing structures through functors and natural transformations. It is tempting to imagine Langlands correspondences as functorial relationships or equivalences between categories.

Gap

This is currently only a guiding intuition. A rigorous categorical formulation must specify the categories, functors, and compatibility conditions. Without those, it remains a helpful but informal picture.

§30.9.ii Algebraic topology

The idea of turning arithmetic into topology is not absurd: cohomology, sheaves, stacks, and derived geometry are everywhere in modern number theory. But the naive idea of treating ideals as cells of a CW complex needs an actual construction before it becomes mathematics rather than metaphor.

§30.9.iii Linear algebra and representation theory

Representation theory can be seen as the extension of linear algebra into group theory and symmetry. Eigenvalues, traces, matrices, and vector spaces occur constantly in Galois representations and automorphic forms.

§30.10 Numerical experiments

Computational experiments cannot prove Langlands statements, but they can build intuition. Natural experiments include:

(i) computing initial values of the Ramanujan tau function;

(ii) plotting the oscillation of modular-form coefficients;

(iii) implementing a Hensel-lifting example;

(iv) testing simple finite-dimensional representations in toy cases.

The first object to compute is Ramanujan's discriminant modular form

$$\Delta(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n, \qquad q = e^{2\pi i z}.$$

Its first coefficients are

$$\tau(1) = 1, \quad \tau(2) = -24, \quad \tau(3) = 252, \quad \tau(4) = -1472, \quad \tau(5) = 4830, \quad \tau(6) = -6048.$$

These numbers are already arithmetic, but one should not match them with an arbitrary elliptic curve.

Now take the elliptic curve

$$E: \quad y^2 + y = x^3 - x^2,$$

Cremona label $11a3$, of conductor 11. For a prime $p \neq 11$, define

$$a_p = p + 1 - \#E(\mathbb{F}_p).$$

A direct enumeration gives

p	$\#E(\mathbb{F}_p)$	a_p
2	5	-2
3	5	-1
5	5	1
7	10	-2

For example, over $\mathbb{F}_5$ the affine solutions are only

$$(0,0), (0,4), (1,0), (1,4),$$

together with the point at infinity, so $\#E(\mathbb{F}_5) = 5$ and $a_5 = 1$.

The important correction is that

$$\tau(2), \tau(3), \tau(5), \tau(7) = -24, 252, 4830, -16744$$

do *not* equal

$$a_2, a_3, a_5, a_7 = -2, -1, 1, -2.$$

This mismatch is not a failure of modularity; it is a failure to choose the matching modular form. Δ is a weight-12, level-1 modular form. The elliptic curve of conductor 11 corresponds instead to a weight-2, level-11 newform whose p-th coefficient is a_p. The concrete Langlands-style lesson is therefore sharper than a slogan: once the level, weight, and normalization are correct, Fourier coefficients on the automorphic side match Frobenius traces or point counts on the Galois side.

A short SageMath check is enough to reproduce the finite-field side:

```
E = EllipticCurve([0,-1,1,0,0])   # y^2 + y = x^3 - x^2
for p in [2,3,5,7,13,17,19]:
    Np = E.change_ring(GF(p)).cardinality()
    print(p, Np, p + 1 - Np, E.ap(p))
```

For the Δ-function coefficients one can use a truncated product:

```
R.<q> = PowerSeriesRing(ZZ, default_prec=12)
Delta = q * prod((1 - q^n)^24 for n in range(1,12))
print(Delta)
```

§30.11 Topics to learn next

Several missing foundations are clear: complex analysis for analytic continuation and functional equations of L-functions; measure theory and integration for automorphic forms and harmonic analysis; Galois theory as the classical source of arithmetic symmetry; Sato–Tate phenomena and equidistribution; and random matrix heuristics for zeros of L-functions.

Question

Why should modular forms and Galois representations, which look completely different at first, speak a common language? This question is not answered here, but it is the right surprise to keep.

§30.12 Synthesis

The route of the note is

$$\text{number fields} \to \text{ideals} \to \text{local fields} \to \text{modular forms and } L\text{-functions} \to \text{Langlands}.$$

Some formulations in the original note were too optimistic or imprecise, but the underlying instinct is good: Langlands is not one isolated theorem, but a web of correspondences linking arithmetic, analysis, geometry, and representation theory.

§30.13 Local-global intuition

A recurring theme in algebraic number theory is to understand a global object by all of its local shadows. For a number field K, the local fields K_v arise by completing K at its archimedean and nonarchimedean places. The rational field gives familiar examples:

$$\mathbb{Q} \hookrightarrow \mathbb{R}, \qquad \mathbb{Q} \hookrightarrow \mathbb{Q}_p.$$

The real completion measures ordinary size; the p-adic completion measures divisibility by p. The collection of all completions records many different ways a number can be close to zero.

This local-global idea is one reason adeles and ideles appear. They package all completions at once, allowing arithmetic problems to be studied through harmonic analysis on locally compact groups.

§30.14 Why L-functions enter

An L-function is a generating function that encodes arithmetic data. For example, the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

has the Euler product

$$\zeta(s) = \prod_p \frac{1}{1-p^{-s}}.$$

The sum sees positive integers; the product sees primes. This equality is the prototype for the idea that analytic behavior of a function can reveal arithmetic structure.

In Langlands-style thinking, Galois representations and automorphic forms have associated L-functions. A correspondence is often tested by matching these L-functions.

§30.15 A first Langlands slogan

A very rough slogan is:

$$\text{Galois representations} \quad \leftrightarrow \quad \text{automorphic representations.}$$

The left side is arithmetic and studies symmetries of field extensions. The right side is analytic and representation-theoretic, involving functions on groups over adeles. The Langlands program predicts a deep dictionary between them.

The modularity theorem for elliptic curves over $\mathbb{Q}$ is a famous instance: arithmetic information from an elliptic curve matches a modular form. This is why modular forms enter a note that begins with number fields and ideals.

§30.16 Structural viewpoint

The note is a map, not a proof. Number fields generalize $\mathbb{Q}$; algebraic integers generalize $\mathbb{Z}$; ideals repair failure of unique factorization; local fields isolate one prime at a time; adeles recombine all primes; modular forms and automorphic representations provide analytic symmetry; L-functions act as fingerprints. Langlands is the claim that these fingerprints match across arithmetic and analysis.

§30.17 Ideals as repaired unique factorization

The central obstruction in algebraic number theory is that elements may fail to factor uniquely. Ideals repair this: in the ring of integers $\mathcal{O}_K$ of a number field, nonzero ideals factor uniquely into prime ideals,

$$\mathfrak{a} = \prod_i \mathfrak{p}_i^{e_i}.$$

The class group measures the gap between principal ideals and all ideals. Class number one means ideal factorization descends all the way back to element factorization.

§30.18 Local fields and arithmetic microscopes

Completion at a prime ideal gives a local field $K_{\mathfrak{p}}$. This is an arithmetic microscope: it sees divisibility and congruence near one prime at a time. Global statements are often proved by comparing all local completions.

The product formula and local-global principles explain why p-adic numbers naturally appear next to number fields.

§30.19 Modular forms and Galois representations

A modular form has Fourier expansion

$$f(q) = \sum_{n \geq 0} a_n q^n.$$

The coefficients a_p often encode arithmetic information. In the Langlands viewpoint, modular forms correspond to Galois representations, so analytic objects and symmetry actions on algebraic numbers describe the same data.

§30.20 Arithmetic symmetry as the bridge to Langlands

The path from algebraic number theory to Langlands is a passage from factorization to symmetry. Ideals repair arithmetic, local fields isolate primes, L-functions package global data, and Langlands predicts that arithmetic Galois symmetries are mirrored by automorphic analytic objects.

Finite Fields and Polynomial Factorization

N-20221002

§31.1 Why finite fields matter

This note is a more systematic essay on finite fields and factorization. The central question is: how do polynomials factor when the coefficients live in a finite field rather than in $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$?

This question is not only theoretical. It appears in coding theory, cryptography, computational algebra, and the construction of algebraic extensions. The note correctly starts from the definition of a field, because factorization depends heavily on the coefficient domain.

§31.2 Fields

A field F is a set with addition and multiplication such that:

(i) $(F, +)$ is an abelian group with identity 0;

(ii) $F \setminus \{0\}$ is an abelian group under multiplication with identity 1;

(iii) multiplication distributes over addition.

The exclusion of 0 from the multiplicative group is essential. If 0 had a multiplicative inverse, then from $0 \cdot a = 0$ one would get contradictions such as $1 = 0$. So the phrase "every nonzero element has an inverse" is not a technical nuisance; it is what makes division possible.

Examples are

$$\mathbb{Q}, \qquad \mathbb{R}, \qquad \mathbb{C},$$

and, for a prime p,

$$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}.$$

The requirement that p be prime is crucial. If p is composite, zero divisors appear. For example in $\mathbb{Z}/6\mathbb{Z}$,

$$2 \cdot 3 \equiv 0 \pmod 6,$$

even though neither factor is zero.

§31.3 Field extensions

A field extension K/F means F is embedded in a larger field K. The point of passing to K is often to make more polynomial equations solvable. For example,

$$x^2 + 1$$

has no root in $\mathbb{R}$, but it splits in $\mathbb{C}$:

$$x^2 + 1 = (x - i)(x + i).$$

If K is finite-dimensional as a vector space over F, its dimension is the degree of the extension:

$$[K : F] = \dim_F K.$$

The note's intuition is good: finite extensions are tractable because linear algebra can be used. Infinite extensions require much heavier tools.

§31.4 Constructing finite fields

For a prime p, $\mathbb{F}_p$ is the simplest finite field. To construct fields with p^n elements, use irreducible polynomials. If

$$f(x) \in \mathbb{F}_p[x]$$

is irreducible of degree n, then

$$\mathbb{F}_p[x]/(f(x))$$

is a field with p^n elements.

For example, over $\mathbb{F}_2$, the polynomial

$$f(x) = x^3 + x + 1$$

has no root in $\mathbb{F}_2$, hence is irreducible of degree 3. Therefore

$$\mathbb{F}_2[x]/(x^3 + x + 1)$$

is a field with $2^3 = 8$ elements.

In this quotient field, if α denotes the class of x, then

$$\alpha^3 + \alpha + 1 = 0,$$

so

$$\alpha^3 = \alpha + 1$$

in characteristic 2. Every element can be uniquely written as

$$a + b\alpha + c\alpha^2, \qquad a, b, c \in \mathbb{F}_2.$$

§31.5 Important properties of finite fields

The standard facts are:

(i) finite fields have order p^n;

(ii) for every prime power p^n, there is a finite field of that order, unique up to isomorphism;

(iii) the nonzero elements $\mathbb{F}_{p^n}^\times$ form a cyclic group of order $p^n - 1$;

(iv) the Frobenius map

$$\operatorname{Frob}(a) = a^p$$

is an automorphism of $\mathbb{F}_{p^n}$.

The cyclicity of the multiplicative group is surprisingly powerful. It means every nonzero element is a power of one primitive element. This is why finite field computations often become exponent arithmetic modulo $p^n - 1$.

§31.6 Polynomial factorization over finite fields

In $\mathbb{F}_q[x]$, every nonzero polynomial factors uniquely into irreducible polynomials, up to order and multiplication by nonzero constants. This is analogous to unique prime factorization of integers.

However, a polynomial need not split into linear factors over its base finite field. For example, a polynomial may be irreducible over $\mathbb{F}_p$ but split over an extension field. This is exactly like $x^2 + 1$ over $\mathbb{R}$, except that the extension fields are finite and highly structured.

§31.7 Example over $\mathbb{F}_3$

Consider
$$f(x) = x^4 + x^2 + 2 \in \mathbb{F}_3[x].$$
First test roots:
$$f(0) = 2 \neq 0,$$
$$f(1) = 1 + 1 + 2 = 4 \equiv 1 \pmod 3,$$
$$f(2) = 2^4 + 2^2 + 2 = 16 + 4 + 2 = 22 \equiv 1 \pmod 3.$$
So f has no linear factor. It might still factor as a product of two quadratics. A systematic check over $\mathbb{F}_3$ shows that no such factorization exists, so f is irreducible over $\mathbb{F}_3$.

This illustrates the basic workflow: root testing only detects linear factors. For degree 4, one must also check quadratic factors.

§31.8 Algorithms

The note lists several factorization algorithms:

(i) trial division, useful for very small degrees;

(ii) Berlekamp's algorithm, which converts factorization into linear algebra over finite fields;

(iii) Cantor–Zassenhaus, a randomized algorithm widely used in computer algebra systems.

The high-level idea behind these algorithms is that finite fields make the Frobenius map computable. Since every element of $\mathbb{F}_q$ satisfies
$$a^q = a,$$
the polynomial
$$x^q - x$$
contains all elements of $\mathbb{F}_q$ as roots. This identity becomes a machine for detecting factors.

§31.9 Applications

Finite-field factorization appears in:

(i) error-correcting codes;

(ii) cryptographic protocols;

(iii) construction of extension fields;

(iv) computational number theory;

(v) symbolic algebra systems.

The note's final reflection is that finite fields are small enough for computation but structured enough to retain deep algebraic behavior. That is why they are so central.

§31.10 Irreducibility over finite fields

A polynomial can factor differently depending on the coefficient field. For example, x^2+1 is irreducible over $\mathbb{R}$ only if one insists on real linear factors, but over $\mathbb{C}$ it splits. Over finite fields the same phenomenon becomes arithmetic.

For $\mathbb{F}_p$, a quadratic polynomial

$$ax^2 + bx + c$$

factors iff its discriminant

$$\Delta = b^2 - 4ac$$

is a square in $\mathbb{F}_p$. Thus irreducibility is controlled by quadratic residues. This links polynomial factorization directly to modular arithmetic.

§31.11 Constructing extensions

If $f(x) \in F[x]$ is irreducible of degree n, then

$$F[x]/(f(x))$$

is a field extension of F of degree n. The class of x becomes an element α satisfying $f(\alpha) = 0$. This is the algebraic meaning of "adjoining a root."

For finite fields, this construction produces fields with p^n elements:

$$\mathbb{F}_{p^n} \cong \mathbb{F}_p[x]/(f(x))$$

for any irreducible f of degree n. Every finite field has prime-power size, and for each prime power there is a unique finite field up to isomorphism.

§31.12 Frobenius map

In characteristic p, the Frobenius map is

$$\varphi(a) = a^p.$$

It is a field homomorphism because

$$(a+b)^p = a^p + b^p$$

in characteristic p. In a finite field, Frobenius is an automorphism. Its fixed field inside $\mathbb{F}_{p^n}$ is $\mathbb{F}_p$.

The polynomial

$$x^{p^n} - x$$

has all elements of $\mathbb{F}_{p^n}$ as roots, and it factors over $\mathbb{F}_p$ as the product of all monic irreducible polynomials whose degrees divide n. This is a central organizing fact for finite-field factorization.

§31.13 What survives later

Finite fields turn algebra into arithmetic. Roots of polynomials become elements of finite extensions, irreducibility becomes a modular phenomenon, and Frobenius gives a canonical symmetry. This is why finite fields appear in coding theory and cryptography: their algebra is rich but completely finite and computable.

§31.14 How the pieces fit

A finite field is where group theory, ring theory, linear algebra, and number theory meet. The additive group of $\mathbb{F}_{p^n}$ is an n-dimensional vector space over $\mathbb{F}_p$. The nonzero elements form a cyclic multiplicative group of order $p^n - 1$. Polynomial quotients construct the field, and Frobenius supplies a canonical symmetry.

Thus a finite field computation often has three simultaneous readings:

operation	structure being used
addition	$\mathbb{F}_p$-linear algebra
multiplication	cyclic unit group and polynomial reduction
Frobenius $x \mapsto x^p$	Galois symmetry

§31.15 Frobenius and the first Galois group

In characteristic p, the Frobenius map

$$F : x \mapsto x^p$$

is a field homomorphism because

$$(x+y)^p = x^p + y^p.$$

In $\mathbb{F}_{p^n}$, it is an automorphism, and

$$F^n = 1.$$

The Galois group is cyclic:

$$\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle F \rangle \cong \mathbb{Z}/n\mathbb{Z}.$$

This is the cleanest first example of a Galois group: roots are moved by raising to powers of p, and the fixed field of Frobenius is $\mathbb{F}_p$.

§31.16 Irreducible polynomials and orbit size

An element $\alpha \in \mathbb{F}_{q^n}$ has conjugates

$$\alpha, \quad \alpha^q, \quad \alpha^{q^2}, \quad \dots$$

under Frobenius. Its minimal polynomial over $\mathbb{F}_q$ is

$$\prod_{i=0}^{d-1} (x - \alpha^{q^i}),$$

where d is the size of the Frobenius orbit. Thus irreducible polynomials over finite fields are Frobenius orbits in disguise.

This explains why the degrees of irreducible factors of a polynomial are controlled by Frobenius cycles.

§31.17 Counting irreducible polynomials

Every element of $\mathbb{F}_{q^n}$ is a root of

$$x^{q^n} - x.$$

Over $\mathbb{F}_q$, this polynomial is the product of all monic irreducible polynomials whose degrees divide n. If N_d is the number of monic irreducible polynomials of degree d, then

$$q^n = \sum_{d|n} d N_d.$$

By Möbius inversion,

$$N_n = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}.$$

This formula turns factorization into enumeration. The factor d appears because each degree-d irreducible polynomial contributes d roots.

§31.18 Square-free and distinct-degree factorization

Finite-field factorization algorithms exploit Frobenius. The polynomial

$$x^{q^m} - x$$

has as roots exactly the elements of $\mathbb{F}_{q^m}$. Therefore

$$\gcd(f(x), x^{q^m} - x)$$

extracts from f the product of irreducible factors whose degrees divide m.

This is the idea behind distinct-degree factorization. First remove repeated factors by checking $\gcd(f, f')$. Then use Frobenius powers to separate factors by degree. Finally use algorithms such as Cantor–Zassenhaus to split equal-degree parts.

§31.19 Berlekamp's algorithm as linear algebra

Berlekamp's algorithm turns factorization into a nullspace computation. Over $\mathbb{F}_q$, consider the map on the quotient algebra $A = \mathbb{F}_q[x]/(f)$

$$B : g \mapsto g^q - g.$$

Elements in the kernel satisfy $g^q = g$, and this kernel contains information about decomposing f into factors.

Thus the algorithm uses the finite-dimensional $\mathbb{F}_q$-vector space structure of the quotient ring. This is a concrete instance of a general computational theme: algebraic structure becomes linear algebra after passing to a finite quotient.

§31.20 Coding theory connection

Finite fields are essential in coding theory because code symbols can be added, multiplied, and evaluated polynomially. Reed–Solomon codes evaluate polynomials over $\mathbb{F}_q$ at distinct field points:

$$f \longmapsto (f(a_1), \ldots, f(a_m)).$$

A nonzero polynomial of degree $< k$ has at most $k-1$ roots, so two different low-degree polynomials cannot agree at too many evaluation points. This gives error detection and correction.

Here the same polynomial-root principle used in factorization becomes a robustness principle for information transmission.

§31.21 Algebraic geometry shadow

The identity

$$\#\mathbb{A}^1(\mathbb{F}_q) = q$$

and the polynomial $x^q - x$ already hint at a larger story: equations over finite fields define finite sets of points, and Frobenius acts on their solutions. In higher-dimensional algebraic geometry, counting points over $\mathbb{F}_{q^n}$ reveals cohomological information.

Thus elementary finite-field polynomial factorization is a first step toward modern arithmetic geometry: finite fields provide a world where algebraic equations, symmetries, and point counts interact exactly.

§31.22 Frobenius structure behind finite fields

Finite fields are not just small fields. They are finite vector spaces, cyclic multiplicative groups, quotient rings, Frobenius dynamical systems, and computational domains all at once. Irreducible polynomials build extensions; Frobenius organizes roots; gcds with $x^{q^m} - x$ detect factor degrees; and the same algebra supports coding theory and cryptography.

Elliptic Curves from Smooth Cubics to Finite Groups

N-20221120

§32.1 A cubic becomes an elliptic curve only after two additions

Over a field k of characteristic different from 2 and 3, a short Weierstrass equation has the form

$$E: \quad y^2 = x^3 + ax + b.$$

Two extra requirements turn this equation into an elliptic curve. First, the cubic must be nonsingular. Second, one chooses a distinguished point that will serve as the identity of a group law.

The affine equation does not visibly contain such an identity point. Homogenizing gives the projective cubic

$$Y^2Z = X^3 + aXZ^2 + bZ^3.$$

On the line at infinity $Z = 0$, the equation forces $X = 0$, leaving the single point

$$\mathcal{O} = [0 : 1 : 0].$$

This point is not attached by hand after the geometry is finished. It is already present in the projective closure, and every vertical affine line meets the cubic there.

§32.2 The discriminant is exactly the obstruction to smoothness

Let

$$f(x) = x^3 + ax + b.$$

An affine point (x, y) on the curve is singular when both partial derivatives of

$$F(x, y) = y^2 - f(x)$$

vanish:

$$2y = 0, \qquad 3x^2 + a = 0.$$

Thus a singular point must have $y = 0$, while x is a common root of f and f'. Equivalently, the cubic polynomial has a repeated root.

For a depressed cubic $x^3 + ax + b$, the resultant of f and f' is, up to sign,

$$4a^3 + 27b^2.$$

Hence the curve is nonsingular precisely when

$$\boxed{4a^3 + 27b^2 \neq 0.}$$

The conventional discriminant is

$$\Delta = -16(4a^3 + 27b^2).$$

At the point at infinity, the projective partial derivative with respect to Z is nonzero, so $\mathcal{O}$ is smooth as well.

The condition is geometric, not decorative. At a node or cusp there is no single tangent direction with the behavior required by the chord-and-tangent construction. Smoothness is what allows the cubic to carry a regular group law.

§32.3 A line determines the third point, and Vieta determines its coordinates

Take two affine points

$$P = (x_1, y_1), \qquad Q = (x_2, y_2)$$

with $x_1 \neq x_2$. The line through them is

$$y = m(x - x_1) + y_1, \qquad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Substituting into the cubic gives a polynomial equation of degree three in x. Two roots are already known: x_1, x_2. If the third intersection has coordinate x_R, comparison of the quadratic coefficient gives

$$x_1 + x_2 + x_R = m^2.$$

Therefore

$$x_R = m^2 - x_1 - x_2.$$

The third point itself is

$$R = (x_R, m(x_R - x_1) + y_1).$$

The group sum is defined by reflecting this point across the x-axis:

$$P + Q = -R.$$

Thus

$$\boxed{x_{P+Q} = m^2 - x_1 - x_2,}$$

$$\boxed{y_{P+Q} = m(x_1 - x_{P+Q}) - y_1.}$$

The formulas are Vieta's relations applied to the line–cubic intersection, not independent identities to memorize.

§32.4 Tangency, inverses, and the point at infinity complete the rule

When $P = Q$, the secant slope is replaced by the tangent slope. Implicit differentiation of

$$y^2 = x^3 + ax + b$$

gives

$$2y\frac{dy}{dx} = 3x^2 + a.$$

Hence, when $y_1 \neq 0$,

$$m = \frac{3x_1^2 + a}{2y_1}.$$

The same coordinate formulas then compute $2P$.

The reflection

$$-(x,y) = (x,-y)$$

is the group inverse. The line through P and $-P$ is vertical and meets the projective cubic at $\mathcal{O}$, so

$$P + (-P) = \mathcal{O}.$$

If $y = 0$, the tangent is vertical; consequently

$$2P = \mathcal{O},$$

and P is a point of order two.

Finally,

$$P + \mathcal{O} = P.$$

Projective geometry has removed the exceptional case: a vertical line still has its third intersection, but that intersection lies at infinity.

§32.5 A rational example shows both addition and doubling

Consider

$$E: \quad y^2 = x^3 - x + 1$$

over $\mathbb{Q}$. The points

$$P = (0,1), \qquad Q = (1,1)$$

lie on the curve. Their joining line is horizontal, so $m = 0$. Hence

$$x_{P+Q} = 0^2 - 0 - 1 = -1,$$

$$y_{P+Q} = 0(0 - (-1)) - 1 = -1.$$

Therefore

$$P + Q = (-1,-1).$$

A direct substitution verifies

$$(-1)^2 = (-1)^3 - (-1) + 1 = 1.$$

For doubling P,

$$m = \frac{3 \cdot 0^2 - 1}{2 \cdot 1} = -\frac{1}{2}.$$

Thus

$$x_{2P} = \frac{1}{4},$$

$$y_{2P} = -\frac{1}{2}\left(0 - \frac{1}{4}\right) - 1 = -\frac{7}{8}.$$

Indeed,

$$\left(-\frac{7}{8}\right)^2 = \left(\frac{1}{4}\right)^3 - \frac{1}{4} + 1 = \frac{49}{64}.$$

The geometry and the rational arithmetic agree exactly.

§32.6 Divisors explain why the chord rule is associative

The coordinate formulas make closure and inverses visible, but a direct verification of

$$(P+Q)+R = P+(Q+R)$$

would be unpleasant. The reason for associativity is better seen through divisors.

On the projective cubic, let L be a line meeting E at P, Q, R, counted with multiplicity. The line at infinity meets the Weierstrass cubic at $\mathcal{O}$ with multiplicity three. The ratio of the two linear forms therefore defines a rational function on E whose divisor is

$$P + Q + R - 3\mathcal{O}.$$

A principal divisor represents zero in the degree-zero Picard group, so

$$[P - \mathcal{O}] + [Q - \mathcal{O}] + [R - \mathcal{O}] = 0 \qquad \text{in } \operatorname{Pic}^0(E).$$

Since the geometric rule defines $P + Q = -R$, it is exactly the rule required for

$$P \longmapsto [P - \mathcal{O}]$$

to respect addition.

For a smooth genus-one curve with chosen point $\mathcal{O}$, this map is an isomorphism

$$E \cong \operatorname{Pic}^0(E).$$

The Picard group is an abelian group because divisor classes add associatively. The chord-and-tangent operation inherits that associativity. Thus divisor theory does not merely restate the picture in advanced language; it supplies the group axiom that the picture alone does not prove.

§32.7 The same formulas work over finite fields

Let $p > 3$ be prime and suppose $\Delta \not\equiv 0 \pmod p$. The set

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\}$$

forms a finite abelian group. The addition formulas are unchanged, but division means multiplication by a modular inverse.

For example, on

$$E: \quad y^2 = x^3 - x + 1 \qquad \text{over } \mathbb{F}_5,$$

the quadratic residues are $0, 1, 4$. Checking each x gives

x	0	1	2	3	4
$x^3 - x + 1$	1	1	2	0	1
$\#\{y : y^2 = x^3 - x + 1\}$	2	2	0	1	2.

Including $\mathcal{O}$,

$$\#E(\mathbb{F}_5) = 2 + 2 + 0 + 1 + 2 + 1 = 8.$$

Hasse's theorem predicts

$$\left|\#E(\mathbb{F}_5) - (5+1)\right| \le 2\sqrt{5},$$

and the count 8 satisfies this bound.

§32.8 One point generates the entire eight-element example

Take
$$P = (0, 1) \in E(\mathbb{F}_5).$$
For doubling,
$$m = \frac{-1}{2} \equiv 4 \cdot 3 \equiv 2 \pmod 5,$$
so
$$2P = (4, 1).$$
Doubling again,
$$m = \frac{3 \cdot 4^2 - 1}{2} \equiv \frac{2}{2} \equiv 1 \pmod 5,$$
and therefore
$$4P = (3, 0).$$
Because a point with $y = 0$ has order two,
$$8P = \mathcal{O}.$$
Since $4P \neq \mathcal{O}$, the order of P is exactly eight. The group has eight elements and contains an element of order eight, hence
$$E(\mathbb{F}_5) \cong \mathbb{Z}/8\mathbb{Z}.$$
This small example makes scalar multiplication concrete rather than treating the finite group as a black box.

§32.9 Scalar multiplication creates the computational asymmetry

For a positive integer n, scalar multiplication is
$$[n]P = P + \cdots + P.$$
It is computed efficiently by binary expansion. To evaluate $[13]P$, for example, one forms
$$P, \; 2P, \; 4P, \; 8P$$
and combines the terms corresponding to
$$13 = 8 + 4 + 1.$$
This double-and-add method requires only logarithmically many group operations.

The inverse problem asks for n given P and $Q = [n]P$. On suitably chosen large finite-field subgroups, no comparably efficient general method is known; this is the elliptic-curve discrete logarithm problem. The cryptographic use therefore rests on a precise asymmetry: projective algebra makes the forward group operation explicit and fast, while recovering the scalar remains difficult.

The path from the cubic equation to that asymmetry is now continuous. The discriminant guarantees a smooth projective curve, the point at infinity supplies an identity, line intersections give the addition formulas, divisor classes prove associativity, and reduction modulo a prime turns the same geometry into a finite abelian group.

When Addition Starts Floating: Theta-Links, Log-Links, and Arithmetic Deformation

N-20251113

§33.1 What has to be explained

The previous note ended with a deliberately strange operation. At a bad split multiplicative valuation v, one has a Tate parameter q_v, and the arithmetic-deformation viewpoint asks us to take seriously transformations of the shape

$$q_v \longmapsto q_v^{m^2}, \qquad u \longmapsto u \quad (u \in \mathcal{O}_{F_v}^\times).$$

This is not a ring homomorphism. It is a transformation of multiplicative data. The central question of this note is therefore precise:

How can one deform multiplicative theta-data, lose ordinary additive compatibility, and still recover enough structure to prove an inequality?

The answer in Fesenko's survey passes through nonarchimedean theta-functions, generalized Kummer theory, theta-links, log-links, log-shells, multiradial reconstruction, and indeterminacies. The goal here is not to pretend to define the full IUT apparatus. The goal is to keep the mathematical objects visible and to explain why each object is forced by the previous one.

§33.2 The nonarchimedean theta-function

Let L be a local field of characteristic zero with finite residue field, and let C_L be the completion of an algebraic closure of L. Let $q \in L$ be a nonzero element of the maximal ideal of $\mathcal{O}_L$. In the application to elliptic curves, q is the Tate parameter q_v.

Definition 33.2.1 (Theta-function of type au^m)**.** A holomorphic function $f : C_L^\times \to C_L$, defined over L, is a theta-function of type au^m, where $a \in C_L^\times$ and $m \in \mathbb{Z}$, if

$$f(u) = au^m f(qu).$$

A meromorphic function invariant under $u \mapsto qu$ descends to a function on the Tate quotient

$$C_L^\times/\langle q\rangle.$$

The basic nonarchimedean theta-function used in the survey is

$$\theta(u) = \sum_{n\in\mathbb{Z}} (-1)^n q^{n(n-1)/2} u^n.$$

It also has the product expansion

$$\theta(u) = (1-u)\prod_{n\geq 1}(1-q^n)(1-q^n u)(1-q^n u^{-1}),$$

which is the nonarchimedean analogue of the Jacobi triple product expression. The function is chosen because its zeros and poles interact well with the Tate curve and with the theta-values used in IUT.

Remark 33.2.2 (Why a theta-function appears) — The theta-function is not added to the story for analogy with complex analysis alone. On a Tate curve, elliptic functions with period q can be expressed in terms of theta-functions. Thus theta-functions are a natural language for functions on the local analytic uniformization

$$E(L) \cong L^\times/\langle q\rangle.$$

Fesenko also records a modified theta-function, written here schematically as $\ddot{\Theta}(u)$, whose special values at points separated by periods from $\pm i$ satisfy a relation of the form

$$q^{m^2/2}\ddot{\Theta}(i\sqrt{q^m}) = \ddot{\Theta}(i).$$

After choosing a $2l$-th root of q, this leads to theta-values of the shape

$$\Theta(\sqrt{-q^m}) = q^{m^2}, \qquad 1 \le m \le \frac{l-1}{2},$$

up to multiplication by roots of unity of order dividing $2l$.

Remark 33.2.3 (A controlled simplification) — The formula above is the correct structural feature needed here: special theta-values encode powers q^{m^2}. The complete IUT construction keeps track of coverings, labels, roots of unity, Galois actions, and compatible local-global data. Those details are not being suppressed because they are unimportant; they are being postponed because the first conceptual step is to see why theta-values can serve as deformation data.

§33.3 The local theta-link as a non-ring-theoretic map

At a bad reduction valuation of odd residue characteristic, the main version of the theta-link described by Fesenko revolves around a local monoid-theoretic transformation. In simplified notation it sends

$$q \longmapsto \{\Theta(\sqrt{-q^m}) = q^{m^2}\}_{1\le m\le (l-1)/2},$$

and it sends units to units by the identity map, up to the appropriate torsion ambiguity.

Definition 33.3.1 (Local theta-link, structural form)**.** A local theta-link is a monoid-theoretic comparison which, at a bad split multiplicative valuation, keeps the unit part essentially fixed while replacing the Tate-period generator q by a family of theta-values behaving like q^{m^2}. It is a comparison of multiplicative and theta-theoretic data, not a homomorphism of local rings.

Thus the local theta-link acts on two kinds of local data:

$$\mathcal{O}_L^\times / \operatorname{Tor}(\mathcal{O}_L^\times), \qquad \{q^{m^2} : 1 \le m \le (l-1)/2\},$$

and it is glued to global monoid-theoretic data satisfying the product formula.

Proposition 33.3.2 (Why the theta-link is not scheme-theoretic)

The theta-link is not compatible with the full ring structure of L. In particular, it should not be interpreted as a morphism of schemes or as a ring isomorphism between two local fields.

Proof. A ring homomorphism must preserve both multiplication and addition. The theta-link is built from the multiplicative monoid of units, powers of the Tate parameter, and special values of a theta-function. It transports multiplicative/theta data, but the additive operation of the local field is not transported by the displayed map. Fesenko therefore describes this as an arithmetic deformation of the local structure rather than as an ordinary scheme-theoretic morphism. □

This is the first honest meaning of the phrase "addition starts floating". It does not mean that the equation $1 + 1 = 2$ becomes false inside a fixed field. It means that after theta-deformation one is no longer allowed to identify the additive coordinates on the two sides as if a ring homomorphism had carried them across.

§33.4 The bridge supplied by Kummer theory

If the theta-link only destroyed additive compatibility, it would not be useful. The next ingredient explains how monoid-theoretic data can still communicate with Galois and fundamental-group data.

Definition 33.4.1 (Kummer map, schematic form)**.** Let H be an open subgroup of a Galois or arithmetic fundamental group acting on an abelian group M. Kummer theory studies maps of the form

$$M^H \longrightarrow H^1(H, \mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, M)),$$

obtained by considering divisibility of elements of M.

For ordinary intuition, Kummer theory turns the question of extracting roots into a cohomological Galois datum. In IUT, the relevant versions are generalized and truncated. They are applied not merely to elements of fields, but also to monoid-theoretic structures associated with theta-functions, theta-values, number fields, and their completions.

> **Remark 33.4.2** (Kummer theory as a translator) — The theta-link is monoid-theoretic. Mono-anabelian reconstruction works on the side of arithmetic fundamental groups. Kummer theory is one of the translators between these languages. It relates theta-values and monoid data to Galois or fundamental-group structures that can be reconstructed and compared.

A key issue here is cyclotomic rigidity. Various copies of $\widehat{\mathbb{Z}}$, or quotients of such copies, arise from roots of unity, from monoid-theoretic data, and from subquotients of fundamental groups. IUT needs algorithms that identify the relevant cyclotomic structures in a canonical way. Without such rigidity, the roots of unity introduced by extracting l-th or $2l$-th roots would create uncontrolled ambiguity.

§33.5 Two symmetries and the bookkeeping of labels

The theta-link is built after choosing a prime $l > 3$. The l-torsion points of the elliptic curve provide labels modeled on $\mathbb{F}_l$. Fesenko emphasizes two different symmetries.

Definition 33.5.1 (The two label symmetries)**.** The geometric-additive symmetry is

$$\mathbb{F}_l^{\rtimes\pm} = \mathbb{F}_l \rtimes \{\pm 1\},$$

acting on labels by

$$z \longmapsto \pm z + a.$$

The arithmetic-multiplicative symmetry is

$$\mathbb{F}_l^{*\pm} = \mathbb{F}_l^{\times}/\{\pm 1\}.$$

The first symmetry is essentially geometric. It is related to the geometric part of the arithmetic fundamental groups and to Kummer theory around theta-values. It is also used to synchronize conjugacy ambiguities of local Galois groups attached to different labels.

The second symmetry is essentially arithmetic. It is related to the Galois action of number fields and is multiplicative by definition. It helps separate the zero label from the nonzero labels and is more naturally multiradial.

Symmetry	Nature	Role
$\mathbb{F}_l^{\rtimes\pm}$	geometric/additive	conjugate synchronization
$\mathbb{F}_l^{*\pm}$	arithmetic/multiplicative	label bookkeeping and descent

Remark 33.5.2 (Why labels are not cosmetic) — The labels are not merely indices in a long construction. They keep track of different copies of local Galois data and theta-values. If these labels are identified too naively, one loses the distinction between the geometric-additive and arithmetic-multiplicative parts of the construction.

§33.6 Conjugate synchronization and the basepoint problem

Fundamental groups depend on basepoints. In ordinary expositions this dependence is often hidden because different basepoints lead to groups identified up to inner automorphism. In IUT this ambiguity cannot simply be ignored, because the theta-link and log-link compare structures that live in different theaters.

Definition 33.6.1 (Conjugate synchronization, informal)**.** Conjugate synchronization is a system of identifications between local absolute Galois groups attached to different labels, arranged so that one does not introduce uncontrolled conjugacy indeterminacy.

The need for such synchronization is one reason the theory becomes much more rigid than a naive "copy-and-paste" of local fields. The comparison is not allowed to choose arbitrary basepoint identifications. It must respect the label symmetries and the Kummer-theoretic data surrounding the theta-values.

§33.7 The log-link: recovering additive structure from multiplicative structure

The theta-link uses multiplicative theta-values. To regain access to additive structure, Fesenko turns to the nonarchimedean logarithm.

Definition 33.7.1 (Nonarchimedean logarithm)**.** For a local field L, the nonarchimedean logarithm

$$\log : \mathcal{O}_L^\times \longrightarrow L$$

is defined on elements $1-\alpha$, with α in the maximal ideal of $\mathcal{O}_L$, by

$$\log(1-\alpha) = -\sum_{n\geq 1} \frac{\alpha^n}{n},$$

and sends multiplicative representatives of the finite residue field to 0.

The key point is that log transforms multiplication into addition where it is defined. Thus the logarithm lets multiplicative unit data speak an additive language.

Definition 33.7.2 (Log-link, structural form)**.** A log-link is the comparison which shifts multiplicative unit data through the nonarchimedean logarithm, so that additive structures can be reconstructed from logarithmic images of multiplicative structures.

Fesenko explains the relation between the two links as follows. The multiplicative structures on the two sides of the theta-link are related through value-group data. The additive structures are related through unit-group data after applying the log-link.

> **Remark 33.7.3** (Why the logarithm is not optional) — Theta-values naturally act on multiplicative data, not directly on the additive group of the local field. The logarithm creates a setting where multiplicative unit data has an additive image. This is why the log-link is not an auxiliary simplification; it is part of the reconstruction mechanism.

§33.8 The log-theta-lattice

The theta-link and log-link do not appear as isolated arrows. They are organized into a two-dimensional pattern called the log-theta-lattice.

Definition 33.8.1 (Log-theta-lattice, schematic)**.** The log-theta-lattice is a two-dimensional array of theaters. Vertical arrows are log-links, and horizontal arrows are theta-links. A horizontal theta-link at one level depends on the log-link structure immediately below it.

The picture is:

vertical arrows: log-links

$$\begin{tikzcd}
\mathcal{H}_{0,2} \arrow[r] & \mathcal{H}_{1,2} \arrow[r] & \mathcal{H}_{2,2} \\
\mathcal{H}_{0,1} \arrow[r] \arrow[u] & \mathcal{H}_{1,1} \arrow[r] \arrow[u] & \mathcal{H}_{2,1} \arrow[u] \\
\mathcal{H}_{0,0} \arrow[r] \arrow[u] & \mathcal{H}_{1,0} \arrow[r] \arrow[u] & \mathcal{H}_{2,0} \arrow[u]
\end{tikzcd}$$

horizontal arrows: theta-links

This diagram is only a guide. Fesenko notes that the main results use just two neighboring infinite vertical lines and the horizontal arrow between the lattice points $(0,0)$ and $(1,0)$. The important idea is that the theory does not compare a single source and target once. It studies a controlled pattern of comparisons.

§33.9 The log-shell: a common container for incompatible links

A technical difficulty now appears. The log-link does not commute with the theta-link. If one expected a clean commutative square, one would be disappointed. IUT instead introduces a common structure that can hold the relevant images even when elementwise compatibility fails.

Definition 33.9.1 (Log-shell)**.** For a nonarchimedean local field L, the nonarchimedean part of the log-shell is a compact subgroup of the form

$$(p^*)^{-1} \log(\mathcal{O}_L^\times),$$

where $p^* = p$ if p is odd and $2^* = 4$. At a complex archimedean place, the corresponding log-shell is the closed ball of radius π.

The log-shell should not be imagined as an error bar in a numerical approximation. It is a structural container. The relevant Kummer isomorphisms are not compatible with the log-link at the level of individual elements, but their images are contained in the log-shell. This is exactly the kind of weakened compatibility that arithmetic deformation needs.

> **Remark 33.9.2** (Compatibility without equality) — In ordinary algebra one often tries to prove that a diagram commutes element by element. The log-shell records a weaker but controlled form of compatibility: one compares inclusions of certain subsets or regions rather than exact equality of individual transported elements.

§33.10 Mono-anabelian reconstruction and multiradiality

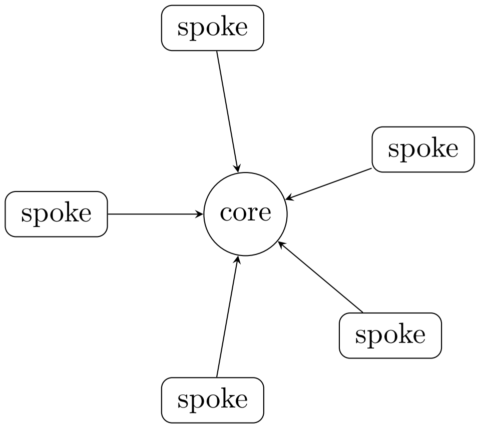
The words “reconstruction” and “multiradiality” are easy to mystify, but their roles are concrete.

Definition 33.10.1 (Mono-anabelian reconstruction)**.** Mono-anabelian reconstruction is the construction of ring-theoretic data from a single topological group that is known to be isomorphic to an arithmetic fundamental group of a suitable hyperbolic curve or orbicurve. The reconstruction is algorithmic and functorial in the sense needed for localization and change of base field.

The difference from ordinary anabelian comparison is important. Classical anabelian geometry often compares two geometric objects through an isomorphism of their fundamental groups. Mono-anabelian geometry asks for algorithms that start from one suitable group and recover the ring structure associated to it.

Definition 33.10.2 (Multiradial algorithm)**.** A reconstruction algorithm is multiradial if an object constructed from one spoke of a wheel-like system is described in terms that still make sense from the viewpoint of other spokes. Equivalently, it is compatible with simultaneous execution at multiple spokes.

Here the “wheel and spokes” language is only an analogy, but a useful one. A spoke corresponds to one theater or one radial perspective. The center corresponds to a core object reconstructed from data that can be described from more than one spoke.



multiradial descriptions must remain meaningful from several spokes at once

Remark 33.10.3 (Why indeterminacy appears) — A multiradial description is stronger than a one-spoke description. To make such descriptions possible, IUT allows certain controlled indeterminacies. These are not defects added after the fact; they are part of how the theory avoids making illegal identifications between different theaters.

§33.11 The three indeterminacies

Fesenko singles out three indeterminacies that enter the computation of volume deformation.

Definition 33.11.1 (The three indeterminacies, structural form)**.** The three main indeterminacies are:

1. Ind1: an indeterminacy related to permutation symmetries of Galois and arithmetic fundamental groups along vertical lines of the log-theta-lattice;
2. Ind2: an indeterminacy related to the horizontal theta-link and to the action of a compact group on $\log(\mathcal{O}_L^\times)$;
3. Ind3: an indeterminacy arising from upper semi-compatibility of Kummer isomorphisms with log-links in a single vertical line.

The third phrase, "upper semi-compatibility", is deliberately weaker than commutativity of a diagram. Instead of asking for equality of transported elements, one asks for controlled inclusion relations among the relevant subsets.

Proposition 33.11.2 (Why indeterminacies are mathematically productive)

The indeterminacies do not merely obscure the comparison. They make it possible to compare data across distinct theaters without falsely identifying the full ring structures on the two sides.

Proof. The theta-link is not ring-theoretic, and the log-link does not commute with it. Therefore an elementwise comparison of ordinary rings would impose more structure than the construction provides. The indeterminacies specify exactly which ambiguity is allowed: permutation ambiguity, horizontal theta-link ambiguity, and weakened Kummer/log compatibility. A volume computation can then be performed after enlarging the target by these controlled ambiguities. □

This is where the intuitive phrase “floating addition” becomes mathematically safer. The additive structure is not discarded into chaos. It is reconstructed through logarithmic images and fundamental-group algorithms, while the unavoidable ambiguity is measured by the three indeterminacies.

§33.12 What the second note adds

We can now summarize the actual mechanism more accurately than by saying “IUT uses many universes”.

1. The Tate parameter q gives local multiplicative data at bad reduction valuations.
2. Nonarchimedean theta-functions produce distinguished values behaving like q^{m^2}.
3. The theta-link transports this theta/multiplicative data but is not a ring morphism.
4. Generalized Kummer theory relates monoid-theoretic data to Galois and arithmetic fundamental-group data.
5. The two label symmetries separate the geometric-additive side from the arithmetic-multiplicative side.
6. The log-link uses $\log : \mathcal{O}_L^\times \to L$ to recover additive information from multiplicative units.
7. The log-shell supplies a common container when the log-link and theta-link fail to commute elementwise.
8. Mono-anabelian reconstruction and multiradial algorithms make it possible to describe reconstructed objects from several theaters at once.
9. The three indeterminacies specify the allowed ambiguity and enter the final volume estimate.

The next note should therefore not treat the controversy around IUT as a piece of mathematical news. The foundational issue is already visible here: once ordinary ring-theoretic identification is forbidden, the theory must replace it by theta-links, log-links, Kummer correspondences, and multiradial reconstruction. The controversy is ultimately about whether this replacement is strong enough, not about whether the vocabulary is unusual.

Teichmueller Theory Without One Universe: Classical Deformation, Hodge Theaters, and Identification

N-20251114

§34.1 Why the word Teichmueller matters

The word "Teichmueller" in inter-universal Teichmueller theory should not be treated as a poetic label. It points to a deformation problem. In classical Teichmueller theory, one fixes a topological surface and studies how complex structures on that surface vary. In IUT, the analogy is not that arithmetic curves literally form a classical Teichmueller space. The analogy is that some structures are treated as rigid reference data, while other structures are deformed and measured.

This note compares the two situations. The comparison is useful only if it is kept precise. Classical Teichmueller theory varies complex structures inside a common topological background. IUT tries to compare arithmetic data across distinct theaters where ordinary ring-theoretic identification is not allowed. The controversial strength of the theory is already visible at this level: it replaces the ordinary demand for one common universe by a system of reconstructed, multiradial, and indeterminacy-controlled comparisons.

§34.2 Classical Teichmueller theory: fixed topology, variable complex structure

Let S be an oriented topological surface of finite type. To remember which complex surface is supposed to be a deformation of S, one uses a marking.

Definition 34.2.1 (Marked Riemann surface)**.** A marked Riemann surface modeled on S is a pair

$$(X, f),$$

where X is a Riemann surface and

$$f : S \longrightarrow X$$

is an orientation-preserving homeomorphism, or more generally a quasiconformal map, regarded up to homotopy.

The marking is the key device. Without it, one has only a collection of complex curves. With it, different complex structures can still be compared as structures on the same underlying topological surface.

Definition 34.2.2 (Teichmueller space)**.** The Teichmueller space $T(S)$ is the set of equivalence classes of marked Riemann surfaces (X, f), where

$$(X, f) \sim (Y, g)$$

if there is a biholomorphism $h : X \to Y$ such that $h \circ f$ is homotopic to g.

Thus classical Teichmueller theory separates two layers:

topological surface S and complex structure on S.

The topology is the reference frame. The complex structure is what moves.

Remark 34.2.3 (The role of marking) — A marking is an authorized identification. It tells us when two points of Teichmueller space represent two complex structures on the same underlying surface. This will be the main contrast with IUT: IUT cannot simply use a single marking to identify all ring structures across its theaters.

§34.3 Distortion is measured in one universe

Teichmueller theory does not only parametrize complex structures. It measures deformation by quasiconformal distortion.

Definition 34.3.1 (Quasiconformal distortion)**.** Let $f : X \to Y$ be a quasiconformal map between Riemann surfaces. In a local coordinate it has distributional derivatives f_z and $f_{\bar{z}}$. Its Beltrami coefficient is

$$\mu_f = \frac{f_{\bar{z}}}{f_z},$$

and the maximal dilatation is

$$K(f) = \frac{1 + \|\mu_f\|_\infty}{1 - \|\mu_f\|_\infty}.$$

The Teichmueller distance is then defined by minimizing distortion among maps in the correct homotopy class:

$$d_T((X, f), (Y, g)) = \frac{1}{2} \inf_h \log K(h),$$

where $h : X \to Y$ ranges over quasiconformal maps such that $h \circ f$ is homotopic to g.

The important point is structural: this distance makes sense because the comparison is made inside one clear framework. Both surfaces are marked by the same S. The phrase "same underlying topological surface" has a precise meaning.

Proposition 34.3.2 (Classical deformation has a common reference object)

In classical Teichmueller theory, the comparison of two complex structures is mediated by the fixed surface S. The marking supplies a legitimate way to say that the two complex structures are structures on the same object.

Proof. A point of $T(S)$ is not merely a Riemann surface X; it is a pair (X, f). Given two points (X, f) and (Y, g), the maps f and g identify both surfaces with the same topological source S, up to homotopy. Therefore quasiconformal maps $h : X \to Y$ can be tested against the homotopy condition $h \circ f \simeq g$. The fixed S is the common reference object. □

§34.4 Why the arithmetic analogy cannot be literal

In the arithmetic setting, the objects are not Riemann surfaces equipped with varying complex structures. The basic geometric objects are hyperbolic curves, elliptic curves with punctures or orbifold quotients, number fields, local fields, arithmetic fundamental groups, and Galois groups. The rigid structure is no longer a topological surface. It is closer to a package of fundamental-group-like data from which ring structures can be reconstructed.

Definition 34.4.1 (Arithmetic fundamental-group reference data, schematic)**.** For a hyperbolic curve X over a number field K, the arithmetic fundamental group fits into an exact sequence

$$1 \longrightarrow \pi_1^{\mathrm{geom}}(X) \longrightarrow \pi_1(X) \longrightarrow G_K \longrightarrow 1.$$

In the IUT entrance, such group-theoretic data acts as a rigid reference structure from which arithmetic and ring-theoretic data may be reconstructed by mono-anabelian methods.

The analogy with classical Teichmueller theory is therefore shifted:

classical theory	:	fixed topological type, variable complex structure,
IUT direction	:	rigid group-theoretic data, deformed ring-theoretic presentations.

This is only an analogy, but it is a mathematically meaningful one.

> **Remark 34.4.2** (What should not be inferred) — The word "Teichmueller" does not mean that IUT constructs a Teichmueller space of number fields in the classical sense. It means that the theory is organized around deformation, rigidity, and comparison of structures, but the structures being compared are arithmetic and group-theoretic rather than complex-analytic.

§34.5 Hodge theaters as arithmetic stages

The language of theaters is one of the points where IUT becomes unfamiliar. A theater should not be imagined as a parallel universe in a science-fiction sense. It is a package of data within which certain arithmetic, geometric, and group-theoretic structures can be discussed internally.

Definition 34.5.1 (Hodge theater, working schematic)**.** In this note, a Hodge theater means a package of arithmetic data containing several mutually related layers: local fields, number-field data, arithmetic fundamental groups, monoid-theoretic data, and theta-data. The full definition in IUT is much more elaborate. The working point is that a theater provides an internal stage on which ring structures, Galois data, theta-values, and reconstruction algorithms coexist.

The word "theater" is useful because IUT does not simply compare two rings by a ring isomorphism. It compares data internal to one theater with data internal to another theater through special links.

Definition 34.5.2 (Inter-universal comparison, schematic)**.** An inter-universal comparison is a comparison between data belonging to distinct Hodge theaters, where one does not assume a global ring-theoretic identification between the theaters. The comparison is mediated by structures such as theta-links, log-links, Kummer correspondences, and mono-anabelian reconstruction.

In ordinary algebraic geometry, the most comfortable comparison has the form

$$R \cong R'$$

or at least a morphism of schemes. In IUT, the central horizontal comparison is not of that kind. The theta-link is precisely a link that deforms ring-theoretic structure rather than preserving it.

§34.6 Etale-like and Frobenius-like directions

Fesenko's survey uses the distinction between structures that behave like rigid etale or fundamental-group data and structures that behave more like monoid-theoretic or Frobenius-type data. This is one of the cleanest ways to understand why ordinary scheme morphisms are not enough.

Definition 34.6.1 (Etale-like and Frobenius-like structures, schematic)**.** An etale-like structure is a structure on the fundamental-group side: it is functorial, rigid, and invariant enough to be used for reconstruction. A Frobenius-like structure is a monoid-theoretic structure used to build deformation links, especially when no ordinary Frobenius morphism of number fields exists.

The terminology is suggestive. Over a field of positive characteristic, the Frobenius map $x \mapsto x^p$ is a genuine morphism. For number fields, there is no such global Frobenius endomorphism. IUT instead builds categorical and monoid-theoretic analogues that can play a deformation role.

> **Remark 34.6.2** (The arithmetic reason this is radical) — The theory tries to get a Frobenius-like deformation effect in a setting where the usual Frobenius morphism does not exist. This is part of the reason the comparison cannot remain inside ordinary scheme-theoretic geometry.

§34.7 The theta-link is the point where one universe breaks

The local model already appeared in the previous two notes:

$$q_v \longmapsto q_v^{m^2}, \qquad u \longmapsto u.$$

This operation is meaningful on the multiplicative/monoid side, but it is not a ring homomorphism. It therefore cannot serve as an identification of local fields as rings.

Proposition 34.7.1 (No common ring is carried by the theta-link)

If a theta-link is interpreted as the assignment that carries a Tate parameter q to theta-values behaving like q^{m^2} while keeping unit data on the monoid side, then it cannot be regarded as a ring-theoretic isomorphism between the source and target local structures.

Proof. A ring-theoretic isomorphism must preserve addition and multiplication simultaneously. The theta-link is constructed from multiplicative monoids, unit groups, Tate parameters, and theta-values. Its displayed local behavior is not an additive operation. Therefore treating it as a ring isomorphism would add a compatibility that the construction is not designed to provide. □

This is the exact mathematical sense in which the "one universe" picture breaks. If one tries to force the two sides of the theta-link into the same ring-theoretic universe, one loses the deformation that the theta-link was built to express.

§34.8 The log-link does not restore naive equality

The log-link exists because the theta-link moves multiplicative data while the final Diophantine inequality needs additive and volume information. The nonarchimedean logarithm

$$\log : \mathcal{O}_L^\times \longrightarrow L$$

translates unit data into additive data. This does not mean that the original ring structure has been transported unchanged.

Definition 34.8.1 (Recovered additive data, schematic)**.** Recovered additive data means additive structure obtained after applying logarithmic and mono-anabelian reconstruction procedures to multiplicative or group-theoretic data. It is not the same thing as an additive structure transported by a ring isomorphism.

The distinction is small in words but large in mathematics. Classical Teichmueller theory deforms complex structures while markings keep track of the same underlying topological surface. IUT deforms arithmetic structures in a way that prevents one from identifying the two sides as the same ring. Additive information must therefore be reconstructed rather than simply carried over.

> **Remark 34.8.2** (Why the log-shell is needed) — The log-link and the theta-link do not give a clean commutative square of elementwise identifications. The log-shell provides a controlled container for the resulting ambiguity. It is a way of replacing exact equality by a bounded or contained region in which the relevant images live.

§34.9 Classical marking versus multiradial reconstruction

The most useful comparison between classical Teichmueller theory and IUT is not a slogan about deformation. It is a comparison between two ways of controlling identity.

The comparison can be kept in five pairs.

1. Classical theory chooses a fixed topological surface S; IUT treats arithmetic fundamental-group-like data as rigid and reconstructive.
2. Classical theory uses a marking $f : S \to X$; IUT uses mono-anabelian algorithms to recover ring-theoretic data from group-theoretic input.
3. Classical theory moves the complex structure on S; IUT moves a ring-theoretic or monoid-theoretic presentation of arithmetic data.
4. Classical theory measures distortion by quasiconformal maps; IUT creates and measures arithmetic deformation by theta-links and log-links.

5. Classical comparison remains inside one marked topological setting; IUT comparison moves across distinct Hodge theaters.

This list shows why IUT is not merely difficult notation. The role played by marking in classical Teichmueller theory is not replaced by one simple arithmetic marking. Instead, it is replaced by a much more elaborate system: reconstruction algorithms, label symmetries, Kummer theory, log-shells, and indeterminacies.

Definition 34.9.1 (Multiradial identification, schematic)**.** A multiradial identification is not a direct equality of objects in one common ring universe. It is an identification of reconstructed structures that remains meaningful when viewed from several theaters or spokes at once.

This is why multiradiality is conceptually central. If a construction only makes sense from one spoke, it may depend on a hidden choice of coordinates. A multiradial construction is designed to survive the change of theater.

§34.10 The mathematical source of the controversy

This note should not become a news summary. The mathematical tension can be stated without naming any episode. It is a tension about legitimate identification.

Proposition 34.10.1 (The identification burden)

Once ordinary ring-theoretic identification is forbidden across theaters, every later comparison must justify itself by some replacement mechanism: mono-anabelian reconstruction, Kummer correspondence, theta-link compatibility, log-link compatibility, multiradiality, or a controlled indeterminacy.

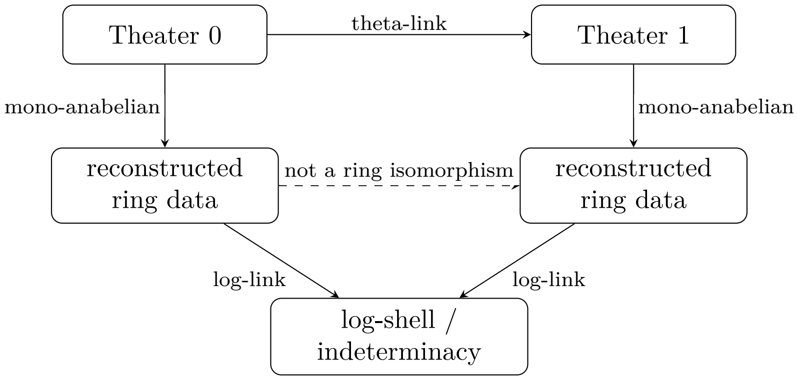
Proof. Without a ring isomorphism, there is no default rule saying that an element, sum, unit, valuation, or theta-value in one theater is the same object as an element, sum, unit, valuation, or theta-value in another. Therefore each permitted comparison must be constructed and checked. The structures listed in the statement are precisely the devices introduced to make such comparisons possible without collapsing all theaters into one ordinary ring-theoretic setting. □

The controversial foundational move is therefore not merely "there are many universes". A vague multiverse would have no mathematical force. The real move is this: do not identify rings directly; instead recover and compare selected structures by controlled algorithms. The difficulty is that the replacement mechanisms must be strong enough to support the final log-volume inequality. If they are too weak, the deformation cannot be measured. If they are interpreted as ordinary equalities, the deformation disappears.

Remark 34.10.2 (Why a one-universe reading is dangerous) — A one-universe reading tries to place all theaters into a single background in which ordinary ring-theoretic equality is available. That may make the theory look more familiar, but it changes the object under study. The theta-link is designed precisely to avoid being a ring-theoretic identification.

§34.11 A compact model of the logic

The logical structure of the comparison can be summarized as follows:



The dashed arrow is the temptation that the theory resists. The comparison does not proceed by declaring the two reconstructed rings equal. It proceeds by controlling how their relevant images sit inside the logarithmic and theta-theoretic structures.

§34.12 What this third note establishes

The relationship with classical Teichmueller theory can now be stated cleanly.

1. Classical Teichmueller theory fixes a topological surface and deforms complex structures on it.

2. The marking is the mechanism that makes comparison legitimate in the classical theory.

3. IUT does not have a single arithmetic marking that identifies all ring structures across theaters.

4. Instead, IUT uses Hodge theaters, theta-links, log-links, mono-anabelian reconstruction, Kummer theory, multiradiality, and indeterminacies.

5. The phrase "inter-universal" means that comparisons are made without collapsing the theaters into one ordinary ring-theoretic universe.

6. The deepest mathematical pressure point is therefore the legitimacy and strength of these replacement identifications.

The theory is difficult not only because it contains many objects, but because it changes the usual rule of comparison. Classical Teichmueller theory asks how complex structures vary when the underlying topological object is held fixed. IUT asks how arithmetic structures can be deformed when the ordinary ring structure is not allowed to serve as a common background. That is the mathematical content behind the word "inter-universal".

V

Functions and Inequalities

Part V: Contents

Designing Inequality Proofs: Equality Cases, Homogeneity, and Cyclic Structure

N-20220417

§35.1 Finding the likely answer is only half of a proof

An inequality problem often has an exploratory phase and a certification phase. During exploration one tests special cases, guesses the equality configuration, normalizes variables, and searches for a familiar pattern. A complete proof must then establish the bound for every admissible input and verify that equality can actually occur.

For an optimization statement, the two obligations are

$$F(x) \geq L$$

for all allowed x, and

$$F(x_0) = L$$

for at least one allowed x_0. The first proves that L is a lower bound; the second proves it is sharp.

A simple example is

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

The symmetric point $a = b = c$ suggests the equality case. The certification is the identity

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}((a-b)^2 + (b-c)^2 + (c-a)^2).$$

The right-hand side is nonnegative, and it vanishes exactly when

$$a = b = c.$$

The equality case is not a decorative sentence after the proof. It confirms that the proposed bound is the best possible and checks that no step has introduced unnecessary slack.

The same habit matters in existence and classification problems. "Find all" requires both producing candidates and excluding everything else. "Find the minimum" requires both a universal lower bound and an example attaining it. The proof strategy should be designed around both obligations from the beginning.

§35.2 Homogeneity tells us what may be normalized

An expression $F(a, b, c)$ is homogeneous of degree d if

$$F(ta, tb, tc) = t^d F(a, b, c)$$

for every positive t. If both sides of an inequality have the same degree, scaling all variables does not change its truth. One degree of freedom may then be removed by a normalization such as

$$a + b + c = 1, \qquad abc = 1, \qquad a^2 + b^2 + c^2 = 1.$$

The best choice is the one that simplifies the structure already present.

For example, the inequality

$$(a+b+c)^2 \geq 3(ab+bc+ca)$$

is homogeneous of degree two. Normalizing $a+b+c=1$ turns it into

$$ab+bc+ca \leq \frac{1}{3}.$$

Using

$$1=(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)$$

and

$$a^2+b^2+c^2 \geq ab+bc+ca$$

gives the result immediately.

If a condition says

$$abc=1,$$

the substitution

$$a=\frac{x}{y}, \qquad b=\frac{y}{z}, \qquad c=\frac{z}{x}$$

builds the constraint automatically. This substitution is useful only when the resulting cyclic ratios simplify the expression; otherwise the normalization $abc=1$ may be better used directly.

Homogeneity is therefore not a command to normalize mechanically. It is permission to choose a convenient cross-section of the scaling orbits.

§35.3 Cyclic and symmetric expressions carry different information

For three variables, cyclic summation means

$$\sum_{\text{cyc}} f(a,b,c)=f(a,b,c)+f(b,c,a)+f(c,a,b).$$

A symmetric sum includes every distinct permutation. For instance,

$$\sum_{\text{cyc}} a^2b=a^2b+b^2c+c^2a,$$

whereas

$$\sum_{\text{sym}} a^2b=a^2b+a^2c+b^2a+b^2c+c^2a+c^2b.$$

A symmetric expression is unchanged by every permutation of the variables. A cyclic expression remembers an orientation. The two orientations differ by

$$\begin{aligned} a^2b+b^2c+c^2a-(ab^2+bc^2+ca^2) \\ =-(a-b)(b-c)(c-a). \end{aligned}$$

Thus a cyclic sum becomes equal to its reverse when two variables coincide, but not in general.

This identity is a useful diagnostic. If a proposed proof replaces a cyclic expression by a symmetric one, it may be discarding the orientation term

$$(a-b)(b-c)(c-a).$$

Sometimes that loss is harmless because the symmetric estimate is strong enough. Sometimes the sign of the orientation term is exactly what an ordering assumption such as

$$a \geq b \geq c$$

was meant to control.

The notation should therefore follow the invariance of the problem. A cyclic problem often rewards cyclic substitutions and ordered cases; a symmetric homogeneous problem may invite smoothing, majorization, or a reduction to elementary symmetric polynomials.

§35.4 Equality cases guide the choice of inequality

Suppose a positive symmetric expression appears to attain equality at

$$a = b = c.$$

That suggests methods whose equality conditions also force equal variables: AM–GM, Jensen with strict convexity or concavity, Cauchy in a proportional configuration, or smoothing two variables toward their average.

Consider

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} \geq a + b + c, \qquad a, b, c > 0.$$

Set

$$A = \frac{a^3}{bc}, \qquad B = \frac{b^3}{ca}, \qquad C = \frac{c^3}{ab}.$$

To produce the term a, apply AM–GM to

$$A, A, B, C.$$

Their product is

$$A^2BC = a^4,$$

so

$$\frac{2A + B + C}{4} \geq a.$$

Cyclically,

$$\frac{A + 2B + C}{4} \geq b,$$
$$\frac{A + B + 2C}{4} \geq c.$$

Adding gives

$$A + B + C \geq a + b + c.$$

Equality in the first AM–GM step requires

$$A = B = C,$$

and the same condition appears cyclically. For positive variables this gives

$$a = b = c.$$

The proof was designed backward from the desired right-hand terms and the expected equality configuration. The repeated copy of A was chosen so that the geometric mean became exactly a.

§35.5 Denominators often reveal an Engel-form Cauchy step

Fractions with positive denominators frequently suggest the Engel form

$$\sum_{i=1}^{n} \frac{x_i^2}{y_i} \geq \frac{(x_1 + \cdots + x_n)^2}{y_1 + \cdots + y_n}.$$

The numerators need not initially be squares; one may rewrite

$$\frac{a}{b+c} = \frac{a^2}{ab+ac}.$$

This gives a short proof of Nesbitt's inequality:

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}, \qquad a, b, c > 0.$$

Indeed,

$$\begin{aligned} \sum_{\text{cyc}} \frac{a}{b+c} &= \sum_{\text{cyc}} \frac{a^2}{ab+ac} \\ &\geq \frac{(a+b+c)^2}{2(ab+bc+ca)}. \end{aligned}$$

Since

$$(a+b+c)^2 \geq 3(ab+bc+ca),$$

we obtain

$$\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}.$$

The equality conditions also match. Engel equality requires

$$\frac{a}{ab+ac} = \frac{b}{bc+ab} = \frac{c}{ca+bc},$$

which reduces to

$$a = b = c.$$

The second inequality has the same equality case, so no slack is lost between the two steps.

A denominator pattern is therefore not merely an obstacle. It often announces which quantities should become the y_i in a Cauchy quotient.

§35.6 Smoothing explains why equal variables often extremize

Fix the sum

$$x + y = s.$$

Replacing x, y by their average

$$\frac{s}{2}, \frac{s}{2}$$

is called a smoothing step. For a convex function f,

$$f(x) + f(y) \geq 2f\left(\frac{x+y}{2}\right).$$

Thus smoothing decreases a convex symmetric sum. For a concave function, the direction reverses.

The elementary identity

$$x^2 + y^2 - \frac{s^2}{2} = \frac{(x-y)^2}{2}$$

shows this directly for $f(t) = t^2$. With the sum fixed, the square sum is minimized when the variables are equal.

The product behaves oppositely:

$$xy = \frac{s^2 - (x-y)^2}{4} \le \frac{s^2}{4}.$$

Hence smoothing increases the product. Iterating the argument yields AM–GM for three variables with fixed sum:

$$abc \le \left(\frac{a+b+c}{3}\right)^3.$$

A useful calculus diagnostic reaches the same candidate. To maximize

$$\log(abc) = \log a + \log b + \log c$$

subject to

$$a + b + c = s,$$

Lagrange multipliers give

$$\frac{1}{a} = \frac{1}{b} = \frac{1}{c},$$

so the only interior critical point is

$$a = b = c = \frac{s}{3}.$$

Because log is strictly concave and the product tends to zero at the boundary, this point is the unique global maximum. The calculus calculation discovers the candidate; AM–GM or concavity supplies the global certificate.

§35.7 Triangle substitutions encode the boundary conditions

If a, b, c are side lengths of a nondegenerate triangle, then

$$a < b + c, \qquad b < c + a, \qquad c < a + b.$$

The substitution

$$a = y + z, \qquad b = z + x, \qquad c = x + y, \qquad x, y, z > 0$$

builds these inequalities automatically. Conversely,

$$x = \frac{b+c-a}{2}, \quad y = \frac{c+a-b}{2}, \quad z = \frac{a+b-c}{2}.$$

As a simple application, Heron's semiperimeter differences become

$$s - a = x, \qquad s - b = y, \qquad s - c = z,$$

where

$$s = x + y + z.$$

The area formula

$$\Delta^2 = s(s-a)(s-b)(s-c)$$

becomes

$$\Delta^2 = (x+y+z)xyz.$$

A geometric constraint has turned into positivity of three independent variables.

The substitution is useful when the expression repeatedly contains

$$b+c-a, \quad c+a-b, \quad a+b-c.$$

It is not automatically beneficial for every triangle inequality. As with normalization, the substitution should simplify the actual algebra rather than merely certify that the solver recognizes a standard trick.

§35.8 Schur's inequality shows how ordering controls a cyclic sign

Schur's inequality of degree three states

$$a^3+b^3+c^3+3abc \geq \sum_{\text{sym}} a^2b$$

for nonnegative a, b, c. Equivalently,

$$\sum_{\text{cyc}} a(a-b)(a-c) \geq 0.$$

Because the expression is symmetric, assume without loss of generality that

$$a \geq b \geq c.$$

Then group the first two terms:

$$\begin{aligned}
a(a-b)(a-c) &+ b(b-c)(b-a) \\
&= (a-b)\big(a(a-c) - b(b-c)\big) \\
&= (a-b)^2(a+b-c).
\end{aligned}$$

The remaining term is

$$c(c-a)(c-b) = c(a-c)(b-c).$$

Therefore

$$\sum_{\text{cyc}} a(a-b)(a-c) = (a-b)^2(a+b-c) + c(a-c)(b-c) \geq 0.$$

Every factor is nonnegative under the chosen ordering. Equality occurs when

$$a=b=c$$

or when one variable is zero and the other two are equal.

This proof illustrates a general pattern. A symmetric inequality permits an ordering assumption. The ordered variables control signs that would otherwise be ambiguous, and a cyclic expression can then be regrouped into visibly nonnegative pieces.

§35.9 A complete proof keeps the equality case alive

Consider the homogeneous inequality

$$\frac{a^2}{b+c}+\frac{b^2}{c+a}+\frac{c^2}{a+b}\geq\frac{a+b+c}{2},\qquad a,b,c>0.$$

The denominators suggest Engel form, and the symmetric right-hand side suggests equality at $a=b=c$. One line gives

$$\begin{aligned}\sum_{\text{cyc}}\frac{a^2}{b+c}&\geq\frac{(a+b+c)^2}{(b+c)+(c+a)+(a+b)}\\&=\frac{(a+b+c)^2}{2(a+b+c)}\\&=\frac{a+b+c}{2}.\end{aligned}$$

The Cauchy equality condition is

$$\frac{a}{b+c}=\frac{b}{c+a}=\frac{c}{a+b},$$

which for positive variables forces

$$a=b=c.$$

The candidate survives the only inequality step, so the bound is sharp.

This small example contains the full design process. Homogeneity permits normalization if desired; symmetry predicts the equality configuration; the denominator pattern identifies Engel form; and the equality condition verifies sharpness. A good inequality proof is not a list of named weapons. It is a sequence of transformations chosen so that the structure of the problem and the structure of the theorem line up without losing the extremal case.

Nonlinear Recurrences, Fixed Points, and Metric-Space Exercises

N-20220718

§36.1 Fixed-point iteration

The note begins with iterative sequences. If a function f has a fixed point x_0, meaning

$$f(x_0) = x_0,$$

then the recurrence

$$a_{n+1} = f(a_n)$$

may converge to x_0, depending on the behavior of f. The graphical picture is the intersection of $y = f(x)$ and $y = x$.

The elementary idea is this: if $a_n \to L$ and f is continuous, then

$$L = \lim a_{n+1} = \lim f(a_n) = f(L),$$

so any limit must be a fixed point. But finding a fixed point is not the same as proving convergence. A recurrence can have fixed points and still oscillate or diverge.

§36.2 Möbius recurrences

The page considers recurrences of the form

$$a_{n+1} = \frac{pa_n + q}{ra_n + t}.$$

These are fractional linear or Möbius transformations. Their fixed points solve

$$x = \frac{px + q}{rx + t},$$

so

$$rx^2 + (t - p)x - q = 0.$$

Suppose x_1, x_2 are two distinct fixed points. Then a standard calculation gives

$$\frac{a_{n+1} - x_1}{a_{n+1} - x_2} = \lambda \frac{a_n - x_1}{a_n - x_2}$$

for a constant λ. Thus the new sequence

$$b_n = \frac{a_n - x_1}{a_n - x_2}$$

is geometric.

This is the key idea in the note: convert a nonlinear recurrence into a geometric progression by using the fixed points as coordinates. This is not accidental. Möbius transformations act naturally on the projective line, and the cross-ratio-like expression above is the coordinate that linearizes the action.

§36.3 Example

For

$$a_1 = \frac{1}{2}, \qquad a_{n+1} = \frac{5a_n - 1}{a_n + 3},$$

the fixed-point equation is

$$x = \frac{5x - 1}{x + 3},$$

so

$$x^2 - 2x + 1 = 0, \qquad x = 1.$$

Here the two fixed points coincide, so the previous method degenerates. One then studies

$$a_{n+1} - 1 = \frac{4a_n - 4}{a_n + 3} = \frac{4(a_n - 1)}{a_n + 3}.$$

Taking reciprocals gives

$$\frac{1}{a_{n+1} - 1} = \frac{a_n + 3}{4(a_n - 1)} = \frac{1}{4} + \frac{1}{a_n - 1}.$$

Thus

$$b_n = \frac{1}{a_n - 1}$$

is arithmetic, not geometric. This is the repeated-root analogue of the fixed-point method.

§36.4 Second-order linear recurrences

The next topic is

$$a_{n+2} = pa_{n+1} + qa_n.$$

Try $a_n = \lambda^n$. The characteristic equation is

$$\lambda^2 - p\lambda - q = 0.$$

If it has two distinct roots λ_1, λ_2, then

$$a_n = A\lambda_1^n + B\lambda_2^n.$$

If the root is repeated, then the second independent solution is $n\lambda^n$, so

$$a_n = (A + Bn)\lambda^n.$$

This is the recurrence analogue of solving a linear differential equation with constant coefficients.

§36.5 Dedekind cuts and metric exercises

The uploaded pages also continue exercises on Dedekind cuts and metric spaces. For Dedekind cuts X, Y, the sum is

$$X + Y = \{x + y : x \in X,\ y \in Y\}.$$

The notes check that this is again a cut: it is nonempty, not all of $\mathbb{Q}$, downward closed, and has no largest rational.

For a metric space (X, d), if $Y \subseteq X$, the subspace metric is

$$d_Y(y_1, y_2) = d(y_1, y_2).$$

The note also checks that Euclidean distance on $\mathbb{R}^n$

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$

is a metric. The only nontrivial axiom is the triangle inequality, which ultimately follows from Cauchy–Schwarz.

§36.6 Fixed points of recurrence maps

A recurrence $x_{n+1} = f(x_n)$ is an iteration problem. A possible limit L must satisfy

$$L = f(L).$$

Thus the first step is to solve the fixed-point equation. The second step is to decide whether the iteration actually converges to that fixed point.

A common local test is $|f'(L)| < 1$. Then nearby points are pulled toward L. If $|f'(L)| > 1$, nearby points are pushed away. This explains why fixed points are not automatically limits.

§36.7 Monotone bounded sequences

Many elementary recurrence proofs use the theorem: a monotone bounded real sequence converges. The method is to prove by induction that the sequence stays in an interval, then prove monotonicity. Completeness of $\mathbb{R}$ is the hidden reason the limit exists.

§36.8 After the examples

The important common thread is change of coordinates. A Möbius recurrence becomes geometric after measuring position relative to fixed points. A repeated fixed point becomes arithmetic after taking a reciprocal. A Dedekind cut becomes a real number after using order-theoretic structure. A subset becomes a metric space after inheriting distance. The mathematical move is the same: choose the representation in which the structure becomes visible.

§36.9 Fixed points and stability

For an iteration $x_{n+1} = F(x_n)$, a fixed point satisfies $F(x_*) = x_*$. The behavior near x_* is controlled by $F'(x_*)$ in one dimension: if $|F'(x_*)| < 1$, the fixed point is locally attracting.

§36.10 Möbius recurrences and matrices

A recurrence of the form

$$x_{n+1} = \frac{ax_n + b}{cx_n + d}$$

is a fractional-linear transformation. It corresponds to the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

acting on the projective line. Iterating the recurrence corresponds to taking powers of this matrix.

§36.11 Second-order recurrences

A linear recurrence

$$a_{n+2} = pa_{n+1} + qa_n$$

is governed by the characteristic equation

$$\lambda^2 = p\lambda + q.$$

Distinct roots give combinations of geometric sequences. This is diagonalization of the shift dynamics.

§36.12 Recurrences as one-dimensional dynamics

Nonlinear and linear recurrences are dynamical systems. Fixed points organize stability, Möbius recurrences are projective matrix dynamics, and second-order linear recurrences are solved by eigenvalue methods.

Probability, Conditional Probability, and Substitution Tricks in Inequalities

N-20220721

§37.1 Random experiments and sample spaces

This note begins with the language of probability. A random experiment is a process that can be repeated under the same conditions, whose possible outcomes are known, but whose actual outcome is uncertain. Examples include tossing a coin, drawing a ball from a box, or choosing a card.

The set of all possible basic outcomes is the sample space, denoted

$$\Omega.$$

A subset $A \subseteq \Omega$ is an event. If the sample space is finite and all outcomes are equally likely, then

$$P(A) = \frac{|A|}{|\Omega|}.$$

The note stresses that events are sets. This is why union, intersection, complement, and inclusion–exclusion immediately become probability tools:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

§37.2 Classical model and Venn diagrams

One example has seven labeled balls, with three properties represented by sets A_1, A_2, A_3. The note uses a Venn diagram and factorial counts to compute a probability such as

$$P = \frac{7! - 3 \cdot 6! + 3 \cdot 5! - 4!}{7!}.$$

The exact numbers matter less than the method. If conditions overlap, count by inclusion–exclusion. A Venn diagram is not a proof by itself, but it helps prevent double-counting.

§37.3 Conditional probability

The conditional probability of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

when $P(A) > 0$. The note includes examples where one must be careful not to confuse $P(B \mid A)$ with $P(B)$.

Independence is the condition

$$P(A \cap B) = P(A)P(B),$$

or equivalently

$$P(B \mid A) = P(B).$$

This means that knowing A occurred does not change the probability of B.

A common trap appears in nested events. If $A \subseteq B$, then

$$P(B \mid A) = 1,$$

but

$$P(A \mid B) = \frac{P(A)}{P(B)}$$

may be much smaller. Conditional probability is not symmetric.

§37.4 A transition to substitution tricks

The note then changes direction and discusses substitutions used in inequalities. A recurring constraint is

$$abc = 1.$$

A standard substitution is

$$a = \frac{x}{y}, \qquad b = \frac{y}{z}, \qquad c = \frac{z}{x}.$$

Another useful substitution is

$$x = a + \frac{1}{a}, \qquad y = b + \frac{1}{b}, \qquad z = c + \frac{1}{c}.$$

Such substitutions are not just algebraic decoration. They encode hidden positivity and symmetry. The note points out that many complicated-looking inequalities become simpler after this change of variables.

§37.5 A classical symmetric equation

The pages discuss equations such as

$$xyz = x + y + z + 2.$$

This can be rewritten as

$$xyz + 1 = (x+1)(y+1)(z+1) - xy - yz - zx - x - y - z.$$

A better-known substitution is

$$x = a + \frac{1}{a}, \quad y = b + \frac{1}{b}, \quad z = c + \frac{1}{c},$$

or trigonometric substitutions when the variables resemble

$$2\cos A, \quad 2\cos B, \quad 2\cos C.$$

The note is circling around the idea that symmetric algebraic equations often become geometry or trigonometry after the right substitution.

§37.6 Inequality examples

One example asks to prove

$$\frac{a}{b+c+a} + \frac{b}{c+a+b} + \frac{c}{a+b+c} \geq \frac{3}{2}$$

in a normalized form. The exact expression on the page is a contest inequality, and the solution uses standard tools such as Cauchy, AM–GM, or substitution.

The point is that conditions like $abc = 1$, $x+y+z =$ constant, or $x^2+y^2+z^2+2xyz = 1$ should not be treated as random constraints. They are invitations to choose a special parametrization.

§37.7 Jensen, trigonometry, and contest style

Later examples involve trigonometric inequalities in a triangle. A triangle condition $A + B + C = \pi$ naturally suggests identities such as

$$\cos^2 A + \cos^2 B + \cos^2 C + 2\cos A\cos B\cos C = 1.$$

This identity is one of the reasons that substitutions by cosines appear in olympiad inequalities.

§37.8 The hidden structure

This note looks like probability followed by inequalities, but both topics are about choosing the right sample space. In probability, the sample space is literal: events are subsets of Ω. In inequalities, the "sample space" is the domain allowed by the constraint. A good substitution changes that domain into something more natural, just as choosing a good probability model turns a counting problem into a set problem.

§37.9 Conditional probability as changing measure

Conditional probability is not just a formula. It changes the universe from Ω to the event B:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The denominator renormalizes the measure so that B has probability 1.

This is why Venn diagrams help: conditioning restricts attention to one region and rescales its area.

§37.10 Bayes formula and inverse problems

Bayes' formula

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_j \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}$$

turns likelihoods into posterior probabilities. It is a finite version of inference: update prior weights after observing evidence.

The formula is especially useful when B is easier to compute conditional on each case A_i than directly.

§37.11 Inequality substitutions as coordinate choices

The substitution tricks in the inequality part should be read as coordinate changes. Symmetric expressions often simplify under

$$u = x + y, \qquad v = xy,$$

or under trigonometric substitutions that encode constraints. The good substitution is the one that turns the constraint into a simple domain.

§37.12 Changing measures and changing coordinates

Probability and inequality substitutions share a habit: change the viewpoint before computing. Conditioning changes the measure; Bayes reverses conditional information; substitutions choose coordinates adapted to the constraint. This last sentence is worth making more precise, because it is the higher-level bridge hidden inside the elementary examples.

Definition 37.12.1 (Conditioning as a new probability measure)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $B \in \mathcal{F}$ satisfy $\mathbb{P}(B) > 0$. The conditional probability given B is the probability measure

$$\mathbb{P}_B(A) = \mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Equivalently,

$$\mathbb{P}_B(A) = \int_A \frac{\mathbf{1}_B}{\mathbb{P}(B)}\, d\mathbb{P}.$$

Thus $\mathbb{P}_B$ is absolutely continuous with respect to $\mathbb{P}$, with Radon–Nikodym derivative

$$\frac{d\mathbb{P}_B}{d\mathbb{P}} = \frac{\mathbf{1}_B}{\mathbb{P}(B)}.$$

This is the clean meaning of the phrase "conditioning changes the measure". The set Ω may stay formally the same, but the weights assigned to its parts have changed. Everything outside B receives weight zero, and the mass inside B is rescaled so that the total mass is again one.

Proposition 37.12.2 (Conditional expectation as integration in the new measure)
For every integrable random variable X,

$$\mathbb{E}[X \mid B] = \int_\Omega X\, d\mathbb{P}_B = \frac{1}{\mathbb{P}(B)} \int_B X\, d\mathbb{P}.$$

Proof. By the definition of $\mathbb{P}_B$ and its density with respect to $\mathbb{P}$,

$$\int_\Omega X\, d\mathbb{P}_B = \int_\Omega X \frac{\mathbf{1}_B}{\mathbb{P}(B)}\, d\mathbb{P} = \frac{1}{\mathbb{P}(B)} \int_B X\, d\mathbb{P}.$$

This is precisely the usual formula for conditional expectation on the event B. $\square$

Bayes' formula is also clearer from this viewpoint. Suppose $A_1, \dots, A_n$ form a partition of Ω. After observing B, the old weights $\mathbb{P}(A_i)$ are replaced by

$$\mathbb{P}_B(A_i) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_j \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}.$$

The numerator is the old weight multiplied by the likelihood of the observation. The denominator is just the normalization which makes the new weights add to one. So Bayes' formula is not a trick for reversing arrows; it is a reweighting rule.

The same pattern appears in substitutions for inequalities. A substitution changes how we integrate, compare, or optimize over a domain. If $T : U \to V$ is a change of variables, then a function on V can be pulled back to U by

$$f \longmapsto f \circ T.$$

For integrals, the measure also changes through the Jacobian:

$$\int_V f(y)\,dy = \int_U f(T(x))\,|\det DT(x)|\,dx.$$

For inequalities, even when no explicit integral is present, the same philosophy remains: choose coordinates in which the constraint and the symmetry become visible.

Remark 37.12.3 (The shared template) — Conditioning replaces $\mathbb{P}$ by the weighted measure $\mathbb{P}_B$. Bayes' formula replaces prior weights by posterior weights. A substitution replaces the original coordinates by coordinates adapted to the constraint. In all three cases, the computation becomes easier because the background viewpoint has changed before the algebra begins.

Set Exercises, Functions, and Elementary Problem-Solving

N-20220801

§38.1 Subsets and characteristic functions

The note begins with the idea of counting subsets. If a set A has n elements, then it has

$$2^n$$

subsets. One way to see this is to use the characteristic function

$$\chi_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

Choosing a subset $S \subseteq A$ is the same as choosing a function

$$\chi_S : A \to \{0, 1\}.$$

There are 2 choices for each element of A, hence 2^n choices in total.

This is a useful conceptual improvement over simply memorizing 2^n. A subset is a yes/no decision for every element.

§38.2 Set laws

The page records De Morgan's laws:

$$C_U(A \cap B) = C_U A \cup C_U B,$$

$$C_U(A \cup B) = C_U A \cap C_U B.$$

It also recalls inclusion–exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The exercises ask for set descriptions such as:

$$\{2n : 1 \leq n \leq 5\},$$

$$\{x : x \leq 5\},$$

and graphs such as

$$\{(x, y) : y = x^2 - x - 1\}.$$

This is a good reminder that sets can be finite lists, solution sets of inequalities, or geometric loci.

§38.3 Venn diagram exercises

One exercise gives

$$U = A \cup B = \{1, 2, 3, 4\}, \qquad A \cap C_U B = \{1, 2\},$$

and asks for B. Since $A \setminus B = \{1, 2\}$, the remaining elements $3, 4$ must lie in B, so

$$B = \{3, 4\}.$$

Another exercise has

$$A = \{x : 1 < x < 7\}, \qquad B = \{x : x^2 - 4x - 5 \le 0\}.$$

Since

$$x^2 - 4x - 5 = (x - 5)(x + 1),$$

we get

$$B = [-1, 5].$$

Therefore

$$A \cap C_{\mathbb{R}} B = (5, 7).$$

The note's drawings are not decorative. They are there to force the habit of separating "inside A, inside B, inside both, inside neither."

§38.4 Inclusion–exclusion with literature examples

One problem describes a survey of 100 students reading classical novels. If 80 read either X or Y, 60 read X, and 40 read both, then the number reading Y is found by

$$|X \cup Y| = |X| + |Y| - |X \cap Y|.$$

Thus

$$80 = 60 + |Y| - 40,$$

so

$$|Y| = 60.$$

Again, the important point is not the story, but the relation among union, intersection, and individual counts.

§38.5 Set constructions with squares

The note includes exercises with sets such as

$$A = \{a : a = x^2 - y^2,\ x, y \in \mathbb{Z}\},$$

and

$$B = \{a : a = x^2 - y^2,\ x, y \in \mathbb{Q}\}.$$

Since

$$x^2 - y^2 = (x + y)(x - y),$$

every difference of squares is a product of two numbers of the same parity in the integer case. This explains why expressions such as

$$2k - 1$$

are easy to represent, while some even numbers require more care.

For rationals, the factorization is more flexible, and quotients of such expressions can often remain in B. The page is training the habit of factoring before trying random examples.

§38.6 Floor functions

A floor-function example studies values of

$$y = \lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 3x \rfloor, \qquad 1 \le y \le 100.$$

The natural move is to write

$$x = k + t, \qquad k \in \mathbb{Z}, \quad 0 \le t < 1.$$

Then the three floor terms change only when t crosses multiples of $1/2$ or $1/3$. Hence the possible values in one period come from the intervals determined by

$$0, \quad \frac{1}{3}, \quad \frac{1}{2}, \quad \frac{2}{3}, \quad 1.$$

This is much cleaner than trying to graph the whole function at once.

§38.7 Functions and monotonicity

The later pages switch to functions. Several problems ask for domains, ranges, monotonicity, and maximum or minimum values. Typical tools include:

(i) rewriting a square root or logarithm condition as a domain restriction;

(ii) using monotonicity to locate maxima and minima;

(iii) drawing a rough graph to understand intervals;

(iv) using substitution when a composite expression repeats.

One example studies a piecewise or iterated function and asks how many cycles appear. The drawing in the note shows the value bouncing among intervals, which is an early form of dynamical thinking.

§38.8 Characteristic functions and subsets

A subset $S \subseteq A$ is the same data as a function

$$\chi_S : A \to \{0, 1\}.$$

Each element chooses either 0 or 1, so a set with n elements has 2^n subsets. This is more than a counting trick: it turns subsets into functions, which is the beginning of a very useful viewpoint.

§38.9 Functions as rules with domains

A function is not merely a formula. It consists of a domain, codomain, and assignment rule. The same formula can define different functions if the domain or codomain changes. This is why image, preimage, injectivity, and surjectivity must always be stated relative to a specified domain and codomain.

§38.10 The method behind the trick

This note is a collection of elementary set and function exercises, but the hidden agenda is representational flexibility. A set can be represented by a list, a predicate, a Venn diagram, or a characteristic function. A function can be represented by a formula, a graph, a domain/range description, or an iteration rule. Much of mathematics consists of switching to the representation in which the problem becomes visible.

§38.11 Set operations as Boolean algebra

For a fixed universe U, the power set $\mathcal{P}(U)$ is a Boolean algebra:

$$A \vee B = A \cup B, \qquad A \wedge B = A \cap B, \qquad \neg A = U \setminus A.$$

De Morgan's laws become algebraic identities:

$$\neg(A \vee B) = \neg A \wedge \neg B, \qquad \neg(A \wedge B) = \neg A \vee \neg B.$$

Thus Venn-diagram manipulation is the visual form of Boolean algebra.

§38.12 Characteristic functions and algebra of predicates

A subset $A \subseteq U$ is equivalently a predicate or a characteristic function

$$\mathbf{1}_A : U \to \{0, 1\}.$$

Set operations become operations on 0-1 functions:

$$\mathbf{1}_{A\cap B} = \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A^c} = 1 - \mathbf{1}_A,$$

and

$$\mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

For finite U, cardinality is summation:

$$|A| = \sum_{x\in U} \mathbf{1}_A(x).$$

This converts set identities into ordinary algebraic identities.

§38.13 Preimages as Boolean homomorphisms

For a function $f : X \to Y$, the preimage operation

$$f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X)$$

preserves Boolean operations:

$$f^{-1}(B \cup C) = f^{-1}(B) \cup f^{-1}(C),$$

$$f^{-1}(B \cap C) = f^{-1}(B) \cap f^{-1}(C), \qquad f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

Images do not preserve intersections or complements in this exact way. This asymmetry explains why topology defines continuity using preimages of open sets.

§38.14 Sigma-algebras and measurable events

Probability does not always use the full power set. A sigma-algebra $\mathcal{A}$ is a family of subsets closed under complement and countable union. Its elements are measurable events. A probability measure assigns numbers to these events so that disjoint unions add:

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

Thus the elementary set operations $\cup, \cap, ^c$ become the grammar of probability and measure theory.

§38.15 Stone duality intuition

Stone duality says that Boolean algebras can be represented as algebras of clopen subsets of topological spaces. Given a Boolean algebra B, its Stone space consists of ultrafilters on B. Each element $b \in B$ determines a clopen set

$$\widehat{b} = \{\mathfrak{u} : b \in \mathfrak{u}\}.$$

This shows that elementary set algebra, propositional logic, and certain zero-dimensional topological spaces are different faces of one structure.

§38.16 Difference of squares as algebraic structure

The exercise involving

$$x^2 - y^2 = (x + y)(x - y)$$

is a reminder that set-builder notation often hides algebraic structure. Over the integers, the two factors $x + y$ and $x - y$ have the same parity. Over the rationals, the parity obstruction disappears. The same formula therefore defines different sets depending on the ambient number system.

So even elementary set-builder problems force a structural question: what universe are the variables living in?

§38.17 Sets as algebra, logic, and measurement

The note's set and function exercises are unified by representation. A subset may be a list, predicate, characteristic function, measurable event, or Boolean-algebra element. A function is not just a formula but a map with domain and codomain, and its preimage operation is the structurally stable one. Most mistakes in these exercises come from forgetting which representation is being used.

§38.18 Boolean algebra beyond Venn diagrams

The original exercises with unions, intersections, complements, and Venn diagrams are the concrete beginning of Boolean algebra. A Boolean algebra is a structure with operations

$$a \vee b, \qquad a \wedge b, \qquad \neg a, \qquad 0, \qquad 1$$

satisfying the familiar laws of union, intersection, complement, empty set, and universal set. For a fixed universe U, the power set $\mathcal{P}(U)$ is the model example:

$$A \vee B = A \cup B, \qquad A \wedge B = A \cap B, \qquad \neg A = U \setminus A.$$

The elementary identities

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \qquad (A \cup B)^c = A^c \cap B^c$$

are therefore not isolated set tricks. They are algebraic laws in a Boolean algebra.

§38.19 Characteristic functions

Every subset $A \subseteq U$ has a characteristic function

$$\mathbf{1}_A : U \to \{0,1\}, \qquad \mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Set operations become arithmetic or logical operations:

$$\mathbf{1}_{A\cap B} = \mathbf{1}_A \mathbf{1}_B, \qquad \mathbf{1}_{A\cup B} = \max(\mathbf{1}_A, \mathbf{1}_B), \qquad \mathbf{1}_{A^c} = 1 - \mathbf{1}_A.$$

For finite sets,

$$|A| = \sum_{x\in U} \mathbf{1}_A(x).$$

This converts Venn-diagram counting into algebra. Inclusion–exclusion is just the expansion of a product of factors $1 - \mathbf{1}_{A_i}$.

§38.20 Sigma-algebras as observable event systems

In probability and measure theory, one does not always allow every subset of a universe. A sigma-algebra $\mathcal{A}$ on U is a collection of subsets satisfying

$$U \in \mathcal{A}, \qquad A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}, \qquad A_1, A_2, \ldots \in \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

This is a Boolean algebra closed under countable unions. The elements of $\mathcal{A}$ are the measurable sets or events.

The elementary set laws still hold, but the important new question is: which subsets are allowed to be measured? Probability begins by assigning a number $P(A)$ to each $A \in \mathcal{A}$ such that

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n)$$

for disjoint events. Thus basic set operations become the grammar of probability.

§38.21 Stone duality as topology of logic

Stone duality says, roughly, that every Boolean algebra can be represented as an algebra of clopen subsets of a topological space. For a Boolean algebra B, its Stone space consists of ultrafilters on B. Each element $b \in B$ determines a set

$$\widehat{b} = \{\mathfrak{u} : \ b \in \mathfrak{u}\}.$$

These sets behave like the subsets of a space.

This tells us that elementary set algebra is not merely a notation for collections. It can encode topology. Logical operations, set operations, and certain topological spaces are three faces of the same structure.

§38.22 Set identities persist as Boolean and measurable algebra

The original exercises train manipulation of $\cup, \cap, ^c, \setminus$. At the advanced level, the same manipulations become Boolean algebra, characteristic-function algebra, measurable-event algebra, and even topological representation theory through Stone duality. Nothing in the elementary calculations is wasted: every identity is a structural law that survives into probability, logic, and topology.

§38.23 From Venn diagrams to algebra of information

A Venn diagram is a visual partition of a universe into atoms determined by a finite family of sets. For two sets, the atoms are

$$A \cap B, \quad A \cap B^c, \quad A^c \cap B, \quad A^c \cap B^c.$$

For n sets, there are at most 2^n atoms. Every Boolean expression in the sets is a union of some of these atoms.

This gives a systematic method for elementary set identities: reduce both sides to the same atoms. In logic, the same method is truth tables. In probability, the atoms are elementary events. In database theory, they are the records satisfying combinations of predicates.

Thus the elementary operation of shading regions is already the algebra of information partitions. A set is a yes/no question about elements; a finite family of sets is a finite collection of yes/no questions; the atoms are the complete answer patterns.

§38.24 Why sigma-algebras restrict what can be asked

In finite examples one usually allows every subset. In analysis, allowing every subset of $\mathbb{R}$ is incompatible with countably additive length if one accepts the usual choice principles. Sigma-algebras solve this by specifying which questions are measurable.

The elementary complement and union laws are retained, but countability enters:

$$A_1, A_2, \ldots \in \mathcal{A} \quad \Longrightarrow \quad \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

This is exactly what is needed to discuss limiting events such as "infinitely many of these events occur." The original finite set exercises therefore become the finite visible model of measurable-event logic.

Set Theory Exercises, Closures, Functions, and Algebraic Side Notes

N-20220806

§39.1 Images, preimages, and finite set examples

The first page contains set-theoretic exercises. For example, let

$$A = \{2, 0, 1, 3\}$$

and define

$$B = \{x : x \in A,\ 2 - x^2 \notin A\}.$$

Checking each element:

$$x = -2 \Rightarrow 2 - x^2 = -2 \notin A,$$
$$x = 0 \Rightarrow 2 - x^2 = 2 \in A,$$
$$x = -1 \Rightarrow 2 - x^2 = 1 \in A,$$
$$x = -3 \Rightarrow 2 - x^2 = -7 \notin A.$$

The page concludes with a finite set B after direct inspection. The point is that a set defined by a condition is not mysterious: test the condition element by element when the universe is small.

§39.2 Closure under addition

The next exercise asks for a subset $S \subseteq \mathbb{R}$ closed under addition:

$$x, y \in S \quad \Rightarrow \quad x + y \in S.$$

Examples include:

$$\mathbb{Z}, \qquad \mathbb{Q}, \qquad \mathbb{R}, \qquad [0, \infty).$$

A finite example such as $\mathbb{Z}/6\mathbb{Z}$ also works if addition is understood modulo 6. The note correctly distinguishes ordinary addition from addition modulo an equivalence relation.

A related problem says: if $S_1, S_2 \subseteq \mathbb{R}$ are closed under addition and are proper in a suitable sense, prove there is an element c outside their union. This resembles the Dedekind-cut intuition: two proper additive-closed pieces cannot cover the entire real line unless one of them already absorbs too much structure.

§39.3 A combinatorial maximal-set problem

Another problem considers

$$M \subseteq \{1, 2, \ldots, 2011\}$$

with the property that for any two elements $a, b \in M$, at least one of

$$a \mid b, \qquad b \mid a, \qquad |a - b| = \max M$$

holds. The note first explores powers of 2:

$$1, 2, 2^2, \ldots, 2^{10},$$

since divisibility is automatic along a chain. It then observes that if one tries to add a number not on the same divisibility chain, the condition quickly fails.

The key structural idea is that divisibility partially orders the integers. A set in which every pair is comparable under divisibility is a chain. Powers of a fixed prime give the cleanest example of a long chain. This is an early encounter with posets and extremal combinatorics.

§39.4 Functional and algebraic exercises

The later pages mix several algebraic problems. One recurring theme is solving equations by rewriting them into factored or symmetric forms. For instance, equations involving

$$x^2 + y^2, \qquad xy, \qquad x + y$$

are usually better handled through substitutions like

$$u = x + y, \qquad v = xy.$$

This is the elementary symmetric-polynomial viewpoint.

Another page discusses functions and constraints of the form

$$f(A \cap B), \quad f(A \cup B), \quad f^{-1}(C).$$

The essential warning is that images and preimages behave differently:

$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$$

always, while

$$f(A \cap B) = f(A) \cap f(B)$$

need not hold unless f is injective on the relevant sets.

§39.5 Matrix and coordinate side computations

Several pages contain coordinate computations and small matrix reductions. These are not central enough to deserve separate chapters, but they reinforce a common habit: reduce a system to normal form. Whether one is checking membership in a set, solving a function equation, or reducing a matrix, the practical goal is to isolate independent parameters.

§39.6 Images and preimages behave differently

For a function $f : X \to Y$, the image of a set $A \subseteq X$ is

$$f(A) = \{f(a) : a \in A\},$$

while the preimage of $B \subseteq Y$ is

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

Preimages preserve unions, intersections, and complements exactly. Images preserve unions but not intersections in general. This asymmetry is why topology defines continuity using preimages of open sets.

§39.7 Closure operations

Many set exercises secretly ask whether an operation is a closure operator. A closure operator should be extensive, monotone, and idempotent:

$$A \subseteq \overline{A}, \qquad A \subseteq B \Rightarrow \overline{A} \subseteq \overline{B}, \qquad \overline{\overline{A}} = \overline{A}.$$

Recognizing these three properties helps organize otherwise scattered set-manipulation problems.

§39.8 The reusable pattern

This note is scattered, but its center is closure. A set may be closed under addition, a family of elements may be closed under a divisibility relation, a function preimage is closed under Boolean operations, and a solution set may be closed under linear combinations. Recognizing closure is one of the first steps toward algebraic structure.

§39.9 Closure operators

Many exercises in this note ask whether a set is stable under an operation. The abstract pattern is a closure operator. On a partially ordered set P, a closure operator is a map

$$c : P \to P$$

satisfying

$$x \le c(x), \qquad x \le y \Rightarrow c(x) \le c(y), \qquad c(c(x)) = c(x).$$

These are extensivity, monotonicity, and idempotence.

Examples include topological closure $A \mapsto \overline{A}$, linear span $S \mapsto \operatorname{span}(S)$, subgroup generation $S \mapsto \langle S \rangle$, and logical consequence from assumptions.

§39.10 Generated objects as intersections

When a problem asks for the smallest object containing S and closed under some rule, the general construction is

$$\langle S \rangle = \bigcap \{C : S \subseteq C,\ C \text{ is closed under the rule}\}.$$

This works because intersections of closed objects are closed in many common settings: intersections of subgroups are subgroups, intersections of subspaces are subspaces, and arbitrary intersections of closed sets are closed.

Thus ”generated by” is not a vague phrase. It means minimality under inclusion plus closure under specified operations.

§39.11 Fixed points of closure

An object is closed when it is a fixed point of the closure operator:

$$c(x) = x.$$

The closed objects form a structured family. For example, topologically closed subsets are exactly fixed points of the topological closure operator; subspaces are fixed points of span; subgroups are fixed points of subgroup generation.

This viewpoint also explains iterative procedures. Start with a set, repeatedly add everything forced by the rule, and stop when no new elements appear. In finite settings this must stabilize; in infinite settings one may need transfinite or lattice-theoretic methods.

§39.12 Images, preimages, and logic

For $f : X \to Y$, preimage preserves all Boolean operations, while image only preserves unions in general. The failure

$$f(A \cap B) \subseteq f(A) \cap f(B)$$

can be strict: two different points may have the same image. Equality for all subsets is closely related to injectivity.

This is the set-theoretic reason that preimages are central in topology, measure theory, and logic. A predicate on Y pulls back to a predicate on X, and logical connectives are preserved exactly.

§39.13 Posets and extremal chains

The maximal-set problem based on divisibility is naturally a problem in a partially ordered set. Write

$$a \preceq b \iff a \mid b.$$

A family in which every pair is comparable is a chain. Powers of a fixed prime form a chain:

$$1, p, p^2, p^3, \ldots.$$

Extremal questions about such families lead toward Dilworth-type and Sperner-type theorems, where one bounds the size of chains or antichains in a poset.

§39.14 Galois connections

A Galois connection translates order conditions between two posets. One common form is

$$f(p) \leq q \iff p \leq g(q).$$

Composites of adjoint maps often produce closure operators. In linear algebra, taking annihilators reverses inclusion and double annihilators produce closure-like operations.

This explains why constraints and solution sets are dual. A set of equations determines a solution set; a solution set determines equations vanishing on it. Applying both directions gives a closure operation.

§39.15 Tarski fixed-point theorem

On a complete lattice, every monotone map has fixed points. Tarski's theorem says the set of fixed points is itself a complete lattice. This is the abstract version of repeatedly applying a monotone rule until it stabilizes.

The elementary finite-set exercises show the finite shadow of this theorem: if a process only adds elements inside a finite universe, it eventually reaches a fixed point.

§39.16 Closure, order, and fixed-point thinking

The scattered exercises are unified by closure and order. Sets closed under addition, divisibility chains, preimage stability, generated substructures, and repeated operations that stabilize are all examples of order-theoretic thinking. The advanced structure is not decoration; it explains why the same proof patterns appear across algebra, topology, logic, and combinatorics.

§39.17 Closure operators across mathematics

The closure-style exercises in the original note can be organized by the notion of a closure operator. On a partially ordered set P, a closure operator is a map

$$c : P \to P$$

satisfying

$$x \le c(x), \qquad x \le y \Rightarrow c(x) \le c(y), \qquad c(c(x)) = c(x).$$

These are called extensivity, monotonicity, and idempotence.

Many familiar constructions have this form:

$$\begin{aligned} &A \mapsto \overline{A} && \text{in topology,} \\ &S \mapsto \operatorname{span}(S) && \text{in linear algebra,} \\ &S \mapsto \langle S \rangle && \text{in group theory.} \end{aligned}$$

The elementary operation "generate the smallest object containing a given set" is exactly closure.

§39.18 Closed sets as fixed points

An element x is closed if

$$c(x) = x.$$

Thus closed sets are fixed points of the closure operator. The family of closed elements is closed under arbitrary meets. In set language, topologically closed sets are stable under arbitrary intersections; subspaces generated by vector sets are stable under intersections; subgroups generated by sets are stable under intersections.

This explains a standard construction:

$$c(S) = \bigcap \{C : \ S \subseteq C, \ C \text{ closed}\}.$$

The closure of S is the intersection of all closed objects containing it.

§39.19 Galois connections and closure

A Galois connection between posets P and Q is a pair of order-reversing or order-preserving maps that translate inequalities in one side into inequalities on the other. A common form is

$$f(p) \le q \quad \Longleftrightarrow \quad p \le g(q).$$

The composite $g \circ f$ is then a closure operator.

In linear algebra, for a subset S of vectors, one can define the annihilator

$$S^\perp = \{\varphi \in V^* : \ \varphi(s) = 0 \text{ for all } s \in S\}.$$

Taking annihilators reverses inclusion and creates closure-like double-perp operations.

Thus the elementary habit of taking complements, generated sets, or common solution sets points toward a deep pattern: objects can be studied through the constraints they impose, and closure is often a double-constraint operation.

§39.20 Tarski fixed points and monotone iteration

On a complete lattice, every monotone map has a fixed point. This is Tarski's fixed-point theorem. A complete lattice is a partially ordered set in which every subset has a supremum and an infimum.

For a monotone map $F : L \to L$, the set of fixed points

$$\{x \in L : \ F(x) = x\}$$

is itself a complete lattice. This theorem appears in logic, semantics of programming languages, and inductive definitions.

The elementary idea of repeatedly applying a closure process until nothing changes is therefore a finite visible version of a fixed-point theorem for ordered structures.

§39.21 Closure operators organize repeated stabilization

Whenever the original note asks for a smallest set satisfying some property, an invariant collection, or a repeated operation that stabilizes, the advanced structure is closure. The same pattern appears in topology, algebra, logic, and theoretical computer science. The elementary set manipulations are the computational entrance to closure operators and fixed-point mathematics.

§39.22 Generation as an intersection construction

Many elementary problems ask for the smallest set satisfying a property. The general construction is:

$$\langle S \rangle = \bigcap \{C : S \subseteq C, \ C \text{ has the desired closure property}\}.$$

This works because the intersection of closed objects is closed. For subspaces, intersections of subspaces are subspaces. For subgroups, intersections of subgroups are subgroups. For topological closed sets, arbitrary intersections are closed.

This explains why "generated by" is such a stable phrase across mathematics. The generated object is not guessed; it is forced by closure under operations and minimality under inclusion.

§39.23 Closure and logic of consequences

A closure operator can also be interpreted as logical consequence. If S is a set of assumptions, $c(S)$ is the set of all consequences of S. Extensivity says assumptions are consequences of themselves. Monotonicity says more assumptions give more consequences. Idempotence says taking consequences twice adds nothing new.

This logic viewpoint connects the original set exercises to algebraic logic and model theory. In algebra, closure under operations generates structures; in logic, closure under inference generates theories.

Set Counting, Modular Patterns, and Mathematical Induction

N-20220808

§40.1 A set-counting problem

The first page studies finite sets $A_1, A_2, \ldots, A_{32}$ and asks how many subsets can satisfy certain intersection conditions. The handwritten solution builds a toy example in which each set contains a fixed number of elements and one carefully controls pairwise intersections.

The general strategy is inclusion–exclusion. If every set has m elements and pairwise overlaps have controlled size, then the size of the union is not simply the sum of sizes. One must subtract overlaps and add back higher overlaps. The note's construction eventually counts a large number of possible sets, obtaining a count of the form

$$31 + 28 \cdot 30 = 871.$$

The exact model is less important than the idea: when many sets overlap, one should count by layers, not by naive addition.

§40.2 A modular arithmetic construction

Another problem constructs two sets:

$$A = \{1, 4, 6, 7, 9, \ldots, 1998\},$$

and a reversed or complementary set near 2000. The note observes that the pattern is governed by residues modulo 11. Since

$$2000 = 11 \cdot 181 + 9,$$

one can write elements as

$$11t + r.$$

The allowed residues are something like

$$r \in \{1, 4, 6, 7, 9\}.$$

The real insight is that forbidden differences correspond to forbidden residue differences. Instead of checking hundreds of numbers, one checks a small table modulo 11. This is one of the most common olympiad uses of modular arithmetic.

§40.3 Why not every construction is finished

The page explicitly says "problem reserved" in a few places. That is an honest part of mathematical notes. The important thing is to record what was tried and why it seemed plausible. Here the attempt is to remove residues that conflict pairwise and leave a maximal collection. Whether the construction is optimal requires a separate upper-bound argument.

This distinction is crucial: constructing an example proves a lower bound, but proving maximality requires an upper bound.

§40.4 Sequence exercises

Later pages return to sequence problems. Examples include arithmetic and geometric sequences, partial sums, and recurrence-like identities. One problem has

$$a_n = a_1 + (n-1)d$$

and conditions such as

$$a_1 + a_2 + \cdots + a_n = S_n.$$

The correct habit is to translate everything into a_1, d, n, solve, and then check the requested value.

Another problem uses a geometric progression and asks for a common ratio or a sum. The formula

$$S_n = a_1 \frac{q^n - 1}{q - 1}$$

again appears, with the warning that $q = 1$ is a separate case.

§40.5 Mathematical induction

The most substantial later pages discuss induction. The note distinguishes several forms:

(i) ordinary induction: prove $P(1)$, then prove $P(n) \Rightarrow P(n+1)$;

(ii) strong induction: prove $P(1), \ldots, P(n) \Rightarrow P(n+1)$;

(iii) induction with a shifted start: begin at $n = m$ instead of 1.

The proof skeleton is:

(i) base case;

(ii) induction hypothesis;

(iii) induction step;

(iv) conclusion.

The note emphasizes that the induction hypothesis must be used precisely. One cannot simply assume the desired statement for $n + 1$.

§40.6 Examples of induction

One exercise proves a formula for a sum, such as

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

The induction step is

$$1 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Another exercise proves a divisibility statement. For such problems, the usual technique is to write the $n+1$ case in terms of the n case plus a visibly divisible remainder.

The page also contains a problem of the form

$$4^{2k+1} + 3^{k+1}$$

or similar, where the induction step uses modular or factorization identities. The guiding rule is: when proving divisibility by induction, preserve a factor rather than expand everything.

§40.7 Induction structure

A proof by induction has two parts: the base case and the inductive step. The inductive step is not simply checking the next case; it is proving

$$P(n) \Rightarrow P(n+1)$$

for arbitrary n. This is why writing the induction hypothesis explicitly matters.

Strong induction allows the step to use all previous cases:

$$P(1), P(2), \ldots, P(n) \Rightarrow P(n+1).$$

It is useful when the object of size $n+1$ breaks into smaller pieces of varying sizes.

§40.8 Modular patterns

When a problem depends on remainders, the natural universe is $\mathbb{Z}/m\mathbb{Z}$. Instead of listing many cases, one should identify the period. If a sequence satisfies

$$a_{n+m} = a_n,$$

then all behavior is determined by the first m terms. This is the organizing idea behind many modular counting exercises.

§40.9 A note on scope

This note is about finite structure. Set problems use inclusion–exclusion and modular residues; sequence problems use closed forms; induction proves that a pattern persists for all n. The conceptual chain is:

$$\text{example} \quad \Rightarrow \quad \text{pattern} \quad \Rightarrow \quad \text{proof by induction or upper bound.}$$

A mature solution must know which stage it is in. Guessing a pattern is not proving it; constructing an example is not proving optimality; verifying a few cases is not induction.

§40.10 Construction versus upper bound

A combinatorial extremal problem has two halves. A construction proves a lower bound: it shows that at least N objects are possible. An upper bound proves that no construction can exceed N. Only when the two match is the extremal value determined.

The note's "problem reserved" comments are therefore mathematically honest. Producing a residue-class construction is not the same as proving optimality. The missing step is usually an invariant, counting argument, or compression method that forbids larger examples.

§40.11 Residue classes as finite quotient spaces

When a construction is governed by residues modulo m, the natural universe is

$$\mathbb{Z}/m\mathbb{Z}.$$

A set of integers with forbidden differences becomes a subset of this finite cyclic group with forbidden difference set D. Equivalently, it is an independent set in the Cayley graph whose edges connect residues differing by elements of D.

This transforms a long integer-checking problem into a finite graph problem on residues. The allowed residues are not a random pattern; they form a configuration avoiding forbidden differences in a quotient group.

§40.12 Induction as well-founded recursion

Ordinary induction works because $\mathbb{N}$ is well-ordered. To prove $P(n)$ for all n, it is enough to prove a base case and a step $P(n) \Rightarrow P(n+1)$. Strong induction uses the stronger hypothesis that all smaller cases are known.

The common generalization is well-founded induction. If every nonempty set of objects has a minimal element under a relation $<$, then one may prove $P(x)$ by assuming $P(y)$ for all $y < x$. This is why induction works on integers, trees, expressions, and recursively defined sets.

§40.13 Finite differences in divisibility induction

Many divisibility inductions become cleaner with finite differences. If

$$f(n+1) - f(n)$$

is divisible by m for all n, and $f(n_0)$ is divisible by m, then every later $f(n)$ is divisible by m. This is the discrete fundamental theorem of calculus:

$$f(n) = f(n_0) + \sum_{k=n_0}^{n-1} (f(k+1) - f(k)).$$

So induction is often just summing a difference identity.

§40.14 Floor functions and lattice points

A floor expression records how many integer thresholds have been crossed. Sums such as

$$\sum_{k=0}^{m-1} \left\lfloor \frac{ak+b}{m} \right\rfloor$$

count lattice points under a rational-slope line. This connects floor-function exercises to lattice-point enumeration.

In higher dimensions, Ehrhart theory studies

$$L_P(n) = |nP \cap \mathbb{Z}^d|,$$

the number of lattice points in dilates of a polytope. For integral polytopes, $L_P(n)$ is a polynomial in n. Thus one-variable floor sums are small shadows of a general theory.

§40.15 Pigeonhole and averaging

Many finite counting arguments are averaging arguments. If N objects are distributed among r boxes, then some box has at least

$$\left\lceil \frac{N}{r} \right\rceil$$

objects. This is the quantitative pigeonhole principle.

In extremal problems, one often defines boxes by residues, intervals, or local configurations, then proves that too many objects force a forbidden collision inside one box.

§40.16 Finite structure from quotients and induction

The note's finite examples are unified by quotienting and persistence. Modular constructions live in finite quotient groups, induction proves that a pattern persists, finite differences turn induction into summation, floor functions count lattice points, and extremal constructions require matching lower and upper bounds.

Basic Inequalities, Cauchy, Triangle Inequality, and Coordinate Geometry

N-20220810

§41.1 Basic inequalities

The first page is a compact inequality note. It begins with the elementary observation that squares are nonnegative. For real numbers,

$$(a-b)^2 \geq 0,$$

so

$$a^2 + b^2 \geq 2ab,$$

with equality iff $a = b$. For three variables,

$$a^2 + b^2 + c^2 \geq ab + bc + ca,$$

because

$$2(a^2 + b^2 + c^2 - ab - bc - ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0.$$

The handwritten page marks equality conditions carefully: equality occurs when all variables involved in the squared differences coincide.

§41.2 AM–GM

For $a, b \geq 0$,

$$\frac{a+b}{2} \geq \sqrt{ab},$$

with equality iff $a = b$. Equivalently,

$$a + b \geq 2\sqrt{ab}.$$

The note also records the three-variable form

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}, \qquad a, b, c \geq 0,$$

and uses it repeatedly in examples.

A typical application is to expressions such as

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc} \quad \Longrightarrow \quad a+b+c \geq 3\sqrt[3]{abc}.$$

The useful habit is to look for terms whose product is fixed. AM–GM then turns a fixed product into a lower bound for a sum.

§41.3 Cauchy's inequality

The note states Cauchy's inequality in two forms. The vector form is

$$(a_1b_1 + \cdots + a_nb_n)^2 \le (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2).$$

The Engel or Titu Andreescu form is

$$\frac{x_1^2}{a_1} + \cdots + \frac{x_n^2}{a_n} \ge \frac{(x_1 + \cdots + x_n)^2}{a_1 + \cdots + a_n}, \qquad a_i > 0.$$

This second form is often what the handwritten examples use. It is obtained by applying Cauchy to

$$\left(\sum_i \frac{x_i^2}{a_i}\right)\left(\sum_i a_i\right) \ge \left(\sum_i x_i\right)^2.$$

§41.4 Triangle inequality and its variants

The page then discusses absolute values:

$$|a+b| \le |a| + |b|, \qquad |a-b| \le |a-c| + |c-b|.$$

The second formula is the triangle inequality with an intermediate point c. The note also uses the reverse triangle inequality

$$||a| - |b|| \le |a-b|.$$

A useful proof is

$$|a| = |a-b+b| \le |a-b| + |b|, \qquad |b| = |b-a+a| \le |a-b| + |a|.$$

Together these imply the reverse inequality.

The handwritten warning is that absolute value inequalities are not merely algebraic signs. They encode distances on the real line.

§41.5 Worked AM–GM examples

One example asks for a lower bound of expressions of the form

$$x + \frac{1}{x}, \qquad x > 0.$$

By AM–GM,

$$x + \frac{1}{x} \ge 2.$$

Similarly,

$$x + y + z \ge 3\sqrt[3]{xyz}$$

when $x, y, z > 0$. If a problem fixes xyz, this gives a direct lower bound for the sum.

Another common pattern is

$$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} \ge 3, \qquad x, y, z > 0,$$

because the product of the three terms is 1.

§41.6 A Cauchy example

For positive a, b, c, a typical inequality is

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

This is exactly Engel form of Cauchy. The handwritten page uses this kind of manipulation to turn a sum of fractions into one clean square divided by one clean denominator.

§41.7 Coordinate geometry at the end of the note

The last page shifts to coordinate geometry. The diagrams show points, vectors, and lines, with formulas for distances and slopes. The essential formulas are:

$$AB = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2},$$

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

and the point-slope form of a line

$$y - y_0 = m(x - x_0).$$

A recurring idea is that geometric claims can be turned into algebraic equalities. For example, perpendicularity of two nonvertical lines is

$$m_1 m_2 = -1,$$

while parallelism is

$$m_1 = m_2.$$

Distance and slope are the coordinate versions of length and direction.

§41.8 What the examples reveal: three structures

This note is about choosing the right inequality language. Squares prove elementary inequalities, AM–GM handles fixed products, Cauchy handles dot products and sums of fractions, and triangle inequalities handle distances. The advanced version of the same habit is to identify the structure behind the inequality before calculating.

Elementary inequality	Higher structure	Equality meaning
Cauchy–Schwarz	inner product / projection / Gram matrix	vectors are dependent
Triangle inequality	normed vector space	same direction in strictly convex cases
AM–GM	convexity or concavity of log	all variables equal
Positive square	positive semidefinite cone	quadratic form degenerates

§41.9 Cauchy–Schwarz and Gram determinants

Cauchy–Schwarz says

$$|\langle x, y\rangle| \le \|x\|\|y\|.$$

It follows from the nonnegativity of

$$\|x - ty\|^2 \ge 0$$

or, equivalently, from positivity of the Gram matrix

$$G(x, y) = \begin{pmatrix} \langle x, x\rangle & \langle x, y\rangle \\ \langle y, x\rangle & \langle y, y\rangle \end{pmatrix}.$$

The inequality $\det G(x, y) \ge 0$ is exactly Cauchy–Schwarz. Equality means G has rank one, i.e. the two vectors are linearly dependent.

In Hilbert spaces, the same inequality controls Fourier coefficients:

$$|\langle f, e_n\rangle| \le \|f\|_2\|e_n\|_2.$$

This is why an inequality first met in finite sums becomes indispensable in functional analysis.

§41.10 Triangle inequality and normed spaces

The triangle inequality

$$\|x + y\| \le \|x\| + \|y\|$$

defines the geometry of a normed vector space. The usual absolute value, Euclidean length, sup norm, and L^p norms are all instances. Minkowski's inequality is precisely the triangle inequality in L^p:

$$\|f + g\|_p \le \|f\|_p + \|g\|_p, \qquad \|f\|_p = \left(\int |f|^p\right)^{1/p}.$$

Thus a contest-level inequality about square roots and sums is structurally the same statement as the triangle inequality for functions.

§41.11 AM–GM, convexity, and cones

For positive variables, AM–GM follows from concavity of log:

$$\log\left(\frac{1}{n}\sum_i x_i\right) \ge \frac{1}{n}\sum_i \log x_i.$$

The equality condition says all variables are equal, meaning there is no spread.

Many inequalities also describe membership in a convex cone. For instance,

$$A \succeq 0 \quad\iff\quad x^T Ax \ge 0 \text{ for all } x$$

defines the cone of positive semidefinite matrices. Basic "square is nonnegative" arguments are the one-dimensional shadow of this cone viewpoint.

Inequality Exercises: Cauchy, AM–GM, Vectors, and Optimization

N-20220811

§42.1 How to read this note

This note is a compact inequality practice sheet. The solutions are handwritten, but the recurring pattern is clear: almost every problem is asking us to recognize which standard inequality is hiding behind the expression. The useful question is not "which formula should I memorize?", but rather "what structure does the expression already have?"

If there are fractions with variable denominators, Cauchy or Engel form is natural. If there are products such as abc, AM–GM is natural. If there are square roots of quadratic expressions, think of vector lengths and the triangle inequality. If the problem asks for a maximum or minimum under a quadratic constraint, expect equality cases in Cauchy–Schwarz.

§42.2 A fraction problem with a constraint

One problem asks to handle expressions of the form

$$\frac{4}{4-x^2}+\frac{9}{9-y^2}$$

under constraints such as $xy=-1$ and bounded intervals for x, y. The handwritten solution uses a Cauchy-style lower bound:

$$\frac{4}{4-x^2}+\frac{9}{9-y^2}\geq\frac{(2+3)^2}{(4-x^2)+(9-y^2)}.$$

Then the constraint controls the denominator. The important point is that the numerator $4, 9$ are squares, so the expression is inviting the Cauchy inequality

$$\frac{a^2}{u}+\frac{b^2}{v}\geq\frac{(a+b)^2}{u+v}.$$

This is one of the most common contest moves: when a sum of fractions has square numerators, try to combine them by Cauchy rather than simplifying separately.

§42.3 A weighted square-root comparison

Another exercise defines

$$P=\sqrt{ab}+\sqrt{cd},$$

and

$$Q=\sqrt{ma+nc}\sqrt{\frac{b}{m}+\frac{d}{n}},$$

where the variables are positive. Expanding Q^2 gives

$$\begin{aligned} Q^2 &= (ma+nc)\left(\frac{b}{m}+\frac{d}{n}\right) \\ &= ab+cd+\frac{mad}{n}+\frac{ncb}{m}. \end{aligned}$$

By AM–GM,

$$\frac{mad}{n}+\frac{ncb}{m} \geq 2\sqrt{abcd}.$$

Hence

$$Q^2 \geq ab+cd+2\sqrt{abcd} = (\sqrt{ab}+\sqrt{cd})^2 = P^2.$$

So $P \leq Q$, with equality exactly when

$$\frac{mad}{n} = \frac{ncb}{m}.$$

The original note asks for the condition for strict inequality; it is precisely the negation of this equality condition.

§42.4 Maximizing a radical expression

The note then asks for the maximum of

$$f(x) = 2\sqrt{x-3}+\sqrt{5-x}, \qquad 3 \leq x \leq 5.$$

One way is calculus. We have

$$f'(x) = \frac{1}{\sqrt{x-3}} - \frac{1}{2\sqrt{5-x}}.$$

Setting $f'(x)=0$ gives

$$2\sqrt{5-x} = \sqrt{x-3},$$

so

$$4(5-x) = x-3, \qquad x = \frac{23}{5}.$$

At this point,

$$f\left(\frac{23}{5}\right) = 2\sqrt{\frac{8}{5}}+\sqrt{\frac{2}{5}} = \sqrt{10}.$$

But the handwritten note also asks whether this can be done without derivatives. The answer is yes, by Cauchy–Schwarz:

$$\left(2\sqrt{x-3}+\sqrt{5-x}\right)^2 \leq (4+1)((x-3)+(5-x)) = 10.$$

Equality holds when

$$\frac{2}{1} = \frac{\sqrt{x-3}}{\sqrt{5-x}},$$

which again gives $x = 23/5$. This non-calculus proof is more elegant because it explains the form of the answer.

§42.5 Equality in Cauchy–Schwarz

A vector problem gives

$$a^2 + b^2 + c^2 = 25, \qquad x^2 + y^2 + z^2 = 36, \qquad ax + by + cz = 30.$$

By Cauchy–Schwarz,

$$(ax + by + cz)^2 \le (a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = 25 \cdot 36 = 900.$$

Since $ax + by + cz = 30$, equality holds. Therefore

$$(a, b, c) = k(x, y, z).$$

Comparing norms gives

$$k = \frac{5}{6}.$$

Thus

$$\frac{a + b + c}{x + y + z} = \frac{5}{6}.$$

This is an excellent example of why equality cases matter. The inequality is not just bounding a number; it tells us the exact geometry of the vectors.

§42.6 Triangle-angle inequalities

Let A, B, C be the angles of a triangle. Then

$$A + B + C = \pi.$$

By Cauchy–Schwarz,

$$(A + B + C)\left(\frac{1}{A} + \frac{1}{B} + \frac{1}{C}\right) \ge (1 + 1 + 1)^2 = 9.$$

Hence

$$\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \ge \frac{9}{\pi}.$$

Similarly,

$$A^2 + B^2 + C^2 \ge \frac{(A + B + C)^2}{3} = \frac{\pi^2}{3}.$$

Both equalities occur only for

$$A = B = C = \frac{\pi}{3}.$$

The note's implicit lesson is that symmetric triangle inequalities often have equality at the equilateral triangle.

§42.7 Two spheres and a linear equation

Another problem asks to solve a system of two sphere equations. The trick is to subtract them. If two equations are

$$(x - a_1)^2 + (y - b_1)^2 + (z - c_1)^2 = R_1^2,$$

$$(x - a_2)^2 + (y - b_2)^2 + (z - c_2)^2 = R_2^2,$$

subtracting cancels $x^2 + y^2 + z^2$ and leaves a plane. The intersection of two spheres is therefore a circle, a point, or empty; algebraically it becomes a sphere-plane problem.

The handwritten solution then uses Cauchy–Schwarz to bound a linear expression such as

$$-8x + 6y - 24z.$$

The equality condition determines the direction of the vector (x, y, z). Again the method is geometric: a linear function is maximized on a sphere in the direction of its coefficient vector.

§42.8 A product inequality

If

$$(1 + a)(1 + b)(1 + c) = 8, \qquad a, b, c > 0,$$

then the note proves

$$abc \leq 1.$$

Indeed,

$$\begin{aligned} 8 &= 1 + a + b + c + ab + bc + ca + abc \\ &\geq 1 + 3\sqrt[3]{abc} + 3\sqrt[3]{(ab)(bc)(ca)} + abc \\ &= 1 + 3t + 3t^2 + t^3 = (1 + t)^3, \end{aligned}$$

where $t = \sqrt[3]{abc}$. Thus $1 + t \leq 2$, so $t \leq 1$, and therefore $abc \leq 1$.

§42.9 Vector-style radical inequality

The pages also contain an inequality of the form

$$\sqrt{x^2 + xy + y^2} + \sqrt{x^2 + xz + z^2} \geq \sqrt{y^2 + yz + z^2}.$$

This is best understood as a disguised triangle inequality. One rewrites each quadratic as a squared Euclidean norm of a two-dimensional vector. For example,

$$x^2 + xy + y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2.$$

Once each radical is interpreted as a vector length, the inequality becomes

$$\|u\| + \|v\| \geq \|u + v\|.$$

This is a typical high-level move: turn algebraic square roots into geometry.

§42.10 The deeper habit

The note is not merely a list of inequality tricks. It repeatedly uses the same three ideas: Cauchy detects equality of vectors, AM–GM detects equality of positive quantities, and radical inequalities often hide triangle inequalities. The more mature way to remember the sheet is therefore:

$$\text{fractions} \to \text{Cauchy}, \qquad \text{products} \to \text{AM–GM}, \qquad \text{radicals} \to \text{norms}.$$

§42.11 Equality cases as structural information

In inequality problems, equality conditions are not afterthoughts. They often reveal the correct substitution or normalization. For Cauchy–Schwarz,

$$\left(\sum_i a_i b_i\right)^2 = \left(\sum_i a_i^2\right)\left(\sum_i b_i^2\right)$$

holds exactly when a and b are proportional. If a proposed solution cannot recover the equality case, it is probably not using the natural structure.

§42.12 Optimization on constrained domains

Many radical or fraction inequalities are optimization problems on a constrained set. A constraint such as $x+y+z=1$ suggests compactness if variables are nonnegative. Then a maximum exists, and one can use convexity, Cauchy, or Lagrange multipliers to locate it.

This separates two tasks: first prove existence or reduce the domain; then identify the inequality that gives the optimum.

§42.13 Vectors and geometric inequalities

Vector methods convert algebraic inequalities into length comparisons. If an expression can be written as a dot product, Cauchy applies. If it is a sum of lengths, triangle inequality or Minkowski applies.

The power of vectors is that equality becomes a geometric condition: parallel vectors, collinearity, or degenerate triangles.

§42.14 Equality cases as geometric diagnostics

The exercise sheet is about choosing the right geometry. Cauchy handles dot products, AM–GM handles balanced positive products, convexity handles averages, and Lagrange multipliers handle smooth constrained extrema. Equality cases are the map showing which method is natural.

§42.15 Equality cases as geometry

The original inequality exercises repeatedly ask when equality holds. This is not a side detail; equality describes the geometry of the inequality. In Cauchy–Schwarz,

$$\left(\sum a_i b_i\right)^2 \le \left(\sum a_i^2\right)\left(\sum b_i^2\right),$$

equality holds exactly when the vectors (a_i) and (b_i) are linearly dependent. In AM–GM,

$$\frac{x_1+\cdots+x_n}{n} \ge \sqrt[n]{x_1\cdots x_n},$$

equality holds when all variables are equal. These equality sets often reveal the hidden symmetry of a problem.

§42.16 Lagrange multipliers and inequalities

Many inequality exercises are optimization problems with constraints. If one wants to extremize $f(x_1, \ldots, x_n)$ under $g(x_1, \ldots, x_n) = 0$, the smooth interior candidates satisfy

$$\nabla f = \lambda \nabla g.$$

For example, optimizing a quadratic form under $\|x\| = 1$ leads to

$$Ax = \lambda x,$$

so eigenvalues are extrema of quadratic forms on the unit sphere. This is the Rayleigh quotient principle:

$$\lambda_{\min} \le \frac{x^T A x}{x^T x} \le \lambda_{\max}$$

for symmetric A.

§42.17 Semidefinite viewpoint

Some inequalities are equivalent to positive semidefiniteness. To prove

$$ax^2 + 2bxy + cy^2 \ge 0$$

for all x, y, one checks

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \succeq 0.$$

That means

$$a \ge 0, \qquad ac - b^2 \ge 0.$$

This connects elementary quadratic inequalities to semidefinite programming, where one optimizes linear functions over the cone of positive semidefinite matrices.

§42.18 Classical inequalities encode convex optimization

The contest techniques in the source—Cauchy, AM–GM, substitutions, and equality checking—should be read as finite-dimensional optimization. The advanced language does not replace the computations; it explains why the same few patterns recur: projection, convexity, eigenvalues, and positive semidefinite forms.

Classical Inequalities from Convexity, Duality, and Norm Geometry

N-20221012

§43.1 A convex graph lies above each supporting tangent

A differentiable function f on an interval is convex when its slopes increase. Equivalently, for every point x_0, the graph lies above the tangent line:

$$\boxed{f(x) \geq f(x_0) + f'(x_0)(x - x_0).}$$

This one-dimensional picture already contains Bernoulli's inequality. Let

$$f(x) = (1 + x)^r, \qquad x > -1, \qquad r \geq 1.$$

Since

$$f''(x) = r(r-1)(1+x)^{r-2} \geq 0,$$

the function is convex. Its tangent at $x = 0$ is

$$f(0) + f'(0)x = 1 + rx.$$

Therefore

$$\boxed{(1 + x)^r \geq 1 + rx.}$$

For an integer $r = n$, this is the classical Bernoulli inequality. If $r > 1$, equality holds only at $x = 0$.

The proof matters more than the formula. Bernoulli is not an isolated binomial estimate; it is the first example of a general principle: convexity converts local first-order information into a global lower bound.

§43.2 Jensen averages the supporting-line inequality

Let f be differentiable and convex, and let

$$\lambda_i \geq 0, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

Set

$$m = \sum_{i=1}^{n} \lambda_i x_i.$$

The supporting line at m gives

$$f(x_i) \geq f(m) + f'(m)(x_i - m).$$

Multiply by λ_i and sum. The linear terms cancel because

$$\sum_i \lambda_i (x_i - m) = 0.$$

Hence

$$\boxed{f\left(\sum_i \lambda_i x_i\right) \le \sum_i \lambda_i f(x_i).}$$

This is Jensen's inequality.

If f is strictly convex and all weights are positive, equality requires

$$x_1 = \cdots = x_n.$$

For a concave function the inequality reverses.

Take the concave function

$$f(x) = \log x$$

on $(0, \infty)$, with equal weights $1/n$. Jensen gives

$$\frac{1}{n}\sum_{i=1}^{n} \log a_i \le \log\left(\frac{a_1 + \cdots + a_n}{n}\right).$$

Exponentiating yields AM–GM:

$$\boxed{\sqrt[n]{a_1 \cdots a_n} \le \frac{a_1 + \cdots + a_n}{n}.}$$

Equality holds exactly when all a_i are equal.

The same tangent-line mechanism therefore proves both an additive estimate such as Bernoulli and a multiplicative estimate such as AM–GM; the difference lies in the chosen convex or concave function.

§43.3 Smoothing leads from Jensen to majorization

Jensen compares a list with its weighted average. Majorization compares two lists with the same total sum and asks which one is more spread out.

For decreasing rearrangements

$$x_1^{\downarrow} \ge \cdots \ge x_n^{\downarrow}, \qquad y_1^{\downarrow} \ge \cdots \ge y_n^{\downarrow},$$

we say that x majorizes y, written $x \succ y$, when

$$\sum_{i=1}^{k} x_i^{\downarrow} \ge \sum_{i=1}^{k} y_i^{\downarrow} \qquad (1 \le k < n),$$

and the total sums are equal.

The elementary move behind this order is smoothing. Keep the mean m fixed and compare

$$m + t, \qquad m - t.$$

For differentiable convex f, define

$$g(t) = f(m + t) + f(m - t), \qquad t \ge 0.$$

Then

$$g'(t) = f'(m + t) - f'(m - t) \ge 0$$

because f' is increasing. Moving two entries closer together decreases the sum of their convex images.

A finite sequence of such balancing moves transforms a majorizing vector into a more even one. This gives the content of Karamata's inequality:

$$x \succ y \quad \Longrightarrow \quad \sum_i f(x_i) \geq \sum_i f(y_i)$$

for convex f.

For example,

$$(4, 1, 1) \succ (2, 2, 2).$$

With $f(t) = t^2$,

$$4^2 + 1^2 + 1^2 = 18 \geq 2^2 + 2^2 + 2^2 = 12.$$

With $f(t) = e^t$, the same ordering gives

$$e^4 + 2e \geq 3e^2.$$

Majorization formalizes the intuition that convex functions penalize unevenness.

§43.4 Young's inequality is a tangent bound in dual exponents

Let

$$p, q > 1, \qquad \frac{1}{p} + \frac{1}{q} = 1.$$

Young's inequality states that for $a, b \geq 0$,

$$\boxed{ab \leq \frac{a^p}{p} + \frac{b^q}{q}.}$$

A direct derivation shows where the conjugate exponents come from. Fix a and consider

$$\Phi(b) = ab - \frac{b^q}{q}.$$

Its derivative is

$$\Phi'(b) = a - b^{q-1}.$$

The maximum occurs at

$$b = a^{1/(q-1)} = a^{p-1},$$

because

$$q - 1 = \frac{1}{p-1}.$$

At this point,

$$\Phi(b) = a^p - \frac{a^p}{q} = \frac{a^p}{p}.$$

Thus

$$ab - \frac{b^q}{q} \leq \frac{a^p}{p},$$

which is Young's inequality.

Equality holds when

$$b = a^{p-1},$$

or equivalently

$$a^p = b^q.$$

The two powers are paired so that each is the optimal supporting term for the other. This duality is what later makes Hölder's inequality exact.

§43.5 Hölder follows by normalizing Young's inequality

For vectors $x = (x_i)$ and $y = (y_i)$, define

$$\|x\|_p = \left(\sum_i |x_i|^p\right)^{1/p}, \qquad \|y\|_q = \left(\sum_i |y_i|^q\right)^{1/q}.$$

If either norm is zero, Hölder is trivial. Otherwise set

$$X_i = \frac{|x_i|}{\|x\|_p}, \qquad Y_i = \frac{|y_i|}{\|y\|_q}.$$

Young's inequality gives

$$X_i Y_i \le \frac{X_i^p}{p} + \frac{Y_i^q}{q}.$$

Summing and using

$$\sum_i X_i^p = 1, \qquad \sum_i Y_i^q = 1$$

yields

$$\sum_i X_i Y_i \le \frac{1}{p} + \frac{1}{q} = 1.$$

Therefore

$$\boxed{\sum_i |x_i y_i| \le \|x\|_p \|y\|_q.}$$

This is Hölder's inequality.

Equality requires equality in Young for every nonzero component:

$$X_i^p = Y_i^q.$$

Thus the normalized mass distributions $(|x_i|^p)$ and $(|y_i|^q)$ must agree.

For $p = q = 2$, Hölder becomes Cauchy–Schwarz:

$$\left|\sum_i x_i y_i\right| \le \left(\sum_i x_i^2\right)^{1/2} \left(\sum_i y_i^2\right)^{1/2}.$$

The equality condition reduces to linear dependence of the two vectors.

§43.6 Hölder identifies the dual norm exactly

Hölder shows that every y with $\|y\|_q = 1$ satisfies

$$\left|\sum_i x_i y_i\right| \le \|x\|_p.$$

The bound is attained. For $x \neq 0$, define

$$y_i = \frac{\operatorname{sgn}(x_i)|x_i|^{p-1}}{\|x\|_p^{p-1}}.$$

Since

$$(p-1)q = p,$$

we have

$$\sum_i |y_i|^q = \frac{\sum_i |x_i|^p}{\|x\|_p^p} = 1.$$

Moreover,

$$\sum_i x_i y_i = \frac{\sum_i |x_i|^p}{\|x\|_p^{p-1}} = \|x\|_p.$$

Hence

$$\boxed{\|x\|_p = \sup_{\|y\|_q=1} \sum_i x_i y_i.}$$

This is the dual-norm formula. An ℓ^p norm is recovered by testing the vector against all unit vectors in the conjugate ℓ^q norm. Hölder is therefore not only an inequality; it identifies the exact linear functionals that measure ℓ^p geometry.

The case $p = 2$ is self-dual. The maximizing vector is parallel to x, which is the familiar equality condition in Cauchy–Schwarz.

§43.7 Minkowski is the triangle inequality produced by duality

Using the dual-norm formula,

$$\begin{aligned}
\|x+z\|_p &= \sup_{\|y\|_q=1} \sum_i (x_i+z_i)y_i \\
&\le \sup_{\|y\|_q=1} \sum_i x_i y_i + \sup_{\|y\|_q=1} \sum_i z_i y_i \\
&= \|x\|_p + \|z\|_p.
\end{aligned}$$

Thus

$$\boxed{\left(\sum_i |x_i+z_i|^p\right)^{1/p} \le \left(\sum_i |x_i|^p\right)^{1/p} + \left(\sum_i |z_i|^p\right)^{1/p}.}$$

This is Minkowski's inequality.

For $p = 2$, it is the Euclidean triangle inequality. For general $p > 1$, it proves that ℓ^p is a normed vector space.

Equality in the strictly convex ℓ^p norm occurs when the two nonzero vectors point in the same nonnegative direction:

$$x = \lambda z, \qquad \lambda \ge 0.$$

The algebraic equality condition is the norm-geometric statement that travelling along two segments is as short as travelling along their sum only when the segments are aligned.

§43.8 Cauchy's Engel form is projection geometry in disguise

Cauchy–Schwarz can be rearranged into the Engel form

$$\boxed{\sum_{i=1}^n \frac{a_i^2}{b_i} \ge \frac{(a_1+\cdots+a_n)^2}{b_1+\cdots+b_n}} \qquad (b_i > 0).$$

Apply Cauchy to the vectors

$$\left(\frac{a_1}{\sqrt{b_1}}, \ldots, \frac{a_n}{\sqrt{b_n}}\right)$$

and

$$(\sqrt{b_1}, \ldots, \sqrt{b_n}).$$

Their inner product is $a_1 + \cdots + a_n$, giving the formula.

Equality holds when the vectors are proportional:

$$\frac{a_i}{\sqrt{b_i}} = \lambda\sqrt{b_i},$$

so

$$\frac{a_i}{b_i} = \lambda$$

for every i.

The same result can be read as an optimization problem. Among all vectors with prescribed weighted sum, the squared weighted norm is minimized by the vector parallel to the constraint direction. The frequently used contest lemma is therefore a projection theorem in a weighted Euclidean space.

§43.9 Power means form one monotone scale

For positive $a_1, \ldots, a_n$ and $r \neq 0$, define

$$M_r(a) = \left(\frac{1}{n}\sum_{i=1}^{n} a_i^r\right)^{1/r}.$$

For $0 < r < s$, the function

$$t \longmapsto t^{s/r}$$

is convex. Jensen gives

$$\left(\frac{1}{n}\sum_i a_i^r\right)^{s/r} \leq \frac{1}{n}\sum_i a_i^s.$$

Taking the s-th root yields

$$M_r(a) \leq M_s(a).$$

As $r \to 0$, write

$$a_i^r = e^{r\log a_i} = 1 + r\log a_i + o(r).$$

Then

$$\log M_r = \frac{1}{r}\log\left(\frac{1}{n}\sum_i a_i^r\right) \longrightarrow \frac{1}{n}\sum_i \log a_i.$$

Hence

$$M_0(a) = \sqrt[n]{a_1\cdots a_n}.$$

For negative orders,

$$M_{-r}(a_1, \ldots, a_n) = \frac{1}{M_r(a_1^{-1}, \ldots, a_n^{-1})}.$$

Combining these observations gives the full monotonicity:

$$r < s \quad\Longrightarrow\quad M_r \leq M_s.$$

The classical chain is therefore one theorem:

$$M_{-1} \leq M_0 \leq M_1 \leq M_2,$$

or

$$\text{harmonic} \leq \text{geometric} \leq \text{arithmetic} \leq \text{quadratic}.$$

Strict inequality holds unless all variables are equal.

§43.10 Chebyshev is a covariance identity

Suppose

$$a_1 \le \cdots \le a_n, \qquad b_1 \le \cdots \le b_n.$$

Chebyshev's sum inequality states

$$\frac{1}{n}\sum_i a_i b_i \ge \left(\frac{1}{n}\sum_i a_i\right)\left(\frac{1}{n}\sum_i b_i\right).$$

The difference has the exact form

$$\frac{1}{n}\sum_i a_i b_i - \left(\frac{1}{n}\sum_i a_i\right)\left(\frac{1}{n}\sum_i b_i\right)$$
$$= \frac{1}{2n^2}\sum_{i,j}(a_i - a_j)(b_i - b_j).$$

Every summand is nonnegative because the two sequences have the same ordering. Hence the difference is nonnegative.

If one sequence is increasing and the other decreasing, every product changes sign and the inequality reverses. Chebyshev is therefore a finite covariance statement: similarly ordered variables have nonnegative covariance.

Equality occurs when one sequence is constant, or more generally when every pair with a nonzero difference in one sequence has zero difference in the other.

§43.11 The log-sum inequality turns convexity into relative entropy

Let $a_i, b_i > 0$, and set

$$A = \sum_i a_i, \qquad B = \sum_i b_i.$$

The function

$$\phi(t) = t\log t$$

is convex on $(0, \infty)$. Apply Jensen with weights

$$\lambda_i = \frac{b_i}{B}$$

and points

$$x_i = \frac{a_i}{b_i}.$$

Since

$$\sum_i \lambda_i x_i = \frac{A}{B},$$

Jensen gives

$$\sum_i \frac{b_i}{B}\frac{a_i}{b_i}\log\frac{a_i}{b_i} \ge \frac{A}{B}\log\frac{A}{B}.$$

Multiplying by B yields the log-sum inequality:

$$\boxed{\sum_i a_i \log\frac{a_i}{b_i} \ge A\log\frac{A}{B}.}$$

Equality holds exactly when the ratios

$$\frac{a_i}{b_i}$$

are all equal.

For probability distributions p, q, we have $A = B = 1$, so

$$D(p\|q) = \sum_i p_i \log \frac{p_i}{q_i} \geq 0.$$

The quantity $D(p\|q)$ is relative entropy or Kullback–Leibler divergence. It vanishes exactly when $p = q$.

The route from a tangent line to relative entropy is continuous. Convexity gives Jensen; the same averaging principle yields AM–GM and majorization; convex conjugacy gives Young; normalization produces Hölder; dual norms produce Minkowski; and the convex function $t \log t$ measures the cost of replacing one probability distribution by another. The named inequalities are not a list of unrelated tricks but consequences of how convex functions and dual norms control averaging, pairing, and distance.

Function Exercises and Inequality Estimates

N-20221105

§44.1 A short note about elementary methods

The handwritten heading includes the sentiment that a person who understands mathematics sees the solution more clearly than someone who only imitates formulas. The actual page is a small exercise sheet: periodicity, quadratic minima, logarithmic comparison, parameter ranges, and an inequality with $abc = 1$. I keep the exercise style, but expand the reasoning so that the methods are visible.

§44.2 Periodicity from a transformed argument

The first problem has a function f satisfying a relation of the form

$$f(x+4) = f(x), \qquad g(x+4) = g(x) + 2,$$

and asks about the periodicity of a composite or transformed function. The handwritten solution notices that the increment in x produces the same value of f and a controlled increment of g. The general principle is:

$$f(x+T) = f(x) \quad \Longrightarrow \quad f \text{ has period } T.$$

If a formula contains $x + 8$, use the period twice. For example,

$$f(x+8) = f((x+4)+4) = f(x+4) = f(x).$$

The important point is that a period is not necessarily the least positive period. A function with period 4 automatically has period $8, 12, \ldots$. Many errors come from confusing "a period" with "the fundamental period."

§44.3 A quadratic minimum

The second problem studies

$$y = x^2 + x + \frac{5}{3}.$$

Complete the square:

$$y = \left(x + \frac{1}{2}\right)^2 + \frac{17}{12}.$$

Hence

$$y \geq \frac{17}{12},$$

with equality at

$$x = -\frac{1}{2}.$$

This is the cleanest form of a quadratic problem. Differentiation gives the same result, but completing the square better reveals the geometry: a parabola shifted horizontally and vertically.

§44.4 A logarithmic comparison

Another problem compares functions involving logarithms, for example

$$f(x) = \log_a(x+1)$$

and a related expression $g(x) - f(x)$. The handwritten solution rewrites the difference using logarithm rules:

$$\log_a u - \log_a v = \log_a \frac{u}{v}.$$

Then the sign of the difference depends on whether the base a is greater than 1 or between 0 and 1. This is the crucial trap:

$$a > 1 \quad \Rightarrow \quad \log_a x \text{ is increasing,}$$

whereas

$$0 < a < 1 \quad \Rightarrow \quad \log_a x \text{ is decreasing.}$$

So a logarithmic inequality must always track the base.

§44.5 A parameterized polynomial

The fourth problem considers a parameter p and a function with a specified zero or extremum. The handwritten computation substitutes the relevant value and reduces to an equation in p. The broad template is:

(i) if x_0 is a zero, impose $f(x_0) = 0$;

(ii) if x_0 is an extremum of a differentiable function, impose $f'(x_0) = 0$;

(iii) if a graph is tangent to a line, impose both value equality and slope equality.

This page is mainly training the habit of translating a verbal condition into equations.

§44.6 An inequality under $abc = 1$

The last problem states: for $a, b, c > 0$, $abc = 1$, prove an inequality of the form

$$\frac{1}{a^3(cb+c)} + \frac{1}{b^3(ca+c)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

The handwritten solution uses the condition

$$abc = 1$$

to replace reciprocal expressions by symmetric sums such as

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = ab + bc + ca.$$

Then Cauchy–Schwarz or AM–GM gives a lower bound.

The useful lesson is that the constraint $abc = 1$ should not sit unused. It usually allows conversion between variables and reciprocals. A standard move is also

$$a = \frac{x}{y}, \qquad b = \frac{y}{z}, \qquad c = \frac{z}{x},$$

which automatically enforces $abc = 1$. Whether one uses this substitution or direct reciprocal rewriting depends on the shape of the expression.

§44.7 The conceptual turn

The exercises look miscellaneous, but the same strategy recurs: normalize the expression before attacking it. Periodicity reduces arguments modulo a period; completing the square normalizes a quadratic; logarithm laws turn products into sums; parameter conditions become equations; $abc = 1$ turns reciprocal expressions into symmetric ones. Elementary problem solving is often the art of finding the right normalization.

§44.8 Periodicity from transformed arguments

A relation such as $f(x+a) = f(x)$ is invariance under translation by a. More complicated relations, for example involving $f(c-x)$, combine translation and reflection. The group generated by these transformations controls the function's repeated values.

Thus many functional-equation exercises are really symmetry problems on the domain.

§44.9 Quadratic and logarithmic estimates

A quadratic minimum is controlled by completing the square:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

A logarithmic comparison often uses monotonicity or concavity of log. In both cases, the point is to expose the structural feature: convexity for quadratics, monotone concavity for logarithms.

§44.10 Functional equations and orbit reduction

If a functional equation lets one transform x into simpler inputs, then the domain splits into orbits under the generated transformations. A fundamental domain is a set of representatives for these orbits.

Solving the equation means assigning values on the fundamental domain and transporting them consistently along the orbit.

§44.11 Functions through symmetry and normal form

Function exercises are unified by structure: periodicity is group invariance, quadratics are convex normal forms, logarithmic inequalities come from monotonicity and concavity, and functional equations reduce values along orbits.

Function Exercises: Domains, Ranges, Dirichlet Function, Parameters, and Functional Equations

N-20221217

§45.1 Structure of the exercise sheet

The note is a function exercise sheet, not a single theorem. It contains domain problems, range problems, a Dirichlet-function example, parameter problems, zero-counting by graphs, and functional equations. The repaired version keeps the exercise logic: each problem is a training example for a specific method.

§45.2 Domain problems

For a function domain, the first rule is to list all legality conditions. Square roots require nonnegative radicands, logarithms require positive arguments and valid bases, denominators cannot be zero, and iterated functions require every intermediate input to lie in the next domain.

For example,

$$y = \frac{\sqrt{x(3-x)}}{\lg(x-3)^2}$$

requires

$$x(3-x) \geq 0, \qquad (x-3)^2 > 0, \qquad \lg(x-3)^2 \neq 0.$$

The first gives $0 \leq x \leq 3$. The second excludes $x = 3$. The third excludes $(x-3)^2 = 1$, so $x = 2$ or 4, but only 2 lies in the current interval. Hence

$$D = [0, 3) \setminus \{2\}.$$

The important step is not the answer; it is the intersection of all restrictions.

For an expression such as

$$\sqrt{a^x - kb^x}$$

in a denominator, the condition is strict:

$$a^x - kb^x > 0.$$

Since $b^x > 0$, divide by b^x:

$$\left(\frac{a}{b}\right)^x > k.$$

Now the inequality depends on whether $a/b > 1$ or $0 < a/b < 1$. This is a recurring trap in exponential and logarithmic domain problems.

§45.3 Floor functions

The exercise sheet also uses the Gauss or floor function $[x]$. The correct method is interval splitting:

$$[x] = m \iff m \le x < m + 1.$$

A floor-function domain problem is therefore a finite or countable case split. One should not treat $[x]$ as a smooth algebraic expression.

§45.4 Range problems

For

$$y = \sqrt{x-4} + \sqrt{15-3x},$$

the domain is $4 \le x \le 5$. Put $u = x - 4$, so $0 \le u \le 1$ and

$$y = \sqrt{u} + \sqrt{3(1-u)}.$$

By Cauchy–Schwarz,

$$y^2 \le (1+3)(u+1-u) = 4,$$

so $y \le 2$. At the endpoints the smaller value is 1, and the maximum is achieved when the Cauchy equality condition holds. Thus the range is $[1, 2]$.

For

$$y = x + \sqrt{1-2x},$$

let $t = \sqrt{1-2x} \ge 0$. Then

$$x = \frac{1-t^2}{2}, \qquad y = -\frac{1}{2}(t-1)^2 + 1.$$

Hence the range is

$$(-\infty, 1].$$

This substitution method is the natural way to remove a radical while preserving the domain.

§45.5 Dirichlet function

The Dirichlet function is

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

It is bounded, because $0 \le D(x) \le 1$. It is not monotone on any interval, because every interval contains both rational and irrational numbers. It has every nonzero rational period T, since

$$x \in \mathbb{Q} \iff x + T \in \mathbb{Q}$$

for $T \in \mathbb{Q}$. It has no irrational period, because adding an irrational changes rationality.

This exercise is included because it teaches that functions can be extremely discontinuous while still having simple algebraic properties.

§45.6 Graph transformations and zero counting

The problem involving f and

$$g(x) = b - f(2 - x)$$

asks for the values of b for which $f(x) - g(x)$ has four zeros. The transformation $x \mapsto 2 - x$ reflects the graph across the vertical line $x = 1$; then $b-$ reflects vertically and shifts. The number of zeros of $f - g$ is the number of intersections of two transformed graphs.

The method is graphical but not vague. The number of intersections changes only when a moving graph becomes tangent, passes through a vertex/corner, or crosses an endpoint. Thus parameter intervals are found by checking these critical positions.

§45.7 Logarithmic parameter equation

For an equation such as

$$\ln(ax) = 2\ln(x + 1),$$

first impose the domain:

$$ax > 0, \qquad x + 1 > 0.$$

Then combine logarithms:

$$\ln(ax) = \ln((x + 1)^2)$$

so

$$ax = (x + 1)^2.$$

This becomes a quadratic in x. Having exactly one real solution means a discriminant or tangency condition, but the domain must still be checked. The sheet is training this two-step habit: algebraic transformation plus domain verification.

§45.8 Functional equations with period-like recurrence

Suppose f is defined on $[0, 1)$ by

$$f(x) = 2x - x^2$$

or another explicit expression, and for all real x satisfies

$$f(x) + f(x + 1) = 1.$$

Then values outside $[0, 1)$ are reduced by repeatedly using

$$f(x + 1) = 1 - f(x).$$

If $a = \log_2 3$, to compute $f(a) + f(2a) + f(3a)$, first determine the integer intervals containing $a, 2a, 3a$, reduce their fractional parts into $[0, 1)$, and then apply the recurrence each time.

This is the same idea as reducing modulo a period, except that shifting by 1 may flip the value rather than preserve it.

§45.9 A reciprocal functional equation

The sheet also has a functional equation of the form

$$f(x) = f(1)x + f(2)\frac{1}{x} - 1$$

for nonzero x. On $(0, +\infty)$, this becomes a problem about minimizing

$$Ax + \frac{B}{x} - 1.$$

The AM–GM inequality gives

$$Ax + \frac{B}{x} \geq 2\sqrt{AB}$$

when $A, B > 0$. The equality condition determines the point where the minimum occurs. This is a typical example where a functional equation reduces to a standard inequality.

§45.10 Quadratic iteration problem

For a quadratic

$$f(x) = x^2 + 2ax + 8,$$

conditions involving both $f(x) \leq 0$ and $f(f(x)) \leq 8$ should be treated by introducing the image variable $u = f(x)$. Then the second condition becomes $f(u) \leq 8$. The problem is about how the range of the first inequality is mapped by the quadratic again. Completing the square and tracking intervals is safer than expanding $f(f(x))$ blindly.

§45.11 Max and absolute value

The exercise about

$$M(x) = \max\{f(x), g(x)\}$$

uses the identity

$$\max\{u, v\} = \frac{u + v + |u - v|}{2}.$$

Thus maximum and minimum operations can be encoded using absolute value. This is useful when deciding whether a function built by max/min remains elementary.

§45.12 Parameter problems and critical values

When a parameter changes the number of solutions, the change usually happens at a critical value: a tangent contact, an endpoint contact, or a collision of two roots. This is why discriminants and graph tangencies appear so often.

For a quadratic equation depending on a parameter, the condition for exactly one real root is usually

$$\Delta = 0.$$

For a graphical equation, the same condition appears geometrically as tangency. The algebraic and geometric languages are two versions of the same idea.

§45.13 Functional equations as reduction rules

A functional equation such as

$$f(x+1) = 1 - f(x)$$

should be used as a reduction rule. It tells us how to move an input back to a fundamental interval. After two shifts,

$$f(x+2) = 1 - f(x+1) = f(x),$$

so the function becomes periodic with period 2. This kind of hidden periodicity is easy to miss if one treats the equation only algebraically.

§45.14 What the pattern suggests

The page is not just a set of answers. It teaches a toolbox: domains are legality intersections; ranges use substitution and inequalities; floor functions require interval splitting; Dirichlet's function uses density of rationals and irrationals; parameter problems are tangency/discriminant problems; functional equations reduce values to a fundamental interval; max and min can be represented by absolute values. The common theme is that a function is not merely a formula but a domain, graph, transformation rule, and set of constraints.

§45.15 Domains as constraint intersections

A domain problem is a constraint-satisfaction problem. Each operation contributes a legality condition: square roots impose inequalities, logarithms impose positivity, denominators impose nonvanishing, and compositions impose intermediate-domain conditions. The domain is the intersection of all these constraints.

This viewpoint prevents a common error: simplifying an expression before recording its original restrictions. For example, an expression with a denominator may algebraically simplify, but the excluded zeros of the original denominator still matter unless the problem explicitly defines the simplified expression as a new function.

§45.16 Dirichlet function through density and measure

The Dirichlet function

$$D(x) = \mathbf{1}_{\mathbb{Q}}(x)$$

is discontinuous at every real number. At any point a, every neighborhood contains rationals and irrationals, so along rational sequences D tends to 1, while along irrational sequences it tends to 0.

It is not Riemann integrable on any interval, because every subinterval has supremum 1 and infimum 0. But it is Lebesgue integrable, and its integral is 0, since $\mathbb{Q}$ has measure zero:

$$\int_0^1 \mathbf{1}_{\mathbb{Q}}(x)\,dx = 0.$$

Thus the same function separates two integration theories: Riemann integration sees oscillation on every interval; Lebesgue integration sees that the exceptional set is small in measure.

§45.17 Thomae's function as a nearby comparison

A useful comparison is Thomae's function

$$T(x) = \begin{cases} \frac{1}{q}, & x = \frac{p}{q} \text{ in lowest terms,} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is continuous at every irrational point and discontinuous at every rational point. Unlike the Dirichlet function, it is Riemann integrable with integral 0.

This comparison shows that dense discontinuities alone do not determine integrability. The size and height distribution of the discontinuities matter.

§45.18 Parameter problems as bifurcation diagrams

When a parameter changes the number of solutions, the change occurs at critical values. Algebraically, these are discriminant-zero cases, endpoint hits, or degeneracies. Geometrically, they are tangencies or changes in intersection pattern.

For a family $F(x, b) = 0$, a typical critical point satisfies

$$F(x, b) = 0, \qquad \frac{\partial F}{\partial x}(x, b) = 0.$$

The first equation says the graph intersects the axis; the second says the intersection is tangent, so two roots can merge or disappear. This is the elementary version of a bifurcation diagram.

§45.19 Functional equations as group actions on the domain

A relation such as

$$f(x + 1) = 1 - f(x)$$

should be read as an action of the translation group on the input line, together with an action on values. Applying the shift twice gives

$$f(x + 2) = f(x),$$

so the function descends to a period-two structure after accounting for the value flip.

The interval $[0, 1)$ is a fundamental domain: every real input can be moved into it by integer shifts. The functional equation tells how the value changes during that reduction. This is the same logic behind periodic functions, modular arithmetic, and covering spaces.

§45.20 Max, min, and nonsmooth points

The identity

$$\max\{u, v\} = \frac{u + v + |u - v|}{2}$$

turns a maximum into an expression involving absolute value. It also explains where nonsmoothness can occur: at points where $u = v$, the absolute value has a corner unless the crossing is tangential in a compatible way.

In optimization language, maximum functions are piecewise-smooth. Their derivatives are governed by the active branch. At a switching point, one may need one-sided derivatives or subgradients rather than an ordinary derivative.

§45.21 Ranges as images and compactness

A range problem asks for the image $f(D)$. If D is compact and f is continuous, then $f(D)$ is compact. In $\mathbb{R}$, that means closed and bounded. This is why many range problems on closed intervals reduce to finding extrema.

If the domain is not compact, endpoints may be missing or infinity may be approached. Then the range must be described with open or half-open endpoints. This is not a cosmetic detail: it records whether a value is achieved or only approached.

§45.22 Function exercises through topology and dynamics

These function exercises are training a structural habit. A function is not just a formula; it is a formula together with a domain, a range, transformations, constraints, and possible exceptional sets. Domain questions are intersections of constraints, range questions are image problems, parameter questions track critical values, and functional equations reduce inputs by a symmetry or group action.

Function Graphs, Symmetry Centers, and Transformations

N-20230107

§46.1 A note on graph problems

This handwritten note studies function graphs and transformations: centers of symmetry, symmetry about vertical lines, absolute values, and graph-based zero counting. The important habit is to translate a visual symmetry into an algebraic identity.

§46.2 Centers of symmetry

A graph $y = f(x)$ has center of symmetry (a, b) if

$$f(a+t) + f(a-t) = 2b$$

whenever both sides are defined. Thus for a rational function like

$$f(x) = \frac{1}{x+1} + \frac{1}{x+2},$$

one shifts $x = a+t$ and looks for cancellation between t and $-t$. The vertical asymptotes also help: the center of symmetry often lies halfway between paired asymptotes.

§46.3 Symmetry about a line

A graph is symmetric about the vertical line $x = c$ iff

$$f(c+t) = f(c-t).$$

The note uses this for functions such as

$$f(x) = (1-x^2)(x^2+ax+b).$$

If the graph is symmetric about $x = -2$, substitute $x = -2+t$ and $x = -2-t$, then equate the two expressions. This is safer than trying to see the answer from the expanded graph.

§46.4 Absolute value transformations

Expressions involving

$$|x-a|, \qquad ||x|-2|, \qquad |f(x)|$$

should be read as folding operations. The graph of $y = |f(x)|$ reflects the part of f below the x-axis upward. The graph of $y = f(|x|)$ copies the right half of the graph to the left symmetrically.

The problem involving

$$|||x|-2|-1| + |||y|-2|-1| = 1$$

is best attacked by symmetry. First solve in the first quadrant, then reflect across the coordinate axes. The curve is made of translated line segments, so the required string length is a sum of segment lengths rather than an area computation.

§46.5 Zero counting by graphs

A later exercise combines a periodic/symmetric function f with

$$g(x) = |x\cos(\pi x)|.$$

The number of zeros of $g(x) - f(x)$ is the number of intersections of two graphs. The correct method is to use symmetry and monotonicity on subintervals instead of solving a transcendental equation directly.

§46.6 How to find a center of symmetry in practice

For a rational function, a center of symmetry is often hidden in the denominator structure. Suppose two vertical asymptotes are paired around a point $x = a$. Then a is a natural candidate for the horizontal coordinate of the center. After shifting $x = a + t$, test whether

$$f(a+t) + f(a-t)$$

is constant. If it equals $2b$, then the center is (a, b).

This method is better than expanding everything at once. The shift recenters the graph so that symmetry becomes ordinary oddness or evenness around the origin. In many problems, the algebra simplifies only after this recentering.

§46.7 Absolute value as folding

The note's absolute-value examples should be read visually. The transformation

$$y = |f(x)|$$

keeps the part of the graph above the x-axis and reflects the part below upward. The transformation

$$y = f(|x|)$$

copies the right half of the original graph to the left. Nested absolute values such as

$$||x| - 2|$$

are successive folds of the number line. Each fold creates new symmetry and new corner points.

For a curve such as

$$|||x| - 2| - 1| + |||y| - 2| - 1| = 1,$$

one should not immediately expand into many cases. First locate the breakpoints $x = 0, \pm 1, \pm 2, \pm 3$, solve the shape in one symmetric region, and reflect it. The algebraic expression is complicated, but the geometry is a repeated folding of straight lines.

§46.8 Parameter motion and intersection counting

Many graph problems ask for the number of roots of

$$f(x) - g(x) = 0.$$

This is the number of intersections of two graphs. When a parameter moves one graph, the number of intersections is stable except at critical moments: tangency, passing through a corner, or passing through an endpoint of a piece.

Thus parameter problems should be solved by first finding the exceptional parameter values. Between two exceptional values, the number of intersections cannot change. This converts an apparently algebraic zero-counting problem into a geometric classification problem.

§46.9 Graph transformations as group actions

The higher viewpoint is that many graph tricks are actions on the plane. Let G be a group of transformations of $\mathbb{R}^2$. If $g \in G$ and $\Gamma_f = \{(x, f(x)) : x \in D\}$ is the graph of f, then g sends Γ_f to another curve. When the image is again a graph, it gives a transformed function.

Definition 46.9.1 (Symmetry of a graph)**.** A transformation $g : \mathbb{R}^2 \to \mathbb{R}^2$ is a symmetry of the graph Γ_f if

$$g(\Gamma_f) = \Gamma_f.$$

In this language, a graph property is an invariance property.

For example, reflection across the vertical line $x = c$ is

$$R_c(x, y) = (2c - x, y).$$

The graph is invariant under R_c iff

$$f(c + t) = f(c - t).$$

Thus vertical symmetry is evenness after translating the origin to c.

Similarly, the half-turn about (a, b) is

$$H_{a,b}(x, y) = (2a - x, 2b - y).$$

The graph is invariant under this half-turn iff

$$f(a + t) + f(a - t) = 2b.$$

Proposition 46.9.2 (Symmetry identities)

Let f be defined on a domain symmetric about a. Then:

(i) Γ_f is symmetric about $x = a$ iff $f(a + t) = f(a - t)$;

(ii) Γ_f has center (a, b) iff $f(a + t) + f(a - t) = 2b$.

Proof. For (i), reflection sends $(a + t, f(a + t))$ to $(a - t, f(a + t))$. This lies on the graph exactly when $f(a - t) = f(a + t)$. For (ii), the half-turn sends $(a + t, f(a + t))$ to $(a - t, 2b - f(a + t))$. This lies on the graph exactly when $f(a - t) = 2b - f(a + t)$. □

§46.10 Absolute value as folding, not algebraic noise

The transformation
$$y = |f(x)|$$
keeps the part of the graph above the x-axis and reflects the lower part upward. The transformation
$$y = f(|x|)$$
forgets the left half of the original input and copies the right half across the y-axis.

> **Remark 46.10.1** (A quotient-like intuition) — An absolute value identifies two points t and $-t$. Thus nested absolute values are repeated foldings of the line. This is why expressions such as
> $$|||x| - 2| - 1|$$
> are better studied by locating breakpoints and symmetries than by expanding all cases blindly.

§46.11 Zero counting as a stability problem

Many problems ask for the number of roots of
$$f(x) - g_t(x) = 0$$
as a parameter t changes. Geometrically, this is the number of intersections of two curves. The number is stable under small changes of t unless a special event occurs.

The common events are:

$$\text{tangency}, \qquad \text{endpoint contact}, \qquad \text{crossing a corner or discontinuity}.$$

For a smooth one-parameter equation
$$F(x,t) = 0,$$
a tangency or repeated-root event is detected by
$$F(x,t) = 0, \qquad F_x(x,t) = 0.$$
This is the calculus version of the geometric statement that two curves touch without crossing transversely.

> **Remark 46.11.1** (The conceptual payoff) — A center of symmetry is a half-turn invariance. A vertical axis of symmetry is reflection invariance. An absolute value is a folding operation. A zero-counting problem is an intersection-stability problem. The advantage of this viewpoint is that it tells us what can change and what cannot change before any heavy algebra begins.

Functions through Fibers, Iteration, and Functional Equations

N-20230116

§47.1 A function includes its source and target

A function is data
$$f : X \to Y$$
consisting of a domain X, a codomain Y, and a rule assigning to every $x \in X$ exactly one value $f(x) \in Y$. A formula by itself is not enough. The same expression
$$x \longmapsto x^2$$
defines different functions when its domain or codomain changes. For example,
$$\mathbb{R} \to \mathbb{R}, \qquad \mathbb{R} \to [0, \infty), \qquad \mathbb{C} \to \mathbb{C}$$
have different surjectivity and inverse-image behavior even though the displayed formula is identical.

Two functions are equal when they have the same domain, the same codomain, and the same value at every input. This convention prevents hidden changes of target from being treated as harmless notation.

If $X = \{1, \ldots, n\}$ and $Y = \{1, \ldots, m\}$, then a function $f : X \to Y$ is completely recorded by the list
$$(f(1), \ldots, f(n)).$$
Each of the n entries has m choices, so the number of such functions is
$$m^n.$$
This simple count is the starting point for more refined counts of injections, surjections, and functions satisfying iteration equations.

§47.2 Images and preimages obey different algebraic laws

For $A \subseteq X$, the direct image is
$$f(A) = \{f(a) : a \in A\}.$$
For $B \subseteq Y$, the inverse image is
$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$
The notation $f^{-1}(B)$ does not require f to be invertible; it denotes a subset of the domain.

Inverse image preserves all ordinary set operations. For any family $(B_i)_{i\in I}$ of subsets of Y,

$$f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i),$$

$$f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i),$$

and

$$f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

Each identity follows by checking membership. For example,

$$\begin{aligned} x \in f^{-1}(Y \setminus B) &\iff f(x) \notin B \\ &\iff x \notin f^{-1}(B). \end{aligned}$$

Thus inverse image is a Boolean-algebra homomorphism between power sets.

Direct image preserves arbitrary unions,

$$f\left(\bigcup_{i} A_i\right) = \bigcup_{i} f(A_i),$$

but for intersections one only has

$$f(A \cap C) \subseteq f(A) \cap f(C).$$

The inclusion can be strict. Let $f(x) = x^2$, $A = \{-1\}$, and $C = \{1\}$. Then

$$A \cap C = \varnothing,$$

while

$$f(A) \cap f(C) = \{1\}.$$

Equality for all subsets $A, C \subseteq X$ holds precisely when f is injective.

§47.3 Direct and inverse image form a Galois connection

Order the power sets $\mathcal{P}(X)$ and $\mathcal{P}(Y)$ by inclusion. Both direct image and inverse image are monotone:

$$A_1 \subseteq A_2 \Longrightarrow f(A_1) \subseteq f(A_2),$$

$$B_1 \subseteq B_2 \Longrightarrow f^{-1}(B_1) \subseteq f^{-1}(B_2).$$

Their deeper relation is

$$\boxed{f(A) \subseteq B \quad\iff\quad A \subseteq f^{-1}(B).}$$

The two statements say the same thing element by element: every point of A is sent into B. In the language of ordered sets, this equivalence is a Galois connection, or adjunction,

$$f(-) \dashv f^{-1}(-).$$

The formula has a useful extremal interpretation. Among all subsets $B \subseteq Y$ whose inverse image contains A, the direct image $f(A)$ is the smallest. Dually, among all subsets $A \subseteq X$ whose direct image lies in B, the inverse image $f^{-1}(B)$ is the largest.

Thus the two operations are not merely convenient notation: each solves an optimization problem in the inclusion order.

Taking $B = f(A)$ gives

$$A \subseteq f^{-1}(f(A)).$$

The right-hand side adds every point lying in the same fiber as some point of A. Define

$$c(A) = f^{-1}(f(A)).$$

Then c is a closure operator on $\mathcal{P}(X)$: it is extensive,

$$A \subseteq c(A),$$

monotone, and idempotent,

$$c(c(A)) = c(A).$$

Its fixed points are exactly the subsets that are unions of whole fibers of f. Such sets are often called *saturated*.

In the other direction,

$$f(f^{-1}(B)) = B \cap f(X) \subseteq B.$$

Define

$$i(B) = f(f^{-1}(B)).$$

Then i is contractive, monotone, and idempotent. Its fixed points are precisely the subsets of the actual image $f(X)$. The two composites therefore remove different kinds of irrelevant information: c forgets distinctions inside fibers, while i discards target points that the function never reaches.

For a concrete example, take

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$

For every $A \subseteq \mathbb{R}$,

$$f^{-1}(f(A)) = A \cup (-A).$$

A set becomes saturated by adding the reflected point $-a$ whenever it contains a. For every $B \subseteq \mathbb{R}$,

$$f(f^{-1}(B)) = B \cap [0, \infty).$$

Negative target values disappear because no real number squares to them. This example makes the two failures visible: noninjectivity enlarges A along fibers, while nonsurjectivity shrinks B to the image.

The inclusion

$$A \subseteq f^{-1}(f(A))$$

is an equality for every A exactly when f is injective. The equality

$$f(f^{-1}(B)) = B$$

holds for every B exactly when f is surjective. The same adjunction also explains why continuity and measurability are defined using inverse images: inverse images preserve all unions, intersections, and complements without requiring either injectivity or surjectivity. The closure operator c now leads naturally to the next question: what are the fibers whose unions it is forcing us to take?

§47.4 Fibers measure information loss

For $y \in Y$, the fiber over y is

$$f^{-1}(y) = \{x \in X : f(x) = y\}.$$

A function is injective exactly when every fiber has at most one element, and surjective exactly when every fiber is nonempty.

The fibers partition the domain after empty fibers are discarded. Define

$$x \sim_f x' \quad \iff \quad f(x) = f(x').$$

Then $\sim_f$ is an equivalence relation, and f factors canonically as

$$X \twoheadrightarrow X/{\sim_f} \xrightarrow{\cong} f(X) \hookrightarrow Y.$$

The first map collapses each fiber to one point. The middle map sends a fiber class to its common value. The last map includes the image into the declared codomain.

This factorization separates two kinds of failure to be bijective. Noninjectivity occurs in the quotient step, where distinct inputs are identified. Nonsurjectivity occurs in the final inclusion, where some target points are missed.

For finite X, counting by fibers gives

$$|X| = \sum_{y \in f(X)} |f^{-1}(y)|.$$

This elementary identity is the discrete prototype of decomposing a space or measure along fibers.

§47.5 Counting injections, surjections, and bijections

Let $|X| = n$ and $|Y| = m$. An injection $X \to Y$ exists only when $n \le m$. Its values can be chosen successively without repetition, so the number of injections is

$$m(m-1)\cdots(m-n+1) = \frac{m!}{(m-n)!}.$$

When $n = m$, every injection is automatically surjective, and the number of bijections is

$$n!.$$

Surjections can be counted by inclusion–exclusion. There are m^n total functions. For each target point y, let E_y be the set of functions that miss y. If a specified set of j target points is missed, the function has only $m - j$ possible values at each input. Therefore the number of surjections is

$$\boxed{\sum_{j=0}^{m} (-1)^j \binom{m}{j} (m-j)^n.}$$

For example, the number of surjections from a four-element set onto a three-element set is

$$3^4 - \binom{3}{1} 2^4 + \binom{3}{2} 1^4 = 81 - 48 + 3 = 36.$$

The same number is $3!S(4,3)$, where $S(n,m)$ is a Stirling number of the second kind: first partition the domain into m nonempty fibers, then label those fibers by the target elements.

§47.6 A finite function is a directed dynamical system

Let $f : X \to X$ with X finite. Draw one directed edge

$$x \longrightarrow f(x)$$

from every vertex. Each vertex has out-degree one, though its in-degree may be arbitrary.

Theorem 47.6.1 (Structure of a finite functional graph)
Every connected component contains exactly one directed cycle, with directed rooted trees feeding into the vertices of that cycle.

Proof. Starting at any $x \in X$, the sequence

$$x, f(x), f^2(x), \ldots$$

must repeat because X is finite. Once a value repeats, the subsequent orbit is periodic and therefore enters a directed cycle.

A connected component cannot contain two disjoint directed cycles. Since every vertex has exactly one outgoing edge, a forward orbit can enter only one cycle. If two cycles lay in one weakly connected component, any undirected path between them would force some vertex to have two different forward destinations, contradicting the uniqueness of its outgoing edge. □

The nonperiodic vertices form trees oriented toward the cycle. Thus every finite orbit has a transient part followed by a periodic part.

A finite self-map is injective exactly when every vertex has in-degree one. In that case there are no attached trees, and every component is a pure cycle. This proves again that, for finite sets of equal size,

$$\text{injective} \iff \text{surjective} \iff \text{permutation}.$$

§47.7 Iteration equations become restrictions on cycle lengths

Suppose

$$f^r = \mathrm{id}_X\,.$$

Then f has the inverse f^{r-1}, so it is a permutation. On a cycle of length d, the map f^r is the identity exactly when

$$d \mid r.$$

Hence the equation $f^r = \mathrm{id}$ means that every cycle length divides r.

For an involution,

$$f^2 = \mathrm{id},$$

all cycles have length one or two. To count involutions on an n-element set, choose $2k$ elements to belong to transpositions and pair them. This gives

$$\boxed{\sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{2^k k!(n-2k)!}}.$$

For $n = 4$, the count is

$$1 + 6 + 3 = 10:$$

one identity, six permutations with one transposition, and three products of two disjoint transpositions.

Now consider the idempotent equation

$$f^2 = f.$$

Every point in the image is fixed, because if $y = f(x)$, then

$$f(y) = f(f(x)) = f(x) = y.$$

Conversely, any function that fixes every point of its image is idempotent. Its functional graph consists of fixed points with depth-one trees feeding into them.

If the image has size r, first choose its r fixed points and then send each of the remaining $n - r$ elements to any image point. Therefore the number of idempotent self-maps on a nonempty n-element set is

$$\boxed{\sum_{r=1}^{n} \binom{n}{r} r^{n-r}.}$$

For $n = 3$, this gives

$$\binom{3}{1}1^2 + \binom{3}{2}2^1 + \binom{3}{3}3^0 = 3 + 6 + 1 = 10.$$

The equation is thus not merely an algebraic string; it specifies an exact graph shape and leads to an exact enumeration.

§47.8 Left inverses, right inverses, and what they prove

Let $f : X \to Y$. If there is a map $g : Y \to X$ with

$$g \circ f = \mathrm{id}_X,$$

then f is injective. Indeed,

$$f(x) = f(x') \Longrightarrow g(f(x)) = g(f(x')) \Longrightarrow x = x'.$$

Such a g is a left inverse of f.

If there is a map $h : Y \to X$ with

$$f \circ h = \mathrm{id}_Y,$$

then f is surjective, because every $y \in Y$ equals $f(h(y))$. Such an h is a right inverse.

Conversely, an injective function with nonempty domain admits a left inverse after assigning arbitrary values away from its image. A surjective function admits a right inverse only after choosing one element from every fiber; for arbitrary families this assertion is equivalent to a form of the axiom of choice. In finite sets the choices can always be made directly.

A map with both a left and a right inverse is bijective, and the two inverses coincide. If

$$g \circ f = \mathrm{id}_X, \qquad f \circ h = \mathrm{id}_Y,$$

then

$$g = g \circ (f \circ h) = (g \circ f) \circ h = h.$$

§47.9 Functional equations transport values along orbits

Many functional equations involve a transformation $T : X \to X$ of the input. An equation such as

$$F(Tx) = \Phi(F(x))$$

propagates values along the orbit

$$x, Tx, T^2x, \dots .$$

If the orbit closes after r steps, consistency requires

$$F(x) = F(T^r x) = \Phi^r(F(x)).$$

Thus cycles in the input action impose algebraic constraints on possible values.

For example, solve

$$f(x+1) = f(x) + 2$$

on $\mathbb{R}$. Subtract the particular solution $2x$ and define

$$p(x) = f(x) - 2x.$$

Then

$$p(x+1) = f(x+1) - 2x - 2 = f(x) - 2x = p(x).$$

Hence every solution has the form

$$\boxed{f(x) = 2x + p(x),}$$

where p is an arbitrary one-periodic function. The equation determines the change along each translation orbit but leaves one period of initial data free.

By contrast, suppose a transformation satisfies $T^r = \mathrm{id}$ and one asks for

$$f(Tx) = f(x) + c$$

with real-valued f. Iterating around the cycle gives

$$f(x) = f(T^r x) = f(x) + rc,$$

so necessarily $c = 0$. The obstruction comes from consistency around a closed orbit.

§47.10 The additive Cauchy equation is rigid only with regularity

Consider

$$f(x+y) = f(x) + f(y) \qquad (x, y \in \mathbb{R}).$$

Without any regularity assumption, this equation does not force the familiar form $f(x) = cx$. It first forces only $\mathbb{Q}$-linearity.

Set $y = 0$ to get $f(0) = 0$. Then

$$0 = f(x + (-x)) = f(x) + f(-x),$$

so $f(-x) = -f(x)$. Repeated addition gives

$$f(nx) = nf(x) \qquad (n \in \mathbb{Z}).$$

For a positive integer q,

$$qf(x/q) = f(x),$$

so

$$f(rx) = rf(x) \qquad (r \in \mathbb{Q}).$$

In particular,

$$f(q) = qf(1) \qquad (q \in \mathbb{Q}).$$

Now assume f is continuous at one point a. Since

$$f(h) = f(a+h) - f(a),$$

continuity at a implies continuity at 0, and additivity then implies continuity everywhere. Choose rationals $q_n \to x$. We obtain

$$f(x) = \lim_{n\to\infty} f(q_n) = \lim_{n\to\infty} q_n f(1) = xf(1).$$

Thus

$$\boxed{f(x) = cx}$$

with $c = f(1)$.

The same conclusion follows from several weaker regularity hypotheses, including measurability or boundedness on a sufficiently large set. Without regularity, choose a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$, prescribe values arbitrarily on the basis, and extend $\mathbb{Q}$-linearly. The resulting additive functions are generally discontinuous everywhere. The lesson is that an algebraic functional equation and an analytic regularity condition play distinct roles.

§47.11 Cancellation gives the categorical meaning of injection and surjection

In the category **Set**, a function $f : X \to Y$ is injective exactly when it is left-cancellable:

$$f \circ g_1 = f \circ g_2 \quad \Longrightarrow \quad g_1 = g_2$$

for all maps $g_1, g_2 : Z \to X$. This is the definition of a monomorphism.

To see the converse, suppose f is not injective and choose distinct $x, x' \in X$ with $f(x) = f(x')$. Taking Z to be a one-point set and sending its point to x or x' gives two different maps whose composites with f agree. Hence f is not monic.

Similarly, a function in **Set** is surjective exactly when it is right-cancellable:

$$h_1 \circ f = h_2 \circ f \quad \Longrightarrow \quad h_1 = h_2.$$

This is the definition of an epimorphism. If f misses a point $y \in Y$, two maps out of Y can agree on $f(X)$ but differ at y, so f is not epic.

These identifications depend on the category. In the category of commutative rings with identity, the inclusion

$$\mathbb{Z} \hookrightarrow \mathbb{Q}$$

is an epimorphism although it is not surjective. Any two ring homomorphisms out of $\mathbb{Q}$ that agree on $\mathbb{Z}$ must agree on every fraction. The set-theoretic notions are therefore prototypes of categorical cancellation, not universal descriptions of it.

In **Set**, however, the preceding viewpoints now fit together exactly. The fiber relation records which inputs a map identifies; the quotient–image factorization separates that information loss from the target points that are missed; monomorphism and epimorphism express the same two phenomena through cancellation. Finite iteration and real functional equations then study what happens when such maps are composed repeatedly, but the basic structural data remain the fibers, the image, and the way composition preserves or destroys distinguishability.

More Function Exercises: Periodicity, Piecewise Definitions, Images, and Functional Equations

N-20230117

§48.1 Continuing the function chapter

This note continues the function exercises from the previous day. The pages mix handwritten solutions with printed exercises. The recurring theme is to translate each problem into one of several languages:

formula, graph, image/preimage, iteration, functional equation.

Many errors in such problems come from using a formula while ignoring its domain or from treating a functional equation as if it held only for special values.

§48.2 Piecewise functions

The first page includes piecewise-defined functions. For example, a function may be defined differently on intervals such as

$$x < 0, \qquad x = 0, \qquad x > 0.$$

When studying continuity or solving equations involving such a function, one must first decide which case the input belongs to. If the input itself is an expression such as $f(x)$ or $x + 1$, the relevant case may change during the calculation.

The safe method is:

(i) identify the possible regions of the input;

(ii) solve the equation in each region;

(iii) check that the solution actually belongs to the assumed region.

§48.3 Periodicity from functional equations

A relation such as

$$f(x+1) = 1 - f(x)$$

implies

$$f(x+2) = 1 - f(x+1) = f(x).$$

Thus the function is periodic with period 2. The note uses this style of argument in several places: apply the same equation repeatedly until the input returns to a familiar form.

If instead

$$f(x+T) = f(x)$$

for all x, then T is a period. The smallest positive period, if it exists, is called the fundamental period.

§48.4 Graph transformations

The pages include small sketches of graphs. The standard transformations are:

$$
\begin{aligned}
y &= f(x-a) \quad \text{shift right by } a,\\
y &= f(x)+b \quad \text{shift up by } b,\\
y &= -f(x) \quad \text{reflect across the } x\text{-axis},\\
y &= f(-x) \quad \text{reflect across the } y\text{-axis}.
\end{aligned}
$$

Absolute values fold graphs:

$$y = |f(x)|$$

reflects the negative part upward, while

$$y = f(|x|)$$

copies the right half to the left.

§48.5 Image of an interval

For a continuous function on an interval, the image is also an interval. This is the intermediate value principle in set language. If the function is monotone on the interval, then the endpoints determine the image directly.

The note's exercises often ask for the range of an expression. The method is to rewrite it as a function of one variable, find monotonicity or extrema, and then translate the result back into an interval.

§48.6 Functional equations and substitution

The handwritten solutions repeatedly substitute special values such as $x = 0$, $y = 0$, or $y = x$. This is a standard strategy. For a functional equation in two variables,

$$F(x, y, f(x), f(y), f(x+y)) = 0,$$

choosing simple values isolates pieces of the function.

For example, if

$$f(x+y) = f(x) + f(y) + c,$$

then setting $x = y = 0$ determines $f(0)$, and subtracting a constant can reduce the equation to the additive Cauchy equation.

§48.7 Counting functions between finite sets

The printed exercise page also touches finite functions. If $|A| = m$ and $|B| = n$, then the number of functions $A \to B$ is

$$n^m,$$

because each element of A independently chooses one image in B.

If one requires injectivity and $m \le n$, the number is

$$n(n-1)\cdots(n-m+1).$$

If one requires bijectivity and $m = n$, the number is

$$n!.$$

§48.8 A compact bridge

The note is a training page for reading functions flexibly. A function may be a formula, a graph, a rule on a finite set, or an unknown object constrained by an equation. Each representation suggests a different method: casework for piecewise definitions, transformations for graphs, counting for finite domains, and substitution for functional equations.

§48.9 Piecewise functions as glued maps

A piecewise function is built by gluing formulas on regions. The important checks occur at boundaries: continuity, differentiability, or jump behavior depends on whether the pieces match there.

§48.10 Periodicity from generated transformations

Relations such as $f(x+a) = f(x)$ or $f(c-x) = f(x)$ generate translations and reflections of the line. The resulting group action partitions the domain into orbits.

§48.11 Images of intervals

For continuous functions, the image of an interval is an interval. If the interval is compact, the image is compact, so maxima and minima are attained. This is the topological reason range problems often reduce to endpoint and critical-point checks.

§48.12 Local definitions and global function behavior

Piecewise definitions, periodicity, graph transformations, images, and functional equations all ask how local rules glue into global behavior over the domain.

§48.13 Functional equations as dynamics

A functional equation often specifies a transformation rule rather than a direct formula. For example, equations involving $f(x+1)$, $f(2x)$, or $f(f(x))$ are naturally studied by iteration. The orbit of x under a transformation T is

$$x,\ T(x),\ T^2(x),\ \ldots.$$

A functional equation may prescribe how f behaves along these orbits.

For instance, if

$$f(x+1) = f(x),$$

then f is constant along orbits of the translation $T(x) = x+1$. Periodicity is invariance under a group action of $\mathbb{Z}$.

§48.14 Cauchy's equation and regularity

Cauchy's equation is

$$f(x+y) = f(x) + f(y).$$

If f is continuous, measurable, or bounded on an interval, then

$$f(x) = cx.$$

But without regularity assumptions, there are wild solutions constructed using a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$.

This shows why elementary functional equations often include hidden conditions such as continuity or monotonicity. These conditions are not technical decorations; they exclude pathological solutions arising from the axiom of choice.

§48.15 Symbolic dynamics

Piecewise-defined functions and iteration can lead to symbolic dynamics. If an interval map sends subintervals across one another, one can encode an orbit by a sequence of symbols recording which subinterval contains each iterate.

The shift map

$$\sigma((a_0, a_1, a_2, \dots)) = (a_1, a_2, a_3, \dots)$$

is a central model. Complicated interval dynamics can often be compared to shifts on symbol spaces.

Thus a piecewise function graph in an elementary note can be the beginning of chaos, coding of orbits, and topological entropy.

§48.16 Rigidity from regularity

Many functional equations have many algebraic solutions but few regular solutions. The pattern is:

Equation	Algebraic freedom	Regularity conclusion
$f(x+y) = f(x) + f(y)$	Hamel-linear maps	$f(x) = cx$ under continuity
$f(xy) = f(x) + f(y)$	Logarithmic freedom over the multiplicative group	$f(x) = c \log x$ under continuity
$f(x+T) = f(x)$	Arbitrary values on a fundamental domain	Periodic structure under smoothness

The original piecewise and functional-equation exercises should therefore be read as local training for a general principle: algebraic equations become meaningful only after the allowed regularity class is specified.

§48.17 Piecewise functions and partitions

A piecewise function is a function defined after partitioning the domain into regions. The mathematical issue is not only which formula applies, but how the pieces meet. Discontinuities occur at boundaries of the partition; differentiability requires matching first-order behavior across boundaries.

For example, if

$$f(x) = \begin{cases} f_1(x), & x < a, \\ f_2(x), & x \geq a, \end{cases}$$

then continuity at a requires

$$\lim_{x \to a^-} f_1(x) = f_2(a).$$

Differentiability additionally requires matching slopes. This is a local gluing condition.

§48.18 Functional equations and invariants

A functional equation often says that f is invariant or equivariant under transformations. The equation

$$f(x+T)=f(x)$$

says f is invariant under the translation action of $T\mathbb{Z}$. The equation

$$f(ax)=bf(x)$$

says f is homogeneous with respect to scaling.

Thus solving a functional equation often means finding the invariants of a group or semigroup action. This perspective makes the original examples more natural: periodicity, parity, and scaling are all symmetry constraints.

Fixed Points, Convergent Iteration, and Compactness Methods

N-20230316

§49.1 A fixed point does not by itself explain an iteration

A fixed point of a map $T : X \to X$ is a point x^* satisfying

$$T(x^*) = x^*.$$

If an iteration

$$x_{n+1} = T(x_n)$$

converges to x and T is continuous, then

$$T(x) = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = x.$$

Thus every convergent orbit ends at a fixed point. The converse is false: the existence of a fixed point says nothing by itself about whether an orbit converges to it.

For example,

$$T(x) = -x$$

has the fixed point 0, but an orbit starting from $x_0 \neq 0$ alternates between x_0 and $-x_0$. The map

$$T(x) = x$$

has every point fixed, so there is no unique target for iteration. A useful fixed-point theorem must therefore specify which mechanism supplies existence, uniqueness, and convergence.

§49.2 Contractions make successive points geometrically closer

Let (X, d) be a metric space. A map $T : X \to X$ is a contraction when there is a constant $0 < c < 1$ such that

$$d(Tx, Ty) \le c\, d(x, y)$$

for all $x, y \in X$.

Starting from x_0, define

$$x_{n+1} = T(x_n).$$

The contraction estimate gives

$$d(x_{n+1}, x_n) \le c\, d(x_n, x_{n-1}) \le \cdots \le c^n d(x_1, x_0).$$

For $m > n$, the triangle inequality then yields

$$\begin{aligned} d(x_m, x_n) &\le \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \\ &\le d(x_1, x_0) \sum_{k=n}^{m-1} c^k \\ &\le \frac{c^n}{1-c} d(x_1, x_0). \end{aligned}$$

The orbit is therefore Cauchy. This is the key analytic fact; all later conclusions depend on where that Cauchy sequence is allowed to converge.

§49.3 Completeness supplies the point that the estimates approach

A metric space is complete when every Cauchy sequence converges to a point of the space.

Theorem 49.3.1 (Banach contraction principle)
Let (X, d) be a nonempty complete metric space and let $T : X \to X$ be a contraction. Then T has a unique fixed point x^*. For every $x_0 \in X$, the iterates $x_{n+1} = T(x_n)$ converge to x^*.

Proof. The previous estimate shows that (x_n) is Cauchy, so completeness gives a limit $x^* \in X$. A contraction is Lipschitz and hence continuous, so

$$T(x^*) = T\left(\lim x_n\right) = \lim T(x_n) = \lim x_{n+1} = x^*.$$

If x and y are fixed points, then

$$d(x, y) = d(Tx, Ty) \le c\, d(x, y).$$

Since $c < 1$, this forces $d(x, y) = 0$, hence $x = y$. □

Completeness cannot be omitted. On the incomplete space $(0, 1)$, the map

$$T(x) = \frac{x}{2}$$

is a contraction with constant $1/2$ and maps $(0, 1)$ into itself. Its iterates satisfy

$$x_n = 2^{-n}x_0 \longrightarrow 0,$$

but $0 \notin (0, 1)$. The contraction has no fixed point in the declared space. The estimates have located a missing boundary point.

§49.4 The proof gives two practical stopping bounds

Let x^* be the fixed point. Passing $m \to \infty$ in the Cauchy estimate gives the a priori bound

$$\boxed{d(x_n, x^*) \le \frac{c^n}{1-c} d(x_1, x_0).}$$

It predicts in advance how many iterations are sufficient.

There is also an a posteriori bound that uses the most recent step. Since

$$d(x_{n+k+1}, x_{n+k}) \le c^k d(x_{n+1}, x_n),$$

we obtain

$$\begin{aligned} d(x_n, x^*) &\le \sum_{k=0}^{\infty} d(x_{n+k+1}, x_{n+k}) \\ &\le \frac{1}{1-c} d(x_{n+1}, x_n). \end{aligned}$$

Thus

$$\boxed{d(x_n, x^*) \le \frac{d(x_{n+1}, x_n)}{1 - c}.}$$

This gives a rigorous stopping rule using quantities already computed by the iteration.

§49.5 The equation $x = \cos x$ is a complete example

Consider

$$T(x) = \cos x$$

on the interval $[0, 1]$. Since

$$\cos 1 \le \cos x \le 1,$$

the map sends $[0, 1]$ into itself. By the mean value theorem,

$$|\cos x - \cos y| \le \sin 1\, |x - y|$$

for $x, y \in [0, 1]$. Because $\sin 1 < 1$, the map is a contraction with

$$c = \sin 1 \approx 0.84147.$$

The interval is complete, so there is a unique solution

$$x^* = \cos x^*$$

in $[0, 1]$, and every initial value in the interval converges to it.

Starting from $x_0 = 1$,

$$x_1 = \cos 1, \qquad x_2 = \cos(\cos 1), \qquad \dots$$

The limit is

$$x^* \approx 0.7390851332.$$

If two successive iterates differ by less than $(1 - c)10^{-6}$, the a posteriori estimate guarantees an absolute error below 10^{-6}. The theorem therefore supplies not only existence and uniqueness but also a certified numerical algorithm.

The choice of fixed-point formulation matters. The equation $f(x) = 0$ can often be rewritten as $x = T(x)$ in many ways. One formulation may be contractive while another diverges. Fixed-point iteration is therefore partly the art of choosing a map with a small Lipschitz constant on an invariant complete set.

§49.6 Picard iteration moves the theorem to a function space

Consider the initial-value problem

$$y'(t) = f(t, y(t)), \qquad y(t_0) = y_0.$$

Integrating gives the equivalent equation

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds.$$

On the Banach space $C([t_0 - h, t_0 + h])$ with the supremum norm, define

$$(Ty)(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds.$$

Assume that f is Lipschitz in its second variable:

$$|f(t,u) - f(t,v)| \le L|u-v|.$$

Then

$$\begin{aligned}\|Ty - Tz\|_\infty &\le \sup_t \int_{t_0}^t |f(s,y(s)) - f(s,z(s))|\,ds \\ &\le Lh\,\|y-z\|_\infty.\end{aligned}$$

If $h < 1/L$, the operator is a contraction. One also chooses a closed ball of functions on which T maps the ball into itself; this follows from a bound on $|f|$ on a suitable rectangle around (t_0, y_0).

Banach's theorem now gives a unique fixed point of T, hence a unique local solution of the differential equation. The iterates

$$y_{n+1}(t) = y_0 + \int_{t_0}^t f(s, y_n(s))\,ds$$

are the Picard approximations. The same geometric contraction estimate that solved $x = \cos x$ now proves existence and uniqueness in an infinite-dimensional space.

§49.7 Topological existence does not provide Banach's algorithm

Banach's theorem is not the only route to a fixed point. Brouwer's theorem says that every continuous map from a nonempty compact convex subset of $\mathbb{R}^n$ to itself has a fixed point. No metric contraction is required.

The difference in conclusions is substantial. The identity map on a closed ball satisfies Brouwer's hypotheses and has every point fixed. A rotation of a disk fixes its center but preserves distances rather than shrinking them. Brouwer gives existence, but not uniqueness, a convergence rate, or even a canonical iteration that approaches a fixed point.

This distinction matters in applications. If the goal is a certified numerical method, the Banach hypotheses are valuable precisely because they are stronger. If the map is continuous on a compact convex region but cannot be made contractive, Brouwer may still prove that a solution exists, though a separate method is needed to locate it.

§49.8 Schauder replaces finite-dimensional compactness by a compact map

Closed bounded sets in infinite-dimensional Banach spaces are usually not compact, so Brouwer cannot be applied directly. Schauder's fixed-point theorem uses compactness of the operator instead.

Theorem 49.8.1 (Schauder fixed-point theorem)
Let C be a nonempty closed convex subset of a Banach space. If $T : C \to C$ is continuous and $T(C)$ is relatively compact, then T has a fixed point.

A typical source of compactness is an integral operator. On $C([0,1])$, let

$$(Tu)(x) = g(x) + \int_0^1 K(x,s)F(s,u(s))\,ds,$$

where K, F, and g are continuous and a closed ball C is chosen so that $T(C) \subseteq C$. Uniform bounds on Tu and uniform continuity of K give equicontinuity of $T(C)$. Arzelà–Ascoli then shows that $T(C)$ is relatively compact.

Schauder therefore yields a solution of the integral equation even when no Lipschitz estimate makes T contractive. What is lost is uniqueness and convergence of the naive iteration. Several fixed points may exist.

§49.9 Krasnoselskii separates the shrinking and compact parts

Many nonlinear equations produce an operator that is neither contractive nor compact as a whole but decomposes as

$$T = A + B.$$

Krasnoselskii's theorem applies on a closed bounded convex set C when A is a contraction, B is continuous and compact, and

$$Ax + By \in C$$

for all $x, y \in C$ under the relevant version of the theorem.

The decomposition reflects two different mechanisms. The map A controls metric instability; the map B supplies compactness. One way to see the proof strategy is to fix $y \in C$ and solve

$$x = Ax + By.$$

For fixed y, the right-hand side is contractive in x, so Banach gives a unique solution $x = R(y)$. Compactness of B is then used to show that the induced map R has a fixed point by a Schauder-type argument.

This theorem is not merely a longer list entry after Banach and Schauder. It is designed for equations whose linear or dissipative part shrinks distances while an integral or smoothing term is compact.

When an equation has been rewritten as $x = T(x)$, the next step is to identify which of these mechanisms is actually available.

If one can prove

$$d(Tx, Ty) \leq c\, d(x, y), \qquad c < 1,$$

then Banach gives existence, uniqueness, global convergence of iteration, and explicit error bounds. If only continuity and finite-dimensional compact convex invariance are available, Brouwer gives existence. In an infinite-dimensional problem, relative compactness of the image points toward Schauder. A natural decomposition into a contraction plus a compact map points toward Krasnoselskii.

The hypotheses are therefore not technical decorations around the same conclusion. They encode different reasons a fixed point must exist, and those reasons determine what else one is entitled to claim about uniqueness, approximation, and stability.

VI

Combinatorics and Models

Part VI: Contents

Counting Principles, Binomial Identities, and Mixed Problem Solving

N-20220720

§50.1 Counting principles

The note begins with the standard counting principles:

Addition principle. If a task can be done in disjoint cases with $m_1, m_2, \ldots, m_n$ possibilities, then the total number is $m_1 + \cdots + m_n$.

Multiplication principle. If a task is done in stages with $m_1, m_2, \ldots, m_n$ choices at each stage, then the total number is $m_1 m_2 \cdots m_n$.

Inclusion–exclusion. For sets $A_1, \ldots, A_n$, one adds single sizes, subtracts pairwise intersections, adds triple intersections, and so on.

A Venn diagram in the note reviews the three-set formula:

$$\begin{aligned}|A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.\end{aligned}$$

§50.2 A divisibility example

One problem asks how many positive integers not exceeding 2021 are neither multiples of 6 nor multiples of 8. The excluded set is

$$A \cup B,$$

where A is the set of multiples of 6, and B is the set of multiples of 8. Since

$$\operatorname{lcm}(6, 8) = 24,$$

we have

$$|A| = 336, \qquad |B| = 252, \qquad |A \cap B| = 84.$$

Thus the number excluded is

$$336 + 252 - 84 = 504,$$

and the answer is

$$2021 - 504 = 1517.$$

This is exactly what inclusion–exclusion is for: avoid double-counting the common multiples.

§50.3 Permutations and combinations

The page distinguishes arrangements and selections:

$$A_n^m = \frac{n!}{(n-m)!}, \qquad C_n^m = \frac{n!}{m!(n-m)!}.$$

A sequence problem asks for the number of 10-term sequences with $x + y = 10$ and $x - y = 4$, giving $x = 7, y = 3$. Therefore the number of sequences is

$$\binom{10}{3} = 120.$$

The thought process is: once the number of one kind of symbol is fixed, the sequence is determined by choosing its positions.

§50.4 Binomial theorem

The binomial theorem is

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

Several standard identities follow immediately:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n,$$

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0,$$

and by differentiating

$$(1+x)^n$$

and setting $x = 1$,

$$\sum_{k=0}^{n} k \binom{n}{k} = n2^{n-1}.$$

The page writes these identities as examples of how a coefficient formula becomes a counting tool.

§50.5 Polynomial coefficients

A worked example asks for the constant term or a particular coefficient in an expansion such as

$$(2x-2y)^5.$$

The term involving $x^{5-k}y^k$ is

$$\binom{5}{k} (2x)^{5-k} (-2y)^k.$$

Thus the coefficient of x^2y^3 comes from $k = 3$:

$$\binom{5}{3} 2^2 (-2)^3 = -320.$$

The note's point is that coefficient problems become simple once the target exponent determines k.

§50.6 Further contest examples

The later pages mix counting with algebra, inequalities, and geometry. There are examples involving choosing books from different subjects, solving inequalities by sign charts, and geometry problems involving similar triangles. The unifying skill is case management: one must decide whether the problem is a sum of cases, a product of stages, or a coefficient extraction.

§50.7 Binomial identities by stories

The identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

comes from choosing a committee of size k from n people. Either a distinguished person is not chosen, or that person is chosen. This kind of proof is often clearer than algebraic manipulation because it explains what the two terms mean.

§50.8 When to use multiplication and when to use addition

The addition principle applies to disjoint alternatives. The multiplication principle applies to successive independent choices. Many counting mistakes come from using multiplication when cases overlap, or using addition when a task has stages. A good habit is to write a sentence: "first choose ..., then choose ..." for multiplication, and "either ..., or ..." for addition.

§50.9 A broader reading

Elementary counting is a first meeting with combinatorial species and generating functions. Addition corresponds to disjoint union, multiplication corresponds to Cartesian product, and binomial coefficients are the coefficients of a generating function. This is why the binomial theorem is not just algebra: it is also a counting machine.

§50.10 Counting principles as operations on combinatorial classes

The addition and multiplication principles are not merely classroom rules. They are the first operations on combinatorial classes. If $\mathcal{A}$ and $\mathcal{B}$ are disjoint classes, then

$$|\mathcal{A} \sqcup \mathcal{B}| = |\mathcal{A}| + |\mathcal{B}|.$$

If an object is built by choosing one object from $\mathcal{A}$ and one from $\mathcal{B}$, then

$$|\mathcal{A} \times \mathcal{B}| = |\mathcal{A}||\mathcal{B}|.$$

Thus addition corresponds to disjoint union, and multiplication corresponds to Cartesian product.

This viewpoint is useful because it explains why formulas should follow from constructions. Before computing a number, identify whether the object is a disjoint alternative, a staged product, a quotient by symmetry, or a coefficient in a generating function.

§50.11 Inclusion–exclusion as Möbius inversion

The formula

$$|A \cup B| = |A| + |B| - |A \cap B|$$

is a small instance of Möbius inversion on the Boolean lattice. For a finite family $A_1, \dots, A_n$, the number of elements lying in at least one A_i is

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{\varnothing \neq S \subseteq [n]} (-1)^{|S|+1} \left|\bigcap_{i \in S} A_i\right|.$$

The alternating signs are the Möbius function of the subset poset:

$$\mu(S,T) = (-1)^{|T|-|S|}.$$

So inclusion–exclusion is not a trick for Venn diagrams; it is inversion of the operation that replaces exact membership data by cumulative membership data.

§50.12 Indicator functions and probability language

For a finite universe U, a set $A \subseteq U$ has indicator function

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

The two-set inclusion–exclusion identity is the pointwise identity

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

Summing over U gives the cardinality formula.

This is also the beginning of probability. If X is uniformly chosen from U, then

$$\mathbb{P}(X \in A) = \frac{|A|}{|U|} = \mathbb{E}[\mathbf{1}_A(X)].$$

Counting and expectation become the same operation after normalization.

§50.13 Generating functions for binomial identities

The binomial theorem says

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

The coefficient of x^k counts subsets of size k. Differentiating and setting $x = 1$ gives

$$\sum_{k=0}^{n} k \binom{n}{k} = n 2^{n-1},$$

which counts pairs (S, s) where $S \subseteq [n]$ and $s \in S$: choose s first, then choose the rest of S.

This illustrates the right use of generating functions: algebraic operations on series encode operations on counted objects.

§50.14 Counting up to symmetry: Burnside's lemma

If a finite group G acts on a finite set X, the number of orbits is

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

This is Burnside's lemma. It explains why one cannot always divide by the number of symmetries: objects with nontrivial stabilizers have smaller orbits.

Elementary examples include necklaces, colorings of polygons, and arrangements considered up to rotation or reflection. The advanced lesson is that symmetry changes the unit of counting from elements to orbits.

§50.15 Counting through structure and symmetry

The counting principles become powerful when read structurally. Addition is disjoint union, multiplication is product, inclusion–exclusion is Möbius inversion, binomial coefficients are generating-function coefficients, and symmetry counting is orbit counting. The elementary skill is to decide which structure the problem is really using before doing arithmetic.

Counting by Logic, Projection, Inversion, and Generating Series

N-20221205

§51.1 Set operations are logical operations written geometrically

For a finite universe U, the indicator of a set $A \subseteq U$ is

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then

$$|A| = \sum_{x \in U} \mathbf{1}_A(x).$$

Union, intersection, and complement become pointwise arithmetic:

$$\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B,$$

$$\mathbf{1}_{A^c} = 1 - \mathbf{1}_A,$$

$$\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

De Morgan's law follows immediately:

$$1 - \mathbf{1}_{A \cup B} = (1 - \mathbf{1}_A)(1 - \mathbf{1}_B).$$

The set identity

$$(A \cup B)^c = A^c \cap B^c$$

is therefore the logical identity

$$\neg(P \vee Q) = (\neg P) \wedge (\neg Q).$$

This translation is more than notation. It allows a set identity to be verified pointwise and then summed to obtain a counting identity. The same habit will later turn inclusion–exclusion into an algebraic inversion formula.

§51.2 Addition counts disjoint cases; multiplication counts successive choices

If a finite collection is partitioned into disjoint cases C_i, then

$$\left| \bigsqcup_i C_i \right| = \sum_i |C_i|.$$

If an object is built through successive independent choices, with m_i options at stage i, then the number of constructions is

$$\prod_i m_i.$$

The first principle counts alternatives; the second counts stages.

Arranging r distinct objects chosen from n gives

$$n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

If order is forgotten, every r-element subset has $r!$ internal orders, so

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The symmetry

$$\binom{n}{r} = \binom{n}{n-r}$$

comes from the complement bijection: choosing the selected elements is equivalent to choosing the omitted ones.

The pigeonhole principle is an averaging consequence of the same counting. If N objects occupy r boxes, some box contains at least

$$\left\lceil \frac{N}{r} \right\rceil$$

objects; otherwise the total would be smaller than N.

§51.3 The binomial theorem counts choices inside a product

Expanding

$$(a+b)^n = (a+b)\cdots(a+b)$$

requires choosing either a or b from each factor. A term $a^{n-r}b^r$ occurs once for every choice of the r factors that contribute b. Hence

$$\boxed{(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r}b^r.}$$

Setting $a = b = 1$ gives

$$\sum_{r=0}^{n} \binom{n}{r} = 2^n.$$

This is also the count of all subsets of an n-element set: each element independently receives the decision “in” or “out.”

Pascal’s identity

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$$

comes from partitioning r-subsets according to whether they contain one distinguished element. The identity is a decomposition of the counted set, not merely a factorial manipulation.

§51.4 Double counting is counting one incidence set through two projections

Let X be a finite set of pairs. The two coordinate projections partition X in different ways, and equality of the resulting totals produces an identity.

For example, count pairs

$$(S, s), \qquad S \subseteq [n], \quad s \in S.$$

Counting first by S gives

$$|X| = \sum_{k=0}^{n} k \binom{n}{k}.$$

Counting first by s, choose the distinguished element in n ways and then choose any subset of the remaining $n-1$ elements:

$$|X| = n2^{n-1}.$$

Thus

$$\boxed{\sum_{k=0}^{n} k \binom{n}{k} = n2^{n-1}.}$$

Vandermonde's identity is another projection count. Split a population into disjoint groups of sizes r and s. Choosing n people altogether can be done directly in

$$\binom{r+s}{n}$$

ways, or by choosing k from the first group and $n-k$ from the second:

$$\boxed{\sum_{k} \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}.}$$

Naming the incidence object explains the summation index and the range automatically.

§51.5 Inclusion–exclusion is a pointwise product expansion

For sets $A_1, \ldots, A_m$, the indicator of the complement of their union is

$$\mathbf{1}_{(\cup_i A_i)^c} = \prod_{i=1}^{m} (1 - \mathbf{1}_{A_i}).$$

Expanding and subtracting from one gives

$$\begin{aligned} \mathbf{1}_{\cup_i A_i} = \sum_i \mathbf{1}_{A_i} - \sum_{i<j} \mathbf{1}_{A_i \cap A_j} \\ + \sum_{i<j<k} \mathbf{1}_{A_i \cap A_j \cap A_k} - \cdots . \end{aligned}$$

Summing over U yields the full inclusion–exclusion formula

$$\left| \bigcup_{i=1}^{m} A_i \right| = \sum_{\varnothing \neq S \subseteq [m]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

The alternating signs arise from expanding a product, and every element is checked locally according to how many sets contain it.

As a quick application, the number of surjections from an n-element set onto an m-element set is

$$\sum_{j=0}^{m}(-1)^j\binom{m}{j}(m-j)^n.$$

Start with all m^n functions and exclude those missing at least one target value. Intersections correspond to missing a specified set of j values, leaving $(m-j)^n$ functions.

§51.6 Boolean Möbius inversion explains the alternating signs structurally

Let functions f, g be defined on subsets of $[m]$, with

$$g(S) = \sum_{T\subseteq S} f(T).$$

This cumulative transform is the zeta transform of the Boolean lattice. Möbius inversion recovers the exact data:

$$\boxed{f(S) = \sum_{T\subseteq S}(-1)^{|S|-|T|}g(T).}$$

The Boolean Möbius function is therefore

$$\mu(T,S) = (-1)^{|S|-|T|} \qquad (T\subseteq S).$$

For $m=3$, the top value is

$$\begin{aligned} f(123) = g(123) - g(12) - g(13) - g(23) \\ + g(1) + g(2) + g(3) - g(\varnothing). \end{aligned}$$

The familiar inclusion–exclusion signs are exactly the coefficients of the inverse transform.

To verify the formula, substitute the definition of g. The coefficient of a fixed $f(R)$ is

$$\sum_{R\subseteq T\subseteq S}(-1)^{|S|-|T|} = (1-1)^{|S|-|R|},$$

which is 1 when $R=S$ and 0 otherwise. Inversion works because all unwanted lower-level contributions cancel.

§51.7 Ordinary generating functions turn decomposition into multiplication

For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$A(x) = \sum_{n\geq 0} a_n x^n.$$

The notation

$$[x^n]A(x) = a_n$$

extracts a coefficient. If an object of size n splits into an object of size k and one of size $n-k$, the number of pairs is the convolution

$$c_n = \sum_{k=0}^{n} a_k b_{n-k},$$

and

$$C(x) = A(x)B(x).$$

The multiplication principle has become multiplication of series.

Vandermonde's identity follows algebraically from

$$(1+x)^r(1+x)^s = (1+x)^{r+s}.$$

Extracting $[x^n]$ from the left gives

$$\sum_k \binom{r}{k}\binom{s}{n-k},$$

while the right gives $\binom{r+s}{n}$. The same identity now has both an incidence-set proof and a coefficient proof.

§51.8 Generating functions solve recurrences and coin-change problems

Let $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}.$$

Set

$$F(x) = \sum_{n\geq 0} F_n x^n.$$

Multiplying the recurrence by x^n and summing for $n \geq 2$ gives

$$F(x) - x = xF(x) + x^2F(x).$$

Therefore

$$\boxed{F(x) = \frac{x}{1-x-x^2}.}$$

Factoring the denominator recovers Binet's formula; alternatively, the rational function generates the sequence without solving each recurrence anew.

For unlimited coins of values $1, 2, 5$, choosing $a, b, c \geq 0$ coins contributes value

$$a + 2b + 5c.$$

The generating function is

$$\frac{1}{(1-x)(1-x^2)(1-x^5)}.$$

The coefficient of x^n counts solutions of

$$a + 2b + 5c = n.$$

For $n = 5$, the four representations are

$$1+1+1+1+1, \quad 2+1+1+1, \quad 2+2+1, \quad 5,$$

so

$$[x^5]\frac{1}{(1-x)(1-x^2)(1-x^5)} = 4.$$

A product of geometric series encodes independent multiplicities of allowed parts.

§51.9 Averaging and random choice can prove existence

The pigeonhole principle says that some box is at least as full as the average. The probabilistic method applies the same idea to a random object: if the expected value of a statistic is M, some object has value at least M.

Let a graph have m edges. Place each vertex independently into one of two parts with probability $1/2$. An edge crosses the cut exactly when its endpoints land in different parts, which occurs with probability $1/2$. If X is the number of crossing edges, linearity of expectation gives

$$\mathbb{E}[X] = \sum_e \mathbb{P}(e \text{ crosses}) = \frac{m}{2}.$$

Therefore at least one cut satisfies

$$X \geq \frac{m}{2}.$$

No independence between edge indicators is required for linearity of expectation.

The argument is constructive in logic even when it does not exhibit the cut: if every cut had fewer than $m/2$ crossing edges, the average could not equal $m/2$.

§51.10 Stirling's formula reveals the scale of the central binomial coefficient

The largest coefficient in the n-th row of Pascal's triangle lies near $n/2$. For even $n = 2m$, Stirling's approximation

$$m! \sim \sqrt{2\pi m}\left(\frac{m}{e}\right)^m$$

gives

$$\begin{aligned}\binom{2m}{m} &= \frac{(2m)!}{(m!)^2} \\ &\sim \frac{\sqrt{4\pi m}(2m/e)^{2m}}{2\pi m(m/e)^{2m}} \\ &= \frac{4^m}{\sqrt{\pi m}}.\end{aligned}$$

Equivalently,

$$\boxed{\binom{n}{\lfloor n/2\rfloor} \sim 2^n\sqrt{\frac{2}{\pi n}}.}$$

Thus the largest layer of the Boolean lattice contains about $2^n/\sqrt{n}$ subsets. The exact identity

$$\sum_k \binom{n}{k} = 2^n$$

counts the whole lattice; the asymptotic formula describes how that mass concentrates around the middle ranks.

§51.11 Species record how labelled structures move under relabelling

A combinatorial species F assigns to each finite label set U a set $F[U]$ of structures, and every bijection

$$\sigma : U \to V$$

transports structures through a bijection

$$F[\sigma] : F[U] \to F[V]$$

compatible with composition. The definition remembers not only how many structures exist, but how they behave under relabelling.

The exponential generating function is

$$F(x) = \sum_{n \geq 0} |F[n]| \frac{x^n}{n!}.$$

For the species SET, there is one set structure on every label set, so

$$\mathrm{SET}(x) = \sum_{n \geq 0} \frac{x^n}{n!} = e^x.$$

For the species SEQ, a label set of size n has $n!$ linear orders, so

$$\mathrm{SEQ}(x) = \sum_{n \geq 0} n! \frac{x^n}{n!} = \frac{1}{1-x}.$$

For $n \geq 1$, a labelled cycle has $(n-1)!$ cyclic orders, hence

$$\mathrm{CYC}(x) = \sum_{n \geq 1} (n-1)! \frac{x^n}{n!} = \sum_{n \geq 1} \frac{x^n}{n} = -\log(1-x).$$

Every permutation decomposes uniquely into an unordered set of disjoint cycles. Species composition therefore gives

$$\mathrm{PERM} = \mathrm{SET}(\mathrm{CYC}).$$

At the level of exponential generating functions,

$$\begin{aligned} \mathrm{PERM}(x) &= \exp(\mathrm{CYC}(x)) \\ &= \exp\big(-\log(1-x)\big) \\ &= \frac{1}{1-x}. \end{aligned}$$

The coefficient of $x^n/n!$ is $n!$, recovering the number of permutations. Addition, product, and composition of labelled structures have become algebraic operations on generating series because relabelling symmetry is built into the objects from the beginning.

Set Problems, Extremal Counting, and Venn-Diagram Reasoning

N-20221207

§52.1 A set construction problem

The first problem defines a finite set

$$A = \{a_1 < a_2 < \cdots < a_k\}$$

of positive integers and builds two derived sets, one from sums and one from differences. The handwritten solution records the important idea: use the union $S \cup T$, and do not try to identify every element exactly. Bound its size from below and above.

The sum set contains at least the ordered chain

$$2a_1,\ a_1 + a_2,\ 2a_2, \ldots,\ a_{k-1} + a_k,\ 2a_k,$$

so

$$|S| \geq 2k - 1.$$

The difference set contains

$$0,\ a_2 - a_1,\ a_3 - a_1, \ldots,\ a_k - a_1,$$

so

$$|T| \geq k.$$

Under the condition of the problem, $S \cap T = \varnothing$, hence

$$|S \cup T| = |S| + |T| \geq 3k - 1.$$

On the other hand, all these numbers lie between 0 and $2a_k$, so

$$|S \cup T| \leq 2a_k + 1.$$

Thus

$$3k - 1 \leq 2a_k + 1.$$

If $a_k \leq 2021$, then

$$k \leq \frac{2 \cdot 2021 + 2}{3} = 1348.$$

The construction

$$A = \{674, 675, \ldots, 2021\}$$

attains this bound, so the maximum is

$$|A| = 1348.$$

§52.2 What this problem teaches

The note's blue comment says that using $S \cup T$ is a good way to avoid proving a complicated extremal statement directly. This is the key lesson. When a problem asks for the largest possible size of a set, it often helps to build auxiliary sets whose sizes are forced by the structure.

A lower bound comes from a construction. An upper bound comes from counting constraints. A complete solution needs both.

§52.3 A Venn-diagram necessary-and-sufficient problem

Another problem defines set difference

$$A - B = \{x : x \in A,\ x \notin B\}$$

and asks for the relation among A, B, C when

$$(A - B) \cup (B - A) \subseteq C.$$

The question asks whether this is equivalent to a condition involving

$$A \subseteq (C - B) \cup (B - C)$$

or

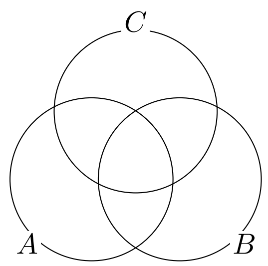
$$A \cap B \cap C = \varnothing.$$

The handwritten solution first draws a Venn diagram, then rewrites the pieces algebraically. The symmetric difference

$$(A - B) \cup (B - A)$$

is the part where membership in A and B differs. If this is contained in C, then outside C the sets A and B must agree.

The correct way to solve such problems is to analyze each of the eight Venn regions. The note records a false start and then corrects it, which is valuable: set identities are easy to get wrong if one reasons only verbally.



A Venn diagram is a region bookkeeping device.

Figure 52.1: For three sets one should reason region by region, not by visual guesswork alone.

§52.4 Difference identities

One useful identity appearing in the handwritten reasoning is

$$B - C = B \cap C^c.$$

Therefore

$$B - C - (A \cap B) = B \cap C^c \cap (A \cap B)^c.$$

Using De Morgan,

$$(A \cap B)^c = A^c \cup B^c,$$

so

$$B \cap C^c \cap (A^c \cup B^c) = (B \cap C^c \cap A^c) \cup (B \cap C^c \cap B^c).$$

The second term is empty, hence

$$B - C - (A \cap B) = B \cap A^c \cap C^c.$$

This is exactly the kind of algebraic cleanup that a Venn diagram suggests but does not replace.

§52.5 Coordinate and algebraic exercises

The later pages contain a mixture of algebraic and coordinate exercises. They include problems about quadratic expressions, tangent or chord diagrams, and conditions that must be translated into equations. The handwritten computations repeatedly use the same pattern:

(i) introduce the unknowns explicitly;

(ii) translate each condition into an equation or inequality;

(iii) reduce by substitution or by completing the square;

(iv) check equality or boundary cases.

Several sketches show lines touching curves or intersecting regions. Rather than reproducing them as raw images, the important content is the method: a picture tells us which algebraic condition to impose, such as tangency, parallelism, or containment.

§52.6 A note on proof writing

The red handwritten reflection says that a purely diagrammatic method is intuitive but can be unreliable. The mature solution uses the diagram to discover the statement, then uses set algebra to prove it. This is an excellent general lesson. Geometry and diagrams are for finding the path; formal manipulations are for making the path safe.

§52.7 Extremal set problems: why constructions matter

In an extremal counting problem, an upper bound is only half the solution. One must also give a construction showing the bound is sharp. The note's sum-set and difference-set arguments follow this pattern: first prove that the derived sets must contain enough elements; then exhibit a set A for which the lower bound is actually attained.

This is a common combinatorial habit. Inequalities show impossibility beyond a threshold. Constructions show that the threshold is real.

§52.8 Venn diagrams versus proofs

Venn diagrams are excellent for discovering set identities, but the proof should eventually be written elementwise. For example,

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

is proved by taking an arbitrary x and translating membership:

$$x \in A \setminus (B \cup C) \iff x \in A,\ x \notin B,\ x \notin C.$$

This is exactly the condition for membership in the right-hand side. The diagram gives intuition; the element chase gives proof.

§52.9 A wider frame

This note is about extremal and logical reasoning. The extremal problem uses auxiliary sets and cardinality bounds. The Venn problem uses Boolean algebra and region decomposition. Both are early forms of a common method: replace a vague configuration by a finite list of regions or quantities that can be counted exactly. This is the same spirit behind combinatorial proofs, measure decompositions, and even sigma-algebra calculations later in analysis.

§52.10 Extremal problems: upper bound plus construction

The sum-set/difference-set problem has the standard shape of extremal combinatorics. One proves an upper bound by showing that certain auxiliary objects must fit inside a finite container. Then one gives a construction meeting the bound.

This two-part structure is essential:

$$\text{upper bound} + \text{matching construction} = \text{exact extremal value.}$$

A construction alone proves possibility; an upper bound alone proves limitation. A complete extremal solution needs both.

§52.11 Additive combinatorics viewpoint

For a finite set $A \subseteq \mathbb{Z}$, the sum set and difference set are

$$A + A = \{a + a' : a, a' \in A\}, \qquad A - A = \{a - a' : a, a' \in A\}.$$

The size of these sets measures additive structure. Arithmetic progressions have unusually small sumsets. Random sets usually have much larger sumsets.

A central theorem in additive combinatorics is Freiman's philosophy: if $|A + A|$ is small compared with $|A|$, then A must resemble a generalized arithmetic progression. The elementary problem is a small concrete case of studying what set structure is forced by sum and difference constraints.

§52.12 Cauchy–Davenport shadow

In a prime cyclic group $\mathbb{Z}/p\mathbb{Z}$, the Cauchy–Davenport theorem says

$$|A+B| \ge \min(p,\ |A|+|B|-1).$$

This is the modular analogue of the simple lower bound that a sumset of ordered integer sets contains a chain of at least $|A|+|B|-1$ elements.

The theorem shows that the lower-bound logic in the note is not accidental. Sumsets are forced to grow unless the ambient group wraps around or the set has special structure.

§52.13 Boolean atoms and Venn regions

For three sets A, B, C, the universe is partitioned into at most eight atoms:

$$A^{\epsilon_1} \cap B^{\epsilon_2} \cap C^{\epsilon_3}, \qquad \epsilon_i \in \{0,1\},$$

where $A^1 = A$ and $A^0 = A^c$. Every Boolean expression in A, B, C is a union of some of these atoms.

This gives a systematic proof method: reduce both sides of a set identity to the same atoms. Venn diagrams are pictures of this atomic decomposition; elementwise proofs are the formal version.

§52.14 Symmetric difference as addition mod two

The symmetric difference

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

behaves like addition modulo 2 on characteristic functions:

$$\mathbf{1}_{A\triangle B} = \mathbf{1}_A + \mathbf{1}_B \pmod 2.$$

Thus the power set $\mathcal{P}(U)$ becomes a vector space over $\mathbb{F}_2$ under symmetric difference. Intersection acts like multiplication.

This explains why expressions such as $(A-B)\cup(B-A)$ have algebraic structure: they are not just set differences, but mod-two sums of predicates.

§52.15 Sperner and intersecting-family context

Many extremal set problems ask for the largest family avoiding a containment or disjointness pattern. Sperner's theorem says that an antichain $\mathcal{F} \subseteq \mathcal{P}([n])$ satisfies

$$|\mathcal{F}| \le \binom{n}{\lfloor n/2 \rfloor}.$$

The largest example is the middle layer.

For intersecting families, the Erdős–Ko–Rado theorem says that if $\mathcal{F} \subseteq \binom{[n]}{k}$ is pairwise intersecting and $n \ge 2k$, then

$$|\mathcal{F}| \le \binom{n-1}{k-1}.$$

The extremal example is a star: all k-sets containing a fixed element.

These theorems are advanced versions of the note's method: identify the forbidden configuration, prove an upper bound, and classify sharp examples.

§52.16 Shifting and stability

Shifting transforms a set family into a more ordered one without changing its size or destroying the relevant forbidden configuration. Repeated shifting often reduces an extremal problem to a canonical family.

A stability theorem asks more than the maximum: if a family is close to maximum size, must it be structurally close to an extremal example? This is the research-level continuation of checking equality cases in an elementary inequality.

§52.17 Extremal sets, Boolean algebra, and stability

The note combines two important habits. Extremal counting uses auxiliary sets, upper bounds, and sharp constructions. Venn reasoning uses Boolean atoms and set algebra. At a higher level these become additive combinatorics, extremal set theory, vector-space structure over $\mathbb{F}_2$, and stability methods.

Set-Theoretic and Combinatorial Exercises after the First Lecture

N-20221209

§53.1 What this page is doing

This note is a handwritten exercise continuation of the set-and-counting lecture. The problems involve floor functions, set-builder notation, binomial identities, lattice or coordinate sets, and counting arguments. The common goal is to learn how to translate a compact set expression into something countable.

§53.2 A floor-function example

One example asks for a set defined by inequalities and floor functions. The handwritten solution separates the problem into intervals because

$$\lfloor x \rfloor = k \quad \Longleftrightarrow \quad k \le x < k + 1.$$

This is the essential method. Any problem involving $\lfloor x \rfloor$ should first be split into integer strips.

For instance, if a condition contains

$$3\lfloor x \rfloor - 5 < 0,$$

then

$$\lfloor x \rfloor < \frac{5}{3},$$

so

$$\lfloor x \rfloor \le 1.$$

One then intersects this with the remaining condition, such as a quadratic inequality. The final answer is an interval or a finite union of intervals.

§53.3 A binomial counting identity

Another exercise invokes the binomial theorem. The handwritten note uses the idea that terms in a sum can be counted by choosing which positions have a particular property. A typical identity is

$$\sum_{j=0}^{n} \binom{n}{j} = 2^n,$$

or, with signs,

$$\sum_{j=0}^{n} (-1)^j \binom{n}{j} = 0.$$

The important thought is not the formula alone. The coefficient $\binom{n}{j}$ counts ways of choosing j positions among n, and the binomial theorem packages all choices simultaneously.

§53.4 Solving a system by discriminants

One problem gives a two-variable system whose equations can be transformed into a quadratic with a parameter p. The handwritten solution examines the discriminant

$$\Delta \geq 0$$

to decide when real solutions exist.

This is a recurring analytic-algebraic pattern: if a system can be reduced to a quadratic equation, existence of real solutions is controlled by the discriminant. The solution set changes when the discriminant becomes zero; this is the boundary between two real roots and no real roots.

§53.5 A prime-number exercise

Another exercise involves primes p, q. The handwritten reasoning uses parity and basic divisibility. The standard approach is:

(i) reduce the equation modulo a small integer such as $2, 3, 4$;

(ii) use the fact that the only even prime is 2;

(iii) separate small exceptional cases before treating odd primes.

This is the same habit developed in earlier divisibility notes: do not solve a Diophantine equation blindly. First use congruences to force the shape of the variables.

§53.6 A nested-set exercise

The note includes a Russian sentence asking to "describe the next set" or "write the following set in ordinary language." The examples involve finite sets such as

$$\varnothing, \qquad \{a_1\}, \qquad \{a_1, a_2\}, \qquad \ldots$$

and collections of subsets. The intended lesson is the difference between an element and a subset:

$$a \in A \quad \text{versus} \quad \{a\} \subseteq A.$$

Confusing these two is one of the most common early set-theory mistakes.

§53.7 A recurrence of sets

One exercise defines sets recursively, for example

$$A_{k+1} = A_k \cup \{x_{k+1}\}$$

or by adding all new subsets involving a new element. The handwritten solution proves the pattern by induction: assume the formula for A_k, then show the construction gives A_{k+1}.

This is the set-theoretic analogue of a recurrence relation. Instead of numbers changing with k, collections of objects change with k.

§53.8 An intersection-counting problem

The final exercise on the second page asks whether a selected element belongs to an intersection of many sets. The handwritten solution builds sets $B(A_i)$ and counts their sizes, arriving at a large product-like number. The exact numerical expression is less important than the logic:

(i) translate membership into a condition;

(ii) count how many sets contain a fixed element;

(iii) use complements or intersections to isolate the desired elements.

This is a typical finite-set counting method: count incidence pairs in two ways.

§53.9 Set-builder notation as a translation problem

A complicated set-builder expression should be read in stages. First identify the universe of discourse. Then translate each condition into an inequality, divisibility condition, or membership condition. Finally decide whether the problem is asking for the set itself, its size, or a proof of equality.

For example, a condition involving $\lfloor x \rfloor$ is not smooth; it divides the real line into half-open intervals. A condition involving binomial coefficients often asks for a counting interpretation rather than algebraic simplification.

§53.10 Recursive set construction

When a set is defined recursively, the first few stages should be computed explicitly. This reveals whether the construction is increasing, periodic, stabilizing, or producing genuinely new elements. After that, induction is the natural proof method.

The general pattern is:

$$S_0 \subseteq S_1 \subseteq S_2 \subseteq \cdots$$

if the rule is monotone. If the universe is finite, an increasing chain must eventually stabilize. This is a useful idea behind many finite combinatorial processes.

§53.11 Further direction

The exercises are elementary, but they are training the language of discrete mathematics. Floor functions partition the real line into strips; binomial coefficients count choices; discriminants detect existence; recursive set definitions invite induction; incidence counting turns membership questions into arithmetic. The high-level skill is always the same: replace notation by a structure that can actually be counted.

§53.12 Floor functions as lattice-point counts

A floor expression such as

$$\left\lfloor \frac{an+b}{m} \right\rfloor$$

counts how many integer thresholds lie below a rational expression. Sums of floor functions often count lattice points below a line. For instance,

$$\sum_{k=0}^{m-1} \left\lfloor \frac{ak}{m} \right\rfloor$$

counts integer points under a rational-slope diagonal in a rectangle.

This links elementary floor exercises to Ehrhart theory, where one studies

$$L_P(n) = |nP \cap \mathbb{Z}^d|$$

for a polytope P. For integral polytopes, $L_P(n)$ is a polynomial in n. Floor-sum exercises are one-dimensional shadows of lattice-point enumeration.

§53.13 Finite differences and binomial basis

For a sequence $a(n)$, define

$$\Delta a(n) = a(n+1) - a(n).$$

If $a(n)$ is a polynomial of degree d, then Δa has degree $d-1$, and

$$\Delta^{d+1} a(n) = 0.$$

The binomial basis is adapted to this:

$$\Delta \binom{n}{k} = \binom{n}{k-1}.$$

Thus integer-valued polynomials are naturally expressed in the basis

$$\binom{n}{0}, \quad \binom{n}{1}, \quad \binom{n}{2}, \quad \ldots$$

rather than ordinary powers. This explains why binomial coefficients repeatedly appear in discrete counting formulas.

§53.14 Recursive set construction and fixed points

A recursive set construction such as

$$S_{k+1} = F(S_k)$$

should first be checked for monotonicity. If $S_k \subseteq S_{k+1}$ inside a finite universe, then the chain

$$S_0 \subseteq S_1 \subseteq S_2 \subseteq \cdots$$

must stabilize. The final set is a fixed point:

$$F(S) = S.$$

This is the finite version of closure-operator and fixed-point thinking.

§53.15 Incidence counting

If a problem asks how many sets contain a fixed element, it often helps to count incidence pairs:

$$I = \{(x, A) : x \in A\}.$$

Counting by x gives

$$|I| = \sum_x \#\{A : x \in A\},$$

while counting by A gives

$$|I| = \sum_A |A|.$$

Equating the two is double counting. Many intersection and membership problems are incidence problems in disguise.

§53.16 Polynomial method

The polynomial method proves combinatorial facts by encoding conditions into polynomial vanishing. The elementary fact is that a nonzero polynomial of degree d over a field has at most d roots. If a low-degree polynomial is forced to vanish at too many points, it must be the zero polynomial.

Lagrange interpolation gives an explicit reconstruction formula:

$$P(x) = \sum_{i=1}^{n} P(a_i) \prod_{j \neq i} \frac{x - a_j}{a_i - a_j}.$$

This is the advanced extension of discriminant and coefficient arguments: existence questions can be turned into degree and vanishing constraints.

§53.17 Group actions and corrected division

When counting objects up to symmetry, one must account for stabilizers. If a finite group G acts on X, orbit-stabilizer says

$$|Gx| = \frac{|G|}{|\operatorname{Stab}(x)|}.$$

Burnside's lemma says

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

This is why symmetry counting cannot always be done by simply dividing by $|G|$.

§53.18 Counting by lattices, incidences, and symmetries

The exercises train translation. Floor functions translate to lattice-point counts, binomial identities to finite differences and coefficient extraction, recursive set definitions to fixed points, membership counts to incidence pairs, and symmetry counts to group actions. The mature solution first finds the structure being counted, then chooses the appropriate algebraic language.

Binomial Theorem, Absolute-Value Estimates, and Set Exercises

N-20221210

§54.1 Binomial theorem

The first page begins with the binomial theorem:

$$(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}.$$

The handwritten derivation emphasizes selection. When expanding

$$(a+b)(a+b)\cdots(a+b),$$

one chooses either a or b from each factor. A term $a^i b^{n-i}$ occurs when a is chosen from exactly i of the n factors, and this can be done in $\binom{n}{i}$ ways.

The same idea proves the recurrence

$$\binom{n+1}{i} = \binom{n}{i} + \binom{n}{i-1},$$

because in choosing i objects from $n+1$, either the new object is not chosen or it is chosen.

§54.2 Absolute value of binomial sums

The note then estimates expressions such as

$$|a+b|^n.$$

Using the triangle inequality and the binomial theorem,

$$|a+b|^n \le (|a|+|b|)^n = \sum_{i=0}^{n} \binom{n}{i} |a|^i |b|^{n-i}.$$

The important habit is to move absolute values inside after expanding only when one has a legitimate inequality. The handwritten page marks the triangle inequality as the justification.

§54.3 A summation exercise

One exercise asks for a sum of the form

$$\sum_{i=0}^{n} \binom{n}{i}.$$

Putting $a = b = 1$ in the binomial theorem gives

$$2^n = \sum_{i=0}^{n} \binom{n}{i}.$$

Similarly, putting $a = 1, b = -1$ gives

$$0 = (1-1)^n = \sum_{i=0}^{n} (-1)^{n-i} \binom{n}{i},$$

which shows that the sum of even-indexed binomial coefficients equals the sum of odd-indexed ones when $n > 0$.

§54.4 Set exercises

The printed pages list elementary set exercises. The notes use standard notation:

$$A \cup B, \qquad A \cap B, \qquad A \setminus B, \qquad A^c.$$

One recurring task is to translate a verbal condition into set operations. For example,

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

This is proved elementwise:

$$x \in A \setminus (B \cup C) \iff x \in A, \ x \notin B, \ x \notin C.$$

The exercises also ask for necessary and sufficient conditions for equalities such as

$$A \cup B = A, \qquad A \cap B = B.$$

Both are equivalent to

$$B \subseteq A.$$

§54.5 Counting subsets with properties

For a finite set X with $|X| = n$, the number of subsets is

$$2^n.$$

If one imposes parity conditions, the count is often half of the total. For example, the number of subsets of an n-element set with even cardinality is

$$2^{n-1},$$

when $n \geq 1$. This follows from

$$\sum_{k \text{ even}} \binom{n}{k} = \sum_{k \text{ odd}} \binom{n}{k} = 2^{n-1}.$$

The printed exercises include conditions involving positive and negative numbers, odd and even elements, and sums of elements. The general method is to split the set into independent categories and multiply the numbers of choices.

§54.6 Set-builder notation

The note also works with sets such as

$$U = \{x : x = 2n,\ n \in \mathbb{Z}\}, \qquad V = \{x : x = 3n,\ n \in \mathbb{Z}\}.$$

Here U is the set of even integers and V is the set of multiples of 3. Then

$$U \cap V = \{x : x = 6n,\ n \in \mathbb{Z}\}$$

and

$$U \cup V$$

is the set of integers divisible by 2 or by 3.

The point is to read set-builder notation as a logical predicate.

§54.7 Binomial coefficients count the same choices in two languages

The note links binomial coefficients with set counting. The binomial theorem counts choices from factors; subset counting counts choices from elements. The same coefficients $\binom{n}{k}$ appear in both languages. Set identities should be proved elementwise, while counting identities should be proved by telling a story about what is being chosen.

§54.8 Binomial identities as counting in two languages

The binomial theorem has two simultaneous meanings. Algebraically, it is coefficient extraction:

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Combinatorially, it counts subsets by size. The coefficient of x^k is the number of ways to choose k marked factors, equivalently a k-element subset of an n-element set.

This double meaning is why identities should be checked in two ways whenever possible. For example, Vandermonde's identity

$$\sum_k \binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}$$

is coefficient extraction from

$$(1+x)^r(1+x)^s = (1+x)^{r+s},$$

but it also counts choosing n elements from a disjoint union of an r-set and an s-set: choose k from the first part and $n-k$ from the second.

§54.9 Binomial transform and Möbius inversion

The binomial transform of a sequence (a_k) is

$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k.$$

The inverse is

$$a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k.$$

This is not a coincidence; it is Möbius inversion on the Boolean lattice of subsets.

Indeed, if $B(S) = \sum_{T \subseteq S} A(T)$, then

$$A(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} B(T).$$

When $A(T)$ only depends on $|T|$, this collapses exactly to the binomial inversion formula above. Thus inclusion–exclusion, binomial inversion, and subset counting are the same mechanism at different resolutions.

§54.10 Finite differences and the binomial basis

The polynomials

$$\binom{x}{0}, \quad \binom{x}{1}, \quad \binom{x}{2}, \quad \dots$$

form a natural basis for polynomial functions on integers. The finite-difference operator

$$\Delta f(x) = f(x+1) - f(x)$$

acts especially simply:

$$\Delta \binom{x}{k} = \binom{x}{k-1}.$$

This mirrors ordinary calculus, where differentiation lowers powers:

$$\frac{d}{dx} x^k = kx^{k-1}.$$

The binomial basis is therefore the right coordinate system for discrete calculus. Identities involving $\binom{n}{k}$ often become transparent after applying Δ or summing finite differences.

§54.11 Probability distribution behind binomial coefficients

If $X \sim \mathrm{Bin}(n, p)$, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The binomial theorem says the probabilities sum to one:

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n = 1.$$

The probability generating function is

$$G_X(t) = \mathbb{E}[t^X] = (1 - p + pt)^n.$$

Differentiating gives

$$\mathbb{E}[X] = G'_X(1) = np,$$

and a second derivative computation gives

$$\mathrm{Var}(X) = np(1-p).$$

So the same binomial expansion that counts subsets also describes the sum of n independent Bernoulli trials.

§54.12 Asymptotic shape: from binomial coefficients to Gaussians

For large n, the row $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$ has a bell shape. Probabilistically, this is the central limit theorem applied to

$$X = X_1 + \cdots + X_n, \qquad X_i \sim \mathrm{Bernoulli}(p).$$

After centering and scaling,

$$\frac{X - np}{\sqrt{np(1-p)}}$$

approaches the standard normal distribution.

For $p = 1/2$, the largest coefficient is near $k = n/2$. Stirling's formula gives the classical estimate

$$\binom{n}{\lfloor n/2 \rfloor} \sim \frac{2^n}{\sqrt{\pi n/2}}.$$

Thus a finite algebraic row of Pascal's triangle already contains the first shadow of probability limit theory.

§54.13 Generating functions and labelled structures

For labelled combinatorial structures, the exponential generating function is

$$A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!}.$$

The factor $n!$ compensates for labels. Products of exponential generating functions correspond to splitting a labelled set into parts. The exponential formula says that if a structure is a set of connected components, then

$$A(x) = \exp(C(x)),$$

where $C(x)$ counts connected components.

This is the advanced version of the elementary rule "split the set into independent choices and multiply." The multiplication principle becomes a structural statement about labelled decomposition.

§54.14 Combinatorial species and relabelling

A combinatorial species is a functor from finite sets and bijections to sets. For each finite label set U, a species F assigns a set $F[U]$ of structures on U. A bijection $U \to V$ transports structures from $F[U]$ to $F[V]$.

The species of subsets sends U to $\mathcal{P}(U)$. Choosing a subset is functorial under relabelling: if labels are renamed, chosen elements are renamed with them. The binomial theorem is the decategorified counting shadow of this functorial behavior.

§54.15 Binomial algebra from sets to species

The binomial theorem is not merely an expansion formula. It is a meeting point of algebra, subset counting, probability, finite differences, generating functions, and functorial relabelling. The elementary exercises train the same habit in many forms: translate a formula into choices, translate a set identity into element logic, and translate a sum into a structural decomposition.

Local Information, Global Structure: Pigeonhole, Ramsey, Radon, and Helly

N-20230118

§55.1 Pigeonhole is an averaging theorem with a design problem hidden inside

If N objects are placed into r boxes, some box contains at least

$$\left\lceil \frac{N}{r} \right\rceil$$

objects. The proof is the contrapositive: if every box held at most q objects, the total would be at most rq.

The theorem is easy; choosing the boxes is the real work. Suppose nine elements are each recorded by membership in three sets A, B, C. Every element receives a bit pattern

$$(\mathbf{1}_A(x), \mathbf{1}_B(x), \mathbf{1}_C(x)) \in \{0, 1\}^3.$$

There are only

$$2^3 = 8$$

possible patterns. Among nine elements, two have the same pattern and therefore belong to exactly the same subcollection of A, B, C.

The conclusion is not merely "two objects coincide." A carefully chosen finite state space forces two complicated objects to become indistinguishable with respect to all recorded tests.

§55.2 Partial sums turn a divisibility problem into a finite-state collision

Let $a_1, \ldots, a_n$ be integers. There is always a nonempty consecutive block whose sum is divisible by n.

Consider the partial sums

$$s_k = a_1 + \cdots + a_k, \qquad k = 1, \ldots, n,$$

and reduce them modulo n. If some

$$s_k \equiv 0 \pmod{n},$$

then $a_1 + \cdots + a_k$ is the required block. Otherwise the n partial sums occupy only the $n-1$ nonzero residue classes, so two satisfy

$$s_i \equiv s_j \pmod{n}, \qquad i < j.$$

Subtracting gives

$$a_{i+1} + \cdots + a_j = s_j - s_i \equiv 0 \pmod{n}.$$

The objects are partial histories, the boxes are residue states, and a collision cancels the common prefix. This pattern appears throughout additive combinatorics and automata: encode a running process in finitely many states, then use repetition to extract a structured segment.

§55.3 Repeated pigeonhole becomes sequential compactness on an interval

Let (x_n) be an infinite sequence in $[0, 1]$. Divide the interval into two closed halves. At least one half contains infinitely many terms; call it I_1. Divide I_1 into two halves and choose one, I_2, that again contains infinitely many terms. Continuing gives nested intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots$$

with

$$|I_k| = 2^{-k}.$$

Choose indices

$$n_1 < n_2 < \cdots$$

so that

$$x_{n_k} \in I_k.$$

The nested interval theorem gives a unique point

$$x \in \bigcap_{k=1}^{\infty} I_k.$$

For $j \geq k$, both x_{n_j} and x lie in I_k, so

$$|x_{n_j} - x| \leq 2^{-k}.$$

Hence

$$x_{n_j} \to x.$$

This is Bolzano–Weierstrass on $[0, 1]$. The infinite conclusion is built by iterating a finite pigeonhole step at finer and finer scales. In a compact metric space, finite covers by small balls play the role of interval subdivisions.

§55.4 Ramsey theory forces a pattern, not merely a repeated label

Color every edge of the complete graph K_6 red or blue. Choose a vertex v. Among the five edges incident to v, at least three have the same color; suppose

$$va, vb, vc$$

are red.

If any of

$$ab, bc, ca$$

is red, that edge together with v forms a red triangle. If none is red, then all three are blue and abc is a blue triangle. Thus every red–blue coloring of K_6 contains a monochromatic triangle:

$$R(3, 3) \leq 6.$$

The bound is sharp. Color the edges of a five-cycle red and its five diagonals blue. The red graph is a cycle and has no triangle; the blue graph is another five-cycle and also has no triangle. Therefore

$$R(3,3) > 5,$$

so

$$\boxed{R(3,3) = 6.}$$

Pigeonhole forced three edges of one color around a vertex. The graph structure then amplified that local repetition into a monochromatic triangle. Ramsey theory studies how large an ambient configuration must be before a specified organized substructure becomes unavoidable.

§55.5 Intervals give the one-dimensional Helly theorem

Let

$$I_i = [a_i, b_i] \qquad (i = 1, \ldots, m)$$

be closed intervals, and assume every pair intersects. Set

$$a = \max_i a_i, \qquad b = \min_i b_i.$$

Choose indices p, q with

$$a = a_p, \qquad b = b_q.$$

Since $I_p \cap I_q \neq \varnothing$,

$$a_p \leq b_q,$$

so $a \leq b$. Every interval contains the segment $[a, b]$, and therefore

$$\bigcap_{i=1}^{m} I_i \neq \varnothing.$$

Pairwise intersection is enough in one dimension because the order lets one reduce all left endpoints to their maximum and all right endpoints to their minimum. In higher dimensions there is no total order of boundaries; convexity and affine dependence replace it.

§55.6 Radon's theorem is an affine dependence split into two convex combinations

Theorem 55.6.1 (Radon)
Any $d+2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets whose convex hulls intersect.

Let the points be $p_1, \ldots, p_{d+2}$. The lifted vectors

$$(p_i, 1) \in \mathbb{R}^{d+1}$$

are linearly dependent, so there are scalars λ_i, not all zero, with

$$\sum_i \lambda_i p_i = 0, \qquad \sum_i \lambda_i = 0.$$

Let

$$I = \{i : \lambda_i > 0\}, \qquad J = \{i : \lambda_i < 0\}.$$

Both are nonempty. Put

$$s = \sum_{i \in I} \lambda_i = -\sum_{j \in J} \lambda_j > 0.$$

Then

$$\sum_{i \in I} \frac{\lambda_i}{s} p_i = \sum_{j \in J} \frac{-\lambda_j}{s} p_j.$$

Both sides are convex combinations. Their common value lies in

$$\operatorname{conv}\{p_i : i \in I\} \cap \operatorname{conv}\{p_j : j \in J\}.$$

The Radon partition is the geometric form of a minimal affine dependence. In oriented-matroid language, such a signed dependence is the prototype of a circuit.

§55.7 Helly's theorem follows by applying Radon to carefully chosen witnesses

Theorem 55.7.1 (Finite Helly)

Let $C_1, \ldots, C_m$ be convex subsets of $\mathbb{R}^d$, with $m \geq d+1$. If every $d+1$ of them have nonempty intersection, then

$$\bigcap_{i=1}^{m} C_i \neq \varnothing.$$

We prove the theorem by induction on m. The case $m = d+1$ is the hypothesis. Assume $m > d+1$ and the result holds for $m-1$ sets.

For each $i = 1, \ldots, d+2$, the family obtained by omitting C_i has $m-1$ members and still satisfies the local intersection hypothesis. By induction, choose

$$x_i \in \bigcap_{j \neq i} C_j.$$

Apply Radon's theorem to $x_1, \ldots, x_{d+2}$. There is a partition

$$I \sqcup J = [d+2]$$

and a point

$$x \in \operatorname{conv}\{x_i : i \in I\} \cap \operatorname{conv}\{x_j : j \in J\}.$$

Fix a set C_k. If $k \in I$, then every x_j with $j \in J$ belongs to C_k, because $j \neq k$. Convexity gives

$$x \in C_k.$$

If $k \in J$, use the convex hull indexed by I. If $k > d+2$, every x_i lies in C_k. Thus x belongs to every C_k, completing the induction.

The proof shows exactly where the number $d+1$ comes from: Radon's affine dependence begins at $d+2$ points.

§55.8 Carathéodory reduces a large convex representation to $d+1$ points

Theorem 55.8.1 (Carathéodory)
If $x \in \operatorname{conv} S \subset \mathbb{R}^d$, then x lies in the convex hull of at most $d+1$ points of S.

Write

$$x = \sum_{i=1}^{m} \alpha_i p_i, \qquad \alpha_i > 0, \qquad \sum_i \alpha_i = 1,$$

with $m > d+1$. The points are affinely dependent, so there are λ_i, not all zero, satisfying

$$\sum_i \lambda_i p_i = 0, \qquad \sum_i \lambda_i = 0.$$

After changing all signs if necessary, some $\lambda_i > 0$. Let

$$t = \min_{\lambda_i > 0} \frac{\alpha_i}{\lambda_i}.$$

Then the new coefficients

$$\alpha_i' = \alpha_i - t\lambda_i$$

remain nonnegative, still sum to one, and give the same point x. At least one coefficient becomes zero. Repeating removes points until at most $d+1$ remain.

Helly compresses an intersection certificate to small subfamilies. The preceding convex-hull theorem compresses a representation to at most $d+1$ points. Both reductions are driven by affine dependence.

§55.9 Fractional Helly tolerates failed local tests

The ordinary Helly theorem assumes that every $(d+1)$-tuple intersects. Fractional Helly allows failures.

For each d and $\alpha > 0$, there is $\beta = \beta(d, \alpha) > 0$ such that the following holds. If at least an α-fraction of the $(d+1)$-tuples in a family of n convex sets in $\mathbb{R}^d$ have nonempty intersection, then some point belongs to at least βn members of the family. One may take

$$\beta = 1 - (1-\alpha)^{1/(d+1)}.$$

In one dimension, the sets are intervals. Intersecting pairs are edges of an interval graph, while intervals containing a common point form a clique. If a positive fraction of all pairs intersect, the theorem guarantees a clique containing a positive fraction of all intervals. Thus numerous local successes cannot be dispersed arbitrarily; they concentrate around a common witness.

The conclusion is weaker than total intersection but robust under noise. It is an extremal version of Helly rather than a different slogan about local-to-global behavior.

§55.10 The nerve records exactly which subfamilies intersect

For a finite family $\mathcal{F} = \{U_i\}_{i \in I}$, its nerve is the simplicial complex

$$N(\mathcal{F}) = \{J \subseteq I : \bigcap_{j \in J} U_j \neq \varnothing\}.$$

Vertices represent sets, edges represent pairwise intersections, triangles represent triple intersections, and higher simplices represent higher common intersections.

Take three open disks arranged in a ring so that every pair intersects but the triple intersection is empty. The nerve consists of three vertices and three edges but no filled triangle. It is the boundary of a triangle, homotopy equivalent to a circle. The union of the disks also has a central hole.

The nerve theorem makes this precise: if every nonempty finite intersection in the family is contractible, then the union is homotopy equivalent to the nerve. Such a family is called a good cover.

A topological Helly theorem follows under acyclicity hypotheses. For a finite family of open subsets of $\mathbb{R}^d$, assume every nonempty finite intersection is contractible, or more generally has the required vanishing reduced homology. If every $d + 1$ members intersect, then the whole family intersects. Convex sets satisfy the hypotheses because every nonempty convex intersection is contractible.

The three-disk ring explains what the topology is detecting. Pairwise data alone allow a one-dimensional hole; filling the triple intersection fills the missing simplex in the nerve. Topological Helly theorems control which holes an intersection pattern could create in dimension d, replacing linear convexity by precise homological conditions.

Combinatorics: Identities, Generating Functions, Catalan Numbers, and More

N-20240323

§56.1 Binomial identities

The note opens with several binomial identities and, more importantly, their combinatorial meanings. The basic identity

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n$$

counts all subsets of an n-element set: choosing a subset means choosing its size i and then choosing i elements.

Pascal's identity

$$\binom{n}{i} + \binom{n}{i+1} = \binom{n+1}{i+1}$$

can be proved by fixing a distinguished element in an $(n+1)$-element set. An $(i+1)$-subset either contains the distinguished element or it does not. The two cases give the two terms on the left.

Vandermonde's identity

$$\sum_{i} \binom{m}{i}\binom{n}{r-i} = \binom{m+n}{r}$$

counts choosing r balls from two boxes with m and n balls. If i balls are chosen from the first box, then $r-i$ are chosen from the second.

Another useful identity is

$$\binom{n}{r}\binom{n-r}{k-r} = \binom{n}{k}\binom{k}{r}.$$

Both sides count a pair (R, K) with $R \subseteq K \subseteq [n]$, $|R| = r$, and $|K| = k$.

§56.2 Good subsets

A nonempty subset $A \subseteq \{1, 2, \ldots, n\}$ is called good if

$$|A| \le \min_{x \in A} x.$$

Let a_n be the number of good subsets. The problem asks one to prove

$$a_{n+2} = a_{n+1} + a_n + 1.$$

A clean proof splits good subsets of $[n+2]$ into cases.

First, if $n+2 \notin A$, then A is just a good subset of $[n+1]$, giving a_{n+1} possibilities.

Second, if $A = \{n+2\}$, this gives one extra subset.

Third, suppose $n+2 \in A$ and $|A| \geq 2$. Remove $n+2$ and subtract 1 from every remaining element. If

$$A' = A \setminus \{n+2\},$$

then the condition becomes

$$|A'| + 1 \leq \min A'.$$

After shifting $B = \{x - 1 : x \in A'\}$, this is exactly

$$|B| \leq \min B,$$

so B is a good subset of $[n]$. This gives a_n possibilities. Hence

$$a_{n+2} = a_{n+1} + a_n + 1.$$

The handwritten page also gives a summation proof by fixing $|A| = k$. If $|A| = k$, then every element must be at least k, so one chooses the remaining data from a tail interval. The recurrence then follows by repeated Pascal identities.

§56.3 Generating functions

For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$f(x) = \sum_{n\geq 0} a_n x^n.$$

The note emphasizes a key principle: coefficients can encode counting problems.

For example, the number of nonnegative integer solutions of

$$x_1 + x_2 + \cdots + x_k = n$$

is the coefficient of x^n in

$$(1 + x + x^2 + \cdots)^k = \frac{1}{(1-x)^k}.$$

Using the binomial expansion,

$$\frac{1}{(1-x)^k} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n,$$

so the number of solutions is

$$\binom{n+k-1}{k-1}.$$

The notes also use generating functions to separate parity or color constraints. A typical trick is to multiply several series, each recording the allowed choices for one box or color, and then read off the coefficient of the target power.

§56.4 A probability example with repeated trials

One example considers repeated trials and asks for a probability after n operations. The handwritten solution introduces a generating function and then rewrites nested binomial sums using exponentials. The important method is not the final closed form alone but the transformation:

$$\text{complicated convolution} \quad \longrightarrow \quad \text{product/composition of generating functions.}$$

This is a standard way to turn repeated independent operations into algebra. Binomial coefficients record choices; exponential series record independent counts; coefficient extraction recovers the desired probability.

§56.5 Catalan numbers

The note then turns to Catalan numbers. One model is the number of ways to triangulate a convex $(n+2)$-gon:

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

For example, a hexagon has 14 triangulations.

Let h_n denote the number of triangulations of an $(n+2)$-gon. Choose the triangle incident with a fixed side. If its third vertex cuts the polygon into two smaller polygons, the numbers of triangulations multiply. Therefore

$$h_n = \sum_{i=0}^{n-1} h_i h_{n-1-i}, \qquad h_0 = 1.$$

If

$$H(x) = \sum_{n\geq 0} h_n x^n,$$

then the recurrence gives

$$H(x) = 1 + xH(x)^2.$$

Solving the quadratic and choosing the branch with $H(0) = 1$ gives

$$H(x) = \frac{1-\sqrt{1-4x}}{2x}, \qquad h_n = C_n.$$

This derivation explains why Catalan numbers appear in triangulations, parentheses, Dyck paths, and many noncrossing structures: all of them are recursively split into a left and a right part.

§56.6 Convex sets and convex hulls

The geometry part defines a convex set S by the condition that for any $x, y \in S$ and any $t \in [0,1]$,

$$tx + (1-t)y \in S.$$

Equivalently, S contains the line segment between any two of its points. The note contrasts convex and non-convex examples.

For finitely many points $x_1, \dots, x_k$, their convex hull is

$$\left\{ \sum_{i=1}^{k} t_i x_i : \ t_i \geq 0, \ \sum_{i=1}^{k} t_i = 1 \right\}.$$

The handwritten proof uses induction on k: the case $k = 2$ is the definition of a segment; the induction step writes the convex combination as a combination of one point and a convex combination of the remaining points.

§56.7 Separating convex sets

The note sketches a separation idea. For two disjoint convex sets, one tries to find a point pair realizing minimal distance and then uses the perpendicular bisector of the segment between them as a separating line. In symbols, if $u \in S_1$ and $v \in S_2$ are closest points, then the hyperplane perpendicular to $v - u$ near the midpoint separates the two sets.

In finite-dimensional Euclidean geometry, this is the intuitive core of the separating hyperplane theorem. The handwritten diagrams show projections onto a line and inequalities involving squared distances. The algebra expands

$$\|u + t(a - u) - v\|^2$$

for small t and extracts a sign condition by letting $t \to 0$.

§56.8 Planar graphs and Euler-type counting

The later pages move toward graph theory and planar configurations. The recurring idea is that a planar drawing imposes counting restrictions. Vertices, edges, and faces cannot vary independently; they are constrained by Euler-type formulas.

This connects naturally with the previous convex-geometry material: triangulations, planar graphs, polygon decompositions, and convex hulls are different languages for controlling incidence data.

§56.9 Graphs and adjacency matrices

A graph G consists of a vertex set $V = \{v_1, \ldots, v_n\}$ and an edge relation. For a simple graph, the adjacency matrix $A(G) = (a_{ij})$ is defined by

$$a_{ij} = \begin{cases} 1, & \text{if there is an edge from } v_i \text{ to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

For a multigraph, a_{ij} records the number of edges from v_i to v_j.

The source quotes the standard theorem: the (i, j)-entry of $A(G)^\ell$ equals the number of walks of length ℓ from v_i to v_j. The proof is matrix multiplication. For $\ell = 2$,

$$(A^2)_{ij} = \sum_k a_{ik} a_{kj},$$

which counts all intermediate vertices v_k. The general case follows by induction.

§56.10 The deeper habit

This note collects several ways of counting: direct binomial identities, recurrence, generating functions, Catalan decomposition, convex geometry, and graph adjacency matrices. The common habit is to replace a raw counting problem by a structure-preserving encoding: subsets become binomial coefficients, recursions become generating functions, triangulations become Catalan recurrences, convex hulls become linear combinations, and walks become matrix powers.

§56.11 Generating functions as algebraic counting

A generating function packages a sequence into a formal series. Products encode decomposition of objects, and coefficient extraction reads off counts:

$$[x^n] A(x) B(x) = \sum_{k=0}^{n} a_k b_{n-k}.$$

§56.12 Catalan structures

Catalan numbers count many recursively decomposable objects:

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Their generating function satisfies

$$C(x) = 1 + xC(x)^2.$$

The quadratic equation records the root-subtree decomposition.

§56.13 Convex hulls and separation

The separating hyperplane theorem says disjoint convex sets can often be separated by a linear functional. This is the geometric dual of optimization: a linear functional witnesses non-intersection.

§56.14 Generating functions, Catalan recursion, and spectra

Binomial identities, generating functions, Catalan numbers, probability trials, and convex separation all share one habit: encode the structure so that algebraic or linear operations reveal the count or obstruction.

§56.15 Catalan numbers as recursive geometry

Catalan numbers count many objects: correctly matched parentheses, triangulations of polygons, rooted binary trees, Dyck paths, and noncrossing partitions. A standard formula is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

The recursion

$$C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$$

comes from choosing the first return of a Dyck path or the root split of a binary tree.

The generating function $C(x) = \sum_{n\geq 0} C_n x^n$ satisfies

$$C(x) = 1 + xC(x)^2.$$

Solving gives

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}.$$

Thus an elementary recursion becomes an algebraic generating function.

§56.16 Lattice paths and reflection principle

Many Catalan formulas are proved by the reflection principle. The number of paths from $(0,0)$ to (n,n) that never go above the diagonal is

$$\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n}.$$

The subtracted term counts bad paths by reflecting the part up to the first diagonal crossing.

This is a model example of an involutive or bijective proof. Rather than evaluating a sum, one constructs a symmetry pairing bad objects with easier objects.

§56.17 Graph walks and adjacency matrices

If G is a finite graph with adjacency matrix A, then

$$(A^k)_{ij}$$

counts walks of length k from vertex i to vertex j. This turns walk counting into matrix powers.

If A is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_n$, the growth of walks is controlled by the largest eigenvalues. In a regular connected graph, the largest eigenvalue is the degree, and the second-largest eigenvalue controls expansion and mixing.

Thus an elementary graph-walk count leads directly to spectral graph theory.

§56.18 Random walks and Markov chains

Normalize the adjacency matrix of a d-regular graph by

$$P = \frac{1}{d}A.$$

Then P is the transition matrix of a random walk. The distribution after k steps is

$$\pi_k = \pi_0 P^k.$$

Eigenvalues of P determine convergence to stationarity. If the graph is connected and non-bipartite, the walk converges to the uniform distribution.

The same object therefore has two readings: A^k counts deterministic walks, while P^k describes probabilities. The note's graph-walk calculations are the finite beginning of Markov-chain theory.

§56.19 Catalan recursion and parenthesization

Polygon triangulations and parenthesizations are the same combinatorial structure. A triangulation of an $(n+2)$-gon corresponds to a way of multiplying $n+1$ factors with binary parentheses. Choosing the triangle incident to a fixed edge splits the polygon into two smaller polygons, producing the Catalan recursion.

This is the combinatorics behind the associahedron, a polytope whose vertices are parenthesizations and whose edges are single reassociations:

$$(ab)c \leftrightarrow a(bc).$$

The elementary Catalan count therefore touches the geometry of moduli spaces of bracketings.

§56.20 Adjacency spectra and expansion

The theorem that $(A^k)_{ij}$ counts walks becomes much stronger when combined with eigenvalues. If a graph is d-regular with adjacency eigenvalues

$$d = \lambda_1 \geq |\lambda_2| \geq \cdots ,$$

then $|\lambda_2|$ controls how quickly random walks mix and how well the graph expands.

Expander graphs are sparse graphs that behave like highly connected graphs. They are central in theoretical computer science, coding theory, and combinatorics. The elementary walk-counting matrix is the first tool for studying them.

From Extremal Triangles to Euler Characteristic and Degree

N-20240330

§57.1 Maximum-area triangles live on the convex hull

Let S be a set of n points in the plane, not all collinear. A triangle determined by S is maximal when its area is the largest area attained by any triangle with vertices in S.

Fix two vertices a, b. The area of abp is

$$\operatorname{area}(abp) = \frac{1}{2}|a-b| \operatorname{dist}(p, \overleftrightarrow{ab}).$$

As a function of p, the signed height above the line ab is affine. Its maximum and minimum over the convex hull of S occur at hull vertices. Therefore, if p lies strictly inside the convex hull, replacing it by a suitable hull vertex does not decrease the area and, unless the height was already extremal along an entire hull edge, increases it.

Under the standing assumption that no three points are collinear, every vertex of a maximum-area triangle is therefore a vertex of the convex hull. Interior points may be deleted without changing the family of maximal triangles. The counting problem reduces to the vertices

$$v_0, v_1, \ldots, v_{m-1}$$

of a strictly convex m-gon, where $m \leq n$. The proof below establishes the upper bound required by the original problem in this general-position setting.

§57.2 Two maximal triangles must alternate around the boundary

The key fact is an alternation lemma.

Proposition 57.2.1 (Alternation of maximal triangles)
Let Δ and Δ' be two maximum-area triangles whose vertices lie on a convex polygon. Every side of Δ' meets some side of Δ.

Proof. Write

$$\Delta = a_1a_2a_3.$$

Through each a_i, draw the line parallel to the opposite side $a_{i+1}a_{i+2}$, with indices modulo three. These three lines bound a larger triangle T.

Every point of the original set lies in T. If a point lay beyond, say, the line through a_1 parallel to a_2a_3, then its distance from the line a_2a_3 would exceed the height of a_1. Replacing a_1 by that point would produce a triangle of larger area.

The central triangle Δ divides $T \setminus \Delta$ into three corner triangles T_1, T_2, T_3, each a translate of $-\Delta$ and therefore of the same area as Δ. Let the vertices of Δ' lie in these four regions.

If all three lie in one T_i, equality of areas forces $\Delta' = T_i$; its sides are parallel to and meet the corresponding boundary sides of Δ. If one vertex lies in each T_i, the three connecting sides cross the three sides of Δ.

The remaining possibility places two vertices, say p, q, in one corner triangle and the third vertex r in another. A line through p parallel to qr separates one vertex of Δ farther from the line qr than p. Replacing p by that vertex produces a triangle with base qr and strictly greater height, contradicting maximality of pqr.

Thus the nonalternating configuration is impossible, proving the claim. $\square$

The proposition is stronger than "all vertices lie on the hull." It says that two extremal triples cannot occupy unrelated arcs of the cyclic order; their vertices must interlace strongly enough that their sides encounter one another.

§57.3 Cyclic lifts turn alternation into the bound n

Index the vertices of the convex m-gon periodically by all integers:

$$v_{i+m} = v_i.$$

Call $(i, j, k) \in \mathbb{Z}^3$ a proper triple when

$$i < j < k < i + m$$

and $v_i v_j v_k$ is a maximal triangle. Order triples coordinatewise:

$$(i, j, k) \le (i', j', k') \quad \iff \quad i \le i',\ j \le j',\ k \le k'.$$

The alternation proposition implies that any two proper triples are comparable. To see the cyclic-order step, place the six lifted indices on the real line. If neither

$$(i, j, k) \le (i', j', k')$$

nor the reverse inequality holds, their coordinate order changes sign at least twice: one lifted vertex of the second triangle lies after the corresponding vertex of the first, while another lies before it. After a cyclic relabelling, two consecutive vertices of one triangle then lie strictly in the same open boundary arc between two consecutive vertices of the other. The side joining those two vertices is contained in the corresponding sector of the convex polygon and cannot meet any side of the other triangle. This contradicts the alternation proposition. Therefore all proper triples form a chain $C \subset \mathbb{Z}^3$.

Fix one proper triple (a, b, c), and consider the half-open subchain

$$\begin{aligned} C_0 = \{(i, j, k) \in C : {} & (a, b, c) \le (i, j, k), \\ & (i, j, k) < (a + m, b + m, c + m)\}. \end{aligned}$$

Here the final strict inequality means coordinatewise $\le$, but not equality of all three coordinates.

Every geometric maximal triangle has exactly three proper representatives in C_0, corresponding to choosing each of its three vertices in turn as the first cyclic vertex. On the other hand, distinct integer points in a coordinatewise chain have strictly increasing coordinate sum. Inside the displayed half-open interval, that sum ranges over fewer than $3m$ integer increments. Hence

$$|C_0| \le 3m.$$

If N is the number of maximal triangles, then

$$3N = |C_0| \leq 3m,$$

so

$$\boxed{N \leq m \leq n.}$$

The bound is sharp. For a regular m-gon with m not divisible by three, the m cyclically shifted triangles whose vertices are as evenly spaced as possible all have maximum area.

§57.4 Euler's formula follows by adding non-tree edges one at a time

Let a connected graph be embedded in the sphere without edge crossings. Choose a spanning tree T. The tree has

$$E_T = V - 1$$

edges and only one face on the sphere, so

$$V - E_T + F = V - (V - 1) + 1 = 2.$$

Now add the remaining edges of the graph one at a time. Each new edge joins two vertices already connected by the tree-and-previous-edges subgraph. Drawn without crossings, it lies inside one existing face and splits that face into two. Thus

$$E \longmapsto E + 1, \qquad F \longmapsto F + 1,$$

while V is unchanged. The quantity $V - E + F$ remains constant.

Therefore every connected planar embedding satisfies

$$\boxed{V - E + F = 2.}$$

For a graph with c connected components in the sphere,

$$V - E + F = 1 + c.$$

The proof identifies the invariant move: a non-tree edge creates exactly one independent cycle and exactly one new face.

Subdivision also preserves the alternating count. Inserting a new vertex into an edge changes

$$(V, E, F) \longmapsto (V + 1, E + 1, F),$$

while drawing a diagonal across a face changes

$$(V, E, F) \longmapsto (V, E + 1, F + 1).$$

Both leave $V - E + F$ unchanged. Euler characteristic therefore belongs to the underlying cell decomposition, not to the accidental fineness of the drawing.

§57.5 Incidence counting leaves only five regular polyhedra

Suppose every face of a regular convex polyhedron is a p-gon and exactly q faces meet at every vertex. Counting edge–face incidences and vertex–edge incidences gives

$$pF = 2E, \qquad qV = 2E.$$

Substitute

$$F = \frac{2E}{p}, \qquad V = \frac{2E}{q}$$

into Euler's formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2.$$

Since $E > 0$,

$$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}.$$

With $p, q \geq 3$, the only possibilities are

$$(p, q) = (3, 3), (3, 4), (4, 3), (3, 5), (5, 3).$$

They correspond to

(p, q)	V	E	F
$(3, 3)$	4	6	4
$(3, 4)$	6	12	8
$(4, 3)$	8	12	6
$(3, 5)$	12	30	20
$(5, 3)$	20	30	12.

Duality interchanges vertices and faces while preserving edges, so it swaps (p, q) with (q, p). The cube and octahedron are dual, as are the dodecahedron and icosahedron; the tetrahedron is self-dual. The classification is therefore an interaction between Euler characteristic, local incidence, and duality.

§57.6 Polygon gluings compute Euler characteristics of quotient surfaces

A topological quotient identifies points rather than cosets. Begin with a polygon as one two-cell and glue its boundary edges according to labels.

For the torus, use a square with boundary word

$$aba^{-1}b^{-1}.$$

All four vertices are identified, the four sides become two one-cells, and the interior is one two-cell. Hence

$$(V, E, F) = (1, 2, 1), \qquad \chi(T^2) = 0.$$

For the real projective plane, glue a two-gon with boundary word

$$aa.$$

There is one vertex, one edge, and one face:

$$\chi(\mathbb{R}P^2) = 1 - 1 + 1 = 1.$$

For the Klein bottle, use the square word

$$aba^{-1}b.$$

Again there is one vertex, two edges, and one face, so

$$\chi(K) = 0.$$

The torus and Klein bottle have the same Euler characteristic but are not homeomorphic; Euler characteristic is powerful but not complete.

Three quotient constructions should not be conflated:

$$G/N$$

is a quotient group whose well-defined multiplication requires $N \trianglelefteq G$;

$$X/\sim$$

is a quotient topological space with the quotient topology; and

$$H_k(X) = Z_k(X)/B_k(X)$$

is an algebraic quotient of cycles by boundaries. They share the act of identifying equivalent elements, but the structures being preserved are different.

§57.7 A chain-complex cancellation proves the homological Euler formula

For a finite cell complex, let C_k be the vector space spanned by its k-cells, with boundary maps

$$\cdots \longrightarrow C_{k+1} \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \longrightarrow \cdots$$

satisfying

$$\partial_k \partial_{k+1} = 0.$$

Set

$$Z_k = \ker \partial_k, \qquad B_k = \operatorname{im} \partial_{k+1}, \qquad H_k = Z_k/B_k.$$

Then

$$\dim C_k = \dim Z_k + \operatorname{rank} \partial_k,$$

and

$$\dim Z_k = \dim H_k + \operatorname{rank} \partial_{k+1}.$$

Therefore

$$\dim C_k = \dim H_k + \operatorname{rank} \partial_k + \operatorname{rank} \partial_{k+1}.$$

Taking the alternating sum over k, the two boundary-rank sums cancel after shifting the index:

$$\boxed{\sum_k (-1)^k \dim C_k = \sum_k (-1)^k \dim H_k.}$$

The left side is the alternating cell count; the right side is the alternating sum of Betti numbers

$$b_k = \dim H_k.$$

For the sphere,

$$(b_0, b_1, b_2) = (1, 0, 1), \qquad \chi(S^2) = 2.$$

For the torus,

$$(b_0, b_1, b_2) = (1, 2, 1), \qquad \chi(T^2) = 0.$$

Over $\mathbb{Q}$, the projective plane has Betti numbers

$$(1, 0, 0),$$

so $\chi(\mathbb{R}P^2) = 1$. Its integral homology contains torsion, but torsion does not affect the alternating ranks.

§57.8 Degree on the circle is winding made algebraic

For

$$f_n : S^1 \to S^1, \qquad f_n(e^{it}) = e^{int},$$

the point travels around the target n times as t runs from 0 to 2π. Its degree is

$$\deg f_n = n.$$

For a smooth map and a regular value y, degree can be computed by oriented preimages:

$$\deg f = \sum_{x \in f^{-1}(y)} \operatorname{sign}(df_x).$$

For f_n, a regular target point has $|n|$ preimages, each with the sign of n.

The homological definition generalizes to every sphere. Since

$$H_d(S^d; \mathbb{Z}) \cong \mathbb{Z},$$

a continuous map $f : S^d \to S^d$ induces multiplication by one integer:

$$f_*([S^d]) = (\deg f)[S^d].$$

Functoriality gives

$$\deg(g \circ f) = \deg g \, \deg f.$$

The multiplicativity is simply the multiplication of the two induced endomorphisms of $\mathbb{Z}$.

§57.9 Nonzero degree forces a map to hit every target point

Suppose $f : S^d \to S^d$ omits a point p. Then it factors through

$$S^d \setminus \{p\} \cong \mathbb{R}^d,$$

whose top homology vanishes:

$$H_d(\mathbb{R}^d) = 0.$$

Consequently the induced map on $H_d(S^d)$ is zero, so

$$\deg f = 0.$$

Taking the contrapositive,

$$\boxed{\deg f \neq 0 \quad \Longrightarrow \quad f \text{ is surjective.}}$$

Degree is therefore an existence invariant. It does not locate a preimage, but it prevents all preimages from disappearing under continuous deformation.

As a simple application, every map homotopic to the identity has degree one and is surjective. No such map can be continuously deformed into a constant map, whose degree is zero.

§57.10 The no-retraction argument yields Brouwer's fixed-point theorem

There is no continuous retraction

$$r : B^{d+1} \to S^d$$

from the closed ball to its boundary. If $i : S^d \hookrightarrow B^{d+1}$ is the inclusion and $r \circ i = \mathrm{id}_{S^d}$, then on top homology

$$r_* \circ i_* = \mathrm{id}_{H_d(S^d)} .$$

But

$$H_d(B^{d+1}) = 0,$$

so i_* is the zero map, making the composition zero. This contradiction proves the no-retraction theorem.

Now let

$$F : B^{d+1} \to B^{d+1}$$

be continuous and suppose it has no fixed point. Put

$$y = F(x), \qquad d = x - y \neq 0.$$

The ray starting at y and passing through x is

$$y + td, \qquad t \geq 0.$$

Its intersection with the boundary satisfies

$$\|y + td\|^2 = 1,$$

or

$$\|d\|^2 t^2 + 2(y \cdot d)t + \|y\|^2 - 1 = 0.$$

The forward intersection is obtained from the positive root

$$t(x) = \frac{-y \cdot d + \sqrt{(y \cdot d)^2 + \|d\|^2(1 - \|y\|^2)}}{\|d\|^2}.$$

Define

$$r(x) = y + t(x)d.$$

Because $d \neq 0$, the formula is continuous and satisfies $\|r(x)\| = 1$. If $x \in S^d$, then $t = 1$ is the forward root, so

$$r(x) = x.$$

Thus a fixed-point-free F would construct a forbidden retraction. Therefore every continuous self-map of the closed ball has a fixed point.

The passage from cell counts to this existence theorem is now explicit. Euler characteristic survives subdivision because boundary ranks cancel in a chain complex. Degree records the action on the surviving top homology class. A nonzero class obstructs both omission of a target point and retraction of a ball onto its boundary.

Self-Avoiding Walks I: Connective Constants, Bridges, Polygons, and Critical Exponents

N-20251126

§58.1 What the model is trying to measure

A self-avoiding walk is the simplest discrete model of a polymer chain with excluded volume. The walk is allowed to move from one lattice point to a neighbouring lattice point, but it is not allowed to visit the same vertex twice. Thus the model keeps the most basic feature of a real chain: two monomers cannot occupy the same place.

Let $\mathbb{Z}^d$ be the hypercubic lattice. An n-step walk is a sequence

$$\omega = (\omega_0, \omega_1, \ldots, \omega_n),$$

where $\omega_i \in \mathbb{Z}^d$, $\omega_0 = 0$, and

$$|\omega_{i+1} - \omega_i| = 1.$$

It is self-avoiding if

$$\omega_i \neq \omega_j \qquad (i \neq j).$$

Write c_n for the number of n-step self-avoiding walks starting at the origin.

The ordinary simple random walk has exactly $(2d)^n$ possible n-step paths. For self-avoiding walks, there is no such closed formula, because each new step depends on the entire past of the path. This is the first reason the subject is difficult: the model is local in its rule but non-local in its memory.

> **Remark 58.1.1 —** The phrase "self-avoiding" sounds like a small constraint, but it changes the problem qualitatively. The number of walks is no longer obtained by multiplying independent choices, and the endpoint distribution is no longer governed by the usual central limit theorem in low dimensions.

§58.2 Submultiplicativity and the connective constant

The first invariant is the exponential growth rate of c_n. Concatenation gives the basic inequality

$$c_{m+n} \leq c_m c_n.$$

Indeed, if an $(m+n)$-step self-avoiding walk is cut after m steps, then the first part is an m-step self-avoiding walk, and the translated second part is an n-step self-avoiding walk. Not every pair can be concatenated without intersection, so this gives an inequality rather than an equality.

By Fekete's lemma, the limit

$$\mu = \lim_{n\to\infty} c_n^{1/n}$$

exists. This number is called the connective constant. Equivalently,

$$\log \mu = \lim_{n\to\infty} \frac{1}{n} \log c_n.$$

The rough meaning is

$$c_n \approx \mu^n \times \text{a subexponential correction.}$$

Definition 58.2.1. The connective constant μ is the exponential growth rate of self-avoiding walks on the lattice. It depends on the lattice, not only on the dimension.

The reciprocal

$$z_c = \mu^{-1}$$

is the critical point of the generating function

$$\chi(z) = \sum_{n\geq 0} c_n z^n.$$

For $z < z_c$, the series converges. For $z > z_c$, it diverges. Thus the connective constant is both a combinatorial growth constant and the location of a statistical-mechanical phase transition.

§58.3 Bridges and why they help

Counting all self-avoiding walks directly is hard because a walk can wander in all directions. A bridge is a self-avoiding walk with a preferred direction. For example, in the first coordinate direction, a bridge is a walk whose first coordinate starts at its minimum and ends at its maximum:

$$\omega_0^{(1)} < \omega_i^{(1)} \leq \omega_n^{(1)} \qquad (1 \leq i \leq n).$$

The precise convention varies slightly, but the guiding idea is stable: a bridge progresses from left to right without going below its starting level in that coordinate.

The following picture is the right mental picture for the Hammersley–Welsh argument. A half-space walk is not yet a bridge, but it can be chopped into alternating bridge-like pieces. The dotted reflected part indicates the trick: reflect suitable tails of the walk so that the pieces line up into one bridge.

Let b_n denote the number of n-step bridges. Bridges are useful because they concatenate better than arbitrary walks. If one bridge is followed by another translated bridge, the result is still a bridge. Hence

$$b_{m+n} \geq b_m b_n.$$

This supermultiplicativity implies that the bridge growth rate exists:

$$\lim_{n\to\infty} b_n^{1/n} = \sup_n b_n^{1/n}.$$

A central fact is that bridges have the same exponential growth rate as all self-avoiding walks:

$$\lim_{n\to\infty} b_n^{1/n} = \mu.$$

The Hammersley–Welsh method gives a much sharper comparison. Its philosophy is to decompose a general self-avoiding walk into bridge-like pieces by reflecting parts of

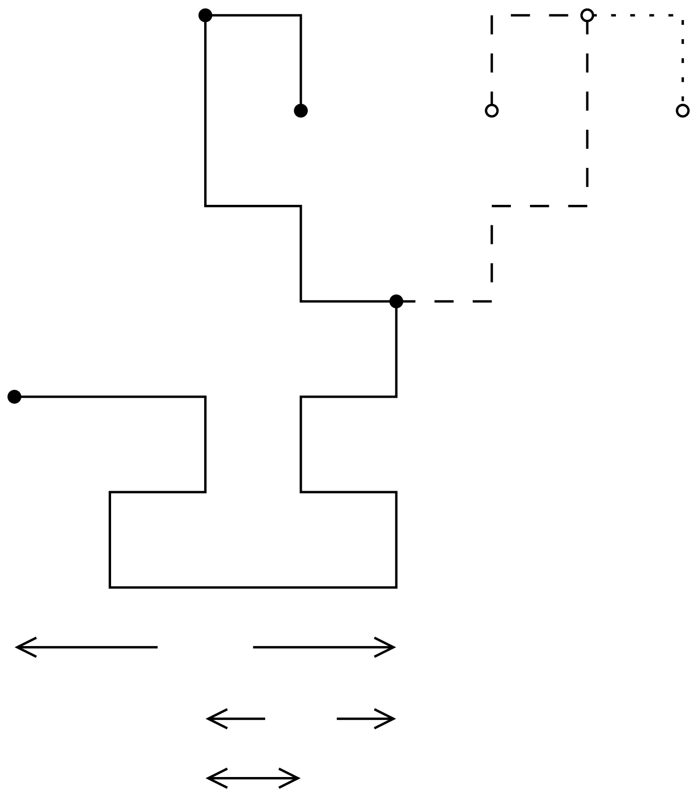


Figure 58.1: a half-space walk decomposed into bridges and reflected into a single bridge.

the walk across suitable hyperplanes. This does not give a bijection with bridges, but it controls the loss by a subexponential factor. In a typical form one obtains a bound of the shape

For a completely general self-avoiding walk, one first cuts it at a suitable extremal point. The picture below shows this as a decomposition into two half-space walks. This is why the proof goes through half-space walks before it reaches bridges.

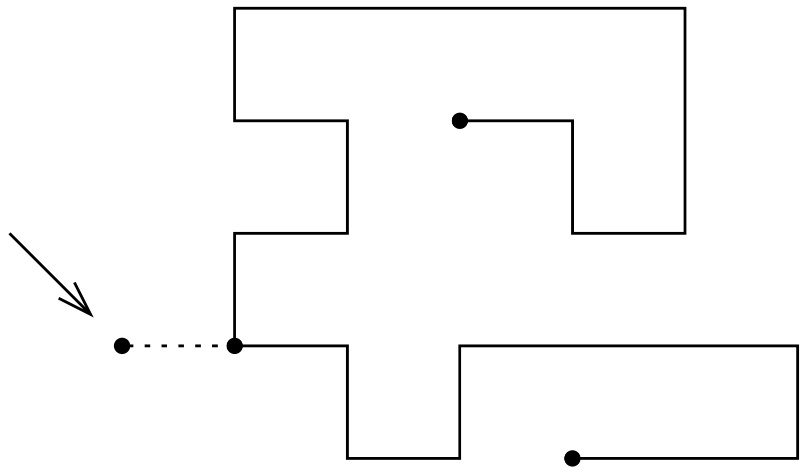


Figure 58.2: decomposing a self-avoiding walk into two half-space walks.

$$c_n \le \exp\{C\sqrt{n}\}\, b_{n+1}$$

for a constant C. The exact constant is not the point here. The point is that the correction is subexponential, so it does not affect μ.

> **Remark 58.3.1** — This is a good example of the style of rigorous self-avoiding-walk arguments. One rarely enumerates walks exactly. Instead one reorganises them into classes that have better concatenation properties, and then proves that the reorganisation loses only subexponentially many choices.

§58.4 Polygons

A self-avoiding polygon is a closed self-avoiding path, considered up to translation and often up to choice of starting point and orientation. If p_n denotes the number of n-edge self-avoiding polygons, then $p_n = 0$ when n has the wrong parity on bipartite lattices such as $\mathbb{Z}^d$.

Polygons have the same connective constant:

$$\lim_{n\to\infty} p_n^{1/n} = \mu,$$

with the usual convention of restricting to the allowed lengths. Intuitively, a long polygon is like a long self-avoiding walk whose two ends happen to meet. The closing condition changes the polynomial correction, not the exponential growth rate.

The polygon viewpoint is important because it connects self-avoiding walks with loop models and planar statistical mechanics. In two dimensions, loops are not just a counting device; they interact with conformal geometry.

The bridge picture has an analogue for polygons. The following construction turns two bridges into a longer self-avoiding walk closely related to a polygon-counting lower bound. The useful point is again not exact enumeration. It is a controlled way to build large objects out of smaller ones while keeping self-avoidance visible.

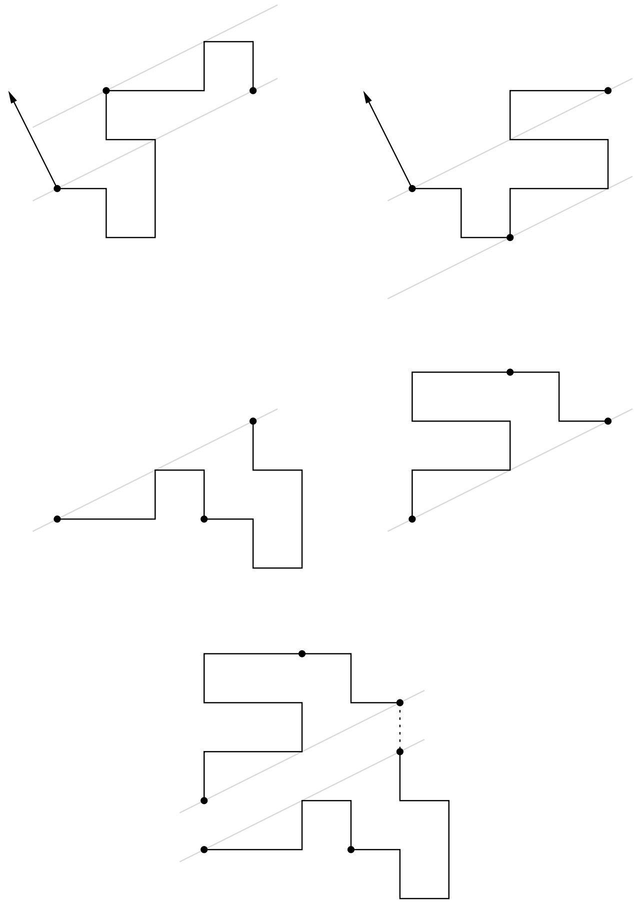


Figure 58.3: bridge data used in a polygon lower-bound construction.

§58.5 Critical exponents

The connective constant describes only the exponential part of c_n. The more delicate question is the subexponential correction. The expected form is

$$c_n \sim A\mu^n n^{\gamma-1},$$

where γ is a critical exponent. Another exponent ν describes the typical size of an n-step walk:

$$\mathbb{E}(|\omega_n|^2) \approx Dn^{2\nu}.$$

The exponent ν measures how swollen the polymer is. For simple random walk one has $\nu = 1/2$. Self-avoidance pushes the walk away from itself, so in low dimensions ν is expected to be larger.

There is also a two-point function

$$G_z(x) = \sum_{\omega:0\to x} z^{|\omega|},$$

where the sum is over self-avoiding walks from 0 to x. Summing over endpoints gives the susceptibility

$$\chi(z) = \sum_{x\in\mathbb{Z}^d} G_z(x) = \sum_{n\geq 0} c_n z^n.$$

Near z_c, one expects

$$\chi(z) \approx (z_c - z)^{-\gamma}.$$

So the same exponent γ can be read from either the coefficient asymptotics of c_n or the singularity of the susceptibility.

§58.6 How dimension changes the story

The dimension d controls how strongly a walk intersects itself. This is why the self-avoiding walk behaves differently in different dimensions.

In dimension 1, the model is trivial: after the first step, the walk cannot turn around without revisiting a vertex. Thus there are only two n-step self-avoiding walks for $n \geq 1$.

In dimension 2, self-avoidance is strong. The expected scaling limit is related to Schramm–Loewner evolution with parameter 8/3. The predicted exponents are

$$\nu = \frac{3}{4}, \qquad \gamma = \frac{43}{32}.$$

These values are part of the conformal-invariance picture, but the full square-lattice scaling limit remains a major open problem.

In dimension 3, the model is physically central but mathematically difficult. The exponents are not expected to be mean-field, and no exact solution is known.

Dimension 4 is critical. The leading power laws are mean-field-like, but logarithmic corrections appear. This is the dimension where renormalisation group methods become essential.

In dimensions $d \geq 5$, the model has mean-field behaviour. The lace expansion proves, in suitable settings, that self-avoiding walks behave diffusively:

$$\nu = \frac{1}{2}, \qquad \gamma = 1.$$

The walk is still self-avoiding, but high-dimensional space gives it enough room that self-intersections become perturbative rather than dominant.

§58.7 The broader lesson

The first layer of the theory is the following dictionary:

combinatorial object	analytic quantity
c_n	$\chi(z) = \sum c_n z^n$
μ	$z_c = 1/\mu$
bridges	controlled concatenation
polygons	closed-loop version of the model
$G_z(x)$	endpoint-resolved two-point function

The connective constant is the exponential growth rate. Critical exponents describe what remains after this exponential growth has been factored out. Bridges and polygons are not side topics: they are the combinatorial tools that make the basic growth theory work.

Self-Avoiding Walks II: The Hexagonal Lattice, Holomorphic Observables, and the Exact Connective Constant

N-20251127

§59.1 Why the hexagonal lattice is special

For most lattices, the connective constant of self-avoiding walks is not known exactly. The hexagonal lattice is exceptional. Duminil-Copin and Smirnov proved that its connective constant is

$$\mu_{\text{hex}} = \sqrt{2+\sqrt{2}}.$$

Equivalently, the critical step weight is

$$x_c = \frac{1}{\mu_{\text{hex}}} = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

This result is not an enumeration trick in the elementary sense. It comes from a discrete complex-analytic observable. The proof is one of the cleanest examples of the modern two-dimensional philosophy: at the critical point, a lattice model may possess a discrete holomorphic structure that forces exact identities.

> **Remark 59.1.1** — The number $\sqrt{2+\sqrt{2}}$ should not be memorised as a miracle constant. It is the value for which the parafermionic observable satisfies a local cancellation identity on the hexagonal lattice.

§59.2 A domain version of the counting problem

Instead of counting walks in the whole plane immediately, one works in a finite domain cut from the hexagonal lattice. Fix a boundary point a. For another boundary point or mid-edge z, consider self-avoiding walks γ starting at a and ending at z, staying inside the domain. Their weighted sum has the form

$$\sum_{\gamma: a\to z} x^{|\gamma|} \times \text{a complex phase depending on the winding of } \gamma.$$

The length weight $x^{|\gamma|}$ remembers the usual self-avoiding-walk generating function. The complex phase is the new ingredient.

The winding of a planar path records how much the direction of the path turns. On the hexagonal lattice the allowed turns are highly constrained, and this makes it possible to tune the phase so that local contributions cancel.

The first picture for this part is especially helpful. On the left, the domain is made of hexagonal-lattice cells and the boundary is described by mid-edges. On the right, the same picture explains winding: the path is not only counted by its length, but also by the total amount it turns.

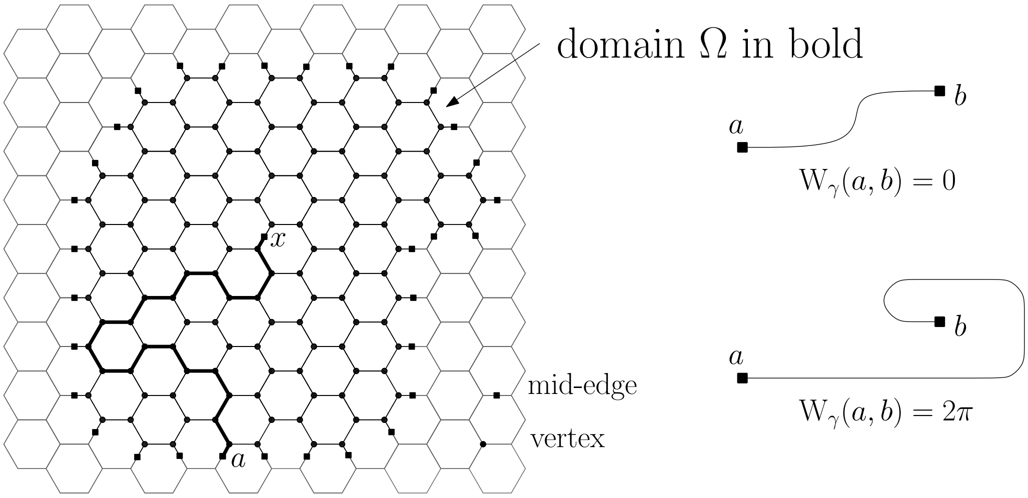


Figure 59.1: a hexagonal-lattice domain and the winding of a self-avoiding walk.

§59.3 The parafermionic observable

The observable can be remembered schematically as

$$F(z) = \sum_{\gamma: a \to z} x^{|\gamma|} e^{-i\sigma W(\gamma)},$$

where $W(\gamma)$ is the total winding and σ is a spin parameter. The exact theorem uses the special choice

$$\sigma = \frac{5}{8}, \qquad x = x_c = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

At this value, the observable satisfies a discrete Cauchy–Riemann type relation around each vertex.

The word "holomorphic" here should be read carefully. The observable is defined on a lattice, so it is not holomorphic in the classical analytic sense. Rather, it satisfies a local linear relation that becomes the Cauchy–Riemann equation in a scaling limit. This is why the observable is often called discrete holomorphic or preholomorphic.

Remark 59.3.1 — The observable is not merely a generating function. The phase remembers geometry. Without the winding phase, the local cancellation identity would not hold.

§59.4 From local cancellation to a global identity

The local identity is summed over all vertices in a finite domain. Interior terms cancel, leaving only boundary contributions. This is the discrete analogue of using Green's theorem or Cauchy's theorem: local holomorphicity turns an interior statement into a boundary statement.

The cancellation is proved by grouping local contributions. The picture below shows the two types of groupings: pairs of walks that visit all three adjacent mid-edges, and triplets obtained by extending or erasing the last step. At the special values of x and σ, each such group has total contribution zero.

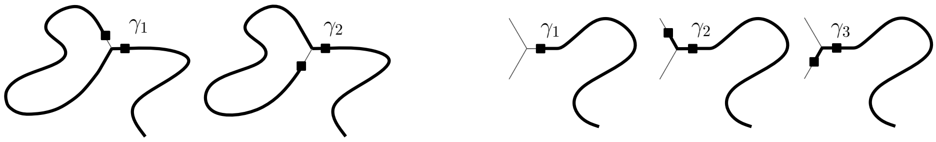


Figure 59.2: pair and triplet cancellations in the proof of the local observable identity.

For a suitable trapezoidal domain on the hexagonal lattice, the resulting identity relates generating functions of walks ending on different parts of the boundary. Very schematically it has the form

Here is the actual domain picture used in the lecture notes. The boundary pieces are labelled separately because the final identity distinguishes walks according to which part of the boundary they hit.

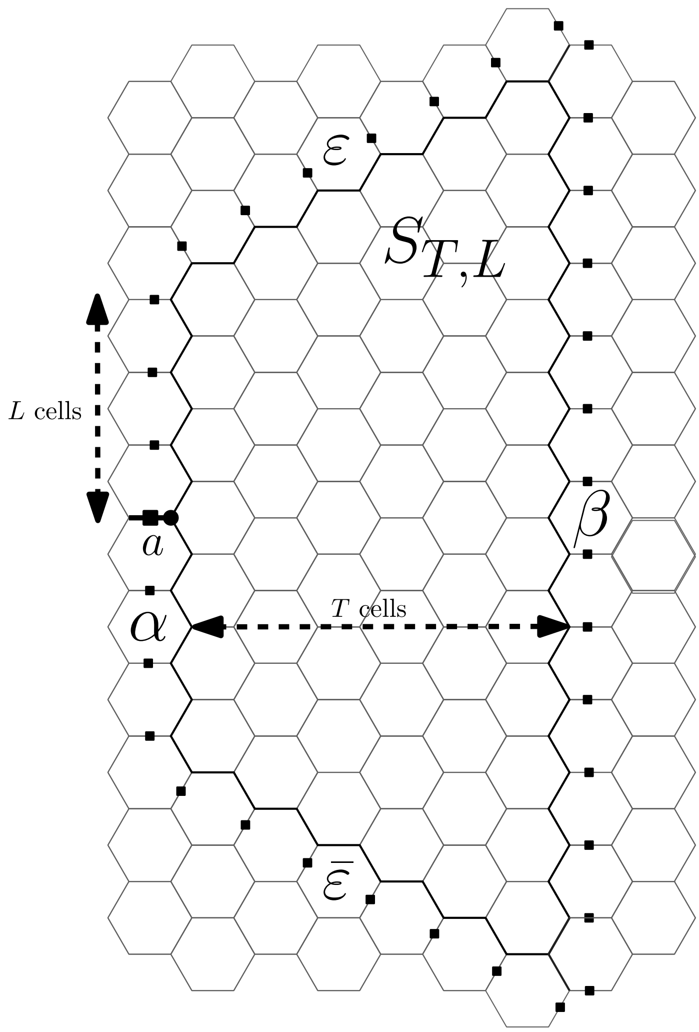


Figure 59.3: the trapezoidal domain $S_{T,L}$ and its boundary pieces.

$$1 = A_T(x_c) + B_T(x_c) + E_T(x_c),$$

where the terms count walks from the marked boundary edge to various boundary arcs, with explicit positive coefficients after taking the correct real part.

The important structural fact is positivity. Once the boundary identity has positive terms, it gives inequalities for ordinary generating functions. Letting the size of the domain tend to infinity then produces information about convergence and divergence at the critical value.

§59.5 The upper bound on the connective constant

To prove

$$\mu_{\text{hex}} \le \sqrt{2+\sqrt{2}},$$

one shows that the self-avoiding-walk generating function cannot converge beyond

$$x_c = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

Equivalently, if $x > x_c$, then the full generating function must diverge. The boundary identity supplies enough positive mass at x_c to force criticality at or before this value.

The proof is subtle because finite-domain identities do not immediately determine the infinite-volume connective constant. One has to pass from boundary generating functions to walks in the full lattice. The Hammersley–Welsh bridge technology from the previous note is part of the background that makes this passage possible.

§59.6 The lower bound on the connective constant

The other inequality

$$\mu_{\text{hex}} \ge \sqrt{2+\sqrt{2}}$$

amounts to showing that the generating function still converges below x_c. In practice one proves that for $x < x_c$, the weighted sum of self-avoiding walks is finite. This uses the same domain identities together with limiting arguments and standard consequences of submultiplicativity.

Combining the two bounds gives the exact value

$$\mu_{\text{hex}} = \sqrt{2+\sqrt{2}}.$$

Theorem 59.6.1 (Connective constant of the hexagonal lattice)
For self-avoiding walks on the hexagonal lattice,

$$\lim_{n\to\infty} c_n^{1/n} = \sqrt{2+\sqrt{2}}.$$

§59.7 Relation with SLE

The same observable also points toward the expected scaling limit. The conjectural statement is that two-dimensional self-avoiding walk, after suitable rescaling, converges to $\mathrm{SLE}_{8/3}$. This would imply many predicted critical exponents, such as

$$\nu = \frac{3}{4}, \qquad \gamma = \frac{43}{32}.$$

The exact connective constant result is much more modest than a full scaling-limit theorem, but it belongs to the same conceptual world. The local observable is designed so that at criticality the lattice model begins to look conformally natural.

Remark 59.7.1 — The theorem gives the exact exponential growth constant on the hexagonal lattice. It does not by itself prove the $\mathrm{SLE}_{8/3}$ scaling limit. The former is a rigorous theorem; the latter is the guiding two-dimensional conjectural picture.

§59.8 Loop models and the same mechanism

Self-avoiding walks can be viewed as the $n = 0$ member of the $O(n)$ loop model. In the loop model, configurations consist of nonintersecting loops, and each loop receives a weight n. Formally setting $n = 0$ kills configurations with closed loops and leaves a single open path, which is the self-avoiding walk.

This viewpoint explains why holomorphic observables appear. For special values of the loop weight and the step weight, the model has a parafermionic observable satisfying local relations. The hexagonal self-avoiding walk is the most famous case, but the mechanism belongs to a broader family of two-dimensional lattice models.

The phase diagram situates this in the $O(n)$ phase diagram. For this note, the picture should be read only as orientation: self-avoiding walk corresponds to the $n = 0$ edge of the loop-model story, while the exactly solvable point belongs to a broader two-dimensional critical curve.

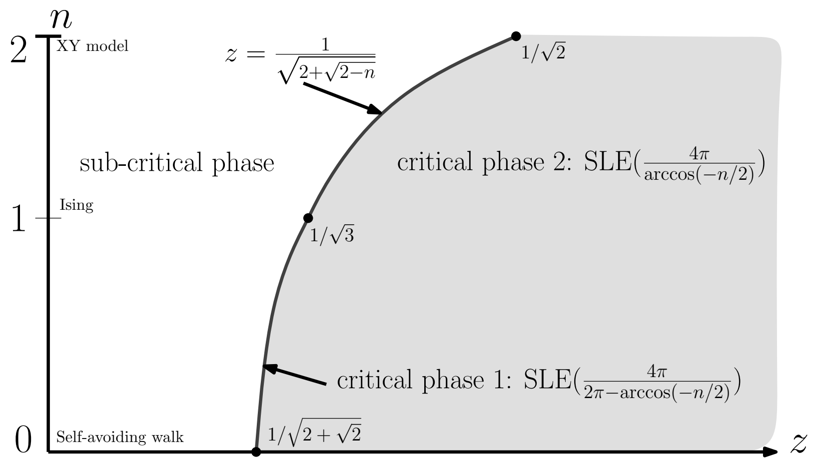


Figure 59.4: phase diagram for $O(n)$ loop models.

§59.9 From example to method

The proof of the hexagonal connective constant has a clear conceptual shape:

$$\begin{gathered}\text{choose a winding-weighted observable}\\ \Downarrow\\ \text{tune } x \text{ and } \sigma \text{ so local cancellations hold}\\ \Downarrow\\ \text{sum over a finite domain}\\ \Downarrow\\ \text{obtain positive boundary identities}\\ \Downarrow\\ \text{pass to the infinite lattice and identify } \mu.\end{gathered}$$

The striking part is that an exact enumerative constant is obtained through a complex-analytic idea rather than through exact enumeration of c_n.

Self-Avoiding Walks III: Lace Expansion, Functional Integrals, and the Critical Dimension Four

N-20251128

§60.1 The high-dimensional philosophy

In low dimensions, self-avoidance dominates the geometry of the walk. In high dimensions, the walk has more room, and self-intersections become less frequent. The expected result is that the self-avoiding walk behaves like simple random walk at large scales:

$$|\omega_n| \text{ is typically of order } n^{1/2}.$$

This is called mean-field behaviour.

The rigorous tool that proves this in dimensions $d \geq 5$, for nearest-neighbour self-avoiding walk, is the lace expansion. It is a systematic inclusion–exclusion method that turns the global self-avoidance constraint into a perturbative correction to the simple random walk equation.

§60.2 The two-point function

The central analytic object is the two-point function

$$G_z(x) = \sum_{\omega:0\to x} z^{|\omega|},$$

where the sum is over self-avoiding walks from 0 to x. The susceptibility is

$$\chi(z) = \sum_{x\in\mathbb{Z}^d} G_z(x).$$

The critical point is $z_c = 1/\mu$.

For simple random walk, the two-point function satisfies a renewal equation. If D is the one-step distribution, then the generating function C_z satisfies

$$C_z = \delta_0 + zD * C_z.$$

Equivalently, in Fourier space,

$$\widehat{C}_z(k) = \frac{1}{1 - z\widehat{D}(k)}.$$

For self-avoiding walk, this exact renewal equation fails because the future path is constrained by the past. The lace expansion repairs the equation by adding correction terms.

§60.3 The lace expansion as corrected random walk

The output of the lace expansion can be remembered schematically as

$$G_z = \delta_0 + zD * G_z + \Pi_z * G_z,$$

where Π_z is the lace-expansion correction. This formula is not meant as a definition of Π_z; it is the structural form to remember. The term $zD * G_z$ is what simple random walk would have. The term $\Pi_z * G_z$ records the accumulated effect of forbidding self-intersections.

The correction terms are represented by small diagrams. The following two examples should be read as diagrammatic shorthand rather than literal pictures of the walk. A line represents a two-point function, and extra chords represent constraints coming from self-intersection corrections.

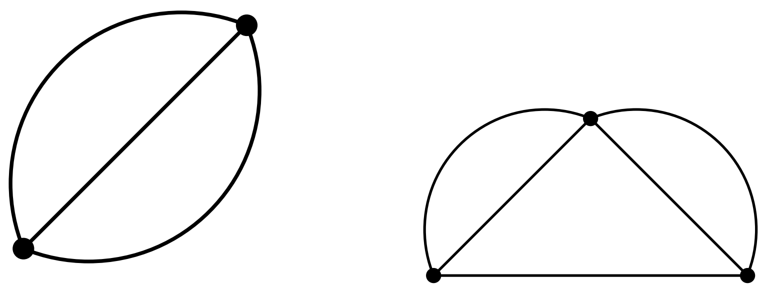


Figure 60.1: typical lace-expansion diagrams.

In Fourier space the relation becomes morally

$$\widehat{G}_z(k) = \frac{1}{1 - z\widehat{D}(k) - \widehat{\Pi}_z(k)}.$$

Thus the proof of mean-field behaviour becomes a problem of showing that Π_z is small and regular enough near criticality.

Remark 60.3.1 — The lace expansion is often intimidating because the combinatorics of laces is elaborate. Conceptually, however, its role is simple: it isolates the first random-walk term and packages all self-intersection corrections into a controlled perturbation.

§60.4 Bubble diagrams and diagrammatic estimates

The perturbative parameter is measured by diagrammatic sums. A basic example is the bubble diagram

The word "bubble" is literal in the diagrammatic notation. One sums over two independent-looking connections between the same endpoints. Finiteness of the corresponding sum is what makes the high-dimensional expansion behave like a small perturbation.

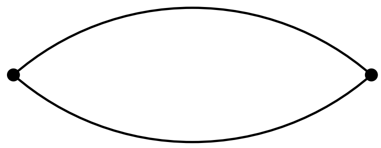


Figure 60.2: the bubble diagram.

$$B(z) = \sum_{x \in \mathbb{Z}^d} G_z(x)^2.$$

For simple random walk at criticality, this sum is finite exactly above the critical dimension 4. This is the first signal that dimension 4 is the borderline.

In dimensions $d > 4$, one can prove estimates of the following kind:

$$\sum_x G_{z_c}(x)^2 < \infty.$$

Such bounds imply that the lace-expansion correction behaves perturbatively. They lead to the mean-field critical exponents

$$\gamma = 1, \qquad \nu = \frac{1}{2}.$$

More concretely, the susceptibility diverges like

$$\chi(z) \asymp \frac{1}{1 - z/z_c},$$

and the mean-square displacement grows linearly in n.

§60.5 Differential inequalities

One route to controlling the susceptibility is through differential inequalities. The generating function $\chi(z)$ is increasing in z, and differentiating it gives information about the average length of a walk under the z-weighted measure. The lace expansion supplies inequalities that compare $\chi'(z)$ to powers of $\chi(z)$.

The mean-field picture corresponds to the ordinary differential equation

$$\chi'(z) \approx C\chi(z)^2.$$

Solving this heuristic equation gives a simple pole at the critical point, hence $\gamma = 1$. The rigorous proof replaces the heuristic approximation by upper and lower differential inequalities with controlled error terms.

§60.6 Functional integral representations

The lecture notes also explain a different representation of the model. A weakly self-avoiding walk can be written using a self-intersection local time. In continuous time, one can weight a walk by

$$\exp\left(-g \sum_x L_x^2\right),$$

where L_x is the local time spent at x and $g > 0$ penalises self-intersections. The strict self-avoiding walk corresponds morally to the limit where self-intersections are forbidden completely.

There is a supersymmetric functional integral representation in which walk quantities are encoded by integrals involving bosonic and fermionic variables. The point is not that one wants to compute these integrals explicitly. The point is that the representation turns the probabilistic walk model into a field-theoretic object suitable for renormalisation group analysis.

The renormalisation-group part organises space into blocks and then into polymers made of blocks. This terminology is unfortunately overloaded: these "polymers" are not the original self-avoiding walks. They are bookkeeping regions used to localise the effective interaction at each scale.

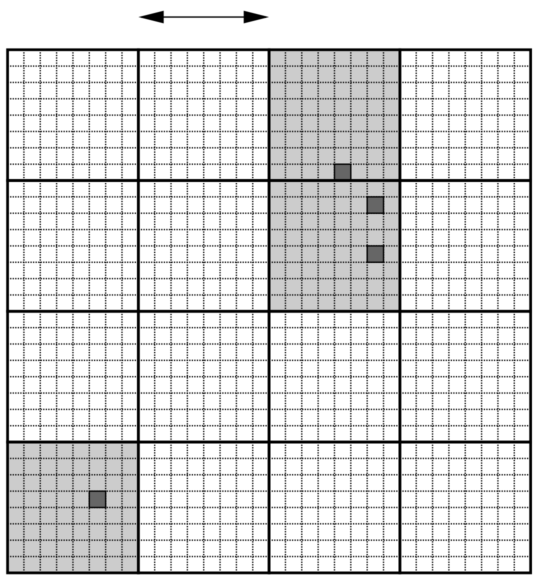


Figure 60.3: a polymer at one scale and its closure at the next scale.

Remark 60.6.1 — The fermionic variables are not an extra physical particle in the polymer. They are an algebraic device that makes cancellations exact. In this representation, supersymmetry is a bookkeeping symmetry for the walk model.

§60.7 Dimension four and logarithmic corrections

Dimension 4 is the upper critical dimension for self-avoiding walk. Above four dimensions, the bubble diagram is finite and the lace expansion gives mean-field behaviour. At four dimensions, the relevant diagram is marginal: it fails only logarithmically. This is why the leading powers look mean-field but acquire logarithmic corrections.

For the continuous-time weakly self-avoiding walk in $d = 4$, the renormalisation group tracks how coupling constants change across length scales. One decomposes the covariance into scale pieces and integrates out fluctuations scale by scale. The effective interaction evolves under a map of the form

$$Z_0 \mapsto Z_1 \mapsto Z_2 \mapsto \cdots .$$

The essential phenomenon is that the self-interaction coupling is marginally irrelevant: it flows slowly to zero. The walk becomes asymptotically Gaussian, but only after logarithmic corrections are taken into account.

A typical conclusion has the form

$$\chi(g,\nu) \sim A\,\varepsilon^{-1}(\log\varepsilon^{-1})^{1/4}$$

as the killing parameter $\varepsilon \downarrow 0$, with model-dependent constants. The exact formulation depends on the version of the weakly self-avoiding walk, but the memory aid is stable: four dimensions means mean-field powers multiplied by logarithms.

§60.8 How the three methods fit together

The three major techniques in these lectures have different roles:

method	dimension or setting	main output
bridges and polygons	all dimensions	growth constants and subexponential bounds
holomorphic observable	$d = 2$, hexagonal lattice	exact connective constant
lace expansion	$d > 4$	mean-field critical behaviour
renormalisation group	$d = 4$	logarithmic corrections

This is a useful mental map because the methods are not interchangeable. The two-dimensional proof uses special planar structure. The high-dimensional proof uses small intersection probabilities. The four-dimensional proof needs a scale-by-scale analysis because the perturbation is exactly marginal at first order.

§60.9 The conceptual turn

The lace expansion should be remembered as the high-dimensional counterpart of the two-dimensional holomorphic observable. Both are ways of making the self-avoidance constraint tractable, but they exploit opposite regimes. In two dimensions, geometry is rigid and special identities appear. In high dimensions, geometry is loose and self-intersections become a small perturbation.

The critical dimension 4 is the boundary between these worlds. Above it, self-avoiding walk is close to simple random walk. At it, the same statement remains morally true, but logarithms record the slow decay of the self-interaction.

VII

Linear Algebra

Part VII: Contents

Prop Presentations for Linear Algebra: Diagrams, Matrices, and Relations

N-20210621

§61.1 First orientation

The small diagram that first caught my attention looked almost too simple: a few white nodes, a few grey nodes, some scalar labels, and many wires. The surprising claim was that this was not merely a mnemonic for linear algebra. It was a presentation of linear algebra as a graphical calculus.

The original fascination was this:

ordinary linear algebra writes maps as grids of numbers;
graphical linear algebra writes them as processes made of wires and nodes.

This note decodes that special symbolic system. The point is not to admire the pictures, but to understand why the pictures form a proof system.

The guiding slogan is:

matrices are compiled diagrams, and diagrams are structured linear processes.

Once this is understood, the later movement toward linear relations, Lagrangian relations, circuits, and stabilizer calculi becomes much less mysterious.

§61.2 Props as typed wiring diagrams

The right ambient language is a prop.

Definition 61.2.1 (Prop)**.** A prop is a strict symmetric monoidal category whose objects are natural numbers

$$0, 1, 2, \ldots$$

and whose monoidal product on objects is addition.

The object n should be read as n parallel wires. A morphism

$$f : n \to m$$

is a process with n input wires and m output wires.

There are two basic operations on morphisms.

First, vertical composition plugs outputs into inputs:

$$f : n \to m, \qquad g : m \to p, \qquad g \circ f : n \to p.$$

Second, the monoidal product puts diagrams side by side:

$$f : n \to m, \qquad g : n' \to m', \qquad f \oplus g : n + n' \to m + m'.$$

This makes a prop feel like a typed programming language for wiring diagrams:

prop notation	programming / wiring intuition
n	n wires
$f : n \to m$	process with n inputs and m outputs
$g \circ f$	run f, then run g
$f \oplus g$	run f and g in parallel

The prop $\mathbf{Mat}_k$ has natural numbers as objects and matrices over k as morphisms. A morphism

$$n \to m$$

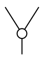
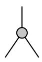
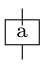
is an $m \times n$ matrix, regarded as a linear map

$$k^n \to k^m.$$

Composition is matrix multiplication, and the monoidal product is block direct sum.

§61.3 The alphabet: copy, discard, add, zero, and scalars

The graphical calculus $\mathbf{cb}_k$ is generated by a small alphabet. In the diagrams below, time flows upward: inputs are at the bottom and outputs are at the top.

symbol	type	linear meaning	intuition
 copy Δ	$1 \to 2$	$x \mapsto (x, x)$	duplicate one wire into two copies
 discard ϵ	$1 \to 0$	$x \mapsto *$	forget the value on a wire
 add μ	$2 \to 1$	$(x, y) \mapsto x + y$	combine two wires by addition
 zero η	$0 \to 1$	$* \mapsto 0$	create the additive zero
 scalar a	$1 \to 1$	$x \mapsto ax$	multiply a wire by $a \in k$

One must be careful about the word "copy." In ordinary vector spaces, there is no natural linear map $V \to V \oplus V$ for every abstract vector space V unless one has fixed the diagonal map. In this prop, one wire represents the free rank-one module k. The copy generator is the linear map

$$k \to k \oplus k, \qquad x \mapsto (x, x).$$

Arbitrary matrices are then built by composing and tensoring these elementary maps.

§61.4 Two colours: a comonoid and a monoid

The white structure is the copying structure:

$$\Delta : 1 \to 2, \qquad \epsilon : 1 \to 0.$$

Algebraically, it is a commutative comonoid. The equations say that copying twice does not depend on which copy is copied first, that the two copies can be swapped, and that discarding a copied output returns the original wire.

The grey structure is the adding structure:

$$\mu : 2 \to 1, \qquad \eta : 0 \to 1.$$

Algebraically, it is a commutative monoid. The equations say that adding three terms does not depend on bracketing, that order does not matter, and that adding zero changes nothing.

A good way to remember the two colours is:

colour	structure	operation
white	comonoid	copy / discard
grey	monoid	add / zero

The calculus becomes interesting because these two structures are not independent. They interact through the bialgebra law.

§61.5 The bialgebra law

The central equation says:

$$\text{copy after add} = extadd after copying both inputs.$$

In ordinary algebra, this is the statement

$$\Delta(x+y) = (x+y, x+y) = (x,x) + (y,y).$$

The two sides can be read as dataflow programs.

Left side:

```
input:  (x, y)
step 1: add       -> x + y
step 2: copy      -> (x + y, x + y)
output: (x + y, x + y)
```

Right side:

```
input:  (x, y)
step 1: copy x and copy y      -> (x, x, y, y)
step 2: swap the middle wires  -> (x, y, x, y)
step 3: add pairwise           -> (x + y, x + y)
output: (x + y, x + y)
```

This is the first moment where the topology of wires matters. The swap of the middle wires is not decoration. It is the wiring needed to pair the first copy of x with the first copy of y, and the second copy of x with the second copy of y.

> **Remark 61.5.1** — The bialgebra law is the graphical form of distributivity. It says that copying respects addition. Without this law, the white and grey nodes would be two unrelated gadgets; with it, they become the algebraic core of linear algebra.

§61.6 Scalars as ring operations

A scalar bead labelled $a \in k$ is the one-wire map

$$[a] : 1 \to 1, \qquad x \mapsto ax.$$

The scalar equations force these beads to behave like elements of the ring k.

Stacking scalar beads corresponds to multiplication:

$$[b] \circ [a] = [ba].$$

If k is commutative, this is just $[ab]$.

Parallel scalar branches combined by the add/copy structure express addition:

$$[a] + [b] = [a + b].$$

The identity scalar is a plain wire:

$$[1] = \mathrm{id}_1.$$

The zero scalar factors through discard and zero:

$$[0] = \eta \circ \epsilon.$$

So the calculus has two layers:

structural layer	copy, discard, add, zero
coefficient layer	scalar beads labelled by elements of k

Together these layers generate matrices.

§61.7 How a matrix becomes a diagram

Consider the matrix

$$A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} : k^2 \to k^2.$$

It sends

$$(x_1, x_2) \longmapsto (y_1, y_2), \qquad y_1 = 2x_1 + 3x_2, \qquad y_2 = 5x_1 + 7x_2.$$

The diagrammatic recipe is:

columns	input wires
rows	output wires
A_{ij}	scalar bead on the branch from input j to output i
copy nodes	fan one input out to all rows
add nodes	sum all contributions landing at one output

In code-like form:

```
def compile_matrix(A):
    # A has m rows and n columns.
    # Input wire j carries x_j.
    # Output wire i should carry sum_j A[i][j] * x_j.
    for each input wire j:
        copy x_j into m branches
    for each pair (i, j):
        put scalar bead A[i][j] on branch j -> i
    for each output wire i:
        add all branches landing at i
```

For the example above, the diagram computes

$$y_1 = 2x_1 + 3x_2,$$
$$y_2 = 5x_1 + 7x_2.$$

This is why diagrams can represent arbitrary matrices. Copy creates enough branches; scalars weight the branches; adders collect contributions.

§61.8 Completeness: a proof system for matrices

The key theorem can be stated as follows.

Theorem 61.8.1 (Completeness of the matrix prop)
For a suitable ring k, the graphical theory $\mathbf{cb}_k$ presents the prop $\mathbf{Mat}_k$ of matrices over k under direct sum.

This theorem has two parts.

Soundness says that every equation derivable by diagram rewriting is true as a matrix equation. If two diagrams can be transformed into each other using the graphical rules, then they denote the same matrix.

Completeness says the converse: if two diagrams denote the same matrix, then the graphical rules are strong enough to prove them equal.

Thus the diagrams are not merely suggestive notation. They form a proof system for linear algebra:

$$\text{diagram equality} \quad \Longleftrightarrow \quad \text{matrix equality.}$$

§61.9 Why the calculus hides indices and exposes structure

Matrix notation is extremely efficient, but it hides the process structure behind indices. The product of two matrices is written

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

The summation index k is an internal wire: output from one process becomes input to another. Diagrammatic notation makes that internal connection visible.

The comparison is:

matrix notation	diagram notation
entries and indices	wires and nodes
matrix multiplication	vertical plugging
block direct sum	side-by-side placement
transpose-like operations	reflection or reversal, depending on convention
trace / feedback	connecting an output back to an input

It would be too strong to say that $\mathbf{Mat}_k$ is completely basis-free: wire counts still refer to finite free modules k^n. The better statement is:

the calculus hides indices and exposes compositional structure.

This is exactly what makes it useful in networks, signal flow graphs, circuits, tensor contractions, and other settings where the shape of composition matters more than individual coordinates.

§61.10 From matrices to linear relations

A matrix is a total linear function

$$k^n \to k^m.$$

A linear relation from k^n to k^m is more general. It is a linear subspace

$$R \subseteq k^n \oplus k^m.$$

Instead of saying each input has exactly one output, a relation says which input-output pairs are compatible.

For example, the equation

$$x + y = z$$

defines a linear relation among x, y, z. It is not best understood as a function until one chooses which variables are inputs and which are outputs.

This is why graphical calculi naturally move beyond matrices. Wires and nodes are good at expressing constraints. A relation can be composed by connecting shared variables and existentially hiding the internal wires. This is the relational analogue of matrix multiplication.

The broader graphical-linear-algebra story uses interacting Hopf-algebra structures to present such relations. The matrix calculus $\mathbf{cb}_k$ is the entry point; linear relations are the next layer.

§61.11 From linear relations to Lagrangian relations

The later part of the story enters symplectic linear algebra. A symplectic vector space is a vector space V equipped with a nondegenerate skew-symmetric form

$$\omega : V \times V \to k.$$

The standard example is k^{2n} with form represented by the block matrix

$$J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}.$$

For a subspace $W \subseteq V$, define its symplectic orthogonal by

$$W^\omega = \{v \in V : \omega(v, w) = 0 \text{ for all } w \in W\}.$$

A subspace is Lagrangian if

$$W = W^\omega.$$

Definition 61.11.1 (Lagrangian relation, informal)**.** A Lagrangian relation is a linear relation that is Lagrangian inside the appropriate symplectic input-output space.

The conceptual shift is:

linear relation	linear constraint
Lagrangian relation	symplectic or phase-space-compatible constraint

This is why the calculus becomes relevant to physics. Phase space, circuits, and stabilizer-like systems are naturally symplectic.

§61.12 Affine shifts and stabilizers

The affine version allows translations. Instead of linear subspaces passing through the origin, one allows affine subspaces:

$$v + W.$$

Diagrammatically, this means adding generators that can shift a relation by a constant.

This affine Lagrangian-relational world connects to several process theories:

domain	role of the calculus
electrical circuits	constraints between currents and potentials
signal flow graphs	linear dynamical interconnection
Spekkens' toy theory	epistemic states as affine Lagrangian relations
qudit stabilizer circuits	symplectic/affine structure over finite fields

For this note, these applications should be read as the research-level endpoint rather than the entry point. The entry point is still the alphabet: copy, discard, add, zero, and scalars.

§61.13 Relation with ZX-calculus

The resemblance to ZX-calculus is real but should be stated carefully. Both calculi use coloured nodes, wires, and bialgebra-like interaction laws. Both are props or prop-like symmetric monoidal graphical languages. Both replace long index manipulations by diagrammatic rewriting.

The difference is the intended semantics:

graphical linear algebra	ZX-calculus
matrices, relations, linear constraints	quantum processes
copy/add structures	complementary observables
linear algebra and circuits	categorical quantum mechanics

Thus $\mathbf{cb}_k$ is not simply ZX-calculus. It is better seen as a training ground for understanding why ZX-like diagrammatic calculi work: algebraic equations are replaced by local topological rewrites.

§61.14 Broader perspective

The main lesson is not that matrices can be drawn prettily. The main lesson is that a small symbolic system can present all matrix algebra.

The core dictionary is:

diagrammatic object	linear-algebraic meaning
n wires	k^n
diagram $n \to m$	$m \times n$ matrix
vertical composition	matrix multiplication
side-by-side composition	block direct sum
copy	$x \mapsto (x, x)$
add	$(x, y) \mapsto x + y$
scalar bead a	$x \mapsto ax$

The bialgebra law is the heart of the system:

copying a sum is the same as summing the copies.

The completeness theorem says that this is not only notation but a genuine proof system:

diagrammatic equality is matrix equality.

After this is understood, the path toward linear relations, Lagrangian relations, affine shifts, circuits, and stabilizer calculi is no longer a jump. It is the natural extension of the same idea: algebra can be presented as topology of wires.

Exercises on $\mathbb{R}^n$, $\mathbb{C}^n$, Vector Spaces, and Subspaces

N-20220407

§62.1 Complex arithmetic

The first page is a worked set of exercises from the beginning of linear algebra, where $\mathbb{C}$ is treated as a field.

If $a, b \in \mathbb{R}$ are not both zero, then

$$\frac{1}{a+bi} = \frac{a-bi}{a^2+b^2} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i.$$

Thus the required real numbers c, d in

$$\frac{1}{a+bi} = c + di$$

are

$$c = \frac{a}{a^2+b^2}, \qquad d = -\frac{b}{a^2+b^2}.$$

A cube root of 1 other than 1 is

$$\omega = \frac{-1+\sqrt{3}i}{2},$$

since

$$\omega^2 + \omega + 1 = 0, \qquad \omega^3 = 1.$$

The two nontrivial cube roots are

$$\frac{-1+\sqrt{3}i}{2}, \qquad \frac{-1-\sqrt{3}i}{2}.$$

Two square roots of i are obtained from polar form:

$$i = e^{i\pi/2}, \qquad \sqrt{i} = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}, \qquad -\sqrt{i} = -\frac{1+i}{\sqrt{2}}.$$

§62.2 Field axioms in $\mathbb{C}$

If

$$\alpha = a + bi, \qquad \beta = c + di, \qquad \lambda = x + yi,$$

then commutativity and associativity of addition follow from those of real numbers. For example,

$$\alpha + \beta = (a+c) + (b+d)i = (c+a) + (d+b)i = \beta + \alpha.$$

Similarly,

$$\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$$

follows by expanding both sides and using distributivity in $\mathbb{R}$.

Every $\alpha \in \mathbb{C}$ has an additive inverse, namely $-\alpha$. If $\alpha \neq 0$, its multiplicative inverse is

$$\alpha^{-1} = \frac{\overline{\alpha}}{|\alpha|^2}.$$

§62.3 Vector space identities

In a vector space V over a field F, one proves

$$-(-v) = v$$

because $-v + v = 0$, so the additive inverse of $-v$ is v.

If $a \in F$, $v \in V$, and $av = 0$, then either $a = 0$ or $v = 0$. Indeed, if $a \neq 0$, multiply by a^{-1}:

$$v = a^{-1}(av) = a^{-1}0 = 0.$$

Given $v, w \in V$, the equation

$$v + 3x = w$$

has the unique solution

$$x = \frac{1}{3}(w - v),$$

provided $3 \neq 0$ in the field.

§62.4 A non-example with infinity

The page asks whether

$$\mathbb{R} \cup \{\infty\} \cup \{-\infty\}$$

can be made into a real vector space with the suggested arithmetic. It cannot. One obstruction is associativity. If one declares

$$\infty + (-\infty) = 0,$$

then

$$(\infty + \infty) + (-\infty) = \infty + (-\infty) = 0,$$

while

$$\infty + (\infty + (-\infty)) = \infty + 0 = \infty.$$

Thus associativity fails.

§62.5 Subspaces

A subset $U \subseteq V$ is a subspace iff it is nonempty and closed under addition and scalar multiplication. Equivalently,

$$x, y \in U,\ a, b \in F \quad \Rightarrow \quad ax + by \in U.$$

Examples from the page include:

(i) $\{(x_1, x_2, x_3, x_4) \in F^4 : x_3 = 5x_4 + b\}$ is a subspace iff $b = 0$.

(ii) The continuous real-valued functions on $[0, 1]$ form a subspace of $\mathbb{R}^{[0,1]}$.

(iii) The differentiable real-valued functions on $\mathbb{R}$ form a subspace of $\mathbb{R}^{\mathbb{R}}$.

(iv) The differentiable functions f on $(0, 3)$ satisfying $f'(2) = b$ form a subspace iff $b = 0$.

(v) The complex sequences converging to 0 form a subspace of $\mathbb{C}^{\infty}$.

§62.6 Bridge to later ideas

These exercises are elementary, but they encode several structural distinctions that reappear throughout algebra and geometry.

Elementary exercise	Later structure
checking closure under linear combinations	subspaces and kernels
conditions like $x_3 = 5x_4 + b$	linear subspace versus affine translate
real versus complex scalar multiplication	scalar restriction and complex linearity
solving constraints with free variables	quotients and cosets of kernels
coordinates of vectors	dual basis and linear functionals

For example, multiplication by $a+bi$ on $\mathbb{C}$, viewed as a real vector space, is represented by

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Its determinant is $a^2 + b^2$, the squared modulus. Thus complex scalar multiplication already contains the geometry of rotation and scaling.

§62.7 How to read basic vector-space exercises

The purpose of these exercises is not to check axioms forever. The real skill is to recognize when an object is already known to be a vector space and when a proposed subset passes the subspace test. The fastest mental template is:

$$0 \in U, \qquad u, v \in U \Rightarrow u + v \in U, \qquad \lambda u \in U.$$

The examples $f'(2) = b$, $x_3 = 5x_4 + b$, and sequences converging to zero are all testing the same issue: affine conditions usually fail to be subspaces unless the constant term is zero. In later linear algebra this becomes the distinction between a linear subspace and an affine translate.

§62.8 Beyond real and complex scalar structures

The elementary note introduces vector spaces through closure under addition and scalar multiplication. The reason this structure is so rigid is that the scalars form a field. A vector space over a field k is a module over k, and field scalars make every linearly independent set extend to a basis. This is why every finite-dimensional vector space has a dimension and why all bases have the same number of vectors.

If the scalars are changed from a field to a ring, the same axioms define a module. Modules over rings need not have bases. For example, $\mathbb{Z}/2\mathbb{Z}$ is a module over $\mathbb{Z}$, but it is not a free module: no element can serve as an integer-basis vector because $2x = 0$ for every x.

This explains why elementary linear algebra is unusually clean. It is not merely because the definitions are simple; it is because fields remove torsion and many divisibility obstacles.

§62.9 Quotient spaces

If $U \subseteq V$ is a subspace, the quotient space V/U consists of cosets

$$v + U = \{v + u : u \in U\}.$$

Two vectors represent the same coset iff their difference lies in U. The quotient forgets directions inside U and keeps only the remaining information.

The dimension formula is

$$\dim(V/U) = \dim V - \dim U.$$

This is the structural version of “free variables after constraints.” Solving a linear system often means identifying a particular affine coset of the null space. The quotient-space language makes this precise.

§62.10 Dual spaces and coordinates

The dual space is

$$V^* = \mathrm{Hom}(V, k),$$

the vector space of linear functionals $V \to k$. If $e_1, \dots, e_n$ is a basis of V, the dual basis $e^1, \dots, e^n$ is defined by

$$e^i(e_j) = \delta^i_j.$$

Coordinates are evaluations by dual basis vectors:

$$v = \sum_i x_i e_i, \qquad x_i = e^i(v).$$

Thus a coordinate is not the vector itself; it is the value of a linear functional on the vector.

This distinction becomes essential for tensors, differential forms, gradients, and coordinate changes. The elementary row/column distinction is the first shadow of the distinction between vectors and covectors.

§62.11 Vector-space exercises are module theory in a favorable case

Subspaces, bases, linear independence, and dimension are not merely bookkeeping. They are the field-module case of a broader algebraic story. Quotients explain constraints; duals explain coordinates; modules explain why vector spaces are the friendly special case. The original exercises train the stable finite-dimensional core from which these advanced notions grow.

§62.12 Why fields make linear algebra work

A useful way to reread the original vector-space exercises is to ask which steps secretly use division. When we prove that a nonzero vector can be rescaled, that a pivot can be normalized to 1, or that a linearly independent list can be extended to a basis, we use the fact that every nonzero scalar has an inverse. This is exactly the field condition.

Over a general ring, the same-looking statements may fail. The equation $2x = 1$ has no solution over $\mathbb{Z}$, so one cannot normalize a pivot 2 by division. The module $\mathbb{Z}/2\mathbb{Z}$ has torsion because $2x = 0$ for every element. This is why row reduction over fields is clean, while integer linear algebra needs Smith normal form.

The elementary notion of a basis also becomes more delicate over rings. A free module has a basis, but not every module is free. Therefore the theorem

$$\dim V = \text{number of basis vectors}$$

is not just a counting convenience; it is a privilege of vector spaces over fields. This explains why linear algebra is often the first genuinely structural algebra course: it gives a setting where generation, independence, coordinates, and quotienting all behave perfectly.

§62.13 The universal property of quotient spaces

The quotient V/U is not merely a set of cosets. It is characterized by a universal property. Let $q : V \to V/U$ be the quotient map. If a linear map $T : V \to W$ kills U, meaning $U \subseteq \ker T$, then there is a unique linear map $\bar{T} : V/U \to W$ such that

$$T = \bar{T} \circ q.$$

This says that any linear map unable to distinguish vectors differing by elements of U must factor through the quotient.

This universal property is the conceptual reason quotient spaces appear in homology, representation theory, and functional analysis. In homology, boundaries are declared trivial, so cycles differing by boundaries become the same class. In linear algebra exercises, free variables often describe a coset of a kernel; quotient language explains what information survives after ignoring kernel directions.

Invertible Matrices, Adjoints, Row Reduction, and Rotations

N-20220411

§63.1 Invertibility

A square matrix A is invertible if there exists a matrix A^{-1} such that

$$A^{-1}A = I, \qquad AA^{-1} = I.$$

Some immediate consequences recorded in the note are:

(i) If A is invertible, then Gaussian elimination has a pivot in every column and every row.

(ii) A may have many distinct powers, but it has only one inverse.

(iii) If A is invertible, then the equation $Ax = b$ has the unique solution

$$x = A^{-1}b.$$

(iv) If the homogeneous equation $Ax = 0$ has a nonzero solution, then A is not invertible.

The last point is fundamental: invertibility is equivalent to trivial kernel for a square matrix.

§63.2 The 2×2 formula

For a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

we have

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

provided

$$ad - bc \neq 0.$$

Thus the determinant is the obstruction to invertibility.

§63.3 Diagonal matrices

If

$$A = \operatorname{diag}(d_1, \dots, d_n),$$

then A is invertible iff every $d_i \neq 0$, and

$$A^{-1} = \operatorname{diag}(d_1^{-1}, \dots, d_n^{-1}).$$

This is the simplest case where the inverse is completely visible.

§63.4 Products of inverses

For invertible matrices,
$$(AB)^{-1} = B^{-1}A^{-1}.$$
More generally,
$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}.$$
The order reverses because composition reverses when undoing operations.

§63.5 Adjugate formula

Let $A = (a_{ij})$. Delete row i and column j to get the minor M_{ij}. The cofactor is
$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$
The adjugate matrix is the transpose of the cofactor matrix:
$$\operatorname{adj}(A) = (C_{ji}).$$
Then
$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$
when $\det A \neq 0$.

The note calls this formula computationally expensive but conceptually important. It proves that entries of the inverse are rational functions of entries of A.

§63.6 Row reduction for inverses

The practical method is row reduction:
$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$
For example, for
$$M = \begin{pmatrix} 1 & 4 \\ 2 & 9 \end{pmatrix},$$
one has
$$\det M = 9 - 8 = 1,$$
and hence
$$M^{-1} = \begin{pmatrix} 9 & -4 \\ -2 & 1 \end{pmatrix}.$$
The note also illustrates obtaining this inverse by elementary row operations on the augmented matrix.

§63.7 Rotation matrices

The final inserted page discusses planar rotations. The rotation matrix is
$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

It sends

$$\begin{pmatrix} x \\ y \end{pmatrix}$$

to

$$\begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}.$$

Its inverse is

$$R_\theta^{-1} = R_{-\theta} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

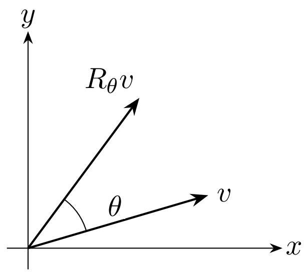


Figure 63.1: A planar rotation acts linearly on vectors and is represented by R_θ.

§63.8 Beyond the first calculation

Invertible matrices are automorphisms of finite-dimensional vector spaces. Gaussian elimination, determinants, adjugates, and block decompositions are different ways to detect or compute the same object: a reversible linear transformation.

§63.9 Invertibility as isomorphism

A matrix is invertible exactly when the associated linear map is an isomorphism. This means three equivalent things in finite dimensions:

$$\ker A = 0, \qquad \operatorname{im} A = V, \qquad \det A \neq 0.$$

Row reduction tests the same fact by constructing inverse operations.

§63.10 Adjugate and exterior powers

The adjugate formula

$$A^{-1} = \frac{1}{\det A}\operatorname{adj}(A)$$

comes from cofactors. Conceptually, determinants describe the action of A on top exterior power, while cofactors encode how $(n-1)$-dimensional volume elements transform.

§63.11 Rotations as orthogonal matrices

A rotation matrix R satisfies

$$R^T R = I, \qquad \det R = 1.$$

The first condition preserves lengths and angles; the second preserves orientation. Thus rotations are the orientation-preserving part of the orthogonal group.

§63.12 Invertibility, volume, and orthogonal motion

Inverse formulas, row reduction, adjugates, diagonal matrices, and rotations are all different faces of isomorphism: solve uniquely, preserve or rescale volume, and understand what the map does independently of coordinates.

§63.13 Inverse as undoing a linear process

The inverse matrix should be read as an operation that undoes a linear process. The formula

$$(AB)^{-1} = B^{-1}A^{-1}$$

then becomes obvious: if one first applies B and then A, one must undo A first and then undo B. Row reduction, adjugates, and determinants are three different languages for this same reversibility.

The rotation matrix is a particularly good example because the inverse is visible geometrically. Rotating by θ is undone by rotating by $-\theta$, so

$$R_\theta^{-1} = R_{-\theta} = R_\theta^T.$$

This is the first glimpse of orthogonal matrices, where algebraic inverse equals geometric transpose.

§63.14 Invertible matrices as a Lie group

The set of all invertible $n \times n$ matrices over $\mathbb{R}$ is

$$GL(n, \mathbb{R}) = \{A : \det A \neq 0\}.$$

It is a group under multiplication and also an open subset of $\mathbb{R}^{n^2}$. Therefore it is a Lie group: a group with a smooth manifold structure.

The tangent space at the identity is the Lie algebra

$$\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R}).$$

The exponential map

$$\exp X = I + X + \frac{X^2}{2!} + \cdots$$

connects infinitesimal generators to invertible transformations. Elementary matrix powers and exponentials are thus the entry point to continuous symmetry.

§63.15 Determinant as a volume form

The determinant is not merely a formula for invertibility. It is the factor by which a linear map scales oriented volume:

$$\mathrm{Vol}(Av_1, \dots, Av_n) = \det(A)\,\mathrm{Vol}(v_1, \dots, v_n).$$

The multiplicative identity

$$\det(AB) = \det A \det B$$

says that volume scaling composes multiplicatively.

This is why row operations affect determinants in exactly the familiar ways. Swapping rows reverses orientation, scaling a row scales volume, and adding one row to another shears volume without changing it.

§63.16 Polar decomposition

Every invertible real matrix has a polar decomposition

$$A = QP,$$

where Q is orthogonal and P is symmetric positive definite. One can take

$$P = (A^T A)^{1/2}, \qquad Q = AP^{-1}.$$

Geometrically, P stretches along orthogonal principal axes, and Q rotates or reflects. This separates deformation into "pure strain" and "rigid motion."

The elementary inverse/rotation examples become clearer through this decomposition. Orthogonal matrices preserve lengths; positive definite matrices perform anisotropic stretching; a general invertible matrix is both.

§63.17 Numerical stability of inverses

Although A^{-1} exists when $\det A \neq 0$, computing it may be unstable. The condition number

$$\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

measures how much errors may be amplified. A tiny determinant is not by itself the best diagnostic; singular values describe directional collapse.

Thus the elementary criterion $\det A \neq 0$ answers existence, while numerical linear algebra asks sensitivity.

§63.18 Row operations and the general linear group

Gaussian elimination writes an invertible matrix as a product of elementary matrices. Geometrically, elementary matrices are simple transformations: shears, scalings, and swaps. Thus an invertible linear transformation can be built from elementary geometric moves. This is the computational skeleton behind the group $GL(n)$.

The determinant map

$$\det : GL(n, \mathbb{R}) \to \mathbb{R}^\times$$

is a group homomorphism. Its kernel is the special linear group

$$SL(n, \mathbb{R}) = \{A : \det A = 1\}.$$

This group consists of volume-preserving linear transformations. The elementary determinant tests in the original note therefore separate invertible maps by how they scale volume.

For rotations, the relevant subgroup is

$$O(n) = \{Q : Q^T Q = I\}, \qquad SO(n) = O(n) \cap SL(n, \mathbb{R}).$$

Orthogonal matrices preserve the inner product, hence all lengths and angles. The chain

$$SO(n) \subset O(n) \subset GL(n)$$

organizes the original examples: rotations are the most rigid, orthogonal maps allow reflections, and invertible maps allow arbitrary linear distortion without collapse.

§63.19 Exponential map and infinitesimal motion

A smooth path $A(t)$ in $GL(n)$ with $A(0) = I$ has derivative $X = A'(0)$, an arbitrary matrix. Conversely, a matrix X generates the path

$$A(t) = e^{tX}.$$

If X is skew-symmetric, then e^{tX} is orthogonal. Indeed,

$$\frac{d}{dt}(e^{tX})^T e^{tX} = 0$$

when $X^T = -X$. This gives a precise meaning to "skew-symmetric matrices generate rotations."

The elementary rotation matrices in the note are therefore not isolated formulas; they are one-parameter subgroups of $GL(n)$.

Indices, Dual Maps, Adjoints, Trace, and Singular Values

N-20220412

§64.1 Indices carry type information before they abbreviate sums

Let V have basis $e_1, \dots, e_n$, with dual basis $e^1, \dots, e^n$. The Kronecker delta is

$$\delta_i{}^j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

It is the coordinate representation of the identity map:

$$I(e_j) = e_i \delta_i{}^j = e_j.$$

The position of an index records which kind of slot it occupies. A vector has components x^j, a covector has components α_i, and contraction pairs one of each type:

$$\alpha(x) = \alpha_i x^i.$$

Einstein convention suppresses the summation sign when an index occurs once upstairs and once downstairs:

$$\alpha_i x^i = \sum_i \alpha_i x^i.$$

An index that survives once in every term is free and determines the output type. A repeated index is dummy and may be renamed. Expressions such as

$$A_i{}^j x^i$$

are ill typed if i is intended to be the output index, because it occurs twice while j remains unmatched. Index notation is valuable precisely because many algebraic mistakes become visible as type errors.

§64.2 Matrix multiplication is contraction of an output with an input

Let

$$B : U \to V, \qquad A : V \to W.$$

Choose bases and write

$$B(e_k) = B_j{}^k v_j, \qquad A(v_j) = A_i{}^j w_i.$$

Then

$$\begin{aligned}(A \circ B)(e_k) &= B_j{}^k A(v_j) \\ &= A_i{}^j B_j{}^k w_i.\end{aligned}$$

Thus

$$(AB)_i{}^k = A_i{}^j B_j{}^k.$$

The repeated index j is the intermediate vector-space direction that is produced by B and consumed by A.

The identity tensor acts neutrally because

$$A_i{}^j \delta_j{}^k = A_i{}^k, \qquad \delta_i{}^j A_j{}^k = A_i{}^k.$$

Trace is another contraction:

$$\operatorname{tr} A = A_i{}^i.$$

Matrix multiplication and trace are therefore not unrelated recipes; both pair an output index with a compatible input index.

§64.3 A change of basis transforms both indices and leaves contractions unchanged

Let P be the change-of-basis matrix, so coordinate columns satisfy

$$[x]_{\text{old}} = P[x]_{\text{new}}.$$

An endomorphism matrix transforms by similarity:

$$A' = P^{-1}AP.$$

In indices,

$$A_i'^j = (P^{-1})_i{}^r A_r{}^s P_s{}^j.$$

Contracting the free indices gives

$$\begin{aligned} A_i'^i &= (P^{-1})_i{}^r A_r{}^s P_s{}^i \\ &= A_r{}^s P_s{}^i (P^{-1})_i{}^r \\ &= A_r{}^s \delta_s{}^r \\ &= A_r{}^r. \end{aligned}$$

Hence

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A.$$

The diagonal sum is invariant because it is a complete contraction: the basis-change matrix and its inverse cancel along the closed index loop.

By contrast, an individual diagonal entry $A_i{}^i$ with i fixed is not invariant. Only the sum over the repeated index has intrinsic meaning.

§64.4 Transpose is the matrix of the dual map

Let

$$A : V \to W$$

and define the dual map

$$A^\vee : W^* \to V^*, \qquad A^\vee(\varphi) = \varphi \circ A.$$

Choose bases e_j of V and f_i of W, with

$$A(e_j) = \sum_i A_{ij} f_i.$$

For the dual basis vector f^i,

$$\begin{aligned} A^\vee(f^i)(e_j) &= f^i(Ae_j) \\ &= A_{ij}. \end{aligned}$$

Therefore

$$A^\vee(f^i) = \sum_j A_{ij} e^j.$$

The matrix of $A^\vee$ has entry A_{ij} in row j, column i, so it is

$$[A^\vee] = A^T.$$

The reversal law now follows without entry manipulation. For

$$U \xrightarrow{B} V \xrightarrow{A} W,$$

one has

$$(A \circ B)^\vee = B^\vee \circ A^\vee.$$

In matrices,

$$(AB)^T = B^T A^T.$$

Transposition reverses order because a functional is pulled backward through a composite map.

§64.5 An adjoint is a dual map returned to vectors by an inner product

The transpose needs only vector spaces and duality. An adjoint additionally needs inner products. Let

$$A : V \to W$$

be a linear map between finite-dimensional complex inner-product spaces. The adjoint

$$A^\dagger : W \to V$$

is characterized by

$$\langle Ax, y\rangle_W = \langle x, A^\dagger y\rangle_V.$$

Riesz representation identifies each covector with a unique vector, so the adjoint is the composite

$$W \xrightarrow{\text{Riesz}} W^* \xrightarrow{A^\vee} V^* \xrightarrow{\text{Riesz}^{-1}} V.$$

In orthonormal bases,

$$[A^\dagger] = \overline{A}^T = A^*.$$

The operations should be kept distinct:

operation	matrix
transpose	A^T
entrywise conjugate	$\overline{A}$
conjugate transpose	$\overline{A}^T$
adjoint in orthonormal bases	A^*.

Over $\mathbb{R}$, conjugation disappears and the adjoint matrix in orthonormal bases is the transpose.

§64.6 In nonorthonormal coordinates the Gram matrices enter the adjoint

Suppose the chosen bases have Gram matrices

$$G_V = (\langle e_i, e_j \rangle_V), \qquad G_W = (\langle f_i, f_j \rangle_W).$$

For coordinate columns,

$$\langle x, z \rangle_V = x^* G_V z, \qquad \langle y, w \rangle_W = y^* G_W w.$$

The defining identity for the adjoint becomes

$$x^* A^* G_W y = x^* G_V [A^\dagger] y$$

for all x, y. Hence

$$\boxed{[A^\dagger] = G_V^{-1} A^* G_W.}$$

The formula $A^\dagger = A^*$ is therefore a feature of orthonormal coordinates, not the definition of adjointness.

For example, on $\mathbb{R}^2$ use the inner product with

$$G = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

and the linear map

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Although $A^T = A$, its adjoint in this metric is

$$\begin{aligned} A^\dagger &= G^{-1} A^T G \\ &= \begin{pmatrix} 0 & 4 \\ 1/4 & 0 \end{pmatrix}. \end{aligned}$$

The coordinate swap is not self-adjoint when the second coordinate has four times the metric weight.

§64.7 Rank is the dimension of the image and the size of a minimal factorization

For

$$A : V \to W,$$

define

$$\operatorname{rank} A = \dim \operatorname{im} A.$$

The first isomorphism theorem gives

$$V / \ker A \cong \operatorname{im} A.$$

Thus rank is the dimension of the effective quotient after directions killed by A have been removed.

If rank $A = r$, choose a basis of the column space and place it in a matrix

$$B \in k^{m\times r}.$$

Express every column of A in that basis and collect the coordinates in

$$C \in k^{r\times n}.$$

Then

$$A = BC, \qquad \operatorname{rank} B = \operatorname{rank} C = r.$$

Conversely, any factorization through an r-dimensional space has rank at most r. Hence rank is the smallest intermediate dimension through which the map can factor.

This factorization also proves equality of row and column rank. Since $A = BC$, every row of A is a linear combination of rows of C, so

$$\operatorname{row}(A) \subseteq \operatorname{row}(C).$$

Because B has full column rank, it has a left inverse L with

$$LB = I_r.$$

Hence

$$C = LA,$$

so every row of C is a linear combination of rows of A. Thus

$$\operatorname{row}(A) = \operatorname{row}(C),$$

and its dimension is r, the same as the column rank.

§64.8 Trace is contraction, and cyclicity is a legal index relabelling

For endomorphisms A, B,

$$\begin{aligned}\operatorname{tr}(AB) &= (AB)_i{}^i \\ &= A_i{}^j B_j{}^i \\ &= B_j{}^i A_i{}^j \\ &= \operatorname{tr}(BA).\end{aligned}$$

The middle equality only renames dummy indices and commutes scalar entries. Consequently,

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr} A.$$

For a rank-one operator

$$A = uv^*,$$

one has

$$\boxed{\operatorname{tr}(uv^*) = v^*u.}$$

Indeed,

$$(uv^*)_{ij} = u_i\overline{v_j},$$

so

$$\operatorname{tr}(uv^*) = \sum_i u_i\overline{v_i} = v^*u.$$

If u is a unit vector, the orthogonal projector onto its span is $P = uu^*$, and

$$\operatorname{tr} P = 1.$$

More generally, an orthogonal projector has eigenvalues $0, 1$, hence

$$\operatorname{tr} P = \operatorname{rank} P.$$

Trace counts the dimension of a projected subspace because it sums the projector's surviving modes.

§64.9 The Hilbert–Schmidt inner product leads to Schatten norms

Matrices themselves form an inner-product space with

$$\langle A, B\rangle_{HS} = \operatorname{tr}(A^*B).$$

Its norm is the Frobenius norm:

$$\begin{aligned}\|A\|_F^2 &= \operatorname{tr}(A^*A)\\ &= \sum_{i,j} |a_{ij}|^2.\end{aligned}$$

If the singular values of A are

$$\sigma_1(A), \ldots, \sigma_r(A),$$

then the Schatten norms are

$$\|A\|_{S_p} = \left(\sum_i \sigma_i(A)^p\right)^{1/p}, \qquad 1 \le p < \infty,$$

and

$$\|A\|_{S_\infty} = \max_i \sigma_i(A).$$

The three common cases are

$$\|A\|_{S_1} = \text{nuclear norm}, \qquad \|A\|_{S_2} = \|A\|_F, \qquad \|A\|_{S_\infty} = \|A\|_{op}.$$

For

$$A = \begin{pmatrix} 0 & 3 \\ 4 & 0 \end{pmatrix},$$

one has

$$A^*A = \begin{pmatrix} 16 & 0 \\ 0 & 9 \end{pmatrix},$$

so the singular values are $4, 3$. Therefore

$$\|A\|_{S_1} = 7, \qquad \|A\|_{S_2} = 5, \qquad \|A\|_{S_\infty} = 4.$$

Rank counts how many singular values are nonzero; Schatten norms measure their size with different emphases.

The Hilbert–Schmidt product also gives the estimate

$$|\operatorname{tr}(A^*B)| \le \|A\|_F \|B\|_F,$$

which is ordinary Cauchy–Schwarz applied in the matrix space.

§64.10 Categorical trace closes an output wire into an input wire

Let V be finite dimensional with basis e_i and dual basis e^i. The coevaluation map is

$$\mathrm{coev} : k \to V \otimes V^*, \qquad 1 \longmapsto \sum_i e_i \otimes e^i.$$

The evaluation map is

$$\mathrm{ev} : V^* \otimes V \to k, \qquad \varphi \otimes v \longmapsto \varphi(v).$$

For an endomorphism $A : V \to V$, compose

$$k \xrightarrow{\mathrm{coev}} V \otimes V^* \xrightarrow{A \otimes I} V \otimes V^* \xrightarrow{\tau} V^* \otimes V \xrightarrow{\mathrm{ev}} k,$$

where $\tau(v \otimes \varphi) = \varphi \otimes v$. Acting on 1, the composite gives

$$\begin{aligned} 1 &\longmapsto \sum_i e_i \otimes e^i \\ &\longmapsto \sum_i Ae_i \otimes e^i \\ &\longmapsto \sum_i e^i \otimes Ae_i \\ &\longmapsto \sum_i e^i(Ae_i). \end{aligned}$$

If

$$Ae_i = A_j{}^i e_j,$$

then

$$\sum_i e^i(Ae_i) = \sum_i A_i{}^i = \operatorname{tr} A.$$

The abstract operation of closing an output back into an input reproduces the ordinary matrix trace exactly. Basis invariance is built into the evaluation–coevaluation pairing, while the coordinate calculation reveals the same contraction $A_i{}^i$ encountered at the beginning of the chapter.

Permutations, Inversions, and Determinants

N-20220725

§65.1 Inversions and signs

The note begins with inversions of a permutation. For a permutation

$$\sigma = (\sigma_1, \dots, \sigma_n),$$

an inversion is a pair $i < j$ such that

$$\sigma_i > \sigma_j.$$

The number of inversions is denoted $\tau(\sigma)$. The sign of the permutation is

$$\operatorname{sgn}(\sigma) = (-1)^{\tau(\sigma)}.$$

For example, for 426315, the inversions are counted by checking how many later entries are smaller than each entry. The note obtains

$$\tau(426315) = 8.$$

In determinant expansions, the sign of each term is exactly the sign of a permutation. Thus a term like

$$a_{21}a_{53}a_{16}a_{42}a_{65}a_{34}$$

corresponds to a row/column permutation, and its sign is determined by its inversion number.

§65.2 Definition of determinant

Definition 65.2.1 (Determinant)**.** The determinant is the alternating multilinear function of the columns of a square matrix normalized by

$$\det I = 1.$$

It measures oriented volume scaling.

The determinant of an $n \times n$ matrix $A = (a_{ij})$ is

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$

This formula is not meant for computation in large dimensions. Its value is conceptual: it explains why row swaps change sign, why repeated rows give zero, and why the determinant is multilinear in rows and columns.

§65.3 Basic determinant properties

The note records the following standard properties:

(i) Transposition does not change the determinant:

$$\det A^T = \det A.$$

(ii) Swapping two rows changes the sign.

(iii) Pulling a scalar out of one row pulls it out of the determinant.

(iv) The determinant is additive in each row separately.

(v) Adding a multiple of one row to another row does not change the determinant.

From these one immediately gets the zero properties:

(i) if a row is zero, then the determinant is zero;

(ii) if two rows are proportional, then the determinant is zero.

§65.4 Sarrus and triangular matrices

For a 3×3 matrix, Sarrus' rule gives a convenient visual computation. But the note also emphasizes that triangular matrices are easier: if A is upper or lower triangular, then

$$\det A = a_{11}a_{22}\cdots a_{nn}.$$

This is why row reduction is so powerful. We use row operations to simplify a determinant into triangular form while tracking sign changes and scalar factors.

§65.5 Cofactor expansion

The cofactor of a_{ij} is

$$C_{ij} = (-1)^{i+j}M_{ij},$$

where M_{ij} is the determinant obtained by deleting row i and column j. Expanding along row i,

$$\det A = \sum_{j=1}^{n} a_{ij}C_{ij}.$$

A page in the note works through determinant computations by choosing rows or columns with many zeros. This is the main practical advice: do not expand randomly; expand where the determinant is sparse.

§65.6 A structured determinant

A common determinant in the note has the form

$$D_n = \begin{vmatrix} x & a & a & \cdots & a \\ a & x & a & \cdots & a \\ \vdots & \vdots & \vdots & & \vdots \\ a & a & a & \cdots & x \end{vmatrix}.$$

This matrix can be written

$$D_n = (x-a)I + aJ,$$

where J is the all-one matrix. The vector $(1, \ldots, 1)$ is an eigenvector with eigenvalue

$$x + (n-1)a,$$

and any vector whose coordinates sum to 0 has eigenvalue

$$x - a.$$

Therefore

$$D_n = (x + (n-1)a)(x-a)^{n-1}.$$

The note obtains this by row and column operations; the eigenvalue argument is the higher-level explanation.

§65.7 The determinant line and exterior powers

The previous computational rules all point to one structure. Let V be an n-dimensional vector space. The top exterior power

$$\Lambda^n V$$

is one-dimensional and is called the determinant line. A nonzero element of this line is a volume form; a choice up to positive scalar is an orientation.

Definition 65.7.1 (Determinant via the top exterior power)**.** For a linear map $A : V \to V$, the induced map

$$\Lambda^n A : \Lambda^n V \to \Lambda^n V$$

acts on a one-dimensional space, hence is multiplication by a scalar. That scalar is $\det A$.

In a basis $e_1, \ldots, e_n$, if the columns of A are $v_1, \ldots, v_n$, then

$$v_1 \wedge \cdots \wedge v_n = (\det A) e_1 \wedge \cdots \wedge e_n.$$

This single identity explains the main determinant rules:

(i) multilinearity comes from multilinearity of the wedge product;

(ii) repeated or dependent columns wedge to zero;

(iii) swapping two columns changes the sign;

(iv) multiplicativity follows from $\Lambda^n(AB) = (\Lambda^n A)(\Lambda^n B)$.

§65.8 Why permutation signs are forced

The sign of a permutation is not an arbitrary convention. It is the unique homomorphism

$$S_n \longrightarrow \{\pm 1\}$$

that sends every transposition to -1. The inversion number gives a concrete formula:

$$\operatorname{sgn}(\sigma) = (-1)^{\#\text{ inversions of }\sigma}.$$

Because the determinant is alternating, permuting the columns by σ must multiply the volume form by this sign. This is why the Leibniz formula has exactly the sign pattern it has.

§65.9 Minors as exterior coordinates

The same exterior-power viewpoint also explains minors. If $A : V \to W$, then

$$\Lambda^k A : \Lambda^k V \to \Lambda^k W$$

is represented, in the induced exterior bases, by the $k \times k$ minors of A. Therefore

$$\operatorname{rank} A < k \quad \iff \quad \Lambda^k A = 0 \quad \iff \quad \text{all } k \times k \text{ minors vanish.}$$

Thus minors are not arbitrary subdeterminants; they are coordinates of an exterior-power map.

§65.10 Cramer's rule from volume ratios

Cramer's rule replaces one column of A by b. Geometrically, solving $Ax = b$ asks for the coordinates of b in the parallelepiped basis given by the columns of A. The coordinate x_i is the ratio of two oriented volumes:

$$x_i = \frac{\det A_i(b)}{\det A}.$$

This interpretation makes the formula less mysterious: a coordinate is measured by replacing the corresponding basis vector and comparing volumes.

Linear Maps, Matrix Representations, and Tensor Invariance

N-20230209

§66.1 From vector identities to linear representations

The previous vector-identity note explains the cross product, scalar triple product, exterior algebra, and Hodge star. This note uses that example as motivation, but its main subject is different: how coordinate arrays represent invariant linear or multilinear objects.

A linear map $T : V \to W$ is an object independent of coordinates. A matrix is what appears only after choosing a basis of V and a basis of W. Similarly, a tensor is not the displayed array of numbers; the array is a coordinate representation whose transformation law tells us what object is being represented.

Remark 66.1.1 (Division of labor between the two notes) — The note N-20230205 is about why three-dimensional vector identities are really statements about metric, orientation, wedge product, and Hodge star. The present note is about basis change, matrices, invariants, and tensor transformation laws. This separation prevents the same vector triple product discussion from being repeated twice.

§66.2 Coordinates as isomorphisms

A basis is not just a list of vectors; it is an isomorphism from an abstract vector space to a coordinate space. If $B = (b_1, \dots, b_n)$ is a basis of V, then

$$[\,]_B : V \longrightarrow k^n$$

sends a vector to its coordinate column in that basis. The vector v is intrinsic; the column $[v]_B$ depends on B.

This distinction is the source of all basis-change formulas. A matrix is not the linear map itself. A matrix is the expression of a linear map after choosing coordinate isomorphisms.

Definition 66.2.1 (Matrix of a linear map)**.** Let $T : V \to W$ be a linear map, let B be a basis of V, and let C be a basis of W. The matrix of T in these bases is the matrix $[T]_{C,B}$ satisfying

$$[T(v)]_C = [T]_{C,B}[v]_B$$

for all $v \in V$. Equivalently,

$$[T]_{C,B} = [\,]_C \circ T \circ [\,]_B^{-1}.$$

The last formula is often the cleanest one. It says that the bottom arrow in coordinate space is obtained by transporting the actual map T through coordinate isomorphisms.

§66.3 The basis-change formula

Suppose B, B' are two bases of V, and C, C' are two bases of W. Let

$$[v]_B = P[v]_{B'}, \qquad [w]_C = Q[w]_{C'}.$$

Thus P converts new V-coordinates into old V-coordinates, and Q converts new W-coordinates into old W-coordinates.

Proposition 66.3.1 (Matrix transformation rule)
If

$$A = [T]_{C,B}, \qquad A' = [T]_{C',B'},$$

then

$$A' = Q^{-1}AP.$$

Proof. Starting with new coordinates $[v]_{B'}$, first convert to old coordinates:

$$[v]_B = P[v]_{B'}.$$

Apply the old matrix A:

$$[T(v)]_C = A[v]_B = AP[v]_{B'}.$$

Finally convert the output from old C-coordinates to new C'-coordinates. Since $[w]_C = Q[w]_{C'}$, we have $[w]_{C'} = Q^{-1}[w]_C$. Hence

$$[T(v)]_{C'} = Q^{-1}AP[v]_{B'}.$$

Therefore $A' = Q^{-1}AP$. □

For an endomorphism $T : V \to V$, if the same basis change is used on domain and codomain, then $P = Q$ and

$$A' = P^{-1}AP.$$

This is similarity. Similar matrices are not two different operators; they are the same operator written in two coordinate systems.

§66.4 A worked basis-change example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard-basis matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

Use the new basis

$$b_1 = (1, 0), \qquad b_2 = (1, 1).$$

The change-of-basis matrix whose columns are the new basis vectors in the old basis is

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$A' = P^{-1}AP = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

The operator did not change. The new basis reveals that b_1 and b_2 are eigenvectors, so the matrix becomes diagonal.

This example also explains the meaning of diagonalization: it is not a magical simplification of numbers; it is the search for a coordinate system adapted to the invariant directions of the operator.

§66.5 Which quantities are invariant?

For an endomorphism, changing basis replaces A by $P^{-1}AP$. Any expression unchanged under this replacement belongs to the operator rather than to the coordinate matrix.

Proposition 66.5.1 (Similarity invariants)
If $A' = P^{-1}AP$, then

$$\operatorname{tr}(A') = \operatorname{tr}(A), \qquad \det(A') = \det(A),$$

and A and A' have the same characteristic polynomial.

Proof. For trace,

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(APP^{-1}) = \operatorname{tr}(A),$$

using $\operatorname{tr}(XY) = \operatorname{tr}(YX)$. For determinant,

$$\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \det(A).$$

For the characteristic polynomial,

$$\det(\lambda I - P^{-1}AP) = \det(P^{-1}(\lambda I - A)P) = \det(\lambda I - A).$$

□

Rank is invariant for maps between different spaces as well, because left and right multiplication by invertible matrices do not change the dimension of the image. Trace and determinant, however, are naturally invariants of endomorphisms, not arbitrary maps $V \to W$.

§66.6 Covectors and the inverse-transpose rule

A covector is a linear functional $\alpha : V \to k$. It pairs with a vector to give a scalar:

$$\alpha(v) \in k.$$

The scalar must not depend on coordinates.

Assume again that

$$[v]_B = P[v]_{B'}.$$

Write a_B and $a_{B'}$ for the coordinate columns of α in the dual bases, so that

$$\alpha(v) = a_B^T[v]_B = a_{B'}^T[v]_{B'}.$$

Substituting $[v]_B = P[v]_{B'}$, we get

$$a_B^T P[v]_{B'} = a_{B'}^T[v]_{B'}$$

for all v, hence
$$a_{B'} = P^T a_B.$$
Equivalently, if one writes coordinate changes in the opposite direction, the familiar inverse-transpose appears. The important point is not the slogan "covectors transform oppositely", but the invariant pairing

$$\alpha(v)$$

that forces the formula.

§66.7 Bilinear forms are not linear operators

A common source of confusion is that both a linear operator and a bilinear form may be represented by square matrices. Their basis-change laws are different.

Let $g : V \times V \to k$ be a bilinear form. In basis B, write

$$g(v, w) = [v]_B^T G[w]_B.$$

If $[v]_B = P[v]_{B'}$, then
$$g(v, w) = [v]_{B'}^T P^T GP[w]_{B'}.$$
Thus the matrix of the bilinear form in the new basis is

$$G' = P^T GP.$$

Compare this with the operator rule $A' = P^{-1}AP$. A matrix representing a bilinear form changes by congruence; a matrix representing an endomorphism changes by similarity.

Example 66.7.1 (The Euclidean metric in a non-orthonormal basis)
In the standard basis of $\mathbb{R}^2$, the Euclidean inner product has matrix I. For the basis
$$b_1 = (1, 0), \qquad b_2 = (1, 1),$$
with
$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$
the metric matrix becomes
$$G' = P^T IP = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The inner product is still the same geometric object, but because the new basis is not orthonormal, its coordinate matrix is no longer the identity.

This is why statements such as "$A^T = A$" must be interpreted carefully. A symmetric matrix may represent a symmetric bilinear form in a chosen basis, or it may represent a self-adjoint operator relative to a chosen inner product. These are related but not identical ideas.

§66.8 Symmetric and skew parts

For an ordinary square matrix over a field of characteristic not 2, one can write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first part is symmetric, and the second part is skew-symmetric. But the coordinate-free meaning depends on what A represents.

If A represents a bilinear form, the symmetric part corresponds to

$$g_s(v, w) = \frac{g(v, w) + g(w, v)}{2},$$

and the skew part corresponds to

$$g_a(v, w) = \frac{g(v, w) - g(w, v)}{2}.$$

The skew part is an alternating 2-form.

If A represents a linear operator on an inner-product space, then self-adjointness is the condition

$$\langle Av, w\rangle = \langle v, Aw\rangle.$$

In an orthonormal basis this becomes $A^T = A$. In a non-orthonormal basis with metric matrix G, it becomes

$$A^T G = GA.$$

So the naive transpose test is valid only after the metric and basis have been fixed.

§66.9 Tensor slots and transformation laws

A tensor is best understood by its slots. The basic examples are:

object	tensor type
vector	V
covector	V^*
bilinear form	$V^* \otimes V^*$
linear operator	$V \otimes V^*$

The components transform according to which slots are vector slots and which slots are covector slots.

If $[v]_B = P[v]_{B'}$, then schematically:

object	components	basis-change law
vector	x	$x' = P^{-1}x$
covector	a	$a' = P^T a$
operator	A	$A' = P^{-1}AP$
bilinear form	G	$G' = P^T GP$

The formulas differ because the objects have different tensor types. Treating every square array as the same kind of object is the conceptual mistake tensor notation is designed to prevent.

§66.10 Tensor products and universal property

A bilinear map

$$B : V \times W \to U$$

is equivalently a linear map

$$\widetilde{B} : V \otimes W \to U$$

satisfying

$$\widetilde{B}(v \otimes w) = B(v, w).$$

This is not just notation. It says that the tensor product is the linear object that represents bilinear dependence.

For example, matrix multiplication can be viewed as contraction of repeated indices:

$$(AB)^i{}_j = A^i{}_k B^k{}_j.$$

The repeated index k is summed. The output has one upper and one lower index, so it is again an operator-type tensor.

Trace is another contraction:

$$\operatorname{tr}(A) = A^i{}_i.$$

This explains why trace is coordinate-invariant for an endomorphism: it contracts one vector slot with one covector slot, producing a scalar.

§66.11 Coordinate-free reading of the original matrix facts

The scattered formulas involving A^{-1}, A^T, and A^{-T} should now be read as transformation laws:

(i) $P^{-1}AP$ belongs to linear operators under basis change;

(ii) P^TGP belongs to bilinear forms and quadratic forms;

(iii) inverse-transpose behavior belongs to covectors and dual bases;

(iv) trace is contraction of an operator-type tensor;

(v) determinant is the induced scalar on the determinant line.

The mature lesson is not merely that "coordinates may change". The real lesson is that each mathematical object has a type, and its type determines the correct transformation law. Tensor notation is useful because it keeps this type information visible.

Tensors in Nonorthogonal Coordinates: Dual Bases, Metrics, and Contraction

N-20230217

§67.1 Why coordinates become interesting when they are not orthogonal

In an orthonormal basis, vector calculus can hide a great deal of structure. The same list of numbers may be used for a vector and for the linear functional obtained by taking its dot product. Upper and lower indices look interchangeable. The identity matrix also serves as the metric matrix, and the dual basis is visually indistinguishable from the original basis.

All of these coincidences disappear as soon as the coordinate directions are tilted or stretched. That is not an inconvenience to be avoided. It is the setting in which tensor notation begins to explain what is genuinely geometric and what is merely a consequence of a particularly friendly basis.

This chapter therefore has a narrower task than a general introduction to tensors. The earlier notes treat exterior algebra, Hodge duality, basis change, and tensor invariance in their own right. Here the focus is concrete: how does one compute honestly in a nonorthogonal frame, and why do dual bases, the metric, index raising, and contraction appear together?

The basic lesson is that a geometric object does not change when coordinates change, but its components must change in a precisely compensating way.

§67.2 Tensor products package bilinear dependence

Suppose

$$B : V \times W \longrightarrow U$$

is bilinear. It is linear in each argument separately, but it is not a linear map on the Cartesian product in the usual sense. The tensor product is constructed so that bilinear data can be handled by ordinary linear algebra.

There is a vector space $V \otimes W$ and a bilinear map

$$\otimes : V \times W \longrightarrow V \otimes W$$

with the following universal property: for every bilinear map B, there is a unique linear map

$$\widetilde{B} : V \otimes W \longrightarrow U$$

such that

$$\widetilde{B}(v \otimes w) = B(v, w).$$

Equivalently,

$$\mathrm{Bil}(V \times W, U) \cong \mathrm{Lin}(V \otimes W, U).$$

The symbol $v \otimes w$ is therefore not an ordered pair. It is a generator subject to the relations required by bilinearity:

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$
$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$
$$(cv) \otimes w = v \otimes (cw) = c(v \otimes w).$$

If $e_1, \ldots, e_m$ is a basis of V and $f_1, \ldots, f_n$ is a basis of W, then

$$e_i \otimes f_j, \qquad 1 \le i \le m, \quad 1 \le j \le n,$$

form a basis of $V \otimes W$. Hence

$$\dim(V \otimes W) = mn.$$

A general element has the form

$$T = T^{ij} e_i \otimes f_j.$$

The repeated indices are summed. The array (T^{ij}) is a coordinate description; the tensor T is the basis-independent object being described.

Not every tensor is a single pure tensor $v \otimes w$. For example,

$$e_1 \otimes f_1 + e_2 \otimes f_2$$

generally cannot be factored into one product. This is analogous to the difference between a rank-one matrix and a general matrix.

§67.3 Vectors and covectors are different kinds of objects

A vector belongs to V. A covector belongs to the dual space

$$V^* = \mathrm{Lin}(V, \mathbb{R}).$$

A covector consumes a vector and returns a scalar.

Given a basis

$$e_1, \ldots, e_n$$

of V, the dual basis

$$\varepsilon^1, \ldots, \varepsilon^n$$

of V^* is defined by

$$\varepsilon^i(e_j) = \delta^i_j.$$

If

$$v = v^i e_i,$$

then

$$v^i = \varepsilon^i(v).$$

Thus the dual basis is the device that extracts coordinates.

A covector α is written

$$\alpha = \alpha_i \varepsilon^i,$$

and its action on v is

$$\alpha(v) = \alpha_i v^i.$$

This pairing is a contraction: one lower index meets one upper index and disappears.

The placement of indices is not decorative. Under a change of basis, vector components and covector components transform in opposite ways so that the scalar $\alpha_i v^i$ remains unchanged. The general transformation laws were developed in the preceding basis-change note; here we will see their concrete geometric meaning through a nonorthogonal frame.

§67.4 A nonorthogonal basis exposes the dual basis

Consider $\mathbb{R}^2$ with the basis

$$g_1 = \begin{bmatrix}1\\0\end{bmatrix}, \qquad g_2 = \begin{bmatrix}1\\1\end{bmatrix}.$$

The coordinate directions meet at 45°, not 90°.

Let

$$v = \begin{bmatrix}3\\2\end{bmatrix}.$$

Writing

$$v = v^1 g_1 + v^2 g_2$$

gives

$$\begin{bmatrix}3\\2\end{bmatrix} = v^1 \begin{bmatrix}1\\0\end{bmatrix} + v^2 \begin{bmatrix}1\\1\end{bmatrix}.$$

Therefore

$$v^1 = 1, \qquad v^2 = 2.$$

These are the contravariant components of v in the basis (g_i).

Now seek vectors g^1, g^2 satisfying

$$g^i \cdot g_j = \delta^i_j.$$

A direct calculation gives

$$g^1 = \begin{bmatrix}1\\-1\end{bmatrix}, \qquad g^2 = \begin{bmatrix}0\\1\end{bmatrix}.$$

Indeed,

$$g^1 \cdot g_1 = 1, \quad g^1 \cdot g_2 = 0, \quad g^2 \cdot g_1 = 0, \quad g^2 \cdot g_2 = 1.$$

The coordinate coefficients can now be recovered by dot products with the reciprocal vectors:

$$v^1 = g^1 \cdot v = 1, \qquad v^2 = g^2 \cdot v = 2.$$

The reciprocal vectors are the Euclidean representatives of the abstract dual covectors. The dual basis lives naturally in V^*; the inner product identifies each covector with a vector, allowing us to draw it in the same plane.

This is the first important distinction:

dual basis is intrinsic to a chosen basis,

whereas

reciprocal vectors also use the metric.

Without an inner product, the dual basis still exists, but there is no canonical way to turn its covectors into vectors.

§67.5 The metric is the Gram matrix of the frame

In the basis (g_i), define

$$g_{ij} = g_i \cdot g_j.$$

For the example above,

$$(g_{ij}) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

This is the Gram matrix of the basis. It records lengths and angles of the coordinate directions.

If

$$u = u^i g_i, \qquad v = v^j g_j,$$

then

$$u \cdot v = g_{ij} u^i v^j.$$

The familiar formula

$$u \cdot v = \sum_i u^i v^i$$

is valid only when $g_{ij} = \delta_{ij}$.

The inverse matrix is

$$(g^{ij}) = (g_{ij})^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

It reconstructs the reciprocal basis:

$$g^i = g^{ij} g_j.$$

For example,

$$g^1 = 2g_1 - g_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

$$g^2 = -g_1 + g_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The inverse relations are

$$g^{ik} g_{kj} = \delta^i_j.$$

They express that the metric and inverse metric undo one another when used to change index type.

§67.6 Raising and lowering indices is the metric acting as a map

The metric is not merely a table of dot products. It defines a linear map

$$\flat : V \longrightarrow V^*, \qquad v^\flat(w) = g(v, w).$$

In differential geometry this is called the flat musical isomorphism. Its inverse is

$$\sharp : V^* \longrightarrow V.$$

In components,

$$v_i = g_{ij} v^j,$$

which is called lowering an index. Conversely,

$$v^i = g^{ij} v_j,$$

raises the index.

For the vector in the running example,

$$(v^1, v^2) = (1, 2).$$

Lowering gives

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

These numbers can also be read directly:

$$v_1 = v \cdot g_1 = 3, \qquad v_2 = v \cdot g_2 = 5.$$

They are not the coefficients used to assemble v from the basis vectors. They are the values of the covector $v^\flat$ on those basis vectors.

Raising them again returns the original coordinates:

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The same example can now verify the transformation laws rather than merely state them. Replace the frame (g_1, g_2) by

$$h_1 = g_1 + g_2, \qquad h_2 = g_2.$$

If

$$P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

then the column relation is

$$\begin{bmatrix} h_1 & h_2 \end{bmatrix} = \begin{bmatrix} g_1 & g_2 \end{bmatrix} P.$$

Vector components therefore transform by the inverse matrix:

$$[v]_h = P^{-1}[v]_g = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Covector components transform in the opposite direction:

$$[v^\flat]_h = P^T [v^\flat]_g = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix}.$$

The metric matrix becomes

$$G_h = P^T G_g P = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}.$$

A direct check closes the diagram:

$$G_h [v]_h = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \end{bmatrix} = [v^\flat]_h.$$

The object v and the covector $v^\flat$ did not change. Three component arrays changed together so that lowering an index still commutes with change of frame.

In an orthonormal basis, both component lists happen to coincide because $g_{ij} = \delta_{ij}$. Tensor notation refuses to build that coincidence into the definitions, which is why it remains valid in arbitrary coordinates and on curved spaces.

§67.7 Tensor type records what the object consumes and produces

A tensor of type (r,s) can be viewed as a multilinear map

$$T : \underbrace{V^* \times \cdots \times V^*}_{r\text{ copies}} \times \underbrace{V \times \cdots \times V}_{s\text{ copies}} \longrightarrow \mathbb{R}.$$

Equivalently, it belongs to

$$V^{\otimes r} \otimes (V^*)^{\otimes s}.$$

A vector is a tensor of type $(1,0)$. A covector has type $(0,1)$. A bilinear form such as the metric has type $(0,2)$:

$$g = g_{ij}\, \varepsilon^i \otimes \varepsilon^j.$$

A linear map $A : V \to V$ can be represented as a tensor of type $(1,1)$:

$$A = A^i{}_j\, e_i \otimes \varepsilon^j.$$

It consumes a vector through the covector slot and produces a vector through the vector slot:

$$Av = A^i{}_j v^j e_i.$$

This notation explains why a matrix can represent different kinds of objects. The same rectangular array could be the components of a bilinear form, a linear map, or a tensor with another slot arrangement. Its transformation law and index type tell us which object it represents.

The metric can change tensor type. For example, from a $(1,1)$-tensor $A^i{}_j$, lowering the upper index gives

$$A_{ij} = g_{ik} A^k{}_j.$$

Whether A_{ij} is symmetric is therefore a metric-dependent statement about the corresponding operator. In Euclidean orthonormal coordinates this subtlety is hidden because lowering an index leaves the numerical array unchanged.

§67.8 Contraction creates invariants by closing one input-output pair

Contraction pairs one upper and one lower index. For a $(1,1)$-tensor,

$$A^i{}_j,$$

contracting the indices gives

$$A^i{}_i.$$

This is the trace:

$$\operatorname{tr} A = A^i{}_i.$$

Why is the result basis-independent? Under a basis change, one index transforms with the change-of-basis matrix and the other with its inverse. When the indices are contracted, these factors cancel. The invariant scalar is what remains after the coordinate dependence closes on itself.

The inner product is also a contraction:

$$u \cdot v = g_{ij} u^i v^j.$$

First the metric lowers one vector index, then the resulting covector is paired with the other vector.

For a rank-two contravariant tensor T^{ij}, contraction with the metric gives

$$g_{ij}T^{ij}.$$

For a differential operator in Euclidean coordinates, this pattern produces the Laplacian from the Hessian:

$$\Delta f = \delta^{ij}\partial_i\partial_j f.$$

On a curved Riemannian manifold, the same idea persists, but partial derivatives must be replaced by covariant derivatives and the metric varies from point to point.

Contraction is therefore more than compact notation. It is one of the principal mechanisms by which tensors produce coordinate-independent scalars and lower-rank tensors.

§67.9 Orientation and volume are not contained in the metric alone

In three-dimensional Euclidean space, the scalar triple product

$$u \cdot (v \times w)$$

measures oriented volume. In a basis g_1, g_2, g_3, the basis volume is

$$J = g_1 \cdot (g_2 \times g_3).$$

The reciprocal vectors can then be written

$$g^1 = \frac{g_2 \times g_3}{J}, \qquad g^2 = \frac{g_3 \times g_1}{J}, \qquad g^3 = \frac{g_1 \times g_2}{J}.$$

The denominator appears because the cross product creates a normal vector with the wrong scale; division by the oriented volume normalizes it to evaluate the corresponding basis vector as 1.

The Levi–Civita symbol is

$$\varepsilon_{ijk} = \begin{cases} 1, & (i,j,k) \text{ is an even permutation of } (1,2,3), \\ -1, & (i,j,k) \text{ is an odd permutation of } (1,2,3), \\ 0, & \text{two indices coincide.} \end{cases}$$

In a right-handed orthonormal basis,

$$(u \times v)_i = \varepsilon_{ijk} u^j v^k.$$

In general coordinates, however, the bare alternating symbol

$$[i_1 \cdots i_n] \in \{-1, 0, 1\}$$

is not by itself the component array of an ordinary tensor. If $\varepsilon^1, \ldots, \varepsilon^n$ is an oriented coordinate coframe, the metric volume form is

$$\mathrm{vol}_g = \sqrt{\det(g_{ij})}\, \varepsilon^1 \wedge \cdots \wedge \varepsilon^n.$$

Equivalently, the covariant Levi–Civita tensor has components

$$\epsilon_{i_1\cdots i_n} = \sqrt{\det(g_{ij})}\,[i_1\cdots i_n].$$

The square root supplies the coordinate-volume scale, while orientation chooses the sign. Keeping the alternating symbol, the tensor, and the differential form distinct prevents the common mistake of assigning tensorial transformation laws to a fixed array of zeros and signs.

This is also why the cross product is special. It requires a metric to identify vectors and covectors, an orientation to choose a sign, and dimension three so that two-dimensional area elements can be identified with vectors. The exterior product $u \wedge v$ is more fundamental and exists in every dimension; the cross product is its three-dimensional metric-dependent avatar.

A reciprocal lattice is a concrete use of the same dual geometry. Given primitive crystal vectors a_1, a_2, a_3, define reciprocal vectors b^i by

$$b^i \cdot a_j = 2\pi\delta^i_j.$$

In three dimensions,

$$b^1 = 2\pi\frac{a_2 \times a_3}{a_1 \cdot (a_2 \times a_3)},$$

with cyclic formulas for b^2, b^3. A reciprocal vector

$$k = h_i b^i, \qquad h_i \in \mathbb{Z},$$

defines planes of constant phase

$$k \cdot x = 2\pi m.$$

Because $k \cdot a_j = 2\pi h_j$, translating by a lattice vector changes the phase by an integer multiple of 2π. Diffraction indices are therefore covector-like data: they measure lattice directions rather than assemble a physical-space displacement.

§67.10 Moving frames turn the same algebra into geometry

On a manifold, the basis vectors may vary from point to point. A local frame is written

$$e_1(x), \ldots, e_n(x),$$

and its dual coframe satisfies

$$\theta^i(e_j) = \delta^i_j.$$

A vector field and a one-form take the forms

$$X = X^i e_i, \qquad \alpha = \alpha_i \theta^i.$$

The metric is

$$g = g_{ij}\,\theta^i \otimes \theta^j.$$

At each point, raising and lowering indices is exactly the finite-dimensional computation already performed above.

The new difficulty is differentiation. Differentiating the components alone is not enough because the frame itself changes:

$$\partial_k(X^i e_i) = (\partial_k X^i)e_i + X^i \partial_k e_i.$$

A connection packages the derivatives of the basis into coefficients $\Gamma^i{}_{jk}$, producing the covariant derivative

$$\nabla_j X^i = \partial_j X^i + \Gamma^i{}_{jk} X^k.$$

For a covector, the correction has the opposite sign:

$$\nabla_j \alpha_i = \partial_j \alpha_i - \Gamma^k{}_{ji} \alpha_k.$$

The signs differ for the same reason vector and covector components transform oppositely.

In moving-frame notation one writes

$$\nabla e_j = \omega^i{}_j \otimes e_i,$$

where the matrix of one-forms $\omega^i{}_j$ records how the frame turns. For the Levi–Civita connection in an orthonormal frame,

$$\omega_{ij} + \omega_{ji} = 0,$$

so infinitesimal frame change is skew-symmetric, just like an infinitesimal rotation. The torsion-free first structure equation is

$$d\theta^i + \omega^i{}_j \wedge \theta^j = 0.$$

This is not a new layer of index decoration: it says that the variation of the coframe and the connection correction cancel in the geometric derivative.

This is the natural continuation of reciprocal bases. Once coordinates vary with position, tensor notation does not merely keep formulas tidy; it ensures that differentiation produces another tensor rather than a coordinate artifact.

§67.11 A curvilinear example: polar coordinates

In the Euclidean plane, polar coordinates are

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The coordinate basis vectors are

$$e_r = \frac{\partial(x,y)}{\partial r} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix},$$

$$e_\theta = \frac{\partial(x,y)}{\partial\theta} = \begin{bmatrix} -r\sin\theta \\ r\cos\theta \end{bmatrix}.$$

They are orthogonal but not both unit length:

$$e_r \cdot e_r = 1, \qquad e_r \cdot e_\theta = 0, \qquad e_\theta \cdot e_\theta = r^2.$$

Thus

$$(g_{ij}) = \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix}, \qquad (g^{ij}) = \begin{bmatrix} 1 & 0 \\ 0 & r^{-2} \end{bmatrix}.$$

The reciprocal basis vector corresponding to e_θ is

$$e^\theta = r^{-2} e_\theta.$$

The area form is
$$\sqrt{\det g}\, dr \wedge d\theta = r\, dr \wedge d\theta.$$
The familiar Jacobian factor r is therefore not an isolated integration trick. It is the square root of the metric determinant.

For a scalar field f, the differential is
$$df = (\partial_r f)\, dr + (\partial_\theta f)\, d\theta.$$
Raising its index with the inverse metric gives the gradient:
$$\nabla f = (\partial_r f) e_r + r^{-2}(\partial_\theta f) e_\theta.$$
Since the unit angular vector is $\widehat{e}_\theta = r^{-1} e_\theta$, this is the familiar formula
$$\nabla f = (\partial_r f)\widehat{e}_r + \frac{1}{r}(\partial_\theta f)\widehat{e}_\theta.$$
The factor $1/r$ is a metric effect, not a memorized correction.

§67.12 A deformation makes the distinction physical

In continuum mechanics, one compares a reference body $\mathcal{B}$ with its deformed configuration $\mathcal{S}$. A deformation is a map
$$\varphi : \mathcal{B} \to \mathcal{S},$$
and its derivative at a material point X is the deformation gradient
$$F = D\varphi(X) : T_X\mathcal{B} \to T_{\varphi(X)}\mathcal{S}.$$
A small material vector dX is pushed forward to
$$dx = F\, dX.$$
A covector transforms in the opposite direction. If α measures current tangent vectors, its pullback to the reference configuration is
$$F^T \alpha,$$
because
$$(F^T \alpha)(dX) = \alpha(F dX).$$
This identity is the physical version of the vector–covector distinction developed earlier: vectors are transported forward, while measurements of vectors are pulled back.

The current Euclidean metric can also be pulled back to the reference body. In coordinates the result is the right Cauchy–Green tensor
$$C = F^T F.$$
For a reference vector a, the squared length after deformation is
$$\|Fa\|^2 = (Fa)^T (Fa) = a^T C a.$$
Thus C stores all changes of lengths and angles while acting entirely on the reference tangent space.

Consider the simple shear

$$\varphi(X, Y) = (X + \gamma Y, Y).$$

Its deformation gradient is

$$F = \begin{bmatrix} 1 & \gamma \\ 0 & 1 \end{bmatrix},$$

so

$$C = F^T F = \begin{bmatrix} 1 & \gamma \\ \gamma & 1 + \gamma^2 \end{bmatrix}.$$

The reference basis vectors behave differently:

$$Fe_1 = e_1, \qquad Fe_2 = \gamma e_1 + e_2.$$

The first direction keeps unit length, while

$$\|Fe_2\|^2 = 1 + \gamma^2.$$

Their current inner product is

$$(Fe_1) \cdot (Fe_2) = \gamma.$$

These three quantities are exactly the entries of C. The determinant satisfies

$$\det F = 1,$$

so the shear preserves area even though it changes lengths and angles. A single tensor therefore separates shape change from volume change more clearly than a list of coordinate formulas.

Stress makes the index placement even more consequential. The Cauchy stress σ acts in the current configuration: after using the current metric to identify a unit normal covector with a normal vector n, the traction is

$$t = \sigma n.$$

The first Piola–Kirchhoff stress P, by contrast, accepts reference-area data and produces a current force. Its two slots belong to different configurations. The relation

$$P = J\sigma F^{-T}, \qquad J = \det F,$$

contains exactly the required pullback of the area covector.

For the simple shear, $J = 1$. If, only for illustration, the current Cauchy stress is the identity matrix, then

$$P = F^{-T} = \begin{bmatrix} 1 & 0 \\ -\gamma & 1 \end{bmatrix}.$$

Although $\sigma = I$, the reference-to-current stress array is not the identity because its input lives in a different tangent space. Writing the two tensors with the same undecorated pair of indices would hide this distinction.

The same lesson was already visible in the nonorthogonal coordinate example. There,

$$(v^1, v^2) = (1, 2)$$

and

$$(v_1, v_2) = (3, 5)$$

represented the same geometric vector in contravariant and covariant form. Mechanics adds another layer: the upper or lower index must also remember whether it belongs to the reference or current configuration. Tensor notation is therefore not decorative bookkeeping. It prevents illegal pairings, records which metric performs an identification, and makes clear which quantities are pushed forward, pulled back, or compared only after a deformation map has connected their spaces.

Notes on MML: Vectors, the Minus-One Trick, Linear Maps, Basis Change, Kernel, and Image

N-20251108

§68.1 Vectors beyond arrows

The first page stresses that vectors are not only geometric arrows. Polynomials can form a vector space: they can be added and scaled. Audio signals can also be treated as vectors: an audio signal is represented by a list of numbers, and adding or scaling those lists gives another signal.

The handwritten remark connects this to embeddings. In machine learning, word vectors or sentence vectors are not geometric arrows in the elementary sense, but they live in vector spaces and obey vector operations. Their meaning comes from the representation learned by the model, not from a literal physical arrow.

The lesson is that a vector is an element of a vector space. Geometry is only one model.

§68.2 The minus-one trick

The source discusses a practical trick for reading solutions of a homogeneous system

$$Ax = 0, \qquad A \in \mathbb{R}^{k\times n}.$$

Assume A is in reduced row echelon form and has no zero rows. Pivot columns correspond to leading ones; free columns correspond to variables that can be chosen freely.

The minus-one trick augments the row-reduced matrix into an $n\times n$ matrix by inserting rows with a single -1 in the diagonal positions corresponding to free variables. Then the columns containing -1 on the diagonal directly give basis vectors for the null space.

The point is not magic. It is a compact way of doing the usual free-variable substitution. Setting one free variable to 1 and the others to 0 gives one kernel basis vector; the inserted -1 row records that choice.

§68.3 Example of the trick

The note's example starts with

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}.$$

The free variables are in columns 2 and 5. Augmenting gives

$$\tilde{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

The columns with diagonal -1 give

$$x = \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}.$$

This is the same solution one obtains by solving by free variables, but the trick makes the basis easy to read.

§68.4 Linear mappings

A mapping $\Phi : V \to W$ between vector spaces is linear if

$$\Phi(\lambda x + \psi y) = \lambda\Phi(x) + \psi\Phi(y)$$

for all $x, y \in V$ and scalars λ, ψ.

The note lists special cases:

- an isomorphism is linear and bijective;
- an endomorphism is a linear map $V \to V$;
- an automorphism is a bijective endomorphism;
- the identity map $\mathrm{id}_V : V \to V$ is the identity automorphism.

The handwritten annotations translate these as "same structure," "same domain and codomain," and "self-isomorphism."

§68.5 Complex numbers as a real vector space

The example

$$\Phi : \mathbb{R}^2 \to \mathbb{C}, \qquad \Phi(x_1, x_2) = x_1 + ix_2$$

is linear over $\mathbb{R}$:

$$\Phi(x + y) = \Phi(x) + \Phi(y), \qquad \Phi(\lambda x) = \lambda\Phi(x).$$

It is also bijective, so it is an isomorphism of real vector spaces.

The note marks this as a homomorphism. The important distinction is that it preserves the vector-space addition and scalar multiplication. It is not claiming that every additional structure is automatically preserved.

§68.6 Finite-dimensional isomorphism theorem

The theorem quoted from Axler says that finite-dimensional vector spaces V and W are isomorphic iff

$$\dim V = \dim W.$$

This is one of the most important simplifications in finite-dimensional linear algebra: after choosing bases, all n-dimensional vector spaces over the same field look like $\mathbb{F}^n$.

But this isomorphism is not canonical unless a basis is chosen. The abstract vector space and its coordinate representation are related but not identical.

§68.7 Basis change

Let $\Phi : V \to W$ be linear. Suppose V has old basis

$$B = (b_1, \dots, b_n)$$

and new basis

$$\widetilde{B} = (\widetilde{b}_1, \dots, \widetilde{b}_n).$$

Similarly, let W have old basis

$$C = (c_1, \dots, c_m)$$

and new basis

$$\widetilde{C} = (\widetilde{c}_1, \dots, \widetilde{c}_m).$$

If A_Φ is the matrix of Φ with respect to B, C, and $\widetilde{A}_\Phi$ is the matrix with respect to $\widetilde{B}, \widetilde{C}$, then

$$\widetilde{A}_\Phi = T^{-1} A_\Phi S.$$

Here S converts coordinates from the new basis of V to the old basis of V, and T converts coordinates from the new basis of W to the old basis of W.

The diagram in the source page says exactly this: to use the old matrix, first translate the input coordinates into the old basis, then apply A_Φ, then translate the output coordinates into the new output basis.

§68.8 Equivalence and similarity

Two rectangular matrices $A, \widetilde{A} \in \mathbb{R}^{m \times n}$ are equivalent if there exist invertible matrices $S \in \mathbb{R}^{n \times n}$ and $T \in \mathbb{R}^{m \times m}$ such that

$$\widetilde{A} = T^{-1} A S.$$

This corresponds to changing the basis in both domain and codomain.

Two square matrices $A, \widetilde{A} \in \mathbb{R}^{n \times n}$ are similar if

$$\widetilde{A} = S^{-1} A S.$$

This corresponds to changing basis for a linear map $V \to V$ using the same coordinate change in domain and codomain.

The handwritten table emphasizes the difference:

Concept	Situation	Invariant meaning
Equivalence	linear map $V \to W$	change bases in domain and codomain
Similarity	linear map $V \to V$	same operator in a new basis

§68.9 Kernel and image

For a linear map $\Phi : V \to W$, the kernel is

$$\ker \Phi = \{v \in V : \Phi(v) = 0_W\},$$

and the image is

$$\operatorname{Im} \Phi = \{w \in W : \exists v \in V,\ \Phi(v) = w\}.$$

The kernel records what is collapsed to zero. The image records what outputs are reachable.

The source diagram shows the domain on the left, codomain on the right, a yellow region for the kernel, and a blue/yellow region for the image. The note also records the injectivity criterion:

$$\Phi \text{ is injective} \iff \ker \Phi = \{0_V\}.$$

§68.10 Categorical side remarks

The margin notes compare kernel, image, and cokernel with categorical language. In ordinary linear algebra,

$$\ker f \to A \to B$$

means the kernel is a subobject of the domain. The image is a subobject of the codomain. The cokernel, when defined, is a quotient-like object on the codomain side.

This is a useful preview: linear algebra already contains many category-theoretic patterns, even before they are named abstractly.

§68.11 The general pattern

This note links computational linear algebra with structural thinking. Row reduction gives the null space; the minus-one trick packages free-variable substitution. Linear maps become matrices only after choosing bases. Basis change explains why the same map can have different matrices. Kernel and image describe what a map destroys and what it reaches. These ideas are also the language behind machine-learning embeddings, where vectors need not be arrows but still live in vector spaces with meaningful linear structure.

§68.12 Vectors as elements of a linear structure

A vector is not essentially an arrow; it is an element of a vector space. Coordinates appear only after choosing a basis. The same vector has different coordinate columns in different bases.

§68.13 The minus-one trick as a kernel graph

The phrase "minus-one trick" should be understood carefully. In this note it is not primarily a homogeneous-coordinate trick and it is not an affine shift. It is a compact way of reading the kernel of a row-reduced homogeneous system.

Proposition 68.13.1 (Kernel graph form of an RREF system)
After possibly reordering variables, suppose a homogeneous system has row-reduced matrix

$$\begin{bmatrix} I_r & A \end{bmatrix}.$$

Write the unknown vector as

$$x = (x_p, x_f),$$

where $x_p \in k^r$ are pivot variables and $x_f \in k^{n-r}$ are free variables. Then the solution space is

$$\ker \begin{bmatrix} I_r & A \end{bmatrix} = \left\{ \begin{pmatrix} -Au \\ u \end{pmatrix} : u \in k^{n-r} \right\}.$$

Thus the kernel is the graph of the linear map

$$u \longmapsto -Au.$$

Proof. The equation is

$$x_p + Ax_f = 0.$$

Hence $x_p = -Ax_f$. If we write $u = x_f$, every solution has the displayed form, and every displayed vector clearly satisfies the equation. □

The inserted -1 rows in the practical trick are a bookkeeping device for the free variables. Choosing $u = e_j$, the j-th standard basis vector of the free-variable space, gives one kernel basis vector:

$$\begin{pmatrix} -Ae_j \\ e_j \end{pmatrix}.$$

So the trick is best read as a way of writing the graph basis of a kernel.

Remark 68.13.2 (Do not confuse two uses of extra coordinates) — Homogeneous coordinates do turn affine transformations into linear transformations one dimension higher. That is a real and useful idea. But the minus-one trick in this row-reduction context is different: it does not linearize an affine map; it packages the free-variable choices in a homogeneous linear system.

§68.14 Kernel-image factorization

For a linear map $T : V \to W$, the kernel records lost directions and the image records reachable directions. Rank-nullity says

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

§68.15 Coordinates, kernels, and affine linearization

The review should distinguish vectors from coordinates, affine maps from linear maps, and a matrix from the linear transformation it represents. Kernel, image, and basis change are the structural core.

§68.16 Exact sequences behind rank-nullity

For a linear map $T : V \to W$, the kernel and image fit into the exact sequence

$$0 \to \ker T \to V \xrightarrow{T} \operatorname{im} T \to 0.$$

Exactness means that the image of each arrow equals the kernel of the next. Taking dimensions gives

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

This is rank-nullity.

The elementary theorem is therefore the dimension shadow of an exact sequence. Exact sequences become one of the central languages of algebra, topology, and geometry.

§68.17 Quotient description of the image

The first isomorphism theorem says

$$V/\ker T \cong \operatorname{im} T.$$

The quotient collapses all vectors that differ by a null vector. What remains is precisely the information visible after applying T.

This gives a clean meaning to free variables and solution spaces. The kernel is what the map forgets; the image is what survives.

§68.18 Cokernel and obstructions

The cokernel of $T : V \to W$ is

$$\operatorname{coker} T = W/\operatorname{im} T.$$

It measures the failure of T to be onto. A system

$$Tx = b$$

is solvable exactly when b maps to zero in the cokernel.

Thus kernel measures non-uniqueness, while cokernel measures obstruction to existence. This is the conceptual form of the elementary distinction between infinitely many solutions and no solution.

§68.19 Kernel, image, and quotient survive basis changes

The minus-one trick, basis conversion, and kernel/image computations in the note are all examples of one structure: a linear map forgets some directions, preserves some information, and may fail to reach some outputs. Exact sequences and quotients express this without depending on a chosen matrix.

§68.20 Row reduction as choosing a splitting

When a matrix is row-reduced, the pivot variables are expressed in terms of free variables. Abstractly, this chooses a splitting of the domain into a complement of the kernel plus the kernel:

$$V \cong U \oplus \ker T.$$

The map T is injective on the chosen complement U and maps U isomorphically onto $\operatorname{im} T$.

This splitting is not canonical: different row-reduction choices or different bases may choose different complements. The exact sequence

$$0 \to \ker T \to V \to \operatorname{im} T \to 0$$

is canonical; a particular parametric solution is a coordinate choice.

§68.21 Cokernel as the missing equation space

For $T : V \to W$, the cokernel $W/\operatorname{im} T$ measures which right-hand sides are not reachable. In matrix terms, left null vectors produce compatibility conditions. If $y \in \ker T^T$, then any solvable equation $Tx = b$ must satisfy

$$y^T b = 0.$$

Thus the left null space is the dual of the obstruction space.

This gives a precise interpretation of consistency conditions in linear systems: they are not extra tricks, but equations defining whether b vanishes in the cokernel.

Determinants and Rank: From Alternating Volume to Elimination and Rank Strata

N-20251110

§69.1 Why one scalar can encode both volume and independence

A determinant problem often looks like a request to simplify a large array. The useful starting point is more geometric. If the columns of

$$A = (a_1, \dots, a_n)$$

are vectors in k^n, then $\det A$ is the oriented volume scale of the parallelotope they span. Three properties determine this quantity uniquely:

(i) it is linear in each column while the others are fixed;

(ii) it changes sign when two columns are exchanged;

(iii) $\det I = 1$.

Alternation immediately gives

$$a_i = a_j \text{ for some } i \neq j \quad \Longrightarrow \quad \det A = 0.$$

More generally,

$$\det A \neq 0 \quad \Longleftrightarrow \quad a_1, \dots, a_n \text{ are linearly independent.}$$

Thus determinant computation and rank detection are not separate topics. A nonzero determinant says that no direction has collapsed; a zero determinant says that the image volume has fallen into a lower-dimensional subspace.

Elementary column operations now have conceptual explanations. Adding a multiple of one column to another leaves the wedge volume unchanged. Scaling one column scales the volume by the same factor. Exchanging two columns reverses orientation. Every safe determinant trick is an application of these three rules.

§69.2 Cofactors and minors are coordinates of exterior powers

A linear map $A : V \to V$ induces maps

$$\Lambda^r A : \Lambda^r V \to \Lambda^r V.$$

In a basis, the entries of $\Lambda^r A$ are the $r \times r$ minors of A. At the top degree, $\Lambda^n V$ is one-dimensional and

$$\Lambda^n A = (\det A) I.$$

This is the coordinate-free source of multiplicativity:

$$\det(AB) = \det A \det B,$$

because

$$\Lambda^n(AB) = \Lambda^n A \circ \Lambda^n B.$$

For $r = n-1$, the minors assemble into the adjugate. If C_{ij} is the cofactor of a_{ij}, then

$$\operatorname{adj}(A) = (C_{ji})$$

and

$$A \operatorname{adj}(A) = \operatorname{adj}(A) A = (\det A) I.$$

The diagonal entries of this product are ordinary Laplace expansions. An off-diagonal entry is the determinant obtained by replacing one row by another already present, hence it vanishes.

The same exterior-power bookkeeping gives the Cauchy–Binet formula. For

$$X \in k^{m\times n}, \qquad Y \in k^{n\times m}, \qquad m \le n,$$

one has

$$\boxed{\det(XY) = \sum_{|S|=m} \det(X_{[:,S]}) \det(Y_{[S,:]}).}$$

General Laplace expansion is a special case: selecting a group of rows and summing products of complementary minors is exactly composition on an exterior power.

§69.3 Vandermonde determinants measure uniqueness of interpolation

Let $\mathcal{P}_{n-1}$ be the vector space of polynomials of degree at most $n-1$. Evaluation at points $x_1, \dots, x_n$ defines a linear map

$$\operatorname{ev} : \mathcal{P}_{n-1} \longrightarrow k^n, \qquad p \longmapsto (p(x_1), \dots, p(x_n)).$$

In the monomial basis $1, t, \dots, t^{n-1}$, its matrix is the Vandermonde matrix

$$V = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Its determinant is

$$\boxed{\det V = \prod_{1\le i<j\le n} (x_j - x_i).}$$

The factorization can be proved by observing that the determinant is alternating in the x_i, hence divisible by every $x_j - x_i$, and that both sides have the same total degree and leading coefficient.

This formula has a direct meaning:

$$\det V \ne 0 \quad \Longleftrightarrow \quad x_1, \dots, x_n \text{ are distinct}$$

$$\iff \quad \text{values at the } x_i \text{ determine } p \in \mathcal{P}_{n-1} \text{ uniquely.}$$

Lagrange interpolation is simply the inverse of this evaluation map.

The square of the Vandermonde product,

$$\prod_{i<j}(x_i - x_j)^2,$$

is the discriminant of a monic polynomial with roots x_i, up to the conventional sign. It vanishes exactly when two roots collide. The same determinant therefore detects both failure of interpolation and repeated roots.

§69.4 Finite differences expose hidden polynomial dependence

Suppose columns are formed by evaluating a polynomial sequence at consecutive integers. Repeated column subtraction is the matrix version of the finite-difference operator

$$(\Delta f)(x) = f(x+1) - f(x).$$

If f has degree d, then Δf has degree $d-1$, and

$$\Delta^{d+1} f = 0.$$

This explains why many determinants built from low-degree polynomial data vanish after a few adjacent differences.

For example, consider

$$D = \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0^2 & 1^2 & 2^2 & 3^2 \\ 0^2+0 & 1^2+1 & 2^2+2 & 3^2+3 \end{pmatrix}.$$

The last row is the sum of the second and third rows, so $D = 0$. Repeated differences reveal the same fact without spotting the relation immediately: every row is generated by polynomials of degree at most two, so four evaluation rows cannot be independent.

A nonzero finite-difference determinant is equally informative. For a polynomial p of exact degree $n-1$ with leading coefficient c, the determinant

$$\det\big(p_i(j)\big)_{0\le i,j<n}$$

can often be reduced to a triangular matrix whose diagonal contains successive leading finite differences. The determinant records the product of the new independent polynomial degrees. The row operations are not a bag of tricks; they are a change from monomials to the Newton finite-difference basis.

§69.5 Block elimination produces the Schur complement

Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

with A square and invertible. Multiplying on the left by

$$L = \begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$$

performs block Gaussian elimination:

$$LM = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

Since $\det L = 1$,

$$\boxed{\det M = \det A \, \det(D - CA^{-1}B).}$$

The matrix

$$S = D - CA^{-1}B$$

is the Schur complement of A.

Its role is clearer in a linear system. From

$$Ax + By = u, \qquad Cx + Dy = v,$$

the first equation gives

$$x = A^{-1}(u - By).$$

Substitution leaves the effective equation

$$Sy = v - CA^{-1}u.$$

The same object controls determinant, invertibility, and elimination of the x-variables.

The hypothesis that A is invertible cannot be silently discarded. If A is singular, the displayed formula is not defined; one must pivot to another block, use a limiting argument only when justified, or return to rank and minors.

§69.6 Low-rank updates reduce a large determinant to a small one

Suppose A is invertible and the perturbation has rank one:

$$A + uv^T.$$

Factor out A:

$$A + uv^T = A(I + A^{-1}uv^T).$$

Sylvester's determinant identity

$$\det(I_m + UV) = \det(I_r + VU)$$

for $U \in k^{m\times r}$, $V \in k^{r\times m}$ gives, with $r = 1$,

$$\boxed{\det(A + uv^T) = \det A \, (1 + v^T A^{-1} u).}$$

This is the matrix determinant lemma.

Consider the arrowhead matrix

$$M = \begin{pmatrix} d_1 & 0 & \cdots & 0 & u_1 \\ 0 & d_2 & \cdots & 0 & u_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & d_n & u_n \\ v_1 & v_2 & \cdots & v_n & \alpha \end{pmatrix},$$

where every $d_i \neq 0$. Taking

$$A = \operatorname{diag}(d_1, \ldots, d_n)$$

in the Schur complement formula yields

$$\det M = \left(\prod_{i=1}^{n} d_i\right)\left(\alpha - \sum_{i=1}^{n} \frac{u_i v_i}{d_i}\right).$$

A determinant of order $n+1$ has become one scalar correction to the diagonal product.

For a rank-r update,

$$\det(A + UV^T) = \det A \, \det(I_r + V^T A^{-1} U).$$

The computational gain is structural: only the coupled r-dimensional directions need a new determinant.

§69.7 Recursive determinants are second-order dynamical systems

Let

$$D_n = \det \begin{pmatrix} a & b & 0 & \cdots & 0 \\ c & a & b & \ddots & \vdots \\ 0 & c & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b \\ 0 & \cdots & 0 & c & a \end{pmatrix}.$$

Expansion along the first row gives

$$D_n = aD_{n-1} - bcD_{n-2}, \qquad D_0 = 1, \quad D_1 = a.$$

Rather than treating each order as a new determinant, package the recurrence as

$$\begin{pmatrix} D_n \\ D_{n-1} \end{pmatrix} = \begin{pmatrix} a & -bc \\ 1 & 0 \end{pmatrix} \begin{pmatrix} D_{n-1} \\ D_{n-2} \end{pmatrix}.$$

The 2×2 transfer matrix carries the entire family.

If the characteristic roots of

$$r^2 - ar + bc = 0$$

are distinct, say r_+, r_-, then

$$D_n = \frac{r_+^{n+1} - r_-^{n+1}}{r_+ - r_-}.$$

When the roots coincide, the solution contains a polynomial factor in n. Repeated-root behavior of the transfer matrix is the same phenomenon as a Jordan block in a recurrence. A patterned determinant has become a small linear dynamical system.

§69.8 Cramer's rule lives only on the invertible open set

For a square system

$$Ax = b$$

with $\det A \neq 0$, the unique solution is

$$x = A^{-1}b.$$

Using

$$A^{-1} = \frac{\operatorname{adj}(A)}{\det A}$$

gives Cramer's formula

$$x_j = \frac{\det A_j}{\det A},$$

where A_j is obtained by replacing column j by b.

The formula is elegant but its domain is important. It describes the open set

$$\{A : \det A \neq 0\}$$

of invertible matrices. On the hypersurface $\det A = 0$, ratios of determinants cease to be the right language. Solvability is instead the image condition

$$b \in \operatorname{im} A,$$

or equivalently

$$\operatorname{rank} A = \operatorname{rank}[A \mid b].$$

If the common rank is $r < n$, the solution set is an affine space

$$x_0 + \ker A$$

of dimension $n - r$. Rank is therefore not a fallback after determinant tricks fail; it is the correct invariant on the singular locus.

§69.9 Minors cut out the geometry of rank failure

A matrix satisfies

$$\operatorname{rank} A \leq r$$

exactly when all $(r+1) \times (r+1)$ minors vanish. Thus matrices of rank at most r form an algebraic set, the determinantal variety

$$\mathcal{D}_r = \{A \in k^{m \times n} : \operatorname{rank} A \leq r\}.$$

For 2×3 matrices of rank at most one, write

$$A = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}.$$

The defining equations are its three 2×2 minors:

$$ae - bd = 0, \qquad af - cd = 0, \qquad bf - ce = 0.$$

Away from the zero matrix, every such matrix factors as

$$A = uv^T, \qquad u \in k^2 \setminus \{0\}, \quad v \in k^3 \setminus \{0\}.$$

The factorization has one scaling redundancy

$$(u, v) \sim (\lambda u, \lambda^{-1} v),$$

so the rank-one stratum has dimension

$$2 + 3 - 1 = 4.$$

This agrees with the general formula

$$\dim \mathcal{D}_r = r(m + n - r).$$

The same example lets us see the singular locus rather than merely name it. On the open chart where $a \neq 0$, the equations

$$ae - bd = 0, \qquad af - cd = 0$$

solve uniquely for

$$e = \frac{bd}{a}, \qquad f = \frac{cd}{a}.$$

The third minor then vanishes automatically:

$$bf - ce = b\frac{cd}{a} - c\frac{bd}{a} = 0.$$

Thus (a, b, c, d) are four local coordinates near every rank-one point with $a \neq 0$. If a different entry is nonzero, the same argument uses the corresponding chart. Every nonzero rank-one matrix is therefore a smooth point of this four-dimensional variety.

At the zero matrix, all first derivatives of the three defining minors vanish. Their Jacobian has rank zero there, so the equations have no nontrivial linear part and the tangent space is the entire six-dimensional ambient space. The origin is singular. In this example,

$$\operatorname{Sing}(\mathcal{D}_1) = \mathcal{D}_0 = \{0\}.$$

More generally, the singular locus of the rank-at-most-r determinantal variety is the lower stratum $\mathcal{D}_{r-1}$.

For square matrices, $\det A = 0$ is the boundary where one direction collapses; deeper rank strata record how many independent directions have collapsed. Cofactors, Schur complements, determinant lemmas, and rank conditions are therefore different local descriptions of the same geometry of elimination and loss of dimension.

Inner-Product Geometry and Spectral Decomposition: From QR to Quadratic Forms

N-20251202

§70.1 Diagonalization asks whether the space splits into invariant lines

A matrix $A \in k^{n\times n}$ is diagonalizable when there is an invertible matrix P such that

$$P^{-1}AP = D$$

is diagonal. If the columns of P are $v_1, \dots, v_n$, then

$$Av_i = \lambda_i v_i.$$

Thus diagonalization is the decomposition

$$k^n = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_s}$$

into eigenspaces. The question is not merely whether eigenvalues exist, but whether their eigenspaces supply enough independent directions.

Several properties that are often confused are logically independent. The zero matrix is diagonalizable but not invertible. The Jordan block

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is invertible but not diagonalizable. Repeated eigenvalues cause no difficulty for the identity matrix, whereas the same repeated eigenvalue leaves J with only one eigendirection. Rank, invertibility, eigenvalue multiplicity, and diagonalizability answer different questions.

§70.2 The minimal polynomial detects the missing eigendirections

The minimal polynomial $m_A(t)$ is the monic polynomial of least degree satisfying

$$m_A(A) = 0.$$

If $A = PDP^{-1}$, then m_A must vanish on every diagonal entry of D, so it is a product of distinct linear factors:

$$m_A(t) = \prod_{i=1}^{s} (t - \lambda_i).$$

Conversely, suppose

$$m_A(t) = \prod_{i=1}^{s} (t - \lambda_i)$$

with distinct roots. Define the Lagrange polynomials

$$\ell_i(t) = \prod_{j \neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

They satisfy

$$\sum_i \ell_i(t) = 1 \pmod{m_A(t)}.$$

Hence

$$I = \sum_i \ell_i(A).$$

Moreover,

$$(A - \lambda_i I)\ell_i(A) = 0,$$

so $\ell_i(A)V \subseteq E_{\lambda_i}$. Every vector is therefore a sum of eigenvectors, and A is diagonalizable.

Thus

$$\boxed{A \text{ is diagonalizable} \iff m_A \text{ splits with no repeated root.}}$$

For the Jordan block above,

$$m_J(t) = (t-1)^2.$$

The repeated factor records the nilpotent motion along the generalized eigendirection.

§70.3 An inner product turns decomposition into best approximation

An inner product adds lengths and angles to a vector space. For a subspace $W \subset \mathbb{R}^n$, the orthogonal projection of x onto W is the unique vector $p \in W$ for which

$$x - p \perp W.$$

If $Q \in \mathbb{R}^{n \times r}$ has orthonormal columns spanning W, then

$$p = QQ^T x.$$

Indeed, $QQ^T x \in W$, while

$$Q^T(x - QQ^T x) = Q^T x - Q^T x = 0.$$

The matrix

$$P_W = QQ^T$$

is symmetric and idempotent:

$$P_W^T = P_W, \qquad P_W^2 = P_W.$$

Orthogonality also proves optimality. For any $w \in W$,

$$x - w = (x - p) + (p - w)$$

is an orthogonal decomposition, so

$$\|x - w\|^2 = \|x - p\|^2 + \|p - w\|^2 \geq \|x - p\|^2.$$

Projection is therefore the closest point in the subspace. Least squares, Fourier approximation, and Gram–Schmidt all begin with this Pythagorean identity.

§70.4 Gram–Schmidt factors a basis into orthogonal directions and coefficients

Given independent vectors $a_1, \dots, a_r$, define

$$u_1 = a_1,$$
$$u_j = a_j - \sum_{i<j} \frac{\langle a_j, u_i \rangle}{\langle u_i, u_i \rangle} u_i.$$

Each u_j is orthogonal to the previous vectors, and independence guarantees $u_j \neq 0$. Normalizing gives orthonormal vectors

$$q_j = \frac{u_j}{\|u_j\|}.$$

For the columns

$$a_1 = (1, 0, 0, 1)^T, \quad a_2 = (1, 1, 0, 1)^T, \quad a_3 = (1, 1, 1, 0)^T,$$

one obtains

$$q_1 = \frac{1}{\sqrt{2}}(1, 0, 0, 1)^T, \qquad q_2 = (0, 1, 0, 0)^T,$$
$$q_3 = \frac{1}{\sqrt{6}}(1, 0, 2, -1)^T.$$

Writing $A = (a_1, a_2, a_3)$ and $Q = (q_1, q_2, q_3)$, the coefficients

$$r_{ij} = \langle q_i, a_j \rangle$$

form the upper-triangular matrix

$$R = \begin{pmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{6}/2 \end{pmatrix},$$

and

$$\boxed{A = QR, \qquad Q^T Q = I.}$$

QR factorization is Gram–Schmidt written as a matrix identity: Q stores the orthonormal geometry, while R stores how the original columns were assembled from it.

§70.5 Householder reflections reveal why stable QR avoids repeated subtraction

Classical Gram–Schmidt can lose accuracy when columns are nearly dependent, because it subtracts nearly equal vectors. Modified Gram–Schmidt reduces the damage by updating the residual after each projection. Householder QR takes a different route: it reflects an entire column onto a coordinate axis.

For a nonzero vector x, choose

$$v = x - \alpha e_1, \qquad \alpha = \operatorname{sign}(x_1)\|x\|,$$

and define

$$H = I - 2\frac{vv^T}{v^T v}.$$

Then

$$H^T = H, \qquad H^2 = I,$$

so H is an orthogonal reflection, and $Hx = \alpha e_1$.

Consider

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}.$$

For the first column $x = (3,4)^T$, take $\alpha = 5$ and $v = (-2,4)^T$. Then

$$H = \begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix},$$

and

$$HA = \begin{pmatrix} 5 & 4 \\ 0 & -3 \end{pmatrix} = R.$$

Since $H^T = H$,

$$A = QR, \qquad Q = H.$$

The reflection zeros an entire subcolumn at once and preserves norms exactly in exact arithmetic. This geometric operation is the basis of stable QR algorithms.

§70.6 Orthogonal maps preserve geometry, and reflections generate them

A real matrix U is orthogonal when

$$U^T U = I.$$

It preserves inner products:

$$\langle Ux, Uy\rangle = x^T U^T U y = \langle x, y\rangle.$$

Hence it preserves lengths, angles, and orthogonality. Its determinant satisfies

$$(\det U)^2 = 1,$$

so

$$\det U = \pm 1.$$

The sign records orientation: rotations have determinant $+1$, while a single reflection has determinant -1.

Householder matrices show that reflections are not merely examples. Every orthogonal matrix is a product of reflections. QR factorization repeatedly uses this fact to rotate or reflect successive column spaces into coordinate position without changing their metric geometry.

Over $\mathbb{C}$, the corresponding maps are unitary:

$$U^* U = I.$$

Complex conjugation in the Hermitian inner product is essential; without it, $\langle x, x\rangle$ need not be positive.

§70.7 The spectral theorem follows from one Rayleigh extremum and induction

Let $A = A^T$ be real symmetric. On the unit sphere, the Rayleigh quotient

$$R_A(x) = x^T Ax$$

attains a maximum because the sphere is compact. At a maximizing unit vector v, Lagrange multipliers give

$$2Av = 2\lambda v,$$

so v is an eigenvector.

Its orthogonal complement is invariant. If $w \perp v$, then

$$\langle Aw, v\rangle = \langle w, Av\rangle = \lambda\langle w, v\rangle = 0.$$

Thus A restricts to a symmetric operator on $v^\perp$. Repeating the argument inductively produces an orthonormal eigenbasis.

Therefore

$$\boxed{A = A^T \quad \Longrightarrow \quad A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)Q^T}$$

for an orthogonal matrix Q. The proof explains both parts of the theorem: symmetry makes the orthogonal complement invariant, while compactness supplies the first eigenvector.

The extreme Rayleigh values are the extreme eigenvalues:

$$\lambda_{\min}(A) = \min_{\|x\|=1} x^T Ax, \qquad \lambda_{\max}(A) = \max_{\|x\|=1} x^T Ax.$$

The spectral theorem is therefore also an optimization theorem.

§70.8 Spectral projectors turn diagonalization into functional calculus

Suppose a diagonalizable matrix has distinct eigenvalues $\lambda_1, \ldots, \lambda_s$. The Lagrange polynomials from the minimal-polynomial argument give projectors

$$P_i = \prod_{j\neq i} \frac{A - \lambda_j I}{\lambda_i - \lambda_j}.$$

They satisfy

$$P_i^2 = P_i, \qquad P_iP_j = 0\ (i \neq j), \qquad \sum_i P_i = I,$$

and $P_iV = E_{\lambda_i}$. Hence

$$A = \sum_i \lambda_i P_i.$$

For any function f defined on the spectrum, set

$$f(A) = \sum_i f(\lambda_i)P_i.$$

This is finite-dimensional functional calculus.

If $A^2 = I$, the eigenvalues are ± 1, and

$$P_+ = \frac{I + A}{2}, \qquad P_- = \frac{I - A}{2}.$$

Therefore

$$\begin{aligned} e^{tA} &= e^t P_+ + e^{-t} P_- \\ &= \cosh t\, I + \sinh t\, A. \end{aligned}$$

A matrix exponential has been computed without multiplying matrices repeatedly. The projectors separate the invariant modes first, and the scalar function is then applied to each mode.

§70.9 Similarity and congruence answer different classification questions

Similarity

$$A \longmapsto P^{-1} A P$$

changes the basis of a linear operator and preserves eigenvalues. Congruence

$$A \longmapsto P^T A P$$

changes variables in a quadratic form

$$q(x) = x^T A x, \qquad x = Py.$$

It need not preserve eigenvalues.

Consider

$$q(x, y) = 2x^2 + 4xy - y^2.$$

Completing the square gives

$$q(x, y) = 2(x + y)^2 - 3y^2.$$

With

$$u = x + y, \qquad v = y,$$

the form becomes

$$q = 2u^2 - 3v^2.$$

Thus it has one positive and one negative square. Sylvester's law of inertia says that the triple

$$(n_+, n_-, n_0)$$

of positive, negative, and zero squares is invariant under every invertible congruence. Here the inertia is

$$(1, 1, 0).$$

Orthogonal diagonalization is a special congruence that also happens to be a similarity for symmetric matrices. It preserves Euclidean geometry as well as the quadratic form's spectral data; a general completing-square substitution preserves only the inertia classification.

§70.10 Positive definiteness has spectral, minor, and factorization tests

For a symmetric matrix A, the following are equivalent:

$$x^T Ax > 0 \quad (x \neq 0),$$

$$\lambda_i(A) > 0 \quad \text{for every } i,$$

$$\Delta_1 > 0, \dots, \Delta_n > 0,$$

where Δ_k are the leading principal minors, and

$$A = LL^T$$

for an invertible lower-triangular matrix L with positive diagonal. These are the quadratic-form, spectral, Sylvester, and Cholesky criteria.

For

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix},$$

the leading principal minors are

$$\Delta_1 = 2, \qquad \Delta_2 = 3, \qquad \Delta_3 = 4.$$

Hence $A \succ 0$. Cholesky elimination gives

$$L = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ -1/\sqrt{2} & \sqrt{3/2} & 0 \\ 0 & -\sqrt{2/3} & 2/\sqrt{3} \end{pmatrix},$$

and a direct multiplication verifies

$$A = LL^T.$$

The factorization makes positivity immediate:

$$x^T Ax = \|L^T x\|^2 > 0$$

for $x \neq 0$.

§70.11 Principal axes organize conics, Hessians, and PCA

A quadratic polynomial in two variables can be written

$$F(x) = x^T Ax + b^T x + c, \qquad A = A^T.$$

An orthogonal change aligns the coordinates with eigenvectors of A, removing cross terms. A translation then completes squares and moves the center when one exists. Rotation and translation perform different jobs: the first changes directions, the second changes the origin.

The same matrix appears as a Hessian in optimization. Near a critical point p,

$$f(p+h) = f(p) + \frac{1}{2}h^T \nabla^2 f(p)h + o(\|h\|^2).$$

The inertia of the Hessian distinguishes a local minimum, maximum, or saddle when the Hessian is nonsingular.

PCA applies the same spectral geometry to data. Consider the centered points

$$(2,1),\ (1,1),\ (-1,-1),\ (-2,-1).$$

Their covariance matrix is

$$C = \frac{1}{4}\sum_{i=1}^{4} x_i x_i^T = \begin{pmatrix} 5/2 & 3/2 \\ 3/2 & 1 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{7 \pm 3\sqrt{5}}{4}.$$

For λ_+, an eigenvector can be chosen proportional to

$$\begin{pmatrix} 1 \\ (\sqrt{5}-1)/2 \end{pmatrix}.$$

Projection onto this direction preserves the largest possible average squared variation. PCA is therefore the Rayleigh-quotient extremum of the covariance quadratic form, not a separate algorithmic miracle.

§70.12 The Hilbert-space extension keeps projection but may lose eigenbases

In a Hilbert space H, every closed subspace M still has an orthogonal projection. To see why closedness matters, take a minimizing sequence $m_n \in M$ for the distance from x to M. The parallelogram identity shows that (m_n) is Cauchy; completeness of M gives a limit $m \in M$, and the usual first-variation argument gives

$$x - m \perp M.$$

Thus

$$H = M \oplus M^\perp.$$

Finite-dimensional least squares is the first instance of this projection theorem.

The spectral theorem also survives, but not always as an eigenbasis. On

$$H = L^2([0,1]),$$

consider the self-adjoint multiplication operator

$$(Mf)(t) = tf(t).$$

If $Mf = \lambda f$, then

$$(t-\lambda)f(t) = 0$$

almost everywhere. The set $\{t = \lambda\}$ has measure zero, so $f = 0$ in L^2. The operator has no nonzero eigenvectors, even though its spectrum is the whole interval $[0,1]$.

The correct infinite-dimensional statement uses a projection-valued measure:

$$M = \int_{[0,1]} \lambda \, dE(\lambda).$$

Finite diagonal matrices are the discrete case of this integral. Orthogonal projection, self-adjointness, and functional calculus remain; what can disappear is a basis made from individual eigenvectors.

Rank, Projections, and Long-Term Matrix Dynamics

N-20260103

§71.1 The first useful question is often which relation the matrix satisfies

A large matrix power or rank identity rarely becomes easier by multiplying entries. The decisive information is often an algebraic relation:

$$P^2 = P, \qquad N^k = 0, \qquad A^2 + c_1 A + c_0 I = 0, \qquad A = UV^T.$$

Each relation compresses the matrix into a small algebra. An idempotent splits the space into image and kernel. A nilpotent has a finite geometric series. A polynomial relation reduces every high power to lower powers. A low-rank factorization says that only a few independent directions pass through the map.

This chapter therefore organizes matrix problems by structure rather than by the surface form of an exercise. Rank describes which directions survive, projections describe how the space splits, polynomial identities describe repeated iteration, and spectral projectors describe what remains after transient modes decay.

§71.2 Rank factorization exposes the quotient–image structure

Let $A : k^n \to k^m$ have rank r. Choose a basis $c_1, \dots, c_r$ of its column space and place these columns in

$$B \in k^{m\times r}.$$

Every column of A has unique coordinates in this basis; collect those coordinates in

$$C \in k^{r\times n}.$$

Then

$$\boxed{A = BC,}$$

with

$$\operatorname{rank} B = \operatorname{rank} C = r.$$

This is a rank factorization. It is the matrix form of

$$k^n \longrightarrow k^n/\ker A \cong \operatorname{im} A \hookrightarrow k^m.$$

For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$$

has rank two. Its first two columns are independent, and the third satisfies

$$c_3 = -c_1 + 2c_2.$$

Thus

$$A = \underbrace{\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 1 \end{pmatrix}}_{B} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}}_{C}.$$

Equivalently,

$$A = c_1(1, 0, -1) + c_2(0, 1, 2),$$

a sum of two rank-one matrices.

The factorization is not unique. For any $S \in GL_r(k)$,

$$A = (BS)(S^{-1}C).$$

The ambiguity is exactly a change of coordinates in the intermediate r-dimensional image space.

§71.3 The four fundamental subspaces are paired by orthogonality

For a real matrix $A \in \mathbb{R}^{m\times n}$, the row space and null space are orthogonal complements:

$$\operatorname{row}(A) = (\ker A)^{\perp}.$$

Indeed, $Ax = 0$ says that x is orthogonal to every row. Similarly,

$$\operatorname{col}(A) = (\ker A^T)^{\perp}.$$

Hence

$$\mathbb{R}^n = \operatorname{row}(A) \oplus \ker A,$$
$$\mathbb{R}^m = \operatorname{col}(A) \oplus \ker A^T.$$

These are the four fundamental subspaces.

Positivity gives the useful identities

$$\ker(A^TA) = \ker A, \qquad \ker(AA^T) = \ker A^T.$$

For example,

$$A^TAx = 0 \implies x^TA^TAx = \|Ax\|^2 = 0,$$

so $Ax = 0$. Therefore

$$\operatorname{rank}(A^TA) = \operatorname{rank}(AA^T) = \operatorname{rank} A.$$

A rank identity has been reduced to equality of null spaces, and that equality is certified by a norm square.

If two homogeneous systems $Ax = 0$ and $Bx = 0$ have the same solution space, then

$$\operatorname{row}(A) = (\ker A)^{\perp} = (\ker B)^{\perp} = \operatorname{row}(B).$$

Thus equality of solution sets and equality of row spaces are the same statement viewed from opposite sides.

§71.4 At corank one, the adjugate becomes a rank-one null-space certificate

For every square matrix,

$$A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = (\det A)I.$$

If $\operatorname{rank} A = n-1$, then $\det A = 0$ but at least one $(n-1)$-minor is nonzero, so

$$\mathrm{adj}(A) \neq 0.$$

The identity

$$A\,\mathrm{adj}(A) = 0$$

shows that every column of $\mathrm{adj}(A)$ lies in $\ker A$, which is one-dimensional. Hence all columns are proportional and

$$\operatorname{rank}\mathrm{adj}(A) = 1.$$

Likewise, every row lies in $\ker A^T$. If

$$Au = 0, \qquad v^T A = 0,$$

with nonzero u, v, then

$$\mathrm{adj}(A) = cuv^T$$

for some nonzero scalar c.

For

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

one has

$$\mathrm{adj}(A) = \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \begin{pmatrix} -2 & 1 \end{pmatrix}.$$

The same null vector appears on the right and left because A is symmetric. Any nonzero column of the adjugate solves $Ax = 0$.

If $\operatorname{rank} A \leq n-2$, every $(n-1)$-minor vanishes and

$$\mathrm{adj}(A) = 0.$$

The adjugate therefore distinguishes corank one from deeper singularity.

§71.5 Idempotents are projections, but not always orthogonal projections

Suppose

$$P^2 = P.$$

Every vector splits as

$$x = Px + (x - Px).$$

The first term belongs to $\operatorname{im} P$, while

$$P(x - Px) = Px - P^2x = 0,$$

so the second belongs to $\ker P$. Their intersection is zero, hence

$$V = \operatorname{im} P \oplus \ker P.$$

The matrix acts as identity on its image and zero on its kernel. Its minimal polynomial divides

$$t(t-1),$$

so it is diagonalizable with eigenvalues $0, 1$.

The projection is orthogonal exactly when

$$P^T = P.$$

Without symmetry it is oblique. For example,

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

satisfies $P^2 = P$. It projects onto the x-axis along the line

$$\ker P = \operatorname{span}(1, 1).$$

The image and kernel are complementary but not orthogonal.

If P, Q are idempotent over a field of characteristic not two, then

$$P + Q \text{ is idempotent} \quad \Longleftrightarrow \quad PQ = QP = 0.$$

Indeed, expanding $(P+Q)^2 = P + Q$ gives

$$PQ + QP = 0.$$

Multiplying on the left and right by P shows $PQ = QP$; the displayed sum then gives $2PQ = 0$. Geometrically, each image lies in the other's kernel, so the two projections occupy noninteracting components.

§71.6 Nilpotence turns infinite-looking formulas into finite sums

If $N^k = 0$, then

$$(I - N)(I + N + \cdots + N^{k-1}) = I - N^k = I.$$

Therefore

$$\boxed{(I - N)^{-1} = I + N + \cdots + N^{k-1}.}$$

The inverse is a polynomial because the geometric series terminates.

For the nilpotent Jordan block

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0,$$

one has

$$(I - N)^{-1} = I + N + N^2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

More generally,

$$(\lambda I + N)^m = \sum_{j=0}^{k-1} \binom{m}{j} \lambda^{m-j} N^j.$$

A Jordan block therefore contributes an exponential factor λ^m multiplied by polynomials in m. The nilpotent part measures the failure of pure eigendirection scaling.

§71.7 The minimal polynomial reduces every high power

Cayley–Hamilton says that a matrix satisfies its characteristic polynomial. The sharper computational object is the minimal polynomial. If

$$m_A(t) = t^d + c_{d-1}t^{d-1} + \cdots + c_0,$$

then

$$A^d = -c_{d-1}A^{d-1} - \cdots - c_0 I,$$

and every higher power reduces to a linear combination of

$$I, A, \ldots, A^{d-1}.$$

Equivalently, compute the remainder of t^n modulo $m_A(t)$ and substitute A.

Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

Its minimal polynomial is

$$m_A(t) = (t-1)(t-2).$$

The degree-one remainder $r_n(t) = \alpha t + \beta$ of t^n satisfies

$$r_n(1) = 1, \qquad r_n(2) = 2^n.$$

Thus

$$r_n(t) = (2^n - 1)t + (2 - 2^n),$$

and

$$\boxed{A^n = (2^n - 1)A + (2 - 2^n)I.}$$

Even A^{100} is no harder than evaluating two scalar powers.

§71.8 Polynomial Chinese remainders produce primary projectors

Suppose

$$m_A(t) = p_1(t)^{e_1} \cdots p_s(t)^{e_s}$$

with pairwise coprime factors. The polynomial Chinese remainder theorem provides idempotent classes $e_i(t)$ satisfying

$$e_i \equiv 1 \pmod{p_i^{e_i}}, \qquad e_i \equiv 0 \pmod{p_j^{e_j}} \quad (j \neq i).$$

Substituting A gives commuting projectors

$$E_i = e_i(A), \qquad \sum_i E_i = I, \qquad E_i E_j = 0.$$

Their images are the primary components

$$\ker p_i(A)^{e_i}.$$

For a concrete example, suppose

$$m_A(t) = (t-1)^2(t+2).$$

Polynomials with the required congruences are

$$e_1(t) = \frac{(t+2)(4-t)}{9}, \qquad e_2(t) = \frac{(t-1)^2}{9}.$$

Indeed,

$$e_1 \equiv 1 \pmod{(t-1)^2}, \qquad e_1 \equiv 0 \pmod{t+2},$$

and $e_2 = 1 - e_1$. Therefore

$$E_1 = \frac{(A+2I)(4I-A)}{9}, \qquad E_2 = \frac{(A-I)^2}{9}$$

project onto the generalized 1- and -2-primary subspaces.

This is the structural source of generalized eigenspace decomposition. The vector space becomes a module over $k[t]$, with t acting as A; coprime polynomial factors split that module into independent primary pieces.

§71.9 Jordan asymptotics decide whether matrix powers converge

On a Jordan block

$$J = \lambda I + N, \qquad N^k = 0,$$

one has

$$J^n = \sum_{j=0}^{k-1} \binom{n}{j} \lambda^{n-j} N^j.$$

This formula gives precise long-term behavior.

- If $|\lambda| < 1$, every term tends to zero despite the polynomial factor.
- If $|\lambda| > 1$, some vectors grow exponentially.
- If $|\lambda| = 1$ and $N \neq 0$, polynomial growth prevents boundedness.
- A semisimple eigenvalue 1 contributes a persistent projection.
- Any other eigenvalue on the unit circle produces oscillation rather than convergence.

Consequently, A^n converges exactly when every eigenvalue lies inside the unit disk except possibly the eigenvalue 1, and the eigenvalue 1 is semisimple. In that case,

$$\lim_{n\to\infty} A^n = P_1,$$

where P_1 is the spectral projector onto $\ker(A-I)$ along the direct sum of the decaying primary components.

The phrase "only the eigenvalue 1 remains" is therefore incomplete. What remains is a specific projection determined by the decomposition of the whole space.

§71.10 A two-state Markov chain displays the limiting projector explicitly

Let

$$T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \qquad 0 < a, b < 1.$$

Each row sums to one, so T evolves row probability vectors. Its eigenvalues are

$$1, \qquad \rho = 1 - a - b,$$

with $|\rho| < 1$. The stationary distribution is

$$\pi = \left(\frac{b}{a+b}, \frac{a}{a+b} \right),$$

because $\pi T = \pi$.

Let

$$P = \mathbf{1}\,\pi = \begin{pmatrix} b/(a+b) & a/(a+b) \\ b/(a+b) & a/(a+b) \end{pmatrix}.$$

Then

$$P^2 = P, \qquad TP = PT = P,$$

and $I - P$ is the projector onto the decaying mode. Since T acts by ρ there,

$$\boxed{T^n = P + \rho^n (I - P).}$$

Hence

$$T^n \longrightarrow P.$$

Every initial probability row converges to π. The Markov limit is a spectral projector, while the convergence rate is the magnitude of the subdominant eigenvalue.

§71.11 Strict diagonal dominance is a fast invertibility certificate

Suppose

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \qquad (i = 1, \ldots, n).$$

If $Ax = 0$ had a nonzero solution, choose k with

$$|x_k| = \max_i |x_i| > 0.$$

The k-th equation gives

$$a_{kk} x_k = -\sum_{j \neq k} a_{kj} x_j.$$

Therefore

$$|a_{kk}||x_k| \leq \sum_{j \neq k} |a_{kj}||x_j| \leq \left(\sum_{j \neq k} |a_{kj}| \right) |x_k|,$$

contradicting strict diagonal dominance. Thus A is invertible.

The same argument is the zero-eigenvalue case of Gershgorin's theorem. Every eigenvalue lies in at least one disk

$$|z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}|.$$

Strict diagonal dominance keeps all disks away from zero. Before computing a determinant or inverse, the row geometry can already certify that no kernel direction exists.

§71.12 The Drazin inverse separates invertible dynamics from nilpotent transients

A singular matrix has no ordinary inverse, but its primary decomposition separates the nonzero spectrum from the generalized zero eigenspace. On the nonzero primary components, A is invertible; on the zero component, it is nilpotent. The Drazin inverse A^D acts as the ordinary inverse on the first part and as zero on the second.

If the nilpotent zero component has index k, the Drazin inverse is characterized by

$$AA^D = A^D A,$$

$$A^D A A^D = A^D,$$

$$A^{k+1} A^D = A^k.$$

For

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

the nilpotent block has index two and

$$A^D = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The Moore–Penrose pseudoinverse is different:

$$A^+ = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The pseudoinverse is determined by orthogonal least-squares geometry, whereas the Drazin inverse is determined by the algebraic decomposition into invertible and nilpotent dynamics.

This distinction completes the path from rank to iteration. Rank factorization identifies the active image, idempotents split complementary components, primary projectors isolate spectral modes, powers reveal persistent and transient behavior, and the Drazin inverse inverts exactly the modes that are dynamically invertible while discarding the nilpotent transient part.

Spectral Calculus and Positive Matrix Geometry: From Basis Changes to Lyapunov Equations

N-20260105

§72.1 Coordinates change, but the operator does not

Let V be a finite-dimensional vector space. A basis turns a linear map $T : V \to V$ into a matrix, but a different basis gives a different matrix for the same operator. Suppose the ordered bases α and β are related by

$$(\beta_1, \dots, \beta_n) = (\alpha_1, \dots, \alpha_n)P.$$

For every vector v,

$$[v]_\alpha = P[v]_\beta.$$

If $A = [T]_\alpha$ and $B = [T]_\beta$, then

$$B = P^{-1}AP.$$

Thus similarity is not an arbitrary matrix manipulation. It is the coordinate shadow of keeping the operator fixed while changing the basis.

Quantities preserved by similarity belong to the operator rather than to its coordinates. The characteristic polynomial, eigenvalues, trace, determinant, rank, and minimal polynomial are all similarity invariants. This is why spectral methods can replace long coordinate computations: they work with data that survive every basis change.

§72.2 A complete transition-matrix calculation

Let $\varepsilon = (\varepsilon_1, \varepsilon_2, \varepsilon_3)$ be a reference basis and define

$$\alpha_1 = \varepsilon_1, \qquad \alpha_2 = \varepsilon_1 + \varepsilon_2, \qquad \alpha_3 = \varepsilon_1 + \varepsilon_2 + \varepsilon_3,$$

while

$$\beta_1 = \varepsilon_1 + \varepsilon_2, \qquad \beta_2 = 2\varepsilon_2 + \varepsilon_3, \qquad \beta_3 = \varepsilon_1 + 3\varepsilon_3.$$

The basis matrices relative to ε are

$$P_\alpha = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad P_\beta = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}.$$

Since

$$(\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3)P,$$

one has

$$P = P_\alpha^{-1}P_\beta = \begin{pmatrix} 0 & -2 & 1 \\ 1 & 1 & -3 \\ 0 & 1 & 3 \end{pmatrix}.$$

A coordinate column transforms by

$$[v]_\alpha = P[v]_\beta.$$

The question "which vectors have the same coordinate column in the ε- and α-bases?" is also structural. Such a column t must satisfy

$$t = P_\alpha t.$$

Thus it lies in the fixed subspace of the transition matrix. Here

$$(P_\alpha - I)t = 0$$

forces

$$t_2 = t_3 = 0,$$

so the common-coordinate vectors are exactly the multiples of ε_1. A basis exercise has become an eigenspace calculation for eigenvalue 1.

§72.3 Polynomial relations turn matrix questions into scalar questions

Suppose

$$Av = \lambda v.$$

For every polynomial p,

$$p(A)v = p(\lambda)v.$$

Consequently, if a matrix satisfies

$$p(A) = 0,$$

then every eigenvalue of A is a root of p.

This simple observation explains several common shortcuts. If

$$A^k = 0,$$

then every eigenvalue is zero, so for an $n \times n$ matrix

$$\det(I + 3A) = \prod_{i=1}^{n}(1 + 3\lambda_i) = 1.$$

The matrix need not be diagonalizable; the determinant conclusion uses only the eigenvalues counted with algebraic multiplicity.

If a real matrix satisfies

$$A^2 + aA + bI = 0$$

and

$$a^2 - 4b < 0,$$

then it has no real eigenvector. Indeed, a real eigenvalue λ would satisfy

$$\lambda^2 + a\lambda + b = 0,$$

contradicting the negative discriminant. The matrix may still have complex eigenvalues and become diagonalizable over $\mathbb{C}$; the obstruction is specifically to real eigendirections.

§72.4 The minimal polynomial decides diagonalizability

The characteristic polynomial lists eigenvalues with algebraic multiplicity, but the minimal polynomial records the largest Jordan blocks. A matrix is diagonalizable over a field precisely when its minimal polynomial splits into distinct linear factors.

For example, if

$$A^3 = A,$$

then

$$A(A - I)(A + I) = 0.$$

Over a field of characteristic different from 2, the polynomial

$$x(x - 1)(x + 1)$$

has three distinct roots. The minimal polynomial divides this square-free polynomial, so A is diagonalizable and

$$V = \ker A \oplus \ker(A - I) \oplus \ker(A + I).$$

This conclusion is stronger than merely saying that the possible eigenvalues are $-1, 0, 1$.

Repeated eigenvalues require more care. If an $n \times n$ matrix is known to be diagonalizable and λ has algebraic multiplicity m, then

$$\dim \ker(A - \lambda I) = m.$$

Thus a parameter problem with a repeated eigenvalue can often be solved by imposing a rank condition on $A - \lambda I$. The rank is not an unrelated trick; it measures the missing eigenspace dimension.

§72.5 Spectral mapping also controls inverses and adjugates

If A is invertible and $Av = \lambda v$, then

$$A^{-1}v = \lambda^{-1}v.$$

Since

$$\operatorname{adj}(A) = \det(A)A^{-1},$$

the same eigenvector satisfies

$$\operatorname{adj}(A)v = \frac{\det A}{\lambda}v.$$

Thus the eigenvalues of the adjugate are the products of all eigenvalues except one:

$$\frac{\lambda_1 \cdots \lambda_n}{\lambda_i} = \prod_{j \neq i} \lambda_j.$$

Trace and determinant are the first two symmetric spectral statistics:

$$\operatorname{tr} A = \sum_i \lambda_i, \qquad \det A = \prod_i \lambda_i.$$

They can be read from diagonal entries, from a triangular form, or from the eigenvalues because these are different coordinate descriptions of the same invariants.

A row-sum shortcut fits the same pattern. If every row sum of A equals c, then

$$A\mathbf{1} = c\mathbf{1}, \qquad \mathbf{1} = (1, \ldots, 1)^T.$$

The aggregate direction $\mathbf{1}$ is an invariant one-dimensional subspace. The shortcut works because the matrix preserves a visible symmetry direction.

§72.6 The symmetric spectral theorem turns spectrum into geometry

For a real symmetric matrix A, there is an orthogonal matrix Q such that

$$A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)Q^T.$$

The eigenspaces are mutually orthogonal, and every geometric statement can be checked along independent principal directions.

Suppose a symmetric 3×3 matrix has eigenvalues

$$-1, 1, 1$$

and a -1-eigenvector

$$u = \frac{1}{\sqrt{2}}(0, 1, 1)^T.$$

The orthogonal complement $u^\perp$ is the $+1$-eigenspace. Therefore A is reflection across $u^\perp$:

$$A = I - 2uu^T.$$

Computing the rank-one term gives

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}.$$

This reconstructs the matrix without guessing two additional eigenvectors. The spectral decomposition is the geometry of a reflection.

§72.7 Functional calculus applies scalar functions along eigendirections

For a symmetric matrix

$$A = Q\,\mathrm{diag}(\lambda_i)Q^T$$

and a function f defined on its spectrum, set

$$f(A) = Q\,\mathrm{diag}(f(\lambda_i))Q^T.$$

This defines powers, exponentials, logarithms, and roots without choosing coordinates beyond the orthonormal eigenbasis.

Every real number has a unique real odd root. Hence, for any positive integer k, define

$$B = Q\,\mathrm{diag}(\sqrt[2k+1]{\lambda_1}, \ldots, \sqrt[2k+1]{\lambda_n})Q^T.$$

Then B is real symmetric and

$$B^{2k+1} = A.$$

Even roots behave differently. A real symmetric matrix has a real symmetric square root exactly when

$$A \succeq 0.$$

In that case it has a unique positive semidefinite square root, although other symmetric square roots may choose signs independently on eigenspaces. For a positive definite matrix, one may similarly define

$$A^t, \qquad \log A, \qquad e^A$$

by applying the scalar functions to positive eigenvalues.

§72.8 Rayleigh quotients define the Loewner order

For symmetric A, the Rayleigh quotient is

$$R_A(x) = \frac{x^T Ax}{x^T x}.$$

Writing $x = Qy$ gives

$$R_A(x) = \frac{\sum_i \lambda_i y_i^2}{\sum_i y_i^2}.$$

It is a weighted average of the eigenvalues, so

$$\lambda_{\min}(A) \leq R_A(x) \leq \lambda_{\max}(A).$$

The extreme values occur exactly on the corresponding eigenspaces.

For symmetric matrices, define the Loewner order by

$$A \succeq B \quad \iff \quad x^T(A - B)x \geq 0 \quad \text{for every } x.$$

Thus

$$A \succeq B \quad \iff \quad \lambda_{\min}(A - B) \geq 0.$$

If

$$\lambda_{\min}(A) > \lambda_{\max}(B),$$

then for every unit vector x,

$$x^T(A - B)x \geq \lambda_{\min}(A) - \lambda_{\max}(B) > 0,$$

so

$$A - B \succ 0.$$

The eigenvalue comparison is exactly an order statement about quadratic forms.

§72.9 Positive definite matrices are ellipsoids and a convex cone

A symmetric matrix A is positive definite when

$$x^T Ax > 0 \qquad (x \neq 0).$$

Spectrally, this means

$$\lambda_i(A) > 0$$

for all i. Geometrically,

$$E_A = \{x : x^T Ax \leq 1\}$$

is an ellipsoid whose semi-axis lengths are

$$\lambda_i(A)^{-1/2}.$$

If $A \succeq B \succ 0$, then

$$E_A \subseteq E_B.$$

Larger quadratic forms produce smaller ellipsoids.

The positive semidefinite matrices form the convex cone

$$S_+^n = \{A = A^T : A \succeq 0\}.$$

Indeed, if $A, B \succeq 0$ and $s, t \geq 0$, then

$$x^T(sA + tB)x = s\,x^T Ax + t\,x^T Bx \geq 0.$$

Sylvester's criterion, which tests positivity through leading principal minors, is a coordinate certificate for membership in the interior of this cone.

§72.10 A block matrix reveals its spectrum after one orthogonal change

Let $B = B^T$ and consider

$$M = \begin{pmatrix} I & B \\ B & I \end{pmatrix}.$$

Use the orthogonal matrix

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}.$$

A direct multiplication gives

$$U^T M U = \begin{pmatrix} I + B & 0 \\ 0 & I - B \end{pmatrix}.$$

Therefore

$$M \succ 0 \iff I + B \succ 0 \text{ and } I - B \succ 0.$$

Since the eigenvalues of $I \pm B$ are $1 \pm \lambda_i(B)$,

$$\boxed{M \succ 0 \iff |\lambda_i(B)| < 1 \text{ for every } i.}$$

The block congruence trick is really a decomposition into symmetric and antisymmetric subspaces

$$(x, x), \qquad (x, -x).$$

§72.11 Noncommutativity is where scalar order intuition fails

For positive numbers,

$$0 < a \leq b \implies a^2 \leq b^2.$$

The analogous matrix statement is false without commutativity. Take

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Both are positive definite, and

$$B - A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \succeq 0.$$

However,

$$B^2 - A^2 = \begin{pmatrix} 6 & 5 \\ 5 & 4 \end{pmatrix}$$

has determinant

$$24 - 25 = -1,$$

so it is indefinite. Thus $A \preceq B$ does not imply $A^2 \preceq B^2$.

Functions that preserve Loewner order on every matrix size are called operator monotone. The inverse function is order reversing:

$$0 \prec A \preceq B \quad \Longrightarrow \quad B^{-1} \preceq A^{-1}.$$

To see this, conjugate by $B^{-1/2}$:

$$0 \prec B^{-1/2} A B^{-1/2} \preceq I.$$

Taking reciprocal eigenvalues reverses the inequality, and conjugating back gives the result. The proof works because inversion can be reduced to an ordinary spectral inequality after a congruence.

§72.12 A noncommuting positive-matrix inequality still reduces to one spectrum

For positive definite A, B, consider

$$(A + B)(A^{-1} + B^{-1}).$$

It need not be symmetric. Nevertheless, it is similar to a symmetric positive definite matrix. Set

$$X = A^{-1/2} B A^{-1/2} \succ 0.$$

Conjugation by $A^{1/2}$ gives

$$A^{-1/2}(A + B)(A^{-1} + B^{-1})A^{1/2} = 2I + X + X^{-1}.$$

If $\lambda > 0$ is an eigenvalue of X, the corresponding eigenvalue of the right-hand side is

$$2 + \lambda + \lambda^{-1} \geq 4.$$

Hence every eigenvalue of the original product is real and at least 4.

The successful reduction does not require A and B to commute. It uses a carefully chosen similarity that packages their relative position into the single positive matrix X.

§72.13 Sylvester equations are entrywise only in the right basis

Consider

$$AX + XB = C.$$

If A and B are diagonalizable with eigenvalues α_i and β_j, the equation becomes entrywise in their eigenbases:

$$(\alpha_i + \beta_j) Y_{ij} = K_{ij}.$$

Thus the solution is unique when

$$\alpha_i + \beta_j \neq 0$$

for every pair, equivalently when

$$\sigma(A) \cap \sigma(-B) = \varnothing.$$

This is the spectral separation condition for the Sylvester operator

$$\mathcal{S}(X) = AX + XB.$$

For the symmetric equation

$$AX + XA = C$$

with $A \succ 0$, uniqueness follows because all sums $\lambda_i(A) + \lambda_j(A)$ are positive. If also $C \succ 0$, the solution has the integral representation

$$\boxed{X = \int_0^\infty e^{-tA} C e^{-tA}\, dt.}$$

Indeed,

$$\frac{d}{dt}(e^{-tA} C e^{-tA}) = -Ae^{-tA} C e^{-tA} - e^{-tA} C e^{-tA} A.$$

Integrating from 0 to ∞ gives

$$AX + XA = C,$$

because $e^{-tA} \to 0$. Moreover, for $v \neq 0$,

$$v^T X v = \int_0^\infty (e^{-tA} v)^T C (e^{-tA} v)\, dt > 0.$$

Thus the unique solution is positive definite. The integral proves positivity without requiring A and C to commute.

§72.14 Lyapunov equations turn spectral decay into an energy function

Let a linear system satisfy

$$x'(t) = Mx(t).$$

Assume every eigenvalue of M has negative real part. For any $Q \succ 0$, define

$$P = \int_0^\infty e^{tM^T} Q e^{tM}\, dt.$$

The exponential decay makes the integral converge, and the same differentiation argument yields

$$M^T P + PM = -Q.$$

The matrix P is positive definite because

$$x^T P x = \int_0^\infty (e^{tM} x)^T Q (e^{tM} x)\, dt > 0$$

for $x \neq 0$.

Now set

$$V(x) = x^T P x.$$

Along a trajectory,

$$\begin{aligned}\frac{d}{dt}V(x(t)) &= x(t)^T(M^TP + PM)x(t)\\ &= -x(t)^TQx(t) < 0\end{aligned}$$

away from the origin. The solution of a matrix equation has become a decreasing energy that proves asymptotic stability.

This is where the chapter's themes meet. Similarity isolates coordinate-independent spectral data; symmetric functional calculus turns that data into matrix functions and quadratic geometry; Loewner order records positivity; and the Lyapunov integral converts spectral decay into a concrete positive quadratic form.

Two-Dimensional Linear Maps, Spectral Geometry, and Singular Value Decomposition

N-20260303

§73.1 Concept map of the chapter

The source note begins with a map of the main linear-algebra concepts used later in the machine-learning-oriented discussion. Determinants test invertibility and also appear before eigenvalues; eigenvalues determine eigenvectors; eigenvectors construct orthogonal matrices in favorable cases; orthogonal matrices are used in diagonalization; diagonalization and singular value decomposition then become the computational language for dimensionality reduction.

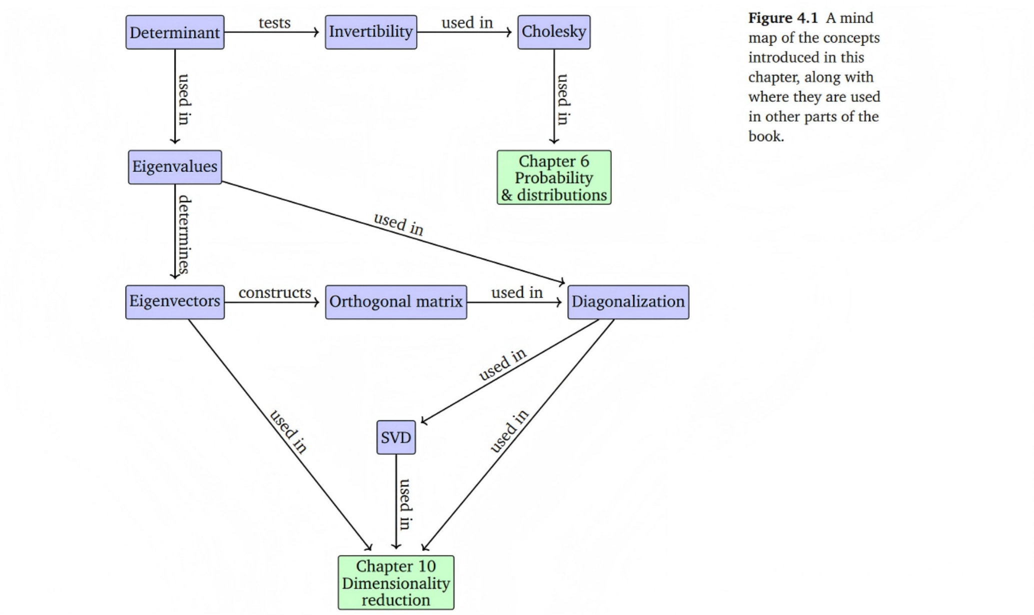


Figure 73.1: A concept map linking determinant, invertibility, eigenvalues, eigenvectors, orthogonal matrices, diagonalization, Cholesky factorization, SVD, and dimensionality reduction.

The diagram should not be read as a strict dependency graph. It is a guide to how these objects are reused: determinant and eigenvalues are algebraic invariants; eigenvectors and orthogonal matrices give coordinate systems; SVD is the robust replacement when diagonalization is unavailable or when the matrix is rectangular.

§73.2 Five model transformations

A 2×2 matrix is not only an array of four numbers; it is a rule that moves every point of the plane. The determinant records signed area scaling, while eigenvectors record directions that survive the transformation without turning:

$$|A(e_1), A(e_2)| = \det A, \qquad Av = \lambda v.$$

The source figure compares five concrete linear maps by applying each to a grid of colored points. The left column is the input, the right column is the transformed output, and the middle column marks eigenvector directions and eigenvalue data where they are real.

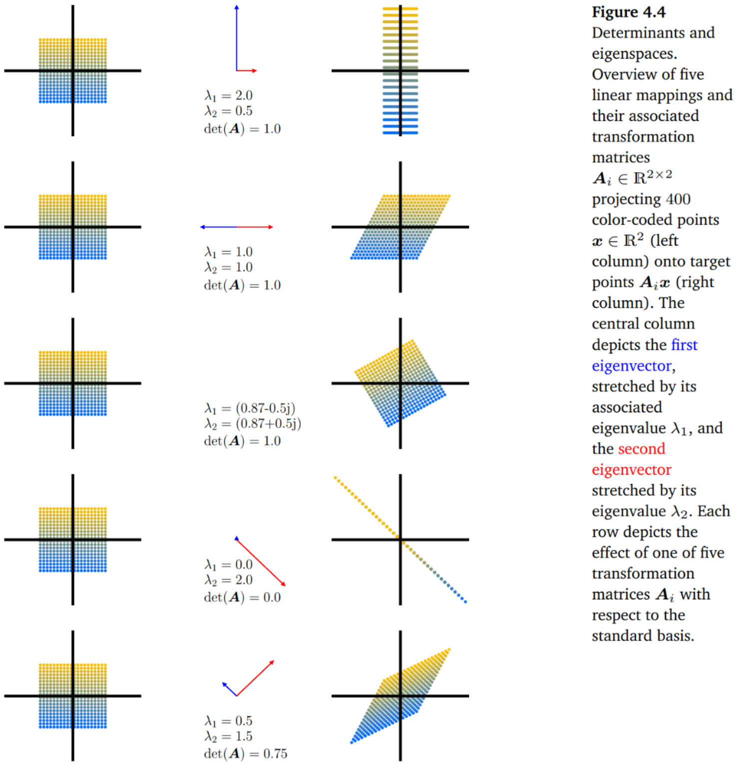


Figure 73.2: Five planar linear mappings and their eigenspace behavior. Each row shows colored input points, eigenvalue/eigenvector information, and the transformed output points.

The five rows illustrate the standard possibilities: axis-aligned stretching, shear, rotation with complex eigenvalues, collapse onto a line, and symmetric stretching. The determinant and the eigenvalues answer different questions. The determinant says how area changes; eigenvectors say which directions are preserved.

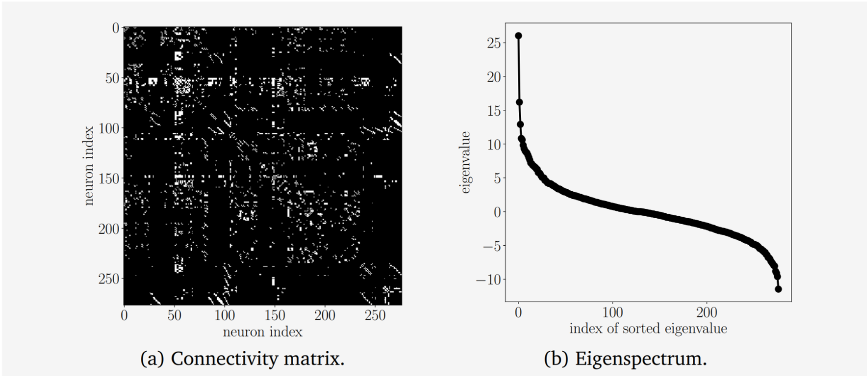
§73.3 A spectral example: a biological neural network

The note then moves from geometry to data. For the *C. elegans* neural network, one may form a connectivity matrix. Since the directed connection matrix is generally not symmetric, one often studies a symmetrized matrix, for example

$$A_{\text{sym}} = A + A^T.$$

The eigenvalues of this matrix describe dominant modes of connectivity.

Figure 4.5 Caenorhabditis elegans neural network (Kaiser and Hilgetag, 2006).(a) Symmetrized connectivity matrix; (b) Eigenspectrum.



(a) Connectivity matrix. (b) Eigenspectrum.

Figure 73.3: The symmetrized connectivity matrix of the C. elegans neural network and its sorted eigenspectrum.

This example is a useful warning: eigenvectors are not only arrows in the plane. In a network or a dataset, an eigenvector can indicate a coherent global mode, and the corresponding eigenvalue measures its strength.

§73.4 SVD as a sequence of geometric operations

Eigenvalue decomposition is powerful but fragile: it is best behaved for square matrices with enough eigenvectors, and especially for symmetric matrices. Singular value decomposition is more stable. Every real matrix $A \in \mathbb{R}^{m\times n}$ admits

$$A = U\Sigma V^T,$$

where U and V are orthogonal and Σ has nonnegative diagonal entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$. Geometrically, V^T changes the input coordinates, Σ stretches by the singular values, and U changes the output coordinates.

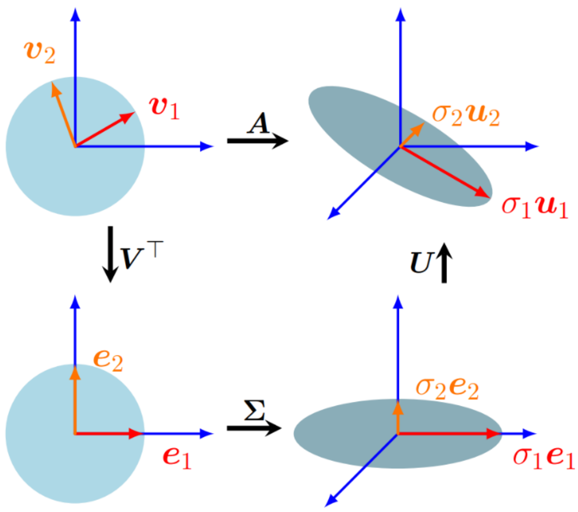


Figure 73.4: Geometric interpretation of SVD: the unit disk is first expressed in the right-singular-vector coordinates, then stretched by the singular values, and finally rotated into the output coordinates.

Thus SVD does not ask for directions that remain invariant under A. Instead, it asks for orthogonal input directions that are sent to orthogonal output axes. The unit circle becomes an ellipse, and the semi-axis lengths are the singular values.

§73.5 SVD and latent structure in data

A rating table can be viewed as a matrix. Rows may represent movies and columns may represent users. The raw entries are ratings, but the singular vectors can reveal latent directions such as genre preference or taste profile.

$$\begin{array}{rc} & \begin{array}{ccc}\text{Ali} & \text{Beatrix} & \text{Chandra}\end{array} \\ \begin{array}{r}\text{Star Wars}\\ \text{Blade Runner}\\ \text{Amelie}\\ \text{Delicatessen}\end{array} & \begin{bmatrix}5&4&1\\5&5&0\\0&0&5\\1&0&4\end{bmatrix}\end{array} = \begin{bmatrix}-0.6710 & 0.0236 & 0.4647 & -0.5774\\ -0.7197 & 0.2054 & -0.4759 & 0.4619\\ -0.0939 & -0.7705 & -0.5268 & -0.3464\\ -0.1515 & -0.6030 & 0.5293 & -0.5774\end{bmatrix}$$

$$\begin{bmatrix}9.6438 & 0 & 0\\ 0 & 6.3639 & 0\\ 0 & 0 & 0.7056\\ 0 & 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}-0.7367 & -0.6515 & -0.1811\\ 0.0852 & 0.1762 & -0.9807\\ 0.6708 & -0.7379 & -0.0743\end{bmatrix}$$

Figure 4.10 Movie ratings of three people for four movies and its SVD decomposition.

Figure 73.5: A movie-rating matrix and its SVD decomposition. Dominant singular directions highlight latent structure in user and movie preferences.

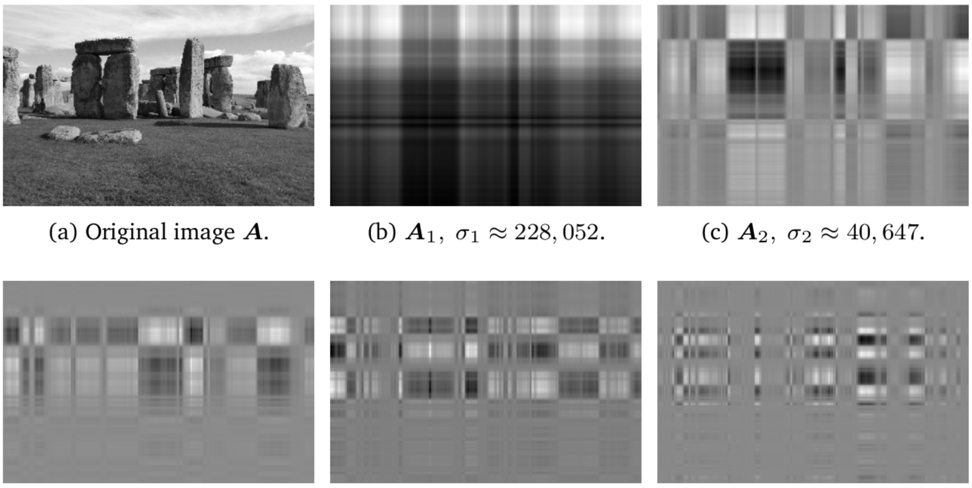
The labels placed on singular directions are interpretations, not part of the theorem. The theorem says that the first singular directions give the most efficient orthogonal explanation of the matrix energy.

§73.6 Rank-one blocks and image compression

SVD decomposes a matrix into rank-one pieces:

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

For an image matrix, each term $\sigma_i u_i v_i^T$ is a weighted rank-one visual pattern.



(a) Original image $\boldsymbol{A}$. (b) $\boldsymbol{A}_1,\ \sigma_1 \approx 228,052$. (c) $\boldsymbol{A}_2,\ \sigma_2 \approx 40,647$.

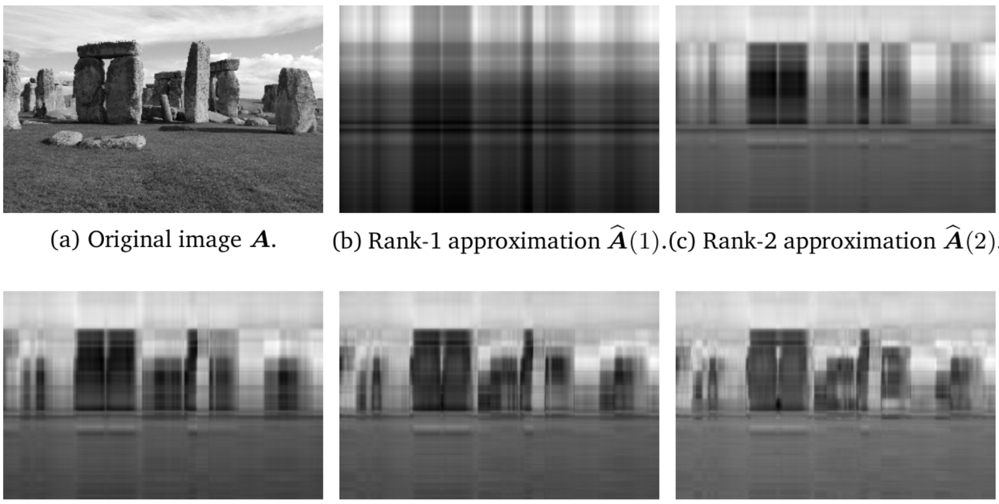
(d) $\boldsymbol{A}_3,\ \sigma_3 \approx 26,125$. (e) $\boldsymbol{A}_4,\ \sigma_4 \approx 20,232$. (f) $\boldsymbol{A}_5,\ \sigma_5 \approx 15,436$.

Figure 73.6: The first rank-one SVD components of an image. Each component captures one weighted visual pattern rather than a recognizable full image.

Truncated SVD keeps only the first k terms:

$$\widehat{A}(k) = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

The Eckart–Young theorem says that this is the best rank-k approximation in the spectral norm and the Frobenius norm.



(a) Original image $\boldsymbol{A}$. (b) Rank-1 approximation $\widehat{\boldsymbol{A}}(1)$. (c) Rank-2 approximation $\widehat{\boldsymbol{A}}(2)$.

(d) Rank-3 approximation $\widehat{\boldsymbol{A}}(3)$. (e) Rank-4 approximation $\widehat{\boldsymbol{A}}(4)$. (f) Rank-5 approximation $\widehat{\boldsymbol{A}}(5)$.

Figure 73.7: Low-rank image reconstruction from the first few singular components. As the rank increases, the approximation recovers more recognizable structure.

This example explains why SVD is useful beyond pure linear algebra: it gives a principled compression method by separating large-scale structure from smaller-scale details.

§73.7 A taxonomy of matrix properties

The final source figure is a taxonomy of real matrices. It distinguishes square from nonsquare matrices, singular from regular matrices, defective from nondefective matrices, normal from non-normal matrices, and the special positions of symmetric, positive definite, orthogonal, diagonal, and identity matrices.

Figure 4.13 A functional phylogeny of matrices encountered in machine learning.

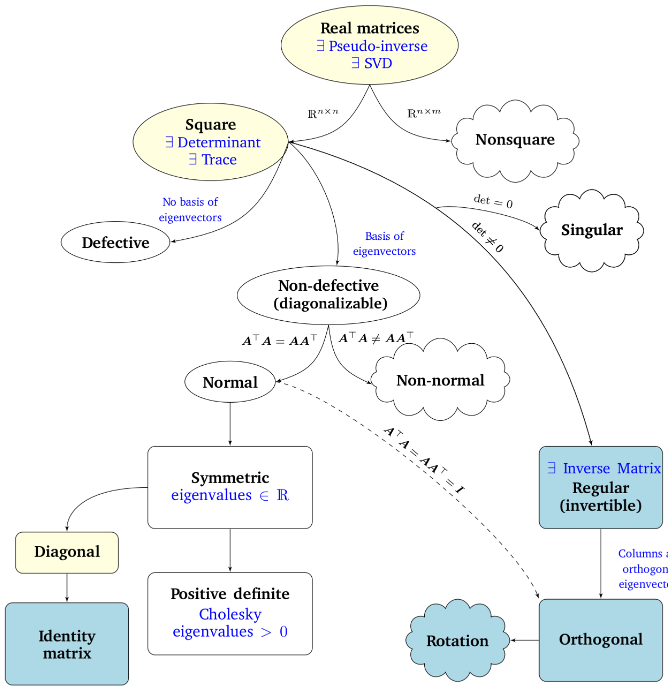


Figure 73.8: A functional phylogeny of real matrices encountered in machine learning, indicating which tools apply to which matrix classes.

The diagram prevents a common error: applying the wrong theorem to the wrong class. Determinants and eigenvalues belong naturally to square matrices; orthogonal diagonalization belongs to symmetric real matrices; pseudoinverses and SVD work for arbitrary real matrices.

§73.8 Choose the invariant appropriate to the matrix question

The unifying theme is that a matrix should be read through the invariant appropriate to the question. The determinant measures signed volume scaling; the trace and eigenvalues measure intrinsic square-matrix behavior; rank measures the dimension of the image; singular values measure metric distortion; and the pseudoinverse measures least-squares inverse behavior when an exact inverse is unavailable.

Similarity

$$A' = P^{-1}AP$$

preserves the linear operator but changes coordinates on the same space. A general matrix representation

$$\widetilde{A} = T^{-1}AS$$

may change both input and output coordinates. SVD chooses these input and output coordinates orthogonally, which is why it is numerically stable.

For the examples in this note, the practical reading rule is:

question	tool
area/orientation	$\det A$
invariant directions	eigenvectors
collapse or information loss	$\operatorname{rank} A$
principal stretching	singular values
best low-rank compression	truncated SVD

This is why the images are not decorative. They show the same matrix simultaneously as an algebraic object, a geometric transformation, and a data-processing device.

Orthogonal Projection, Least Squares, and the Geometry of Approximation

N-20260305

§74.1 The nearest point is found by a right angle

Suppose a vector y does not lie in a subspace W. There are infinitely many vectors in W, so what could make one of them the best approximation to y?

The answer is visible before it is algebraic. If $p \in W$ is the closest point, then the error

$$r = y - p$$

must be perpendicular to every direction in W. Otherwise one could move a little inside W in a direction having positive inner product with r, shortening the distance.

This gives the projection principle:

$$p = \operatorname{proj}_W y \qquad \Longleftrightarrow \qquad p \in W, \quad y - p \perp W.$$

The right angle is not a decorative consequence. It is the condition that singles out the nearest point.

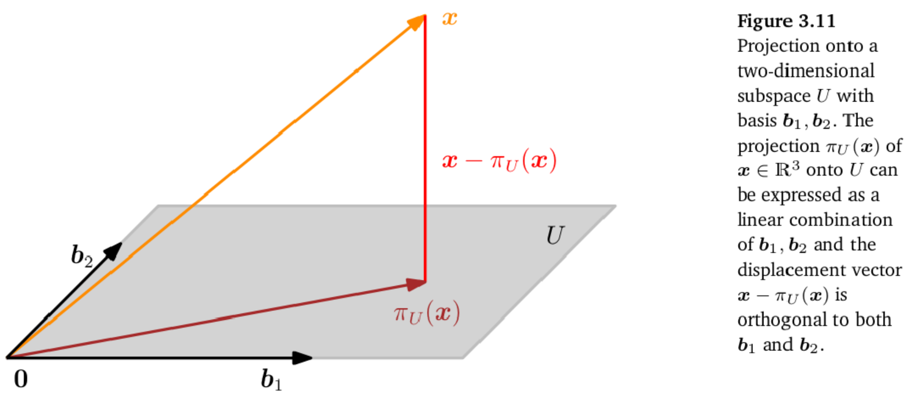


Figure 74.1: The best approximation is characterized by an orthogonal residual. Moving within the subspace cannot reduce the remaining error.

§74.2 Projection onto one direction

Let $u \neq 0$. A candidate point on the line $\operatorname{span}\{u\}$ has the form $p = \alpha u$. Orthogonality of the error says

$$\langle y - \alpha u, u \rangle = 0.$$

Hence

$$\alpha = \frac{\langle y, u \rangle}{\langle u, u \rangle}, \qquad \operatorname{proj}_u y = \frac{\langle y, u \rangle}{\langle u, u \rangle} u.$$

If u is a unit vector, this simplifies to

$$\operatorname{proj}_u y = \langle y, u \rangle u.$$

Take

$$y = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then

$$\langle y, u \rangle = 5, \qquad \langle u, u \rangle = 5,$$

so

$$p = u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad r = y - p = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Indeed,

$$r^T u = 2 - 2 = 0.$$

The decomposition

$$y = p + r$$

separates the part visible inside the line from the part the line cannot represent.

§74.3 A basis turns the geometric condition into equations

Let the columns of

$$B = \begin{bmatrix} b_1 & \cdots & b_m \end{bmatrix} \in \mathbb{R}^{n \times m}$$

form a basis of W. Every candidate approximation is Bc for some coefficient vector c. The residual condition

$$y - Bc \perp W$$

means that it is orthogonal to each basis vector. In matrix form,

$$B^T (y - Bc) = 0.$$

Therefore

$$B^T B c = B^T y.$$

If the columns of B are independent, the Gram matrix $B^T B$ is positive definite and invertible, giving

$$c = (B^T B)^{-1} B^T y$$

and

$$\boxed{\operatorname{proj}_W y = B(B^T B)^{-1} B^T y.}$$

This formula looks more complicated than the one-dimensional version only because the basis vectors need not be orthogonal. The Gram matrix records their mutual angles and corrects for the fact that the coordinate directions interfere with one another.

§74.4 The projection matrix remembers the subspace, not the basis

Define
$$P = B(B^T B)^{-1} B^T.$$
Then
$$Py = \operatorname{proj}_W y.$$
Two identities reveal its geometry:
$$P^T = P, \qquad P^2 = P.$$

Symmetry says the projection is orthogonal. Idempotence says that once a vector has been projected, projecting again does nothing.

Although the formula uses a basis, the matrix depends only on W. If $C = BR$ is another basis matrix with R invertible, then
$$C(C^T C)^{-1} C^T = BR(R^T B^T BR)^{-1} R^T B^T = B(B^T B)^{-1} B^T.$$
Coordinates changed, but the geometric operation did not.

There is also a complementary projection:
$$I - P.$$
It sends y to the residual $r = y - Py$, which lies in $W^\perp$. Thus
$$\mathbb{R}^n = W \oplus W^\perp, \qquad y = Py + (I - P)y.$$

§74.5 Least squares is projection written as a fitting problem

Now consider a system
$$Ax \approx b$$
with more equations than unknowns. Exact equality may be impossible because $b \notin R(A)$, the column space of A. The least-squares problem asks for
$$\min_x \|Ax - b\|_2.$$

But every vector Ax lies in $R(A)$. Therefore we are simply looking for the point in the column space closest to b.

At the minimizer $\widehat{x}$, the residual
$$r = b - A\widehat{x}$$
must be orthogonal to every column of A. Hence
$$A^T r = 0,$$
which gives the normal equations
$$\boxed{A^T A\widehat{x} = A^T b.}$$

The phrase “normal equations” comes from this normality: the residual is normal, or perpendicular, to the allowable model space.

§74.6 A complete line-fitting example

Suppose three measured points are

$$(0, 1), \qquad (1, 2), \qquad (2, 2).$$

We fit a line

$$y \approx \alpha + \beta x.$$

The design matrix and observation vector are

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives

$$\widehat{\alpha} = \frac{7}{6}, \qquad \widehat{\beta} = \frac{1}{2}.$$

Thus the fitted line is

$$y = \frac{7}{6} + \frac{1}{2}x.$$

The predicted data are

$$A\widehat{x} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix},$$

and the residual is

$$r = b - A\widehat{x} = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}.$$

A direct check gives

$$A^T r = \begin{bmatrix} -1/6 + 1/3 - 1/6 \\ 0 + 1/3 - 2/6 \end{bmatrix} = 0.$$

The fit is not exact, but no infinitesimal change in either the intercept or slope can improve the squared error to first order.

§74.7 The Pythagorean proof of optimality

Let $p = Py$ be the orthogonal projection of y onto W. For any other $w \in W$, write

$$y - w = (y - p) + (p - w).$$

The first term lies in $W^\perp$, while the second lies in W. Therefore they are orthogonal, and Pythagoras gives

$$\|y - w\|^2 = \|y - p\|^2 + \|p - w\|^2.$$

Hence

$$\|y - w\| \geq \|y - p\|,$$

with equality only when $w = p$.

This proof explains both existence and uniqueness in finite dimensions. It also shows why the residual condition is stronger than a derivative calculation: the whole excess error is exactly the squared distance from w to the projection.

For least squares, the same identity becomes

$$\|b - Ax\|^2 = \|b - A\widehat{x}\|^2 + \|A(x - \widehat{x})\|^2.$$

Every competing parameter vector pays the irreducible residual plus an additional non-negative model-space error.

§74.8 Orthonormal coordinates remove the Gram matrix

If the columns of Q are orthonormal, then

$$Q^T Q = I$$

and projection simplifies to

$$P = QQ^T.$$

The coefficients are simply inner products:

$$c = Q^T y.$$

This is one reason orthonormal bases are so useful: geometry and coordinates agree without a correction matrix.

For a full-column-rank matrix A, a QR factorization writes

$$A = QR,$$

where $Q^T Q = I$ and R is upper triangular. Then

$$\|Ax - b\|_2 = \|QRx - b\|_2.$$

Multiplying by an orthogonal completion of Q separates the reachable and unreachable parts of b, and the least-squares solution is obtained from

$$Rx = Q^T b.$$

This is computationally preferable to explicitly forming $A^T A$. The condition number is squared by the normal equations:

$$\kappa_2(A^T A) = \kappa_2(A)^2.$$

Thus the normal equations are conceptually illuminating but can be numerically fragile. QR preserves the projection geometry without amplifying the conditioning in this way.

§74.9 Rank deficiency changes uniqueness, not the geometry

If the columns of A are dependent, the fitted vector $A\widehat{x}$ can still be unique even though the coefficient vector is not. Indeed, all least-squares minimizers produce the same projection

$$A\widehat{x} = Pb,$$

but two parameter vectors may differ by an element of the null space:

$$x_1 - x_2 \in N(A).$$

This distinction is easy to miss in applications. The model prediction can be completely determined while the internal coefficients remain ambiguous. Collinear features in regression are a familiar example.

The Moore–Penrose inverse, studied later in more detail, chooses the minimizer orthogonal to $N(A)$. It is therefore the least-squares solution with minimum Euclidean norm:

$$\widehat{x} = A^+ b.$$

Here the pseudoinverse is a canonical coordinate choice, while the geometric heart of the problem remains projection onto $R(A)$.

§74.10 Weighted least squares changes the notion of perpendicular

Sometimes residual coordinates should not be treated equally. If W is symmetric positive definite, define

$$\langle u, v\rangle_W = u^T W v, \qquad \|u\|_W^2 = u^T W u.$$

Weighted least squares minimizes

$$\|Ax - b\|_W^2 = (Ax - b)^T W (Ax - b).$$

Differentiation or weighted orthogonality gives

$$A^T W (A\widehat{x} - b) = 0,$$

so

$$A^T W A \widehat{x} = A^T W b.$$

The algebra resembles ordinary least squares, but the geometry has changed. The residual is no longer Euclidean-orthogonal to the column space; it is orthogonal in the W-inner product. Equivalently, after writing $W = C^T C$, one solves the ordinary least-squares problem

$$\min_x \|CAx - Cb\|_2.$$

This is how measurement confidence, covariance information, and anisotropic geometry enter the same projection framework.

§74.11 Projection beyond finite-dimensional matrices

The finite-dimensional picture survives in Hilbert spaces. If H is a Hilbert space and $M \subseteq H$ is a closed subspace, then every $y \in H$ has a unique decomposition

$$y = p + r, \qquad p \in M, \quad r \in M^\perp.$$

The point p is the unique best approximation to y from M.

The word "closed" matters. Without it, an infimum distance may exist without being attained inside the subspace. Completeness and closedness are the infinite-dimensional replacements for the compact geometric intuition that works automatically in finite dimensions.

Fourier series are a central example. Truncating a Fourier expansion is orthogonal projection onto a finite-dimensional space of trigonometric modes. Polynomial approximation, finite elements, and Galerkin methods use the same principle with different trial spaces. The matrix normal equations are therefore one coordinate shadow of a much broader approximation theorem.

§74.12 What should remain after the formulas

Projection and least squares are one story. A model can reproduce only vectors in its column space. The best approximation is obtained by dropping a perpendicular to that space. A basis converts the perpendicularity condition into normal equations; an orthonormal basis makes the coordinates transparent; QR turns that geometry into a stable algorithm. Rank deficiency separates uniqueness of predictions from uniqueness of parameters, and weighted inner products alter what "nearest" means without changing the logical structure.

The enduring test is simple:

$$\text{best approximation} \quad \Longleftrightarrow \quad \text{residual orthogonal to every allowed correction.}$$

That sentence is the bridge from a diagram with a right angle to regression, Fourier analysis, numerical PDEs, and Hilbert-space approximation.

Matrix Computation I: Norms, Amplification, and Numerical Size

N-20260311

§75.1 A matrix can be small entrywise and large as an operator

A matrix is not merely a rectangular table of numbers. It is a map

$$x \longmapsto Ax,$$

and numerical analysis cares about what that map can do to perturbations.

Suppose the input is changed from x to $x + \delta x$. The output changes by

$$A(x + \delta x) - Ax = A\delta x.$$

To say anything useful, we need a notion of size for both δx and $A\delta x$. A vector norm measures the size of data; an operator norm measures the largest factor by which a matrix can enlarge that size.

This distinction matters. The matrix

$$A = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$$

has entries no larger than 1, yet in the all-ones direction it multiplies Euclidean length by n. Looking at the largest entry alone misses the collective action of the columns.

§75.2 Different norms ask different questions

For $x = (x_1, \ldots, x_n)^T$, the standard norms are

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \qquad \|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2},$$

$$\|x\|_\infty = \max_i |x_i|.$$

More generally,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}, \qquad 1 \le p < \infty.$$

These norms emphasize different features. The 1-norm measures total mass. The 2-norm is rotationally invariant and fits inner-product geometry. The ∞-norm records the worst coordinate. For

$$x = (1, 1, \ldots, 1)^T,$$

we have

$$\|x\|_1 = n, \qquad \|x\|_2 = \sqrt{n}, \qquad \|x\|_\infty = 1.$$

None of these answers is more correct in isolation. They answer different questions about what counts as a large perturbation.

In finite dimensions all norms are equivalent: for any two norms $\|\cdot\|_a$ and $\|\cdot\|_b$, there exist constants $c, C > 0$ such that

$$c\|x\|_a \le \|x\|_b \le C\|x\|_a.$$

Thus they define the same convergent sequences, but the constants may depend strongly on dimension. Numerical estimates therefore do depend on the chosen norm even when qualitative convergence does not.

§75.3 Hölder duality explains the familiar inequalities

Let $1 \le p, q \le \infty$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Hölder's inequality states

$$|y^T x| \le \|y\|_q \|x\|_p.$$

The exponent q is called the conjugate exponent of p.

This inequality says that a linear functional $x \mapsto y^T x$ has size $\|y\|_q$ when inputs are measured in p-norm. More precisely,

$$\max_{\|x\|_p=1} |y^T x| = \|y\|_q.$$

For $1 < p < \infty$, equality is achieved by choosing the phases of x_i to match y_i and setting

$$|x_i| \propto |y_i|^{q-1}.$$

For $p = 1$, the maximizing vector concentrates on a coordinate where $|y_i|$ is largest; hence the dual of ℓ^1 is ℓ^∞. For $p = \infty$, choose each coordinate with maximal allowed magnitude and matching sign; the dual is ℓ^1.

The Cauchy–Schwarz inequality is simply the case $p = q = 2$:

$$|y^T x| \le \|y\|_2 \|x\|_2.$$

This dual viewpoint is useful because matrix norms can be read as collections of linear functionals acting on columns or rows.

§75.4 An induced norm measures worst-case amplification

Given norms on the input and output spaces, define

$$\|A\| = \max_{x \ne 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|.$$

This is the induced, or subordinate, operator norm. By definition,

$$\|Ax\| \le \|A\| \, \|x\|.$$

It is the smallest constant for which this inequality holds for every x.

Induced norms are automatically submultiplicative:

$$\|AB\| = \max_{x\neq 0} \frac{\|ABx\|}{\|x\|} \le \max_{x\neq 0} \frac{\|A\|\,\|Bx\|}{\|x\|} \le \|A\|\|B\|.$$

Thus a product of linear maps cannot amplify more than the product of their individual worst-case factors.

The standard formulas are

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \qquad \|A\|_\infty = \max_i \sum_j |a_{ij}|.$$

The first is the largest absolute column sum; the second is the largest absolute row sum.

To see the 1-norm formula, write

$$\|Ax\|_1 \le \sum_i \sum_j |a_{ij}|\,|x_j| = \sum_j \left(\sum_i |a_{ij}|\right) |x_j|.$$

This is bounded by

$$\left(\max_j \sum_i |a_{ij}|\right) \|x\|_1.$$

Equality is obtained by putting all mass on a column with maximal sum and choosing the sign or phase appropriately.

§75.5 A worked comparison of three operator norms

Let

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 2 \end{bmatrix}.$$

The absolute column sums are

$$3, \qquad 4,$$

so

$$\|A\|_1 = 4.$$

The absolute row sums are

$$3, \qquad 4,$$

so

$$\|A\|_\infty = 4.$$

For the Euclidean norm, compute

$$A^T A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_\pm = \frac{13 \pm 5}{2} = 9, 4.$$

Therefore

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = 3.$$

The three values differ because they optimize over different unit balls. The 1- and ∞-balls have corners aligned with coordinate directions; the Euclidean ball is rotationally symmetric.

The right singular vector corresponding to singular value 3 is a direction in which A stretches Euclidean length exactly by a factor of 3. Other input directions stretch less.

§75.6 The spectral norm comes from singular values

The Euclidean operator norm satisfies

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2.$$

Squaring gives

$$\|Ax\|_2^2 = x^T A^T Ax.$$

The maximum of this Rayleigh quotient over the unit sphere is the largest eigenvalue of $A^T A$. Hence

$$\boxed{\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} = \sigma_{\max}(A).}$$

This formula explains why singular values, rather than eigenvalues, govern Euclidean amplification. Eigenvalues describe invariant directions satisfying $Ax = \lambda x$. Singular vectors solve a different optimization problem: which input direction produces the longest output?

For a nonnormal matrix these questions can diverge dramatically. Consider

$$N_M = \begin{bmatrix} 0 & M \\ 0 & 0 \end{bmatrix}.$$

Both eigenvalues are zero, so the spectral radius is zero, yet

$$\|N_M\|_2 = M.$$

The matrix is nilpotent, but it can cause a large one-step transient amplification. This is why eigenvalues alone are not a complete measure of numerical size.

§75.7 Unitary transformations preserve Euclidean geometry

If $U^T U = I$, then

$$\|Ux\|_2^2 = x^T U^T Ux = \|x\|_2^2.$$

Thus orthogonal or unitary matrices are changes of orthonormal coordinates. They rotate or reflect without changing length.

Consequently,

$$\|UAV\|_2 = \|A\|_2$$

for unitary U, V. Singular value decomposition writes

$$A = U\Sigma V^T,$$

so the Euclidean operator norm is simply the largest diagonal entry of Σ.

The Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2}$$

is also unitarily invariant. Since

$$\|A\|_F^2 = \operatorname{tr}(A^T A),$$

we obtain

$$\|UAV\|_F^2 = \operatorname{tr}(V^T A^T U^T UAV) = \operatorname{tr}(V^T A^T AV) = \operatorname{tr}(A^T A).$$

Equivalently,

$$\|A\|_F^2 = \sum_i \sigma_i(A)^2.$$

The spectral norm asks for the single strongest direction. The Frobenius norm combines energy across all singular directions. They satisfy

$$\|A\|_2 \le \|A\|_F \le \sqrt{r}\,\|A\|_2,$$

where $r = \operatorname{rank} A$.

§75.8 Schatten norms place spectral and Frobenius size in one family

For $1 \le p < \infty$, define the Schatten p-norm by applying a vector p-norm to the singular values:

$$\|A\|_{S_p} = \left(\sum_i \sigma_i(A)^p\right)^{1/p}.$$

Then

$$\|A\|_{S_2} = \|A\|_F, \qquad \|A\|_{S_\infty} = \|A\|_2.$$

The case $p = 1$ is the nuclear norm,

$$\|A\|_* = \sum_i \sigma_i(A).$$

It is used as a convex surrogate for rank because it penalizes the total singular-value mass.

These norms separate two notions that are often conflated. Entrywise sparsity concerns how many coordinates are nonzero. Low rank concerns how many singular directions are active. The 1-norm of entries encourages the former; the nuclear norm encourages the latter.

§75.9 Norms turn perturbation statements into inequalities

If x is perturbed by δx, then

$$\|A(x + \delta x) - Ax\| \le \|A\|\,\|\delta x\|.$$

This is the basic Lipschitz estimate for a linear map.

For a product,

$$F(x) = W_L W_{L-1} \cdots W_1 x,$$

we have

$$\|F(x + \delta x) - F(x)\|_2 \le \left(\prod_{\ell=1}^{L} \|W_\ell\|_2\right) \|\delta x\|_2.$$

The bound can be pessimistic because the maximally amplified direction of one layer may not align with that of the next. Still, it gives a rigorous worst-case control and explains why spectral normalization constrains the Lipschitz constant of a network layer.

Norms also bound truncation and rounding errors. If a computed matrix is $A + E$, then

$$\|(A + E)x - Ax\| \le \|E\|\,\|x\|.$$

The norm of the perturbation is meaningful only relative to the chosen geometry. A small Frobenius perturbation controls average singular energy, while a small spectral perturbation controls every Euclidean input direction.

§75.10 The distance to singularity appears in the smallest singular value

For an invertible square matrix,

$$\sigma_{\min}(A) = \min_{\|x\|_2=1} \|Ax\|_2.$$

This is the least amount by which A stretches any unit direction. If it is small, there is a direction that is almost collapsed.

The Euclidean distance from A to the set of singular matrices is exactly

$$\sigma_{\min}(A)$$

when distance is measured by the spectral norm. Indeed, if u_n, v_n are singular vectors for $\sigma_{\min}$, then

$$E = -\sigma_{\min} u_n v_n^T$$

has norm $\sigma_{\min}$ and makes

$$(A+E)v_n = 0.$$

No smaller perturbation can make the matrix singular, because

$$\|(A+E)x\|_2 \geq \|Ax\|_2 - \|E\|_2\|x\|_2 \geq \sigma_{\min}(A) - \|E\|_2$$

for unit x.

Thus the smallest singular value is not merely one more number in an SVD table. It measures how close invertibility is to failure.

§75.11 A norm is useful only when its role is stated

Several quantities called "matrix size" coexist because they serve different purposes.

The maximum entry answers a local coordinate question. The Frobenius norm records total entrywise or singular-value energy. An induced norm controls worst-case amplification for a chosen notion of vector size. The spectral norm is the Euclidean Lipschitz constant. The nuclear norm measures total singular mass and is sensitive to low-rank structure. The smallest singular value measures the weakest direction and the distance to singularity.

The practical habit should therefore be to complete the sentence:

"This norm is being used to control *what*?"

Without that clause, a numerical bound may be formally correct but conceptually irrelevant. With it, norms become the language that connects geometry, perturbations, algorithms, and stability.

Matrix Computation II: Spectral Radius, Stability, and Conditioning

N-20260312

§76.1 One step and many steps are different questions

A matrix norm asks how much a vector can be amplified in one application:

$$\|Ax\| \leq \|A\|\|x\|.$$

The spectral radius

$$\rho(A) = \max_{\lambda \in \sigma(A)} |\lambda|$$

asks about the exponential rate that remains after the matrix is applied many times.

These quantities agree for normal matrices in the Euclidean norm, but not in general. A nonnormal matrix can produce enormous short-time growth even when all its eigenvalues lie well inside the unit disk. Conversely, a large one-step norm does not by itself imply that powers grow forever.

This chapter separates four issues that are often compressed into the word "stability":

$$\begin{array}{rcl} \|A\| & : & \text{one-step amplification,} \\ \rho(A) & : & \text{asymptotic exponential growth,} \\ \kappa(A) & : & \text{sensitivity of solving or inversion,} \\ \text{backward stability} & : & \text{whether an algorithm solves a nearby problem.} \end{array}$$

A stable algorithm cannot make an ill-conditioned problem well-conditioned, and a well-conditioned problem can still be mishandled by an unstable algorithm.

§76.2 Spectral radius is not a norm

If $Ax = \lambda x$ with $x \neq 0$, then every induced norm satisfies

$$|\lambda|\|x\| = \|Ax\| \leq \|A\|\|x\|.$$

Hence

$$\rho(A) \leq \|A\|.$$

The inequality is fundamental, but ρ itself is not a matrix norm.

The nonzero nilpotent matrix

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

has

$$\rho(N) = 0.$$

Thus positive definiteness fails. Moreover, with

$$N_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad N_2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$$

we have

$$\rho(N_1) = \rho(N_2) = 0, \qquad \rho(N_1 + N_2) = 1.$$

So the triangle inequality fails as well.

The spectral radius is nevertheless the right asymptotic quantity. If A is diagonalizable,

$$A = S\Lambda S^{-1},$$

then

$$A^k = S\Lambda^k S^{-1}.$$

The eigenvalues determine the exponential factors λ_i^k, while the geometry of S determines how strongly the corresponding modes can interfere at finite times.

§76.3 Gelfand's formula extracts the exponential rate

For every matrix norm,

$$\boxed{\rho(A) = \lim_{k\to\infty} \|A^k\|^{1/k}.}$$

This is Gelfand's formula.

The lower bound is immediate from

$$\rho(A^k) = \rho(A)^k \le \|A^k\|,$$

which gives

$$\rho(A) \le \|A^k\|^{1/k}.$$

For the opposite direction, put A into Jordan form. On a Jordan block

$$J = \lambda I + N, \qquad N^m = 0,$$

we have

$$J^k = \sum_{j=0}^{m-1} \binom{k}{j} \lambda^{k-j} N^j.$$

The binomial factors grow only polynomially in k, while $|\lambda|^k$ carries the exponential rate. Taking the k-th root makes every fixed polynomial factor tend to 1. The largest $|\lambda|$ therefore survives, yielding the formula.

Gelfand's formula should not be confused with the power method. The power method follows a particular vector

$$x, Ax, A^2x, \ldots$$

and tries to isolate a dominant eigenvector. It can fail if the starting vector misses the dominant invariant direction, if dominant eigenvalues tie in modulus, or if nonnormal transient behavior obscures the asymptotic regime. Gelfand's formula is a statement about the norm of the whole operator power, not a promise about one trajectory.

§76.4 Jordan blocks explain the boundary

For a 2×2 Jordan block

$$J = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix},$$

direct calculation gives

$$J^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}.$$

Three regimes are visible.

If $|\lambda| < 1$, both λ^k and $k\lambda^{k-1}$ tend to zero. Exponential decay eventually defeats the polynomial factor. If $|\lambda| > 1$, powers grow exponentially. If $|\lambda| = 1$, the off-diagonal term grows like k, even though every eigenvalue has modulus one.

Consequently,

$$A^k \to 0 \quad \Longleftrightarrow \quad \rho(A) < 1.$$

The strict inequality matters. Eigenvalues on the unit circle are harmless only when the associated Jordan blocks have size one and the remaining geometry is controlled.

The same calculation explains why a matrix can look stable by eigenvalues and still exhibit large transient effects. Let

$$B = \begin{bmatrix} 0.9 & 10 \\ 0 & 0.9 \end{bmatrix}.$$

Then

$$B^k = \begin{bmatrix} 0.9^k & 10k\,0.9^{k-1} \\ 0 & 0.9^k \end{bmatrix}.$$

Compare it with the normal matrix $C = 0.9I$, which has the same eigenvalues.

k	$\|C^k\|_2$	lower bound for $\|B^k\|_2$
1	0.9000	10.0000
5	0.5905	32.8050
10	0.3487	38.7420
20	0.1216	27.0170
50	0.00515	2.8632

The lower bound comes from applying B^k to e_2. Both matrices converge to zero, but only one does so monotonically in operator norm. Eigenvalues describe the destination; nonnormality can make the route violently indirect.

§76.5 Neumann series turns repeated feedback into an inverse

If $\|A\| < 1$, then

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k.$$

Indeed,

$$(I - A)\sum_{k=0}^{m} A^k = I - A^{m+1},$$

and $A^{m+1} \to 0$.

The truncation error satisfies

$$(I - A)^{-1} - \sum_{k=0}^{m} A^k = A^{m+1}(I - A)^{-1}.$$

Therefore
$$\left\|(I-A)^{-1}-\sum_{k=0}^{m}A^k\right\| \le \frac{\|A\|^{m+1}}{1-\|A\|}.$$

Take
$$A=\begin{bmatrix}0.2 & 0.1\\ 0 & 0.2\end{bmatrix}.$$

Its induced ∞-norm is 0.3. The exact inverse is

$$(I-A)^{-1}=\begin{bmatrix}1.25 & 0.15625\\ 0 & 1.25\end{bmatrix}.$$

The cubic truncation is

$$I+A+A^2+A^3=\begin{bmatrix}1.248 & 0.152\\ 0 & 1.248\end{bmatrix}.$$

Its actual ∞-norm error is
$$0.00625,$$

while the general bound gives
$$\frac{0.3^4}{1-0.3}\approx 0.01157.$$

The bound is not exact, but it is guaranteed without diagonalizing the matrix.

The same idea defines the resolvent

$$R(z,A)=(zI-A)^{-1}.$$

Whenever $|z|>\|A\|$,
$$(zI-A)^{-1}=\frac{1}{z}\sum_{k=0}^{\infty}\left(\frac{A}{z}\right)^k.$$

The resolvent packages the behavior of all powers into one analytic matrix-valued function.

§76.6 Condition number is the geometry of inversion

For an invertible matrix,
$$\kappa(A)=\|A\|\|A^{-1}\|.$$

In the Euclidean norm,
$$\kappa_2(A)=\frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}.$$

The image of the unit sphere under A is an ellipsoid. The longest semiaxis has length $\sigma_{\max}$, and the shortest has length $\sigma_{\min}$. Inversion must expand the shortest output direction by $1/\sigma_{\min}$. A highly elongated ellipsoid therefore corresponds to a sensitive inverse problem.

Suppose only the right-hand side is perturbed:

$$Ax=b,\qquad A(x+\delta x)=b+\delta b.$$

Then
$$A\delta x=\delta b,\qquad \delta x=A^{-1}\delta b.$$

Using

$$\|b\| = \|Ax\| \le \|A\|\|x\|,$$

we obtain the relative error bound

$$\boxed{\frac{\|\delta x\|}{\|x\|} \le \kappa(A)\frac{\|\delta b\|}{\|b\|}.}$$

The condition number is a worst-case amplification factor. It does not say that every perturbation is amplified maximally; the dangerous perturbations align with the smallest singular directions.

§76.7 A tiny data change can reverse the solution

Let

$$A_\varepsilon = \begin{bmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 2+\varepsilon \end{bmatrix}.$$

The exact solution is

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Now perturb only the second observation by ε:

$$\delta b = \begin{bmatrix} 0 \\ \varepsilon \end{bmatrix}.$$

Since

$$A_\varepsilon^{-1} = \frac{1}{\varepsilon}\begin{bmatrix} 1+\varepsilon & -1 \\ -1 & 1 \end{bmatrix},$$

we get

$$\delta x = A_\varepsilon^{-1}\delta b = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

For $\varepsilon = 10^{-4}$, the relative data perturbation is about

$$3.54 \times 10^{-5},$$

while the relative solution perturbation is 1. The amplification is roughly 2.8×10^4.

The matrix is not singular; it is nearly singular. Its two columns are almost parallel, so the data barely distinguish the two parameters. No algorithm can recover information that the model geometry has nearly erased.

In this example,

$$\kappa_2(A_\varepsilon) \sim \frac{4}{\varepsilon}.$$

For $\varepsilon = 10^{-4}$, the condition number is about 4×10^4, consistent with the observed amplification.

§76.8 Forward error, residual, and backward error

Given an approximate solution $\widehat{x}$, the residual is

$$r = b - A\widehat{x}.$$

The forward error is

$$x - \widehat{x} = A^{-1}r.$$

Thus a small residual implies a small forward error only when A^{-1} is not too large.

Backward error asks a different question: what is the smallest perturbation of the data for which $\widehat{x}$ is exact? If only the right-hand side is perturbed, then

$$A\widehat{x} = b - r,$$

so $-r$ is exactly the required perturbation. A backward-stable algorithm produces the exact solution of a nearby problem.

The distinction can be summarized as follows:

$$\begin{array}{ccl} \text{small backward error} & \Rightarrow & \text{the algorithm behaved stably,} \\ \text{small condition number} & \Rightarrow & \text{nearby problems have nearby solutions,} \\ \text{both together} & \Rightarrow & \text{small forward error.} \end{array}$$

If the condition number is enormous, a backward-stable computation may still have a large forward error. That is not a contradiction; it is the correct response to an ill-conditioned problem.

When both A and b are perturbed, a standard first-order estimate is

$$\frac{\|\delta x\|}{\|x\|} \lesssim \kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|b\|}\right).$$

A rigorous finite-perturbation bound contains the denominator

$$1 - \kappa(A)\frac{\|\delta A\|}{\|A\|},$$

which must remain positive. The denominator records the possibility that the perturbed matrix crosses into singularity.

§76.9 Eigenvalues can be sensitive too

If A is normal, each eigenvalue of $A + E$ lies within $\|E\|_2$ of the spectrum of A. For a diagonalizable nonnormal matrix

$$A = S\Lambda S^{-1},$$

the Bauer–Fike theorem gives the weaker bound

$$\min_{\lambda \in \sigma(A)} |\mu - \lambda| \le \kappa_2(S)\|E\|_2$$

for every eigenvalue μ of $A + E$.

The factor $\kappa_2(S)$ measures how far the eigenvector basis is from orthonormal. Nearly parallel eigenvectors make eigenvalues highly sensitive.

Take

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 0.01 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

Then

$$A = S\Lambda S^{-1} = \begin{bmatrix} 1 & 100 \\ 0 & 2 \end{bmatrix},$$

and

$$\kappa_2(S) \approx 200.$$

Perturb by

$$E = \begin{bmatrix} 0 & 0 \\ 10^{-4} & 0 \end{bmatrix}.$$

The characteristic polynomial of $A + E$ is

$$(1-\mu)(2-\mu) - 0.01,$$

so the eigenvalues are approximately

$$0.9901, \qquad 2.0099.$$

A perturbation of norm 10^{-4} has moved the eigenvalues by about 10^{-2}. Bauer–Fike permits movement up to roughly

$$200 \times 10^{-4} = 0.02.$$

The sensitivity comes from the eigenvectors, not from the separation of the displayed diagonal entries alone.

§76.10 Pseudospectrum records nearby spectra

For $\varepsilon > 0$, the ε-pseudospectrum can be defined by

$$\sigma_\varepsilon(A) = \{z : \|(zI - A)^{-1}\|_2 > \varepsilon^{-1}\} \cup \sigma(A).$$

Equivalently, it is the set of eigenvalues of matrices $A + E$ with $\|E\|_2 < \varepsilon$.

For a normal matrix, the resolvent norm is exactly

$$\|(zI - A)^{-1}\|_2 = \frac{1}{\operatorname{dist}(z, \sigma(A))}.$$

Its pseudospectrum is simply a union of small disks around the eigenvalues. For a nonnormal matrix, the resolvent can be much larger.

Consider

$$B = \begin{bmatrix} 0.8 & 10 \\ 0 & 0.8 \end{bmatrix}.$$

For $z \neq 0.8$, write $d = z - 0.8$. Then

$$(zI - B)^{-1} = \begin{bmatrix} d^{-1} & 10d^{-2} \\ 0 & d^{-1} \end{bmatrix}.$$

Therefore

$$\|(zI - B)^{-1}\|_2 \geq \frac{10}{|d|^2}.$$

z	$\lvert z-0.8\rvert$	resolvent lower bound
1.0	0.2	250
1.2	0.4	62.5
1.6	0.8	15.625

With $\varepsilon = 0.02$, the threshold is $1/\varepsilon = 50$. The point $z = 1.2$ therefore lies in the 0.02-pseudospectrum, even though the only eigenvalue is 0.8. The pseudospectrum has expanded far beyond a radius-0.02 disk.

This expansion is the frequency-domain counterpart of transient growth. A large resolvent means that a small forcing or perturbation at a particular complex frequency can produce a large response. It also warns that computed eigenvalues of a nonnormal matrix may be dominated by tiny perturbations.

§76.11 A practical diagnostic order

When a computation involving powers, inverses, or eigenvalues looks unstable, several questions should be asked in order.

First, identify the quantity of interest. Powers A^k, solutions $A^{-1}b$, and eigenvalues have different condition theories. Next, inspect structure: normality, symmetry, positive definiteness, sparsity, and rank can radically change the relevant bounds. Then compare spectral radius with an operator norm. A large gap suggests nonnormal transient behavior. For solving, estimate $\kappa(A)$ and report a residual or backward error rather than relying on residual size alone. For eigenvalue sensitivity, inspect eigenvector conditioning or the resolvent; a pseudospectrum may be more informative than a list of eigenvalues.

The conceptual map is

$$\text{powers} \longleftrightarrow \rho(A) \text{ and Jordan structure,}$$

$$\text{inverse} \longleftrightarrow \sigma_{\min}(A) \text{ and } \kappa(A),$$

$$\text{nearby spectra} \longleftrightarrow \|(zI-A)^{-1}\| \text{ and pseudospectrum.}$$

The same matrix may be harmless for one question and difficult for another. Stability begins by deciding which question is actually being asked.

Matrix Computation III: Canonical Forms, Spectral Localization, and Eigenvalue Iteration

N-20260318

§77.1 Before computing an eigenvalue, decide what can change

A matrix changes when the basis changes. If

$$B = S^{-1}AS,$$

then the entries of B may look nothing like those of A, yet both matrices represent the same linear operator in different coordinates. A useful theory of eigenvalues therefore begins by separating coordinate accidents from invariant structure.

Three questions guide this chapter.

First, what algebraic data survive every change of basis? Canonical forms answer this. Second, before carrying out an expensive computation, where can the eigenvalues possibly lie? Gershgorin's theorem gives a cheap geometric answer. Third, once a spectral region or dominant mode has been identified, how can one actually approximate the corresponding eigenvector? Power and inverse iterations turn localization into computation.

These are different levels of information:

$$\text{canonical structure} \longrightarrow \text{spectral localization} \longrightarrow \text{numerical approximation}.$$

Trying to use one level as a substitute for another causes confusion. Jordan form explains exact structure but is usually a poor numerical algorithm. Gershgorin discs are robust and cheap but rarely exact. Iteration computes selected eigenpairs but only under separation and stability assumptions.

§77.2 The polynomial matrix $\lambda I - A$ contains more than the characteristic polynomial

The characteristic polynomial

$$\chi_A(\lambda) = \det(\lambda I - A)$$

records the eigenvalues with algebraic multiplicity. It does not, by itself, reveal how many Jordan blocks occur or how large they are. That missing information is already present in the whole polynomial matrix

$$\lambda I - A,$$

not merely in its determinant.

Work over the principal ideal domain $\mathbb{C}[\lambda]$. Row and column operations are allowed when they are invertible over this ring: exchange two rows or columns, add a polynomial multiple of one to another, or multiply by a nonzero constant. These operations produce polynomial matrices

$$U(\lambda)(\lambda I - A)V(\lambda)$$

with unimodular U, V, meaning their determinants are nonzero constants.

Every polynomial matrix of rank r can be reduced to Smith normal form

$$\operatorname{diag}\big(d_1(\lambda), \dots, d_r(\lambda), 0, \dots, 0\big),$$

where the nonzero polynomials are monic and satisfy

$$d_1 \mid d_2 \mid \cdots \mid d_r.$$

The polynomials d_i are the invariant factors.

If $D_k(\lambda)$ denotes the monic greatest common divisor of all nonzero $k \times k$ minors, then

$$d_1 = D_1, \qquad d_k = \frac{D_k}{D_{k-1}}.$$

The determinant is only the last determinantal divisor:

$$D_n = d_1 d_2 \cdots d_n = \chi_A(\lambda).$$

The smaller minors are what remember how the characteristic polynomial is distributed among cyclic pieces.

§77.3 Invariant factors are a module decomposition

Let $V = \mathbb{C}^n$, and let the indeterminate λ act on vectors by

$$\lambda \cdot v = Av.$$

This turns V into a $\mathbb{C}[\lambda]$-module. The structure theorem for finitely generated modules over a principal ideal domain gives

$$V \cong \bigoplus_i \mathbb{C}[\lambda]/(d_i(\lambda)),$$

where the nonconstant invariant factors are those of $\lambda I - A$.

This description explains why Smith form belongs naturally beside canonical matrix forms. A cyclic summand

$$\mathbb{C}[\lambda]/(d(\lambda))$$

is generated by one vector

$$v, Av, A^2 v, \dots,$$

subject to the relation $d(A)v = 0$. In a suitable cyclic basis, the matrix of A on this summand is the companion matrix of d. Taking the direct sum over invariant factors gives the rational canonical form.

The largest invariant factor is the minimal polynomial:

$$m_A(\lambda) = d_n(\lambda),$$

while their product is the characteristic polynomial. Thus

$$m_A \mid \chi_A.$$

The minimal polynomial records the strongest polynomial relation satisfied by the operator and controls every polynomial expression in A, including high powers and matrix functions.

§77.4 From invariant factors to Jordan blocks

Over $\mathbb{C}$, factor each invariant factor into powers of linear terms. The resulting powers

$$(\lambda - \mu)^s$$

are called elementary divisors. Each elementary divisor corresponds to one Jordan block

$$J_s(\mu).$$

For example, suppose the elementary divisors are

$$(\lambda - 2)^2, \qquad (\lambda - 2), \qquad (\lambda - 3).$$

Then the Jordan form is

$$J_2(2) \oplus J_1(2) \oplus J_1(3).$$

The characteristic polynomial is

$$(\lambda - 2)^3(\lambda - 3),$$

but the minimal polynomial is only

$$(\lambda - 2)^2(\lambda - 3).$$

The exponent 2 of $\lambda - 2$ in the minimal polynomial records the largest Jordan block for eigenvalue 2, not the total multiplicity of that eigenvalue.

A concrete polynomial-matrix example makes the mechanism visible. Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Then

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 3 \end{bmatrix}.$$

Because one entry is the unit -1, the gcd of the entries is $D_1 = 1$. Among the 2×2 minors appear both

$$(\lambda - 2)^2 \qquad \text{and} \qquad -(\lambda - 3),$$

whose gcd is 1. Hence $D_2 = 1$. Finally,

$$D_3 = (\lambda - 2)^2(\lambda - 3).$$

The invariant factors are therefore

$$1, \quad 1, \quad (\lambda - 2)^2(\lambda - 3),$$

and the elementary divisors are

$$(\lambda - 2)^2, \qquad (\lambda - 3).$$

They recover the block structure $J_2(2) \oplus J_1(3)$.

§77.5 Jordan chains turn algebraic multiplicity into dynamics

A Jordan chain for eigenvalue μ is a sequence

$$v_1, v_2, \dots, v_s$$

satisfying

$$(A - \mu I)v_1 = 0,$$
$$(A - \mu I)v_{j+1} = v_j.$$

The first vector is an eigenvector; the others are generalized eigenvectors. In this basis the operator acts by

$$Av_1 = \mu v_1, \qquad Av_{j+1} = \mu v_{j+1} + v_j.$$

Write a Jordan block as

$$J = \mu I + N, \qquad N^s = 0.$$

Then

$$J^k = \sum_{j=0}^{s-1} \binom{k}{j} \mu^{k-j} N^j.$$

The exponential term μ^k is decorated by polynomial factors in k. This is the precise origin of terms such as

$$k\mu^{k-1}, \qquad \binom{k}{2}\mu^{k-2}.$$

For

$$J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

we obtain

$$J^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Every eigenvalue equals 1, yet powers grow linearly. The Jordan chain carries dynamical information that the list of eigenvalues alone cannot express.

This also explains why Jordan form is mathematically revealing but numerically delicate. A tiny perturbation can split a multiple eigenvalue and destroy a large Jordan block. Computing exact block sizes from floating-point data is therefore ill-conditioned. Stable numerical algorithms usually prefer Schur form,

$$A = QTQ^*,$$

where Q is unitary and T is upper triangular. The diagonal of T gives the eigenvalues while unitary similarity avoids an ill-conditioned eigenvector basis.

§77.6 Gershgorin reads spectral information directly from rows

Canonical forms describe exact similarity invariants, but they are not a cheap first diagnostic. Gershgorin's theorem locates every eigenvalue using only row sums.

For $A = (a_{ij})$, define

$$R_i = \sum_{j \neq i} |a_{ij}|.$$

The i-th Gershgorin disc is

$$G_i = \{z \in \mathbb{C} : |z - a_{ii}| \le R_i\}.$$

Every eigenvalue lies in

$$\bigcup_i G_i.$$

The proof is a one-line maximum-coordinate argument. If

$$Ax = \lambda x$$

and $|x_i| = \max_j |x_j|$, then

$$(\lambda - a_{ii})x_i = \sum_{j \ne i} a_{ij} x_j.$$

Therefore

$$|\lambda - a_{ii}| \le \sum_{j \ne i} |a_{ij}| \frac{|x_j|}{|x_i|} \le R_i.$$

The chosen coordinate may depend on the eigenvector, so the theorem gives a union of discs rather than one fixed disc.

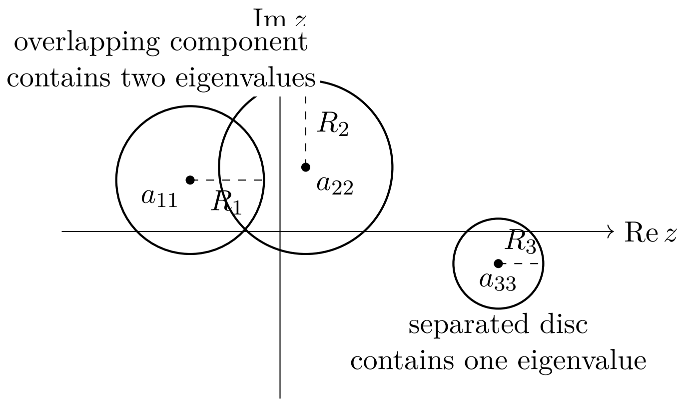


Figure 77.1: A separated component containing exactly m Gershgorin discs contains exactly m eigenvalues, counted with algebraic multiplicity.

§77.7 Separated discs count eigenvalues, not merely contain them

A stronger form of the theorem says that if a connected component of

$$\bigcup_i G_i$$

contains exactly m discs and is disjoint from the remaining discs, then it contains exactly m eigenvalues, counted with algebraic multiplicity.

The idea is deformation. Consider matrices

$$A(t) = D + t(A - D), \qquad 0 \le t \le 1,$$

where D is the diagonal part of A. At $t = 0$, every eigenvalue is a diagonal entry, and each disc has radius zero. As t increases, the discs expand continuously. If one cluster stays separated from the rest, eigenvalues cannot cross the gap between components. The number inside that component is therefore conserved.

For example, let

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

The first two discs are both the interval-like discs centered at 4 with radius 1, while the third is the point 9. Hence two eigenvalues lie in the component around 4, and one lies at 9. Direct calculation gives

$$\sigma(A) = \{3, 5, 9\},$$

exactly matching the count.

This separated-component principle is particularly useful for proving that a matrix is nonsingular. If every Gershgorin disc avoids the origin, then $0 \notin \sigma(A)$. Strict diagonal dominance is the familiar special case

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|.$$

§77.8 Diagonal scaling can reveal localization hidden by coordinates

Gershgorin discs depend on the chosen basis even though eigenvalues do not. This is not a defect; it is an opportunity. A diagonal similarity

$$B = D^{-1} A D$$

preserves the spectrum while changing off-diagonal magnitudes:

$$b_{ij} = d_i^{-1} a_{ij} d_j.$$

Consider

$$A = \begin{bmatrix} 1 & 100 \\ 0.01 & 2 \end{bmatrix}.$$

The row discs have radii 100 and 0.01, giving a very poor picture. Choose

$$D = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$D^{-1} A D = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Now both radii are 1. The eigenvalues have not moved, but the coordinates have been balanced so that the row equations display their coupling more honestly.

Choosing an optimal diagonal scaling is itself a nontrivial problem. The example nonetheless illustrates a broad numerical principle: poor coordinates can hide useful structure, and similarity transformations can expose it without changing the underlying operator.

§77.9 Power iteration extracts a dominant direction

Suppose A is diagonalizable with eigenpairs (λ_i, v_i), ordered so that

$$|\lambda_1| > |\lambda_2| \geq \cdots .$$

Write the initial vector as

$$x_0 = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n, \qquad c_1 \neq 0.$$

Then

$$A^k x_0 = \lambda_1^k \left(c_1 v_1 + \sum_{j \geq 2} c_j \left(\frac{\lambda_j}{\lambda_1} \right)^k v_j \right).$$

After normalization, the smaller ratios decay, so the direction approaches v_1. The asymptotic rate is governed by

$$\left| \frac{\lambda_2}{\lambda_1} \right| .$$

Take

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

with eigenvalues 3 and 1. Starting from

$$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} v_1 + \frac{1}{2} v_2,$$

we obtain

$$A^k x_0 = \frac{1}{2} \begin{bmatrix} 3^k + 1 \\ 3^k - 1 \end{bmatrix}.$$

The ratio of the two coordinates tends to 1, and the normalized vector approaches the dominant direction $(1,1)^T$.

A convenient eigenvalue estimate is the Rayleigh quotient

$$\rho(x) = \frac{x^* A x}{x^* x}.$$

For Hermitian A, if x_k approaches a normalized eigenvector, then $\rho(x_k)$ approaches its eigenvalue. The residual

$$r_k = A x_k - \rho(x_k) x_k$$

provides an a posteriori check: an exact eigenvector has zero residual.

§77.10 Power iteration has precise failure modes

The method is not guaranteed merely because a matrix has eigenvalues.

If the initial vector has no component in the dominant eigenspace, the dominant mode never appears. If two eigenvalues tie in modulus, the direction may fail to converge. For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

has two eigenvalues of equal modulus. Starting from a generic vector, the second coordinate changes sign at every step, so the normalized sequence oscillates.

Nonnormality creates another complication. Even when one eigenvalue is asymptotically dominant, an ill-conditioned eigenvector basis can produce large transient mixtures before the asymptotic regime becomes visible. A small residual is therefore more informative than a visually stable sequence of normalized coordinates.

Finally, normalization prevents overflow but does not repair loss of information. If roundoff repeatedly suppresses a weak component or if the dominant eigenvalue is poorly separated, convergence may stagnate.

§77.11 Inverse iteration turns a target into a dominant mode

Power iteration naturally finds the eigenvalue of largest modulus. To find an eigenvalue near a chosen shift μ, apply power iteration to

$$(A - \mu I)^{-1}.$$

Its eigenvalues are

$$\frac{1}{\lambda_j - \mu}.$$

The eigenvalue of A closest to μ becomes the dominant eigenvalue of the inverse operator.

The practical iteration is not to form the inverse. Instead, solve

$$(A - \mu I)y_{k+1} = x_k$$

and normalize

$$x_{k+1} = \frac{y_{k+1}}{\|y_{k+1}\|}.$$

If the shift stays fixed, factor $A - \mu I$ once and reuse the triangular solves. This makes shifted inverse iteration efficient when several steps are required.

The apparent paradox is that the method becomes stronger as μ approaches an eigenvalue, while the linear system becomes more ill-conditioned. The near singularity is exactly what amplifies the desired eigendirection. Stable pivoting, residual monitoring, and an appropriate stopping criterion are therefore essential.

§77.12 Rayleigh quotient iteration lets the shift move

A natural refinement updates the shift using the current Rayleigh quotient:

$$\mu_k = \frac{x_k^* A x_k}{x_k^* x_k},$$

then solves

$$(A - \mu_k I)y_{k+1} = x_k$$

and normalizes.

For Hermitian matrices and a sufficiently good initial vector, Rayleigh quotient iteration converges cubically. This remarkable speed comes from two facts: the Rayleigh quotient already approximates the eigenvalue to second order in the angular error, and inverse iteration strongly amplifies the nearest eigendirection.

Take again

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

If

$$x_0 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

then

$$\mu_0 = x_0^T A x_0 = \frac{14}{5} = 2.8,$$

already close to the dominant eigenvalue 3. Solving with $A - 2.8I$ strongly favors the $(1,1)^T$ direction, and the next Rayleigh quotient is dramatically more accurate.

The method should not be described as universally cubic. Symmetry or normality, a nearby simple eigenvalue, and a sufficiently accurate starting vector are important. For a general nonnormal matrix, convergence can be less regular and the moving linear systems can become difficult.

§77.13 Large matrices lead from vectors to Krylov subspaces

Power iteration uses the one-dimensional space spanned by the newest vector. A more informative approach retains the whole Krylov space

$$\mathcal{K}_k(A, x_0) = \operatorname{span}\{x_0, Ax_0, \dots, A^{k-1}x_0\}.$$

Instead of asking one vector to represent all spectral information, project A onto this growing subspace.

For Hermitian matrices, the Lanczos process constructs an orthonormal basis of the Krylov space and produces a small tridiagonal matrix. Its eigenvalues, called Ritz values, approximate selected eigenvalues of A. For general matrices, Arnoldi iteration produces an upper Hessenberg projection.

These methods are the large-scale descendants of power iteration. They retain the same polynomial idea—vectors in $\mathcal{K}_k$ have the form $p_{k-1}(A)x_0$—but optimize over many polynomials rather than repeatedly using only $p(t) = t^k$. This is why Krylov eigensolvers can resolve several eigenvalues and converge much faster than plain power iteration.

§77.14 The three viewpoints should remain distinct

Smith form and invariant factors answer what survives exact algebraic equivalence. Jordan chains explain generalized eigenvectors, polynomial factors in powers, and the minimal polynomial. Schur form is usually the numerically stable replacement when floating-point computation is required.

Gershgorin discs answer a cheaper question: where can the spectrum lie, and can separated clusters be counted before computing them? Diagonal scaling shows that localization may improve after a better choice of coordinates.

Power iteration, shifted inverse iteration, and Rayleigh quotient iteration answer a computational question: how can a selected eigendirection be amplified? Their success depends on spectral separation, a nonzero starting component, stable linear solves, and residual checks. Krylov methods generalize the same idea from one direction to an adaptively constructed subspace.

The natural workflow is therefore

$$\begin{array}{c}\text{understand invariant structure}\\ \downarrow\\ \text{localize the spectrum}\\ \downarrow\\ \text{choose and verify an iteration.}\end{array}$$

Canonical forms tell us what the operator is. Localization tells us where to look. Iteration tells us how to reach the part we want.

Matrix Decompositions: LU, Cholesky, Rank Factorization, and SVD

N-20260602

§78.1 One matrix, several kinds of hidden structure

A matrix is often presented as a rectangular array of numbers, but an algorithm rarely wants to work with that array in its original form. It would rather expose a structure that the matrix is hiding. A triangular matrix can be solved from one end to the other. A positive definite matrix carries an inner-product geometry. A low-rank matrix really passes through a smaller space. An arbitrary linear map stretches some orthogonal directions more than others.

These observations lead to four decompositions:

$$PA = LU, \qquad A = LL^*, \qquad A = FG, \qquad A = U\Sigma V^*.$$

They are not four versions of the same trick. Each answers a different question.

Factorization	Question it answers	Structure it exposes
$PA = LU$	How can elimination be reused?	successive triangular solves
$A = LL^*$	What extra benefit comes from positivity?	a Gram-type square root
$A = FG$	Through how many coordinates does the map really pass?	exact rank
$A = U\Sigma V^*$	Which orthogonal directions are stretched most?	singular directions and scales

The point of this chapter is therefore not to memorize four formulas. It is to watch four different pieces of linear algebra become algorithms.

§78.2 Elimination remembers its work

Consider a square system

$$Ax = b.$$

Gaussian elimination replaces rows of A until the matrix becomes upper triangular. Every elimination step is multiplication on the left by a lower triangular elementary matrix. If

$$E_{n-1} \cdots E_2 E_1 A = U,$$

then

$$A = E_1^{-1} E_2^{-1} \cdots E_{n-1}^{-1} U.$$

The product of the inverse elimination matrices is lower triangular. This is the origin of

$$A = LU.$$

The factor L is therefore a record of what elimination did. Its entries below the diagonal are the multipliers used to cancel entries of A; U is the echelon-shaped matrix left behind.

Proposition 78.2.1 (Existence without row exchanges)
Let $A \in \mathbb{C}^{n\times n}$. If every leading principal minor

$$\Delta_k = \det A_{1:k,1:k}, \qquad k = 1, \dots, n,$$

is nonzero, then there is a factorization

$$A = LU,$$

where L is unit lower triangular and U is upper triangular.

The condition has a direct computational meaning: elimination reaches a nonzero pivot at every stage without exchanging rows. It is not a mysterious determinant test added after the algorithm; it is the determinant form of saying that the algorithm never gets stuck.

The factors are unique under this normalization. Indeed, if

$$A = LU = \widetilde{L}\widetilde{U},$$

then

$$L^{-1}\widetilde{L} = U\widetilde{U}^{-1}.$$

The left side is unit lower triangular and the right side is upper triangular. A matrix of both kinds must be the identity, so $L = \widetilde{L}$ and $U = \widetilde{U}$.

§78.3 A complete LU calculation

Take

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}.$$

The first pivot is 2. To clear the first column, use the multipliers

$$m_{21} = 2, \qquad m_{31} = -1.$$

After

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 + R_1,$$

we obtain

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix}.$$

The second multiplier is

$$m_{32} = \frac{8}{-8} = -1.$$

Thus

$$R_3 \leftarrow R_3 + R_2$$

produces

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

The multipliers, placed in the corresponding positions below a diagonal of ones, give

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}.$$

A direct multiplication confirms that $A = LU$.

Now solve

$$Ax = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}.$$

Instead of eliminating again, first solve

$$Ly = b.$$

Forward substitution gives

$$y_1 = 5, \qquad 2y_1 + y_2 = -2, \qquad -y_1 - y_2 + y_3 = 9,$$

so

$$y = \begin{bmatrix} 5 \\ -12 \\ 2 \end{bmatrix}.$$

Then solve

$$Ux = y.$$

Backward substitution gives

$$x_3 = 2, \qquad -8x_2 - 2x_3 = -12, \qquad 2x_1 + x_2 + x_3 = 5,$$

and hence

$$x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

This is why factorization matters when several right-hand sides are present. The expensive elimination is performed once. Each new vector b requires only one forward and one backward triangular solve.

§78.4 When the first pivot betrays us

The matrix

$$A_0 = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -2 & -3 \\ 2 & 3 & 1 \end{bmatrix}$$

is invertible, but elimination cannot begin with its upper-left entry. Swapping the first and third rows gives

$$PA_0 = \begin{bmatrix} 2 & 3 & 1 \\ 1 & -2 & -3 \\ 0 & 2 & 1 \end{bmatrix},$$

where

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Elimination now yields

$$PA_0 = LU$$

with

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & -\frac{4}{7} & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & -\frac{7}{2} & -\frac{7}{2} \\ 0 & 0 & -1 \end{bmatrix}.$$

Row exchanges are also useful when a pivot is nonzero but dangerously small. For example,

$$B_\varepsilon = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}, \qquad 0 < \varepsilon \ll 1,$$

would use the multiplier $1/\varepsilon$ if elimination proceeded in the displayed order. That large multiplier creates subtraction between large nearby numbers and can amplify rounding error. Partial pivoting exchanges the rows and uses the entry 1 instead.

Partial pivoting chooses the largest available entry in magnitude from the current column. It keeps all elimination multipliers at most 1. This is one reason Gaussian elimination with partial pivoting is reliable in ordinary numerical work. The statement needs one caveat: specially constructed matrices can still exhibit a large growth factor. Pivoting greatly improves stability, but it does not turn every matrix into a harmless one.

There is also a distinction that should not be blurred:

$$\text{algorithmic stability} \qquad \text{versus} \qquad \text{conditioning of the problem.}$$

Pivoting controls the computation. It cannot repair a linear system whose solution is intrinsically extremely sensitive to perturbations.

§78.5 Positive definiteness buys a square root

Suppose now that A is Hermitian positive definite:

$$A = A^*, \qquad x^*Ax > 0 \quad (x \neq 0).$$

Then no row exchange is needed, and symmetry allows us to use essentially one triangular factor instead of two.

Theorem 78.5.1 (Cholesky decomposition)
Every Hermitian positive definite matrix has a unique factorization

$$A = LL^*,$$

where L is lower triangular with positive diagonal entries.

The proof reveals more than the formula. Write

$$A = \begin{bmatrix} a & c^* \\ c & B \end{bmatrix}.$$

Positive definiteness gives $a > 0$, so set

$$\ell_{11} = \sqrt{a}, \qquad w = \frac{c}{\sqrt{a}}.$$

The trailing matrix that remains is the Schur complement

$$S = B - ww^*.$$

It is again positive definite. To see this, for any nonzero y, choose

$$t = -\frac{c^* y}{a}.$$

Then

$$\begin{bmatrix} t \\ y \end{bmatrix}^* A \begin{bmatrix} t \\ y \end{bmatrix} = y^* S y > 0.$$

The problem has therefore reduced to a smaller positive definite matrix. Repeating the argument constructs L recursively.

This induction explains why every square root appearing in Cholesky is real and positive. Positive definiteness is exactly the hypothesis preventing the algorithm from taking the square root of a negative pivot.

§78.6 A Cholesky solve from beginning to end

Consider

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 5 & 1 \\ 2 & 1 & 5 \end{bmatrix}.$$

The first Cholesky entry is

$$\ell_{11} = \sqrt{4} = 2.$$

The entries below it are

$$\ell_{21} = \frac{2}{2} = 1, \qquad \ell_{31} = \frac{2}{2} = 1.$$

The next diagonal entry is

$$\ell_{22} = \sqrt{5 - 1^2} = 2,$$

and

$$\ell_{32} = \frac{1 - 1 \cdot 1}{2} = 0.$$

Finally,

$$\ell_{33} = \sqrt{5 - 1^2 - 0^2} = 2.$$

Thus

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}, \qquad A = LL^T.$$

For

$$Ax = \begin{bmatrix} 8 \\ 8 \\ 8 \end{bmatrix},$$

solve first

$$Ly = b.$$

This gives

$$y = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}.$$

Then solve

$$L^T x = y,$$

which gives

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Cholesky uses about half the leading-order arithmetic and half the off-diagonal storage of a general LU factorization. The gain is not a clever coding accident; it comes from using symmetry instead of computing the same information twice.

The hypothesis cannot be discarded. For

$$C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix},$$

the first step would give $\ell_{11} = 1$ and $\ell_{21} = 2$, but then

$$\ell_{22}^2 = 1 - 2^2 = -3.$$

The failure reflects the negative eigenvalue of C. Cholesky is not merely an LU decomposition with matching factors; it is a certificate of positive definiteness.

There is a geometric interpretation as well. If $A = LL^*$, then

$$x^* A x = \|L^* x\|_2^2.$$

Thus a positive definite quadratic form is Euclidean length after an invertible change of coordinates. Equivalently, A is a Gram matrix of independent vectors.

§78.7 Rank factorization removes redundant coordinates

A rectangular matrix may contain exact redundancy rather than a difficult pivot. If

$$A \in \mathbb{C}^{m\times n}, \qquad \operatorname{rank} A = r,$$

then there are matrices

$$F \in \mathbb{C}^{m\times r}, \qquad G \in \mathbb{C}^{r\times n}$$

with

$$\operatorname{rank} F = \operatorname{rank} G = r$$

such that

$$A = FG.$$

This is a rank factorization.

The map represented by A now visibly passes through an r-dimensional space:

$$\mathbb{C}^n \xrightarrow{G} \mathbb{C}^r \xrightarrow{F} \mathbb{C}^m.$$

Because G has full row rank, it is onto. Because F has full column rank, it is one-to-one. Consequently,

$$\ker A = \ker G, \qquad \operatorname{im} A = \operatorname{im} F.$$

The factorization separates two jobs: G discards the directions that carry no information, and F places the remaining coordinates into the output space.

For example, let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

The first two columns are independent, while the third satisfies

$$A_{:3} = -A_{:1} + 2A_{:2}.$$

Choose

$$F = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Then $A = FG$, and both factors have rank 2.

The factors are not unique. For every invertible $S \in \mathbb{C}^{r\times r}$,

$$A = (FS)(S^{-1}G).$$

This freedom is a change of coordinates in the intermediate space. What is intrinsic is not a particular pair (F, G), but the fact that the map factors through an r-dimensional object. In coordinate-free language, this is the familiar passage

$$\mathbb{C}^n / \ker A \cong \operatorname{im} A.$$

§78.8 SVD chooses orthogonal coordinates on both sides

Rank factorization exposes the correct dimension, but it does not say which coordinates in that dimension are geometrically best. The singular value decomposition makes a special choice: it uses orthonormal coordinates in both the domain and codomain.

Theorem 78.8.1 (Singular value decomposition)
For every $A \in \mathbb{C}^{m\times n}$, there are unitary matrices U and V such that

$$A = U\Sigma V^*,$$

where Σ is rectangular diagonal with entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$$

and all remaining entries zero. The number r is rank A.

The construction begins with the positive semidefinite matrix A^*A. Choose an orthonormal eigenbasis

$$A^*Av_i = \sigma_i^2 v_i.$$

For every nonzero singular value, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

Then

$$Av_i = \sigma_i u_i,$$

and the vectors u_i are orthonormal because

$$u_i^* u_j = \frac{v_i^* A^* A v_j}{\sigma_i \sigma_j} = \delta_{ij}.$$

Complete the two orthonormal families to bases. This yields $A = U\Sigma V^*$.

The equation

$$Av_i = \sigma_i u_i$$

is the most useful way to read the SVD. The input direction v_i is sent to the output direction u_i and stretched by σ_i. Directions corresponding to zero singular values lie in the kernel.

§78.9 A tilted ellipse and its best rank-one shadow

Let

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix}.$$

Its two columns are orthogonal. The first has length 3, and the second has length 1. Define

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Then

$$Ae_1 = 3u_1, \qquad Ae_2 = u_2.$$

Therefore an SVD is

$$A = \underbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}}_{\Sigma} \underbrace{I}_{V^T}.$$

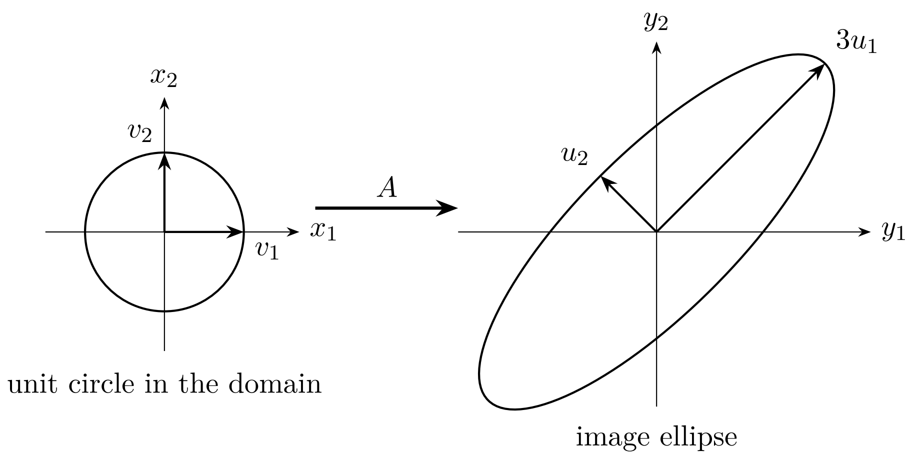


Figure 78.1: The right singular vectors give orthogonal input directions; the left singular vectors give the principal axes of the image ellipse.

The compact SVD is also a rank factorization:

$$A = U_r \Sigma_r V_r^* = (U_r \Sigma_r) V_r^*.$$

Unlike an arbitrary rank factorization, it orders the intermediate coordinates by geometric importance.

This ordering leads to a remarkable optimality statement.

Theorem 78.9.1 (Eckart–Young, spectral-norm form)
If
$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^*, \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0,$$
then among all matrices B of rank at most k, the truncated SVD
$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*$$
minimizes $\|A - B\|_2$, and
$$\|A - A_k\|_2 = \sigma_{k+1}.$$

For the matrix above, the best rank-one approximation is
$$A_1 = 3u_1 e_1^T = \frac{3}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$
The discarded part is
$$A - A_1 = u_2 e_2^T,$$
so
$$\|A - A_1\|_2 = 1.$$
No other rank-one matrix can do better. This is why singular values appear in compression, denoising, latent-factor models, and reduced-order simulation: truncation is not merely plausible; for these norms it is optimal.

Small singular values carry a second message. If A is invertible, then
$$\|A^{-1}\|_2 = \frac{1}{\sigma_{\min}(A)}.$$
A direction that is almost collapsed by A must be enormously expanded by A^{-1}. Thus the same SVD that provides a low-rank approximation also diagnoses instability in solving linear systems.

§78.10 Choosing the factorization by the question

The four decompositions now fit together without becoming interchangeable.

Use $PA = LU$ when a general square system must be solved, especially for several right-hand sides. The essential advantage is reusable elimination. Use Cholesky when the matrix is Hermitian positive definite. It is faster, stores less, and certifies a geometric property of the quadratic form. Use a rank factorization when exact redundancy is the main issue and one wants an explicit passage through the image dimension. Use the SVD when orthogonal geometry, conditioning, or optimal low-rank approximation matters enough to justify its greater cost.

A decomposition should therefore be read as a mathematical answer to a computational question:

$$\begin{array}{ccl} \text{elimination} & \rightsquigarrow & \text{triangular factors,} \\ \text{positive quadratic geometry} & \rightsquigarrow & \text{Cholesky,} \\ \text{quotient by the kernel} & \rightsquigarrow & \text{rank factorization,} \\ \text{orthogonal principal directions} & \rightsquigarrow & \text{SVD.} \end{array}$$

The formulas are useful because each one exposes exactly the structure needed by the next calculation. That is the sense in which a matrix decomposition is an algorithm rather than a decorative identity.

Generalized Inverses, Projections, and Least-Squares Geometry

N-20260603

§79.1 What an inverse was supposed to do

For a square nonsingular matrix, the inverse performs two jobs at once. It solves every equation

$$Ax = b$$

and it undoes the action of A:

$$A^{-1}A = I, \qquad AA^{-1} = I.$$

Once A is rectangular or rank deficient, these two jobs separate. Some vectors b may not lie in the column space, so an exact solution does not exist. If the null space is nontrivial, several vectors x may produce the same output, so an exact solution need not be unique.

The Moore–Penrose inverse does not pretend that these failures have disappeared. It makes two geometric choices:

(i) replace an unreachable target by its orthogonal projection onto the column space;

(ii) among all inputs producing that projected target, choose the one orthogonal to the null space.

The resulting vector

$$x^+ = A^+b$$

is therefore canonical even when ordinary inversion is impossible.

The familiar cases are recovered as special formulas:

matrix type	formula for A^+	interpretation
$m = n = \operatorname{rank} A$	A^{-1}	ordinary inverse
$m > n,\ \operatorname{rank} A = n$	$(A^*A)^{-1}A^*$	full-column-rank least squares
$m < n,\ \operatorname{rank} A = m$	$A^*(AA^*)^{-1}$	full-row-rank minimum norm
$\operatorname{rank} A < \min(m, n)$	$V\Sigma^+U^*$	rank-deficient case

The last line is the real definition in computational practice; the first three are consequences of it.

§79.2 The four fundamental subspaces reveal the answer

Let

$$A : \mathbb{C}^n \longrightarrow \mathbb{C}^m.$$

The domain and codomain split orthogonally as

$$\mathbb{C}^n = R(A^*) \oplus N(A),$$

and

$$\mathbb{C}^m = R(A) \oplus N(A^*).$$

These are the four fundamental subspaces of the matrix.

The restriction

$$A\big|_{R(A^*)} : R(A^*) \longrightarrow R(A)$$

is a bijection. It is injective because

$$R(A^*) \cap N(A) = \{0\},$$

and the two spaces have the same dimension, namely $\operatorname{rank} A$. Thus A has a genuine inverse after we discard precisely the directions on which it vanishes and restrict the target to the vectors it can actually reach.

This gives a coordinate-free description of the pseudoinverse. Decompose

$$b = b_R + b_\perp, \qquad b_R \in R(A), \quad b_\perp \in N(A^*).$$

Then A^+b is the unique vector in $R(A^*)$ satisfying

$$A(A^+b) = b_R.$$

The orthogonal component $b_\perp$ is ignored because no input can produce it.

The two central products are now transparent:

$$AA^+ = P_{R(A)}, \qquad A^+A = P_{R(A^*)}.$$

The first projects targets onto the reachable output space. The second projects inputs away from the null-space ambiguity.

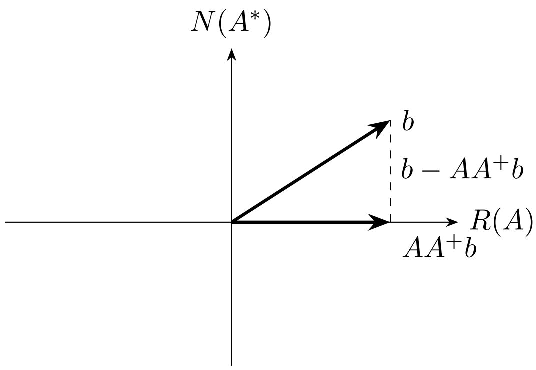


Figure 79.1: The pseudoinverse first replaces b by the closest reachable vector AA^+b.

§79.3 Why the four Penrose equations appear

The Moore–Penrose inverse is usually introduced as the unique matrix X satisfying

$$AXA = A, \qquad XAX = X,$$

$$(AX)^* = AX, \qquad (XA)^* = XA.$$

Read without context, these equations look ceremonial. The projection picture explains every one of them.

If $X = A^+$, then AX should be the orthogonal projection onto $R(A)$. An orthogonal projection is idempotent and self-adjoint. The relation

$$AXA = A$$

says that AX fixes every vector already in $R(A)$, while

$$(AX)^* = AX$$

forces the projection to be orthogonal rather than oblique.

Similarly, XA should be the orthogonal projection onto $R(A^*)$. The equations

$$XAX = X, \qquad (XA)^* = XA$$

encode that requirement on the input side.

Dropping some of the four conditions produces broader families of generalized inverses. There are 15 nonempty subsets of the four equations, which explains the traditional count of generalized-inverse classes. The count itself is less important than the distinction: algebraic retraction conditions alone allow many choices, while the two adjoint conditions select the orthogonal geometry and make the answer unique.

Proposition 79.3.1

The Moore–Penrose inverse, if it exists, is unique.

One way to see uniqueness is to use the two projection identities. Any matrix satisfying the four equations must act as the inverse of A from $R(A)$ to $R(A^*)$, must vanish on $N(A^*)$, and must return values orthogonal to $N(A)$. Those requirements determine its value on the orthogonal decomposition

$$\mathbb{C}^m = R(A) \oplus N(A^*).$$

§79.4 SVD turns the definition into a calculation

Suppose

$$A = U\Sigma V^*$$

is a singular value decomposition, with positive singular values

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Write

$$\Sigma = \begin{bmatrix} \operatorname{diag}(\sigma_1, \ldots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$A^+ = V\Sigma^+U^*,$$

where

$$\Sigma^+ = \begin{bmatrix} \operatorname{diag}(\sigma_1^{-1}, \ldots, \sigma_r^{-1}) & 0 \\ 0 & 0 \end{bmatrix}^T.$$

Only the nonzero singular values are inverted.

The formula is easier to remember mode by mode. Since

$$Av_i = \sigma_i u_i,$$

the pseudoinverse must satisfy

$$A^+ u_i = \frac{1}{\sigma_i} v_i \qquad (i \le r),$$

and

$$A^+ u = 0 \qquad (u \in N(A^*)).$$

Thus A^+ literally reverses the active singular directions and does nothing to unreachable directions.

The projection formulas follow at once:

$$AA^+ = U_r U_r^*, \qquad A^+ A = V_r V_r^*.$$

Here the columns of U_r form an orthonormal basis of $R(A)$, and the columns of V_r form an orthonormal basis of $R(A^*)$.

A rank factorization also gives a useful formula. If

$$A = FG,$$

where $F \in \mathbb{C}^{m\times r}$ has full column rank and $G \in \mathbb{C}^{r\times n}$ has full row rank, then

$$A^+ = G^*(GG^*)^{-1}(F^*F)^{-1}F^*.$$

The middle two inverses are only $r \times r$. This expression makes the factor-through-the-image viewpoint explicit, although SVD is usually the cleaner numerical language when rank is uncertain.

§79.5 One rank-one matrix contains the whole geometry

Let

$$u = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A = uv^T = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 0 \end{bmatrix}.$$

For a nonzero rank-one matrix uv^*,

$$(uv^*)^+ = \frac{vu^*}{\|u\|_2^2 \|v\|_2^2}.$$

Since

$$\|u\|_2^2 = 5, \qquad \|v\|_2^2 = 2,$$

we obtain

$$A^+ = \frac{1}{10}\begin{bmatrix} 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}.$$

The two products are

$$AA^+ = \frac{1}{5}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \frac{uu^T}{u^T u},$$

and

$$A^+A = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = \frac{vv^T}{v^Tv}.$$

They are the orthogonal projections onto $\operatorname{span}(u)$ and $\operatorname{span}(v)$, respectively.

Now take

$$b = \begin{bmatrix}1\\0\\1\end{bmatrix}.$$

The closest reachable vector is

$$\widehat{b} = AA^+b = \frac{1}{5}\begin{bmatrix}1\\2\\0\end{bmatrix}.$$

The residual is

$$r = b - \widehat{b} = \begin{bmatrix}\frac{4}{5}\\ -\frac{2}{5}\\ 1\end{bmatrix},$$

and indeed

$$u^Tr = 0.$$

The pseudoinverse solution is

$$x^+ = A^+b = \begin{bmatrix}\frac{1}{10}\\ \frac{1}{10}\end{bmatrix}.$$

Every vector satisfying

$$x_1 + x_2 = \frac{1}{5}$$

produces $\widehat{b}$, but x^+ is the one with smallest Euclidean norm. The equality of its two coordinates is not accidental: it is the orthogonal projection of the origin onto that affine line.

This single example displays all four subspaces. The column space is $\operatorname{span}(u)$; the left null space is its orthogonal complement. The row space is $\operatorname{span}(v)$; the null space consists of vectors proportional to $(1, -1)^T$. The pseudoinverse moves only between the two one-dimensional active spaces.

§79.6 Exact solutions, least squares, and minimum norm

For an arbitrary b, the vector

$$x^+ = A^+b$$

minimizes

$$\|Ax - b\|_2.$$

Indeed,

$$Ax^+ = AA^+b = P_{R(A)}b,$$

which is the closest point in the column space to b.

All least-squares minimizers have the form

$$x = A^+b + (I - A^+A)z, \qquad z \in \mathbb{C}^n.$$

The second term lies in $N(A)$, so it does not change Ax. Since

$$A^+b \in R(A^*) = N(A)^\perp,$$

the two terms are orthogonal. Therefore

$$\|x\|_2^2 = \|A^+b\|_2^2 + \|(I - A^+A)z\|_2^2.$$

This proves that A^+b is the unique minimum-norm least-squares solution.

Exact solvability is detected by

$$AA^+b = b.$$

If this holds, the same formula describes every exact solution:

$$x = A^+b + (I - A^+A)z.$$

Thus one expression covers the overdetermined, underdetermined, and rank-deficient cases. What changes is only whether the projection AA^+b equals b and whether the null-space term is present.

The common identities

$$(A^+)^+ = A, \qquad (A^*)^+ = (A^+)^*, \qquad \operatorname{rank} A^+ = \operatorname{rank} A$$

are immediate from the SVD. They are consequences of the singular-direction picture, not a separate collection of tricks.

§79.7 Why inverse-like algebra can fail

Ordinary inverses reverse products:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The corresponding pseudoinverse identity is false in general. Let

$$A = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad (AB)^+ = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

On the other hand,

$$A^+ = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad B^+ = B,$$

so

$$B^+A^+ = \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} \neq (AB)^+.$$

The failure is geometric. Each pseudoinverse inserts orthogonal projections onto row and column spaces. In a product, the active output space of B need not align with the active input space of A. Reversing the two pseudoinverses introduces projections in the wrong places. Reverse-order laws do hold under additional range and commutation hypotheses, but they should never be assumed from notation alone.

This is a useful general warning: the pseudoinverse behaves like an inverse only on the active singular subspaces. Algebraic identities that move through kernels or orthogonal complements require extra compatibility.

§79.8 Updating a pseudoinverse one column at a time

Recomputing an SVD after every new data column may be wasteful. Greville's recursion updates the pseudoinverse when

$$A_{k+1} = \begin{bmatrix} A_k & a \end{bmatrix}.$$

Set

$$d = A_k^+ a, \qquad c = a - A_k d = (I - A_k A_k^+)a.$$

The vector c is the part of the new column orthogonal to the old column space. If $c \neq 0$, define

$$b^* = \frac{c^*}{c^* c}.$$

Then

$$A_{k+1}^+ = \begin{bmatrix} A_k^+ - db^* \\ b^* \end{bmatrix}.$$

For a small example, begin with

$$A_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A_1^+ = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

and append

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Then

$$d = A_1^+ a = 1, \qquad c = a - A_1 d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad b^* = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Therefore

$$A_2^+ = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

But

$$A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

is invertible, and the update has recovered its ordinary inverse exactly.

If $c = 0$, the new column already lies in the old column space and a different scalar correction is used. That branch is not a technical nuisance; it records whether the incoming data adds a genuinely new direction.

§79.9 Iteration sees the pseudoinverse as a limit

The pseudoinverse need not be formed explicitly in a large problem. Consider Landweber iteration

$$x_{k+1} = x_k + \alpha A^*(b - Ax_k),$$

with

$$0 < \alpha < \frac{2}{\sigma_1^2}.$$

This is gradient descent on

$$\frac{1}{2}\|Ax - b\|_2^2.$$

In singular-vector coordinates, each active component evolves independently. If

$$x_k = \sum_i \xi_{i,k} v_i,$$

then

$$\xi_{i,k+1} = (1 - \alpha\sigma_i^2)\xi_{i,k} + \alpha\sigma_i u_i^* b.$$

The step-size condition makes

$$|1 - \alpha\sigma_i^2| < 1$$

for every nonzero singular value. Hence

$$\xi_{i,k} \longrightarrow \frac{u_i^* b}{\sigma_i}.$$

If the iteration starts from $x_0 = 0$, the null-space components remain zero, so

$$x_k \longrightarrow A^+ b.$$

The minimum-norm choice is therefore encoded by the initialization: gradient descent never invents a component in a direction that the data cannot see.

This observation is useful beyond numerical linear algebra. In an underdetermined linear model, the optimization dynamics themselves can impose an implicit bias toward the minimum-norm solution.

§79.10 Small singular values and the need for regularization

The formula

$$A^+ b = \sum_{i=1}^{r} \frac{u_i^* b}{\sigma_i} v_i$$

shows a danger. If σ_i is tiny, even a small component of noise along u_i is magnified by $1/\sigma_i$. The pseudoinverse is canonical, but canonical does not mean numerically safe.

Tikhonov regularization replaces exact least squares by

$$\min_x \big(\|Ax - b\|_2^2 + \lambda \|x\|_2^2\big), \qquad \lambda > 0.$$

The solution is

$$x_\lambda = (A^* A + \lambda I)^{-1} A^* b.$$

Using the SVD,

$$x_\lambda = \sum_{i=1}^{r} \frac{\sigma_i}{\sigma_i^2 + \lambda} (u_i^* b) v_i.$$

Compare the two spectral filters:

$$\frac{1}{\sigma_i} \qquad \text{and} \qquad \frac{\sigma_i}{\sigma_i^2 + \lambda}.$$

For large singular values they are close. For small singular values the regularized factor remains controlled instead of exploding.

The geometric story is now complete. The pseudoinverse inverts A exactly on the active subspaces, projects away what cannot be reached, and removes null-space ambiguity. Iterative methods approach the same object without forming it, while regularization deliberately softens the inversion when the smallest singular directions are dominated by noise. These are not separate techniques attached to one symbol; they are different responses to the same geometry of range, kernel, and singular scale.

Solving Linear Systems: Elimination, Iteration, and Energy

N-20260604

§80.1 One equation, three ways of thinking

The equation

$$Ax = b$$

looks like a single problem, but numerically it can mean three rather different things.

If the matrix is moderate in size and used several times, it is natural to *factor* it. If the matrix is huge and sparse, it may be better to improve a guess one step at a time. If A is symmetric positive definite, the equation is also the stationarity condition of a convex quadratic energy. These are not three unrelated collections of formulas. They are three views of the same linear map:

eliminate variables, propagate corrections, minimize energy.

A useful solver therefore starts by asking what structure is available. Dense or sparse? Symmetric or nonsymmetric? Positive definite or indefinite? One right-hand side or many? The answer often matters more than the nominal matrix size.

§80.2 Triangular systems are the basic currency

Nearly every direct method tries to reduce a general system to triangular ones. If L is lower triangular, then the first equation contains only x_1, the second contains x_1, x_2, and so on. Forward substitution gives

$$x_i = \frac{1}{l_{ii}} \left(b_i - \sum_{j<i} l_{ij} x_j \right).$$

For an upper triangular matrix U, backward substitution runs in the opposite direction:

$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j>i} u_{ij} x_j \right).$$

The arithmetic cost is only $O(n^2)$. The expensive part of a direct solve is not substitution but producing the triangular factors. This explains why an LU or Cholesky factorization is especially valuable when many vectors b must be solved against the same matrix: the cubic work is paid once, while each new right-hand side needs only triangular solves.

§80.3 Gaussian elimination is a factorization in disguise

Consider

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}.$$

The first pivot is 2. Replacing

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 + R_1$$

gives

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right].$$

One more elimination step,

$$R_3 \leftarrow R_3 + R_2,$$

produces

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Back substitution yields

$$x_3 = 2, \qquad x_2 = 1, \qquad x_1 = 1.$$

The multipliers used during elimination were $2, -1, -1$. Placed in a unit lower triangular matrix, they form

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}, \qquad A = LU.$$

Thus elimination is not merely a sequence of row tricks. It constructs a reusable representation of the linear map.

§80.4 Why pivoting is part of the algorithm

Exact arithmetic only objects to a zero pivot. Floating-point arithmetic also objects to a very small one. For example,

$$A = \begin{bmatrix} 10^{-12} & 1 \\ 1 & 1 \end{bmatrix}$$

is perfectly invertible, yet elimination without row exchange uses the multiplier 10^{12}. A tiny rounding error in the first row can then be magnified enormously.

Partial pivoting swaps the largest available entry in the current column into the pivot position. Algebraically the result is

$$PA = LU,$$

where P is a permutation matrix. Pivoting does not change the problem; it changes the order in which information is eliminated so that intermediate numbers remain under control.

This distinction is worth keeping: conditioning belongs to the original problem, whereas numerical stability belongs to the algorithm. A well-conditioned system can still be damaged by a poor elimination order, and a stable algorithm cannot make an intrinsically ill-conditioned system insensitive.

§80.5 Structure can make elimination cheaper

A general dense factorization costs $O(n^3)$, but special matrices often require much less.

If A is symmetric positive definite, Cholesky factorization writes

$$A = LL^T.$$

Only one triangular factor is needed, symmetry halves the storage, and the work is about half that of a general LU factorization. An LDL^T factorization separates scaling from elimination:

$$A = LDL^T,$$

with L unit lower triangular and D diagonal or block diagonal.

For a tridiagonal matrix,

$$A = \begin{bmatrix} d_1 & c_1 & & & \\ a_1 & d_2 & c_2 & & \\ & a_2 & d_3 & \ddots & \\ & & \ddots & \ddots & c_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix},$$

elimination preserves the band. The resulting Thomas algorithm needs only $O(n)$ work and storage. This is a first example of a general numerical principle: preserve sparsity whenever possible, because unnecessary fill-in can cost more than the original equations.

§80.6 When a sequence of guesses is better than a factorization

For a very large sparse matrix, even storing the factors may be too expensive. Iterative methods instead generate

$$x^{(0)}, x^{(1)}, x^{(2)}, \ldots$$

and hope that these vectors approach the solution.

Write

$$A = M - N.$$

Then

$$Mx = Nx + b,$$

so a natural iteration is

$$Mx^{(k+1)} = Nx^{(k)} + b.$$

Equivalently,

$$x^{(k+1)} = Bx^{(k)} + c, \qquad B = M^{-1}N, \quad c = M^{-1}b.$$

If x_* is the exact solution and $e^{(k)} = x^{(k)} - x_*$, then

$$e^{(k+1)} = Be^{(k)}, \qquad e^{(k)} = B^k e^{(0)}.$$

Therefore convergence for every initial guess is equivalent to

$$\rho(B) < 1.$$

This one equation explains why stationary iterations are spectral problems rather than merely recursive formulas.

§80.7 Jacobi and Gauss–Seidel seen on the same example

Let

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The exact solution is

$$x_* = \begin{bmatrix} 1/10 \\ 3/5 \end{bmatrix}.$$

Starting from $x^{(0)} = (0, 0)^T$, Jacobi updates both coordinates using only the old iterate:

$$x_1^{(k+1)} = \frac{1 - x_2^{(k)}}{4}, \qquad x_2^{(k+1)} = \frac{2 - 2x_1^{(k)}}{3}.$$

The first two iterates are

$$x^{(1)} = \begin{bmatrix} 1/4 \\ 2/3 \end{bmatrix}, \qquad x^{(2)} = \begin{bmatrix} 1/12 \\ 1/2 \end{bmatrix}.$$

Gauss–Seidel immediately reuses the newest value of x_1:

$$x_1^{(k+1)} = \frac{1 - x_2^{(k)}}{4}, \qquad x_2^{(k+1)} = \frac{2 - 2x_1^{(k+1)}}{3}.$$

Thus

$$x^{(1)} = \begin{bmatrix} 1/4 \\ 1/2 \end{bmatrix}, \qquad x^{(2)} = \begin{bmatrix} 1/8 \\ 7/12 \end{bmatrix}.$$

Both converge here, but Gauss–Seidel moves more quickly because each sweep contains a small amount of implicit coupling.

Strict diagonal dominance,

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|,$$

is a convenient sufficient condition for convergence of both methods. It is not the only condition, and failure of diagonal dominance does not automatically imply divergence. The exact question remains $\rho(B) < 1$.

§80.8 Residuals are visible; errors are not

During an iteration the true error

$$e^{(k)} = x^{(k)} - x_*$$

is usually unknown, because knowing x_* would make the computation unnecessary. What can be computed is the residual

$$r^{(k)} = b - Ax^{(k)}.$$

Since

$$Ae^{(k)} = -r^{(k)},$$

we have

$$e^{(k)} = -A^{-1} r^{(k)}.$$

Hence a small residual implies a small error only after the inverse is controlled. In norm form,

$$\|e^{(k)}\| \le \|A^{-1}\| \, \|r^{(k)}\|.$$

For an ill-conditioned matrix, a visually tiny residual can still hide a noticeable solution error. Stopping criteria should therefore be interpreted together with scaling and condition estimates, not as universal certificates.

§80.9 Positive definite systems are convex optimization problems

Suppose $A = A^T$ is positive definite. Define

$$\phi(x) = \frac{1}{2}x^T A x - b^T x.$$

Then

$$\nabla\phi(x) = Ax - b, \qquad \nabla^2\phi(x) = A \succ 0.$$

Thus ϕ is strictly convex, and its unique minimizer satisfies

$$Ax = b.$$

Completing the square gives an even clearer identity:

$$\phi(x) - \phi(x_*) = \frac{1}{2}(x - x_*)^T A (x - x_*) = \frac{1}{2}\|x - x_*\|_A^2,$$

where

$$\|v\|_A = \sqrt{v^T A v}.$$

Solving the linear system is therefore exactly the same as driving the A-energy error to zero.

Steepest descent follows the negative residual direction. With

$$r_k = b - Ax_k,$$

choose

$$x_{k+1} = x_k + \alpha_k r_k.$$

Minimizing $\phi(x_k + \alpha r_k)$ along this line gives

$$\alpha_k = \frac{r_k^T r_k}{r_k^T A r_k}.$$

The method always decreases the energy, but for elongated level sets it tends to zigzag. The elongation is governed by the condition number $\kappa_2(A)$.

§80.10 Conjugate gradients remembers previous progress

Steepest descent repeatedly chooses the locally best direction, yet may undo part of earlier progress. Conjugate gradients avoids this by constructing search directions p_k that are A-orthogonal:

$$p_i^T A p_j = 0 \qquad (i \neq j).$$

Starting from x_0, set

$$r_0 = b - Ax_0, \qquad p_0 = r_0.$$

Then iterate

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}, \qquad x_{k+1} = x_k + \alpha_k p_k,$$

$$r_{k+1} = r_k - \alpha_k A p_k, \qquad \beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k},$$

$$p_{k+1} = r_{k+1} + \beta_k p_k.$$

After k steps, x_k lies in the affine Krylov space

$$x_0 + \mathcal{K}_k(A, r_0), \qquad \mathcal{K}_k(A, r_0) = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1} r_0\},$$

and minimizes the energy over that space. In exact arithmetic, conjugate gradients reaches the exact solution in at most n steps. Its practical value is that useful accuracy often arrives much sooner, especially when the eigenvalues are clustered.

A standard bound is

$$\frac{\|x_k - x_*\|_A}{\|x_0 - x_*\|_A} \le 2 \left(\frac{\sqrt{\kappa_2(A)} - 1}{\sqrt{\kappa_2(A)} + 1} \right)^k.$$

This makes the role of conditioning explicit: poor geometry slows the polynomial approximation carried out by the Krylov method.

§80.11 Preconditioning changes the geometry, not the solution

A preconditioner M is a matrix that is easy to solve with and approximates A. Instead of attacking

$$Ax = b$$

directly, one solves an equivalent system such as

$$M^{-1} A x = M^{-1} b.$$

The solution is unchanged, but the spectrum may become much more favorable.

For symmetric positive definite problems, a symmetric preconditioner can be understood through

$$M^{-1/2} A M^{-1/2} y = M^{-1/2} b, \qquad x = M^{-1/2} y.$$

Good preconditioning clusters the eigenvalues of the transformed matrix and reduces its condition number. Diagonal scaling, incomplete Cholesky factors, and multigrid operators are increasingly sophisticated realizations of the same idea: solve an easier nearby problem to correct the hard one.

§80.12 Choosing a method without turning the chapter into a menu

The methods above are best remembered through the obstacle each one removes.

A dense system with several right-hand sides rewards a stable factorization such as pivoted LU. Positive definiteness makes Cholesky cheaper and supplies an energy interpretation. Tridiagonal or banded structure should be preserved rather than destroyed.

Very large sparse systems favor iteration because matrix–vector products are cheap while fill-in is expensive. Stationary methods are simple smoothers, but their speed is tied to one fixed iteration matrix. Krylov methods adapt a polynomial to the observed residuals and are usually far more effective. Preconditioning is the decisive step when the original coordinates make convergence slow.

The deeper unity is that every solver tries to expose directions in which the equation is easy. Elimination removes them one by one, stationary iteration repeatedly damps them, and Krylov methods build a subspace tailored to them. The matrix is the same; the algorithm decides how its geometry is revealed.

VIII

Calculus

Part VIII: Contents

Quadratic Parameters, Tangency, and Constrained Extrema

N-20220315

§81.1 A scattered algebraic note

The original page is titled "some scattered things." The common theme is that elementary-looking algebraic problems often hide a parameter, a tangency condition, or an extremum under constraint. I rewrite the problems in a more systematic form.

§81.2 A quadratic with parameters

Let

$$f(x) = ax^2 + bx + c, \qquad a < b < c,$$

and suppose the graph passes through $A = (1, 0)$ and $B = (m, -a)$. The condition $f(1) = 0$ gives

$$a + b + c = 0.$$

The condition $f(m) = -a$ is equivalent to saying that $f(x) + a$ has $x = m$ as a root:

$$am^2 + bm + c + a = 0.$$

Using $c = -a - b$, this becomes

$$am^2 + bm - b = 0, \qquad am^2 = -b(m - 1).$$

The axis of symmetry is

$$x_0 = -\frac{b}{2a}.$$

Thus the parameter m and the axis x_0 are tied by

$$x_0 = \frac{m^2}{2m - 2}.$$

Equivalently,

$$m^2 = (2m - 2)x_0, \qquad m^2 - 2x_0 m + 2x_0 = 0.$$

Solving for m gives

$$m = x_0 \pm \sqrt{x_0^2 - 2x_0}.$$

The original calculation uses this relation to examine endpoints and stationary points of m as a function of x_0. The conclusion is that the extremal parameter occurs at

$$x_0 = 0 \quad \text{or} \quad x_0 = -\frac{1}{2},$$

and the relevant value is

$$m = \frac{\sqrt{5} - 1}{2}.$$

This is a small example of a recurring trick: convert the algebraic parameter into a geometric parameter, usually the axis, vertex, or tangency point.

§81.3 A parabola and a hyperbola

The second problem considers the parabola

$$y = x^2 - 2ax + a^2 - 4 = (x - a)^2 - 4$$

and the rectangular hyperbola

$$y = \frac{k}{x}, \qquad k > 0,$$

in the first quadrant. The intersection equation is

$$x^2 - 2ax + a^2 - 4 = \frac{k}{x}, \qquad x > 0,$$

or

$$x^3 - 2ax^2 + (a^2 - 4)x - k = 0.$$

Questions about “only one intersection” are therefore tangency or root-counting questions for this cubic, but the geometry is usually cleaner: one restricts to the right side of the parabola, because the branch k/x lies in the first quadrant.

> **Remark 81.3.1 —** The hidden principle is that a graph-intersection problem may become either a polynomial root-counting problem or a tangency problem. The latter is often more geometric and less computational.

§81.4 A constrained extremum

The third problem asks for extrema of

$$x + y$$

under the constraint

$$x^2 + y^3 = 2.$$

Using Lagrange multipliers for

$$F(x, y, \lambda) = x + y + \lambda(x^2 + y^3 - 2),$$

we get

$$1 + 2\lambda x = 0, \qquad 1 + 3\lambda y^2 = 0, \qquad x^2 + y^3 = 2.$$

Eliminating λ gives

$$2x = 3y^2.$$

Hence

$$x = \frac{3}{2}y^2.$$

Substitution into the constraint gives

$$\frac{9}{4}y^4 + y^3 = 2,$$

so the stationary points come from the quartic

$$9y^4 + 4y^3 - 8 = 0.$$

The numerical plot in the original note marks approximately

$$A = (1.15195, 2.02829), \qquad B = (1.82947, 0.72509),$$

for the function

$$h(x) = x + (2 - x^2)^{1/3}.$$

These are the local maximum and local minimum along the real branch.

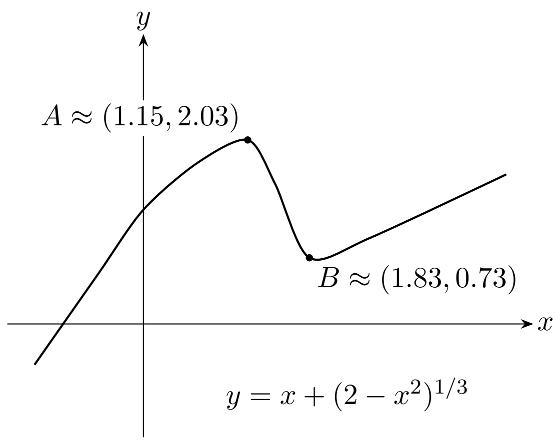


Figure 81.1: Reconstruction of the graph used for the constrained extremum problem.

§81.5 Tangency as a double-root condition

A parameter problem involving a quadratic often asks when a line is tangent to a parabola or when an equation has exactly one solution. Algebraically, this is a discriminant condition:

$$\Delta = b^2 - 4ac = 0.$$

Geometrically, the line touches the parabola at one point. The same idea appears in constrained extrema: a level curve just touches a constraint curve at an optimum.

§81.6 Completing the square

For a quadratic

$$ax^2 + bx + c, \qquad a \neq 0,$$

completing the square gives

$$a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

This form reveals the vertex, minimum or maximum value, and symmetry axis. It is usually more informative than the expanded form.

§81.7 A wider frame

The three examples are connected by the same idea: reduce a complicated statement to a structural condition. The axis of a quadratic, the tangency of two graphs, and the Lagrange multiplier equations are all ways of saying that an extremum occurs when a first-order freedom disappears.

§81.8 Tangency as multiplicity

A line is tangent to a curve when the intersection equation has a repeated root. If a family is written as

$$F(x,t)=0,$$

then a tangency or transition value often satisfies

$$F(x,t)=0, \qquad \frac{\partial F}{\partial x}(x,t)=0.$$

For a quadratic, this is the discriminant condition $\Delta = 0$. The same idea persists for higher-degree equations: a repeated root is detected by a common factor of F and F_x.

§81.9 Constrained extrema and normal directions

For a constraint $g(x,y)=0$, the Lagrange multiplier condition

$$\nabla f = \lambda \nabla g$$

says that at an interior constrained extremum, the gradient of f has no component tangent to the constraint curve. Equivalently, the differential df vanishes on the tangent space of the constraint.

This is the coordinate-free meaning of the method: extrema occur where the objective's first-order change is normal to every allowed first-order motion.

§81.10 Degeneracy and second-order data

After first derivatives vanish, the next question is second-order behavior. For an unconstrained two-variable function, the Hessian quadratic form determines the local model:

$$f(a+h)=f(a)+\frac{1}{2}h^T H_f(a)h + o(\|h\|^2).$$

For constrained problems, the Hessian must be restricted to tangent directions. Degenerate cases occur when this quadratic form has zero directions, forcing higher-order analysis.

§81.11 Multiplicity, normality, and local geometry

Quadratic parameter problems, tangency, and constrained extrema are one theme: detect where the number or type of solutions changes. Algebra sees this as multiplicity or discriminant zero; geometry sees it as tangency; optimization sees it as vanishing first variation plus second-order testing.

A Review of Calculus: Sequences, Limits, Derivatives, and Integrals

N-20220319

§82.1 Sequences and convergence

After defining real numbers, the first analytic object to understand is a sequence. Intuitively,

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$$

should converge to 0, while

$$1, 2, 3, 4, \ldots$$

should diverge. The difficulty is to turn this intuition into a definition that is precise.

Definition 82.1.1. A sequence is a function

$$a : \mathbb{N} \to \mathbb{R}$$

or more generally $a : \mathbb{N} \to \mathbb{C}$. We write a_n instead of $a(n)$, and denote the sequence by (a_n) or $(a_n)_{n=1}^{\infty}$.

Definition 82.1.2. Let (a_n) be a real sequence and let $\ell \in \mathbb{R}$. We say $a_n \to \ell$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that whenever $n \geq N$,

$$|a_n - \ell| < \varepsilon.$$

Equivalently,

$$(\forall \varepsilon > 0)(\exists N)(\forall n \geq N)\ |a_n - \ell| < \varepsilon.$$

Example 82.1.3

Let

$$x_n = \frac{n}{n+1}.$$

Then $x_n \to 1$. Indeed,

$$|x_n - 1| = \frac{1}{n+1}.$$

Given $\varepsilon > 0$, choose $N > 1/\varepsilon - 1$. Then $n > N$ implies $|x_n - 1| < \varepsilon$.

§82.2 Limits of functions

Definition 82.2.1. Let $A \subseteq \mathbb{R}$, $f : A \to \mathbb{R}$, and a be a limit point of A. We write

$$\lim_{x \to a} f(x) = \ell$$

if

$$(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in A)\big(0 < |x - a| < \delta \Rightarrow |f(x) - \ell| < \varepsilon\big).$$

The condition $0 < |x - a|$ is important: the value of $f(a)$ is irrelevant, and f does not even have to be defined at a.

§82.3 Derivatives

Definition 82.3.1. The derivative of f at x is

$$\frac{df}{dx}(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists.

Theorem 82.3.2 (Chain rule)
If $f(x) = F(g(x))$, then

$$\frac{df}{dx} = \frac{dF}{dg}\frac{dg}{dx}.$$

Theorem 82.3.3 (Product rule)
If $f(x) = u(x)v(x)$, then

$$f'(x) = u'(x)v(x) + u(x)v'(x).$$

Theorem 82.3.4 (Quotient rule)
If $f(x) = u(x)/v(x)$, then

$$f'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{v(x)^2}.$$

Theorem 82.3.5 (Leibniz rule)
If $f = uv$, then

$$f^{(n)}(x) = \sum_{r=0}^{n} \binom{n}{r} u^{(r)}(x)v^{(n-r)}(x).$$

§82.4 Small-o and big-O

Definition 82.4.1. As $x \to x_0$, write $f(x) = o(g(x))$ if

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.$$

Write $f(x) = O(g(x))$ if $f(x)/g(x)$ remains bounded near x_0.

Thus $o(g)$ means much smaller than g, while $O(g)$ means no larger than a constant multiple of g. Clearly,

$$f = o(g) \Rightarrow f = O(g).$$

§82.5 Integration

An integral is a limit of sums. The usual Riemann integral is modeled by

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_{n=0}^{N} f(x_n)\Delta x.$$

For an interval partition with $\Delta x = (b-a)/N$ and $x_n = a + n\Delta x$, this approximates the area under the graph.

If f is differentiable, then over a small interval the area is

$$f(x_n)\Delta x + O((\Delta x)^2).$$

Summing and letting $N \to \infty$ gives the usual integral.

Theorem 82.5.1 (Fundamental theorem of calculus)
Let

$$F(x) = \int_a^x f(t)\,dt.$$

Then, under the usual continuity hypothesis on f,

$$F'(x) = f(x).$$

Remark 82.5.2 — This is a review note rather than a full proof-based analysis text. The next revision should add precise hypotheses for every theorem, especially differentiability and continuity assumptions.

§82.6 Why the epsilon definition is shaped this way

The definition of convergence has the order of quantifiers

$$\forall \varepsilon > 0\ \exists N\ \forall n \geq N.$$

This order is essential. We are not allowed to choose one fixed tolerance; the sequence must eventually enter every tolerance window around the limit. The number N may depend on ε, because smaller error demands later terms.

For example, to prove $1/n \to 0$, given $\varepsilon > 0$, choose $N > 1/\varepsilon$. Then $n \geq N$ implies

$$\left| \frac{1}{n} - 0 \right| < \varepsilon.$$

The proof is not a calculation of the limit; it is a demonstration that every requested precision can eventually be met.

§82.7 Continuity by sequences

A function f is continuous at a if

$$x_n \to a \quad \Longrightarrow \quad f(x_n) \to f(a)$$

for every sequence (x_n) in the domain. This sequence criterion is often easier to use than epsilon-delta directly. To prove discontinuity, it suffices to find one sequence $x_n \to a$ such that $f(x_n) \not\to f(a)$.

For example, the Dirichlet function is discontinuous everywhere. Near any a, choose rational and irrational sequences both converging to a. Their function values approach different limits.

§82.8 Derivative as best linear approximation

The derivative is often introduced as a slope:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

A more structural reading is that differentiability means

$$f(a+h) = f(a) + f'(a)h + o(h).$$

Thus the derivative is the coefficient of the best linear approximation near a. This viewpoint generalizes directly to multivariable calculus, where the derivative becomes a linear map.

§82.9 Integral as accumulation

The definite integral can be understood as the limit of accumulated small contributions. For continuous f, Riemann sums

$$\sum_i f(\xi_i)\Delta x_i$$

converge to the area-signed accumulation. The fundamental theorem of calculus says that differentiation and integration are inverse operations in the sense that

$$\frac{d}{dx} \int_a^x f(t)\, dt = f(x).$$

This is not merely an area formula; it says local rate and global accumulation determine one another.

§82.10 Convergence as topology

The elementary epsilon definition in this note is the first appearance of a topology. On the real line, saying

$$x_n \to x$$

means that every interval around x eventually contains all x_n. In a metric space (X, d) the same sentence becomes

$$\forall \varepsilon > 0, \quad \exists N, \quad n > N \Rightarrow d(x_n, x) < \varepsilon.$$

Thus the elementary sequence definition is not a special trick for numbers; it is the definition of convergence once a notion of distance has been chosen.

This immediately explains why different spaces have different convergence. In $C[0,1]$ with the sup norm,

$$\|f_n - f\|_\infty = \max_{x\in[0,1]} |f_n(x) - f(x)| \to 0$$

means uniform convergence. With the L^1 norm,

$$\|f_n - f\|_1 = \int_0^1 |f_n(x) - f(x)|\,dx \to 0,$$

thin spikes may disappear in integral size while still being tall. The same elementary sentence "goes to zero" therefore depends on the surrounding space.

§82.11 Derivative as a linear map

In one-variable calculus, the derivative is introduced as a number:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}.$$

The higher-level meaning is that $f'(a)h$ is the best first-order linear approximation to the change in f:

$$f(a+h) = f(a) + f'(a)h + o(h).$$

For a map $F : \mathbb{R}^n \to \mathbb{R}^m$, the derivative at a is a linear map $DF(a)$ satisfying

$$F(a+h) = F(a) + DF(a)h + o(|h|).$$

The Jacobian matrix is only the coordinate representation of this linear map.

§82.12 Limits and derivatives test topology and linearization

The elementary limit and derivative computations in the original note should be read in two layers. At the calculation layer, they train epsilon estimates and algebraic simplification. At the structural layer, they introduce topology and linearization. A limit asks whether a function is continuous with respect to the chosen topology; a derivative asks whether the function is locally a linear map plus a smaller error.

A Review of Calculus: Mean Value Theorems and Continuity

N-20220320

§83.1 From the mean value theorem to Cauchy's form

This note continues the calculus review. The main topic is the mean value theorem and its Cauchy form.

Theorem 83.1.1 (Mean value theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Geometrically, there is a point at which the tangent slope equals the slope of the secant line through the endpoints.

Theorem 83.1.2 (Cauchy's mean value theorem)
Let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $g'(x) \neq 0$ on (a, b), then there exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Equivalently,

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Remark 83.1.3 — This is the version sometimes called Cauchy's mean value theorem. It is the natural two-function generalization of the ordinary mean value theorem and is often the cleanest route to l'Hospital-type arguments.

§83.2 Continuity at a point

The note then returns to continuity.

Definition 83.2.1. Let f be a real-valued function defined in a neighborhood of $a \in \mathbb{R}$. The function f is continuous at a if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently, for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

Remark 83.2.2 — The intuitive statement is that when x is close to a, the value $f(x)$ is close to $f(a)$. The epsilon-delta form is the precise version of that intuition.

§83.3 How this note fits the previous one

The previous review note introduced limits, derivatives, big-O/small-o notation, and integration. This continuation adds the mean value theorem and continuity, which are the bridge between local derivative information and global information about a function on an interval.

The original text layer for this note was short and duplicated in places, so the missing proof details are filled in explicitly below. The key point is that both the ordinary mean value theorem and Cauchy's mean value theorem are consequences of Rolle's theorem: one subtracts a secant line in the first case, and a suitable linear combination of two functions in the second.

§83.4 Rolle's theorem as the engine

Theorem 83.4.1 (Rolle's theorem)

Let f be continuous on $[a, b]$, differentiable on (a, b), and suppose $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

The mean value theorem is usually proved from Rolle's theorem. Rolle says that if $f(a) = f(b)$, and f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ such that

$$f'(c) = 0.$$

Geometrically, a curve beginning and ending at the same height must have a horizontal tangent somewhere.

To prove the mean value theorem, subtract the secant line. Define

$$h(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then $h(a) = h(b)$. Rolle's theorem gives $h'(c) = 0$, which is exactly

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This proof explains the theorem better than the formula alone: the secant line is removed so that the endpoints line up.

§83.5 Cauchy's theorem and L'Hopital's rule

Theorem 83.5.1 (Cauchy's mean value theorem)

Let f, g be continuous on $[a, b]$ and differentiable on (a, b). If $g'(x) \neq 0$ on (a, b), then there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Cauchy's mean value theorem applies Rolle's theorem to a linear combination of two functions. It is the natural source of L'Hopital's rule. Suppose

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$$

and $g'(x) \neq 0$. Applying Cauchy's theorem to f and g on $[a, x]$ gives a point c between a and x such that

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}.$$

If $f'(c)/g'(c)$ tends to a limit as $x \to a$, then so does $f(x)/g(x)$. This derivation shows why L'Hopital's rule has hypotheses; it is not a symbolic cancellation rule.

§83.6 Uniform continuity

Continuity at each point allows δ to depend on both ε and the point. Uniform continuity allows δ to depend only on ε. On a compact interval, every continuous function is uniformly continuous.

This fact is often proved by contradiction using sequences, or by compactness using finite subcovers. It is the hidden reason many limiting arguments on closed intervals work smoothly.

§83.7 Further direction

Rolle, the mean value theorem, and Cauchy's mean value theorem are not isolated formulas. They are mechanisms for turning global information on an interval into local derivative information at some point. This is the central bridge of one-variable calculus: endpoint data forces a tangent slope somewhere in between.

§83.8 Mean value theorems as compactness plus local linearity

Theorem 83.8.1 (Mean value theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Rolle's theorem and the mean value theorem combine two ingredients: continuity on a compact interval and differentiability inside. Compactness gives a maximum and minimum. Differentiability says that an interior extremum has derivative zero.

The result
$$f(b) - f(a) = f'(c)(b - a)$$
means that some instantaneous slope equals the average slope. It is not a numerical accident; it is forced by the existence of extrema on a closed interval.

§83.9 Cauchy's theorem and ratios of differentials

Theorem 83.9.1 (Cauchy's theorem, ratio form)
Under the hypotheses of Cauchy's mean value theorem, some $c \in (a, b)$ satisfies
$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Cauchy's mean value theorem says that for suitable f, g, there is c with
$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$
This is the rigorous ancestor of many quotient-limit rules. L'Hopital's rule is a limiting version in which the interval endpoint moves toward a singular point.

Thus quotient estimates should be read as comparing two functions through their first-order changes.

§83.10 Uniform continuity and why closed intervals matter

Continuity at each point gives a local δ. Uniform continuity asks for one δ that works everywhere. Heine–Cantor says that a continuous function on a compact set is uniformly continuous.

The theorem explains why many epsilon proofs on closed intervals feel easier: compactness lets finitely many local controls become one global control.

§83.11 From local slopes to global control

The mean value theorem is the bridge from local derivative information to global estimates. Rolle gives the zero-slope engine, Cauchy compares two functions, L'Hopital studies singular ratios, and uniform continuity records when local control can be made global.

§83.12 Mean value theorem via compactness

Remark 83.12.1 — The compact interval hypothesis is not decorative: continuity on $[a, b]$ gives maxima and minima, and differentiability inside the interval turns an extremum into a zero derivative through Rolle's theorem.

Rolle's theorem and the mean value theorem are often memorized as calculus facts, but their proof rests on two deeper one-dimensional principles. First, a continuous function on a closed interval attains a maximum and a minimum; this is compactness. Second, differentiability turns information at an extremum into the equation $f'(\xi) = 0$.

For the usual mean value theorem, define

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then $g(a) = g(b)$, so Rolle's theorem gives $g'(\xi) = 0$, hence

$$f(b) - f(a) = f'(\xi)(b - a).$$

The theorem is therefore a statement that a function and its secant line must have a parallel tangent somewhere.

§83.13 Lipschitz estimates and uniqueness

A powerful corollary is the Lipschitz estimate. If $|f'(x)| \leq M$ on an interval, then

$$|f(x) - f(y)| \leq M|x - y|.$$

This turns derivative bounds into metric bounds.

This idea reappears in differential equations. Suppose $y' = F(t, y)$ and F is Lipschitz in y:

$$|F(t, u) - F(t, v)| \leq L|u - v|.$$

If y_1, y_2 are two solutions with the same initial value, then

$$|y_1(t) - y_2(t)| \leq \int_{t_0}^{t} L|y_1(s) - y_2(s)|\, ds.$$

Gronwall's inequality forces $y_1 = y_2$. Thus a tangent-slope theorem becomes the analytic mechanism behind uniqueness of solutions.

§83.14 Taylor formula with remainder

The mean value theorem also generates Taylor's theorem. For example,

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(\xi)}{2}(x - a)^2$$

for some ξ between a and x. The remainder measures the error of replacing a function by its local polynomial model. In numerical analysis this error term controls approximation; in geometry it is the beginning of the language of jets.

Composition and Elimination: Higher Derivatives, Determinants, and Schur Complements

N-20220409

§84.1 Two bookkeeping problems that should not be confused

This note contains two mathematical case studies. The first asks how repeated differentiation behaves under products and composition. Its indexing objects are binomial coefficients, integer partitions, and Bell polynomials. The second asks how linear maps compose and how internal variables can be eliminated from a block system. Its indexing objects are contracted matrix indices and Schur complements.

These subjects are not instances of one theorem. Their genuine common feature is methodological: an apparently unmanageable expression becomes transparent after identifying what is being composed, what is being summed over, and what data remain after internal choices have been removed. Keeping the two cases separate prevents a vague analogy from replacing a proof.

§84.2 Repeated derivations produce the higher Leibniz rule

Let D be a derivation on a commutative algebra, so

$$D(ab) = D(a)b + aD(b).$$

Applying D repeatedly gives

$$D^2(ab) = D^2(a)b + 2D(a)D(b) + aD^2(b),$$

and

$$D^3(ab) = D^3(a)b + 3D^2(a)D(b) + 3D(a)D^2(b) + aD^3(b).$$

The coefficients are not accidental.

Theorem 84.2.1 (Higher Leibniz rule)

For every integer $n \geq 0$,

$$D^n(ab) = \sum_{k=0}^{n} \binom{n}{k} D^k(a)D^{n-k}(b).$$

Proof. The result is clear for $n = 0$. Assume it holds for n. Differentiating once more gives

$$D^{n+1}(ab) = \sum_{k=0}^{n} \binom{n}{k} \big(D^{k+1}(a)D^{n-k}(b) + D^k(a)D^{n-k+1}(b)\big).$$

After shifting the index in the first sum, the coefficient of $D^k(a)D^{n+1-k}(b)$ becomes

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}.$$

This proves the induction step. □

There is also a combinatorial reading. During n differentiations, each derivative must land on either the first factor or the second. Choosing the k stages that land on a gives $\binom{n}{k}$. The formula is therefore a count of derivative-placement histories.

For a product of r factors, the same argument gives the multinomial rule

$$D^n(a_1 \cdots a_r) = \sum_{\alpha_1+\cdots+\alpha_r=n} \frac{n!}{\alpha_1! \cdots \alpha_r!} \prod_{j=1}^{r} D^{\alpha_j}(a_j).$$

This form will reappear when differentiating a determinant row by row.

§84.3 Faà di Bruno records partitions of derivative labels

The ordinary chain rule has one outer derivative and one inner derivative:

$$(f \circ g)' = f'(g)g'.$$

At second and third order,

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g'',$$
$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The coefficients and monomials are controlled by set partitions. Think of the n derivative operations as carrying labels $1, \dots, n$. Labels in the same block ultimately combine into one derivative of g; the number of blocks tells us which derivative of f appears.

For nonnegative integers $k_1, \dots, k_n$, impose

$$k_1 + 2k_2 + \cdots + nk_n = n.$$

Here k_j is the number of blocks of size j. The total number of blocks is

$$m = k_1 + \cdots + k_n.$$

The number of set partitions with these block sizes is

$$\frac{n!}{k_1! \cdots k_n! (1!)^{k_1} \cdots (n!)^{k_n}}.$$

This gives the one-variable Faà di Bruno formula.

Theorem 84.3.1 (Faà di Bruno)

For sufficiently differentiable functions f and g,

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\substack{k_1,\dots,k_n \ge 0 \\ k_1+2k_2+\cdots+nk_n=n}} \frac{n!}{k_1! \cdots k_n!} f^{(k_1+\cdots+k_n)}(g(x)) \prod_{j=1}^{n} \left(\frac{g^{(j)}(x)}{j!} \right)^{k_j}.$$

The same information is often compressed using the partial exponential Bell polynomials

$$B_{n,m}(x_1,\dots,x_{n-m+1}) = \sum \frac{n!}{k_1!\cdots k_{n-m+1}!} \prod_{j=1}^{n-m+1} \left(\frac{x_j}{j!}\right)^{k_j},$$

where the sum satisfies

$$\sum_j k_j = m, \qquad \sum_j jk_j = n.$$

Then

$$(f\circ g)^{(n)} = \sum_{m=1}^{n} f^{(m)}(g) B_{n,m}(g', g'', \dots, g^{(n-m+1)}).$$

Bell polynomials separate the combinatorics of composition from the particular functions being composed.

§84.4 The fourth derivative as a complete partition calculation

The integer partitions of 4 are

$$1+1+1+1, \qquad 2+1+1, \qquad 2+2, \qquad 3+1, \qquad 4.$$

They correspond respectively to the block-size data

$$(k_1,k_2,k_3,k_4) = (4,0,0,0), (2,1,0,0), (0,2,0,0), (1,0,1,0), (0,0,0,1).$$

Substitution into Faà di Bruno gives

$$(f\circ g)^{(4)} = f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2 g'' + 3f''(g)(g'')^2 + 4f''(g)g'g''' + f'(g)g^{(4)}.$$

The coefficients $1, 6, 3, 4, 1$ count set partitions of the five displayed block types. For example, the type $2+1+1$ has

$$\frac{4!}{1!2!(2!)^1} = 6$$

partitions, while the type $2+2$ has

$$\frac{4!}{2!(2!)^2} = 3.$$

Take

$$h(x) = e^{\sin x}.$$

Since every derivative of $f(u) = e^u$ equals e^u,

$$h^{(4)}(x) = e^{\sin x}\big((\cos x)^4 + 6(\cos x)^2(-\sin x) + 3(-\sin x)^2 + 4(\cos x)(-\cos x) + \sin x\big).$$

At $x = 0$, this reduces to

$$h^{(4)}(0) = 1 - 4 = -3.$$

The Taylor series confirms the calculation:

$$e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + O(x^5),$$

so the fourth derivative at the origin is

$$4!\left(-\frac{1}{8}\right) = -3.$$

This check is useful because a missing block type in Faà di Bruno usually appears as an incorrect Taylor coefficient.

§84.5 Jets turn higher differentials into truncated Taylor algebra

Leibniz notation becomes subtle beyond first order because the symbols $dx, d^2x, \dots$ do not behave like ordinary powers of one number. The clean way to see what they encode is to work with Taylor polynomials modulo terms that are too small to matter at the chosen order.

Fix a point a. Two smooth functions have the same n-jet at a when their derivatives through order n agree there. The jet can be represented by the truncated Taylor polynomial

$$j_a^n f(h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \cdots + \frac{f^{(n)}(a)}{n!}h^n,$$

where calculations are performed modulo h^{n+1}. In other words, one works in the truncated polynomial algebra

$$\mathbb{R}[h]/(h^{n+1}).$$

Terms of degree greater than n are discarded because they cannot affect derivatives of order at most n.

This makes composition completely explicit. Suppose $g(a) = b$, and write the second-order jets as

$$j_a^2 g(h) = b + b_1 h + \frac{b_2}{2}h^2,$$
$$j_b^2 f(u) = c + c_1 u + \frac{c_2}{2}u^2.$$

Here

$$b_1 = g'(a), \qquad b_2 = g''(a), \qquad c_1 = f'(b), \qquad c_2 = f''(b).$$

Set

$$u = b_1 h + \frac{b_2}{2}h^2.$$

Since

$$u^2 = b_1^2 h^2 \pmod{h^3},$$

substitution gives

$$\begin{aligned} j_a^2(f \circ g)(h) &\equiv c + c_1 u + \frac{c_2}{2}u^2 \pmod{h^3} \\ &= c + c_1 b_1 h + \frac{c_2 b_1^2 + c_1 b_2}{2}h^2. \end{aligned}$$

Reading off coefficients recovers

$$(f \circ g)'(a) = f'(b)g'(a)$$

and

$$(f \circ g)''(a) = f''(b)(g'(a))^2 + f'(b)g''(a).$$

The second-order chain rule is therefore ordinary polynomial substitution followed by truncation.

At third order, write

$$j_a^3 g(h) = b + b_1 h + \frac{b_2}{2}h^2 + \frac{b_3}{6}h^3$$

and

$$j_b^3 f(u) = c + c_1 u + \frac{c_2}{2}u^2 + \frac{c_3}{6}u^3.$$

Modulo h^4,

$$u^2 = b_1^2 h^2 + b_1 b_2 h^3, \qquad u^3 = b_1^3 h^3.$$

Hence the cubic coefficient of the composition is

$$\frac{c_3 b_1^3 + 3c_2 b_1 b_2 + c_1 b_3}{6},$$

so

$$(f \circ g)'''(a) = f'''(b)(g'(a))^3 + 3f''(b)g'(a)g''(a) + f'(b)g'''(a).$$

The coefficient 3 is no longer mysterious: it comes from the two cross terms in u^2 together with the factorial normalization in the Taylor coefficients.

For a concrete check, take $f(u) = e^u$, $g(x) = \sin x$, and expand at 0. The relevant jets are

$$j_0^3 \sin(h) = h - \frac{h^3}{6}$$

and

$$j_0^3 e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6}.$$

Substitution gives

$$j_0^3(e^{\sin x})(h) = 1 + h + \frac{h^2}{2},$$

because the cubic contributions $-h^3/6$ and $+h^3/6$ cancel. Thus

$$(e^{\sin x})'''\big|_{x=0} = 0,$$

which agrees with the third-order chain rule.

The differential notation is a compressed version of this bookkeeping. If $y = f(x)$, then

$$dy = f'(x)\,dx$$

and

$$d^2y = f''(x)(dx)^2 + f'(x)d^2x.$$

When x itself depends on a parameter t, the first two pieces of its jet are

$$dx = x'(t)dt, \qquad d^2x = x''(t)(dt)^2.$$

Therefore

$$d^2y = \left(f''(x)(x')^2 + f'(x)x''\right)(dt)^2.$$

Only when x is chosen as an affine coordinate in the parameter does d^2x vanish. Under a nonlinear reparametrization it carries genuine second-order information. Jets explain precisely why treating d^2x as though it were merely $(dx)^2$ loses part of the chain rule.

§84.6 Parametric differentiation is repeated application of one operator

Let a regular plane curve be given by

$$x = x(t), \qquad y = y(t),$$

with $x'(t) \neq 0$. Differentiation with respect to x means applying

$$\frac{d}{dx} = \frac{1}{x'(t)}\frac{d}{dt}.$$

Hence

$$\frac{dy}{dx} = \frac{y'}{x'},$$

and

$$\frac{d^2y}{dx^2} = \frac{x'y'' - y'x''}{(x')^3}.$$

The numerator is a determinant:

$$x'y'' - y'x'' = \det \begin{pmatrix} x' & y' \\ x'' & y'' \end{pmatrix}.$$

Applying the operator once more gives

$$\frac{d^3y}{dx^3} = \frac{(x')^2y''' - 3x'x''y'' + (3(x'')^2 - x'x''')y'}{(x')^5}.$$

The growing expression is another composition problem: one repeatedly composes multiplication by $1/x'$ with d/dt.

For the semicubical parabola

$$x = t^2, \qquad y = t^3, \qquad t > 0,$$

we obtain

$$\frac{dy}{dx} = \frac{3t}{2}, \qquad \frac{d^2y}{dx^2} = \frac{3}{4t}, \qquad \frac{d^3y}{dx^3} = -\frac{3}{8t^3}.$$

Since $y = x^{3/2}$ on this branch, direct differentiation gives exactly the same values. The singular behavior as $t \to 0$ records the failure of x to be a valid local coordinate at the cusp.

§84.7 Determinants inherit a multinomial Leibniz rule

Write an $m \times m$ matrix-valued function by rows:

$$A(t) = \begin{pmatrix} r_1(t) \\ \vdots \\ r_m(t) \end{pmatrix}.$$

Because the determinant is multilinear in its rows,

$$\frac{d}{dt} \det A(t) = \sum_{i=1}^{m} \det(r_1, \dots, r_i', \dots, r_m).$$

Repeated differentiation and the multinomial Leibniz rule give

$$\frac{d^n}{dt^n} \det A(t) = \sum_{\alpha_1 + \cdots + \alpha_m = n} \frac{n!}{\alpha_1! \cdots \alpha_m!} \det\big(r_1^{(\alpha_1)}, \dots, r_m^{(\alpha_m)}\big).$$

Thus the higher derivative of a determinant is organized by distributing n derivative operations among m rows.

At first order there is also a basis-independent formula. Let $\operatorname{adj}(A)$ denote the adjugate matrix. Cofactor expansion gives

$$\frac{d}{dt} \det A(t) = \operatorname{tr}\big(\operatorname{adj}(A(t))A'(t)\big).$$

If $A(t)$ is invertible, then

$$\operatorname{adj}(A) = \det(A)A^{-1},$$

so

$$\boxed{\frac{d}{dt}\det A(t) = \det A(t)\operatorname{tr}\big(A(t)^{-1}A'(t)\big).}$$

This is Jacobi's formula. Equivalently,

$$\frac{d}{dt}\log\det A(t) = \operatorname{tr}\big(A(t)^{-1}A'(t)\big)$$

whenever the logarithm is defined.

For

$$A(t) = \begin{pmatrix} e^t & t \\ 0 & 1+t \end{pmatrix},$$

we have

$$\det A(t) = e^t(1+t),$$

and therefore

$$\frac{d}{dt}\det A(t) = e^t(2+t).$$

Jacobi's formula gives the same result because

$$\operatorname{tr}(A^{-1}A') = 1 + \frac{1}{1+t}.$$

Multiplication by $\det A = e^t(1+t)$ yields $e^t(2+t)$.

§84.8 Matrix multiplication is composition with one index hidden

Let

$$B : U \to V, \qquad A : V \to W$$

be linear maps. After bases are chosen, their composition is represented by the product AB. Its entries are

$$(AB)_{ij} = \sum_k A_{ik}B_{kj}.$$

The index k is internal: it labels the intermediate coordinate in V, and summation removes it from the final answer. This is a contraction between the input index of A and the output index of B.

Associativity follows because finite sums may be regrouped:

$$\begin{aligned}((AB)C)_{ij} &= \sum_k \left(\sum_\ell A_{i\ell}B_{\ell k}\right) C_{kj} \\ &= \sum_\ell A_{i\ell}\left(\sum_k B_{\ell k}C_{kj}\right) \\ &= (A(BC))_{ij}.\end{aligned}$$

Commutativity fails because reversing the order reverses the composition. Even when both products exist, AB and BA generally represent maps on different spaces or different operations on the same space.

Block multiplication is the same rule with each scalar entry replaced by a compatible linear map. For example,

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix} = \begin{pmatrix} Ax + Bz \\ Cx + Dz \end{pmatrix}.$$

This notation becomes powerful when one block of variables is to be eliminated.

§84.9 Schur complements are exact block elimination

Consider

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A is invertible. Multiplying on the left by a block elimination matrix gives

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & S \end{pmatrix},$$

where

$$\boxed{S = D - CA^{-1}B}$$

is the Schur complement of A in M.

This identity immediately gives

$$\det M = \det A \, \det S.$$

It also gives a block solve. For

$$Ax + Bz = b, \qquad Cx + Dz = c,$$

the first equation gives

$$x = A^{-1}(b - Bz).$$

Substitution into the second equation produces the reduced system

$$Sz = c - CA^{-1}b.$$

After solving for z, one recovers x by back-substitution. The Schur complement is exactly what remains after the variable x has been eliminated.

§84.10 A complete block-elimination example

Take

$$M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad D = (3).$$

Since

$$A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

we obtain

$$CA^{-1}B = \frac{2}{3}, \qquad S = 3 - \frac{2}{3} = \frac{7}{3}.$$

Therefore

$$\det M = \det A\, S = 3 \cdot \frac{7}{3} = 7.$$

Now solve

$$M \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The reduced right-hand side is

$$2 - CA^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2 - \frac{2}{3} = \frac{4}{3}.$$

Hence

$$\frac{7}{3} z = \frac{4}{3}, \qquad z = \frac{4}{7}.$$

Back-substitution gives

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \frac{4}{7} \right) = A^{-1} \begin{pmatrix} 3/7 \\ 0 \end{pmatrix} = \begin{pmatrix} 2/7 \\ -1/7 \end{pmatrix}.$$

A direct multiplication verifies

$$M \begin{pmatrix} 2/7 \\ -1/7 \\ 4/7 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The calculation shows concretely that the reduced scalar $7/3$ contains the full effect of the eliminated two-dimensional block.

§84.11 Block inverse and positive definiteness

If both A and S are invertible, block factorization gives

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}.$$

The formula is best remembered as elimination followed by back-substitution, not as four unrelated blocks.

For a symmetric block matrix

$$M = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}$$

with A positive definite, completing the square gives

$$\begin{aligned} \begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix} &= (x + A^{-1}Bz)^T A (x + A^{-1}Bz) \\ &\quad + z^T (D - B^T A^{-1} B) z. \end{aligned}$$

Consequently,

$$M \succ 0 \quad \Longleftrightarrow \quad A \succ 0 \text{ and } D - B^T A^{-1} B \succ 0.$$

Thus the same Schur complement controls elimination, determinants, inverses, and positivity. In statistics it also appears as a conditional covariance; in constrained optimization it appears after eliminating primal or dual variables.

This block calculation also clarifies the limited but useful comparison with the first half of the note. Repeated differentiation exposes hidden partitions of derivative labels; matrix composition sums over an intermediate coordinate; block elimination removes internal variables and records their net influence in a Schur complement. The operations are not instances of one theorem, but each becomes manageable only after the internal choices are represented exactly. For Faà di Bruno the surviving data are derivatives indexed by partitions, for matrix multiplication they are the external row and column indices, and for block elimination they form the reduced operator $D - CA^{-1}B$. The common method is precise bookkeeping, not a claim that calculus and matrix algebra are secretly the same subject.

Techniques of Integration: Partial Fractions, Parts, and Substitution

N-20220628

§85.1 Linearity

The note begins with the linearity of the indefinite integral:

$$\int (f(x) + g(x) - h(x))\,dx = \int f(x)\,dx + \int g(x)\,dx - \int h(x)\,dx.$$

This is the integration analogue of the linearity of differentiation.

§85.2 Partial fractions

The first worked example is

$$\int \frac{dx}{x^2 - a^2}.$$

Since

$$\frac{1}{x^2 - a^2} = \frac{1}{(x-a)(x+a)} = \frac{1}{2a}\left(\frac{1}{x-a} - \frac{1}{x+a}\right),$$

we get

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$$

A general quadratic denominator is handled by completing the square. For

$$\int \frac{mx+n}{x^2+px+q}\,dx,$$

put

$$t = x + \frac{p}{2}, \qquad x^2 + px + q = t^2 + q - \frac{p^2}{4}.$$

Then write $mx + n = At + B$. The integral splits into

$$A\int \frac{t}{t^2+a^2}\,dt + B\int \frac{dt}{t^2+a^2}$$

or, if the constant term is negative,

$$A\int \frac{t}{t^2-a^2}\,dt + B\int \frac{dt}{t^2-a^2}.$$

This produces logarithms and arctangents depending on the sign of $q - p^2/4$.

A page warns about a common partial-fraction mistake: repeated factors must be treated separately. For example,

$$\int \frac{3x+5}{x^3 - x^2 - x + 1}\,dx$$

has denominator

$$x^3 - x^2 - x + 1 = (x+1)(x-1)^2.$$

Thus the decomposition must be

$$\frac{3x+5}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}.$$

Solving gives

$$A = \frac{1}{2}, \qquad B = -\frac{1}{2}, \qquad C = 4,$$

so

$$\int \frac{3x+5}{x^3 - x^2 - x + 1}\,dx = \frac{1}{2}\ln|x+1| - rac12\ln|x-1| - \frac{4}{x-1} + C.$$

§85.3 Integration by parts

The integration-by-parts formula is

$$\int u\,dv = uv - \int v\,du.$$

For example,

$$\int \ln x\,dx = x\ln x - x + C.$$

A higher-order version repeatedly integrates by parts:

$$\int u\,v^{(n+1)}\,dx = uv^{(n)} - u'v^{(n-1)} + \cdots + (-1)^n u^{(n)}v + (-1)^{n+1}\int u^{(n+1)}v\,dx.$$

This is especially useful for

$$\int P(x)e^x\,dx, \qquad \int P(x)\sin x\,dx, \qquad \int P(x)\cos x\,dx,$$

where P is a polynomial.

One worked example is

$$\int (2x^3 + 3x^2 + 4x + 5)e^x\,dx.$$

Repeated integration by parts gives

$$e^x(2x^3 - 3x^2 + 10x - 5) + C.$$

§85.4 Substitution

The substitution rule is

$$x = \rho(t), \qquad dx = \rho'(t)\,dt, \qquad \int f(x)\,dx = \int f(\rho(t))\rho'(t)\,dt.$$

For instance,

$$\int \sin^3 x\cos x\,dx$$

is solved by setting $t = \sin x$, giving

$$\int t^3\,dt = \frac{t^4}{4} + C = \frac{\sin^4 x}{4} + C.$$

Similarly,

$$\int \sin^2 x\,dx$$

can be done by the trigonometric identity

$$\sin^2 x = \frac{1-\cos 2x}{2},$$

so

$$\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

A trigonometric substitution example is

$$\int \sqrt{a^2-x^2}\,dx.$$

Let $x = a\sin t$. Then $dx = a\cos t\,dt$, and the integral becomes

$$a^2 \int \cos^2 t\,dt.$$

Hence

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C.$$

The standard substitutions are:

$$\sqrt{a^2-x^2} : x = a\sin t, \qquad \sqrt{a^2+x^2} : x = a\tan t, \qquad \sqrt{x^2-a^2} : x = a\sec t.$$

§85.5 Partial fractions as algebra before calculus

Partial fractions are not an integration trick alone. They first require algebraic decomposition. For example, if

$$\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b},$$

then multiplying by the denominator gives an identity of polynomials. The coefficients A, B are determined before any integration happens.

This is why one should factor the denominator and compare degrees first. If the rational function is improper, divide polynomials before decomposing.

§85.6 Integration by parts as product rule reversed

Integration by parts comes from

$$(uv)' = u'v + uv'.$$

Integrating gives

$$\int u\,dv = uv - \int v\,du.$$

The choice of u and dv is guided by which part simplifies after differentiation and which part is easy to integrate. Logarithms and inverse trigonometric functions often become simpler after differentiation.

§85.7 A second lens

The note is an inventory of the classical calculus toolkit. Partial fractions exploit algebraic factorization; integration by parts exploits the product rule; substitution exploits the chain rule. In a more advanced language, all three are ways of choosing coordinates and decompositions so that the differential form becomes elementary.

§85.8 Partial fractions as algebra of poles

Partial fractions decompose a rational function according to its poles. For example, a factor $(x-a)^m$ contributes terms

$$\frac{c_1}{x-a} + \frac{c_2}{(x-a)^2} + \cdots + \frac{c_m}{(x-a)^m}.$$

This is the real-variable shadow of meromorphic function theory, where principal parts record local behavior near poles.

§85.9 Integration by parts as adjointness

The formula

$$\int_a^b u\,v' = uv\big|_a^b - \int_a^b u'v$$

transfers a derivative from one factor to another. In operator language, differentiation is skew-adjoint up to boundary terms. This viewpoint becomes central in Fourier analysis, weak derivatives, and PDE.

§85.10 Substitution as pullback of one-forms

For $x = \phi(t)$, substitution says

$$\int f(x)\,dx = \int f(\phi(t))\phi'(t)\,dt.$$

The invariant object is the one-form $f(x)\,dx$. Under the map ϕ, it pulls back to $f(\phi(t))\phi'(t)\,dt$.

This explains why the derivative factor is necessary: integration is coordinate-independent only when the differential transforms correctly.

§85.11 Integration techniques as structural transformations

The standard integration techniques are structural transformations. Partial fractions isolate poles, integration by parts moves derivatives with boundary terms, and substitution pulls back differential forms. The goal is not formula memorization but recognizing which structure simplifies the integrand.

§85.12 Integration as reverse differentiation

The integration techniques in this note are best remembered as reversed differentiation rules. Partial fractions reverse the derivative of logarithms and arctangents. Integration by parts reverses the product rule. Substitution reverses the chain rule. Trigonometric substitution is a geometric way of replacing a radical by an identity such as $1-\sin^2 t = \cos^2 t$.

A good workflow is therefore: first simplify the algebraic shape of the integrand; next ask which differentiation rule could have produced it; only then choose a technique. This prevents the common mistake of trying substitution, parts, and partial fractions randomly.

§85.13 Substitution as pullback

The elementary substitution rule

$$\int f(g(t))g'(t)\,dt = \int f(x)\,dx$$

is the one-dimensional form of pullback. If $x = g(t)$, then $dx = g'(t)dt$, so the differential form $f(x)dx$ pulls back to

$$g^*(f(x)dx) = f(g(t))g'(t)dt.$$

The derivative is not an artificial correction factor; it is how lengths are transformed by the map g.

In several variables the same principle becomes the Jacobian determinant:

$$dx\,dy = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

Thus elementary substitution is the first instance of the change-of-variables theorem.

§85.14 Integration by parts and adjoints

Integration by parts comes from the product rule:

$$\int_a^b u'v\,dx = [uv]_a^b - \int_a^b uv'\,dx.$$

This formula is also the origin of adjoint differential operators. For $D = d/dx$,

$$\int_a^b (Du)v\,dx = [uv]_a^b - \int_a^b u(Dv)\,dx.$$

Ignoring boundary terms, D is formally skew-adjoint. This observation is central in Sturm–Liouville theory, Fourier analysis, and PDEs.

§85.15 Partial fractions and residues

Partial fractions decompose a rational function into simple local pieces around its poles. In complex analysis, the coefficient of $(z-a)^{-1}$ is the residue, and the residue theorem says

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{a\ inside \gamma} \mathrm{Res}(f,a).$$

Thus the elementary partial-fraction method is a real-variable shadow of a deeper principle: rational functions are controlled by their singularities.

Trigonometric Limits, Derivatives, and Inverse Trigonometric Examples

N-20221107

§86.1 Radians are built into the derivative

The familiar formula

$$\frac{d}{dx}\sin x = \cos x$$

is true only when the variable is measured in radians. This is not a typographical convention. A derivative compares a small change in function value with a small change in the input, and radian measure is the angle coordinate whose small increment agrees with arc length on the unit circle.

To see what changes in degrees, write

$$\sin_{\text{deg}}(x) = \sin\left(\frac{\pi x}{180}\right),$$

where the sine on the right uses radians. The chain rule gives

$$\frac{d}{dx}\sin_{\text{deg}}(x) = \frac{\pi}{180}\cos\left(\frac{\pi x}{180}\right).$$

The extra factor is the derivative of the unit conversion. Radians are the unique constant multiple of degrees for which that factor becomes one.

The analytical reason is the limit

$$\lim_{h\to 0}\frac{\sin h}{h} = 1.$$

The rest of the trigonometric derivative table can be reconstructed from this limit, the addition formulas, and ordinary differentiation rules.

§86.2 The circle squeeze proves the first basic limit

Let $0 < h < \pi/2$. On the unit circle, compare the area of the inscribed triangle, the circular sector, and the tangent triangle. Their areas are respectively

$$\frac{1}{2}\sin h, \qquad \frac{1}{2}h, \qquad \frac{1}{2}\tan h.$$

Therefore

$$\sin h < h < \tan h.$$

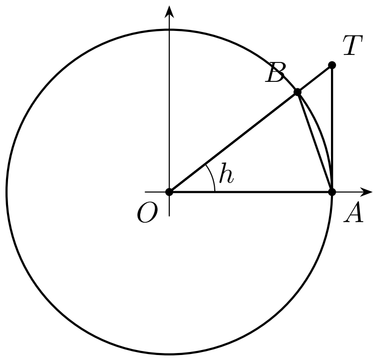


Figure 86.1: The area comparison gives $\sin h < h < \tan h$ for a positive acute angle measured in radians.

Since $\sin h > 0$, division gives

$$1 < \frac{h}{\sin h} < \frac{1}{\cos h},$$

or equivalently

$$\cos h < \frac{\sin h}{h} < 1.$$

As $h \to 0^+$, both outer terms tend to 1. Hence

$$\lim_{h \to 0^+} \frac{\sin h}{h} = 1.$$

The quotient is even because

$$\frac{\sin(-h)}{-h} = \frac{\sin h}{h},$$

so the same limit holds from the left:

$$\boxed{\lim_{h \to 0} \frac{\sin h}{h} = 1.}$$

The conclusion depends on radian measure at the area-of-sector step. In degrees, the sector area would be proportional to $\pi h/180$, and the limit would be $\pi/180$.

§86.3 The second limit and the correct use of small-angle equivalence

The identity

$$1 - \cos h = 2 \sin^2 \frac{h}{2}$$

gives

$$\frac{1 - \cos h}{h} = \frac{h}{2} \left(\frac{\sin(h/2)}{h/2} \right)^2.$$

The factor in parentheses tends to 1, while $h/2 \to 0$. Therefore

$$\boxed{\lim_{h \to 0} \frac{1 - \cos h}{h} = 0.}$$

Dividing by one more factor of h gives

$$\frac{1-\cos h}{h^2} = \frac{1}{2}\left(\frac{\sin(h/2)}{h/2}\right)^2,$$

so

$$\boxed{\lim_{h\to 0}\frac{1-\cos h}{h^2} = \frac{1}{2}.}$$

These limits are often summarized as

$$\sin h \sim h, \qquad 1-\cos h \sim \frac{h^2}{2} \qquad (h\to 0).$$

The symbol $f \sim g$ means $f/g \to 1$. It can safely replace factors inside products and quotients when the surrounding expression respects the limit. It cannot be substituted blindly inside a cancellation.

For example,

$$\frac{\sin h - h}{h}$$

cannot be evaluated by replacing $\sin h$ with h, because the leading terms cancel and the next-order behavior matters. Small-angle equivalence records the first nonzero scale, not a full Taylor expansion.

§86.4 Sine and cosine emerge from their addition laws

Using

$$\sin(x+h) = \sin x\cos h + \cos x \sin h,$$

we obtain

$$\frac{\sin(x+h)-\sin x}{h} = \sin x\frac{\cos h - 1}{h} + \cos x\frac{\sin h}{h}.$$

The two basic limits give

$$\lim_{h\to 0}\frac{\cos h - 1}{h} = 0, \qquad \lim_{h\to 0}\frac{\sin h}{h} = 1.$$

Therefore

$$\boxed{(\sin x)' = \cos x.}$$

Similarly,

$$\cos(x+h) = \cos x\cos h - \sin x\sin h,$$

so

$$\frac{\cos(x+h)-\cos x}{h} = \cos x\frac{\cos h - 1}{h} - \sin x\frac{\sin h}{h}.$$

Hence

$$\boxed{(\cos x)' = -\sin x.}$$

The derivation exhibits a tight dependency chain:

$$\text{radian geometry} \Longrightarrow \frac{\sin h}{h} \to 1 \Longrightarrow \frac{1-\cos h}{h} \to 0 \Longrightarrow \begin{cases}(\sin x)' = \cos x,\\ (\cos x)' = -\sin x.\end{cases}$$

The addition formulas are also required. A separate note compares their geometric, matrix, and analytic proofs; here they are part of the input to the differentiation argument.

Differentiating once more gives the coupled differential equations

$$(\sin x)'' = -\sin x, \qquad (\cos x)'' = -\cos x.$$

Thus sine and cosine are not merely periodic graphs. They are the two normalized solutions of

$$y'' + y = 0$$

with initial data

$$\cos 0 = 1, \quad \cos' 0 = 0, \qquad \sin 0 = 0, \quad \sin' 0 = 1.$$

§86.5 The other four derivatives are generated, not memorized

Once sine and cosine are known, quotient and reciprocal rules produce the remaining functions. Where $\cos x \neq 0$,

$$\tan x = \frac{\sin x}{\cos x},$$

so

$$(\tan x)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \sec^2 x.$$

Where $\sin x \neq 0$,

$$(\cot x)' = \left(\frac{\cos x}{\sin x}\right)' = -\csc^2 x.$$

The reciprocal rules give

$$(\sec x)' = \left(\frac{1}{\cos x}\right)' = \sec x \tan x,$$

$$(\csc x)' = \left(\frac{1}{\sin x}\right)' = -\csc x \cot x.$$

The domains belong to the formulas:

Function	Domain	Derivative	Excluded points
$\sin x$	$\mathbb{R}$	$\cos x$	none
$\cos x$	$\mathbb{R}$	$-\sin x$	none
$\tan x$	$\cos x \neq 0$	$\sec^2 x$	$\pi/2 + k\pi$
$\cot x$	$\sin x \neq 0$	$-\csc^2 x$	$k\pi$
$\sec x$	$\cos x \neq 0$	$\sec x \tan x$	$\pi/2 + k\pi$
$\csc x$	$\sin x \neq 0$	$-\csc x \cot x$	$k\pi$

A derivative formula does not repair a missing function value. At a pole of tangent or secant, the function itself is undefined, so there is no ordinary real derivative to discuss.

§86.6 Inverse functions require branches

A periodic function cannot have a global inverse on all of $\mathbb{R}$. Inverse trigonometric functions are obtained by restricting to intervals on which the original function is one-to-one:

Inverse	Domain	Principal-value range	Derivative domain
$\arcsin x$	$[-1,1]$	$[-\pi/2,\pi/2]$	$(-1,1)$
$\arccos x$	$[-1,1]$	$[0,\pi]$	$(-1,1)$
$\arctan x$	$\mathbb{R}$	$(-\pi/2,\pi/2)$	$\mathbb{R}$

Let

$$y = \arcsin x.$$

Then

$$x = \sin y, \qquad y \in [-\pi/2, \pi/2].$$

For $|x| < 1$, the inverse function theorem applies because

$$\cos y \neq 0.$$

Differentiating gives

$$1 = \cos y\, y'.$$

On the chosen branch, $\cos y \geq 0$, and

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

Therefore

$$\boxed{(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}}.}$$

The positive square root is forced by the principal range. Without the branch information, the sign would be ambiguous.

For $y = \arccos x$, the principal branch lies in $[0, \pi]$, where $\sin y \geq 0$. Differentiating $x = \cos y$ gives

$$\boxed{(\arccos x)' = -\frac{1}{\sqrt{1 - x^2}}.}$$

For $y = \arctan x$,

$$x = \tan y, \qquad y \in (-\pi/2, \pi/2),$$

so

$$1 = \sec^2 y\, y'$$

and

$$\boxed{(\arctan x)' = \frac{1}{1 + x^2}.}$$

§86.7 Endpoint blow-up is a failure of local invertibility

At $x = 1$, the derivative formula for $\arcsin x$ diverges. This is not an algebraic accident. The original sine curve has a horizontal tangent at $y = \pi/2$:

$$\cos \frac{\pi}{2} = 0.$$

The inverse function theorem fails precisely when the derivative of the original map vanishes.

The local shape can be quantified. Put

$$y = \frac{\pi}{2} - t, \qquad t \to 0^+.$$

Then

$$\sin y = \cos t = 1 - \frac{t^2}{2} + o(t^2).$$

If $x = \sin y$, this gives

$$1 - x \sim \frac{t^2}{2}.$$

Therefore

$$\boxed{\frac{\pi}{2} - \arcsin x \sim \sqrt{2(1-x)} \qquad (x \to 1^-).}$$

Differentiating the square-root model predicts the vertical tangent:

$$(\arcsin x)' \sim \frac{1}{\sqrt{2(1-x)}}.$$

The same phenomenon occurs near -1.

The square root in this asymptotic is also what appears when the inverse is continued to complex arguments. Starting from

$$z = \sin w = \frac{e^{iw} - e^{-iw}}{2i},$$

set $u = e^{iw}$. Multiplying by $2iu$ gives the quadratic equation

$$u^2 - 2izu - 1 = 0.$$

Hence

$$u = iz \pm \sqrt{1 - z^2},$$

and solving for w gives a local inverse of the form

$$\boxed{\arcsin z = -i \operatorname{Log}\left(iz + \sqrt{1 - z^2}\right).}$$

Neither the logarithm nor the square root has a single-valued inverse on all of $\mathbb{C} \setminus \{0\}$, so this formula requires compatible branch choices. The points $z = \pm 1$ are special because the two roots of the quadratic merge there:

$$1 - z^2 = 0.$$

They are square-root branch points, and the real vertical tangents are their visible restriction to the interval $[-1, 1]$. Differentiating on any consistent local branch recovers

$$\frac{d}{dz} \arcsin z = \frac{1}{\sqrt{1 - z^2}},$$

with the sign determined by the chosen branch. Thus the real endpoint blow-up, the inverse-function theorem, and the complex branch structure are three descriptions of the same local degeneracy.

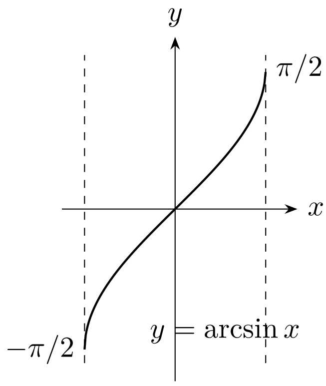


Figure 86.2: The principal inverse has vertical tangents at $x = \pm 1$, where the derivative of sine on the corresponding branch vanishes.

§86.8 Three examples that use different ideas

First consider the composite expression

$$f(x) = \arcsin(2x - x^2).$$

Its real domain is determined before differentiation:

$$-1 \le 2x - x^2 \le 1.$$

Since

$$2x - x^2 = 1 - (x-1)^2 \le 1,$$

the lower inequality gives

$$(x-1)^2 \le 2.$$

Thus

$$x \in [1 - \sqrt{2}, 1 + \sqrt{2}].$$

At interior points where $|2x - x^2| < 1$, the chain rule gives

$$f'(x) = \frac{2 - 2x}{\sqrt{1 - (2x - x^2)^2}}.$$

The apparent zero-over-zero at $x = 1$ should be simplified from the original function or treated by a local limit; blindly substituting into the unsimplified derivative formula hides the cusp-like branch behavior.

For an implicit example, let

$$\sin(x + y) = x.$$

Differentiating along a differentiable solution curve gives

$$\cos(x + y)(1 + y') = 1.$$

Where $\cos(x + y) \ne 0$,

$$y' = \sec(x + y) - 1.$$

The excluded points are exactly where the implicit function theorem cannot solve locally for y as a function of x.

For a parametric example, take

$$x(t) = \cos^3 t, \qquad y(t) = \sin^3 t.$$

Where $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{3\sin^2 t \cos t}{-3\cos^2 t \sin t} = -\tan t.$$

At $t = 0$, both numerator and denominator of the ratio vanish, so the simplified expression must be interpreted by a limit or by local expansion. Parametric differentiation is not permission to divide by a zero velocity component.

A final warning concerns units. If

$$y = \sin(30x^\circ),$$

then in radian notation

$$y = \sin\left(\frac{\pi x}{6}\right),$$

so

$$y' = \frac{\pi}{6}\cos\left(\frac{\pi x}{6}\right).$$

Writing $30\cos(30x)$ mixes degree notation with the radian derivative rule and is wrong.

§86.9 The circle generator independently recovers the derivative system

The elementary argument above began with circle geometry and the limit

$$\frac{\sin h}{h} \longrightarrow 1.$$

There is a second route once matrix exponentials and linear differential equations have been developed independently. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The matrix J performs a positive quarter-turn and satisfies

$$J^2 = -I.$$

Separating even and odd powers in the exponential series gives

$$\begin{aligned} e^{xJ} &= I + xJ + \frac{x^2 J^2}{2!} + \frac{x^3 J^3}{3!} + \cdots \\ &= I\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + J\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) \\ &= I\cos x + J\sin x. \end{aligned}$$

Therefore

$$R(x) = e^{xJ} = \begin{pmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{pmatrix}.$$

Because all scalar multiples of J commute,

$$R(x+y) = e^{(x+y)J} = e^{xJ}e^{yJ} = R(x)R(y).$$

Thus $R : \mathbb{R} \to SO(2)$ is a one-parameter group. Differentiating the exponential series term by term gives

$$R'(x) = JR(x) = R(x)J.$$

If the first column is written

$$u(x) = \begin{pmatrix} c(x) \\ s(x) \end{pmatrix},$$

then

$$u'(x) = Ju(x)$$

means

$$c'(x) = -s(x), \qquad s'(x) = c(x),$$

with initial condition

$$c(0) = 1, \qquad s(0) = 0.$$

Uniqueness for the linear system identifies c and s with cosine and sine. The derivative formulas are therefore the infinitesimal form of the rotation-group law.

This route also explains the word *generator*. The tangent vector to the curve $R(x)$ at the identity is

$$R'(0) = J.$$

Every later rotation is obtained by transporting the same infinitesimal turning rule along the group:

$$R'(x) = JR(x).$$

The matrix J is the Lie-algebra element that generates the entire one-parameter subgroup.

Logically, this is an independent foundation only when the matrix exponential, its differentiation rule, and uniqueness of the linear ODE have been proved without using the trigonometric derivative formulas. Otherwise it remains a valuable consistency check rather than a new proof. The two routes meet at the same system,

$$y'' + y = 0,$$

but arrive there from different starting data: one from the local metric geometry of the circle, the other from exponentiating an infinitesimal rotation.

Derivatives: Local Linearization, Computation Graphs, and Higher-Order Data

N-20221211

§87.1 A tangent line is a local model, not only a slope

The difference quotient

$$\frac{f(a+h)-f(a)}{h}$$

compares the actual change in f with the input change h. If the quotient approaches a finite number L, then

$$f(a+h)-f(a)\approx Lh$$

for small h. The derivative is therefore the coefficient of the best first-order linear model near a.

This viewpoint is stronger than the phrase "slope of the tangent line." A slope describes a graph in the plane. Linearization describes how the function transports small perturbations:

$$h\longmapsto f'(a)h.$$

It survives in several variables, on manifolds, and inside automatic differentiation systems where no graph is ever drawn.

The tangent-line approximation is

$$f(a+h)\approx f(a)+f'(a)h.$$

To make the word "approximately" precise, define a remainder

$$r(h)=f(a+h)-f(a)-Lh.$$

The linear model is first-order accurate when

$$\frac{r(h)}{h}\longrightarrow 0.$$

This is written

$$r(h)=o(h).$$

Thus differentiability at a is equivalently the existence of a number L such that

$$\boxed{f(a+h)=f(a)+Lh+o(h).}$$

The number L is necessarily $f'(a)$.

Indeed, subtract $f(a)$, divide by h, and let $h\to 0$:

$$\frac{f(a+h)-f(a)}{h}=L+\frac{o(h)}{h}\longrightarrow L.$$

Conversely, if the difference quotient tends to L, then

$$r(h)=h\left(\frac{f(a+h)-f(a)}{h}-L\right)=o(h).$$

The limit definition and the linearization definition contain exactly the same information.

§87.2 The algebra of small remainders generates the rules

Suppose f and g are differentiable at a. Write

$$f(a+h) = f(a) + f'(a)h + r_f(h), \qquad r_f(h) = o(h),$$

$$g(a+h) = g(a) + g'(a)h + r_g(h), \qquad r_g(h) = o(h).$$

The sum rule is immediate:

$$(f+g)(a+h) = (f+g)(a) + (f'(a) + g'(a))h + o(h).$$

Hence

$$(f+g)'(a) = f'(a) + g'(a).$$

For the product, multiply the two expansions:

$$\begin{aligned} f(a+h)g(a+h) = f(a)g(a) & \\ &+ (f'(a)g(a) + f(a)g'(a))h \\ &+ f'(a)g'(a)h^2 + f(a)r_g(h) + g(a)r_f(h) \\ &+ f'(a)hr_g(h) + g'(a)hr_f(h) + r_f(h)r_g(h). \end{aligned}$$

Every term after the first-order line is $o(h)$. Therefore

$$\boxed{(fg)' = f'g + fg'.}$$

The product rule is not an arbitrary mnemonic. It is what remains after second-order-small terms are discarded.

If $g(a) \neq 0$, continuity gives $g(a+h) \neq 0$ for sufficiently small h. Applying the product rule to

$$g \cdot \frac{1}{g} = 1$$

gives

$$\left(\frac{1}{g}\right)' = -\frac{g'}{g^2}.$$

Combining this with the product rule yields the quotient rule.

The chain rule has the same interpretation. Suppose $b = f(a)$. Then

$$f(a+h) = b + f'(a)h + o(h).$$

Let

$$k = f'(a)h + o(h).$$

Since $k = O(h)$, differentiability of g at b gives

$$g(b+k) = g(b) + g'(b)k + o(k).$$

Because $o(k) = o(h)$,

$$g(f(a+h)) = g(f(a)) + g'(f(a))f'(a)h + o(h).$$

Thus

$$\boxed{(g \circ f)'(a) = g'(f(a))f'(a).}$$

The chain rule is composition of local linear maps.

§87.3 Powers, exponentials, and logarithms

For a positive integer n, the binomial theorem gives

$$(x+h)^n - x^n = nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

After division by h, every term except the first vanishes as $h \to 0$. Hence

$$(x^n)' = nx^{n-1}.$$

Negative integer powers follow from the reciprocal rule.

For a rational exponent $r = m/n$, one may differentiate

$$y^n = x^m$$

on a real domain where the chosen power is defined and locally single-valued. This gives

$$ny^{n-1}y' = mx^{m-1},$$

so

$$(x^r)' = rx^{r-1}.$$

The domain matters. A denominator with even n restricts real inputs, while an odd denominator permits negative inputs but may still create a singular derivative at zero.

For arbitrary real α, define on $x > 0$

$$x^\alpha = e^{\alpha \ln x}.$$

Then the chain rule gives

$$\frac{d}{dx}x^\alpha = e^{\alpha \ln x}\frac{\alpha}{x} = \alpha x^{\alpha-1}.$$

This proof explains why the unrestricted real-power formula naturally lives on the positive half-line.

The exponential can be characterized by the differential equation

$$y' = y, \qquad y(0) = 1.$$

Its flow property is

$$e^{x+t} = e^x e^t.$$

Differentiating with respect to t at $t = 0$ gives

$$(e^x)' = e^x.$$

More generally,

$$(a^x)' = a^x \ln a$$

for $a > 0$.

The logarithm is the inverse of the exponential. If

$$y = \ln x, \qquad x = e^y,$$

then

$$1 = e^y y' = xy',$$

so

$$\boxed{(\ln x)' = \frac{1}{x}.}$$

The identity

$$\ln(xy) = \ln x + \ln y$$

shows that logarithm converts multiplication into addition. Its differential

$$d(\ln x) = \frac{dx}{x}$$

is the infinitesimal quantity invariant under multiplicative rescaling.

§87.4 The inverse rule and where it fails

Let f be differentiable near a, one-to-one there, and suppose

$$f'(a) \neq 0.$$

Put

$$b = f(a).$$

If the local inverse is differentiable at b, the identity

$$f^{-1}(f(x)) = x$$

and the chain rule give

$$(f^{-1})'(b)f'(a) = 1.$$

Therefore

$$\boxed{(f^{-1})'(b) = \frac{1}{f'(a)}.}$$

The one-dimensional inverse function theorem supplies the missing existence statement: under suitable continuity hypotheses, nonvanishing derivative makes the map locally invertible with a differentiable inverse.

The condition $f'(a) \neq 0$ is sufficient, not logically necessary for every kind of inverse. The function

$$f(x) = x^3$$

is globally one-to-one and has an inverse, even though $f'(0) = 0$. Its inverse

$$f^{-1}(y) = y^{1/3}$$

has no finite derivative at zero. The vertical tangent of the inverse records the collapse of first-order information in the original map.

Inverse trigonometric derivatives are applications rather than a separate theory. Once sine is restricted to

$$[-\pi/2, \pi/2],$$

the inverse rule gives, for $|x| < 1$,

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}.$$

Likewise,

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}, \qquad (\arctan x)' = \frac{1}{1+x^2}.$$

The branch choices and endpoint singularities are developed in the preceding trigonometric note. Here they illustrate the general inverse rule and its failure when the original derivative vanishes.

§87.5 A small derivative library and its dependency map

A derivative table is useful when it remembers how each entry was produced.

Function	Derivative	Structural source
x^α, $x > 0$	$\alpha x^{\alpha-1}$	exponential and logarithm
e^x	e^x	one-parameter flow
$\ln x$	$1/x$	inverse function rule
$\sin x$, $\cos x$	$\cos x$, $-\sin x$	basic limits and addition laws
$\tan x$	$\sec^2 x$	quotient rule
$\arcsin x$	$(1-x^2)^{-1/2}$	branch-restricted inverse rule

The general rules form a compact dependency graph:

$$\text{linearization} \longrightarrow \begin{cases} \text{sum and product rules,} \\ \text{chain rule,} \\ \text{inverse rule,} \end{cases} \longrightarrow \text{elementary derivative library.}$$

This ordering is more reusable than memorizing a long page of formulas. New functions enter the library by exposing their algebraic composition or defining differential equation.

§87.6 Forward mode: propagate a perturbation

Automatic differentiation applies the chain rule to a computation graph without symbolic expansion and without finite-difference approximation. Consider

$$F(x) = \ln\left(1 + \sin^2(x^2)\right).$$

Break the expression into nodes:

$$u = x^2, \qquad v = \sin u, \qquad w = 1 + v^2, \qquad F = \ln w.$$

Forward mode attaches to every value its derivative with respect to the chosen input. Starting from

$$\dot{x} = 1,$$

we propagate

$$\dot{u} = 2x\dot{x} = 2x,$$
$$\dot{v} = \cos u\,\dot{u} = 2x\cos(x^2),$$
$$\dot{w} = 2v\dot{v} = 4x\sin(x^2)\cos(x^2),$$
$$\dot{F} = \frac{\dot{w}}{w} = \frac{4x\sin(x^2)\cos(x^2)}{1+\sin^2(x^2)}.$$

The dot notation represents a tangent perturbation. If the input changes by ε, then to first order each node changes by its dotted value times ε.

This mechanism can be encoded by dual numbers. Introduce a symbol ϵ with

$$\epsilon^2 = 0.$$

Then

$$f(x+\epsilon) = f(x) + f'(x)\epsilon.$$

Ordinary arithmetic on pairs

$$x + \dot{x}\epsilon$$

automatically propagates values and first derivatives. The nilpotent relation discards all terms beyond first order, exactly as the $o(h)$ calculus did analytically.

§87.7 Reverse mode: pull sensitivity backward

Reverse mode evaluates the same graph forward to store intermediate values, then propagates output sensitivity backward. Write

$$\bar{z} = \frac{\partial F}{\partial z}$$

for the adjoint associated with a node z. Begin at the output with

$$\bar{F} = 1.$$

The local derivatives give

$$\begin{aligned}
\bar{w} &= \bar{F}\frac{\partial F}{\partial w} = \frac{1}{w}, \\
\bar{v} &= \bar{w}\frac{\partial w}{\partial v} = \frac{2v}{w}, \\
\bar{u} &= \bar{v}\frac{\partial v}{\partial u} = \frac{2v\cos u}{w}, \\
\bar{x} &= \bar{u}\frac{\partial u}{\partial x} = \frac{4x\sin(x^2)\cos(x^2)}{1+\sin^2(x^2)}.
\end{aligned}$$

The final adjoint $\bar{x}$ is $F'(x)$, matching forward mode.

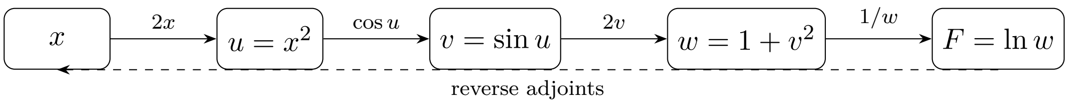


Figure 87.1: Forward mode pushes a tangent from left to right; reverse mode pulls an output covector from right to left through the same local derivatives.

Forward mode is efficient when there are few input directions and many outputs. Reverse mode is efficient for a scalar output with many inputs, which is why it underlies backpropagation. The two modes are not different chain rules. They choose opposite orders for composing the same local linear maps and their duals.

§87.8 Second-order automatic differentiation is jet composition

Forward mode used dual numbers

$$\mathbb{R}[\epsilon]/(\epsilon^2)$$

to retain a value and one first-order perturbation. Keeping one more power gives

$$\mathbb{R}[\epsilon]/(\epsilon^3),$$

which stores second-order local data. Represent an input curve through x by

$$X(\epsilon) = x + v\epsilon + \frac{1}{2}a\epsilon^2.$$

The coefficients v and a are its velocity and acceleration at $\epsilon = 0$.

Taylor expansion followed by the rule $\epsilon^3 = 0$ gives

$$f(X(\epsilon)) = f(x) + f'(x)\left(v\epsilon + \frac{1}{2}a\epsilon^2\right) + \frac{1}{2}f''(x)(v\epsilon)^2 \pmod{\epsilon^3}.$$

Therefore

$$\boxed{f(X(\epsilon)) = f(x) + f'(x)v\epsilon + \frac{1}{2}(f''(x)v^2 + f'(x)a)\epsilon^2.}$$

The first-order coefficient is the transported velocity. The second-order coefficient contains two effects: curvature of the function, $f''(x)v^2$, and acceleration already present in the input curve, $f'(x)a$. This is the same distinction that appears in the second derivative of a composition.

For a concrete computation, take

$$F(x) = e^{\sin x}$$

and seed a unit-speed straight perturbation at $x = 0$:

$$X(\epsilon) = \epsilon, \qquad v = 1, \qquad a = 0.$$

First propagate through sine:

$$\sin\epsilon = \epsilon \pmod{\epsilon^3},$$

because the first discarded term is $-\epsilon^3/6$. Then propagate through the exponential:

$$e^\epsilon = 1 + \epsilon + \frac{1}{2}\epsilon^2 \pmod{\epsilon^3}.$$

The three stored coefficients give

$$F(0) = 1, \qquad F'(0) = 1, \qquad F''(0) = 1.$$

No symbolic second derivative of the full composite was formed; each node merely propagated a truncated local polynomial.

This procedure is precisely composition of second jets. The second jet of f at x can be represented by

$$j_x^2 f(h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 \pmod{h^3}.$$

Composing functions means substituting one truncated polynomial into another and discarding terms of degree three or higher. At higher orders the coefficient bookkeeping is governed by set partitions and the Faà di Bruno formula; the dedicated higher-derivative chapter develops that combinatorics in full.

The same idea works in several variables. For a smooth scalar function

$$f : \mathbb{R}^n \to \mathbb{R}$$

and a direction v, substitute

$$x + \epsilon v$$

into its second-order expansion:

$$f(x + \epsilon v) = f(x) + \epsilon\nabla f(x)^T v + \frac{1}{2}\epsilon^2 v^T \nabla^2 f(x) v \pmod{\epsilon^3}.$$

Thus one second-order forward sweep computes a Hessian quadratic form $v^T Hv$ without constructing every entry of H. Mixed terms can be recovered by polarization:

$$u^T Hv = \frac{1}{2}\big((u+v)^T H(u+v) - u^T Hu - v^T Hv\big).$$

Higher-order automatic differentiation is therefore not an unrelated extension of the chain rule. It is ordinary computation performed in a richer truncated algebra that retains more of the local Taylor model.

§87.9 Differentials are covectors

For a function $f : \mathbb{R} \to \mathbb{R}$, the differential at a is the linear functional

$$df_a(h) = f'(a)h.$$

In the coordinate x, this is written

$$df = f'(x)\,dx.$$

The symbol dx is the basic covector that sends an increment h to h. It is not necessary to treat dy/dx as an ordinary fraction in order to justify the notation.

If g is another function, then

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

In pullback notation,

$$d(g \circ f) = f^*(dg).$$

This is the chain rule for one-forms. Reverse-mode automatic differentiation follows the same direction: an output covector is pulled backward through the transpose or adjoint of each derivative map.

For a function of several variables, the differential is no longer multiplication by one number. If

$$f : \mathbb{R}^n \to \mathbb{R},$$

then

$$df_x(h) = \nabla f(x)^T h.$$

The gradient vector depends on the chosen inner product, while the differential is intrinsically a covector. The one-dimensional notation hides this distinction because every linear map $\mathbb{R} \to \mathbb{R}$ is represented by a scalar.

§87.10 Differentiable does not mean smoothly differentiable

Continuity is necessary for differentiability, but it is not sufficient. The function

$$f(x) = |x|$$

is continuous at zero, yet

$$\lim_{h\to 0^+} \frac{|h|}{h} = 1, \qquad \lim_{h\to 0^-} \frac{|h|}{h} = -1.$$

No single linear map approximates the function from both sides.

A derivative may exist without varying continuously. Define

$$g(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

At zero,

$$\frac{g(h) - g(0)}{h} = h \sin(1/h) \longrightarrow 0,$$

so $g'(0) = 0$. For $x \neq 0$,

$$g'(x) = 2x \sin(1/x) - \cos(1/x),$$

which has no limit as $x \to 0$. The function is differentiable everywhere but not continuously differentiable at zero.

These examples separate three notions:

$$\text{continuity}, \qquad \text{differentiability}, \qquad C^1 \text{ regularity}.$$

Each additional level controls more than the existence of a tangent at one point.

§87.11 Fréchet differentiability asks for one uniform linear model

The formula

$$f(a + h) = f(a) + f'(a)h + o(h)$$

contains the multidimensional definition almost unchanged. For a map between normed spaces,

$$F : X \to Y,$$

we seek a bounded linear map

$$DF(a) : X \to Y$$

such that

$$\boxed{\|F(a + h) - F(a) - DF(a)h\|_Y = o(\|h\|_X).}$$

This is the Fréchet derivative. The estimate must hold uniformly as h approaches zero from every direction, not along one line at a time.

Consider

$$F(x, y) = \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix}.$$

At $(0, 0)$, use

$$e^h = 1 + h + O(h^2), \qquad \cos k = 1 + O(k^2), \qquad \sin k = k + O(k^3).$$

Then

$$e^h \cos k = 1 + h + O(h^2 + k^2),$$
$$e^h \sin k = k + O(|hk| + |k|^3).$$

Hence

$$F(h, k) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} h \\ k \end{pmatrix} + O(h^2 + k^2),$$

so

$$DF(0, 0) = I.$$

The Jacobian is not merely a table of partial derivatives; it is the linear map that simultaneously approximates the function for every small vector (h, k).

Directional derivatives alone do not guarantee such a map. Define

$$G(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Along a direction $v = (a, b)$,

$$G(ta, tb) = t\frac{a^3}{a^2+b^2}.$$

Therefore every directional derivative at the origin exists and equals

$$D_v G(0,0) = \frac{a^3}{a^2+b^2}.$$

But this expression is not linear in (a, b). For example,

$$D_{(1,0)}G = 1, \qquad D_{(0,1)}G = 0, \qquad D_{(1,1)}G = \frac{1}{2},$$

whereas linearity would require the last value to be 1. No single linear map can represent all directional changes, so G is not Fréchet differentiable at the origin.

Once the Fréchet derivative exists, the chain rule becomes literal composition:

$$D(H \circ F)(a) = DH(F(a)) \circ DF(a).$$

After bases are chosen, this is Jacobian multiplication. Forward mode applies the composite map to tangent vectors; reverse mode applies its dual maps to covectors. The differential df_a is the scalar-valued covector version, while jets retain the higher-order terms discarded by the first linear model.

The chapter therefore ends with the same object with which it began: the best local linear map. Elementary differentiation rules explain how it behaves under algebra and composition; inverse-function formulas ask when it can be inverted; automatic differentiation transports it through a graph; and Fréchet differentiability specifies what it means for one approximation to control all nearby directions at once.

Derivatives, Tangents, Extremal Problems, Young–Holder–Minkowski, and Jensen

N-20230310

§88.1 The derivative from secants to tangents

The note begins with the geometric idea of derivative. Given two nearby points on the curve,

$$(x_0, f(x_0)), \qquad (x_0 + \Delta x, f(x_0 + \Delta x)),$$

the secant slope is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

If, as $\Delta x \to 0$, this quotient tends to a finite limit, then the curve has a definite tangent line at x_0. The derivative is

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

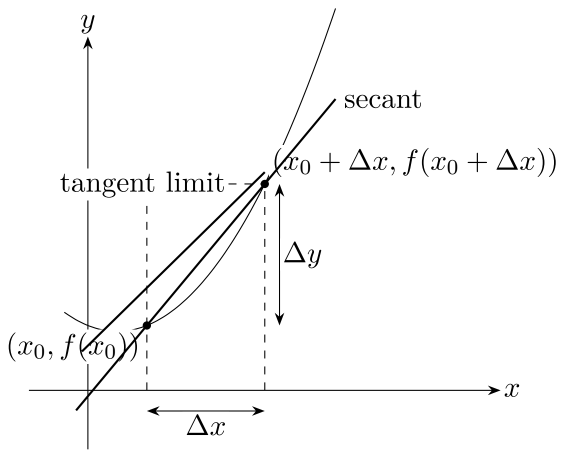
Equivalently,

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

The tangent line at x_0 is therefore

$$y - f(x_0) = f'(x_0)(x - x_0).$$

The source page stresses the visual transition: the secant line approaches a unique limiting line. If that limiting line is not vertical and the slope is finite, we call the slope the derivative.



§88.2 Derivative notation and local definition

The definition requires f to be defined on a punctured neighborhood of x_0, and usually on a small open interval around it. The derivative at x_0 may be denoted by any of

$$f'(x_0), \qquad \left.\frac{df}{dx}\right|_{x=x_0}, \qquad y'(x_0), \qquad \left.\frac{dy}{dx}\right|_{x=x_0}.$$

If f is differentiable at every point of an open interval (a,b), then $x \mapsto f'(x)$ is the derivative function on that interval.

The note emphasizes a small logical point: before asking whether f is differentiable at a point, one must first know that the function is defined near that point. Differentiability is a local property.

§88.3 Rules of differentiation

If f and g are differentiable at x_0, then sums, scalar multiples, products, and quotients obey:

$$(af+bg)' = af' + bg', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g(x_0) \neq 0.$$

The chain rule says

$$(f \circ g)'(x_0) = f'(g(x_0))g'(x_0).$$

The inverse-function rule says that if f and f^{-1} are inverse functions, $f'(x_0) \neq 0$, and $y_0 = f(x_0)$, then

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

These rules are not simply a table to memorize. Each rule is a disguised statement about first-order linear approximation. For example,

$$f(x_0+h) = f(x_0) + f'(x_0)h + o(h), \qquad g(x_0+h) = g(x_0) + g'(x_0)h + o(h),$$

and multiplying these expansions gives the product rule.

§88.4 Elementary derivative formulas

The table in the note includes the standard formulas:

$$C' = 0, \qquad (x^a)' = ax^{a-1}, \qquad (a^x)' = a^x \ln a, \qquad (\log_a x)' = \frac{1}{x \ln a}.$$

For trigonometric functions,

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x, \qquad (\tan x)' = \sec^2 x, \qquad (\cot x)' = -\csc^2 x,$$

$$(\sec x)' = \sec x \tan x, \qquad (\csc x)' = -\csc x \cot x.$$

For inverse trigonometric functions,

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \qquad (\arccos x)' = -\frac{1}{\sqrt{1-x^2}}, \qquad (\arctan x)' = \frac{1}{1+x^2}.$$

§88.5 Extrema, Fermat's lemma, and critical points

A point x_0 is a local maximum if $f(x) \le f(x_0)$ near x_0, and a local minimum if $f(x) \ge f(x_0)$ near x_0. If $f'(x_0) = 0$, then x_0 is called a stationary or critical point.

Fermat's lemma says: if f is differentiable at x_0 and has a local extremum there, then

$$f'(x_0) = 0.$$

The note writes a warning beside this theorem: the converse is false. For example,

$$f(x) = x^3$$

has $f'(0) = 0$, but 0 is not an extremum. Fermat's lemma only provides candidates for extrema. One must then check sign changes, second derivatives, endpoints, or non-differentiable points.

If $f''(x_0) > 0$ and $f'(x_0) = 0$, then x_0 is a local minimum. If $f''(x_0) < 0$, it is a local maximum. The reason is that f'' describes whether f' is increasing or decreasing near the critical point.

§88.6 A derivative proof of a basic inequality

The note proves, for $x > 0$,

$$x^\alpha - \alpha x + \alpha - 1 \le 0, \qquad 0 < \alpha < 1,$$

and

$$x^\alpha - \alpha x + \alpha - 1 \ge 0, \qquad \alpha < 0 \text{ or } \alpha > 1.$$

Let

$$F(x) = x^\alpha - \alpha x + \alpha - 1.$$

Then

$$F'(x) = \alpha(x^{\alpha-1} - 1), \qquad F(1) = 0.$$

For $0 < \alpha < 1$, F increases on $(0, 1)$ and decreases on $(1, \infty)$, so $x = 1$ is a maximum and $F(x) \le 0$. For $\alpha < 0$ or $\alpha > 1$, the monotonicity is reversed, and $x = 1$ is a minimum, so $F(x) \ge 0$.

This small inequality is the engine behind Young's inequality.

§88.7 Young's inequality

Let $a, b > 0$ and let p, q satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

For $p > 1$, Young's inequality says

$$a^{1/p} b^{1/q} \le \frac{a}{p} + \frac{b}{q},$$

with equality iff $a = b$. The proof substitutes

$$x = \frac{a}{b}, \qquad \alpha = \frac{1}{p}$$

into the previous derivative inequality, then multiplies by b.

The source also records the reversed version for $p < 1$, where the inequality sign changes. The important conceptual point is that an inequality between weighted geometric and arithmetic means is reduced to the shape of one single-variable function.

§88.8 Holder's inequality

For $x_i, y_i \geq 0$ and $1/p + 1/q = 1$, Holder's inequality is

$$\sum_{i=1}^{n} x_i y_i \leq \left(\sum_{i=1}^{n} x_i^p\right)^{1/p} \left(\sum_{i=1}^{n} y_i^q\right)^{1/q}, \qquad p > 1.$$

The proof in the source normalizes

$$a_i = \frac{x_i^p}{\sum_j x_j^p}, \qquad b_i = \frac{y_i^q}{\sum_j y_j^q}$$

and applies Young's inequality termwise. After adding the inequalities, the normalized sum becomes 1, giving Holder.

A handwritten note observes that the integral form follows the same idea:

$$\int_a^b f(x)g(x)\,dx \leq \left(\int_a^b |f(x)|^p\,dx\right)^{1/p} \left(\int_a^b |g(x)|^q\,dx\right)^{1/q}.$$

The comment says this feels almost trivial once the finite-sum proof is understood: one replaces sums by integrals and uses the same normalization logic.

For $p = q = 2$, Holder becomes the Cauchy–Bunyakovsky–Schwarz inequality.

§88.9 Minkowski's inequality

For $p > 1$, Minkowski's inequality says

$$\left(\sum_{i=1}^{n}(x_i + y_i)^p\right)^{1/p} \leq \left(\sum_{i=1}^{n} x_i^p\right)^{1/p} + \left(\sum_{i=1}^{n} y_i^p\right)^{1/p}.$$

The proof applies Holder to the two sums

$$\sum_i x_i(x_i + y_i)^{p-1}, \qquad \sum_i y_i(x_i + y_i)^{p-1}.$$

Because $q = p/(p-1)$, we have

$$((x_i + y_i)^{p-1})^q = (x_i + y_i)^p.$$

After factoring the common term and dividing by it, one obtains Minkowski.

The handwritten margin notes that for $p = 2$ this is exactly the triangle inequality for Euclidean length:

$$\sqrt{(x_1 + y_1)^2 + \cdots + (x_n + y_n)^2} \leq \sqrt{x_1^2 + \cdots + x_n^2} + \sqrt{y_1^2 + \cdots + y_n^2}.$$

§88.10 Convex functions and Jensen's inequality

The note defines a convex function by the inequality

$$f(\alpha_1 x_1 + \alpha_2 x_2) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2), \qquad \alpha_1 + \alpha_2 = 1, \quad \alpha_i \geq 0.$$

The handwritten annotation explains the geometry: the point on the chord between $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lies above the graph.

Equivalently, for $x_1 < x < x_2$,

$$\frac{f(x) - f(x_1)}{x - x_1} \leq \frac{f(x_2) - f(x)}{x_2 - x}.$$

This slope inequality implies that if f is differentiable, then f' is increasing; if f is twice differentiable, then convexity corresponds to $f'' \geq 0$.

Jensen's inequality generalizes the two-point definition:

$$f\left(\sum_{i=1}^{n} \alpha_i x_i\right) \leq \sum_{i=1}^{n} \alpha_i f(x_i), \qquad \alpha_i \geq 0, \quad \sum_i \alpha_i = 1.$$

The source proves this by induction. Let $\beta = \alpha_2 + \cdots + \alpha_n = 1 - \alpha_1$ and normalize the remaining weights by α_i/β. Then use the two-point convexity inequality followed by the induction hypothesis.

§88.11 Typical exercise themes

The last pages list exercises. They are not random; they are organized around the derivative as a tool for inequality and extrema.

Examples include:

(i) determining when a line and cubic have three distinct intersections;

(ii) finding monotonicity intervals of $f(x) = \sqrt{x} - \ln(x + a)$;

(iii) maximizing $\sin^2 x \cos x$ on $(0, \pi/2)$;

(iv) comparing numbers such as $0.1e^{0.1}$, $1/9$, and $-\ln 0.9$ using auxiliary functions;

(v) proving $e^x \geq 1 + x$ and $\ln x \leq x - 1$;

(vi) using Jensen or logarithms to prove entropy-like inequalities such as

$$\sum_{i=1}^{2^n} p_i \log_2 p_i \geq -n, \qquad \sum_i p_i = 1;$$

(vii) proving product inequalities of the form

$$\prod_{i=1}^{n} a_i^{b_i} \leq 1$$

under weighted-sum hypotheses;

(viii) minimizing expressions such as

$$\frac{9}{\sin^2 x} + \frac{1}{\cos x \cos y \sin^2 y}.$$

The common method is to build an auxiliary function, compute derivatives, locate monotonicity or convexity, and then read off the desired inequality.

§88.12 The reusable pattern

This note shows how the derivative becomes more than a tangent slope. It gives a language for extrema, monotonicity, convexity, and inequalities. Young, Holder, Minkowski, and Jensen all look like different inequalities, but the note reveals a chain:

$$\text{single-variable derivative inequality} \to \text{Young} \to \text{Holder} \\ \to \text{Minkowski}, \qquad \text{convexity} \to \text{Jensen}.$$

Thus calculus becomes a proof technology for analysis and inequalities.

§88.13 Derivative as local linear approximation

The derivative is the coefficient of the best first-order model:

$$f(a+h) = f(a) + f'(a)h + o(h).$$

This viewpoint explains the product, quotient, and chain rules: they are rules for composing and multiplying first-order approximations.

In several variables the same idea becomes the Fréchet derivative, a linear map rather than a single number.

§88.14 Extrema and first variation

Fermat's lemma says that if f has a local extremum at an interior point and is differentiable there, then $f'(a) = 0$. The derivative is the first variation; if a small positive and negative move are both allowed, the linear term must vanish.

Second derivative tests then study the next nonzero term in the local expansion.

§88.15 Young, Holder, and convex duality

Young's inequality

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}, \qquad \frac{1}{p} + \frac{1}{q} = 1,$$

is a convex-duality statement. It is the pointwise engine behind Holder's inequality. Minkowski then gives the triangle inequality for L^p norms.

Thus the inequality part of the note connects differential calculus to convex analysis and normed spaces.

§88.16 Local linearity and convex control

The chapter links two central ideas: derivatives provide local linear models, and inequalities describe convex or norm geometry. Tangents, extrema, Young–Holder–Minkowski, and Jensen all express how local or averaged information controls global behavior.

Calculus Review I: Limits, Completeness, Asymptotic Scales, and Local Approximation

N-20251112

§89.1 Cauchy–Schwarz and Minkowski are error-control mechanisms

For vectors $x, y \in \mathbb{R}^n$, the Cauchy–Schwarz inequality is

$$|\langle x, y\rangle| \le \|x\|_2 \|y\|_2.$$

If $y \neq 0$, consider the nonnegative quadratic

$$0 \le \|x - ty\|_2^2 = \|x\|_2^2 - 2t\langle x, y\rangle + t^2\|y\|_2^2.$$

Its discriminant is nonpositive, so

$$\langle x, y\rangle^2 \le \|x\|_2^2 \|y\|_2^2.$$

Equality holds exactly when x and y are linearly dependent.

Minkowski's inequality follows:

$$\begin{aligned}\|x + y\|_2^2 &= \|x\|_2^2 + 2\langle x, y\rangle + \|y\|_2^2 \\ &\le \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2.\end{aligned}$$

Hence

$$\boxed{\|x + y\|_2 \le \|x\|_2 + \|y\|_2.}$$

For nonzero vectors, equality holds exactly when x and y point in the same direction.

These inequalities control accumulated errors. If $e_1, \ldots, e_m$ are perturbations, then

$$\left\|\sum_{j=1}^{m} e_j\right\| \le \sum_{j=1}^{m} \|e_j\|.$$

If an error is represented by an inner product, Cauchy–Schwarz converts it into a product of sizes. The same two patterns survive in Hilbert and normed spaces.

§89.2 Limits are statements about eventual membership in neighborhoods

For $a \in \mathbb{R}$, let

$$N_\varepsilon(a) = \{x : |x - a| < \varepsilon\}.$$

A sequence (x_n) converges to a when

$$\forall \varepsilon > 0, \quad x_n \in N_\varepsilon(a) \text{ eventually}.$$

Here "eventually" means that there exists N such that the statement holds for every $n \geq N$.

For a function f,

$$\lim_{x \to x_0} f(x) = L$$

means that for every neighborhood V of L, there is a punctured neighborhood U of x_0 such that

$$x \in U \quad \Longrightarrow \quad f(x) \in V.$$

In epsilon–delta notation,

$$0 < |x - x_0| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The neighborhood families form filters: intersections of finitely many neighborhoods are neighborhoods, and larger sets remain neighborhoods. This is why limits compose. If

$$f(x) \to L \quad \text{and} \quad g(y) \to M \text{ as } y \to L,$$

then the preimage under g of a neighborhood of M contains a neighborhood of L, and the preimage under f of that neighborhood contains a punctured neighborhood of x_0.

§89.3 Sequence limits are unique and respect algebraic operations

Suppose $x_n \to a$ and $x_n \to b$. If $a \neq b$, choose

$$\varepsilon = \frac{|a - b|}{3}.$$

For large n,

$$|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon < |a - b|,$$

a contradiction. Thus limits are unique.

If $x_n \to a$ and $y_n \to b$, then

$$x_n + y_n \to a + b, \qquad x_n y_n \to ab.$$

For the product,

$$|x_n y_n - ab| \leq |x_n||y_n - b| + |b||x_n - a|.$$

A convergent sequence is bounded, so $|x_n| \leq M$ eventually, and both terms tend to zero. If $b \neq 0$, then $y_n \neq 0$ eventually and

$$\frac{x_n}{y_n} \to \frac{a}{b}.$$

The definition of $x_n \to +\infty$ is also eventual:

$$\forall M > 0, \quad x_n > M \text{ eventually}.$$

It is not convergence to a real number, but it becomes ordinary convergence after adjoining points at infinity or applying a compactifying coordinate such as $x \mapsto \arctan x$.

§89.4 Heine's criterion converts function limits into sequence limits

Theorem 89.4.1 (Heine criterion)
Let f be defined on a punctured neighborhood of x_0. Then

$$\lim_{x\to x_0} f(x) = L$$

iff for every sequence $x_n \to x_0$ with $x_n \neq x_0$, one has

$$f(x_n) \to L.$$

Proof. Assume the epsilon–delta limit. Given $\varepsilon > 0$, choose $\delta > 0$ so that

$$0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon.$$

Since $x_n \to x_0$, the hypothesis on x_n holds eventually, so $f(x_n) \to L$.

Conversely, suppose the epsilon–delta condition fails. Then there exists $\varepsilon_0 > 0$ such that for every n one can choose x_n with

$$0 < |x_n - x_0| < \frac{1}{n}, \qquad |f(x_n) - L| \geq \varepsilon_0.$$

Then $x_n \to x_0$, but $f(x_n) \not\to L$, contradicting the sequential assumption. □

To disprove a limit, construct incompatible approaches. For

$$f(x) = \cos\frac{1}{x},$$

take

$$x_n = \frac{1}{2\pi n}, \qquad y_n = \frac{1}{(2n+1)\pi}.$$

Both tend to zero, but

$$f(x_n) = 1, \qquad f(y_n) = -1.$$

Therefore $\lim_{x\to 0} f(x)$ does not exist.

§89.5 Asymptotic notation compares scales, not fixed small numbers

Let $g(x) > 0$ near a limiting point. The notation

$$f = O(g)$$

means that there are constants $C > 0$ and a neighborhood on which

$$|f(x)| \leq Cg(x).$$

The notation

$$f = o(g)$$

means

$$\frac{f(x)}{g(x)} \to 0.$$

We write

$$f = \Theta(g)$$

when both $f = O(g)$ and $g = O(f)$, and

$$f \sim g$$

when

$$\frac{f(x)}{g(x)} \to 1.$$

The implications are

$$f \sim g \Longrightarrow f = \Theta(g), \qquad f = o(g) \Longrightarrow f = O(g),$$

but the converses fail.

Equivalent infinitesimals may be substituted safely in products and quotients when the remaining factors have controlled limits. For example,

$$\sin x \sim x, \qquad 1 - \cos x \sim \frac{x^2}{2} \quad (x \to 0).$$

Hence

$$\lim_{x\to 0} \frac{1-\cos x}{x \sin x} = \lim_{x\to 0} \frac{x^2/2}{x^2} = \frac{1}{2}.$$

Substitution is not valid inside arbitrary sums where cancellation may change the leading order. Although

$$\sqrt{1+x} \sim 1,$$

one cannot replace it by 1 in

$$\sqrt{1+x} - 1.$$

The difference is of smaller scale:

$$\sqrt{1+x} - 1 = \frac{x}{\sqrt{1+x}+1} \sim \frac{x}{2}.$$

§89.6 Squeeze and monotone convergence use order structure

The squeeze theorem says that if

$$a_n \le b_n \le c_n$$

eventually and

$$a_n \to L, \qquad c_n \to L,$$

then $b_n \to L$. Indeed,

$$|b_n - L| \le \max\{|a_n - L|, |c_n - L|\}$$

when L lies between the bounding terms.

A monotone bounded sequence converges because $\mathbb{R}$ has suprema.

Theorem 89.6.1 (Monotone convergence)
If (x_n) is increasing and bounded above, then

$$x_n \to \sup_n x_n.$$

Proof. Let
$$L = \sup\{x_n : n \in \mathbb{N}\}.$$
Given $\varepsilon > 0$, the number $L - \varepsilon$ is not an upper bound. Hence some N satisfies
$$x_N > L - \varepsilon.$$
For $n \geq N$, monotonicity gives
$$L - \varepsilon < x_N \leq x_n \leq L.$$
Thus $|x_n - L| < \varepsilon$. □

The proof exposes the completeness input: without the least-upper-bound property, the candidate L may be missing from the number system.

§89.7 The Cauchy criterion characterizes convergence in complete spaces

A sequence is Cauchy if
$$\forall \varepsilon > 0, \quad |x_m - x_n| < \varepsilon \text{ for all sufficiently large } m, n.$$
Every convergent sequence is Cauchy by the triangle inequality.

For the converse in $\mathbb{R}$, first note that a Cauchy sequence is bounded. Choose N so that
$$|x_n - x_N| < 1 \quad (n \geq N).$$
The tail lies in a bounded interval, and the finitely many earlier terms are also bounded.

By Bolzano–Weierstrass, a bounded sequence has a convergent subsequence
$$x_{n_k} \to L.$$
Given $\varepsilon > 0$, choose K so that
$$|x_m - x_n| < \frac{\varepsilon}{2} \quad (m, n \geq K)$$
and choose k with $n_k \geq K$ and
$$|x_{n_k} - L| < \frac{\varepsilon}{2}.$$
Then for every $n \geq K$,
$$|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon.$$
Thus $x_n \to L$.

The same statement fails in $\mathbb{Q}$. Let
$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}.$$
Then (q_n) is Cauchy in the rational metric, but it has no limit in $\mathbb{Q}$. Completion adds the missing limit $\sqrt{2}$.

A Banach space is exactly a normed vector space in which every Cauchy sequence converges. Completeness is therefore not a technical appendix; it is what turns internal error control into the existence of a limit.

§89.8 Continuity at each point is weaker than uniform continuity

A function $f : X \to Y$ between metric spaces is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad d_X(x, y) < \delta \Longrightarrow d_Y(f(x), f(y)) < \varepsilon$$

for all $x, y \in X$. The same δ must work everywhere.

The function $f(x) = x^2$ is continuous on $\mathbb{R}$ but not uniformly continuous. Set

$$x_n = n, \qquad y_n = n + \frac{1}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \to 0,$$

but

$$|x_n^2 - y_n^2| = 2 + \frac{1}{n^2} \not\to 0.$$

On compact domains, continuity becomes uniform.

Theorem 89.8.1 (Heine–Cantor)
A continuous function from a compact metric space to a metric space is uniformly continuous.

Proof. If uniform continuity failed, there would be $\varepsilon_0 > 0$ and pairs x_n, y_n with

$$d(x_n, y_n) < \frac{1}{n}, \qquad d(f(x_n), f(y_n)) \geq \varepsilon_0.$$

Compactness gives a convergent subsequence $x_{n_k} \to x$. Since

$$d(y_{n_k}, x) \leq d(y_{n_k}, x_{n_k}) + d(x_{n_k}, x) \to 0,$$

one also has $y_{n_k} \to x$. Continuity then forces both image subsequences to converge to $f(x)$, contradicting their separation by ε_0. $\square$

Uniform continuity preserves Cauchy sequences and is therefore the natural hypothesis for extending a map to a completion.

§89.9 A derivative is a first-order approximation with a controlled remainder

The derivative at a can be written as

$$f(a + h) = f(a) + f'(a)h + r(h), \qquad \frac{r(h)}{h} \to 0.$$

Equivalently,

$$f(a + h) = f(a) + f'(a)h + o(h).$$

This is stronger than the existence of a slope formula: it says the nonlinear error is negligible compared with the increment.

The differential

$$df_a(h) = f'(a)h$$

is the linear map giving the best first-order approximation. In several variables, the correct extension is the Fréchet derivative:

$$f(a+h) = f(a) + Df(a)h + o(\|h\|).$$

The derivative is therefore a local linear model, not merely a quotient of small numbers.

The chain rule follows by composing the two local models. If

$$g(a+h) = g(a) + g'(a)h + o(h)$$

and

$$f(g(a)+k) = f(g(a)) + f'(g(a))k + o(k),$$

then $k = g'(a)h + o(h) = O(h)$, so

$$f(g(a+h)) = f(g(a)) + f'(g(a))g'(a)h + o(h).$$

Thus

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

§89.10 Taylor's theorem is a finite jet plus a quantified remainder

If $f \in C^{n+1}$ near a, Taylor's theorem gives

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x).$$

The Lagrange remainder has the form

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}$$

for some ξ between a and x.

The integral remainder is

$$\boxed{R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt.}$$

It follows by repeated use of the fundamental theorem of calculus. For example,

$$\begin{aligned} f(x) - f(a) &= \int_a^x f'(t_1)\,dt_1 \\ &= f'(a)(x-a) + \int_a^x \int_a^{t_1} f''(t_2)\,dt_2\,dt_1, \end{aligned}$$

and exchanging the order of integration produces the kernel $(x-t)$. Iterating gives the general formula.

If

$$|f^{(n+1)}(t)| \le M$$

between a and x, then

$$|R_n(x)| \le \frac{M|x-a|^{n+1}}{(n+1)!}.$$

For e^x on $[0,1]$, the degree-n Maclaurin approximation satisfies

$$\left| e^x - \sum_{k=0}^{n} \frac{x^k}{k!} \right| \le \frac{e}{(n+1)!}.$$

Taylor expansion is therefore an asymptotic statement with a certified error, not permission to replace every smooth function by an infinite series.

§89.11 Rotation matrices generate trigonometric identities

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

The matrix exponential separates into even and odd powers:

$$\begin{aligned} e^{tJ} &= I\cos t + J\sin t \\ &= \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}. \end{aligned}$$

It preserves the Euclidean quadratic form:

$$(e^{tJ})^T e^{tJ} = I.$$

The group law

$$e^{(s+t)J} = e^{sJ}e^{tJ}$$

gives the addition formulas by matrix multiplication:

$$\cos(s+t) = \cos s\cos t - \sin s\sin t,$$

$$\sin(s+t) = \sin s\cos t + \cos s\sin t.$$

Thus the identities are coordinates of the rotation group, not an isolated table to memorize.

§89.12 Lorentz boosts generate hyperbolic identities

Let

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

Then

$$\begin{aligned} e^{tK} &= I\cosh t + K\sinh t \\ &= \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}. \end{aligned}$$

Let

$$\eta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$(e^{tK})^T \eta e^{tK} = \eta.$$

Therefore the boost preserves

$$x^2 - y^2.$$

Applying it to $(1,0)^T$ yields

$$\begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix},$$

so invariance gives

$$\boxed{\cosh^2 t - \sinh^2 t = 1.}$$

Again the group law

$$e^{(s+t)K} = e^{sK} e^{tK}$$

produces

$$\cosh(s+t) = \cosh s \cosh t + \sinh s \sinh t,$$

$$\sinh(s+t) = \sinh s \cosh t + \cosh s \sinh t.$$

Circular and hyperbolic functions are parallel because both arise from one-parameter matrix groups. The sign difference comes from

$$J^2 = -I, \qquad K^2 = I,$$

and from the quadratic forms they preserve: Euclidean x^2+y^2 versus Lorentzian x^2-y^2.

Calculus Review II: Monotonicity, Integration Methods, Applications, and More

N-20251210

§90.1 Strict monotonicity from the derivative

If f is differentiable on an interval I and

$$f'(x) > 0 \quad (x \in I),$$

then f is strictly increasing on I. If

$$f'(x) < 0 \quad (x \in I),$$

then f is strictly decreasing.

The proof uses the Lagrange mean value theorem. For $x_2 > x_1$,

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1)$$

for some $\xi \in (x_1, x_2)$. If $f'(\xi) > 0$, then the difference is positive.

§90.2 First derivative test for extrema

Suppose f is continuous near x_0 and differentiable in the punctured neighborhood. If the sign of f' changes from negative to positive at x_0, then $f(x_0)$ is a local minimum. If the sign changes from positive to negative, then $f(x_0)$ is a local maximum.

The page records the compact criterion:

$$\begin{aligned} (x - x_0)f'(x) > 0 \quad &\Longrightarrow \quad f(x_0) \text{ is a local minimum,} \\ (x - x_0)f'(x) < 0 \quad &\Longrightarrow \quad f(x_0) \text{ is a local maximum.} \end{aligned}$$

§90.3 Second derivative test

If

$$f'(x_0) = 0, \qquad f''(x_0) \neq 0,$$

then

$$\begin{aligned} f''(x_0) > 0 \quad &\Longrightarrow \quad x_0 \text{ is a local minimum,} \\ f''(x_0) < 0 \quad &\Longrightarrow \quad x_0 \text{ is a local maximum.} \end{aligned}$$

The handwritten note adds the mnemonic that $f'' > 0$ looks like a smile and $f'' < 0$ looks like a frown.

§90.4 Concavity and inflection points

A curve segment is concave up if it lies above its tangent lines, and concave down if it lies below its tangent lines. In the usual calculus language,

$$f''(x) > 0$$

indicates concave up, while

$$f''(x) < 0$$

indicates concave down.

An inflection point is a point where the concavity changes. A necessary condition, when f'' exists, is

$$f''(x_0) = 0.$$

But this is not sufficient. The note highlights that $f''(x_0) = 0$ or nonexistence of $f''(x_0)$ only gives candidates; one must check whether the sign of f'' changes across x_0.

The scanned page distinguishes several types: smooth horizontal inflection, smooth non-horizontal inflection, and sharp or vertical tangent behavior. The visual test is concavity change, not merely whether the tangent exists.

§90.5 Asymptotes

The notes classify asymptotes.

A vertical asymptote occurs when

$$\lim_{x \to x_0} f(x) = \pm\infty.$$

Then $x = x_0$ is a vertical asymptote.

A horizontal asymptote occurs when

$$\lim_{x \to +\infty} f(x) = c \quad \text{or} \quad \lim_{x \to -\infty} f(x) = c.$$

Then $y = c$ is a horizontal asymptote.

For an oblique asymptote

$$y = ax + b,$$

we require

$$\lim_{x \to \infty} [f(x) - (ax + b)] = 0.$$

The page derives

$$a = \lim_{x \to \infty} \frac{f(x)}{x},$$

and then

$$b = \lim_{x \to \infty} (f(x) - ax).$$

These formulas are highlighted in the source.

§90.6 General procedure for sketching a curve

The pages give a curve-sketching checklist:

(i) determine the domain;

(ii) check symmetry and periodicity;

(iii) compute $f'(x)$ and $f''(x)$;

(iv) find critical points and possible inflection points;

(v) determine intervals of monotonicity and concavity;

(vi) find asymptotes;

(vii) locate intercepts or special points if useful;

(viii) draw the graph according to this information.

This is not a mechanical ritual. The derivative table tells how the curve moves; the second derivative table tells how it bends; asymptotes determine the large-scale shape.

§90.7 Basic antiderivative table

The integration section records standard formulas:

$$\int 0\,dx = C, \qquad \int x^\mu\,dx = \frac{x^{\mu+1}}{\mu+1} + C \quad (\mu \neq -1),$$
$$\int \frac{1}{x}\,dx = \ln|x| + C, \qquad \int a^x\,dx = \frac{a^x}{\ln a} + C.$$

Trigonometric formulas include

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C,$$
$$\int \sec^2 x\,dx = anx + C, \qquad \int \csc^2 x\,dx = -\cot x + C,$$
$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C, \qquad \int \frac{dx}{1+x^2} = \arctan x + C.$$

The page also lists logarithmic forms such as

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln|x + \sqrt{x^2+a^2}| + C.$$

§90.8 Substitution

For a composite function, the substitution rule is

$$\int f(arphi(v))arphi'(v)\,dv = \int f(u)\,du, \qquad u = arphi(v).$$

Equivalently,

$$\int f(u)\,du = \int f(arphi(v))arphi'(v)\,dv.$$

The scanned page calls the two directions the first and second substitution methods. The first recognizes a derivative already present; the second introduces a new variable to simplify the integral.

§90.9 Integration by parts

From
$$d(uv) = u\,dv + v\,du,$$
one obtains
$$\int u\,dv = uv - \int v\,du.$$
The page presents integration by parts as the other major method after substitution. Its use depends on choosing u so that differentiating it simplifies the expression and choosing dv so that it can be integrated.

§90.10 Rational functions and partial fractions

A rational function is
$$\frac{P(x)}{Q(x)},$$
where P, Q are polynomials and $Q \neq 0$. If $\deg P \geq \deg Q$, first divide polynomials to write it as a polynomial plus a proper rational function.

The pages list the elementary partial fraction forms. A real linear factor $(x-a)^k$ contributes
$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}.$$
An irreducible quadratic factor $(x^2+px+q)^k$ with $p^2-4q<0$ contributes
$$\frac{M_1x+N_1}{x^2+px+q} + \cdots + \frac{M_kx+N_k}{(x^2+px+q)^k}.$$
Thus integration of rational functions reduces to two basic integral types:
$$\int \frac{dx}{(x-a)^k}, \qquad \int \frac{Mx+N}{(x^2+px+q)^k}\,dx.$$

§90.11 Quadratic denominator reduction

For the second type, split the numerator as a derivative part plus a remainder:
$$Mx+N = \frac{M}{2}(2x+p) + \left(N - \frac{M}{2}p\right).$$
Then
$$\int \frac{Mx+N}{(x^2+px+q)^k}\,dx = \frac{M}{2}I_k^1 + \left(N - \frac{M}{2}p\right)I_k^2,$$
where
$$I_k^1 = \int \frac{d(x^2+px+q)}{(x^2+px+q)^k},$$
and
$$I_k^2 = \int \frac{dx}{(x^2+px+q)^k}.$$
Complete the square by setting
$$t = x + \frac{p}{2}, \qquad a^2 = q - \frac{p^2}{4} > 0.$$

Then

$$x^2 + px + q = t^2 + a^2.$$

For $k = 1$,

$$I_1^2 = \frac{1}{a}\arctan\frac{t}{a} + C.$$

For higher k, the recurrence shown on the page is

$$I_{k+1}^2 = \frac{1}{2ka^2}\left(\frac{t}{(t^2+a^2)^k} + (2k-1)I_k^2\right).$$

§90.12 Trigonometric rational functions

The notes next handle rational functions of $\sin x$ and $\cos x$. Use the tangent half-angle substitution

$$t = an\frac{x}{2}, \qquad -\pi < x < \pi.$$

Then

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad anx = \frac{2t}{1-t^2},$$

and

$$dx = \frac{2\,dt}{1+t^2}.$$

Thus

$$\int R(\sin x, \cos x)\,dx = \int R\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right)\frac{2\,dt}{1+t^2},$$

which becomes a rational integral in t.

§90.13 Definite integrals and Riemann sums

The page defines the definite integral. Divide $[a, b]$ by

$$a = x_0 < x_1 < \cdots < x_n = b,$$

let

$$\Delta x_i = x_i - x_{i-1},$$

and choose sample points $\xi_i \in [x_{i-1}, x_i]$. The Riemann sum is

$$\sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

If the limit exists as the mesh

$$\lambda = \max_i \Delta x_i$$

tends to zero, then

$$\int_a^b f(x)\,dx$$

is the definite integral.

The pages list sufficient conditions for integrability: continuous functions, monotone bounded functions, and bounded functions with finitely many discontinuities are integrable on closed intervals.

§90.14 Basic properties of definite integrals

The standard properties are:

$$\int_a^b [\alpha f(x) + \beta g(x)]\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx,$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx, \qquad \int_a^a f(x)\,dx = 0,$$

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

If $f(x) \geq 0$ on $[a,b]$, then

$$\int_a^b f(x)\,dx \geq 0.$$

If $f \leq g$, then

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

§90.15 Integral mean value theorem

Theorem 90.15.1 (Integral mean value theorem)
If f is continuous on $[a,b]$, then there exists $c \in [a,b]$ such that

$$\int_a^b f(x)\,dx = f(c)(b-a).$$

If f is continuous on $[a,b]$, then there exists $\xi \in [a,b]$ such that

$$\int_a^b f(x)\,dx = f(\xi)(b-a).$$

If g does not change sign, the weighted form is

$$\int_a^b f(x)g(x)\,dx = f(\xi) \int_a^b g(x)\,dx.$$

The note uses this to interpret integrals as average values.

§90.16 Applications: area, volume, and arc length

Area between curves is computed by

$$A = \int_a^b [f(x) - g(x)]\,dx$$

when $f \geq g$. If horizontal slicing is better,

$$A = \int_c^d [x_{right}(y) - x_{left}(y)]\,dy.$$

The volume of a solid of revolution around the x-axis is

$$V = \pi \int_a^b [f(x)]^2\,dx.$$

The shell method has the form

$$V = 2\pi \int_a^b x f(x)\,dx$$

when rotating a region around the y-axis.

Arc length of $y = f(x)$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx.$$

For a parametric curve $x = x(t), y = y(t)$,

$$L = \int_\alpha^\beta \sqrt{(x'(t))^2 + (y'(t))^2}\,dt.$$

§90.17 Improper integrals

The last pages include reminders about improper integrals. If the interval is unbounded,

$$\int_a^{+\infty} f(x)\,dx = \lim_{R\to+\infty} \int_a^R f(x)\,dx.$$

If the integrand has a singularity at b, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx.$$

Convergence depends on the existence of these limits.

A standard comparison is the p-integral:

$$\int_1^\infty \frac{dx}{x^p}$$

converges iff $p > 1$, while

$$\int_0^1 \frac{dx}{x^p}$$

converges iff $p < 1$.

§90.18 Broader perspective

This note connects derivative information with the global shape of a curve, then connects antiderivatives with accumulation. Derivatives decide monotonicity, extrema, concavity, inflection points, and asymptotes. Integration methods reduce complicated antiderivatives to known forms. Definite integrals turn limits of sums into area, volume, length, and average value. The same calculus ideas are being used both for qualitative geometry and quantitative computation.

§90.19 Monotonicity and derivative sign

If $f' \ge 0$ on an interval, the mean value theorem gives

$$f(y) - f(x) = f'(c)(y - x) \ge 0 \qquad (x < y).$$

So monotonicity is not a separate rule; it is a global consequence of local derivative sign.

Strict monotonicity needs care: $f' > 0$ everywhere is sufficient, but not necessary. Functions can be strictly increasing even when the derivative vanishes at isolated or many points.

§90.20 Convexity and supporting lines

Convexity can be read geometrically: the graph lies below chords and above supporting tangent lines. If f is differentiable and convex, then

$$f(y) \geq f(x) + f'(x)(y - x).$$

This inequality is the foundation of many optimization methods.

§90.21 Integration methods as inverse rules

Substitution reverses the chain rule, integration by parts reverses the product rule, and partial fractions use algebraic decomposition before integration. These methods are not unrelated tricks; each integration technique recognizes which differentiation rule produced the integrand.

§90.22 Error terms and numerical integration

Midpoint and trapezoidal error formulas depend on second derivatives. They express how curvature controls the difference between an integral and a simple area approximation. In numerical analysis, this becomes a systematic study of quadrature rules and convergence rates.

§90.23 Local derivative data and global shape

Curve sketching, monotonicity, extrema, concavity, asymptotes, integration methods, and numerical errors all use derivatives to convert local information into global shape or accumulated quantity.

Calculus Review III: Convexity, Integral Inequalities, Wallis Formula, and More

N-20260106

§91.1 Convexity and the Hermite–Hadamard inequality

The first blackboard block considers a function f continuous on $[a, b]$ and twice differentiable on (a, b) with

$$f''(x) > 0.$$

This means the graph is convex. The visible statement is the Hermite–Hadamard inequality:

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a}\int_a^b f(x)\,dx \leq \frac{f(a)+f(b)}{2}.$$

The left inequality says that the value at the midpoint is no larger than the average value of a convex function. The right inequality says that the average value is no larger than the average of the endpoint values.

The geometric proof shown on the board uses two facts. First, a convex curve lies below the chord joining its endpoints:

$$f(x) \leq \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b).$$

Integrating this gives the upper bound. Second, symmetry around the midpoint and convexity imply

$$f(x) + f(a+b-x) \geq 2f\left(\frac{a+b}{2}\right).$$

Integrating from a to b gives the lower bound.

The handwritten comment points to the geometric picture: the tangent line is below a convex graph, while the chord is above it.

§91.2 Kantorovich-type integral inequality

Another framed problem assumes that f is continuous and bounded between two positive constants:

$$0 < \lambda_1 \leq f(x) \leq \lambda_2, \qquad x \in [a, b].$$

The goal is the estimate

$$\left(\int_a^b f(x)\,dx\right)\left(\int_a^b \frac{dx}{f(x)}\right) \leq \frac{(\lambda_1+\lambda_2)^2}{4\lambda_1\lambda_2}(b-a)^2.$$

The board proof starts from

$$(f(x)-\lambda_1)(f(x)-\lambda_2) \leq 0.$$

After expanding and dividing by $f(x) > 0$, one obtains

$$f(x) + \frac{\lambda_1\lambda_2}{f(x)} \le \lambda_1 + \lambda_2.$$

Integrating gives

$$A + \lambda_1\lambda_2 B \le (\lambda_1 + \lambda_2)(b-a),$$

where

$$A = \int_a^b f(x)\,dx, \qquad B = \int_a^b \frac{dx}{f(x)}.$$

Since for positive numbers u, v one has $uv \le ((u+v)/2)^2$, we get

$$\lambda_1\lambda_2 AB \le \frac{(\lambda_1+\lambda_2)^2}{4}(b-a)^2.$$

This is the desired inequality.

The note's important warning is that the usual Cauchy inequality gives the lower bound

$$\left(\int_a^b f\right)\left(\int_a^b \frac{1}{f}\right) \ge (b-a)^2,$$

whereas this problem asks for an upper bound under a two-sided bound on f.

§91.3 Integral mean value with odd symmetry

A lower-left blackboard block considers a function defined on a symmetric interval and transforms two integrals. The visible pattern is

$$\int_0^x f(t)\,dt + \int_0^{-x} f(t)\,dt = \int_0^x [f(t) - f(-t)]\,dt.$$

If f is differentiable near 0, then the integral mean value theorem gives a number $heta \in (0,1)$ such that

$$\int_0^x [f(t) - f(-t)]\,dt = x\,[f(hetax) - f(-hetax)].$$

When $xo0$, the quotient comparison uses

$$f(hetax) - f(-hetax) \approx 2hetaxf'(0),$$

while

$$\int_0^x [f(t) - f(-t)]\,dt \approx \int_0^x 2tf'(0)\,dt = x^2 f'(0).$$

Thus the natural limiting value is

$$hetao\frac{1}{2}.$$

The board notes this as a typical use of the integral mean value theorem plus local linearization.

§91.4 A lower bound involving f''/f

Another problem assumes

$$f \in C^2[a,b], \qquad f(x) > 0exton(a,b), \qquad f(a) = f(b) = 0.$$

The blackboard proves an inequality of the form

$$\int_a^b \left| \frac{f''(x)}{f(x)} \right| dx > \frac{4}{b-a}.$$

The idea is to let x_0 be a point where f attains its positive maximum. By the mean value theorem, there exist

$$\xi_1 \in (a, x_0), \qquad \xi_2 \in (x_0, b)$$

such that

$$f'(\xi_1) = \frac{f(x_0)}{x_0 - a}, \qquad f'(\xi_2) = -\frac{f(x_0)}{b - x_0}.$$

Since $f(x) \leq f(x_0)$, one has

$$\int_a^b \left| \frac{f''(x)}{f(x)} \right| dx \geq \frac{1}{f(x_0)} \int_a^b |f''(x)| \, dx.$$

The total variation of f' is at least the drop from $f'(\xi_1)$ to $f'(\xi_2)$, so

$$\frac{1}{f(x_0)} \int_a^b |f''(x)| \, dx \geq \frac{1}{x_0 - a} + \frac{1}{b - x_0}.$$

Finally,

$$\frac{1}{x_0 - a} + \frac{1}{b - x_0} \geq \frac{4}{b-a}.$$

The proof is a good example of turning a global integral estimate into a maximum-point argument.

§91.5 Symmetry of integrals about the midpoint

The second page records a common symmetry fact. If

$$f(x) = f(a + b - x),$$

then

$$\int_a^b f(x) \, dx = 2 \int_a^{(a+b)/2} f(x) \, dx.$$

Indeed, substitute

$$u = a + b - x$$

in the right half of the integral. This midpoint symmetry is used repeatedly in the board calculations.

More generally, for any integrable f,

$$\int_a^b f(x) \, dx = \int_a^b f(a + b - x) \, dx,$$

and therefore

$$\int_a^b f(x) \, dx = \frac{1}{2} \int_a^b [f(x) + f(a + b - x)] \, dx.$$

This averaging trick is useful whenever $f(x) + f(a + b - x)$ simplifies.

§91.6 Midpoint and trapezoidal error terms

The board also derives the error term for the midpoint rule. Let

$$m = \frac{a+b}{2}.$$

Taylor's formula around m gives

$$f(x) = f(m) + f'(m)(x-m) + \frac{1}{2}f''(\xi_x)(x-m)^2.$$

After integrating, the odd term vanishes, and the second derivative can be pulled out by the integral mean value theorem. Thus there exists $\xi \in (a,b)$ such that

$$\int_a^b f(x)\,dx = (b-a)f(m) + \frac{(b-a)^3}{24}f''(\xi).$$

Similarly, the trapezoidal rule error has the form

$$\int_a^b f(x)\,dx = \frac{b-a}{2}[f(a)+f(b)] - \frac{(b-a)^3}{12}f''(\xi)$$

for some $\xi \in (a,b)$. The board comments that confusing these constants is a common mistake: the midpoint coefficient is $1/24$, while the trapezoidal coefficient is $1/12$ with the opposite sign.

§91.7 First special integral

One board example evaluates

$$I = \int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx.$$

Set

$$x = ant, \qquad 0 \le t \le \frac{\pi}{4}.$$

Then $dx/(1+x^2) = dt$, and

$$I = \int_0^{\pi/4} \ln(1+ant)\,dt.$$

Use the substitution

$$t \mapsto \frac{\pi}{4} - t.$$

Since

$$1 + an\left(\frac{\pi}{4} - t\right) = \frac{2}{1+ant},$$

one obtains

$$I = \int_0^{\pi/4} [\ln 2 - \ln(1+ant)]\,dt.$$

Adding the two expressions for I gives

$$2I = \frac{\pi}{4}\ln 2,$$

hence

$$I = \frac{\pi}{8}\ln 2.$$

The trick is not the tangent substitution alone; it is pairing the interval by the involution $t \leftrightarrow \pi/4 - t$.

§91.8 Positivity of a Fresnel-type integral

Another board example proves

$$\int_0^{\sqrt{2\pi}} \sin x^2\, dx > 0.$$

Let

$$t = x^2, \qquad dx = \frac{dt}{2\sqrt{t}}.$$

Then

$$\int_0^{\sqrt{2\pi}} \sin x^2\, dx = \frac{1}{2}\int_0^{2\pi} \frac{\sin t}{\sqrt{t}}\, dt.$$

Split the interval into $[0,\pi]$ and $[\pi, 2\pi]$. In the second part use $t = u+\pi$; since $\sin(u+\pi) = -\sin u$, the sum becomes

$$\frac{1}{2}\int_0^{\pi} \sin u \left(\frac{1}{\sqrt{u}} - \frac{1}{\sqrt{u+\pi}}\right) du.$$

For $u \in (0,\pi)$, both factors are positive, so the integral is positive.

The board circles the comparison of the two weights. The oscillation alone is not enough; the positive half has larger weight because $1/\sqrt{t}$ decreases.

§91.9 A symmetry integral with arcsine

The lower-left board example evaluates

$$I = \int_0^1 \frac{\arcsin\sqrt{x}}{\sqrt{x(1-x)}}\, dx.$$

There are two neat ways. First,

$$d(\arcsin\sqrt{x}) = \frac{dx}{2\sqrt{x(1-x)}}.$$

Hence

$$I = 2\int_0^1 \arcsin\sqrt{x}\, d(\arcsin\sqrt{x}) = \left[\arcsin\sqrt{x}\right]^2\Big|_0^1 = \frac{\pi^2}{4}.$$

The board also uses symmetry:

$$\arcsin\sqrt{x} + \arcsin\sqrt{1-x} = \frac{\pi}{2}.$$

Averaging x and $1-x$ gives

$$I = \frac{1}{2}\int_0^1 \frac{\arcsin\sqrt{x} + \arcsin\sqrt{1-x}}{\sqrt{x(1-x)}}\, dx = \frac{\pi}{4}\int_0^1 \frac{dx}{\sqrt{x(1-x)}}.$$

The last integral equals π, so again $I = \pi^2/4$.

§91.10 A logarithmic sine integral

The lower-right board example is

$$I = \int_0^{\pi/2} \sin x \ln(\sin x)\, dx.$$

Let $u = \cos x$. Then $du = -\sin x\, dx$ and

$$I = \int_0^1 \ln\sqrt{1-u^2}\, du = \frac{1}{2}\int_0^1 \ln(1-u^2)\, du.$$

Since

$$\ln(1-u^2) = \ln(1-u) + \ln(1+u),$$

we use

$$\int_0^1 \ln(1-u)\, du = -1,$$

and

$$\int_0^1 \ln(1+u)\, du = 2\ln 2 - 1.$$

Therefore

$$I = \ln 2 - 1.$$

This example trains the habit of choosing $u = \cos x$ when the factor $\sin x\, dx$ appears.

§91.11 Integrals of $1/(a + b\cos x)$

The final page records two standard results. If $a > 0$ and $|b| < a$, then on the interval $[-\pi/2, \pi/2]$,

$$\int_{-\pi/2}^{\pi/2} \frac{dx}{a + b\cos x} = \frac{2}{\sqrt{a^2 - b^2}} \arccos\frac{b}{a}.$$

Using

$$\arccos\frac{b}{a} = 2\arctan\sqrt{\frac{a-b}{a+b}},$$

this can also be written as

$$\frac{4}{\sqrt{a^2-b^2}} \arctan\sqrt{\frac{a-b}{a+b}}.$$

On the interval $[0, \pi]$, the standard formula is

$$\int_0^{\pi} \frac{dx}{a + b\cos x} = \frac{\pi}{\sqrt{a^2-b^2}}, \qquad a > 0,\ |b| < a.$$

The note remarks that this follows directly from the universal tangent half-angle substitution, and that symmetry over the full half-period makes the final result simpler.

§91.12 Wallis integral and Wallis product

The last block recalls the Wallis integral

$$I_n = \int_0^{\pi/2} \sin^n x \, dx.$$

Integration by parts gives the recurrence

$$I_n = \frac{n-1}{n} I_{n-2} \qquad (n \geq 2).$$

Indeed,

$$I_n = \int_0^{\pi/2} \sin^{n-1} x \sin x \, dx,$$

choose

$$u = \sin^{n-1} x, \qquad dv = \sin x \, dx.$$

The boundary term vanishes and the remaining integral becomes

$$\frac{n-1}{n} I_{n-2}.$$

Comparing even and odd subsequences I_{2n} and I_{2n+1} and letting $no\infty$ gives the Wallis product.

§91.13 A wider frame

This note is about integral problem-solving as pattern recognition. Convexity gives chord and midpoint inequalities. Bounded positive functions give product estimates. Symmetry turns an integral into an average with its reflected version. Taylor's formula gives quadrature error terms. Special substitutions, especially $x = ant$ and $t = x^2$, turn difficult integrals into symmetric or sign-controlled expressions. Wallis' recurrence shows that integration by parts can convert a family of integrals into a product formula.

§91.14 Convexity as second-order positivity

For twice differentiable functions, convexity on an interval follows from

$$f''(x) \geq 0.$$

The Hermite–Hadamard inequality

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) \, dx \leq \frac{f(a)+f(b)}{2}$$

compares midpoint value, average value, and endpoint average for convex functions.

§91.15 Integral inequalities as weighted averages

Many integral inequalities are Jensen-type statements with a probability measure. If $w \geq 0$ and $\int w = 1$, then

$$f\left(\int xw(x) \, dx\right) \leq \int f(x)w(x) \, dx$$

for convex f. This is Jensen's inequality in measure form.

§91.16 Symmetry and cancellation

Integrals over symmetric intervals often vanish because an odd part cancels:

$$\int_{-a}^{a} f_{\mathrm{odd}}(x)\,dx = 0.$$

More generally, symmetry lets one project a function onto invariant and anti-invariant parts. This is the same idea behind Fourier even/odd decompositions.

§91.17 Wallis formula and asymptotic products

Wallis-type integrals encode asymptotic behavior of products and beta functions. The recurrence for

$$I_n = \int_0^{\pi/2} \sin^n x\,dx$$

leads to ratios of products and ultimately to approximations involving π. This is an early bridge from calculus to asymptotic analysis.

§91.18 Integral inequalities through convexity and asymptotics

The integral inequalities in this review are not isolated estimates. They are consequences of convexity, weighted averages, symmetry, error analysis, and asymptotic product behavior. The advanced habit is to identify which structural feature of the integrand controls the inequality.

IX

Multivariable Calculus

Part IX: Contents

Fréchet Differentiability, Hessians, and Constrained Tests

N-20220405

§92.1 Fréchet differentiability and complex differentiability

The first part of the note asks why complex differentiability is so much stronger than ordinary real differentiability. In one real direction, a limit may exist; in the complex plane, however, the difference quotient must have the same limit along every direction.

Consider

$$f(z) = \overline{z}, \qquad z = a + bi.$$

The complex derivative would be

$$f'(z) = \lim_{h \to 0} \frac{\overline{z+h} - \overline{z}}{h} = \lim_{h \to 0} \frac{\overline{h}}{h}.$$

Along the real direction $h = c \to 0$,

$$\frac{\overline{h}}{h} = \frac{c}{c} = 1.$$

Along the imaginary direction $h = di \to 0$,

$$\frac{\overline{h}}{h} = \frac{-di}{di} = -1.$$

Along the diagonal direction $h = c + ci$,

$$\frac{\overline{h}}{h} = \frac{c - ci}{c + ci} = -i.$$

The directional limits disagree, so $f(z) = \overline{z}$ is not complex differentiable. This is the basic reason complex differentiability is rigid.

If $f(z) = u(x, y) + iv(x, y)$, the Cauchy–Riemann equations are

$$u_x = v_y, \qquad u_y = -v_x.$$

When these hold with suitable regularity, the complex derivative is

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

§92.2 Fréchet derivative

Let V, W be normed vector spaces, and let $U \subseteq V$ be open. A function $f : U \to W$ is Fréchet differentiable at $x \in U$ if there exists a bounded linear operator $A : V \to W$ such that

$$\lim_{\|h\| \to 0} \frac{\|f(x+h) - f(x) - Ah\|_W}{\|h\|_V} = 0.$$

Equivalently,

$$f(x+h) = f(x) + Ah + o(\|h\|).$$

The operator A is unique and is denoted

$$Df(x) = A.$$

The note compares this with the Gâteaux differential. If $k = \varepsilon h$, then

$$f(x + \varepsilon h) = f(x) + \varepsilon(Df)h + h\,o(\varepsilon),$$

and, in the limit, one gets a directional derivative. But the Fréchet derivative is stronger: it requires one linear map to approximate the function uniformly in all directions.

§92.3 Visualizing complex maps

The page shows conformal grid pictures: a rectangular grid is sent to curved orthogonal families under maps such as

$$f(z) = z^2, \qquad f(z) = \frac{1}{z}, \qquad f(z) = \sin z.$$

The key geometric point is that holomorphic maps with nonzero derivative preserve angles locally. They may bend the grid, but they preserve the orthogonality of the two coordinate families.

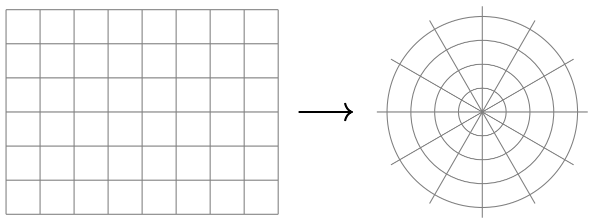


Figure 92.1: Schematic reconstruction: a rectangular grid is deformed into orthogonal curve families by a holomorphic map.

§92.4 Hessian matrices and Taylor expansion

The second main topic is the relationship between Hessians, Taylor expansion, and extrema. For a smooth function $f : \mathbb{R}^n \to \mathbb{R}$, Taylor expansion at a critical point p gives

$$f(p + h) = f(p) + \frac{1}{2}h^T H_f(p)h + O(\|h\|^3).$$

Thus the quadratic form associated to the Hessian controls the local behavior.

For unconstrained extrema, positive definiteness of $H_f(p)$ gives a local minimum, negative definiteness gives a local maximum, and indefiniteness gives a saddle. For constrained extrema the situation is subtler, because one must restrict the second variation to the tangent space of the constraint.

§92.5 Bordered Hessian

For a constrained problem

$$\text{extremize } f(x) \quad \text{subject to } g_1(x) = \cdots = g_m(x) = 0,$$

one introduces the Lagrangian

$$\mathcal{L}(x,\lambda) = f(x) - \sum_{i=1}^{m} \lambda_i g_i(x).$$

The bordered Hessian packages the second derivatives of $\mathcal{L}$ together with the first derivatives of the constraints:

$$H_B = \begin{pmatrix} 0 & Dg \\ Dg^T & D_x^2\mathcal{L} \end{pmatrix}.$$

The sign pattern of certain principal minors then diagnoses constrained maxima and minima.

§92.6 Example

The note considers the inequality problem

$$a, b, c > 0, \qquad ab + bc + ca = a + b + c,$$

and asks to prove

$$\sqrt{ab} + \sqrt{bc} + \sqrt{ca} - \sqrt{abc} \geq 2.$$

Let

$$f(a,b,c) = \sqrt{ab} + \sqrt{bc} + \sqrt{ca} - \sqrt{abc},$$

and

$$g(a,b,c) = a + b + c - ab - bc - ca.$$

The Lagrangian is

$$F(a,b,c,\lambda) = f(a,b,c) + \lambda g(a,b,c).$$

The first-order equations are

$$\begin{cases} \dfrac{\sqrt{abc} - \sqrt{ab} + \sqrt{ac}}{2a} + \lambda(1 - b - c) = 0, \\ \dfrac{\sqrt{abc} + \sqrt{ab} + \sqrt{bc}}{2b} + \lambda(1 - a - c) = 0, \\ \dfrac{\sqrt{abc} + \sqrt{ac} + \sqrt{bc}}{2c} + \lambda(1 - a - b) = 0, \\ a + b + c - ab - bc - ca = 0. \end{cases}$$

The symmetric point

$$a = b = c = 1$$

is a critical point. At that point the bordered Hessian has the schematic form

$$H_B = \begin{pmatrix} 0 & -1 & -1 & -1 \\ -1 & -\frac{1}{4} & 0 & 0 \\ -1 & 0 & -\frac{1}{4} & 0 \\ -1 & 0 & 0 & -\frac{1}{4} \end{pmatrix}.$$

The relevant minors have the required sign pattern, so the symmetric point gives the constrained extremum. Since

$$f(1,1,1) = 2,$$

we obtain the desired inequality.

§92.7 Finite fields and Galois theory

A final aside in the note jumps to finite fields and Galois theory. The basic facts listed are:

(i) the finite field $\mathbb{F}_p$ has arithmetic modulo p;

(ii) polynomial factorization over $\mathbb{F}_p$ can differ dramatically from factorization over $\mathbb{Q}$;

(iii) adjoining roots creates finite extensions such as $\mathbb{F}_{p^n}$;

(iv) quotient rings $\mathbb{F}_p[x]/(q(x))$ construct finite fields when q is irreducible.

Thus the same algebraic idea reappears: to solve an equation, enlarge the field, then study the symmetries of the roots.

§92.8 Why complex analysis meets Hessians

The original pages feel discontinuous because they jump from complex differentiability to Hessians and bordered Hessians. There is nevertheless a shared theme: first-order information tells us what linear approximation is allowed, and second-order information tells us what happens after the first-order obstruction vanishes.

For $f(z) = \overline{z}$, the real derivative exists as a real-linear map, but complex differentiability asks for multiplication by one complex number. The failure is exactly the failure of a real-linear map to commute with multiplication by i. In modern language, complex differentiability means the derivative is $\mathbb{C}$-linear, not merely $\mathbb{R}$-linear.

For constrained extrema, the first-order condition removes the tangent freedom, so the Hessian test must be performed only on tangent directions to the constraint. This is why the bordered Hessian appears: it is not a mysterious determinant recipe, but a bookkeeping device for second variation under first-order constraints.

§92.9 Frechet derivative as linear approximation

For a map $F : V \to W$ between normed spaces, differentiability at a means there is a bounded linear map L such that

$$F(a + h) = F(a) + Lh + o(\|h\|).$$

This L is the Frechet derivative. It is stronger and cleaner than checking directional derivatives separately, because it requires one linear map to approximate the function uniformly in all directions.

§92.10 Hessian as second derivative

For scalar-valued functions, the Hessian is the derivative of the gradient. It records the quadratic part of the Taylor expansion:

$$f(a + h) = f(a) + Df(a)h + \frac{1}{2}h^T H_f(a)h + o(\|h\|^2).$$

At a critical point, the linear term vanishes, so the Hessian controls the first visible local shape.

Lagrange Multipliers, Hessians, Bordered Hessians, and Degenerate Extrema

N-20220501

§93.1 Why this note was too compressed before

The note is not just "Lagrange multipliers plus Hessian." It is a long reflection on how to decide extrema of multivariable functions. It contains the method, the derivation, examples, doubts, bordered Hessian, degeneracy, higher-order tests, numerical methods, MATLAB, and links to optimization theory. The repaired version preserves that exploration.

§93.2 Lagrange multipliers: the basic idea

Suppose we want to find extrema of

$$f(x_1, \dots, x_n)$$

under the constraint

$$g(x_1, \dots, x_n) = 0.$$

The Lagrange function is

$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) - \lambda g(x_1, \dots, x_n).$$

The stationarity equations are

$$\frac{\partial L}{\partial x_i} = 0, \qquad i = 1, \dots, n, \qquad \frac{\partial L}{\partial \lambda} = 0.$$

Equivalently,

$$\nabla f = \lambda \nabla g, \qquad g = 0.$$

Geometrically, at a constrained extremum, the level surface of f is tangent to the constraint surface. Therefore their normal directions are parallel. This explains why gradients are proportional.

The original Chinese note says that this method is both surprising and confusing: surprising because it transforms a constrained problem into a stationarity system, confusing because the multiplier seems to appear out of nowhere. The geometric explanation removes much of the mystery: λ is the scalar measuring how the two normal vectors compare.

§93.3 Example: minimizing distance to a line

Consider

$$f(x, y) = x^2 + y^2$$

under the constraint

$$x + y = 1.$$

Set

$$L(x, y, \lambda) = x^2 + y^2 - \lambda(x + y - 1).$$

The equations are

$$2x - \lambda = 0, \qquad 2y - \lambda = 0, \qquad x + y = 1.$$

The first two give $x = y$, and therefore $x = y = 1/2$. This is the closest point of the line $x + y = 1$ to the origin.

The note remarks that this example is simple but educational. For nonlinear constraints, the stationarity system can become hard to solve; in practice one may need numerical methods such as Newton iteration. The Lagrange method gives equations, not automatic closed forms.

§93.4 Why the method is valid

A more rigorous proof uses the implicit function theorem. If $\nabla g(p) \neq 0$, then near p the constraint $g = 0$ is a smooth hypersurface. A tangent vector v to the constraint satisfies

$$\nabla g(p) \cdot v = 0.$$

If p is a constrained extremum of f, then the derivative of f along every tangent direction must vanish:

$$\nabla f(p) \cdot v = 0 \qquad \text{for all tangent } v.$$

Thus $\nabla f(p)$ is perpendicular to the tangent space. The normal space is spanned by $\nabla g(p)$, so

$$\nabla f(p) = \lambda \nabla g(p).$$

For several constraints $g_1 = \cdots = g_m = 0$, the condition becomes

$$\nabla f = \sum_{i=1}^{m} \lambda_i \nabla g_i,$$

assuming the constraint gradients are independent.

This is where the note's later warning about singular constraints enters: if the gradients of constraints are linearly dependent, the usual bordered-Hessian test may fail.

§93.5 Hessian matrix and the second derivative test

For a twice differentiable function $f(x_1, \ldots, x_n)$, the Hessian matrix at x_0 is

$$H_f(x_0) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x_0) \right)_{i,j}.$$

If the mixed partials are continuous, the Hessian is symmetric. This lets us use the linear algebra of symmetric matrices.

At an unconstrained critical point $\nabla f(x_0) = 0$:

(i) if $H_f(x_0)$ is positive definite, x_0 is a local minimum;

(ii) if $H_f(x_0)$ is negative definite, x_0 is a local maximum;

(iii) if $H_f(x_0)$ is indefinite, x_0 is a saddle point;

(iv) if $H_f(x_0)$ is singular or semidefinite, the second derivative test is inconclusive.

The proof comes from Taylor expansion:

$$f(x_0 + h) = f(x_0) + \frac{1}{2}h^T H_f(x_0)h + o(\|h\|^2)$$

when the gradient term vanishes. Thus the sign of the quadratic form controls the local shape.

§93.6 Sylvester criterion and eigenvalues

For a real symmetric matrix, positive definiteness can be tested by eigenvalues or by leading principal minors. Sylvester's criterion says that A is positive definite iff all leading principal minors are positive. Negative definiteness has alternating signs. Eigenvalues give an equivalent test: positive definite means all eigenvalues are positive; indefinite means there are both positive and negative eigenvalues.

The note explicitly connects this to linear algebra. A Hessian is not just an array of second derivatives; it is a quadratic form. The eigenvectors are principal directions of curvature, and the eigenvalues measure curvature along those directions.

§93.7 Bordered Hessian

For constrained extrema, the ordinary Hessian of f alone is not enough because we are only allowed to move along directions tangent to the constraint. The note introduces the bordered Hessian as a way to add constraint-gradient information to the second derivative test.

For constraints $g_i(x) = 0$, define the Lagrangian

$$L = f - \sum_{i=1}^{m} \lambda_i g_i.$$

A common bordered Hessian has the block form

$$B = \begin{pmatrix} 0_{m\times m} & (Dg)^T \\ Dg & H_x L \end{pmatrix},$$

where Dg records gradients of the constraints and $H_x L$ is the Hessian with respect to the original variables. Different texts use slightly different sign conventions and ordering of rows, so one must check the convention before applying a sign rule.

The intuitive meaning is: the border records which directions are forbidden by the constraints, and the Hessian block records curvature of the Lagrangian in the remaining directions.

§93.8 Example revisited with a bordered Hessian

For the simple constraint $x + y = 1$, the constraint gradient is

$$\nabla g = (1, 1).$$

If $f(x, y) = x^2 + y^2$, then

$$H_f = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The tangent direction to $x + y = 1$ is spanned by $(1, -1)$. Along that direction,

$$(1, -1)H_f(1, -1)^T = 4 > 0,$$

so the constrained critical point is a constrained minimum.

This tangent-space calculation is often clearer than memorizing signs of bordered determinants. The bordered Hessian is a compact determinant method for doing the same restriction systematically.

§93.9 Degenerate cases and higher-order tests

The note explicitly worries about cases where the Hessian is singular or the bordered Hessian test is inconclusive. In such cases, second-order information is not enough. One must examine higher-order Taylor terms:

$$f(x_0 + h) = f(x_0) + P_k(h) + o(\|h\|^k),$$

where P_k is the first nonzero homogeneous term after the constant term. If P_k has a fixed sign, one may still obtain an extremum; if it changes sign, one gets a saddle-type point.

A standard example is $f(x, y) = x^4 + y^4$, whose Hessian at the origin is zero, but the origin is still a strict local minimum. Conversely, $f(x, y) = x^4 - y^4$ also has zero Hessian at the origin but has a saddle. Thus degeneracy forces us to look beyond quadratic approximation.

§93.10 Saddle points and geometric visualization

A saddle point is a critical point that is neither a local maximum nor a local minimum. If the Hessian has both positive and negative eigenvalues, then some directions curve upward and others downward. The note suggests using contour plots to detect saddle behavior. This is a good habit: level curves reveal local shape far more clearly than raw formulas.

The classical picture is $f(x, y) = x^2 - y^2$. Along the x-axis it increases; along the y-axis it decreases. The origin is therefore a saddle.

§93.11 Optimization beyond equality constraints

The note mentions Kuhn–Tucker conditions, penalty methods, and interior-point methods. These belong to constrained optimization with inequalities. If constraints include

$$h_j(x) \leq 0,$$

then Lagrange multipliers are supplemented by nonnegativity and complementary slackness:

$$\mu_j \geq 0, \qquad \mu_j h_j(x) = 0.$$

This says that an inequality constraint contributes a force only when it is active. This is the natural extension of the Lagrange multiplier idea.

Penalty methods instead replace constraints by large penalty terms, for example

$$f(x) + \rho \sum_j h_j(x)^2.$$

Interior-point methods keep the iterate inside the feasible region using barrier functions. All of these methods share the same philosophy: convert a constrained problem into a system that can be treated with unconstrained or nearly unconstrained tools.

§93.12 Global topology and extrema

The note speculates about global topology. This is not random. Local Hessian information tells us what happens near a critical point, but global existence of maxima and minima depends on compactness. A continuous function on a compact set attains maximum and minimum. Thus topology enters before calculus even begins.

There are also deeper links: Morse theory studies how critical points of smooth functions reflect the topology of a manifold. The Hessian at a nondegenerate critical point gives the Morse index, and the collection of critical points helps reconstruct the space. This is the high-level version of the note's intuition that extrema are connected with global structure.

§93.13 Computation and MATLAB

The note ends with computational reflection. For high-dimensional functions, one can use symbolic or numerical software to compute gradients, Hessians, eigenvalues, and bordered Hessians. A typical workflow is:

(i) define f and constraints g_i;

(ii) solve $\nabla f = \sum \lambda_i \nabla g_i$ together with $g_i = 0$;

(iii) compute the Hessian of the Lagrangian;

(iv) restrict to the tangent space or use a bordered Hessian;

(v) inspect eigenvalues or determinant signs.

The note's MATLAB idea is therefore mathematically meaningful: computation is not replacing theory, but automating the linear algebra after the theoretical conditions have been derived.

§93.14 Second variation on the tangent space

For constrained extrema, the cleanest conceptual test is to restrict the Hessian of the Lagrangian to tangent directions. If the constraint is $g(x) = 0$, then tangent vectors v satisfy

$$\nabla g(x_0) \cdot v = 0.$$

The second variation is

$$v^T H_x L(x_0, \lambda) v.$$

If this quadratic form is positive definite on the tangent space, the point is a constrained local minimum; if negative definite, a constrained local maximum.

This explains what the bordered Hessian is doing: it is a determinant device for testing the same restricted quadratic form.

§93.15 From constrained extrema to second variation and Morse data

This note is about the transition from calculus to optimization and differential geometry. Lagrange multipliers are normal-vector equations. Hessians are quadratic forms. Bordered Hessians restrict second variation to tangent directions. Degenerate cases require higher-order terms. Inequality constraints lead to KKT theory. Compactness guarantees global extrema, and Morse theory relates critical points to topology. The original confusion is productive: it points exactly toward the deeper structure behind the computational rules.

§93.16 Lagrange multipliers as cotangent annihilators

For constraints $g_1 = \cdots = g_k = 0$, the tangent space consists of directions annihilated by all dg_i. At a constrained extremum,

$$df \in \operatorname{span}(dg_1, \ldots, dg_k).$$

This is the coordinate-free form of Lagrange multipliers.

§93.17 Bordered Hessian and restricted second variation

The second-order test should be applied only to tangent directions of the constraint. The bordered Hessian is a coordinate device for computing the quadratic form of the Lagrangian on that tangent space.

If the restricted quadratic form is positive definite, the point is a constrained local minimum; if negative definite, a constrained local maximum.

§93.18 Degenerate extrema and higher order

When the second variation has zero directions, the Hessian test is inconclusive. One must examine higher-order terms or use normal forms. This is the optimization analogue of a multiple root in algebra: first-order and second-order information may not separate the cases.

§93.19 First and second variations under constraints

Constrained extrema are governed by tangent and cotangent spaces. Multipliers express first-order normality, bordered Hessians test second-order behavior along constraints, and degeneracy signals the need for higher-order local geometry.

What Does "Product" Mean? Spatial Vectors, Cross Products, and Universal Products

N-20220728

§94.1 One word can hide different mathematical types

In elementary mathematics, the word "product" often means the output of a multiplication. With vectors, however, several constructions carry that name:

$$a \cdot b \in \mathbb{R}, \qquad a \times b \in \mathbb{R}^3, \qquad (a, b) \in V \times W.$$

These expressions do not merely compute different values. They have different *types*. The dot product returns a scalar, the cross product returns a vector, and a Cartesian product is a new space whose elements are ordered pairs.

The distinction becomes even sharper in category theory. A categorical product is not primarily a binary operation on elements. It is an object characterized by projection maps and a universal mapping property. The phrase "product of X and Y" then means an object that packages maps into X and Y in the most economical possible way.

This chapter follows the word through two settings. Spatial vector geometry shows how metric and orientation create the dot and cross products. Categorical products show how a construction can instead be characterized by the maps it receives. The comparison is useful only if the two meanings are developed separately before they are placed side by side.

§94.2 Coordinates describe a vector only after a basis is chosen

Let V be a three-dimensional real vector space. A basis

$$e_1, e_2, e_3$$

means that every vector $v \in V$ has a unique expression

$$v = v^1 e_1 + v^2 e_2 + v^3 e_3.$$

The triple (v^1, v^2, v^3) is not the vector itself; it is the coordinate list of the vector in this basis. Changing the basis changes the list while leaving the geometric vector unchanged.

In an orthonormal basis of Euclidean space, vector addition and scalar multiplication act componentwise:

$$(a^1, a^2, a^3) + (b^1, b^2, b^3) = (a^1 + b^1, a^2 + b^2, a^3 + b^3),$$

$$\lambda(a^1, a^2, a^3) = (\lambda a^1, \lambda a^2, \lambda a^3).$$

The zero vector has coordinates $(0, 0, 0)$. Nonzero vectors a, b are parallel exactly when

$$a = \lambda b$$

for some nonzero scalar λ. Three vectors form a basis exactly when no one lies in the plane spanned by the other two.

A coordinate list therefore answers a decomposition problem: which scalar multiples of the basis vectors assemble the given vector? Later, an ordered pair in a categorical product will answer a different packaging problem: which two component maps assemble a map into the product?

§94.3 The dot product adds metric information

A bare vector space has addition and scalar multiplication, but no intrinsic lengths or angles. An inner product adds that geometry. In an orthonormal basis,

$$a \cdot b = a^1 b^1 + a^2 b^2 + a^3 b^3.$$

The norm is

$$\|a\| = \sqrt{a \cdot a},$$

and for nonzero vectors the angle $\theta \in [0, \pi]$ is determined by

$$\cos\theta = \frac{a \cdot b}{\|a\|\|b\|}.$$

Perpendicularity becomes the algebraic condition

$$a \cdot b = 0.$$

The dot product also computes projection. To approximate v by a multiple of a nonzero vector u, minimize

$$\|v - tu\|^2.$$

Expanding gives

$$\|v - tu\|^2 = \|v\|^2 - 2t(u \cdot v) + t^2\|u\|^2.$$

The quadratic is minimized at

$$t = \frac{u \cdot v}{u \cdot u}.$$

Thus

$$\mathrm{proj}_u(v) = \frac{u \cdot v}{u \cdot u} u.$$

The formula is not merely a coordinate trick: the inner product measures the component of v along u.

For example, with

$$u = (1, 1, 0), \qquad v = (2, 0, 1),$$

we have

$$u \cdot v = 2, \qquad u \cdot u = 2,$$

so

$$\mathrm{proj}_u(v) = u = (1, 1, 0).$$

The residual

$$v - u = (1, -1, 1)$$

is orthogonal to u, as the minimizing condition predicts.

§94.4 The cross product packages area and orientation

In an oriented Euclidean three-space, the cross product of a and b is the unique vector $a \times b$ satisfying three conditions:

(i) it is perpendicular to both a and b;

(ii) its length is
$$\|a \times b\| = \|a\|\|b\| \sin\theta;$$

(iii) its direction follows the chosen orientation.

The length is the area of the parallelogram spanned by the two vectors. The sign convention distinguishes a right-handed frame from a left-handed one. Reversing the inputs reverses the orientation:

$$a \times b = -(b \times a).$$

In a positively oriented orthonormal basis,

$$a \times b = \begin{pmatrix} a^2b^3 - a^3b^2 \\ a^3b^1 - a^1b^3 \\ a^1b^2 - a^2b^1 \end{pmatrix}.$$

Take
$$a = (1, 2, 0), \qquad b = (0, 1, 3).$$

Then
$$a \times b = (6, -3, 1).$$

A direct check gives
$$a \cdot (a \times b) = 6 - 6 = 0,$$
$$b \cdot (a \times b) = -3 + 3 = 0.$$

Moreover,
$$\|a \times b\|^2 = 36 + 9 + 1 = 46.$$

The Gram determinant of a, b is

$$\det \begin{pmatrix} a \cdot a & a \cdot b \\ b \cdot a & b \cdot b \end{pmatrix} = \det \begin{pmatrix} 5 & 2 \\ 2 & 10 \end{pmatrix} = 46,$$

confirming that the squared area equals $\|a \times b\|^2$.

The cross product therefore depends on more structure than vector addition: it uses the metric to define perpendicularity and length, the orientation to choose a sign, and the fact that the dimension is three to turn an oriented area into a perpendicular vector.

§94.5 The scalar triple product measures oriented volume

Pairing the cross product with a third vector gives

$$[a, b, c] = a \cdot (b \times c).$$

This scalar is the oriented volume of the parallelepiped spanned by a, b, c. In coordinates,

$$a \cdot (b \times c) = \det \begin{pmatrix} a^1 & a^2 & a^3 \\ b^1 & b^2 & b^3 \\ c^1 & c^2 & c^3 \end{pmatrix}.$$

Swapping two vectors reverses the sign. The value vanishes exactly when the three vectors are coplanar.

For

$$a = (1, 0, 1), \quad b = (0, 2, 0), \quad c = (1, 1, 3),$$

we obtain

$$b \times c = (6, 0, -2),$$

so

$$a \cdot (b \times c) = 4.$$

The parallelepiped has volume 4, and the positive sign says that the ordered triple has the chosen orientation.

The determinant, cross product, and scalar triple product are thus not three unrelated formulas. They are three ways of reading the same oriented-volume structure in three dimensions.

§94.6 A cross product can generate a rotation

Fix $\omega \in \mathbb{R}^3$. The map

$$v \longmapsto \omega \times v$$

is linear, so in coordinates it is represented by the skew-symmetric matrix

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}.$$

Skew-symmetry follows from

$$u \cdot (\omega \times v) = -v \cdot (\omega \times u).$$

If a rigid body rotates with angular velocity ω, the instantaneous velocity of a point at position r is

$$r'(t) = \omega \times r(t) = [\omega]_\times r(t).$$

The solution is

$$r(t) = e^{t[\omega]_\times} r(0).$$

Suppose ω is a unit vector. Let

$$K = [\omega]_\times.$$

The vector triple-product identity gives

$$K^2 v = \omega \times (\omega \times v) = \omega(\omega \cdot v) - v.$$

If

$$P = \omega\omega^T$$

is projection onto the rotation axis, then

$$K^2 = P - I, \qquad K^3 = -K.$$

The exponential series therefore collapses to

$$\boxed{e^{tK} = I + \sin t\, K + (1 - \cos t)K^2.}$$

This is Rodrigues’ rotation formula. It fixes the component parallel to ω and rotates the perpendicular plane by angle t.

For the axis $\omega = e_3$,

$$K = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and Rodrigues’ formula gives

$$e^{tK} = \begin{pmatrix} \cos t & -\sin t & 0 \\ \sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The same operation that first measured oriented area now appears as the infinitesimal generator of spatial rotation.

§94.7 Why three dimensions are exceptional

The elementary coordinate formula can make the cross product look universal, but its target being another vector is special. Two vectors naturally span an oriented parallelogram. The algebraic object recording that area is a bivector, written

$$a \wedge b \in \Lambda^2 V.$$

In dimension three,

$$\dim \Lambda^2 V = \binom{3}{2} = 3 = \dim V.$$

Once a metric and orientation are chosen, oriented area elements can be identified with perpendicular vectors. The cross product is the result of that identification.

In dimension four,

$$\dim \Lambda^2 V = \binom{4}{2} = 6,$$

so a general oriented two-dimensional area has six independent components and cannot be represented by one ordinary four-vector. In dimension two, $\Lambda^2 V$ is one-dimensional, so the oriented area is naturally a scalar multiple of the plane’s area form rather than another vector in the plane.

The following note develops this exterior-algebra explanation and derives vector identities from wedge product, contraction, and the Hodge star. Here the dimension count already tells us why the familiar binary cross product should not be copied blindly into arbitrary dimensions.

§94.8 Cartesian products package two independent pieces of data

Now change settings. For sets X and Y, the Cartesian product is

$$X \times Y = \{(x, y) : x \in X,\ y \in Y\}.$$

Its elements are ordered pairs. The projections

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y$$

recover the two components:

$$\pi_X(x, y) = x, \qquad \pi_Y(x, y) = y.$$

A function

$$h : Z \to X \times Y$$

is therefore the same information as a pair of functions

$$f : Z \to X, \qquad g : Z \to Y.$$

From h, take

$$f = \pi_X \circ h, \qquad g = \pi_Y \circ h.$$

Conversely, from f, g, define

$$\langle f, g\rangle(z) = (f(z), g(z)).$$

These constructions are inverse to one another.

For example, a parametrized plane curve

$$h(t) = (\cos t, \sin t)$$

is exactly the pairing of its coordinate functions:

$$h = \langle \cos, \sin\rangle.$$

The Cartesian product does not multiply the coordinates. It stores them side by side while allowing each to be recovered independently.

§94.9 The universal property removes dependence on ordered pairs

A category has objects, morphisms between objects, identity morphisms, and associative composition. In the category **Set**, objects are sets and morphisms are functions. In $\mathbf{Vect}_{\mathbb{R}}$, objects are real vector spaces and morphisms are linear maps. In **Grp**, objects are groups and morphisms are group homomorphisms.

The elementwise definition of $X \times Y$ makes sense in **Set**, but category theory asks for a characterization using only morphisms.

Definition 94.9.1 (Categorical product)**.** A product of objects X, Y is an object P with morphisms

$$\pi_X : P \to X, \qquad \pi_Y : P \to Y$$

such that for every object Z and every pair of morphisms

$$f : Z \to X, \qquad g : Z \to Y,$$

there exists a unique morphism

$$\langle f, g\rangle : Z \to P$$

with

$$\pi_X \circ \langle f, g\rangle = f, \qquad \pi_Y \circ \langle f, g\rangle = g.$$

$$\begin{tikzcd}
& Z \arrow[d, dashed, "\langle f,g\rangle"] \arrow[ddl, bend right, "f"'] \arrow[ddr, bend left, "g"] & \\
& P \arrow[dl, "\pi_X"] \arrow[dr, "\pi_Y"'] & \\
X & & Y
\end{tikzcd}$$

Existence says that the two component maps can be packaged into one map. Uniqueness says that no additional hidden data are present. Any map into P is completely determined by its two projections.

In **Set**, the ordinary ordered-pair construction satisfies this property. If

$$h : Z \to X \times Y$$

has projections f, g, then for every z,

$$h(z) = (\pi_X h(z), \pi_Y h(z)) = (f(z), g(z)).$$

Thus

$$h = \langle f, g\rangle$$

and the mediating map is unique.

§94.10 The universal property makes products unique

Suppose

$$(P, \pi_X, \pi_Y) \quad \text{and} \quad (Q, \rho_X, \rho_Y)$$

are both products of X, Y. The product property of P, applied to the maps ρ_X, ρ_Y, gives a unique map

$$u : Q \to P$$

with

$$\pi_X u = \rho_X, \qquad \pi_Y u = \rho_Y.$$

Similarly there is a unique map

$$v : P \to Q$$

with

$$\rho_X v = \pi_X, \qquad \rho_Y v = \pi_Y.$$

Now both $u \circ v$ and id_P have the same projections:

$$\pi_X uv = \pi_X, \qquad \pi_Y uv = \pi_Y.$$

Uniqueness in the product property forces

$$u \circ v = \mathrm{id}_P.$$

Likewise,

$$v \circ u = \mathrm{id}_Q.$$

Therefore P and Q are uniquely isomorphic in a way that preserves their projections.

This is why category theory says "the product" even though many concrete models may exist. The universal property determines the object up to a unique structure-preserving relabelling.

§94.11 Products in vector spaces and groups

In $\mathbf{Vect}_{\mathbb{R}}$, the product of V and W is the vector space

$$V \times W$$

with componentwise operations:

$$(v, w) + (v', w') = (v + v', w + w'),$$

$$\lambda(v, w) = (\lambda v, \lambda w).$$

The projections are linear. Given linear maps

$$f : U \to V, \qquad g : U \to W,$$

the pairing

$$\langle f, g\rangle(u) = (f(u), g(u))$$

is linear and is the unique map with the required projections.

For finite-dimensional spaces,

$$\dim(V \times W) = \dim V + \dim W.$$

Choosing bases of V and W concatenates them into a basis of the product. A vector in $V \times W$ is therefore decomposed into a V-component and a W-component, just as a spatial vector is decomposed into scalar coordinates after a basis is chosen. The similarity lies in packaging components; the universal property, not a coordinate formula, defines the product.

In $\mathbf{Grp}$, the product of groups G, H has elements (g, h) and multiplication

$$(g, h)(g', h') = (gg', hh').$$

The projections are homomorphisms, and a pair of homomorphisms into G, H combines uniquely into a homomorphism into $G \times H$.

These examples show why the categorical definition is useful: the same diagram characterizes products in settings whose elements carry different algebraic structures.

The three main constructions in this chapter can now be compared without relying on the shared English word:

construction	input	output
dot product	$V \times V$	$\mathbb{R}$
cross product	$\mathbb{R}^3 \times \mathbb{R}^3$	$\mathbb{R}^3$
categorical product	X, Y	an object P with projections

The categorical product $V \times V$ may serve as the *domain* of the dot or cross product, but neither bilinear operation is itself a categorical product. The type signatures already prevent the confusion:

$$\cdot : V \times V \to \mathbb{R}, \qquad \times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3.$$

A categorical product is characterized by maps *into* it, while a dot or cross product is a particular map *out of* a product object.

The difference can also be seen as information preservation. An element of $V \times V$ retains both inputs because the projections recover them exactly:

$$\pi_1(u, v) = u, \qquad \pi_2(u, v) = v.$$

The cross product deliberately discards information. For example,

$$e_1 \times e_2 = e_3$$

and

$$(2e_1) \times \left(\frac{1}{2}e_2\right) = e_3.$$

The same output comes from different ordered pairs, so no pair of projections from the output vector can reconstruct the original inputs. The dot product loses still more: every orthogonal pair has output zero. These operations are valuable precisely because they extract a selected geometric invariant rather than preserving all component data.

In categorical language, the cross product is a morphism

$$V \times V \longrightarrow V$$

whose domain happens to be a product object. A product cone consists instead of an object together with projection morphisms *from* that object. Reversing these arrows changes the mathematical question completely.

The dependencies are equally different. The dot product depends on a metric. The cross product depends on metric, orientation, and three-dimensionality. The categorical product depends only on the morphisms of the category and its universal property. Understanding a construction therefore means more than knowing a formula: one must know its type, the structure it requires, and the property that makes it canonical.

Vector Identities from Exterior Algebra: Wedge, Contraction, and the Hodge Star

N-20230205

§95.1 Why the vector triple product is easy to misremember

The identity
$$u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$$
is often compressed into the mnemonic "BAC minus CAB." The mnemonic remembers the order of the letters, but it does not explain why the result lies in the v, w-plane or why the two coefficients have those signs.

The geometry already determines most of the answer. Since
$$v \times w$$
is perpendicular to the plane spanned by v, w, the vector
$$u \times (v \times w)$$
is perpendicular to $v \times w$ and therefore lies in that plane. Hence
$$u \times (v \times w) = Av + Bw$$
for some scalars A, B.

A quick test fixes the coefficients. If $u = v$ and $v \perp w$, then
$$v \times (v \times w) = -\|v\|^2 w.$$
Thus the coefficient of w must involve $-u \cdot v$. If $u = w$ with $v \perp w$, then
$$w \times (v \times w) = \|w\|^2 v,$$
so the coefficient of v must involve $u \cdot w$.

This reasoning reveals the projection structure, but it still relies on familiar cross-product behavior. The rest of the chapter builds a framework in which the identity follows from two systematic operations: contraction and Hodge duality.

§95.2 The scalar triple product is a volume form

The scalar triple product is
$$[u, v, w] = u \cdot (v \times w).$$
It equals the oriented volume of the parallelepiped spanned by the ordered triple. In a positively oriented orthonormal basis,
$$[u, v, w] = \det \begin{pmatrix} u^1 & u^2 & u^3 \\ v^1 & v^2 & v^3 \\ w^1 & w^2 & w^3 \end{pmatrix}.$$

The expression is multilinear and alternating. If two inputs are exchanged, the sign changes; if the vectors are linearly dependent, the value is zero.

These properties are better packaged by the Euclidean volume form

$$\mathrm{vol} = e^1 \wedge e^2 \wedge e^3,$$

where e^1, e^2, e^3 is the dual coframe. Then

$$\mathrm{vol}(u, v, w) = u \cdot (v \times w).$$

The cross product can therefore be characterized without coordinates: $v \times w$ is the unique vector whose inner product with every u equals the oriented volume of u, v, w.

For example, let

$$u = (1, 0, 1), \quad v = (0, 2, 0), \quad w = (1, 1, 3).$$

Then

$$v \times w = (6, 0, -2),$$

and

$$u \cdot (v \times w) = 4.$$

The sign is positive because the ordered triple agrees with the chosen orientation, and the absolute value is the geometric volume.

§95.3 The wedge product records an oriented plane before choosing a normal

Given vectors u, v in a vector space V, their wedge product

$$u \wedge v \in \Lambda^2 V$$

is a bivector. It represents the oriented parallelogram spanned by the pair. The defining properties are bilinearity and alternation:

$$(au + bv) \wedge w = a(u \wedge w) + b(v \wedge w),$$

$$u \wedge v = -v \wedge u, \qquad u \wedge u = 0.$$

The vanishing condition

$$u \wedge v = 0$$

holds exactly when u, v are linearly dependent.

In a three-dimensional basis e_1, e_2, e_3, the bivectors

$$e_1 \wedge e_2, \qquad e_2 \wedge e_3, \qquad e_3 \wedge e_1$$

form a basis of $\Lambda^2 V$. If

$$u = u^i e_i, \qquad v = v^j e_j,$$

then

$$\begin{aligned} u \wedge v = {} & (u^1 v^2 - u^2 v^1) e_1 \wedge e_2 \\ & + (u^2 v^3 - u^3 v^2) e_2 \wedge e_3 \\ & + (u^3 v^1 - u^1 v^3) e_3 \wedge e_1. \end{aligned}$$

The three coefficients are the familiar 2×2 minors appearing in the cross product. At this stage, however, the result is an oriented area element, not yet a perpendicular vector.

That distinction matters in other dimensions. In four dimensions,

$$\dim \Lambda^2 V = \binom{4}{2} = 6,$$

so an oriented plane carries six independent components and cannot be encoded by an ordinary four-vector. The wedge product is the dimension-independent object; the cross product is a special identification available in three-dimensional metric geometry.

§95.4 The metric turns vectors into covectors

The Hodge star is most naturally defined on differential forms, so first use the Euclidean metric to convert vectors into covectors. For $u \in V$, define

$$u^\flat(v) = u \cdot v.$$

This is a one-form. The inverse operation is denoted by $\sharp$:

$$(u^\flat)^\sharp = u.$$

In an orthonormal basis the coordinate lists happen to agree, but the types do not. A vector is an input to a one-form; a one-form is a linear measurement of vectors.

The metric also induces an inner product on k-forms. On decomposable two-forms,

$$\left\langle u^\flat \wedge v^\flat, w^\flat \wedge z^\flat \right\rangle = \det \begin{pmatrix} u \cdot w & u \cdot z \\ v \cdot w & v \cdot z \end{pmatrix}.$$

In particular,

$$\|u^\flat \wedge v^\flat\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2.$$

The right-hand side is the Gram determinant of u, v, so the norm of the wedge product is exactly the area of their parallelogram.

Thus the metric does two jobs: it lets us compare forms by an inner product, and it identifies vectors with one-forms. Orientation will supply the remaining step that turns a two-form into a one-form.

§95.5 The Hodge star converts complementary dimensions

Let V be an oriented Euclidean n-space with volume form vol. The Hodge star is the linear map

$$* : \Lambda^k V^* \to \Lambda^{n-k} V^*$$

defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \,\mathrm{vol}$$

for all k-forms α, β.

In an oriented orthonormal coframe of three-space,

$$*(e^1 \wedge e^2) = e^3, \qquad *(e^2 \wedge e^3) = e^1, \qquad *(e^3 \wedge e^1) = e^2.$$

The cross product is precisely

$$\boxed{(u \times v)^\flat = *(u^\flat \wedge v^\flat).}$$

The wedge product forms the oriented area, the Hodge star selects the complementary one-form using metric and orientation, and $\sharp$ converts that one-form back into a vector.

This formula immediately gives perpendicularity. Pairing with u,

$$\begin{aligned} u \cdot (u \times v)\,\mathrm{vol} &= u^\flat \wedge *(u \times v)^\flat \\ &= u^\flat \wedge u^\flat \wedge v^\flat = 0. \end{aligned}$$

The same argument works with v. Since the Hodge star preserves norms,

$$\|u \times v\| = \|u^\flat \wedge v^\flat\|,$$

so the cross-product norm equals the parallelogram area.

The ordinary right-hand rule is therefore not an extra mnemonic attached to a coordinate determinant. It is the visible consequence of the chosen orientation in the definition of the Hodge star.

§95.6 Lagrange's identity is an inner product of bivectors

Because the Hodge star is an isometry,

$$\begin{aligned} (u \times v) \cdot (w \times z) &= \left\langle *(u^\flat \wedge v^\flat), *(w^\flat \wedge z^\flat) \right\rangle \\ &= \left\langle u^\flat \wedge v^\flat, w^\flat \wedge z^\flat \right\rangle. \end{aligned}$$

Using the induced inner product on two-forms yields

$$\boxed{(u \times v) \cdot (w \times z) = (u \cdot w)(v \cdot z) - (u \cdot z)(v \cdot w).}$$

This is Lagrange's identity.

Setting $w = u$, $z = v$ gives

$$\|u \times v\|^2 = \|u\|^2 \|v\|^2 - (u \cdot v)^2.$$

The nonnegativity of the left-hand side gives the Cauchy–Schwarz inequality

$$|u \cdot v| \leq \|u\| \|v\|.$$

Equality holds exactly when the wedge product vanishes, that is, when u, v are linearly dependent.

What looks like a special identity about a three-dimensional operation is therefore the determinant formula for the inner product of two oriented area elements.

§95.7 Contraction removes one vector from an alternating form

For a vector u, the interior product or contraction

$$\iota_u : \Lambda^k V^* \to \Lambda^{k-1} V^*$$

is defined by inserting u into the first slot:

$$(\iota_u \alpha)(v_2, \ldots, v_k) = \alpha(u, v_2, \ldots, v_k).$$

For one-forms α, β,

$$\iota_u(\alpha \wedge \beta) = \alpha(u)\beta - \beta(u)\alpha.$$

Taking $\alpha = v^\flat$, $\beta = w^\flat$ gives

$$\iota_u(v^\flat \wedge w^\flat) = (u \cdot v)w^\flat - (u \cdot w)v^\flat.$$

In an oriented Euclidean three-space, the identity

$$*(u^\flat \wedge *\eta) = -\iota_u \eta$$

holds for every two-form η. Apply it to

$$\eta = v^\flat \wedge w^\flat.$$

Then

$$\begin{aligned}(u \times (v \times w))^\flat &= *\left(u^\flat \wedge (v \times w)^\flat\right) \\ &= *\left(u^\flat \wedge *(v^\flat \wedge w^\flat)\right) \\ &= -\iota_u(v^\flat \wedge w^\flat) \\ &= (u \cdot w)v^\flat - (u \cdot v)w^\flat.\end{aligned}$$

Applying $\sharp$ gives

$$\boxed{u \times (v \times w) = (u \cdot w)v - (u \cdot v)w.}$$

The BAC–CAB identity is no longer a mnemonic. It is the contraction rule for an oriented area form, translated back into vectors by the Hodge star.

§95.8 The Levi–Civita symbol is the coordinate form of volume

In a positively oriented orthonormal basis, define

$$\varepsilon_{ijk} = \begin{cases} 1, & (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (i, j, k) \text{ is an odd permutation,} \\ 0, & \text{two indices coincide.} \end{cases}$$

These are the components of the volume form in that basis. The cross product becomes

$$(u \times v)_i = \varepsilon_{ijk} u_j v_k.$$

The repeated indices are summed, while i remains free and labels the output component.

The key contraction is

$$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}.$$

Therefore

$$\begin{aligned}(u \times (v \times w))_i &= \varepsilon_{ijk} u_j \varepsilon_{klm} v_l w_m \\ &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) u_j v_l w_m \\ &= v_i(u_j w_j) - w_i(u_j v_j).\end{aligned}$$

This is the component proof of the same contraction identity.

The notation must be interpreted carefully. The array with entries $0, \pm 1$ is tied to an oriented orthonormal basis. In arbitrary curvilinear coordinates, the components of the geometric volume form include the metric factor

$$\sqrt{\det g}.$$

Treating the permutation symbol as though it were an unchanged tensor array in every coordinate system leads to missing Jacobian factors.

§95.9 The same operators produce curl and divergence

The Hodge star is not limited to static vector identities. On oriented Euclidean three-space, a vector field X becomes a one-form $X^\flat$. Its exterior derivative

$$d(X^\flat)$$

is a two-form measuring infinitesimal circulation. Applying the Hodge star and converting back to a vector gives curl:

$$\boxed{(\operatorname{curl} X)^\flat = *d(X^\flat).}$$

Consider the rotation field

$$X(x, y, z) = (-y, x, 0).$$

Then

$$X^\flat = -y\,dx + x\,dy.$$

Taking the exterior derivative,

$$\begin{aligned} d(X^\flat) &= -dy \wedge dx + dx \wedge dy \\ &= 2\,dx \wedge dy. \end{aligned}$$

Since

$$*(dx \wedge dy) = dz,$$

we obtain

$$(\operatorname{curl} X)^\flat = 2\,dz,$$

so

$$\operatorname{curl} X = (0, 0, 2).$$

The field rotates around the z-axis with constant vorticity.

Divergence uses the complementary construction:

$$d(*X^\flat) = (\operatorname{div} X)\operatorname{vol}.$$

For the radial field

$$Y = (x, y, z),$$

we have

$$Y^\flat = x\,dx + y\,dy + z\,dz$$

and

$$*Y^\flat = x\,dy \wedge dz - y\,dx \wedge dz + z\,dx \wedge dy.$$

Differentiating gives

$$d(*Y^\flat) = 3\,dx \wedge dy \wedge dz,$$

so

$$\operatorname{div} Y = 3.$$

The progression is now continuous. Wedge product records oriented area, the metric and Hodge star translate complementary dimensions, contraction produces vector identities, and the exterior derivative turns the same algebra into differential operators on fields. The familiar cross product is one three-dimensional expression of a calculus that continues beyond coordinates and beyond constant vectors.

Local Linear Geometry: A Mapmaker's View of Derivatives and Tangent Spaces

N-20240413

§96.1 The mapmaker's problem

Suppose one wants to draw a map of a curved surface. A small enough region of the Earth can be treated as flat, even though the Earth is not flat. This is not a lie; it is a local approximation.

Differential geometry begins from the same idea. Curved objects are understood by replacing them, near one point, with linear objects. The derivative is the machine that performs this replacement.

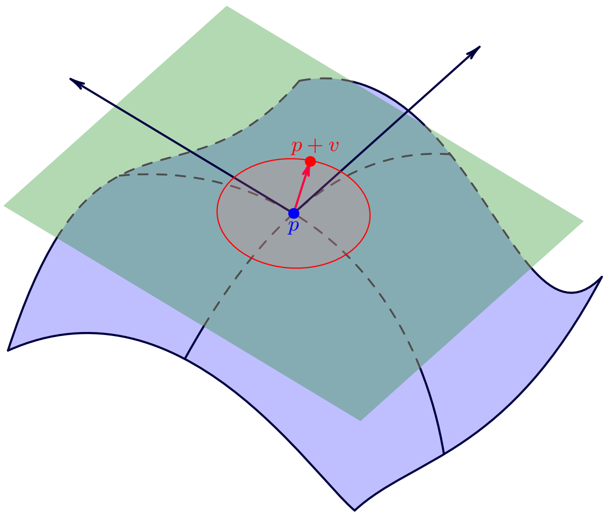


Figure 96.1: The tangent line is the first linear shadow of a curve. Higher-order details are deliberately ignored.

The guiding principle for this chapter is:

to see a curved object clearly, first learn its best linear shadow.

This principle sounds elementary, but it is the starting point for manifolds, tangent spaces, differential forms, and eventually curvature.

§96.2 Derivative first, partial derivatives second

Let

$$f : \mathbb{R}^n \to \mathbb{R}^m.$$

At a point p, the derivative should be thought of as a linear map

$$Df_p : \mathbb{R}^n \to \mathbb{R}^m$$

satisfying

$$f(p+h) = f(p) + Df_p(h) + o(\|h\|).$$

This formula says that $Df_p(h)$ is the best first-order prediction of how f changes when the input moves by h.

The Jacobian matrix is only a coordinate display of this linear map. If $f = (f_1, \dots, f_m)$, then

$$J_f(p) = \begin{bmatrix} \partial f_1/\partial x_1 & \cdots & \partial f_1/\partial x_n \\ \vdots & \ddots & \vdots \\ \partial f_m/\partial x_1 & \cdots & \partial f_m/\partial x_n \end{bmatrix}_p .$$

The matrix is useful, but the linear map is the concept.

A useful correction. Students often learn multivariable calculus as a list of partial derivatives. Geometry reverses the priority: the total derivative is the object, and partial derivatives are its coordinates.

§96.3 The chain rule is not a formula trick

If

$$\mathbb{R}^n \xrightarrow{f} \mathbb{R}^m \xrightarrow{g} \mathbb{R}^k,$$

then

$$D(g \circ f)_p = Dg_{f(p)} \circ Df_p.$$

In coordinates this becomes matrix multiplication. Geometrically it says something simpler: if f is locally approximated by Df_p, and g is locally approximated by $Dg_{f(p)}$, then their composition is locally approximated by the composition of the two linear maps.

This is why the chain rule is one of the first genuinely structural theorems in calculus. It says that differentiation respects composition.

§96.4 A small laboratory: polar coordinates

Consider

$$F(r, \theta) = (r\cos\theta, r\sin\theta).$$

Then

$$DF_{(r,\theta)} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}.$$

The two columns are

$$\frac{\partial F}{\partial r} = (\cos\theta, \sin\theta), \qquad \frac{\partial F}{\partial \theta} = (-r\sin\theta, r\cos\theta).$$

The first vector points outward. The second vector points around the circle. Thus a coordinate change is not only a substitution rule; it turns coordinate motion into tangent motion.

Notice the factor r in the angular direction. Moving by $d\theta$ near the origin is physically shorter than moving by the same $d\theta$ far away. The derivative records this distortion.

§96.5 Tangent planes from equations

A graph $z = f(x, y)$ has tangent plane

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

A level surface

$$F(x, y, z) = c$$

has tangent plane

$$\nabla F(p) \cdot (X - p) = 0.$$

Both formulas are versions of the same principle: differentiate the defining data and keep only the first-order part.

For the sphere

$$S^2 = \{x \in \mathbb{R}^3 : \|x\|^2 = 1\},$$

let $p \in S^2$. If $\gamma(t) \subset S^2$ and $\gamma(0) = p$, then

$$\gamma(t) \cdot \gamma(t) = 1.$$

Differentiating gives

$$p \cdot \gamma'(0) = 0.$$

So

$$T_pS^2 = \{v \in \mathbb{R}^3 : p \cdot v = 0\}.$$

The tangent plane is perpendicular to the radius vector. This is the most concrete form of the slogan "differentiate the constraint."

§96.6 A manifold is a space with local Euclidean windows

A smooth manifold is a space that can be covered by coordinate windows

$$\varphi : U \to \mathbb{R}^n$$

whose overlaps are smoothly compatible. A chart is not the manifold itself. It is a local language for talking about it.

The circle, sphere, torus, and projective spaces all share this feature: no single coordinate system describes them perfectly, but many local coordinate systems can be patched together.

Why charts matter. Charts let us import calculus into spaces that are not globally Euclidean. Differentiation, tangent vectors, and smooth maps are all defined by moving into a chart, doing the Euclidean calculation, and checking that the answer does not depend on the chart in a bad way.

§96.7 Two pictures of tangent vectors

There are two useful definitions of a tangent vector at p.

The first is dynamic: a tangent vector is the velocity of a curve through p:

$$v = \gamma'(0), \qquad \gamma(0) = p.$$

The second is operational: a tangent vector differentiates functions:

$$v(f) = \frac{d}{dt}\bigg|_{t=0} f(\gamma(t)).$$

The second definition is more intrinsic. It does not require the manifold to be sitting inside a larger Euclidean space.

A smooth map $F : M \to N$ sends tangent vectors forward:

$$dF_p : T_pM \to T_{F(p)}N.$$

This is the derivative on manifolds.

§96.8 Regular values: one clean machine for making manifolds

Suppose

$$F : \mathbb{R}^n \to \mathbb{R}^k$$

is smooth. If c is a regular value, meaning DF_p is surjective for every $p \in F^{-1}(c)$, then

$$F^{-1}(c)$$

is a smooth manifold of dimension $n - k$.

This theorem is useful because many geometric objects are naturally described by equations. Spheres, matrix groups, and many surfaces arise this way. The theorem says that if the equations cut cleanly, the solution set is a manifold.

§96.9 One chart in detail: stereographic projection

The sphere is the first place where one really feels why a single coordinate system is not enough. Remove the north pole

$$N = (0, 0, 1)$$

from S^2. Stereographic projection sends a point $(X, Y, Z) \in S^2 \setminus \{N\}$ to the point where the line through N and (X, Y, Z) meets the plane $Z = 0$:

$$(X, Y, Z) \longmapsto \left(\frac{X}{1 - Z}, \frac{Y}{1 - Z}\right).$$

The inverse map is

$$(u, v) \longmapsto \left(\frac{2u}{u^2 + v^2 + 1}, \frac{2v}{u^2 + v^2 + 1}, \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1}\right).$$

This formula is worth seeing once because it makes the word "chart" concrete. The sphere is not being flattened globally; one point is missing, and the remaining part is described by ordinary coordinates (u, v).

A second chart, obtained by projecting from the south pole, covers the missing point. The transition map between the two charts is smooth away from the origin. This is the atlas idea in its most visible form.

§96.10 Vector fields: tangent spaces glued along a space

A tangent vector lives at one point. A vector field chooses a tangent vector at every point:

$$p \longmapsto V(p) \in T_pM.$$

On the plane, one may draw a vector field as little arrows attached to points. On a manifold, this picture is still correct, but each arrow belongs to a different tangent space.

For example, on the unit circle $S^1 \subset \mathbb{R}^2$, the vector field

$$V(x, y) = (-y, x)$$

is tangent to the circle because

$$(x, y) \cdot (-y, x) = 0.$$

It rotates counterclockwise around the circle. This example shows a useful habit: to check whether a vector field is tangent to a constraint, check whether it preserves the constraint to first order.

§96.11 Pushforward in a concrete example

Let

$$F : \mathbb{R}^2 \to \mathbb{R}^2, \qquad F(x, y) = (x^2 - y^2, 2xy).$$

This is the complex squaring map $z \mapsto z^2$ written in real coordinates. Its derivative is

$$DF_{(x,y)} = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}.$$

At the point $(1, 0)$, a tangent vector (a, b) is sent to

$$(2a, 2b).$$

At the point $(0, 1)$, it is sent to

$$(-2b, 2a).$$

The same coordinate vector can be stretched and rotated differently depending on the base point. This is why the derivative of a nonlinear map is a family of linear maps, one for each point.

§96.12 What curvature will later add

This chapter only keeps first-order information. A tangent plane cannot distinguish a sphere from a plane at the point of tangency. Curvature begins when one asks how tangent spaces change from point to point.

So the local-linear story is not the end of geometry. It is the first layer. The later layers ask:

How do tangent spaces vary?
How do we compare vectors at different points?
How does a manifold bend?

Those questions require connections, metrics, and curvature. But without the tangent-space layer, none of them can even be stated cleanly.

The local-linear picture becomes useful through the chain

$$\text{derivative} \longrightarrow \text{linear approximation} \longrightarrow \text{tangent space} \longrightarrow \text{manifold calculus.}$$

In a calculation this chain has a definite order. First choose a chart or a defining equation. Then replace the nonlinear object by its first-order part: for a parametrized surface this means differentiating the parametrization; for a level set $F^{-1}(c)$ this means studying dF_p. The tangent space is the kernel of the linear constraint when the manifold is cut out by equations, and it is the span of coordinate velocity vectors when the manifold is parametrized.

The stereographic-projection example and the regular-value criterion are two versions of the same habit. A chart lets one compute on $\mathbb{R}^n$; a regular value tells one that the kernel of a derivative has the expected dimension. Curvature, connections, and metrics will later add second-order and comparison data, but the first check remains the same: before asking how a manifold bends, identify the tangent space in which the infinitesimal question lives.

Point-Set Language in $\mathbb{R}^n$: Neighborhoods, Boundaries, and Compactness

N-20260304

§97.1 Why point-set language appears

In one-variable calculus, the domain is often an interval, so the topology is mostly hidden. In several variables, a domain may have holes, corners, isolated points, missing boundary pieces, or several connected components. Pictures help, but pictures can lie. The reliable language is the language of balls, neighborhoods, boundary points, and sequences.

Throughout this note, points are in $\mathbb{R}^n$, and distance means Euclidean distance:

$$\rho(P, Q) = \|P - Q\|_2 = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}.$$

§97.2 Balls and neighborhoods

Definition 97.2.1. For $P_0 \in \mathbb{R}^n$ and $\delta > 0$, the open ball of radius δ around P_0 is

$$B_\delta(P_0) = \{P \in \mathbb{R}^n : \rho(P, P_0) < \delta\}.$$

A set U is a **neighborhood** of P_0 if it contains some open ball $B_\delta(P_0)$.

A neighborhood is the formal version of "all points sufficiently close to P_0." This phrase is the engine behind limits, continuity, differentiability, and local extrema.

Example 97.2.2
In $\mathbb{R}^2$, the disk

$$D = \{(x, y) : x^2 + y^2 < 1\}$$

is a neighborhood of $(0, 0)$, because it contains $B_{1/2}(0, 0)$. The punctured disk

$$D \setminus \{(0, 0)\}$$

is not a neighborhood of $(0, 0)$, although every small ball around the origin meets it.

§97.3 Interior, exterior, and boundary

Definition 97.3.1. Let $E \subseteq \mathbb{R}^n$. A point P is

(i) an **interior point** of E if some ball around P is contained in E;

(ii) an **exterior point** of E if some ball around P is disjoint from E;

(iii) a **boundary point** of E if every ball around P meets both E and $\mathbb{R}^n \setminus E$.

The set of all boundary points is denoted ∂E.

These three classes partition $\mathbb{R}^n$. The boundary is where local membership cannot be decided by zooming in.

Example 97.3.2

For
$$E = \{(x, y) : x^2 + y^2 < 1\},$$
the interior is E itself, the exterior is $\{(x, y) : x^2 + y^2 > 1\}$, and the boundary is the circle
$$\partial E = \{(x, y) : x^2 + y^2 = 1\}.$$
For the closed disk $\{x^2 + y^2 \leq 1\}$, the boundary is the same circle. Boundary is not the same as "points excluded from the set."

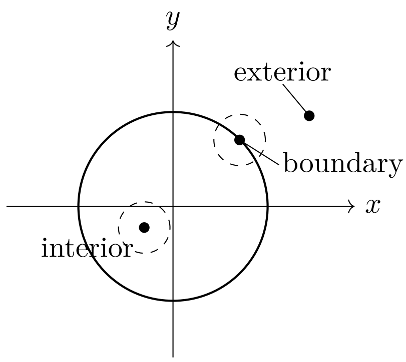


Figure 97.1: Interior points have a small ball inside the set; boundary points never do.

§97.4 Open and closed sets

Definition 97.4.1. A set $E \subseteq \mathbb{R}^n$ is **open** if every point of E is an interior point. It is **closed** if its complement is open.

The two words are not opposites in the everyday sense. A set can be both open and closed, or neither open nor closed.

Example 97.4.2

In $\mathbb{R}$, the interval $(0, 1)$ is open but not closed, $[0, 1]$ is closed but not open, $[0, 1)$ is neither open nor closed, and both $\varnothing$ and $\mathbb{R}$ are both open and closed.

Proposition 97.4.3

A set $E \subseteq \mathbb{R}^n$ is closed if and only if it contains all its boundary points.

Proof skeleton. If E is closed, its complement is open, so no boundary point can lie in the complement; hence every boundary point lies in E. Conversely, if E contains all boundary points, then any point outside E is not an interior point of E and not a boundary point, so it is exterior. Thus the complement is open. □

§97.5 Accumulation points and closure

Definition 97.5.1. A point P is an **accumulation point** of E if every punctured ball

$$B_\delta(P) \setminus \{P\}$$

contains a point of E. The **closure** of E, denoted $\overline{E}$, is the union of E with all its accumulation points.

Proposition 97.5.2 (Sequential test)

For $E \subseteq \mathbb{R}^n$, a point P lies in $\overline{E}$ if and only if there is a sequence (P_k) in E such that $P_k \to P$.

Proof skeleton. If $P \in \overline{E}$, choose $P_k \in E \cap B_{1/k}(P)$, using $P_k = P$ if $P \in E$ and necessary. Then $P_k \to P$. Conversely, if $P_k \in E$ and $P_k \to P$, then every ball around P contains some P_k, so P is in the closure. □

This gives the most useful closed-set criterion:

$$E \text{ is closed} \quad \Longleftrightarrow \quad \text{every convergent sequence in } E \text{ has its limit in } E.$$

§97.6 Boundedness and compactness

Definition 97.6.1. A set $E \subseteq \mathbb{R}^n$ is **bounded** if it is contained in some ball $B_R(0)$. It is **compact** if every open cover of E has a finite subcover.

In $\mathbb{R}^n$, compactness has a simpler working form.

Theorem 97.6.2 (Heine–Borel)

A subset $K \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

For computation, compactness is an existence machine. It makes ”largest,” ”smallest,” and ”convergent subsequence” statements true.

Theorem 97.6.3 (Extreme value theorem)

If $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum on K.

Example 97.6.4

The function $f(x, y) = x^2 + y^2$ has a maximum and a minimum on the closed disk $x^2 + y^2 \leq 1$. It has a minimum but no maximum on the open disk $x^2 + y^2 < 1$. The missing boundary is exactly where the supremum would be reached.

§97.7 Connectedness and path-connectedness

Definition 97.7.1. A set E is **connected** if it cannot be written as the union of two nonempty separated pieces. It is **path-connected** if for any two points $P, Q \in E$, there is a continuous curve $\gamma : [0, 1] \to E$ with $\gamma(0) = P$ and $\gamma(1) = Q$.

Path-connectedness implies connectedness. In the elementary domains of multivariable calculus, the two often coincide, but they are not the same in general.

A **region** in many calculus texts usually means an open connected subset of $\mathbb{R}^n$, sometimes together with a piecewise smooth boundary when integration is involved.

§97.8 What this vocabulary controls

The words in this note are not decorative definitions. They decide which theorems are legal:

open set	local perturbations stay inside the domain
closed set	limits of internal sequences remain inside
compact set	continuous functions attain extrema
boundary	where constraints and integration surfaces live
connectedness	where one can move without jumping

When later notes discuss limits, tangent planes, extrema, or multiple integrals, these are the hidden hypotheses that make the calculations legitimate.

Multivariable Limits, Continuity, Differentiability, and the Total Derivative

N-20260306

§98.1 What changes from one variable to many

In one variable, approaching a point means moving from the left or from the right. In several variables, there are infinitely many ways to approach a point: lines, curves, spirals, parabolas, and irregular paths. This is why multivariable limits are stricter than they first appear.

Definition 98.1.1. For $f : D \subseteq \mathbb{R}^n \to \mathbb{R}^m$, the statement

$$\lim_{x \to a} f(x) = L$$

means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < \|x - a\| < \delta,\ x \in D \quad \Longrightarrow \quad \|f(x) - L\| < \varepsilon.$$

The definition requires all sufficiently close points, not only points on convenient lines.

§98.2 How to prove and disprove limits

There are two basic strategies.

To prove a limit, dominate the expression by a function of $r = \|x - a\|$ that tends to zero:

$$|f(x) - L| \le g(\|x - a\|), \qquad g(r) \to 0.$$

To disprove a limit, find two paths approaching the same point that give different limiting values.

Example 98.2.1 (Two paths are enough to disprove)

Let

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Along the x-axis, $f(x,0) = 0$. Along the line $y = x$,

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Thus the limit at the origin does not exist.

Example 98.2.2 (Lines are not enough)
Let

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^4 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Along every line $y = mx$, the value tends to 0. But along $y = x^2$,

$$f(x, x^2) = \frac{x^4}{2x^4} = \frac{1}{2}.$$

So checking all straight lines still does not prove a two-variable limit.

§98.3 Continuity

Definition 98.3.1. A map $f : D \to \mathbb{R}^m$ is **continuous** at $a \in D$ if

$$\lim_{x \to a} f(x) = f(a).$$

It is continuous on D if it is continuous at every point of D.

Continuity is not just a pointwise property; it interacts strongly with compactness and connectedness. A continuous real-valued function on a compact set attains maxima and minima; a continuous image of a connected set is connected.

§98.4 Partial and directional derivatives

For $f : \mathbb{R}^2 \to \mathbb{R}$, the partial derivatives at (x_0, y_0) are

$$f_x(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$
$$f_y(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}.$$

For a unit vector u, the directional derivative is

$$D_u f(a) = \lim_{t \to 0} \frac{f(a + tu) - f(a)}{t}.$$

Partial derivatives measure coordinate-direction changes. Directional derivatives measure changes along chosen lines. Neither, by itself, guarantees differentiability.

Example 98.4.1
For the function

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

both partial derivatives at the origin exist and equal 0, because the function is zero on both axes. But the function is not continuous at the origin, hence cannot be differentiable there.

§98.5 Differentiability as linear approximation

The correct higher-dimensional analogue of differentiability is not ”all partial derivatives exist.” It is ”there is a good linear approximation.”

Definition 98.5.1. A map $F : \mathbb{R}^n \to \mathbb{R}^m$ is **differentiable** at a if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{h \to 0} \frac{\|F(a+h) - F(a) - Lh\|}{\|h\|} = 0.$$

The linear map L is the **total derivative** or **Fréchet derivative**, denoted $DF(a)$.

In coordinates, $DF(a)$ is the Jacobian matrix

$$DF(a) = \left(\frac{\partial F_i}{\partial x_j}(a) \right)_{i,j}.$$

For scalar-valued f, the derivative is represented by the row vector $\nabla f(a)^T$, so

$$f(a+h) = f(a) + \nabla f(a) \cdot h + o(\|h\|).$$

Proposition 98.5.2

If F is differentiable at a, then F is continuous at a. Moreover, for every direction u, the directional derivative exists and is

$$D_u F(a) = DF(a)u.$$

Proof. Differentiability gives

$$F(a+h) - F(a) = DF(a)h + r(h), \qquad \frac{\|r(h)\|}{\|h\|} \to 0.$$

Both $DF(a)h$ and $r(h)$ tend to 0, so continuity follows. For the directional derivative, put $h = tu$ and divide by t. □

Theorem 98.5.3 (A useful sufficient condition)

If all first partial derivatives of F exist in a neighborhood of a and are continuous at a, then F is differentiable at a.

The safe hierarchy is therefore

$$C^1 \implies \text{differentiable} \implies \text{continuous} \implies \text{partial derivatives need not exist.}$$

Existence of partial derivatives alone sits outside this chain.

§98.6 Tangent planes and normal vectors

For a graph $z = f(x, y)$, differentiability at (x_0, y_0) gives the tangent plane

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0),$$

where $z_0 = f(x_0, y_0)$. This is just the linear approximation written geometrically.

For an implicit surface $F(x, y, z) = 0$, if $\nabla F(P_0) \neq 0$, the tangent plane is

$$\nabla F(P_0) \cdot (P - P_0) = 0.$$

The gradient is normal to the level surface.

Example 98.6.1

For the sphere $F(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$, at $P_0 = (0, 0, 1)$,

$$\nabla F(P_0) = (0, 0, 2).$$

The tangent plane is therefore

$$(0, 0, 2) \cdot (x, y, z - 1) = 0,$$

so $z = 1$.

§98.7 Mixed partial derivatives

For a surface $z = f(x, y)$, the mixed partial derivative

$$f_{xy} = \partial_x(\partial_y f)$$

asks how the y-direction slope changes when one moves in the x-direction. Under the usual smoothness hypotheses, the order does not matter.

Theorem 98.7.1 (Clairaut)

If f_{xy} and f_{yx} are continuous in a neighborhood of a point, then

$$f_{xy} = f_{yx}$$

at that point.

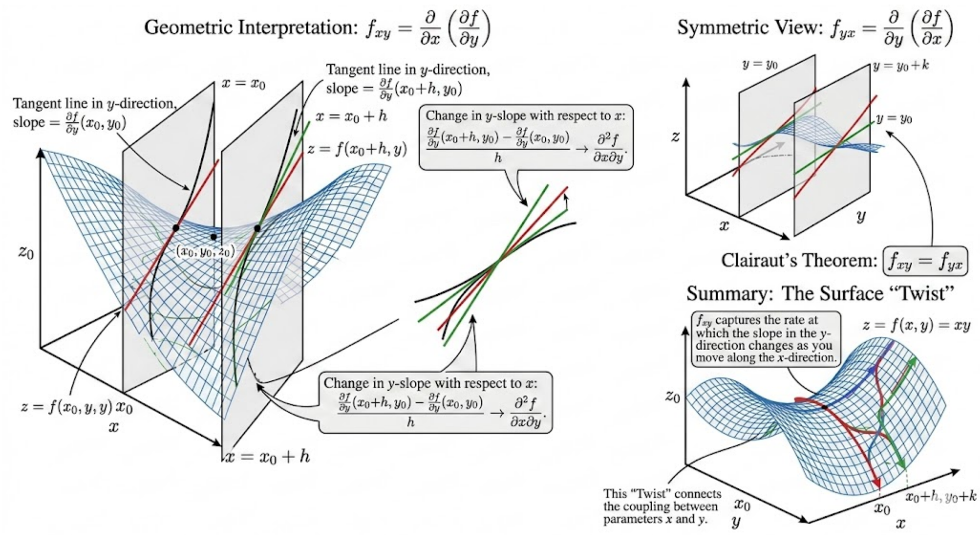


Figure 98.1: A mixed partial derivative measures how one coordinate slope changes as we move in the other coordinate direction.

§98.8 The main lesson

The word "derivative" in several variables means "linear map that captures first-order change." Partial derivatives are entries of that map, directional derivatives are evaluations of that map, tangent planes are pictures of that map, and the Jacobian is its matrix representation.

Multivariable Chain Rule, Total Differentials, and Implicit Differentiation

N-20260313

§99.1 The chain rule is composition of linear maps

The chain rule is not a collection of unrelated formulas. It says that first-order approximations compose.

Theorem 99.1.1 (Chain rule)
Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at x, and let $G : \mathbb{R}^m \to \mathbb{R}^k$ be differentiable at $F(x)$. Then $G \circ F$ is differentiable at x, and

$$D(G \circ F)(x) = DG(F(x))\, DF(x).$$

Proof skeleton. Write

$$F(x+h) = F(x) + DF(x)h + r(h), \qquad \|r(h)\| = o(\|h\|).$$

Then apply the differentiability expansion for G at $F(x)$:

$$G(F(x)+u) = G(F(x)) + DG(F(x))u + s(u), \qquad \|s(u)\| = o(\|u\|).$$

Substitute $u = DF(x)h + r(h)$. Since $u = O(\|h\|)$, the remainder is still $o(\|h\|)$. The linear part is $DG(F(x))DF(x)h$. □

This theorem is the source of all ordinary partial-derivative chain rules.

§99.2 One input variable, two intermediate variables

Suppose

$$z = f(u,v), \qquad u = u(x), \qquad v = v(x).$$

Then

$$\frac{dz}{dx} = f_u \frac{du}{dx} + f_v \frac{dv}{dx}.$$

In matrix form, this is

$$\frac{dz}{dx} = \begin{pmatrix} f_u & f_v \end{pmatrix} \begin{pmatrix} u_x \\ v_x \end{pmatrix}.$$

The row vector is Df, the column vector is the derivative of (u,v), and their product is the derivative of the composite.

Example 99.2.1

Let $z = u^2 + uv$, $u = x^2$, and $v = \sin x$. Then

$$z_u = 2u + v, \qquad z_v = u, \qquad u' = 2x, \qquad v' = \cos x.$$

Hence

$$\frac{dz}{dx} = (2u + v)2x + u\cos x.$$

Substituting $u = x^2$, $v = \sin x$ gives

$$\frac{dz}{dx} = 2x(2x^2 + \sin x) + x^2 \cos x.$$

§99.3 Two input variables

Suppose

$$z = f(u, v), \qquad u = u(x, y), \qquad v = v(x, y).$$

Then

$$\begin{pmatrix} z_x & z_y \end{pmatrix} = \begin{pmatrix} f_u & f_v \end{pmatrix} \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Equivalently,

$$z_x = f_u u_x + f_v v_x, \qquad z_y = f_u u_y + f_v v_y.$$

Example 99.3.1

Let $z = e^u \cos v$, where $u = x^2 + y^2$ and $v = x - y$. Then

$$f_u = e^u \cos v, \qquad f_v = -e^u \sin v,$$

and

$$\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} 2x & 2y \\ 1 & -1 \end{pmatrix}.$$

Thus

$$z_x = e^u \cos v \cdot 2x - e^u \sin v, \qquad z_y = e^u \cos v \cdot 2y + e^u \sin v.$$

§99.4 Total differentials

The notation

$$dz = f_x\,dx + f_y\,dy$$

is best read as a compact way to write a linear approximation. If (x, y) changes by (dx, dy), then the first-order change in $z = f(x, y)$ is

$$dz \approx f_x\,dx + f_y\,dy.$$

For a vector-valued map F, the same statement is

$$dF = J_F\,dx.$$

The differential notation is safe when one remembers that it represents a linear map, not ordinary fraction arithmetic.

§99.5 Implicit differentiation in one equation

Suppose a relation
$$F(x,y)=0$$
defines y locally as a differentiable function of x. Differentiating the identity $F(x,y(x))=0$ gives
$$F_x+F_y\frac{dy}{dx}=0.$$
If $F_y\neq 0$, then
$$\frac{dy}{dx}=-\frac{F_x}{F_y}.$$

Example 99.5.1

For the circle $F(x,y)=x^2+y^2-1=0$, where $y\neq 0$,
$$\frac{dy}{dx}=-\frac{2x}{2y}=-\frac{x}{y}.$$
This is the slope of the tangent line to the circle.

§99.6 Implicit differentiation for systems

The same idea becomes matrix algebra. Suppose
$$F(x,u)=0,\qquad F:\mathbb{R}^p\times\mathbb{R}^q\to\mathbb{R}^q,$$
and u is locally a function of x. Differentiating gives
$$F_x\,dx+F_u\,du=0.$$
If F_u is invertible, then
$$du=-F_u^{-1}F_x\,dx,$$
so
$$Du(x)=-F_u^{-1}F_x.$$

Theorem 99.6.1 (Implicit function theorem, working form)

Let $F:\mathbb{R}^p\times\mathbb{R}^q\to\mathbb{R}^q$ be C^1, and suppose
$$F(a,b)=0,\qquad \det F_u(a,b)\neq 0.$$
Then near (a,b), the equation $F(x,u)=0$ determines u as a C^1 function of x, and
$$Du(a)=-F_u(a,b)^{-1}F_x(a,b).$$

The theorem says when the formal linearized computation is legitimate. The condition $\det F_u\neq 0$ means the constraint can be solved in the u-directions.

§99.7 A shape table

Keeping track of shapes prevents many mistakes:

map	derivative shape	meaning
$f : \mathbb{R}^n \to \mathbb{R}$	$1 \times n$	row gradient
$F : \mathbb{R}^n \to \mathbb{R}^m$	$m \times n$	Jacobian
$G \circ F$	$k \times n$	$(k \times m)(m \times n)$
$F(x, u) = 0$	$q \times p,\ q \times q$	F_x, F_u

The formulas work because the linear maps have compatible domains and codomains.

§99.8 Singular linearization marks the boundary of implicit solvability

The invertibility condition in the implicit function theorem cannot simply be omitted. Consider

$$F(x, y) = y^2 - x.$$

At the origin,

$$F(0, 0) = 0, \qquad F_y(0, 0) = 0.$$

The zero set is

$$x = y^2,$$

or, for $x \geq 0$, the two branches

$$y = \sqrt{x}, \qquad y = -\sqrt{x}.$$

No neighborhood of the origin in the zero set is the graph of a single function $y = u(x)$, and the two branch derivatives become unbounded as $x \downarrow 0$. The formal formula

$$u'(x) = -\frac{F_x}{F_y} = \frac{1}{2y}$$

therefore signals its own breakdown at $y = 0$.

The same zero set can be solved smoothly in the other direction:

$$x = y^2,$$

because

$$F_x(0, 0) = -1 \neq 0.$$

Implicit solvability is directional. The Jacobian block that is invertible determines which variables may be expressed as functions of the others.

A Problem-Solving Guide for Multivariable Limits and Local Geometry

N-20260314

§100.1 Purpose of this guide

This note is a compact problem-solving companion to the previous multivariable calculus notes. It should not replace the definitions. Its job is to say what to try first, what each method proves, and where each method fails.

§100.2 Classifying points relative to a set

For a set $G \subseteq \mathbb{R}^n$, classify a point P by inspecting small balls around it:

classification	ball test
interior	$\exists r > 0,\ B_r(P) \subseteq G$
exterior	$\exists r > 0,\ B_r(P) \cap G = \varnothing$
boundary	$\forall r > 0,\ B_r(P) \cap G \neq \varnothing$ and $B_r(P) \cap G^c \neq \varnothing$

Example 100.2.1

For

$$G = \{(x, y) : 0 < x^2 + y^2 < 1\},$$

the origin is not in G, but it is a boundary point. Every ball around the origin contains points with $0 < x^2 + y^2 < 1$, and also contains the origin itself, which lies outside G.

§100.3 Limit templates

To prove a limit at the origin, try to convert the expression into polar form

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

If the expression becomes

$$r^\alpha g(\theta)$$

where $\alpha > 0$ and g is bounded, then the limit is 0.

Example 100.3.1

For

$$f(x,y) = \frac{x^2y}{x^2+y^2},$$

we have

$$f(r\cos\theta, r\sin\theta) = r\cos^2\theta\sin\theta.$$

Since $|\cos^2\theta\sin\theta| \le 1$, the limit is 0.

To disprove a limit, find paths giving different values. The common path choices are:

$$y = 0, \quad x = 0, \quad y = mx, \quad y = x^2, \quad x = y^2.$$

Caution

Path testing can disprove a limit, but it usually cannot prove one. Even checking every straight line is not enough, because a curved path may still detect different behavior.

§100.4 Degree comparison for rational expressions

For rational expressions near the origin, compare the lowest total degrees of numerator and denominator only after checking whether the denominator is comparable to a power of r. The estimate

$$x^2 + y^2 = r^2$$

is reliable. The expression $x^4 + y^2$, however, is anisotropic: along $y = x^2$, the two terms have the same size.

Example 100.4.1

The function

$$f(x,y) = \frac{x^2y}{x^4+y^2}$$

looks as if the numerator has degree 3 and the denominator has degree 2, but the denominator changes scale depending on direction. Along $y = x^2$, the limit is $1/2$. Degree-counting alone fails.

§100.5 Differentiability checklist

For a scalar function $f(x,y)$, use the following hierarchy:

$$C^1 \implies \text{differentiable} \implies \text{continuous}.$$

Partial derivatives alone do not imply continuity. Directional derivatives in every direction alone do not imply differentiability.

To prove differentiability at a, the definition asks for

$$f(a+h) = f(a) + L(h) + o(\|h\|).$$

The candidate linear map is usually

$$L(h) = \nabla f(a) \cdot h.$$

Then one must estimate the remainder.

§100.6 Tangent and normal templates

For a graph $z = f(x, y)$, the tangent plane at (x_0, y_0, z_0) is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

For an implicit surface $F(x, y, z) = 0$, the normal vector is ∇F, and the tangent plane is

$$\nabla F(P_0) \cdot (P - P_0) = 0.$$

For a space curve given by the intersection

$$F(x, y, z) = 0, \qquad G(x, y, z) = 0,$$

a tangent direction is

$$\nabla F(P_0) \times \nabla G(P_0),$$

provided the two normals are not parallel.

Example 100.6.1

The curve

$$x^2 + y^2 + z^2 = 1, \qquad z = x$$

at a point P_0 has tangent direction

$$\nabla(x^2 + y^2 + z^2 - 1)(P_0) \times \nabla(z - x)(P_0).$$

This works because the curve lies on both surfaces, so its tangent vector must be perpendicular to both surface normals.

§100.7 Local extrema template

For an unconstrained interior extremum of a differentiable function, first solve

$$\nabla f = 0.$$

Then classify candidates using the Hessian when possible. For two variables,

$$H = \begin{pmatrix} A & B \\ B & C \end{pmatrix}, \qquad D = AC - B^2.$$

At a critical point,

condition	classification
$D > 0,\ A > 0$	local minimum
$D > 0,\ A < 0$	local maximum
$D < 0$	saddle
$D = 0$	inconclusive

For a constraint $g(x, y) = 0$, first try

$$\nabla f = \lambda \nabla g.$$

Geometrically, the level curve of f is tangent to the constraint curve at a constrained extremum.

§100.8 Common failure modes

(i) Checking axes proves almost nothing about a two-variable limit.

(ii) Checking all lines still may not prove a limit.

(iii) Existence of partial derivatives is not differentiability.

(iv) The Hessian test classifies only critical points, and it is inconclusive when $D = 0$.

(v) Lagrange multipliers require checking regularity and comparing all candidate points, including boundary or endpoint cases when the constraint has them.

The common idea is local replacement: balls replace vague closeness, radial estimates replace path guessing, and linear maps replace nonlinear functions to first order.

Extrema of Two-Variable Functions and the Hessian Test

N-20260325

§101.1 Local extrema

Let $f : G \subseteq \mathbb{R}^2 \to \mathbb{R}$. A point $P_0 \in G$ is a **local maximum** if there is a neighborhood U of P_0 such that

$$f(P) \leq f(P_0) \qquad (P \in U \cap G).$$

It is a **local minimum** if the inequality is reversed. If the inequality is strict for all nearby $P \neq P_0$, the extremum is strict.

The word "local" matters. A point can be best among nearby points without being best on the whole domain.

§101.2 First-order necessary condition

Theorem 101.2.1 (Fermat theorem in several variables)
If f is differentiable at an interior point $P_0 \in G$ and P_0 is a local extremum, then

$$\nabla f(P_0) = 0.$$

Proof. For every direction u, the one-variable function

$$\varphi(t) = f(P_0 + tu)$$

has a local extremum at $t = 0$. Hence $\varphi'(0) = 0$. But

$$\varphi'(0) = \nabla f(P_0) \cdot u.$$

This vanishes for every u, so $\nabla f(P_0) = 0$. □

A point with $\nabla f = 0$ is called a **critical point**. Critical points are candidates, not conclusions.

§101.3 Quadratic model and the Hessian

At a critical point P_0, the Taylor expansion begins with the quadratic term:

$$f(P_0 + h) = f(P_0) + \frac{1}{2}h^T H(P_0)h + o(\|h\|^2),$$

where

$$H(P_0) = \begin{pmatrix} f_{xx}(P_0) & f_{xy}(P_0) \\ f_{yx}(P_0) & f_{yy}(P_0) \end{pmatrix}.$$

When the second partial derivatives are continuous, $f_{xy} = f_{yx}$, so the Hessian is symmetric.

Thus the classification problem becomes a problem about a quadratic form.

§101.4 Second derivative test

Let

$$A = f_{xx}(P_0), \qquad B = f_{xy}(P_0), \qquad C = f_{yy}(P_0), \qquad D = AC - B^2.$$

Theorem 101.4.1 (Second derivative test)

Assume $\nabla f(P_0) = 0$ and the second partial derivatives are continuous near P_0. Then:

condition	conclusion
$D > 0,\ A > 0$	P_0 is a strict local minimum
$D > 0,\ A < 0$	P_0 is a strict local maximum
$D < 0$	P_0 is a saddle point
$D = 0$	the test is inconclusive

Proof skeleton. The quadratic part is

$$q(h,k) = Ah^2 + 2Bhk + Ck^2.$$

A symmetric 2×2 matrix is positive definite exactly when $A > 0$ and $D = AC - B^2 > 0$. It is negative definite exactly when $A < 0$ and $D > 0$. If $D < 0$, the quadratic form takes both positive and negative values. The Taylor expansion shows that the sign of the quadratic term controls the sign of $f(P_0 + h) - f(P_0)$ for sufficiently small h, except in the inconclusive case $D = 0$. □

Example 101.4.2

For

$$f(x,y) = x^2 + 3y^2,$$

the only critical point is $(0,0)$, and

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}.$$

Here $D = 12 > 0$ and $A = 2 > 0$, so $(0,0)$ is a strict local minimum.

Example 101.4.3

For

$$f(x,y) = x^2 - y^2,$$

the origin is critical, but

$$H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \qquad D = -4 < 0.$$

The origin is a saddle point. Indeed, along the x-axis the function is positive, while along the y-axis it is negative.

Example 101.4.4 (Inconclusive case)

For $f(x,y) = x^4 + y^4$, the Hessian at the origin is the zero matrix, so $D = 0$. The test is inconclusive, but the origin is still a strict local minimum. For $g(x,y) = x^3 - y^3$, the Hessian at the origin is also zero, but the origin is not an extremum. This is why $D = 0$ really means "use another method."

§101.5 A picture of the Hessian test

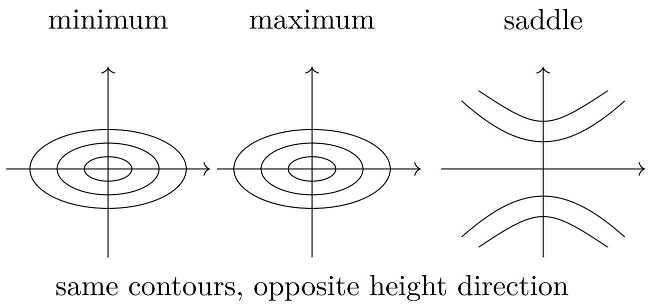


Figure 101.1: Near a nondegenerate critical point, the Hessian determines the local quadratic shape.

§101.6 Constrained extrema

Suppose one wants to optimize $f(x,y)$ subject to a constraint

$$g(x,y) = 0.$$

At a regular point of the constraint curve, $\nabla g \neq 0$, so ∇g is normal to the constraint. At a constrained extremum, moving along the constraint produces no first-order change in f. Thus ∇f must also be normal to the constraint.

Theorem 101.6.1 (Lagrange multiplier condition)

If P_0 is a constrained local extremum of f on $g = 0$, and $\nabla g(P_0) \neq 0$, then there exists λ such that

$$\nabla f(P_0) = \lambda \nabla g(P_0).$$

For several constraints $g_1 = \cdots = g_k = 0$, the gradient of f lies in the span of the constraint gradients:

$$\nabla f = \lambda_1 \nabla g_1 + \cdots + \lambda_k \nabla g_k.$$

Example 101.6.2

Find extrema of $f(x, y) = x + y$ on the circle $x^2 + y^2 = 1$. The equations are

$$(1, 1) = \lambda(2x, 2y), \qquad x^2 + y^2 = 1.$$

Thus $x = y$, and $2x^2 = 1$, so

$$(x, y) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \quad \text{or} \quad (x, y) = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$

The corresponding values are $\sqrt{2}$ and $-\sqrt{2}$, the maximum and minimum.

§101.7 Endpoint and boundary warnings

The equation $\nabla f = 0$ applies only to interior unconstrained extrema. The equation $\nabla f = \lambda \nabla g$ applies only at regular constrained points. If the domain has corners, endpoints, singular constraint points, or boundary pieces, they must be checked separately.

A reliable workflow is:

1. find interior critical points;
2. find constrained or boundary candidates;
3. check singular points and endpoints;
4. compare values when the problem asks for absolute extrema.

X

Vector Calculus

Part X: Contents

Definite Integrals, Line Integrals, Work, Green-Type Thinking, and Vector Operations

N-20230123

§102.1 Indefinite and definite integrals

The note begins by distinguishing indefinite and definite integrals. An indefinite integral is an antiderivative family:

$$\int f(x)\,dx = F(x) + C, \qquad F'(x) = f(x).$$

A definite integral is a number:

$$\int_a^b f(x)\,dx.$$

Geometrically, it is signed area under the graph. Analytically, it is the limit of Riemann sums.

The fundamental theorem of calculus connects the two:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

when $F' = f$.

§102.2 Area by slicing

The page includes diagrams of shaded regions. The standard method is to slice a region into thin strips. If vertical strips are convenient, then

$$A = \int_a^b (y_{top}(x) - y_{bottom}(x))\,dx.$$

If horizontal strips are more convenient, then

$$A = \int_c^d (x_{right}(y) - x_{left}(y))\,dy.$$

The note's sketches show that choosing the direction of slicing is part of the solution, not a fixed rule.

§102.3 Work as an integral

The physical meaning of a definite integral appears in work. If a force $F(x)$ acts along a line, then the work from $x = a$ to $x = b$ is

$$W = \int_a^b F(x)\,dx.$$

If the force is constant, this reduces to $W = F(b-a)$. If the force changes with position, integration adds the small contributions

$$dW = F(x)\,dx.$$

This is the same accumulation idea as area, but with a physical interpretation.

§102.4 Line integrals

The note then introduces line integrals. If a curve is parametrized by

$$r(t) = (x(t), y(t)), \qquad a \le t \le b,$$

then the scalar line element is

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2}\,dt.$$

A scalar line integral is

$$\int_C f\,ds = \int_a^b f(x(t), y(t))\sqrt{(x'(t))^2 + (y'(t))^2}\,dt.$$

For a vector field $F = (P, Q)$, the work integral is

$$\int_C F \cdot dr = \int_C P\,dx + Q\,dy = \int_a^b [P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)]\,dt.$$

§102.5 Green-type intuition

The handwritten diagrams show a small oriented boundary and a vector field. The idea is that circulation around a boundary can be related to a double integral over the region. In the plane, Green's theorem says

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Even if the note does not prove the theorem fully, it records the key intuition: local rotation accumulated over an area produces circulation on the boundary.

§102.6 Polar coordinates

The note uses polar coordinates in examples. The change of variables is

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The area element becomes

$$dA = r\,dr\,d\theta.$$

The factor r is the Jacobian. Geometrically, a small polar rectangle has side lengths approximately dr and $r\,d\theta$.

Thus a disk of radius R has area

$$\int_0^{2\pi}\int_0^R r\,dr\,d\theta = \pi R^2.$$

§102.7 Spherical coordinates

The later page contains a spherical-coordinate diagram. A common convention is

$$x = \rho \sin\varphi \cos\theta, \qquad y = \rho \sin\varphi \sin\theta, \qquad z = \rho \cos\varphi.$$

The volume element is

$$dV = \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta.$$

The factor $\rho^2 \sin\varphi$ comes from the three small lengths:

$$d\rho, \qquad \rho \, d\varphi, \qquad \rho \sin\varphi \, d\theta.$$

This explains the diagram on the final page: the small spherical patch has dimensions determined by radius and angle.

§102.8 Dot product and cross product

The note also summarizes vector operations. The dot product is

$$a \cdot b = |a||b| \cos\theta.$$

It measures projection and appears in work:

$$dW = F \cdot dr.$$

The cross product has magnitude

$$|a \times b| = |a||b| \sin\theta$$

and direction given by orientation. It measures the area of the parallelogram spanned by a and b.

In coordinates,

$$a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

§102.9 After the examples: what object is being integrated?

The examples develop integration as accumulation over different geometric objects. The clean way to organize them is not by dimension alone, but by the type of object being integrated.

Integral	Object being integrated	Geometric meaning
$\int_a^b f(x)\,dx$	density on an oriented interval	signed accumulation
$\int_C f\,ds$	scalar density along a curve	mass/length-weighted accumulation
$\int_C P\,dx + Q\,dy$	one-form	work or circulation
$\iint_D f\,dA$	area density	accumulation over a region
$\iiint_\Omega f\,dV$	volume density	weighted volume or mass

This table prevents a common confusion. A line integral of $f\,ds$ and a work integral $P\,dx + Q\,dy$ both run along a curve, but they integrate different geometric objects. The first uses arclength and ignores orientation; the second is a pullback of a one-form and changes sign when the orientation is reversed.

Green's theorem then becomes a boundary-to-interior conversion:

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D (\partial_x Q - \partial_y P)\,dA.$$

It says that circulation around the boundary is curl accumulated over the interior.

§102.10 Line integrals as pullbacks

For a parametrized curve $\gamma : [a,b] \to \mathbb{R}^n$, a line integral of a one-form

$$\omega = P_1 dx_1 + \cdots + P_n dx_n$$

is

$$\int_\gamma \omega = \int_a^b \sum_i P_i(\gamma(t))\gamma_i'(t)\,dt.$$

This is the pullback of the one-form to the interval:

$$\gamma^*\omega = \sum_i P_i(\gamma(t))\gamma_i'(t)dt.$$

Thus the elementary formula $Pdx + Qdy$ is already differential-form notation.

§102.11 Exact forms and potentials

A one-form ω is exact if $\omega = df$ for some potential f. Then

$$\int_\gamma df = f(\gamma(b)) - f(\gamma(a)).$$

This is the fundamental theorem of line integrals.

A form is closed if $d\omega = 0$. In the plane, for $\omega = Pdx + Qdy$, this means

$$Q_x - P_y = 0.$$

On simply connected domains, closed one-forms are exact. On domains with holes, this may fail.

§102.12 Conservation laws

In mechanics, work is a line integral

$$W = \int_C F \cdot dr.$$

If $F = -\nabla V$ is conservative, then

$$W = V(A) - V(B)$$

depends only on endpoints. Conservation of energy is therefore a statement about exact one-forms.

This connects elementary work integrals to topology: path independence is not only about derivatives; it also depends on whether the domain has holes.

§102.13 De Rham preview

De Rham cohomology measures the failure of closed forms to be exact:

$$H^1_{dR}(M) = \frac{\{\text{closed 1-forms}\}}{\{\text{exact 1-forms}\}}.$$

For the punctured plane, the form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed but not exact. Its integral around the unit circle is 2π.

The elementary line integral around a circle is therefore a measurement of a topological hole.

§102.14 Work integrals and gauge freedom

If a force field is conservative, $F = -\nabla V$, then work depends only on endpoints. The potential V is not unique: adding a constant changes nothing. This is the simplest form of gauge freedom. The physically meaningful object is the differential dV, not the absolute zero of potential.

In electromagnetism, potentials have richer gauge freedom. A vector potential A may be changed by a gradient without changing the magnetic field $B = \nabla \times A$. The elementary fact that potentials are determined only up to constants is the first example of this broader principle.

§102.15 Polar and spherical adapted charts

The original integral examples using polar or spherical coordinates should be read as choosing charts adapted to symmetry. A chart is a coordinate map from a parameter domain to the geometric region. The integration element is obtained by pulling back the metric or volume form.

Thus polar coordinates are not merely a trick for circular regions. They are a coordinate chart on the punctured plane adapted to the action of rotations. Spherical coordinates are adapted to radial symmetry in three dimensions.

Newton's Shell Theorem: Solid Angle, Potential, and Gauss Law

N-20230201

§103.1 First orientation

The elementary statement is familiar: the gravitational field produced by a uniform spherical shell is zero at every point inside the shell. But this result is not a trivial symmetry statement. If the field point is not the center, the near side of the shell is closer and should pull more strongly, while the far side has more area visible in the same direction. The theorem says that these two effects cancel exactly.

This note rewrites the old solution into three coherent languages:

solid-angle cancellation,
scalar potential computation,
Gauss law and harmonic potential.

The point is not only to remember that the answer is zero. The point is to understand why the inverse-square law, spherical geometry, and potential theory fit together so perfectly.

§103.2 Statement of Newton's shell theorem

Let a spherical shell of radius R have uniform surface density σ. Its total mass is

$$M = 4\pi R^2\sigma.$$

Let $\mathbf{x} \in \mathbb{R}^3$, and write $r = |\mathbf{x}|$, with the shell centered at the origin.

Theorem 103.2.1 (Newton's shell theorem)
The gravitational field of the shell is

$$\mathbf{g}(\mathbf{x}) = \begin{cases} \mathbf{0}, & 0 \le r < R, \\ -\dfrac{GM}{r^2}\dfrac{\mathbf{x}}{r}, & r > R. \end{cases}$$

Equivalently, outside the shell the field is the same as if all mass were concentrated at the center, while inside the shell the field is zero.

The old note focused only on the interior part. It is better to prove both halves together, because the same potential calculation gives both at once.

§103.3 Why symmetry alone is not enough

At the center of the shell, cancellation is immediate: every mass element has an opposite mass element at the same distance. Away from the center, this argument fails. A point P inside the shell is closer to one side than the other.

The correct local cancellation uses three facts at once:

the same narrow cone cuts two opposite surface patches,
patch area grows like distance squared,
gravity decays like distance squared.

There is also a small geometric correction: the patch may be tilted relative to the cone. This is the obliquity factor. Omitting it makes the proof feel like a slogan rather than a proof.

§103.4 Solid-angle proof in full detail

Let P be an interior point. Draw a line through P, meeting the shell at two points A and B. Around this line take a very narrow cone with vertex P and solid angle $d\Omega$. It cuts two small patches of the shell near A and B.

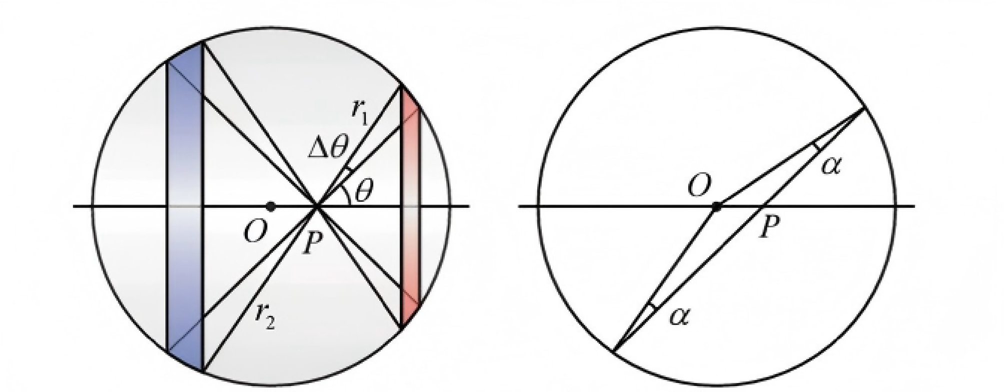


Figure 103.1: The geometric cancellation proof: the same narrow cone from an interior point cuts two opposite shell patches, and the equal angle geometry supplies the common obliquity factor.

Let

$$s_A = PA, \qquad s_B = PB.$$

Let β_A and β_B be the angles between the ray and the outward normal to the shell, taken with absolute cosine. For a small surface patch, solid angle and area satisfy

$$d\Omega = \frac{|\cos\beta|}{s^2}\,dS,$$

so

$$dS = \frac{s^2}{|\cos\beta|}\,d\Omega.$$

Therefore

$$dS_A = \frac{s_A^2}{|\cos\beta_A|}\,d\Omega, \qquad dS_B = \frac{s_B^2}{|\cos\beta_B|}\,d\Omega.$$

The missing geometric point is that the two obliquity factors are equal. Since $OA = OB = R$, the triangle OAB is isosceles. Thus the base angles at A and B are equal. Because P lies on the chord AB, the angles between the chord direction and the radii at A and B have the same absolute cosine:

$$|\cos\beta_A| = |\cos\beta_B|.$$

The mass of a patch is $dm = \sigma dS$. Its gravitational field magnitude at P is

$$|d\mathbf{g}| = G\frac{dm}{s^2} = G\sigma\frac{dS}{s^2}.$$

Substituting the area formula gives

$$|d\mathbf{g}_A| = G\sigma\frac{d\Omega}{|\cos\beta_A|}, \qquad |d\mathbf{g}_B| = G\sigma\frac{d\Omega}{|\cos\beta_B|}.$$

These magnitudes are equal, and the directions are opposite along the same line through P. Hence the two contributions cancel.

Pairing all such cones through P gives

$$\mathbf{g}(P) = \mathbf{0}.$$

The proof is now precise: the inverse-square law cancels the square growth of apparent area, and spherical geometry cancels the obliquity factor.

§103.5 Potential proof: doing the integral

The force integral is a vector integral. The potential is scalar, so it is cleaner. Let the field point have distance r from the center, and put it on the positive z-axis. A point of the shell at polar angle θ has distance

$$\ell(\theta) = \sqrt{R^2 + r^2 - 2Rr\cos\theta}$$

from the field point.

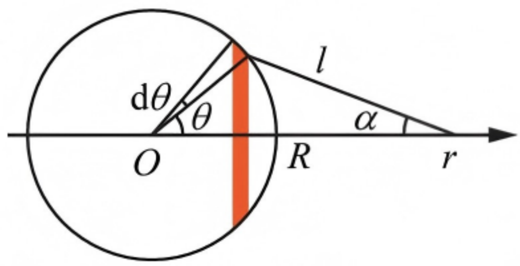


Figure 103.2: Geometry of the shell integral: the distance ℓ from the field point to a surface element and the projection angle used when passing from scalar potential to radial force.

The gravitational potential is

$$V(r) = -G\int\frac{dm}{\ell}.$$

For the shell,

$$dm = \sigma R^2 \sin\theta\, d\theta\, d\phi.$$

Hence

$$V(r) = -G\sigma R^2 \int_0^{2\pi}\int_0^{\pi} \frac{\sin\theta\, d\theta\, d\phi}{\sqrt{R^2 + r^2 - 2Rr\cos\theta}}.$$

Integrating over ϕ and putting $u = \cos\theta$, we obtain

$$V(r) = -2\pi G\sigma R^2 \int_{-1}^{1} \frac{du}{\sqrt{R^2 + r^2 - 2Rru}}.$$

Now compute the elementary integral:

$$\begin{aligned} \int_{-1}^{1} \frac{du}{\sqrt{R^2 + r^2 - 2Rru}} &= \left[-\frac{1}{Rr}\sqrt{R^2 + r^2 - 2Rru} \right]_{-1}^{1} \\ &= \frac{(R + r) - |R - r|}{Rr}. \end{aligned}$$

Therefore

$$V(r) = -2\pi G\sigma R^2 \cdot \frac{(R + r) - |R - r|}{Rr}.$$

Since $M = 4\pi R^2\sigma$, this becomes

$$V(r) = \begin{cases} -\dfrac{GM}{R}, & 0 \le r < R, \\ -\dfrac{GM}{r}, & r > R. \end{cases}$$

Now recover the field by

$$\mathbf{g} = -\nabla V.$$

For a radial potential, this means

$$\mathbf{g}(r) = -\frac{dV}{dr}\,\hat{\mathbf{r}}.$$

Inside the shell, $V(r)$ is constant, so

$$\mathbf{g} = \mathbf{0}.$$

Outside the shell,

$$V(r) = -\frac{GM}{r},$$

so

$$\mathbf{g}(r) = -\frac{GM}{r^2}\hat{\mathbf{r}}.$$

This proves both halves of Newton's shell theorem.

§103.6 The direct force integral

The direct force calculation is possible, but it is less elegant. By symmetry only the radial component survives. The radial component from a ring at polar angle θ is proportional to the projection of the force onto the radial axis.

One obtains

$$g_r(r) = 2\pi G\sigma R^2 \int_0^{\pi} \frac{(R\cos\theta - r)\sin\theta}{(R^2 + r^2 - 2Rr\cos\theta)^{3/2}}\, d\theta.$$

The sign convention here takes the positive direction away from the center. This integral is exactly

$$-\frac{dV}{dr}.$$

So the potential method does not avoid the physics; it packages the same computation in a scalar form and differentiates only at the end.

This explains why potential theory is the natural language for shell problems. Instead of separately tracking force directions, one integrates a scalar quantity whose gradient gives the field.

§103.7 Gauss-law viewpoint

There is a still shorter proof once the field is known to be spherically symmetric. The gravitational Gauss law is

$$\nabla \cdot \mathbf{g} = -4\pi G\rho,$$

or, in integral form,

$$\oint_S \mathbf{g} \cdot d\mathbf{S} = -4\pi G M_{\text{enc}}.$$

For a spherical shell, symmetry implies that the field at radius r is radial and has constant magnitude on each sphere:

$$\mathbf{g}(r) = g(r)\hat{\mathbf{r}}.$$

Taking S to be the sphere of radius r, the flux is

$$\oint_S \mathbf{g} \cdot d\mathbf{S} = 4\pi r^2 g(r).$$

Thus

$$4\pi r^2 g(r) = -4\pi G M_{\text{enc}}(r),$$

so

$$g(r) = -\frac{G M_{\text{enc}}(r)}{r^2}.$$

Inside the shell, $M_{\text{enc}}(r) = 0$, hence $g(r) = 0$. Outside the shell, $M_{\text{enc}}(r) = M$, hence

$$g(r) = -\frac{GM}{r^2}.$$

The nuance is important: Gauss law alone gives the result quickly only because spherical symmetry has already reduced the field to one radial function. The uniform spherical shell provides that symmetry.

§103.8 Harmonic potential

In an empty region, the gravitational potential satisfies Laplace's equation:

$$\Delta V = 0.$$

Inside the shell there is no mass. Moreover the shell is spherically symmetric, so the interior potential must be radial:

$$V = V(r).$$

For a radial function in $\mathbb{R}^3$, Laplace's equation becomes

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dV}{dr}\right) = 0.$$

Integrating once gives

$$r^2 V'(r) = C.$$

If $C \neq 0$, then $V'(r) = C/r^2$, which is singular at $r = 0$. The potential inside the shell should be regular at the center, so $C = 0$. Therefore

$$V'(r) = 0,$$

and the potential is constant inside.

This is the PDE version of the theorem. The absence of mass gives harmonicity; spherical symmetry reduces the harmonic equation to an ODE; regularity at the center forces the constant solution.

§103.9 Why the inverse-square law is special

The solid-angle proof shows exactly why the exponent 2 matters. For a patch cut out by a fixed solid angle,

$$dS \sim s^2\, d\Omega$$

up to the common obliquity factor. Gravity contributes a factor $1/s^2$. These cancel.

If the force law were $1/s^p$, the patch contribution would scale like

$$\frac{dS}{s^p} \sim s^{2-p} d\Omega.$$

The near and far patches would have equal magnitudes for all distances only when

$$p = 2.$$

Thus the shell theorem is not a generic fact about attractive forces. It is a special feature of inverse-square fields.

The other hypotheses also matter. If the shell is not uniform, the two paired patches do not have masses proportional only to area. If the surface is not spherical, the obliquity factors do not match in the same way. The theorem is the result of three matching structures:

$$\text{uniform density} + extspherical geometry + \text{inverse-square law}.$$

§103.10 Application: a uniform solid sphere

A uniform solid sphere can be regarded as a stack of thin spherical shells. At a point of radius r, all shells outside radius r contribute zero field. Only the mass enclosed inside radius r matters.

If the solid sphere has volume density ρ, then

$$M(r) = \frac{4\pi}{3}\rho r^3.$$

Thus the interior field is

$$g(r) = -\frac{GM(r)}{r^2} = -\frac{4\pi G\rho}{3}r.$$

So the field inside a uniform ball grows linearly with distance from the center.

This has a famous dynamical consequence. If one imagines a straight tunnel through a uniform spherical planet and ignores friction and rotation, a particle moving along the tunnel satisfies

$$\ddot{r} = -\frac{4\pi G\rho}{3}r.$$

This is simple harmonic motion. The shell theorem turns a gravitational problem into the equation of a spring.

§103.11 Multipole viewpoint

There is also a useful advanced memory aid. The Newtonian kernel has the expansion

$$\frac{1}{|x-y|} = \sum_{\ell=0}^{\infty} \frac{r_<^{\ell}}{r_>^{\ell+1}} P_\ell(\cos\gamma),$$

where

$$r_< = \min(|x|, |y|), \qquad r_> = \max(|x|, |y|).$$

For a uniformly distributed spherical shell, integrating over the angles kills all higher spherical harmonics. Only the monopole term survives outside. That is why the exterior field sees only total mass.

Inside the shell, the same expansion leaves a constant potential. A constant potential has zero gradient, hence zero field. This is the harmonic-analysis version of the same theorem.

§103.12 A broader reading

The shell theorem should not be remembered as "obvious by symmetry." For an off-center interior point, ordinary pairwise symmetry is not enough. The real mechanism is:

spherical geometry controls apparent area and obliquity,
the inverse-square law cancels the distance-squared factor,
potential theory turns cancellation into harmonicity,
and Gauss law turns it into flux conservation.

The geometric proof explains the local cancellation. The potential proof computes the whole field. Gauss law explains the flux structure. Laplace's equation explains why the interior potential must be constant.

Together these four views make Newton's shell theorem more than a trick: it is a meeting point of geometry, analysis, and field theory.

From Oscillation to Stable Limits: Riemann and Lebesgue Integration

N-20230311

§104.1 The difficult question is whether integration survives a limit

For a single continuous function on a compact interval, integration is rarely mysterious. The difficulty appears when analysis constructs a function as a limit:

$$f_n(x) \longrightarrow f(x).$$

One then wants to exchange two limiting operations:

$$\int \lim_{n\to\infty} f_n \, d\mu \stackrel{?}{=} \lim_{n\to\infty} \int f_n \, d\mu.$$

Pointwise convergence alone does not justify this exchange. A sequence may carry a fixed amount of area farther and farther away, or compress it into narrower and taller spikes.

Riemann integration begins by controlling oscillation on small intervals. Lebesgue integration changes the basic pieces from intervals to measurable sets and builds the integral so that monotone approximation is automatic. The two theories therefore differ less in the picture of rectangles than in the limiting processes they are designed to support.

§104.2 Riemann integration controls oscillation on intervals

Let $f : [a, b] \to \mathbb{R}$ be bounded and let

$$P : a = x_0 < x_1 < \cdots < x_m = b$$

be a partition. On $I_i = [x_{i-1}, x_i]$, define

$$M_i = \sup_{x\in I_i} f(x), \qquad m_i = \inf_{x\in I_i} f(x).$$

The upper and lower Darboux sums are

$$U(P, f) = \sum_{i=1}^{m} M_i |I_i|, \qquad L(P, f) = \sum_{i=1}^{m} m_i |I_i|.$$

Their difference is

$$U(P, f) - L(P, f) = \sum_{i=1}^{m} \omega(f; I_i)|I_i|,$$

where

$$\omega(f; I_i) = M_i - m_i$$

is the oscillation of f on the interval.

The Darboux criterion says

$$f \text{ is Riemann integrable} \iff \forall\varepsilon > 0\ \exists P : U(P, f) - L(P, f) < \varepsilon.$$

Thus the theory succeeds when intervals on which the function oscillates substantially can be made to have small total length.

A function may have discontinuities and still satisfy this condition. If

$$f(x) = \begin{cases} 1, & x = c, \\ 0, & x \neq c, \end{cases}$$

then every lower sum is 0. Only one interval of a partition contains c, and its length can be made arbitrarily small, so the upper sums also approach 0.

§104.3 Dense oscillation defeats every interval partition

Consider the Dirichlet function on $[a, b]$:

$$D(x) = \mathbf{1}_{\mathbb{Q}}(x).$$

Every nonempty interval contains rational and irrational points. Hence on every partition interval,

$$\sup D = 1, \qquad \inf D = 0.$$

Therefore

$$U(P, D) = b - a, \qquad L(P, D) = 0$$

for every partition. No refinement reduces the oscillation gap.

The Thomae function gives a subtler comparison. Define on $[0, 1]$

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

It is discontinuous at every rational and continuous at every irrational. Nevertheless it is Riemann integrable with integral 0.

Fix $\varepsilon > 0$. There are only finitely many reduced fractions in $[0, 1]$ with denominator at most $2/\varepsilon$. Cover these finitely many points by intervals of total length less than $\varepsilon/2$. Outside those intervals,

$$T(x) < \frac{\varepsilon}{2}.$$

A partition adapted to the cover has upper sum bounded by

$$1 \cdot \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \cdot 1 = \varepsilon.$$

Thus dense discontinuities alone are not the obstruction; what matters is the size and strength of the oscillation.

Lebesgue's criterion makes this exact: a bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has Lebesgue measure zero. The theorem is stated using measure because the Riemann criterion was already measuring how much interval length supports persistent oscillation.

§104.4 Measure spaces decide which sets may be counted

Lebesgue integration begins with a measure space

$$(X, \mathcal{M}, \mu).$$

The sigma-algebra $\mathcal{M}$ specifies the measurable subsets of X, and the measure assigns sizes to them. Countable additivity means that for pairwise disjoint measurable sets E_k,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

A function $f : X \to \overline{\mathbb{R}}$ is measurable when sets of the form

$$\{x : f(x) > t\}$$

are measurable for every real t. This condition ensures that the regions on which the function lies in a prescribed range have a defined size.

Indicator functions connect sets and functions:

$$\int_X \mathbf{1}_E \, d\mu = \mu(E).$$

For the Dirichlet function on an interval,

$$D = \mathbf{1}_{\mathbb{Q}\cap[a,b]}.$$

Since the rationals are countable and every singleton has Lebesgue measure zero,

$$\mu(\mathbb{Q} \cap [a, b]) = 0.$$

Therefore

$$\int_a^b D(x)\, dx = 0.$$

The function that defeats every interval partition becomes simple once its exceptional set can be measured directly.

§104.5 Simple functions build the nonnegative integral from below

A nonnegative simple function has the form

$$s = \sum_{j=1}^{r} a_j \mathbf{1}_{E_j}, \qquad a_j \geq 0,$$

where the measurable sets E_j are pairwise disjoint. Define

$$\int_X s \, d\mu = \sum_{j=1}^{r} a_j \mu(E_j).$$

For a nonnegative measurable function f, define

$$\boxed{\int_X f \, d\mu = \sup\left\{\int_X s \, d\mu : 0 \leq s \leq f,\ s \text{ simple}\right\}.}$$

The integral is therefore designed around monotone approximation from below.

For example, on $[0,1]$, let

$$s_n(x) = 2^{-n}\lfloor 2^n x \rfloor.$$

Then s_n is simple,

$$0 \le s_n(x) \le x, \qquad s_n(x) \uparrow x.$$

Its integral is

$$\begin{aligned}\int_0^1 s_n(x)\,dx &= \sum_{k=0}^{2^n-1} \frac{k}{2^n}\cdot\frac{1}{2^n}\\ &= \frac{2^n(2^n-1)}{2\cdot 2^{2n}}\\ &= \frac{1}{2}-\frac{1}{2^{n+1}}.\end{aligned}$$

Taking the supremum gives

$$\int_0^1 x\,dx = \frac{1}{2}.$$

Here the value is not imported from Riemann integration; it follows directly from measurable step approximations.

For a real-valued measurable function, write

$$f = f^+ - f^-, \qquad f^+ = \max(f,0), \qquad f^- = \max(-f,0).$$

The function is Lebesgue integrable when

$$\int f^+ < \infty, \qquad \int f^- < \infty,$$

or equivalently when $\int |f| < \infty$.

§104.6 Monotone convergence is built into the definition

Suppose

$$0 \le f_1 \le f_2 \le \cdots, \qquad f_n(x) \to f(x).$$

Since $f_n \le f$, monotonicity gives

$$\lim_{n\to\infty} \int f_n\,d\mu \le \int f\,d\mu.$$

For the reverse inequality, take a simple function $0 \le s \le f$ and a number $0 < c < 1$. Define

$$E_n = \{x : f_n(x) \ge cs(x)\}.$$

The sets E_n increase and cover the region on which $s > 0$. Hence

$$\int f_n\,d\mu \ge c\int_{E_n} s\,d\mu \longrightarrow c\int s\,d\mu.$$

Letting $c \uparrow 1$ and then taking the supremum over all simple $s \le f$ yields

$$\boxed{\int f\,d\mu = \lim_{n\to\infty}\int f_n\,d\mu.}$$

This is the Monotone Convergence Theorem.

The proof works because nonnegative mass only accumulates. No positive and negative parts can cancel, and no contribution already present in f_n is later removed.

§104.7 Fatou and dominated convergence control general limits

For nonnegative measurable functions f_n, define

$$g_k(x) = \inf_{n \geq k} f_n(x).$$

Then

$$g_k \uparrow \liminf_{n\to\infty} f_n.$$

Monotone convergence gives

$$\int \liminf f_n \, d\mu = \lim_{k\to\infty} \int g_k \, d\mu.$$

Since $g_k \leq f_n$ for every $n \geq k$,

$$\int g_k \, d\mu \leq \inf_{n\geq k} \int f_n \, d\mu.$$

Therefore

$$\boxed{\int \liminf f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.}$$

This is Fatou's lemma.

Now suppose $f_n \to f$ almost everywhere and

$$|f_n| \leq g$$

for some integrable g. Apply Fatou to the nonnegative sequences

$$g + f_n \quad \text{and} \quad g - f_n.$$

The first gives

$$\int g + \int f \leq \int g + \liminf \int f_n,$$

so

$$\int f \leq \liminf \int f_n.$$

The second gives

$$\int g - \int f \leq \int g - \limsup \int f_n,$$

so

$$\limsup \int f_n \leq \int f.$$

Consequently,

$$\boxed{\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu.}$$

This is the Dominated Convergence Theorem.

A basic example is

$$f_n(x) = x^n \qquad (0 \leq x \leq 1).$$

Then $f_n(x) \to 0$ for every $x < 1$, and the exceptional point $x = 1$ has measure zero. Since $|f_n| \leq 1$, dominated convergence gives

$$\int_0^1 x^n \, dx \longrightarrow 0,$$

consistent with the exact value $1/(n+1)$.

§104.8 Escape and concentration explain why domination is needed

On $\mathbb{R}$, let

$$f_n = \mathbf{1}_{[n,n+1]}.$$

For each fixed x, eventually $x \notin [n, n+1]$, so

$$f_n(x)o0.$$

Yet

$$\int_{\mathbb{R}} f_n \, dx = 1$$

for every n. The mass has escaped to infinity. No single integrable function can dominate all translated indicators.

On $[0, 1]$, let

$$h_n = n\mathbf{1}_{[0,1/n]}.$$

Then $h_n(x)o0$ for every $x > 0$, but

$$\int_0^1 h_n \, dx = 1.$$

The mass has concentrated near 0. Any pointwise dominating function would need to be at least of order $1/x$ near zero and therefore could not be integrable.

These examples show what the hypothesis $|f_n| \leq g \in L^1$ prevents. It supplies a fixed integrable envelope that forbids both translation of unit mass to infinity and compression of unit mass into arbitrarily narrow spikes.

§104.9 Improper convergence and Lebesgue integrability are different

The improper integral

$$\int_1^\infty \frac{\sin x}{x} \, dx$$

converges by cancellation. Integration by parts on $[1, R]$ gives

$$\int_1^R \frac{\sin x}{x} \, dx = \left[-\frac{\cos x}{x} \right]_1^R - \int_1^R \frac{\cos x}{x^2} \, dx.$$

The boundary term has a limit and the remaining integral converges absolutely.

However,

$$\int_1^\infty \frac{|\sin x|}{x} \, dx = \infty.$$

Indeed, on a fixed positive fraction of each period, $|\sin x|$ is bounded below by a positive constant, and the resulting comparison is with the harmonic series.

Thus $\sin x/x$ is not Lebesgue integrable on $[1, \infty)$, because Lebesgue integrability of a signed function requires

$$\int |f| < \infty.$$

This is not a weakness. Absolute integrability makes the integral insensitive to rearrangement and places the function in a normed space. Conditional improper integration preserves cancellation along a specified exhaustion of the domain, which is a different structure.

§104.10 The spaces L^p record quantitative integrability

For $1 \le p < \infty$, define

$$L^p(X) = \left\{ f : \int_X |f|^p \, d\mu < \infty \right\} / \sim,$$

where functions equal almost everywhere are identified. The norm is

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p}.$$

The quotient by almost-everywhere equality is forced by integration: changing a function on a null set changes neither its integral nor its L^p norm. The Dirichlet function and the zero function therefore represent the same element of $L^p([a, b])$.

If $1/p + 1/q = 1$, Hölder's inequality gives

$$\int_X |fg| \, d\mu \le \|f\|_p \|g\|_q.$$

Thus the integral defines a continuous pairing between L^p and L^q. In the case $p = q = 2$,

$$\langle f, g \rangle = \int_X f\overline{g} \, d\mu$$

is an inner product, and integration becomes part of Hilbert-space geometry.

Dominated convergence also has a norm interpretation. If $f_n \to f$ almost everywhere and $|f_n| \le g \in L^p$, then

$$|f_n - f|^p \le (2g)^p.$$

Applying dominated convergence to $|f_n - f|^p$ yields

$$\|f_n - f\|_p o 0.$$

A pointwise limit has become convergence in a function-space norm because one integrable envelope controls the entire sequence.

§104.11 Probability reveals the role of uniform integrability

A probability space is a measure space

$$(\Omega, \mathcal{F}, \mathbb{P})$$

with $\mathbb{P}(\Omega) = 1$. A random variable is a measurable function, and expectation is integration:

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}.$$

For an event A,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A).$$

Suppose $X_n \to X$ almost surely and $|X_n| \le Y$ with $\mathbb{E}[Y] < \infty$. Dominated convergence gives

$$\mathbb{E}[X_n] \to \mathbb{E}[X].$$

A single dominating random variable is sufficient, but it is stronger than necessary. What the proof really needs is uniform control of the tails.

A family $\{X_n\} \subset L^1$ is *uniformly integrable* if

$$\lim_{K\to\infty} \sup_n \mathbb{E}\left[|X_n|\mathbf{1}_{\{|X_n|>K\}}\right] = 0.$$

This condition says that no member of the family can hide a fixed amount of expected mass at arbitrarily large values.

Domination implies uniform integrability. If $|X_n| \le Y \in L^1$, then

$$\{|X_n| > K\} \subseteq \{Y > K\},$$

and therefore

$$\mathbb{E}\left[|X_n|\mathbf{1}_{\{|X_n|>K\}}\right] \le \mathbb{E}\left[Y\mathbf{1}_{\{Y>K\}}\right].$$

The right-hand side tends to zero by dominated convergence and is independent of n.

The concentrating sequence

$$X_n = n\mathbf{1}_{(0,1/n]}$$

on $(0,1]$ converges almost surely to zero, but

$$\mathbb{E}[X_n] = 1.$$

It is not uniformly integrable. For any fixed K, choose $n > K$. Then $|X_n| > K$ on its whole support, so

$$\mathbb{E}\left[|X_n|\mathbf{1}_{\{|X_n|>K\}}\right] = 1.$$

The supremum over n never decreases. Uniform integrability detects exactly the mass concentration that pointwise convergence misses.

Vitali's convergence theorem gives the corresponding exchange principle: if $X_n \to X$ in probability and $\{X_n\}$ is uniformly integrable, then

$$\mathbb{E}|X_n - X| \to 0.$$

In particular,

$$\mathbb{E}[X_n] \to \mathbb{E}[X].$$

The mechanism is a bounded-tail decomposition. Choose K so that the large-value tails contribute little uniformly. On the region where $|X_n|$ and $|X|$ are at most K, convergence in probability controls the integral after separating the set on which $|X_n - X|$ exceeds a small threshold. The tails are then controlled by uniform integrability.

The development has therefore returned to the question that motivated the two integration theories. Riemann integration controls spatial oscillation well enough to integrate a single bounded function on an interval. Lebesgue integration reorganizes the construction around measurable sets and monotone approximation, making precise when limits may pass through integrals, norms, and expectations. Dominated convergence supplies one fixed envelope; uniform integrability isolates the more general tail condition that the exchange actually requires.

Differential Forms and Stokes: The Geometry of Boundary Measurements

N-20240427

§105.1 A theorem looking for its natural language

The fundamental theorem of calculus says

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

Green's theorem, the divergence theorem, and Stokes' theorem look more complicated, but they share the same shape: integrate a derivative over a region, and the answer lives on the boundary.

Differential forms are the language in which this pattern becomes one theorem:

$$\int_M d\omega = \int_{\partial M} \omega.$$

The purpose of this chapter is not to glorify notation. It is to make clear why dx, dy, pullback, orientation, and boundary signs belong together.

§105.2 One-forms: measuring motion

A tangent vector tells us a direction of motion. A covector measures such a direction. At a point p, a covector is a linear map

$$\alpha : T_pM \to \mathbb{R}.$$

A 1-form is a smoothly varying covector.

In the plane, a typical 1-form is

$$\omega = P(x,y)\,dx + Q(x,y)\,dy.$$

If

$$v = a\frac{\partial}{\partial x} + b\frac{\partial}{\partial y},$$

then

$$\omega(v) = Pa + Qb.$$

Thus a 1-form is not a tiny arrow. It is a measuring rule for arrows.

§105.3 The pullback experiment

Let $\gamma(t) = (x(t), y(t))$ be a parametrized curve. Pulling back the form

$$\omega = P(x,y)\,dx + Q(x,y)\,dy$$

to the interval gives

$$\gamma^*\omega = \big(P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)\big)\,dt.$$

Therefore

$$\int_\gamma \omega = \int_a^b \big(P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)\big)\,dt.$$

This is the usual line integral. The conceptual advantage is that the curve integral is reduced to ordinary integration after pullback.

§105.4 Two-forms: measuring oriented area

The wedge product is antisymmetric:

$$dx \wedge dy = -dy \wedge dx, \qquad dx \wedge dx = 0.$$

A 2-form such as

$$f(x, y)\,dx \wedge dy$$

measures signed area with density f. The sign changes if the orientation is reversed.

This is why forms are naturally tied to oriented geometry. A form does not merely ask "how large?" It asks "how large, with which orientation?"

§105.5 Exterior derivative as one operator

For a function f,

$$df = f_x\,dx + f_y\,dy.$$

For a 1-form $\omega = P\,dx + Q\,dy$,

$$d\omega = (Q_x - P_y)\,dx \wedge dy.$$

The rule

$$d^2 = 0$$

is the compact source of several familiar identities: gradients have zero curl, curls have zero divergence, and exact forms are closed.

§105.6 A familiar theorem reappears

Apply Stokes to a planar region D and a 1-form $\omega = P\,dx + Q\,dy$:

$$\int_D d\omega = \int_{\partial D} \omega.$$

Writing this in coordinates gives

$$\iint\limits_D (Q_x - P_y)\,dx\,dy = \int_{\partial D} P\,dx + Q\,dy.$$

This is Green's theorem. The point is not that Green's theorem becomes shorter, but that its boundary nature becomes unavoidable.

§105.7 Orientation is not decoration

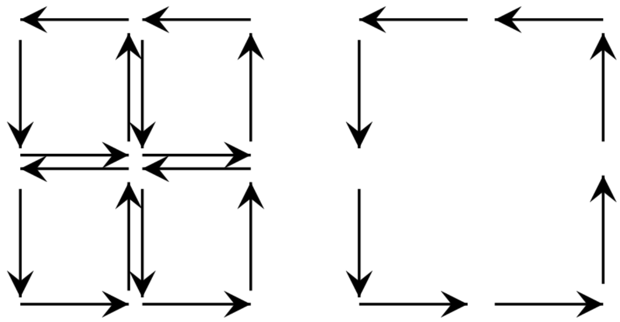


Figure 105.1: The orientation of a patch forces an induced orientation on its boundary.

The boundary of an oriented region inherits an orientation. If this convention is changed, Stokes' theorem changes sign. So orientation is not a drawing preference; it is part of the data needed for the theorem to be true.

§105.8 Why arc length is a different kind of geometry

It is tempting to treat ds as a 1-form, but arc length is not linear in the velocity:

$$ds(\gamma') = \|\gamma'\|\, dt.$$

A 1-form is linear in the tangent vector. A norm is not. This distinction separates two kinds of geometry:

differential forms	oriented measurement and integration
Riemannian metrics	lengths, angles, and distances

Both are important, but they should not be confused.

§105.9 Closed forms can detect holes

The distinction between closed and exact forms is one of the first places where analysis notices topology. On $\mathbb{R}^2 \setminus \{0\}$, consider

$$\omega = \frac{-y\, dx + x\, dy}{x^2 + y^2}.$$

One computes that

$$d\omega = 0.$$

So ω is closed. But it is not exact on $\mathbb{R}^2 \setminus \{0\}$. Indeed, along the unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$, we get

$$\gamma^*\omega = dt,$$

and therefore

$$\int_\gamma \omega = 2\pi.$$

If $\omega = df$, the integral around a closed loop would be 0. The nonzero integral detects the missing origin.

This is a beautiful lesson: a differential form can carry topological information.

§105.10 A table of classical shadows

The same theorem appears in several costumes:

dimension	form	classical theorem
1	f	$\int_a^b df = f(b) - f(a)$
2	$P\,dx + Q\,dy$	Green's theorem
3	flux form	divergence theorem
3	circulation form	classical Stokes theorem

The table is not meant as a replacement for calculation. It is meant to prevent the common mistake of treating these theorems as unrelated facts.

§105.11 Flux as a form

In $\mathbb{R}^3$, a vector field

$$\mathbf{F} = (P, Q, R)$$

corresponds to the 2-form

$$\omega = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy.$$

Then

$$d\omega = (P_x + Q_y + R_z)\,dx \wedge dy \wedge dz.$$

Thus the divergence theorem is exactly Stokes' theorem for this 2-form:

$$\int_M d\omega = \int_{\partial M} \omega.$$

This is a good example of how vector calculus becomes cleaner when translated into forms.

§105.12 The moral of pullback

Integration should not depend on a lucky parametrization. If a surface is parametrized by

$$\Phi : U \to M,$$

then an integral over M is computed by pulling the form back to U:

$$\int_M \omega = \int_U \Phi^*\omega.$$

The pullback automatically inserts the Jacobian-like factors and orientation signs. This is why forms are the natural objects to integrate: they know how to move backward along maps.

This naturality is exactly what Stokes' theorem needs. Differential forms are quantities that can be pulled back and integrated, the exterior derivative is compatible with pullback, and the integral of that derivative over a body becomes a measurement on the induced boundary orientation.

Double Integrals: Riemann Sums, Area, Symmetry, and Average Value

N-20260403

§106.1 From area sums to double integrals

A double integral is the two-dimensional version of adding many small contributions. If a quantity has density $f(x, y)$ over a plane region D, then a small cell of area ΔA_i contributes approximately

$$f(\xi_i, \eta_i)\Delta A_i.$$

Adding all cells and refining the partition gives the integral.

Definition 106.1.1 (Riemann double integral)**.** Let $D \subseteq \mathbb{R}^2$ be a bounded region and let $f : D \to \mathbb{R}$ be bounded. Partition D into small subregions ΔD_i, choose sample points $(\xi_i, \eta_i) \in \Delta D_i$, and form

$$S(P, f) = \sum_i f(\xi_i, \eta_i)\Delta A_i.$$

If $S(P, f)$ tends to a limit independent of the partition and sample points as the mesh tends to 0, then the limit is denoted

$$\iint\limits_D f(x, y)\, dA.$$

The notation dA or $d\sigma$ represents an infinitesimal area element. It is not an independent variable; it tells us what geometric measure is being used.

§106.2 Geometric meanings

The meaning depends on the integrand.

integrand	meaning of $\iint_D f\, dA$
$f = 1$	area of D
$f \geq 0$	volume under $z = f(x, y)$
f sign-changing	signed volume
$f = \rho$	mass of a lamina with density ρ

Example 106.2.1 (Area as an integral)
For the disk $D = \{(x, y) : x^2 + y^2 \le R^2\}$,

$$\operatorname{area}(D) = \iint_D 1\, dA.$$

Using polar coordinates later gives

$$\int_0^{2\pi} \int_0^R r\, dr\, d\theta = \pi R^2.$$

§106.3 Basic properties

The following properties are the main algebra of double integrals.

Proposition 106.3.1
Assume the displayed integrals exist.

(i) Linearity:
$$\iint_D (af + bg)\, dA = a \iint_D f\, dA + b \iint_D g\, dA.$$

(ii) Additivity over regions: if $D = D_1 \cup D_2$ and the overlap has area zero, then
$$\iint_D f\, dA = \iint_{D_1} f\, dA + \iint_{D_2} f\, dA.$$

(iii) Monotonicity: if $f \le g$ on D, then
$$\iint_D f\, dA \le \iint_D g\, dA.$$

(iv) Estimate: if $m \le f \le M$ on D, then
$$m \operatorname{area}(D) \le \iint_D f\, dA \le M \operatorname{area}(D).$$

Proof idea. Each statement is first true for finite Riemann sums. Passing to the limit preserves addition, scalar multiplication, and inequalities. The estimate comes from multiplying $m \le f(\xi_i, \eta_i) \le M$ by ΔA_i and summing. $\square$

§106.4 Existence of the integral

For most calculus computations, the following sufficient condition is enough.

Theorem 106.4.1
If f is continuous on a bounded closed elementary region D, then f is Riemann integrable on D.

The theorem is often used silently. It is the reason that ordinary continuous examples can be computed by choosing convenient coordinates and orders of integration.

Remark 106.4.2 — The statement is not saying that only continuous functions are integrable. Piecewise continuous functions and many functions with mild discontinuities are also integrable. Continuity is simply the safest elementary hypothesis.

§106.5 Symmetry before computation

Symmetry is one of the fastest ways to evaluate an integral.

Proposition 106.5.1 (Odd cancellation)
Let D be symmetric under a measure-preserving transformation $T : D \to D$. If

$$f(T(x, y)) = -f(x, y),$$

then

$$\iint_D f\, dA = 0.$$

If $f \circ T = f$, then the integral over D can often be reduced to a fundamental half or sector of D.

Example 106.5.2
Let D be the disk centered at the origin. Since D is symmetric under $(x, y) \mapsto (-x, y)$,

$$\iint_D x e^{y^2}\, dA = 0.$$

The integrand is odd in x, while the region is unchanged by reflecting across the y-axis.

§106.6 Mean value theorem for double integrals

Theorem 106.6.1 (Mean value theorem)
Let D be compact and connected, and let $f : D \to \mathbb{R}$ be continuous. Then there exists a point $P_0 \in D$ such that

$$\iint_D f\, dA = f(P_0)\, \mathrm{area}(D).$$

Equivalently, the average value

$$f_{\text{avg}} = \frac{1}{\mathrm{area}(D)} \iint_D f\, dA$$

is attained somewhere on D.

Proof skeleton. By compactness, f attains a minimum m and maximum M on D. The estimate property gives

$$m \le f_{\text{avg}} \le M.$$

Since D is connected and f is continuous, the image $f(D)$ is an interval. Therefore the intermediate value theorem gives a point P_0 where $f(P_0) = f_{\text{avg}}$. □

§106.7 A worked average-value example

Example 106.7.1

Let $D = [0,1] \times [0,2]$ and $f(x,y) = x + y$. Then

$$\iint_D (x+y)\,dA = \int_0^1 \int_0^2 (x+y)\,dy\,dx = \int_0^1 (2x+2)\,dx = 3.$$

Since $\operatorname{area}(D) = 2$, the average value is $3/2$. The mean-value theorem says there is a point in the rectangle with $x + y = 3/2$, for example $(1/2, 1)$.

§106.8 Relation with product measure

The elementary Riemann picture divides a region into small rectangles. In measure-theoretic language, this is integration with respect to two-dimensional Lebesgue measure:

$$\iint_D f(x,y)\,dA = \int_{\mathbb{R}^2} f(x,y)\mathbf{1}_D(x,y)\,d\lambda^2.$$

This viewpoint separates two choices:

(i) the function being accumulated;

(ii) the measure that tells how large each piece of the region is.

For nonnegative measurable functions, Tonelli's theorem justifies iterated integration even when the value may be infinite. For sign-changing functions, Fubini's theorem requires integrability of $|f|$. The elementary rule "switch the order of integration" is therefore backed by real hypotheses.

§106.9 A real Fubini warning: conditional two-dimensional accumulation

It is tempting to treat order-switching as a harmless algebraic move. The advanced point is sharper: for sign-changing functions, order-switching is licensed by absolute integrability. Without that hypothesis, different limiting procedures can see different cancellations.

A standard improper example on the square near the origin is

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}, \qquad (x,y) \in (0,1]^2.$$

For fixed $x > 0$,

$$\int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dy = \left[\frac{y}{x^2+y^2}\right]_0^1 = \frac{1}{1+x^2}.$$

Hence

$$\int_0^1 \int_0^1 f(x,y)\,dy\,dx = \frac{\pi}{4}.$$

But for fixed $y > 0$,

$$\int_0^1 \frac{x^2 - y^2}{(x^2+y^2)^2}\,dx = \left[-\frac{x}{x^2+y^2}\right]_0^1 = -\frac{1}{1+y^2},$$

so

$$\int_0^1 \int_0^1 f(x,y)\,dx\,dy = -\frac{\pi}{4}.$$

The two answers differ because the singularity at $(0,0)$ is not absolutely integrable. This is the concrete content of the distinction between Tonelli's theorem and Fubini's theorem.

§106.10 Product measure behind rectangular sums

The phrase "divide the region into rectangles" hides a construction. Rectangles $A \times B$ are assigned area by

$$\lambda^2(A \times B) = \lambda(A)\lambda(B).$$

This rule first defines area on finite unions of rectangles. One then extends it to the sigma-algebra generated by rectangles. The double integral is integration with respect to this product measure.

This explains why iterated integrals work first for indicator functions of rectangles:

$$\int \mathbf{1}_{A\times B}(x,y)\,d(\mu \times \nu) = \mu(A)\nu(B) = \int_A \left(\int_B 1\,d\nu\right) d\mu.$$

Linearity handles simple functions, monotone convergence handles nonnegative measurable functions, and absolute integrability handles signed functions. Thus the elementary Riemann-sum picture is not abandoned in measure theory; it is completed.

§106.11 Symmetry as projection onto invariant functions

If a finite group G acts on D by area-preserving transformations, averaging over the group defines the Reynolds operator

$$\mathcal{R}f = \frac{1}{|G|}\sum_{g\in G} f \circ g.$$

Because the action preserves area,

$$\iint_D f\,dA = \iint_D \mathcal{R}f\,dA.$$

So integration only sees the invariant part of the function. Odd functions cancel because their invariant part is zero. This is the same idea that later appears in representation theory as projection onto the trivial representation, but here it is already visible in a disk or rectangle symmetry problem.

For example, on the disk, the rotation group kills all angular Fourier modes except the constant mode:

$$f(r,\theta) = a_0(r) + \sum_{n\geq 1} (a_n(r)\cos n\theta + b_n(r)\sin n\theta)$$

gives

$$\iint_D f\,dA = \int_0^R \int_0^{2\pi} a_0(r)\,r\,d\theta\,dr.$$

The elementary symmetry shortcut is the first case of harmonic decomposition.

§106.12 Average value as a pushforward distribution

For continuous $f : D \to \mathbb{R}$, the average value can be viewed probabilistically. Put the uniform probability measure on D:

$$\mathbb{P}(E) = \frac{\mathrm{area}(E)}{\mathrm{area}(D)}.$$

Then

$$f_{\mathrm{avg}} = \mathbb{E}[f].$$

The pushforward measure $f_*\mathbb{P}$ on $\mathbb{R}$ records how much area of D lies at each range of values of f. The mean value theorem says that if D is connected and f is continuous, this expectation lies inside the interval $f(D)$, hence equals $f(P_0)$ for some point.

This interpretation is useful later: integration can be studied either by integrating over the original region or by pushing measure forward through a function and integrating over its values.

§106.13 How the double-integral setup is checked

A double integral begins as a limit of weighted area sums, so the first data to identify are the region D, the area element, and the function being averaged over that region. Linearity and additivity come directly from cutting the region into pieces; this is why splitting a domain along a change in boundary formula is legitimate rather than cosmetic.

The examples in this note give three practical tests. If the region has a symmetry and the integrand is odd under that symmetry, the corresponding contribution vanishes because the area measure is preserved. If the question asks for an average value, rewrite

$$\iint_D f\,dA = f_{\mathrm{avg}}\,\mathrm{area}(D)$$

and check connectedness and continuity before invoking the mean value theorem. If the computation is rearranged by Fubini or by a change of variables, the real issue is not the order of symbols but whether the slices describe the same region and whether the Jacobian accounts for area distortion. This is the point at which Tonelli, Fubini, and change of variables turn the geometric setup into a justified calculation.

Computing Double Integrals: Slicing, Polar Coordinates, and Change of Variables

N-20260405

§107.1 The real task: describe the region

In double-integral problems, the hard part is usually not the antiderivative. It is translating a plane region into inequalities. Once the region is described correctly, Fubini's theorem turns the double integral into ordinary one-variable integration.

Definition 107.1.1. A region D is **vertically simple** if it has the form

$$D = \{(x, y) : a \le x \le b,\ \varphi_1(x) \le y \le \varphi_2(x)\}.$$

It is **horizontally simple** if it has the form

$$D = \{(x, y) : c \le y \le d,\ \psi_1(y) \le x \le \psi_2(y)\}.$$

For a vertically simple region,

$$\iint_D f(x, y)\, dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y)\, dy\, dx.$$

For a horizontally simple region,

$$\iint_D f(x, y)\, dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy.$$

§107.2 Changing the order of integration

Changing order means describing the same set with the other variable outermost.

Example 107.2.1

Consider

$$I = \int_0^1 \int_{x^2}^{x} f(x, y)\, dy\, dx.$$

The region is

$$0 \le x \le 1, \qquad x^2 \le y \le x.$$

For a horizontal slice, $0 \le y \le 1$, and the inequalities become

$$y \le x \le \sqrt{y}.$$

Thus

$$I = \int_0^1 \int_y^{\sqrt{y}} f(x, y)\, dx\, dy.$$

A reliable order-changing workflow is:

1. write the original region as inequalities;
2. draw or infer its projection on the new outer axis;
3. solve the old inequalities for the new inner variable;
4. split the region if one pair of bounds is not enough.

§107.3 Polar coordinates

Polar coordinates are useful when circles, disks, sectors, or expressions $x^2 + y^2$ appear:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r \geq 0.$$

The area element is

$$dA = r\,dr\,d\theta.$$

The factor r is not a decoration; it is the Jacobian determinant of the map $(r, \theta) \mapsto (x, y)$.

Example 107.3.1
For the upper half disk $x^2 + y^2 \leq 4$, $y \geq 0$, the polar description is

$$0 \leq r \leq 2, \qquad 0 \leq \theta \leq \pi.$$

Therefore

$$\iint_D (x^2 + y^2)\,dA = \int_0^\pi \int_0^2 r^2 \cdot r\,dr\,d\theta = \pi \cdot 4 = 4\pi.$$

§107.4 Jacobian as local area scaling

Let

$$T(u, v) = (x(u, v), y(u, v)).$$

The derivative matrix is

$$DT = \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}.$$

A tiny rectangle in the (u, v)-plane is sent approximately to a parallelogram spanned by the two columns of DT. Its area is multiplied by

$$|\det DT| = \left|\frac{\partial(x, y)}{\partial(u, v)}\right|.$$

Theorem 107.4.1 (Change of variables)
Let $T : D' \to D$ be a one-to-one C^1 change of variables, except possibly on harmless boundary pieces, and suppose its Jacobian does not vanish in the interior. Then

$$\iint_D f(x, y)\,dx\,dy = \iint_{D'} f(x(u, v), y(u, v)) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\,dv.$$

§107.5 Why the absolute value appears

For scalar area, orientation is irrelevant. A coordinate map may preserve or reverse orientation, but both cases scale ordinary area by a positive factor. Hence the absolute value.

For oriented forms, the sign is kept:

$$dx \wedge dy = \det \frac{\partial(x,y)}{\partial(u,v)}\, du \wedge dv.$$

This explains the difference between scalar area integrals and oriented circulation or flux formulas.

§107.6 A linear change example

Example 107.6.1

Let

$$x = u + v, \qquad y = u - v.$$

Then

$$\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2.$$

Thus $dA = 2\,du\,dv$. If a diamond-shaped region in the (x,y)-plane is hard to describe but becomes a rectangle in (u,v)-coordinates, this coordinate change turns geometric complexity into a constant Jacobian.

§107.7 Choosing coordinates

The best coordinate system simplifies both the integrand and the boundary.

signal in problem	candidate coordinates	reason
$x^2 + y^2$	polar	radial expression
circle or sector	polar	simple bounds
parallelogram	linear change	constant Jacobian
curved family of level sets	adapted variables	region straightens

§107.8 Slicing as pushforward of area

Slicing a region vertically means integrating over fibers $x =$ constant and then integrating over x. In measure language, the map

$$\pi_x : D \to \mathbb{R}, \qquad (x,y) \mapsto x$$

pushes area measure on D down to a measure on the x-axis. The inner integral computes the contribution of each fiber, and the outer integral adds the fibers.

This viewpoint separates three related operations:

operation	map used	geometric effect
Fubini slicing	$(x,y) \mapsto x$	integrate over fibers
change of variables	$(u,v) \mapsto (x,y)$	pull back area by a chart
coarea	$F(x,y) \mapsto t$	integrate over level sets

Ordinary order-changing is the classroom version of this general fiber-integration principle.

§107.9 Exterior algebra proof of the Jacobian factor

The Jacobian determinant is not merely a remembered formula. It is the coefficient by which a coordinate change pulls back the oriented area form.

For

$$x = x(u, v), \qquad y = y(u, v),$$

we have

$$dx = x_u\, du + x_v\, dv, \qquad dy = y_u\, du + y_v\, dv.$$

Taking wedge products,

$$\begin{aligned} dx \wedge dy &= (x_u\, du + x_v\, dv) \wedge (y_u\, du + y_v\, dv) \\ &= (x_u y_v - x_v y_u)\, du \wedge dv. \end{aligned}$$

Thus

$$dx \wedge dy = \det \frac{\partial(x, y)}{\partial(u, v)}\, du \wedge dv.$$

Scalar area forgets orientation, so it uses the absolute value. Oriented integration keeps the sign. This is why the same determinant appears both in change of variables and in Green/Stokes type formulas.

§107.10 Polar coordinates as a singular chart

The polar map

$$T(r, \theta) = (r \cos\theta, r \sin\theta)$$

has Jacobian r. It is not one-to-one globally: all values of θ represent the origin when $r = 0$, and θ is periodic. Calculus usually ignores this because the bad set has area zero, but geometrically it matters.

This is the first hint that coordinates are local. On a manifold, one generally cannot describe everything by one global coordinate chart. Spherical coordinates have the same issue at the poles; longitude is not defined there. Integration survives because measure-zero coordinate singularities do not affect scalar integrals, while topology remembers them.

§107.11 Coarea formula in the simplest radial case

Changing variables is not the only way to decompose an integral. The coarea formula decomposes a domain by level sets. For the distance function

$$r(x, y) = \sqrt{x^2 + y^2}, \qquad |\nabla r| = 1 \quad ((x, y) \neq 0),$$

coarea says, informally,

$$\int_{\mathbb{R}^2} g(x, y)\, dA = \int_0^\infty \left(\int_{r^{-1}(t)} g\, dS \right) dt.$$

If $g(x, y) = h(r)$ is radial, the inner integral over the circle of radius t is $2\pi t h(t)$, so

$$\int_{\mathbb{R}^2} h(r)\, dA = \int_0^{\infty} 2\pi t h(t)\, dt.$$

This is the polar-coordinate formula reinterpreted as integration over level curves of r. The elementary formula $dA = r\, dr\, d\theta$ and the advanced coarea formula are describing the same geometry from two sides.

§107.12 When a clever substitution is actually a quotient

Sometimes a change of variables collapses a family of curves into a coordinate. For example, the variables

$$u = x + y, \qquad v = x - y$$

are adapted to diagonal lines. A region bounded by lines $x + y =$ constant and $x - y =$ constant becomes a rectangle in (u, v)-space. This is not just a trick: the new variables are invariants of two transverse foliations of the plane.

In harder problems, one chooses variables because level sets of the integrand or boundary are natural. The computation succeeds when the coordinate fibers match the geometry already present in the problem.

§107.13 How a change of variables earns its bounds

Fubini should be read as a slicing statement: a two-dimensional region is decomposed into one-dimensional fibres over a projection. A change of variables is stronger: it replaces the region by a coordinate domain and pays for the replacement with a Jacobian determinant. In polar coordinates this payment is the familiar factor r, because a small rectangle in (r, θ)-space becomes an annular sector whose area is approximately $r\, dr\, d\theta$.

The examples show how to decide whether a substitution is natural. Linear maps are useful when the boundary lines are already expressed by linear combinations such as $x+y$ and $x-y$. Polar coordinates are useful when the boundary or integrand is organized by distance from a point. More generally, the variables should be chosen so that level curves of the new coordinates match the visible families of curves in the problem. After that choice, the bounds are obtained by drawing the image region in the new variables; they are not copied from the old coordinates by syntax.

Triple Integrals: Solids, Cylindrical Coordinates, and Spherical Coordinates

N-20260406

§108.1 Triple integrals as weighted volume

A triple integral accumulates a scalar field over a spatial region $\Omega \subseteq \mathbb{R}^3$:

$$\iiint_{\Omega} f(x, y, z)\, dV.$$

If $f = 1$, the integral is volume. If $f = \rho$ is a mass density, the integral is mass. If f is source density, the integral is total source.

Definition 108.1.1. An xy-simple solid has the form

$$\Omega = \{(x, y, z) : (x, y) \in D,\ z_1(x, y) \le z \le z_2(x, y)\}.$$

Then

$$\iiint_{\Omega} f\, dV = \iint_{D} \int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, dz\, dA.$$

There are analogous yz-simple and zx-simple descriptions. If no single description works, split the solid.

§108.2 Rectangular-coordinate example

Example 108.2.1

Let Ω be the tetrahedron

$$x \ge 0, \quad y \ge 0, \quad z \ge 0, \quad x + y + z \le 1.$$

One description is

$$0 \le x \le 1, \qquad 0 \le y \le 1 - x, \qquad 0 \le z \le 1 - x - y.$$

Thus

$$\mathrm{vol}(\Omega) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\, dz\, dy\, dx = \frac{1}{6}.$$

§108.3 Cylindrical coordinates

Cylindrical coordinates keep the vertical coordinate and use polar coordinates in the horizontal plane:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

The volume element is

$$dV = r\,dr\,d\theta\,dz.$$

They are natural for cylinders, cones around the z-axis, and solids where x^2+y^2 appears.

Example 108.3.1

For the cylinder $x^2 + y^2 \le R^2$, $0 \le z \le h$,

$$V = \int_0^{2\pi} \int_0^R \int_0^h r\,dz\,dr\,d\theta = \pi R^2 h.$$

§108.4 Spherical coordinates

Spherical coordinates are

$$x = \rho \sin\varphi \cos\theta, \qquad y = \rho \sin\varphi \sin\theta, \qquad z = \rho \cos\varphi,$$

where usually $\rho \ge 0$, $0 \le \varphi \le \pi$, and $0 \le \theta < 2\pi$. The volume element is

$$dV = \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta.$$

The factor has a geometric interpretation: small coordinate lengths are approximately

$$d\rho, \qquad \rho\,d\varphi, \qquad \rho \sin\varphi\,d\theta.$$

Multiplying them gives $\rho^2 \sin\varphi$.

Example 108.4.1

The ball $x^2 + y^2 + z^2 \le R^2$ has volume

$$\int_0^{2\pi} \int_0^{\pi} \int_0^R \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{4}{3}\pi R^3.$$

§108.5 Coordinate-choice table

geometry	coordinates	typical bounds
box or graph over a plane	rectangular	$z_1(x,y) \le z \le z_2(x,y)$
cylinder, cone around z-axis	cylindrical	r, θ, z
ball, spherical shell, cone from origin	spherical	ρ, φ, θ

Coordinates should be chosen because they simplify the solid, not because they are fashionable.

§108.6 Mass, center of mass, and moments

If $\rho(x,y,z)$ is density, then

$$M = \iiint\limits_{\Omega} \rho\,dV.$$

The center of mass is

$$\bar{x} = \frac{1}{M} \iiint\limits_{\Omega} x\rho\,dV, \qquad \bar{y} = \frac{1}{M} \iiint\limits_{\Omega} y\rho\,dV, \qquad \bar{z} = \frac{1}{M} \iiint\limits_{\Omega} z\rho\,dV.$$

Moments of inertia are obtained by integrating squared distance to an axis. For example, about the z-axis,

$$I_z = \iiint_{\Omega} (x^2 + y^2)\rho \, dV.$$

§108.7 Symmetry

Before computing a triple integral, check symmetry. If Ω is symmetric under $x \mapsto -x$ and f is odd in x, then

$$\iiint_{\Omega} f(x, y, z) \, dV = 0.$$

If f is even, integrate over half the solid and double. In three dimensions, symmetry can remove a whole family of terms.

Example 108.7.1

On the ball centered at the origin,

$$\iiint_{x^2+y^2+z^2 \le 1} xz^2 \, dV = 0,$$

because the region is symmetric under $x \mapsto -x$, while the integrand changes sign.

§108.8 Three ways to understand volume factors

The Jacobian factors in cylindrical and spherical coordinates can be read in three compatible ways.

Viewpoint	What it explains
linear approximation	DT sends a small box to a parallelepiped
volume form	$dx \wedge dy \wedge dz$ pulls back by $\det DT$
metric tensor	volume density is $\sqrt{\det g}$ in curvilinear coordinates

For scalar volume, orientation is forgotten and one uses $|\det DT|$. For oriented integration, the sign is kept. Cylindrical and spherical volume elements are special cases of these principles.

§108.9 Toward divergence theorem

Triple integrals of divergence appear in Gauss' theorem:

$$\iiint_{\Omega} \nabla \cdot F \, dV = \iint_{\partial\Omega} F \cdot n \, dS.$$

Before learning the theorem, $\iiint_{\Omega} \nabla \cdot F \, dV$ can be read as total source in the volume. After the theorem, the same total source is measured by boundary flux.

§108.10 Metric tensor derivation of spherical volume

The spherical factor $\rho^2 \sin\varphi$ can be derived from the metric. In spherical coordinates,

$$ds^2 = d\rho^2 + \rho^2\, d\varphi^2 + \rho^2 \sin^2\varphi\, d\theta^2.$$

Thus the metric matrix is diagonal:

$$g = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & \rho^2 \sin^2\varphi \end{pmatrix}.$$

The Riemannian volume density is $\sqrt{\det g}$, hence

$$\sqrt{\det g} = \rho^2 \sin\varphi.$$

So

$$dV = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta.$$

This is the same factor as the Jacobian determinant, but the metric viewpoint generalizes directly to curved surfaces and Riemannian manifolds.

§108.11 Divergence as infinitesimal volume expansion

Divergence is not just a formal sum of partial derivatives. If a vector field F generates a flow Φ_t, then

$$\mathcal{L}_F(dV) = (\nabla \cdot F)dV,$$

where $\mathcal{L}_F$ is the Lie derivative. In words: divergence is the infinitesimal rate at which the flow expands volume.

For $F(x, y, z) = (x, y, z)$, the flow is $\Phi_t(p) = e^t p$. A region is scaled by e^t in each of three directions, so its volume is scaled by e^{3t}. The infinitesimal expansion rate is 3, and indeed

$$\nabla \cdot F = 1 + 1 + 1 = 3.$$

This makes the divergence theorem conceptually natural: total infinitesimal volume production inside Ω should equal flux through $\partial\Omega$.

§108.12 Pushforward of densities and probability

The triple-integral change-of-variables formula is also the density transformation formula in probability. If X has density f_X and $Y = T(X)$, then for an invertible smooth T,

$$f_Y(y) = f_X(T^{-1}y) \left|\det DT^{-1}(y)\right|.$$

For example, in $\mathbb{R}^3$, a radially symmetric density can be pushed forward by the radius map $R = |X|$. The shell of radius r has area $4\pi r^2$, so the radial density contains the same factor that appears in spherical coordinates:

$$f_R(r) = 4\pi r^2 f_X(r) \quad \text{for a radial density } f_X.$$

Thus the spherical Jacobian is not just a calculus artifact; it is the volume growth of shells.

§108.13 Volume forms and orientation

The standard oriented volume form is

$$dx \wedge dy \wedge dz.$$

Under a coordinate map $T(u, v, w)$, it pulls back as

$$T^*(dx \wedge dy \wedge dz) = \det DT\, du \wedge dv \wedge dw.$$

Scalar volume uses $|\det DT|$, but oriented integration keeps $\det DT$. This distinction becomes essential in Stokes' theorem, where signs encode boundary orientation.

Triple integrals are therefore the scalar shadow of a more structured object: integration of top-degree forms.

§108.14 How the solid determines the coordinates

A triple integral is weighted volume, but the coordinate system decides how that volume is sliced. Rectangular coordinates are best when the solid is bounded by graphs over coordinate planes. Cylindrical coordinates are best when the projection to the xy-plane is circular or when the integrand depends on $x^2 + y^2$. Spherical coordinates are best when the boundary is organized by distance from the origin or by cones through the origin.

The volume factors are not separate formulas to memorize. They are Jacobians:

$$dV = dx\, dy\, dz, \qquad dV = r\, dr\, d\theta\, dz, \qquad dV = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta.$$

The metric-tensor and volume-form derivations explain why these factors appear and why orientation matters once one passes from scalar volume to Stokes-type formulas. In a computation the safe order is therefore: describe the solid geometrically, choose coordinates whose coordinate surfaces match that description, compute or recall the Jacobian, and only then write the iterated bounds.

Volumes, Spatial Regions, and Order of Integration

N-20260408

§109.1 Volume as an integral

The volume of a solid is the integral of the constant density 1. If a solid lies between two graphs

$$z = z_1(x, y), \qquad z = z_2(x, y)$$

over a plane region D, then

$$V = \iint_D [z_2(x, y) - z_1(x, y)]\, dA.$$

If the solid is better described directly in three dimensions, then

$$V = \iiint_\Omega 1\, dV.$$

Example 109.1.1

The solid under $z = 4 - x^2 - y^2$ and above the disk $x^2 + y^2 \le 4$ has volume

$$V = \iint_{x^2+y^2\le 4} (4 - x^2 - y^2)\, dA.$$

In polar coordinates this becomes

$$V = \int_0^{2\pi} \int_0^2 (4 - r^2) r\, dr\, d\theta = 8\pi.$$

§109.2 Describing spatial regions

A spatial region can be simple with respect to different coordinate planes. An xy-simple description has the form

$$\Omega = \{(x, y, z) : (x, y) \in D,\ z_1(x, y) \le z \le z_2(x, y)\}.$$

Then

$$\iiint_\Omega f\, dV = \iint_D \int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, dz\, dA.$$

The same solid may be yz-simple or zx-simple instead. The best description is the one in which the inner bounds are simplest.

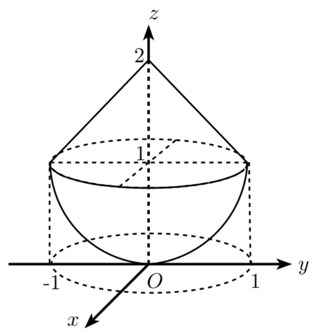


Figure 109.1: A composite solid with a hemispherical part and a conical part. The picture is used to decide cross-sections and height bounds before writing the integral.

§109.3 A reconstruction workflow

When an integral or picture is given, reconstruct the solid before integrating.

(i) Identify the bounding surfaces.

(ii) Find the projection or shadow on a coordinate plane.

(iii) Decide which variable has clear lower and upper surfaces.

(iv) Split the shadow when the top, bottom, left, or right boundary changes formula.

(v) Only then write the integral.

Most wrong triple integrals have correct antiderivatives but wrong regions.

§109.4 Changing order of integration

Changing the order of integration means choosing a new projection and new fibers. For example, an integral written as

$$\int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} \int_{z_1(x,y)}^{z_2(x,y)} f \, dz \, dy \, dx$$

uses vertical segments above an xy-region. Reordering may require projecting onto the xz- or yz-plane instead.

Caution

Do not change the order by simply permuting differentials. The bounds describe a region. A new order requires a new description of the same region.

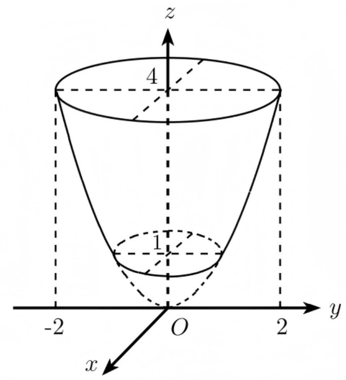


Figure 109.2: A rotational solid with circular cross-sections at different heights. Such diagrams are read by projecting to the base plane and then recovering the vertical bounds.

§109.5 Cross-sections

Sometimes the cleanest volume method is to integrate cross-sectional area:

$$V = \int_a^b A(t)\, dt.$$

Here $A(t)$ is the area of the slice of the solid by a plane such as $z = t$. This is the one-variable version of slicing.

Example 109.5.1

For a right circular cone of height h and base radius R, the slice at height z has radius $R(1 - z/h)$, so

$$A(z) = \pi R^2 \left(1 - \frac{z}{h}\right)^2.$$

Thus

$$V = \int_0^h \pi R^2 \left(1 - \frac{z}{h}\right)^2 dz = \frac{1}{3}\pi R^2 h.$$

§109.6 Top minus bottom

For a solid between surfaces, the double-integral formula

$$V = \iint_D (z_{\text{top}} - z_{\text{bottom}})\, dA$$

is often faster than a full triple integral. But this formula is only legal after the top and bottom are correctly identified over the same base region D.

§109.7 Cylindrical and spherical descriptions

If the solid has rotational symmetry, rectangular projections may be awkward. Cylindrical coordinates describe a solid by inequalities in r, θ, z, and spherical coordinates describe it by inequalities in ρ, φ, θ. The cost is the Jacobian:

$$dV = r\, dr\, d\theta\, dz, \qquad dV = \rho^2 \sin\varphi\, d\rho\, d\varphi\, d\theta.$$

The coordinate system should match the surface equations. A sphere centered at the origin asks for spherical coordinates; a cylinder around the z-axis asks for cylindrical coordinates.

§109.8 Absolute extrema over solids

Volume setup also appears in average-value problems. If f is continuous on a solid Ω, its average value is

$$f_{\text{avg}} = \frac{1}{\operatorname{vol}(\Omega)} \iiint_{\Omega} f\, dV.$$

When Ω is compact and connected, this average value is attained somewhere in Ω. This is the three-dimensional analogue of the mean-value theorem for integrals.

§109.9 Fibre integration: top-minus-bottom as a pushforward

The formula

$$V = \iint_{D} (z_{\text{top}} - z_{\text{bottom}})\, dA$$

has a precise structural meaning. Let

$$\pi : \Omega \to D, \qquad \pi(x, y, z) = (x, y)$$

be projection onto the base. The fibre over (x, y) is the vertical segment

$$\pi^{-1}(x, y) = \{z : z_{\text{bottom}}(x, y) \le z \le z_{\text{top}}(x, y)\}.$$

Its length is

$$\ell(x, y) = z_{\text{top}}(x, y) - z_{\text{bottom}}(x, y).$$

The volume is the integral of fibre length over the base:

$$\operatorname{vol}(\Omega) = \int_{D} \ell(x, y)\, dA.$$

This is the elementary version of integration along fibres. In differential geometry and topology, fibre integration pushes a form on a total space down to a form on the base.

§109.10 Cavalieri principle and layer-cake formula

Cross-sectional integration

$$V = \int_{a}^{b} A(t)\, dt$$

is Cavalieri's principle in analytic form. A more advanced cousin is the layer-cake formula. For a nonnegative function f,

$$\int_{D} f\, dA = \int_{0}^{\infty} \operatorname{area}(\{(x, y) \in D : f(x, y) > t\})\, dt.$$

Geometrically, the volume under $z = f(x, y)$ can be computed by stacking horizontal slices instead of vertical columns.

For example, for $f(x, y) = 1 - x^2 - y^2$ on the unit disk, the level set $f > t$ is the disk $x^2 + y^2 < 1 - t$, so

$$\int_{D} f\, dA = \int_{0}^{1} \pi(1 - t)\, dt = \frac{\pi}{2}.$$

This equals the usual polar-coordinate computation. The point is not the answer but the shift: one can integrate by columns, by horizontal layers, or by fibres of a map.

§109.11 Why regions must be split: projections can change topology

A projection can make a simple solid look complicated. Suppose a vertical line over a base point intersects the solid in two disjoint intervals rather than one. Then there is no single pair $z_1(x, y) \leq z \leq z_2(x, y)$ describing the fibre. The correct response is to split the base into pieces on which the fibre structure is stable.

This is a low-dimensional version of a general principle: maps are easiest to integrate over where their fibres vary regularly. Critical values and singular fibres force decompositions. In advanced language, one tries to partition the base so that the projection behaves like a fibration on each piece.

§109.12 Order of integration as choosing a foliation

Writing

$$\iiint_{\Omega} f\, dz\, dy\, dx$$

means that Ω is being decomposed into one-dimensional fibres parallel to the z-axis, then those fibres are organized over a region in the xy-plane. A different order chooses a different family of fibres. Thus changing order is choosing a different foliation of the same solid.

This language sounds advanced, but it is exactly what the pictures are doing. A good diagram shows which family of slices has the simplest geometry.

§109.13 Reconstructing the solid before integrating

For the problems in this note, the integral sign is less important than the solid it describes. An order such as

$$dz\, dy\, dx$$

means that vertical line fibres are being placed over a region in the xy-plane. Changing the order means choosing a different family of fibres, so the projection may change and the projected region may need to be split.

A reliable setup has four checks. First draw or describe the solid independently of the given bounds. Second choose the projection plane whose shadow has the simplest boundary. Third write the inner bounds as the entrance and exit points of a fibre through the solid. Fourth check whether the shadow changes formula across the projection; if it does, split the integral. The layer-cake and Cavalieri interpretations are useful because they keep the same discipline: volume is assembled from slices, and each slice must correspond to an actual cross-section of the solid.

Line Integrals and Surface Integrals of the First and Second Kind

N-20260416

§110.1 Two meanings of integration on curves

There are two common line integrals, and they answer different questions.

A first-kind line integral integrates a scalar density along length:

$$\int_C f\,ds.$$

A second-kind line integral integrates a vector field along an oriented curve:

$$\int_C P\,dx + Q\,dy + R\,dz = \int_C F \cdot dr.$$

The first is about amount distributed along a curve. The second is about work or circulation.

§110.2 Line integrals with respect to arc length

Let C be parametrized by $r(t) = (x(t), y(t), z(t))$, $a \le t \le b$. Then

$$ds = \|r'(t)\|\;dt,$$

and

$$\int_C f\,ds = \int_a^b f(r(t))\,\|r'(t)\|\;dt.$$

This integral does not depend on orientation.

Example 110.2.1

For the circle $r(t) = (R\cos t, R\sin t)$, $0 \le t \le 2\pi$, and $f(x,y) = x^2 + y^2$,

$$\int_C f\,ds = \int_0^{2\pi} R^2 \cdot R\,dt = 2\pi R^3.$$

§110.3 Line integrals of vector fields

For $F = (P, Q, R)$,

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t)\,dt.$$

In coordinates,

$$\int_C P\,dx + Q\,dy + R\,dz = \int_a^b (Px'(t) + Qy'(t) + Rz'(t))\,dt.$$

Reversing orientation changes the sign.

Example 110.3.1

Let $F = (-y, x)$ and let C be the unit circle traversed counterclockwise, $r(t) = (\cos t, \sin t)$. Then

$$F(r(t)) = (-\sin t, \cos t), \qquad r'(t) = (-\sin t, \cos t),$$

so

$$\int_C F \cdot dr = \int_0^{2\pi} 1\, dt = 2\pi.$$

This is circulation around the origin.

§110.4 Surface integrals of scalar fields

Let a surface S be parametrized by $r(u, v)$, $(u, v) \in D$. The surface area element is

$$dS = \|r_u \times r_v\|\, du\, dv.$$

The first-kind surface integral is

$$\iint_S f\, dS = \iint_D f(r(u, v))\, \|r_u \times r_v\|\, du\, dv.$$

It integrates scalar density over area and does not require an orientation.

Example 110.4.1

For the graph $z = g(x, y)$, use $r(x, y) = (x, y, g(x, y))$. Then

$$r_x \times r_y = (-g_x, -g_y, 1), \qquad dS = \sqrt{1 + g_x^2 + g_y^2}\, dx\, dy.$$

§110.5 Flux integrals

A flux integral measures how much a vector field crosses an oriented surface. If S is parametrized by $r(u, v)$, then

$$\iint_S F \cdot n\, dS = \iint_D F(r(u, v)) \cdot (r_u \times r_v)\, du\, dv,$$

where the direction of $r_u \times r_v$ determines the orientation.

Reversing the normal reverses the sign. This is the surface analogue of reversing a line integral of a vector field.

§110.6 First kind versus second kind

object	first kind	second kind
curve	$\int_C f\, ds$	$\int_C F \cdot dr$
surface	$\iint_S f\, dS$	$\iint_S F \cdot n\, dS$

First-kind integrals use metric size: length or area. Second-kind integrals use orientation: direction along a curve or normal direction through a surface.

§110.7 Parametrization invariance

A geometric integral should not depend on the particular parametrization. For $\int_C f\,ds$, reparametrizing a curve changes $\|r'(t)\|\,dt$ in exactly the compensating way. For $\int_C F\cdot dr$, orientation-preserving reparametrizations leave the value unchanged, while orientation reversal changes sign.

For surfaces, changing parameters multiplies the area element by the absolute Jacobian in first-kind integrals. In flux integrals, the sign of the Jacobian records whether the orientation is preserved or reversed.

§110.8 Differential forms viewpoint

A vector-field line integral in $\mathbb{R}^3$ can be read as the integral of a one-form

$$\omega = P\,dx + Q\,dy + R\,dz.$$

A flux integral corresponds to the two-form

$$\omega_F = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy.$$

Parametrization formulas are pullback formulas for these forms.

This viewpoint explains why orientation is built into second-kind integrals but not into scalar length or area integrals.

§110.9 Orientability warning

Flux through a surface requires a consistent choice of normal. Spheres and tori are orientable; a Möbius strip is not. On a non-orientable surface, a globally defined flux integral needs extra care or must be treated locally.

§110.10 Densities versus differential forms

The first-kind line integral and the second-kind line integral look similar only because both are written with an integral sign. Their transformation laws are different.

For a curve $r(t)$, arclength is a density:

$$ds = \|r'(t)\|\ dt.$$

The absolute value makes it insensitive to orientation. By contrast, a one-form pulls back as

$$r^*(P\,dx + Q\,dy + R\,dz) = (P(r(t))x'(t) + Q(r(t))y'(t) + R(r(t))z'(t))\,dt.$$

If t is reversed, dt changes sign, so the integral changes sign. This is the exact mathematical distinction between density and form.

§110.11 Flux and the Hodge star

In Euclidean three-space, a vector field $F = (P, Q, R)$ can be converted into a two-form by the Hodge star:

$$F \cdot dS \quad \longleftrightarrow \quad \omega_F = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy.$$

If $r(u, v)$ parametrizes a surface, then

$$\iint_S F \cdot n \, dS = \iint_D r^* \omega_F.$$

Expanding this pullback gives exactly the cross-product formula

$$F(r(u, v)) \cdot (r_u \times r_v) \, du \, dv.$$

Thus the vector-calculus formula is not arbitrary: it is the coordinate expression of pulling back a two-form.

§110.12 Boundary orientation is not a sign convention

For Stokes-type theorems, the orientation of a boundary is forced by the orientation of the interior. In the plane, positive orientation means the region stays on the left as one traverses the boundary. On an oriented surface, the boundary orientation is determined by the right-hand rule.

This matters because the theorem

$$\int_{\partial M} \omega = \int_M d\omega$$

can only be correct when the two orientations are compatible. A wrong sign in a flux or circulation problem is often not a computational accident; it means the boundary orientation and interior orientation were not matched.

§110.13 Non-orientability as a real obstruction

A scalar surface integral $\iint_S f \, dS$ makes sense on a Möbius strip because area does not need a global normal. A flux integral $\iint_S F \cdot n \, dS$ needs a consistent normal direction. The Möbius strip has no such global choice.

This is a topological obstruction, not a defect of parametrization. Locally one can always choose a normal; globally the choices may return reversed after going once around the strip. Vector calculus is already detecting orientability.

§110.14 What the parametrization is responsible for

First-kind integrals integrate scalar densities against geometric measure. For a curve this means that the parametrization contributes the speed $\|r'(t)\|$; for a surface it contributes the area factor $\|r_u \times r_v\|$. Changing parameters should not change the value, because these factors exactly compensate for stretching or compressing the parameter domain.

Second-kind integrals keep orientation. A line integral of work uses the directed tangent displacement dr, and a flux integral uses the oriented normal vector $r_u \times r_v$ or a chosen unit normal n. This is why reversing orientation changes the sign of work and flux but not the sign of scalar length or area. The Möbius-strip example marks the boundary of the usual formula: scalar surface area is still meaningful, but a global flux integral is not available until a consistent normal direction has been chosen.

Green's Theorem, Conservative Fields, and the Planar Stokes Principle

N-20260417

§111.1 Green's theorem

Green's theorem relates circulation around a boundary to local rotation inside the region.

Theorem 111.1.1 (Green's theorem)
Let $D \subseteq \mathbb{R}^2$ be a positively oriented region with piecewise smooth simple closed boundary $C = \partial D$. If P and Q have continuous first partial derivatives on a neighborhood of D, then

$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

The left side is circulation along the boundary. The right side is total scalar curl over the region.

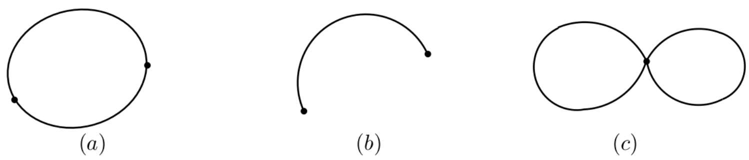


Figure 111.1: Simple closed curves and non-simple closed curves in the statement of Green's theorem.

§111.2 Proof idea: rectangles and cancellation

For a rectangle $[a, b] \times [c, d]$, the theorem follows from the fundamental theorem of calculus. The $Q\,dy$ part detects horizontal change in Q, and the $P\,dx$ part detects vertical change in P with the opposite sign.

For a region subdivided into many rectangles, internal edges are traversed twice in opposite directions. Their line integrals cancel. Only the outer boundary remains.

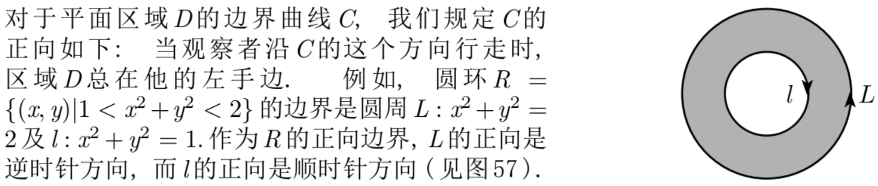


Figure 111.2: For a vertically simple region, Green's theorem is proved by reducing to one-variable fundamental theorem computations.

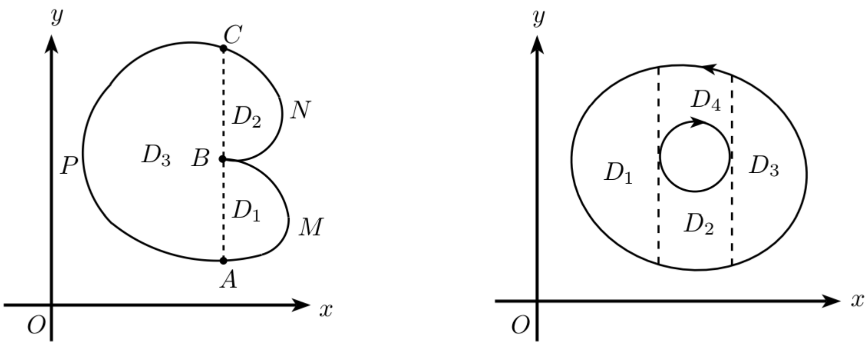


Figure 111.3: Subdivision by auxiliary arcs: internal boundary contributions cancel, leaving only the outer boundary.

§111.3 Conservative fields and path independence

A vector field $F = (P, Q)$ is **conservative** on a region U if there is a potential function ϕ such that

$$\nabla\phi = F.$$

Then

$$\int_C P\,dx + Q\,dy = \phi(B) - \phi(A)$$

for every path C from A to B. Thus the integral depends only on endpoints.

Proposition 111.3.1
If $F = \nabla\phi$, then $P_y = Q_x$.

Proof. If $P = \phi_x$ and $Q = \phi_y$, then under the usual continuity hypotheses,

$$P_y = \phi_{xy} = \phi_{yx} = Q_x.$$

□

The converse requires a topological hypothesis.

Theorem 111.3.2 (Conservative-field test)
If $U \subseteq \mathbb{R}^2$ is simply connected and $P_y = Q_x$ on U, then $F = (P, Q)$ is conservative on U.

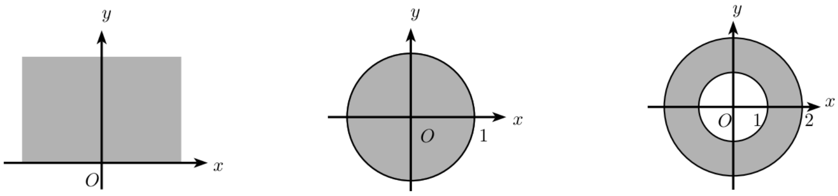


Figure 111.4: Simply connected and multiply connected plane regions. The distinction matters for path independence and conservative fields.

§111.4 The punctured-plane warning

The standard example is

$$F(x,y) = \left(-\frac{y}{x^2+y^2}, \frac{x}{x^2+y^2}\right)$$

on $\mathbb{R}^2 \setminus \{0\}$. A direct computation gives $P_y = Q_x$ away from the origin, but around the unit circle $r(t) = (\cos t, \sin t)$,

$$F(r(t)) = (-\sin t, \cos t), \qquad r'(t) = (-\sin t, \cos t),$$

so

$$\oint_{S^1} F \cdot dr = \int_0^{2\pi} 1\, dt = 2\pi.$$

Therefore no global potential exists on the punctured plane. The hole is detected by circulation.

§111.5 Area by Green's theorem

Green's theorem can compute area. Choose

$$P = -\frac{y}{2}, \qquad Q = \frac{x}{2}.$$

Then $Q_x - P_y = 1$, so

$$\text{area}(D) = \iint_D 1\, dA = \frac{1}{2}\oint_{\partial D} x\, dy - y\, dx.$$

Example 111.5.1

For the unit circle $x = \cos t$, $y = \sin t$,

$$\frac{1}{2}\int_0^{2\pi} (xy' - yx')\, dt = \frac{1}{2}\int_0^{2\pi} 1\, dt = \pi.$$

§111.6 Differential forms notation

Let

$$\omega = P\, dx + Q\, dy.$$

Then

$$d\omega = (Q_x - P_y)\, dx \wedge dy.$$

Green's theorem becomes

$$\int_{\partial D} \omega = \int_D d\omega.$$

This is Stokes' theorem in the plane.

The algebraic slogan is: integrate a form over the boundary, or integrate its derivative over the interior.

§111.7 Closed forms, exact forms, and topology

A one-form ω is **closed** if $d\omega = 0$. It is **exact** if $\omega = d\phi$ for some function ϕ. Exact forms are always closed, because $d^2 = 0$. The converse depends on the domain.

On simply connected plane regions, closed one-forms are exact. On the punctured plane, the form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed but not exact, because

$$\int_{S^1} \omega = 2\pi.$$

§111.8 Connection with de Rham cohomology

The first de Rham cohomology group is

$$H^1_{\mathrm{dR}}(M) = \frac{\ker(d : \Omega^1(M) \to \Omega^2(M))}{\operatorname{im}(d : \Omega^0(M) \to \Omega^1(M))}.$$

It measures closed one-forms modulo exact one-forms. In elementary vector calculus language, it measures the obstruction to finding potentials for curl-free fields.

§111.9 Winding number hidden in the punctured-plane example

For the one-form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2},$$

write $x = r\cos\theta$, $y = r\sin\theta$. Then formally

$$\omega = d\theta.$$

The word "formally" matters: θ is not a globally single-valued function on $\mathbb{R}^2 \setminus \{0\}$. Around a closed loop γ,

$$\frac{1}{2\pi}\int_\gamma \omega$$

is the winding number of γ about the origin.

For the unit circle, the winding number is 1. For a loop going twice around, it is 2. For a loop not enclosing the origin, it is 0. Thus the line integral is not merely detecting that a potential is missing; it measures how the curve winds around the hole.

§111.10 From Green's theorem to residues

The same punctured-plane form is connected to complex analysis. Since

$$\frac{dz}{z} = \frac{dx + i\,dy}{x + iy},$$

one computes

$$\operatorname{Im} \frac{dz}{z} = \frac{x\,dy - y\,dx}{x^2 + y^2} = \omega.$$

Thus

$$\int_{S^1} \omega = 2\pi$$

is the imaginary part of

$$\int_{S^1} \frac{dz}{z} = 2\pi i.$$

The residue theorem is a complex-analytic refinement of the same topological fact: a closed integral can detect a puncture and its winding number.

§111.11 Cohomology as bookkeeping for failed potentials

The statement "curl-free implies conservative" is not universally true; it is true when the relevant cohomology vanishes. In de Rham notation,

$$H^1_{\mathrm{dR}}(M) = \frac{\text{closed one-forms}}{\text{exact one-forms}}.$$

A nonzero class is represented by a closed form that is not the differential of any global function.

For $M = \mathbb{R}^2 \setminus \{0\}$, the class of ω above generates $H^1_{\mathrm{dR}}(M) \cong \mathbb{R}$. The integral over S^1 is a period pairing:

$$([\omega], [S^1]) \longmapsto \int_{S^1} \omega.$$

So the elementary warning about holes is really the first example of pairing cohomology with cycles.

§111.12 Boundary cancellation and the algebra $\partial^2 = 0$

The rectangle-subdivision proof of Green's theorem says that every internal edge appears twice with opposite orientations. Algebraically this is the principle

$$\partial^2 = 0 : \quad \text{the boundary of a boundary cancels.}$$

De Rham theory has the dual identity

$$d^2 = 0.$$

Stokes' theorem connects them:

$$\int_{\partial C} \omega = \int_C d\omega.$$

If one applies this again, both sides become zero for structural reasons. Thus Green's theorem is not an isolated vector-calculus trick; it is the two-dimensional visible face of a chain/cochain duality.

§111.13 What Green's theorem checks in practice

Green's theorem converts a boundary integral into an area integral:

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

In a computation this requires more than recognizing the formula. The boundary must be positively oriented, the vector field must be defined on a region containing D, and holes in the domain must be accounted for by extra boundary components with the correct orientation.

The punctured-plane example explains why this check matters. The form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed away from the origin, but its integral around S^1 is nonzero, so it is not globally exact on $\mathbb{R}^2 \setminus \{0\}$. Green's theorem, the identity $d^2 = 0$, and the cancellation of internal boundaries are therefore three views of the same mechanism: local derivative data controls boundary circulation only after the topology of the region has been included.

Surface Integrals, Gauss Theorem, and Stokes Theorem

N-20260615

§112.1 Why this note is split this way

Surface integration is where several notations begin to look dangerously similar. The guiding principle of this note is that every integral should be identified by the geometric object it integrates over and by the degree of the quantity being integrated. A scalar surface integral measures density against area; a flux integral measures a two-dimensional oriented flow; Gauss turns a two-dimensional boundary flux into a three-dimensional divergence integral; Stokes turns a one-dimensional boundary circulation into a two-dimensional curl integral.

This is not a taxonomy for its own sake. It is the first place where the general formula

$$\int_{\partial M} \omega = \int_M d\omega$$

starts to become visible behind vector calculus.

§112.2 Four similar-looking integrals

The first task is to separate four objects:

$$\iint_S f(x,y,z)\,dS, \qquad \iint_S P\,dy\,dz + Q\,dz\,dx + R\,dx\,dy,$$

$$\iiint_\Omega \nabla\cdot F\,dV, \qquad \oint_{\partial S} P\,dx + Q\,dy + R\,dz.$$

The first is a scalar surface integral: it integrates a scalar density over area. The second is a flux integral: it integrates a vector field through an oriented surface. The third is the volume integral of divergence. The fourth is circulation along an oriented boundary curve.

The common mistake is to treat all of them as "surface integral formulas." They are not the same mathematical object. Their relation is supplied by Gauss' theorem and Stokes' theorem.

§112.3 First-kind surface integrals

Let S be a smooth surface parametrized by

$$r(u,v) = (x(u,v), y(u,v), z(u,v)), \qquad (u,v)\in D.$$

Then

$$dS = |r_u \times r_v|\,du\,dv, \qquad \iint_S f\,dS = \iint_D f(r(u,v))|r_u\times r_v|\,du\,dv.$$

This is the surface analogue of the first-kind line integral $\int_C f\,ds$.

For a graph $z = z(x, y)$,

$$r(x, y) = (x, y, z(x, y)), \qquad r_x \times r_y = (-z_x, -z_y, 1),$$

so

$$dS = \sqrt{1 + z_x^2 + z_y^2}\, dx\, dy.$$

For a graph $x = x(y, z)$ or $y = y(z, x)$, analogous formulas hold:

$$dS = \sqrt{1 + x_y^2 + x_z^2}\, dy\, dz, \qquad dS = \sqrt{1 + y_z^2 + y_x^2}\, dz\, dx.$$

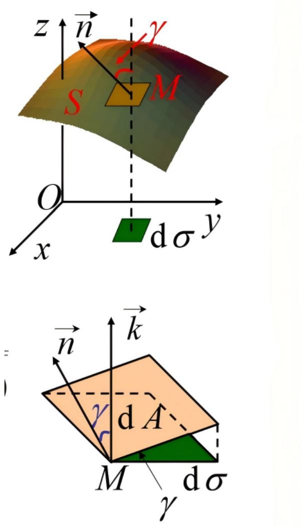


Figure 112.1: A surface area element and its projection. The factor between dS and the projected area element is determined by the angle between the surface normal and the projection direction.

§112.4 Template surfaces

The original note lists the common templates, and they should be kept because they prevent repeated derivations.

For a plane $z = ax + by + c$,

$$dS = \sqrt{1 + a^2 + b^2}\, dx\, dy.$$

For a sphere $x^2 + y^2 + z^2 = R^2$, spherical coordinates give

$$dS = R^2 \sin\varphi\, d\varphi\, d\theta.$$

For a cylinder $x = R\cos\theta$, $y = R\sin\theta$, $z = z$,

$$dS = R\, d\theta\, dz.$$

For a cone $z = \sqrt{x^2 + y^2}$, use $x = r\cos\theta$, $y = r\sin\theta$, $z = r$, so

$$|r_r \times r_\theta| = \sqrt{2}\, r.$$

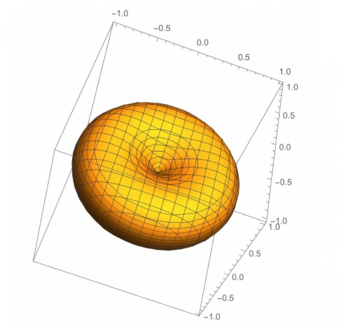


Figure 112.2: A torus as a standard example of a parametrized surface.

§112.5 Example pattern: spherical cap

For a spherical cap on $x^2 + y^2 + z^2 = R^2$, the scalar integral of f should normally be written in spherical coordinates. If the cap is cut by $z = h$, then $z = R\cos\varphi$ gives

$$0 \le \varphi \le \arccos(h/R).$$

Thus

$$\iint_S f\,dS = \int_0^{2\pi} \int_0^{\arccos(h/R)} f(R\sin\varphi\cos\theta, R\sin\varphi\sin\theta, R\cos\varphi)R^2\sin\varphi\,d\varphi\,d\theta.$$

The important part of the example is not the final antiderivative. It is recognizing the right parameter domain.

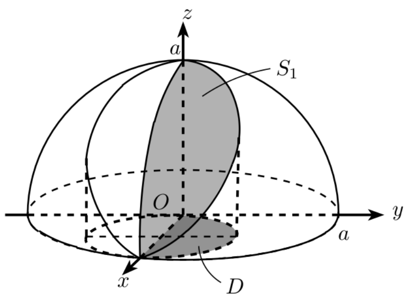


Figure 112.3: A spherical surface patch and its projection domain. This is the geometric model behind many spherical-cap surface integrals.

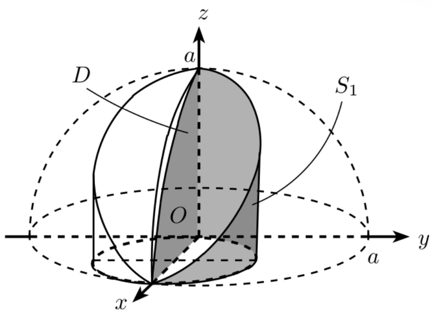


Figure 112.4: A related spherical surface cut out by a vertical boundary. The dashed curves indicate the hidden projection and bounding geometry.

§112.6 Second-kind surface integrals as flux

Let $F = (P, Q, R)$. The flux through an oriented surface S is

$$\iint_S F \cdot n\, dS.$$

In differential notation this is

$$\iint_S P\, dy\, dz + Q\, dz\, dx + R\, dx\, dy.$$

For a parametrization $r(u, v)$,

$$\iint_S F \cdot n\, dS = \iint_D F(r(u, v)) \cdot (r_u \times r_v)\, du\, dv,$$

where the order of u, v determines orientation.

For a graph $z = z(x, y)$ with upward orientation,

$$n\, dS = (-z_x, -z_y, 1)\, dx\, dy,$$

so

$$\iint_S F \cdot n\, dS = \iint_D [-Pz_x - Qz_y + R]\, dx\, dy.$$

For downward orientation, change the sign.

§112.7 Model flux computations: closed, capped, singular, and sheeted

The useful flux examples are not a random list of surfaces; they represent four different decisions about orientation and theorem choice.

§112.7.i Closed sphere: Gauss beats parametrization

Let $F(x, y, z) = (x, y, z)$ and let S be the sphere $x^2 + y^2 + z^2 = R^2$ with outward orientation. Directly, the outward unit normal is $n = x/R$, so

$$F \cdot n = R, \qquad dS = R^2 \sin\varphi\, d\varphi\, d\theta.$$

Hence

$$\iint_S F \cdot n \, dS = R \cdot 4\pi R^2 = 4\pi R^3.$$

Gauss gives the same answer faster:

$$\nabla \cdot F = 3, \qquad \iiint_{B_R} 3 \, dV = 3 \cdot \frac{4}{3}\pi R^3 = 4\pi R^3.$$

The theorem choice is dictated by closedness of the surface.

§112.7.ii Open surface: close it and subtract the cap

For the upper hemisphere S of radius R, outward orientation, Gauss cannot be applied to S alone because it is not closed. Close it by adding the disk D in the plane $z = 0$. For $F = (x, y, z)$, the flux through D is zero because $F \cdot n = 0$ there. Therefore the hemisphere flux equals the full closed half-ball flux:

$$\iint_S F \cdot n \, dS = \iiint_{z \geq 0,\ x^2+y^2+z^2 \leq R^2} 3 \, dV = 2\pi R^3.$$

For another field, the cap flux might not vanish; subtracting it is the whole point of the method.

§112.7.iii Cone: orientation is the computation

For the cone $z = r$, $0 \leq r \leq a$, use

$$r(r, \theta) = (r\cos\theta, r\sin\theta, r).$$

Then

$$r_r \times r_\theta = (-r\cos\theta, -r\sin\theta, r),$$

which has positive z-component. Thus it is the upward orientation. If a problem asks for outward orientation of the cone as the side of a solid below its tip, the sign may be the opposite. Here the formula is easy; choosing the correct normal is the real work.

§112.7.iv Two sheets over one projection

For a sphere cut by projection onto the xy-plane, upper and lower sheets have opposite normal components. A flux written as

$$\iint_S R \, dx \, dy$$

really means integrating the two-form $R \, dx \wedge dy$. On the upper sheet, upward orientation contributes $+R \, dx \, dy$; on the lower sheet, the same geometric outward orientation may contribute $-R \, dx \, dy$. Thus sheets may add or cancel depending on the chosen orientation and the component of the flux form.

This is the differential-form reason that projection formulas are dangerous when treated as scalar area formulas.

§112.8 Gauss theorem

Theorem 112.8.1 (Gauss divergence theorem)
For a compact region Ω with oriented boundary $\partial\Omega$,

$$\int_{\partial\Omega} F \cdot n\, dS = \int_{\Omega} \nabla \cdot F\, dV.$$

Let Ω be a solid with outward boundary $\partial\Omega$. For $F = (P, Q, R)$,

$$\iint\limits_{\partial\Omega} F \cdot n\, dS = \iiint\limits_{\Omega} (P_x + Q_y + R_z)\, dV.$$

The divergence

$$\nabla \cdot F = P_x + Q_y + R_z$$

measures local source strength. Gauss' theorem says that total source inside equals outward flux through the boundary.

Use Gauss when the surface is closed, or when it can be closed by adding a simple cap. If the required surface is not closed, compute the closed flux and subtract the flux across the added part. If the problem asks for inward orientation, multiply the outward result by -1.

§112.9 Singular fields and missing points

The note includes the important example type where the field is singular at a point. Gauss' theorem cannot be applied across a singularity. The usual remedy is to remove a small sphere around the singular point, apply Gauss' theorem to the punctured region, and then let the radius tend to zero. For inverse-square radial fields, the flux through the small sphere often remains nonzero. This is the analytic origin of point charges and delta-type sources.

§112.10 Stokes theorem

Theorem 112.10.1 (Stokes theorem)
For an oriented surface S with boundary ∂S,

$$\int_{\partial S} F \cdot dr = \int_{S} (\nabla \times F) \cdot n\, dS.$$

Let S be an oriented surface with boundary ∂S. Then

$$\oint_{\partial S} F \cdot dr = \iint\limits_{S} (\nabla \times F) \cdot n\, dS.$$

Here

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix}.$$

The right-hand rule fixes orientation: if the thumb points along n, the fingers curl in the positive boundary direction.

Use Stokes when the boundary curve is complicated but some spanning surface is simple, or when the curl is much simpler than the original vector field. The surface can be changed as long as the boundary and orientation are preserved.

§112.11 Stokes model computations: area vector and spanning-surface freedom

§112.11.i Triangle in a plane

Suppose ∂S is the positively oriented boundary of a triangle with vertices A, B, C, and suppose $\nabla \times F = c$ is constant. Stokes gives

$$\oint_{\partial S} F \cdot dr = \iint_S c \cdot n \, dS = c \cdot \vec{A}_S,$$

where the oriented area vector is

$$\vec{A}_S = \frac{1}{2}(B - A) \times (C - A).$$

So the whole line integral collapses to one vector dot product. The triangle is not merely a convenient region; it packages orientation and area into $\vec{A}_S$.

§112.11.ii Ellipse as a chosen spanning surface

If the boundary curve is an ellipse lying in a plane and $\nabla \times F$ is simple, Stokes lets us choose the planar elliptical disk rather than parameterizing the boundary. If the ellipse has semiaxes a, b and unit normal n, then its oriented area vector is

$$\vec{A} = \pi ab\, n.$$

When $\nabla \times F = c$ is constant,

$$\oint_{\partial S} F \cdot dr = c \cdot (\pi ab\, n).$$

This illustrates the true power of Stokes: the line integral depends on the boundary, but the chosen surface can be any surface spanning it, provided orientation is consistent.

§112.11.iii Why surface replacement is legal

If two oriented surfaces S_1 and S_2 have the same boundary, then applying Stokes to the closed surface $S_1 \cup (-S_2)$ gives

$$\iint_{S_1} (\nabla \times F) \cdot n \, dS - \iint_{S_2} (\nabla \times F) \cdot n \, dS = 0,$$

as long as the field is smooth in the region between them. If there is a singularity between the two surfaces, this argument can fail. This is the Stokes-side analogue of the singular-source warning in Gauss' theorem.

§112.12 Decision table

Object	Meaning	Best tool
$\iint_S f\,dS$	scalar density on surface	parametrization or graph dS
$\iint_S F \cdot n\,dS$	flux through oriented surface	graph formula or Gauss
closed flux	total outward flow	Gauss theorem
line circulation	work/circulation along boundary	direct line integral or Stokes

§112.13 Conceptual endpoint

The high-level statement is the boundary principle:

$$\int_{\partial M} \omega = \int_M d\omega.$$

Green, Gauss, and Stokes are all versions of this formula. The old vector-calculus notation hides the unity; differential forms reveal it. The derivative $d\omega$ records local change inside a region, and the boundary integral records the accumulated effect at the edge.

§112.14 The de Rham complex behind the vector formulas

The vector identities become less mysterious if the objects are placed by degree:

degree	form in $\mathbb{R}^3$	vector-calculus name
0	f	scalar potential
1	$P\,dx + Q\,dy + R\,dz$	work/circulation form
2	$P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy$	flux form
3	$g\,dx \wedge dy \wedge dz$	volume density

The exterior derivative moves one row down:

$$d : \Omega^0 \to \Omega^1 \to \Omega^2 \to \Omega^3.$$

In vector notation, these three moves are gradient, curl, and divergence. The identities

$$\nabla \times (\nabla f) = 0, \qquad \nabla \cdot (\nabla \times F) = 0$$

are both the single identity

$$d^2 = 0.$$

This is why conservative fields, curl-free forms, flux through closed surfaces, and singular sources are connected rather than separate topics.

§112.15 General Stokes theorem

Theorem 112.15.1 (General Stokes theorem)
For a compact oriented manifold M with boundary and a differential form ω,

$$\int_M d\omega = \int_{\partial M} \omega.$$

The vector-calculus theorems in this note are all cases of the general Stokes theorem. If M is an oriented smooth manifold with boundary and ω is a compactly supported differential form of degree one less than $\dim M$, then

$$\int_{\partial M} \omega = \int_M d\omega.$$

Green's theorem, Stokes' theorem, and Gauss' theorem correspond to different degrees and dimensions:

Green	$M \subseteq \mathbb{R}^2$	$\omega = P dx + Q dy,$
Stokes	M a surface in $\mathbb{R}^3$	$\omega = P dx + Q dy + R dz,$
Gauss	$M \subseteq \mathbb{R}^3$	$\omega = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy.$

The apparently different vector formulas are one theorem written in different degrees.

§112.16 Singular sources and distributions

The note mentions singular fields and missing points. The advanced language for point sources is distribution theory. The Dirac delta δ_0 is not an ordinary function but satisfies

$$\int \delta_0 \varphi = \varphi(0)$$

for test functions φ.

For the inverse-square field in $\mathbb{R}^3$,

$$F(x) = \frac{x}{|x|^3},$$

one has classically $\nabla \cdot F = 0$ away from the origin, but distributionally

$$\nabla \cdot F = 4\pi\delta_0.$$

This reconciles the nonzero flux through spheres with zero ordinary divergence away from the singularity.

§112.17 Conservation laws

Many PDEs have conservation-law form

$$\partial_t \rho + \nabla \cdot J = 0.$$

Integrating over a region Ω and applying Gauss' theorem gives

$$\frac{d}{dt} \int_\Omega \rho \, dV = - \iint_{\partial\Omega} J \cdot n \, dS.$$

Change inside equals negative outward flux. Thus Gauss' theorem is the geometric mechanism behind conservation of mass, charge, and probability.

§112.18 Currents

A current generalizes an oriented submanifold by acting on differential forms. An oriented curve C defines a current

$$T_C(\omega) = \int_C \omega.$$

The boundary of a current is defined by

$$\partial T(\omega) = T(d\omega).$$

This extends Stokes' theorem to singular chains, weak surfaces, and geometric measure theory.

The elementary surface and curve integrals in the note are therefore the smooth case of a much more flexible integration theory over generalized geometric objects.

§112.19 Gauss and Stokes as dual local-to-global laws

Gauss' theorem and Stokes' theorem are both local-to-global principles. Gauss says that local source density accumulates to boundary flux. Stokes says that local rotation accumulates to boundary circulation. In differential-form notation, both are one formula:

$$\int_{\partial M} \omega = \int_M d\omega.$$

The difference lies only in the degree of the form and the dimension of the manifold.

This unification matters because it prevents memorizing vector identities as separate facts. The object being integrated determines the theorem: a one-form over a boundary curve gives curl through a surface; a two-form over a closed surface gives divergence through a volume.

§112.20 Weak forms and PDE

In PDE, integration by parts and Stokes-type formulas define weak solutions. Instead of requiring derivatives to exist pointwise, one tests against smooth functions. For example, a weak divergence equation is read through

$$\int_\Omega F \cdot \nabla\varphi \, dV = -\int_\Omega (\nabla \cdot F)\varphi \, dV$$

up to boundary terms.

Thus vector-calculus integral theorems are not merely computational shortcuts; they are the foundation of weak formulations, conservation laws, and modern PDE theory.

XI

Complex Analysis

Part XI: Contents

Motion on the Unit Circle, Roots of Unity, and Finite Fourier Analysis

N-20220321

§113.1 The differential equation $Z' = iZ$ produces uniform circular motion

Consider the complex-valued initial-value problem

$$Z'(t) = iZ(t), \qquad Z(0) = 1.$$

Its solution is

$$Z(t) = e^{it}.$$

Multiplication by i sends

$$x + iy \longmapsto -y + ix,$$

which is a counterclockwise quarter-turn. Thus the velocity $Z'(t) = iZ(t)$ is always perpendicular to the radius vector.

The equation also proves that the radius is constant. Since

$$|Z|^2 = Z\overline{Z},$$

one has

$$\begin{aligned}\frac{d}{dt}|Z(t)|^2 &= Z'(t)\overline{Z(t)} + Z(t)\overline{Z'(t)} \\ &= iZ(t)\overline{Z(t)} - iZ(t)\overline{Z(t)} = 0.\end{aligned}$$

Because $|Z(0)| = 1$,

$$|Z(t)| = 1$$

for every t. Moreover,

$$|Z'(t)| = |iZ(t)| = 1,$$

so the particle moves around the unit circle with unit speed.

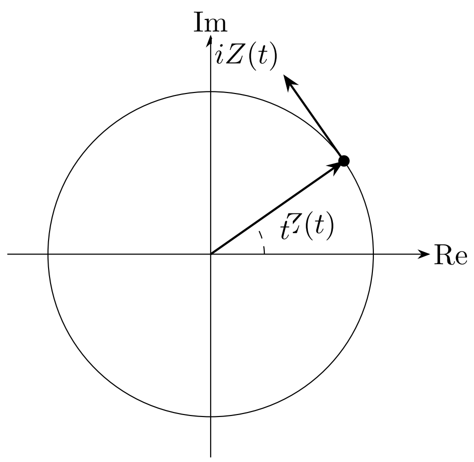


Figure 113.1: The equation $Z' = iZ$ rotates the radius vector by 90°, producing tangent velocity of constant length.

§113.2 The exponential map wraps a line onto a circle

The map
$$\exp_i : \mathbb{R} \longrightarrow S^1, \qquad t \longmapsto e^{it}$$
is a homomorphism from addition to multiplication:
$$e^{i(s+t)} = e^{is}e^{it}.$$
It is surjective because every unit complex number has the form
$$\cos t + i \sin t.$$
Its kernel is
$$\ker \exp_i = 2\pi\mathbb{Z}.$$
Therefore the first isomorphism theorem gives an isomorphism of groups
$$\boxed{\mathbb{R}/2\pi\mathbb{Z} \cong S^1.}$$
Angles differing by a full number of turns are not merely numerically similar; they represent the same coset in the quotient.

Locally, $t \mapsto e^{it}$ is one-to-one, but globally it wraps the real line around the circle infinitely many times. For a continuous loop
$$\gamma : [0,1] \to S^1, \qquad \gamma(0) = \gamma(1),$$
one can choose a continuous lift $\theta : [0,1] \to \mathbb{R}$ with
$$\gamma(t) = e^{i\theta(t)}.$$
The endpoint difference satisfies
$$\theta(1) - \theta(0) = 2\pi k$$
for an integer k, the winding number of the loop. The kernel of the exponential map is what makes this integer possible.

§113.3 Taking an n-th root means dividing angles and choosing a branch

Let
$$w = \rho e^{i\theta}, \qquad \rho > 0.$$
The equation
$$z^n = w$$
has the n solutions
$$z_k = ho^{1/n} e^{i(\theta+2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1.$$
The radii are fixed by
$$|z|^n = \rho,$$
while the arguments solve
$$n \arg z \equiv \theta \pmod{2\pi}.$$

The solutions form a regular n-gon on the circle of radius $\rho^{1/n}$.

For example, the fourth roots of -4 are

$$\sqrt{2}\, e^{i(\pi+2\pi k)/4}, \qquad k = 0, 1, 2, 3.$$

They occur at angles

$$\frac{\pi}{4}, \quad \frac{3\pi}{4}, \quad \frac{5\pi}{4}, \quad \frac{7\pi}{4}.$$

The multivalued appearance of complex roots is therefore the geometry of lifting one point through the covering map $z \mapsto z^n$.

§113.4 The roots of unity form a finite cyclic group

Let

$$\zeta_n = e^{2\pi i/n}.$$

The n-th roots of unity are

$$\mu_n = \{1, \zeta_n, \ldots, \zeta_n^{n-1}\}.$$

Multiplication adds exponents modulo n:

$$\zeta_n^a \zeta_n^b = \zeta_n^{a+b}.$$

Hence

$$\boxed{\mathbb{Z}/n\mathbb{Z} \longrightarrow \mu_n, \qquad [k] \longmapsto \zeta_n^k}$$

is a group isomorphism.

The order of ζ_n^k is

$$\frac{n}{\gcd(n,k)}.$$

Indeed, $(\zeta_n^k)^m = 1$ exactly when

$$n \mid km.$$

The smallest positive such m is $n/\gcd(n,k)$.

For $n = 12$, the powers ζ_{12}^k with

$$\gcd(k, 12) = 1$$

are the primitive roots:

$$\zeta_{12}, \quad \zeta_{12}^5, \quad \zeta_{12}^7, \quad \zeta_{12}^{11}.$$

Exact rotational period has become an elementary gcd calculation.

§113.5 Polynomial divisibility becomes evaluation on a finite subgroup

Let

$$\omega = e^{2\pi i/3}.$$

The roots of

$$x^2 + x + 1$$

are ω and ω^2. Therefore

$$x^2 + x + 1 \mid x^{2n} + x^n + 1$$

exactly when the latter polynomial vanishes at ω.

If $n \equiv 0 \pmod 3$, then $\omega^n = 1$ and

$$\omega^{2n} + \omega^n + 1 = 3.$$

If $n \equiv 1 \pmod 3$, then

$$\omega^{2n} + \omega^n + 1 = \omega^2 + \omega + 1 = 0.$$

If $n \equiv 2 \pmod 3$, then

$$\omega^{2n} + \omega^n + 1 = \omega + \omega^2 + 1 = 0.$$

Thus

$$\boxed{x^2 + x + 1 \mid x^{2n} + x^n + 1 \iff 3 \nmid n.}$$

This is the first prototype of a character test. Powers of ω remember only the exponent modulo three, so evaluation converts a polynomial identity into arithmetic on a finite cyclic group.

§113.6 Character orthogonality filters one congruence class

For $m \in \mathbb{Z}$, the geometric series gives

$$\sum_{j=0}^{n-1} \zeta_n^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

Equivalently,

$$\frac{1}{n} \sum_{j=0}^{n-1} \zeta_n^{j(k-r)} = \begin{cases} 1, & k \equiv r \pmod n, \\ 0, & k \not\equiv r \pmod n. \end{cases}$$

This is the roots-of-unity filter.

Let

$$P(x) = \sum_{k \geq 0} a_k x^k.$$

The sum of the coefficients with indices congruent to $r \pmod n$ is

$$\sum_{k \equiv r\,(n)} a_k = \frac{1}{n} \sum_{j=0}^{n-1} \zeta_n^{-jr} P(\zeta_n^j).$$

For example, let

$$P(x) = (1+x)^8$$

and ask for the sum of coefficients whose exponent is divisible by four. With $\zeta_4 = i$,

$$\begin{aligned} a_0 + a_4 + a_8 &= \frac{1}{4}(P(1) + P(i) + P(-1) + P(-i)) \\ &= \frac{1}{4}(256 + (1+i)^8 + 0 + (1-i)^8). \end{aligned}$$

Since

$$1 \pm i = \sqrt{2} e^{\pm i\pi/4},$$

one has

$$(1+i)^8 = (1-i)^8 = 16.$$

Therefore

$$a_0 + a_4 + a_8 = 72,$$

matching

$$\binom{8}{0} + \binom{8}{4} + \binom{8}{8} = 1 + 70 + 1.$$

The filter replaces a restricted coefficient sum by a short list of evaluations.

§113.7 The discrete Fourier transform is the full set of character coordinates

For a function

$$f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C},$$

define

$$\widehat{f}(m) = \sum_{k=0}^{n-1} f(k)\zeta_n^{-mk}.$$

The inverse formula is

$$\boxed{f(k) = \frac{1}{n}\sum_{m=0}^{n-1} \widehat{f}(m)\zeta_n^{mk}.}$$

To verify it, substitute the definition:

$$\begin{aligned}
\frac{1}{n}\sum_m \widehat{f}(m)\zeta_n^{mk} &= \frac{1}{n}\sum_m \sum_{j=0}^{n-1} f(j)\zeta_n^{m(k-j)} \\
&= \sum_{j=0}^{n-1} f(j)\left(\frac{1}{n}\sum_m \zeta_n^{m(k-j)}\right) \\
&= f(k).
\end{aligned}$$

The roots-of-unity filter is exactly the orthogonality step in this proof.

For $n = 4$, take

$$f = (1, 2, 0, -1).$$

Using $\zeta_4 = i$,

$$\begin{aligned}
\widehat{f}(0) &= 1 + 2 + 0 - 1 = 2, \\
\widehat{f}(1) &= 1 + 2(-i) + (-1)i = 1 - 3i, \\
\widehat{f}(2) &= 1 - 2 + 0 + 1 = 0, \\
\widehat{f}(3) &= 1 + 2i - i = 1 + 3i.
\end{aligned}$$

Thus

$$\widehat{f} = (2, 1 - 3i, 0, 1 + 3i).$$

Applying the inverse formula recovers all four original entries. The transform is not merely evaluation at roots of unity; orthogonality makes those evaluations a complete coordinate system.

§113.8 Fourier characters diagonalize every circulant operator

Let S be the cyclic shift on $\mathbb{C}^n$:

$$S(x_0, x_1, \ldots, x_{n-1}) = (x_{n-1}, x_0, \ldots, x_{n-2}).$$

For each m, define the Fourier vector

$$v_m = (1, \zeta_n^{-m}, \zeta_n^{-2m}, \ldots, \zeta_n^{-(n-1)m})^T.$$

Then

$$Sv_m = \zeta_n^m v_m.$$

Hence the Fourier basis diagonalizes the shift.

Every circulant matrix is a polynomial in S:

$$C = c_0 I + c_1 S + \cdots + c_{n-1} S^{n-1}.$$

Therefore

$$Cv_m = \lambda_m v_m,$$

where

$$\lambda_m = c_0 + c_1 \zeta_n^m + \cdots + c_{n-1} \zeta_n^{m(n-1)}.$$

The eigenvalues are the DFT of the first row, up to the chosen sign convention.

For the four-cycle adjacency matrix

$$C = S + S^{-1},$$

the eigenvalues are

$$\lambda_m = \zeta_4^m + \zeta_4^{-m} = 2\cos\frac{2\pi m}{4},$$

so

$$\sigma(C) = \{2, 0, -2, 0\}.$$

A graph operator with coupled coordinates becomes four independent scalar multipliers in the Fourier basis.

Circular convolution is diagonalized for the same reason. If

$$(f * g)(r) = \sum_{k \in \mathbb{Z}/n\mathbb{Z}} f(k) g(r - k),$$

then

$$\widehat{f * g}(m) = \widehat{f}(m)\widehat{g}(m).$$

Convolution in physical coordinates is multiplication in character coordinates.

§113.9 Cyclotomic factors separate roots by exact period

The factorization

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x)$$

groups roots of unity according to exact order. For $n = 12$,

$$\begin{aligned} x^{12} - 1 &= \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x) \\ &= (x-1)(x+1)(x^2+x+1)(x^2+1) \\ &\quad \cdot (x^2 - x + 1)(x^4 - x^2 + 1). \end{aligned}$$

The last factor contains exactly the primitive twelfth roots. This is enough for the present chapter: finite Fourier analysis uses all characters of the cyclic group, while cyclotomic factorization separates those characters by exact order. The arithmetic of the factors themselves belongs to the dedicated cyclotomic chapter.

§113.10 A primitive fifth root already explains the constructible pentagon

Let

$$\zeta = e^{2\pi i/5}$$

and set

$$u = \zeta + \zeta^{-1} = 2\cos\frac{2\pi}{5}.$$

Because

$$1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0,$$

dividing by ζ^2 gives

$$\zeta^2 + \zeta + 1 + \zeta^{-1} + \zeta^{-2} = 0.$$

Now

$$\zeta^2 + \zeta^{-2} = u^2 - 2,$$

so

$$u^2 + u - 1 = 0.$$

The positive root is

$$u = \frac{\sqrt{5} - 1}{2}.$$

Therefore

$$\cos\frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}.$$

The central angle of a regular pentagon is obtained using only a square root, which is why the pentagon is constructible by straightedge and compass.

This small calculation is the appropriate arithmetic endpoint here. The finite cyclic group supplies the roots, Fourier characters use all their powers as coordinates, and the real combination $\zeta + \zeta^{-1}$ collapses a complex symmetry orbit to the quadratic number controlling the pentagon.

Complex Differentiability, Cauchy–Riemann Equations, and Holomorphic Rigidity

N-20220323

§114.1 One difference quotient must survive every planar direction

Let $U \subseteq \mathbb{C}$ be open and let $f : U \to \mathbb{C}$. The complex derivative at $z_0 \in U$ is

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h},$$

provided the limit exists. The increment h approaches zero through the whole plane. This is much stronger than asking for two one-sided real limits.

Write

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y).$$

If f is complex differentiable at $z_0 = x_0 + iy_0$, then the quotient has the same limit along the real and imaginary axes.

Along $h = t \in \mathbb{R}$,

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

Along $h = it$,

$$\begin{aligned} f'(z_0) &= \lim_{t \to 0} \frac{u(x_0, y_0 + t) - u(x_0, y_0) + i\big(v(x_0, y_0 + t) - v(x_0, y_0)\big)}{it} \\ &= v_y(x_0, y_0) - iu_y(x_0, y_0). \end{aligned}$$

Equality gives the Cauchy–Riemann equations

$$\boxed{u_x = v_y, \qquad u_y = -v_x.}$$

The derivative can then be written in either form:

$$f' = u_x + iv_x = v_y - iu_y.$$

§114.2 Cauchy–Riemann is sufficient under real differentiability

The equations alone at one point do not automatically imply complex differentiability. A clean sufficient hypothesis is real differentiability.

Proposition 114.2.1

Suppose $u, v : \mathbb{R}^2 \to \mathbb{R}$ are differentiable at (x_0, y_0) and satisfy the Cauchy–Riemann equations there. Then $f = u+iv$ is complex differentiable at $z_0 = x_0+iy_0$.

Proof. Let $h = \Delta x + i\Delta y$. Real differentiability gives

$$\Delta u = u_x \Delta x + u_y \Delta y + r_1(h),$$
$$\Delta v = v_x \Delta x + v_y \Delta y + r_2(h),$$

where

$$\frac{|r_1(h)| + |r_2(h)|}{|h|} \to 0.$$

Using $u_x = v_y$ and $u_y = -v_x$,

$$\begin{aligned}\Delta f &= \Delta u + i\Delta v \\ &= (u_x + iv_x)(\Delta x + i\Delta y) + r_1(h) + ir_2(h).\end{aligned}$$

Hence

$$\frac{\Delta f}{h} = u_x + iv_x + \frac{r_1(h) + ir_2(h)}{h} \longrightarrow u_x + iv_x.$$

□

If the first partial derivatives are continuous in a neighborhood, real differentiability is automatic. Thus the familiar C^1 version is:

$$f \text{ holomorphic} \iff u_x = v_y, \quad u_y = -v_x.$$

§114.3 Wirtinger derivatives isolate the holomorphic part

Define

$$\partial_z = \frac{1}{2}\left(\partial_x - i\partial_y\right), \qquad \partial_{\bar z} = \frac{1}{2}\left(\partial_x + i\partial_y\right).$$

Since

$$x = \frac{z + \bar z}{2}, \qquad y = \frac{z - \bar z}{2i},$$

these operators treat z and $\bar z$ as formally independent coordinates:

$$\partial_z z = 1, \quad \partial_z \bar z = 0, \quad \partial_{\bar z} z = 0, \quad \partial_{\bar z} \bar z = 1.$$

For $f = u + iv$,

$$2\partial_{\bar z} f = (u_x - v_y) + i(v_x + u_y).$$

Therefore the Cauchy–Riemann equations are exactly

$$\boxed{\partial_{\bar z} f = 0.}$$

Under real differentiability, this is equivalent to holomorphicity.

The operator $\partial_{\bar z}$ measures failure of holomorphicity. For example,

$$\partial_{\bar z} \bar z = 1,$$

so $\bar z$ is nowhere holomorphic. Also

$$|z|^2 = z\bar z, \qquad \partial_{\bar z}|z|^2 = z.$$

Hence $|z|^2$ is complex differentiable only at $z = 0$, and it is not holomorphic on any neighborhood of that point.

§114.4 The Jacobian must be a rotation followed by a scaling

The real derivative of $f = u + iv$ is the Jacobian matrix

$$Df = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}.$$

Under the Cauchy–Riemann equations,

$$Df = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a = u_x, \quad b = v_x.$$

This is precisely the real matrix of multiplication by

$$f'(z_0) = a + ib.$$

If $f'(z_0) \neq 0$, write

$$f'(z_0) = \rho e^{i\theta}.$$

Then

$$Df(z_0) = \rho \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Infinitesimally, f scales every direction by ρ and rotates every direction by θ. Thus a holomorphic map with nonzero derivative preserves oriented angles. This is the local meaning of conformality.

At a critical point $f'(z_0) = 0$, the first-order map collapses. For instance, $f(z) = z^m$ near 0 multiplies angles by m but has zero linear part when $m \geq 2$.

§114.5 Cauchy's integral formula turns boundary values into derivatives

The rigidity of holomorphic functions does not follow from the Cauchy–Riemann equations alone by a short algebraic argument. The decisive analytic input is Cauchy's integral formula.

If f is holomorphic on a neighborhood of the closed disk

$$\overline{D(a, R)},$$

then for $|z - a| < R$,

$$\boxed{f(z) = \frac{1}{2\pi i} \int_{|\zeta - a| = R} \frac{f(\zeta)}{\zeta - z}\, d\zeta.}$$

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{|\zeta - a| = R} \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta.$$

One complex derivative therefore forces derivatives of every order.

The same formula yields the Cauchy estimate. If

$$M_R = \max_{|\zeta - a| = R} |f(\zeta)|,$$

then

$$|f^{(n)}(a)| \leq \frac{n! M_R}{R^n}.$$

Boundary control becomes quantitative control of all derivatives in the interior.

§114.6 Holomorphic functions are analytic

For $|z-a|<R$, expand the kernel as a geometric series:

$$\begin{aligned}\frac{1}{\zeta - z} &= \frac{1}{\zeta - a}\frac{1}{1-\frac{z-a}{\zeta-a}} \\ &= \sum_{n=0}^{\infty}\frac{(z-a)^n}{(\zeta-a)^{n+1}}.\end{aligned}$$

The series converges uniformly on the integration circle for $|z-a|<R$. Substitution into Cauchy's formula gives

$$f(z)=\sum_{n=0}^{\infty}c_n(z-a)^n,$$

where

$$c_n=\frac{1}{2\pi i}\int_{|\zeta-a|=R}\frac{f(\zeta)}{(\zeta-a)^{n+1}}\,d\zeta=\frac{f^{(n)}(a)}{n!}.$$

Thus

$$\boxed{f(z)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(z-a)^n.}$$

Holomorphicity implies analyticity. This is the sharp contrast with smooth real functions such as

$$g(x)=\begin{cases}e^{-1/x^2}, & x\neq 0,\\ 0, & x=0,\end{cases}$$

whose Taylor series at 0 vanishes identically although the function does not.

§114.7 Zeros are isolated unless the whole function vanishes

Suppose f is holomorphic near a and not identically zero. Its Taylor series has a first nonzero coefficient:

$$f(z)=c_m(z-a)^m+c_{m+1}(z-a)^{m+1}+\cdots, \qquad c_m\neq 0.$$

Factor

$$f(z)=(z-a)^m g(z), \qquad g(a)=c_m\neq 0.$$

By continuity, g is nonzero near a, so a is an isolated zero of multiplicity m.

This proves the identity theorem. If two holomorphic functions agree on a set with a limit point inside a connected domain, their difference has non-isolated zeros and must vanish identically.

The chain of rigidity is now explicit:

$$\begin{aligned}\partial_{\bar z} f = 0 &\longrightarrow \text{Cauchy formula} \longrightarrow \text{power series},\\ &\longrightarrow \text{isolated zeros and unique continuation.}\end{aligned}$$

The operator $\partial_{\bar z}$ is the local differential equation; Cauchy's formula supplies the global analytic mechanism that makes its solutions rigid.

Contour Integrals: Parametrization, Exactness, Winding, and Residues

N-20220813

§115.1 A contour integral is a limit of transformed tangent increments

A contour is a piecewise C^1 path

$$\gamma : [a, b] \to \mathbb{C}.$$

For a continuous function f on the image of γ, define

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

The right-hand side is an ordinary complex-valued Riemann integral.

Let

$$a = t_0 < t_1 < \cdots < t_n = b$$

and choose $\tau_j \in [t_{j-1}, t_j]$. The Riemann sum is

$$\sum_{j=1}^{n} f(\gamma(\tau_j))\big(\gamma(t_j) - \gamma(t_{j-1})\big).$$

Each directed increment is multiplied by f, hence scaled and rotated before the increments are added. A contour integral is not an area under a graph; it is a limit of transformed tangent vectors.

The basic estimate follows immediately:

$$\left|\int_\gamma f(z)\,dz\right| \le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\,dt \le L(\gamma) \max_{z\in\gamma} |f(z)|,$$

where

$$L(\gamma) = \int_a^b |\gamma'(t)|\,dt.$$

This ML-estimate is the main error-control tool for contour deformation and limiting arguments.

§115.2 Orientation, concatenation, and reparametrization

The reverse contour is

$$\overline{\gamma}(t) = \gamma(a + b - t).$$

Substitution gives

$$\boxed{\int_{\overline{\gamma}} f(z)\,dz = -\int_\gamma f(z)\,dz.}$$

If $\gamma_1(b_1) = \gamma_2(a_2)$, the concatenated contour $\gamma_1 * \gamma_2$ traverses the first and then the second. Splitting the parameter interval yields

$$\boxed{\int_{\gamma_1 * \gamma_2} f(z)\,dz = \int_{\gamma_1} f(z)\,dz + \int_{\gamma_2} f(z)\,dz.}$$

Let $\phi : [c, d] \to [a, b]$ be a piecewise C^1, increasing surjection with $\phi(c) = a$, $\phi(d) = b$. Then

$$\begin{aligned}\int_{\gamma\circ\phi} f(z)\,dz &= \int_c^d f(\gamma(\phi(s)))\gamma'(\phi(s))\phi'(s)\,ds \\ &= \int_a^b f(\gamma(t))\gamma'(t)\,dt.\end{aligned}$$

Thus the integral is unchanged by orientation-preserving reparametrization. If ϕ reverses orientation, the sign changes.

For the unit circle, all three parametrizations

$$e^{it}, \quad 0 \le t \le 2\pi; \qquad e^{2\pi it}, \quad 0 \le t \le 1; \qquad e^{2\pi it^5}, \quad 0 \le t \le 1$$

give the same oriented contour integral.

§115.3 The complex integral is the pullback of a one-form

Write

$$z = x + iy, \qquad f(z) = u(x, y) + iv(x, y), \qquad dz = dx + i\,dy.$$

Then

$$\begin{aligned}f(z)\,dz &= (u + iv)(dx + i\,dy) \\ &= (u\,dx - v\,dy) + i(v\,dx + u\,dy).\end{aligned}$$

Therefore

$$\int_\gamma f(z)\,dz = \int_\gamma (u\,dx - v\,dy) + i\int_\gamma (v\,dx + u\,dy).$$

A complex contour integral is two real line integrals coupled by the complex multiplication rule.

In differential-form notation, $f(z)\,dz$ is a complex-valued one-form. The parametrized formula is exactly pullback:

$$\int_\gamma f(z)\,dz = \int_{[a,b]} \gamma^*(f(z)\,dz).$$

This explains reparametrization invariance: integration is applied to the pulled-back one-form on the parameter interval.

§115.4 Primitives are equivalent to path independence

Let $D \subseteq \mathbb{C}$ be a domain and let $f : D \to \mathbb{C}$ be continuous.

Theorem 115.4.1

The following are equivalent.

(i) There is a holomorphic function $F : D \to \mathbb{C}$ with $F' = f$.

(ii) The integral of f depends only on the endpoints of the contour.

(iii) Every closed contour γ in D satisfies

$$\int_\gamma f(z)\,dz = 0.$$

Proof. If $F' = f$, the chain rule gives

$$\frac{d}{dt}F(\gamma(t)) = f(\gamma(t))\gamma'(t).$$

Hence

$$\int_\gamma f(z)\,dz = F(\gamma(b)) - F(\gamma(a)),$$

which proves (i)⇒(ii), and (ii)⇒(iii) is immediate.

Assume (iii). Fix $z_0 \in D$, and for $z \in D$ define

$$F(z) = \int_{\gamma_{z_0,z}} f(\zeta)\,d\zeta,$$

where $\gamma_{z_0,z}$ is any contour from z_0 to z. If two contours are chosen, following one and reversing the other gives a closed contour, so (iii) makes F well defined.

For small h, use the same contour to z and then the line segment from z to $z+h$:

$$\frac{F(z+h) - F(z)}{h} = \int_0^1 f(z+th)\,dt.$$

By continuity, the right-hand side tends to $f(z)$. Thus $F'(z) = f(z)$. □

The topology of the domain enters in condition (iii): local primitives may exist while a global primitive fails because closed loops can carry nonzero periods.

§115.5 Cauchy–Goursat turns holomorphicity into exactness on simple domains

Suppose $f = u + iv$ has continuous first partial derivatives on a region containing a positively oriented simple closed contour $\gamma = \partial\Omega$. Using the decomposition of $f\,dz$ and Green's theorem,

$$\begin{aligned}\operatorname{Re}\int_\gamma f(z)\,dz &= \int_\gamma u\,dx - v\,dy \\ &= \iint_\Omega (-v_x - u_y)\,dx\,dy,\end{aligned}$$

and

$$\begin{aligned}\operatorname{Im}\int_\gamma f(z)\,dz &= \int_\gamma v\,dx + u\,dy \\ &= \iint_\Omega (u_x - v_y)\,dx\,dy.\end{aligned}$$

The Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

make both integrands zero. Therefore

$$\int_\gamma f(z)\,dz = 0.$$

Goursat's theorem removes the assumption of continuous partial derivatives: complex differentiability alone is enough. On a simply connected domain, every closed contour is deformable through the domain to a point, and Cauchy–Goursat implies

$$\int_\gamma f(z)\,dz = 0.$$

By the preceding theorem, every holomorphic function on a simply connected domain has a primitive.

The conclusion fails when the domain contains a hole around a singularity. The model obstruction is $1/z$ on $\mathbb{C}^*$.

§115.6 Powers on a circle isolate one Fourier mode

For the circle traversed k times,

$$\gamma_k(t) = e^{ikt}, \qquad 0 \le t \le 2\pi,$$

one has

$$\begin{aligned}\int_{\gamma_k} z^m\,dz &= \int_0^{2\pi} e^{imkt} ike^{ikt}\,dt \\ &= ik\int_0^{2\pi} e^{ik(m+1)t}\,dt.\end{aligned}$$

If $m \neq -1$, the oscillation completes an integer number of periods and cancels. If $m = -1$, the integrand is the constant ik. Hence

$$\boxed{\int_{\gamma_k} z^m\,dz = \begin{cases} 2\pi ik, & m = -1, \\ 0, & m \neq -1. \end{cases}}$$

The exceptional exponent -1 is the coefficient that survives contour averaging. This is the local algebra behind residues.

§115.7 Winding number is the period of $dz/(z-a)$

Let γ be a closed contour avoiding a. Define

$$\mathrm{Ind}(\gamma, a) = \frac{1}{2\pi i}\int_\gamma \frac{dz}{z-a}.$$

For a circle centered at a traversed k times, the preceding computation gives

$$\mathrm{Ind}(\gamma, a) = k.$$

In general the index is an integer, equal to the endpoint change of a lifted argument of $\gamma(t) - a$.

It is invariant under homotopies that avoid a. For a smooth homotopy $\Gamma(t,s)$, differentiate under the integral sign:

$$\begin{aligned}\frac{d}{ds}\int_0^1 \frac{\partial_t\Gamma(t,s)}{\Gamma(t,s)-a}\,dt &= \int_0^1 \partial_t\left(\frac{\partial_s\Gamma(t,s)}{\Gamma(t,s)-a}\right)dt\\ &= 0,\end{aligned}$$

because the loop has the same initial and terminal point. Approximation or covering-space lifting gives the general continuous-homotopy statement.

Thus the integral depends on the homotopy class of the loop in $\mathbb{C}\setminus\{a\}$, not on its detailed shape.

§115.8 Closed need not mean exact on a punctured domain

On $\mathbb{R}^2\setminus\{0\}$, consider the real one-form

$$\alpha = \frac{1}{2\pi}\frac{-y\,dx + x\,dy}{x^2+y^2}.$$

A direct calculation gives

$$d\alpha = 0.$$

So α is closed. On the unit circle $x=\cos t$, $y=\sin t$,

$$\alpha = \frac{dt}{2\pi},$$

and therefore

$$\int_{S^1}\alpha = 1.$$

If $\alpha = dF$ were exact, every closed-loop integral would vanish. Hence α is closed but not exact.

This period represents the generator of

$$H^1_{\mathrm{dR}}(\mathbb{C}^*)\cong\mathbb{R}.$$

The complex form dz/z contains the same obstruction:

$$\frac{1}{2\pi i}\int_\gamma \frac{dz}{z} = \mathrm{Ind}(\gamma,0).$$

The failure of a global logarithm on $\mathbb{C}^*$ is the same phenomenon. Locally,

$$d(\log z) = \frac{dz}{z},$$

but no single-valued logarithm exists around a loop of nonzero winding.

§115.9 Cauchy's integral formula recovers interior values from the boundary

Let f be holomorphic on a neighborhood of a positively oriented simple closed contour γ and its interior, and let z lie inside. Then

$$\boxed{f(z) = \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{\zeta - z}\,d\zeta.}$$

To see the mechanism, write

$$\frac{f(\zeta)}{\zeta - z} = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}.$$

The first term has a removable singularity at $\zeta = z$, with value $f'(z)$, so its closed-contour integral vanishes. The second term contributes

$$f(z) \int_\gamma \frac{d\zeta}{\zeta - z} = 2\pi i\, f(z)\, \mathrm{Ind}(\gamma, z).$$

For a simple positively oriented contour containing z, the index is 1.

Differentiating under the integral sign gives

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta.$$

For example,

$$\int_{|\zeta|=2} \frac{e^\zeta}{(\zeta - 1)^3}\, d\zeta = \frac{2\pi i}{2!} e = \pi i e.$$

§115.10 Residues convert local Laurent coefficients into global integrals

If f has an isolated singularity at a, its Laurent expansion is

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n.$$

The residue is

$$\mathrm{Res}(f, a) = c_{-1}.$$

All other powers integrate to zero around a small circle, while $(z - a)^{-1}$ contributes $2\pi i$. Therefore

$$\int_{|z-a|=r} f(z)\, dz = 2\pi i\, \mathrm{Res}(f, a).$$

For a contour γ avoiding finitely many poles $a_1, \ldots, a_N$, the residue theorem is

$$\boxed{\int_\gamma f(z)\, dz = 2\pi i \sum_{j=1}^{N} \mathrm{Ind}(\gamma, a_j)\, \mathrm{Res}(f, a_j).}$$

Local Laurent coefficients are paired with global winding numbers.

Consider

$$f(z) = \frac{z + 1}{z(z - 1)}.$$

The residues are

$$\mathrm{Res}(f, 0) = -1, \qquad \mathrm{Res}(f, 1) = 2.$$

Hence

$$\int_{|z|=2} f(z)\, dz = 2\pi i(-1 + 2) = 2\pi i.$$

For the smaller contour $|z| = 1/2$, only the pole at 0 is enclosed, so

$$\int_{|z|=1/2} f(z)\, dz = -2\pi i.$$

The integrand has not changed; the answer changes because the winding data relative to the poles has changed.

Numerical Series, Power Series, Taylor Series, and Improper Integrals

N-20260616

§116.1 Why these topics belong together

The long Markdown note moves from numerical series to function series, then to power series and improper integrals. The common theme is convergence: when does an infinite summation or an infinite accumulation define a finite object, and when may we manipulate it like a finite expression?

§116.2 Numerical series

A numerical series is

$$\sum_{n=1}^{\infty} a_n, \qquad s_N = \sum_{n=1}^{N} a_n.$$

It converges if the partial sums s_N converge. The first necessary test is

$$a_n \to 0.$$

If this fails, the series diverges. If it holds, nothing is decided.

The geometric series is the base model:

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \qquad |r| < 1.$$

The harmonic series shows why terms tending to zero is insufficient:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

A standard proof groups terms:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots .$$

Each block after the first contributes at least $1/2$.

§116.3 Positive-term series

For $a_n \geq 0$, convergence is controlled by size.

Comparison test: if $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges. If $a_n \geq b_n \geq 0$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Limit comparison: if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c, \qquad 0 < c < \infty,$$

then $\sum a_n$ and $\sum b_n$ have the same convergence behavior.

The p-series is the main comparison model:

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges iff $p > 1$.

The examples in the original note include

$$\sum \sin\frac{1}{n}, \qquad \sum \ln\left(1+\frac{1}{n^2}\right), \qquad \sum \left(\sqrt{n+1}-\sqrt{n}\right).$$

The first behaves like $\sum 1/n$ and diverges. The second behaves like $\sum 1/n^2$ and converges. The third behaves like $\sum 1/(2\sqrt{n})$ and diverges.

§116.4 Ratio and root tests

The ratio test is suited to factorials and exponentials:

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|.$$

If $L < 1$, the series converges absolutely; if $L > 1$, it diverges; if $L = 1$, the test is inconclusive.

The root test is suited to expressions of the form $(\cdots)^n$:

$$L = \limsup_{n\to\infty} \sqrt[n]{|a_n|}.$$

Again $L < 1$ gives absolute convergence, $L > 1$ gives divergence, and $L = 1$ is inconclusive.

Typical original examples include $\sum a^n/n^s$, factorials in the numerator or denominator, and powers with variable bases. The rule is not to memorize which test belongs to which problem, but to inspect which part of a_n changes fastest.

§116.5 Alternating and arbitrary-sign series

Absolute convergence means

$$\sum |a_n| < \infty.$$

Conditional convergence means $\sum a_n$ converges but $\sum |a_n|$ diverges.

The alternating-series test says that if $b_n \downarrow 0$, then

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n$$

converges. The alternating harmonic series is the standard example.

Dirichlet's test: if partial sums $A_N = \sum_{n=1}^{N} a_n$ are bounded and $b_n \downarrow 0$, then $\sum a_n b_n$ converges. This handles examples such as

$$\sum_{n=1}^{\infty} \frac{\cos n}{n}.$$

Abel's test is a neighboring result where one factor has a convergent series and the other is monotone bounded.

§116.6 Function series and uniform convergence

A function series is

$$\sum_{n=1}^{\infty} u_n(x).$$

Pointwise convergence asks for each fixed x. Uniform convergence asks whether the tail becomes small simultaneously for all x in a set:

$$\sup_{x\in E} |s_N(x) - s(x)| \to 0.$$

Uniform convergence is what allows continuity and integration to pass to the limit. Without it, a limit of continuous functions may fail to be continuous.

The Weierstrass M-test says that if

$$|u_n(x)| \le M_n, \qquad \sum M_n < \infty,$$

then $\sum u_n$ converges uniformly and absolutely.

§116.7 Power series

A power series centered at x_0 is

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n.$$

There exists a radius of convergence R such that the series converges absolutely for $|x - x_0| < R$ and diverges for $|x - x_0| > R$. Endpoints require separate testing.

A common formula is

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|,$$

when the limit exists. Missing powers require care: if the series contains $(x - x_0)^{2n}$, one should apply the ratio/root test directly rather than blindly reading a coefficient sequence as though every power were present.

Inside the interval of convergence, power series may be differentiated and integrated term by term:

$$\left(\sum a_n(x - x_0)^n\right)' = \sum n a_n(x - x_0)^{n-1}.$$

The radius of convergence remains the same, though endpoints may change.

§116.8 Taylor series

The Taylor series at a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

At $a = 0$, it is a Maclaurin series. Standard expansions include

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \qquad |x| < 1.$$

Many exam examples are indirect expansions: rewrite the function in terms of known series, substitute, integrate, differentiate, or multiply series.

§116.9 Improper integrals

Improper integrals appear when the interval is infinite or the integrand is unbounded:

$$\int_a^\infty f(x)\,dx = \lim_{R\to\infty} \int_a^R f(x)\,dx,$$

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$$

if f is singular at b.

The two fundamental p-templates are

$$\int_1^\infty \frac{dx}{x^p} \quad \text{converges iff } p > 1,$$

$$\int_0^1 \frac{dx}{x^p} \quad \text{converges iff } p < 1.$$

Logarithmic refinements include

$$\int_e^\infty \frac{dx}{x(\ln x)^p},$$

which converges iff $p > 1$.

For sign-changing improper integrals, absolute convergence means $\int |f| < \infty$. Conditional convergence often comes from oscillation, as in Dirichlet-type integrals.

§116.10 Unified viewpoint

Series and improper integrals ask the same question: can infinitely many small contributions accumulate to a finite value? Positive objects are controlled by size; sign-changing objects are controlled by cancellation plus size. Power series add a new idea: convergence depends on the point x, producing a domain where infinite algebraic manipulation becomes legitimate.

§116.11 Uniform convergence as norm convergence

For a function series $\sum u_n(x)$, uniform convergence on E is convergence in the sup norm:

$$\|s_N - s\|_{\infty,E} = \sup_{x\in E} |s_N(x) - s(x)| \to 0.$$

The Weierstrass M-test is simply comparison in this norm. If $|u_n(x)| \le M_n$ and $\sum M_n < \infty$, then the tails satisfy

$$\sup_x \left| \sum_{n=N}^\infty u_n(x) \right| \le \sum_{n=N}^\infty M_n \to 0.$$

Thus the elementary test is really completeness of a normed function space at work.

§116.12 Analytic functions and convergence radius

A power series

$$\sum a_n(z - z_0)^n$$

defines a holomorphic function inside its disk of convergence. The radius is controlled by the nearest complex singularity, not merely by real-variable behavior. This explains why real Taylor series sometimes stop converging at a point where the real function looks harmless: the obstruction may live off the real axis.

For example,

$$\frac{1}{1 + x^2}$$

has real Taylor series at 0

$$1 - x^2 + x^4 - \cdots ,$$

with radius 1, because the complex singularities are at $x = \pm i$.

§116.13 Dominated convergence and improper integrals

Improper integrals and parameter-dependent integrals become cleaner in measure language. If $f_n \to f$ pointwise and $|f_n| \le g$ with g integrable, dominated convergence gives

$$\int f_n \to \int f.$$

This justifies many limiting operations that elementary calculus handles case by case.

A useful contrast is between absolute convergence and conditional cancellation. The integral

$$\int_1^\infty \frac{\sin x}{x}\, dx$$

converges conditionally by oscillation, but

$$\int_1^\infty \left| \frac{\sin x}{x} \right| dx$$

diverges. This is the integral analogue of conditional series.

§116.14 Tauberian flavor

Power series also encode sequences. Given a_n, define

$$F(r) = \sum_{n=0}^{\infty} a_n r^n, \qquad 0 < r < 1.$$

Abelian theorems say that ordinary convergence of $\sum a_n$ often implies limiting behavior of $F(r)$ as $r \to 1^-$. Tauberian theorems ask for converses under additional hypotheses. This is the research-level continuation of the elementary question: how does convergence of coefficients relate to behavior of the generating function near the boundary?

Fourier Series and Orthogonal Expansion of Functions

N-20260617

§117.1 Why Fourier series is a separate theme

The Fourier part of the original long note deserves its own chapter. It is not merely another convergence test. It is the beginning of representing functions by an orthogonal basis, which is the infinite-dimensional analogue of decomposing a vector in an orthonormal basis.

§117.2 Orthogonality of trigonometric functions

On $[-\pi, \pi]$,

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, dx = 0 \quad (m \neq n),$$

$$\int_{-\pi}^{\pi} \sin mx \sin nx \, dx = 0 \quad (m \neq n), \qquad \int_{-\pi}^{\pi} \sin mx \cos nx \, dx = 0.$$

Also,

$$\int_{-\pi}^{\pi} \cos^2 nx \, dx = \int_{-\pi}^{\pi} \sin^2 nx \, dx = \pi.$$

This is why the Fourier coefficients are projections.

§117.3 Fourier coefficients

For a 2π-periodic function,

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx),$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \qquad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

For a_0,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx.$$

§117.4 Dirichlet convergence theorem

Theorem 117.4.1 (Dirichlet convergence theorem)

If a periodic function is piecewise smooth, its Fourier series converges at each point to the average of the left and right limits.

Under standard piecewise smooth hypotheses, the Fourier series converges at x to

$$\frac{f(x+0)+f(x-0)}{2}.$$

At a point of continuity, this equals $f(x)$. At a jump, it equals the midpoint of the jump. This is why Fourier series may represent a function correctly except at discontinuities while still showing endpoint averaging.

§117.5 Even and odd functions

If f is even, then all sine coefficients vanish and the Fourier series is a cosine series. If f is odd, all cosine coefficients vanish and the series is a sine series. This is not just a computational shortcut; it says symmetry determines which basis vectors can appear.

For example, $f(x) = x$ on $[-\pi, \pi]$ is odd, so only sine terms occur. The standard result is

$$x \sim 2\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n}.$$

For $f(x) = |x|$, only cosine terms occur, and evaluating the series at special points gives sums of reciprocal squares over odd integers.

§117.6 Half-range expansions

On $[0, l]$, a function may be extended oddly or evenly to $[-l, l]$. Odd extension gives a sine series; even extension gives a cosine series. The choice should match the boundary behavior one wants. In boundary-value problems for PDEs, sine series naturally encode zero boundary values, while cosine series encode even reflection or Neumann-type behavior.

For period $2l$, the basis becomes

$$\cos\frac{n\pi x}{l}, \qquad \sin\frac{n\pi x}{l}.$$

The coefficients are

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx, \qquad b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}\,dx.$$

§117.7 Concrete coefficient models

The standard examples are valuable only when they are computed, not merely named.

§117.7.i The sawtooth function $f(x) = x$

On $[-\pi, \pi]$, the function $f(x) = x$ is odd, so only sine terms occur. The coefficient is

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} x\sin nx\,dx.$$

Integration by parts gives

$$\int_0^{\pi} x\sin nx\,dx = -\frac{\pi(-1)^n}{n},$$

so

$$b_n = 2\frac{(-1)^{n+1}}{n}.$$

Hence

$$x \sim 2\sum_{n=1}^{\infty}(-1)^{n+1}\frac{\sin nx}{n}.$$

At $x = \pi$, the periodic extension jumps from π to $-\pi$, so the Fourier series converges to 0, not to π.

§117.7.ii The even function $f(x) = |x|$

Since $|x|$ is even, only cosine terms occur. One has

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} |x|\,dx = \pi,$$

and for $n \geq 1$,

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{2}{\pi}\frac{(-1)^n - 1}{n^2}.$$

Thus only odd n contribute:

$$|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\cos(2k+1)x}{(2k+1)^2}.$$

Putting $x = 0$ gives

$$0 = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2},$$

so

$$\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

This is a typical Fourier-series trick: evaluate a function expansion at a point where the convergence value is known.

§117.7.iii A square wave

For the 2π-periodic square wave equal to 1 on $(0, \pi)$ and -1 on $(-\pi, 0)$, the function is odd, so

$$b_n = \frac{2}{\pi}\int_0^{\pi}\sin nx\,dx = \frac{2}{\pi}\frac{1 - (-1)^n}{n}.$$

Thus

$$f(x) \sim \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right).$$

At the jump points the series converges to 0, the midpoint of the jump.

§117.8 Dirichlet kernel: what partial sums really do

The N-th Fourier partial sum is not a vague approximation; it is convolution with the Dirichlet kernel:

$$S_N f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t)\, dt,$$

where

$$D_N(t) = \sum_{n=-N}^{N} e^{int} = \frac{\sin((N+1/2)t)}{\sin(t/2)}.$$

The kernel has total mass 2π, but its L^1-norm grows. This growth is one reason Fourier partial sums can behave badly near discontinuities.

The Dirichlet convergence theorem is therefore not magic: near a point x, the oscillatory kernel samples values of f from both sides. At a jump, the symmetric sampling yields the midpoint $(f(x+0) + f(x-0))/2$.

§117.9 Fejer means: averaging partial sums improves convergence

The Cesaro average of Fourier partial sums is

$$\sigma_N f = \frac{1}{N+1} \sum_{k=0}^{N} S_k f.$$

It is convolution with the Fejer kernel

$$F_N(t) = \frac{1}{N+1} \left(\frac{\sin((N+1)t/2)}{\sin(t/2)} \right)^2.$$

Unlike the Dirichlet kernel, $F_N \geq 0$ and has total mass 2π. This positivity makes Fejer's theorem work: for continuous periodic f, $\sigma_N f$ converges uniformly to f.

This is a concrete lesson about convergence methods. The same Fourier coefficients can be summed by raw partial sums or by averaged partial sums, and averaging changes the analytic behavior without changing the underlying frequency data.

§117.10 Hilbert-space viewpoint

The conceptual endpoint is that Fourier series is linear algebra in the Hilbert space $L^2[-\pi, \pi]$. The inner product is

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x)\, dx.$$

The trigonometric system is an orthogonal basis in an appropriate sense. Fourier coefficients are coordinates, Parseval's identity is the Pythagorean theorem, and convergence questions ask which topology of function space is being used.

§117.11 Fourier series in Hilbert space

Let

$$L^2[-\pi, \pi]$$

be the space of square-integrable functions with inner product

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)\overline{g(x)}\, dx.$$

The functions

$$e_n(x) = \frac{1}{\sqrt{2\pi}} e^{inx}, \qquad n \in \mathbb{Z},$$

form an orthonormal basis in the Hilbert-space sense. The Fourier coefficient is

$$\widehat{f}(n) = \langle f, e_n\rangle.$$

Thus Fourier series is orthogonal projection onto an infinite orthonormal basis.

§117.12 Parseval identity

For $f \in L^2[-\pi, \pi]$, Parseval's identity says

$$\|f\|_2^2 = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2.$$

This is the Pythagorean theorem in infinite dimensions. In real sine-cosine notation, it becomes a statement relating integrals of f^2 to sums of squares of Fourier coefficients.

The elementary coefficient computations in the note are therefore coordinates of a vector in Hilbert space.

§117.13 Gibbs phenomenon

At jump discontinuities, Fourier series converges to the midpoint of the jump, but partial sums overshoot near the discontinuity. The overshoot does not vanish in height as the number of terms grows; it only concentrates closer to the jump. This is the Gibbs phenomenon.

This explains why Fourier series can converge correctly pointwise while still showing visible oscillations near discontinuities. It is a warning that convergence mode matters: pointwise, uniform, and L^2 convergence behave differently.

§117.14 Sturm–Liouville and Fourier bases

The functions $\sin nx$ and $\cos nx$ are eigenfunctions of

$$-\frac{d^2}{dx^2}.$$

For example,

$$-\frac{d^2}{dx^2} \sin nx = n^2 \sin nx.$$

With suitable boundary conditions, this operator is self-adjoint, and its eigenfunctions form an orthogonal basis.

Fourier series is therefore one example of Sturm–Liouville expansion. Other boundary conditions and intervals produce other orthogonal systems.

§117.15 Heat and wave equations

For the heat equation on an interval,

$$u_t = u_{xx},$$

expand

$$u(x,t) = \sum_n a_n(t) \sin nx.$$

Then each coefficient satisfies

$$a_n'(t) = -n^2 a_n(t), \qquad a_n(t) = a_n(0)e^{-n^2 t}.$$

High-frequency modes decay faster. For the wave equation,

$$u_{tt} = u_{xx},$$

Fourier modes oscillate in time.

Thus Fourier series is not only a representation of functions; it diagonalizes differential operators and solves PDEs.

§117.16 Convergence modes of Fourier series

Fourier series can converge in different senses. For $f \in L^2$, partial sums converge to f in the L^2 norm:

$$\|S_N f - f\|_2 \to 0.$$

For smoother functions, convergence may be pointwise or uniform. At jumps, Dirichlet's theorem gives midpoint convergence, while Gibbs oscillation persists near the discontinuity.

This distinction is essential. A Fourier series may be excellent for energy approximation while not uniformly accurate near jumps. Thus the elementary endpoint rules and half-range expansions should always be paired with a statement of the convergence mode.

§117.17 Smoothness is decay of coefficients

Fourier coefficients measure regularity. If f is smooth, repeated integration by parts gives decay. For example, in complex notation,

$$\widehat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx}\, dx.$$

If f is C^1 and periodic with matching endpoint values, then

$$\widehat{f}(n) = \frac{1}{in} \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) e^{-inx}\, dx,$$

so $|\widehat{f}(n)| = O(1/|n|)$ if f' is integrable. More derivatives give faster decay.

This explains the difference between the examples above. The square wave has jumps, so coefficients decay like $1/n$. The function $|x|$ is continuous but has a corner, so its nonzero coefficients decay like $1/n^2$. Smoothness in physical space becomes decay in frequency space.

§117.18 Sobolev norm from Fourier coefficients

On the circle, Sobolev spaces can be described by Fourier coefficients:

$$\|f\|_{H^s}^2 = \sum_{n\in\mathbb{Z}} (1+n^2)^s |\widehat{f}(n)|^2.$$

Large positive s demands fast coefficient decay and therefore smoothness. Negative s permits rough objects, even distributions.

This is the precise functional-analytic version of the heuristic: high frequencies measure oscillation and roughness. Fourier series turns differential regularity into weighted square summability.

§117.19 Fourier transform as the non-periodic limit

Fourier series decomposes periodic functions into discrete frequencies. On the real line, the Fourier transform decomposes non-periodic functions into a continuum of frequencies:

$$\widehat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-ix\xi}\,dx.$$

The inverse transform reconstructs f under suitable hypotheses. The derivative becomes multiplication by frequency:

$$\widehat{f'}(\xi) = i\xi\widehat{f}(\xi).$$

This is why Fourier methods solve constant-coefficient differential equations: they diagonalize differentiation.

§117.20 Distributions and weak Fourier series

Some important objects, such as the Dirac comb

$$\sum_{k\in\mathbb{Z}} \delta(x-2\pi k),$$

are not ordinary functions but distributions. Fourier analysis extends to them by duality against test functions. This viewpoint explains why formal series sometimes work beyond classical convergence: they are meaningful in a weak or distributional sense.

First-Order and Higher-Order Differential Equations

N-20260618

§118.1 Why the differential-equation part is one note

The last part of the original long Markdown note covers first-order equations and higher-order linear equations. These belong together because the central idea is the same: transform an equation for an unknown function into a form that can be integrated, linearized, or solved as an operator equation.

§118.2 Basic concepts

A differential equation relates an unknown function to its derivatives. The order is the highest derivative appearing. A general first-order equation has the form

$$F(x, y, y') = 0.$$

A solution is a function $y(x)$ satisfying the equation on an interval. A general solution contains arbitrary constants; an initial condition selects a particular solution.

§118.3 Separable equations

A separable equation has

$$\frac{dy}{dx} = g(x)h(y).$$

If $h(y) \neq 0$, then

$$\frac{dy}{h(y)} = g(x)\,dx.$$

Integrating gives an implicit solution. One must also check equilibrium solutions coming from $h(y) = 0$, because division by $h(y)$ can lose them.

§118.4 Homogeneous first-order equations

A homogeneous first-order equation has the form

$$y' = F\left(\frac{y}{x}\right).$$

Set $v = y/x$, so $y = vx$ and

$$y' = v + xv'.$$

Then

$$v + x\frac{dv}{dx} = F(v),$$

which is separable:

$$\frac{dv}{F(v) - v} = \frac{dx}{x}.$$

Some equations become homogeneous after translating the origin. If the linear expressions in x, y meet at a point, shift that point to the origin.

§118.5 First-order linear equations

A first-order linear equation is

$$y' + p(x)y = q(x).$$

The integrating factor is

$$\mu(x) = e^{\int p(x)\,dx}.$$

Multiplying the equation by μ gives

$$(\mu y)' = \mu q.$$

Thus

$$y = e^{-\int p}\left(\int q e^{\int p}\,dx + C\right).$$

The integrating factor is not a trick; it conjugates the operator $D+p(x)$ into the ordinary derivative D.

§118.6 Bernoulli equations

A Bernoulli equation has

$$y' + p(x)y = q(x)y^n.$$

For $n \neq 0, 1$, set

$$u = y^{1-n}.$$

Then

$$u' = (1-n)y^{-n}y',$$

and the equation becomes linear in u. Again, check special solutions lost by division.

§118.7 Exact equations

An equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is exact if there is a potential Φ with

$$d\Phi = M\,dx + N\,dy.$$

The test is

$$M_y = N_x$$

on a suitable simply connected region. Then the solution is

$$\Phi(x,y) = C.$$

If the equation is not exact, one may look for an integrating factor, often depending only on x or only on y.

§118.8 Reducible higher-order equations

If a second-order equation does not explicitly contain y, set $p = y'$ and reduce the order. If it does not explicitly contain x, set $p = p(y) = y'$, so

$$y'' = \frac{dp}{dx} = \frac{dp}{dy}\frac{dy}{dx} = p\frac{dp}{dy}.$$

These substitutions are not random: they use a missing variable as a symmetry.

§118.9 Second-order linear equations

A second-order linear equation has

$$y'' + p(x)y' + q(x)y = f(x).$$

The associated homogeneous equation has a solution space. If y_1, y_2 are independent homogeneous solutions, then

$$y_h = C_1y_1 + C_2y_2.$$

A general nonhomogeneous solution is

$$y = y_h + y_p.$$

This is the affine-space principle: one particular solution plus the kernel of a linear operator.

The Wronskian

$$W(y_1, y_2) = y_1y_2' - y_1'y_2$$

measures independence. If it is nonzero at one point for solutions of a linear homogeneous equation, the two solutions form a fundamental system.

§118.10 Constant-coefficient homogeneous equations

For

$$y'' + ay' + by = 0,$$

try $y = e^{rx}$. The characteristic equation is

$$r^2 + ar + b = 0.$$

Distinct real roots r_1, r_2 give

$$y = C_1e^{r_1x} + C_2e^{r_2x}.$$

A repeated root r gives

$$y = (C_1 + C_2x)e^{rx}.$$

Complex roots $\alpha \pm i\beta$ give

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x).$$

§118.11 Constant-coefficient nonhomogeneous equations

For
$$L[y] = f(x), \qquad L = D^2 + aD + b,$$
the method of undetermined coefficients works when f is built from polynomials, exponentials, sines, and cosines. If the trial function overlaps with a homogeneous solution, multiply by a sufficient power of x.

Variation of parameters works more generally. If y_1, y_2 are fundamental solutions, seek
$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x).$$
The auxiliary condition $u_1'y_1 + u_2'y_2 = 0$ produces a solvable linear system for u_1', u_2'.

§118.12 Initial-value problems

An initial-value problem specifies
$$y(x_0) = a, \qquad y'(x_0) = b.$$
For regular linear equations, existence and uniqueness say that these data select exactly one solution. This is conceptually important: a second-order equation needs two pieces of initial data because its solution space is two-dimensional.

§118.13 Functional-analytic viewpoint

Let
$$L[y] = y'' + p(x)y' + q(x)y.$$
Then a differential equation is an operator equation
$$L[y] = f.$$
The homogeneous solutions form $\ker L$. A nonhomogeneous solution set, if nonempty, is an affine translate of $\ker L$. Initial conditions are linear functionals that choose one element from that affine space.

This viewpoint explains why linear differential equations behave like linear algebra in function space. The derivative operator replaces a matrix; the Wronskian replaces a determinant; fundamental solutions replace a basis; Green functions, when introduced later, are kernels for inverse operators.

§118.14 Differential equations as operators

The elementary note classifies equations by solvable forms: separable, homogeneous, linear, Bernoulli, exact, constant-coefficient. The higher-level unity is that a differential equation is an equation in a function space. For example,
$$L[y] = f, \qquad L = \frac{d^2}{dx^2} + p(x)\frac{d}{dx} + q(x).$$
The homogeneous solutions form the kernel $\ker L$. A nonhomogeneous solution set, when nonempty, is an affine translate
$$y_p + \ker L.$$
This is the same algebraic structure as a linear system $Ax = b$.

§118.15 Green functions

For a linear operator L, a Green function $G(x,\xi)$ is a kernel that represents the inverse problem. Formally,

$$L_x G(x,\xi) = \delta(x-\xi),$$

where δ is the Dirac delta distribution. Then a solution of

$$L[y] = f$$

can often be written as

$$y(x) = \int G(x,\xi) f(\xi)\, d\xi$$

with boundary conditions built into G.

This connects elementary particular-solution methods to integral operators. Undetermined coefficients works for special forcing terms; Green functions describe the response to arbitrary forcing as a superposition of responses to point sources.

§118.16 Semigroups and evolution

For first-order evolution equations, one often writes

$$\frac{du}{dt} = Au.$$

If A were a number, the solution would be $u(t) = e^{tA}u(0)$. In a Banach or Hilbert space, A may be an unbounded operator, and the solution is described by a semigroup

$$T(t) = e^{tA}, \qquad T(t+s) = T(t)T(s).$$

The heat equation is the model example: the Laplacian generates a smoothing semigroup. The elementary exponential solution of constant-coefficient ODEs is the finite-dimensional shadow of this theory.

§118.17 Sturm–Liouville and Fourier expansion

A second-order operator of the form

$$L[y] = -(p(x)y')' + q(x)y$$

with suitable boundary conditions is often self-adjoint. Its eigenvalue problem

$$L[y] = \lambda w(x) y$$

produces orthogonal eigenfunctions. Fourier sine and cosine series are special cases of this principle.

Thus the elementary method of solving $y'' + \lambda y = 0$ under boundary conditions is not an isolated trick. It is the beginning of spectral theory for differential operators.

First-Order Partial Differential Equations and Characteristics

N-20260619

§119.1 What a first-order PDE asks for

A partial differential equation is an equation for an unknown function of several variables and its partial derivatives. A first-order scalar equation for $u = u(x, t)$ has the schematic form

$$F(x, t, u, u_x, u_t) = 0.$$

The simplest useful class is the linear transport equation

$$u_t + au_x = b(x, t),$$

where a is a constant. It is called a transport equation because it moves the initial profile without changing its shape, except for the forcing term b.

The geometric method for first-order equations is the method of characteristics. Instead of trying to understand u on the whole (x, t)-plane at once, one looks for curves along which the PDE becomes an ordinary differential equation.

Definition 119.1.1 (Characteristic curves)**.** For the constant-coefficient transport equation

$$u_t + au_x = b(x, t),$$

a characteristic is a curve $x = x(t)$ satisfying

$$\frac{dx}{dt} = a.$$

Along such a curve,

$$\frac{d}{dt}u(x(t), t) = u_x(x(t), t)x'(t) + u_t(x(t), t) = au_x + u_t = b(x(t), t).$$

Thus a first-order PDE has been replaced by two ordinary differential equations:

$$\frac{dx}{dt} = a, \qquad \frac{du}{dt} = b(x(t), t).$$

The first equation describes the characteristic line, and the second describes the value of the solution along that line.

§119.2 The unforced transport equation

Consider

$$u_t - cu_x = 0.$$

This is the sign convention used by the first-order wave example in the original notes. It has the form $u_t + au_x = 0$ with $a = -c$, so the characteristics satisfy

$$\frac{dx}{dt} = -c.$$

Hence

$$x(t) = x_0 - ct, \qquad x_0 = x + ct.$$

Along each such line,

$$\frac{d}{dt}u(x(t), t) = u_t - cu_x = 0.$$

Therefore u is constant on the line $x + ct = x_0$.

Theorem 119.2.1 (Solution of the constant-speed transport equation)
Let f be a differentiable function. The solution of

$$u_t - cu_x = 0, \qquad u(x, 0) = f(x)$$

is

$$u(x, t) = f(x + ct).$$

Proof. The initial point of the characteristic through (x, t) is $(x + ct, 0)$. Since u is constant along the characteristic,

$$u(x, t) = u(x + ct, 0) = f(x + ct).$$

A direct check gives

$$u_t = cf'(x + ct), \qquad u_x = f'(x + ct),$$

so $u_t - cu_x = 0$. □

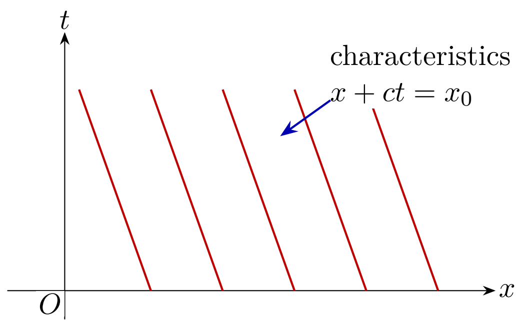


Figure 119.1: Characteristics for $u_t - cu_x = 0$. The solution is constant along each red line and the initial profile moves to the left as t increases.

§119.3 Example: translating a parabola

Let

$$u_t - cu_x = 0, \qquad u(x, 0) = x^2 - 3.$$

The theorem gives

$$u(x, t) = (x + ct)^2 - 3.$$

For each fixed time t, the graph in the (x, u)-plane is the same parabola shifted left by ct.

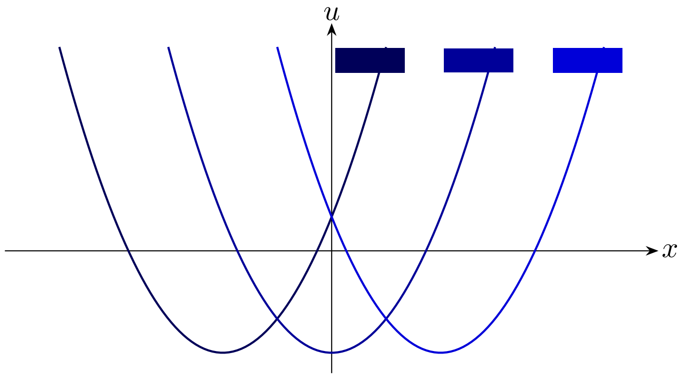


Figure 119.2: Time slices of $u(x,t) = (x+ct)^2 - 3$, shown as translated copies of the initial profile.

§119.4 Forced transport

Now consider

$$u_t + au_x = b(x,t), \qquad u(x,0) = f(x).$$

The characteristic through (x,t) has the form

$$x(s) = x - a(t-s), \qquad 0 \le s \le t,$$

because $x(t) = x$. Along this path,

$$\frac{d}{ds}u(x(s),s) = b(x(s),s).$$

Integrating from 0 to t gives the formula

$$u(x,t) = f(x-at) + \int_0^t b(x-a(t-s),s)\,ds.$$

Example 119.4.1 (A forced first-order wave equation)
Solve

$$u_t + 5u_x = e^{-t}, \qquad u(x,0) = e^{-x^2}.$$

Here $a = 5$, so

$$x(s) = x - 5(t-s).$$

The forcing depends only on time, and therefore

$$\int_0^t e^{-s}\,ds = 1 - e^{-t}.$$

The initial point is $x - 5t$. Hence

$$u(x,t) = e^{-(x-5t)^2} + 1 - e^{-t}.$$

Equivalently, if one writes the characteristic label as $x_0 = x - 5t$, then $u = f(x_0) + 1 - e^{-t}$. The sign of the last term is fixed by differentiating the proposed solution back into the PDE.

§119.5 What can go wrong

For first-order equations, the initial curve must meet the characteristic curves transversely. If the data are prescribed on a characteristic, then the PDE may fail to determine a unique solution. For constant-speed transport, prescribing $u(x,0) = f(x)$ is safe because the initial line $t = 0$ intersects each characteristic once. This is the first place where PDEs differ from ordinary differential equations: the geometry of the set on which data are given is part of the problem.

§119.6 General linear first-order equations

The constant-coefficient equation is only the first model. A more general linear first-order equation is

$$a(x,t)u_x + b(x,t)u_t = c(x,t)u + d(x,t).$$

The characteristic curves now satisfy

$$\frac{dx}{ds} = a(x,t), \qquad \frac{dt}{ds} = b(x,t).$$

Along such a curve, the value of u satisfies

$$\frac{du}{ds} = c(x(s),t(s))u + d(x(s),t(s)).$$

Thus even when the coefficients vary, the method is still the same: solve a characteristic system in the independent variables, then solve an ordinary differential equation for the dependent variable along the characteristic.

Example 119.6.1 (Variable speed transport)
Consider

$$u_t + xu_x = 0, \qquad u(x,0) = f(x).$$

The characteristics solve

$$\frac{dx}{dt} = x,$$

so $x(t) = x_0 e^t$. The initial footpoint through (x,t) is therefore $x_0 = xe^{-t}$. Since u is constant along characteristics,

$$u(x,t) = f(xe^{-t}).$$

A direct check gives

$$u_t = -xe^{-t}f'(xe^{-t}), \qquad xu_x = xe^{-t}f'(xe^{-t}),$$

and hence $u_t + xu_x = 0$.

§119.7 Quasilinear equations

A first-order equation is called quasilinear if it is linear in the derivatives but the coefficients may depend on u:

$$a(x,t,u)u_x + b(x,t,u)u_t = c(x,t,u).$$

The characteristic system is now

$$\frac{dx}{ds} = a(x,t,u), \qquad \frac{dt}{ds} = b(x,t,u), \qquad \frac{du}{ds} = c(x,t,u).$$

This is already a nonlinear system of ordinary differential equations. In this sense, first-order nonlinear PDEs are not an entirely new object: they are families of coupled ODEs whose curves fill the plane.

Example 119.7.1 (Inviscid Burgers equation)
The equation

$$u_t + uu_x = 0$$

has characteristic speed u itself. Along a characteristic,

$$\frac{du}{dt} = u_t + uu_x = 0,$$

so u is constant. If $u(x,0) = f(x)$, then the characteristic starting from x_0 has

$$x = x_0 + f(x_0)t, \qquad u = f(x_0).$$

Thus the implicit solution is obtained from

$$x_0 = x - ut, \qquad u = f(x - ut).$$

This formula is useful only as long as the map $x_0 \mapsto x_0 + f(x_0)t$ remains one-to-one.

§119.8 Crossing characteristics and shocks

For linear transport with constant speed, characteristics never cross. For nonlinear equations such as Burgers' equation, faster characteristics can overtake slower ones. When this happens, the classical solution becomes multi-valued if one continues the characteristic construction naively.

Differentiate the characteristic map for Burgers' equation:

$$x = x_0 + f(x_0)t.$$

The map stops being locally invertible when

$$\frac{\partial x}{\partial x_0} = 1 + f'(x_0)t = 0.$$

Thus if $f'(x_0) < 0$ somewhere, characteristic crossing occurs at time

$$t_* = -\frac{1}{\min f'(x_0)}$$

provided the minimum is negative. This is the first appearance of shock formation.

In an introductory PDE course, one often stops here and says that classical differentiable solutions break down. A more advanced treatment replaces them by weak solutions and adds an entropy condition to select the physically relevant shock. The point for this note is more elementary: first-order PDEs are controlled by the geometry of characteristic curves, and loss of uniqueness in that geometry is visible before one writes any distribution theory.

§119.9 Initial curves and well-posedness

Let initial data be prescribed on a curve Γ in the (x,t)-plane. For the characteristic method to determine a solution near Γ, the curve must not be tangent to the characteristic direction. For

$$a(x,t)u_x + b(x,t)u_t = c(x,t,u),$$

the characteristic direction in the (x,t)-plane is (a,b). If Γ has tangent vector τ, then the non-characteristic condition is

$$\tau \not\parallel (a,b).$$

Equivalently, the characteristic curves must cross the data curve rather than slide along it.

Example 119.9.1 (Bad data curve)

For $u_t + u_x = 0$, the characteristics are lines $x - t = \text{const.}$. If data are prescribed on the line $x - t = 0$, then the data are given along a characteristic. The equation says the solution is constant on that line, but it does not determine what happens on neighboring characteristic lines. Therefore the problem is not locally well-posed from that data curve.

§119.10 Conservation-law viewpoint

Some first-order equations are written in conservation form:

$$u_t + \partial_x F(u) = 0.$$

For smooth solutions, this is the same as

$$u_t + F'(u)u_x = 0.$$

The characteristic speed is $F'(u)$. Burgers' equation corresponds to $F(u) = u^2/2$. Conservation form becomes important when shocks appear, because it still has meaning after differentiability is lost.

Integrating over an interval $[a,b]$, we obtain

$$\frac{d}{dt}\int_a^b u(x,t)\,dx = F(u(a,t)) - F(u(b,t)).$$

Thus the amount of u in the interval changes only by flux through the boundary. This is the conceptual reason that conservation laws are the natural nonlinear continuation of transport equations.

The Second-Order Wave Equation and d'Alembert's Formula

N-20260620

§120.1 The equation and its physical meaning

The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx},$$

where $u(x, t)$ is the vertical displacement of a string and $c > 0$ is the wave speed. The equation says that acceleration is proportional to curvature. This is physically reasonable: translating the whole string upward produces no restoring force, and a straight slanted string has no net restoring force; only bending produces acceleration.

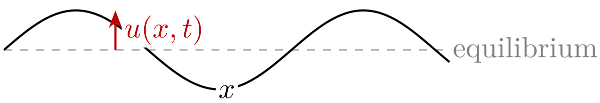


Figure 120.1: A vibrating string. The vertical displacement $u(x, t)$ is measured from the equilibrium line.

The equation is called hyperbolic because its highest-order part has the sign pattern of $\partial_t^2 - c^2\partial_x^2$. This name is about the algebraic type of the differential operator, not about hyperbolas as curves.

§120.2 Factorization into transport operators

The wave operator factors as

$$\partial_t^2 - c^2\partial_x^2 = (\partial_t + c\partial_x)(\partial_t - c\partial_x).$$

Since the two constant-coefficient operators commute, the wave equation can be read as a pair of first-order transport equations. Any function of $x + ct$ solves the equation, and any function of $x - ct$ also solves it.

Theorem 120.2.1 (d'Alembert form)
Every twice differentiable solution of

$$u_{tt} = c^2 u_{xx}$$

on a simply connected region in the (x, t)-plane can locally be written as

$$u(x, t) = F(x + ct) + G(x - ct).$$

The term $F(x + ct)$ travels left, and $G(x - ct)$ travels right.

Proof. Set

$$\xi = x + ct, \qquad \eta = x - ct.$$

By the chain rule,

$$\partial_x = \partial_\xi + \partial_\eta, \qquad \partial_t = c\partial_\xi - c\partial_\eta.$$

A direct calculation gives

$$u_{tt} - c^2 u_{xx} = -4c^2 u_{\xi\eta}.$$

Thus the wave equation is $u_{\xi\eta} = 0$. Integrating once in η gives $u_\xi = A(\xi)$, and integrating in ξ gives $u = F(\xi) + G(\eta)$. □

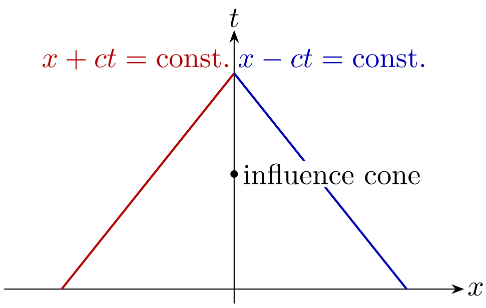


Figure 120.2: The two characteristic families of the wave equation. A point receives information from both left-moving and right-moving characteristic directions.

§120.3 Initial-value problem on the whole line

Suppose

$$u(x,0) = \phi(x), \qquad u_t(x,0) = \psi(x).$$

From $u = F(x+ct) + G(x-ct)$, the first condition gives

$$F(x) + G(x) = \phi(x).$$

The second condition gives

$$cF'(x) - cG'(x) = \psi(x).$$

Solving these two equations yields the d'Alembert formula

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)\, ds.$$

This formula shows finite propagation speed: the value at (x,t) depends only on the initial data in the interval $[x-ct, x+ct]$.

Example 120.3.1 (Zero initial velocity)
Let

$$\phi(x) = \frac{1}{1+x^2}, \qquad \psi(x) = 0.$$

Then

$$u(x,t) = \frac{1}{2}\left[\frac{1}{1+(x+ct)^2} + \frac{1}{1+(x-ct)^2}\right].$$

The initial bump splits into two half-sized bumps travelling in opposite directions.

§120.4 Finite strings and Fourier series

For a string fixed at $x = 0$ and $x = L$, the boundary conditions are

$$u(0,t) = u(L,t) = 0.$$

The spatial eigenfunctions satisfying these boundary conditions are

$$\sin \frac{n\pi x}{L}, \qquad n = 1, 2, 3, \ldots.$$

Separation of variables gives normal modes

$$u_n(x,t) = \sin \frac{n\pi x}{L} \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right).$$

A general solution is a Fourier sine series built from these modes:

$$u(x,t) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{L} \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right).$$

This is the first bridge from the Fourier-series note to PDEs: boundary conditions choose the spatial basis, while the PDE determines the time oscillation of each coefficient.

§120.5 Energy conservation

The wave equation has a conserved energy. On the whole line, assume u and its derivatives decay sufficiently fast as $|x| \to \infty$. Define

$$E(t) = \frac{1}{2} \int_{-\infty}^{\infty} \left(u_t(x,t)^2 + c^2 u_x(x,t)^2 \right) dx.$$

Then

$$\frac{dE}{dt} = \int_{-\infty}^{\infty} (u_t u_{tt} + c^2 u_x u_{xt})\, dx.$$

Using $u_{tt} = c^2 u_{xx}$,

$$\frac{dE}{dt} = c^2 \int_{-\infty}^{\infty} (u_t u_{xx} + u_x u_{xt})\, dx.$$

Integrating the first term by parts gives

$$\int u_t u_{xx}\, dx = -\int u_{tx} u_x\, dx,$$

so the two terms cancel and $E'(t) = 0$.

Theorem 120.5.1 (Uniqueness by energy)

For the wave equation on the whole line with sufficiently decaying smooth solutions, the initial data $u(x,0)$ and $u_t(x,0)$ determine at most one solution.

Proof. If u and v are two solutions with the same initial data, set $w = u - v$. Then w solves the homogeneous wave equation with zero initial displacement and zero initial velocity. Its energy at $t = 0$ is zero. Since energy is conserved, $E_w(t) = 0$ for all t. Hence $w_t = 0$ and $w_x = 0$, so w is constant in space and time. The initial data give $w = 0$. □

Energy is often more robust than an explicit formula. On domains where d'Alembert's formula is unavailable, energy still gives uniqueness and stability.

§120.6 Inhomogeneous wave equation and Duhamel's principle

Consider

$$u_{tt} - c^2 u_{xx} = F(x,t), \qquad u(x,0) = 0, \qquad u_t(x,0) = 0.$$

The forcing F can be viewed as continuously injecting infinitesimal initial velocities. Duhamel's principle gives

$$u(x,t) = \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} F(\xi,s)\,d\xi\,ds.$$

This formula is the forced analogue of d'Alembert's formula. The triangular integration region is the past light cone of (x,t).

Example 120.6.1 (Localized forcing)

If F is supported near a point (x_*, s_*), then it can influence (x,t) only if

$$|x - x_*| \le c(t - s_*), \qquad s_* \le t.$$

This is finite propagation speed in another form. Information does not instantly appear everywhere; it travels along the characteristic cone.

§120.7 Boundary conditions on a finite interval

For a finite string $0 < x < L$, the physical boundary condition depends on how the endpoint is constrained.

- Fixed endpoints give Dirichlet conditions:

$$u(0,t) = u(L,t) = 0.$$

- Free endpoints give Neumann conditions:

$$u_x(0,t) = u_x(L,t) = 0.$$

- A damped or elastic endpoint can lead to mixed conditions such as

$$\alpha u + \beta u_x = 0$$

  at an endpoint.

The boundary condition selects the spatial eigenfunctions. Dirichlet conditions produce sine modes, while Neumann conditions produce cosine modes. This explains why half-range Fourier expansions are not only computational tricks: they encode boundary behavior.

§120.8 Deriving the Fourier coefficients

For the fixed string, write

$$u(x,t) = \sum_{n=1}^{\infty} q_n(t) \sin\frac{n\pi x}{L}.$$

Substituting into $u_{tt} = c^2 u_{xx}$ gives

$$q_n''(t) + \left(\frac{n\pi c}{L}\right)^2 q_n(t) = 0.$$

Hence

$$q_n(t) = A_n \cos\frac{n\pi ct}{L} + B_n \sin\frac{n\pi ct}{L}.$$

If

$$u(x,0) = \phi(x), \qquad u_t(x,0) = \psi(x),$$

then

$$A_n = \frac{2}{L}\int_0^L \phi(x)\sin\frac{n\pi x}{L}\,dx,$$

and

$$\frac{n\pi c}{L}B_n = \frac{2}{L}\int_0^L \psi(x)\sin\frac{n\pi x}{L}\,dx.$$

Thus

$$B_n = \frac{2}{n\pi c}\int_0^L \psi(x)\sin\frac{n\pi x}{L}\,dx.$$

The role of orthogonality is explicit: it decouples one PDE into infinitely many harmonic oscillators.

§120.9 Standing waves and travelling waves

The functions

$$\sin\frac{n\pi x}{L}\cos\frac{n\pi ct}{L}$$

are standing waves. Their nodes are fixed in space. By contrast, d'Alembert's formula writes the solution as travelling waves. These are not contradictory descriptions. A standing wave is the superposition of a left-moving and a right-moving travelling wave of the same frequency.

For example,

$$\sin(kx)\cos(kct) = \frac{1}{2}\left[\sin(k(x+ct)) + \sin(k(x-ct))\right].$$

The Fourier description is best adapted to bounded intervals and boundary conditions; the travelling-wave description is best adapted to the whole line and propagation.

§120.10 The wave equation with damping

A simple damped string model is

$$u_{tt} + \gamma u_t = c^2 u_{xx}, \qquad \gamma > 0.$$

The energy is no longer conserved. Multiplying by u_t and integrating gives

$$\frac{dE}{dt} = -\gamma\int u_t^2\,dx \le 0.$$

Thus damping removes kinetic energy. This example shows how a small change in the PDE changes the qualitative behavior: the undamped wave equation transports energy, while the damped wave equation dissipates it.

The Heat Equation, Diffusion, and Similarity Solutions

N-20260621

§121.1 Diffusion versus wave motion

The one-dimensional heat equation is

$$T_t = \kappa T_{xx}, \qquad \kappa > 0.$$

Here $T(x,t)$ is temperature and κ is the thermal diffusivity. The equation says that the rate of change of temperature is proportional to curvature in the temperature profile. Peaks have negative curvature and decay; troughs have positive curvature and rise. Unlike the wave equation, diffusion is dissipative rather than oscillatory.

The heat equation is parabolic. One practical consequence is infinite propagation speed in the ideal model: a localized heat distribution immediately becomes nonzero everywhere, although it may be extremely small far away.

§121.2 Heat kernel on the whole line

The fundamental solution on the real line is the heat kernel

$$G(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{x^2}{4\kappa t}\right), \qquad t > 0.$$

If $T(x,0) = f(x)$, then the solution is the convolution

$$T(x,t) = (G(\cdot,t) * f)(x) = \int_{-\infty}^{\infty} G(x-s,t) f(s)\, ds.$$

The kernel has total mass one:

$$\int_{-\infty}^{\infty} G(x,t)\, dx = 1,$$

so heat is redistributed rather than created. The width of the Gaussian is of order $\sqrt{\kappa t}$, which is the natural diffusion length scale.

§121.3 Similarity solution on a semi-infinite rod

Consider a semi-infinite rod $x > 0$ initially at temperature zero, with the end held at temperature one after $t = 0$:

$$T(x,0) = 0, \qquad T(0,t) = 1, \qquad T(x,t) \to 0 \quad (x \to \infty).$$

The heat equation is invariant under the scaling

$$x \mapsto \lambda x, \qquad t \mapsto \lambda^2 t.$$

This suggests the similarity variable

$$\eta = \frac{x}{2\sqrt{\kappa t}}, \qquad T(x,t) = \theta(\eta).$$

The derivatives are

$$T_x = \frac{1}{2\sqrt{\kappa t}}\theta'(\eta), \qquad T_{xx} = \frac{1}{4\kappa t}\theta''(\eta), \qquad T_t = -\frac{\eta}{2t}\theta'(\eta).$$

Substitution into $T_t = \kappa T_{xx}$ gives

$$\theta'' + 2\eta\theta' = 0.$$

Let $v = \theta'$. Then

$$v' + 2\eta v = 0, \qquad (e^{\eta^2}v)' = 0,$$

so

$$\theta' = Ae^{-\eta^2}, \qquad \theta = A\int_0^\eta e^{-s^2}\,ds + B.$$

Using

$$\operatorname{erf}(\eta) = \frac{2}{\sqrt{\pi}}\int_0^\eta e^{-s^2}\,ds, \qquad \operatorname{erfc}(\eta) = 1 - \operatorname{erf}(\eta),$$

the boundary conditions give

$$T(x,t) = \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right).$$

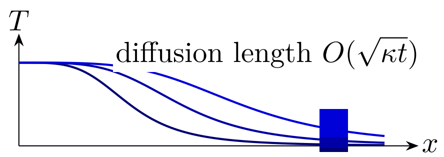


Figure 121.1: A qualitative picture of the semi-infinite rod solution. As time grows, the transition layer spreads over length scale $\sqrt{\kappa t}$.

§121.4 Fourier series for a finite rod

For a finite rod $0 < x < L$ with fixed zero temperature at the endpoints,

$$T(0,t) = T(L,t) = 0,$$

separation of variables gives

$$T(x,t) = X(x)A(t).$$

Substitution yields

$$\frac{A'}{\kappa A} = \frac{X''}{X} = -\lambda.$$

The boundary value problem

$$X'' + \lambda X = 0, \qquad X(0) = X(L) = 0$$

has eigenfunctions

$$X_n(x) = \sin\frac{n\pi x}{L}, \qquad \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

Thus each Fourier mode decays exponentially:

$$T(x,t) = \sum_{n=1}^{\infty} b_n e^{-\kappa(n\pi/L)^2 t} \sin\frac{n\pi x}{L}.$$

High-frequency modes have larger n^2 and therefore decay faster. This is the analytic meaning of smoothing by the heat equation.

§121.5 Maximum principle

A basic qualitative theorem for the heat equation is the maximum principle.

Theorem 121.5.1 (Maximum principle, informal form)

For a smooth solution of $T_t = \kappa T_{xx}$ on a closed space-time rectangle, the maximum and minimum occur either at the initial time or on the spatial boundary, unless the solution is constant.

This theorem matches the physical intuition that heat does not create a new interior temperature peak. It also explains why the heat equation behaves differently from the wave equation: diffusion forgets fine details, while waves transport and superpose them.

§121.6 Energy decay on an interval

For the heat equation on $0 < x < L$ with zero boundary values, define the thermal energy norm

$$\|T(\cdot,t)\|_{L^2}^2 = \int_0^L T(x,t)^2\,dx.$$

Then

$$\frac{d}{dt}\frac{1}{2}\int_0^L T^2\,dx = \int_0^L TT_t\,dx = \kappa\int_0^L TT_{xx}\,dx.$$

Integrating by parts and using $T(0,t) = T(L,t) = 0$,

$$\frac{d}{dt}\frac{1}{2}\int_0^L T^2\,dx = -\kappa\int_0^L T_x^2\,dx \le 0.$$

Thus the L^2-size of the temperature distribution decreases in time. This is the heat-equation analogue of the energy identity for the wave equation, but with a crucial difference: wave energy is conserved, while heat energy decays.

§121.7 The maximum principle with proof idea

The maximum principle says that heat cannot create a new interior maximum. Here is the local calculation behind it. Suppose T has a strict interior maximum at (x_0, t_0) with $t_0 > 0$. At that point,

$$T_x(x_0,t_0) = 0, \qquad T_{xx}(x_0,t_0) \le 0.$$

The heat equation gives

$$T_t(x_0,t_0) = \kappa T_{xx}(x_0,t_0) \le 0.$$

Thus the value cannot be increasing at an interior spatial maximum. A rigorous proof perturbs the function by a small term such as $-\varepsilon t$ to rule out equality cases.

Theorem 121.7.1 (Uniqueness for the heat equation)
For the heat equation on a bounded interval with fixed boundary and initial data, there is at most one smooth solution.

Proof. Let T_1 and T_2 be two solutions and set $w = T_1 - T_2$. Then $w_t = \kappa w_{xx}$ with zero initial data and zero boundary data. By the maximum principle, $w \le 0$. Applying the same argument to $-w$ gives $w \ge 0$. Hence $w = 0$. ☐

§121.8 Smoothing and the heat semigroup

On the whole line, the solution operator is convolution with the heat kernel:

$$T(\cdot, t) = G_t * f.$$

The family G_t satisfies the semigroup identity

$$G_t * G_s = G_{t+s}.$$

Therefore the solution operators satisfy

$$S(t)S(s) = S(t+s), \qquad S(t)f = G_t * f.$$

This is the infinite-dimensional analogue of the exponential law $e^{tA}e^{sA} = e^{(t+s)A}$. Formally,

$$S(t) = e^{t\kappa\partial_x^2}.$$

The heat equation is the basic example of a smoothing semigroup: even if f is rough, convolution with a Gaussian produces a smooth function for every $t > 0$.

§121.9 Fourier transform solution on the real line

Let

$$\widehat{T}(\xi, t) = \int_{-\infty}^{\infty} T(x, t)e^{-ix\xi}\, dx.$$

Taking the Fourier transform of $T_t = \kappa T_{xx}$ gives

$$\partial_t \widehat{T}(\xi, t) = -\kappa\xi^2 \widehat{T}(\xi, t).$$

For each fixed frequency ξ, this is an ordinary differential equation:

$$\widehat{T}(\xi, t) = e^{-\kappa\xi^2 t}\widehat{f}(\xi).$$

High frequencies are multiplied by a rapidly decaying factor. This is the frequency-space explanation of smoothing.

Inverting the Fourier transform gives the heat-kernel formula. Thus the Gaussian kernel, the semigroup law, and the Fourier multiplier $e^{-\kappa\xi^2 t}$ are three descriptions of the same solution operator.

§121.10 Boundary conditions and conservation

Boundary conditions change what is conserved.

For insulated endpoints, one uses Neumann conditions:

$$T_x(0,t) = T_x(L,t) = 0.$$

Then the total heat is conserved:

$$\frac{d}{dt}\int_0^L T(x,t)\,dx = \kappa\int_0^L T_{xx}\,dx = \kappa[T_x]_0^L = 0.$$

For fixed-temperature endpoints, total heat may enter or leave through the boundary. The PDE describes local diffusion; the boundary condition describes the external thermal reservoir.

§121.11 Steady states and long-time behavior

A steady state satisfies $T_t = 0$, so

$$T_{xx} = 0.$$

On an interval, the steady states are linear functions

$$T_\infty(x) = Ax + B.$$

If the endpoints are held at temperatures $T(0,t) = A_0$ and $T(L,t) = A_L$, then the steady state is

$$T_\infty(x) = A_0 + \frac{A_L - A_0}{L}x.$$

The transient part $T - T_\infty$ satisfies zero boundary conditions and decays by the Fourier sine expansion. This separates the problem into two pieces: first solve the static boundary problem, then solve a homogeneous heat equation for the decaying error.

Example 121.11.1 (Nonzero boundary temperatures)

Let $T(0,t) = 0$, $T(L,t) = 1$, and $T(x,0) = f(x)$. The steady state is x/L. Set

$$v(x,t) = T(x,t) - \frac{x}{L}.$$

Then v satisfies

$$v_t = \kappa v_{xx}, \qquad v(0,t) = v(L,t) = 0,$$

with initial data $v(x,0) = f(x) - x/L$. Hence

$$v(x,t) = \sum_{n=1}^{\infty} b_n e^{-\kappa(n\pi/L)^2 t}\sin\frac{n\pi x}{L},$$

where

$$b_n = \frac{2}{L}\int_0^L \left(f(x) - \frac{x}{L}\right)\sin\frac{n\pi x}{L}\,dx.$$

§121.12 Comparison with the wave equation

The heat and wave equations both use the second spatial derivative, but they use it in different ways:

$$T_t = \kappa T_{xx}, \qquad u_{tt} = c^2 u_{xx}.$$

For heat, curvature determines velocity; for waves, curvature determines acceleration. This is why heat smooths and decays, while waves oscillate and propagate. Fourier modes make the difference transparent:

$$e^{-\kappa n^2 t} \quad \text{versus} \quad \cos(nct),\ \sin(nct).$$

The same spatial eigenfunctions occur, but their time factors encode the qualitative nature of the PDE.

XII

Classical Geometry

Part XII: Contents

A Russian-Style Early Olympiad Game: Coordinates, Packing, and Symmetry

N-20220615

§122.1 What the note contains

This note consists of a short mock or translated olympiad-style problem sheet. The cover page imitates an elementary mathematics olympiad booklet and states the usual rules: a multiple-choice test, one correct answer per problem, no calculators, and figures not necessarily drawn to scale. The mathematically relevant part is a collection of four problems about coordinate geometry, optimization, packing, and symmetric equations.

§122.2 Problem 1: two distances under a linear constraint

In the coordinate plane let

$$A = (0, -2), \qquad B = (2, -2), \qquad M = (x, y), \qquad x + y = 1.$$

The problem asks for the maximum and minimum of

$$AM^2 + BM^2.$$

Since $y = 1 - x$,

$$\begin{aligned} AM^2 + BM^2 &= x^2 + (y+2)^2 + (x-2)^2 + (y+2)^2 \\ &= x^2 + (3-x)^2 + (x-2)^2 + (3-x)^2 \\ &= 4x^2 - 16x + 22 \\ &= 4(x-2)^2 + 6. \end{aligned}$$

Thus the minimum is 6, achieved at $x = 2, y = -1$. There is no maximum on the whole line $x + y = 1$, because the expression tends to infinity as $|x| \to \infty$.

Remark 122.2.1 — The page asks for both maximum and minimum. This is a useful trap: on an unbounded line, a positive quadratic has a minimum but no maximum.

§122.3 Problem 2: parabola and distance

Let $M = (x, y)$ lie on the parabola

$$y = ax^2 + bx, \qquad a < 0, \quad b > 0.$$

The quantity to maximize is

$$OM^2 = x^2 + y^2 = x^2 + (ax^2 + bx)^2.$$

The problem asks for the maximum on intervals such as

$$0 \le x \le -\frac{b}{a}$$

and a larger interval involving b. The first interval is exactly the interval between the two x-intercepts of the downward-opening parabola. Hence the endpoints lie on the x-axis, while the vertex lies at

$$x = -\frac{b}{2a}, \qquad y = -\frac{b^2}{4a} > 0.$$

The derivative method is direct. Put

$$F(x) = x^2 + (ax^2 + bx)^2.$$

Then

$$F'(x) = 2x + 2(ax^2 + bx)(2ax + b).$$

The maximum on a closed interval occurs at an endpoint or at a critical point of this derivative.

§122.4 Problem 3: packing unit circles

The third problem asks about packing n circles of unit diameter into a circle of smallest possible diameter, and into a square of smallest possible side length. This is a circle-packing problem. For small n, the optimal configurations are concrete and geometric; for general n, exact formulas are difficult and often unknown.

A few elementary bounds are immediate. If n unit-diameter circles are packed into a container circle of diameter D, then area gives

$$n \cdot \pi \left(\frac{1}{2}\right)^2 \le \pi \left(\frac{D}{2}\right)^2,$$

so

$$D \ge \sqrt{n}.$$

This bound is not usually sharp because circles leave gaps. For a square of side s, the area bound gives

$$n \cdot \frac{\pi}{4} \le s^2, \qquad s \ge \frac{\sqrt{n\pi}}{2}.$$

Again, the actual packing problem is subtler.

§122.5 Problem 4: a symmetric Diophantine equation

The final problem asks for the least integer solution of

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} = 4.$$

This is a symmetric rational equation. A standard approach is to clear denominators:

$$a(a+c)(a+b) + b(b+c)(a+b) + c(b+c)(a+c) = 4(a+b)(a+c)(b+c).$$

After expansion and symmetrization, one may search for integer solutions or reduce by substituting elementary symmetric polynomials

$$s = a + b + c, \qquad p = ab + bc + ca, \qquad q = abc.$$

The problem belongs to the family of olympiad equations where symmetry is more important than direct expansion.

§122.6 The larger pattern

This short problem sheet moves between three worlds: quadratic optimization, geometric packing, and symmetric algebra. The common olympiad skill is to recognize the structure before calculating: a quadratic may have no maximum on an unbounded domain, a packing problem is governed first by area bounds and then by geometry, and a symmetric equation should be attacked through symmetric functions rather than arbitrary expansion.

§122.7 Coordinate choice and invariants

The olympiad-style coordinate problems are not about coordinates for their own sake. A good coordinate system preserves the invariant that matters and simplifies the constraint. For distance problems, squared distance often removes radicals:

$$d^2 = (x-a)^2 + (y-b)^2.$$

For symmetry problems, placing the origin at the center makes odd terms cancel.

§122.8 Packing and density

Packing unit circles is an elementary entrance to discrete geometry. A packing argument usually has two sides: a construction giving a lower bound and an area or distance obstruction giving an upper bound. In the plane, the optimal infinite equal-circle packing density is

$$\frac{\pi}{\sqrt{12}}.$$

Even when this theorem is far beyond the exercise, the same local logic appears: disks cannot overlap, so centers must be separated.

§122.9 Symmetric Diophantine equations

A symmetric equation in integers should first be rewritten in symmetric variables, such as

$$u = x + y, \qquad v = xy.$$

This reduces the number of independent expressions and often exposes divisibility or parity obstructions.

§122.10 Olympiad structure behind coordinates and symmetry

The sheet trains the olympiad habit of choosing structure before computation: coordinates for constraints, area/distance for packing, symmetry for Diophantine equations, and equality cases for sharpness.

§122.11 What an olympiad sheet trains

This sheet looks elementary, but it trains several habits that reappear later. The distance problem asks us to parametrize a line and reduce geometry to a quadratic. The parabola problem asks us to decide whether a maximum exists on a specified interval. The packing problem teaches the difference between lower bounds and exact solutions. The symmetric equation teaches that expanding blindly is rarely the best first move.

Thus the real notebook value is in the decision process: before computing, decide whether the domain is bounded, whether the object is symmetric, whether a crude bound is sharp, and whether a problem is actually asking for an exact classification.

§122.12 Coordinate geometry as optimization

The original olympiad-style coordinate problems often reduce a geometric claim to an algebraic inequality or an extremum. This is the beginning of optimization geometry. For example, placing points in coordinates turns distances into quadratic functions:

$$|x-y|^2 = (x-y)\cdot(x-y).$$

Then many geometric constraints become quadratic inequalities.

A modern viewpoint is to treat a geometric configuration as a feasible set and a desired length, angle, or area as an objective function. Convexity, symmetry, and compactness then explain why extrema exist and why equality cases are structured.

§122.13 Packing and convex bodies

Packing problems ask how many objects can fit without overlap. In the plane, disk packing leads to density questions. A convex body $K \subseteq \mathbb{R}^n$ is a compact convex set with nonempty interior. A packing by translates of K is a family

$$K + x_i$$

with disjoint interiors.

The density of a lattice packing is

$$\Delta = \frac{\operatorname{vol}(K)}{\operatorname{covol}(\Lambda)}$$

when the translates are indexed by a lattice Λ. Elementary circle or sphere packing problems are finite visible versions of questions about optimal densities.

This connects school geometry to research-level topics such as lattice packing, sphere packing, and the geometry of numbers.

§122.14 Group actions and symmetry reduction

Many coordinate solutions become shorter after using symmetry. A group action formalizes this. If a group G acts on a set X, then points in the same orbit

$$Gx = \{gx : g \in G\}$$

are equivalent under the symmetry.

For a symmetric geometric configuration, one can choose coordinates adapted to the group. For example, rotational symmetry suggests polar coordinates; reflection symmetry suggests placing axes on symmetry lines. The elementary trick of “put the origin at the center” is the coordinate form of quotienting by translations or rotations.

§122.15 Convexity and supporting hyperplanes

A line tangent to a convex curve or a plane tangent to a convex body is a supporting hyperplane. A hyperplane H supports a convex set K if K lies in one closed half-space determined by H and $H \cap K \neq \varnothing$.

Many olympiad inequalities about distances to lines or extreme points can be rephrased using supporting lines. This is the finite Euclidean beginning of convex analysis.

§122.16 Coordinate methods exploit symmetry and convexity

The elementary coordinate transformations, distance formulas, and symmetry placements should be kept exactly because they teach the practical side. The advanced layer explains what they are doing: reducing a geometric problem by symmetry, encoding constraints algebraically, and optimizing over a convex or compact feasible region.

§122.17 Coordinate choice as quotienting symmetry

When an olympiad solution places the origin at a center or aligns an axis with a symmetry line, it is not merely making the algebra shorter. It is quotienting out irrelevant degrees of freedom. Translations remove the absolute position of the configuration; rotations remove absolute direction; scaling removes absolute size when the problem is similarity-invariant.

A clean coordinate setup therefore identifies the true moduli of the configuration. For a triangle up to similarity, for instance, three vertex coordinates contain redundant information because translation, rotation, and scaling do not change the intrinsic shape. The remaining parameters describe shape, not placement.

§122.18 Packing bounds as volume plus local obstruction

Elementary packing arguments often begin with area or volume comparison. But pure volume comparison is rarely sharp because local geometry imposes additional constraints. In circle packing, density depends on how neighboring circles can touch; the hexagonal packing is optimal in the plane.

The higher-dimensional sphere-packing problem asks for the maximal density of non-overlapping equal balls in $\mathbb{R}^n$. In special dimensions, such as 8 and 24, exceptional lattices provide optimal packings. Thus a simple finite packing exercise belongs to a long research line connecting geometry, lattices, modular forms, and optimization.

§122.19 Equality cases as rigid configurations

In geometry problems, equality cases often correspond to symmetric or rigid configurations. If a bound is proved by Cauchy, AM–GM, or convexity, equality conditions translate back into geometric constraints: equal lengths, parallel directions, tangency, or regularity. The advanced habit is to track equality throughout the proof, because it reveals the extremal geometry.

Directed Angles, Reflections, and the Miquel Point

N-20220708

§123.1 Reflecting the orthocenter

Let H be the orthocenter of triangle ABC. Let X be the reflection of H over BC, and let Y be the reflection of H over the midpoint of BC. The lemma in the note states:

(i) X lies on the circumcircle (ABC);

(ii) AY is a diameter of (ABC).

For the first claim, reflecting across BC preserves angles with BC. Since $BH \perp AC$ and $CH \perp AB$, angle chasing gives

$$\angle BXC = \angle BAC.$$

Thus A, B, C, X are concyclic.

For the second claim, the reflection of H through the midpoint of BC has the property that

$$\angle ABY = 90^\circ,$$

so AY is a diameter of the circumcircle by Thales' theorem.

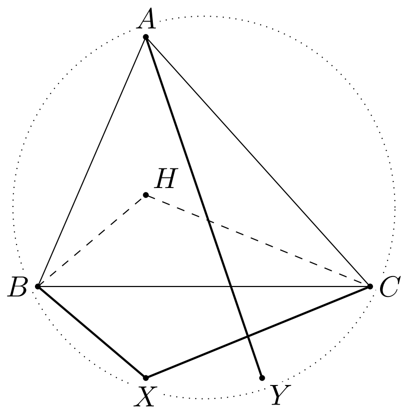


Figure 123.1: Schematic reconstruction of the orthocenter reflection lemma.

§123.2 The incenter–excenter lemma

Let ABC have incenter I. Ray AI meets the circumcircle again at L. Let I_A be the reflection of I over L. Then:

(i) I, B, C, I_A lie on a circle with diameter II_A and center L;

(ii) in particular,

$$LI = LB = LC = LI_A;$$

(iii) the rays BI_A and CI_A bisect the exterior angles of triangle ABC.

The geometry is governed by the well-known fact that the midpoint of the arc BC not containing A is equidistant from B, C, I. Reflecting I across this point gives the corresponding excenter configuration.

§123.3 Directed angles modulo $180°$

The note then explains why ordinary angles create too many cases in cyclic geometry. The cure is to use directed angles modulo 180°. I write

$$\measuredangle ABC$$

for the directed angle from line BA to line BC, modulo 180°.

The basic rules are:

(i) Oblivion: $\measuredangle APA = 0$.

(ii) Anti-reflexivity: $\measuredangle ABC = -\measuredangle CBA$.

(iii) Replacement: if A, B, C are collinear, then

$$\measuredangle PBA = \measuredangle PBC.$$

(iv) Right angles: if $AP \perp BP$, then

$$\measuredangle APB = 90°.$$

(v) Addition:

$$\measuredangle APB + \measuredangle BPC = \measuredangle APC.$$

(vi) Triangle sum:

$$\measuredangle ABC + \measuredangle BCA + \measuredangle CAB = 0.$$

(vii) Isosceles triangles:

$$AB = AC \iff \measuredangle ACB = \measuredangle CBA.$$

(viii) Inscribed angle theorem:
if (ABC) has center P, then

$$\measuredangle APB = 2\measuredangle ACB.$$

(ix) Parallel lines: if $AB \parallel CD$, then

$$\measuredangle ABC + \measuredangle BCD = 0.$$

One warning is important: because angles are modulo 180°, the statement

$$2\measuredangle ABC = 2\measuredangle XYZ$$

does not necessarily imply

$$\measuredangle ABC = \measuredangle XYZ.$$

Therefore one should not casually divide a directed angle congruence by 2.

§123.4 Miquel point of a triangle

Let D, E, F lie on the lines BC, CA, AB, respectively. The Miquel point theorem says that there is a point P lying on all three circles

$$(AEF), \qquad (BFD), \qquad (CDE).$$

Directed-angle proof. Let P be the second intersection of (AEF) and (BFD). Since A, E, F, P are concyclic,

$$\measuredangle EPF = \measuredangle EAF.$$

Since B, F, D, P are concyclic,

$$\measuredangle FPD = \measuredangle FBD.$$

Adding gives

$$\measuredangle EPD = \measuredangle EAF + \measuredangle FBD.$$

Using the collinearities E, A, C, F, A, B, and D, B, C, the right side becomes the angle condition for C, D, E, P to be cyclic. Hence $P \in (CDE)$. $\square$

§123.5 Directed angles reduce case distinctions

Using directed angles modulo 180° allows one to avoid separating many configurations. The cyclicity criterion becomes

$$A, B, C, D \text{ concyclic} \quad \Longleftrightarrow \quad \angle ABC \equiv \angle ADC \pmod{180^\circ}.$$

This remains valid even when points move to extensions of lines. That is why olympiad geometry often uses directed angles once configurations become crowded.

§123.6 Miquel point proof pattern

For a triangle with points on the three sides, one proves the existence of the Miquel point by showing that two circumcircles meet at a point and then checking that this point lies on the third circle. The check is an angle chase: cyclicity in the first two circles gives angle equalities, and the directed-angle sum forces cyclicity in the third.

The proof is not about guessing the point. It is about showing that once two circles meet, the third circle is forced by angle consistency.

§123.7 Further context

The page is an excellent example of why directed angles are not merely notation. They turn many separate configuration cases into one proof. The deeper geometric message is inversive: circles and lines should be treated as one family of generalized circles, and Miquel configurations are natural incidence phenomena in that family.

§123.8 Directed angles as projective bookkeeping

Directed angles modulo 180° turn many cyclic and collinearity statements into algebraic angle sums. A statement like

$$\angle(AB, AC) = \angle(DB, DC)$$

encodes concyclicity without separating acute and obtuse cases.

This is a first step toward projective thinking: incidence and cross-ratio-type data matter more than metric length.

§123.9 Reflections and isogonal structure

Reflecting the orthocenter or a line across an angle bisector preserves angle data. In triangle geometry, many constructions are isogonal: two lines are mirror images with respect to an angle bisector. This explains why incenter/excenter lemmas naturally produce paired angle equalities.

§123.10 Miquel point as a compatibility condition

The Miquel point appears when several cyclic conditions are compatible. Each quadrilateral supplies a circle; directed angle chasing proves that the remaining circle passes through the same point. The deeper structure is consistency of local cyclic constraints.

§123.11 Angle invariants and cyclic compatibility

The geometry here is controlled by angle invariants. Directed angles remove cases, reflections preserve angle data, and the Miquel point records when cyclic conditions glue into one global configuration.

§123.12 Why directed angles are worth learning

The directed-angle pages are a turning point in geometry study. Before directed angles, one often proves the same cyclic statement four times because points move across arcs or outside the triangle. After directed angles modulo 180°, all these cases collapse into one algebraic identity of angles.

The Miquel point theorem is the perfect example. The theorem is not mainly about a mysterious point; it is about converting three cyclicity conditions into a chain of directed-angle equalities. Once the notation is accepted, the proof becomes almost linear.

§123.13 Directed angles and invariants

Directed angles modulo π are a way to make angle chasing stable under orientation and cyclic order. In projective geometry, angles themselves are not generally preserved, but incidence and cross ratios are. The bridge is inversive geometry: circles and angles are preserved by Möbius transformations.

A Möbius transformation has the form

$$z \mapsto \frac{az+b}{cz+d}, \qquad ad - bc \neq 0.$$

It maps generalized circles to generalized circles and preserves oriented angles. Many Miquel-point and cyclic quadrilateral configurations simplify after a Möbius transformation sends one circle or line to a convenient position.

§123.14 Circle pencils

A pencil of circles is a one-parameter family of circles passing through the same two points, tangent at a point, or sharing limiting points. Algebraically, if two circles are given by equations $C_1 = 0$ and $C_2 = 0$, then the pencil is

$$C_1 + \lambda C_2 = 0.$$

Miquel configurations can often be interpreted as statements about intersecting pencils of circles.

This replaces repeated angle chasing by a structural object: the family of all circles compatible with part of the configuration.

§123.15 Cross ratio

For four points on a line or circle, the cross ratio is

$$[a, b; c, d] = \frac{(c-a)(d-b)}{(c-b)(d-a)}$$

in affine coordinates. It is invariant under projective transformations. Many cyclicity and concurrence statements can be reformulated by equality or reality of cross ratios.

The elementary theorem that equal angles imply concyclicity has a projective/inversive continuation: concyclicity and cross-ratio reality are two descriptions of the same configuration under suitable coordinates.

§123.16 Moduli of configurations

A geometric configuration can be studied up to a symmetry group. For instance, four distinct points on a projective line modulo projective transformations are classified by one parameter, the cross ratio. This is a simple example of a moduli problem: classify objects up to equivalence.

The Miquel point in the note is therefore not merely a special point. It belongs to a broader practice of understanding which features of a configuration are invariant under allowed transformations.

§123.17 Miquel configurations and spiral similarity

A Miquel point is often best understood through spiral similarity. If two segments AB and CD are seen under equal oriented angles from a point P, then P is the center of a spiral similarity sending one segment to the other. This combines a rotation and a dilation.

In complex coordinates, a direct similarity has the form

$$z \mapsto az + b, \qquad a \in \mathbb{C}^{\times}.$$

A spiral similarity center can be solved algebraically from equations relating two pairs of points. Thus angle chasing and complex affine transformations are two languages for the same configuration.

§123.18 Inversion as normalization of circles

Inversion centered at O with radius r sends a point P to P' satisfying

$$OP \cdot OP' = r^2.$$

It sends circles and lines to circles and lines, preserving angles. If a configuration contains many circles through a common point, inversion at that point turns them into lines, often reducing Miquel-type statements to elementary parallel or intersection statements.

This is why inversive geometry is not an optional decoration. It is a normalization technique for circle configurations, just as choosing coordinates is a normalization technique for analytic geometry.

Conics and the Dedekind Construction of Addition

N-20220715

§124.1 Conic sections

The note starts with the geometry of conic sections. A cone cut by a plane produces the classical conics:

circle, ellipse, parabola, hyperbola.

An ellipse can be defined as the locus of points P for which the sum of distances to two foci is constant:

$$PF_1 + PF_2 = 2a.$$

With foci $F_1 = (-c, 0)$, $F_2 = (c, 0)$, this gives the standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad c^2 = a^2 - b^2.$$

The eccentricity is

$$e = \frac{c}{a}, \qquad 0 < e < 1.$$

A hyperbola is defined by a constant difference of distances to the two foci:

$$|PF_1 - PF_2| = 2a,$$

with standard equations

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

or

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

depending on the transverse axis.

A parabola is the locus of points equidistant from a focus and a directrix. Its standard equations are

$$y^2 = 2px, \qquad y^2 = -2px, \qquad x^2 = 2py, \qquad x^2 = -2py.$$

For $y^2 = 2px$, the focus is $(p/2, 0)$, the directrix is $x = -p/2$, and the eccentricity is 1.

§124.2 Latus rectum and tangent geometry

For an ellipse, a vertical line through a focus intersects the ellipse in endpoints of a latus rectum. The length is

$$\frac{2b^2}{a}.$$

The tangent at $P = (x_0, y_0)$ to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is
$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1.$$
The note also records the focus-directrix relation. For an ellipse with eccentricity $e = c/a$, the directrices are
$$x = \pm\frac{a^2}{c}.$$
The ratio
$$\frac{PF}{\mathrm{dist}(P, \mathrm{directrix})} = e$$
characterizes the conic.

§124.3 Rational and irrational density

The page then shifts to real analysis and proves that rational and irrational numbers are dense in $\mathbb{R}$. For rationals, given $x \in \mathbb{R}$ and $\varepsilon > 0$, choose $q \in \mathbb{N}$ with
$$\frac{1}{q} < \varepsilon.$$
Then choose an integer p such that
$$p \le qx < p + 1.$$
Thus
$$\left| x - \frac{p}{q} \right| < \frac{1}{q} < \varepsilon.$$
For irrationals, take a rational approximation r to $x - \sqrt{2}/N$, then
$$r + \frac{\sqrt{2}}{N}$$
is irrational and can be made arbitrarily close to x.

§124.4 Vector spaces

The note includes the vector space axioms. Let F be a field and V a set equipped with addition and scalar multiplication:
$$+ : V \times V \to V, \qquad \cdot : F \times V \to V.$$
The axioms are associativity and commutativity of addition, existence of 0, additive inverses, compatibility of scalar multiplication, distributivity, and $1v = v$. With the usual coordinate operations,
$$\mathbb{R}^n$$
is an $\mathbb{R}$-linear space.

§124.5 Dedekind cuts and order

The note returns to Dedekind cuts. We denote by $\mathbb{R}$ the set of Dedekind cuts. For cuts X, Y, define an order by
$$X < Y \iff X \subset Y, \quad X \neq Y.$$
This order is total because if $X \not\subseteq Y$, choose $x \in X \setminus Y$; the downward-closed property of cuts forces $Y \subset X$.

§124.6 Addition of Dedekind cuts

Define
$$X + Y = \{x + y : x \in X,\ y \in Y\}.$$
The zero cut is
$$\bar{0} = \{x \in \mathbb{Q} : x < 0\}.$$
One must check that $X + Y$ is again a Dedekind cut:

(i) it is nonempty and not all of $\mathbb{Q}$;

(ii) it is downward closed;

(iii) it has no greatest element.

The page gives the standard proof. For downward closure, if $z < x+y$, then $z = x+(z-x)$ and $z - x < y$, so $z - x \in Y$. For no greatest element, choose larger elements $x' > x$ in X and $y' > y$ in Y; then $x' + y' > x + y$.

Associativity and commutativity follow directly:
$$(X + Y) + Z = X + (Y + Z), \qquad X + Y = Y + X.$$
Order compatibility is subtler. One needs the lemma: if X is a cut and $n \in \mathbb{N}$, then there exist $x \in X$ and $x' \in X'$ such that
$$0 < x' - x < \frac{1}{n}.$$
This approximation lemma lets one prove
$$X < Y \quad \Rightarrow \quad X + Z < Y + Z.$$

§124.7 Additive inverses

For a cut X, define
$$-X = \{y - x' : y \in \bar{0},\ x' \in X'\}.$$
Then $-X$ is the unique cut Z such that
$$X + Z = \bar{0}.$$
This reproduces the usual negative of a real number. In particular, if $p/q \in \mathbb{Q}$, then
$$-X_{p/q} = X_{-p/q}.$$

§124.8 Conics through focus and directrix

A conic can be defined by the ratio
$$e = \frac{\text{distance to focus}}{\text{distance to directrix}}.$$
If $e < 1$, the conic is an ellipse. If $e = 1$, it is a parabola. If $e > 1$, it is a hyperbola. This unifies the three classical curves by one parameter: eccentricity.

§124.9 Dedekind cuts and addition

The Dedekind construction treats a real number as a cut of rational numbers. Addition is then defined by adding representatives from cuts:

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

The hard part is not writing the formula, but proving that it again satisfies the properties of a cut. This illustrates how arithmetic operations must be reconstructed after the real numbers are built.

§124.10 A compact bridge

This note is a striking mixture: classical conics and the construction of real addition by Dedekind cuts. The hidden connection is that both topics turn geometry into algebra. A conic becomes an equation and a focus-directrix ratio; a real number becomes a cut with algebraic operations.

§124.11 Conics through affine and projective lenses

A conic can be represented by a quadratic equation

$$ax^2 + bxy + cy^2 + dx + ey + f = 0.$$

Affine transformations preserve conic type up to degeneracy, while projective transformations unify ellipses, parabolas, and hyperbolas as sections of a cone.

The elementary focus-directrix and latus-rectum formulas are metric shadows of a more invariant projective object.

§124.12 Density and completion

The density of $\mathbb{Q}$ in $\mathbb{R}$ says every real interval contains rationals. Dedekind cuts use order rather than decimal expansions to construct real numbers. Addition of cuts must be defined so that order and completeness are preserved.

This is the structural reason the note can move from conics to real-number construction: analytic geometry needs a complete ordered field.

§124.13 Dedekind addition as forced operation

If A and B are lower cuts, their sum is

$$A + B = \{a + b : a \in A,\ b \in B\}.$$

The definition is not arbitrary. It is forced by wanting the embedding of rationals into cuts to preserve addition and order.

§124.14 Coordinates, conics, and the completed line

Conic geometry depends on the coordinate field. Dedekind cuts explain what the real line is, and projective/affine transformations explain which features of conics survive coordinate changes.

Conic Exercises: Lines, Ellipses, Parabolas, and Tangents

N-20220716

§125.1 Line and ellipse

The note continues the conic-section exercises. A line and an ellipse can have three possible positional relationships:

$$\begin{aligned} \Delta > 0 &: \text{ two intersections,} \\ \Delta = 0 &: \text{ tangent,} \\ \Delta < 0 &: \text{ no real intersection.} \end{aligned}$$

This is obtained by substituting the line equation into the conic equation and examining the discriminant of the resulting quadratic.

For example, consider

$$y = 2x - 2, \qquad \frac{x^2}{5} + \frac{y^2}{4} = 1.$$

Substitution gives

$$\frac{x^2}{5} + \frac{(2x-2)^2}{4} = 1,$$

so

$$\frac{6}{5}x^2 - 2x = 0.$$

Thus

$$x = 0, \qquad x = \frac{5}{3},$$

and the intersection points are

$$(0, -2), \qquad \left(\frac{5}{3}, \frac{4}{3}\right).$$

§125.2 Tangents from a point to an ellipse

Let

$$C : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

and let $P = (0, 2)$. A line through P has equation

$$y = kx + 2.$$

Substituting into the ellipse gives a quadratic in x. Tangency occurs exactly when the discriminant is zero.

For the example on the page, this produces

$$k = \pm\frac{\sqrt{6}}{3},$$

so the two tangents are

$$y = \frac{\sqrt{6}}{3}x + 2, \qquad y = -\frac{\sqrt{6}}{3}x + 2.$$

§125.3 Line and hyperbola

For the hyperbola

$$\frac{x^2}{2} - \frac{y^2}{2} = 1,$$

and the line

$$y = x - 1,$$

substitution gives

$$x^2 - (x-1)^2 = 2.$$

Hence

$$2x - 1 = 2, \qquad x = \frac{3}{2}, \qquad y = \frac{1}{2}.$$

The line is not tangent; it meets the hyperbola at one point because the equation degenerates to a linear equation after substitution. This is a useful warning: for hyperbolas, "one intersection" need not mean tangency, because the line may be parallel to an asymptotic direction.

§125.4 Parabola tangent through a point

For the parabola

$$x^2 = \frac{1}{2}y,$$

we have

$$y = 2x^2.$$

A line through $P = (2, 6)$ may be written

$$y = kx + 6 - 2k.$$

Tangency means that

$$2x^2 = kx + 6 - 2k$$

has a repeated root. Thus

$$2x^2 - kx - 6 + 2k = 0,$$

and the discriminant condition is

$$k^2 - 4 \cdot 2(-6 + 2k) = 0.$$

Solving gives

$$k = 12, \qquad k = -4.$$

The tangent lines are therefore

$$y = 12x - 18, \qquad y = -4x + 14.$$

§125.5 A parabola and a circle

The second page considers the parabola

$$C : x^2 = 2py, \qquad p > 0,$$

with focus F, and a circle

$$M : x^2 + (y + 4)^2 = 1.$$

The problem asks for p when the shortest distance between F and the circle is 4, and then asks for a maximal triangle area using two tangents from a point P on the circle.

Since the focus of $x^2 = 2py$ is

$$F = \left(0, \frac{p}{2}\right),$$

and the circle has center

$$O = (0, -4)$$

and radius 1, the distance from F to the circle is

$$FO - 1 = \left(\frac{p}{2} + 4\right) - 1 = \frac{p}{2} + 3.$$

Setting this equal to 4 gives

$$p = 2.$$

Hence the parabola is

$$x^2 = 4y.$$

For a tangent point $A = (x_1, x_1^2/4)$ on $x^2 = 4y$, the tangent has equation

$$xx_1 = 2(y + x_1^2/4),$$

or

$$y = \frac{x_1}{2}x - racx_1^2 4.$$

Similarly for a second tangent point $B = (x_2, x_2^2/4)$. The two tangents meet at

$$P = \left(\frac{x_1 + x_2}{2}, \frac{x_1 x_2}{4}\right).$$

Using the condition that P lies on the circle gives an optimization problem for the area of triangle PAB. The computation in the page reduces it to a one-variable expression and obtains the maximum

$$[PAB]_{\max} = 20\sqrt{5}.$$

§125.6 An ellipse with prescribed quadrilateral area

The final exercise considers an ellipse

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b > 0,$$

passing through $A = (0, -2)$. Four vertices form a quadrilateral of area $4\sqrt{5}$. Since $A = (0, -2)$ lies on the ellipse, $b = 2$. The area condition gives

$$2ab = 4\sqrt{5},$$

so

$$a = \sqrt{5}.$$

Therefore

$$E : \frac{x^2}{5} + \frac{y^2}{4} = 1.$$

For the line through $P = (0, -3)$

$$\ell : y = kx - 3,$$

intersecting the ellipse at B, C, the equation is

$$\frac{x^2}{5} + \frac{(kx-3)^2}{4} = 1.$$

This gives a quadratic in x. The condition that the line meet the ellipse in two points is

$$k > 1 \quad \text{or} \quad k < -1.$$

The problem then studies where the chords meet the line $y = -3$ and imposes

$$|PM| + |PN| \le 15.$$

The computation on the page reduces this to

$$|k| \le 3.$$

Combining conditions gives

$$k \in [-3, -1) \cup (1, 3].$$

§125.7 Discriminants distinguish tangency from asymptotic intersection

The page is a concentrated exercise in the algebraic method for conics: substitute a line into a conic, read the discriminant, and interpret the result geometrically. The only subtlety is that different conics behave differently: for ellipses, one intersection usually signals tangency; for hyperbolas, a line parallel to an asymptotic direction can produce a single finite intersection without being tangent.

§125.8 Tangency via discriminants

A line intersecting a conic usually produces a quadratic equation in the line parameter. Tangency occurs when the quadratic has a repeated root:

$$\Delta = 0.$$

This converts a geometric contact condition into algebra. It is the same multiplicity idea that appears in polynomial factorization.

§125.9 Polarity with respect to a conic

For a nondegenerate conic, a point outside the conic determines a polar line through its contact points with tangents. In matrix form, if the conic is

$$X^T AX = 0,$$

then the polar of P is

$$P^T AX = 0.$$

This gives a projective explanation of many tangent-chord relations.

§125.10 Area constraints and affine invariance

Some ellipse problems impose an area or quadrilateral condition. Area changes by the determinant of an affine map. Therefore one can often transform an ellipse to a circle, solve the symmetric problem, then transform back with the determinant factor.

§125.11 Tangency, polarity, and affine normalization

Line-conic exercises are controlled by three ideas: intersection multiplicity, polar duality, and affine normalization. The computations are coordinate expressions of these invariant geometric mechanisms.

Solid Geometry: Points, Lines, Planes, Parallelism, and Perpendicularity

N-20220719

§126.1 Why solid geometry feels different

The note starts with a warning: solid geometry is not just plane geometry with one more coordinate. The difficulty is that objects can miss each other in space. Two lines in the plane either meet or are parallel; in space, they may also be skew. A line can be parallel to a plane without being parallel to every line in that plane. These small distinctions are exactly why the axioms matter.

§126.2 Basic axioms of incidence

Axiom 126.2.1 (Plane incidence axioms)
Two distinct points determine a unique line, and two distinct lines intersect in at most one point. These axioms encode the basic incidence structure before metric notions such as distance and angle are introduced.

The fundamental incidence principles recorded in the note are:

(i) Through three non-collinear points there is one and only one plane.

(ii) If two points of a line lie in a plane, then the whole line lies in that plane.

(iii) If two distinct planes have a common point, then their intersection is exactly one line through that point.

From these one derives several useful consequences:

(i) Through a line and a point not on that line, there is one and only one plane.

(ii) Through two intersecting lines, there is one and only one plane.

(iii) Through two parallel lines, there is one and only one plane.

The proofs are good training in mathematical language. One repeatedly shows existence by choosing suitable points, and uniqueness by reducing to the uniqueness of a plane through three non-collinear points.

§126.3 Relations between two lines

In space, two distinct lines can be:

(i) coplanar and intersecting;

(ii) coplanar and parallel;

(iii) non-coplanar, called skew lines.

The first page underlines that skew lines are the genuinely new case. Drawings can be misleading: two projected lines may cross on paper even when the actual spatial lines do not intersect.

§126.4 A line and a plane

A line l and a plane α have three possible relations:

(i) $l \subset \alpha$;

(ii) l intersects α at exactly one point;

(iii) $l \parallel \alpha$, meaning there is no common point.

The note stresses a common mistake: a line parallel to a plane is not necessarily parallel to every line in the plane. It is parallel to some line in the plane, but it may be skew to many others.

§126.5 Parallel planes and parallel lines

Two planes are parallel if they have no common point. The standard theorem is: if two intersecting lines in one plane are respectively parallel to two intersecting lines in another plane, then the two planes are parallel. Conversely, if two parallel planes are cut by a third plane, the intersection lines are parallel.

This explains many of the little diagrams in the note: the right move is to find the auxiliary plane that contains the relevant lines, then reduce the spatial statement to a planar one.

§126.6 Perpendicularity

A line l is perpendicular to a plane α if l is perpendicular to every line in α through the foot point. The useful test is stronger-looking but easier to check:

Theorem 126.6.1

If a line is perpendicular to two intersecting lines in a plane, then it is perpendicular to the plane.

Once a line is perpendicular to a plane, it is perpendicular to every line in that plane through the foot. This is the basic tool behind many distance and projection problems.

§126.7 Projection theorem

Theorem 126.7.1 (Projection theorem)

In an inner-product space, the closest point in a subspace W to a vector v is the vector $p \in W$ such that

$$v - p \perp W.$$

The note also touches the three-perpendicular theorem. In one common form: let $PO \perp \alpha$, let $AB \subset \alpha$, and let OQ be the projection of PQ onto α. Then perpendicularity of PQ to a line in space can be detected by perpendicularity of its projection in the plane. Informally, right angles survive projection when the projection is taken along a perpendicular direction.

§126.8 A coordinate viewpoint

Although the note is synthetic, it is useful to connect it with vectors. A plane in $\mathbb{R}^3$ can be written

$$ax + by + cz = d,$$

with normal vector

$$n = (a, b, c).$$

A line with direction vector v is perpendicular to the plane iff

$$v \parallel n.$$

A line lies in or is parallel to the plane iff

$$v \cdot n = 0.$$

Thus the synthetic tests are shadows of dot-product computations.

§126.9 Skew lines

In space, two lines may be neither intersecting nor parallel. Such lines are skew. This is the first major difference from plane geometry. A diagram can be misleading because a projection of skew lines may appear to intersect.

To prove two lines are skew, one usually shows they are not parallel and that the system of equations for intersection has no solution. In vector form, lines

$$p + tu, \qquad q + sv$$

intersect iff there exist s, t such that $p + tu = q + sv$.

§126.10 Planes by normal vectors

A plane can be described by a point p_0 and a normal vector n:

$$n \cdot (x - p_0) = 0.$$

This equation is often more useful than a picture. Parallelism of planes becomes parallelism of normal vectors. Perpendicularity of a line and a plane becomes parallelism between the line direction and the plane normal.

§126.11 The organizing idea

Solid geometry trains an important habit: always ask what ambient plane contains the objects you are comparing. Many false arguments in space come from treating skew lines as if they were coplanar. Later, vector algebra solves this with span and dot products; projective geometry solves it with incidence; linear algebra solves it with subspaces and dimensions.

§126.12 Incidence geometry

Solid geometry begins with incidence: which points lie on which lines and planes. Many theorems are consequences of dimension counting. Two distinct points determine a line; three non-collinear points determine a plane.

This is the three-dimensional analogue of affine geometry, where parallelism and intersection are primary relations.

§126.13 Orthogonality as inner-product structure

Perpendicularity is not merely visual; it depends on an inner product. In coordinates,

$$u \perp v \quad \Longleftrightarrow \quad u \cdot v = 0.$$

A line perpendicular to a plane is perpendicular to every direction vector in that plane. This is why normal vectors encode planes efficiently.

§126.14 Projection as least-distance decomposition

Orthogonal projection decomposes a vector into parallel and perpendicular parts:

$$v = \operatorname{proj}_W v + (v - \operatorname{proj}_W v).$$

The perpendicular part is the shortest error. This geometric fact later becomes least squares in linear algebra.

§126.15 Affine incidence plus metric projection

Solid geometry has two layers: affine incidence/parallelism and metric orthogonality/projection. Confusing the layers causes many errors. Lines and planes are affine objects; perpendicularity and distance require inner-product structure.

§126.16 Affine geometry of points, lines, and planes

Solid geometry often begins with points, lines, and planes. In affine geometry, a line through points p, q is

$$p + t(q - p), \qquad t \in k,$$

and a plane through p with independent directions u, v is

$$p + su + tv.$$

Incidence questions become linear-dependence questions among direction vectors.

This explains the elementary tests: parallelism is dependence of direction vectors; coplanarity is rank at most two; intersection of planes is solving a linear system.

§126.17 Projective completion

Parallel lines do not meet in affine geometry, but they meet at points at infinity in projective geometry. Projective space $\mathbb{P}^n$ consists of one-dimensional subspaces of k^{n+1}. A point is written in homogeneous coordinates

$$[x_0 : x_1 : \cdots : x_n].$$

Affine space sits inside projective space as the chart $x_0 \neq 0$.

Adding points at infinity makes many incidence theorems cleaner because parallel and intersecting cases become one case.

§126.18 Grassmannians

The set of all k-dimensional linear subspaces of an n-dimensional vector space is the Grassmannian

$$Gr(k, n).$$

Lines through the origin in k^3 form $Gr(1, 3) = \mathbb{P}^2$. Planes through the origin form $Gr(2, 3)$.

Thus the collection of all possible directions of lines and planes is itself a geometric space. The elementary act of choosing a line direction is a point in a Grassmannian.

§126.19 Exterior algebra and incidence

A line direction spanned by u and a plane direction spanned by v, w can be encoded by wedge products:

$$u, \qquad v \wedge w.$$

Incidence conditions become wedge equations. For example, u lies in the plane spanned by v, w iff

$$u \wedge v \wedge w = 0.$$

This gives a coordinate-free version of determinant tests for coplanarity.

§126.20 Incidence tests become wedge and projective calculations

The original point-line-plane exercises are exactly the practical computations needed for affine and projective geometry. Direction vectors become points in Grassmannians; determinant tests become wedge-product incidence; parallel cases become ordinary intersections after projective completion.

§126.21 Incidence via ranks

In three-dimensional analytic geometry, incidence conditions are rank conditions. Points P_0, P_1, P_2, P_3 are coplanar iff the direction vectors

$$P_1 - P_0, \quad P_2 - P_0, \quad P_3 - P_0$$

are linearly dependent. Equivalently, the determinant formed by these three vectors is zero.

A line intersects a plane when the linear system consisting of the line equation and plane equation is solvable. A line is parallel to a plane when its direction vector lies in the kernel of the plane's normal functional. These are all linear-algebraic conditions.

§126.22 Projective duality of points and planes

In projective three-space, a plane is given by a linear equation

$$a_0X_0 + a_1X_1 + a_2X_2 + a_3X_3 = 0.$$

Thus planes correspond to covectors up to scale, i.e. points in the dual projective space. Point-plane incidence is the pairing

$$\ell(P) = 0.$$

This duality explains why many theorems about points have dual theorems about planes.

The elementary distinction between direction vectors and normal vectors is the affine shadow of vector-covector duality.

Trigonometric Functions, Radians, Identities, and Triangle Formulas

N-20220802

§127.1 An angle is a motion before it is a number

An angle begins with two rays, but the most useful picture is a rotation. Fix an initial ray and rotate it until it reaches a terminal ray. Counterclockwise rotations are positive and clockwise rotations are negative. A full turn changes the amount of rotation without changing the terminal direction, so

$$\theta \sim \theta + 2k\pi, \qquad k \in \mathbb{Z}.$$

This is why an angle can be treated either as a real number recording a motion or as a point of the quotient

$$\mathbb{R}/2\pi\mathbb{Z}$$

recording only the final direction.

Radian measure is not chosen merely to replace degrees by another table of numbers. On a circle of radius r, an arc of length s subtends the angle

$$\theta = \frac{s}{r}.$$

The quotient is independent of the size of the circle. On the unit circle, the numerical value of the angle is exactly the signed arc length travelled. One full revolution therefore has length and angle

$$2\pi.$$

This normalization makes small rotations look linear. For $0 < \theta < \pi/2$, the standard comparison of an inscribed triangle, a circular sector, and a tangent triangle gives

$$\sin\theta < \theta < \tan\theta.$$

Dividing by $\sin\theta$ yields

$$1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}.$$

As $\theta \to 0$, the two outer quantities tend to 1, so

$$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1.$$

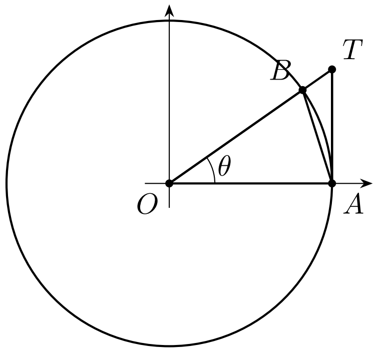


Figure 127.1: The areas of $\triangle OAB$, sector OAB, and $\triangle OAT$ give $\sin\theta < \theta < \tan\theta$.

The limit would contain a conversion constant if the variable were measured in degrees. Radians are therefore the coordinate in which the circle has unit speed at the identity. A later calculus chapter will use this limit to derive the trigonometric derivatives; here it already explains why the geometric and analytic normalizations coincide.

§127.2 Coordinates on the unit circle

A point on the unit circle has the form

$$p(\theta) = (\cos\theta, \sin\theta).$$

This is not merely notation for two unrelated functions. Sine and cosine are the coordinate functions of one moving point. The equation of the circle immediately gives

$$\cos^2\theta + \sin^2\theta = 1.$$

Several identities are visible before any algebra is done. Reversing the direction of travel reflects the point across the horizontal axis:

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta.$$

A half turn changes both coordinates:

$$\cos(\theta+\pi) = -\cos\theta, \qquad \sin(\theta+\pi) = -\sin\theta.$$

A quarter turn exchanges the coordinates up to sign:

$$\cos\left(\theta+\frac{\pi}{2}\right) = -\sin\theta, \qquad \sin\left(\theta+\frac{\pi}{2}\right) = \cos\theta.$$

The familiar quadrant signs are simply the signs of the two coordinates.

The tangent function records the slope of the terminal direction:

$$\tan\theta = \frac{\sin\theta}{\cos\theta}.$$

It is defined whenever the direction is not vertical. Its poles at

$$\theta = \frac{\pi}{2} + k\pi$$

are not defects of the circle; they are defects of using slope as a coordinate. In the real projective line, the vertical direction is an ordinary point at infinity, and tangent becomes a local coordinate on directions rather than a globally defined real function.

The graphs also come directly from the motion. Sine and cosine have period 2π, while a direction and its opposite have the same slope, so tangent has period π:

$$\sin(heta + 2\pi) = \sin\theta, \qquad \cos(heta + 2\pi) = \cos\theta,$$

$$\tan(heta + \pi) = \tan\theta.$$

Special-angle values should be reconstructed from symmetric points of the circle rather than memorized as isolated entries. For example,

$$p\left(\frac{2\pi}{3}\right) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

so

$$\cos\frac{2\pi}{3} = -\frac{1}{2}, \qquad \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \qquad \tan\frac{2\pi}{3} = -\sqrt{3}.$$

§127.3 Rotations contain the addition formulas

The rotation through angle θ is represented by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Its columns are the images of the coordinate basis vectors. Performing a rotation by β and then one by α is the same motion as rotating once by $\alpha + \beta$:

$$R(\alpha)R(\beta) = R(\alpha + \beta).$$

Multiplying the left-hand side gives

$$\begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix}.$$

Comparing entries with $R(\alpha + \beta)$ yields

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Replacing β by $-\beta$ and using parity gives the subtraction formulas.

There is also a direct geometric proof of the cosine difference identity. The dot product of two unit vectors is the cosine of the angle between them:

$$p(\alpha) \cdot p(\beta) = \cos(\alpha - \beta).$$

Expanding the coordinates gives

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta.$$

The matrix proof emphasizes composition; the dot-product proof emphasizes relative angle. They prove the same formula for different reasons.

Dividing the sine and cosine formulas gives

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta},$$

whenever both sides are defined. The denominator condition matters. If

$$1 - \tan\alpha\tan\beta = 0,$$

then the sum direction is vertical and the tangent is not a finite real number.

Setting $\beta = \alpha$ gives

$$\sin 2\alpha = 2\sin\alpha\cos\alpha,$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha.$$

Solving the last two expressions for the squares gives

$$\cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}, \qquad \sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2}.$$

The square-root versions require a sign chosen from the quadrant of $\theta/2$. Omitting that sign is one of the most common ways a correct identity becomes a false formula.

§127.4 Complex multiplication and multiple angles

Identify $(x, y) \in \mathbb{R}^2$ with $x + iy \in \mathbb{C}$. The unit-circle point becomes

$$z(\theta) = \cos\theta + i\sin\theta.$$

The addition formulas say exactly that

$$z(\alpha + \beta) = z(\alpha)z(\beta).$$

With the exponential notation

$$z(\theta) = e^{i\theta},$$

angle addition becomes ordinary multiplication.

Repeated multiplication gives De Moivre's formula:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta), \qquad n \in \mathbb{Z}.$$

For $n = 3$, expanding the left-hand side and comparing real and imaginary parts gives

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta,$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta.$$

Multiple-angle identities are therefore polynomial consequences of multiplication on the circle.

The roots of unity are equally transparent. Solving

$$z^n = 1$$

gives

$$z_k = e^{2\pi ik/n}, \qquad k = 0, 1, \ldots, n-1.$$

They are the vertices of a regular n-gon. Their sum is zero for $n > 1$, either by the geometric-series identity or because the centroid of a regular polygon is the origin.

This chapter uses roots of unity only as the natural endpoint of multiple-angle geometry. Their arithmetic and finite Fourier structure are developed in separate notes.

§127.5 Chebyshev polynomials turn angles into algebra

Define T_n by

$$T_n(\cos\theta) = \cos(n\theta).$$

The addition formula

$$\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos\theta\cos(n\theta)$$

shows that

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x),$$

with

$$T_0(x) = 1, \qquad T_1(x) = x.$$

Thus

$$T_2(x) = 2x^2 - 1,$$
$$T_3(x) = 4x^3 - 3x,$$
$$T_4(x) = 8x^4 - 8x^2 + 1.$$

The roots are obtained without solving a general polynomial equation. Since

$$T_n(\cos\theta) = 0$$

when

$$n\theta = \frac{(2k-1)\pi}{2},$$

we have

$$x_k = \cos\frac{(2k-1)\pi}{2n}, \qquad k = 1, \ldots, n.$$

Because the leading coefficient of T_n is 2^{n-1} for $n \geq 1$,

$$T_n(x) = 2^{n-1}\prod_{k=1}^{n}\left(x - \cos\frac{(2k-1)\pi}{2n}\right).$$

A trigonometric zero set has become an exact polynomial factorization.

Chebyshev polynomials also solve an extremal problem. On $[-1, 1]$,

$$|T_n(x)| \leq 1.$$

After normalizing the leading coefficient, no monic polynomial of degree n has smaller maximum absolute value on this interval. The alternating values

$$T_n\left(\cos\frac{k\pi}{n}\right) = (-1)^k$$

force this minimax behavior. This is why the same polynomials appear in interpolation nodes, approximation theory, and stable numerical algorithms. Multiple-angle formulas are not merely simplification tricks; they encode an optimal oscillation pattern.

§127.6 Triangle formulas are inner-product geometry

Let a Euclidean triangle have side lengths a, b, c opposite angles A, B, C. Place two side vectors u, v at the vertex with angle C, with

$$\|u\| = a, \qquad \|v\| = b.$$

The third side is $u - v$, so

$$c^2 = \|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2u \cdot v.$$

Since

$$u \cdot v = ab\cos C,$$

we obtain the law of cosines:

$$\boxed{c^2 = a^2 + b^2 - 2ab\cos C.}$$

The Pythagorean theorem is the special case $C = \pi/2$.

The area K can be written using any included angle:

$$K = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C.$$

Dividing these expressions gives

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

To identify the common value, place the triangle on its circumcircle. A chord of length a subtending inscribed angle A has length

$$a = 2R\sin A,$$

so

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.}$$

Heron's formula is also an inner-product identity. Using the vectors u, v with lengths b, c and included angle A, the Gram determinant is

$$\det\begin{pmatrix} u \cdot u & u \cdot v \\ v \cdot u & v \cdot v \end{pmatrix} = b^2c^2 - (bc\cos A)^2 = 4K^2.$$

The law of cosines gives

$$bc\cos A = \frac{b^2 + c^2 - a^2}{2}.$$

Therefore

$$16K^2 = 4b^2c^2 - (b^2 + c^2 - a^2)^2.$$

Factoring the right-hand side yields

$$16K^2 = (a + b + c)(-a + b + c)(a - b + c)(a + b - c).$$

With

$$s = \frac{a + b + c}{2},$$

we recover

$$\boxed{K = \sqrt{s(s - a)(s - b)(s - c)}.}$$

The four factors are not mysterious. They are the triangle inequalities written symmetrically.

§127.7 The ambiguous triangle is a real geometric bifurcation

Three sides determine a Euclidean triangle up to congruence, and two sides with the included angle do the same. Two sides with a nonincluded angle can behave differently.

Take

$$a = 7, \qquad b = 10, \qquad A = 30^\circ.$$

The sine law gives

$$\sin B = \frac{b \sin A}{a} = \frac{5}{7}.$$

There are two angles in $(0, \pi)$ with this sine:

$$B_1 = \arcsin \frac{5}{7}, \qquad B_2 = \pi - B_1.$$

Both satisfy

$$A + B_j < \pi,$$

so both define triangles. Their third angles are

$$C_1 = \pi - A - B_1, \qquad C_2 = B_1 - A,$$

and their third sides are

$$c_j = \frac{a \sin C_j}{\sin A}.$$

Numerically,

$$B_1 \approx 45.58^\circ, \qquad B_2 \approx 134.42^\circ,$$

producing one obtuse triangle and one narrow acute triangle.

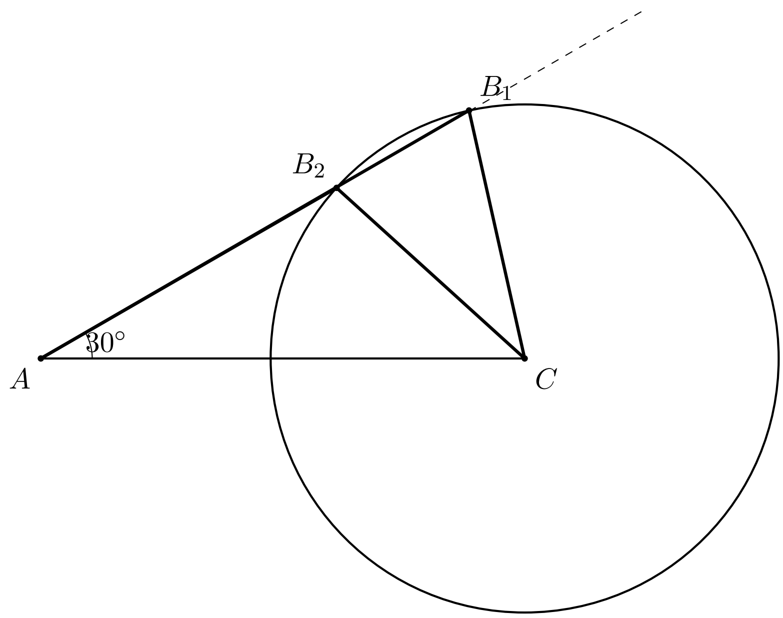


Figure 127.2: A ray from A can meet the circle centered at C twice. The two intersections are the two SSA triangles.

The ambiguity is not caused by a defective formula. It is the geometry of a circle intersecting a ray in two points. The cases of zero, one, or two triangles correspond to no intersection, tangency, or two intersections.

§127.8 The circle group explains why frequencies add

The rotation matrices form the group

$$SO(2) = \{Q \in \mathbb{R}^{2\times 2} : Q^T Q = I,\ \det Q = 1\}.$$

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since $J^2 = -I$, its exponential separates into even and odd powers:

$$e^{\theta J} = I\cos\theta + J\sin\theta = R(\theta).$$

Thus

$$SO(2) = \{e^{\theta J} : \theta \in \mathbb{R}\}.$$

The matrix J is the infinitesimal generator of rotations. Differentiating this identity gives

$$R'(\theta) = JR(\theta) = R(\theta)J.$$

This yields the coupled system

$$\cos'\theta = -\sin\theta, \qquad \sin'\theta = \cos\theta.$$

Logically, this is an alternative analytic construction once the matrix exponential has been defined; it is not a substitute for the elementary limit proof used in an introductory calculus development.

Under the identification $SO(2) \cong S^1$, the continuous one-dimensional complex characters are

$$\chi_n(e^{i\theta}) = e^{in\theta}, \qquad n \in \mathbb{Z}.$$

A short reason is that a character lifted to $\mathbb{R}$ must have the form $e^{ic\theta}$, and periodicity at 2π forces $c \in \mathbb{Z}$.

Multiplying characters adds their indices:

$$\chi_m\chi_n = \chi_{m+n}.$$

Taking real and imaginary parts gives the product-to-sum identities. For example,

$$\cos m\theta \cos n\theta = \frac{1}{2}\cos((m+n)\theta) + \frac{1}{2}\cos((m-n)\theta).$$

A product of oscillations becomes a sum of two frequencies because multiplication in the character group adds frequency labels.

The same characters are orthogonal on the circle:

$$\int_{-\pi}^{\pi} e^{im\theta}\overline{e^{in\theta}}\,d\theta = \begin{cases} 2\pi, & m = n, \\ 0, & m \neq n. \end{cases}$$

This is the minimal bridge from elementary trigonometry to Fourier analysis. The deeper theory belongs elsewhere, but the reason trigonometric modes are natural is already visible: they are the irreducible characters of the circle group.

§127.9 Changing curvature changes triangle trigonometry

Euclidean triangle formulas are not universal laws of all geometries. On the unit sphere, triangle sides are great-circle arc lengths. The spherical law of cosines is

$$\boxed{\cos c = \cos a \cos b + \sin a \sin b \cos C.}$$

For a right angle $C = \pi/2$, this becomes

$$\cos c = \cos a \cos b,$$

which is visibly different from the Euclidean Pythagorean theorem.

A particularly clean example is the spherical octant triangle with vertices at the north pole and two equatorial points separated by 90°. Its three side lengths are all $\pi/2$, and its three angles are also all $\pi/2$. Hence

$$A + B + C = \frac{3\pi}{2}.$$

The angular excess is

$$A + B + C - \pi = \frac{\pi}{2}.$$

The triangle occupies one eighth of the unit sphere, whose area is 4π, so its area is also $\pi/2$. This is a special case of the spherical excess formula

$$\text{area} = A + B + C - \pi.$$

It is the first visible shadow of Gauss–Bonnet: angle excess measures positive curvature.

The Euclidean law of cosines reappears at small scale. Replace a, b, c by $\varepsilon a, \varepsilon b, \varepsilon c$ in the spherical formula. Using

$$\cos(\varepsilon x) = 1 - \frac{\varepsilon^2 x^2}{2} + O(\varepsilon^4),$$

$$\sin(\varepsilon x) = \varepsilon x + O(\varepsilon^3),$$

we obtain

$$c^2 = a^2 + b^2 - 2ab \cos C + O(\varepsilon^2).$$

After zooming in, curvature disappears to first order and the geometry becomes Euclidean.

In hyperbolic geometry the corresponding formula is

$$\cosh c = \cosh a \cosh b - \sinh a \sinh b \cos C.$$

Its small-scale expansion gives the same Euclidean law, but hyperbolic triangles have angle sum less than π. Spherical, Euclidean, and hyperbolic trigonometry are therefore the positive-, zero-, and negative-curvature versions of one question: how do distances and directions interact in the ambient geometry?

§127.10 What holds the system together

The identities in this chapter arise from a small number of structures rather than from a long table.

Arc length gives the radian coordinate and the local relation

$$\sin\theta \sim \theta.$$

The unit circle gives sine, cosine, parity, periodicity, and the Pythagorean identity. Composition in $SO(2)$ gives the addition formulas. Complex multiplication gives De Moivre's formula, roots of unity, and multiple angles. Chebyshev polynomials translate those angles into algebra. Dot products and Gram determinants give the laws of cosines, sines, and Heron's formula. Characters of the circle explain why frequencies add and why Fourier modes are orthogonal. Finally, spherical and hyperbolic laws show which parts of trigonometry depend on curvature.

This organization also separates nearby topics. A proof-comparison chapter can ask how many logically independent routes lead to the addition formulas. A calculus chapter can build derivatives from the basic limit and analyze inverse branches. A Fourier chapter can develop completeness and convergence. The role of the present chapter is different: it presents trigonometry itself as a coherent geometry of rotations, triangles, polynomial recurrences, and constant curvature.

Analytic Geometry through Normals, Line Pencils, and Quadratic Forms

N-20220911

§128.1 A line is an affine level set, not a collection of unrelated formulas

A line in the plane can be written

$$H = \{x \in \mathbb{R}^2 : n \cdot x = c\}, \qquad n \neq 0.$$

The vector $n = (a, b)$ is normal to the line, and a direction vector is

$$d = (-b, a).$$

If $p \in H$, every point of the line has the parametric form

$$x = p + td, \qquad t \in \mathbb{R}.$$

Conversely, eliminating t gives

$$a(x_1 - p_1) + b(x_2 - p_2) = 0.$$

This single description includes vertical lines, where slope notation fails.

The familiar equations are coordinate choices for the same object. If $b \neq 0$,

$$ax + by = c \iff y = -\frac{a}{b}x + \frac{c}{b}.$$

If a line passes through p_1, p_2, its direction is $p_2 - p_1$. If it meets the coordinate axes away from the origin, dividing by its intercepts produces

$$\frac{x}{a} + \frac{y}{b} = 1.$$

None of these forms is universally best; the normal form survives every orientation, while slope form is efficient only in the nonvertical chart.

In homogeneous coordinates, the line is represented by the coefficient triple

$$\ell = [a : b : -c],$$

and incidence with a projective point $X = [x : y : z]$ is the bilinear equation

$$ax + by - cz = 0.$$

The homogeneous form will later make points at infinity and polar lines ordinary parts of the calculation.

§128.2 Angles and Euclidean distance come from the normal vector

Two lines with normals n_1, n_2 are parallel exactly when the normals are proportional. They are perpendicular exactly when

$$n_1 \cdot n_2 = 0,$$

because the direction of each line is a quarter-turn of its normal.

For

$$H = \{x : n \cdot x = c\}$$

and a point x_0, the signed displacement in the normal direction is

$$\frac{n \cdot x_0 - c}{\|n\|_2}.$$

Hence

$$\boxed{\operatorname{dist}_2(x_0, H) = \frac{|n \cdot x_0 - c|}{\|n\|_2}.}$$

To derive it, let

$$p = x_0 - \frac{n \cdot x_0 - c}{\|n\|_2^2} n.$$

Then $n \cdot p = c$, so $p \in H$, and $x_0 - p$ is normal to the line. Any other point of H differs from p by a tangent vector, so the Pythagorean theorem makes p the closest point.

For parallel lines

$$n \cdot x = c_1, \qquad n \cdot x = c_2,$$

the same formula gives

$$\operatorname{dist}_2(H_1, H_2) = \frac{|c_1 - c_2|}{\|n\|_2}.$$

The denominator is not a memorized normalization constant; it compensates for rescaling the defining equation.

§128.3 Distance to a hyperplane is governed by the dual norm

The Euclidean formula has a normed-space version. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$, and let φ be a nonzero linear functional. Its dual norm is

$$\|\varphi\|_* = \sup_{\|u\| \le 1} |\varphi(u)|.$$

For the hyperplane

$$H = \{x : \varphi(x) = c\},$$

one has

$$\boxed{\operatorname{dist}(x_0, H) = \frac{|\varphi(x_0) - c|}{\|\varphi\|_*}.}$$

Indeed, for every $x \in H$,

$$|\varphi(x_0) - c| = |\varphi(x_0 - x)| \le \|\varphi\|_* \|x_0 - x\|,$$

which gives the lower bound. In finite dimensions, the supremum defining the dual norm is attained by a unit vector u; moving from x_0 in the signed u-direction reaches equality.

For $\varphi(x, y) = ax + by$, the dual of ℓ^1 is ℓ^∞, so

$$\operatorname{dist}_1((x_0, y_0), H) = \frac{|ax_0 + by_0 - c|}{\max(|a|, |b|)}.$$

The dual of ℓ^∞ is ℓ^1, giving

$$\operatorname{dist}_\infty((x_0, y_0), H) = \frac{|ax_0 + by_0 - c|}{|a| + |b|}.$$

The familiar square root in the Euclidean formula is therefore a special case of dual-norm geometry.

§128.4 Circles are quadratic level sets, and subtraction reveals their common geometry

A circle with center $m = (a, b)$ and radius r is

$$\|x - m\|_2^2 = r^2.$$

Expanding gives

$$x^2 + y^2 + Dx + Ey + F = 0,$$

while completing squares recovers

$$m = \left(-\frac{D}{2}, -\frac{E}{2}\right),$$

$$r^2 = \frac{D^2 + E^2 - 4F}{4}.$$

The sign of this final quantity distinguishes a real circle, a point circle, and no real locus.

For a point p, define its power with respect to the circle by

$$\operatorname{Pow}_C(p) = \|p - m\|^2 - r^2.$$

If a line through p meets the circle at a, b, then

$$\operatorname{Pow}_C(p) = \overline{pa}\,\overline{pb}$$

using directed lengths. This follows by substituting the line parametrization $p + td$ into the circle equation: the product of the two parameter roots is the constant term divided by the leading coefficient.

For two circles, the locus of equal powers is their radical axis. Subtracting

$$C_1 : x^2 + y^2 - 4x = 0$$

and

$$C_2 : x^2 + y^2 - 2y - 3 = 0$$

cancels the quadratic terms and gives the line

$$-4x + 2y + 3 = 0.$$

The radical axis is linear because it compares two quadratic functions with the same leading form.

§128.5 A fractional range problem is a pencil of lines through one point

Consider

$$\mu = \frac{x}{y-3}$$

under the constraint

$$|x| + |y| \leq 1.$$

For fixed μ, rewrite the equation as

$$x = \mu(y-3).$$

This is a line through

$$P = (0, 3).$$

The range problem asks which lines in the pencil through P meet the diamond

$$K = \{(x, y) : |x| + |y| \leq 1\}.$$

Because K is convex and $P \notin K$, the extreme parameters occur at supporting lines from P to K.

The support function of the ℓ^1-unit ball is

$$h_K(n) = \max_{x \in K} n \cdot x = \|n\|_\infty.$$

The line can be written

$$(1, -\mu) \cdot (x, y) = -3\mu.$$

It meets K precisely when

$$|3\mu| \leq h_K(1, -\mu) = \max(1, |\mu|).$$

If $|\mu| > 1$, this is impossible. If $|\mu| \leq 1$, it becomes

$$3|\mu| \leq 1.$$

Therefore

$$\boxed{\mu \in \left[-\frac{1}{3}, \frac{1}{3}\right].}$$

The endpoints correspond to the supporting lines through the vertices $(1, 0)$ and $(-1, 0)$. Convex duality has not replaced the elementary picture; it has explained why the endpoints are support data.

§128.6 Ratios on a graph are radial coordinates

For a point $(x, f(x))$ with $x \neq 0$, the ratio

$$\frac{f(x)}{x}$$

is the slope of the ray from the origin to the graph point. Counting solutions of

$$\frac{f(x)}{x} = m$$

is therefore the same as counting intersections between the graph and the line

$$y = mx.$$

A rational expression has become a radial-projection problem.

This viewpoint also identifies tangency thresholds. As m varies, the number of intersections can change only when the line becomes tangent to the graph or passes through a singular point. Algebraically, solving

$$f(x) = mx$$

together with

$$f'(x) = m$$

detects those critical slopes. The graphical and derivative descriptions are the same incidence event viewed globally and locally.

§128.7 A perpendicular moving-line problem is a circle problem

Let fixed points be

$$A = (-3, 0), \qquad B = (1, 6).$$

Suppose lines through A and B meet at P and are perpendicular. Then

$$\angle APB = 90^\circ,$$

so Thales' theorem places P on the circle with diameter AB.

The right triangle gives

$$PA^2 + PB^2 = AB^2.$$

Hence

$$PA\,PB \le \frac{PA^2 + PB^2}{2} = \frac{AB^2}{2}.$$

Since

$$AB^2 = (1+3)^2 + 6^2 = 52,$$

one obtains

$$\boxed{PA\,PB \le 26.}$$

Equality occurs when $PA = PB$, namely at the intersections of the diameter circle with the perpendicular bisector of AB.

The coordinate version would introduce two slopes and a quadratic constraint. The circle formulation identifies the locus first, after which the optimization is a one-line inequality.

§128.8 A general conic is an affine quadratic form

Write an affine conic as

$$F(x) = x^T Q x + 2\ell^T x + f = 0, \qquad Q = Q^T.$$

An orthogonal change of coordinates diagonalizes Q, removing the cross term without changing Euclidean angles. If Q is invertible, translating by

$$x = u - Q^{-1}\ell$$

removes the linear term. Rotation handles directions; translation handles the center.

Consider
$$xy - x - y = 0.$$
Set
$$u = \frac{x+y}{\sqrt{2}}, \qquad v = \frac{x-y}{\sqrt{2}}.$$
Then
$$xy = \frac{u^2 - v^2}{2}, \qquad x + y = \sqrt{2}\,u.$$
The equation becomes
$$u^2 - v^2 - 2\sqrt{2}\,u = 0,$$
or
$$\boxed{(u - \sqrt{2})^2 - v^2 = 2.}$$
The original rectangular-looking equation is a hyperbola whose principal axes are the eigenvectors of the quadratic matrix
$$Q = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

§128.9 Rank and the line at infinity distinguish conic types

Homogenizing a conic gives
$$X^T A X = 0, \qquad X = (x, y, z)^T,$$
for a symmetric 3×3 matrix A. The conic is degenerate when A loses rank. For example,
$$xy = 0$$
is the union of two lines and has a rank-deficient homogeneous matrix.

The intersection with the line at infinity $z = 0$ records affine directions.

- An ellipse
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2$$
has no real point at infinity.
- A parabola
$$y^2 = 2pxz$$
meets the line at infinity at one double point $[1 : 0 : 0]$.
- A hyperbola
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z^2$$
has two real points at infinity, corresponding to its asymptotic directions.

For

$$xy - x - y = 0,$$

the projective closure is

$$XY - XZ - YZ = 0.$$

At infinity,

$$Z = 0 \quad \Longrightarrow \quad XY = 0,$$

so the two points are

$$[1 : 0 : 0], \qquad [0 : 1 : 0].$$

They are the projective endpoints of the two asymptote directions revealed by the standard hyperbola form.

§128.10 Polarity turns the conic matrix into its tangent geometry

Let a nondegenerate projective conic be

$$F(X) = X^T AX = 0, \qquad A = A^T.$$

For a point p, its polar line is

$$p^T AX = 0.$$

If p lies on the conic, this polar is the tangent line. Indeed, the differential is

$$dF_p(H) = 2p^T AH.$$

The tangent condition at p is

$$p^T A(X - p) = 0.$$

Since $p^T Ap = 0$, this reduces to

$$p^T AX = 0.$$

For the conic

$$XY - XZ - YZ = 0,$$

use the symmetric matrix

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix},$$

which represents twice the equation. The affine point $(2, 2)$ corresponds to

$$p = [2 : 2 : 1]$$

and lies on the conic. Since

$$Ap = \begin{pmatrix} 1 \\ 1 \\ -4 \end{pmatrix},$$

its polar, hence its tangent, is

$$X + Y - 4Z = 0.$$

In the affine chart $Z = 1$,

$$\boxed{x + y = 4.}$$

The same answer follows from the gradient of $xy-x-y$ at $(2, 2)$, but the polar calculation remains valid in homogeneous coordinates and transforms naturally with the conic under projective changes of coordinates.

Plane Geometry I: Ceva, Menelaus, Ptolemy, Simson Line, and Euler Line

N-20230119

§129.1 Why this geometry chapter matters

The note is a chapter-style plane geometry note. It does not merely list named theorems; it gives statements, pictures, proofs, and comments about when each theorem is useful. The main pattern is:

$$\text{Ceva detects concurrence,}\quad \text{Menelaus detects collinearity,}$$
$$\text{Ptolemy detects cyclicity.}$$

Then Simson line and Euler line enter as richer configurations involving circles, projections, centers, and ratios.

§129.2 Ceva's theorem

Theorem 129.2.1 (Ceva's theorem)
For points on the sides of a triangle, the corresponding cevians are concurrent exactly when the signed product of the three side ratios is 1.

Let D, E, F lie on BC, CA, AB or their extensions in triangle ABC. Ceva's theorem says that AD, BE, CF are concurrent iff

$$\frac{BD}{DC}\cdot\frac{CE}{EA}\cdot\frac{AF}{FB}=1$$

using directed lengths.

The proof in the note uses areas. Suppose the cevians meet at O. Since triangles with bases on the same line and the same altitude have areas proportional to their bases,

$$\frac{BD}{DC}=\frac{[ABD]}{[ACD]}=\frac{[OBD]}{[OCD]}.$$

Subtracting the smaller triangles gives

$$\frac{[ABO]}{[ACO]}=\frac{BD}{DC}.$$

Similarly,

$$\frac{[BCO]}{[BAO]}=\frac{CE}{EA},\qquad \frac{[CAO]}{[CBO]}=\frac{AF}{FB}.$$

Multiplying the three identities cancels all area factors and gives the Ceva product 1.

The inverse theorem is often the useful direction: if the product of ratios is 1, then the three cevians are concurrent. In olympiad geometry, this is the standard tool for proving three lines meet at one point.

§129.3 Menelaus' theorem

Theorem 129.3.1 (Menelaus' theorem)
For points on the three sidelines of a triangle, the points are collinear exactly when the signed product of the three side ratios is -1.

Menelaus is the collinearity counterpart of Ceva. If a line meets the sidelines of triangle ABC at D, E, F, then

$$\frac{AE}{EC} \cdot \frac{CD}{DB} \cdot \frac{BF}{FA} = 1$$

again with directed lengths.

The proof in the original note draws a parallel through a vertex. For instance, draw through B a line parallel to the transversal and compare similar triangles. The ratios on the sides of the triangle become ratios on a pair of parallel lines. After substitution, the product becomes 1.

The memory rule is useful but not enough: Ceva and Menelaus look similar, so one must remember what they prove. Ceva: three cevians concurrent. Menelaus: three points collinear.

§129.4 Ptolemy's theorem

Theorem 129.4.1 (Ptolemy's theorem)
For a cyclic quadrilateral with side lengths a, b, c, d and diagonals e, f,

$$ef = ac + bd.$$

For a cyclic quadrilateral $ABCD$, Ptolemy's theorem states

$$AC \cdot BD = AB \cdot CD + AD \cdot BC.$$

The note also records the converse: if a convex quadrilateral satisfies this equality, then it is cyclic.

A classical proof chooses a point or diagonal construction so that two pairs of triangles become similar. The products $AB \cdot CD$ and $AD \cdot BC$ then appear as pieces of $AC \cdot BD$. What matters is that cyclicity converts equal angles into similarity.

Ptolemy is metric: it relates lengths, not only ratios. That is why it is more rigid than Ceva or Menelaus.

§129.5 Generalized Ptolemy inequality

For any convex quadrilateral $ABCD$, the generalized Ptolemy inequality says

$$AC \cdot BD \leq AB \cdot CD + AD \cdot BC,$$

with equality iff the quadrilateral is cyclic.

The note gives a complex-plane proof. Assign complex numbers a, b, c, d to the vertices. The identity

$$(a - b)(c - d) + (a - d)(b - c) = (a - c)(b - d)$$

is exact. Taking moduli and applying the triangle inequality gives

$$|a-c||b-d| \leq |a-b||c-d| + |a-d||b-c|.$$

This is precisely Ptolemy's inequality. Equality in the triangle inequality corresponds to the relevant complex numbers having the same argument, which geometrically means the points are concyclic in the required order.

This proof is important because it changes the language of geometry: a quadrilateral becomes four complex numbers, and a metric inequality becomes the ordinary triangle inequality.

§129.6 Simson line

Simson's theorem says: if P lies on the circumcircle of triangle ABC, and perpendiculars are dropped from P to the three sides (or their extensions), then the three feet are collinear. The common line is the Simson line of P.

The proof in the note uses cyclic quadrilaterals. Let the feet be D, E, F. Because $PE \perp AB$ and $PF \perp AC$, quadrilaterals such as $AEPF$ are cyclic. Similar cyclic-angle relations show that the angle between two of the foot segments is supplementary to the angle between another pair. Hence the three feet lie on a line.

The inverse theorem is also true. The practical use is: when a problem contains many perpendicular feet from a point to the sides of a triangle, look for a circumcircle point and a Simson line.

§129.7 Euler line and Euler formula

For triangle ABC, let O be the circumcenter, G the centroid, and H the orthocenter. Euler's theorem says that O, G, H are collinear and

$$OG : GH = 1 : 2,$$

equivalently $GH = 2OG$. This common line is the Euler line.

One proof uses midpoints and similar triangles. Let D be the midpoint of BC. Since G is the centroid, $AG : GD = 2 : 1$. Using the facts that the circumcenter lies on perpendicular bisectors and the orthocenter lies on altitudes, one constructs similar triangles showing that the point on OH with the centroid ratio is exactly G.

The note also states Euler's formula relating the circumradius R, inradius r, and the distance $d = OI$ between circumcenter and incenter:

$$R^2 - d^2 = 2Rr.$$

This immediately gives Euler's inequality

$$R \geq 2r,$$

with equality iff the triangle is equilateral. The proof in the note uses the angle bisector through the incenter, intersections with the circumcircle, and similar triangles to derive a product relation. The key point is that the inequality is not a random radius estimate; it is the nonnegativity of d^2.

§129.8 Why the pattern matters

This chapter is a small map of classical geometry. Ceva and Menelaus are ratio theorems, almost projective in flavor. Ptolemy is metric and cyclic. Simson line turns perpendicular projections into collinearity. Euler line organizes triangle centers, and Euler's formula links local radii with the global placement of centers. The proofs constantly switch languages: area, similarity, complex numbers, cyclic angles, and centers. That switching is the real skill of plane geometry.

§129.9 An invariant map for the theorem list

The theorems in this chapter are not all of the same type. Each belongs to a different invariance principle.

Theorem	Main structure	What is invariant or detected
Ceva	affine/projective incidence	concurrence of cevians through side ratios
Menelaus	affine/projective incidence	collinearity of side points through signed ratios
Ptolemy	cyclic metric geometry	cyclicity through a length identity
Simson line	cyclic projection geometry	collinearity from perpendicular projections
Euler line	mixed affine/metric geometry	alignment of centroid, circumcenter, orthocenter

Remark 129.9.1 (Why this classification matters) — A geometric theorem becomes easier to use when one knows which transformations preserve it. Incidence statements survive projective transformations. Ratio statements survive affine changes on a line or projective normalization after coordinates are chosen. Perpendicularity, circumcenters, and orthocenters depend on the Euclidean metric and therefore do not survive arbitrary affine or projective transformations.

Proposition 129.9.2 (Mixed nature of the Euler line)
The centroid is affine in the vertices, while the circumcenter and orthocenter depend on perpendicularity. Therefore the Euler line is not a purely affine theorem and not a purely projective theorem.

Proof. The centroid has the coordinate expression

$$G = \frac{A+B+C}{3},$$

so affine maps send the centroid of ABC to the centroid of the image triangle. By contrast, the circumcenter is defined using perpendicular bisectors, and the orthocenter is defined using altitudes. Perpendicularity is not preserved by a general affine transformation. Hence the Euler line mixes an affine point with metric points. □

This map keeps the elementary proofs in place while explaining why different tools appear. Area ratios prove Ceva, similar triangles prove Menelaus, the triangle inequality in the complex plane proves Ptolemy's inequality, cyclic angles prove Simson, and metric center relations prove Euler-type formulas.

§129.10 Barycentric coordinates

For a triangle ABC, barycentric coordinates write a point P as

$$P = \frac{\alpha A + \beta B + \gamma C}{\alpha + \beta + \gamma}.$$

The ratios in Ceva and Menelaus are naturally expressed through barycentric coordinates. For instance, a point on side BC has coordinates $(0 : \beta : \gamma)$, and the side ratio is encoded by β/γ.

Ceva's theorem becomes a product condition on ratios because three cevians are concurrent exactly when their barycentric coordinate equations have a common solution.

§129.11 Projective transformations

Ceva, Menelaus, and many incidence theorems are projective: they remain true under projective transformations. A projective transformation sends lines to lines and preserves incidence, but not lengths or angles.

This explains why ratio and cross-ratio methods are so powerful. They use quantities that survive projective changes. One may move a complicated triangle configuration to a simpler projective position, prove the statement there, and transfer it back.

§129.12 Ptolemy and inversive geometry

Ptolemy's theorem characterizes cyclic quadrilaterals:

$$AC \cdot BD = AB \cdot CD + BC \cdot AD$$

for a cyclic quadrilateral $ABCD$. Inversive geometry explains why cyclic configurations are flexible. Inversion in a circle sends generalized circles to generalized circles and preserves angles.

Many circle theorems, including Simson line configurations, become simpler after an inversion sends one circle to a line or normalizes the circumcircle.

§129.13 Elliptic and hyperbolic analogues

Classical Euclidean triangle geometry has analogues in spherical and hyperbolic geometry. Ceva-type and Menelaus-type theorems survive with trigonometric replacements. For example, in spherical geometry, side lengths and angles interact through sine laws rather than linear ratios.

This shows which parts of theorems are affine/projective and which depend on Euclidean metric. Ratio statements tend to be projective; angle and length statements depend more strongly on the geometry.

§129.14 Classical geometry theorems separate by invariance type

The original list of Ceva, Menelaus, Ptolemy, Simson, and Euler-line facts is not just a catalogue. It spans three geometries: affine ratio geometry, projective incidence geometry, and inversive circle geometry. The advanced reconstruction should keep the elementary proofs while identifying which invariance principle makes each theorem natural.

§129.15 Ceva and Menelaus as dual affine statements

Ceva concerns concurrence of cevians; Menelaus concerns collinearity of points on sides. In barycentric coordinates, these become dual linear-algebraic conditions. Concurrence means three lines have a common solution. Collinearity means three points satisfy one linear equation.

This duality is why the product of ratios appears in both theorems, with reciprocal-looking signs. The elementary ratio-chasing proof is a coordinate proof in the affine-/projective plane.

§129.16 Euler line and affine covariance

The centroid, circumcenter, orthocenter, and nine-point center belong to metric triangle geometry, but some constructions behave well under affine maps and others do not. The centroid is affine: it is the average of vertices. Orthocenter and circumcenter depend on perpendicularity, so they are metric.

The Euler line therefore mixes affine and metric structures. Understanding which points survive affine transformations clarifies why some triangle centers are more universal than others.

§129.17 Simson line through projection

The Simson line theorem says that projections of a point on the circumcircle onto the three sides of a triangle are collinear. This can be interpreted through oriented angles and cyclic quadrilaterals, but also through projective degeneration: as a point moves on the circumcircle, a family of pedal triangles degenerates exactly when the point lies on the circumcircle.

Thus the theorem is not just a miracle of angle chasing; it records a degeneracy condition in a moving geometric construction.

Special Proofs of Trigonometric Addition Formulas

N-20230220

§130.1 Why prove the same identity more than once?

The addition formulas are familiar:

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

A second proof is worthwhile only when it reveals something the first proof hides. A Euclidean proof shows how angle and signed area enter. A rotation-matrix proof identifies the formulas with a group law. A complex-exponential proof packages both identities into one multiplicative equation. A projective calculation explains why the tangent formula has a denominator and what happens when it vanishes.

These proofs also differ in logical cost. Some begin from synthetic Euclidean geometry. Some assume a classification of plane rotations. Some require power series or differential equations. The aim of this chapter is therefore not to accumulate tricks, but to compare proof architectures:

$$\text{assumptions} \longrightarrow \text{mechanism} \longrightarrow \text{conclusion} \longrightarrow \text{possible generalization.}$$

Throughout, put

$$u(\theta) = (\cos\theta, \sin\theta).$$

Sine and cosine are initially the coordinates of the unit-circle point at directed angle θ.

§130.2 A chord-length proof of the cosine difference formula

Take two points

$$P = u(\alpha), \qquad Q = u(\beta)$$

on the unit circle. The angle at the center between the two radii is $\alpha - \beta$, up to orientation. The law of cosines in the isosceles triangle OPQ gives

$$|P - Q|^2 = 1^2 + 1^2 - 2\cos(\alpha - \beta).$$

Thus

$$|P - Q|^2 = 2 - 2\cos(\alpha - \beta).$$

The same squared distance can be computed from coordinates:

$$|P - Q|^2 = (\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2.$$

Expanding and using

$$\cos^2\theta + \sin^2\theta = 1$$

gives

$$|P - Q|^2 = 2 - 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta).$$

Comparing the two expressions yields

$$\boxed{\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta.}$$

Replacing β by $-\beta$ and using the parity of sine and cosine gives the cosine addition formula.

This proof does not use rotation matrices or complex numbers. It does use the Euclidean distance formula, the Pythagorean identity, and the law of cosines. If the law of cosines was itself proved from the dot-product identity

$$u\cdot v=|u||v|\cos\angle(u,v),$$

then the proof is really an inner-product proof in geometric clothing. That is not a defect, but it should be stated honestly.

§130.3 Signed area proves the sine difference formula

The determinant of two plane vectors is their signed parallelogram area:

$$\det(P,Q)=\det\begin{pmatrix}\cos\alpha & \cos\beta\\ \sin\alpha & \sin\beta\end{pmatrix}.$$

Computing the determinant gives

$$\det(P,Q)=\cos\alpha\sin\beta-\sin\alpha\cos\beta.$$

Geometrically, P and Q are unit vectors whose directed angle from P to Q is $\beta-\alpha$. The signed parallelogram area is therefore

$$\det(P,Q)=\sin(\beta-\alpha).$$

Consequently,

$$\boxed{\sin(\beta-\alpha)=\sin\beta\cos\alpha-\cos\beta\sin\alpha.}$$

Changing signs or interchanging α and β produces all sine addition and subtraction formulas.

Orientation is essential here. Ordinary unsigned area would give only

$$|\sin(\beta-\alpha)|.$$

The determinant remembers whether the second vector lies counterclockwise or clockwise from the first. The sign in the addition formula is therefore an orientation statement, not an arbitrary convention.

The cosine and sine proofs can be combined in one identity. In an oriented Euclidean plane,

$$P\cdot Q+i\det(P,Q)=e^{i(\beta-\alpha)}.$$

The real part records metric information; the imaginary part records oriented area.

§130.4 The rotation proof begins from a group law

Let $R(\theta)$ be the unique orientation-preserving Euclidean isometry fixing the origin and sending the first basis vector to $u(\theta)$. Its second column must be the positive quarter-turn of the first, so

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Rotating first by β and then by α adds directed angles. Hence

$$R(\alpha)R(\beta) = R(\alpha+\beta).$$

This statement is geometric: it says that composition in the rotation group corresponds to addition of angle parameters.

Matrix multiplication gives

$$R(\alpha)R(\beta) = \begin{pmatrix} \cos\alpha\cos\beta - \sin\alpha\sin\beta & -\cos\alpha\sin\beta - \sin\alpha\cos\beta \\ \sin\alpha\cos\beta + \cos\alpha\sin\beta & -\sin\alpha\sin\beta + \cos\alpha\cos\beta \end{pmatrix}.$$

Comparing with $R(\alpha+\beta)$ proves both addition formulas at once.

The proof is short, but it is not assumption-free. One must know that rotations compose by adding directed angles and that the displayed matrix represents the geometric rotation independently of the formulas being proved. If the matrix formula for $R(\theta)$ was itself derived by applying the addition formulas to a general vector, the reasoning would be circular.

The structural advantage is substantial. The formulas are now matrix coefficients of a representation

$$R : \mathbb{R}/2\pi\mathbb{Z} \longrightarrow SO(2).$$

The identity

$$R(\alpha+\beta) = R(\alpha)R(\beta)$$

is a homomorphism law, and the trigonometric identities are its coordinate expansion.

§130.5 An analytic Euler proof with no hidden circularity

Euler's formula is often used in one line:

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Then

$$e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$$

immediately yields the addition formulas by comparing real and imaginary parts. This is valid only if Euler's formula and the exponential law were established without already using the addition formulas.

One noncircular route begins from power series. Define

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \qquad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

Absolute convergence justifies regrouping terms, and separating even and odd powers gives
$$e^{iz} = \cos z + i \sin z.$$
The Cauchy product of the exponential series proves
$$e^{z+w} = e^z e^w.$$
Therefore
$$\cos(\alpha + \beta) + i \sin(\alpha + \beta) = (\cos\alpha + i \sin\alpha)(\cos\beta + i \sin\beta).$$
Multiplying the right-hand side and comparing components proves the formulas.

A second noncircular analytic route uses differential equations. Let $z(t)$ solve
$$z'(t) = iz(t), \qquad z(0) = 1.$$
Uniqueness implies the flow law
$$z(s + t) = z(s)z(t).$$
Writing $z = x + iy$ gives
$$x' = -y, \qquad y' = x, \qquad x(0) = 1, \quad y(0) = 0.$$
If cosine and sine are defined as the solutions of this system, then
$$z(t) = \cos t + i \sin t,$$
and the flow law again yields the addition formulas.

These analytic proofs generalize well because exponentials and flows exist in many algebras and Lie groups. Their cost is that they depend on analysis rather than elementary circle geometry.

§130.6 Where the circular Euler proof enters

There is another common derivation of Euler's formula. One starts with the trigonometric addition formulas, proves that
$$\theta \longmapsto \cos\theta + i \sin\theta$$
is multiplicative, and then identifies this multiplicative map with $e^{i\theta}$. This is an excellent derivation of Euler's formula from trigonometry.

It cannot then be reversed and advertised as an independent proof of the addition formulas. The logical loop would be
$$\text{addition formulas} \Longrightarrow \text{Euler formula} \Longrightarrow \text{addition formulas}.$$
The final implication is correct, but the chain as a whole proves nothing new.

The issue is not that a theorem may never be used in both directions. Equivalent statements often illuminate each other. The issue is whether one is claiming an independent foundation. To break the loop, Euler's formula must enter from a separate construction such as power series, an ODE, or the complex exponential defined by functional analysis.

§130.7 Tangent lives on a projective line

Once the sine and cosine formulas are known, division gives

$$\tan(\alpha+\beta) = \frac{\tan\alpha + an\beta}{1 - an\alpha an\beta},$$

provided the expressions are finite. A direct slope calculation explains the fractional-linear shape.

Represent a nonvertical direction by a vector $(1,m)^T$, where m is its slope. Rotating it through α gives

$$R(\alpha)\begin{pmatrix}1\\ m\end{pmatrix} = \begin{pmatrix}\cos\alpha - m\sin\alpha\\ \sin\alpha + m\cos\alpha\end{pmatrix}.$$

The new slope is

$$m' = \frac{\sin\alpha + m\cos\alpha}{\cos\alpha - m\sin\alpha} = \frac{\tan\alpha + m}{1 - man\alpha}.$$

Putting $m = \tan\beta$ gives the tangent addition formula.

If

$$1 - man\alpha = 0,$$

the rotated direction is vertical. The denominator does not signal a breakdown of rotation; it says that the affine slope chart has reached the point at infinity of the projective line

$$\mathbb{RP}^1.$$

The transformation

$$m \longmapsto \frac{\tan\alpha + m}{1 - man\alpha}$$

is a Möbius transformation on that projective line. This is why proving tangent first is usually awkward: tangent is not a global coordinate, whereas sine and cosine remain finite everywhere.

§130.8 A dependency graph for the proofs

The four routes can be displayed as a directed acyclic graph only after the foundational choices are fixed.

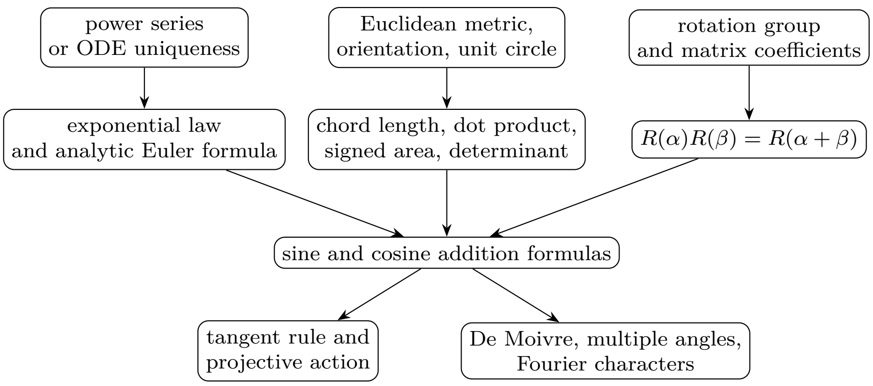


Figure 130.1: Three independent routes to the addition formulas. A geometrically derived Euler formula would lie downstream of the central box and cannot be moved upstream without circularity.

The graph also clarifies equivalence. Once the addition formulas are known, they can reconstruct the rotation homomorphism and the multiplicativity of $\cos\theta + i\sin\theta$. Those reverse arrows are mathematically useful, but they belong to an equivalence diagram, not to a foundational proof DAG.

§130.9 The same group template produces hyperbolic identities

The rotation matrix is built from an element whose square is $-I$. Hyperbolic functions arise from a generator whose square is I. Let

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

Then

$$e^{tK} = I\cosh t + K\sinh t = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}.$$

Write this matrix as $H(t)$. The exponential law gives

$$H(s+t) = H(s)H(t).$$

Multiplying matrices yields

$$\boxed{\cosh(s+t) = \cosh s\cosh t + \sinh s\sinh t,}$$

$$\boxed{\sinh(s+t) = \sinh s\cosh t + \cosh s\sinh t.}$$

The sign difference from circular cosine is explained by

$$K^2 = I \qquad \text{instead of} \qquad J^2 = -I.$$

Moreover,

$$H(t)^T \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} H(t) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $H(t)$ preserves the Minkowski quadratic form $x^2 - y^2$, just as $R(t)$ preserves the Euclidean form $x^2 + y^2$. Circular and hyperbolic addition laws are two real two-dimensional representations of one-parameter groups.

Spherical trigonometry does not arise by replacing every symbol in this calculation. Rotations about different axes in three dimensions generally do not commute, and the sides of a spherical triangle are not one additive parameter along a fixed subgroup. Genuine generalization requires identifying the preserved structure, not merely changing sin into another named function.

§130.10 Which proof should be used?

The answer depends on what has already been developed and what one wants next.

Proof route	Main prerequisites	What becomes transparent	Best continuation
Chord and area	Euclidean congruence, distance, orientation	metric and sign	triangle geometry
Rotation matrices	classification and composition of rotations	group law and representation	linear algebra, Lie groups
Power series or ODE	convergence or uniqueness theory	exponential flow	complex analysis, differential equations
Projective slope	rotation action on directions	tangent denominator and infinity	projective geometry

For formal verification, the analytic series route has explicit algebraic subgoals but requires substantial infrastructure for convergence and rearrangement. The matrix route is short once a library already contains rotations as orientation-preserving isometries. The synthetic route may look elementary on paper but demands careful encoding of directed angles, congruence, and orientation. A proof assistant rewards explicit dependencies; it does not automatically make the visually shortest proof the formally shortest one.

The important lesson is not that one proof wins. The geometric proof identifies the quantities being measured. The matrix proof identifies the composition law. The analytic proof exposes a flow and generalizes to exponentials in other algebras. The projective calculation explains a singular denominator. Together they show how one identity can sit at the intersection of geometry, algebra, analysis, and representation theory without those subjects becoming logically indistinguishable.

Trigonometric Systems, Multiple-Angle Formulas, and a Determinant Recurrence

N-20230223

§131.1 Correcting a trigonometric mistake

The first handwritten line says that an earlier calculation from February 20 contained a mistake. The attempted determinant manipulation involved vectors built from angles α and β, and a factor such as $\tan(\alpha - \beta)$. The note records that the determinant-style computation did not work and that the reason for the error was not immediately clear.

This is worth preserving. A failed determinant manipulation is not useless; it tells us that the problem probably has hidden dependencies among the trigonometric variables. Before choosing a linear algebra method, one must check whether the equations are genuinely linear in the chosen unknowns.

§131.2 A trigonometric system

The main problem asks for $\cos\alpha\cos\beta$ under conditions of the form

$$\sin\alpha + \sin\beta = \frac{1}{2}, \qquad \cos\alpha - \cos\beta = \frac{1}{3}.$$

The note introduces

$$a = \sin\alpha, \qquad b = \sin\beta, \qquad c = \cos\alpha, \qquad d = \cos\beta.$$

Then the equations become

$$a + b = \frac{1}{2}, \qquad c - d = \frac{1}{3}, \qquad a^2 + c^2 = 1, \qquad b^2 + d^2 = 1.$$

This transforms a trigonometric problem into an algebraic system.

§131.3 A rational-parametrization idea

The handwritten solution tries to express the four variables as sums of rational parts and radicals:

$$a = q_1 + r_1, \qquad b = q_2 - r_1, \qquad c = q_3 + r_2, \qquad d = q_4 + r_2,$$

with

$$q_1 + q_2 = \frac{1}{2}, \qquad q_3 - q_4 = \frac{1}{3}.$$

The two circle equations become quadratic equations in the radical parts. Comparing rational and irrational parts can force relations among the q_i and r_i.

The note finds relations such as

$$q_1^2 + q_3^2 = q_2^2 + q_4^2,$$

and a linear relation between r_1 and r_2. One conclusion written on the page is that the signs may be controlled by introducing one radical and then solving back for c, d. The method is experimental, and the note explicitly says that another method was not successful.

§131.4 A direct algebraic route

A cleaner derivation is to use sum-to-product formulas. We have

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} = \frac{1}{2},$$

and

$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} = \frac{1}{3}.$$

Dividing gives

$$-\tan\frac{\alpha-\beta}{2} = \frac{2}{3},$$

provided the common factor is nonzero. This determines the half-difference angle. Then one can recover products such as

$$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2}.$$

The direct method shows why the algebraic system has hidden trigonometric structure.

§131.5 Multiple-angle formulas

The second page records standard multiple-angle formulas. One form is

$$\sin(nx) = \sum_{k=0}^{n}\binom{n}{k}\cos^k x\,\sin^{n-k} x\,\sin\left(\frac{(n-k)\pi}{2}\right),$$

and

$$\cos(nx) = \sum_{k=0}^{n}\binom{n}{k}\cos^k x\,\sin^{n-k} x\,\cos\left(\frac{(n-k)\pi}{2}\right).$$

These come from expanding

$$(\cos x + i\sin x)^n = \cos(nx) + i\sin(nx).$$

Thus De Moivre's formula is the conceptual source of the binomial-looking expressions.

The recurrence formulas are

$$\sin(nx) = 2\sin((n-1)x)\cos x - \sin((n-2)x),$$

$$\cos(nx) = 2\cos((n-1)x)\cos x - \cos((n-2)x),$$

and

$$\tan(nx) = \frac{\tan((n-1)x)+\tan x}{1-\tan((n-1)x)\tan x}.$$

The sine and cosine recurrences are Chebyshev-type recurrences.

§131.6 The determinant recurrence

The final page asks one to prove a determinant formula. The displayed matrix is tridiagonal, with diagonal entries involving $2\cos\theta$ and off-diagonal entries equal to 1, with a modified first entry involving $\cos\theta$. The intended proof is expansion along the last row or last column.

If D_n denotes the determinant of the $n \times n$ tridiagonal matrix with diagonal $2\cos\theta$ and off-diagonal 1, then it satisfies

$$D_n = 2\cos\theta\, D_{n-1} - D_{n-2}.$$

This is the same recurrence as the multiple-angle formulas. Therefore D_n can be expressed in terms of $\sin((n+1)\theta)/\sin\theta$ or a related Chebyshev polynomial, depending on the exact boundary convention.

The note's blue comment says that proving the determinant is a cosine expression can be done by the recurrence. This is exactly the right idea: the matrix determinant and the trigonometric formula are the same recurrence with the same initial values.

§131.7 From technique to structure

This note is about recognizing when a problem should be treated as trigonometry, algebra, or linear algebra. A failed determinant trick reveals hidden constraints. Sum-to-product formulas solve the angle system more naturally. Multiple-angle identities come from De Moivre's formula. Tridiagonal determinants satisfy the same recurrence as Chebyshev polynomials. The common theme is that recurrence relations are a bridge between trigonometry and matrices.

§131.8 Trigonometric systems as algebraic systems

A trigonometric system can often be converted into algebra by setting

$$u = \cos x, \qquad v = \sin x, \qquad u^2 + v^2 = 1.$$

This turns identities into polynomial relations on the unit circle. The constraint must always be kept; otherwise extraneous algebraic solutions appear.

§131.9 Rational parametrization of the circle

The substitution

$$t = \tan\frac{x}{2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}$$

rationally parametrizes the unit circle except one point. It turns many trigonometric equations into rational equations.

§131.10 Multiple-angle formulas and Chebyshev polynomials

The identity

$$\cos(nx) = T_n(\cos x)$$

defines the Chebyshev polynomial T_n. Thus multiple-angle formulas are polynomial dynamics on the interval $[-1, 1]$.

§131.11 Determinant recurrences

A determinant recurrence often comes from expanding along a row or column. The resulting sequence is commonly governed by a second-order linear recurrence, whose characteristic roots explain closed forms.

§131.12 Trigonometry through algebraic parametrization

The note joins trigonometry and algebra: systems become polynomial constraints, half-angle substitution gives rational parametrization, multiple angles give Chebyshev polynomials, and determinant recurrences reveal linear recurrence structure.

Projective Geometry as the Art of Removing Exceptions

N-20240518

§132.1 The annoyance of parallel lines

Euclidean geometry has a small but persistent exception: two lines meet unless they are parallel. Projective geometry removes the exception by adding points at infinity.

This is not a cosmetic choice. It changes the grammar of geometry. Statements about incidence become cleaner, and perspective drawing becomes natural.

The guiding sentence is:

projective geometry sacrifices distance to save incidence.

§132.2 Homogeneous coordinates are ratios

The projective line over a field k is

$$\mathbb{P}^1(k) = \{[x : y] : (x, y) \neq (0, 0)\}/\sim,$$

where

$$[x : y] = [\lambda x : \lambda y] \qquad (\lambda \neq 0).$$

The pair $[x : y]$ is not a point in k^2. It is a line through the origin in k^2.

When $y \neq 0$, the point becomes $[x/y : 1]$. The missing point $[1 : 0]$ is the point at infinity.

Similarly,

$$\mathbb{P}^2(k) = \{[x : y : z] : (x, y, z) \neq (0, 0, 0)\}/\sim.$$

The chart $z \neq 0$ is the ordinary affine plane, and the equation $z = 0$ is the line at infinity.

§132.3 A coordinate scene: where parallel lines meet

The affine lines

$$y = 0, \qquad y = 1$$

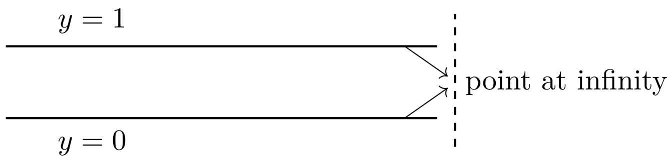
are parallel. Homogenizing them gives

$$y = 0, \qquad y - z = 0.$$

On the line at infinity $z = 0$, both equations force $y = 0$, so the intersection is

$$[1 : 0 : 0].$$

Thus projective geometry does not say that parallel lines visibly meet in the affine plane. It says that the affine plane was missing one direction point.



The diagram is symbolic: the point at infinity is not an ordinary point far to the right. It is a direction added to complete the incidence relation.

§132.4 Duality: points and lines speak the same language

A line in $\mathbb{P}^2$ has equation

$$ax + by + cz = 0.$$

The coefficients $[a : b : c]$ are homogeneous. Thus lines themselves have projective coordinates.

Incidence is the bilinear equation

$$ax + by + cz = 0.$$

This symmetry produces projective duality: many theorems remain true after swapping point and line.

For instance:

$$\text{two points determine a line} \quad \longleftrightarrow \quad \text{two lines determine a point.}$$

Projective geometry is full of such paired statements.

§132.5 Transformations and what survives

An invertible linear map $A : k^3 \to k^3$ induces a projective transformation

$$[v] \longmapsto [Av].$$

It preserves incidence but not length or angle. A circle may become an ellipse under a perspective projection, but a line remains a line.

This is the geometric lesson: once one changes the allowed transformations, one changes the invariants. Euclidean geometry cares about distances and angles. Projective geometry cares about incidence, alignment, cross-ratio, and projective equivalence.

§132.6 A first invariant: the cross-ratio

For four points a, b, c, d on a projective line, the cross-ratio is written in an affine coordinate as

$$[a, b; c, d] = \frac{(c - a)(d - b)}{(c - b)(d - a)}.$$

It is preserved by projective transformations of the line.

This formula is less important here than the role it plays. It is a warning that projective geometry is not invariant-free. It simply has different invariants from Euclidean geometry.

§132.7 Conics as one projective family

A projective conic is given by a homogeneous quadratic equation

$$X^T AX = 0, \qquad X = (x, y, z)^T.$$

In the affine chart $z = 1$, it becomes an ordinary quadratic equation in x and y. Ellipses, parabolas, and hyperbolas are different affine views of projective conics.

This explains why conics look artificially split in elementary analytic geometry. The affine chart cuts a projective object and then classifies the slice.

§132.8 The projective plane as all lines through the origin

There is a concrete model that removes much of the mystery. A point $[x : y : z] \in \mathbb{P}^2$ is a line through the origin in k^3. A projective line in $\mathbb{P}^2$ is a plane through the origin in k^3.

Then incidence becomes containment:

$$\text{projective point on projective line} \\ \Longleftrightarrow \\ \text{line in } k^3 \text{ inside plane in } k^3.$$

This model is often the easiest way to remember why two projective lines meet. Two planes through the origin in k^3 usually meet in a line through the origin, which is a projective point.

§132.9 Desargues as a projective-flavored theorem

Projective geometry is naturally comfortable with theorems about incidence. Desargues' theorem says roughly: if two triangles are perspective from a point, then the intersections of corresponding sides lie on a line.

The exact diagram is less important here than the type of statement. It uses points lying on lines, lines meeting at points, and no measurements. This is projective geometry's native habitat.

One reason projective geometry feels more elegant than Euclidean geometry is that many such incidence theorems survive projection. A perspective drawing may distort lengths wildly, but it preserves collinearity and concurrence.

§132.10 Projective transformations in coordinates

A projective transformation of the projective line is represented by a matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

with nonzero determinant. In affine coordinate $t = x/y$, it acts by a fractional linear transformation

$$t \longmapsto \frac{at + b}{ct + d}.$$

This explains why the cross-ratio is built from ratios of differences: it is designed to survive these fractional linear transformations.

The point at infinity is not an exception in this formula. It is exactly what is needed to make the transformation defined as a map of the whole projective line.

§132.11 The conic as a disguised projective line

A nonsingular conic is projectively equivalent to $\mathbb{P}^1$. For example, the conic

$$xz - y^2 = 0$$

has the parametrization

$$[s : t] \longmapsto [s^2 : st : t^2].$$

This formula sends a point of $\mathbb{P}^1$ to a point on the conic, because

$$s^2t^2 - (st)^2 = 0.$$

So a conic is not merely a quadratic curve; projectively, it behaves like a line wrapped into the plane by quadratic coordinates.

This fact is a useful bridge to later algebraic geometry: parametrizations, homogeneous equations, and projective spaces start to work together.

The conic calculation also shows what projective completion is doing throughout the chapter. Coordinates are replaced by ratios, missing affine directions become ordinary projective points, and the admissible transformations are exactly those that preserve incidence and cross-ratio rather than Euclidean length.

Transformations of Conics and the Erlangen View of Geometry

N-20250211

§133.1 The organizing question

This chapter studies conic sections through transformations. The guiding question is not only how an ellipse, parabola, or hyperbola is described by an equation, but what kind of geometry regards two such objects as the same.

The hierarchy is:

$$\text{Euclidean} \subset \text{similarity} \subset \text{affine} \subset \text{projective} \subset \text{conformal/inversive phenomena}.$$

Each enlargement forgets some invariants and preserves others. This is the spirit of Klein's Erlangen program: a geometry is the study of properties invariant under a chosen transformation group.

§133.2 Affine transformations

Definition 133.2.1. An affine transformation of $\mathbb{R}^n$ is a map

$$T(x) = Ax + b,$$

where A is a linear map and $b \in \mathbb{R}^n$. If $A \in GL(n, \mathbb{R})$, then T is an affine automorphism.

Affine transformations preserve straight lines, parallelism, ratios along a line, convexity, and barycentric combinations. They do not preserve lengths, angles, or circles.

Example 133.2.2

An ellipse can be sent to a circle by an affine transformation. If

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

then the affine map

$$(x, y) \mapsto (x/a, y/b)$$

sends it to the unit circle.

§133.3 Conics in matrix form

A plane conic may be written as

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

In homogeneous coordinates $[x : y : z]$, this becomes

$$Ax^2 + Bxy + Cy^2 + Dxz + Eyz + Fz^2 = 0.$$

Equivalently,

$$X^T Q X = 0, \qquad X = (x, y, z)^T,$$

where Q is a symmetric 3×3 matrix.

Under a projective transformation $X \mapsto MX$, the conic transforms by

$$Q \mapsto M^{-T} Q M^{-1}.$$

Thus projective classification of conics becomes a question about symmetric bilinear forms up to change of coordinates.

A concrete matrix is the best antidote to the vague sentence that all nondegenerate conics are projectively equivalent. The parabola

$$C_1 : \quad y = x^2$$

has homogeneous equation

$$X^2 - YZ = 0,$$

so it is represented by

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \end{pmatrix}.$$

The unit circle

$$C_2 : \quad X^2 + Y^2 = Z^2$$

is represented by

$$A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Now take

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

A direct multiplication gives

$$M^T A_1 M = A_2.$$

Indeed, if $U = (u, v, w)^T$ and $X = MU = (u, v + w, -v + w)^T$, then

$$X^T A_1 X = u^2 - (v + w)(-v + w) = u^2 + v^2 - w^2 = U^T A_2 U.$$

Thus the projective change of coordinates $U = M^{-1} X$ sends the parabola equation to the circle equation.

This is not an affine transformation. The inverse matrix is

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The unique real point at infinity of the parabola,

$$[0 : 1 : 0],$$

is sent to

$$M^{-1}[0 : 1 : 0] = [0 : 1 : 1],$$

a finite real point on the circle. Meanwhile the usual circular points at infinity of the real circle are the complex points

$$[1 : i : 0], \qquad [1 : -i : 0].$$

So the projective transformation has moved the affine line at infinity itself; this is exactly why a parabola and a circle can be the same projective conic while remaining different in affine geometry. The difference between them is absolute for affine geometry, but only a coordinate choice for projective geometry.

§133.4 Projective geometry

Projective geometry adds points at infinity. In the real projective plane,

$$\mathbb{P}^2(\mathbb{R}) = \{[x : y : z] \mid (x, y, z) \neq 0\}/\sim,$$

where

$$[x : y : z] = [\lambda x : \lambda y : \lambda z], \qquad \lambda \neq 0.$$

The affine plane is the chart $z \neq 0$, and the line at infinity is $z = 0$.

Definition 133.4.1. A projective transformation of $\mathbb{P}^2$ is induced by an invertible matrix

$$M \in GL(3, \mathbb{R}),$$

acting by

$$[X] \mapsto [MX].$$

Scalar multiples of M induce the same projective transformation. Thus the transformation group is

$$PGL(3, \mathbb{R}).$$

Projective transformations preserve incidence and cross-ratio, but not parallelism, distance, or angle. In projective geometry, all nondegenerate conics over an algebraically closed field are projectively equivalent.

§133.5 Cross-ratio

For four collinear points A, B, C, D, the cross-ratio is denoted

$$(A, B; C, D).$$

In an affine coordinate it has the form

$$(a, b; c, d) = \frac{(c - a)(d - b)}{(c - b)(d - a)}.$$

Theorem 133.5.1

Projective transformations preserve cross-ratio.

Remark 133.5.2 — The cross-ratio is the prototype of a projective invariant. Once one understands it, projective geometry stops looking like distorted Euclidean geometry and becomes an invariant theory of incidence.

§133.6 Projective duality and conics

Projective duality interchanges points and lines. A nondegenerate conic defines a duality: to a point p, one associates its polar line with respect to the conic. In matrix form, if the conic is $X^TQX = 0$, then the polar line of p is

$$p^TQX = 0.$$

Remark 133.6.1 — The same quadratic form that defines the conic also turns points into lines. The conic is therefore not just a curve; it is a device for dualizing projective geometry.

§133.7 Inversion

Definition 133.7.1. Inversion in the circle of radius r centered at the origin is the map

$$x \mapsto \frac{r^2}{\|x\|^2}x.$$

In the complex plane, inversion in the unit circle is essentially

$$z \mapsto \frac{1}{\bar{z}}.$$

Inversion sends lines and circles to lines and circles, with the convention that lines are circles through infinity. It is conformal away from the center of inversion: it preserves angles but not distances.

§133.8 Möbius transformations

A Möbius transformation has the form

$$z \mapsto \frac{az+b}{cz+d}, \qquad ad - bc \neq 0.$$

It acts naturally on the Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}.$$

The group of Möbius transformations is

$$PGL(2, \mathbb{C}).$$

Möbius transformations send generalized circles to generalized circles and preserve cross-ratios. They are simultaneously projective transformations of $\mathbb{P}^1(\mathbb{C})$ and conformal automorphisms of the Riemann sphere.

§133.9 Conic sections

Classically, conics arise as plane sections of a cone. Analytically, they are degree-two plane curves. Projectively, a conic is a homogeneous quadratic equation in $\mathbb{P}^2$.

Geometry	Transformations	Conic classification
Euclidean	Rigid motions	Metric shape matters
Similarity	Rotations, translations, scalings	Eccentricity-like shape remains
Affine	Invertible affine maps	Ellipses are equivalent to circles
Projective	$PGL(3)$	Nondegenerate conics are equivalent over algebraically closed fields

§133.10 The Erlangen program

Klein's Erlangen program says that geometry should be understood through a group of transformations and the invariants under that group. Euclidean geometry studies invariants under Euclidean motions; affine geometry studies invariants under affine transformations; projective geometry studies invariants under projective transformations.

> **Remark 133.10.1** — This viewpoint explains why the same object can look different in different geometries. A circle is special in Euclidean geometry, but not in affine geometry. Parallel lines are meaningful in affine geometry, but not in projective geometry. Cross-ratio is invisible in elementary Euclidean geometry, but central in projective geometry.

§133.11 A hierarchy of invariants

Transformation group	Preserved	Forgotten
Euclidean	Distance, angle, incidence	Coordinates
Similarity	Angle, ratios of lengths	Absolute scale
Affine	Incidence, parallelism, ratios on a line	Angle and length
Projective	Incidence, cross-ratio	Parallelism and affine ratio
Möbius	Generalized circles, angles, cross-ratio on $\widehat{\mathbb{C}}$	Euclidean size

§133.12 Quadratic forms and matrix language

A conic in the projective plane can be written as

$$X^T Q X = 0,$$

where $X = (x : y : z)^T$ is a homogeneous coordinate vector and Q is a symmetric matrix. Passing to homogeneous coordinates is not cosmetic. It lets ellipses, parabolas, and hyperbolas be treated uniformly and allows points at infinity to participate in the geometry.

Under a projective change of coordinates $X = MX'$, the conic matrix transforms as

$$Q' = M^T Q M$$

or equivalently $Q \mapsto M^{-T} Q M^{-1}$, depending on whether one transforms coordinates or points. The equation changes, but the geometric conic is the same object in new coordinates.

§133.13 Affine classification versus projective classification

In affine geometry, ellipses, parabolas, and hyperbolas are genuinely different because the line at infinity is fixed. The way a conic meets the line at infinity distinguishes the types: an ellipse has no real points at infinity, a parabola is tangent to the line at infinity, and a hyperbola meets it in two real points.

In projective geometry over an algebraically closed field, all nonsingular conics are projectively equivalent. Projective geometry forgets metric shape and focuses on incidence and cross-ratio-type data.

§133.14 Polar lines

For a nonsingular conic $X^T QX = 0$, a point P has a polar line

$$P^T QX = 0.$$

If P lies on the conic, the polar line is the tangent line at P. If P lies outside, the polar encodes the chord of contact of tangents from P.

The pole-polar relation is one of the best examples of why projective conic theory is richer than coordinate classification. It converts points to lines and lines to points in a way controlled by the conic.

§133.15 Transformation groups determine the geometry being studied

The transformation-centered view naturally leads to modern geometry:

(i) algebraic geometry studies varieties and morphisms, with projective space as a basic ambient object;

(ii) differential geometry studies invariants under diffeomorphisms or isometries;

(iii) complex geometry studies holomorphic maps;

(iv) representation theory studies symmetries through linear actions;

(v) category theory abstracts the entire pattern by treating structure-preserving maps as primary.

§133.16 How to choose the right geometry

The Erlangen viewpoint becomes useful only when it tells us which tools are allowed. For conic problems, a practical guide is:

Question being asked	Natural geometry	Reason
length, angle, focus, eccentricity	Euclidean/similarity	metric data is essential
midpoints, parallelism, area ratios	affine	affine maps preserve barycentric structure
incidence, tangency, conic equivalence	projective	the line at infinity may move
cross-ratio or four points on a line	projective	cross-ratio is projectively invariant
angle with generalized circles	conformal/inversive	inversion preserves angles and generalized circles

Remark 133.16.1 (A common mistake) — Saying that two conics are projectively equivalent does not mean they are the same in Euclidean or affine geometry. A parabola and a circle may be projectively equivalent, but the projective map can move the line at infinity. Once that happens, affine notions such as parallelism and the distinction between ellipse/parabola/hyperbola need not be preserved.

§133.17 Using the transformation group in a conic problem

A conic in homogeneous coordinates is

$$X^T AX = 0.$$

If the problem asks only for incidence or tangency, one may use a projective change of coordinates; algebraically this replaces A by a congruent matrix $P^T AP$. The tangent at a point X is read from the bilinear form as

$$X^T AY = 0,$$

so the same matrix controls the conic, its tangent lines, and the pole–polar correspondence.

The restrictions become important when the question contains metric words. A projective map may send a circle to a parabola because it can move the line at infinity, so it does not preserve Euclidean length, angle, focus, or eccentricity. An affine map keeps parallelism and ratios on parallel lines but may change angles. Inversion and conformal maps preserve angles and generalized circles, but not distances. Thus in an actual problem the first step is to mark which data must survive: cross-ratio and incidence suggest projective coordinates; parallelism suggests affine coordinates; angles with circles suggest inversive geometry; lengths force a Euclidean or similarity argument.

XIII

Curves and Structures

Part XIII: Contents

Riemann Surfaces as One-Dimensional Complex Manifolds

N-20220212

§134.1 Structure beyond topology

The starting question is: what does it mean for a space to carry a structure? A topological manifold has a topology; a smooth manifold has a smooth structure; a complex manifold has a complex structure.

A topology tells us which sets are open, which sequences or nets converge, and which maps are continuous. A smooth structure adds the ability to speak about differentiability in local coordinates. A complex structure is stricter: coordinate changes must be holomorphic.

Example 134.1.1

The interval $(0, 2) \subset \mathbb{R}$ is a topological 1-manifold. One chart may be the identity map

$$\varphi_1(x) = x.$$

Another chart can still be a homeomorphism while failing to be smooth. This shows why topology alone is not enough to define smoothness.

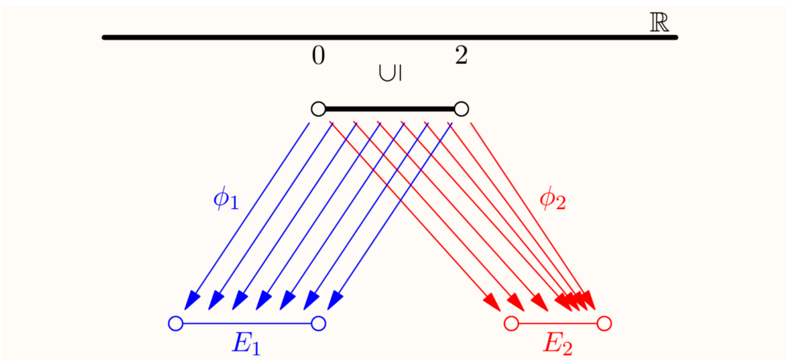


Figure 134.1: Two coordinate descriptions of the interval used to separate topology from smooth structure.

Remark 134.1.2 — The lesson is that the same underlying topological space may admit different additional structures. Smoothness is not merely a property of the set; it is encoded by charts and transition maps.

§134.2 From smooth structure to complex structure

A smooth surface embedded in $\mathbb{R}^3$ can be described locally by diffeomorphisms to open subsets of $\mathbb{R}^2$. But this embedded definition is not sufficiently general: for example, the Klein bottle cannot be embedded in $\mathbb{R}^3$.

The abstract definition of a smooth manifold therefore uses charts

$$\varphi_i : U_i \to E_i \subseteq \mathbb{R}^n$$

whose transition maps are smooth. The same philosophy gives complex manifolds by replacing $\mathbb{R}^n$ with $\mathbb{C}^n$ and smooth transition maps with holomorphic transition maps.

Remark 134.2.1 — Because holomorphic functions are automatically analytic, a complex structure is very rigid. This is why a Riemann surface is much more than a real two-dimensional smooth manifold.

§134.3 Definition of a Riemann surface

Definition 134.3.1. A Riemann surface is a connected, Hausdorff, second-countable topological space X equipped with an atlas $\{(U_i, \varphi_i)\}$, where

$$\varphi_i : U_i \xrightarrow{\sim} E_i \subseteq \mathbb{C}$$

is a homeomorphism onto an open subset of $\mathbb{C}$, and all transition maps

$$\varphi_j \circ \varphi_i^{-1}$$

are holomorphic on their domains.

Each φ_i is a complex chart. A local coordinate near $p \in X$ is the complex number

$$z = \varphi_i(x).$$

If $\varphi_i(p) = 0$, the coordinate is centered at p.

Definition 134.3.2. More generally, a complex n-manifold is a Hausdorff space locally modeled on open subsets of $\mathbb{C}^n$, with holomorphic transition maps.

Every complex n-manifold is naturally a smooth real $2n$-manifold.

§134.4 Examples

Example 134.4.1 (Open subsets of the plane)

Every connected open subset $U \subseteq \mathbb{C}$ is a Riemann surface, with the identity chart.

Example 134.4.2 (The Riemann sphere)

The Riemann sphere $\mathbb{C}_\infty$ is the complex plane together with a point at infinity. One way to construct its complex structure is to cover it by two charts: one chart with coordinate z, and another chart near infinity with coordinate

$$t = 1/z.$$

On the overlap, the transition function is

$$z = 1/t,$$

which is holomorphic away from 0. Thus the two charts glue into a Riemann surface.

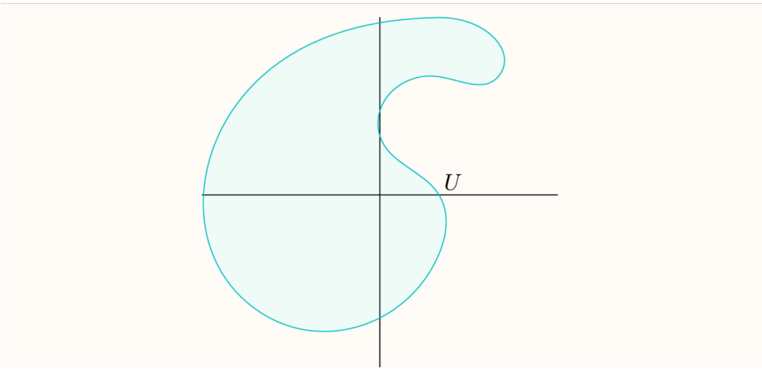


Figure 134.2: Stereographic charts used to define the complex structure on the Riemann sphere.

Caution

The sign choice in the stereographic chart matters. If the transition map became $z = 1/\bar{t}$, it would be antiholomorphic rather than holomorphic. The complex structure depends on getting this orientation right.

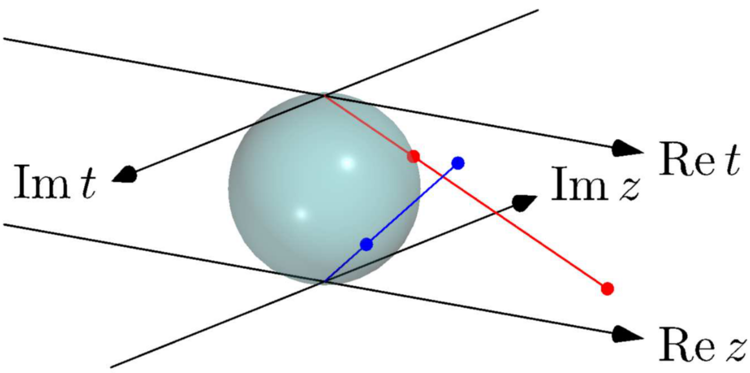


Figure 134.3: The second stereographic chart and the transition coordinate near infinity.

Example 134.4.3 (A complex torus)
Let $L = \mathbb{Z}[i] \subset \mathbb{C}$, a lattice under addition. The quotient

$$\mathbb{C}/L$$

is a complex torus. Locally, the quotient map from $\mathbb{C}$ gives complex coordinates, and different choices of lifts differ by translations

$$z \mapsto z + a, \qquad a \in L,$$

which are holomorphic.

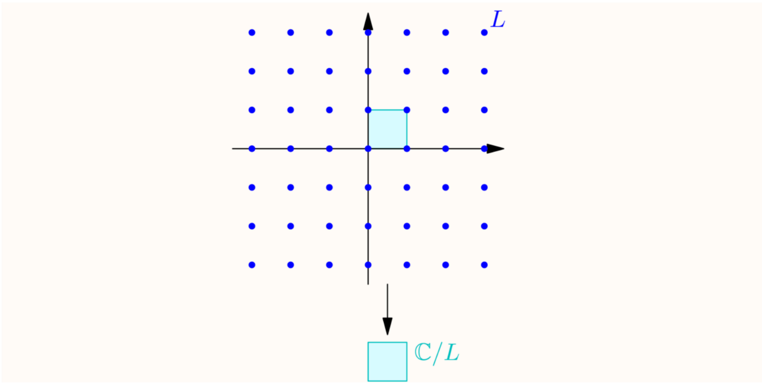


Figure 134.4: The quotient picture of a complex torus, with opposite sides identified.

Remark 134.4.4 — The complex torus is compact. Any holomorphic function on it is constant, but meromorphic functions on it are much more interesting.

Example 134.4.5
A disjoint union of two Riemann spheres is not a Riemann surface under the convention used here, because a Riemann surface is required to be connected. If one drops connectedness, each connected component is a Riemann surface.

§134.5 Holomorphic maps between Riemann surfaces

Definition 134.5.1. Let X, Y be Riemann surfaces. A map $f : X \to Y$ is holomorphic at $p \in X$ if, for charts

$$\varphi_1 : U_1 \to E_1, \qquad \varphi_2 : U_2 \to E_2$$

with $p \in U_1$ and $f(p) \in U_2$, the local expression

$$\varphi_2 \circ f \circ \varphi_1^{-1}$$

is holomorphic near $\varphi_1(p)$. The map is holomorphic if this holds at every point.

Example 134.5.2 (The cubic map)

The map

$$f : \mathbb{C} \to \mathbb{C}, \qquad f(z) = z^3$$

is holomorphic. In the usual global coordinate on $\mathbb{C}$, this is just a polynomial, hence holomorphic everywhere. At a point $a \neq 0$, the derivative

$$f'(a) = 3a^2$$

is nonzero, so the inverse function theorem gives a local holomorphic inverse near a. At 0, however, the derivative vanishes. The local model is exactly

$$z \mapsto z^3,$$

so the map has multiplicity 3 at the origin. This is the first concrete place where "holomorphic map" and "branched covering" begin to touch.

Example 134.5.3 (Inclusion into the Riemann sphere)

The inclusion

$$\mathbb{C} \hookrightarrow \mathbb{C}_\infty$$

is a holomorphic map of Riemann surfaces. On the finite chart of $\mathbb{C}_\infty$, this map is simply

$$z \mapsto z,$$

so it is holomorphic in the most literal possible sense. The point of the example is not the formula itself, but the target: the Riemann sphere adds one extra point ∞, and the ordinary plane sits inside it as the finite chart. Thus a map into $\mathbb{C}_\infty$ may be checked by using the coordinate z away from infinity and the coordinate $w = 1/z$ near infinity.

§134.6 Meromorphic functions as maps to the sphere

The Riemann sphere lets us treat poles as honest values. A meromorphic function

$$f : X \to \mathbb{C}$$

can be regarded as a holomorphic map

$$g : X \to \mathbb{C}_\infty$$

by sending poles to ∞.

Proposition 134.6.1

Nonconstant meromorphic functions $X \to \mathbb{C}$ correspond to holomorphic maps $X \to \mathbb{C}_\infty$ not identically equal to ∞.

Idea. Away from a pole this is clear. Near a pole, use the coordinate $t = 1/z$ near infinity on $\mathbb{C}_\infty$. If f has a pole, then $1/f$ is holomorphic and vanishes there, so the map into the sphere is holomorphic in the t-coordinate. □

§134.7 Degree and multiplicity

For nonconstant holomorphic maps between compact Riemann surfaces, fibres behave surprisingly well when counted with multiplicity.

Proposition 134.7.1

Let X, Y be compact Riemann surfaces and let $f : X \to Y$ be nonconstant holomorphic. For each $y \in Y$, count the points in $f^{-1}(y)$ with multiplicity. The resulting number is finite and independent of y. This number is the degree of f, denoted $\deg(f)$.

Remark 134.7.2 — This is much stronger than what one expects from arbitrary smooth maps. The rigidity of holomorphic maps makes the fibre count locally constant, and connectedness makes it globally constant.

Example 134.7.3 (The power map has degree k)

The map

$$z \mapsto z^k$$

extends to a holomorphic map $\mathbb{C}_\infty \to \mathbb{C}_\infty$ of degree k. On the finite chart this is a polynomial. Near infinity, use the coordinate $w = 1/z$. Then

$$z^k = \frac{1}{w^k}, \qquad \frac{1}{z^k} = w^k,$$

so the extended map is also holomorphic at ∞. For a generic value $a \in \mathbb{C}^*$, the equation

$$z^k = a$$

has k distinct solutions. Counting multiplicity, the same number remains true over the exceptional values 0 and ∞. This is why the degree is k, not merely because the formula contains the exponent k.

Definition 134.7.4. Let $f : X \to Y$ be a nonconstant holomorphic map and $p \in X$. The multiplicity of f at p is the unique integer $m \geq 1$ such that, after choosing suitable local coordinates centered at p and $f(p)$, the map is locally of the form

$$z \mapsto z^m.$$

Remark 134.7.5 — The guiding slogan is: every nonconstant holomorphic map locally looks like a power map. This is one of the reasons Riemann surfaces are so much cleaner than arbitrary smooth surfaces.

Idea of the local normal form. Choose local coordinates centered at p and $f(p)$, so the local expression becomes a nonconstant holomorphic function F with $F(0) = 0$. Its Taylor expansion has the form

$$F(z) = a_m z^m + a_{m+1} z^{m+1} + \cdots, \qquad a_m \neq 0.$$

Thus

$$F(z) = z^m u(z),$$

where u is holomorphic and $u(0) \neq 0$. After shrinking the coordinate neighborhood, u has a holomorphic m-th root v. Replacing the source coordinate z by $zv(z)$, the map becomes exactly

$$z \mapsto z^m.$$

The integer m is uniquely determined by the order of vanishing of F at 0, hence does not depend on the chosen coordinates. □

§134.8 Local power form is not global

The local normal form $z \mapsto z^m$ must be read with the word *local* emphasized. A useful test case is

$$U = \mathbb{C}^*, \qquad F : U \to \mathbb{C}^*, \qquad F(z) = z^2.$$

At $p = 1$, the derivative is nonzero, so in the local-normal-form sense the multiplicity is $m = 1$. Nearby, one can choose a local inverse

$$z = \sqrt{w}$$

with $w = F(z)$. But this square-root coordinate cannot be chosen globally on $\mathbb{C}^*$. If the target point travels once around the loop

$$w(t) = e^{2\pi i t}, \qquad 0 \leq t \leq 1,$$

then analytic continuation of the local branch starting at $\sqrt{1} = 1$ ends at -1, not 1. This sign change is monodromy.

There is a small trap here. If one instead moves the source point as

$$p(t) = e^{2\pi i t},$$

then $F(p(t)) = e^{4\pi i t}$ winds twice around the target origin, so the square root changes sign twice and returns to the original branch. The monodromy is seen by following a local inverse over a loop in the target, not by confusing the covering coordinate with a single global coordinate downstairs.

Thus the map $z \mapsto z^2$ is an unbranched two-fold covering of $\mathbb{C}^*$, while the missing point 0 is where the corresponding map on $\mathbb{C}$ would branch. To make $\sqrt{w}$ single-valued one must either cut the base, for example by removing a ray, or pass to a covering space. This is the global meaning of the failure: local coordinate charts glue with a monodromy representation, and on a non-simply-connected region those gluings need not collapse to one global coordinate.

§134.9 Holomorphic transition maps

A Riemann surface is locally modeled on open subsets of $\mathbb{C}$, but the crucial condition is on overlaps. If two charts

$$\varphi : U \to V \subseteq \mathbb{C}, \qquad \psi : U' \to V' \subseteq \mathbb{C}$$

overlap, then the transition map

$$\psi \circ \varphi^{-1} : \varphi(U \cap U') \to \psi(U \cap U')$$

must be holomorphic. This is what makes complex analysis possible on the surface without choosing one global coordinate.

The condition is stronger than smoothness. Holomorphic maps are rigid: they preserve angles, satisfy Cauchy–Riemann equations, and are locally power series. Thus a Riemann surface is not just a two-dimensional smooth surface; it is a surface with complex-analytic coordinate changes.

§134.10 Examples beyond the plane

The complex plane $\mathbb{C}$ is a Riemann surface with one global chart. The Riemann sphere

$$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$$

requires two charts: one coordinate z near finite points and another coordinate $w = 1/z$ near infinity. The transition map $w = 1/z$ is holomorphic away from zero, so the charts are compatible.

A complex torus is obtained from $\mathbb{C}$ by quotienting by a lattice

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2.$$

The quotient

$$\mathbb{C}/\Lambda$$

is topologically a torus but also carries a complex structure inherited from the plane. This example is the bridge from elementary complex analysis to elliptic curves.

§134.11 Meromorphic functions

On a compact Riemann surface, holomorphic functions are often forced to be constant by the maximum principle. Meromorphic functions are therefore more flexible. A meromorphic function is holomorphic except at isolated poles. Locally near a pole, it has a Laurent expansion with finitely many negative terms.

Thinking of meromorphic functions as holomorphic maps to the Riemann sphere is useful:

$$f : X \to \widehat{\mathbb{C}}.$$

A pole is simply a point mapped to ∞. This viewpoint turns singularities into ordinary values on a compact target.

§134.12 Riemann surfaces retain complex structure beyond topology

The main transition in the note is from topology to complex geometry. A topological surface only knows neighborhoods up to homeomorphism. A smooth surface knows differentiable coordinate changes. A Riemann surface knows holomorphic coordinate changes, so complex analysis can be done intrinsically. This is why Riemann surfaces are simultaneously one-dimensional complex manifolds and two-dimensional real manifolds.

§134.13 Complex structure is stronger than topology

A Riemann surface is topological dimension two but complex dimension one. The complex charts have holomorphic transition maps, so local coordinates remember angle and orientation. This is why every holomorphic map between Riemann surfaces is far more rigid than a continuous map.

§134.14 Meromorphic functions and branched coverings

A nonconstant meromorphic function on a compact Riemann surface is a holomorphic map to the Riemann sphere:

$$f : X \to \mathbb{P}^1(\mathbb{C}).$$

Its degree counts preimages with multiplicity. Branch points are where local behavior looks like

$$z \mapsto z^e.$$

The integer e records ramification.

§134.15 Riemann–Hurwitz

For a holomorphic map $f : X \to Y$ of degree d, Riemann–Hurwitz says

$$2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X} (e_p - 1).$$

This formula ties topology, degree, and branching into one invariant statement.

§134.16 Complex curves through topology and ramification

Riemann surfaces are where topology, complex analysis, and algebraic geometry first coincide. Holomorphic charts give rigidity, meromorphic functions give maps to the sphere, divisors record zeros and poles, and ramification translates analytic multiplicity into topological genus.

Tarski's Circle-Squaring Problem: From Polygon Dissection to Borel Equidecomposition

N-20221214

§135.1 The route of the note

This note is about Tarski's circle-squaring problem, but the source pages do not treat it as a single isolated puzzle. They move through a sequence of increasingly subtle notions:

$$\begin{gathered}\text{Tarski circle-squaring} \longrightarrow \text{Hilbert's third problem}\\ \longrightarrow \text{measure and paradoxical decomposition}\\ \longrightarrow \text{Laczkovich's theorem} \longrightarrow \text{Borel circle squaring.}\end{gathered}$$

The figures below are kept only when the visual configuration itself matters. The surrounding theorem statements, comments, formulas, and handwritten annotations have been incorporated into the prose.

Tarski asked in 1925 whether a disk and a square of equal area in the plane are equidecomposable: can the disk be partitioned into finitely many pieces, and can those pieces be moved by isometries to partition the square?

§135.2 Equidecomposability and dissection congruence

Let a group G act on a set X. Two subsets $A, B \subseteq X$ are G-equidecomposable if there are finite partitions

$$A = A_1 \sqcup \cdots \sqcup A_m, \qquad B = B_1 \sqcup \cdots \sqcup B_m,$$

and elements $g_i \in G$ such that

$$g_i(A_i) = B_i.$$

For plane polygons, one usually lets G be the group of Euclidean isometries. In that setting the classical language is dissection congruence: cut one polygon into finitely many polygonal pieces and rearrange them to obtain the other.

The first example is the familiar parallelogram-to-rectangle cut.

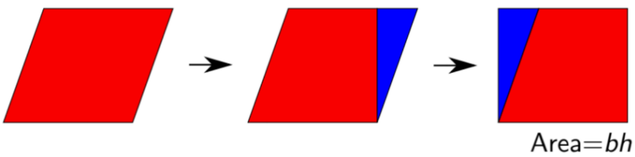


Figure 135.1: A parallelogram dissected and rearranged into a rectangle.

Area is obviously preserved by dissection congruence. The Wallace–Bolyai–Gerwien theorem says that, for plane polygons, area is also sufficient.

Theorem 135.2.1 (Wallace–Bolyai–Gerwien)
Any two plane polygons of the same area are dissection congruent.

One small point in the note is transitivity. If polygon P is dissection congruent to Q, and Q is dissection congruent to R, then P is dissection congruent to R. One refines the two dissections so that the intermediate polygon Q is cut compatibly, then composes the rearrangements.

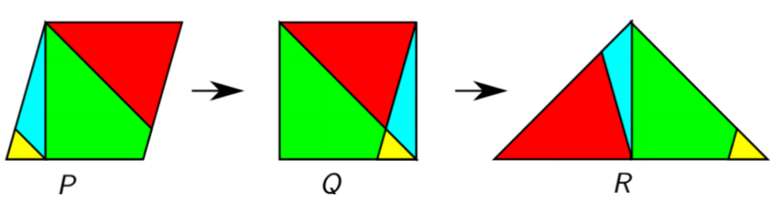


Figure 135.2: The transitivity intuition: a dissection from P to Q and one from Q to R compose.

The proof of the theorem is constructive. First chop a polygon into triangles.

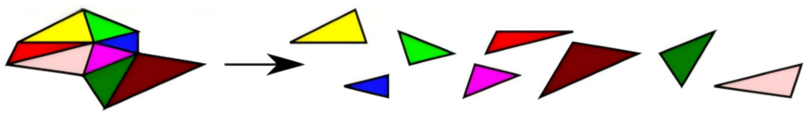


Figure 135.3: The first reduction: chop a polygon into triangles.

Then dissect each triangle into a parallelogram and then a rectangle; dissect each rectangle into a square; combine squares by the Pythagorean dissection.

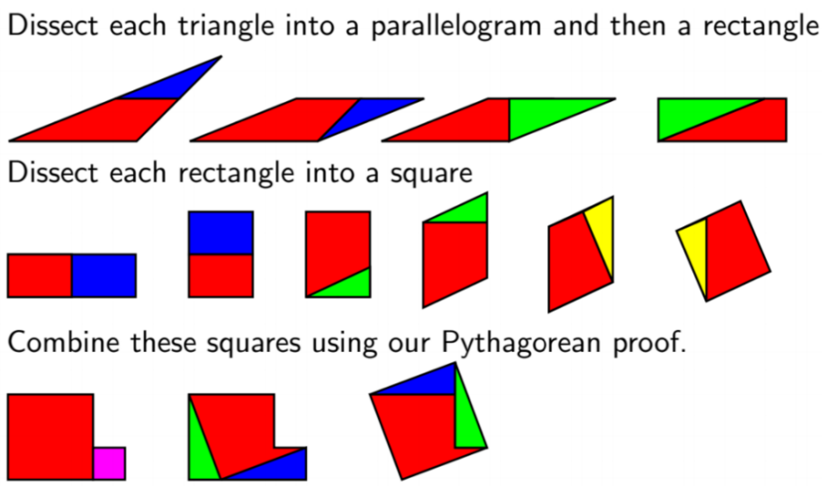


Figure 135.4: The visual chain: triangle to parallelogram/rectangle, rectangles to squares, then combine squares.

The theorem establishes the baseline: for tame plane polygons, equal area exactly characterizes finite dissection congruence.

§135.3 Hilbert's third problem and the Dehn invariant

Hilbert's third problem asks whether equal volume similarly characterizes dissection congruence of three-dimensional polytopes. Dehn's theorem says no: a cube and a regular tetrahedron of the same volume are not dissection congruent.

The extra obstruction is the Dehn invariant. If a polyhedron has edge lengths ℓ_i and corresponding dihedral angles θ_i, the invariant has the form

$$D(P) = \sum_i \ell_i \otimes \theta_i$$

with values in a tensor product such as $\mathbb{R} \otimes_{\mathbb{Z}} \mathbb{R}/(2\pi\mathbb{Z})$, depending on conventions. The invariant records length-angle data that volume does not see. The handwritten note correctly marks the tensor product as a point worth noticing: this is not elementary area bookkeeping anymore.

Sydler's theorem says that volume and Dehn invariant together are complete in dimension three. Thus the two-dimensional polygon theorem has no naive three-dimensional analogue.

§135.4 Measure-theoretic background

The source page then discusses measure. Vitali sets imply that for every $n \geq 1$, there is no extension of Lebesgue measure to all subsets of $\mathbb{R}^n$ that is both countably additive and invariant under isometries.

If countable additivity is weakened to finite additivity, the dimension matters. For $n \geq 3$, the Banach–Tarski paradox shows that there is no such finitely additive isometry-invariant measure on all bounded sets compatible with volume. For $n \leq 2$, Banach measures exist. The note emphasizes the reason: in dimension at least three, the isometry group contains enough free-group behavior to allow paradoxical decompositions; in dimensions one and two it does not.

For Tarski's problem in the plane, this means that equal area is still a necessary condition. A disk and square that are equidecomposable by isometries must have the same Lebesgue measure, even if the pieces themselves are nonmeasurable.

§135.5 Tarski's circle-squaring problem

The central question is then:

> Are a disk and a square in $\mathbb{R}^2$ of the same area equidecomposable?

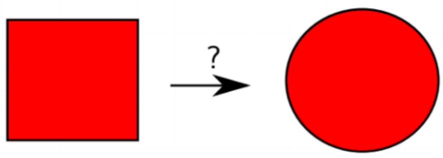


Figure 135.5: The central visual question: equal-area disk and square.

At first sight the boundary seems to obstruct the construction. A disk has circular boundary; a square has straight boundary. This intuition is correct for a stricter notion, but not for arbitrary equidecomposition.

§135.6 Scissors congruence and why nice pieces fail

The note distinguishes equidecomposability from scissors congruence. In the scissors-congruence version, the pieces must be tame: each piece is homeomorphic to a disk and has boundary given by curves of finite length. Under such restrictions, the disk and square are not scissors congruent.

The invariant mentioned in the source is:

$$\text{convex circular perimeter} - \text{concave circular perimeter}.$$

When a curved circular arc is created inside a dissection, it is paired with an opposite concave arc, so the internal contributions cancel. The remaining outer boundary contribution is preserved. A disk has a nonzero circular-boundary contribution, while a square has none. Hence a disk and square cannot be scissors congruent in this tame sense.

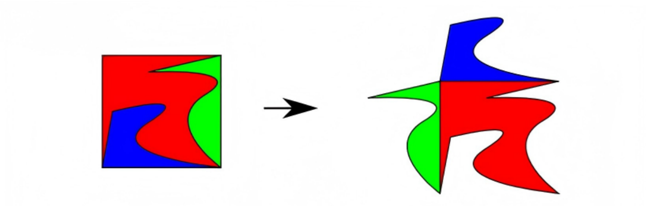


Figure 135.6: Tame scissors pieces preserve circular-boundary information.

This is the first major subtlety: the positive answer to Tarski's problem requires pieces much wilder than ordinary scissors pieces.

§135.7 Laczkovich's positive theorem

Laczkovich proved in 1990 that Tarski's circle-squaring problem has a positive answer. In 1992 he proved a broader theorem. A simplified statement is:

Theorem 135.7.1 (Laczkovich)
Let $A, B \subseteq \mathbb{R}^k$ be bounded sets with the same positive Lebesgue measure. If the boundaries of A and B have upper Minkowski dimension strictly less than k, then A and B are equidecomposable by translations.

For a disk and square in the plane, the boundaries are one-dimensional, so the hypothesis applies.

The proof does not draw a literal jigsaw. It converts the problem into a bounded matching problem in a graph generated by translations.

§135.8 First idea: work in the torus

Scale and translate A and B so that they lie inside $[0, 1)^k$. Identify this cube with the k-torus

$$\mathbb{T}^k = (\mathbb{R}/\mathbb{Z})^k.$$

Then A and B are equidecomposable by translations as subsets of the torus iff, after possibly using more pieces, they are equidecomposable by translations in $\mathbb{R}^k$.

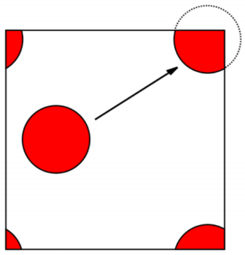


Figure 135.7: Working on the torus: translations wrap around the fundamental cube.

This torus reduction matters because translations become an action on a compact group. Compactness and periodic boundary conditions make the orbit structure manageable.

§135.9 Random translations and a free action

Choose a sufficiently large integer d and random translations

$$u_1, \ldots, u_d \in \mathbb{T}^k.$$

They define an action of $\mathbb{Z}^d$ on $\mathbb{T}^k$ by

$$(n_1, \ldots, n_d) \cdot x = n_1 u_1 + \cdots + n_d u_d + x.$$

For almost every choice of the u_i, this action is free: no nonzero element of $\mathbb{Z}^d$ fixes a point. Thus each orbit can be visualized as a copy of $\mathbb{Z}^d$.

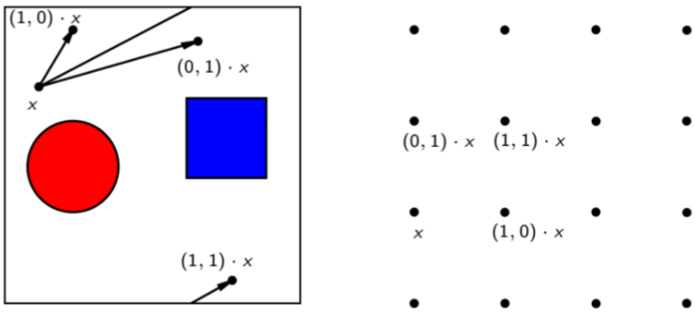


Figure 135.8: Random translations generate lattice-like orbits on the torus.

§135.10 The graph and bounded-distance bijection

Define a graph G with vertex set $\mathbb{T}^k$. Two points x, y are adjacent if there exists $g \in \mathbb{Z}^d$ with

$$g \cdot x = y, \qquad \|g\|_\infty = 1.$$

So edges correspond to single generator moves.

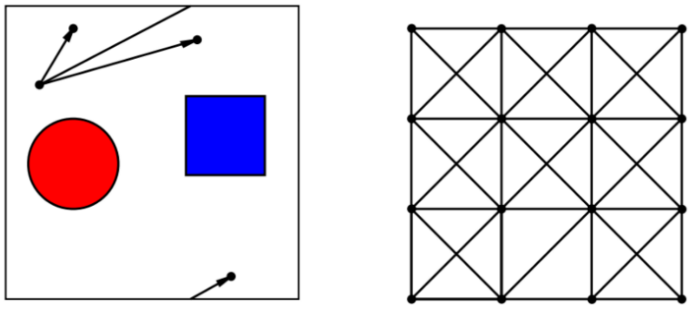


Figure 135.9: The generator graph: locally it looks like a lattice with bounded moves.

To show A and B are equidecomposable, it suffices to find a Borel bijection

$$f : A \to B$$

of bounded distance in G: there is some fixed N such that

$$d_G(x, f(x)) \le N \qquad \text{for all } x \in A.$$

If such an f exists, then only finitely many group elements g can occur. Put

$$A_g = \{x \in A : f(x) = g \cdot x\}.$$

The sets A_g partition A, and the translated sets $g \cdot A_g$ partition B. This gives the desired finite equidecomposition.

This is the conceptual heart of the proof:

$$\text{finite translation equidecomposition} \quad \Longleftarrow \quad \text{bounded-distance orbit matching.}$$

§135.11 Orbit matching

Inside a single orbit of the $\mathbb{Z}^d$ action, the problem becomes matching red A-points to blue B-points by bounded paths.

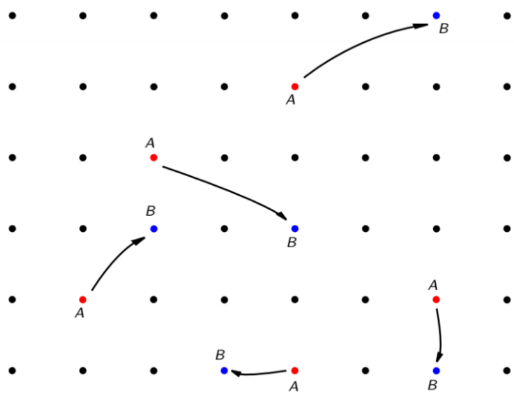


Figure 135.10: Within one orbit, equidecomposition becomes a matching problem.

A bounded matching can exist only if large finite regions of the orbit contain approximately the same number of A-points and B-points. This leads to discrepancy estimates.

§135.12 Discrepancy estimates

For an orbit point x, define a finite box

$$S_N(x) = \{(n_1, \dots, n_d) \cdot x : 0 \leq n_i < N\}.$$

Ergodic intuition suggests

$$|S_N(x) \cap A| \approx \lambda(A) N^d.$$

The handwritten note warns that this is only a heuristic: ordinary ergodic theorems alone are not strong enough. Laczkovich needs a uniform quantitative estimate from Diophantine approximation and discrepancy theory.

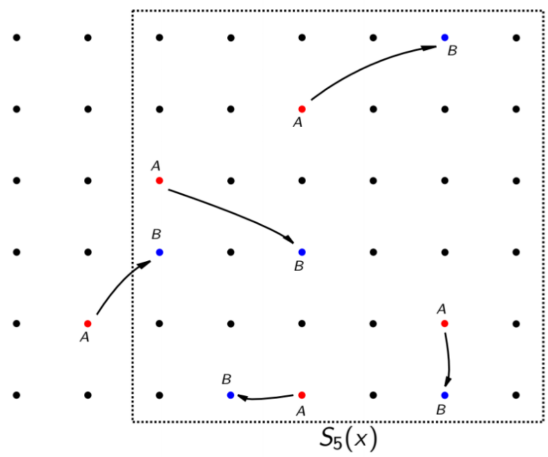


Figure 135.11: A finite orbit box $S_N(x)$; discrepancy compares observed counts with expected measure.

The key lemma states that there exist $\varepsilon > 0$ and M such that for all x and N,

$$\Big| |S_N(x) \cap A| - \lambda(A) N^d \Big| \leq M N^{d-1-\varepsilon},$$

and similarly for B:

$$\Big| |S_N(x) \cap B| - \lambda(B) N^d \Big| \leq M N^{d-1-\varepsilon}.$$

Since $\lambda(A) = \lambda(B)$, large boxes contain nearly equal numbers of A and B points. The error term grows more slowly than the volume term N^d, which is exactly what makes matching possible.

§135.13 From Laczkovich to Borel circle squaring

Laczkovich's original proof used the axiom of choice. Later work by Marks and Unger showed that the pieces can be chosen Borel. This is much stronger: the pieces are still extremely complicated, but they are definable in the Borel hierarchy.

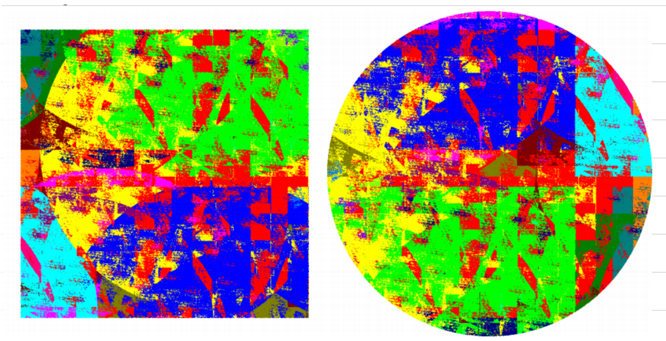


Figure 135.12: A visual impression of the Borel circle-squaring pieces.

The final handwritten note says the main components are: Dehn invariant, group action, torus embedding, traversal/matching, and measure/discrepancy theory. That summary is accurate. The proof is not one trick; it is a combination of several fields.

§135.14 Bridge to later ideas

The same question changes answer depending on the category of pieces. Plane polygons are controlled by area. Three-dimensional polyhedra require the Dehn invariant. Tame planar scissors pieces preserve circular-boundary data, so a disk cannot become a square. But arbitrary or Borel pieces are flexible enough for Laczkovich's theorem and Borel circle squaring.

Thus Tarski's circle-squaring problem is a crossroads of geometry, measure theory, group actions, discrepancy estimates, graph matching, and descriptive set theory. The important lesson is not only that the answer is yes, but why the yes lives far beyond ordinary scissors geometry.

§135.15 Scissors congruence and invariants

Polygon dissection problems ask whether one shape can be cut into finitely many pieces and rearranged into another. Area is an invariant, but in higher dimensions it is not enough; Hilbert's third problem needs the Dehn invariant.

§135.16 Measurable versus nonmeasurable decompositions

Tarski-style paradoxes show that arbitrary set-theoretic decompositions can violate geometric intuition. Requiring pieces to be measurable or Borel changes the problem drastically because measure becomes a meaningful invariant.

§135.17 Circle squaring and group actions

Laczkovich's theorem uses translations and carefully controlled pieces to equidecompose a disk and a square of equal area. The modern refinements ask for regularity of the pieces, such as Borel or measurable structure.

§135.18 Equidecomposition under measure and group action

Circle-squaring is not just a geometric puzzle. It sits at the meeting point of group actions, measure theory, regularity of sets, and invariants of dissection.

From Self-Duality to Spectral Curves: A Computable Route through Hitchin's Equations

N-20230801

§136.1 Why the story begins in four dimensions

Hitchin's equations live on a Riemann surface, but their form is much easier to remember once one sees the four-dimensional equation from which they descend. Let $P \to M$ be a principal bundle over an oriented Riemannian four-manifold, and let $\mathcal{A}$ be a connection. Its curvature is

$$F_{\mathcal{A}} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}.$$

The Hodge star acts on two-forms and satisfies $*^2 = 1$, so every curvature two-form splits as

$$F_{\mathcal{A}} = F_{\mathcal{A}}^+ + F_{\mathcal{A}}^-, \qquad *F_{\mathcal{A}}^{\pm} = \pm F_{\mathcal{A}}^{\pm}.$$

An anti-self-dual connection satisfies

$$F_{\mathcal{A}}^+ = 0, \qquad \text{equivalently} \qquad F_{\mathcal{A}} + *F_{\mathcal{A}} = 0.$$

This first-order equation is stronger than the Yang–Mills equation

$$d_{\mathcal{A}}^* F_{\mathcal{A}} = 0.$$

Indeed, the Bianchi identity $d_{\mathcal{A}} F_{\mathcal{A}} = 0$ and anti-self-duality give

$$d_{\mathcal{A}}^* F_{\mathcal{A}} = - * d_{\mathcal{A}} * F_{\mathcal{A}} = *d_{\mathcal{A}} F_{\mathcal{A}} = 0.$$

Thus the four-dimensional equation already combines geometry, gauge symmetry, and a variational problem. The two-dimensional Hitchin system appears when two directions are suppressed but their connection components are retained as fields.

§136.2 Dimensional reduction produces three coupled equations

Work locally on

$$M = \Sigma \times \mathbb{R}^2$$

with oriented coordinates (x, y, u, v), and assume that all fields are independent of u, v. Write the four-dimensional connection as

$$\mathcal{A} = A_x\, dx + A_y\, dy + \phi_1\, du + \phi_2\, dv.$$

The first two components form a connection

$$A = A_x\, dx + A_y\, dy$$

on Σ; the suppressed components ϕ_1, ϕ_2 become adjoint-valued functions on Σ.

The curvature components are now explicit:

$$F_{xy} = \partial_x A_y - \partial_y A_x + [A_x, A_y],$$

$$F_{xu} = D_x\phi_1, \qquad F_{xv} = D_x\phi_2,$$

$$F_{yu} = D_y\phi_1, \qquad F_{yv} = D_y\phi_2, \qquad F_{uv} = [\phi_1, \phi_2].$$

With the chosen orientation, the anti-self-duality equations are equivalent, up to the harmless global sign convention between self-dual and anti-self-dual, to

$$F_{xy} + [\phi_1, \phi_2] = 0,$$

$$D_x\phi_1 - D_y\phi_2 = 0, \qquad D_x\phi_2 + D_y\phi_1 = 0.$$

The first equation balances curvature against the commutator of the two reduced fields. The last two are a covariant Cauchy–Riemann system.

Introduce the complex coordinate

$$z = x + iy$$

and package the reduced fields into a Higgs field

$$\Phi = \frac{1}{2}(\phi_1 - i\phi_2)\, dz.$$

After choosing a Hermitian metric and writing Φ^{*h} for the adjoint, the three real equations become the two Hitchin equations

$$\boxed{F_A + [\Phi, \Phi^{*h}] = 0, \qquad \bar\partial_A\Phi = 0.}$$

Different normalizations of z, the metric, or the factor $1/2$ move constants between the two terms, but the structure is invariant: a real curvature equation and a complex holomorphicity equation.

§136.3 Gauge symmetry explains why the solution space is a quotient

A unitary gauge transformation g changes the pair by

$$A \longmapsto gAg^{-1} - dg\, g^{-1}, \qquad \Phi \longmapsto g\Phi g^{-1}.$$

Consequently,

$$F_A \longmapsto gF_Ag^{-1}, \qquad [\Phi, \Phi^{*h}] \longmapsto g[\Phi, \Phi^{*h}]g^{-1},$$

and

$$\bar\partial_A\Phi \longmapsto g(\bar\partial_A\Phi)g^{-1}.$$

The equations are therefore gauge invariant. Two solutions related by a gauge transformation describe the same geometric object, so the moduli space is not merely a zero set; it is a quotient

$$\mathcal{M}_H = \{(A, \Phi) : \text{Hitchin equations}\}/\mathcal{G}.$$

The quotient matters. A raw solution space contains whole gauge orbits, while stabilizers can create singular points. This is the first reason that deformation theory and stability cannot be postponed to an ornamental final section.

There is also a symplectic reading. The affine space of pairs (A, Φ) carries a natural hyperkähler structure, and the unitary gauge group acts by hyperkähler isometries. In one complex structure, the real moment map is

$$\mu_{\mathbb{R}}(A, \Phi) = F_A + [\Phi, \Phi^{*h}],$$

while the complex moment map is

$$\mu_{\mathbb{C}}(A, \Phi) = \bar{\partial}_A \Phi.$$

Thus Hitchin's moduli space is an infinite-dimensional hyperkähler quotient. The moment-map language does not replace the PDE; it explains why a metric equation, a holomorphic equation, and a quotient appear together.

§136.4 The complex equation turns the pair into a Higgs bundle

The operator $\bar{\partial}_A$ defines a holomorphic structure on the underlying complex bundle E. The equation

$$\bar{\partial}_A \Phi = 0$$

then says that

$$\Phi \in H^0(\Sigma, \mathrm{End}(E) \otimes K),$$

where K is the canonical bundle. A pair (E, Φ) of this kind is a Higgs bundle.

Not every holomorphic Higgs bundle corresponds to a solution of the real equation. The missing condition is stability. For a bundle E, define its slope by

$$\mu(E) = \frac{\deg E}{\operatorname{rk} E}.$$

A Higgs bundle is stable when every proper nonzero holomorphic subbundle $F \subset E$ satisfying

$$\Phi(F) \subset F \otimes K$$

obeys

$$\mu(F) < \mu(E).$$

Only Φ-invariant subbundles are tested, because a subbundle ignored by Φ is not a subsystem of the Higgs pair.

The Hitchin–Kobayashi correspondence says, in the appropriate degree-zero setting, that a polystable Higgs bundle admits a Hermitian metric solving

$$F_A + [\Phi, \Phi^{*h}] = 0.$$

The analytic quotient by unitary gauge and the algebraic quotient by complex gauge are therefore two descriptions of the same well-behaved moduli problem. Stability is not an extra adjective attached after the equations; it is the algebraic condition that replaces the missing compactness of complex gauge orbits.

§136.5 A local rank-two ansatz turns the matrix equation into one scalar PDE

The real equation becomes concrete in rank two. Choose a local holomorphic frame in which

$$\Phi = \begin{pmatrix} 0 & Q \\ 1 & 0 \end{pmatrix} dz$$

for a local quadratic-differential coefficient Q, and choose a diagonal Hermitian metric

$$h = \begin{pmatrix} e^{-u} & 0 \\ 0 & e^{u} \end{pmatrix}.$$

The adjoint of the matrix part of Φ is

$$\Phi^{*h} = \begin{pmatrix} 0 & e^{2u} \\ \overline{Q}e^{-2u} & 0 \end{pmatrix} d\bar{z}.$$

Hence

$$[\Phi, \Phi^{*h}] = \begin{pmatrix} |Q|^2 e^{-2u} - e^{2u} & 0 \\ 0 & e^{2u} - |Q|^2 e^{-2u} \end{pmatrix} dz \wedge d\bar{z}.$$

The Chern curvature of the diagonal metric is

$$F_h = \begin{pmatrix} -\partial\bar{\partial}u & 0 \\ 0 & \partial\bar{\partial}u \end{pmatrix}.$$

Therefore the first Hitchin equation reduces, with this convention, to the scalar equation

$$\boxed{u_{z\bar{z}} + e^{2u} - |Q|^2 e^{-2u} = 0.}$$

An overall factor changes if the area form is normalized differently, but the competition between the two exponentials is intrinsic. Away from zeros of Q, a shift of u converts the equation into a sinh–Gordon type equation. The abstract matrix PDE has become a nonlinear scalar equation whose singular behavior is tied to the zeros of the spectral data.

§136.6 A solution also produces a family of flat connections

The Hitchin equations connect Higgs bundles to representations because they imply flatness of a one-parameter family. For $\lambda \in \mathbb{C}^*$, define

$$\nabla_\lambda = D_A + \lambda^{-1}\Phi + \lambda\Phi^{*h}.$$

Its curvature contains three different powers of λ:

$$F_{\nabla_\lambda} = \lambda^{-1} D_A^{0,1}\Phi + (F_A + [\Phi, \Phi^{*h}]) + \lambda D_A^{1,0}\Phi^{*h}.$$

The complex equation kills the λ^{-1} term, its adjoint kills the λ term, and the real equation kills the middle term. Thus

$$F_{\nabla_\lambda} = 0$$

for every $\lambda \neq 0$.

A flat connection has monodromy, giving a representation of the fundamental group. This is the operational core of nonabelian Hodge theory:

$$\begin{aligned}\text{harmonic bundle} &\longleftrightarrow \text{polystable Higgs bundle},\\ \text{flat connection} &\longleftrightarrow \text{representation of } \pi_1(\Sigma).\end{aligned}$$

The arrows are not merely analogies. The harmonic metric is what combines a holomorphic Higgs pair with its adjoint to manufacture the flat family.

§136.7 The Hitchin map records global eigenvalue data

For a rank-n Higgs field, the coefficients of

$$\det(\eta I - \Phi) = \eta^n + a_1\eta^{n-1} + \cdots + a_n$$

are invariant under conjugation. Since Φ is a K-valued endomorphism,

$$a_i \in H^0(\Sigma, K^i).$$

They define the Hitchin map

$$h : \mathcal{M}_H \longrightarrow \bigoplus_{i=1}^{n} H^0(\Sigma, K^i).$$

For trace-free rank two, the characteristic polynomial is

$$\eta^2 - q, \qquad q \in H^0(\Sigma, K^2),$$

so the Hitchin base is simply $H^0(K^2)$.

The equation

$$\eta^2 = q$$

defines a curve $\widetilde{\Sigma}$ inside the total space of K. Here η is the tautological one-form: at a point $(x, \xi) \in K$, its value is the cotangent vector ξ. Over a point where $q(x) \neq 0$, the two square roots of $q(x)$ are the two eigenvalues of $\Phi(x)$. At a simple zero of q, the sheets meet and the projection

$$\pi : \widetilde{\Sigma} \to \Sigma$$

branches. The spectral curve is therefore the global replacement for diagonalizing a matrix point by point.

§136.8 A global rank-two model requires a theta characteristic

The local companion matrix contains entries of different bundle types, so it should not be placed globally on $\mathcal{O} \oplus \mathcal{O}$. Choose instead a theta characteristic

$$S^2 \cong K$$

and set

$$E = S \oplus S^{-1}.$$

Then the Higgs field

$$\Phi_q = \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix} : E \longrightarrow E \otimes K$$

is globally well defined. Indeed, the lower-left entry belongs to

$$\mathrm{Hom}(S, S^{-1} \otimes K) \cong S^{-2} \otimes K \cong \mathcal{O},$$

while the upper-right entry belongs to

$$\mathrm{Hom}(S^{-1}, S \otimes K) \cong S^{2} \otimes K \cong K^{2}.$$

The determinant is trivial, the trace is zero, and

$$\det(\eta I - \Phi_q) = \eta^2 - q.$$

This is the rank-two Hitchin section. The bundle bookkeeping is not cosmetic: it is what turns a local matrix mnemonic into a global Higgs bundle.

§136.9 A genus-two spectral curve can be calculated completely

Take the genus-two hyperelliptic curve

$$\Sigma : \quad y^2 = (x^2 - 1)(x^2 - 4)(x^2 - 9).$$

A basis of holomorphic one-forms is

$$\omega_0 = \frac{dx}{y}, \qquad \omega_1 = x\frac{dx}{y}.$$

Consider the quadratic differential

$$q = (x^2 - 2)\left(\frac{dx}{y}\right)^2.$$

It is holomorphic. Near either point at infinity, with $t = 1/x$, one has

$$\frac{dx}{y} \sim -t\,dt, \qquad x^2 - 2 \sim t^{-2},$$

so q is asymptotic to a nonzero multiple of dt^2.

The zeros occur at $x = \pm\sqrt{2}$. These are not branch values of the hyperelliptic map, so each value of x gives two points of Σ. Thus q has four simple zeros, matching

$$\deg K^2 = 4g - 4 = 4.$$

The spectral curve is

$$\widetilde{\Sigma} : \quad \eta^2 = q.$$

It is a double cover of Σ branched at those four zeros. Riemann–Hurwitz gives

$$2g(\widetilde{\Sigma}) - 2 = 2\big(2g(\Sigma) - 2\big) + 4 = 2 \cdot 2 + 4 = 8,$$

so

$$\boxed{g(\widetilde{\Sigma}) = 5.}$$

The slogan "the spectral curve is a branched cover" has now produced a concrete genus.

§136.10 Line bundles on the spectral curve reconstruct the Higgs field

Let $\mathcal{L}$ be a line bundle on $\widetilde{\Sigma}$. Pushing forward gives a rank-two bundle

$$E = \pi_* \mathcal{L}.$$

Multiplication by the tautological eigenvalue defines

$$\eta : \mathcal{L} \longrightarrow \mathcal{L} \otimes \pi^* K.$$

After applying π_* and the projection formula,

$$\pi_*(\mathcal{L} \otimes \pi^* K) \cong (\pi_* \mathcal{L}) \otimes K,$$

this becomes a Higgs field

$$\Phi : E \longrightarrow E \otimes K.$$

Thus the nonabelian rank-two object is reconstructed from abelian line-bundle data on the eigenvalue cover.

For fixed determinant, the line bundle is constrained by a norm condition. The generic Hitchin fiber is the Prym variety

$$\mathrm{Prym}(\widetilde{\Sigma}/\Sigma) = \ker\Big(\mathrm{Nm} : \mathrm{Jac}(\widetilde{\Sigma}) \to \mathrm{Jac}(\Sigma)\Big)^0 .$$

Its dimension is

$$g(\widetilde{\Sigma}) - g(\Sigma) = 5 - 2 = 3.$$

On the other hand,

$$\dim H^0(\Sigma, K^2) = 3g - 3 = 3.$$

The base and generic fiber therefore have equal complex dimension. This is the dimension count behind algebraic complete integrability: the generic Prym fibers are half-dimensional Lagrangian abelian varieties.

§136.11 The deformation complex computes the tangent space

To understand a moduli space locally, vary both the holomorphic structure and the Higgs field. An infinitesimal deformation is a pair

$$(a, \varphi) \in \Omega^{0,1}(\mathrm{End}_0 E) \oplus \Omega^{1,0}(\mathrm{End}_0 E).$$

An infinitesimal complex gauge transformation $s \in \Omega^0(\mathrm{End}_0 E)$ changes the pair by

$$D_0 s = (\bar{\partial}_E s, [\Phi, s]).$$

Linearizing the holomorphicity equation gives

$$D_1(a, \varphi) = \bar{\partial}_E \varphi + [a, \Phi].$$

Because $\bar{\partial}_E \Phi = 0$, one has $D_1 D_0 = 0$. The infinitesimal moduli problem is therefore controlled by the complex

$$\Omega^0(\mathrm{End}_0 E) \xrightarrow{D_0} \Omega^{0,1}(\mathrm{End}_0 E) \oplus \Omega^{1,0}(\mathrm{End}_0 E) \xrightarrow{D_1} \Omega^{1,1}(\mathrm{End}_0 E).$$

Equivalently, in holomorphic language one uses the two-term complex

$$\mathcal{C}^\bullet : \operatorname{End}_0 E \xrightarrow{[\Phi,\cdot]} \operatorname{End}_0 E \otimes K.$$

Its hypercohomology has a direct interpretation:

$$\mathbb{H}^0(\mathcal{C}^\bullet) = \text{infinitesimal automorphisms},$$

$$\mathbb{H}^1(\mathcal{C}^\bullet) = \text{tangent space}, \qquad \mathbb{H}^2(\mathcal{C}^\bullet) = \text{obstructions}.$$

For a stable trace-free Higgs bundle, the automorphism space is zero and, by duality, the obstruction space vanishes at a smooth point. Hence the tangent dimension is minus the Euler characteristic of the complex.

Since $\operatorname{End}_0 E$ has rank 3 and degree 0, Riemann–Roch gives

$$\chi(\operatorname{End}_0 E) = 3(1-g).$$

Also,

$$\chi(\operatorname{End}_0 E \otimes K) = 3(2g-2) + 3(1-g) = 3(g-1).$$

Therefore

$$\chi(\mathcal{C}^\bullet) = 3(1-g) - 3(g-1) = -6(g-1),$$

and

$$\boxed{\dim_{\mathbb{C}} T_{(E,\Phi)}\mathcal{M}_H = 6g-6.}$$

For genus two this is 6, exactly the sum of the three-dimensional Hitchin base and the three-dimensional Prym fiber. The local deformation calculation and the global spectral description meet at the same dimension.

The agreement is more than a dimension check. It shows how the different descriptions solve different parts of the same problem. Four-dimensional anti-self-duality explains the origin of the equations. Dimensional reduction identifies the Higgs field as connection data in suppressed directions. Gauge theory supplies the real moment-map equation and the analytic quotient. Stability produces a workable algebraic quotient. The flat family records monodromy and representations. The Hitchin map extracts invariant eigenvalue data, while the spectral curve converts a nonabelian bundle into a line bundle on a branched cover. Finally, the deformation complex determines whether the quotient is smooth and computes its tangent dimension.

None of these viewpoints is an optional advanced appendix. The local scalar equation explains what the PDE asks a metric to do; the genus-two cover makes the spectral construction calculable; and the equality

$$6g - 6 = (3g-3) + (3g-3)$$

shows why the Hitchin fibration has exactly the right size to be an integrable system.

Plane Curves as Algebraic Creatures: Equations, Rings, and Singularities

N-20240608

§137.1 The curve is not only the drawing

A plane curve often begins as a picture. Algebraic geometry asks for something more durable: an equation.

Given a polynomial $f(x, y)$, define

$$C = V(f) = \{(x, y) : f(x, y) = 0\}.$$

The drawing of C may be helpful, but the equation controls its functions, singularities, and intersections.

A curve is therefore a small laboratory where geometry and algebra constantly translate into each other.

§137.2 The function ring of a curve

For $C = V(f)$, define

$$k[C] = k[x, y]/(f).$$

This is the coordinate ring of the curve. Two polynomials define the same function on C if their difference is a multiple of f.

For the parabola $C = V(y - x^2)$,

$$k[C] = k[x, y]/(y - x^2) \cong k[x].$$

So although the parabola is drawn curved, its polynomial function theory is the same as that of a line.

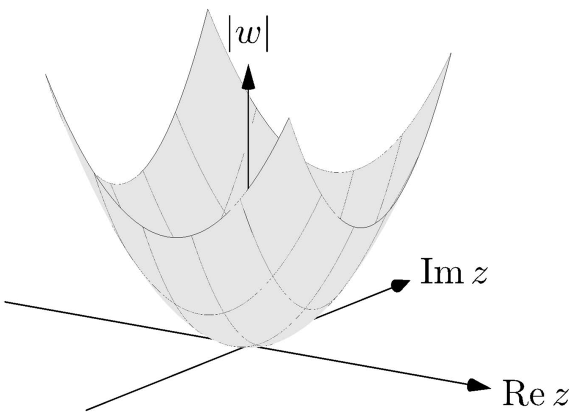


Figure 137.1: A parabola is geometrically curved in the plane, but its coordinate ring is as simple as $k[x]$.

§137.3 When factorization becomes geometry

If

$$f = gh,$$

then

$$V(f) = V(g) \cup V(h).$$

Thus factorization of equations corresponds to decomposition of curves.

Example:

$$xy = 0$$

is the union of the two coordinate axes. The equation is reducible, and the curve is reducible.

This is the first piece of the algebra-geometry dictionary: algebraic factorization can mean geometric splitting.

§137.4 Smooth points and singular points

A point $p = (a, b) \in V(f)$ is singular if

$$f(a, b) = 0, \qquad f_x(a, b) = 0, \qquad f_y(a, b) = 0.$$

At such a point, the linear part of the equation vanishes. The usual tangent line no longer comes from the first derivative.

The cusp

$$y^2 = x^3$$

has a singularity at the origin. The node

$$y^2 = x^2(x + 1)$$

also has a singularity at the origin, but it looks like two branches crossing. Both are singular, but they are not singular in the same way.

§137.5 Tangent cones: the first nonzero shadow

When the linear part vanishes, look at the lowest-degree nonzero part of f. If it is f_d, then

$$f_d = 0$$

is the tangent cone.

For

$$f(x, y) = y^2 - x^2 - x^3,$$

the lowest-degree part is

$$y^2 - x^2 = (y - x)(y + x).$$

So the tangent cone at the origin consists of the two lines

$$y = x, \qquad y = -x.$$

This detects the two branches of the node.

§137.6 The cusp as a ring

For the cusp $C = V(y^2 - x^3)$, the parametrization

$$x = t^2, \qquad y = t^3$$

gives a ring map

$$k[x, y]/(y^2 - x^3) \longrightarrow k[t], \qquad x \mapsto t^2, \quad y \mapsto t^3.$$

Its image is $k[t^2, t^3]$, not all of $k[t]$. This failure to become the full polynomial ring reflects the singularity. The cusp is close to a line, but not quite a smooth line.

This example is more memorable than the definition: singularities change the algebra of functions.

§137.7 Intersections: one visible point may count twice

The line $y = 0$ and the parabola $y = x^2$ meet at the origin. Substituting $y = 0$ into $y = x^2$ gives

$$x^2 = 0.$$

The intersection has multiplicity 2. Geometrically, the line is tangent to the parabola.

This explains why intersection theory counts multiplicities. A tangent intersection is not the same as a transverse crossing.

§137.8 Projective closure repairs missing intersections

Affine curves may hide points at infinity. Homogenization fills them in. For

$$f(x, y) = y - x^2,$$

the homogenization is

$$F(x, y, z) = yz - x^2.$$

The projective closure is $V(F) \subset \mathbb{P}^2$. In the chart $z = 1$ it is the original parabola. At infinity $z = 0$, it contains the point

$$[0 : 1 : 0].$$

This is the same philosophy as projective geometry: complete the affine picture so that geometry stops losing information at the boundary of the chart.

§137.9 Bezout as a promise of clean accounting

A first form of Bezout's theorem says that two projective plane curves of degrees m and n meet in mn points, counted with multiplicity, over an algebraically closed field, provided they have no common component.

One should not first treat this as a formula to memorize. It is a statement that projective geometry and multiplicity create the correct accounting system for intersections.

§137.10 Normalization: repairing a singular curve

The cusp $y^2 = x^3$ is parametrized by

$$x = t^2, \qquad y = t^3.$$

The parameter line with coordinate t is smooth. The map from the line to the cusp is almost one-to-one, but at the cusp it repairs the singularity by replacing the bad point with a smooth parameter.

Algebraically, the coordinate ring of the cusp is

$$k[t^2, t^3] \subset k[t].$$

Passing to $k[t]$ is a normalization. One can think of it as adding the missing function t, which the singular coordinate ring did not contain.

This is a useful slogan:

normalization separates or smooths hidden branches of a curve.

For a node, normalization separates the two crossing branches. For a cusp, it smooths a single branch with a bad tangent behavior.

§137.11 Resultants: eliminating a variable

Intersections can also be studied by eliminating variables. If two curves are given by

$$f(x, y) = 0, \qquad g(x, y) = 0,$$

one may eliminate y to get a polynomial condition on x. This is the idea behind the resultant.

For example, intersect

$$y = x^2, \qquad y = mx + b.$$

Substitution gives

$$x^2 - mx - b = 0.$$

The discriminant

$$m^2 + 4b$$

tells whether the line meets the parabola in two distinct points, one tangent point, or no real points. Over an algebraically closed field, the quadratic always has two roots counted with multiplicity.

This elementary calculation is a small preview of Bezout's theorem.

§137.12 A projective calculation with a line and a conic

Take the conic

$$xz - y^2 = 0$$

and the line

$$z = 0.$$

Substituting $z = 0$ into the conic gives

$$-y^2 = 0.$$

Thus $y = 0$, and the intersection point is

$$[1 : 0 : 0].$$

The square indicates multiplicity 2. Geometrically, the line at infinity is tangent to this projective conic.

This example is compact but important: projective closure not only adds the missing point; it also remembers how the curve touches the line at infinity.

§137.13 Real pictures and algebraically closed accounting

Over $\mathbb{R}$, the circle

$$x^2 + y^2 + 1 = 0$$

has no real points. Over $\mathbb{C}$, it has many points. This is why algebraic geometry often works over algebraically closed fields when stating clean intersection theorems.

Real drawings are valuable, but they can hide complex points. Bezout's theorem counts in the algebraic world where equations have all their roots.

§137.14 How to read a plane curve from its equation

For the examples in this note, the useful dictionary is concrete:

equation	zero set being studied
factorization	decomposition into components
coordinate ring	polynomial functions restricted to the curve
vanishing of F_x, F_y	possible singular point
homogenization	points added on the line at infinity

Thus a quick analysis of a plane curve should not stop after drawing its real graph. One first factors the polynomial if possible, then solves $F = F_x = F_y = 0$ to find singularities, and finally homogenizes if an intersection count looks wrong in the affine plane. The line–conic example shows why this last step matters: an intersection can disappear to infinity, and over $\mathbb{R}$ it can also be hidden by the absence of real points. Bezout's theorem becomes reliable only after the curve is viewed projectively and, when necessary, over an algebraically closed field.

Divisors on Curves: The Bookkeeping of Zeros, Poles, and Functions

N-20240914

§138.1 A ledger for meromorphic functions

A meromorphic function on a compact Riemann surface has zeros and poles. Instead of listing them in prose, we record them in a divisor:

$$D = \sum_p n_p[p].$$

Positive coefficients record zeros or permissions. Negative coefficients record poles or debts.

This is the central metaphor of the chapter:

a divisor is a ledger for allowed behavior at points.

§138.2 Principal divisors: the ledger of one function

For a nonzero meromorphic function f, its principal divisor is

$$(f) = \sum_p \mathrm{ord}_p(f)[p].$$

On the Riemann sphere, the function $z - a$ has divisor

$$(z - a) = [a] - [\infty].$$

The function

$$\frac{z-a}{z-b}$$

has divisor

$$\left(\frac{z-a}{z-b}\right) = [a] - [b].$$

Zeros and poles balance:

$$\deg(f) = 0.$$

This is the simplest global constraint on rational functions.

§138.3 Reading a divisor as permission

Given a divisor D, define

$$L(D) = \{f \text{ meromorphic} : (f) + D \geq 0\} \cup \{0\}.$$

This is the vector space of functions whose poles are no worse than D allows.

For example, on $\mathbb{CP}^1$, take

$$D = n[\infty].$$

Then $L(D)$ consists of polynomials of degree at most n:

$$1, z, z^2, \ldots, z^n.$$

So

$$\ell(D) = n + 1.$$

This example should be kept close. It is the friendly face of the general theory.

§138.4 Line bundles: permission becomes geometry

A line bundle is a family of one-dimensional vector spaces varying over a curve. Locally it looks like

$$U \times \mathbb{C},$$

but the local pieces may glue nontrivially.

A divisor D determines a line bundle $\mathcal{O}(D)$. Its sections are the geometric version of functions satisfying the permission rule encoded by D.

Thus divisors are not just notation for zeros and poles. They produce spaces of sections, and those spaces are the objects one can count.

§138.5 The canonical divisor

Holomorphic 1-forms also have zeros. Their divisor class is called the canonical divisor K. For a compact Riemann surface of genus g,

$$\deg K = 2g - 2.$$

The formula already links analysis and topology. A differential form knows something about the number of handles of the surface.

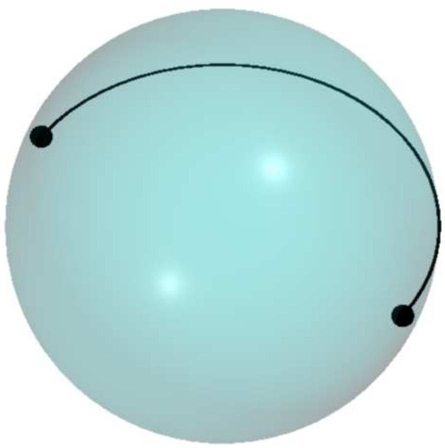


Figure 138.1: A curve is simultaneously analytic and topological; divisors help transfer information between these views.

§138.6 Riemann–Roch without intimidation

The theorem says
$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$
It counts the functions allowed by D, with a correction term involving the canonical divisor.

On $\mathbb{CP}^1$, this becomes the elementary fact that polynomials of degree at most n form an $(n + 1)$-dimensional space. In that case $g = 0$, and for $D = n[\infty]$ with $n \geq 0$, the correction term disappears.

So Riemann–Roch is deep, but its first example is not mysterious. It extends a familiar polynomial dimension count to arbitrary compact Riemann surfaces.

§138.7 Hurwitz as topology with branch points

If $f : X \to Y$ is a nonconstant holomorphic map of compact Riemann surfaces of degree d, then
$$2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X} (e_p - 1).$$
The correction term records ramification. Locally, ramification means the map looks like
$$z \mapsto z^{e_p}.$$
Thus the formula says that global topology changes according to local branching.

§138.8 A divisor game on the sphere

Let us play the simplest possible game. On $\mathbb{CP}^1$, ask for meromorphic functions with at most a double pole at infinity and no other poles. The answer is
$$L(2[\infty]) = \mathrm{span}\{1, z, z^2\}.$$
If we also require a zero at 0, we are asking for
$$L(2[\infty] - [0]).$$
The allowed functions are now
$$az + bz^2,$$
so the dimension drops from 3 to 2.

This is how divisors should feel: adding a pole allowance increases freedom, while imposing a zero condition removes freedom.

§138.9 Degree as a first estimate

The degree of a divisor
$$D = \sum_p n_p[p]$$
is
$$\deg D = \sum_p n_p.$$

A positive degree suggests more allowed poles than required zeros, so one expects more functions. A negative degree usually means the conditions are too strict.

This intuition is made precise in many cases. On a compact Riemann surface, if $\deg D < 0$, then

$$L(D) = 0.$$

Indeed, a nonzero meromorphic function has principal divisor of degree 0, so $(f)+D \geq 0$ would force a nonnegative divisor of negative total degree, impossible.

§138.10 Canonical divisors through examples

On $\mathbb{CP}^1$, the differential dz has a double pole at infinity. In the coordinate $w = 1/z$,

$$dz = -w^{-2}\, dw.$$

Thus the canonical divisor has degree -2, matching

$$2g - 2 = -2$$

for genus 0.

On a torus, $g = 1$, so $\deg K = 0$. This matches the existence of a nowhere-vanishing holomorphic differential. The formula $\deg K = 2g - 2$ is therefore not merely symbolic; it reflects the geometry of different surfaces.

§138.11 Why Riemann–Roch has a correction term

The naive guess would be that the dimension of $L(D)$ is roughly

$$\deg D + 1 - g.$$

Riemann–Roch says this guess is corrected by

$$\ell(K - D).$$

This correction measures hidden differentials compatible with the opposite divisor condition. It is the reason the theorem is both a counting formula and a duality statement.

The formula therefore has two personalities:

$$\begin{array}{r|c} \text{counting side} & \ell(D) \text{ counts functions} \\ \text{duality side} & \ell(K-D) \text{ counts differential obstructions} \end{array}$$

This correction term returns the discussion to the original divisor problem. A zero is an imposed local condition, a pole is controlled permission, and the global question is which meromorphic functions can satisfy the entire ledger at once. Riemann–Roch answers that question by balancing the expected degree count against the differential obstruction space $L(K - D)$.

Zariski Geometry: When Functions Become the Space

N-20241026

§139.1 Starting from functions, not pictures

In elementary geometry, one first draws a space and then studies functions on it. Algebraic geometry often reverses the direction: start with polynomial functions, and let the space be what those functions define.

Given an ideal

$$I \subset k[x_1, \ldots, x_n],$$

we define the zero set

$$V(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}.$$

Conversely, a set $X \subset k^n$ has an ideal of functions vanishing on it:

$$I(X) = \{f : f(x) = 0 \text{ for all } x \in X\}.$$

This back-and-forth is the first algebra-geometry dictionary.

§139.2 Coordinate rings: a space through its functions

If $X = V(I)$, the coordinate ring is

$$k[X] = k[x_1, \ldots, x_n]/I(X).$$

It is the ring of polynomial functions on X.

For the parabola $X = V(y - x^2)$,

$$k[X] = k[x, y]/(y - x^2) \cong k[x].$$

For the hyperbola $X = V(xy - 1)$,

$$k[X] = k[x, y]/(xy - 1) \cong k[x, x^{-1}].$$

The hyperbola is algebraically the affine line with the origin removed. The drawing is curved, but the function ring reveals a simpler identity.

§139.3 Nullstellensatz: the dictionary becomes reliable

Over an algebraically closed field, Hilbert's Nullstellensatz says

$$I(V(I)) = \sqrt{I}.$$

This means that polynomial equations and radical ideals determine each other.

The theorem is not just a technical result. It is a guarantee that algebraic information can faithfully describe geometric zero sets, once nilpotent ambiguity is removed.

§139.4 The Zariski topology is coarse on purpose

Closed sets in the Zariski topology are algebraic sets. On the affine line, proper closed sets are finite. Therefore nonempty open sets are very large.

This topology is not designed to model physical closeness. It is designed to model algebraic dependence. If a polynomial vanishes on a large enough open set, it is forced to vanish much more widely than a continuous function would be.

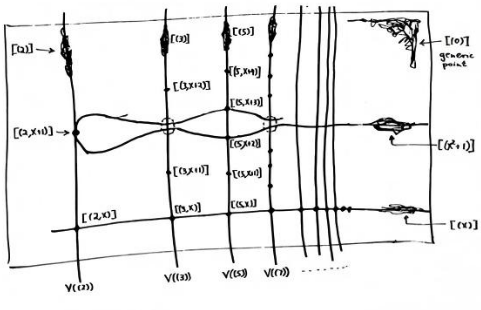


Figure 139.1: Pictures can guide intuition, but the real geometry is carried by equations and functions.

§139.5 Distinguished opens and localization

For $f \in k[X]$, the distinguished open set $D(f)$ is the region where $f \neq 0$. On this region, f should be invertible.

For example, in $\mathbb{A}^1$, the open set $D(x)$ has ring of functions

$$k[x, x^{-1}].$$

This is localization. It is the algebraic operation of allowing denominators that do not vanish on the open set.

This idea is one of the most important technical moves in algebraic geometry: local geometry is controlled by localized rings.

§139.6 Sheaves: local data with a gluing rule

Regular functions are local. A function may be represented by different fractions on different open sets, as long as they agree on overlaps.

A sheaf formalizes this behavior. If a space is covered by opens U_i, and functions f_i on U_i agree on every overlap $U_i \cap U_j$, then they glue to a unique function on the union.

This is not abstraction for its own sake. It is the ordinary behavior of functions, stated carefully enough to work in algebraic geometry.

§139.7 Stalks: looking through a microscope

The stalk at a point stores the local behavior near that point. For regular functions, a stalk consists of germs of functions: two functions are the same germ if they agree on some neighborhood of the point.

In algebraic geometry, stalks are often local rings. They are the algebraic microscopes through which one studies smoothness, singularities, and local equations.

§139.8 A local example: two axes crossing

Consider the curve

$$X = V(xy) \subset \mathbb{A}^2.$$

Geometrically it is the union of the two coordinate axes. Algebraically its coordinate ring is

$$k[X] = k[x, y]/(xy).$$

Away from the origin, each branch looks like an ordinary line. At the origin, the two branches meet, and the local ring remembers both directions.

On the open set $D(x)$, the element x is invertible. Since $xy = 0$, multiplying by x^{-1} gives $y = 0$. Thus on $D(x)$, only the x-axis branch remains. Similarly, on $D(y)$, only the y-axis branch remains.

This is a small but useful lesson: localization is not just algebraic manipulation. It literally focuses attention on the part of the variety where a chosen function does not vanish.

§139.9 Why the strange topology helps

The Zariski topology feels strange because its open sets are large. But large opens are exactly what make polynomial continuation possible. If two regular functions agree on a nonempty open subset of an irreducible variety, they often must agree much more widely.

This is why the topology is useful for algebraic geometry even though it is poor for measuring distance. It is built for rigidity, not for analysis.

§139.10 Products and unions through rings

Algebraic operations on rings often reverse geometric operations. The product of equations gives a union:

$$V(fg) = V(f) \cup V(g).$$

The sum of ideals gives an intersection:

$$V(I + J) = V(I) \cap V(J).$$

This reversal is a persistent theme. Geometry and algebra face opposite directions: more equations mean smaller spaces, while larger ideals mean fewer points.

For instance, in $k[x, y]$, the ideals (x) and (y) define the two coordinate axes. Their sum (x, y) defines the origin, the intersection of the two axes.

§139.11 Radical ideals and forgotten nilpotents

The ideals (x) and (x^2) have the same classical zero set in $\mathbb{A}^1$: both vanish only at 0. But the quotient rings differ:

$$k[x]/(x), \qquad k[x]/(x^2).$$

The second ring contains a nonzero nilpotent element.

Classical varieties often pass to radical ideals and forget nilpotents. Schemes later restore them. This is one reason the transition from varieties to schemes is not artificial: nilpotent structure is already waiting in simple equations.

§139.12 Regular functions versus continuous functions

On a usual topological space, continuous functions are flexible. They can be changed locally with bump functions. Regular functions in algebraic geometry are rigid. They are controlled by polynomial or rational formulas.

This rigidity is why algebraic geometry can extract strong global conclusions from local algebra. The price is that the topology becomes coarse and unintuitive from an analytic point of view.

§139.13 Morphisms as pullback of functions

A map of affine varieties

$$F : X \to Y$$

induces a pullback of coordinate rings

$$F^* : k[Y] \to k[X].$$

A function on Y becomes a function on X by composition with F.

This reverses arrows:

$$X \longrightarrow Y \qquad \rightsquigarrow \qquad k[Y] \longrightarrow k[X].$$

This arrow reversal is one of the conceptual gates into modern algebraic geometry. Spaces are studied through their function rings, and maps of spaces become maps of rings in the opposite direction.

§139.14 A compact dictionary

The essential translations are:

geometric phrase	algebraic phrase
zero set	$V(I)$
functions on X	$k[X]$
remove $f = 0$	localize at f
local behavior at p	stalk at p
glue local functions	sheaf condition

The point of Zariski geometry is not to make topology strange. It is to let functions define the correct topology for polynomial geometry.

Schemes Gently: A Guided Tour of Spectrum, Nilpotents, and Local Rings

N-20250118

§140.1 A new kind of point

Classical algebraic geometry studies points satisfying polynomial equations. Scheme theory asks us to enlarge the meaning of point.

For a ring R, define

$$\operatorname{Spec} R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is prime}\}.$$

This is the spectrum of R. Its points are prime ideals.

At first this feels like a category mistake: ideals are algebraic objects, not points. But this change is exactly what allows geometry to remember generic behavior, arithmetic information, and infinitesimal thickening.

§140.2 The affine line spectrum

For $R = k[x]$, the maximal ideals

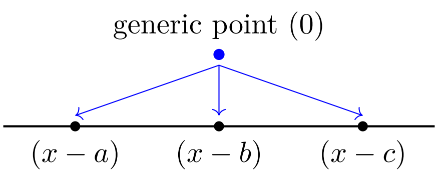
$$(x - a)$$

correspond to ordinary points $a \in k$. But there is also the prime ideal (0), the generic point.

The closure of the generic point is the whole space:

$$\overline{\{(0)\}} = \operatorname{Spec} k[x].$$

This is not Hausdorff behavior, but it is meaningful. The generic point represents the whole irreducible line, while $(x - a)$ represents a closed point.



The arrows are suggestive: the generic point specializes to closed points.

§140.3 Closed sets and distinguished opens

For an ideal $I \subset R$, define

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} R : I \subseteq \mathfrak{p}\}.$$

These are closed sets.

For $f \in R$, define

$$D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}.$$

This is the region where f is allowed to be inverted. Distinguished opens are the basic open sets of affine schemes.

§140.4 Localization is zooming in

At a prime ideal $\mathfrak{p}$, the local ring is

$$R_{\mathfrak{p}}.$$

It is obtained by inverting every element not in $\mathfrak{p}$. The interpretation is simple: near $\mathfrak{p}$, functions not vanishing there should be units.

For $R = k[x]$ and $\mathfrak{p} = (x - a)$, the ring

$$k[x]_{(x-a)}$$

contains rational functions whose denominators do not vanish at a. This is the algebraic version of functions defined near a.

§140.5 The structure sheaf

A spectrum is not only a topological space. It carries functions. On a distinguished open $D(f)$, the structure sheaf satisfies

$$\mathcal{O}_{\operatorname{Spec} R}(D(f)) = R_f.$$

The stalk at $\mathfrak{p}$ is

$$\mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}} = R_{\mathfrak{p}}.$$

These two formulas are enough to make affine schemes computational rather than mystical.

§140.6 Nilpotents: invisible but geometric

Consider

$$R = k[x]/(x^2).$$

The element x is nonzero but satisfies $x^2 = 0$. Topologically, $\operatorname{Spec} R$ has only one point. Algebraically, it remembers an infinitesimal thickening of that point.

A useful picture is a point with a first-order tangent fuzz. The fuzz cannot be seen as a second classical point, but it can be detected by functions.

This is one of the reasons schemes are powerful: they keep information that ordinary point sets discard.

§140.7 The arithmetic curve

The spectrum of the integers has points

$$(p)$$

for prime numbers p, together with the generic point (0). An integer n vanishes at exactly the primes dividing it:

$$V(n) = \{(p) : p \mid n\}.$$

Thus divisibility becomes geometry.

This is one of the most charming features of schemes: number theory begins to resemble geometry, not metaphorically but structurally.

§140.8 Maximal ideals are not enough

It is tempting to keep only maximal ideals, because over an algebraically closed field they correspond to ordinary closed points. But prime ideals contain information about irreducible subspaces.

In $k[x, y]$, the prime ideal (x) is not a closed point. It corresponds to the whole y-axis. The maximal ideal $(x, y - a)$ is a closed point on that axis.

Thus prime ideals let a scheme contain both points and subvarieties as points of different levels of generality. This is strange at first, but it is exactly what makes generic points possible.

§140.9 Specialization as movement from generic to concrete

In $\operatorname{Spec} R$, one says that $\mathfrak{q}$ specializes to $\mathfrak{p}$ when

$$\mathfrak{q} \subseteq \mathfrak{p}.$$

The smaller prime is more generic; the larger prime is more special.

For $k[x]$,

$$(0) \subset (x - a),$$

so the generic point specializes to the closed point $(x - a)$. This order relation is the topological shadow of ideal containment.

§140.10 Residue fields

At a prime $\mathfrak{p}$, the local ring $R_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$. The residue field is

$$\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}.$$

For a closed point $(x - a) \subset k[x]$, the residue field is k. For the generic point (0), the residue field is $k(x)$, the field of rational functions.

This is another way to see the meaning of generic points: at the generic point of the affine line, the coordinate x is not specialized to a number; it remains a variable.

§140.11 A tiny nonreduced calculation

Let

$$R = k[\epsilon]/(\epsilon^2).$$

A function on $\operatorname{Spec} R$ can have the form

$$a + b\epsilon.$$

The term $b\epsilon$ is invisible if one only looks at ordinary points, because ϵ is nilpotent. But algebraically it records first-order infinitesimal information.

This is the simplest model of a dual number. It is also a first hint of why tangent vectors and infinitesimal thickenings are naturally scheme-theoretic.

§140.12 Reading $\operatorname{Spec} R$ in examples

The dictionary from this note should be used with actual rings in mind:

$\operatorname{Spec} R$	prime ideals, including non-maximal ones
$D(f)$	the part where f becomes invertible
R_f	functions allowed to divide by powers of f
$R_{\mathfrak{p}}$	functions near the prime $\mathfrak{p}$
$\kappa(\mathfrak{p})$	the residue field visible at $\mathfrak{p}$
nilpotents	infinitesimal information not seen by closed points

For $\operatorname{Spec} \mathbb{Z}$, the generic point (0) and the closed points (p) are both needed: localizing at (p) records arithmetic near one prime, while the generic point records information over $\mathbb{Q}$. For $k[\epsilon]/(\epsilon^2)$, there is only one underlying prime, but the function $a + b\epsilon$ still contains a first-order term. These two examples explain why schemes keep more than the set of maximal ideals: they retain specialization, residue fields, and nilpotent thickening.

Projective Schemes and Gluing: Building Projective Space from Affine Windows

N-20250301

§141.1 Projective geometry returns with functions attached

Earlier projective geometry used homogeneous coordinates to add points at infinity and remove affine exceptions. Scheme theory adds another layer: every affine window comes with a ring of functions, and the windows must be glued compatibly.

The construction that does this is Proj. It is to graded rings what Spec is to ordinary rings.

§141.2 The projective line as the model

The best first example is $\mathbb{P}^1$. Take two affine lines:

$$U_0 = \operatorname{Spec} R[t], \qquad U_1 = \operatorname{Spec} R[s].$$

On the region where both coordinates are nonzero, glue them by

$$s = \frac{1}{t}.$$

The result is the projective line. The point is not the formula itself, but the construction style:

projective space is made by gluing affine charts.

§141.3 Graded rings and homogeneous data

Let

$$S = S_0 \oplus S_1 \oplus S_2 \oplus \cdots$$

be a graded ring. Projective geometry only respects homogeneous equations, because

$$[x_0 : \cdots : x_n] = [\lambda x_0 : \cdots : \lambda x_n].$$

The construction $\operatorname{Proj} S$ uses homogeneous prime ideals not containing the irrelevant ideal.

This definition sounds heavier than the picture, but it encodes a familiar rule: equations in projective space must be homogeneous.

§141.4 Projective space as Proj

The scheme-theoretic projective space is

$$\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n],$$

where the variables have degree 1.

The standard affine open where $x_i \neq 0$ is written

$$D_+(x_i).$$

On it, the ratios

$$\frac{x_j}{x_i}$$

serve as affine coordinates.

Thus $\mathbb{P}^n$ is covered by $n+1$ affine schemes. Projective geometry is global, but it is computed through affine windows.

§141.5 The projective plane through three windows

For $\mathbb{P}^2$, the standard opens are

$$D_+(x), \qquad D_+(y), \qquad D_+(z).$$

On $D_+(z)$, use

$$X = \frac{x}{z}, \qquad Y = \frac{y}{z}.$$

On $D_+(x)$, use

$$U = \frac{y}{x}, \qquad V = \frac{z}{x}.$$

Each chart is affine. The projective plane is the result of gluing these affine pieces by ratio changes on overlaps.

This is the practical meaning of projective schemes: use affine algebra locally, then glue.

§141.6 A conic across charts

Consider the projective conic

$$xz - y^2 = 0.$$

On the chart $z \neq 0$, set $X = x/z$ and $Y = y/z$. The equation becomes

$$X = Y^2.$$

On the chart $x \neq 0$, set $U = y/x$ and $V = z/x$. The equation becomes

$$V = U^2.$$

The same projective curve has multiple affine faces. A single affine chart may hide what happens at infinity, but the projective object keeps all charts at once.

§141.7 Projective closure with scheme memory

Given an affine equation $f(x, y) = 0$, homogenizing gives a projective equation

$$F(x, y, z) = 0.$$

Classically, this adds points at infinity. Scheme-theoretically, it also keeps track of nilpotent or embedded structure if such structure is present.

So Proj is not merely a notation for old projective varieties. It is the projective version of the local-function viewpoint developed in scheme theory.

§141.8 What Proj adds

The earlier projective note emphasized incidence and transformations. The Proj viewpoint emphasizes functions and gluing.

homogeneous coordinates	homogeneous primes
affine charts	standard affine opens
projective equations	homogeneous ideals
points at infinity	glued affine schemes

These are not competing languages. They are layers of the same geometry.

§141.9 Why the irrelevant ideal is excluded

For $S = R[x_0, \dots, x_n]$, the ideal

$$S_+ = (x_0, \dots, x_n)$$

is called the irrelevant ideal. It corresponds to the forbidden point where all homogeneous coordinates vanish at once:

$$[0 : \cdots : 0],$$

which is not a projective point.

Thus $\operatorname{Proj} S$ uses homogeneous prime ideals that do not contain all positive-degree elements. This technical-looking exclusion is simply the algebraic version of the rule that homogeneous coordinates cannot all be zero.

§141.10 Standard opens as spectra

For a homogeneous element $f \in S$, the standard open $D_+(f)$ is affine. Its coordinate ring is the degree-zero part of the localization:

$$\mathcal{O}(D_+(f)) = (S_f)_0.$$

This formula explains why ratios appear in projective coordinates. Localizing at x_i allows division by powers of x_i, and taking degree zero keeps only scale-invariant expressions such as

$$\frac{x_j}{x_i}.$$

So affine coordinates on projective charts are not arbitrary. They are degree-zero fractions.

§141.11 A second conic chart calculation

For the conic

$$xz - y^2 = 0,$$

the chart $y \neq 0$ uses coordinates

$$A = \frac{x}{y}, \qquad B = \frac{z}{y}.$$

The equation becomes

$$AB = 1.$$

So in this chart the conic looks like a hyperbola. In another chart it looked like a parabola. This is a concrete reminder that affine appearances depend on the chosen window.

The projective curve is the object obtained by holding all these windows together.

§141.12 Line bundles are already nearby

Projective geometry naturally leads to line bundles. On $\mathbb{P}^n$, the twisting sheaves $\mathcal{O}(d)$ organize homogeneous polynomials of degree d. A homogeneous equation of degree d is better thought of as a section of $\mathcal{O}(d)$ than as an ordinary global function.

This explains why projective varieties have fewer global regular functions than affine varieties. Projective geometry shifts attention from functions to sections of line bundles.

This remark points back to divisors and Riemann–Roch: projective geometry, line bundles, and spaces of sections are parts of the same story.

The twisting sheaves therefore complete the passage from homogeneous coordinates to projective schemes. The same graded algebra that builds $\operatorname{Proj} S$ also supplies its affine windows and its line bundles: projective geometry has been reconstructed from homogeneous algebra and gluing while retaining the local functions invisible in a purely incidence-based picture.

Nikonov's Seminar Prelude: Knots, Diagrams, Probes, and the Space Between the Lines

N-20250705

§142.1 First orientation

This note has a slightly unusual origin. It was not written after a finished course or after a fully digested paper. It grew out of a post-Gaokao attempt to understand an online seminar by I. M. Nikonov on partial tribrackets. The seminar was exciting, but the words arrived too quickly: quasigroups, biquandloids, thickened surfaces, crossings, arcs, regions, and probes all appeared before I had a stable picture of what they were supposed to mean.

The first reading memo I made after the seminar was useful as a record of confusion. Its main focus was correct: it noticed that Nikonov's paper is not merely introducing another knot invariant, but trying to explain why the pieces of a knot diagram should have topological meaning. However, that memo was too blog-like and too top-heavy for a mathematical notebook. It mixed motivation, informal analogies, and technical claims in one stream. Here I rewrite the same impulse into a more usable note.

The rule of this note is therefore modest: do not try to read the whole paper. Instead, understand the first conceptual move. Nikonov starts with an old tension in knot theory:

a knot is a topological object, but we calculate using diagrams.

The paper asks whether the diagrammatic pieces themselves–arcs, semiarcs, regions, and crossings–can be described topologically.

§142.2 The topological object and its planar shadow

Topologically, a knot is not first of all a picture on paper. It is an embedded circle in three-dimensional space:

$$K : S^1 \hookrightarrow \mathbb{R}^3,$$

or, more generally, a tangle in the thickening $F \times I$ of a surface F. The equivalence relation is ambient isotopy: the whole surrounding space is allowed to deform, carrying one knot to the other.

The story begins with the familiar planar shadow of a knot. A diagram is what we can draw, but the actual knot lives one dimension higher.

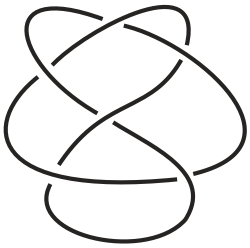


Figure 142.1: From Nikonov's paper: a knot diagram. It is a planar record of a three-dimensional object.

This distinction matters. The diagram has crossings, but the knot in space has no literal self-intersection. At a crossing, one strand passes over the other. The diagram remembers this by marking over/under information.

§142.3 Some basic knot-theory language

A few standard definitions make the paper much less mysterious.

Definition 142.3.1 (Knot and ambient isotopy)**.** A knot is a tame embedding $K : S^1 \hookrightarrow S^3$, usually drawn in $\mathbb{R}^3 \subset S^3$. Two knots K_0, K_1 are equivalent if there is an ambient isotopy $H_t : S^3 \to S^3$ with $H_0 = \mathrm{id}$ and $H_1(K_0) = K_1$.

The word "ambient" is important. We do not merely slide points of the curve along the curve. We deform the whole surrounding three-dimensional space. This is why a knot is a topological object and not just a parametrized curve.

Definition 142.3.2 (Tangle in a thickened surface)**.** Let F be an oriented compact surface and $I = [0, 1]$. A tangle in $F \times I$ is a properly embedded compact one-dimensional manifold, possibly with boundary on $\partial F \times I$. A knot is the closed one-component special case.

The thickened-surface language is the natural setting of Nikonov's paper. Instead of drawing everything in the plane, one works over a surface F, and the knot or tangle floats in the slab $F \times I$.

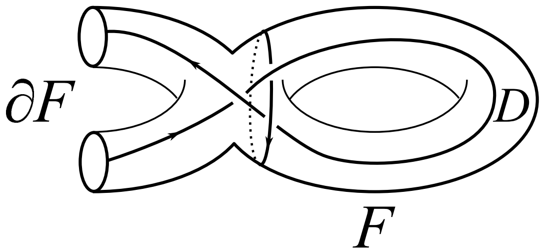


Figure 142.2: From Nikonov's paper: a tangle diagram on a surface.

Definition 142.3.3 (Diagram)**.** A diagram of a tangle is the projection of the tangle to F, with crossing data recording which strand passes over the other. Thus a diagram is a two-dimensional shadow plus over/under information.

The basic theorem justifying all diagrammatic knot theory is Reidemeister's theorem.

Theorem 142.3.4 (Reidemeister theorem)

Two tame knot diagrams represent ambient isotopic knots if and only if they are related by a finite sequence of planar isotopies and Reidemeister moves.

This theorem is the bridge between topology and combinatorics. It says that if a construction on diagrams is unchanged under the three Reidemeister moves, then it really defines a knot invariant.

The combinatorial definition says that a knot can be represented by a diagram modulo Reidemeister moves. These are the local moves that change the drawing without changing the underlying knot.

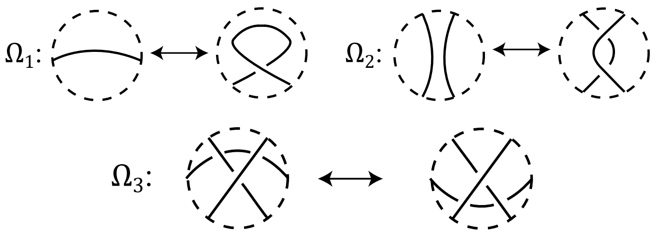


Figure 142.3: From Nikonov's paper: Reidemeister moves. These are the local equivalences that make diagrams represent knots.

So the first dictionary is:

$$\text{knot in space} \quad \longleftrightarrow \quad \text{diagram modulo Reidemeister moves.}$$

This is powerful because diagrams are finite and computable. But it also creates the problem that motivates the paper: if a calculation assigns labels to arcs or regions of a diagram, what exactly are those arcs and regions as topological objects?

A few famous theorems live on the topological side of this dictionary. They are not the main subject of the note, but they explain the background against which the diagrammatic approach became so attractive.

Definition 142.3.5 (Knot complement and knot group)**.** The complement of a knot $K \subset S^3$ is $S^3 \setminus N(K)$, where $N(K)$ is a small tubular neighbourhood of the knot. The knot group is

$$\pi_1(S^3 \setminus K),$$

or equivalently the fundamental group of the complement after removing a tubular neighbourhood.

The complement is often more informative than the curve itself. A loop in the complement asks: can this loop be contracted to a point without crossing the knot? The knot group packages the obstruction algebraically.

Theorem 142.3.6 (Schubert prime decomposition, informal form)
Every nontrivial knot decomposes as a connected sum of prime knots, and this decomposition is unique up to order.

Here the connected sum operation is the knot-theoretic analogue of multiplication: cut out a small arc from each knot and splice the loose ends together. Prime knots are the knots that cannot be decomposed further.

Another landmark is the geometric decomposition picture. Very roughly, many knot complements split into pieces that are torus-like, satellite-like, or hyperbolic. The figure-eight knot is the standard first example of a hyperbolic knot. This is the "God's-eye" topological world mentioned in the original reading memo: elegant, deep, and hard to compute with directly.

§142.4 The pieces of a diagram

A diagram is not just a curve with crossings. It has several kinds of local pieces. Four basic types are especially important here:

arcs, semiarcs, crossings, regions.

These are the pieces that appear in coloring rules, parity theories, skein relations, and many diagrammatic constructions.

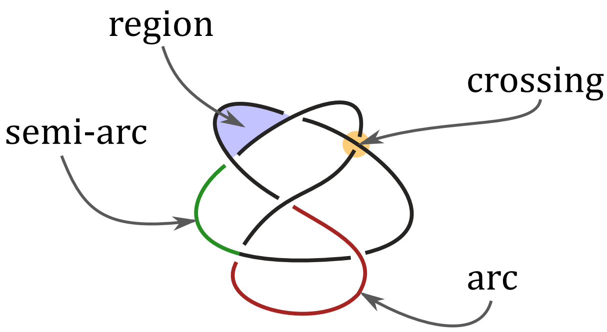


Figure 142.4: From Nikonov's paper: the elementary pieces of a knot diagram.

The naive view is that these are pieces of ink on the page. An arc is a piece of curve between undercrossings. A semiarc is a smaller piece between consecutive crossings. A region is a component of the complement of the diagram. A crossing is a vertex with over/under data.

This is good enough for drawing, but it is not yet good enough for topology. A Reidemeister move can create or destroy a crossing, split an arc, merge two regions, or change how pieces correspond. If one wants to build invariants by labeling these pieces, one needs a way to say when a piece before the move is the same as a piece after the move.

This is exactly where many elementary knot invariants already give hints. For example, the linking number of a two-component link is computed from signs of mixed crossings. A quandle coloring assigns algebraic labels to arcs. A shadow or region coloring assigns labels to complementary regions. These methods work only because the labels transform correctly under Reidemeister moves.

So there are two levels:

$$\begin{gathered}\text{first choose pieces of a diagram}\\ \Downarrow\\ \text{then assign algebraic labels to those pieces}\\ \Downarrow\\ \text{prove the labels survive Reidemeister moves.}\end{gathered}$$

Nikonov's paper attacks the first level. Before asking which labels to assign, ask what the pieces are.

This is the hidden question behind the paper:

Can a diagram element be defined without choosing a particular diagram?

§142.5 Nikonov's bridge: diagram elements as probes

Nikonov's answer is to replace a two-dimensional piece of a diagram by a three-dimensional path in a thickened surface. Work in

$$F \times I, \qquad I = [0,1],$$

where the knot or tangle lies between the bottom surface $F \times \{0\}$ and the top surface $F \times \{1\}$. A diagram is obtained by projecting down to F.

The key idea is that a diagram element can be detected by a probe: a path travelling through the thickened surface and interacting with the knot in a controlled way.

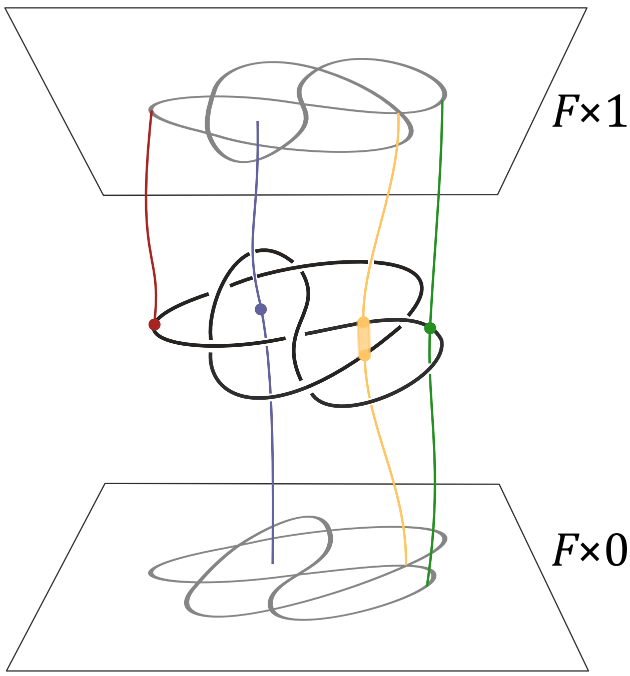


Figure 142.5: From Nikonov's paper: topological interpretation of diagram elements by probes in $F \times I$.

The slogan is:

diagram element	topological probe
arc	a path from the knot to the top surface
semiarc	a bottom-to-top path meeting the knot once
crossing	a bottom-to-top path meeting the knot twice
region	a bottom-to-top path avoiding the knot

This is the first place where the seminar became more understandable to me. The word "arc" is no longer merely a segment in a planar picture. It is represented by a way of reaching the knot from the ambient three-dimensional space. The word "region" is no longer just an empty component on paper. It is represented by a path through the complement of the knot.

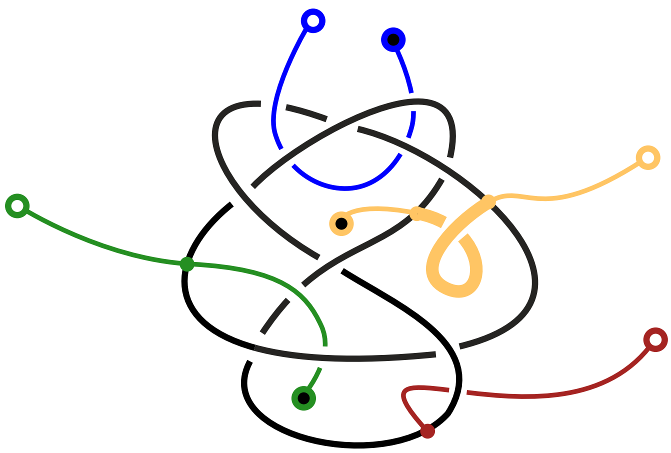


Figure 142.6: From Nikonov's paper: probe diagrams for arc, semiarc, region, and crossing elements.

The great advantage of this formulation is that isotopy becomes natural. If the tangle moves in $F \times I$, the probes can move with it. Thus the correspondence of diagram elements under Reidemeister moves is no longer a purely combinatorial bookkeeping rule; it is the shadow of a topological motion.

§142.6 Invariants, coinvariants, and labels

The first seminar memo paid special attention to the distinction between invariants and coinvariants. This distinction is worth keeping, but it is better to phrase it in a cleaner way.

An invariant is something that gives the same value for every diagram of the same knot. For example, a polynomial invariant or a numerical invariant should not depend on which diagram we draw.

A coinvariant is subtler. It may give different-looking objects for different diagrams, but these objects are canonically isomorphic once we account for the moves between diagrams. A vector space is a good mental model: its dimension is an invariant, while the vector space itself, together with the way it changes under coordinate changes, is closer to a coinvariant.

For diagram elements, the relevant object is often not one number but a whole set of possible classes. The set of arc probes, region probes, or crossing probes should be thought of as a universal object from which more concrete labels can be obtained.

In this language, Nikonov's first main conceptual claim is:

isotopy classes of probes give universal homotopical coinvariants of diagram elements.

This sentence is dense, but the intuition is simple. Instead of assigning a label to an arc immediately, first ask what the arc is intrinsically. The answer is a class of probes. Any later coloring rule is a way of reading algebraic information from these probe classes.

> **Remark 142.6.1 —** The word "universal" should be read in the usual algebraic way. A universal object is not one more label among many labels. It is the object from which all compatible labels should factor. Thus the universal arc object is like the raw space of possible arc labels before choosing a particular quandle, polynomial, or coefficient set.

This is also why the paper talks about coinvariants rather than only invariants. A set of probes may change its concrete presentation when the diagram changes, but the induced object is naturally identified through the isotopy or through the homotopy class of Reidemeister moves. The object is stable, even when its drawing is not literally identical.

§142.7 What this note is not trying to do

This note does not attempt to understand the whole paper. The paper continues into quandles, partial ternary quasigroups, biquandloids, crossoids, and a multicrossing complex. Those structures are not independent tricks. They are different algebraic shadows of the same topological idea:

diagram pieces become probes, and probes have isotopy or homotopy classes.

For a first reading, the main thing to remember is this:

$$\begin{gathered}\text{topological knot or tangle in } F \times I\\ \Downarrow\\ \text{planar diagram with arcs, regions, crossings}\\ \Downarrow\\ \text{topological reinterpretation of those pieces as probes}\\ \Downarrow\\ \text{algebraic colorings extracted from homotopy classes of probes.}\end{gathered}$$

This is why the seminar about partial tribrackets led back to the beginning of knot theory. To understand a ternary operation on regions, one first has to understand what a region is. Nikonov's paper answers: a region is not merely a blank patch in a diagram; it is a topological probe through the complement of the knot.

Nikonov's Paper at One Crossing: Region Probes, Partial Tribrackets, and What the Seminar Was About

N-20250707

§143.1 Reading only one small part of a long paper

This note is deliberately selective. Nikonov's paper is long, and it is not reasonable to digest all of it just because one seminar made the subject look interesting. The first seminar memo had the right instinct: the seminar topic, partial tribrackets, is best understood by looking at the part of the paper where regions of diagrams are turned into topological objects and then into algebraic operations.

So here I focus on one small path through the paper:

$$\begin{array}{c}\text{regions of a diagram}\\ \Downarrow\\ \text{region probes}\\ \Downarrow\\ \text{partial ternary operation at one crossing}\end{array}$$

The point is not to master all definitions. The point is to understand why the word "partial" appears and why a tribracket has anything to do with a knot diagram.

§143.2 Where this sits inside Nikonov's paper

The paper has two large movements. The first movement is topological: arcs, regions, semiarcs, and crossings of diagrams are reinterpreted as isotopy classes of probes. The second movement is algebraic: homotopy classes of these probes naturally carry coloring structures.

The local dictionary is:

diagram element	algebraic shadow
arc	quandle
region	partial ternary quasigroup / partial tribracket
semiarc	biquandloid
crossing	crossoid

This note chooses the second row. That is the row closest to the seminar topic.

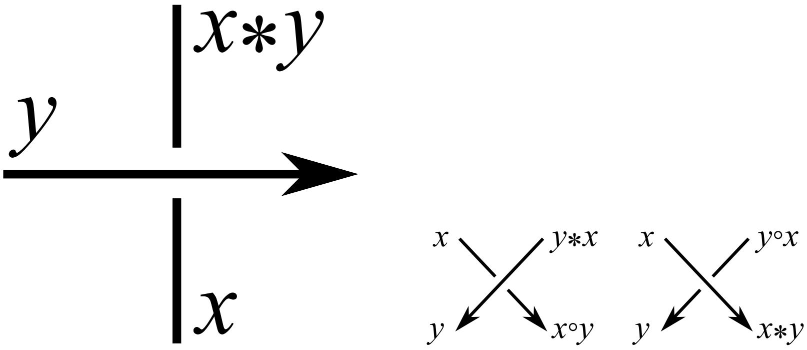


Figure 143.1: From Nikonov's paper: arc and semiarc colorings are the neighbouring examples around the region story.

The figures above are not the main object of this note, but they help orient the reader. Quandles and biquandles are familiar algebraic gadgets for diagram colorings. Nikonov's point is that the less familiar partial tribracket and crossoid fit into the same topological pattern.

§143.3 A region is not just a blank patch

In an ordinary knot diagram, a region is a connected component of the plane or surface after removing the diagram. Informally, it is a blank patch between strands. But this definition depends on the drawing.

Nikonov replaces the patch by a topological path. In a thickened surface $F \times I$, a region probe is a path from the bottom $F \times \{0\}$ to the top $F \times \{1\}$ that does not intersect the knot. In other words, it is a way to pass through the empty space around the knot.

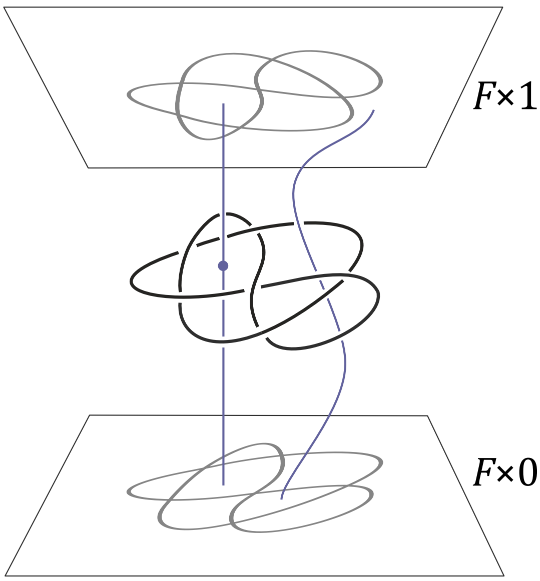


Figure 143.2: From Nikonov's paper: a region probe passes through the complement of the knot.

This is the most important mental shift. In the diagram, a region looks passive. In the thickened surface, a region is represented by an actual route through space. If the knot moves by isotopy, the route can move too. Thus a region can be followed through Reidemeister moves without pretending that it is only a two-dimensional patch of paper.

For a quick analogy, imagine the knot as a collection of wires suspended in a slab of transparent jelly. A region probe is a thin needle pushed from the bottom of the jelly to the top without touching any wire. Moving the knot deforms the possible needle paths. The algebra of regions is really an algebra of these possible needle paths.

§143.4 At a crossing, four regions want to talk to each other

A tribracket is a ternary operation. Instead of combining two inputs, it combines three:

$$[a, b, c] = d.$$

In knot diagrams, the reason for a ternary operation is visual. Around one crossing there are four adjacent regions. If three of their colors are known, the coloring rule determines the fourth.

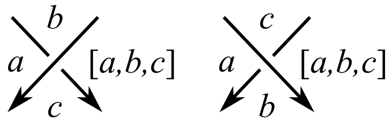


Figure 143.3: From Nikonov's paper: the tribracket coloring rule near a crossing.

This is the cleanest way to remember the algebra. A quandle is suited to arc colorings because arcs meet at a crossing in a binary-looking way. A tribracket is suited to region colorings because four regions meet around a crossing, so knowing three should determine the fourth.

The operation is designed so that the coloring survives Reidemeister moves. That is the usual diagrammatic requirement: if two diagrams represent the same knot, their coloring data should correspond.

Definition 143.4.1 (Horizontal tribracket, informal version)**.** A tribracket is a set X with a ternary operation

$$[\ ,\ ,\ ] : X^3 \to X$$

such that in an equation

$$[a, b, c] = d$$

any three of a, b, c, d determine the fourth, and such that the operation satisfies the compatibility identity forced by the third Reidemeister move.

The first condition is the analogue of invertibility. If a crossing tells us the fourth region from three known regions, then we must also be able to solve backwards when the diagram is read in another local direction.

The second condition is the real knot-theoretic condition. Around a third Reidemeister move, there are two ways of propagating region colors across three nearby crossings. These two ways must give the same final color. In Nikonov's source, this is expressed by the identity

$$[b, [a, b, c], [a, b, d]] = [c, [a, b, c], [a, c, d]] = [d, [a, b, d], [a, c, d]].$$

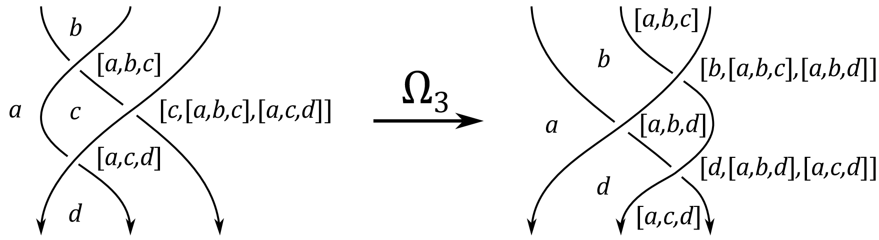


Figure 143.4: From Nikonov's paper: the third Reidemeister move is the source of the tribracket identity.

Theorem 143.4.2 (Coloring invariance for tribrackets)
If X is a tribracket, then tribracket colorings of a diagram correspond bijectively under Reidemeister moves.

This theorem is the usual coloring-invariant mechanism. The algebraic axioms are not arbitrary: they are exactly the local rules needed so that the diagram move does not change the set of colorings.

Example 143.4.3 (Two useful models)
An Alexander-type tribracket has the form

$$[a, b, c] = tb + sc - tsa$$

on a module over $\mathbb{Z}[t^{\pm 1}, s^{\pm 1}]$. A group-based Dehn-type model, in the convention used later for the partial construction, has the form

$$[a, b, c] = ba^{-1}c.$$

The first example is linear and computational; the second example is closer to the topology of paths in a complement.

§143.5 Why the operation is partial

In the plane, it is tempting to think that any three region colors can be inserted into the operation. Nikonov's point is that, on a fixed surface and from the topological viewpoint, not every formal triple of regions is genuinely realizable around a crossing.

A region is represented by a probe. Three regions can be composed at a crossing only when the corresponding probes can be arranged in the correct local configuration. If topology prevents that configuration, then the operation should simply not be defined. This is why the algebra is partial.

The paper includes a useful warning example: a formal composition may look legal if one only stares at symbols, but may fail to be realizable on the actual diagram.

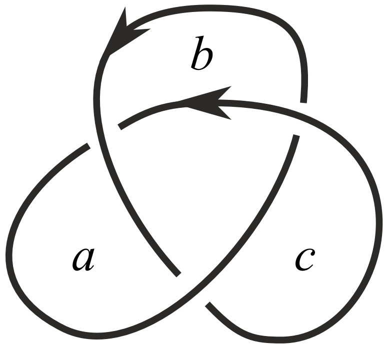


Figure 143.5: From Nikonov's paper: an indicated composition of regions that is not realizable on a classical diagram.

This figure is the whole story in miniature. A purely algebraic operation would like to accept any triple. The topology says: no, only triples that can actually occur as neighbouring regions around a crossing should be allowed. The adjective "partial" is not a technical nuisance; it records genuine geometric obstruction.

Definition 143.5.1 (Partial ternary quasigroup, notebook version)**.** A partial ternary quasigroup is a set X, a compatibility relation $\uparrow$, and a ternary operation

$$[a, b, c]$$

defined only when $b \uparrow a$ and $c \uparrow a$. The axioms say that the output is compatible with the neighbouring inputs, that the operation can be solved uniquely for any missing entry whenever the compatibility conditions make sense, and that the third-Reidemeister identity still holds for compatible quadruples.

For this note, the compatibility relation is the essential new ingredient. One should read

$$b \uparrow a$$

as "the region b is allowed to sit next to the region a in the required local way." In a flat beginner picture this sounds unnecessary, but in a fixed thickened surface it records real topology.

Example 143.5.2 (Alexander numbering)

The paper gives a very small partial example:

$$X = \mathbb{Z}, \qquad a \uparrow b \Longleftrightarrow b = a - 1, \qquad [a, b, c] = a + 2.$$

This is not meant to be the most exciting invariant. It is a memory aid: partiality can be as simple as allowing only neighbouring integer levels to interact.

§143.6 Probe clearing and color rules

The paper also explains how a region probe can be cleared through the diagrammatic situation. The figure below should be read as a topological mechanism behind the coloring rule. We are not just writing $[a, b, c] = d$; we are moving a probe through the thickened surface and watching how its class changes.

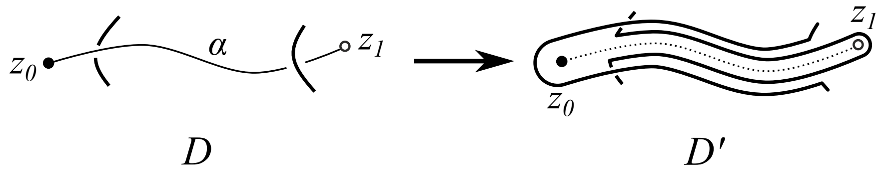


Figure 143.6: From Nikonov's paper: clearing a region probe in the construction of the partial tribracket.

This is the connection with the first note. Diagrammatic elements become probes; homotopy classes of probes become the raw material for algebra; algebraic coloring rules become shadows of probe motion.

The theorem proved in this part of the paper can be read as follows.

Theorem 143.6.1 (Nikonov's fundamental partial tribracket, informal form)
For a tangle $T \subset F \times I$ with diagram D, colorings of D by a partial ternary quasigroup correspond to homomorphisms out of the topological partial tribracket built from homotopy classes of region probes, with the natural $\pi_1(F)$-invariance condition.

The exact statement has more notation, but the meaning is very clean. A diagram coloring is not merely a coloring of a planar picture. It is the same thing as a compatible algebraic map from the topological object made of region probes.

In the proof, "clearing a probe" is the operational step. One pulls the diagram away from the probe so that the probe's value can be read as the color of a region after the transformation. Reidemeister invariance then says that this value does not depend on the particular clearing process.

A small local rule can also be expressed as propagation of colors between neighbouring regions.

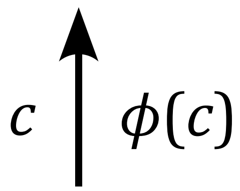


Figure 143.7: From Nikonov's paper: a local propagation rule for neighbouring regions.

This is the part I would keep from the seminar if I had to compress it into one sentence:

> a partial tribracket is the algebra of how region probes interact near crossings.

§143.7 What about crossings themselves?

The paper does not stop at regions. Crossings also become topological objects: a crossing probe is a bottom-to-top path intersecting the tangle twice, with extra framing data on the segment between the two intersection points.

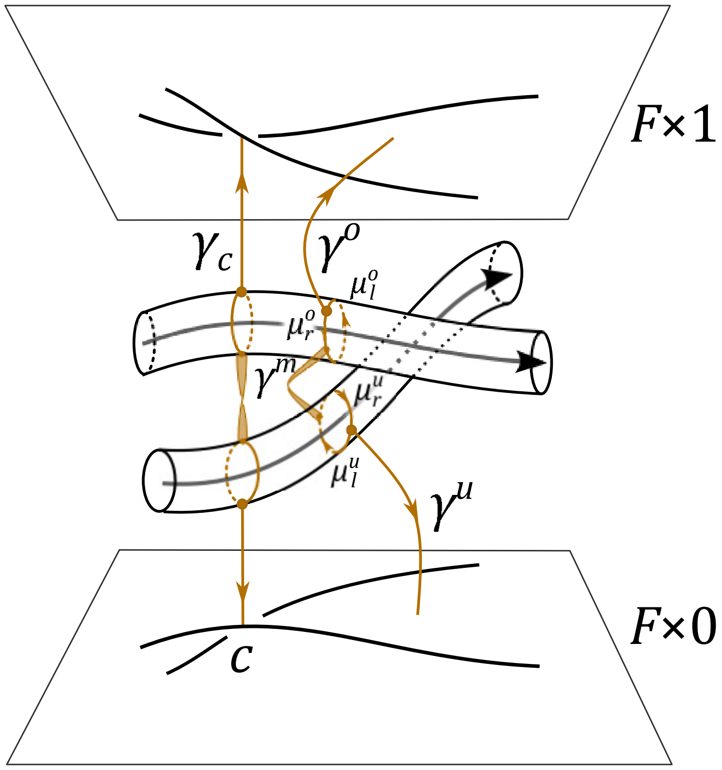


Figure 143.8: From Nikonov's paper: a crossing probe. Crossings are treated as probe classes rather than merely as vertices of a drawing.

This leads toward crossoids, which generalize parity-type structures on knots. I do not try to unfold crossoids here, but the pattern is now visible:

$$\begin{gathered}\text{choose a diagram element}\\ \Downarrow\\ \text{replace it by a probe}\\ \Downarrow\\ \text{take homotopy classes}\\ \Downarrow\\ \text{read off an algebraic coloring structure.}\end{gathered}$$

The partial tribracket is simply the region case of this general machine.

§143.8 The paper in one playful sentence

A knot diagram looks like a flat picture, but Nikonov treats its arcs, regions, and crossings as little test paths sent through a three-dimensional slab. Once those test paths are allowed to move, their homotopy classes begin to carry algebra. Arc probes lead toward quandles; region probes lead toward partial tribrackets; semiarc probes lead toward biquandloids; crossing probes lead toward crossoids.

So the paper is not merely saying "here is another coloring invariant." It is saying something more structural:

the algebraic coloring gadgets are shadows of topological probe spaces.

For a casual post-seminar reading, this is already enough. The rest of the paper builds a much larger machine, but this one crossing explains why the machine is interesting.

XIV

Topology and Homotopy

Part XIV: Contents

Group Actions, Homotopy, and a Rough Road to the Hopf Fibration

N-20210806

§144.1 Purpose of the note

This note was written as preparation for a seminar on homotopy groups and the Hopf fibration. The goal is modest: I only want enough background so that the seminar is not blocked by unfamiliar words. The content is therefore a quick route through group actions, homotopy, cofibrations and fibrations, homotopy groups, and finally the Hopf fibration.

Caution

The original note was written quickly and openly says that errors may remain. I have corrected the most obvious convention problems and phrasing issues, while keeping the exploratory style.

§144.2 Group actions

A group action is one of the first ways in which abstract algebra touches geometry and topology. A group can be understood as acting by symmetries or permutations of a set.

Definition 144.2.1. Let G be a group and X a set. A left action of G on X is a map

$$G \times X \to X, \qquad (g, x) \mapsto g \cdot x,$$

such that

$$e \cdot x = x, \qquad (gh) \cdot x = g \cdot (h \cdot x).$$

Equivalently, it is a homomorphism

$$\varphi : G \to S_X,$$

where S_X is the permutation group of X.

Remark 144.2.2 — One may also use right actions, written $x \cdot g$, satisfying

$$x \cdot 1 = x, \qquad (x \cdot g) \cdot h = x \cdot (gh).$$

The original note switches viewpoints, so I fix the left-action convention unless stated otherwise.

Example 144.2.3

The symmetric group S_n acts on $X = \{1, 2, \ldots, n\}$. Cayley's theorem says that every group G can be realized as a subgroup of the permutation group of its underlying set, by the action of left multiplication.

Example 144.2.4

The symmetry group of a cube acts on the cube and may also be regarded as a group of linear transformations of $\mathbb{R}^3$ preserving the cube.

Example 144.2.5 (Conjugation)

A group G acts on itself by conjugation:

$$g \cdot x = gxg^{-1}.$$

For a fixed $a \in G$, its orbit is

$$\mathrm{Orb}_G(a) = \{gag^{-1} \mid g \in G\},$$

which is the conjugacy class of a. Its stabilizer is

$$\mathrm{Stab}_G(a) = \{g \in G \mid gag^{-1} = a\},$$

which is the centralizer of a.

Definition 144.2.6. For an action of G on X, the stabilizer and orbit of $x \in X$ are

$$\mathrm{Stab}_G(x) = \{g \in G \mid g \cdot x = x\},$$

$$\mathrm{Orb}_G(x) = \{g \cdot x \mid g \in G\}.$$

Theorem 144.2.7 (Orbit-stabilizer)

If G is finite, then

$$|G| = |\,\mathrm{Stab}_G(x)|\,|\,\mathrm{Orb}_G(x)|.$$

Proof. The map $G \to \mathrm{Orb}_G(x)$, $g \mapsto g \cdot x$, has fibres equal to left cosets of $\mathrm{Stab}_G(x)$. Hence the orbit has $[G : \mathrm{Stab}_G(x)]$ elements. □

§144.3 Homotopy

Algebraic topology tries to classify topological spaces and continuous maps using algebraic invariants. Before doing that, one needs a flexible notion of sameness. Homotopy is such a notion.

Definition 144.3.1. Let X, Y be topological spaces and $f, g : X \to Y$ continuous maps. A homotopy from f to g is a continuous map

$$F : X \times I \to Y, \qquad I = [0, 1],$$

such that

$$F(x, 0) = f(x), \qquad F(x, 1) = g(x).$$

When such an F exists, write $f \simeq g$.

Equivalently, a homotopy may be viewed as a map

$$F : X \to Y^I,$$

where $Y^I = \mathrm{Top}(I, Y)$ is the path space, and the endpoint evaluation maps recover f and g.

Proposition 144.3.2

Homotopy is an equivalence relation on maps $X \to Y$, and it is compatible with composition. If $f \simeq g$, then

$$k \circ f \circ h \simeq k \circ g \circ h$$

whenever the compositions are defined.

Definition 144.3.3. Two spaces X and Y have the same homotopy type if there are maps $f : X \to Y$ and $g : Y \to X$ such that

$$g \circ f \simeq \mathrm{id}_X, \qquad f \circ g \simeq \mathrm{id}_Y.$$

§144.4 Cofibrations, fibrations, and deformation retracts

The original note introduces cofibrations and fibrations through lifting diagrams. These diagrams are not decorative: each arrow is one piece of the data in the definition. I spell them out because this is exactly where the abstraction tends to hide the simple geometric idea.

§144.4.i Homotopy extension property

A map $j : A \to X$ is a cofibration if it has the homotopy extension property. The situation is this. We know a map

$$g : X \to Z,$$

and we also know a homotopy on the smaller space

$$G : A \times I \to Z.$$

These two pieces of data must agree at time 0: for every $a \in A$,

$$G(a, 0) = g(j(a)).$$

If this compatibility condition holds, the homotopy extension property asks for a map

$$\overline{G} : X \times I \to Z$$

such that

$$\overline{G}(x,0) = g(x), \qquad \overline{G}(j(a),t) = G(a,t).$$

In other words, a homotopy that has only been prescribed on the subspace A must extend to a homotopy on all of X, while keeping the original time-zero map on X.

The corresponding diagram is

$$\begin{tikzcd}
A \arrow[r, "j"] \arrow[d, "i_0"'] & X \arrow[r, bend left, "g"] \arrow[d, "i_0"] & Z \arrow[d, equal] \\
A \times I \arrow[r, "j\times I"] \arrow[rr, bend right, "G"'] & X \times I \arrow[r, dashed, "\overline{G}"] & Z
\end{tikzcd}$$

Here i_0 denotes insertion at time zero: $a \mapsto (a,0)$ or $x \mapsto (x,0)$, according to the source. The bottom arrow $j \times I$ says that the cylinder over A sits inside the cylinder over X. The two solid maps into Z are the information already given: g on X, and G on $A \times I$. The rightmost equality means that both pieces are maps into the same target Z, not two different copies of Z. The dashed arrow is not part of the data. It is exactly the thing whose existence is required.

This is why the word "extension" is literal. We first define a map on the union

$$X \times \{0\} \cup A \times I.$$

On $X \times \{0\}$ it is g. On $A \times I$ it is G. The compatibility condition $G(a,0) = g(j(a))$ says these two formulas agree on the overlap $A \times \{0\}$. The H.E.P. says this partial map extends across the whole cylinder $X \times I$.

A useful mental picture is: $A \subset X$ is a part of the space whose motion has already been specified. A cofibration is an inclusion good enough that this partial motion can be filled in over the rest of X. Thus cofibrations are the maps along which homotopies extend outward.

§144.4.ii Homotopy lifting property

A fibration is the dual-looking story. Let

$$p : X \to A$$

be a map. The homotopy lifting property says that if a homotopy has been drawn downstairs in A, and if its starting point has already been lifted upstairs to X, then the whole homotopy can be lifted upstairs.

Using path spaces makes the diagram compact. The notation A^I means the space of paths $I \to A$, and X^I means the space of paths $I \to X$. Evaluation at time 0 gives

$$ev_0 : A^I \to A, \qquad ev_0 : X^I \to X.$$

The map $p : X \to A$ also induces a map on path spaces

$$p^I : X^I \to A^I, \qquad \gamma \mapsto p \circ \gamma.$$

Now take a parameter space Z. A map

$$F : Z \to A^I$$

is a Z-indexed family of paths downstairs in A. A map

$$\widetilde{f}_0 : Z \to X$$

chooses a starting lift upstairs. The compatibility condition is

$$p \circ \widetilde{f}_0 = ev_0 \circ F.$$

This simply says: the chosen point upstairs really lies over the initial point of the path downstairs.

The lifting diagram is

$$\begin{tikzcd}
X^I \arrow[r, "ev_0"] \arrow[d, "p^I"'] & X \arrow[d, "p"] \\
A^I \arrow[r, "ev_0"'] & A
\end{tikzcd}
\qquad
\begin{tikzcd}
 & X^I \arrow[d, "{(p^I, ev_0)}"] \\
Z \arrow[ur, dashed, "\overline{F}"] \arrow[r, "{(F, \widetilde{f}_0)}"'] & A^I \times_A X
\end{tikzcd}$$

The square on the left is the structural square. The horizontal map $p^I : X^I \to A^I$ sends an upstairs path to its downstairs projection, and the two ev_0 arrows take initial points. The square commutes because evaluating $p \circ \gamma$ at time 0 gives $p(\gamma(0))$. The diagram on the right packages the actual lifting problem. A pair $(F, \widetilde{f}_0)$ lands in the fiber product $A^I \times_A X$ exactly when $p \circ \widetilde{f}_0 = ev_0 \circ F$. The dashed arrow $\overline{F} : Z \to X^I$ is the desired lifted family of paths.

Unpacking the dashed arrow, for each $z \in Z$ it gives a path

$$\overline{F}(z) : I \to X.$$

The two required equations are

$$p \circ \overline{F}(z) = F(z), \qquad \overline{F}(z)(0) = \widetilde{f}_0(z).$$

So the lifted path projects to the given downstairs path and starts at the chosen upstairs point. This is the exact formal meaning of "lift upward from the base."

The relationship with the earlier definition is now transparent. A cofibration problem begins with a homotopy on a subspace and asks to extend it sideways across a larger cylinder. A fibration problem begins with a homotopy downstairs and asks to lift it vertically to a path or homotopy upstairs. Both are extension problems, but in different directions: H.E.P. fills missing points in the domain; H.L.P. fills missing lifts in the total space.

Definition 144.4.1. A map $j : A \to X$ is a cofibration if it has the homotopy extension property described above.

Definition 144.4.2. A map $p : X \to A$ is a fibration if it has the homotopy lifting property described above.

> **Remark 144.4.3 —** Cofibration and fibration are dual-looking ideas. A cofibration lets homotopies extend outward from a subspace; a fibration lets homotopies lift upward from the base. In the Hopf fibration $S^3 \to S^2$, the word "fibration" means exactly that small motions of a point in the base sphere can be lifted to motions in S^3, once a starting point in the circle fiber has been chosen.

Definition 144.4.4. A deformation retract of X onto a subspace A is a homotopy

$$H : X \times I \to X$$

such that

$$H(x, 0) = x, \qquad H(x, 1) \in A, \qquad H(a, t) = a$$

for all $x \in X$, $a \in A$, and $t \in I$.

Proposition 144.4.5

The pair (X, A) has the homotopy extension property if and only if

$$X \times \{0\} \cup A \times I$$

is a retract of $X \times I$.

§144.5 Spheres and homotopy groups

The n-sphere is

$$S^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1^2 + \cdots + x_{n+1}^2 = 1\}.$$

The n-torus is

$$T^n = (S^1)^n.$$

Thus a point of T^2 may be described by a pair of angles, or equivalently by a point of $S^1 \times S^1$.

> **Caution**
>
> The original note says that a personal translation habit used the wrong word for "equator". I use the standard word *equator* throughout.

Stereographic projection identifies a sphere minus one point with Euclidean space. For example,

$$S^2 \setminus \{\text{north pole}\} \cong \mathbb{R}^2.$$

The equator is the region where the projection is least distorted; away from it, shapes become increasingly distorted.

Definition 144.5.1. Let X be a based space with basepoint x_0. The n-th homotopy group or homotopy set is

$$\pi_n(X, x_0) = [(S^n, *), (X, x_0)],$$

the set of based homotopy classes of based maps $S^n \to X$. For $n \geq 1$, it has a natural group structure.

Using the model

$$S^n \simeq I^n / \partial I^n,$$

the group operation for $n \geq 1$ may be described by pinching the sphere along an equator:

$$S^n \to S^n \vee S^n.$$

Given two based maps $f, g : S^n \to X$, the product first pinches S^n into two copies, then applies f on one copy and g on the other.

Remark 144.5.2 — For $n = 0$, $\pi_0(X)$ records path components and is not usually a group. For $n \geq 1$, $\pi_n(X, x_0)$ is a group; for $n \geq 2$, it is abelian.

§144.6 The Hopf fibration

The Hopf fibration is a map

$$S^3 \to S^2.$$

It is one of the first examples showing that spheres are connected by unexpectedly rich maps.

Describe

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}.$$

A clean formula for the Hopf map is

$$h : S^3 \to \mathbb{C}P^1 \cong S^2, \qquad h(z_1, z_2) = [z_1 : z_2].$$

On the chart $z_1 \neq 0$, this looks like

$$z_2/z_1.$$

The case $z_1 = 0$ is the point at infinity.

Caution

It is tempting to say $z_2/0 = \infty$, and that intuition is useful. The mathematically clean way is to use projective coordinates $[z_1 : z_2]$, which handle infinity automatically.

The fibers over three simple points can be written without any abstraction:

$$F_1 = h^{-1}([1:0]) = \{(e^{it}, 0) : 0 \leq t < 2\pi\},$$

$$F_2 = h^{-1}([0:1]) = \{(0, e^{it}) : 0 \leq t < 2\pi\},$$

and

$$F_3 = h^{-1}([1:1]) = \left\{ \left(\frac{e^{it}}{\sqrt{2}}, \frac{e^{it}}{\sqrt{2}} \right) : 0 \leq t < 2\pi \right\}.$$

To see them inside ordinary three-space, use stereographic projection

$$\Pi(z_1, z_2) = \left(\frac{\operatorname{Re}(z_1)}{1 - \operatorname{Im}(z_2)}, \frac{\operatorname{Im}(z_1)}{1 - \operatorname{Im}(z_2)}, \frac{\operatorname{Re}(z_2)}{1 - \operatorname{Im}(z_2)} \right).$$

Writing $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, this is the projection from the point $(0, i) \in S^3$.

For F_1, substitution gives

$$\Pi(e^{it}, 0) = (\cos t, \sin t, 0),$$

the standard unit circle in the plane $z = 0$. For F_2,

$$\Pi(0, e^{it}) = \left(0, 0, \frac{\cos t}{1 - \sin t}\right),$$

so this fiber becomes the vertical line together with the point at infinity; the missing value $t = \pi/2$ is precisely the projection point. For F_3,

$$\Pi\left(\frac{e^{it}}{\sqrt{2}}, \frac{e^{it}}{\sqrt{2}}\right) = \left(\frac{\cos t/\sqrt{2}}{1 - \sin t/\sqrt{2}}, \frac{\sin t/\sqrt{2}}{1 - \sin t/\sqrt{2}}, \frac{\cos t/\sqrt{2}}{1 - \sin t/\sqrt{2}} \right).$$

This curve lies in the plane $X = Z$. Eliminating t gives

$$2X^2 + (Y - 1)^2 = 2, \qquad Z = X,$$

so it is again a circle, not a random distorted loop.

The important topological fact is that $\Pi(F_1)$ and $\Pi(F_3)$ form a Hopf link. With the orientations induced by increasing t, their linking number is $+1$; reversing one orientation changes the sign. The computation matches the picture: one circle passes once through a spanning disk of the other. The core geometric intuition of the Hopf fibration is that any two distinct circle fibers in S^3 become linked once after stereographic projection to $\mathbb{R}^3$.

There is also a quaternionic picture. Regard S^3 as the unit quaternions and S^2 as the purely imaginary unit quaternions. Fix $p \in S^2$. For a unit quaternion $q \in S^3$, define

$$q \mapsto qpq^{-1}.$$

This gives another description of a map $S^3 \to S^2$, equivalent to the Hopf fibration.

> **Remark 144.6.1 —** The main fact to remember is simple but powerful: the Hopf fibration has total space S^3, base S^2, and fibre S^1. A full understanding needs pictures, but the formula already shows why complex numbers, quaternions, and topology appear together.

§144.7 References to return to

The original note points toward Hatcher's *Algebraic Topology*, Niles Johnson's visual material on the Hopf fibration, and interactive Hopf-fibration demonstrations. Those should be treated as the real next step, because this note is mainly a vocabulary bridge.

The diagrammatic part of the note should be read as a staged program. The homotopy extension property and the homotopy lifting property have now been drawn explicitly above, because those two diagrams are the quickest way to connect the abstract definitions with the words "extend" and "lift". The same visual habit should also be used when returning to the pinch map

$$S^n \to S^n \vee S^n$$

and to the Hopf fibration itself. The definitions and formulas are already in place, but the geometry becomes much more memorable when the pinch map, circle fibers, and linked fibers are kept visible.

§144.8 Group actions as symmetry bookkeeping

A group action of G on X is a map

$$G \times X \to X$$

satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. The orbit of x is

$$Gx = \{g \cdot x : g \in G\},$$

and the stabilizer is

$$G_x = \{g : g \cdot x = x\}.$$

The orbit-stabilizer relation says that symmetry and movement are complementary. Large stabilizer means small orbit; small stabilizer means large orbit.

§144.9 Hopf fibration intuition

The Hopf fibration is a map

$$S^3 o S^2$$

whose fibers are circles. One way to see it is through complex coordinates: points of $S^3 \subset \mathbb{C}^2$ are pairs (z_1, z_2) with $|z_1|^2 + |z_2|^2 = 1$. Multiplying both coordinates by the same unit complex number moves along a circle fiber. The quotient by this circle action gives $\mathbb{CP}^1 \cong S^2$.

The sentence to remember is concrete: the fibers are not merely many disjoint circles; any two of them are linked once. This is why the formula $S^3 \to S^2$ is already a three-dimensional knot-theoretic picture.

Basic Topology I: Metric Spaces, Boundedness, Norms, Open Sets, and Completeness

N-20221218

§145.1 Why the note starts with metrics

This note begins a topology sequence, but it does not begin with arbitrary open sets. It begins with metric spaces, because metric spaces keep the analysis intuition alive. In real analysis, convergence means

$$x_n \to x \quad \iff \quad (\forall \varepsilon > 0)(\exists N)(\forall n > N)\ |x_n - x| < \varepsilon.$$

The first abstraction is to replace $|x_n - x|$ by a general distance $d(x_n, x)$. Thus a sequence in a metric space converges when

$$(\forall \varepsilon > 0)(\exists N)(\forall n > N)\ d(x_n, x) < \varepsilon.$$

The hidden question is: which parts of analysis really require a numerical distance, and which only require a notion of neighborhood? Metric spaces are the intermediate language between analysis and topology.

§145.2 Boundedness: diameter and radius

A metric space M is bounded if there is a constant D such that

$$d(p, q) \le D \qquad \text{for all } p, q \in M.$$

One can think of D as a diameter bound. The note also gives an equivalent radius formulation: fixing a point p, if there is R such that

$$d(p, q) \le R \qquad \text{for all } q \in M,$$

then M is bounded. Indeed, for any $q_1, q_2 \in M$,

$$d(q_1, q_2) \le d(q_1, p) + d(p, q_2) \le 2R.$$

Conversely, a diameter bound immediately gives a radius bound from any chosen point.

The examples are deliberately chosen to break naive intuition: $[0, 1]$ is bounded, $\mathbb{R}^n$ is not, an infinite set with the discrete metric is bounded, but $\mathbb{N}$ with the usual metric is not. Thus boundedness belongs to a metric, not to a bare set.

§145.3 Metric spaces

A metric space is a pair (X, d) where $d : X \times X \to \mathbb{R}$ satisfies:

(i) $d(x, y) \ge 0$;

(ii) $d(x, y) = 0$ iff $x = y$;

(iii) $d(x, y) = d(y, x)$;

(iv) $d(x, z) \leq d(x, y) + d(y, z)$.

A subset $Y \subseteq X$ inherits the subspace metric

$$d_Y(a, b) = d_X(a, b).$$

This is the metric ancestor of subspace topology.

§145.4 Sequences and continuity

A sequence (x_n) converges to x if every small ball around x eventually contains all x_n. In $\mathbb{R}^k$ with the Euclidean metric, convergence is coordinatewise:

$$x_n = (x_n^1, \ldots, x_n^k) \to x = (x^1, \ldots, x^k)$$

iff $x_n^i \to x^i$ for each i. One direction follows from $|x_n^i - x^i| \leq \|x_n - x\|_2$; the other follows by summing the small coordinate errors.

A map $f : X \to Y$ between metric spaces is continuous at x_0 if

$$(\forall \varepsilon > 0)(\exists \delta > 0) \quad d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

In metric spaces, continuity can also be detected by sequences:

$$x_n \to x_0 \Rightarrow f(x_n) \to f(x_0).$$

The later general topology notes warn that sequences no longer detect everything in arbitrary topological spaces.

§145.5 Standard metrics

On $\mathbb{R}^n$ the Euclidean, taxicab, and max metrics are

$$d_2(x, y) = \left(\sum_i |x_i - y_i|^2 \right)^{1/2},$$

$$d_1(x, y) = \sum_i |x_i - y_i|,$$

and

$$d_\infty(x, y) = \max_i |x_i - y_i|.$$

Their balls have different shapes: round disks, diamonds, and squares. Nevertheless they generate the same usual topology on $\mathbb{R}^n$. This is an early hint that topology remembers neighborhoods but forgets exact metric geometry.

The discrete metric on a set X is

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Then $B_{1/2}(x) = \{x\}$. Every subset is open and closed, and a sequence converges iff it is eventually constant.

§145.6 Norms and inner products

A norm on a vector space V satisfies positivity, homogeneity, and the triangle inequality:

$$\|v + w\| \le \|v\| + \|w\|.$$

Every norm gives a metric by

$$d(v, w) = \|v - w\|.$$

The p-norms are

$$\|x\|_p = \left(\sum_i |x_i|^p\right)^{1/p},$$

and $\|x\|_\infty = \max_i |x_i|$.

An inner product gives a norm by

$$\|v\| = \sqrt{\langle v, v\rangle}.$$

The triangle inequality for this norm follows from Cauchy–Schwarz. So Euclidean metric geometry, linear algebra, and inequalities are already intertwined.

§145.7 Open and closed balls

For $x \in X$ and $r > 0$,

$$B_r(x) = \{y : d(x, y) < r\}, \qquad \overline{B}_r(x) = \{y : d(x, y) \le r\}.$$

Open balls are open. If $y \in B_r(x)$, choose

$$\delta = r - d(x, y) > 0.$$

Then $B_\delta(y) \subseteq B_r(x)$ by the triangle inequality. The proof says: every point inside a ball has some remaining margin before hitting the boundary.

§145.8 Open sets, closed sets, and limit points

A set $U \subseteq X$ is open if every point $x \in U$ has some ball $B_r(x)$ contained in U. A set is closed if its complement is open. In metric spaces, a set C is closed iff it contains the limits of all convergent sequences from C.

A point x is a limit point of A if every ball around x meets A in a point other than x. The closure satisfies

$$x \in \overline{A} \iff (\forall r > 0)\ B_r(x) \cap A \ne \varnothing.$$

The boundary is

$$\partial A = \overline{A} \cap \overline{X \setminus A}.$$

Examples matter: every real number is a limit point of $\mathbb{Q}$, and also of the irrational numbers; in the discrete metric, points are isolated by small balls.

§145.9 Cauchy sequences and completeness

A sequence is Cauchy if

$$(\forall \varepsilon > 0)(\exists N)(\forall m, n > N)\ d(x_m, x_n) < \varepsilon.$$

A metric space is complete if every Cauchy sequence converges in the space. The standard contrast is

$$\mathbb{R} \text{ is complete}, \qquad \mathbb{Q} \text{ is not complete}.$$

A rational sequence converging to $\sqrt{2}$ is Cauchy in $\mathbb{Q}$, but it has no rational limit.

Convergence names a destination; Cauchy only says the sequence internally stabilizes. Completeness says internal stabilization is enough to guarantee a destination inside the space.

§145.10 Total boundedness and compactness

The note also points toward total boundedness and compactness. A metric space is totally bounded if for every $\varepsilon > 0$, finitely many ε-balls cover it. Total boundedness is stronger than boundedness: an infinite discrete metric space is bounded, but for $\varepsilon < 1$ it cannot be covered by finitely many ε-balls.

In $\mathbb{R}^n$, compactness is equivalent to closed and bounded, but that is the Heine–Borel theorem, not the definition in arbitrary metric spaces. A better metric-space memory is:

$$\text{compact} \Rightarrow \text{complete and totally bounded}.$$

Conversely, in metric spaces, complete plus totally bounded implies compact.

§145.11 Beyond the first calculation

This note stages the transition from analysis to topology. Metrics let us talk about boundedness, convergence, continuity, balls, closed sets, limit points, Cauchy sequences, and completeness. But later topology keeps only what can be expressed through open sets. Boundedness and completeness will fail to be topological invariants; continuity and connectedness will survive.

§145.12 Equivalent metrics and what topology forgets

On $\mathbb{R}^n$, the usual norms produce different numerical distances but the same open sets. For instance,

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\,\|x\|_\infty\,.$$

Therefore a ball for one norm contains a smaller ball for the other norm, and the two norms generate the same topology.

This inequality is the concrete mechanism behind the slogan "topology forgets exact metric geometry." A square ball and a round ball look different, but each can be squeezed inside the other after changing radius. Thus convergence and continuity agree, while lengths, angles, and diameters do not.

Remark 145.12.1 — The finite-dimensional hypothesis matters. In infinite-dimensional function spaces, norms need not be equivalent. On $C[0,1]$, the sup norm and the L^1-norm lead to very different convergence: narrow spikes can have small L^1-norm but large sup norm.

§145.13 Completion as adding ideal limit points

The completion of a metric space is not just a larger space containing more points. It is a controlled way to turn every Cauchy sequence into a convergent sequence. The construction by Cauchy sequences makes this precise:

$$\widehat{X} = \{\text{Cauchy sequences in } X\}/\sim, \qquad (x_n) \sim (y_n) \iff d(x_n, y_n) \to 0.$$

The distance between two completed points is

$$\widehat{d}([(x_n)], [(y_n)]) = \lim_{n\to\infty} d(x_n, y_n).$$

This is well-defined because the two sequences are Cauchy.

For $\mathbb{Q}$, a Cauchy sequence of rationals approximating $\sqrt{2}$ becomes a genuine point in the completion. The point is not a preferred decimal expansion; it is the whole equivalence class of rational approximations that close in on the same missing limit.

§145.14 Uniform structure hidden inside metric topology

Continuity only cares about open sets. Cauchy sequences care about more: they need a way to compare two moving points x_m and x_n. This information is called a uniform structure. Two metrics may define the same topology but different Cauchy behavior.

The homeomorphism

$$(0,1) \to \mathbb{R}, \qquad x \mapsto \tan \pi(x - 1/2)$$

shows the issue. Topologically the spaces are the same. But the Euclidean metric on $(0,1)$ sees sequences approaching 1 as Cauchy without a limit inside $(0,1)$, while their images in $\mathbb{R}$ go to infinity and are not Cauchy. Thus completeness is not topological; it is uniform/metric.

§145.15 Compactness through total boundedness

In metric spaces,

$$\text{compact} \iff \text{complete and totally bounded.}$$

This statement is more portable than Heine–Borel. The two hypotheses do different jobs. Total boundedness prevents points from spreading out into infinitely many separated regions; completeness prevents Cauchy subsequences from escaping the space.

The contrast with boundedness is useful. An infinite set with the discrete metric $d(x,y) = 1$ is bounded, because all distances are at most 1. But it is not totally bounded: for $\varepsilon < 1$, every ε-ball contains just one point, so no finite family can cover the space.

§145.16 Baire category as a completeness phenomenon

Completeness has consequences beyond sequences. The Baire category theorem says that a complete metric space cannot be written as a countable union of nowhere dense closed sets. In practice, this means complete spaces are too large to be exhausted by thin pieces.

This theorem explains why complete function spaces are powerful. Many existence theorems in analysis prove that a desired object lies in a countable intersection of dense open sets. Completeness then guarantees that the intersection is nonempty and large. Thus the elementary Cauchy condition eventually becomes a tool for proving generic existence.

§145.17 Metric spaces through completion and category

Metric topology is a bridge. It keeps enough numerical structure to discuss boundedness, Cauchy sequences, total boundedness, and completeness, while already pointing toward open-set topology through balls, neighborhoods, closure, and continuity. The advanced lesson is to separate three layers:

$$\text{topological} \quad \subsetneq \quad \text{uniform/metric} \quad \subsetneq \quad \text{geometric.}$$

Continuity lives in the first layer, completeness in the second, and lengths/angles in the third.

Completion, Uniform Structure, and Quotient Topology

N-20221221

§146.1 A Cauchy sequence can describe a point missing from the space

A sequence (x_n) in a metric space (X, d) is Cauchy when for every $\varepsilon > 0$ there is N such that

$$d(x_m, x_n) < \varepsilon \qquad (m, n \geq N).$$

It records points that eventually become mutually indistinguishable at every metric scale. A complete space contains a limit for every such sequence.

The rational sequence

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}$$

is Cauchy in $\mathbb{Q}$, but it has no rational limit. Nothing is wrong with the sequence; the intended limit is absent from the space. Completion constructs a space in which every coherent approximation process has a point to name.

Two Cauchy sequences should represent the same new point when their mutual distance tends to zero:

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad d(x_n, y_n) \longrightarrow 0.$$

Reflexivity and symmetry are immediate. If $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$, then

$$d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n) \longrightarrow 0,$$

so the relation is transitive.

§146.2 The completion metric is a limit of mutual distances

Let $\widehat{X}$ be the set of equivalence classes of Cauchy sequences. Define

$$\widehat{d}([(x_n)], [(y_n)]) = \lim_{n \to \infty} d(x_n, y_n).$$

The scalar sequence on the right is Cauchy because

$$|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_n, y_m).$$

Hence the limit exists in $\mathbb{R}$.

The value does not depend on the chosen representatives. If

$$(x_n) \sim (x'_n), \qquad (y_n) \sim (y'_n),$$

then

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \longrightarrow 0.$$

Thus both representative pairs give the same limit.

Nonnegativity and symmetry pass directly to the limit. If $\widehat{d} = 0$, then $d(x_n, y_n) \to 0$, so the classes are equal. Finally,

$$d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$$

passes to the limit and gives the triangle inequality. Therefore $\widehat{d}$ is a genuine metric.

§146.3 The original space sits isometrically and densely inside its completion

Send $x \in X$ to the constant sequence:

$$\iota(x) = [(x, x, x, \ldots)].$$

Then

$$\widehat{d}(\iota(x), \iota(y)) = d(x, y),$$

so ι is an isometric embedding.

Its image is dense. Let

$$\xi = [(x_n)] \in \widehat{X}.$$

For fixed N,

$$\widehat{d}(\xi, \iota(x_N)) = \lim_{m \to \infty} d(x_m, x_N).$$

Because (x_n) is Cauchy, this quantity tends to zero as $N \to \infty$. Thus

$$\iota(x_N) \longrightarrow \xi.$$

Every completed point is a limit of original points.

For $X = \mathbb{Q}$, the decimal truncations (q_n) of $\sqrt{2}$ define a class in $\widehat{\mathbb{Q}}$. It cannot equal any constant rational class: if it equalled $r \in \mathbb{Q}$, then

$$|q_n - r| \longrightarrow 0,$$

forcing $r = \sqrt{2}$. The new class is exactly the missing irrational point described by rational approximations.

§146.4 A diagonal approximation proves that the completed space is complete

Let (ξ_r) be a Cauchy sequence in $\widehat{X}$. Passing to a subsequence if necessary, arrange that

$$\widehat{d}(\xi_r, \xi_{r+1}) < 2^{-r}.$$

Density of $\iota(X)$ lets us choose $y_r \in X$ with

$$\widehat{d}(\iota(y_r), \xi_r) < 2^{-r}.$$

Then

$$\begin{aligned} d(y_r, y_{r+1}) &\le \widehat{d}(\iota(y_r), \xi_r) + \widehat{d}(\xi_r, \xi_{r+1}) \\ &\quad + \widehat{d}(\xi_{r+1}, \iota(y_{r+1})) \\ &< 2^{-r} + 2^{-r} + 2^{-(r+1)}. \end{aligned}$$

The geometric tail shows that (y_r) is Cauchy in X. Let

$$\xi = [(y_r)] \in \widehat{X}.$$

The construction gives

$$\xi_r \longrightarrow \xi$$

along the chosen subsequence, and since the original sequence was Cauchy, the whole sequence has the same limit. Hence $\widehat{X}$ is complete.

The argument exposes the mechanism: approximate each abstract completed point by an original point with rapidly shrinking error, then use one diagonal Cauchy sequence in X to encode the limit.

§146.5 Uniformly continuous maps extend because they preserve Cauchy information

Let Y be complete and let

$$f : X \to Y$$

be uniformly continuous. If (x_n) is Cauchy, then $(f(x_n))$ is Cauchy. Indeed, for every $\varepsilon > 0$, choose one $\delta > 0$ working simultaneously for all points:

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon.$$

The Cauchy condition eventually places every pair x_m, x_n within δ.

Define

$$\widehat{f} : \widehat{X} \to Y, \qquad \widehat{f}([(x_n)]) = \lim_{n\to\infty} f(x_n).$$

If $(x_n) \sim (y_n)$, uniform continuity gives

$$d_Y(f(x_n), f(y_n)) \longrightarrow 0,$$

so the definition is independent of representatives. It extends f because constant sequences retain their values.

The extension is unique. If $g : \widehat{X} \to Y$ is continuous and agrees with f on X, then for

$$\xi = \lim_n \iota(x_n)$$

one has

$$g(\xi) = \lim_n g(\iota(x_n)) = \lim_n f(x_n) = \widehat{f}(\xi).$$

Thus completion is characterized by the extension property

$$\boxed{\text{uniformly continuous maps into complete spaces extend uniquely.}}$$

Uniform continuity is essential. The continuous map

$$f : (0,1) \to \mathbb{R}, \qquad f(x) = \frac{1}{x}$$

has no continuous extension to the completion $[0,1]$, because it diverges along $x_n = 1/n \to 0$.

§146.6 The universal property makes completion unique

Suppose

$$\iota_X : X \hookrightarrow Y, \qquad j_X : X \hookrightarrow Z$$

are dense isometric embeddings into complete metric spaces. Apply the extension property to the uniformly continuous map

$$j_X \circ \iota_X^{-1} : \iota_X(X) \to Z.$$

It extends uniquely to an isometry

$$F : Y \to Z.$$

Reversing the roles gives

$$G : Z \to Y.$$

Both composites agree with the identity on the dense copy of X. By uniqueness of extension,

$$GF = \mathrm{id}_Y, \qquad FG = \mathrm{id}_Z.$$

Therefore any two completions are uniquely isometric in a way that fixes X.

The quotient of Cauchy sequences is one concrete construction; the universal property is what makes the result canonical.

§146.7 Homeomorphism forgets the uniform scale of closeness

The interval $(0,1)$ and the real line are homeomorphic via

$$h(x) = \tan(\pi(x - 1/2)).$$

Yet $(0,1)$ is bounded and incomplete in its Euclidean metric, while $\mathbb{R}$ is unbounded and complete. Boundedness and completeness are therefore not topological invariants.

The map is not uniformly continuous. Take

$$x_n = 1 - \frac{1}{n}, \qquad y_n = 1 - \frac{2}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \longrightarrow 0,$$

but

$$h(x_n) = \cot\frac{\pi}{n}, \qquad h(y_n) = \cot\frac{2\pi}{n},$$

and their difference grows on the order of n. No single δ controls the map near both endpoints.

Topology records which sets are neighborhoods. Uniform structure additionally records which pairs are uniformly close across the whole space. Completion depends on the latter, because Cauchy sequences are uniform objects.

§146.8 A continuous bijection needs extra hypotheses to be a homeomorphism

The map

$$f : [0, 2\pi) \to S^1, \qquad f(t) = (\cos t, \sin t)$$

is continuous and bijective, but its inverse is not continuous at $(1,0)$. Points approaching $(1,0)$ from the lower semicircle have parameters tending to 2π, while the point itself has parameter 0.

A useful positive theorem is the compact–Hausdorff criterion.

Theorem 146.8.1

If X is compact, Y is Hausdorff, and $f : X \to Y$ is a continuous bijection, then f is a homeomorphism.

Proof. A closed subset $C \subseteq X$ is compact. Its image $f(C)$ is compact in Y, hence closed because Y is Hausdorff. Thus f is a closed map. For any closed $C \subseteq X$,

$$(f^{-1})^{-1}(C) = f(C)$$

is closed, so f^{-1} is continuous. □

The failed half-open-interval example lacks compactness. The theorem identifies exactly which global hypotheses make inverse continuity automatic.

§146.9 Quotient topology is designed to control maps out of a gluing

Let $q : X \to Y$ be a surjection. The quotient topology on Y is defined by

$$U \subseteq Y \text{ open} \quad \Longleftrightarrow \quad q^{-1}(U) \text{ open in } X.$$

It is the strongest topology on Y making q continuous.

Its universal property is the factorization test:

$$g : Y \to Z \text{ is continuous} \quad \Longleftrightarrow \quad g \circ q : X \to Z \text{ is continuous.}$$

Thus a map out of a quotient is constructed upstairs, where the original points are still visible, provided the map is constant on equivalence classes.

A subset $A \subseteq X$ is saturated when it is a union of equivalence classes:

$$A = q^{-1}(q(A)).$$

For arbitrary A,

$$q(A) \text{ is open in } Y \quad \Longleftrightarrow \quad q^{-1}(q(A)) \text{ is open in } X.$$

A quotient map need not send every open set to an open set. On $[0, 1]$, collapse the entire interval $[0, 1/2]$ to one point. The open set $(0, 1/2)$ has saturation $[0, 1/2]$, which is not open, so its image is not open in the quotient.

The quotient topology controls precisely the saturated information that survives the identification.

§146.10 Endpoint gluing becomes the circle by quotient factorization

Let

$$q : [0, 1] \to [0, 1]/(0 \sim 1)$$

be the endpoint quotient, and define

$$F(t) = e^{2\pi i t}.$$

Since

$$F(0) = F(1),$$

the map is constant on equivalence classes. The quotient universal property produces a unique continuous map

$$\overline{F} : [0, 1]/(0 \sim 1) \to S^1$$

with

$$\overline{F} \circ q = F.$$

It is bijective: every point of the circle has a parameter in $[0, 1]$, and the only repeated parameter values are the identified endpoints.

The quotient of the compact interval is compact, while S^1 is Hausdorff. By the compact–Hausdorff theorem,

$$\boxed{[0,1]/(0 \sim 1) \cong S^1.}$$

A neighborhood of the glued point has preimage

$$[0,\varepsilon) \cup (1-\varepsilon, 1],$$

so two one-sided endpoint neighborhoods become one two-sided circular neighborhood. The concrete local picture, the quotient factorization property, and the compactness theorem all describe the same gluing.

Open-Set Calculus: Continuity, Closure, Nets, and Connected Components

N-20221222

§147.1 Continuity is the stability of tests under preimage

Let $f : X \to Y$ be a map between topological spaces. The definition

$$V \subseteq Y \text{ open} \quad \Longrightarrow \quad f^{-1}(V) \subseteq X \text{ open}$$

uses preimages because preimages preserve all Boolean operations:

$$f^{-1}\left(\bigcup_\alpha V_\alpha\right) = \bigcup_\alpha f^{-1}(V_\alpha), \qquad f^{-1}(V \cap W) = f^{-1}(V) \cap f^{-1}(W).$$

Consequently continuous maps compose without an additional theorem:

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)).$$

The open-set definition has equivalent closed-set and neighborhood forms.

Proposition 147.1.1

For a map $f : X \to Y$, the following are equivalent.

(i) f is continuous.

(ii) The preimage of every closed set is closed.

(iii) For every $x \in X$ and every neighborhood V of $f(x)$, there is a neighborhood U of x such that $f(U) \subseteq V$.

Proof. The equivalence of (i) and (ii) follows from

$$f^{-1}(Y \setminus C) = X \setminus f^{-1}(C).$$

Assume (i). If V is a neighborhood of $f(x)$, choose an open set W with

$$f(x) \in W \subseteq V.$$

Then $U = f^{-1}(W)$ is a neighborhood of x and $f(U) \subseteq V$. Conversely, apply (iii) to an open set V and every point of $f^{-1}(V)$. The resulting neighborhoods show that $f^{-1}(V)$ is open. $\square$

Images do not enjoy this stability. The continuous map $x \mapsto x^2$ sends $(-1, 1)$ to $[0, 1)$, which is not open in $\mathbb{R}$.

§147.2 The Sierpiński space classifies open subsets

Let
$$\Sigma = \{0,1\}, \qquad \tau_\Sigma = \{\varnothing, \{1\}, \Sigma\}.$$
This is the Sierpiński space. For a subset $U \subseteq X$, define
$$\chi_U(x) = \begin{cases} 1, & x \in U, \\ 0, & x \notin U. \end{cases}$$
Then
$$\chi_U^{-1}(\{1\}) = U.$$
Since $\{1\}$ is the only nontrivial open test in Σ, one obtains
$$\boxed{U \text{ is open in } X \quad \Longleftrightarrow \quad \chi_U : X \to \Sigma \text{ is continuous.}}$$
Thus open subsets of X are in bijection with continuous maps $X \to \Sigma$. The preimage operation is literally composition:
$$\chi_{f^{-1}(U)} = \chi_U \circ f.$$
This gives a concrete meaning to the contravariance of topology: a map of spaces pulls open tests backward.

§147.3 Closure is an operator from which the topology can be recovered

For $A \subseteq X$, the closure is
$$\overline{A} = \bigcap \{C \subseteq X : C \text{ closed and } A \subseteq C\}.$$
Equivalently,
$$x \in \overline{A} \quad \Longleftrightarrow \quad U \cap A \neq \varnothing \text{ for every open neighborhood } U \ni x.$$
The closure operator satisfies the Kuratowski rules
$$\overline{\varnothing} = \varnothing, \qquad A \subseteq \overline{A}, \qquad \overline{\overline{A}} = \overline{A}, \qquad \overline{A \cup B} = \overline{A} \cup \overline{B}.$$
Monotonicity follows: $A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$.

Conversely, suppose an operation $c : \mathcal{P}(X) \to \mathcal{P}(X)$ satisfies these four rules. Declare C closed exactly when $c(C) = C$. Arbitrary intersections of such fixed points are fixed, because if $C = \bigcap_i C_i$, then monotonicity gives
$$c(C) \subseteq \bigcap_i c(C_i) = C,$$
while extensivity gives the reverse inclusion. Finite unions are fixed by the fourth rule. Hence the fixed points define the closed sets of a topology, and c is its closure operation.

Closure is therefore not merely derived notation; it contains the same information as the topology.

§147.4 Subspace topology is the weakest topology making inclusion continuous

For $Y \subseteq X$, the subspace topology is

$$\tau_Y = \{Y \cap U : U \in \tau_X\}.$$

It is the weakest topology on Y for which the inclusion

$$\iota : Y \hookrightarrow X$$

is continuous. More precisely, for any space Z and map $g : Z \to Y$,

$$g : Z \to Y \text{ is continuous} \quad \iff \quad \iota \circ g : Z \to X \text{ is continuous.}$$

Closure behaves exactly as expected.

Proposition 147.4.1
If $A \subseteq Y \subseteq X$, then

$$\boxed{\overline{A}^{Y} = Y \cap \overline{A}^{X}.}$$

Proof. A point $y \in Y$ lies in $\overline{A}^{Y}$ iff every subspace neighborhood $Y \cap U$ of y meets A. This is equivalent to saying that every neighborhood U of y in X meets A, hence $y \in \overline{A}^{X}$. □

For example, $[0, 1)$ is open in $[0, 2]$ because

$$[0, 1) = [0, 2] \cap (-1, 1),$$

even though it is not open in $\mathbb{R}$.

§147.5 Nets recover closure in every topological space

A directed set is a preorder $(D, \leq)$ such that every two elements have a common upper bound. A net in X is a function

$$x_\bullet : D \to X.$$

It converges to x if for every neighborhood $U \ni x$, there exists $d_0 \in D$ such that

$$d \geq d_0 \quad \Longrightarrow \quad x_d \in U.$$

Sequences are the special case $D = \mathbb{N}$.

Theorem 147.5.1 (Net characterization of closure)
For $A \subseteq X$,

$$x \in \overline{A} \quad \iff \quad \text{some net in } A \text{ converges to } x.$$

Proof. If a net (a_d) in A converges to x, every neighborhood of x eventually contains some a_d, so it meets A.

Conversely, suppose $x \in \overline{A}$. Direct the set $\mathcal{N}(x)$ of neighborhoods of x by reverse inclusion:

$$U \preceq V \quad \Longleftrightarrow \quad V \subseteq U.$$

For each $U \in \mathcal{N}(x)$, choose $a_U \in A \cap U$. Given a neighborhood $W \ni x$, whenever $U \succeq W$ one has $U \subseteq W$, hence $a_U \in W$. Therefore (a_U) converges to x. ☐

Sequences may fail to detect closure. Let $[0, \omega_1]$ carry the order topology, where ω_1 is the first uncountable ordinal, and set

$$A = [0, \omega_1).$$

Every neighborhood of ω_1 contains a tail $(\alpha, \omega_1]$, so $\omega_1 \in \overline{A}$. But a sequence $(\alpha_n) \subset A$ has countable supremum

$$\alpha = \sup_n \alpha_n < \omega_1,$$

and therefore cannot converge to ω_1. The net indexed by ordinals $\alpha < \omega_1$, ordered in the usual way, does converge to ω_1.

§147.6 Connectedness is the absence of a discrete-valued separator

A separation of X is a decomposition

$$X = U \sqcup V$$

into nonempty open sets. A space is connected when no separation exists. Equivalently, the only clopen subsets are $\varnothing$ and X.

Let $D = \{0, 1\}$ have the discrete topology.

Proposition 147.6.1

A space X is connected iff every continuous map

$$f : X \to D$$

is constant.

Proof. If f is nonconstant, then $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty disjoint open sets covering X. Conversely, a separation $X = U \sqcup V$ defines a nonconstant continuous characteristic map taking U to 0 and V to 1. ☐

Continuous images preserve connectedness. If $f(X) = A \sqcup B$ were separated, then $f^{-1}(A)$ and $f^{-1}(B)$ would separate X.

This gives the intermediate value theorem in topological form. Since an interval $I \subseteq \mathbb{R}$ is connected, the image $f(I)$ of a continuous real-valued function is connected, hence an interval. Therefore every value between $f(a)$ and $f(b)$ is attained.

§147.7 Paths give connectedness, but not every connected space is path-connected

A path from x to y is a continuous map

$$\gamma : [0, 1] \to X, \qquad \gamma(0) = x, \quad \gamma(1) = y.$$

Since $[0, 1]$ is connected, its image is connected. If a path-connected space had a separation, no path could join points in the two pieces. Hence

$$\text{path-connected} \quad \Longrightarrow \quad \text{connected.}$$

The converse fails for the topologist's sine curve

$$T = \{(x, \sin(1/x)) : x > 0\} \cup (\{0\} \times [-1, 1]).$$

The graph

$$G = \{(x, \sin(1/x)) : x > 0\}$$

is the continuous image of $(0, \infty)$, so it is connected. Its closure is T, and the closure of a connected set is connected; thus T is connected.

It is not path-connected. Suppose a path $\gamma(t) = (x(t), y(t))$ joined a point of the vertical segment to a point of G. Let

$$t_0 = \sup\{t : x(t) = 0\}.$$

Immediately after t_0, the coordinate $x(t) > 0$ and tends to zero, while

$$y(t) = \sin(1/x(t)).$$

By the intermediate value property of x, the quantity $1/x(t)$ crosses arbitrarily large intervals, forcing $y(t)$ to oscillate between values near 1 and -1 arbitrarily close to t_0. This contradicts continuity of y at t_0.

§147.8 Local path-connectedness repairs the gap between components

The connected component of x is the union of all connected subsets containing x. The path component of x is the set of points joined to x by paths. Every path component lies inside a connected component.

A space is locally path-connected if every point has a neighborhood basis of path-connected open sets.

Theorem 147.8.1

In a locally path-connected space, path components are open and coincide with connected components.

Proof. Let P be a path component and let $x \in P$. Choose a path-connected open neighborhood $U \ni x$. Every point of U can be joined to x, and x can be joined to the chosen basepoint of P. Hence $U \subseteq P$. Thus P is open.

Path components partition X, so the complement of P is a union of other open path components. Therefore P is also closed. A connected component cannot meet two disjoint clopen path components, so it is contained in one path component. The reverse containment always holds, giving equality. $\square$

Open subsets of $\mathbb{R}^n$, manifolds, and CW complexes are locally path-connected. The topologist's sine curve fails precisely at the limiting vertical segment, where arbitrarily small neighborhoods inherit infinitely many oscillating pieces. Local hypotheses explain why ordinary geometric spaces obey the simpler intuition while general topology does not.

Paths to Groups: Homotopy, Coverings, Fundamental Groups, and Presentations

N-20221223

§148.1 Path concatenation is reversible but not strictly associative

A path in a space X is a continuous map

$$\gamma : [0,1] \to X.$$

Its initial and terminal points are

$$s(\gamma) = \gamma(0), \qquad t(\gamma) = \gamma(1).$$

If $t(\gamma) = s(\eta)$, define the concatenation

$$(\gamma * \eta)(u) = \begin{cases} \gamma(2u), & 0 \le u \le \frac{1}{2}, \\ \eta(2u-1), & \frac{1}{2} \le u \le 1. \end{cases}$$

The reverse path is

$$\overline{\gamma}(u) = \gamma(1-u).$$

At the level of parametrized paths, concatenation is not strictly associative. The paths

$$(\alpha * \beta) * \gamma \qquad \text{and} \qquad \alpha * (\beta * \gamma)$$

traverse the same geometric pieces but allocate different subintervals of $[0,1]$ to them. Likewise, $\gamma * \overline{\gamma}$ is not literally the constant path. The desired algebra appears only after quotienting by a suitable deformation relation.

§148.2 Homotopy relative endpoints is a congruence for concatenation

Two paths $\gamma_0, \gamma_1 : [0,1] \to X$ with the same endpoints are homotopic relative endpoints if there is a continuous map

$$H : [0,1] \times [0,1] \to X$$

such that

$$H(u,0) = \gamma_0(u), \qquad H(u,1) = \gamma_1(u),$$

and

$$H(0,v) = s(\gamma_0), \qquad H(1,v) = t(\gamma_0)$$

for every v.

This is an equivalence relation. It is also compatible with concatenation: if

$$\gamma_0 \simeq \gamma_1 \text{ rel } \partial I, \qquad \eta_0 \simeq \eta_1 \text{ rel } \partial I,$$

then
$$\gamma_0 * \eta_0 \simeq \gamma_1 * \eta_1 \text{ rel } \partial I.$$
Indeed, concatenate the two homotopies at each time v. Thus homotopy relative endpoints is a congruence for path composition.

Reparametrization through an increasing map $r : [0,1] \to [0,1]$ with $r(0) = 0$, $r(1) = 1$ does not change the homotopy class. A linear interpolation between $r(u)$ and u gives the homotopy. This removes the associativity defect:
$$[\alpha] * ([\beta] * [\gamma]) = ([\alpha] * [\beta]) * [\gamma].$$
Similarly,
$$[\gamma] * [\overline{\gamma}] = [c_{s(\gamma)}], \qquad [\overline{\gamma}] * [\gamma] = [c_{t(\gamma)}].$$

§148.3 The fundamental groupoid remembers every basepoint at once

The fundamental groupoid $\Pi_1(X)$ has

- points of X as objects;
- homotopy classes relative endpoints of paths as morphisms;
- concatenation as composition.

Every morphism is invertible, with inverse represented by the reversed path. This is why a groupoid, rather than a monoid, is the natural first algebraic object attached to a space.

Fixing a basepoint x_0 and taking endomorphisms gives the fundamental group
$$\pi_1(X, x_0) = \mathrm{Aut}_{\Pi_1(X)}(x_0).$$
Its elements are based loops modulo based homotopy.

If ρ is a path from x_0 to x_1, basepoint change is
$$\rho_\# : \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad [\gamma] \longmapsto [\overline{\rho} * \gamma * \rho].$$
A different choice of ρ changes this isomorphism by an inner automorphism. The groupoid keeps this dependence visible instead of hiding it behind a noncanonical identification.

§148.4 The exponential covering turns loops into integer endpoints

Consider
$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{2\pi i t}.$$
For each $t_0 \in \mathbb{R}$, restriction to
$$\left(t_0 - \frac{1}{2}, t_0 + \frac{1}{2}\right)$$
is a homeomorphism onto $S^1 \setminus \{-p(t_0)\}$. Thus p is a covering map.

Let $\gamma : [0,1] \to S^1$ and choose $a \in \mathbb{R}$ with $p(a) = \gamma(0)$. Subdivide $[0,1]$ so that the image of each subinterval lies in one evenly covered arc. On the first subinterval use the inverse branch whose value at 0 is a; on each later subinterval choose the unique branch agreeing at the common endpoint. This constructs a unique lift

$$\widetilde{\gamma} : [0,1] \to \mathbb{R}, \qquad p \circ \widetilde{\gamma} = \gamma, \qquad \widetilde{\gamma}(0) = a.$$

If γ is a loop based at 1 and $\widetilde{\gamma}(0) = 0$, then

$$e^{2\pi i \widetilde{\gamma}(1)} = 1,$$

so

$$\widetilde{\gamma}(1) \in \mathbb{Z}.$$

Define the winding number by

$$\mathrm{wind}(\gamma) = \widetilde{\gamma}(1).$$

Concatenation adds endpoints of lifts:

$$\mathrm{wind}(\gamma * \eta) = \mathrm{wind}(\gamma) + \mathrm{wind}(\eta).$$

A based homotopy of loops lifts after fixing the initial lift. The terminal lift is a continuous integer-valued function of the homotopy parameter, hence constant. Therefore winding number depends only on the based homotopy class.

§148.5 The covering computation gives $\pi_1(S^1) \cong \mathbb{Z}$

The map

$$W : \pi_1(S^1, 1) \to \mathbb{Z}, \qquad W([\gamma]) = \mathrm{wind}(\gamma)$$

is a homomorphism.

It is surjective because

$$\gamma_n(u) = e^{2\pi i n u}$$

has lift $\widetilde{\gamma}_n(u) = nu$ and winding number n.

It is injective. Suppose $W([\gamma]) = 0$. The lift $\widetilde{\gamma}$ is then a loop in $\mathbb{R}$ based at 0. The linear homotopy

$$\widetilde{H}(u, v) = (1 - v)\widetilde{\gamma}(u)$$

contracts it to the constant loop. Projecting by p gives a based null-homotopy of γ. Hence

$$\ker W = 0.$$

Therefore

$$\boxed{\pi_1(S^1, 1) \cong \mathbb{Z}.}$$

The group operation is addition of winding numbers, and reversal changes the sign.

§148.6 The punctured plane retracts onto the circle

Let

$$\mathbb{C}^* = \mathbb{C} \setminus \{0\}.$$

The map

$$r : \mathbb{C}^* \to S^1, \qquad r(z) = \frac{z}{|z|}$$

is a deformation retraction. Define

$$H(z,v) = \left((1-v) + \frac{v}{|z|}\right) z.$$

The scalar coefficient is positive, so $H(z,v) \neq 0$. Moreover,

$$H(z,0) = z, \qquad H(z,1) = \frac{z}{|z|},$$

and if $|z| = 1$, then $H(z,v) = z$ for all v. Thus the inclusion $S^1 \hookrightarrow \mathbb{C}^*$ and r are homotopy inverses.

Consequently,

$$\boxed{\pi_1(\mathbb{C}^*, 1) \cong \mathbb{Z}.}$$

The missing point is measured exactly by winding number.

§148.7 Continuous maps induce homomorphisms

A based continuous map

$$f : (X, x_0) \to (Y, y_0)$$

sends a loop γ to $f \circ \gamma$. It preserves homotopies and concatenation, hence induces

$$f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0), \qquad f_*([\gamma]) = [f \circ \gamma].$$

Functoriality is immediate:

$$(g \circ f)_* = g_* \circ f_*, \qquad (\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X)}\,.$$

For

$$f_m : S^1 \to S^1, \qquad f_m(z) = z^m,$$

one has

$$f_m \circ \gamma_n(u) = e^{2\pi i m n u}.$$

Under $\pi_1(S^1) \cong \mathbb{Z}$, the induced homomorphism is therefore

$$(f_m)_* : \mathbb{Z} \to \mathbb{Z}, \qquad n \longmapsto mn.$$

This computation distinguishes the maps $z \mapsto z^m$ up to based homotopy.

§148.8 Van Kampen assembles groups from overlapping pieces

Let $X = U \cup V$, where $U, V, U \cap V$ are path-connected open sets containing a basepoint x_0. The Seifert–van Kampen theorem states that the diagram

$$\begin{tikzcd}
\pi_1(U \cap V, x_0) \arrow[r] \arrow[d] & \pi_1(U, x_0) \arrow[d] \\
\pi_1(V, x_0) \arrow[r] & \pi_1(X, x_0)
\end{tikzcd}$$

is a pushout in the category of groups. Concretely,

$$\pi_1(X, x_0) \cong \frac{\pi_1(U, x_0) * \pi_1(V, x_0)}{\langle\langle i_{U*}(\omega) i_{V*}(\omega)^{-1} : \omega \in \pi_1(U \cap V, x_0) \rangle\rangle}.$$

The free product records loops from the two pieces; the normal closure imposes the identifications already visible in the overlap.

For the figure-eight

$$X = S^1 \vee S^1,$$

choose neighborhoods retracting onto the two circles whose intersection is contractible. The overlap contributes no relation, so

$$\boxed{\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} = F(a, b).}$$

The words

$$ab, \quad ba, \quad aba^{-1}b^{-1}$$

represent genuinely different loop classes; no commutativity relation is present.

§148.9 Attaching a two-cell adds one normal relation

Let X be path-connected and attach a disk along a loop

$$\varphi : S^1 \to X.$$

The resulting space

$$X \cup_\varphi D^2$$

contains a new disk across which the attaching loop can contract. Van Kampen gives

$$\pi_1(X \cup_\varphi D^2) \cong \pi_1(X)/\langle\langle[\varphi]\rangle\rangle.$$

Thus cell attachment translates directly into a group presentation.

A torus has a CW structure with one vertex, two one-cells a, b, and one two-cell attached along

$$aba^{-1}b^{-1}.$$

Starting from the figure-eight gives

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} = 1\rangle \cong \mathbb{Z}^2.$$

The two basic loops commute because the square filling of the commutator becomes the two-cell of the torus.

§148.10 Connected coverings of the circle are indexed by subgroups of $\mathbb{Z}$

For each $n \geq 1$, the map

$$p_n : S^1 \to S^1, \qquad p_n(z) = z^n$$

is an n-sheeted connected covering. Its induced subgroup is

$$(p_n)_*\pi_1(S^1) = n\mathbb{Z} \subseteq \mathbb{Z}.$$

Conversely, every subgroup of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 0$. The general classification theorem for sufficiently well-behaved spaces says that connected based coverings correspond to subgroups of the fundamental group. For the circle, the entire classification is already visible in the elementary maps $z \mapsto z^n$.

The progression is now concrete:

$$\text{paths} \longrightarrow \text{homotopy classes} \longrightarrow \text{groupoid and groups} \longrightarrow \text{coverings and presentations.}$$

Algebra enters because deformation classes of reversible paths carry composition, and topology becomes computable because coverings and gluings turn that composition into integers, free words, and relations.

Quotient Topology and Product Topology

N-20221228

§149.1 Interval modulo endpoints

The note begins with the quotient

$$D^1 = [-1, 1], \qquad -1 \sim 1.$$

Identifying the two endpoints gives a circle:

$$D^1/(-1 \sim 1) \cong S^1.$$

A small neighborhood around the glued point corresponds to two half-interval neighborhoods at -1 and 1 upstairs. This is exactly the local reason quotient topology is defined by preimages.

More generally,

$$D^n/S^{n-1} \cong S^n.$$

Collapsing the boundary of a disk to one point produces a sphere.

§149.2 More quotient spaces

The note records the standard square identifications. If a square has opposite vertical sides identified in the same direction, one obtains a cylinder. If the two circular boundary components of that cylinder are then identified in the same direction, one obtains a torus. If one pair of edges is identified with a twist, one obtains a Möbius band or a Klein bottle depending on the full gluing pattern.

Another example is

$$\mathbb{R}/\mathbb{Z} \cong S^1,$$

where $x \sim y$ iff $y - x \in \mathbb{Z}$. This is the real line wrapped around the circle.

§149.3 Definition of quotient topology

Definition 149.3.1 (Quotient topology)**.** Given a surjective map $q : X \to Y$, the quotient topology on Y declares $U \subseteq Y$ open exactly when $q^{-1}(U)$ is open in X.

Let $q : X \to X/\sim$ be the quotient map. A set $U \subseteq X/\sim$ is open iff

$$q^{-1}(U) \text{ is open in } X.$$

This is the finest topology on the quotient making q continuous. The note emphasizes that quotient topology is not guessed by drawing; it is forced by the universal property of the quotient map.

§149.4 Product topology

For spaces X, Y, the product topology on $X \times Y$ is generated by basic open rectangles

$$U \times V, \qquad U \subseteq X \text{ open}, \quad V \subseteq Y \text{ open}.$$

Thus a set is open in $X \times Y$ if it is a union of such rectangles.

The page's picture of a small rectangle in a grid illustrates the basic rule: local neighborhoods in a product are products of local neighborhoods in the factors.

§149.5 Universal property of quotient topology

The quotient topology is not chosen by visual intuition alone. Let $q : X \to Y = X/\sim$ be the quotient map. A map $f : Y \to Z$ is continuous iff

$$f \circ q : X \to Z$$

is continuous. This is the universal property of the quotient topology.

This property explains the definition. We declare $U \subseteq Y$ open precisely when $q^{-1}(U)$ is open in X, because this is exactly what makes continuity out of the quotient detectable before quotienting.

§149.6 Endpoint identification in detail

For $[0, 1]/(0 \sim 1)$, points near the glued point come from two pieces upstairs: a small interval near 0 and a small interval near 1. This is why the quotient looks like a circle rather than an interval with a strange endpoint. The local neighborhoods have been glued.

Similarly, $D^2/S^1 \cong S^2$: collapse the boundary circle of a disk to a single point. The interior of the disk becomes the sphere with one point removed, and the collapsed boundary becomes the missing point.

§149.7 Product topology via projections

The product topology on $X \times Y$ is the coarsest topology making the projection maps

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y$$

continuous. Its basis consists of rectangles

$$U \times V$$

where $U \subseteq X$ and $V \subseteq Y$ are open.

The word "coarsest" matters. We do not add more open sets than necessary. Once all projection preimages $\pi_X^{-1}(U) = U \times Y$ and $\pi_Y^{-1}(V) = X \times V$ are open, finite intersections give $U \times V$, and arbitrary unions give the full product topology.

§149.8 Quotient and product as opposite directions

Product topology builds a larger space from two spaces while preserving maps out to the factors. Quotient topology builds a smaller space by identifying points while preserving maps out of the quotient. Product is controlled by projections; quotient is controlled by the quotient map.

This contrast is a useful memory:

$$\text{product: maps out to factors,} \qquad \text{quotient: maps out after gluing.}$$

Both definitions are forced by continuity requirements rather than by pictures.

§149.9 A second lens

Quotient topology and product topology are dual-looking constructions. A quotient identifies points and defines open sets by pulling back along the quotient map. A product combines spaces and defines open sets by rectangles from the two factors. Both are examples of topology being controlled by maps rather than by drawings alone.

§149.10 Saturated sets and quotient openness

For a quotient map $q : X \to Y$, a subset $A \subseteq X$ is saturated if it is a union of equivalence classes, equivalently

$$A = q^{-1}(q(A)).$$

Open sets in Y correspond exactly to saturated open sets in X. This is a practical way to test quotient topology: first check that the upstairs set is open, then check that it contains whole equivalence classes.

Endpoint gluing works because neighborhoods near the glued point have saturated preimages containing pieces near both endpoints. A one-sided interval near only one endpoint is not saturated, so it does not define a neighborhood downstairs.

§149.11 Quotients can destroy Hausdorffness

Not every quotient of a nice space is nice. For example, identify all rational numbers in $\mathbb{R}$ to one point and all irrational numbers to another point. The quotient has two points, but the preimage of either singleton is dense in $\mathbb{R}$, so the two quotient points cannot be separated by disjoint open neighborhoods.

This warning matters because quotient pictures can be misleading. The quotient topology is defined by preimages, and sometimes that produces spaces with poor separation properties.

§149.12 Product topology and the mapping property

The product topology is characterized by a universal property dual to the quotient one. A map

$$h : Z \to X \times Y$$

is continuous iff both coordinate maps

$$\pi_X \circ h : Z \to X, \qquad \pi_Y \circ h : Z \to Y$$

are continuous. This is why the product topology is the coarsest topology making the projections continuous.

Thus product and quotient are both controlled by maps:

construction	distinguished map	continuity test
quotient	$q : X \to X/{\sim}$	f out of quotient via $f \circ q$
product	π_X, π_Y	h into product via coordinates

§149.13 From squares to surfaces through edge words

The quotient square diagrams can be encoded algebraically. A torus has boundary word

$$aba^{-1}b^{-1},$$

while a Klein bottle has a word of the form

$$aba^{-1}b.$$

The arrows in the picture determine whether a generator appears as a or a^{-1}. This is already the language of CW complexes and fundamental-group presentations.

So the square gluing is not merely visual. It tells us how a two-cell is attached to a one-skeleton, and the attaching word becomes a relation in the fundamental group.

§149.14 Products and quotients need not commute blindly

Although $(\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/\mathbb{Z})$ gives the torus, one should not assume products and quotients always interact without hypotheses. The map

$$X \times Y \to (X/{\sim}) \times (Y/{\approx})$$

passes through the quotient by the product equivalence relation, but whether the resulting topology behaves as expected depends on the quotient maps and spaces involved.

This is another reason universal properties are safer than pictures: they tell us which maps are actually continuous and which topology is being imposed.

§149.15 Products, quotients, and gluing data

Quotient topology is about gluing while controlling maps out of the glued space. Product topology is about pairing spaces while controlling maps into the product. Endpoint identification, $\mathbb{R}/\mathbb{Z}$, torus gluings, and square diagrams are not separate examples; they are the same principle expressed through equivalence classes, saturated sets, and universal properties.

Algebraic Topology and Algebra: Products, Wedge Sums, CW Complexes, Cosets, and Surface Gluings

N-20221229

§150.1 What this note is really doing

This note is a working page where several threads meet: the torus as $(\mathbb{R}/\mathbb{Z})^2$, the product $S^1 \times S^1$, disjoint union and wedge sum, CW complexes, the algebra of cosets, and the square/polygon gluing pictures for surfaces. The original page should not be compressed into a list of definitions. Its real movement is: topology suggests quotient pictures; quotient pictures resemble congruence classes; congruence classes lead to cosets; then the topological quotient of a square returns as the torus and Klein bottle.

The top margin says that there is another intuitive understanding of

$$S^1 \times S^1 \cong T^2.$$

Think of a torus as a vertical circle carried around a horizontal circle. The path traced by that moving circle is the torus. This is not merely a picture: it is a parametrization.

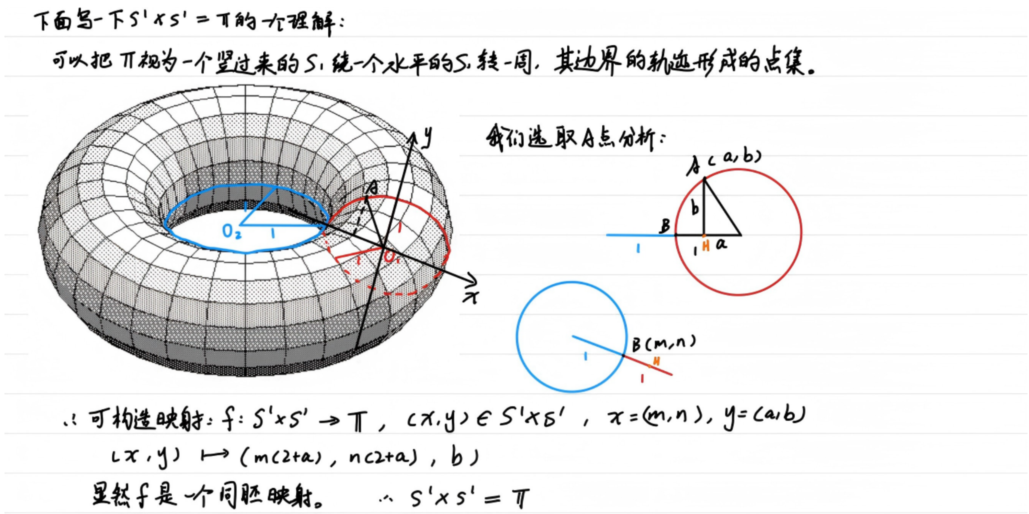


Figure 150.1: The original visual analysis of $S^1 \times S^1$: one parameter goes around the big circle, the other around the small circular fiber.

§150.2 $S^1 \times S^1$ as a parametrized torus

Write a point of the first circle as an angle θ, and a point of the second circle as an angle ϕ. For a torus with major radius R and minor radius r, a standard parametrization is

$$F(\theta, \phi) = ((R + r\cos\phi)\cos\theta, (R + r\cos\phi)\sin\theta, r\sin\phi).$$

This realizes $S^1 \times S^1 \to T^2$. The first angle decides where the circular cross-section is placed around the central hole; the second decides where we are on that cross-section.

The handwritten picture tries to choose a point A and decompose it into two circle parameters. That is exactly the right intuition, but the invariant statement is cleaner: the torus is the space of ordered pairs of circle-points, then realized geometrically by the map above.

§150.3 Disjoint union

For topological spaces X and Y, the disjoint union $X \amalg Y$ keeps the two spaces formally separate. A subset $U \subseteq X \amalg Y$ is open iff

$$U \cap X \text{ is open in } X, \qquad U \cap Y \text{ is open in } Y.$$

If X and Y are nonempty, then $X \amalg Y$ is disconnected. Indeed, X and Y are both open in the disjoint union topology, disjoint, nonempty, and their union is the whole space. This proof is more precise than the vague statement that the spaces are simply "side by side". The topology, not only the picture, gives the separation.

§150.4 Wedge sum

The wedge sum is obtained from a disjoint union by identifying chosen basepoints:

$$X \vee Y = (X \amalg Y)/(x_0 \sim y_0).$$

Thus disjoint union places spaces side by side, while wedge sum fuses them at one point. The prototype is $S^1 \vee S^1$, the figure-eight space.

Example 57.4.4 ($S^1 \vee S^1$ is a figure eight)
Let $X = S^1$ and $Y = S^1$, and let $x_0 \in X$ and $y_0 \in Y$ be any points. Then $X \vee Y$ is a "figure eight": it is two circles fused together at one point.

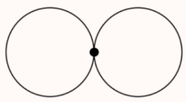


Figure 150.2: The wedge sum $S^1 \vee S^1$ as two circles fused at one point.

This construction matters because loops in the two circles remain visibly independent. Algebraically, the fundamental group of the figure eight is the free group on two generators. So the picture is not a toy: it is one of the first spaces where topology produces noncommutative algebra.

§150.5 CW complexes: building spaces by cells

A CW complex is built inductively. Start with a set of 0-cells. Attach 1-cells by gluing endpoints to the 0-skeleton. Attach 2-cells by gluing the boundary circle of a disk to the 1-skeleton. Continue dimension by dimension:

$$X^0 \subseteq X^1 \subseteq X^2 \subseteq \cdots .$$

A k-cell is a copy of D^k attached along S^{k-1} by an attaching map

$$\varphi_\alpha : S_\alpha^{k-1} \to X^{k-1}.$$

Informally: attach disks along their boundaries.

The note compares two CW structures on D^2. One uses two 0-cells, two 1-cells, and one 2-cell. The other uses one 0-cell, one 1-cell whose two endpoints are glued to that point, and one 2-cell. Both build a disk, but the second is more economical.

Example 57.5.2 (D^2 with $2+2+1$ and $1+1+1$ cells)

(a) First, we start with X^0 having two points e_a^0 and e_b^0. Then, we join them with two 1-cells D^1 (green), call them e_c^1 and e_d^1. The endpoints of each 1-cell (the copy of S^0) get identified with distinct points of X^0; hence $X^1 \cong S^1$. Finally, we take a single 2-cell e^2 (yellow) and weld it in, with its boundary fitting into the copy of S^1 that we just drew. This gives the figure on the left.

(b) In fact, one can do this using just $1+1+1=3$ cells. Start with X^0 having a single point e^0. Then, use a single 1-cell e^1, fusing its two endpoints into the single point of X^0. Then, one can fit in a copy of S^1 as before, giving D^2 as on the right.

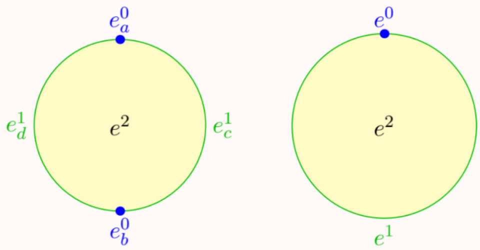


Figure 150.3: Two CW descriptions of D^2: one more literal, one more economical.

The lesson is that a CW decomposition is not unique. It is chosen because it makes later topology and algebra computable.

§150.6 The torus as a quotient of a square

The torus can be formed from a square by identifying opposite edges in the same direction. Walking off the right edge returns one at the corresponding point on the left edge; walking off the top returns one at the bottom. Algebraically,

$$T^2 \cong (\mathbb{R}/\mathbb{Z})^2 \cong S^1 \times S^1.$$

The quotient square viewpoint treats $[0,1] \times [0,1]$ as a fundamental domain for $(\mathbb{R}/\mathbb{Z})^2$.

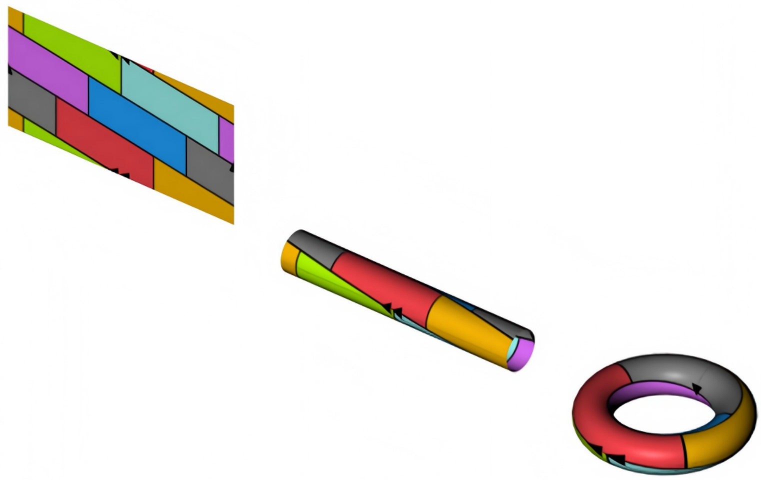


Figure 150.4: The quotient square/cylinder/torus picture: $(\mathbb{R}/\mathbb{Z})^2$ as a glued square.

As a CW complex, the torus has one 0-cell, two 1-cells a, b, and one 2-cell. The boundary of the 2-cell is attached along the word

$$aba^{-1}b^{-1}.$$

This gives the familiar presentation

$$\pi_1(T^2) = \langle a, b \mid aba^{-1}b^{-1} = 1\rangle \cong \mathbb{Z}^2.$$

The commutativity of $\mathbb{Z}^2$ is encoded by the boundary word of the square.

§150.7 Cosets as algebraic quotients

The algebra excerpt in the original pages is not random. It is the algebraic analogue of quotienting. If H is a subgroup of G, right congruence modulo H is

$$a \equiv_r b \pmod{H} \iff ab^{-1} \in H.$$

It is an equivalence relation. Reflexivity uses $aa^{-1} = e \in H$; symmetry uses $(ab^{-1})^{-1} = ba^{-1}$; transitivity uses

$$ac^{-1} = (ab^{-1})(bc^{-1}).$$

The equivalence class of a is the right coset

$$Ha = \{ha : h \in H\}.$$

The map $H \to Ha$, $h \mapsto ha$, is a bijection, so $|Ha| = |H|$. The group G is partitioned into cosets, and two cosets are either equal or disjoint. This is exactly the algebraic version of cutting a space into equivalence classes.

This is why the note can return from cosets to $(\mathbb{R}/\mathbb{Z})$ naturally. Both are quotient constructions: identify things differing by a subgroup.

§150.8 The Klein bottle

The Klein bottle is built like the torus, except one pair of edges is identified in the opposite direction. If one first glues a pair of edges, one gets a cylinder. But gluing the resulting boundary circles in opposite directions cannot be embedded in ordinary three-dimensional space without self-intersection. The usual picture in $\mathbb{R}^3$ is therefore an immersion, not a faithful embedding.

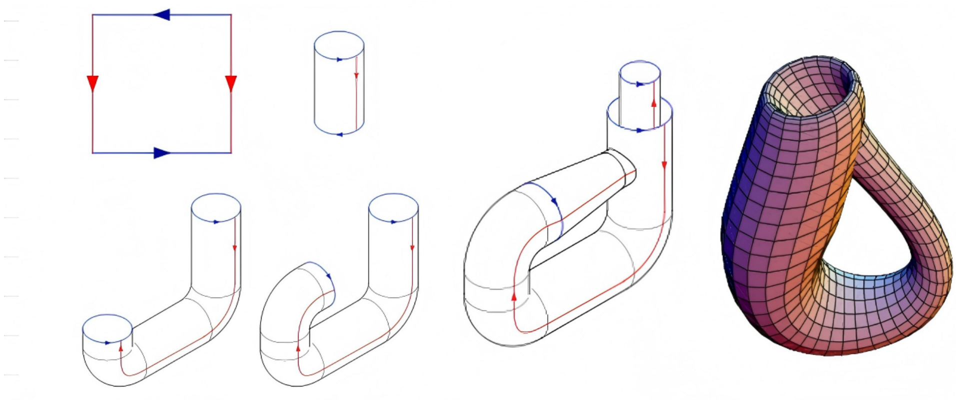


Figure 150.5: The Klein bottle: one edge identification is reversed, so the usual 3D picture self-intersects.

As a CW complex, the Klein bottle again has one 0-cell, two 1-cells, and one 2-cell, but the attaching word changes. A common presentation is

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

The relation says that going around one direction reverses the other. Thus the torus and Klein bottle have the same cell counts but different attaching words, and this difference controls orientability and fundamental group.

§150.9 Polygon schemata for surfaces

The last figure records polygon schemata for surfaces. A polygon with labelled, oriented boundary edges can be glued according to those labels. The resulting quotient is a surface. For example, the torus corresponds to

$$aba^{-1}b^{-1},$$

while an orientable surface of genus g has a boundary word of the form

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

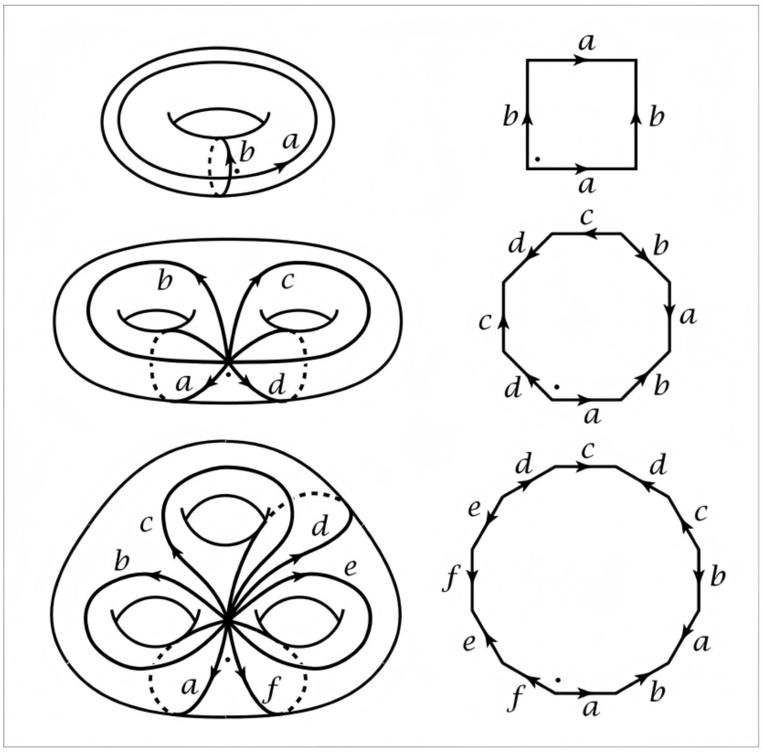


Figure 150.6: Polygon schemata and surface gluings: topology encoded by labelled boundary words.

The handwritten comment says these diagrams took a long time to understand. That is exactly right. The picture compresses several ideas: cut a surface into a polygon, label the boundary, glue equal labels, and read the labels as algebraic generators and relations.

§150.10 Projective spaces as quotient spaces

The section title also mentions real and complex projective spaces. They fit the same quotient philosophy. Real projective space is

$$\mathbb{RP}^n = S^n/(x \sim -x),$$

the space of lines through the origin in $\mathbb{R}^{n+1}$. Complex projective space is the space of complex lines through the origin in $\mathbb{C}^{n+1}$, equivalently

$$\mathbb{CP}^n = S^{2n+1}/S^1.$$

A useful CW fact is that $\mathbb{RP}^n$ has one cell in each dimension $0, 1, \ldots, n$, while $\mathbb{CP}^n$ has one cell in each even real dimension $0, 2, 4, \ldots, 2n$.

§150.11 Cellular boundary words and fundamental groups

The attaching word of the two-cell is not a decorative label. For the torus, the word

$$aba^{-1}b^{-1}$$

becomes the relation saying that a and b commute. For the Klein bottle, the twist changes the relation to one of the form

$$aba^{-1} = b^{-1}.$$

Thus two spaces can have the same numbers of cells but different topology because the attaching maps differ.

§150.12 Why polygon schemata are efficient

A labelled polygon schema compresses a surface into boundary data. The labels record which edges are glued, and the arrows record orientation. Once this is understood, the diagram becomes an algebraic object: reading around the boundary gives a word in generators and inverses.

This is the reason algebraic topology can turn pictures into group presentations and chain complexes.

§150.13 Further context

The repaired version of this note should be read as one continuous story. First, $S^1 \times S^1$ parametrizes the torus. Second, a square with opposite sides identified gives the same torus as a quotient. Third, disjoint union and wedge sum explain how spaces can be combined before quotienting. Fourth, CW complexes systematize this by attaching cells. Fifth, cosets provide the algebraic analogue of quotienting. Finally, torus, Klein bottle, and polygon schemata show that a whole surface can be encoded by a boundary word.

The central lesson is that algebraic topology is not simply topology plus algebra added later. The topology is already algebraic as soon as we record how pieces are glued: arrows, labels, words, equivalence classes, and quotient maps are the bridge.

§150.14 Cellular chain complex of the torus

The torus CW structure with one 0-cell, two 1-cells a, b, and one 2-cell gives a cellular chain complex

$$0 \to C_2 \cong \mathbb{Z} \xrightarrow{\partial_2} C_1 \cong \mathbb{Z}^2 \xrightarrow{\partial_1} C_0 \cong \mathbb{Z} \to 0.$$

Since both ends of each one-cell attach to the single zero-cell, $\partial_1 = 0$. The two-cell is attached by the commutator word

$$aba^{-1}b^{-1}.$$

In cellular homology, the total exponent of a is $1 - 1 = 0$, and the total exponent of b is $1 - 1 = 0$, so $\partial_2 = 0$. Therefore

$$H_0(T^2) \cong \mathbb{Z}, \qquad H_1(T^2) \cong \mathbb{Z}^2, \qquad H_2(T^2) \cong \mathbb{Z}.$$

The same square diagram that gives the fundamental group also gives homology by linearizing the boundary word.

§150.15 Klein bottle: same cells, different boundary

The Klein bottle has the same cell counts as the torus: one 0-cell, two 1-cells, and one 2-cell. But the attaching word is different. With a common convention, it is

$$aba^{-1}b.$$

The total exponent of a is 0, while the total exponent of b is 2. Hence the cellular boundary has the form

$$\partial_2(1) = 2b.$$

So

$$H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, \qquad H_2(K) = 0.$$

The absence of $H_2 \cong \mathbb{Z}$ is the homological shadow of non-orientability.

§150.16 Euler characteristic as cell bookkeeping

For a finite CW complex,

$$\chi(X) = \sum_k (-1)^k c_k,$$

where c_k is the number of k-cells. The torus and Klein bottle both have

$$\chi = 1 - 2 + 1 = 0.$$

Thus Euler characteristic cannot distinguish them. Their difference lies not in the number of cells but in the attaching map of the two-cell. This is the core lesson of CW complexes: cell counts are only the first approximation; attaching maps carry the topology.

§150.17 Orientability from polygon words

In polygon schemata, orientability can be read from edge orientations. For an orientable surface, each label appears once in each direction, as in

$$aba^{-1}b^{-1}.$$

For a non-orientable surface, some label appears twice with the same direction after normalizing the word, as in projective-plane-type words.

This explains why the torus is orientable and the Klein bottle is not. The twist in the edge gluing reverses local orientation after going around a loop.

§150.18 Van Kampen and presentations from gluing

The one-skeleton of the torus is $S^1 \vee S^1$, whose fundamental group is the free group $F(a, b)$. Attaching the two-cell along $aba^{-1}b^{-1}$ imposes the relation

$$aba^{-1}b^{-1} = 1.$$

Thus

$$\pi_1(T^2) = \langle a, b \mid aba^{-1}b^{-1} = 1 \rangle \cong \mathbb{Z}^2.$$

For the Klein bottle, the attaching word imposes a different relation, giving

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle.$$

This is the precise algebraic meaning of the arrows on the edge-identification diagram.

§150.19 Projective spaces through cells

The quotient descriptions

$$\mathbb{RP}^n = S^n/(x \sim -x), \qquad \mathbb{CP}^n = S^{2n+1}/S^1$$

are not just definitions. They explain the CW structures:

$$\mathbb{RP}^n : \text{one cell in every dimension } 0, 1, \dots, n,$$

while

$$\mathbb{CP}^n : \text{one cell in every even dimension } 0, 2, \dots, 2n.$$

For $\mathbb{CP}^n$, this leads to homology groups $\mathbb{Z}$ in even degrees and 0 in odd degrees. Thus a quotient construction immediately produces computable algebraic invariants.

§150.20 Cellular surfaces through homology and presentations

This note is about turning pictures into algebra. A product picture gives $S^1 \times S^1$. A gluing picture gives a quotient. A CW picture gives chain groups and boundary maps. A labelled boundary word gives a fundamental-group presentation and, after abelianization, homology. The advanced lesson is that algebraic topology is not added after the drawings; it is already encoded in the arrows, labels, and attaching maps.

Hopf Fibration, Homotopy, and the Infinite-Stability Reading

N-20240727

§151.1 What this chapter is for

This note records the first route by which the line

$$0 = 1 - 1 = -1 + 1 = 0$$

became mathematically suggestive to me. It should be read as a working interpretation rather than as the final explanation. The important point is to preserve the original path of thought in full detail.

The triggering background was a lecture poster by Peng Keyao, whose abstract seemed to connect elementary arithmetic, the Hopf fibration, homotopy groups, configuration spaces, algebraic K-theory, and the strange equation above. The poster immediately reminded me of an older lecture on the Hopf fibration. At that time the picture of linked circles in S^3 was one of the first images that made topology feel like something deeper than point-set definitions.

But the line itself first looked analytic and algebraic rather than topological. The string

$$1 - 1 = -1 + 1$$

looks like rearrangement, cancellation, and ambiguity of formal sums. So the first reading I tried to build was:

$$\begin{gathered}\text{alternating sums} \longrightarrow \text{regularization}\\ \longrightarrow \text{Eilenberg swindle}\\ \longrightarrow \text{vanishing under infinite stability.}\end{gathered}$$

This note unfolds that reading carefully.

§151.2 The first analytic reflex

The most immediate association is the infinite alternating series

$$1 - 1 + 1 - 1 + \cdots .$$

Its partial sums are

$$1,\ 0,\ 1,\ 0,\ \ldots,$$

so the ordinary limit does not exist.

Definition 151.2.1 (Cesàro summation)**.** Let $s_n = a_0 + \cdots + a_n$ be the partial sums of a series. If the averages

$$\sigma_N = \frac{s_0 + s_1 + \cdots + s_N}{N+1}$$

converge to L, then the series is Cesàro summable to L.

For the alternating series, the Cesàro averages of the partial sums tend to

$$\frac{1}{2}.$$

Thus the expression is not convergent in the usual sense, but it has a regularized value in the Cesàro sense.

Definition 151.2.2 (Abel summation)**.** Given a formal series $\sum_{n\geq 0} a_n$, form its power series

$$F(r) = \sum_{n\geq 0} a_n r^n$$

for $0 < r < 1$. If $F(r)$ has a limit L as $r \to 1^-$, then the original series is Abel summable to L.

For

$$1 - 1 + 1 - 1 + \cdots,$$

one has

$$1 - r + r^2 - r^3 + \cdots = \frac{1}{1+r},$$

so Abel summation again gives $1/2$.

> **Remark 151.2.3** — The same cultural memory also points toward the analytic continuation of the Dirichlet eta function. At the level of this first note, the important point is not the exact analytic formalism, but the habit of reading a formally divergent expression through a more flexible rule.

This is why the line

$$0 = 1 - 1 = -1 + 1 = 0$$

first felt like a statement about how formal algebra changes when one passes from finite arithmetic to limiting procedures. In ordinary arithmetic, the line is trivial. In the world of infinite expressions, however, the order of grouping and the rule of interpretation may change what the expression seems to mean.

Claim 151.2.4 (First analytic slogan)**.** The equation should perhaps be read not as a numerical equality, but as evidence that cancellation becomes unstable once infinite formal processes are allowed.

§151.3 Why the Hopf and K-theory hints changed the search

The analytic interpretation alone did not explain why the meme included the Hopf fibration and K-theory. Those are not the usual tools for assigning values to divergent series. The Hopf fibration suggests homotopy and stable phenomena. K-theory suggests addition, cancellation, and group completion of mathematical objects.

So the first analytic reflex had to be upgraded. Instead of asking only whether

$$1 - 1 + 1 - 1 + \cdots$$

has a regularized value, I began asking whether there is a categorical or K-theoretic situation in which cancellation becomes legitimate because an object is infinitely large or infinitely stable.

That is exactly the kind of situation where the Eilenberg swindle appears.

§151.4 The Eilenberg swindle as the central guess

The Eilenberg swindle is the mechanism that made the analytic reflex look relevant to K-theory.

Theorem 151.4.1 (Eilenberg swindle, working form)
Suppose an additive category has countable direct sums and an object

$$S = A \oplus A \oplus A \oplus \cdots .$$

Then

$$S \cong A \oplus S.$$

Consequently, any Grothendieck-type invariant K_0 compatible with direct sums satisfies

$$[S] = [A] + [S],$$

and therefore forces

$$[A] = 0.$$

Idea. The isomorphism is the shift of the infinite direct sum:

$$A \oplus A \oplus A \oplus \cdots \cong A \oplus (A \oplus A \oplus A \oplus \cdots).$$

The finite summand A is absorbed by the infinite tail. Passing to an additive invariant converts this absorption into a vanishing statement. □

This is the algebraic version of the same psychological picture as the alternating series. One can group an infinite expression in different ways:

$$(1 - 1) + (1 - 1) + \cdots = 0,$$

but also

$$1 + (-1 + 1) + (-1 + 1) + \cdots = 1.$$

The paradox is not that arithmetic is broken. The point is that infinite formal manipulation can erase the distinction between adding one copy and adding no copy, because the tail of the expression absorbs the finite change.

Remark 151.4.2 — This is the reason I originally took the line $0 = 1 - 1 = -1 + 1 = 0$ as a compact symbolic form of infinite absorption. The two middle expressions are different finite presentations, but after entering an infinitely stable world they both collapse back to the same zero object.

§151.5 How this looked like a K-theory statement

In algebraic K-theory, one often begins with a category of objects equipped with a direct-sum operation. The first invariant records formal additive differences.

Definition 151.5.1 (Grothendieck group)**.** Let M be a commutative monoid. Its Grothendieck group $K_0(M)$ is the universal abelian group receiving a monoid homomorphism from M. Concretely, elements may be represented by formal differences

$$[A] - [B],$$

subject to the relation

$$[A \oplus C] - [B \oplus C] = [A] - [B].$$

The expression $1 - 1$ then resembles a formal difference between one positive copy and one negative copy. The Eilenberg-swindle reading was that some sufficiently infinite environment might force such formal differences to vanish.

Proposition 151.5.2 (Vanishing by absorption, heuristic form)

If a category contains an object S with

$$S \cong S \oplus A,$$

then any additive invariant that sees direct sum as addition tends to identify the class of A with zero.

This is why swindle arguments prove vanishing theorems in categories with infinite coproducts or infinite-dimensional stabilization. At the level of intuition, the meme ladder became a short chain: Hopf fibration points toward stable homotopy; stable homotopy points toward group completion in K-theory; group completion suggests infinite stability and the Eilenberg swindle; the final line is read as symbolic infinite cancellation. In this reading, the final line is a “God-tier” arithmetic trick: after enough stabilization, the complicated machinery collapses into the fact that infinity can absorb a finite summand.

§151.6 The role I imagined for the Hopf fibration

The Hopf fibration still needed a place in this interpretation. I imagined it as the geometric entrance into stable homotopy. The Hopf map

$$S^3 \to S^2$$

is a concrete example of a map whose nontriviality is not visible from ordinary dimension-counting. It teaches that topology can remember a hidden process.

From there, stable homotopy suggests that some phenomena persist after suspending or stabilizing. In the original reading, I loosely connected this with the word “stable” in algebra: perhaps after passing to a sufficiently stabilized category, the same kind of hidden structure becomes algebraic cancellation, and the end result is a vanishing statement.

This was not a precise theorem. It was a ladder of associations:

$$\text{Hopf} \quad \rightsquigarrow \quad \text{stable homotopy} \quad \rightsquigarrow \quad \text{stable algebra} \quad \rightsquigarrow \quad \text{swindle}.$$

But as a first notebook explanation, it made the meme feel less arbitrary. The Hopf fibration stood for the beginning of stable topology; the equation stood for the end of a stabilization trick.

§151.7 The meme as a learning ladder

The meme that triggered the reflection should be kept as part of the record. It should not be read as a proof. It is a ladder of associations.

Figure 151.1: The meme that compressed Hopf fibration, stable homotopy, K-theory, and the equation $0 = 1 - 1 = -1 + 1 = 0$ into one ladder.

In the first reading, the ladder was interpreted as follows: In the first reading, the ladder was interpreted as a sequence of working meanings. The Hopf fibration meant hidden topology; homotopy groups meant invariants of deformation; stable homotopy meant persistence under stabilization; K-theory meant organized addition and cancellation; the Eilenberg swindle meant vanishing by infinite stability. This is why the original explanation was not merely a random guess about divergent series. It was trying to reconcile every visible clue in the meme with a known mechanism in K-theory.

§151.8 What the original interpretation claimed

The original interpretation can be stated cleanly as a working thesis.

Theorem 151.8.1 (Original working interpretation, not yet the final reading)
In elementary arithmetic, $0 = 1 - 1 = -1 + 1 = 0$ is trivial. In an infinitely stable categorical setting, however, finite additions and cancellations can be absorbed by an infinite tail. The Eilenberg swindle expresses this absorption. Therefore the equation may be read as a symbolic way to say that certain K-theoretic invariants vanish once infinite direct-sum operations are allowed.

This reading emphasizes three words:

infinity, stability, vanishing.

The line $0 = 1-1 = -1+1 = 0$ then becomes an emblem of the idea that a sophisticated invariant may collapse because the ambient category is too large.

§151.9 A slightly uneasy stopping point

Even at this first stage, there was something slightly unstable about the explanation. The lecture poster mentioned finite sets, configuration spaces, and the Hopf fibration, while the swindle explanation relies heavily on infinite objects. The analytic examples also use infinite series and limiting procedures. This suggested that the original route might be pointing toward the right neighbourhood but not yet at the exact object.

Still, the first interpretation is worth preserving. It captured several genuine connections:

- alternating sums explain why $1 - 1$ and $-1 + 1$ feel like more than ordinary arithmetic;
- the Eilenberg swindle explains how infinite stabilization can trivialize K-theoretic information;
- K-theory really is about direct sum, group completion, and cancellation;
- the Hopf fibration really does point toward stable homotopy.

So this note stops at the original working picture. The central claim of that picture is not that 0 is secretly nonzero. It is that in an infinitely stabilized algebraic world, the distinction between adding and not adding a finite piece can disappear. The later note will separate this first interpretation from the finite homotopical mechanism actually used in the lecture.

A Nontrivial Loop at Zero: FinSet K-Theory and Stable Homotopy

N-20251217

§152.1 Why this second note exists

This note is the reflective version of the earlier summer memo. The first reaction to the line

$$0 = 1 - 1 = -1 + 1 = 0$$

was not the answer eventually given by the lecture. A few days before writing this correction, while I was organizing my homepage, my roommate Su Yuyang saw the same line. His immediate thought was the familiar analytic story around

$$\sum_{k \geq 0} (-1)^k = 1 - 1 + 1 - 1 + \cdots ,$$

whose Cesàro value and Abel value are $1/2$, and which is also connected with the analytic continuation of the Dirichlet eta function. This was also my own first thought. The string $1 - 1 = -1 + 1$ looks like a rearrangement or regularization trick before it looks like topology.

But the meme and the lecture poster clearly featured the Hopf fibration and K-theory. That made the analytic-summation explanation feel too small. Following the same original line of thought, I then searched for a K-theoretic or categorical mechanism in which an apparently impossible cancellation becomes legitimate. This led naturally to the Eilenberg swindle.

The model I had in mind was this. If an object admits an infinite decomposition

$$S = A \oplus A \oplus A \oplus \cdots ,$$

then adding or removing one copy of A may not change S. Formally, one shifts the infinite direct sum:

$$S \cong A \oplus S.$$

In a Grothendieck group this kind of absorption can force

$$[A] = 0.$$

In the language of the meme, I was reading

$$0 = 1 - 1 = -1 + 1 = 0$$

as a symbolic version of infinite stability: by regrouping an infinite alternating expression, one can make the same expression look like 0, 1, or a vanishing object. This was my "Level 6" interpretation: the ultimate trick behind the line is that infinity absorbs finite differences, and therefore certain K-groups or invariants vanish.

That guess was not completely unrelated. It correctly noticed that K-theory is about addition, cancellation, group completion, and stability. It also correctly noticed that an equality such as $0 = 0$ might be hiding a process rather than an elementary arithmetic fact. But the direction was almost the opposite of Prof. Peng's point. My first interpretation emphasized:

infinite objects,
analytic or algebraic cancellation,
triviality and vanishing.

The lecture's point is instead:

finite sets,
homotopy of a finite creation–swap–annihilation process,
nontriviality of a loop.

So the slogan of this note is:

the important object is not the endpoint 0, but the loop based at 0.

The loop is created by making a $+$ and a $-$, swapping them, and annihilating them again:

$$\varnothing \sim \{+, -\} \sim \{-, +\} \sim \varnothing.$$

Under the K-theory of finite sets, this loop becomes a nontrivial element. Under the Barratt–Priddy–Quillen theorem, it corresponds to the stable Hopf class.

§152.2 The lecture starts from ordinary equations

The slide deck begins with familiar equations: Pythagoras, Euler's identity, Stokes' theorem, the Navier–Stokes equation, the Langlands slogan, and then the strange meme equation. This is pedagogically important. It suggests that an equation is not merely a string of symbols with a truth value. An equation may also encode a method, a structure, or a path.

In elementary arithmetic,

$$0 = 1 - 1 = -1 + 1 = 0$$

is boring. Both sides are zero. But the lecture asks us to stop looking only at the value. In a higher structure, there can be more than one way for an object to equal itself. The identity loop at 0 and the creation–swap–annihilation loop at 0 need not be the same.

> **Remark 152.2.1** — This is the conceptual bridge to homotopy type theory and higher category theory. A statement $a = b$ can be represented by a path from a to b. Then two proofs of equality may themselves be compared by a homotopy. The lecture uses this philosophy in a concrete topological model rather than as formal type theory.

§152.3 The Hopf fibration as the first nontrivial map

The first mathematical anchor is the Hopf fibration. Define

$$S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}.$$

The Hopf map is

$$H : S^3 \to \mathbb{CP}^1 \cong S^2, \qquad (z_1, z_2) \mapsto [z_1 : z_2].$$

For every point of S^2, its preimage is a circle. These circles are linked in a highly organized way.

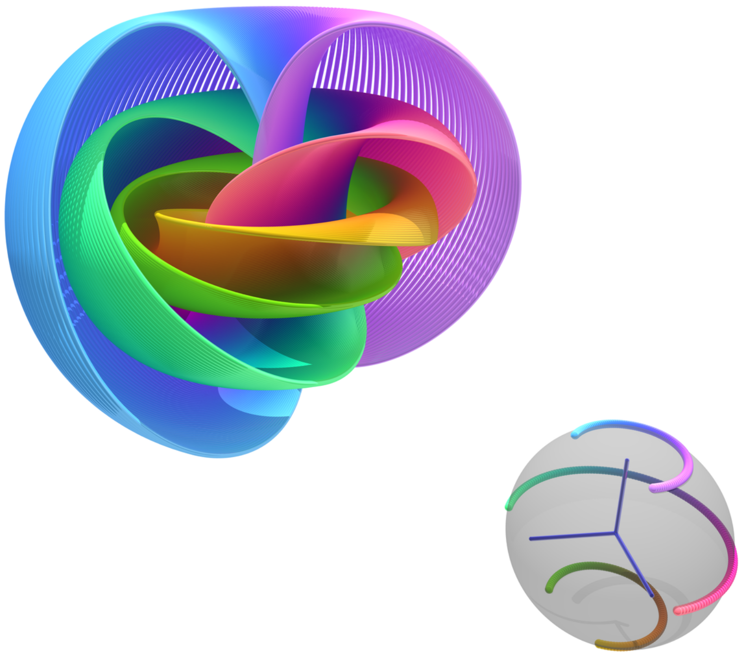


Figure 152.1: the Hopf fibration visualized by linked fibres.

The Hopf fibration matters because it is not null-homotopic. It represents a nontrivial element of

$$\pi_3(S^2).$$

In beginner language: even though S^3 and S^2 are both spheres, the map $H : S^3 \to S^2$ cannot be continuously shrunk to a constant map.

Definition 152.3.1 (Null-homotopic map)**.** A map $f : X \to Y$ is null-homotopic if it is homotopic to a constant map. Equivalently, there is a continuous family of maps $f_t : X \to Y$ with $f_0 = f$ and f_1 constant.

The Hopf map is the first warning: a process can be topologically nontrivial even if it looks, from far away, like a map between simple spaces.

§152.4 Homotopy groups and the first stable class

For a based space X, the group $\pi_n(X)$ is made from based maps $S^n \to X$, modulo based homotopy. The lecture reviews the low-dimensional intuition:

$$\pi_0(X) = \text{connected components}, \qquad \pi_1(X) = \text{loops modulo homotopy}.$$

Higher homotopy groups generalize this to maps from higher-dimensional spheres.

For spheres, the groups are easy below the diagonal:

$$\pi_i(S^n) = 0 \qquad (i < n).$$

On the diagonal, one has the degree:

$$\pi_n(S^n) \cong \mathbb{Z}.$$

But above the diagonal the world becomes subtle. The Hopf map gives the famous example

$$\pi_3(S^2) \neq 0.$$

The stable homotopy groups of spheres are obtained by repeatedly suspending:

$$\pi_i^s(\mathbb{S}) = \lim_{n\to\infty} \pi_{i+n}(S^n).$$

The Hopf map contributes the first stable nontrivial class

$$\eta \in \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2.$$

This class has order two: doing it twice becomes trivial, but doing it once is not trivial.

§152.5 From natural numbers to finite sets

The lecture then changes viewpoint. Instead of defining the natural numbers as primitive symbols, define them as isomorphism classes of finite sets:

$$\mathbb{N} \cong \pi_0(\mathrm{FinSet}).$$

Here a finite set of size n represents the number n. Disjoint union represents addition:

$$A + B := A \sqcup B.$$

To get the integers, one takes a group completion. Informally, one allows formal differences

$$A - B.$$

Thus the number $1 - 1$ is represented by a pair consisting of one positive point and one negative point.

> **Remark 152.5.1** — The corresponding slide phrases this as a move from numbers to finite sets: the natural number n is represented by an n-element set, and addition is represented by disjoint union. This is the first categorification step in the lecture: instead of remembering only the cardinality, one keeps the finite set itself and later also its symmetries.

Definition 152.5.2 (Grothendieck group)**.** Given a commutative monoid M, its Grothendieck group $K_0(M)$ is the universal abelian group receiving a monoid homomorphism from M. For finite sets under disjoint union this gives

$$K_0(\mathrm{FinSet}) \cong \mathbb{Z}.$$

So far, this still only explains the value of $1 - 1$. It does not yet explain why the path

$$0 \to 1 - 1 \to -1 + 1 \to 0$$

can be interesting. For that, we must remember not just the set of isomorphism classes, but the space of finite sets and isomorphisms.

§152.6 The space of finite sets

The slide deck asks us to consider the moduli space

$$\mathcal{F} = \mathrm{FinSet}/\sim.$$

This is not merely a set of cardinalities. It remembers automorphisms of finite sets. A finite set with n elements has automorphism group S_n, the symmetric group.

Remark 152.6.1 — The slide at this point depicts the moduli space of finite sets as a space whose n-point component remembers S_n. In notebook language, the important formula is

$$\mathcal{F} \simeq \coprod_{n\geq 0} BS_n.$$

Thus the component remembers the number n, while loops in that component remember permutations.

One can think of $\mathcal{F}$ as a disjoint union of classifying spaces:

$$\mathcal{F} \simeq \coprod_{n\geq 0} BS_n.$$

This formula is the first place where the hidden loops appear. A point in BS_n remembers a finite set of size n, but loops at that point remember permutations of the set.

Thus the finite set with one positive and one negative point is not just a cardinality-zero object. It can have a nontrivial path coming from swapping the two labelled positions before annihilation.

§152.7 The K-space of finite signed sets

To group-complete finite sets at the level of spaces, the slides introduce a space $K(\mathcal{F})$. Its points are finite ordered configurations whose elements are marked by $+$ or $-$. Adjacent opposite signs may cancel.

Remark 152.7.1 — The slide introduces the K-space by replacing finite sets with signed finite configurations. The picture is simple enough to state in words: points marked $+$ and points marked $-$ may be created and cancelled in opposite-signed pairs, and the empty configuration is used as the base point.

The empty configuration $\varnothing$ is the base point. The configuration $\{+,-\}$ has total signed cardinality zero, but it is not identical to the empty configuration before one performs cancellation. This distinction is the seed of the whole example.

The higher K-groups are defined by

$$K_i(\mathrm{FinSet}) = \pi_i(K(\mathcal{F}), \varnothing).$$

Thus K_0 records components, while K_1 records loops based at the empty configuration.

§152.8 The loop h

Now the lecture constructs the loop

$$h : \varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing.$$

It has three stages.

First, create a cancelling pair:

$$\varnothing \longrightarrow \{+,-\}.$$

Second, swap the two points:

$$\{+,-\} \longrightarrow \{-,+\}.$$

Third, annihilate the cancelling pair:

$$\{-,+\} \longrightarrow \varnothing.$$

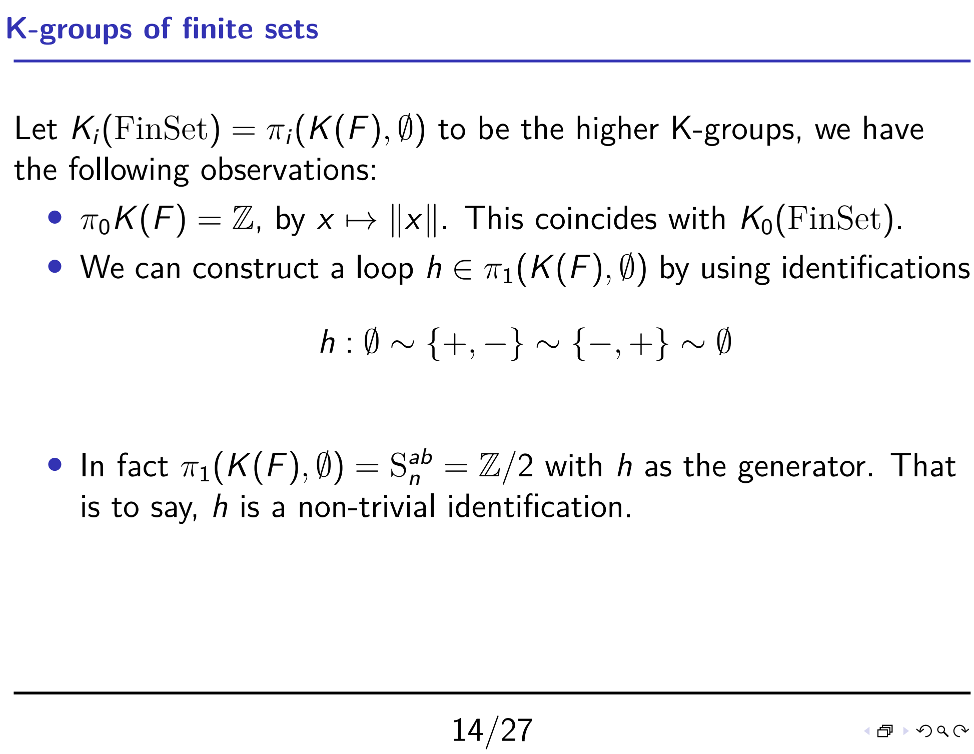


Figure 152.2: From the lecture slides: the loop h represented by creation, swap, and annihilation.

This is the corrected meaning of

$$0 = 1 - 1 = -1 + 1 = 0.$$

It is not the claim that zero equals zero. It is the claim that there is a particular loop at zero. The middle step, the swap, is the nontrivial part.

> **Remark 152.8.1** — If the two points were not moved, the movie would look like a boring creation followed by immediate annihilation. The swap forces a braid-like motion. In configuration space, this motion wraps around the collision locus where the two points would meet.

The slides record the key computation:

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2,$$

and the loop h gives the nontrivial element. Doing the swap twice is null-homotopic, but doing it once is not.

§152.9 Why the first guess was close but reversed

The analytic reading begins with the infinite alternating expression

$$1 - 1 + 1 - 1 + \cdots.$$

Cesàro summation replaces the raw partial sums

$$1,\ 0,\ 1,\ 0, \dots$$

by their averages, which tend to $1/2$. Abel summation studies

$$1 - r + r^2 - r^3 + \cdots = \frac{1}{1+r}$$

and then lets $r \to 1^-$, again obtaining $1/2$. Analytic continuation of the Dirichlet eta function gives another sophisticated version of the same cultural memory.

This is why Su Yuyang's first reaction and my first reaction were natural. The line

$$0 = 1 - 1 = -1 + 1 = 0$$

looks as if it is inviting a discussion of how formal rearrangements, limiting processes, and regularizations change the meaning of an alternating expression.

The Eilenberg swindle pushes that same instinct from analysis into algebra. Suppose a category admits countable direct sums, and an object S satisfies

$$S \cong A \oplus S.$$

Then in a Grothendieck group one obtains

$$[S] = [A] + [S],$$

so formally

$$[A] = 0.$$

This is the mechanism behind many vanishing arguments: an infinite object absorbs finite additions, and the invariant becomes trivial. In my first interpretation, the meme equation was a compressed version of this phenomenon. The "truth" was supposed to be that infinity makes cancellation too flexible, so the sophisticated K-theoretic machinery collapses into an infinite arithmetic trick.

The actual lecture keeps several words from that interpretation but reverses their role. It still cares about K-theory, addition, cancellation, and stability. But it does not use infinity to erase a finite object. It uses finite configurations to produce a nontrivial loop. The contrast is:

first interpretation	lecture's mechanism
infinite alternating series	finite signed configuration
Cesàro/Abel/eta regularization	creation–swap–annihilation path
Eilenberg swindle	$K(\text{FinSet})$
absorption at infinity	nontrivial loop at the base point
vanishing of invariants	generator of $K_1(\text{FinSet}) \cong \mathbb{Z}/2$
triviality	stable Hopf nontriviality

Thus the first guess was close in vocabulary but opposite in direction. It looked for the triviality of the infinite. Prof. Peng's point is the nontriviality of the finite.

§152.10 Finite sets and stable spheres

The surprising theorem behind the example is the Barratt–Priddy–Quillen theorem. In the form used by the lecture, it says that the K-theory of finite sets is the sphere spectrum:

$$K_i(\text{FinSet}) \cong \pi_i^s(\mathbb{S}).$$

Theorem 152.10.1 (Barratt–Priddy–Quillen, lecture form)
The K-theory of finite sets is the sphere spectrum. Equivalently,

$$K_i(\mathrm{FinSet}) \cong \pi_i^s(\mathbb{S}).$$

In the slide deck this theorem is presented as the conceptual bridge: a finite-set loop can detect a stable homotopy class of spheres.

This theorem explains why the finite-set loop h can be compared to the Hopf class. Under BPQ, the nontrivial element of

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2$$

corresponds to

$$\eta \in \pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2.$$

This is the stable Hopf class.

Another way to say this is that $K(\mathcal{F})$ can be seen as a stable configuration space of framed points:

$$K(\mathcal{F}) \simeq \mathrm{Conf}^{\mathrm{fr}}(\mathbb{R}^\infty).$$

The creation–swap–annihilation loop becomes a framed one-dimensional motion. Its framing carries the $\mathbb{Z}/2$-twist.

§152.11 Relation with the classical Hopf map

The classical Hopf fibration represents a nontrivial element of $\pi_3(S^2)$. Stabilizing the Hopf map gives the stable element

$$\eta \in \pi_1^s(\mathbb{S}).$$

The finite-set loop h gives the same class under the BPQ theorem.

Thus the meme ladder is not merely free association. The chain is:

$$\begin{gathered} \text{Hopf fibration } S^3 \to S^2 \\ \Downarrow \\ \text{stable Hopf element } \eta \in \pi_1^s(\mathbb{S}) \\ \Downarrow \\ K_1(\mathrm{FinSet}) \cong \pi_1^s(\mathbb{S}) \\ \Downarrow \\ \text{the loop } h : \varnothing \sim \{+,-\} \sim \{-,+\} \sim \varnothing. \end{gathered}$$

This is the clean mathematical sense in which the arithmetic-looking line is the Hopf map in disguise.

§152.12 Cobordism: the movie becomes a circle

The slides then place the motion in three-dimensional space. As time evolves, the moving point configuration sweeps out a one-dimensional manifold. The creation of the pair, their swap, and their annihilation form a closed loop. Because the points carry signs or framings, the resulting circle is a framed one-manifold.

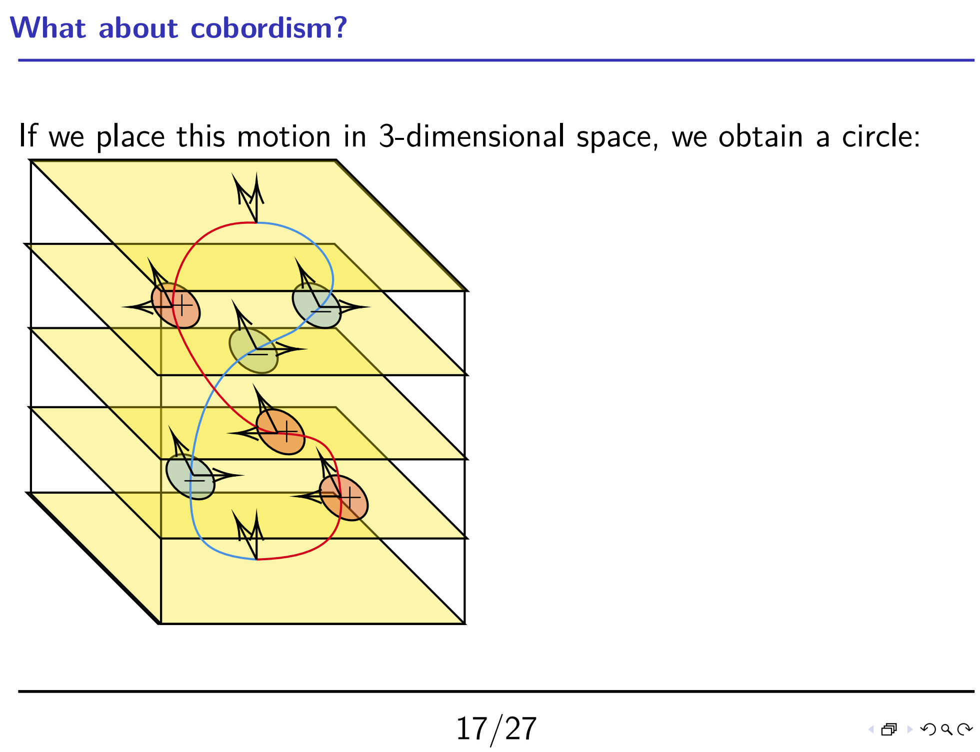


Figure 152.3: From the lecture slides: the creation–swap–annihilation movie becomes a framed circle.

Framed cobordism asks whether this framed circle is the boundary of a framed disk. The answer is no for the loop h. This is another expression of the same nontrivial $\mathbb{Z}/2$-class.

Definition 152.12.1 (Framed cobordism, informal)**.** A framed n-manifold is an n-manifold equipped with a trivialization of its normal bundle. Two framed manifolds are framed cobordant if they form the boundary of a framed manifold one dimension higher.

This is the geometric face of the equation. The path starts and ends at the empty configuration, but its trace is not the boundary of a trivial framed disk. The loop is geometrically nonzero.

§152.13 Algebraic K-theory of a field

After finite sets, the lecture repeats the construction with finite-dimensional vector spaces over a field k. Let

$$\mathcal{V}_k = \mathrm{FinVect}_k/\sim$$

be the moduli space of finite-dimensional vector spaces. Under direct sum, it behaves like a categorified version of addition.

The low K-groups are familiar:

$$K_0(k) \cong \mathbb{Z}, \qquad K_1(k) \cong k^\times.$$

The first says that a vector space is measured by its dimension after group completion. The second says that the determinant detects the abelianized automorphism information:

$$\mathrm{GL}_\infty(k)^{\mathrm{ab}} \cong k^\times.$$

This parallels the finite-set case:

object	K_0	K_1
FinSet	$\mathbb{Z}$	$\mathbb{Z}/2$
$\mathrm{FinVect}_k$	$\mathbb{Z}$	$k^\times$

The lecture now asks for a geometric model of these algebraic K-groups beyond degree one.

§152.14 Intrinsic skein-lasagna

The advanced second half of the slide deck introduces intrinsic skein-lasagna. This is not merely decorative. It is a proposed geometric language for K-groups analogous to how framed cobordism gives a geometric language for stable homotopy.

The first definition is an n-treefold: a space locally homeomorphic to

$$T \times \mathbb{R}^{n-1},$$

where T is a tree. The branching models a controlled kind of singularity.

Given a symmetric monoidal category $(\mathcal{C}, \otimes)$, an n-$(\mathcal{C}, \otimes)$-lasagna is, roughly, a framed n-treefold with labels from $\mathcal{C}$, satisfying compatibility at the branching loci.

Definition 152.14.1 (Intrinsic skein-lasagna, informal)**.** An intrinsic lasagna is a geometric object built from treefold-like pieces, with labels coming from a symmetric monoidal category. The word "intrinsic" means that the object is studied on its own, rather than as something embedded in a previously chosen ambient manifold.

The slide illustrates n-treefolds as spaces locally modelled on $T \times \mathbb{R}^{n-1}$, where T is a tree. For the notebook, the useful point is that branching is allowed but controlled.

The word "intrinsic" is important. Classical skein-lasagna modules often involve embedded objects inside an ambient manifold. Here the lecture emphasizes that one can study the geometry of the lasagna itself, without first choosing an ambient space. If desired, one may imagine the ambient space as $\mathbb{R}^\infty$, but it is no longer part of the primary data.

§152.15 The $(\mathcal{V}_k, \oplus)$-cobordism group

The lecture specializes the lasagna story to

$$(\mathcal{V}_k, \oplus),$$

where $\mathcal{V}_k$ is the category or moduli object of finite-dimensional vector spaces under direct sum. One obtains groups denoted

$$\Omega_n^{\mathcal{V}_k, \oplus}.$$

The examples in low degree match algebraic K-theory:

$$\Omega_0^{\mathcal{V}_k, \oplus} \cong K_0(k) \cong \mathbb{Z},$$

$$\Omega_1^{\mathcal{V}_k, \oplus} \cong K_1(k) \cong k^\times.$$

Remark 152.15.1 — The slide summarizes the first two $(\mathcal{V}_k, \oplus)$-cobordism groups: degree 0 recovers dimension, and degree 1 recovers determinant. In formulas,

$$\Omega_0^{\mathcal{V}_k,\oplus} \cong K_0(k) \cong \mathbb{Z}, \qquad \Omega_1^{\mathcal{V}_k,\oplus} \cong K_1(k) \cong k^\times.$$

This is why vector-space-labelled circles are the K_1 analogue of the finite-set swap loop.

The degree-one case is especially helpful. A circle labelled by an automorphism $f \in \mathrm{GL}(V)$ represents a class, and the determinant gives its value in $k^\times$. A commutator becomes null-cobordant, matching the abelianization of $\mathrm{GL}_\infty(k)$.

This is the lasagna analogue of the finite-set story. In finite sets, a swap gives the nontrivial element of $\mathbb{Z}/2$. In vector spaces, an automorphism gives an element of $k^\times$.

§152.16 Generalized Pontryagin–Thom

The slide deck then states the central claim:

$$K_n(k) \xrightarrow{\cong} \Omega_n^{\mathcal{V}_k,\oplus} \qquad (n \geq 0).$$

This is presented as a generalized Pontryagin–Thom isomorphism.

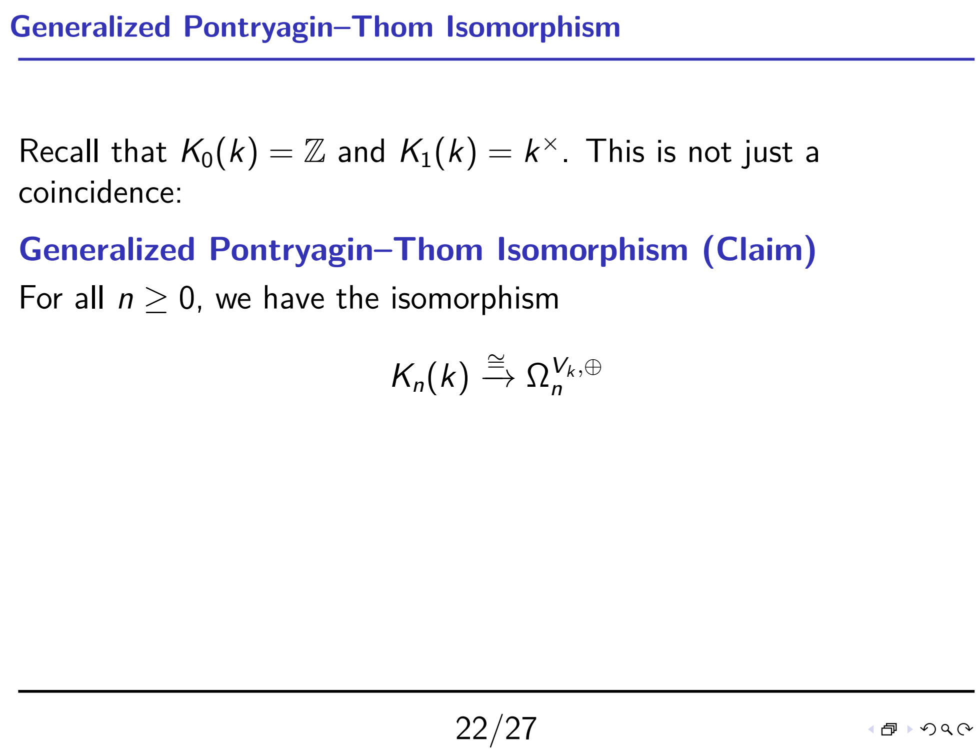


Figure 152.4: From the lecture slides: the generalized Pontryagin–Thom isomorphism identifies $K_n(k)$ with a lasagna-cobordism group.

The analogy is:

classical/stable homotopy	algebraic K-theory side
$\pi_n^s(\mathbb{S})$	$K_n(k)$
Ω_n^{fr}	$\Omega_n^{\mathcal{V}_k,\oplus}$
framed manifolds	vector-space-labelled lasagnas

This is why the lecture moves from elementary arithmetic to Hopf fibration and then to intrinsic skein-lasagna. It is building a geometric model of algebraic K-theory.

§152.17 The K_2 slide and Steinberg relations

In degree two, the lecture recalls the classical presentation

$$K_2(k) \cong k^\times \otimes k^\times / \langle a \otimes (1-a) \rangle.$$

It then considers matrices $f, g \in \mathrm{GL}(k)$ with commutator

$$[f, g] = 1.$$

Such a commuting pair defines a two-dimensional lasagna $C(f, g)$.

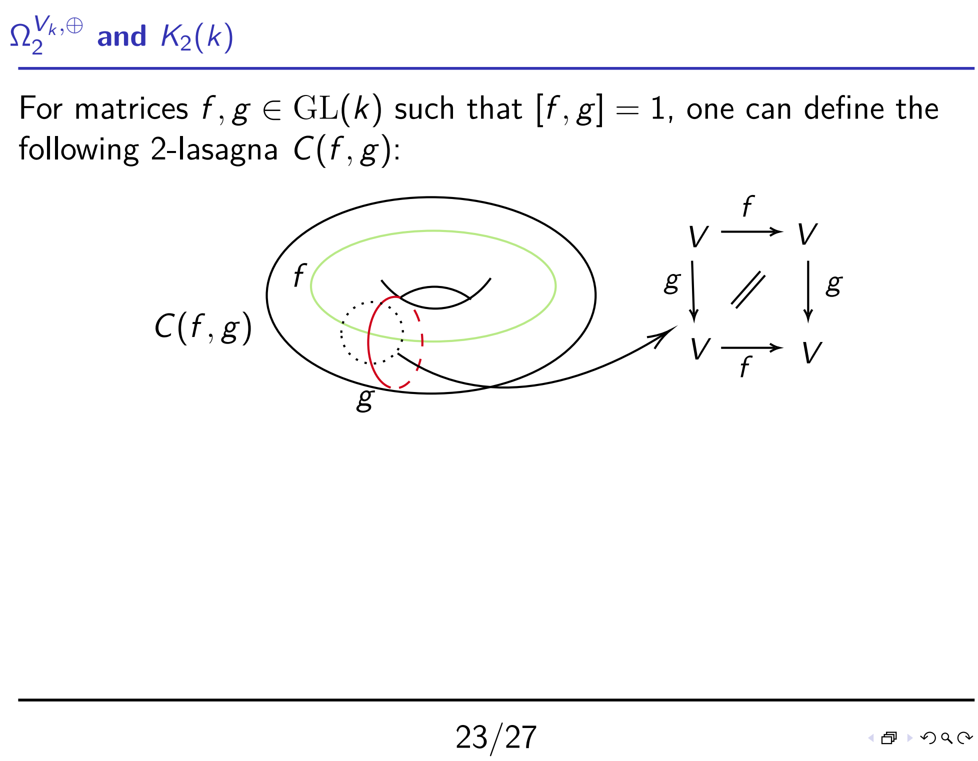


Figure 152.5: From the lecture slides: a two-lasagna $C(f, g)$ associated to a commuting pair.

The next slide gives a more concrete example using elementary matrices. For $i \neq j$, let

$$e_{ij}(a)$$

be the elementary matrix with a in the (i, j)-entry and ones on the diagonal. These matrices satisfy Steinberg relations, such as

$$[e_{ij}(a), e_{kl}(b)] = \begin{cases} e_{il}(ab), & j = k, \\ \mathbf{1}, & \text{otherwise.} \end{cases}$$

The slide then uses Hall–Witt-type commutator identities to produce a nontrivial proof that a certain commutator is the identity. This proof is not just algebraic decoration; it is a blueprint for constructing a bounding three-lasagna.

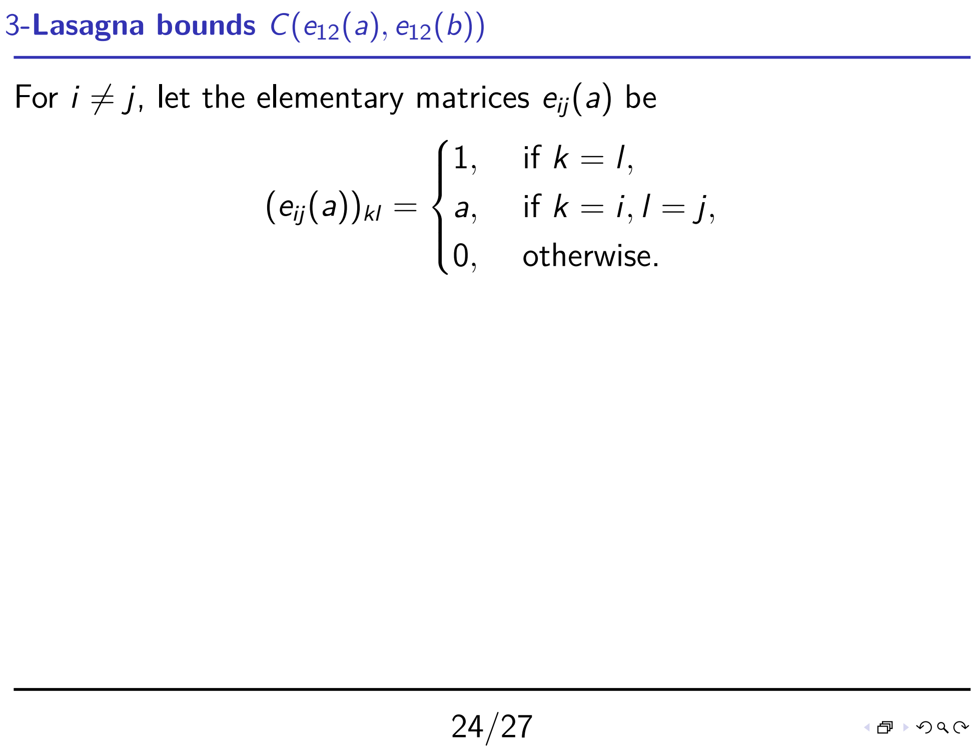


Figure 152.6: From the lecture slides: elementary matrices, Steinberg relations, and the algebraic pattern behind the three-lasagna.

§152.18 The final lasagna diagrams

The final nontrivial slides display a Heegaard splitting and a boundary relation. They are not necessary for understanding the original meme, but they are essential for understanding where the lecture is going. The creation–swap–annihilation loop was the baby example. The lasagna diagrams are the research-level continuation of the same philosophy: algebraic K-theory classes are represented by geometric/cobordism-like objects.

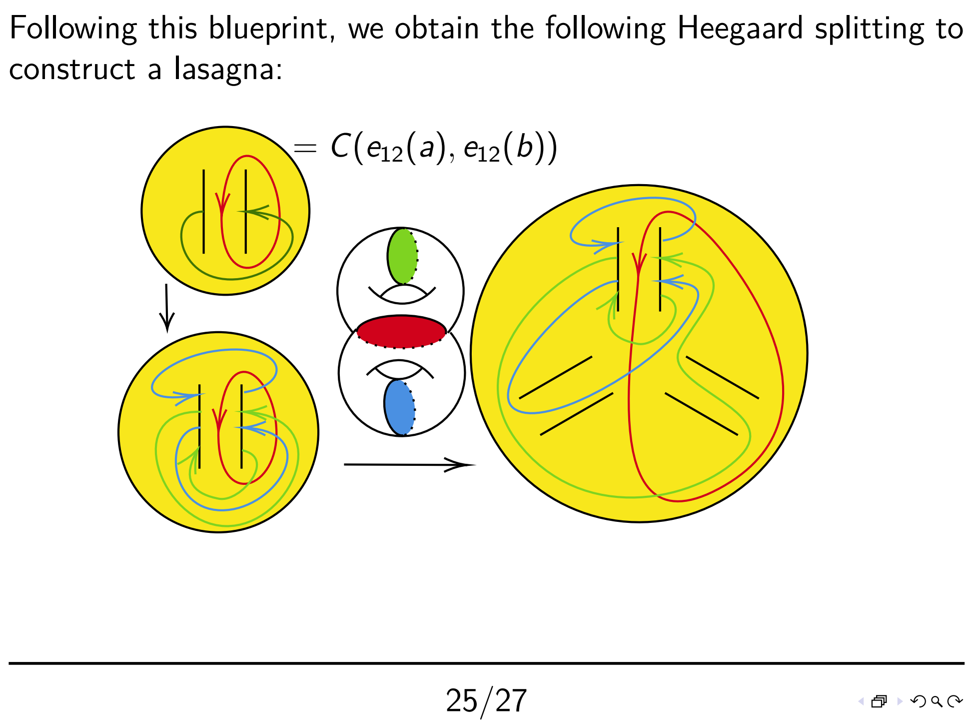


Figure 152.7: From the lecture slides: a Heegaard-splitting picture used to construct a three-lasagna.

At this point the lecture has travelled very far from the original equation. But the path is coherent:

$$\begin{gathered}
\text{equation as path} \\
\Downarrow \\
\text{finite signed configurations} \\
\Downarrow \\
K\text{-theory of finite sets} \\
\Downarrow \\
\text{stable homotopy and framed cobordism} \\
\Downarrow \\
\text{algebraic } K\text{-theory of fields} \\
\Downarrow \\
\text{intrinsic skein-lasagna as a geometric model.}
\end{gathered}$$

§152.19 What the equation finally says

The equation

$$0 = 1 - 1 = -1 + 1 = 0$$

should be read as the name of a loop:

$$h : \varnothing \to \{+, -\} \to \{-, +\} \to \varnothing.$$

It is a loop at zero in the K-space of finite sets. It is not the identity loop. It represents the generator of

$$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2.$$

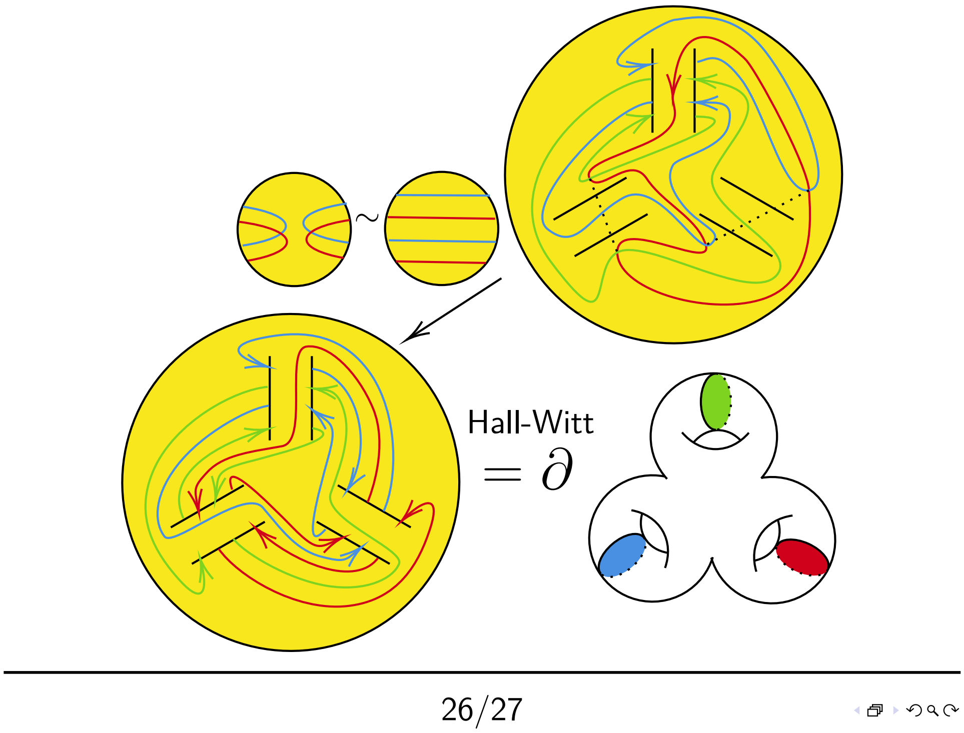


Figure 152.8: From the lecture slides: the boundary relation for the three-lasagna construction.

By Barratt–Priddy–Quillen, this group is

$$\pi_1^s(\mathbb{S}) \cong \mathbb{Z}/2,$$

and h corresponds to the stable Hopf class η.

This is why the Hopf fibration belongs in the same story as a signed finite-set calculation. The shared object is not an ordinary number. It is a stable homotopy class.

§152.20 Structural viewpoint

The two notes should now be read in chronological order rather than as if the first one already knew the second one's answer. The earlier note preserves the first route: a summer memory of the Hopf fibration, a lecture poster, a meme, and the attempt to explain the line through analytic summation, infinite cancellation, and the Eilenberg swindle. This second note records the later correction: the same vocabulary of K-theory and stability is present, but the mathematical emphasis moves from vanishing to a nontrivial finite loop.

The hierarchy is:

first reaction	later understanding
$0 = 0$	a loop from 0 to 0
$1 - 1 + 1 - 1 + \cdots$	one created $+/-$ pair
Cesàro/Abel/eta	finite configuration space
Eilenberg swindle	group completion of finite sets
infinite absorption	creation–swap–annihilation movie
vanishing of invariants	$K_1(\mathrm{FinSet}) \cong \mathbb{Z}/2$
triviality of the infinite	nontriviality of the finite
arithmetic/regularization instinct	stable Hopf class η

This distinction is the real lesson of the reflection. The first idea was not stupid; it was a reasonable search direction because K-theory really does interact with group completion, stability, and swindle arguments. But it placed the weight on the wrong side. It tried to make the equation true by allowing infinitely flexible cancellation. The lecture instead makes the equation interesting by refusing to collapse the finite movie to its endpoint.

The advanced tail of the lecture should also be remembered. The same style of thought continues beyond finite sets. For a field k, algebraic K-groups can be approached through vector-space-labelled lasagnas:

$$K_n(k) \cong \Omega_n^{\mathcal{V}_k,\oplus}.$$

The final slides are not random extra material. They show that the playful equation is the doorway into a geometric model of algebraic K-theory.

So the final lesson is not that 0 is secretly nonzero, and it is not that infinity makes all cancellation meaningless. The lesson is sharper: in higher mathematics, the space of ways for 0 to be 0 can contain a nontrivial loop, and in this example that loop is already visible in the smallest possible finite signed configuration.

XV

Machine Learning

Part XV: Contents

Backpropagation in One MLP Layer: From a Three-Dimensional Tensor to an Outer Product

N-20250926

§153.1 The setting

Consider one middle layer of a multilayer perceptron:

$$z = Wx + b, \qquad y = \sigma(z).$$

Here

$$x \in \mathbb{R}^n, \qquad W \in \mathbb{R}^{m\times n}, \qquad b \in \mathbb{R}^m, \qquad z, y \in \mathbb{R}^m.$$

The activation function σ acts elementwise, for example sigmoid or ReLU. The goal of backpropagation is to compute

$$\frac{\partial L}{\partial W}$$

for a loss function L.

The note uses the common softmax–cross-entropy setting as motivation. If the final layer produces logits z_i, then

$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \qquad L = -\sum_i t_i \ln \hat{y}_i,$$

where t_i is a one-hot label. Backpropagation asks how this scalar loss changes when each weight $W_{j,k}$ changes.

§153.2 The theoretical tensor

Strictly speaking, the derivative of the vector $y \in \mathbb{R}^m$ with respect to the matrix $W \in \mathbb{R}^{m\times n}$ has three indices:

$$\frac{\partial y_i}{\partial W_{j,k}}.$$

The index i chooses the output component, j chooses the row of W, and k chooses the column of W. So the object has shape

$$m \times m \times n.$$

This is the source of the confusion recorded in the note. The derivative is theoretically a third-order tensor, but in neural-network code one usually sees only a matrix gradient. The matrix appears after chain-rule contraction with the upstream gradient.

§153.3 Computing the sparse derivative

Write the linear preactivation component as

$$z_r = \sum_{t=1}^{n} W_{r,t} x_t + b_r.$$

Then

$$\frac{\partial z_r}{\partial W_{j,k}} = \begin{cases} x_k, & r = j, \\ 0, & r \neq j, \end{cases} = \delta_{rj} x_k.$$

This is sparse because z_r depends only on row r of W.

For $y_i = \sigma(z_i)$ with elementwise activation,

$$\frac{\partial y_i}{\partial z_r} = 0 \quad (i \neq r), \qquad \frac{\partial y_i}{\partial z_i} = \sigma'(z_i).$$

Thus

$$\frac{\partial y_i}{\partial W_{j,k}} = \sum_{r=1}^{m} \frac{\partial y_i}{\partial z_r} \frac{\partial z_r}{\partial W_{j,k}} = \frac{\partial y_i}{\partial z_j} x_k.$$

For elementwise activation this is nonzero only when $i = j$.

§153.4 Contraction by the chain rule

The loss L is scalar. Therefore

$$\frac{\partial L}{\partial W_{j,k}} = \sum_{i=1}^{m} \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial W_{j,k}}.$$

Substituting the previous formula gives

$$\frac{\partial L}{\partial W_{j,k}} = \sum_{i=1}^{m} \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial z_j} x_k.$$

The sum over i is exactly the j-th component of the upstream gradient with respect to z:

$$\delta_j = \frac{\partial L}{\partial z_j}.$$

Hence

$$\frac{\partial L}{\partial W_{j,k}} = \delta_j x_k.$$

Stacking all j, k gives the outer product

$$\boxed{\frac{\partial L}{\partial W} = \delta x^T}$$

where

$$\delta = \frac{\partial L}{\partial z}.$$

§153.5 Why the tensor disappears

The original derivative $\partial y/\partial W$ is a tensor with shape (m, m, n). But backpropagation does not need to store it. The chain rule contracts it with the vector $\partial L/\partial y$, leaving a matrix of shape (m, n).

In the note's language:

$$\text{three-dimensional tensor} \xrightarrow{\text{chain-rule summation}} \text{matrix gradient.}$$

This is not a loss of information for gradient descent, because gradient descent only needs the derivative of the scalar loss with respect to each parameter.

§153.6 Contravariant reading of the same calculation

The same computation can be described as a pullback of covectors. A forward layer is a map

$$f : \text{parameters and activations} \longrightarrow \text{outputs.}$$

Its differential sends small forward perturbations to small output perturbations:

$$df : T_xX \to T_{f(x)}Y.$$

But the loss supplies a covector at the output,

$$dL \in T^*_{f(x)}Y,$$

and backpropagation sends it backward by precomposition:

$$f^*(dL) = dL \circ df \in T^*_xX.$$

This is why reverse-mode differentiation is naturally contravariant.

For a batched linear layer in PyTorch convention,

$$X \in \mathbb{R}^{b\times n}, \qquad W \in \mathbb{R}^{m\times n}, \qquad Y = XW^T \in \mathbb{R}^{b\times m},$$

the component formula is

$$Y_{\alpha i} = \sum_{k=1}^{n} X_{\alpha k}W_{ik}.$$

Hence

$$\frac{\partial Y_{\alpha i}}{\partial W_{jk}} = \delta_{ij}X_{\alpha k},$$

a Jacobian tensor of shape (b, m, m, n). Contracting with the upstream gradient $G = \partial L/\partial Y$, of shape (b, m), gives

$$\begin{aligned}\frac{\partial L}{\partial W_{jk}} &= \sum_{\alpha,i} G_{\alpha i}\frac{\partial Y_{\alpha i}}{\partial W_{jk}}\\ &= \sum_{\alpha} G_{\alpha j}X_{\alpha k},\end{aligned}$$

so

$$\frac{\partial L}{\partial W} = G^TX.$$

If the weights are stored instead as $W_{\text{col}} \in \mathbb{R}^{n\times m}$ and $Y = XW_{\text{col}}$, the same formula appears as

$$\frac{\partial L}{\partial W_{\text{col}}} = X^TG.$$

The two expressions differ only by the convention for storing the weight matrix.

§153.7 Outer-product expression

Let
$$\frac{\partial y}{\partial z}$$
be treated as a column vector for elementwise activation, with entries $\sigma'(z_i)$. Then the derivative of y with respect to W can be represented row by row as
$$\frac{\partial y}{\partial W} \sim \begin{bmatrix} \frac{\partial y_1}{\partial z_1} x^T \\ \frac{\partial y_2}{\partial z_2} x^T \\ \vdots \\ \frac{\partial y_m}{\partial z_m} x^T \end{bmatrix} = \left(\frac{\partial y}{\partial z}\right) x^T.$$
For the scalar loss, replace $\partial y/\partial z$ by the backpropagated vector $\delta = \partial L/\partial z$.

§153.8 Numerical example

The note gives a simple example with
$$m = 2, \qquad n = 3, \qquad x = [1, 2, 3]^T, \qquad z = [0.1, -1.0]^T.$$
For sigmoid,
$$\sigma'(0.1) \approx 0.2494, \qquad \sigma'(-1.0) \approx 0.1966.$$
Therefore
$$\frac{\partial y}{\partial W} \sim \begin{bmatrix} 0.2494 \\ 0.1966 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 0.2494 & 0.4988 & 0.7482 \\ 0.1966 & 0.3932 & 0.5898 \end{bmatrix}.$$
Each row is the input vector scaled by the derivative of the corresponding output.

§153.9 Einstein notation proof

The handwritten proof uses indices. Let $y = f(z)$ and $z = Wx$. For a matrix entry $A_{pq} = W_{pq}$,
$$\frac{\partial z_i}{\partial A_{pq}} = \frac{\partial}{\partial A_{pq}} \sum_j A_{ij} x_j = \delta_{ip} x_q.$$
Then
$$\frac{\partial L}{\partial A_{pq}} = \sum_i \frac{\partial L}{\partial y_i} \frac{\partial y_i}{\partial z_p} x_q.$$
For elementwise activation, this becomes
$$\frac{\partial L}{\partial A_{pq}} = \frac{\partial L}{\partial z_p} x_q.$$
This is the component form of $\partial L/\partial W = \delta x^T$.

§153.10 Kronecker product viewpoint

The note also explores vectorization. If $y = f(Wx)$ and we vectorize W, then the Jacobian with respect to $\mathrm{vec}(W)$ can be written using Kronecker products. The important identity is that changing one matrix entry affects only one row of z, which is encoded by a Kronecker delta.

In batch form, if X is a matrix of inputs and Δ is the matrix of upstream gradients, then the gradient becomes

$$\frac{\partial L}{\partial W} = \Delta X^T.$$

This is the same outer-product rule, summed over the batch dimension.

§153.11 Softmax–cross-entropy simplification

For the final layer with softmax and cross entropy, the derivative simplifies to

$$\frac{\partial L}{\partial z} = \hat{y} - t.$$

Therefore the final-layer weight gradient is

$$\frac{\partial L}{\partial W} = (\hat{y} - t)x^T.$$

This is why implementation code often looks much simpler than the theoretical tensor derivation.

§153.12 A wider interpretation

The key lesson is that matrix calculus is compressed tensor calculus. The full derivative of a vector with respect to a matrix is a higher-order object, but backpropagation contracts it immediately with an upstream gradient. The result is an outer product: error signal times input. This is the local rule behind every dense neural-network layer.

§153.13 Backpropagation as cotangent pullback

For a layer $z = Wx + b$, the forward map sends activations forward, while backpropagation sends covectors backward:

$$\bar{x} = W^T \bar{z}.$$

This is the pullback on cotangent vectors.

§153.14 Why the three-dimensional tensor collapses

The derivative of $z_i = \sum_j W_{ij}x_j + b_i$ with respect to W_{ab} is

$$\frac{\partial z_i}{\partial W_{ab}} = \delta_{ia}x_b.$$

Contracting with upstream gradient $\bar{z}_i$ gives

$$\frac{\partial L}{\partial W_{ab}} = \bar{z}_a x_b.$$

Thus the huge derivative tensor disappears after contraction, leaving an outer product.

§153.15 Batch form

For a batch, gradients add over examples:

$$\nabla_W L = \bar{Z} X^T.$$

This is matrix multiplication as accumulated outer products.

§153.16 Backpropagation as sparse tensor contraction

Backpropagation is not magic tensor cancellation. It is cotangent pullback plus contraction of sparse Jacobians, and the familiar weight gradient is the outer product of upstream error with input activation.

Probability Language for Variational Autoencoders

N-20251008

§154.1 Why probability enters generative modeling

A deterministic autoencoder learns a pair of functions

$$x \longmapsto h \longmapsto \widehat{x}.$$

This is useful for compression and representation learning, but it is not yet a probabilistic generative model. A generative model should tell us how likely different data points are and how to sample new ones. Therefore the mathematical object is not merely a map, but a probability distribution.

The VAE begins from the following change of viewpoint:

a data point is not only reconstructed;
it is regarded as a sample from an unknown distribution.

If X denotes the random data point, then the ideal object is the data distribution $p_{\text{data}}(x)$. A model tries to build a tractable family $p_\theta(x)$ and choose θ so that p_θ resembles the data distribution.

The slogan for this chapter is:

generation means sampling from a learned distribution.

Before discussing encoders, decoders, or neural networks, we first need the probability language in which the VAE objective is written.

§154.2 Random variables, distributions, and densities

A random variable X is a measurable quantity whose value is uncertain. Its distribution tells us the probability of events such as $X \in A$. When X has a density $p(x)$ with respect to Lebesgue measure, probabilities are computed by integration:

$$\mathbb{P}(X \in A) = \int_A p(x)\,dx.$$

A density is not itself a probability. The value $p(x)$ may even be larger than 1. What matters is the integral over a region.

The expectation of a function $f(X)$ is

$$\mathbb{E}[f(X)] = \int f(x)p(x)\,dx.$$

For a discrete variable, the corresponding formula is a sum:

$$\mathbb{E}[f(X)] = \sum_x f(x)p(x).$$

The formulas look similar, but the measure matters. In VAEs, latent variables are usually continuous, so integrals over z appear everywhere.

§154.3 Joint, marginal, and conditional distributions

VAEs use latent variables, so we need distributions involving both visible variables and hidden variables. Let X be observed data and Z be a latent variable. Their joint density is

$$p(x, z).$$

The joint density can be factored in two ways:

$$p(x, z) = p(x \mid z)p(z) = p(z \mid x)p(x).$$

The marginal density of X is obtained by integrating out Z:

$$p(x) = \int p(x, z)\, dz.$$

This operation is called marginalization. It means that all possible hidden explanations z are summed or integrated away.

The conditional density is given by Bayes' rule:

$$p(z \mid x) = \frac{p(x, z)}{p(x)} = \frac{p(x \mid z)p(z)}{\int p(x \mid z')p(z')\, dz'}.$$

The three words should be kept distinct:

joint	describe x and z together
marginal	hide one variable by integration
conditional	update one variable after observing the other

§154.4 Bayes' rule as inversion of a generative story

A latent variable model usually starts with a generative story:

$$z \sim p(z), \qquad x \sim p_\theta(x \mid z).$$

This is the easy direction: first choose a hidden cause z, then generate an observation x.

Inference goes in the opposite direction. Given an observation x, we ask which latent values z could have produced it. The posterior is

$$p_\theta(z \mid x) = \frac{p_\theta(x \mid z)p(z)}{p_\theta(x)}.$$

The denominator

$$p_\theta(x) = \int p_\theta(x \mid z)p(z)\, dz$$

is called the evidence or marginal likelihood.

This is the first obstacle behind VAEs. The generative direction is easy to write, but the inverse direction requires an integral that is usually not tractable.

§154.5 Gaussian distributions

The standard VAE uses Gaussian latent variables. The one-dimensional Gaussian density is

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Here μ is the mean and σ^2 is the variance.

The multivariate Gaussian density is

$$\mathcal{N}(z;\mu,\Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(z-\mu)^T\Sigma^{-1}(z-\mu)\right).$$

The matrix Σ is the covariance matrix. It measures not only the variance of each coordinate, but also how coordinates co-vary.

The diagonal case is especially important:

$$\Sigma = \operatorname{diag}(\sigma_1^2,\dots,\sigma_d^2).$$

Then the coordinates are independent under the Gaussian, and the density factorizes:

$$\mathcal{N}(z;\mu,\operatorname{diag}(\sigma^2)) = \prod_{j=1}^{d} \mathcal{N}(z_j;\mu_j,\sigma_j^2).$$

This is why VAE encoders often output a vector of means and a vector of variances rather than a full covariance matrix.

§154.6 Expectation and Monte Carlo approximation

Many VAE formulas contain expectations that cannot be computed exactly. If

$$z \sim q(z),$$

then

$$\mathbb{E}_{z\sim q}[f(z)] = \int f(z)q(z)\,dz.$$

A Monte Carlo approximation uses samples $z^{(1)},\dots,z^{(L)} \sim q$:

$$\mathbb{E}_{z\sim q}[f(z)] \approx \frac{1}{L}\sum_{\ell=1}^{L} f(z^{(\ell)}).$$

This is not a deterministic quadrature rule. It is a random approximation whose error decreases statistically. The key benefit is that the dimension of z can be large: the same formula still makes sense, while grid integration becomes impossible.

§154.7 KL divergence

The Kullback–Leibler divergence compares two distributions q and p:

$$D_{\mathrm{KL}}(q\|p) = \mathbb{E}_q\left[\log\frac{q(Z)}{p(Z)}\right].$$

For continuous densities, this means

$$D_{\mathrm{KL}}(q\|p) = \int q(z)\log\frac{q(z)}{p(z)}\,dz.$$

For discrete distributions, replace the integral by a sum.

The order matters. Usually

$$D_{\mathrm{KL}}(q\|p) \neq D_{\mathrm{KL}}(p\|q).$$

So KL divergence is not a metric distance.

A useful interpretation is coding loss. If data are really drawn from q, but one uses a code optimized for p, then KL measures the expected extra log-loss. In variational inference, we use it to measure how expensive it is to replace the true posterior by an approximate posterior.

The fundamental inequality is

$$D_{\mathrm{KL}}(q\|p) \geq 0,$$

with equality exactly when $q = p$ almost everywhere. This nonnegativity is what turns the ELBO into a lower bound.

§154.8 Jensen's inequality

The other basic ingredient is Jensen's inequality. Since log is concave,

$$\log \mathbb{E}[Y] \geq \mathbb{E}[\log Y]$$

for positive random variables Y.

This inequality is the analytic reason variational lower bounds exist. In the VAE derivation, we will write a difficult likelihood as

$$\log \int q(z) \frac{p(x,z)}{q(z)}\, dz = \log \mathbb{E}_q \left[\frac{p(x,Z)}{q(Z)} \right]$$

and then move the logarithm inside the expectation:

$$\log \mathbb{E}_q \left[\frac{p(x,Z)}{q(Z)} \right] \geq \mathbb{E}_q \left[\log \frac{p(x,Z)}{q(Z)} \right].$$

This is exactly the shape of the ELBO.

§154.9 The next abstraction

The VAE is built from probability, not merely from neural network architecture. The important vocabulary is:

term	role in VAE
$p(z)$	prior over latent variables
$p_\theta(x \mid z)$	decoder likelihood
$p_\theta(x) = \int p_\theta(x \mid z) p(z)\, dz$	marginal likelihood
$p_\theta(z \mid x)$	true posterior
$q_\phi(z \mid x)$	approximate posterior / encoder
$D_{\mathrm{KL}}(q\|p)$	penalty for replacing p by q
$\mathbb{E}_q[f]$	average under the approximate posterior

The next note will use this language to describe a latent variable model. The main tension will be simple: the model is easy to sample from, but hard to invert.

Latent Variable Models: Priors, Decoders, and Intractable Posteriors

N-20251009

§155.1 From autoencoders to latent variable models

A deterministic autoencoder is a pair of maps

$$x \xrightarrow{\text{encoder}} h \xrightarrow{\text{decoder}} \widehat{x}.$$

The code h is a deterministic representation of x. A VAE keeps the encoder–decoder visual shape, but the mathematical object is different. The hidden representation is a random variable Z, and the decoder defines a conditional distribution, not merely a reconstruction function.

The generative story is

$$z \sim p(z), \qquad x \sim p_\theta(x \mid z).$$

Thus a VAE is best understood as a latent variable model equipped with an efficient approximate inference model.

The key distinction is:

deterministic autoencoder	VAE / latent variable model
$h = f_\phi(x)$	$z \sim q_\phi(z \mid x)$
$\widehat{x} = g_\theta(h)$	$x \sim p_\theta(x \mid z)$
reconstruction map	probabilistic generative model

§155.2 The generative story

The model begins with a prior distribution on the latent variable:

$$p(z).$$

In the basic VAE,

$$p(z) = \mathcal{N}(0, I).$$

This prior says that latent variables should live in a simple, continuous, isotropic space.

Given z, the decoder defines the likelihood of an observation:

$$p_\theta(x \mid z).$$

The joint density is then

$$p_\theta(x, z) = p(z)p_\theta(x \mid z).$$

This formula is the mathematical core of the generative model.

The decoder network does not directly output a final sample. It outputs parameters of a distribution. For binary data, one often writes

$$p_\theta(x \mid z) = \text{Bernoulli}(x; \pi_\theta(z)).$$

Here $\pi_\theta(z)$ gives the success probabilities of the Bernoulli coordinates. For real-valued data, one may use

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I).$$

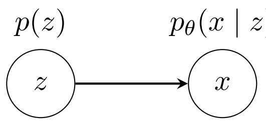
Here the decoder outputs a mean $\mu_\theta(z)$, and the variance is either fixed or parameterized separately.

§155.3 Graphical model notation

The generative model has one directed probabilistic arrow:

$$z \longrightarrow x.$$

A small diagram is:



For a dataset $x^{(1)}, \dots, x^{(N)}$, the story repeats independently:

$$z^{(i)} \sim p(z), \qquad x^{(i)} \sim p_\theta(x \mid z^{(i)}).$$

The parameters θ are shared across all observations.

§155.4 Marginal likelihood

The model likelihood of one observation is obtained by integrating out the latent variable:

$$p_\theta(x) = \int p_\theta(x, z)\, dz = \int p(z) p_\theta(x \mid z)\, dz.$$

For a dataset, maximum likelihood would choose θ by maximizing

$$\sum_{i=1}^{N} \log p_\theta(x^{(i)}).$$

Equivalently, one wants the model distribution $p_\theta(x)$ to assign high probability density to the observed data.

The problem is that the integral

$$\int p(z) p_\theta(x \mid z)\, dz$$

is usually intractable. If $p_\theta(x \mid z)$ is parameterized by a nonlinear neural network, there is no closed-form expression for the integral over z.

This is the first reason variational methods enter.

§155.5 The posterior problem

The posterior distribution of the latent variable is

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)} = \frac{p(z)p_\theta(x \mid z)}{\int p(z')p_\theta(x \mid z')\, dz'}.$$

The numerator is easy to evaluate up to normalization. The denominator is the marginal likelihood, which is the same difficult integral.

Therefore the model has a basic asymmetry:

sampling direction	inference direction
$z \sim p(z),\ x \sim p_\theta(x \mid z)$	$p_\theta(z \mid x)$ is hard

This is the central probabilistic problem behind VAEs:

to train the model, we want $\log p_\theta(x)$,
to understand $\log p_\theta(x)$, we need to reason about z,
but the true posterior $p_\theta(z \mid x)$ is intractable.

§155.6 Why naive Monte Carlo is not enough

One might try to estimate the marginal likelihood by drawing samples from the prior:

$$z^{(\ell)} \sim p(z), \qquad p_\theta(x) \approx \frac{1}{L}\sum_{\ell=1}^{L} p_\theta(x \mid z^{(\ell)}).$$

This is formally reasonable, but in high dimension it is usually inefficient. Most prior samples z do not explain a fixed observation x well. The average is then dominated by rare samples that land in useful regions of latent space.

For inference about a particular x, we would rather sample z values that are already plausible explanations of x. That is the motivation for introducing an approximate posterior.

§155.7 The recognition model

A VAE introduces a family

$$q_\phi(z \mid x)$$

to approximate the true posterior $p_\theta(z \mid x)$. This is often called the recognition model or encoder.

The standard Gaussian encoder has the form

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu_\phi(x), \mathrm{diag}(\sigma_\phi^2(x))).$$

The functions $\mu_\phi(x)$ and $\sigma_\phi(x)$ are outputs of a neural network. Mathematically, they parameterize a distribution over latent variables.

It is important not to confuse the two conditional distributions:

symbol	name	direction
$p_\theta(x \mid z)$	decoder likelihood	$z \to x$
$q_\phi(z \mid x)$	approximate posterior / encoder	$x \to z$

Only $p_\theta(x \mid z)$ belongs to the generative model. The encoder $q_\phi(z \mid x)$ is introduced to make inference and training feasible.

§155.8 Amortized inference

Classical variational inference might introduce separate variational parameters for each data point. For example, for every observation $x^{(i)}$, one could choose its own mean and variance:

$$q_i(z) = \mathcal{N}(z; \mu_i, \mathrm{diag}(\sigma_i^2)).$$

A VAE instead uses one shared inference network:

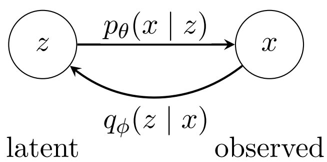
$$x \longmapsto \mu_\phi(x), \qquad x \longmapsto \sigma_\phi(x).$$

This is called amortized inference. The cost of inference is amortized across the dataset: after learning ϕ, the model can quickly produce an approximate posterior for a new x.

The mathematical tradeoff is that the family is restricted. We do not choose arbitrary variational parameters for each observation; we require them to be outputs of the same function.

§155.9 The VAE model diagram

The two directions can be displayed together:



The upper arrow is the probabilistic generative story. The lower arrow is the learned inference shortcut. The VAE objective will force these two directions to be compatible, but they play different roles.

§155.10 Remaining derivation

At this point, the problem is clear. We want to maximize

$$\log p_\theta(x),$$

but

$$p_\theta(x) = \int p(z) p_\theta(x \mid z)\, dz$$

is hard. We also want to use

$$p_\theta(z \mid x),$$

but it contains the same intractable marginal likelihood.

The solution is to introduce $q_\phi(z \mid x)$, then build a lower bound on $\log p_\theta(x)$ that can be optimized. That lower bound is the evidence lower bound, or ELBO.

The next note derives it from two facts:

$$D_{\mathrm{KL}}(q \| p) \geq 0, \qquad \log \mathbb{E}[Y] \geq \mathbb{E}[\log Y].$$

Variational Inference: KL Divergence, Jensen, and the ELBO

N-20251010

§156.1 The approximate posterior as an optimization problem

The true posterior in a latent variable model is

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)}.$$

It is usually intractable because the marginal likelihood

$$p_\theta(x) = \int p_\theta(x, z)\, dz$$

is hard to compute.

Variational inference introduces a tractable family

$$q_\phi(z \mid x)$$

and tries to make it close to the true posterior. A natural measure of closeness is

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

This is the reverse direction from what one might first guess: we place expectation under q_ϕ, because q_ϕ is the distribution from which we can sample and whose density we can evaluate.

The obstacle is that this KL divergence contains $p_\theta(z \mid x)$, which itself contains the intractable $p_\theta(x)$. The central algebraic trick is to rewrite it so that $\log p_\theta(x)$ appears as a constant plus a lower bound.

§156.2 The ELBO identity

Start from the KL divergence:

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)) = \mathbb{E}_{q_\phi(z|x)}\left[\log \frac{q_\phi(z \mid x)}{p_\theta(z \mid x)}\right].$$

Using

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)},$$

we get

$$\begin{aligned} D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)) &= \mathbb{E}_{q_\phi}\left[\log q_\phi(z \mid x) - \log p_\theta(x, z) + \log p_\theta(x)\right] \\ &= \log p_\theta(x) - \mathbb{E}_{q_\phi}\left[\log p_\theta(x, z) - \log q_\phi(z \mid x)\right]. \end{aligned}$$

Define

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x, z) - \log q_\phi(z \mid x)\right].$$

Then the identity becomes

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

Since KL divergence is nonnegative,

$$\mathcal{L}(\theta, \phi; x) \leq \log p_\theta(x).$$

This is why $\mathcal{L}$ is called the evidence lower bound, or ELBO.

> **Remark 156.2.1** — The ELBO is not introduced by magic. It is exactly what remains when the intractable posterior KL is separated from the log marginal likelihood.

§156.3 Jensen's inequality derivation

The same lower bound can be derived directly from Jensen's inequality. Insert any density $q_\phi(z \mid x)$ whose support is compatible with $p_\theta(x, z)$:

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x, z)\, dz \\
&= \log \int q_\phi(z \mid x) \frac{p_\theta(x, z)}{q_\phi(z \mid x)}\, dz \\
&= \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right].
\end{aligned}$$

Because log is concave,

$$\begin{aligned}
\log p_\theta(x) &\geq \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right] \\
&= \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x, z) - \log q_\phi(z \mid x) \right] \\
&= \mathcal{L}(\theta, \phi; x).
\end{aligned}$$

The two derivations emphasize different things. The KL derivation shows the gap between the bound and the true log-likelihood. The Jensen derivation shows why a lower bound exists at all.

§156.4 Reconstruction term and regularization term

Using the latent variable factorization

$$p_\theta(x, z) = p(z) p_\theta(x \mid z),$$

the ELBO becomes

$$\begin{aligned}
\mathcal{L}(\theta, \phi; x) &= \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x \mid z) + \log p(z) - \log q_\phi(z \mid x) \right] \\
&= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).
\end{aligned}$$

This is the familiar VAE decomposition:

$$\boxed{\mathcal{L} = \text{expected log-likelihood} - \text{KL to the prior.}}$$

The first term asks whether latent samples from $q_\phi(z \mid x)$ explain x well under the decoder. The second term asks whether the approximate posterior stays close to the prior.

It is common to call the first term the reconstruction term. But mathematically it is an expected log-likelihood. It becomes a squared-error reconstruction loss only under a Gaussian decoder with fixed variance. It becomes a cross-entropy-like loss under a Bernoulli decoder.

§156.5 The ELBO gap

The identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x))$$

shows that there are two goals mixed together.

Increasing $\mathcal{L}$ can improve the model likelihood $\log p_\theta(x)$. But it also tries to reduce the posterior approximation gap

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

If q_ϕ were flexible enough to equal the true posterior, then the bound would be tight:

$$\mathcal{L}(\theta, \phi; x) = \log p_\theta(x).$$

In practice, q_ϕ is restricted, often to a diagonal Gaussian, so the bound is generally not exact.

§156.6 Closed-form KL for diagonal Gaussians

The standard VAE uses

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu, \mathrm{diag}(\sigma^2)), \qquad p(z) = \mathcal{N}(0, I).$$

For this pair, the KL divergence has a closed form:

$$D_{\mathrm{KL}}(q \| p) = \frac{1}{2} \sum_{j=1}^{d} \left(\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right).$$

Here is the one-dimensional calculation. Let

$$q(z) = \mathcal{N}(\mu, \sigma^2), \qquad p(z) = \mathcal{N}(0, 1).$$

Then

$$\log q(z) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(z-\mu)^2}{2\sigma^2},$$

and

$$\log p(z) = -\frac{1}{2} \log(2\pi) - \frac{z^2}{2}.$$

Therefore

$$\begin{aligned}
D_{\mathrm{KL}}(q \| p) &= \mathbb{E}_q[\log q(z) - \log p(z)] \\
&= -\frac{1}{2} \log \sigma^2 - \frac{1}{2} \mathbb{E}_q \left[\frac{(z-\mu)^2}{\sigma^2} \right] + \frac{1}{2} \mathbb{E}_q[z^2] \\
&= -\frac{1}{2} \log \sigma^2 - \frac{1}{2} + \frac{1}{2}(\mu^2 + \sigma^2) \\
&= \frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2).
\end{aligned}$$

The multivariate diagonal formula follows by summing over independent coordinates.

This is one reason the diagonal Gaussian encoder is practical: the regularization term does not need Monte Carlo estimation.

§156.7 Negative ELBO as a loss

Training usually minimizes the negative ELBO:

$$-\mathcal{L}(\theta, \phi; x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

For a Bernoulli decoder with probabilities $\pi_\theta(z)$,

$$p_\theta(x \mid z) = \prod_i \pi_i(z)^{x_i}(1 - \pi_i(z))^{1-x_i},$$

so

$$-\log p_\theta(x \mid z) = -\sum_i \left[x_i \log \pi_i(z) + (1 - x_i)\log(1 - \pi_i(z))\right].$$

For a Gaussian decoder

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I),$$

we have

$$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2}\|x - \mu_\theta(z)\|^2 + \text{constant}.$$

Thus the common reconstruction losses are not arbitrary engineering choices. They are negative log-likelihoods under different decoder distributions.

§156.8 Empirical lower bounds

The original AEVB experiments plotted the variational lower bound during training. The figure below is useful as a historical reminder: what is being optimized is not raw reconstruction error, but a lower bound $\mathcal{L}$ on marginal likelihood.

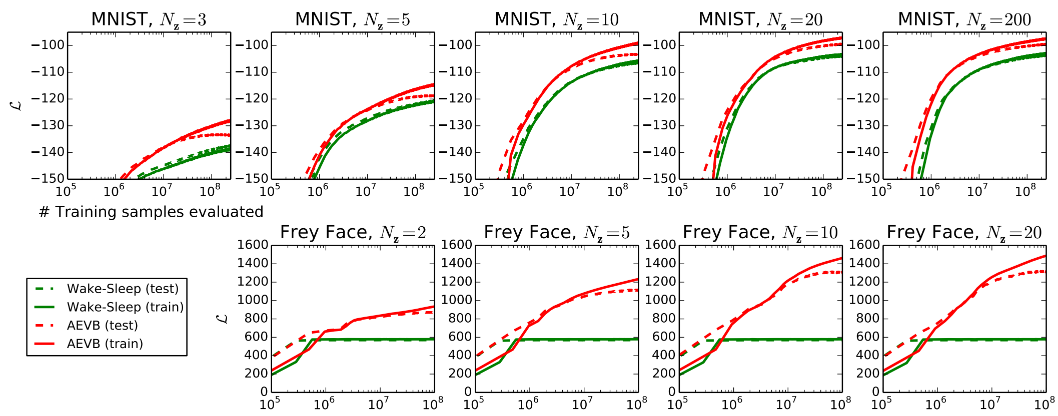


Figure 156.1: Variational lower bound curves in the AEVB experiments.

For the purpose of these notes, the important point is mathematical rather than empirical: the plotted quantity is meaningful because of the identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

§156.9 Information-theoretic reading

The ELBO also resembles a rate–distortion tradeoff. The expected log-likelihood asks the latent variable to preserve enough information to explain x. The KL term penalizes deviations of $q_\phi(z \mid x)$ from the prior $p(z)$.

Loosely:

term	role
$-\mathbb{E}_q[\log p_\theta(x \mid z)]$	distortion / reconstruction cost
$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$	rate / information cost

This interpretation is helpful, but it should not replace the probabilistic derivation. The ELBO is first of all a lower bound on marginal likelihood.

§156.10 Limits of the ELBO

The ELBO is powerful, but it is not magic. Several limitations remain.

First, the variational family may be too small. A diagonal Gaussian $q_\phi(z \mid x)$ cannot represent a complicated multimodal posterior.

Second, the prior may be too simple. The standard normal prior is convenient, but the true aggregated latent structure of data may not be standard Gaussian.

Third, the decoder may be too powerful. If the decoder can model the data without using z, the approximate posterior may collapse toward the prior. This is posterior collapse.

Fourth, maximizing a lower bound is not the same as exactly maximizing $\log p_\theta(x)$. The gap is the posterior KL.

§156.11 What survives later

The VAE objective is not merely a heuristic sum of two losses. It is a variational identity:

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

The ELBO can be written as

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

The first form explains why it is a lower bound. The second form explains why it looks like reconstruction plus regularization.

The remaining question is computational: how can we differentiate an expectation with respect to parameters that also control the sampling distribution? That is the role of the reparameterization trick.

The Reparameterization Trick and Auto-Encoding Variational Bayes

N-20251011

§157.1 The gradient problem

The ELBO for one data point is

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

For the diagonal Gaussian encoder and standard Gaussian prior, the KL term has a closed form. The difficult part is the expectation

$$\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)].$$

We need gradients with respect to both θ and ϕ:

$$\nabla_{\theta,\phi} \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x \mid z)].$$

The subtlety is that ϕ controls the distribution from which z is sampled. The sample z is random, but its randomness depends on the encoder parameters.

The reparameterization trick is the mathematical device that turns this stochastic node into a differentiable transformation of parameter-free noise.

§157.2 Score-function estimator and pathwise estimator

There is a general identity for differentiating expectations:

$$\nabla_\phi \mathbb{E}_{q_\phi(z)}[f(z)] = \mathbb{E}_{q_\phi(z)}[f(z) \nabla_\phi \log q_\phi(z)].$$

This is often called a score-function estimator. It is very general because it does not require z to be a differentiable function of ϕ. But it often has high variance.

The reparameterization trick uses a more geometric idea. Instead of sampling

$$z \sim q_\phi(z \mid x),$$

write

$$z = g_\phi(\epsilon, x), \qquad \epsilon \sim p(\epsilon),$$

where the noise distribution $p(\epsilon)$ does not depend on ϕ.

Then

$$\mathbb{E}_{z \sim q_\phi(z|x)}[f(z)] = \mathbb{E}_{\epsilon \sim p(\epsilon)}[f(g_\phi(\epsilon, x))].$$

Now the parameters appear inside an ordinary differentiable function. Gradients can pass through g_ϕ.

§157.3 Gaussian reparameterization

For the standard VAE encoder,

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu_\phi(x), \operatorname{diag}(\sigma^2_\phi(x))).$$

A sample from this distribution can be written as

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

Here $\odot$ denotes coordinatewise multiplication.

Thus

$$\begin{aligned} &\mathbb{E}_{z\sim q_\phi(z|x)}[\log p_\theta(x \mid z)] \\ &\quad = \mathbb{E}_{\epsilon\sim\mathcal{N}(0,I)} \left[\log p_\theta(x \mid \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)\right]. \end{aligned}$$

The distribution of ϵ is fixed. The dependence on ϕ is now inside μ_ϕ and σ_ϕ, so ordinary backpropagation can be used.

The trick is not a heuristic. It is a change of variables in the sampling procedure.

§157.4 The stochastic ELBO estimator

With L samples $\epsilon^{(1)}, \dots, \epsilon^{(L)} \sim \mathcal{N}(0, I)$, define

$$z^{(\ell)} = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon^{(\ell)}.$$

Then the Monte Carlo estimator of the ELBO is

$$\widehat{\mathcal{L}}(\theta, \phi; x) = \frac{1}{L}\sum_{\ell=1}^{L} \log p_\theta(x \mid z^{(\ell)}) - D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

For minibatches, the dataset objective is approximated by averaging this quantity over the examples in the minibatch.

The original AEVB method is therefore a stochastic-gradient method for maximizing a variational lower bound. The randomness is not removed; it is organized so that differentiation is possible.

§157.5 Why implementations output log variance

Mathematically, the encoder needs $\sigma_\phi(x) > 0$. In practice, one often outputs

$$\log \sigma^2_\phi(x)$$

instead of $\sigma_\phi(x)$ directly. If

$$\ell_\phi(x) = \log \sigma^2_\phi(x),$$

then

$$\sigma_\phi(x) = \exp\left(\frac{1}{2}\ell_\phi(x)\right).$$

This automatically keeps the standard deviation positive.

The Gaussian KL then becomes

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\sum_{j=1}^{d} \left(\mu_j^2 + \exp(\ell_j) - 1 - \ell_j\right).$$

This is the same formula as before, written in terms of log variance.

§157.6 Training as ELBO maximization

The mathematical training loop is:

$$\begin{array}{ll} \text{encoder:} & x \mapsto \mu_\phi(x),\ \log \sigma_\phi^2(x), \\ \text{noise:} & \epsilon \sim \mathcal{N}(0, I), \\ \text{latent sample:} & z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \\ \text{decoder:} & z \mapsto \text{parameters of } p_\theta(x \mid z), \\ \text{objective:} & \text{maximize } \widehat{\mathcal{L}}(\theta, \phi; x). \end{array}$$

If one minimizes a loss instead, the loss is the negative ELBO:

$$-\widehat{\mathcal{L}} = -\frac{1}{L} \sum_{\ell=1}^{L} \log p_\theta(x \mid z^{(\ell)}) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

The first term depends on the decoder likelihood. The second term is analytic for the standard Gaussian encoder.

§157.7 Decoder likelihoods and reconstruction losses

For binary data, a Bernoulli decoder gives

$$p_\theta(x \mid z) = \prod_i \pi_i(z)^{x_i} (1 - \pi_i(z))^{1 - x_i}.$$

The negative log-likelihood is

$$-\log p_\theta(x \mid z) = -\sum_i \left[x_i \log \pi_i(z) + (1 - x_i) \log(1 - \pi_i(z)) \right].$$

This is binary cross entropy.

For real-valued data, a Gaussian decoder with fixed variance gives

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I).$$

Then

$$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2} \|x - \mu_\theta(z)\|^2 + \text{constant}.$$

This is squared error up to scaling and constants.

Thus the word "reconstruction loss" hides a probabilistic choice. Different likelihood models produce different reconstruction terms.

§157.8 Generation after training

After learning θ, generation follows the model direction, not the inference direction:

$$z \sim p(z), \qquad x \sim p_\theta(x \mid z).$$

If the decoder likelihood is Gaussian, one may sample x from that Gaussian or use its mean as a representative output. If the decoder likelihood is Bernoulli, one samples binary coordinates or uses probabilities for visualization.

Latent interpolation uses points such as

$$z_t = (1 - t) z_0 + t z_1, \qquad 0 \le t \le 1.$$

This often produces visually smooth transitions, but it is not a theorem that every linear path in latent space is semantically meaningful. It works when the learned latent geometry is reasonably aligned with the prior and decoder.

§157.9 Empirical marginal likelihood curves

The original experiments compared AEVB with older methods by plotting marginal log-likelihood estimates. The figure is included not as the main point of the theory, but as a reminder that the ELBO machinery was designed to make likelihood-based generative modeling trainable.

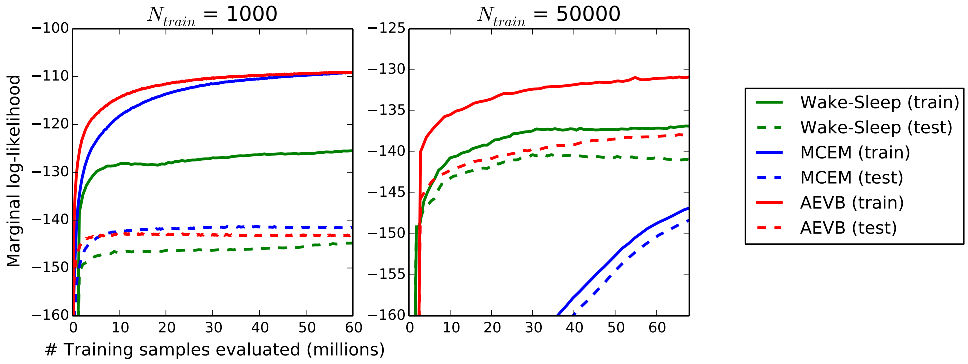


Figure 157.1: Marginal log-likelihood curves in the AEVB experiments.

The mathematical object behind these curves is still the same:

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

Training improves a tractable stochastic estimate of $\mathcal{L}$, and the hope is that this also improves the marginal likelihood.

§157.10 Limitations and extensions

Several limitations are already visible from the mathematics.

First, the posterior family may be too simple. A diagonal Gaussian cannot capture all posterior shapes. Normalizing flows enrich $q_\phi(z \mid x)$ by transforming a simple base distribution through invertible maps.

Second, the prior may be too simple. The standard normal prior is convenient, but the aggregated posterior may have more complicated geometry.

Third, posterior collapse may occur. If the decoder is too expressive, the model may learn

$$q_\phi(z \mid x) \approx p(z),$$

so that z carries little information about x. Then the KL term is small, but the latent representation is weak.

Fourth, the usual ELBO uses one particular weighting between reconstruction and KL. Variants such as β-VAE modify the objective:

$$\mathbb{E}_q[\log p_\theta(x \mid z)] - \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).$$

This changes the rate–distortion tradeoff and can encourage more disentangled latent coordinates, though not automatically.

§157.11 How the pieces fit

The VAE becomes trainable because the stochastic latent sample is rewritten as

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

This makes the expectation differentiable with respect to the encoder parameters:

$$\mathbb{E}_{z \sim q_\phi(z|x)}[f(z)] = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)}[f(\mu_\phi(x) + \sigma_\phi(x) \odot \epsilon)].$$

The full mathematical chain is now visible:

$$\begin{gathered} \text{latent variable model } p(z)p_\theta(x \mid z) \\ \Downarrow \\ \text{intractable posterior } p_\theta(z \mid x) \\ \Downarrow \\ \text{variational approximation } q_\phi(z \mid x) \\ \Downarrow \\ \text{ELBO lower bound} \\ \Downarrow \\ \text{reparameterized stochastic gradients.} \end{gathered}$$

This is why the VAE is more than an autoencoder with noise. It is a method for turning approximate Bayesian inference in a latent variable model into differentiable optimization.

§158.1 Backpropagation as cotangent pullback

The previous backpropagation calculation can be read in a categorical way without making it mystical. A forward neural network is a composite of maps between parameter spaces, activation spaces, and finally the one-dimensional loss space. If

$$L : Y \to \mathbb{R}$$

is the loss, then its differential at the output is a cotangent vector

$$dL_y \in T_y^* Y.$$

Backpropagation pulls this covector backward along the forward computational graph. In this sense the derivative operation used in training is contravariant:

$$f : X \to Y \qquad \Longrightarrow \qquad f^* : T^* Y \to T^* X.$$

Forward maps push tangent perturbations forward; reverse-mode automatic differentiation pulls covectors backward.

For a concrete PyTorch-shaped layer, take

$$X \in \mathbb{R}^{b \times d}, \qquad W \in \mathbb{R}^{n \times d}, \qquad Y = XW^T \in \mathbb{R}^{b \times n}.$$

In indices,

$$Y_{\alpha i} = \sum_{k=1}^{d} X_{\alpha k} W_{ik}.$$

The full Jacobian with respect to the weight matrix has entries

$$\frac{\partial Y_{\alpha i}}{\partial W_{jk}} = \delta_{ij} X_{\alpha k},$$

so its tensor shape is

$$(b, n, n, d).$$

This is the honest tensor object. Neural-network code does not materialize it because the upstream gradient

$$G = \frac{\partial L}{\partial Y} \in \mathbb{R}^{b \times n}$$

immediately contracts it:

$$\begin{aligned} \frac{\partial L}{\partial W_{jk}} &= \sum_{\alpha=1}^{b} \sum_{i=1}^{n} \frac{\partial L}{\partial Y_{\alpha i}} \frac{\partial Y_{\alpha i}}{\partial W_{jk}} \\ &= \sum_{\alpha=1}^{b} G_{\alpha j} X_{\alpha k}. \end{aligned}$$

Therefore

$$\boxed{\frac{\partial L}{\partial W} = G^T X.}$$

If one stores weights in the column convention $W_{\text{col}} \in \mathbb{R}^{d\times n}$ and writes $Y = XW_{\text{col}}$, the same contraction is

$$\frac{\partial L}{\partial W_{\text{col}}} = X^T G.$$

The categorical phrase "pull back the covector" is exactly this matrix multiplication: the high-order Jacobian collapses because it is being paired with a covector from the scalar loss.

§158.2 Geometric data analysis

The original note is an ambitious AI-assisted essay about the role of pure mathematics in machine learning. Its most useful part is not the rhetoric, but the catalogue of mathematical languages that repeatedly appear when data are no longer well described by coordinates alone.

A point cloud is often treated as a finite sample from a hidden space. Topological data analysis makes this idea precise by replacing the cloud with a filtered simplicial complex. For a scale parameter ε, one forms for instance the Vietoris–Rips complex

$$VR_\varepsilon(X) = \{\sigma \subseteq X :\ d(x,y) \le \varepsilon \text{ for all } x, y \in \sigma\}.$$

As ε grows, connected components merge, loops appear and disappear, and higher-dimensional holes may be born and die. Persistent homology records this evolution. The chain complex viewpoint is governed by the boundary identity

$$\partial_{k-1} \circ \partial_k = 0,$$

and homology is

$$H_k = \ker \partial_k / \operatorname{im} \partial_{k+1}.$$

This quotient is the mathematical form of the phrase "holes that are cycles but not boundaries."

§158.3 Manifold hypothesis and embeddings

The manifold hypothesis says that high-dimensional data often concentrate near a lower-dimensional manifold $M \subseteq \mathbb{R}^N$. If M is a smooth d-manifold, Whitney's embedding theorem says that M can be embedded in $\mathbb{R}^{2d}$, and generically in a slightly larger ambient space. In machine learning this gives a conceptual reason why dimensionality reduction is not only compression: it tries to recover the intrinsic coordinates of M.

The practical warning is that an embedding preserves the manifold as a smooth object, but not necessarily distances, probability densities, or task-relevant semantics. PCA preserves a linear subspace model; manifold learning tries to preserve local neighborhoods; representation learning tries to learn coordinates useful for a loss function.

§158.4 Optimal transport and stochastic dynamics

For probability measures μ, ν on a metric space, the p-Wasserstein distance is

$$W_p(\mu,\nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int d(x,y)^p \, d\pi(x,y) \right)^{1/p}.$$

Here $\Pi(\mu,\nu)$ is the set of couplings with marginals μ and ν. The dual viewpoint is equally important. For $p = 1$,

$$W_1(\mu,\nu) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left(\int f \, d\mu - \int f \, d\nu \right).$$

For $p = 2$ there is a useful one-dimensional closed form. Let

$$\mu = \mathcal{N}(a, \sigma_1^2), \qquad \nu = \mathcal{N}(b, \sigma_2^2), \qquad \sigma_1, \sigma_2 > 0.$$

In one dimension the optimal map between these two Gaussians is affine and monotone, so write

$$T(x) = \alpha x + \beta.$$

The condition $T_{\#}\mu = \nu$ forces

$$\alpha = \frac{\sigma_2}{\sigma_1}, \qquad \beta = b - \alpha a.$$

If $X = a + \sigma_1 Z$ with $Z \sim \mathcal{N}(0,1)$, then

$$T(X) = b + \sigma_2 Z.$$

Thus

$$\begin{aligned} W_2^2(\mu,\nu) &= \mathbb{E}\big[(X - T(X))^2\big] \\ &= \mathbb{E}\big[((a-b) + (\sigma_1 - \sigma_2)Z)^2\big] \\ &= (a-b)^2 + (\sigma_1 - \sigma_2)^2. \end{aligned}$$

This formula is small, but it explains a lot of geometric language around generative models: matching Gaussians means matching both their centers and their spreads.

Diffusion models can be read through stochastic differential equations

$$dX_t = b(t, X_t)\, dt + \sigma(t)\, dW_t.$$

The density p_t evolves according to a Fokker–Planck equation, and the reverse-time dynamics involve the score $\nabla_x \log p_t(x)$. Thus score-based generative modeling is an attempt to learn the vector field that reverses a noisy transport of probability mass.

For DDPMs the training objective is usually written as a variational lower bound, or in its simplified form as a denoising mean-square loss. It is not literally a Wasserstein objective. Still, each reverse step compares Gaussian conditional distributions, and for Gaussians the geometry above says that a squared displacement of means and a mismatch of variances are exactly the ingredients of W_2^2. The ELBO uses KL divergences between Gaussian conditionals; when the variance is fixed or separately prescribed, minimizing that KL has the same practical center-matching behavior as minimizing the corresponding Gaussian W_2 term. My preferred reading is therefore: optimal transport gives the geometric picture of local probability movement, while the DDPM ELBO gives the statistically convenient training objective.

§158.5 Symmetry, equivariance, and category-like thinking

If a group G acts on an input space and an output space, a map F is equivariant if

$$F(gx) = gF(x).$$

Convolutional neural networks exploit translation equivariance; graph neural networks exploit permutation equivariance; gauge-equivariant networks try to respect local choices of frame on a bundle.

Here is the matrix calculation behind the slogan. Let the graph be the cycle C_4, with adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix},$$

and let P be the permutation matrix rotating the four vertices by $90°$. Since P is a graph automorphism,

$$PA = AP.$$

The same is true for the usual normalized adjacency $\widehat{A}$, because normalization is built from A and the degree matrix, both invariant under the same rotation:

$$P\widehat{A} = \widehat{A}P.$$

A standard graph neural-network layer is

$$H^{(l+1)} = \sigma(\widehat{A}H^{(l)}W),$$

where σ acts entrywise. If the input features are rotated, then

$$\begin{aligned} F(PH^{(l)}) &= \sigma(\widehat{A}PH^{(l)}W) \\ &= \sigma(P\widehat{A}H^{(l)}W) \\ &= P\sigma(\widehat{A}H^{(l)}W) \\ &= PH^{(l+1)}. \end{aligned}$$

Equivariance is therefore not only a philosophical commitment to symmetry. In this example it is the algebraic fact that the permutation matrix commutes with the graph operator.

The original essay also speculates about topos-theoretic or categorical semantics for AI. A cautious reconstruction is this: category theory is useful when one needs to track structure-preserving maps, functorial changes of representation, and compositional semantics. It does not by itself explain intelligence. Its honest role is to provide a disciplined language for invariance, abstraction, and transport of structure.

§158.6 Further direction

The note is best read as a map of mathematical languages for data: topology sees holes, geometry sees coordinates and curvature, measure theory sees distributions, dynamics sees evolution, algebra sees symmetry, and category theory sees structure-preserving translation. The useful question is not which language is fashionable, but which structure is actually stable under the transformations used by the model.

§158.7 Backpropagation and differential geometry

Backpropagation is reverse-mode differentiation. Geometrically, forward computation pushes tangent vectors forward, while backpropagation pulls cotangent vectors backward through the computation graph.

§158.8 Manifold hypothesis and representation learning

The manifold hypothesis says high-dimensional data may concentrate near lower-dimensional geometric structures. Embeddings, charts, curvature, and local neighborhoods then become relevant to learning representations.

§158.9 Symmetry and equivariance

If a task has a symmetry group G, an equivariant model satisfies

$$f(gx) = gf(x).$$

This reduces sample complexity by building the correct transformation law into the model.

§158.10 Optimal transport and distributions

Optimal transport compares probability measures by the cost of moving mass. It connects geometry, PDE, and machine learning through distances between distributions.

§158.11 Mathematical structures for learning systems

Pure mathematics contributes language for structure: cotangent pullback for gradients, manifolds for data geometry, equivariance for symmetry, optimal transport for distributions, and category-like compositionality for architectures.

§158.12 Rebuilt reading path: what pure mathematics contributes

There are several layers.

(i) **Geometry** studies the shape of the representation space: distances, curvature, geodesics, embeddings, and coordinate charts.

(ii) **Topology** studies features stable under continuous deformation: connected components, loops, voids, and persistence across scales.

(iii) **Measure theory** studies data as distributions rather than isolated points.

(iv) **Dynamical systems** study learning or generation as evolution of states or densities.

(v) **Algebra and category theory** study invariance, symmetry, and composition of structure-preserving maps.

Pure mathematics does not magically explain AI. It supplies languages in which different kinds of stability can be stated precisely.

§158.13 Topological data analysis in usable form

Given a finite metric point cloud X, persistent homology constructs a family of complexes K_ε indexed by a scale parameter. For the Vietoris–Rips complex,

$$\{x_0, \dots, x_k\} \in VR_\varepsilon(X) \iff d(x_i, x_j) \le \varepsilon \text{ for all } i, j.$$

The homology group is

$$H_k(K_\varepsilon) = \ker \partial_k / \operatorname{im} \partial_{k+1}.$$

This quotient is the mathematical form of the phrase "holes that are cycles but not boundaries."

§158.14 Wasserstein geometry and stochastic dynamics

For probability measures μ, ν on a metric space,

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int d(x, y)^p \, d\pi(x, y) \right)^{1/p}.$$

The coupling π is the hidden plan. In generative modeling, one often tries to construct a transport from a simple distribution to a complicated one. Normalizing flows do this by invertible maps, diffusion models by stochastic dynamics, and optimal transport by variational minimization.

A diffusion process has the schematic form

$$dX_t = b(t, X_t) \, dt + \sigma(t) \, dW_t.$$

The reverse process depends on the score field $\nabla_x \log p_t(x)$. Training a score network is therefore an attempt to approximate a time-dependent vector field on probability density.

§158.15 Category language without exaggeration

A category consists of objects and morphisms, with associative composition and identity morphisms. A functor preserves this structure. Category theory is useful when the same construction can be performed across many spaces in a way compatible with maps. For example, homology is functorial:

$$f : X \to Y \implies H_k(f) : H_k(X) \to H_k(Y).$$

The honest categorical contribution is functoriality and compositionality, not a mystical explanation of intelligence.